

Strategic Environment – Joint Planning Process

AFSNCOA Joint Planning Process Primer

01 May 2024

Joint Planning Process Overview

The purpose of **Joint Planning**, as articulated in JP 5-0, is a deliberate process aimed at determining how to implement strategic guidance and/or commander's intent. It is fundamental in guiding the DoD in planning joint campaigns and operations, it is a deliberate process that translates strategic guidance into executable plans and operations. Joint planning is crucial in addressing the multifaceted crises and contingencies that a Joint Force faces, which cut across multiple combatant commands, domains, and functions. The concept of global integration is central to this process, addressing transregional, all-domain, and multifunctional challenges. This strategic approach ensures the Joint Force is organized, employed, and integrated effectively to achieve national objectives within an acceptable level of risk, making joint planning a cornerstone in the foundation of military strategy and operations.

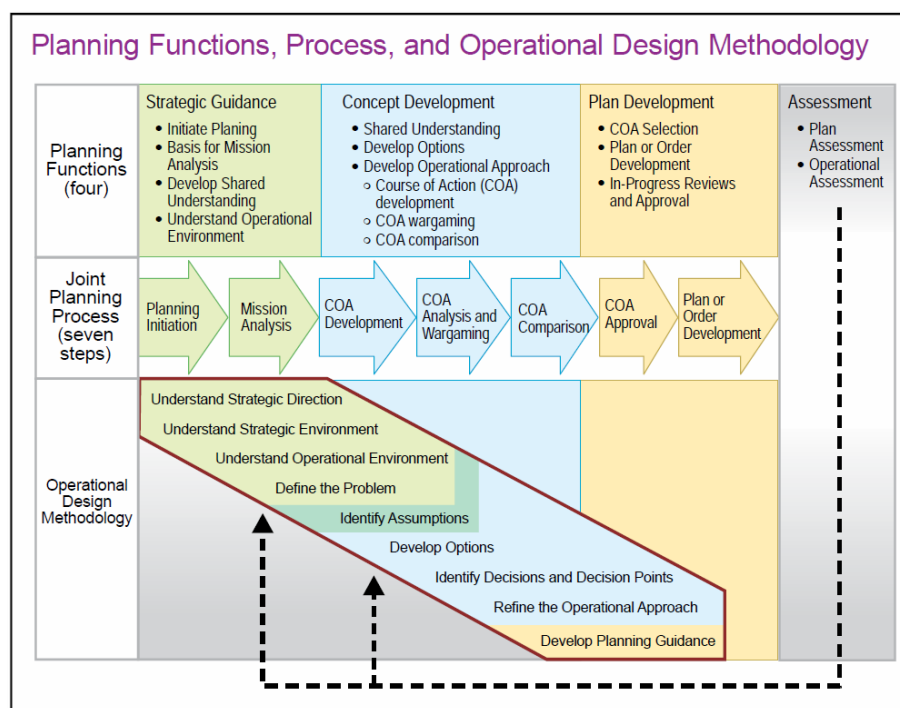


Figure 1 - Planning Functions, Process, and Operational Design Methodology, From: JP 5-0, Figure III-1.

Joint Planning Process Steps

Planning Initiation: This step marks the beginning of the planning process, usually triggered by a perceived need for military action. It involves identifying the problem, receiving the mission, and setting the initial planning guidance. It sets the stage for detailed planning and identifies the resources, constraints, and objectives.

(1) Joint planning begins when an appropriate authority recognizes potential for military capability to be employed in support of national objectives or in response to a potential or actual crisis.

JP 5-0, III-11

Step 1	Planning Initiation
Step 2	Mission Analysis
Step 3	Course of Action (COA) Development
Step 4	COA Analysis and Wargaming
Step 5	COA Comparison
Step 6	COA Approval
Step 7	Plan or Order Development

Figure 2 - JPP Steps, From: JP 5-0, Figure III-4

Integrated Deterrence and Strategic Competition – Joint Planning Process

Mission Analysis: In this phase, planners conduct a detailed examination of the assigned mission to understand the commander's intent, the operational environment, threats, and available resources. This analysis helps in defining the operational problem, identifying limitations, and determining the desired end state. It's crucial for establishing a clear understanding of what needs to be achieved.

When the commander receives a mission tasking, analysis begins with the following questions:

- (a) What is the purpose of the mission? (What problem is the commander being asked to solve or what change to the OE is desired?)*
- (b) What tasks will accomplish the mission?*
- (c) Will the mission achieve the desired objectives?*
- (d) What limitations have been placed on my own forces' actions?*
- (e) What forces are needed to support my operation?*
- (f) How will I know when the mission is accomplished successfully?*
- (g) How will friendly, neutral, and threat networks affect the accomplishment of the mission?*

JP 5-0, III-12 to III-13

Course of Action (COA) Development:

During this step, multiple viable strategies or actions to accomplish the defined mission are developed. Each course of action outlines how forces will be employed, the sequence of actions, and how objectives will be achieved. This creative phase involves brainstorming and leveraging insights from the mission analysis.

(a) A COA is a potential way (solution, method) to accomplish the assigned mission. Staffs develop multiple COAs to provide commanders with options to attain the military end state. A good COA accomplishes the mission within the commander's guidance, provides flexibility to meet unforeseen events during execution, and positions the joint force for future operations. It also gives components the maximum latitude for initiative. All COAs must be suitable, feasible, acceptable, distinguishable, and complete.

JP 5-0, III-32

Testing COA Validity

All COAs selected for analysis must be valid, and the staff should reject COA alternatives that do not meet all five of the following validity criteria:

Suitable—Can accomplish the mission within the commander's guidance. Preliminary tests include (but are not limited to: a. Does it accomplish the mission? b. Does it meet the commander's intent? c. Does it accomplish all the essential tasks? d. Does it meet the conditions for the relevant end state? Etc.

Feasible—Can accomplish the mission within the established time, space, and resource limitations. a. Does the commander have the force structure, posture, transportation, and logistics (e.g., munitions) (means) to execute it? The COA is feasible if it can be executed with the forces, support, and technology available within the constraints of the OE and against expected enemy opposition. Etc.

Acceptable—Must balance cost and risk with the advantage gained. a. Does it contain unacceptable risks? (Is it worth the possible cost?) A COA is considered acceptable if the estimated results justify the risks. The basis of this test consists of an estimation of friendly losses in forces, time, position, and opportunity. b. Does it take into account the limitations placed on the commander

Distinguishable—Must be sufficiently different from other COAs in the following: a. The focus or direction of main effort. b. The scheme of maneuver, including integration and sequencing of actions in the physical domains, information environment (including cyberspace), and electromagnetic spectrum. c. Sequential versus simultaneous maneuvers. Etc.

Complete—Does it answer the questions who, what, where, when, how, and why? The COA must incorporate: a. Objectives, desired effects to be created, and tasks to be performed. b. Major forces and capabilities required, to include the forces and capabilities of international partners. c. Concepts for deployment, employment, and sustainment. d. Time estimates for achieving objectives. e. Military end state and mission success criteria (including the assessment: how the commander will know they have achieved success).

JP 5-0, III-41 to III-42

Integrated Deterrence and Strategic Competition – Joint Planning Process

COA Analysis and Wargaming: Each proposed course of action is thoroughly analyzed and subjected to wargaming, where planners play out the COAs against potential adversarial actions. This helps identify strengths, weaknesses, and potential outcomes of each COA, enabling a deeper understanding of possible operational approaches and their implications.

Wargaming is most effective when it involves the following elements:

- 1. A well-developed, valid COA*
- 2. People making decisions*
- 3. A fair, competitive environment*
- 4. Adjudication*
- 5. Consequences of actions*
- 6. Iteration*

JP 5-0, III-45

*(a) **COA analysis** is the process of closely examining potential COAs to reveal details that enable the commander and staff to tentatively evaluate COA validity and identify the advantages and disadvantages of each proposed friendly COA. The commander and staff analyze each COA separately according to the commander's guidance. COA analysis is a valuable use of time that ensures COAs are valid.*

*(b) **Wargaming** is a primary means for COA analysis. Wargames are representations of conflict or competition in a synthetic environment, in which people make decisions and respond to the consequences of those decisions. COA wargaming is a conscious attempt to visualize the flow of the operation, given joint force strengths and dispositions, adversary capabilities and possible COAs, the OA, and other aspects of the OE.*

JP 5-0, III-45

COA Analysis and Wargaming - The Red Cell (Red Team): A red cell in wargaming plays an essential role in planning exercises by taking on the perspective of an adversary or competitor. Incorporating a red cell offers added benefits of ***Enhanced Critical Thinking & Creativity and Improved Readiness and Adaptability***. The red cell challenges conventional thinking and assumptions of the planning team by adopting the adversary's tactics, techniques, and procedures. This opposition forces the planning team to think more critically and creatively about their strategies and plans, leading to more robust and resilient operational plans. Additionally, engaging a red cell in wargaming exercises exposes the planning and executing teams to a wide range of potential adversary actions and reactions. Often times, completely unanticipated. This exposure prepares planners to adapt more quickly to unexpected changes and challenges in the real-world operational environment.

- a. A robust, well-trained, imaginative, and skilled red cell that aggressively pursues the enemy's point of view during wargaming is essential. By accurately portraying the full range of realistic capabilities and options available to the enemy (to include all aspects of operations in the information environment, which includes cyberspace and some electromagnetic and counter-space capabilities), they help the staff address friendly responses for each enemy COA. For campaign and noncombat operation planning, the red cell provides expected responses to US actions, based on their knowledge and analysis of the OE.*
- b. The red cell should be composed of personnel from the joint force J-2 staff and augmented by other subject matter experts.*
- c. The red cell develops critical decision points, projects enemy and other OE reactions to friendly actions, and estimates impacts and implications on the enemy forces and objectives. By trying to win the wargame, the red cell helps the staff identify weaknesses and vulnerabilities before a real enemy does.*
- d. Given time constraints, as a minimum, the most dangerous and most likely COAs should be wargamed and role-played by the red cell during the wargame and include plausible enemy contested environment actions targeting deployment and distribution operations.*

JP 5-0, III-52

Integrated Deterrence and Strategic Competition – Joint Planning Process

COA Comparison: After wargaming, the different COAs are compared against each other based on criteria such as feasibility, acceptability, and suitability. The comparison helps in assessing the advantages and disadvantages of each COA, leading to a recommendation for the most effective and efficient course of action.

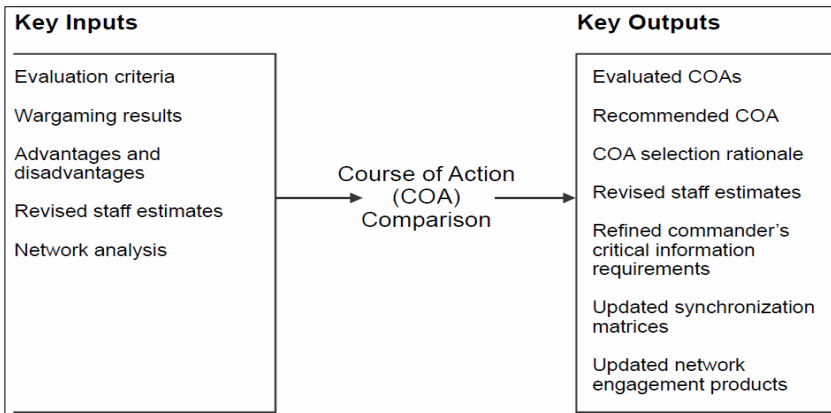


Figure 3 - Figure III-18. Joint Planning Process Step 5: Course of Action Comparison

COA Approval: The recommended COA is presented to the commander or decision-making authority for approval. This step involves briefing on the analysis, comparison, and rationale behind the recommended COA. The decision-maker may approve, disapprove, or modify the proposed course of action.

(a) In this JPP step, the staff briefs the commander on the COA comparison and the analysis and wargaming results, including a review of important supporting information. The staff determines the preferred COA to recommend to the commander.

(b) The nature of the OE or contingency may make it difficult to determine the desired end state until the crisis actually occurs. In these cases, the JFC may choose to present two or more valid COAs for approval by higher authority. A single COA can then be approved when specific circumstances become clear. However, in a crisis, the desired end state should be based on the set of objectives approved by the President or SecDef.

JP 5-0, III-58

Plan or Order Development: Following COA approval, detailed planning is undertaken to develop the operational plan or order that outlines the execution phase. This comprehensive document includes timelines, task assignments, resource allocations, and coordination instructions. It serves as a blueprint for the operation's execution.

(a) The CONOPS clearly and concisely expresses what the JFC intends to accomplish and how it will be done using available resources. It describes how the actions of the joint force components and supporting organizations will be integrated, synchronized, and phased to accomplish the mission, including potential branches and sequels.

JP 5-0, III-62

Integrated Deterrence and Strategic Competition – Joint Planning Process

The CONOPS:

- States the commander's intent.
- Describes the central approach the JFC intends to take to accomplish the mission.
- Provides for the application, sequencing, synchronization, and integration of forces and capabilities in time, space, and purpose (including those of multinational and interagency organizations, as appropriate).
- Describes when, where, and under what conditions the supported commander intends to conduct operations and give or refuse battle, if required.
- Focuses on friendly, allied, partner, and enemy COGs and their associated critical vulnerabilities.
- Provides for controlling the tempo of the operation.
- Visualizes the campaign in terms of the forces and functions involved.
- Relates the joint force's objectives and desired effects to those of the next higher command and other organizations as necessary. This enables assignment of tasks to subordinate and supporting commanders.

JP 5-0, III-62 to III-63

Joint Planning Process Primer – Attachment 1 Reflection Questions

Question 1: Considering the *Joint Planning Process* and this statement, “*The JPP is an orderly, analytical set of logical steps to frame a problem; examine a mission; develop, analyze, and compare alternative COAs; select the best COA; and produce a plan or order*”, reflect on and describe an issue you’re facing in your workplace where the JPP could be applied.

(Optional) Use the SEE format for the question above

State: *An issue I’m currently facing at work where JPP could be applied is _____.*

Elaborate: *This is because:* OR *In other words:*

Exemplify (examples): *For example, at work:*

Illustrate (OPTIONAL): *This is similar to:* OR *This is like:*

Question 2: Describe a core mission or process in your work center where you could apply or have applied a *Red Cell (red teaming)*. What would be some of the benefits?

Question 3: Describe a time when you struggled with vague or missing *Commander’s Intent*. What were some of the negative 2nd & 3rd order effects? What about positive 2nd or 3rd order effects?

Joint Planning Process Primer – Attachment 2 Experiential Note Taker

Student Note Taker: Joint Planning Process

Step 1 - Planning Initiation

Purpose:

Key Considerations:

Joint Planning Process Primer – Attachment 2 Experiential Note Taker

Step 2 - Mission Analysis

Purpose:

Key Considerations:

Mission Analysis Activities: Research and present the key considerations, notes, actions etc associated with your assigned activity. Provide illustrations/examples/references for the flight to use throughout the wargame and capstone.

Determine known facts and develop planning assumptions

Determine specified, implied and essential tasks

Develop mission statement

Develop risk assessment

Develop COA evaluation criteria

Determine CCIRs

Prepare and deliver mission analysis briefing

Joint Planning Process Primer – Attachment 2 Experiential Note Taker

Step 3 - Course of Action (COA) Development

Purpose:

Key Considerations:

Activity: In small teams, practice COA development techniques IAW JP5-0 using this scenario. Ensure you do so by addressing the 5Ws' and how.

Scenario

Scenario: You are a senior enlisted leader in charge of preparing for a highly anticipated Air Show that will showcase the capabilities of your unit's aircraft and personnel. The Air Show is scheduled to take place in two months, and you have been tasked with developing two Courses of Action (COAs) to ensure a successful and impactful event.

Situation: Your unit has recently acquired advanced aircraft with cutting-edge technology, and this Air Show presents a unique opportunity to demonstrate these capabilities to key stakeholders, including military leaders, government officials, industry partners, and the public. The success of the Air Show is crucial not only for showcasing your unit's readiness but also for building positive relationships and garnering support for future initiatives.

Mission: Develop two COAs that outline comprehensive plans for organizing and executing the Air Show, considering factors such as safety, logistics, audience engagement, and showcasing the full range of your unit's capabilities.

COA #1:

COA #2:

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Step 4 - Wargaming

Purpose:

Key Considerations: Referencing JP5-0, Appendix J, consider the following questions as assigned by your instructor.

Joint Planning Process Primer – Attachment 2 Experiential Note Taker

Step 5 - COA Comparison

Purpose:

Key Considerations:

Example Numerical Comparison

		Course of Action					
		COA 1		COA 2		COA 3	
Criteria	Weight	Rating	Product	Rating	Product	Rating	Product
Exploits maneuver	2	3	6	2	4	1	2
Attacks COGs	3	2	6	3	9	1	3
Integrates maneuver and interdiction	2	2	4	3	6	1	2
Exploits deception	2	1	2	2	4	3	6
Provides flexibility	2	1	2	3	6	2	4
CSS (best use of transportation)	1	3	3	2	2	1	1
Total		12		15		9	
Weighted total			23		31		18

NOTE: The higher the number, the better.

- The joint force commander's intent explained that the most important criterion was "attacking the enemy's COGs." Therefore, assign a value of 3 for that criterion and lower numbers for other criteria that the staff devises (**this is the weighing criterion**).
- For attacking the enemy COGs, COA 2 was rated the best (with a number of 3). Therefore, COA 2 = 9, COA 1 = 6, and COA 3 = 3.
- After the relative COA rating is multiplied by the weight given each criterion and the product columns are added, COA 2 (with a score of 31) is rated the most appropriate according to the criteria used to evaluate it.

Legend

COA course of action

COG center of gravity

CSS combat service support

Figure 4 - Example Numerical Comparison, JP5-0, Appendix F-2, Figure F-1

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Step 6 - COA Approval

Purpose:

Key Considerations:

Activity: Create a sample COA decision brief using the information below IAW JP5-0, Chapter 3, page 60.

Scenario

Commanders Intent: Neutralize the enemy threat and control the objective area, denying the enemy any further operational capability.

COA 1: Airborne Assault

Objective: To conduct a rapid airborne assault to seize a key enemy position.

Wargaming Results:

Strengths:

Surprise: The enemy was caught off guard by the sudden airborne assault, allowing forces to establish a foothold quickly

Maneuver: Airborne units were able to maneuver swiftly and adapt to changing battlefield conditions

Weaknesses:

Vulnerability: Enemy anti-air defenses inflicted casualties on our airborne units during insertion

Isolation: Some airborne units became isolated due to scattered drop zones, leading to difficulties in coordination

Lessons Learned:

Importance of suppressing enemy air defenses before airborne insertion.

Need for better coordination and communication among airborne units to prevent isolation.

Overall Assessment: While the airborne assault achieved surprise and maneuver, vulnerabilities during insertion and isolation of units pose challenges to sustained operations.

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COA 2: Amphibious Assault

Objective: To conduct an amphibious assault to establish a beachhead and secure a strategic coastline.

Wargaming Results:

Strengths:

Time: Forces successfully secured the designated beachhead, allowing for rapid deployment of follow-on units.

Force protection: Close naval gunfire support effectively neutralized enemy coastal defenses, facilitating the amphibious landing.

Weaknesses:

Casualties: Forces sustained casualties during the landing due to enemy fire and beach obstacles.

Sustainment: Establishing supply lines from the beachhead to inland positions proved to be challenging, slowing down operational tempo.

Lessons Learned:

Importance of pre-landing bombardment to neutralize enemy coastal defenses.

Need for efficient logistical planning to sustain operations following the initial landing.

Overall Assessment: The amphibious assault successfully established a beachhead, but initial casualties and logistical challenges must be addressed for sustained operations.

COA Evaluation Criteria	WEIGHT	COA 1		COA 2	
		RATING	PRODUCT	RATING	PRODUCT
<i>Surprise</i>	3	1	3	2	6
<i>Maneuver</i>	3	1	3	2	6
<i>Vulnerability</i>	2	2	4	1	2
<i>Isolation</i>	2	2	4	1	2
<i>Time</i>	4	2	8	1	4
<i>Force Protection</i>	2	2	4	1	2
<i>Casualties</i>	1	1	1	2	2
<i>Sustainment</i>	4	1	4	2	8
TOTAL		12	31	12	32

Joint Planning Process Primer – Attachment 2 Experiential Note Taker

Step 7 - Plan or Order Development

Purpose:

Key Considerations:

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Joint Planning Process Primer – Attachment 3 Wargame (MP3) Note Taker

Step 1 - Planning Initiation

Task: Immediately following Module 3 Overview, “Wargame Introduction” gain familiarity with your assigned CCMD using ODIN/DATE World. When doing so, consider the following:

To Note: This list is not all inclusive, it is a general understanding of the information that can aid in developing the mission analysis briefing.

- How will the US achieve its objectives in this region?
 - Resources
 - Capabilities
- Gain familiarity with the area:
 - Maps
 - Access (land, sea, air etc.)
 - Friendly v non-friendly neighboring countries
 - Infrastructure (airport, seaport etc)
 - Geography or terrain
 - Stability of the region and/or key actors

Notes:

Joint Planning Process Primer – Attachment 3 Wargame (MP3) Note Taker

Step 2 - Mission Analysis Briefing

Task: As a flight, formulate a Mission Analysis Briefing IAW the requirements identified in JP5-0, Chapter III, page 30, “Example Mission Analysis Brief.” *ODIN/DATE World is the sole information resource for the briefing. No real-world information will supplement the data contained within the website.* Once complete, elect a representative to brief your instructor, acting as the commander. The briefing will not exceed 15 minutes.

Notes:

Joint Planning Process Primer – Attachment 3 Wargame (MP3) Note Taker

Step 3 – COA Development

Task: As a flight, develop two COAs to meet your commander's intent. When doing so, refer to JP5-0, Chapter III, page 40, "Develop Initial COA Sketches and Statements," as listed below:

- Who (type of forces) will execute the tasks?
- What are the tasks?
- Where will the tasks occur?
- When will the tasks begin?
- What are key/critical decision points?
- How can the commander provide "operational direction" so components can accomplish "tactical actions?"
- Why (for what purpose) will each force conduct its part of the operation?
- How will the commander assess mission accomplishment?

Each COA must be valid as defined by the following criteria (reference JP5-0 chapter 3 pages 40-41):

- Suitable
- Feasible
- Acceptable
- Distinguishable
- Complete

Notes:

Joint Planning Process Primer – Attachment 3 Wargame (MP3) Note Taker

Step 4 – COA Analysis and Wargaming

Task: Capture red teaming results for your respective COA to use during step 5, COA comparison. Focus notes on the actions, reactions, adjudication and counteractions that occur within each turn.

Notes:

Joint Planning Process Primer – Attachment 3 Wargame (MP3) Note Taker

Step 5 – COA Comparison

Task: Using the results of the wargame, home flights will compare their two COAs against their previously established criteria using a weighted numerical comparison matrix. When doing so, refer to JP5-0, Appendix F-2, Figure F-1, “Example Numerical Comparison.”

Notes:

Joint Planning Process Primer – Attachment 3 Wargame (MP3) Note Taker

Step 6 – COA Approval

Task: Once COA comparison is complete, elect a representative to brief the instructor, acting as the commander. When doing so, refer to JP5-0, Chapter 3, Page 60, “Sample Course of Action Decision Briefing Guide.”

Notes:

Joint Planning Process Primer – Attachment 4 Resources

Student Preparation:

Review: JP 5-0, Chapter III, Section B, Joint Planning Process (JPP), pages III-9 to III-63 (pg. 87 - 141) and appendix J (*Located in LMS*)

Read: JP 5-0, Chapter I, Section 1 para b (pg. 33) (*Located in LMS*)

Read: Joint Planning Process Primer (*Located in LMS*)

External Link to JP 5-0: https://irp.fas.org/doddir/dod/jp5_0.pdf

Additional Resources (for increased understanding):

AFDP 1, Page 12, Mission Type Orders

https://www.doctrine.af.mil/Portals/61/documents/AFDP_1/AFDP-1.pdf

Operational Environment Data Integration Network (ODIN), United States Army Training and Doctrine Command (TRADOC)

<https://odin.tradoc.army.mil/DATEWORLD> (*accessible via commercial networks or AFNET*)

PLANNING



U.S. AIR FORCE

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Air Force Doctrine Publication 5-0, *Planning*

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FOREWORD

Air Force Doctrine Publication (AFDP) 5-0, *Planning* provides a planning methodology for Airmen to solve complex problems. All Airmen should be prepared to solve problems in contested, degraded, or operationally limited environments. The Air Force planning process (AFPP) focuses on providing commanders options to accomplish their mission.

Doctrine embodies the fundamental principles by which military forces guide their actions in support of national objectives. It is a body of carefully developed and tested ideas officially approved to provide a common frame of reference for solving military problems. However, the challenge for United States Air Force (USAF) doctrine is to be simultaneously based on past experiences, applicable in the present, and leaning forward toward the future, all in equal measure.

The USAF must prepare for a new reality, one in which freedom of maneuver and freedom of action are increasingly challenged. To deter, compete, and win across the competition continuum, Airmen should be ready to conduct operations in contested, degraded, and operationally limited environments. Therefore, Airmen develop an understanding of their environment, make decisions, and converge effects when disconnected from higher headquarters (HHQ). AFDP 1, *The Air Force*, supports this effort by establishing mission command as the Airmen's philosophy for leadership and framework for the command and control (C2) of airpower. The mission command framework is centralized command, distributed control, and decentralized execution (CC-DC-DE). Mission command provides the responsiveness, flexibility, and initiative necessary to prevail in a future fight. By embracing a planning and orders development culture, USAF Service component or unit staffs (A-staffs) and Airmen at all echelons can accomplish their assigned mission while operating in uncertain, complex, and rapidly changing environments.

The AFPP is intended to provide Airmen with a conceptual framework for planning across the competition continuum. Planners must be proficient in their craft to enable Agile Combat Employment (ACE), set the conditions for victory in the six fights described in the *Air Force Future Operating Concept*, and address problems that typically aid force optimization for great power competition. The processes contained in this publication are equally useful for addressing institutional problems that require a disciplined planning methodology.

This publication should guide the air component in systematically addressing problems. All Airmen are encouraged to read this publication in conjunction with AFDP 3-0, *Operations*. Being familiar with doctrine, policy, and guidance related to the development and production of orders helps prepare us for the challenges we face. Throughout our history, innovative Airmen have developed and adapted technologies and employment methods to meet the challenges our nation has faced. AFDP 5-0, *Planning*, contributes to our distinguished lineage as overcomers.

CHAPTER 1: INTRODUCTION TO PLANNING

Planning is how Airmen address and solve complicated problems. It is the process through which Airmen frame a problem and provide solutions to achieve the full range of military objectives.

Plans and orders production allow commanders to provide subordinates formal, standardized, and authoritative guidance in a joint force recognized format.

Planning is a part of the C2 Process and a key enabler of the USAF's CC-DC-DE framework. Airmen on wing A-Staffs, and at all echelons, are problem solvers who use planning processes to produce clear plans and orders.

It is crucial that Airmen use standardized planning, orders production, and dissemination processes to foster disciplined initiative in the decentralized execution of operations. Commanders enable decentralized execution effective distribution of their intent. Tactically proficient Airmen with a mission statement, commander's intent, supporting logistics, and communication can exercise disciplined initiative and rapidly adapt to changing conditions.

The commander is the central element of all USAF planning. Commanders at all levels enable airpower and contribute to the C2 function. Planning enhances the commander's ability to make sound and timely decisions by increasing situational understanding of problems and providing a methodology to devise practical solutions.

Airmen may use the AFPP to address institutional issues while increasing familiarity with existing joint and Service planning processes. Examples of existing planning methodologies include the joint planning process (JPP), joint planning process for air (JPPA), and sister-Service planning processes. The AFPP enables Airmen to produce Service component staff estimates, base plans (BPLAN), authority-appropriate orders, USAF modified five-paragraph orders, and orders written using the mission-type order (MTO) technique.¹ **Airmen planning at any echelon through the distributed control continuum use the AFPP.²**

Contested, degraded, or operationally limited environments require commanders and staffs that can identify, understand, and methodically address problems through a repeatable process. The AFPP presented in this publication provides Airmen with an adaptable planning process to solve problems regardless of echelon or organization.

PLANNING AS A MAJOR C2 ACTIVITY

Planning the first C2 activity conducted by a commander and their staff in the overall C2 process. The four C2 activities that comprise the C2 process are: planning, preparing, executing, and assessing.³ These activities are both cumulative and continuous. The

¹ For additional information on order formats and guidance, see Chairman of the Joint Chiefs of Staff Manual (CJCSM) 3130.03A, *Planning and Execution Formats and Guidance*.

² For additional information on distributed control continuum, see AFDP 3-0.1 *Command and Control*.

³ For additional information on the C2 activities, see AFDP 3-0.1 *Command and Control*.

outputs of planning provide the foundational guidance for the remainder of the C2 process.

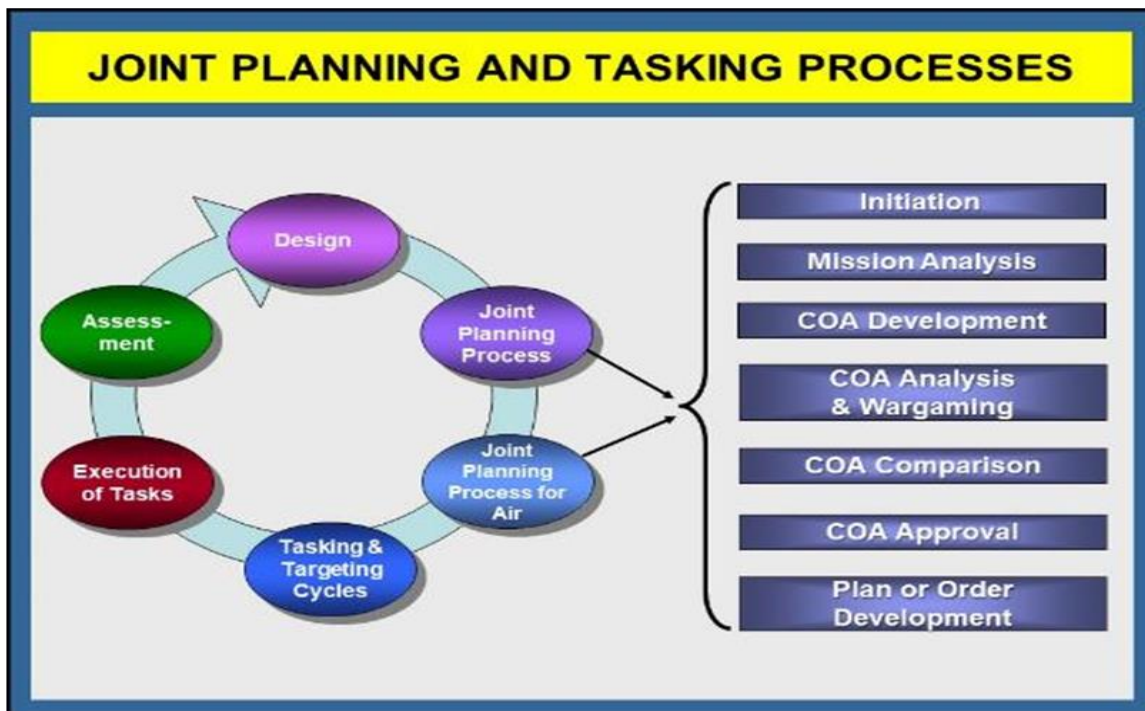
PLANNING METHODS

Several planning processes are described in this publication to highlight selection considerations for each process. **When commanders are tasked with a mission, they choose the right planning method based on the situation, guidance, and the operational environment (OE).** They should consider the level of guidance from HHQ and the type of plan or orders they want to produce. The various planning processes generally include the same steps, what differs is who is doing the planning and what plan is produced. Factors that might influence a commander's choice of planning methodology include:

- ★ HHQ guidance.
- ★ Desired output of the planning process.
- ★ Applicability of the doctrinal planning guidance to address the situation.
- ★ Command authority considerations.
- ★ A-staff's familiarity with specific planning methodologies.
- ★ Size, scope, structure, and/or the dynamic nature of the problem or task.
- ★ Time and resources available for planning.

JOINT PLANNING PROCESS (JPP)

The JPP is the method that guides the planning of joint campaigns and operations. Joint Publication (JP) 5-0, *Joint Planning*, details the JPP, used by a joint force headquarters (HQ) staff, to create or support strategic and theater-level objectives through joint operations. Airmen assigned to joint staffs generally use the JPP. Outputs of the JPP and JPPA are vital to the design efforts in the AFPP. The steps of the JPP and JPPA are identified in the figure, "Joint Planning and Tasking Processes."

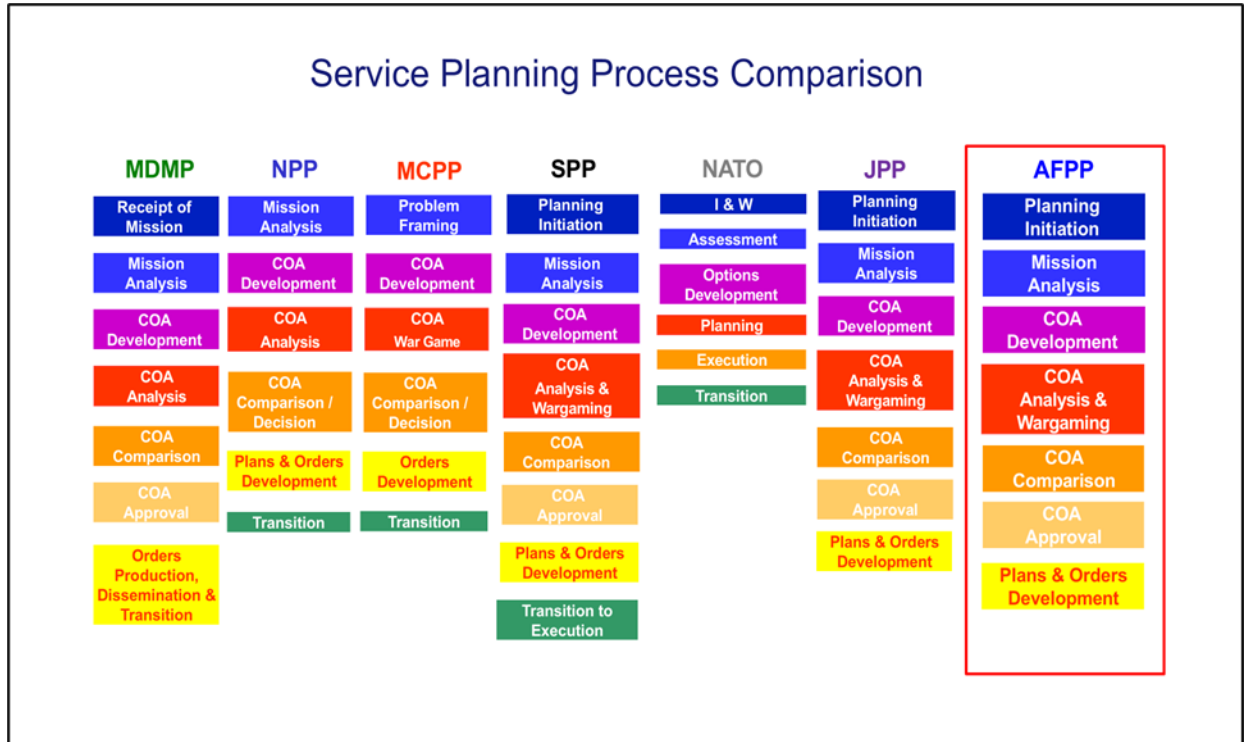


JOINT PLANNING PROCESS FOR AIR (JPPA)

The JPPA is an air-centric application of the JPP utilized by an air component commander and their staff at the operational level of war to guide the employment of joint air capabilities. **JPPA nests under JPP with specific details and considerations for joint air operations, and is captured in JP 3-30, *Joint Air Operations*.** Airmen assigned to a theater combined force air component commander (CFACC) or joint force air component commander (JFACC) generally employ JPPA to create a joint air operations plan (JAOP). The use of JPPA, and the guidance provided in JP 3-30, is ideally suited for Airmen at or in concert with the joint air operations center (JAOC). An air expeditionary task force (AETF) or wing may need to use the JPPA to plan joint air operations depending on the environment.

OTHER SERVICE PLANNING PROCESSES

All US Military Services have their own planning processes that link their planning activities to C2. The Army has the military decision-making process (MDMP), the Navy has the Navy planning process (NPP), the Marines have the Marine Corps planning process (MCP), and the Space Force has the Space Planning Process (SPP). All Services incorporate their Service-specific planning considerations into their process.



AIR FORCE PLANNING PROCESS

The AFPP is a planning method derived from the JPP for use by USAF commanders and staffs at all levels. This publication serves as the principal document for guidance on the application of the AFPP. Airmen utilize the AFPP when the guidance, mission, situation, and environment dictate the need for USAF specific outputs.

The AFPP reflects the extant best practices for planning. The AFPP presents an easily adaptable planning process for Airmen and provides tailored guidance for the planning needs of the USAF. **Airmen employ the AFPP to solve problems. The AFPP is a tool for commander's staffs to methodically develop optimal courses of action to address the range of challenges confronting their units. The result of the AFPP is a commander's plan or order.**

CHAPTER 2: THE AIR FORCE PLANNING PROCESS

The AFPP is a commander led planning methodology that integrates the activities of the staff, subordinate and adjacent units, and other partners to understand the situation and mission, develop, analyze, and compare the course of actions (COA) available, and decide on one that that best accomplishes the mission.

The AFPP can be applied to planning situations encountered by wing A-staffs in garrison or deployed. The steps of the AFPP are outlined in the figure, “Air Force Planning Process”. **This publication guides planners through the AFPP by providing an overview, inputs, activities, and results for each step. For more information on how to progress through the AFPP see Appendix F, “AFPP Kneeboard.”** Inputs, activities, and results for each step will vary based on HHQ guidance, commander’s guidance, and echelon of command. Developed operational design, operational approaches, air component design, and air component approaches can inform planning at subcomponent commands. Material more appropriate for the JPP or JPPA will be annotated with additional remarks.

- ★ The AFPP helps commanders apply thoroughness, clarity, sound judgment, logic, and professional knowledge to understand situations, develop options to solve problems, and reach decisions.
- ★ The AFPP facilitates collaborative and parallel planning as the HHQ solicits input and continuously shares information concerning future operations with subordinate and adjacent units, supported, and supporting units, and other military and civilian partners.
- ★ The AFPP drives preparation. Since time is a factor in all operations, commanders and staffs conduct a time analysis early in the planning process. This analysis helps them determine what actions are required and when those actions should begin to ensure forces are ready before execution. The AFPP is iterative and may require the commander to direct subordinates to start necessary movements, conduct task organization changes, begin information collection operations, and execute preparation activities before completing the plan.

Air Force Planning Process

1. Planning Initiation.
2. Mission Analysis.
3. Course of Action Development.
4. Course of Action Analysis and Wargaming.
5. Course of Action Comparison.
6. Course of Action Approval.
7. Plans and Orders Development.

The steps of the AFPP are detailed in the figure above. As the commander and staff conduct the AFPP they use the air component design elements to create a conceptual framework underpinning planning.

DESIGN

AFPP integrates design elements to produce an air component approach, plans, and orders that reflect the Airman's perspective.

Design and planning are inextricably linked—design is part of planning and planning informs design. Commanders and staffs, across all echelons, use design to define and refine problems that need to be solved. Simultaneously, planning efforts produce potential solutions for the identified problems. This publication specifically articulates operational design and air component design.

Effective planning starts with a joint force commander's (JFC's) operational design. Operational design is the conceptual basis and framework underpinning a campaign, operation plan, and/or orders. "Design is about creating operational vision,"⁴ to define the scope of planning. Design continues throughout planning.

Operational design enables the JFC and staff to comprehend wide-ranging solutions for mission completion and to grasp the uncertainty in a complex operational environment. The air component should actively participate in the JFC's operational design process, particularly during the early stages. Commanders and staffs use the JPPA to create plans for joint air operations while informing operational design and creating support plans and orders for the JFC. Design is integrated in planning efforts. Planners utilize the JFC's operational approach to inform the development of specific courses of action, plans, and orders. Planning and operational design facilitate the transformation of broad guidance into tactical-level tasks.

Operational design provides a methodology consisting of four continually interacting components: define the problem, analyze guidance, understand the environment, and develop a method to solve the problem to create the construct for further planning. Thirteen operational design elements assist the commander and staff in conducting the operational design methodology.⁵ Operational design normally produces planning guidance and an operational approach prior to AFPP initiation. The air component design, air component approach, and AFPP support the JFC's operational design and planning (detailed later in this chapter).

Typically, JFCs provide planning guidance and an operational approach to subordinate commanders before detailed joint planning. **The elements of design are continually evaluated and refined in the major activities of the C2 Process (planning, preparing, executing, and assessing).** Ultimately, a well-articulated and understood design helps define the problem, establish the objective, and set the stage for further planning. The "Air Component Design Elements" figure lists elements initially identified during the first

⁴ Jeffrey M. Reilly, *Operational Design: Distilling Clarity for Decisive Action* (2012).

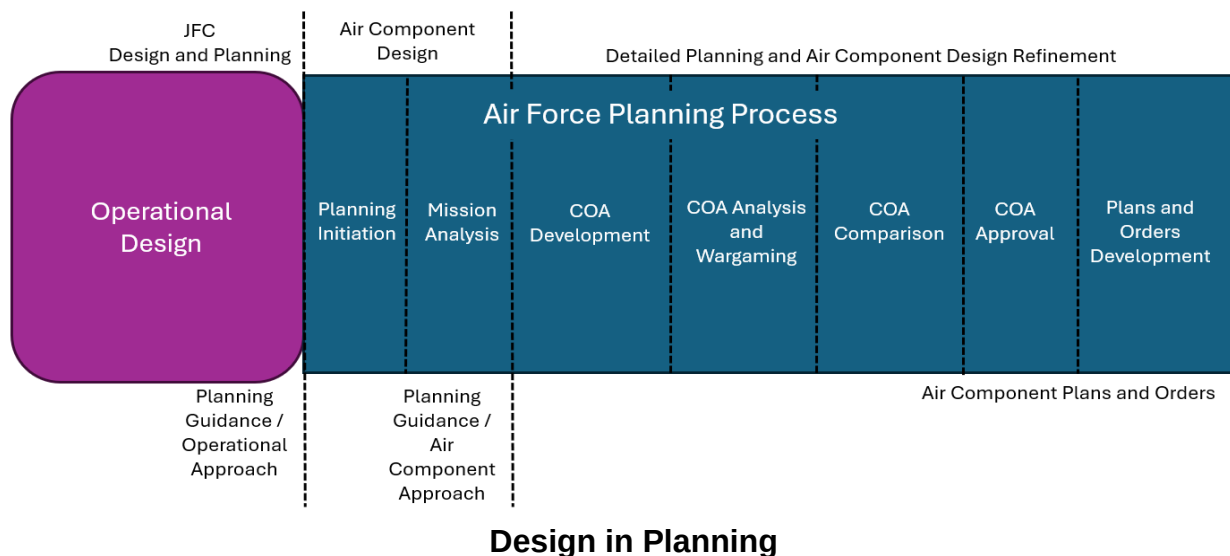
⁵ See JP 5-0, *Joint Planning*, for additional information.

two AFPP steps of planning initiation and mission analysis.

Air Component Design Elements

1. Objective.
2. Desired End State.
3. Root Cause.
4. Effects.
5. Culmination.
6. Physical Characteristics of the OE.
7. Lines of Operation and Lines of Effort.
8. Decisive Points.
9. Direct and Indirect Approach.
10. Operational Reach.
11. Arranging Operations.
12. Anticipation.
13. Forces and Functions.

Air component design is a methodology for applying adaptive thinking to identify and define problems. **The air component design is derived from the JFC's operational design**, and it requires commanders to coordinate their design efforts with higher and lower echelons as well as supporting commanders. Air component design applies to all USAF commander directed planning, regardless of the assigned unit or echelon. A commander constructs an air component approach, scoped to their applicable problem and mission, in the context of the air component design. The "Design in Planning" figure shows the relationship between operational design, air component design, and the AFPP.



Commanders and their staffs use design thinking to produce an air component approach during planning initiation and mission analysis. Sometimes the mission permits HHQ and subordinate units to conduct simultaneous design and planning. Simultaneous planning enables subordinate units to frame their efforts within the HHQ direction, ensuring alignment with the overall strategic objectives. The outputs of design reflect the commander's intent and are pivotal to further planning efforts. The broad description of actions needed to address the defined problem is conveyed with an air component approach. The initial air component approach is completed by the end of mission analysis. Plans and orders are normally produced at the end of the plans and orders development step.

ADAPTIVE THINKING

Adaptive thinking is used during all steps of the AFPP and enables problem identification and framing. Adaptive thinking is the commander and staffs' ability to critically analyze information, think creatively, make decisions, and solve problems dynamically. Thus, it is recommended that planning teams include individuals with diverse skills, knowledge, and experience to ensure the planning process is conducted from multiple viewpoints.

THE AIRMAN'S PERSPECTIVE ON PLANNING

The Airmen's perspective seeks to establish a cohesive and coherent link between objectives, effects, tasks, and actions to ensure all echelons are aligned with strategic objectives and the desired end state.

- ★ The perspective is comprehensive and integrates all domains and environments, all disciplines and partnerships, all levels, and all instruments of national power.
- ★ The perspective is about thinking creatively and reflectively while seeking solutions.
- ★ The perspective integrates all design, planning, execution, and assessment efforts into a unitary whole.
- ★ The perspective focuses on behavior and seeks to achieve objectives most effectively, then to the degree possible, most efficiently.
- ★ The perspective is unique because it places the end state and objective at the forefront of planning.

Civilian Harm Mitigation and Response (CHMR)

Commanders and their staffs have a moral and legal obligation in planning to reduce the risk of harm to civilians and civilian objects. CHMR efforts reflect the importance of protecting and respecting human life and treating civilians with dignity and respect. Prioritizing CHMR in planning only enhances the effectiveness and legitimacy of military operations and contributes to mission success. CHMR considerations should be incorporated during all steps of the AFPP.

- ★ Consider CHMR in the air component design and approach.
- ★ Planning initiation: Identify and address cognitive bias in operational art.
- ★ Mission analysis: emphasize CHMR considerations in the:
 - ★★ Analysis of centers of gravity (COGs).
 - ★★ Completion of specified, implied, and essential tasks.
 - ★★ Analysis of the assigned mission.
 - ★★ Identification and mitigation of risks.
- ★ COA development: identify potential civilian harm incidents, mitigation, and response options.
- ★ COA analysis and wargaming: Consider CHMR during analysis and wargaming of the adversary's most likely and most dangerous COAs.
- ★ COA comparison: consider differences in CHMR for each COA.
- ★ COA approval: discuss CHMR during COA Approval.
- ★ Plans and orders development: clearly describe CHMR considerations and options in the plan or order.

The remaining chapters of this publication discuss the AFPP steps. Each chapter is focused on a single step of the AFPP and is organized into overview, inputs, activities, and results sections.

CHAPTER 3: PLANNING INITIATION

The first step of the AFPP is planning initiation. **Commanders at any echelon initiate the AFPP in response to HHQ guidance and direction.** Traditionally, planning activities are led by the commander and their A3 or A5 staff. However, the commander may appoint any Airmen to serve as part of a planning team. **HHQ products support air component planning initiation by providing either an operational approach or planning guidance and commander's intent.** At lower echelons of command, analysis of this intent may suffice for design. Descriptive guidance and iterative air component design help the commander and staff ensure they understand the problem.⁶

PLANNING INITIATION INPUTS

Planning initiation normally starts in response to HHQ direction. Inputs that inform this step may include planning guidance, operational approaches, planning orders, air component approaches, and operational planning documents. Likewise, a commander may initiate planning in response to local or internal problems that require rigorous and methodological development and evaluation of options.

PLANNING INITIATION ACTIVITIES

When a new mission is identified, commanders and staffs perform the following planning initiation activities:

- ★ Alert the staff and other key participants.
- ★ Document previously identified and relevant air component design elements from operational guidance, as required.
- ★ Gather the tools. Once notified of the new planning requirement, the staff prepares for mission analysis by gathering the necessary tools and examining relevant information about friendly, neutral, and threat networks. These tools include, but are not limited to:
 - ★★ Appropriate manuals, doctrine, military terms, and symbols.
 - ★★ All documents related to the mission and the area of operations (AO), including the HHQ's planning guidance, rules of engagement (ROE), operational approach, orders, air component approach, maps and terrain products, and operational graphics.
 - ★★ HHQ and other organizations' intelligence and assessment products.
 - ★★ Both the unit's and HHQ's standing operating procedures.
- ★ Develop an initial understanding of the OE.

⁶ See AFDP 3-0.1, *Command and Control*, for additional information.

- ★ Begin or update running staff estimates (e.g., logistics estimate, intel estimate).
 - ★★ Estimates are an essential part of the air component design process. Through their estimates, the staff provides expert assessment of the OE and relevant factors affecting effective planning and execution toward the achievement of objectives and attainment of end states. **Airmen running AFPP use staff estimates to inform their planning.**
- ★ Conduct an initial assessment. During this step, the commander and staff conduct an assessment of the time and resources available to plan, prepare, and begin the execution of an operation. This initial assessment:
 - ★★ Identifies and frames the problem to address conditions that must change, or stay the same, to reach the desired system state.
 - ★★ Determines which outside agencies and organizations to contact and incorporate into the planning process.
 - ★★ Assesses the staff's experience and cohesiveness.
 - ★★ Creates a staff planning timeline that outlines how long they can spend on each step of the AFPP. The staff planning timeline indicates what products are due, who is responsible for them, and who receives them. Timelines may be compressed in crisis compared to contingency planning. Artificial intelligence (AI) may be able to expedite or improve some steps of the AFPP.
- ★ Apply adaptive thinking and the Airmen's perspective to the OE and problem.
- ★ Issue the commander's initial guidance. Initial guidance may include the following:
 - ★★ How to abbreviate the AFPP, if required. An abbreviation may involve compressing planning timelines due to not having to produce elements of mission analysis or restraining the number of COAs developed. The abbreviation of the AFPP does not involve eliminating or skipping steps but rather focuses on the allocation of time, resources, and priority of steps.
 - ★★ Necessary coordination, including liaison officers.
 - ★★ Authorization to begin information collection activities.
 - ★★ Locations, dates, and times of planning meetings.
 - ★★ Initial information requirements.
 - ★★ Additional staff tasks.

OPERATIONAL ENVIRONMENT

Planning initiation involves developing an understanding of OE as part of air component design. OE analysis is crucial for planners to define the problem and create an operational approach. This includes all the tangible and intangible factors that influence operations. Planners can gain awareness of the OE by examining guidance from HHQ, intelligence estimates, and country plans. Factors to consider when assessing the OE include physical location, potential actors and stakeholders, weather, geography, sociocultural aspects, infrastructure, communication lines, information environment, and population. Models are often used to analyze the OE, and more details can be found in the mission analysis section of this publication.

Constant analysis is essential to comprehend the OE. This is necessary because the OE is an intricate system with linkages that impact strategic, operational, and tactical levels of warfare. It is important to recognize that our understanding of the OE evolves as we gather more information. HHQ may have already conducted an analysis of the OE to guide subordinate design efforts. Airmen's analysis of the OE should be guided by HHQ, staff estimates and the commander's intent. This understanding can come from various sources, such as higher-level guidance, which includes plans, planning orders (PLANORDs), warning orders (WARNORDs), operation orders (OPORDs), execution orders (EXORDs), planning directives, operational approaches, joint intelligence preparation of the operational environment (JIPOE), intelligence preparation of the battlespace (IPB), and staff estimates.

A challenge in examining the OE is that it exists as a part of a constantly evolving series of complex systems and understanding requires a holistic analysis of the systemic interconnectivity. One way to frame the OE involves the deliberate division of the environment into two interrelated subsystems that foster a shared understanding of the OE's overarching interconnectivity. The two subsystems are the observed system and the desired system.



"In preparing for battle, I have always found that plans are useless, but planning is indispensable."

—Dwight D. Eisenhower
34th President of the United States

Planners using the AFPP focus on the systems related to their mission and problem. The air component is often provided with an understanding of the OE during planning initiation. HHQ guidance provided to the air component allows Airmen to focus on the subsystems and define the problem.

The observed system is an analysis of how the environment currently exists and functions. There is no exact prescription for visualizing the observed system because the observed system is a construct of what one has the capability to see and understand. Planners begin the development of the observed system by examining key factors such

as principal actors, the historical content of the situation, current operations, security, logistics, base infrastructure, communications, command and control, computers, intelligence, surveillance, and reconnaissance, and other elements related to the problem's resolution. The overarching goal of visualizing the observed system is learning how the system currently functions and understanding how the system will change in reaction to an action or series of actions. Based on the projected changes in the observed system, planners have a mechanism to exploit the changes and anticipate adversary operations. An important aspect of the observed system is setting the foundation for understanding the desired system.

The desired system analyzes the air component's objectives and how to change the observed system to achieve those objectives. In addition to objectives, the desired system assesses actors capable of influencing objectives and end states, barriers to achieving objectives, and time.

PLANNING INITIATION RESULTS

Planning initiation may result in a planning timeline, planning team roster, collection of planning guidance, initial understanding of the OE, and identification of necessary design elements. The staff may issue a preparatory order. A preparatory order describes the situation (guidance overview and OE), mission (tasks and purpose), execution (concept of operations), administration and logistics (concept of support), and C2 (command relationships and C2 structure).

CHAPTER 4: MISSION ANALYSIS

During the mission analysis step of the AFPP, the staff analyzes the HHQ's operational approach, plans, or orders to gain an understanding of the mission, intent, OE, resources available, limitations, and specified and implied tasks. This step defines and refines the problem through the study of the specified tasks and identifies the implied tasks as well as the specified and implied tasks that are essential to accomplish the mission and resolve the problem. It focuses the commander and the staff on the problem at hand and lays a foundation for effective planning. **During mission analysis, Airmen identify the elements of air component design and construct an air component approach.**

The staff performs updates, initiates, or reviews JIPOE, develops critical facts and assumptions, and determines the initial commander's critical information requirements (CCIRs). Additionally, IPB should be conducted by intelligence personnel staff within the Service and functional components and support the individual operations with force-specific micro-analysis. Throughout mission analysis, staff members have an open iterative dialogue with the commander to refine and update their guidance.

While the staff conducts their analysis, the commander conducts their own review of HHQ guidance and mission objectives. The commander's analysis and adaptive thinking provide a frame of reference that helps to quickly assess the staff's work.

Mission analysis marks the beginning of the staff estimate and is not a one-time event. Mission analysis is a continuous effort that is constantly re-evaluated as the situation develops and new information is presented. Although some activities occur before others, mission analysis typically involves substantial concurrent processing of information by the commander and staff, particularly in a crisis.

MISSION ANALYSIS INPUTS

Primary inputs to mission analysis are planning initiation results and HHQ operational or air component approaches. When the commander receives a mission tasking, analysis begins with questions like:

- ★ What is the purpose of the mission? (What problem is the commander asked to solve or what change to the OE is desired?)
- ★ What are the specified, implied, and mission essential tasks?
- ★ Will the mission achieve the desired objectives?
- ★ What limitations (i.e., constraints and restraints) have been placed on my own forces' actions?

Planning for In-residence Education

The COVID-19 outbreak impacted living conditions and operations around the world. Air Education and Training Command momentarily paused in-residence education at several institutions due to safety concerns. Gradually, guidance was received to initiate planning to conduct a safe return to in-resident education operations. Air University planning course instructors worked with commanders, staffs, and outside organizations to plan for a safe return to normal operations.

The planning process directed by the instructors was a modified version of the JPP. The use of a heavily modified JPP was the beginning of the AFPP.

Teams developed multiple COAs that informed leadership's decision to return to in-residence operations. Commander selected COAs were molded into an approved plan that was shared among organizations to develop an understanding of the mission, intent, tasks, logistics, and authorities for COVID-19 education operations. The successful planning efforts resulted in schools resuming professional military education in a safe and timely manner.

MISSION ANALYSIS ACTIVITIES

Mission Analysis enables commanders and staffs to understand the problem(s), mission, and situation. **The mission analysis activities are often simultaneously conducted.** During mission analysis, the tasks (specified, implied, and essential) and their purposes should be identified to ensure planning encompasses all requirements, including any limitations on actions that the commander or subordinate forces may take.

Development, and dissemination of an air component approach at the end of mission analysis helps commanders and staffs frame the problem and develop viable approaches to achieve the desired end state. Mission analysis should produce an understanding of the OE, observed system, desired system, and tasks the commander should accomplish in support of the HHQ. The essential tasks derived during mission analysis should be accomplished to enable the end state as expressed in the commander's operational objectives, mission, and intent statements.

Review Commander's Initial Planning Guidance. Planners should have an active dialogue with their commander to understand their initial vision, intent, and view of the problem. The commander and staff should consider the timeframe and scope of the problem. Planning leads provide the following to the rest of the staff:

- ★ JFC's narrative, mission statement, intent statement, and concept of operations (CONOPS).
- ★ Forces and capabilities available for planning.

- ★ Transportation, airlift priority, and airlift allocation.
- ★ Other capabilities available (United States [US], multinational forces and organizations, interagency support).
- ★ The political situation (e.g., host nation [HN], friendly and other foreign governments).
- ★ Time allocation.
 - ★★ Balance the desire for detailed planning against the need for speed. Be keenly aware of the time available until the commander and HHQ decide, issue orders, and begin to execute the operation.
- ★ A timeline reflecting known and assumed operational requirements (e.g., C-day, D-day) to focus the planning groups during planning.
- ★ A timeline identifying the time available to complete the planning activities based on the HHQ delivery requirements.

Review HHQ's Planning Activities and Strategic Guidance. During this mission analysis, planners should examine all HHQ guidance and policy (at least two levels above), level of operational command authority, strategic communications, WARNORDs, planning orders, alert orders, draft plans, other planning group feedback, and commander conversations. Most importantly, they seek to understand their unit's role in the HHQ mission, intent, and CONOPS.

- ★ HHQ orders should provide specific guidance to include a description of the situation, purpose, objectives, mission, tasks, and pertinent limitations.
- ★ If available, analyze the HHQ's (and JFC's) operational approach to gain an appreciation for their understanding and visualization of the situation and problem. Develop an air component approach at the air component level (air operations center [AOC] or Air Force forces [AFFOR] staff). At the AETF and wing level review the air component approach. This provides a basis for a detailed analysis of the OE and of the tasks that may describe the mission and its parameters.

Determine Facts. A fact is a statement of information known to be true and relevant to planning. A fact defines the scope of the problem. If a fact is invalidated, it should result in a new or modified plan. When facts change it may drive contingency planning. Planners should use caution in characterizing information as facts, as some data thought to be facts may be based on incomplete information or open to subjective interpretation based on the observer's perspective. **Several of the facts that inform the AFPP are determined during HHQ mission analysis. Airmen should seek to reference facts identified by HHQ to inform their planning.**

- ★ The JIPOE aids in understanding the complex, interconnected conditions and influences that bear on commander decisions. It enhances a friendly understanding of the OE and adversary intentions and behaviors. Other steps include listing the facts relevant to the situation or mission and submitting requests for information (RFIs)

early when information is needed.

- ★ Identify the adversary's system of key nodes and links (the behavioral, physical, or functional relationship between nodes) to help understand the threat networks.
- ★ Treat HHQ assumptions as facts for planning (seek to verify HHQ assumptions).
- ★ Also consider:
 - ★★ Geographic features and meteorological and oceanographic characteristics.
 - ★★ Population demographics (e.g., ethnic groups, tribes, ideological factions, religious groups and sects, language dialects, age and income groups, gender, public health issues).
 - ★★ Political and socioeconomic factors (e.g., economic system, political or tribal factions).
 - ★★ Infrastructure (e.g., transportation, energy, and information systems).
 - ★★ Law of War requirements as specified in US law, international law, or host-nation agreements (HNA).
 - ★★ Environmental conditions (e.g., earthquakes, volcanic activity, pollution, diseases).
 - ★★ Locations of intergovernmental organizations, foreign embassies, and non-governmental organizations operating within the OA.
 - ★★ Capabilities and intent of individuals, or organizations conducting cyber or information and influence operations.
 - ★★ Prevalence of misinformation, disinformation, or competitor operations in the OE related to joint force activities.

Develop Assumptions. An assumption is a belief about the current situation or future course of events, presumed to be true in the absence of facts. **Valid assumptions should be logical, and realistic, and are essential to continued planning.** It is also important to emphasize that assumptions identify risks to the plan and important decision points for executing branches and sequels. Assumptions encompass four basic categories: time, political, enemy, and friendly forces. The staff may prioritize assumptions for the commander in terms of what causes mission failure, what leads to partial mission failure, and what affects timing and tempo. Assumptions that make the mission fail are the highest priority and essential for identifying the branches and sequels in the plan.

Consider all assumptions in risk analysis. Potential plan changes due to incorrect assumptions should be addressed as branches or sequels. Include assumptions made by HHQ, however, challenge assumptions that appear unrealistic. A method for

identifying assumptions is presented in the figure, “Technique for Identifying Assumptions”. Assumptions should be continually reviewed to ensure validity. Additionally, assumptions should be turned into facts as soon as possible. Tools to validate assumptions include RFI, priority intelligence requirements (PIRs), and friendly force information requirements (FFIRs). PIRs, FFIRs, and RFIs are discussed later in this chapter.

An assumption, proved invalid, may lead to substantial changes in the approved plan. If an issue does not have this level of impact, it should not be an assumption. Never “assume away” potential problems, such as weather or likely adversary capabilities and options. Intelligence and planning staffs should keep commanders apprised of assumptions and knowledge gaps in mission analysis.

Technique for Identifying Assumptions

When identifying assumptions, consider the five A’s:

Awareness: the amount of warning or notification expected.

Assets: types and numbers available (e.g., airlift, allied capabilities, force availability).

Access: basing, overflight, and ability to stage operations.

Additional/other activity: degree and type of involvement of bordering or third-party nations (e.g., what else is going on?).

Authorities: commander-specific (e.g., JFC-delegated, cross-combatant command [CCMD] relationships).

Define and Develop a Problem Statement: The problem can be defined by determining what needs to be reconciled between the observed and desired system. Once the problem is defined, it should be communicated in a clear and concise problem statement.⁷ The statement considers how tension and competition affect the OE by identifying how to transform the observed system to the desired system—before adversaries begin to transform current conditions to their desired end state. The statement broadly describes the requirements for transformation, anticipating changes in the OE while identifying critical transitions. The problem statement provides a focus for planning efforts and helps ensure the conservation of resources during potential execution.

⁷ See JP 5-0, *Joint Planning*, for additional information.

Example Problem Statement

The XX AETF seeks to establish additional operating locations to enable ACE and defeat adversary actions. Adversaries are expected to impede the positioning of necessary resources and capabilities at remote locations. Host nation populations at desired operating locations may suffer negative economic impacts by assisting with positioning efforts. New operating locations must establish air base defense. The new operating locations will ensure a stable balance of power in the region that deters adversary aggression.

Analyze the Most Likely and Most Dangerous Threats to mission success. In AFPP, Airmen should consider the most likely and most dangerous threat that could prevent mission success and reaching the desired system. The threats should relate to the identified problem requiring action to move the observed system to the desired system. **Airmen should consider the most likely and most dangerous threat to mission success during tabletop exercises in COA analysis and wargaming.** Joint staffs at the operational level may focus on the most likely and most dangerous adversary COAs.

Develop Air Component Design. Air component design sets the stage for more detailed planning and is revisited during planning. **Airmen using the AFPP use the air component design to establish a framework for further planning.** The scope of a problem is revealed through air component design and adaptive thinking. The scope and complexity of the problem can vary. Well-structured problems may be addressed with tactics and techniques while medium-structured problems could use doctrine to develop potential solutions. Plans at the strategic level, however, usually deal with ill-structured problems that are interactively complex.

Adaptive thinking that considers CC's guidance, the OE, and **air component design produces an air component approach.** The analysis of air component design to produce an air component approach occurs throughout the AFPP. However, this analysis is particularly focused during planning initiation and mission analysis.

The air component design provides a logical framework comprised of several elements to conduct problem identification and framing. Establishing the air component design framework correlates to activities associated with conducting a comprehensive mission analysis. Furthermore, the air component design allows for the coordination of operations and activities to achieve objectives. Air component design is **iterative** and requires a consistently evolving understanding of the observed system and desired system to define the problem. Importantly, air component design empowers mission command in a contested, degraded, and operationally limited environment by providing a shared vision that analyzes key decisions and risks.

Planning Example: Operation ALLIES REFUGE, Ramstein AB

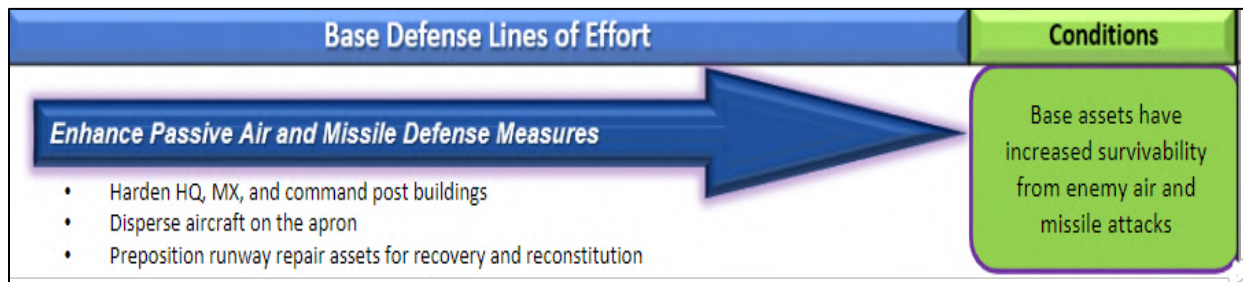
In 2021, Ramstein AB conducted crisis planning to assist in the evacuation of Special Immigrant Visa applicants and other vulnerable Afghans. With little time to prepare, the base cleared out two of its aircraft hangars and set up hundreds of cots to provide temporary shelters. However, planning efforts quickly changed when base leaders realized Ramstein AB would become a primary evacuation hub for US European Command. Ramstein AB enhanced planning efforts to encompass medical, logistics, maintenance, civil engineering, security, and food services personnel to provide necessary services and a haven for thousands of evacuees.

An undertaking of this magnitude required enormous resources, planning, and people. The modified planning efforts necessitated adjustment of non-essential services and several course of action alterations. Commander-selected COAs were developed into an approved plan that was shared among several organizations. The clearly written plan developed a shared understanding of the mission, intent, tasks, logistics, and authorities. The successful planning efforts resulted in a historic evacuee operation and the processing, security, and safety of over 35,000 Afghans.

In order for commanders and staffs to effectively develop their air component approach, they must have a clear understanding of the HHQ guidance, observed system, desired system, problem, and its root cause. This understanding allows them to analyze the operational approaches of HHQ and create their air component approach. **Commanders decide which air component design elements are depicted in their air component approach.** Air component design elements are:

- ★ *Objective*: Clearly defined, decisive, and attainable goal toward which an operation is directed.
- ★ *Desired end state*: The set of required conditions that define the achievement of the commander's objectives.
- ★ *Root cause*: The fundamental underlying reason or factor that is driving the problem, essentially the most critical aspect that needs to be addressed to solve the issue. Teams discuss lines of effort (subordinate problem sets) to address the root cause utilizing the "5 Why Analysis" method by asking "Why" as many times as needed (minimum 5) to identify the root cause of the initial problem statement.

- ★ **Effects:** Physical and/or behavioral state of a system that results from an action, a set of actions, or another effect. Desired effects should link to objectives. Effects should be measurable. Effects should be distinguishable from objectives and tasks as they support conditions for success. For more information on Effects see Appendix C, Actions and Effects.”
- ★ **Culmination:** Point in time and/or space when the operation can no longer maintain momentum.
- ★ **Physical characteristics of the OE:** Tangible factors of the OE to consider are physical size, weather/climate, and geography.
- ★ **Lines of Operation (LOO) and Lines of Effort (LOE):**
 - ★★ **LOO:** Defines the interior or exterior orientation of the force in relation to the enemy. Describe and connect a series of decisive actions that lead to control of a geographic or force-oriented objective.
 - ★★ **LOE:** Links multiple tasks and missions using the logic of purpose, cause and effect, to focus efforts towards establishing operational level objectives that can lead to strategic objectives. An LOE may represent subordinate problem sets of an overarching problem. The Base Defense Lines of Effort example illustrates how tasks can support a line of effort to achieve a condition.



- ★ **Decisive points (DPs):** Key terrain, key event, critical factor, or function that, when acted upon, enables a commander to gain a marked advantage over an enemy or contributes materially to achieving success (e.g., creating a desired effect, or achieving an objective).
- ★ **Direct and Indirect Approach:** The manner in which a commander contends with the problem. The direct approach addresses the problem’s principal strength by applying effort directly against it. An indirect approach addresses the problem by applying effort against key components of the problem to reach an effective solution.
- ★ **Operational reach:** The distance and duration across which a joint force can successfully employ military capabilities.

- ★ *Arranging Operations*: Commanders may consider organizing operations according to depth, timing, and tempo. Some ways to organize complex operations include phases, branches, sequels, and operational pauses.
- ★ *Anticipation*: Consider what might happen and look for indicators of forecasted events.
- ★ *Forces and Functions*: Comprised of all friendly, enemy, and adversary forces, capabilities, and associated functions. The functions include the mechanisms the commander may use to solve the problem (e.g., defeat the enemy or stabilize the situation).

Determine and Plan for Authorities. Commanders and planners at all echelons should be aware of their commander's authorities either in the operational or administrative branches of the chain of command. In the joint force, all authorities reside with the JFC and are delegated to subordinate commanders, with considerations for further delegation to lower echelons, through an appropriate order. Planning for the delegation of **appropriate and legitimate** authorities in contested, degraded, and operationally limited environments should be considered. Mission command requires leaders to ensure subordinates are provided and aware of the authorities they possess in the execution of their responsibilities outlined in the commander's intent. Distributed control enables commanders to delegate authorities to plan, prepare, execute, and assess activities to dispersed locations for an effective span of control and maintain the initiative, particularly in contested environments. Distributed control may be enduring or temporary to allow execution at multiple echelons. Conditions-based authorities are authorities that are often limited in duration and identified in plans approved by authorized commanders to facilitate execution that meets commander's intent. Failure to provide a subordinate with sufficient authority to act may inadvertently encourage them to operate beyond limitations. Alternatively, providing a subordinate with authority that exceeds situation or mission requirements may result in chaotic, uncoordinated operations. Plans and orders should clearly delineate distributed control and authorities.



"It is a given in future conflicts that the joint force will be conducting operations in a contested environment. We must be prepared to execute in a degraded C2 environment where clearly delineated and forward-thinking commander's intent will be a requirement. It is imperative senior leaders provide our commanders with conditions-based authorities delegated to the lowest capable and competent level and empower command by negation to accept the appropriate level of risk, all while working toward moments of clear C2."

—General Charles Q. Brown, Jr., 22nd Chief of Staff, USAF
Remarks as Commander, Pacific Air Forces, 2019

Determine Operational Limitations (Constraints and Restraints). Constraints and restraints are actions required or prohibited by higher authority that limit the commander's freedom of action. Additionally, a constraint and/or restraint can be a restriction imposed by conditions and circumstances. **Restraints are HHQ requirements placed on the command that prohibit action.** An example of a restraint is a prohibition from conducting a movement to a specific location (e.g., a bordering nation). **Constraints are HHQ requirements placed on the command that dictates action.** An example of a constraint is HHQ's direction to fly a certain mission as part of an operation within a specific time, sequence, or phase.⁸

Identify Specified, Implied, and Essential Tasks. For planners to accurately develop a mission statement and air component approach, they should be able to identify the three types of tasks that will be used to solve the problem. Specified tasks are assigned to the commander verbally or in writing by the HHQ. Specified tasks often become essential tasks. Implied tasks are tasks that are revealed by mission analysis but not explicitly stated by HHQ guidance. Implied tasks may NOT be essential to mission accomplishment but should be identified to inform further planning. Determining the essential tasks from the specified and implied tasks ensures that the commander addresses what needs to be accomplished for mission success.

- ★ **Specified tasks** are tasks specifically assigned to the commander verbally or in writing by the HHQ. These are normally written in the 'Tasks to Subordinates' paragraph and/or 'Coordinating Instructions' in the HHQ plan or order.
- ★ **Implied tasks** are tasks that mission analysis reveals the commander must perform (or prepare to perform) but are not stated explicitly in an order or elsewhere. Implied tasks do not include routine tasks or standard operating procedures (SOP) inherent in most operations. Guidance for identifying implied tasks is presented in the Deriving Implied Tasks figure.

Deriving Implied Tasks

To derive implied tasks, focus on analyzing the HHQ mission, intent, concept, and specified tasks, to include specified tasks to other units. As a technique, ask:

- ★ "In this situation, what other major tasks would my echelon normally perform?"
- ★ "What in the HHQ mission, intent, or concept implies a major (but unstated) task for the commander?" For example, if the HHQ has a deter portion of the mission, then contributing to deterrence could be an implied task.
- ★ "What task to another unit, component, or agency is likely to require significant support from the commander?"

⁸ The Law of War always applies and should be considered, along with ROE and Rules for the Use of Force, in evaluating constraints and Restraints. Refer to the Department of Defense Law of War Manual.

- ★ **Mission essential tasks** are a subset of specified and/or implied tasks the commander must accomplish to succeed at the mission. Mission essential tasks are the basis for the commander's mission statement.

Develop Commander's Mission Statement. A mission statement is a short sentence or paragraph that provides a clear statement of the action to be taken and the reason for doing so. The mission statement is written to cover the entire operation. It is the basis for planning and is included in planning guidance, staff estimates, commander's estimates, CONOPS, and the completed plan/order. The mission statement can be revised if initial circumstances change. A mission statement once approved drives the remainder of the AFPP. The commander's mission statement should address:

- ★ *Who:* The entity responsible for the mission.
- ★ *What:* The essential tasks.
- ★ *When:* Often expressed by 'when directed,' 'on order,' or 'be prepared to.'
- ★ *Where:* In the designated geographic area.
- ★ *Why:* The reason for accomplishing the essential tasks.

An example mission statement highlighting the 5 W's is presented in the Example Mission Statement figure.

Example Mission Statement

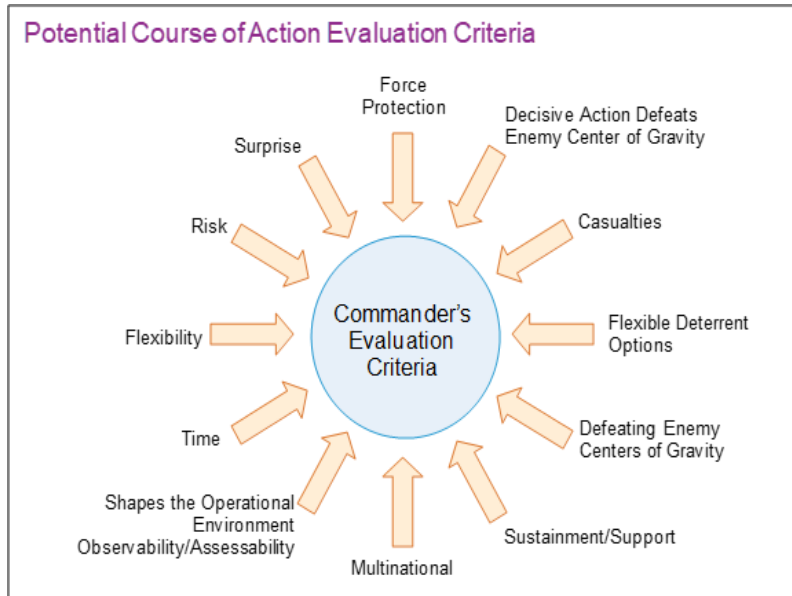
When directed, the 325th Fighter Wing will stand up the crisis action team, move required personnel and equipment to emergency divert locations, harden base facilities at Tyndall Air Force Base, open storm shelters, and establish necessary life support to ensure the safety of personnel and critical resources during hurricane conditions.

Conduct Initial Resource Review. The initial resource review in mission analysis is a first look at all resource requirements e.g., time, personnel, funding, etc.) to determine broad requirements and identify initial shortfalls. This review identifies resources required to accomplish the specified and implied tasks and highlights any disconnects that may exist. Resource shortfalls become a part of the risk discussion.

Develop COA Evaluation Criteria. COA evaluation criteria are the measures used for developing and determining an acceptable COA. The commander should include these criteria in initial planning guidance and may be determined from the commander's intent. Planners should seek to obtain COA evaluation criteria as early in the planning process as possible to inform planning efforts. They help determine supporting objectives, effects, tasks, and if or when to move to the next major operation or phase. These criteria also

become the basis for assessment.⁹

The commander and staff use the COA evaluation criteria during AFPP step five to select the best COA. **Developing initial evaluation criteria during mission analysis or as part of the commander's initial planning guidance helps eliminate bias before COA analysis and comparison.** Evaluation criteria address factors that affect mission success or may cause failure. Commanders adjust criteria and weighting according to their own experience and vision. Suggestions for possible COA criteria are in the figure, "Potential Course of Action Evaluation Criteria".



Develop Commander's Risk Assessment. A risk assessment should be conducted at all levels of command. There are two categories of military risk: risk-to-force (RF) and risk-to-mission (RM). RF is the probability and consequence of current and contingency events causing harm to the provisioning and sustainment of sufficient military resources. RM is the probability and consequence of current and contingency events causing harm to current or future military objectives.¹⁰

Commanders may delegate risk management decisions to lower echelons, however, they maintain the responsibility for overall risk. Airmen and wing A-staffs should be familiar with how to use the joint risk framework and discuss probabilities and consequences of risk events.

Staffs use historical data, intuitive analysis, modeling and simulation tools as available, and judgment to consider mission risk, mitigation, and residual risk. Risks are evaluated and presented by assessing the probability of occurrence and severity of consequence.

Staffs should address assumptions and PIR during risk assessment. PIRs are considered for risks aligned with how they impact the mission. During the risk assessment step, you may encounter additional valid and necessary assumptions. Planners should address the potential opportunities and consequences of assumptions being valid/invalid. Risk articulation is the most important conversation the planning team can have with the commander and the commander will have with HHQ. With every risk, there is an

⁹ See AFDP 3-0.1, *Command and Control*, for additional information.

¹⁰ See CJCSM 3105.01B, *Joint Risk Analysis Methodology*, for additional information.

opportunity to discuss both positive and negative aspects.

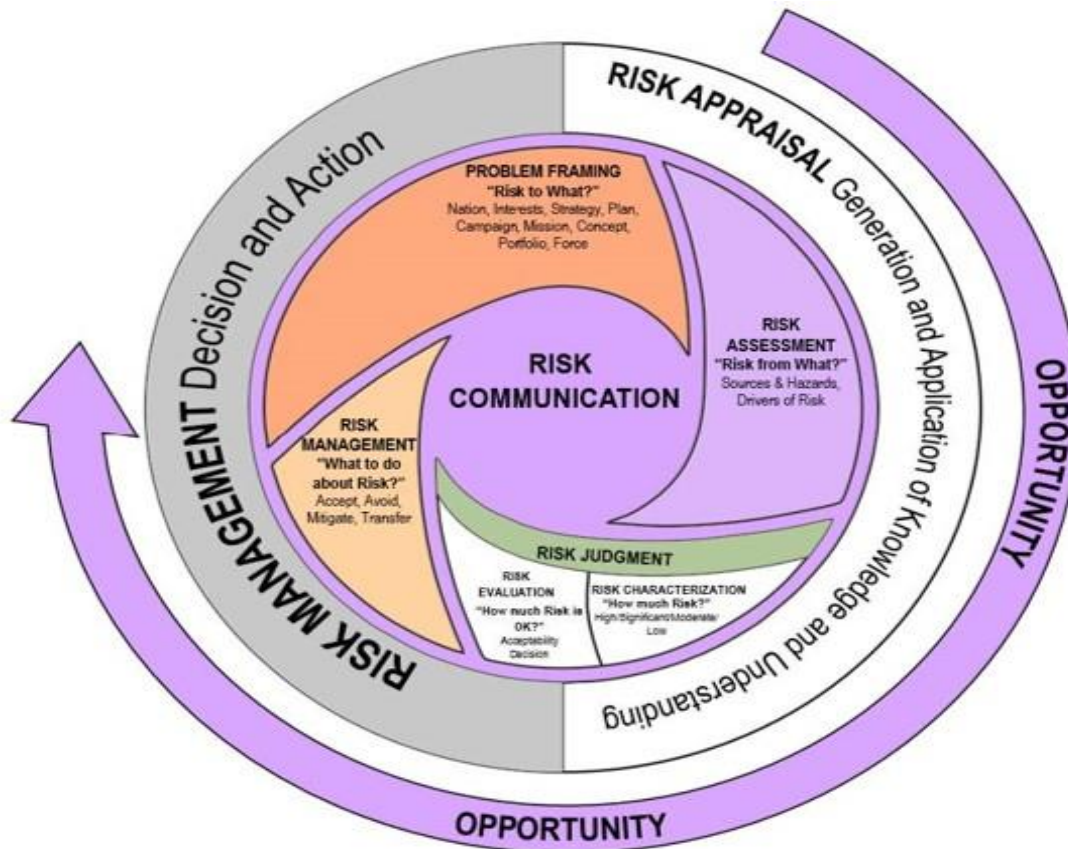
Guidance on determining the probability and consequence of an event is provided in the figure, “Probability and Consequence of Event”.

Probability of Event (P)	Consequence of Event (C) to NMS
Very Likely (~81-100%)	Extreme • RM, Mission Failure, Objectives Unachievable • RF, No Sourcing Solutions Exist for Critical Requirements
Likely (~51-80%)	Major • RM, Objectives Minimally Achieved (consider time, priority) • RF, Shortfalls Exist for Critical Requirements
Unlikely (~21-50%)	Modest • RM, Objectives Mostly Achieved (consider time, priority) • RF, Worldwide Sourcing Solution Exist for Most Requirements
Very Unlikely (~0-20%)	Minor • RM, Mission Success, Objective Achievable • RF, Joint Force Fully Sustained and Requirements Sourced

Probability and Consequence of Event

Risk should be discussed in planning using the joint risk framework. Planners may start identifying risk by asking about what resources are in jeopardy. Upon determination of what is at risk, planners assess what is placing the resource in peril. The risk assessment is provided to the appropriate commander to make a risk decision. Depending on the situation and the risk acceptance authority given to the commander, they may decide to accept, avoid, mitigate, or transfer the risk. Commanders and staffs should consider how risk management decisions may displace risks to other forces. Risk assessments and commander’s risk management decision should be addressed in the commander’s intent statement, CONOPS, or as additional guidance that describes what risks the commander allows subordinates to accept and under what conditions.¹¹ **The joint risk framework figure applies to the JPP, JPPA, and AFPP.**

¹¹ For additional information on risk assessment, see CJCSM 3105.01B, *Joint Risk Analysis Methodology*, and JP 5-0 *Joint Planning*.



The Joint Risk Framework

Determine Initial Commander's Critical Information Requirements. CCIRs are elements of information the commander identifies as being critical to facilitate timely decision-making. CCIRs belong exclusively to the commander and are not static. Commanders refine and update both decision points and associated CCIRs throughout the operation. **CCIRs are identified when using the JPP, JPPA, and AFPP.** When developing CCIRs at subcomponent commands, focus on decision points related to the air component approach. CCIRs should also be coordinated with the commander's servicing judge advocate to ensure compliance with the law and existing rules of engagement.

CCIRs may include:

- ★ *Priority intelligence requirements:* PIRs focus on the adversary and the OE and are often tied to commander's decision points. Airmen determining PIRs focus on the problem's root cause, observed system, and desired system in relation to commander's decision points (AFPP). Commanders at all levels use PIRs to drive ISR operations to support decisions and operations.
- ★ *Friendly force information requirements:* Information regarding the status of friendly forces and supporting capabilities associated with the operation.

- ★ *Essential elements of information (EEIs)*: Intel gaps when PIRs cannot be answered and drive the development of detailed EEIs. EEIs form the basis for RFIs by outlining specific information requirements.
- ★ *Requests for information*: RFIs are requirements for intelligence information or products to support an operation not necessarily related to standing requirements or scheduled intelligence production. RFIs may satisfy PIRs and FFIRs.¹²

Initial CCIR identification during mission analysis can assist in identifying anticipated decision points and related information requirements. At this point, neither a COA nor a CONOPS exists, therefore, the intent is to capture ideas that may become CCIRs.¹³

Determine Decision Points. Decision points often require a commander decision that can be detailed on a decision support matrix (DSM). DSMs are useful for informing commander decisions and conducting COA analysis and wargaming. An example DSM template is found below.¹⁴

Decision Support Matrix							
Decision Point	Named Area of Interest	Target Area of Interest	Event	Decision Criteria	Sources of Decision Criteria	Assets Available	CJTF Actions
					SIGINT HUMINT MASINT IMINT		
Adversary's Assumptions			Adversary's Possible Decisions			Adversary's CCIRs	
Risk		Mission Impact		Probability		Mitigation	
Notes							
Weather, Solar, Sea State, Maneuver Rates...							

Develop Commander's Intent Statement. The commander's intent statement should articulate their vision of the mission's purpose, end state, and risks. A well-framed intent statement provides subordinates with the ability to exercise disciplined initiative. Commanders are encouraged to write their own commander's intent as it is pivotal to mission command. Commander's intent may change based on the scope of the mission

¹² For more information on PIRs, EEIs, and RFIs see Air Force Tactics, Techniques, and Procedures 3-4.14, *Contingency Intelligence Network*.

¹³ Focus on decision points related to the air component approach. Even early in planning, it may be apparent there are critical commander decisions that are required regardless of the COA selected. CCIRs are more applicable to the JPP and JPPA but can still be identified in the AFPP.

¹⁴ For more information on decision points, see JP 5-0, *Joint Planning*.

and with the planning process. The final intent statement is published in the order.

- ★ **Purpose.** Purpose delineates the reason for the military action with respect to the mission of the next higher echelon. The purpose explains why the military action is being conducted. Purpose helps the force pursue the mission without further orders, even when actions do not unfold as planned. Thus, if an unanticipated situation arises, participating commanders understand the purpose of the forthcoming action well enough to act decisively and within the bounds of the higher commander's intent (JP 5-0).
- ★ **End State.** The end state is a set of desired future conditions the commander wants to exist when an operation ends. Commanders may describe the operation's end state by stating the desired system and conditions of the friendly force in relation to the desired conditions of the enemy, terrain, and civil considerations. A clearly defined end state promotes unity of effort among the force and with unified action partners.
- ★ **Risks.** The intent statement outlines the identified risk(s) throughout the planning process. Specifically, highlight risk accepted by the commander. Significant known risks requiring additional approval are combined with constraints and restraints.

Example Commander's Intent Statement

Purpose: The purpose of this operation is to ensure air base defense by hardening facilities, employing deception schemes, establishing defense in depth force protection strategies, supporting ballistic missile defense operations, and supporting air defense sortie generation to ensure unhindered base operations.

End State: At the end of this operation, facilities will be hardened, enemy surveillance attempts degraded, force protection threats mitigated, air and missile defense units supported, and base operations fully functioning.

Risk: There is a high probability that forces will be in danger of adversary fires should air base defense efforts fail, which results in a major consequence. This presents a significant risk to air base forces. There is a likely probability that degradation to base facilities will hinder mission generation capability. This poses a major risk to air base forces' accomplishment of the mission.

Prepare or Update Staff Estimates. A staff estimate is an evaluation of how factors in a functional area support and/or impact the mission. The staff estimate process is central to formulating or updating the plan to accomplish the mission. Staff members continuously refine products as planning continues. Staff estimates focus on collecting information from a given functional area to help the commander and staff understand the situation and conduct mission analysis. The preparation and updating of staff estimates will normally be conducted by joint staffs. Airmen using the AFPP will normally reference staff estimates during planning. In staff estimates, each element:

- ★ Reviews the mission and situation from their staff functional perspective.
- ★ Examines the factors and assumptions for which they are the responsible staff.
- ★ Evaluates how factors in their functional area may influence each COA.
- ★ Provides detailed analysis of how that functional area would best support a given COA.
- ★ Reviews the tentative force structure and determine additional lift requirements.
- ★ Informs HHQ staff of logistics feasibility by leveraging the logistics overview.
- ★ Identifies existing contract availability, point of debarkation (POD), percentage of PODs, inventories, and other distribution infrastructure available.
- ★ Works to identify operational capability requirements in the plan.

Staff Estimate Functional Area Examples:

- ★ Personnel estimate: Manning.
- ★ *Intelligence estimate*: JIPOE, intelligence community factors, collection and processing, exploitation, and dissemination capabilities.
- ★ *Operations estimate*: Offensive ops, air defense, air refueling, airlift, airspace control, C2.
- ★ *Logistics estimate*: Time-phased force deployment data (TPFDD), beddown, fuel, munitions, water, supply.
- ★ *Communications estimate*: Staff, systems, and communication requirements.
- ★ *Information estimate*: Evaluation of how relevant actor perceptions, attitudes, and behaviors affect operations planning and execution.



“There will be challenges and this is much greater than just an AOC going away. Is the person that I am working for describing their priorities clearly? Are those priorities meeting with the realities of the operational environment? Is their feedback mechanism going back to command echelons so they can understand if they need to adjust their priorities?”

—General Mike Minihan, USAF

Remarks as Commander, Air Mobility Command, 2024
Doctrine Podcast: Lessons Learned in Doctrine

The exact format and level of detail vary among commands and staff sections based on theater specifics and other factors. Principal elements normally include mission, situation, functional considerations, and conclusions.

Present Commander’s Mission Analysis Brief. The mission analysis brief is presented to the commander. The mission analysis brief is to ensure the commander and planning staff have a common understanding of the environment, problem, situation, and mission and is essential to forging unity of effort. It focuses on relevant conclusions reached during mission analysis. The brief should include representatives from the entire staff and partner organizations. The “Common Mission Analysis Briefing Elements” figure depicts a common framework for the commander’s mission analysis brief.

Common Mission Analysis Briefing Elements

1. Purpose: Provide draft commander mission, obtain commander guidance.
2. Identify the relevant operating area (an area of interest may also be briefed).
3. Initial situation brief.
4. HHQ guidance, mission, and intent statements.
5. Problem statement, root cause, and subordinate problem sets.
6. Facts.
7. Assumptions.
8. Constraints and restraints.
9. CC's problem and key components.
10. End state.
11. Specified, implied, and essential tasks.
12. Commander's initial risk assessment.
13. Time analysis-Including projection of planning milestones.
14. Initial CCIRs.
15. Initial resource analysis.
16. Other guidance as applicable.
17. Air component approach, as required.
18. Proposed commander mission and intent statements.

NOTE: The briefing elements fall into two categories. Elements 1-10 "set the problem" and usually reflect information already provided. Elements 11-17 provide specific actions and needs.

MISSION ANALYSIS RESULTS

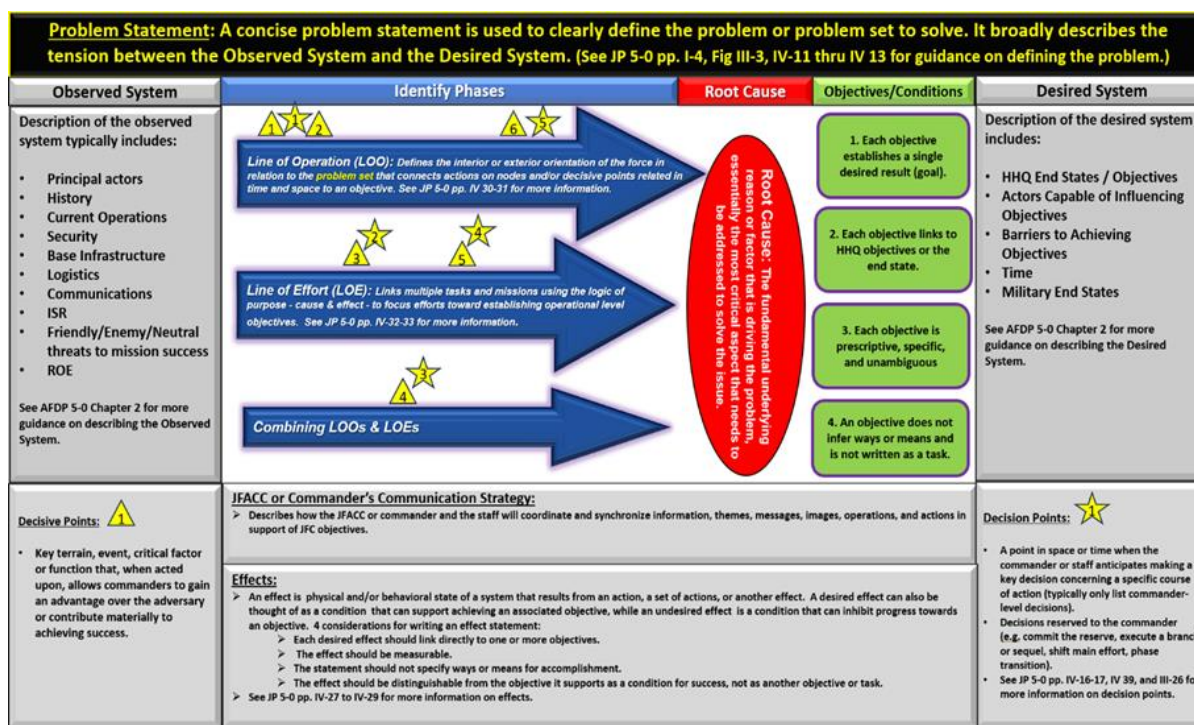
The results of mission analysis are an air component approach, specified, implied, and mission essential tasks, component(s) of the problem that need(s) to be addressed, mission statement, intent statement, updated planning guidance, and initial CCIRs. An additional output of mission analysis is the commander's approval of the staff's understanding of the observed and desired systems and refined guidance for further planning and design refinement.

Publish Commander's Refined Planning Guidance. Following the mission analysis brief, staff note guidance provided by the commander to guide further planning. Mission analysis outcomes, OE changes, and updated HHQ guidance can drive a requirement for refined planning guidance. Refined planning guidance may address the following:

- ★ Approved mission statement.
- ★ Updated commander's intent statement.

- ★ Overall narratives two echelons up.
- ★ Key elements of the OE.
- ★ Key assumptions and operational limitations.
- ★ End state and termination criteria.
- ★ Acceptable or unacceptable levels of risk in key areas.
- ★ The role of agencies and multinational partners in the pending operation and any related special considerations as required.
- ★ Air component approach.

Air Component Approach. The air component approach is a fundamental output of air component design and initially produced at the end of mission analysis. **Air component design elements are normally used to construct the air component approach.** The approach is a commander's description of the broad actions the force can take to achieve an objective, support HHQ objectives, and attain a desired end state. The ultimate purpose of an air component approach is to describe how to achieve a desired system that leads to an enduring advantage for friendly interests. The accomplishment of all objectives should lead to the desired system and set of future conditions. Tactical tasks necessary to support lines of effort or lines of operations should be depicted on the air component approach. An "Air Component Approach Template" figure is provided below. For greater detail, see Appendix D, "Air Component Approach Template."



Air Component Approach Template

HHQs may develop an operational approach that informs the air component approach efforts. An operational approach primarily differs from an air component approach by focusing on the center(s) of gravity.

CHAPTER 5: COURSE OF ACTION DEVELOPMENT

The third step of the AFPP is COA development. A COA is “any sequence of activities that an individual or unit may follow” or “scheme developed to accomplish a mission.” COA development seeks to determine “how” to use capabilities made available to accomplish the “what” derived during mission analysis. The COA is framed around time, space, and purpose. Airmen have a unique perspective on developing COAs focused on generating effects. **Commander involvement is critical.** The necessary degree of commander guidance and involvement depends on the situation.

COA DEVELOPMENT INPUTS

COAs provide potential ways to address an identified problem and create desired changes in the observed system resulting in the desired system. Inputs that inform COA development are the mission analysis results, refined air component design, and a refined air component approach.

Additional Planning Consideration

All major C2 activities may be enhanced with AI and human-machine teaming to optimize operations. First, data synthesis and processing capabilities can help planners make sense of a complex OE. Commanders and their staffs can then use their understanding of the OE to inform their air component design. As planners work to develop plans, AI can provide recommendations that influence COA development, and AI-enabled wargaming systems may allow iterative review of potential COAs in a shortened amount of time to improve the COA approval process. Subsequently, approved plans will lead to streamlined preparation. AI can make logistics and movement recommendations, enhance intelligence exploitation, and accelerate execution decisions. As the OE evolves during execution, AI can aid battle management nodes as they proactively maneuver forces to gain an advantage over the adversary. Finally, AI assisted analysis of friendly effects and adversary reactions can support operational assessment.

NOTE: AI and human-machine planning may enhance planning efforts but should not fully replace planning. An understanding of the mission and OE are cognitive planning products often considered more important than completed plans.

COA DEVELOPMENT ACTIVITIES

COAs should take into consideration the time available (time to plan), the most likely impact on the OE, and the most dangerous impact on the OE. Distinguishability is typically the most challenging facet of developing multiple COAs. Airmen typically build COAs in a "divide and conquer" fashion vs sequentially, often creating similar COAs. Staffs should consider the elements of the 5-paragraph order format as they conduct COA development. For additional information on the five-paragraph order format see Appendix A, "Plans and Order Format" and Appendix E, "Example Modified Five-Paragraph Order." The figure, "COA Development Activities" provides more information on approaches to solving the problem.

COA Development Steps

1. Brainstorm COA(s).
2. Chart or Visualize COA(s).
3. Prepare COA Statement, Sketch, and Task Organization.
4. Present COA Selection Brief.
5. Refine Air Component Design and COA(s).
6. Continue Vertical and Horizontal Parallel Planning.

DEVELOPMENT STEP 1: BRAINSTORM COA(S)

A COA is a broad potential solution to an identified problem. COA brainstorming generates options for follow-on analysis and comparison that satisfy the commander's intent and planning guidance. During COA brainstorming, planners use knowledge, skills, experience, creativity, judgment, and products developed during mission analysis to develop broad concepts. Typical activities in this step include (see "COA Development Activities" figure):

- ★ Review information from mission analysis HHQ operational approach, and air component approach if available.
- ★ Determine COA development technique-simultaneous or sequential.
- ★ Consider the elements of air component design and air component approach.
- ★ Develop alternative COA themes and approaches.
- ★ Analyze forces and capabilities, including force flow issues.
- ★ Identify command relationships.
- ★ Integrate staff estimates and updates.

DEVELOPMENT STEP 2: CHART OR VISUALIZE COA(S)

This step organizes and synchronizes COAs. Graphically depicting the COA in space and time allows the commander to visualize the COA and synchronize it with current operations and requirements. In some cases, a sketch or graphic may be useful.

- ★ Analyze mission analysis data and develop an operational timeline.
- ★ Translate specified and implied tasks into objectives using an effects-based approach by general phases.
 - ★★ Phases, such as "shaping", or "redeployment operations", are used to achieve one or more major objectives. Transitioning to the next phase indicates a change in the emphasis, e.g., "stability ops" to "redeployment ops." A single-minded focus on specific phases can help bind various facets of the problem to focus planners' ideas.¹⁵
- ★ Identify the sequencing of the operation for each COA as appropriate.
- ★ Identify main and supporting efforts by phase.
- ★ Develop initial COA sketches.
- ★ Develop initial COA statement.

DEVELOPMENT STEP 3: PREPARE COA STATEMENT, SKETCH, AND TASK ORGANIZATION

This step creates a detailed narrative, graphical depiction, and a task organization methodology that explains the "5-Ws" and "how" the COA accomplishes the mission.

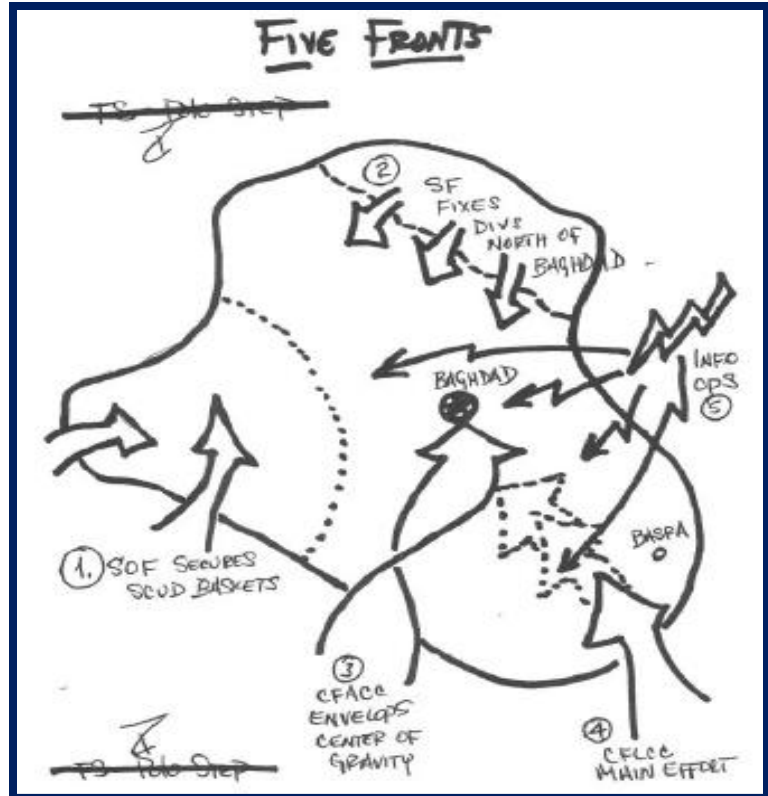
COA Statement. The COA statement is a narrative describing the CONOPS. It is typically written chronologically and addresses all phases of the operation. The statement addresses "who" (type of forces) executes "what" (tasks), expands on "when", shows "where" the commander's mission statement applies, explains the "why", and adds "how". The "how" of the statement could address the means, methods, and priority of effort that makes a COA distinguishable.

¹⁵ For additional information on operational phases and phasing, see JP 3-0, *Joint Campaigns and Operations*.

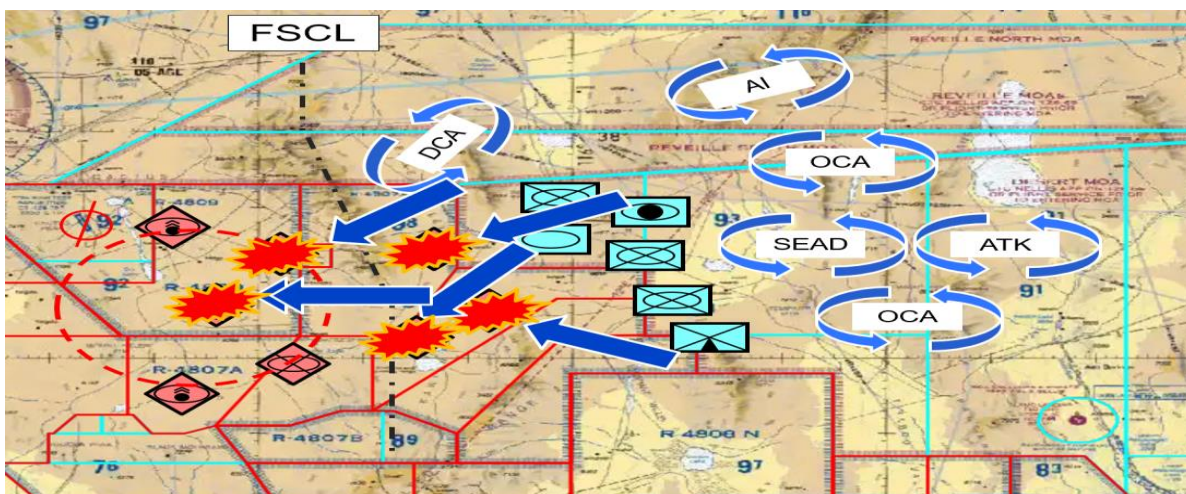
Sketch. COAs are often depicted and briefed using a combination of CONOPS graphics and sketches (the execution portion of the COA). COA sketches visualize how each COA can vary and evolve with maps and charts. An example sketch is the “Five Fronts” sketch from Gen Tommy Frank’s book, *American Soldier*.

Task Organization. The task organization should detail the relationship between tasked forces, the commander’s authorities, the overall purpose of the COA, and supporting organizations. Resourcing (main effort) should be considered in larger operations during task organization construction.

In general, air COAs may vary with respect to ends, ways, means, and/or risks.¹⁶ The staff should see how they can best synchronize (arrange in terms of time, space, and purpose) the actions of all the elements of the force once they begin to visualize COA alternatives.



Five Fronts sketch from Gen Tommy Frank’s book, *American Soldier*



Example COA Sketch

¹⁶ For more information on COA distinguishability, see JP 5-0, *Joint Planning*.

Test The Validity of Each COA. A "valid" COA should be feasible, acceptable, suitable, distinguishable, and complete. A COA should accomplish all HHQ tasks and any additional tasks identified during mission analysis. While definitions are given below, the commander retains the final judgment for COA requirements. In addition to any requirements given by the commander, examine each prospective COA for validity using the following screening criteria:

- ★ *Feasible*: The COA can accomplish the mission within the established time, space, and resource limitations.
- ★ *Acceptable*: The COA should balance cost and risk with the advantage gained.
- ★ *Suitable*: The COA can accomplish the mission within the commander's intent and planning guidance.
- ★ *Distinguishable*: Each COA should differ significantly from the others (such as scheme of maneuver, lines of effort, phasing, use of reserve forces, and task organization).
- ★ *Complete*: A COA should incorporate the following information:
 - ★★ How the decisive operation leads to mission accomplishment.
 - ★★ How shaping operations creates and preserves conditions for the success of the decisive operation or effort.
 - ★★ How sustaining operations enable shaping and decisive operations or efforts.
 - ★★ How to account for offensive, defensive, and stability or civil support tasks.
 - ★★ Tasks to be performed and conditions to be achieved (a COA should accomplish all tasks from HHQ, and additional tasks identified during mission analysis).

DEVELOPMENT STEP 4: Present COA Selection Brief

If required, a COA brief provides the commander an opportunity to provide refined guidance, including further analysis, revisions, wargaming direction, changes to design, and updated COA evaluation criteria. The "Typical COA Briefing Elements" figure guides items commonly discussed during the brief.

Typical COA Briefing Elements

1. Purpose.
2. HHQ mission and intent statements (two echelons higher).
3. Commander's mission and intent.
4. Intelligence estimate and environmental factors (JIPOE).
5. Logistics and sustainment considerations.
6. Timeline, phasing, and sequencing (correlate with previous steps).
7. Broad COA differences.
8. Depiction of each COA (e.g., charts, sketches, and statements).
9. Logistics and sustainment considerations.
10. Shortfalls, additional support, and authorities required.
11. Risks and identified mitigation measures.

DEVELOPMENT STEP 5: REFINE AIR COMPONENT DESIGN AND COA(S)

Continue to refine air component design and COAs. Changes in one COA, situation, or air design element may require corresponding changes to other COAs. Properly managed, air component design refinement is in an iterative process that keeps all products aligned and provides more refined COAs.

DEVELOPMENT STEP 6: CONTINUE VERTICAL AND HORIZONTAL PARALLEL PLANNING

Planners should integrate planning efforts with staff counterparts at HHQ and parallel agencies as appropriate. Airmen at the wing level should ensure that their planning supports the JFACC and/or commander, Air Force forces' (COMAFFOR's) guidance and direction. Expect planning adjustments as additional details become available and permit lower echelons to start planning and generate questions. If applicable, issue a preparatory order or Fragmentary Order (FRAGORD) to subordinates.

Providing sufficient guidance and time to subordinates is important in planning. A time consideration for planning is provided in "The One-Third, Two-Thirds Rule for Air Force Planning" figure.

COA DEVELOPMENT RESULTS

Proper COA development refines air component design and may produce six results: CONOPS, C2 structure and command relationships (COMREL), required capabilities, deployment and sustainment concept, COA statement (narrative description), and valid COAs.

CONOPS. A CONOPS is a verbal or graphic statement of the commander's intent to identify the overall purpose. It includes elements like a high-level view of the environment (e.g., terrain, multidomain considerations, weather, threats), an approach for how forces are to be organized, employed, and coordinated, and an initial list of tasks and responsibilities to subordinate and supporting organizations. Frequently presented as a graphic, the CONOPS is the foundational element that planners can refine and develop the COA into the execution paragraph of the plan or order.

The One-Third, Two-Thirds Rule For Air Force Planning

Planning can be a time-intensive process for both superior and subordinate organizations. Air Force commanders and planners should be aware of the timeline for subordinates to plan. As a general rule when developing a planning timeline divide the time available into thirds and seek to utilize a third of the time available for their staff. The remaining two-thirds of the time should be preserved for subordinate planning. Commanders can publish updates in the OPORD, or following Fragmentary Orders (FRAGORDs), if the situation, timeline, or plan changes.

C2 and Command Relationships (COMREL). This deliverable should narratively identify required authorities (e.g., tactical control [TACON]) or support requirements (e.g., direct support) across the entire COA (and COA phases, if appropriate) to achieve the defined objectives and end state. A simple diagram may accompany the narrative C2 and COMREL. This deliverable forms the basis of the command, signal, and communications paragraph of the resulting order or plan, and where appropriate, informs Annex J of a HHQ CONOPS or plan.¹⁷

Required Capabilities and Support. This deliverable identifies the capabilities and support necessary including additional capabilities and support needed that the unit does not own or control. Higher level assistance may be required to leverage outside agencies or other Service interdependencies.¹⁸

Deployment and Sustainment Concept. This deliverable asks how planners envision "setting the force" and how logistics for sustainment should be performed.

COA Statement. This deliverable summarizes the who, what, when, where, why, and how, emphasizing the decisive action that is necessary to change the situation firmly in the commander's favor. The COA statement should provide details on how to accomplish the mission statement.

¹⁷ See JP 5-0, *Joint Planning*, for additional information.

¹⁸ This deliverable may inform paragraph four in order production or an additional annex in a plan.

CHAPTER 6: COURSE OF ACTION ANALYSIS AND WARGAMING

COA Analysis and Wargaming is a critical step of the AFPP, but it is often overlooked due to planners' inexperience and/or time mismanagement. Wargaming provides the opportunity for planners to analyze the problem to improve the COA(s). This step provides an opportunity to visualize the friendly COAs. It is the best opportunity to analyze COAs against threats to mission success and adversary capabilities before execution. Planners should seek to accomplish some degree of analysis or wargaming regardless of how simple the problem is. **Tabletop exercises focused on the most likely and most dangerous threat to mission success are best suited for use at the AETF and wing level.** Planners using the JPP and JPPA will focus on the adversary's most likely and most dangerous COAs.¹⁹

The Importance of Wargaming

Wargaming may be manually conducted or computer-assisted. Wing A-staffs and Airmen at all echelons may be limited to manual wargaming capabilities. AI and human-machine teaming may enhance future wargaming capabilities and allow for multiple iterations of the wargame in a compressed timeframe. Wargaming's value is underscored by the more than fifty wargame iterations conducted prior to the execution of Operation DESERT STORM. COA refinement is often necessary due to wargaming. Results from each iteration should be analyzed to enhance understanding of the problem, mission, and OE. The value of planning is greater than the plan itself.

Wargaming for non-operational problems. Wargaming is a recorded "what if" session of actions and reactions designed to visualize the flow of the conflict and evaluate each COA in light of the challenges posed by the problem and OE. The plan may not involve adversary forces per se, but all plans entail other factors, interdependencies, and threats or challenges to mission accomplishment. COA analysis and wargaming should be tailored to methodically evaluate these various considerations and their outcomes.

Planners should work with the commander to further define or refine the COA evaluation criteria during this step. Solidifying the evaluation criteria before wargaming enables planners to record the strengths and weaknesses of each air COA according to those criteria. Results from COA analysis and wargaming may require refinements of the air component design and air component approach.

COA ANALYSIS AND WARGAMING INPUTS

COA analysis and wargaming inputs include the COA development results, refined air component design, and a refined air component approach. Planners may need to review or identify COA evaluation criteria to conduct COA wargaming. Additionally, the staff

¹⁹ For additional information on most likely and most dangerous COAs see JP 5-0, *Joint Planning*.

should identify or confirm the wargame methodology that will be used for COA analysis and wargaming.

COA ANALYSIS AND WARGAMING ACTIVITIES

Like other steps, COA analysis and wargaming involve open and honest discussion framed around the problem at hand. COA analysis and wargaming apply to both non-combat and combat operations. It is a deliberate attempt to visualize the flow of the operation while focusing on CCIRs and decision points for the commander. If COA modification occurs during analysis or wargaming, ensure changes are reflected in relevant mission analysis documents. Finally, a modification in one COA may affect other COAs. A suggested method for advancing through COA Analysis and Wargaming is presented in the Wargaming Steps figure.

COA Analysis Steps

1. Prepare for COA Analysis and Wargaming.
2. Conduct Analysis and Wargaming, and Record Results.
3. Refine Selected COA (as required).
4. Report Results.

ANALYSIS STEP 1: PREPARE FOR COA ANALYSIS AND WARGAMING

The most important outcome of this step is to ensure that all materials are made available, and the game is set up and chosen to test the critical events, DPs, LOOs, LOEs, deliberate timeline analysis, and/or operational phasing. Other activities may include:

- ★ Gather tools, materials, and data.
- ★ Define or refine COA evaluation criteria.
- ★ List friendly forces and review friendly COAs.
- ★ List adversary forces and prepare adversary COAs.
- ★ List known critical events and decision points.
- ★ Determine wargame participants and roles.
- ★ Select wargame method (manual or computer-assisted).
- ★ Select a means to represent the operating area, force laydown, and moves.
- ★ Select a method to record and display wargame results.

The Action-Reaction-Counteraction Wargame Methodology

This method may use a dedicated Red Team to roleplay as the adversary against Blue COA(s). It is managed by a neutral wargame director, preferably one who has not participated in COA development activities. The Red Team can advise planners of potential implications associated with each COA, and help explore unintended consequences and likely second- and third-order effects of specific actions.

The wargame proceeds through a situation, action, reaction, and counteraction drill. Conduct each drill until the event being tested is adequately discussed or time is exhausted.

Situation

- ★ Facilitator provides an overview of the wargame describing the specific event to be wargamed along with beginning and ending criteria.
- ★ Red Team presents current disposition of forces IAW specified COA.
- ★ Blue Team presents current disposition of forces IAW specified COA.

Action

- ★ Starts with the initiating event (typically friendly action) and the full range of friendly force operations that comprise that action.

Reaction

- ★ The Red Team identifies the same scope of the adversary reaction.

Counteraction

- ★ Blue Team describes all areas of counteraction. The counteraction can begin the next sequence or a separate, new action can begin the sequence.
- ★ Consider risk assessments and mitigation efforts throughout the wargame.
- ★ Identify branches, sequels, and CCIRs.

NOTE: Wargamed events can and probably will occur simultaneously.

ANALYSIS STEP 2: CONDUCT ANALYSIS/WARGAMING AND RECORD RESULTS

There are many different methods to conduct wargaming. It is rare below the air component level to have access to computer-assisted wargaming. Most wargames can be accomplished or modified to a simple tabletop format. Key to this step is the accurate recording of the wargame results for COA refinement.

Wargaming does not have to be time intensive. When facing time or personnel constraints, focusing on specific elements of the plan or operation can greatly reduce the wargame's duration. A focused key leader discussion may also suffice for wargaming. When choosing what to wargame consider:

- ★ **Tabletop Exercises:** Walk through the plan with a focus on the most likely and most dangerous threat to mission success. The problem and the anticipated changes to the observed system should be discussed.
- ★ **Critical events:** Focus on specific critical events that encompass the essence of the COA or that address events of particular interest to the commander.
- ★ **Decision points:** An identified point in time or space that require a commander's decision. Available options for the commander, to select upon reaching the decision point, should be annotated. The DSM discussed in Chapter Two can be used for annotating DPs and potential adversary or enemy reactions.
- ★ **LOO/LOE method:** Wargame a specific LOO or LOE.
- ★ **Box method:** Wargame all actions within a discrete geographic area.
- ★ **Deliberate timeline analysis:** Wargame all actions within a discrete block of time.
- ★ **Operational phasing:** Phasing organizes an operation into manageable parts. A phase integrates and synchronizes related activities to improve unity of effort during executions. Use campaign or operation phasing as the framework to identify and analyze significant actions and requirements.

During the wargame ensure all relevant data is recorded for later COA refinement and modification. The following items are typically recorded and analyzed during COA analysis and wargaming:

- ★ Timing issues and time required for objectives.
- ★ Additional resources and effort for improvements.
- ★ Potential risks and mitigations.
- ★ COA strengths and weaknesses, especially those related to evaluation criteria.
- ★ Branch and sequel requirements.
- ★ Commander's DPs and CCIRs.

ANALYSIS STEP 3: REFINE OR MODIFY EACH COA

In this step, the results of the wargame are utilized to modify and refine each COA. Airmen are encouraged to review wargame outputs utilizing the seven joint functions as a framework to ensure alignment with the joint force.²⁰ Planners should recognize the possibility that a wargame result could invalidate a potential COA. In these cases, it is the responsibility of the planning team to be transparent about the result and consider removing the COA from further consideration.

²⁰ For additional information on the joint functions, see JP 3-0, *Campaigns and Operations*.

ANALYSIS STEP 4: REPORT RESULTS

If requested, reporting results provides the commander wargame results before COA comparison. If accomplished, results should address:

- ★ COAs wargamed against the most likely and most dangerous threats to mission success.
- ★ Key strengths and weaknesses.
- ★ Potential decision points identified and associated CCIRs.
- ★ Potential branches and sequels identified.
- ★ Key changes required and made to COAs.
- ★ Invalidated COAs identified by the wargame.
- ★ Refined Blue COAs and outstanding issues.
- ★ Wargame conclusions and recommendations.

COA ANALYSIS AND WARGAMING RESULTS

The results of wargaming are refined COAs, commander DPs, refined air component design, and notes on critical events. This helps inform the commander's decisions during execution.

Additionally, the detailed notes from the analysis and wargaming provide background information to interested parties about the advantages and disadvantages of the COAs.

CHAPTER 7: COURSE OF ACTION COMPARISON

The COA comparison step of the AFPP provides the commanders and staffs an opportunity to see how the developed COAs rate against previously approved evaluation criteria. **COAs are not initially evaluated against other COAs.** Rather, each COA is assessed independently against COA evaluation criteria, and selection criteria related to the commander's intent, and then compared to other COAs after independent assessments are complete. COA comparison facilitates the commander's decision-making process by considering the ends, ways, means, and risks associated with each COA.

COA COMPARISON INPUTS

Key to this step is the COA analysis and wargaming results, refined air component design, and a refined air component approach. Planners may need to review or identify the COA evaluation criteria and definitions to conduct the COA comparison. Additionally, the staff should identify or confirm the COA comparison method and potential weighting considerations.

COA COMPARISON ACTIVITIES

Planners should consider the guidance provided in the "COA Comparison Inputs and Outputs" figure prior to starting the step.

Whichever comparison method(s) are used, this step aids in determining which COA best suits the situation and commander's needs.²¹

Some of the COA comparison methods typically used to conduct comparison:

- ✧ Simple plus, minus, neutral assessments.
- ✧ Narrative qualitative comparisons of advantages and disadvantages.
- ✧ Numerically (weighted or not weighted) comparisons with multiple evaluation criteria.

²¹ For additional information on COA comparison methods, see JP 5-0, *Joint Planning*.

Example Numerical Comparison

		Course of Action					
		COA 1		COA 2		COA 3	
Criteria	Weight	Rating	Product	Rating	Product	Rating	Product
Exploits maneuver	2	3	6	2	4	1	2
Attacks COGs	3	2	6	3	9	1	3
Integrates maneuver and interdiction	2	2	4	3	6	1	2
Exploits deception	2	1	2	2	4	3	6
Provides flexibility	2	1	2	3	6	2	4
CSS (best use of transportation)	1	3	3	2	2	1	1
Total		12		15		9	
Weighted total			23		31		18

NOTE: The higher the number, the better.

- The joint force commander's intent explained that the most important criterion was "attacking the enemy's COGs." Therefore, assign a value of 3 for that criterion and lower numbers for other criteria that the staff devises (**this is the weighing criterion**).
- For attacking the enemy COGs, COA 2 was rated the best (with a number of 3). Therefore, COA 2 = 9, COA 1 = 6, and COA 3 = 3.
- After the relative COA **rating** is multiplied by the **weight** given each criterion and the product columns are added, COA 2 (with a score of 31) is rated the most appropriate according to the criteria used to evaluate it.

Legend

COA course of action

COG center of gravity

CSS combat service support

COA COMPARISON RESULTS

Results of COA comparison are individual COA advantages and disadvantages, COA discussion notes, COA recommendations, and staff recommend COAs. Air component design and approach should be refined if necessary.

CHAPTER 8: COURSE OF ACTION APPROVAL

During the AFPP COA approval step, the planning staff work together to develop and present COAs and a recommendation to the commander for approval. The COA decision briefing summarizes the work completed to develop, analyze, and compare COAs. Afterward, the commander may brief HHQ and receive their approval if the COA is part of a larger operation or campaign. The key to COA approval is the commander and planning staff combining their experience, knowledge, skill, and creativity to analyze the data and recommendations. The commander selects a COA from the confluence of adaptive thinking and air component design. A method for progressing through the COA approval step is presented in the “COA Approval Activities” figure.

COA APPROVAL INPUTS

COA approval should be informed by the COA analysis and wargaming results, refined air component design, and a refined air component approach. COA approval is driven by the cumulative AFPP effort.

COA APPROVAL ACTIVITIES

COA Approval Steps

1. Prepare the COA decision briefing.
2. Present the COA decision briefing.
3. Commander selects or modifies COA.
4. Refine selected COA (as required).
5. Prepare briefing, message, or commander’s estimate for HHQ Approval.

APPROVAL STEP 1: PREPARE THE COA DECISION BRIEFING

Tailor the brief to fit the commander’s level of understanding at the time of the brief. If the brief occurs shortly after the COA development brief, or if a wargame brief was given, the COA decision brief may be abbreviated.

APPROVAL STEP 2: PRESENT THE COA DECISION BRIEFING

Typically, the planning lead(s) brief the commander. Principal staff directors and division chiefs should attend. When necessary, capabilities or issues should be addressed at a higher classification level, including a supplemental briefing with appropriately cleared individuals in a suitable facility.

Sample COA Decision Brief Outline

The staff prepares a briefing to the commander providing the following (as applicable):

1. Purpose of the briefing.
2. Update of the OE.
 - a. Opposing situation:
 - i. Adversary order of battle/location and disposition (threats to mission success at the AETF and Wing level).
 - ii. Logistics-summary of ability to support operations.
 - iii. Time and space factors-capability to move to and reinforce initial positions.
 - iv. Combat capability-state of training, readiness, battle experience, physical condition, morale, leadership, motivation, tactical doctrine, discipline, and significant strengths and weaknesses.
 - v. Unknowns-intelligence gaps about the adversary or OE.
 - b. Friendly military situation-implications of other component requirements on air component operations (e.g., ISR, refueling).
 - c. Status of the operational area including infrastructure, geographic terrain, information environment, and social terrain.
3. HHQ commander's mission and intent statements.
4. Operational concepts and COAs developed.
 - a. Any changes from the mission analysis briefing regarding: Assumptions, limitations, adversary and friendly centers of gravity, phasing of the operation, LOOs, and LOEs.
 - b. COAs: COA name, COA statement, COA sketch, COA architecture. Major differences between COAs, and a summary of COAs.
 - i. COA Architecture: Task organization, COMREL, support requirements, additional authorities required, additional forces needed, organization of the operational area.
 - ii. COA Analysis Summary: Wargame results/evaluation criteria/comparison methodology/strengths and weakness of each COA

APPROVAL STEP 3: COMMANDER SELECTS OR MODIFIES COA

Following the COA decision briefing the commander may:

- ★ Concur with the recommendation as presented.
- ★ Select a different COA.

- ★ Direct modification (possibly combining elements of two or more COAs).
- ★ Direct development of additional COAs.
- ★ Request additional information or analysis prior to decision.
- ★ Direct that one of the COAs be used as the basis for a deception plan.
- ★ Direct that a briefing or message be prepared to present the COA to HHQ for approval.
- ★ Defer the decision and consult with selected staff and commanders.

APPROVAL STEP 4: REFINE SELECTED COA (AS REQUIRED)

The staff refines the selected COA in accordance with the commander's guidance and briefs the commander again if required.

APPROVAL STEP 5: PRESENT SELECTED COA FOR HHQ APPROVAL

When a selected COA is part of a HHQ operation or campaign plan, the staff may prepare a commander's estimate to accompany the recommended COA. The commander's estimate provides a concise, narrative statement of how the commander intends to accomplish the mission and provides the necessary focus for plan development. HHQ will use appropriate staffing and approval processes as well as other techniques to ensure plans remain synchronized with HHQ's concept, intent, and campaign plan. Being integrated and aware of HHQ planning can help ensure the commander's selected COA receives approval from the appropriate authority.

COA APPROVAL RESULTS

A COA approval decision that informs the development of a plan or order is the primary result of this step. The elements of air component design are still being refined at the end of COA approval.

CHAPTER 9: PLANS AND ORDERS DEVELOPMENT

The plans and orders development step of AFPP explains the basic steps of writing orders with a special focus on the paragraph format. Ultimately, the order type should convey the commander's intent and direction to their subordinate commanders. While explained in the context of the AFPP, the steps and guidance provided in the chapter apply to all JPP based planning processes. In addition, we discuss the emerging efforts of writing orders utilizing the mission type order (MTO) technique, which enables commanders to distribute control of specified operations and activities to subordinate commanders.

EFFECTIVE PLANS AND ORDERS

Effective plans and orders are simple and direct to reduce misunderstanding. Orders should be brief and clear so that the subordinate units can conduct operations in contested environments if necessary. The situation determines the degree of simplicity required since simple plans executed on time are typically better than detailed plans executed late. Commanders at all echelons weigh the potential benefits of a complex plan against the risk that subordinates may fail to understand it.

As such, the language is direct. An example of this is, "The palletized cargo remains in the assembly area" instead of "The palletized cargo will not accompany the unit." Effective plans and orders directly and positively state what the commander wants the unit and its subordinate units to accomplish and why.

Effective plans and orders are brief and clear. These plans use short words, sentences, and paragraphs. Plans use acronyms unless clarity is hindered. Brief and clear orders:

- ★ Use doctrinally correct terms and symbols.
- ★ Avoid jargon.
- ★ Use plain language when a doctrinal term is not available.
- ★ Eliminate every opportunity for misunderstanding the commander's intended direction.

PLANS AND ORDERS OVERVIEW

Administrative instructions and formats for preparing plans and orders can include:²²

Types of Plans. A plan is a design for a future or anticipated operation. Plans come in many forms, and they vary in scope, complexity, and duration.

Campaigning. Campaigning is the persistent conduct of related operations, activities, and investments that align military actions with the other instruments of national power, supporting global integration across the competition continuum in pursuit of strategic objectives. Campaign and contingency plans support national strategy.

²² See CJCSM 3130.03A, *Planning and Execution Formats and Guidance*, for additional information.

Campaign Plans. JFCs develop and execute campaign plans. Developing and issuing a campaign plan is appropriate when the contemplated simultaneous or sequential military operations exceed the scope of a single major operation. Contingency plans are branches of campaign plans, typically prepared in advance of an anticipated crisis, and they should be modified during execution. Campaign and contingency plans have four levels of details: commander's estimate, base plan (BPLAN), concept plan (CONPLAN), and operation plan (OPLAN). **Doctrinally there is no separate air campaign, rather the JFACC executes joint air operations in support of a JFC's or combatant commander's (CCDR's) campaign plan.**

A campaign is a series of related operations aimed at achieving strategic and operational objectives within a given time and space.²³ CCDRs develop campaigns to support the global campaign and shape the OE in a manner that supports those strategic objectives. They conduct their campaigns primarily through military engagement, operations, posture, and other activities that seek to achieve US national objectives, protect US national interests, and prevent the need to resort to armed conflict while setting conditions to transition to contingency operations if required.

Contingency Plans. Campaign plans establish conditions for contingency operations. Plans for contingencies and conflict are often branches of, or sequels to, base campaign plans. Contingency plans are developed during ongoing steady state conditions, employing the JPP to create one of the same four levels of detail as campaign plans: commander's estimate, BPLAN, CONPLAN, or OPLAN.

Planning during a crisis entails the positioning of forces, or at least the start of that process, often concurrent with early planning. Although existing plans may not align with an emerging crisis, those plans can often be modified for a specific situation to facilitate rapid COA development. Existing planning products for force execution and deployment, particularly TPFDD built for contingencies, can often be modified in time-constrained situations. Crisis planning produces joint OPORDs and other orders associated with the time-sensitive execution of operations.

Observations should be captured during planning and after every operation in the form of lessons learned. Absorbing lessons learned and adapting to them appropriately is critical to operational success. Events should be documented in detail to provide information that improves the planning and execution of future actions. Planners should review after-action reports and other lessons-learned analyses in preparation for the planning process. These reviews ensure planners benefit from prior experience and use what they learn to inform, adapt, and improve steady state, contingency, or crisis planning.

Types of Orders. An order is a communication, written, oral, or by signal, which conveys instructions from a superior to a subordinate. In the context of joint operational planning, some of the types of written orders are:

²³ For additional information on campaigning, see JP 5-0, *Joint Planning*.

- ★ Operation Order (OPORD).
- ★ Fragmentary Order (FRAGORD).
- ★ Planning Order (PLANORD).
- ★ Warning Order (WARNORD).
- ★ Tasking Order (TASKORD).

A Note of Caution on Orders Terminology

The terminology associated with an order has a very specific meaning when used in the joint environment under a CCDR. Commanders must have the appropriate authority to issue certain orders. The guidance provided in this document is associated with their use by a commander under a CCDR command relationship. Commanders should be aware of the guidance provided in CJCSM 3130.03A, Enclosure C when developing their orders both inside and outside of this structure to ensure that the order type is appropriate and does not exceed their authority.

An *OPORD* is a directive issued by a commander to subordinate commanders to affect the coordinated execution of a national command authority operation. Commander's issue OPORDs to direct the execution of long-term operations and the execution of discrete short-term operations within the framework of a long-range OPORD. This type of order can be issued at any echelon of command.

A *FRAGORD* is a modification to an existing order, and it is fragmented by only describing the changes, additions, deletions, and modifications. FRAGORDs differ from OPORDs in the degree of detail provided. This type of order can be issued at any echelon of command.

A *PLANORD* provides essential planning guidance to develop, adapt, or refine an existing plan due to emergent changes in the environment. A PLANORD must be issued by an appropriate commander.

A *TASKORD* is a direct task for subordinate unit execution that must be done as it is imposed by an appropriate authority.

A *WARNORD* is a preliminary notice of an order or action that is to follow. WARNORDs help subordinate units and staffs prepare for new missions by describing the situation, providing initial planning guidance, and directing preparation activities. A WARNORD must be issued by an appropriate commander.²⁴

²⁴ See CJCSM 3130.03A, *Planning and Execution Formats and Guidance*, for additional information.

In addition to OPORDs, FRAGORDs, PLANORDs, and WARNORDs, USAF forces may receive the following types of orders from a joint HQ:

- ★ Alert order.
- ★ Execute order.
- ★ Prepare-to-deploy order.
- ★ Deployment order.
- ★ Redeployment order.

PLANS AND ORDERS DEVELOPMENT INPUTS

The COA approval results, refined air component design, and a refined air component approach, informs the creation of plans and orders. Information from the cumulative AFPP effort can aid in plan or order creation.

PLANS AND ORDERS DEVELOPMENT ACTIVITIES

This section outlines the basic steps for the creation of a five-paragraph order. Not all orders follow the five-paragraph format (e.g., air tasking orders and airspace control orders). However, the five-paragraph form is the Department of Defense (DoD) standard used by the joint staff and unified commands. The USAF has a modified five-paragraph order format as seen in AFDP 1-1, *Mission Command*, and in the figure. For additional information see attached Appendix A, “Plans and Order Formats,” and Appendix E, “Example Modified Five-Paragraph Order.”

If the planning processes have been followed, writing, and assembling the plan or order should not be difficult. Most of the information needed has already been discussed and refined in previous steps of the AFPP. The administrative tasks required to physically produce and synchronize the plan, or order should consume most of the time. The “Orders Development Steps” figure provides a general guide for orders creation.

**USAF Modified
Five-Paragraph Order
Format (JP 5-0)**

- 1. Situation**
- 2. Mission**
- 3. Execution**
- 4. Force Sustainment**
- 5. Command, Signal
and Communications**

Orders Development Steps

1. Develop base paragraphs:
 - a. Review approved information, products, guidance, etc.
 - b. Finalize intent statement and CONOPS.
 - c. Assign production tasks.
 - d. Synchronize with parallel and supporting plans and orders.
2. Publish base order or plan.
3. Develop appropriate annexes.
4. Issue FRAGORDs to disseminate annexes and updates (as required).

ORDERS STEP 1: DEVELOP BASE PARAGRAPHS

Paragraph 1: Situation. The situation paragraph describes the OE including areas of interest and the geographically defined area of operations. The planning work completed to understand the OE provides information for completing this paragraph. Additionally, assumptions the commander and staff made about the mission and problem are captured in the situation paragraph. The other sub-paragraphs within the situation paragraph are the adversary and friendly components. Both sub-paragraphs are intended to describe for the reader the composition, disposition, and strength of the forces and describe how the adversary is expected to employ their capabilities. In the friendly sub-paragraph, the mission and intent of commanders of higher echelon commanders are included so the subordinates can clearly understand how the nesting of tasks, purposes, and missions are related to unity of effort.

Paragraph 2: Mission Statement. The mission statement paragraph is solely comprised of the mission statement. The mission statement is developed during mission analysis and refined once a COA is chosen. The five elements of a mission statement answer these questions:

- ★ Who will execute the operation (unit or organization)?
- ★ What is the unit's essential task (normally a tactical mission task or tactical enabling task)?
- ★ Where will the operation occur (assigned area, objective, engagement areas, or grid coordinates)?
- ★ When will the operation begin (by time or event)?
- ★ Why will the force conduct the operation (for what purpose)?

Paragraph 3: Execution. The execution paragraph describes the commander's approach to achieving mission success-the "how" of an operation. It begins with the commander's intent and CONOPS and continues with tasks directed to subordinate units.

These tasks identify which subordinate force is the main effort for the operation and what forces are supporting, shaping, or sustaining elements to execute the scheme of movement and maneuver that result in mission accomplishment. The execution paragraph includes prioritized tasks. Lastly, the execution paragraph includes coordinating instructions that are specified tasks applicable to all forces. Coordinating instructions usually include constraints and restraints, commander's critical information requirements, and other information that is common to all subordinates and not stated in unit SOP.

- ★ **Commander's Intent Statement.** The commander's intent statement is developed through the planning process. In the order writing process, the commander's intent statement is refined into a clear and concise statement that frames the operation's purpose, desired end state, and risk. It remains nested within higher echelon commander's intent and guidance with an awareness of the larger operational and strategic context. The intent includes purpose, end state, and risk the commander is willing to accept. An intent statement may also include method and mission essential tasks.
- ★ **CONOPS.** The CONOPS is a statement that directs the way subordinate units cooperate to accomplish the mission and establishes the sequence of actions the force uses to achieve the end state. In the CONOPS, the commander describes how the actions of subordinate units fit together to accomplish the overall mission (essential task and purpose) within the assigned area. Commanders and their staffs describe the CONOPS by defining time, space, and resources.

Time refers to the sequencing and synchronization of tasks or operations to attain end state conditions with the least cost in personnel and resources.

Space refers to the establishment of an assigned area: AO, zones, and sectors.

Resources refer to designating the priority and allocation of resourcing based on the assignment of the main effort, supporting efforts, and reserves.

The CONOPS also includes:

- ★ The central approach the commander intends to accomplish the mission.
- ★ Provides for the integration, synchronization, and sequencing of time, space, forces, and resources.
- ★ Provides for the tempo of the operation.
- ★ Relates and nests the objectives and effects to those of the higher commander.

In developing the CONOPS, commanders and staff ensure their concept nests with that of their HHQ. Nesting a concept is a planning technique to achieve unity of purpose whereby each succeeding echelon's CONOPS are aligned by purpose with the higher echelons' CONOPS. An effective CONOPS describes how the forces support a mission of the HHQ and how the actions of subordinate units fit together to accomplish a mission.

Paragraph 4: Force Sustainment. Force sustainment describes all required aspects of sustaining the force during an operation (typically broken down into individual phases of the operation). It includes a description of the quantities of all classes of supply, where they are, and what the resupply plan is across the area of operations. It gives an estimate of resupply rates based on forecasted consumption. The paragraph may also include the concept for maintenance (preventative, repair, and replacement) of key pieces of equipment. Time and quantity requirements should be primary planning factors. It should cover functional areas of logistics, transportation, personnel policies, maintenance, health services, and administration.

- ★ **Logistics.** This paragraph addresses the commander's logistics priorities and intent: basing, combat, general, and geospatial engineering requirements, required contracted support, and environmental considerations. Identify the priority and movement of logistic support for each option and phase of the concept.
- ★ **Personnel.** Identify detailed planning requirements and subordinate taskings. Assign tasks for establishing and operating personnel facilities, managing accurate and timely personnel accountability and strength reporting, and making provisions for staffing them. Discuss the administrative management of participating personnel. Refer to annex E (if published).

Paragraph 5: Command, Signal, and Communications. The purpose of this section is to summarize command arrangements and their respective roles and responsibilities. The section identifies changes to major C2 organizations, the commander's battle rhythm, and the staff's expected duty hours. The locations of headquarters and command posts should be annotated. Additionally, this section discusses the scope of communications systems and procedures required to support the mission. The procedures should detail a primary, alternate, contingency, and emergency (PACE) plan for an organization's internal and external communications.

- ★ **Command.** Include a designation of supported and supporting commanders, coordination instructions, and a listing of the command relationship's organizational structure, expected to exist in support of mission execution. This section describes the location(s) of the command and subordinate commands as well as alternate HQ locations. Additionally, COMREL should be detailed in this section along with the plan for succession of command.
- ★ **Signal and communications.** This section describes the means of communication equipment for the force, (e.g., voice, digital, satellite communications, mesh networks, common operating picture, etc.) and usually defines any shortfalls in communications equipment requirements for the force to operate. It prioritizes the ways communication is conducted during the operation through a PACE plan. The purpose of a good PACE plan is to establish redundancy so that communications are always available. Most units have two PACE plans: one for communications to HHQ and one for subordinate units. A PACE plan for a HHQ is likely established by the HHQ.

ORDERS STEP 2: PUBLISH BASE ORDER

Due to the time involved in creating detailed annexes and appendices, the commander approves, and the staff publishes the base order (five-paragraphs) as soon as possible to allow subordinate units to begin subsequent planning and preparation. Updates to the base order can be published in a FRAGORD on their own or with the annexes developed in step 3. If annexes are intentionally omitted with no plan for further development indicate this on the order.

ORDERS STEP 3: DEVELOP APPROPRIATE ANNEXES

Orders at echelons below the AETF do not normally have annexes. However, the more complex an operation becomes and/or the greater the number of capabilities involved, the greater the need for the detail provided in annexes. In these circumstances, the key to the success of a good order is well-written annexes, appendices, tabs, and exhibits further detailing concepts or subjects that do not need to be specified in the main paragraphs of the order.

Commanders and staffs are not required to develop any or all the annexes listed below. The number and type depend on the commander, echelon of command, and needs of a particular operation. Minimizing the number of annexes keeps the order consistent with completeness and clarity. If the information relating to an annex's subject is brief, the order writer places the information in the base order and omits the annex. Staffs list annexes under an appropriate heading at the end of the document they expand. For example, they list annexes at the end of the base order, appendices at the end of annexes, and so forth.

When an annex is not required, staffs indicate this by stating "omitted." For example, the order writer would state, "Annex R (Reports) omitted." If the situation requires an additional annex not shown below, leaders can add to this structure.

Amplify information in annexes where necessary by appendices to annexes, tabs to appendices, and exhibits to tabs. Ensure annexes, appendices, and associated pieces comply with the general sequential structure outlined in CJCSM 3130.03A, *Planning and Execution Formats and Guidance*. Use only those annexes appropriate to the order. At lower echelons, annexes are typically only used when the BPLAN is limited by classification or level of detail. Following is a list of the annexes and examples of their content. **Do not change or reorder annex numbers.**

A: TASK ORGANIZATION.

- ★ Time-phased Force and Deployment List.
- ★ Shortfall identification.

B: INTELLIGENCE.

- ★ Commander's PIRs and supporting information requirements provide an overall picture of intelligence support to operations.

C: OPERATIONS.

- ★ Appendices address various aspects (e.g., personnel recovery, ROE, etc.).

D: LOGISTICS.

- ★ Broad sustainment and logistics concepts that support operations, synchronized with operations and phases for joint forces over which the commander exercises TACON (e.g., fuels management, subsistence, mortuary affairs, mobility, transportation).

E: PERSONNEL.

- ★ Personnel matters and other support (e.g., personnel processing, legal, postal).

F: PUBLIC AFFAIRS.

- ★ Broad public affairs concepts that support all operations, and the broad public affairs plan synchronized with operations phases.

G: CIVIL-MILITARY OPERATIONS.

- ★ Considerations for military interaction with governmental and non-governmental agencies, multinational forces, and other interorganizational partners.

H: METEOROLOGICAL AND OCEANOGRAPHIC OPERATIONS.

- ★ Broad weather factors affect all commander tasked operations.

I: KNOWLEDGE AND INFORMATION MANAGEMENT.

- ★ Knowledge management (KM) roles responsibilities, KM plan, and collaborative tools.
- ★ Preventive measures taken to mitigate hostile actions against DoD personnel (including family members), resources, facilities, and critical information.

J: COMMAND RELATIONSHIPS.

- ★ C2.
- ★ Command considerations.
- ★ Command relationships.
- ★ Memorandum of understanding (MOU).
- ★ Designation and location of all air-capable command HQ.
- ★ Continuity of operations (COOP) and degraded operations.
- ★ Command, control, and communications considerations.

K: COMMAND, CONTROL, COMMUNICATIONS, COMPUTERS, AND INTELLIGENCE (C4I) SYSTEMS.

- ★ General overview of C4I systems and procedures required to support air operations (e.g., cybersecurity, satellite communications).

L: ENVIRONMENTAL CONSIDERATIONS.

- ★ May contain geospatial information and services.
- ★ Broad geospatial factors affect all commander tasked operations.

N: ASSESSMENTS.

- ★ Commander-specific assessment plan and metrics.

P: HOST-NATION AGREEMENT.

- ★ HNA reliability, presumed support.

Q: HEALTH SERVICES.

- ★ Patient movement, hospitalization, etc.

S: SPECIAL TECHNICAL OPERATIONS (STO).

- ★ Air component guidance based on HHQ Annex S, “Integrated Joint Special Technical Operations” that informs COMAFFOR logistics, employment, assessment, and consequence management of Service-provided STO capabilities, as required.

T: CHEMICAL, BIOLOGICAL, RADIOLOGICAL, AND NUCLEAR (CBRN) RESPONSE.

- ★ Consequence management and CBRN response could be addressed within Annex C.

V: INTERAGENCY-INTERORGANIZATIONAL COORDINATION.

- ★ Interaction and collaboration with other government departments, and nongovernmental, private, and international entities.
- ★ Interagency synchronization matrix.

W: OPERATIONAL CONTRACT SUPPORT.

- ★ Contracted support requirements estimate and contractor management plan.

Y: COMMANDER’ S COMMUNICATIONS STRATEGY.

- ★ Messages and themes to promote or avoid with the target audience.

ORDERS STEP 4: PUBLISH ANNEXES AND SUPPLEMENTAL GUIDANCE AS FRAGORDS

This step publishes and disseminates FRAGORDs for any remaining annexes or to update the base order or plan. This process occurs continually to meet evolving situations and changes in the OE. Receiving subordinate elements should actively confirm receipt of orders, plans, and annexes.

PLANS AND ORDERS DEVELOPMENT RESULTS

Plans and orders development results in a plan or order that communicates the commander's direction and intent. The plan or order should be clearly written in the five-paragraph format. The plan, order, and air component approach may require refinement as the problem, observed system, and desired system change.

MISSION-TYPE ORDER TECHNIQUE

Utilizing the MTO technique is a direct enabler of mission command. Commanders can use the MTO technique to distribute control of specified operations and activities to subordinate commanders in contested, degraded, and operationally limited environments. Alternatively, commanders can provide very detailed and specific orders to subordinates. Orders should provide subordinate commanders with clear intent, a shared understanding of an operation's purpose, and visibility of the wider operational and strategic context. MTO styled orders enable subordinate commanders to exercise flexibility and creativity during C2 activities to meet the changes in the OE and exploitation of emergent opportunities.²⁵ MTO is a technique for writing orders that provides latitude for flexibility in execution while meeting the commander's intent. Not every order utilizes the MTO technique, but the process for orders production does not change based on the technique, only the level of detailed direction contained in the order.

Operating Under MTOs

Capacity is a measure or degree to which an individual or organization is able, has the potential, or has demonstrated the ability to operate according to the principles of mission command. Units or individuals who have trained to and demonstrated an embodiment of the attributes of mission command and practice its principles consistently are able to operate with minimal guidance while still achieving the commander's intent. The MTO technique for writing orders is the USAF's preferred method. Airmen and USAF units should strive to maximize their capacity to ensure the effectiveness of the MTOs.

–AFDP 1-1, *Mission Command*

²⁵ For additional information on C2 activities, see AFDP 3-0.1, *Command and Control*.

Under the mission command philosophy, the MTO technique focuses on giving and creating orders centered on the purpose of the operation rather than details of how to perform assigned tasks. Commanders should utilize the MTO technique by writing orders that delegate decisions and empower subordinate initiative to make decisions based on the commander's guidance rather than constant communications. Subordinates' understanding of the commander's intent (spelled out in paragraph 3a of orders) at every level of command is essential to execution under the mission command philosophy.

Orders written using the MTO technique are as brief and simple as possible. MTO styled orders clearly convey the unit's mission and commander's intent. They summarize the situation (including observed system or anticipated starting conditions), describe the operation's end state, and may provide a CONOPS to accomplish the unit's mission. When assigning tasks to subordinate units, MTOs include all components of a mission statement: who, what, when, where, and why. However, commanders particularly emphasize the purpose (what and why) of the tasks to guide disciplined initiative.

MTOs contain the proper level of detail, they are neither so detailed that they stifle initiative nor so general that they provide insufficient direction. The proper level depends on each situation, and it is not easy to determine. Some phases, and types, of operations require tighter control over subordinate elements than others. A phase of an operation involving an air assault, for example, may require precise synchronization and guidance. On a similar note, nuclear operations are an example of a type of operation that often requires specific guidance for effective execution.²⁶

²⁶ As a rule, the base order contains only the specific information required to guide while allowing subordinates as much freedom of action as possible. Attachments to the plan or order contain details regarding the situation and instructions necessary for synchronization.

APPENDIX A: PLANS AND ORDERS FORMATS

PARAGRAPH FORMAT

Regardless of echelon, order writers show the main five-paragraph headings on written orders. There are several different types of orders and plans discussed in this publication. **Orders and plans should be written according to the Chairman of the Joint Chief's of Staff directive guidance to the maximum extent possible.**²⁷

The example modified five-paragraph order provided in Appendix E is aligned with the USAF modified order format in AFDP 1-1, *Mission Command*.

ACRONYMS AND ABBREVIATIONS

Order writers use acronyms and abbreviations to save time and space if these acronyms and abbreviations do not cause confusion. However, order writers should not sacrifice clarity for brevity. Order writers keep acronyms and abbreviations consistent throughout the order and its attachments. They avoid using acronyms and abbreviations not found in the *Air Force Glossary* or the *DoD Dictionary of Military and Associated Terms*. Before using an acronym or abbreviation, at its first use in the document, order writers spell out the acronym or abbreviation and then place the acronym or abbreviation between parentheses immediately after the term. After this first use, they use the acronym or abbreviation throughout the document.

LOCATION AND DIRECTION DESIGNATIONS

Location and direction designations are important in orders. Order writers describe locations or points on the ground by:

- ★ Providing the map datum used throughout the order.
- ★ Referring to military grid reference system coordinates.
- ★ Referring to longitude and latitude if available maps do not have the military grid reference system.
- ★ Order writers designate directions in one of two ways:
 - ★★ As a point of the compass (for example, north or northeast).
 - ★★ As a magnetic, grid, or true bearing, stating the unit of measure (for example, 85 degrees [magnetic]).

When first mentioning a place or feature on a map, order writers print the name in capital letters exactly as spelled on the map and show its complete grid coordinates (grid zone designator, 100-kilometer grid square, and four-, six-, eight-, or ten-digit grid coordinates) in parentheses after it. When first using a control measure, such as a contact point or a phase line, order writers print the name or designation of the control measure in capital

²⁷ See CJCSM 3130.03A, *Planning and Execution Formats and Guidance*, for additional information.

letters followed by its complete grid coordinates in parentheses. Thereafter, they repeat the coordinates only for clarity.

Order writers describe areas by naming the northernmost (1200) point first and the remaining points in clockwise order. They describe positions from left to right and from front to rear, facing the enemy. To avoid confusion, order writers identify flanks by compass directions, rather than right or left of the friendly force.

If the possibility of confusion exists when describing a route, order writers add a compass direction for clarity (for example, “The route is northwest along the road LAPRAIRIE-DELSON.”). If a particular route already has a planning name, such as the main supply route LION, order writers refer to the route using only that designator.

Order writers designate trails, roads, and railroads by the names of places along them or with grid coordinates. They precede place names with a trail, road, or railroad (for example, “road GRANT-CODY”). Order writers designate the route for a movement by listing a sequence of grids from the start point to the release point. Otherwise, they list the sequence of points from left to right or front to rear, facing the enemy.

Order writers identify riverbanks as north, south, east, or west. In wet gap-crossing operations, they identify riverbanks as either near or far.

NAMING CONVENTIONS

Unit SOP normally designates naming conventions for graphics (for example, assembly areas, phase lines, and objectives). Otherwise, planners select them. For clarity, order writers avoid multiword names, such as “JUNCTION CITY.” Simple names are better than complex ones. To ensure operations security, order writers avoid assigning names that could reveal unit identities, such as the commander’s name or the unit’s home station. They do not name sequential phase lines and objectives in alphabetical order. For memory aids, order writers use sets of names designated by the type of control measure or subordinate unit. For example, a division order might use colors for objective names and minerals for phase line names.

CLASSIFICATION MARKINGS

Classification marking for orders and plans follows DoDM 5200.01 Volume 2 *Marking of Information*, DoDI 5200.48, *Controlled Unclassified Information (CUI)*, and applicable security classification guides.

EXPRESSING UNNAMED DATES AND HOURS

Order writers use specific letters to designate unnamed dates and times in plans and orders. Joint doctrine lists common letters for dates and times as shown in the below table.

Term	Designates
<i>C-day</i>	The unnamed day on which a deployment operation commences or is to commence (JP 5-0).
<i>D-day</i>	The unnamed day on which a particular operation commences or is to commence (JP 3-02).
<i>H-hour</i>	The specific hour on D-day at which a particular operation commences (JP 5-0).
<i>L-hour</i>	The specific hour on C-day at which a deployment operation commences or is to commence (JP 5-0).
<i>P-hour</i>	The specific hour on D-day at which a parachute assault commences with the exit of the first Soldier from an aircraft over a designated drop zone. P-hour may or may not coincide with H-hour planning horizon.

Designated letters for dates and times

EXPRESSING TIME

USAF orders express all times in a plan or order in the ZULU time zone (universal time coordinate). Local time should not be used due to the nature of USAF operations which often span multiple time zones.

The effective time for implementing a plan or order is the same as the date-time group of the order. Order writers express the date and time as a six-digit date-time group. The first two digits indicate the day of the month, the next four digits indicate the time. The letter at the end of the time indicates the time zone. Staffs add the month and year to the date-time group to avoid confusion. For example, a complete date-time group for 6 August 2025 at 1145Z appears as “061145Z August 2025”.

If the effective time of any portion of the order differs from that of the order, staffs identify those portions at the beginning of the coordinating instructions (paragraph 3). For example, order writers may use “Effective only for planning on receipt” or “Task organization effective 261300Z May 2025.”

When using inclusive dates, staffs express them by writing both dates separated by a dash (for example, 6-9 August 2025 or 6 August-6 September 2025). They express times in the 24-hour clock system using four-digit Arabic numbers, including the ZULU time zone indicator.

IDENTIFYING PAGES

Staffs identify pages following the first page of plans and orders with a short title identification heading located two spaces under the classification marking. They include the number (or letter) designation of the plan and the issuing HQ. For example, OPLAN 00-15-9AETF-A (U) or Annex B (Intelligence) to OPLAN 00-15-23rd AD (U). (In this case, AETF-A stands for Air Expeditionary Task Force Afghanistan.)

NUMBERING PAGES

Order writers number the pages of the base order and each attachment separately beginning on the first page of each attachment. They use a combination of alphanumeric designations to identify each attachment.

Order writers use Arabic numerals only to indicate page numbers. They place page numbers after the alphanumeric designation that identifies the attachment. They use Arabic numerals without any proceeding alphanumeric designation for base order page numbers. For example, the designation of the third page of Annex C is C-3. Order writers assign each attachment either a letter or Arabic numeral that corresponds to the letter or number in the attachment's short title. They assign letters to annexes, Arabic numerals to appendices, letters to tabs, and Arabic numerals to exhibits. For example, the designation of the third page of Appendix 5 to Annex C is C-5-3.

Order writers separate elements of the alphanumeric designation with hyphens. For example, the designation of the third page of Exhibit 2 to Tab B to Appendix 5 to Annex C is C-5-B-2-3.

ATTACHMENTS

Attachments (annexes, appendices, tabs, and exhibits) are information management tools that expand on the BPLAN or order. However, even when attachments are used, an effective base order contains enough information to be executed without them.

Staffs refer to attachments by letter or number and title. They use the following naming conventions:

ANNEXES. Staffs designate annexes with capital letters, for example, Annex D (Fires) to OPOD 19-06—9AETF-A.

APPENDICES. Staffs designate appendices with Arabic numbers, for example, Appendix 1 (Intelligence Estimate) to Annex B (Intelligence) to OPOD 19-06-9AETF-A.

TABS. Staffs designate tabs with capital letters, for example, Tab B (Target Synchronization Matrix) to Appendix 3 (Targeting) to Annex D (Fires) to OPOD 19-06-9AETF-A.

EXHIBITS. Staffs designate exhibits with Arabic numbers, for example, Exhibit 1 (Traffic Circulation and Control) to Tab C (Transportation) to Appendix 1 (Logistics) to Annex F (Sustainment) to OPOD 19-06-1 9AETF-A.

If an attachment has a wider distribution than the base order or is issued separately, the attachment requires a complete heading and acknowledgment instructions. When staffs distribute attachments with the base order, these elements are not required.

APPENDIX B: CENTER OF GRAVITY ANALYSIS METHODS

COGS AND LINES OF EFFORT

A practical means of linking objectives, effects, tasks, and actions is through COG analysis and designing lines of effort. **COG identification and analysis is often completed at an operational level HQs.** A COG is the “source of power that provides moral or physical strength, freedom of action, or a will to act.”²⁸ COG analysis should reveal what is critical and relate what is critical to what is vulnerable. Critical vulnerabilities may be attacked, exploited, or protected to decisively affect the enemy and enable friendly action. Analysis of a COG’s critical factors generally yields DPs, which are key terrain, key events, critical factors, or functions that, when acted upon, enable a commander to gain a marked advantage over an enemy. A COG analysis may reveal vulnerabilities for countering an enemy kill chain.

Counter Kill Chain

Advancements in adversary anti-access, area denial capabilities greatly expand the range and lethality of adversary kill chains. These enhanced lethal kill chains limit friendly freedom of action and raise the risk to mission while making achievement of the JFC’s objectives more difficult. Minimizing adversary enhanced denial capabilities and achieving intended friendly effects in the OE has elevated the need to counter adversary kill chains.

To attack friendly forces, an adversary must complete a sequence of actions to implement their kill chain. They must detect friendly forces, communicate via command-and-control networks, decide on engagement options, and then employ fires. Exploiting vulnerabilities in every step of this sequence through joint all domain operations can counter these kill chains. Electromagnetic attacks can disrupt and degrade adversary detection systems. Offensive cyberspace operations can disable communication networks. Long-range fires can destroy C2 nodes. Agile combat employment can defeat adversary targeting and fires employment.

BL: Joint operations leveraging all domains can counter adversary kill chains to generate desired effects, achieve the joint force commander’s objectives, and attain the desired end state.

During design and planning, it is helpful to have a tool that depicts the relationship of effects to COGs, DPs, objectives, and other events and concepts, using the logic of purpose- cause and effect. Such a tool is usually arranged in proper time sequence to

²⁸ See JP 5-0, *Joint Planning*, for additional information.

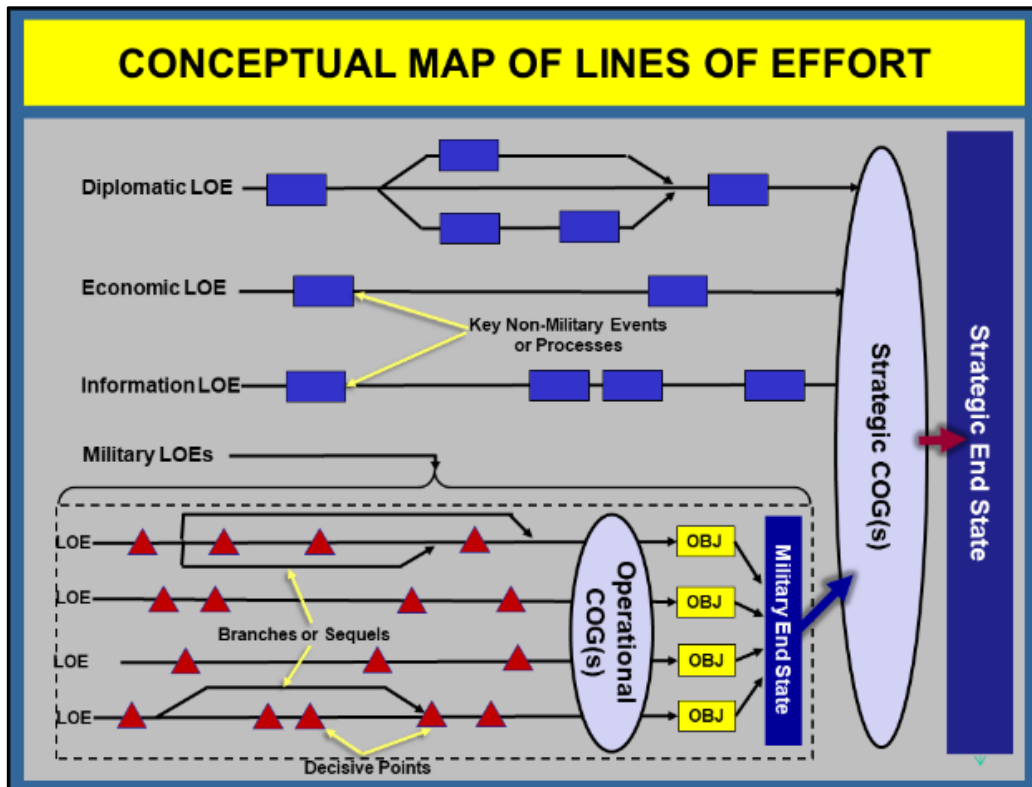
help commanders and strategists visualize how operations evolve and interact over time. LOEs provide just such a tool.

LOEs help visualize COAs, laying them out in time sequence and helping identify where certain effects should be created and where DPs are located in time relative to other events. LOEs are useful when working with interagency and multinational partners, helping commanders and strategists visualize how military means can support all instruments of national and multinational power. The aggregate of the effects of all instruments of power acting together forms a series of LOEs leading directly to the strategic end state. Each LOE can be broken down into constituent objectives, DPs, effects, and actions or tasks. LOEs contribute to accomplishing objectives in other LOEs and may define DPs.

Each LOE can be refined further by including tactical-level objectives, effects, and individual tactical tasks. Given sufficiently sophisticated planning tools, each organization's tasks could be shown as LOEs. Regardless of whether each task is so depicted, it can be useful for each tasked organization to understand how its assigned task contributes to the overall end state. This can aid in understanding the cross-domain effects of given actions. See the figure, "Conceptual Map of Lines of Effort" to visualize LOEs.²⁹

Analyzing COGs provides a means of focusing friendly efforts, both offensively and defensively. There are several tools and techniques available to identify and analyze COGs. Joint doctrine (JP 5-0, Chapter IV) presents one model, but there are others, each with its assumptions, strengths, and weaknesses. All, however, attempt to relate what is critical to what is vulnerable in some helpful way-to identify and prioritize critical, targetable vulnerabilities. Each of the common methods is examined below, with a summary discussing the strengths and weaknesses of each.

²⁹ For additional information on COG analysis and LOEs, see JP 5-0, *Joint Planning*.



Caution: In the same sentence in which he first described a “center of gravity,” Clausewitz made it clear that it was only a metaphor, a picture to help understand the “main thing.”³⁰ The COG analysis techniques—even using COG as an acronym—highlight the extent to which military planning can uncritically employ shortcuts. The process of COG analysis may also lead to a mental image of a static adversary or enemy. The best correction for this oversimplification is to study the opponent thoroughly. Respect that the opponent is capable and willing to fight wherever possible. Accept that the opponent could be employing a strategy that we may find hard to understand. Addressing these challenges can be aided by using red teams.

All models have in common that any COG a commander chooses to affect should always be linked to one or more objectives. If the objective changes, the COG may also change. At the strategic level, a COG could be one or a set of leaders (political or military), an alliance, a military force, critical functions, or national will. A COG is often associated with an adversary’s or enemy’s military capabilities at the operational level, such as a powerful element of the armed forces. Still, it could also include other capabilities in the OE. COGs can emerge or change over time due to the interplay of friendly, hostile, and other forces in the OE. They may be based on the end state, mission, objectives, and the opponent’s strategy.

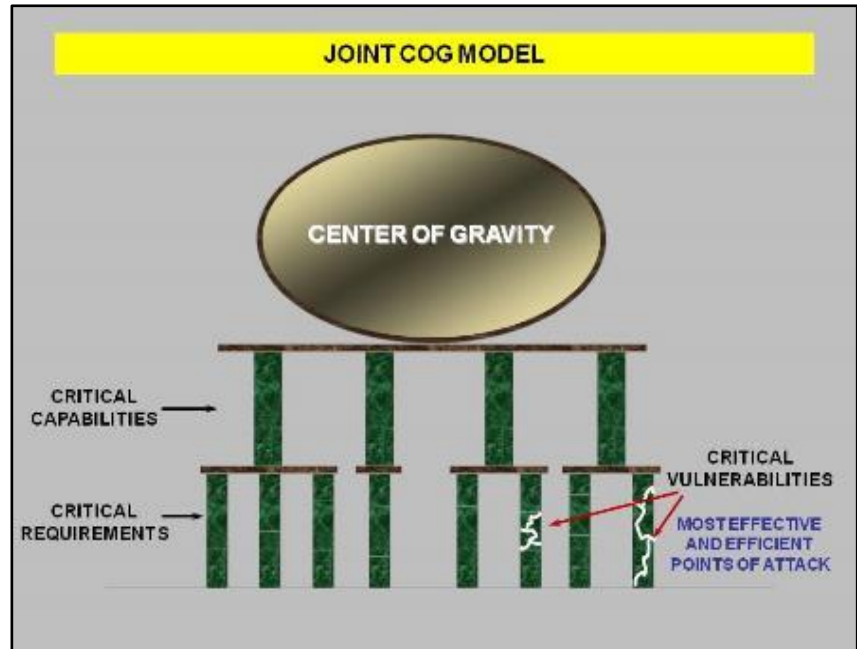
COG analysis occurs as part of JIPOE, mission analysis, or both. Commanders should consider the opponent’s COGs and identify and protect their COGs. Effects-based

³⁰ See Clausewitz, *On War*, for additional information.

approach to operations should orient on creating effects in time and space that decisively affect a COG.

THE JOINT MODEL

The model endorsed in joint doctrine is also known as the Marine Corps model, the “CG-CC-CR-CV Model,” and the “Strange Model,” for its developer, Dr. Joe Strange of the Marine Corps War College. This model uses the abbreviation “CG” for the center of gravity. For all purposes, “CG” and “COG” should be considered synonymous.³¹ This model is depicted in the figure “Joint COG Model,” and an example of its application is given in the figure “Joint COG Model Example (World War II).”



Description. This model starts with the joint definition of a COG as a source of strength, freedom of action, or the will to act. It then analyzes the COG to determine (in order) its:

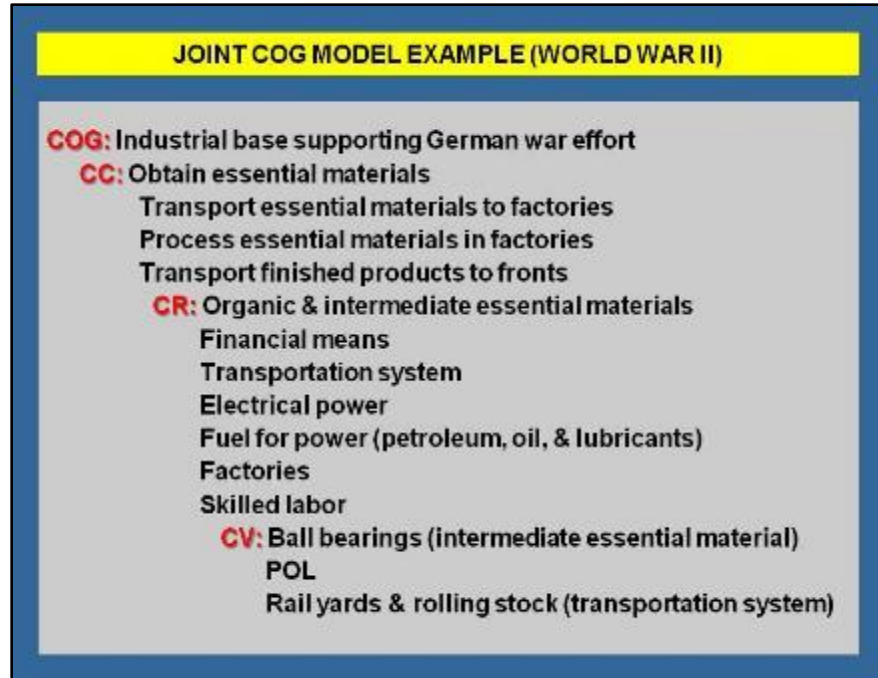
- ★ **Critical capabilities** (CCs): those means that are considered crucial enablers for a COG to function as such (and are essential to the accomplishment of the specified or implied objectives).
- ★ **Critical requirements** (CRs): essential conditions, resources, or means for a CC to be fully operational.
- ★ **Critical vulnerabilities** CRs, or components thereof, are deficient or vulnerable to attack (or other effects) that create decisive or significant effects on the COG.

COGs are nouns-tangible or intangible sources of power. CC can be thought of as verbs-things a COG *does*. CRs are nouns-those *things* a critical capability needs to function as such. CVs are those CR that are vulnerable.

Advantages. This is an intellectually thorough manner of analyzing COGs. It relates critical elements to vulnerabilities via a logical causal chain. It has been endorsed in joint doctrine and is taught in some form in most if not all, Service schools.

³¹ See Dr Joseph Strange, *Centers of Gravity and Critical Vulnerabilities: Building on the Clausewitzian Foundation So That We Can All Speak the Same Language*, for additional information.

Disadvantages. This method can be difficult to “operationalize”-to work through intellectually so that it yields actionable tasks and targets. Effective application of this approach requires a comprehensive and detailed understanding of a competitor’s systems. Doing it properly thus takes time. This model has significant power, but analysts may sometimes find it challenging to derive valid CC or properly determine



vulnerabilities from requirements. Experience has shown that these are the most common points when the model breaks down. As a result, analysts should use care and understand the system they are analyzing. This method also tends to be more labor-and information-intensive than other models.

THE STRATEGIC RING MODEL

This model is also known as the “five-rings model” and as “Warden’s Rings,” after its developer, Col (Ret) John A. Warden III.³²

Description. The basic structure of this model is not of COGs, per se, but of characteristics common to all living organisms. This is depicted in “The Strategic Ring COG Model.” It posits that there are one or more COGs within each ring of the systems, it is thus a straightforward systems analysis tool as much as it is a tool for COG analysis.

The model maintains that there are certain functions necessary for every system to function:

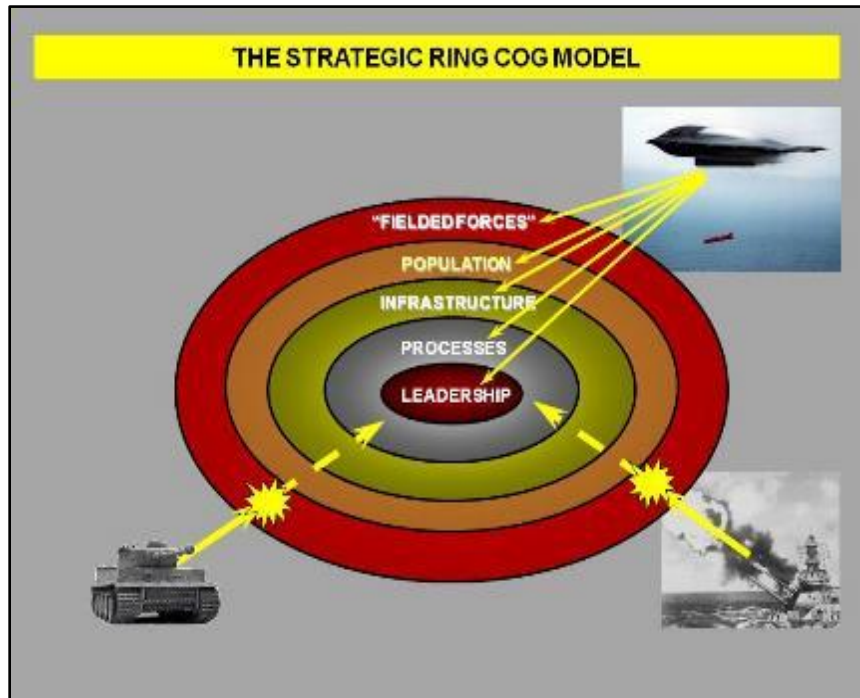
- ★ A C2 and information processing system, such as the leadership and C2 apparatus within a military or the central nervous system of a human body.
- ★ The processes necessary for the system's survival, such as communications, food production and distribution, financing, and manufacturing in a state, or respiration and blood circulation in a living body.
- ★ A system of infrastructure, like the electrical power distribution or transportation

³² See John Warden, *Warden’s Five-Ring System Theory*, for additional information.

systems of a nation or the bone and vascular systems of a body.

- ★ A population, such as the aggregate of individuals within a nation or armed force or the cells within a body.
- ★ A fighting or defense mechanism, such as the fielded armed forces of a nation or the immune system of a body (*NOTE: Col [Ret] Warden chooses to call this ring “fielded forces”*).

Advantages. This model shows the central value of leadership as a COG—it helps demonstrate the value of shock and dislocation on all rings through effects on leadership. It also shows that airpower does not have to fight through enemy fighting mechanisms (fielded forces) to affect the critical opponent systems defended by them, as other forms of military power often do. If used according to Warden’s original logic, one of the more powerful aspects of



the model is that the five-rings are “fractal”—they recapitulate at lower levels. They can be used to describe system behavior at successively lower levels. One can easily analyze any component or subcomponent of a system using the same five rings model.

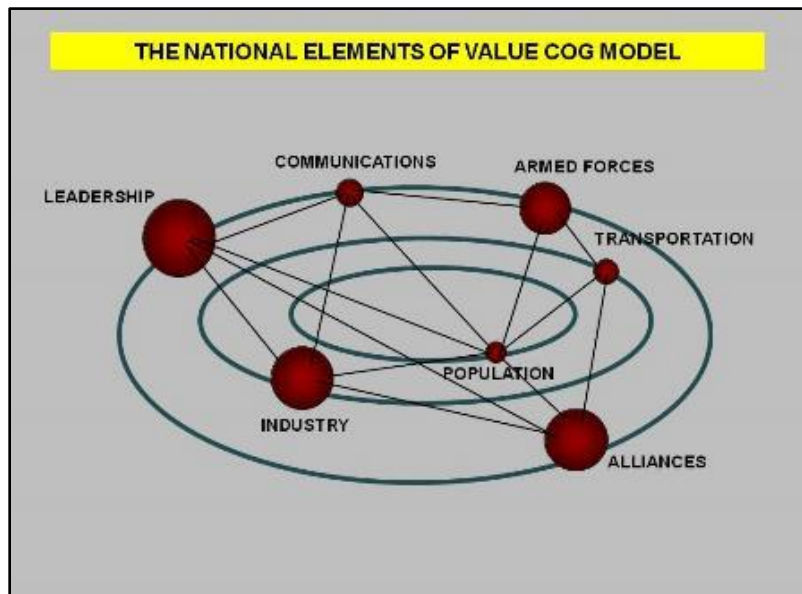
Disadvantages. This model makes no distinction between what is critical and vulnerable, in fact, it confuses the two. Applying blindly (“we’ve drawn our five-rings, and those are our COGs”) can encourage mirror imaging of the opponent’s system and lead to a mechanistic and reductionist inputs-based approach to targeting. Experience has shown that some teams using only this method list the five categories as *the* COGs and immediately begin listing “customary” target sets below them. This is the antithesis of effects-based targeting. Finally, this model considers the subject system in isolation, ignoring its connectivity to external systems and other aspects of the OE. This is the antithesis of a systems approach to COG analysis.

THE NATIONAL ELEMENTS OF VALUE (NEV) MODEL

This is also known as the NEV model and Barlow's Model, after its originator, Col (Ret) Jason Barlow.³³

Description. This model is generically like the strategic rings model but seeks to show a greater degree of interconnectivity and connectivity to external systems. The national elements of value include:

- ★ **Leadership**-The political and military decision-makers within the government.
- ★ **Industry**-All of a country's manufacturing, agriculture, research, technical enterprises, and parts necessary to support them, such as power production, water supply, and raw materials.
- ★ **Population**-A country's ubiquitous features that are important but hard to categorize and quantify, e.g., nationalism, morale, the will of the people, esprit de corps, ethnocentrism, ability to endure hardship, and religious conviction or fervor.
- ★ **Transportation**-All modes.
- ★ **Communications**-The physical means thereof.
- ★ **Alliances**-The friends, trading partners, and neighbors from which a country receives support for continuing the conflict. NEVs are interdependent and self-compensating. They are a critical means of system adaptation, redistribution, and recuperation. The lines connecting NEVs (depicted in the figure, "The National Elements of Value COG Model") are constantly varying in size and texture, as they represent the strength and direction of influence, both formal and informal, and the various lines of command, control, and authority inherent among the elements. Although the NEVs are the same for every country, they vary in importance from country to country and from day to day within a given country. Therefore, it can be assumed that commanders make rational decisions concerning their NEVs.



³³ See Jason Barlow, *Strategic Paralysis: An Airpower Theory for the Present*, for additional information.

Advantages. This model provides a somewhat more sophisticated analysis of the elements of a nation-state than does the strategic ring model. It also accounts for connectivity between elements and entities external to the system.

Disadvantages. The NEV model is designed to evaluate national systems and thus may be of limited value in analyzing non-state actors. Further, like the strategic ring model, it does not provide a means of analyzing individual elements as systems. It thus may have the exact disadvantages the strategic ring model does: oversimplification, a cookie-cutter approach, and a tendency to fit preconceived targeting information to the model rather than letting the model drive targeting decisions.

THE “CARVER” METHOD

This model is used in the special forces community and elements of the Federal government to assess Centers of Gravity and may have some validity for USAF elements (at echelon) to assess localized threats as well as friendly critical assets or capabilities.

Description. “CARVER” stands for “criticality, accessibility, recoverability, vulnerability, effect, and recognizability.” Its elements are used to conduct a comparative assessment of previously identified critical elements, according to the following criteria:

- ★ **Criticality**-How essential is this element to the successful functioning of its parent component, complex, or system?
- ★ **Accessibility**-How susceptible is this element to attack, given its defenses and friendly offensive capabilities?
- ★ **Recoverability**-How quickly and easily can this element recover from inflicted damage or destruction?
- ★ **Vulnerability**-How susceptible is this element to neutralization, damage, or destruction given friendly offensive capabilities?
- ★ **Effect**-What is the confidence that successfully prosecuting this element as planned creates the overall desired effect of the mission?
- ★ **Recognizability**-How easily recognizable is this element (i.e., differentiated from surrounding nodes) considering sensor capabilities, employment conditions (weather, etc.), and time available to analyze the situation and act?

The CARVER method is a means to help analyze which COGs are susceptible to effects, whether friendly, neutral, or enemy, given determination by other methods. At the tactical level, Commanders and planners should rate prospective COGs (or the associated critical vulnerabilities) as objectively as possible according to the six criteria above and then total the scores to indicate which element might be the most susceptible to effects. This may be applied to friendly force assessment (force protection, survivability) or hostile systems lucrative for multidomain attack. CARVER can be an excellent tool for tactical commanders to develop COOP Plans as well as in-depth force protection schema.

Advantages. This method can offer valuable insights, more on which CR is vulnerable or which CV to attack than on what constitutes a COG and its relation to the rest of the opponent's system.

Disadvantages. This is only a partial COG analysis tool and should be used in conjunction with other methods to determine the most lucrative elements for targeting.

SYNTHESIS

Time and staffing permitting, one of the best methods of analyzing COGs is to synthesize the methods described above. One notional means of doing so is to:

- ★ Identify opponent COGs.
 - ★★ Begin with the strategic rings model because of its simplicity.
 - ★★ Apply Barlow's NEV model for greater detail and functional nuance.
 - ★ Identify critical vulnerabilities.
 - ★★ Employ the joint (Strange) model (CG-CC-CR-CV) to determine CVs.
 - ★ Validate and prioritize the identified CVs.
 - ★★ Apply the CARVER method to rank CVs as subjects for action.
 - ★ Re-accomplish the first three steps for friendly COGs. Reassess periodically through COA wargaming and during each iteration of design and planning.
-

APPENDIX C: ACTIONS AND EFFECTS

There are two broad categories of actions relevant at the tactical and operational levels: **kinetic** and **non-kinetic**. Examples of kinetic actions include the use of explosive munitions and directed energy weapons. Examples of non-kinetic actions include using cyberspace weapons, radio broadcasts through operations in the information environment to encourage enemy surrender, and the employment of electromagnetic warfare capabilities.

Notes on the Terms “Lethal,” “Nonlethal,” “Kinetic,” and “Non-kinetic”

The terms “lethal” and “nonlethal” are currently recognized, although not formally defined, in joint doctrine. The existing dictionary definitions of these words describe them adequately. Joint doctrine refers to “lethal or nonlethal effects” (JP 3-0 and JP 3-09, *Joint Fire Support*), as well as “lethal and nonlethal fires.” This volume refers to the effects that both lethal and nonlethal weapons and fires have on targets exactly as joint doctrine does.

Two other terms are in widespread, if informal, use as well: “kinetic” and “non-kinetic,” intended to mean weapons or actions that create effects by releasing physical, kinetic energy, and those that do not. Joint doctrine, however, does not recognize these terms. Since they show no signs of disappearing from common use (including in much tactical doctrine), this publication uses definitions that convey distinct military meaning while keeping them as close as possible to the technical meaning of the terms in physics:

Kinetic: Relating to actions, capabilities, or weapons designed to produce effects using the forces and energy of moving bodies and directed energy. Kinetic actions can have lethal or nonlethal effects. **Non-kinetic:** Relating to actions, capabilities, or weapons designed to produce effects without the direct use of the force or energy of moving objects and directed energy sources. Non-kinetic actions can have lethal or nonlethal effects. **Neither term is used to describe an effect.**

TYPES OF EFFECTS

Four broad effects categories often overlap: *direct*, *indirect*, *intended*, and *unintended*. In addition, there are many subcategories within these categories, especially concerning indirect effects. A few of these subcategories are highlighted in the following sections because of their doctrinal implications. Understanding these types of effects is important to an effects-based approach.

- ★ **Direct Effects** are the results of action with no intervening effect or mechanism between act and outcome. They are also known as first-order effects. They are often immediate and easy to recognize. In most cases, they can be assessed empirically and meaningfully quantified.

- ★ **Indirect Effects.** Direct effects trigger additional outcomes-intermediate effects or causal mechanisms that produce additional outcomes or results. These are indirect effects, sometimes known as second-, third-, or higher-order effects. Indirect effects may be intended or unintended and may be categorized in many other ways (see discussion below). They are usually displaced from direct effects in time and space and often can be hard to quantify or measure empirically. As a result, they are often assessed or evaluated in qualitative terms. Generally, the less direct the effect (the further removed it is in the causal chain or in time from the initial action), the harder it is to predict its results and measure after. Historically, it has proven extremely difficult to anticipate beyond third-order effects with certainty.
- ★ **Intended Effects.** Intended effects are the desired, planned, and anticipated outcomes of an action or set of actions. They can be direct or indirect. Intended effects should always represent a net gain in accomplishing objectives or the end state
- ★ **Unintended Effects.** Unintended effects are outcomes of an action that are not part of the original intent. These effects may be undesired or desired, presenting opportunities for exploitation. Almost all actions produce some unintended effects. These can be direct but are usually indirect. If unplanned, they can also be desirable or undesirable from the friendly point of view, leading to outcomes that help or hinder the achievement of friendly objectives.

There is another aspect of unintended effects that is easy to overlook in planning. Even successful operations carry a cost in terms of lost opportunities. For example, destroying certain C2 or communications nodes to degrade enemy cohesion can remove valuable sources of friendly intelligence or prevent the transmission of surrender guidance by the enemy government. Likewise, destroying transportation nodes like bridges to impede enemy movement may interfere with future friendly schemes of maneuver or recovery efforts after combat has ceased. Effective planning should account for these opportunity costs.

An Airmen's perspective will often highlight alternatives to outright destruction that can create desired effects without removing future opportunities for exploitation or negatively affecting the end state. For instance, in strategic attacks against enemy electrical power, carried out to cripple conflict-sustaining resources and disrupt national leadership functions, planners can use nondestructive weapons to bring down power for a given period or can destroy only a few critical nodes to avoid wholesale destruction of infrastructure that could impede later civil stabilization efforts.

NATURE OF EFFECTS

Within the categories direct, indirect, intended, and unintended, the nature of effects may be further understood to be *physical*, *psychological*, *behavioral*, or *functional*.

- ★ **Physical effects** are the results of actions or effects that physically alter an object or system. Most physical effects are direct, but some may be indirect. Unintended or undesirable physical effects, like collateral damage, can often be major concerns in an operation.

- ★ **Psychological effects** are the results of actions or effects that influence the emotions, motives, and reasoning of individuals, groups, organizations, and governments.
- ★ Certain effects may result in changes in the outward behavior of these actors, which are then known as **behavioral effects**.
- ★ When components of a system act in concert to produce a given function (as when those manning an IADS operate that system), intended behavioral effects may lead to changes in the system's behavior. These changes are known as **functional effects**.

Operational level objectives often entail defeating enemy forces, and defeat inevitably involves a psychological component. Unless the enemy is destroyed outright, all such changes entail a change in the enemy's emotions, motivations, or reasoning. Thus, there is a psychological component to almost every set of effects and may be the most important aspect in generating desired behavioral changes. There are very few instances in history where an enemy, however thoroughly beaten in battle, was utterly denied a means of resistance. Ultimately, collapse entails a series of choices framed by emotion, motivation, and reason. In conflict or coercive operations, defeat is an event that occurs in the mind of the enemy, who chooses to end resistance or aggression and otherwise act (behave) as we desire.

While it is seldom possible to directly measure psychological effects in living systems, behavioral results (and related functional results) can be measured. Nonetheless, the psychological states leading to a particular behavior can be important to understanding causal mechanisms during planning.

TYPES OF INDIRECT EFFECTS

Cumulative and Cascading Effects. Indirect effects can be achieved in a cumulative or cascading manner. Effects that result from the combination of many other effects are *cumulative*. The effects of synergy function the same as elsewhere, the combined, cumulative effect of such actions designed and coordinated in concert should be greater than the sum of each individual action added together.

Cascading effects are indirect effects that ripple through an enemy system, usually affecting other systems. Typically, they flow from higher to lower levels and typically result of attacking critical system nodes that connect systems or sub-systems.

As a practical matter, some of the most desirable effects have both cumulative and cascading aspects. The point at which a military unit "fails" and ceases to act as a coherent fighting force is a typical example. The collapse itself may be triggered by an accumulation of losses (although the precise point at which collapse occurs is often difficult to predict). In addition, however, the unit's collapse may foster significant changes that spread through its component elements, subordinate units, and other connected or related systems. These are cascading effects.

Cascading effects may accomplish desired ends more effectively than cumulative effects. For example, removing critical nodes may ensure a complete collapse or neutralization than a cumulative, attritional approach. They may also achieve ends more efficiently, requiring fewer resources to achieve equivalent effects, thus freeing those resources for other uses. Of course, some systems do not lend themselves to this approach. It may not always be possible to identify or target key nodes but targeting efforts should strive to do so whenever possible.

Cumulative and Cascading Effects

Weakening an enemy unit's combat power by destroying a command vehicle would be part of the **cumulative effects** of attack upon the unit, as would the unit's eventual collapse through attrition of many of its vehicles and personnel. To the extent the attack impacts other units and elements across the enemy system due to the degraded C2 ability, such impacts could be considered **cascading effects**.

In the case of an integrated air defense system (IADS), air superiority may be achieved through the accumulation of effects against the IADS' components and achieving it may cascade into many other desirable effects, giving the air component greater freedom of action, as well as increasing freedom of action for the joint force in other domains.

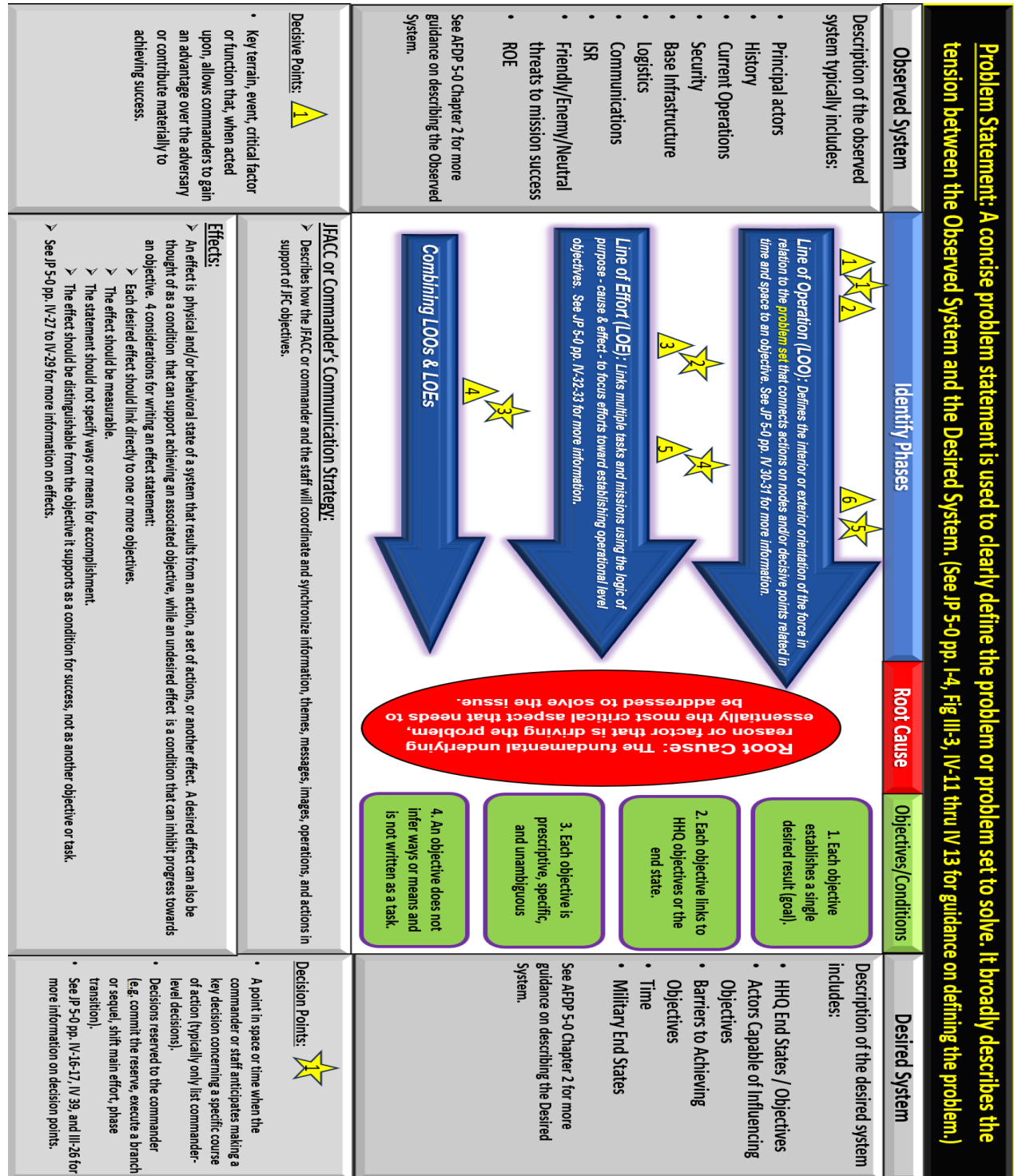
An electrical network, as an integrated complex system, demonstrates a different aspect of cascading effects. Targeting a few critical nodes within the network, then allowing internal system stress to cause successive cascading system-wide failure. Nature has inadvertently caused such effects with US power grids several times and coalition forces were able to achieve them early in Operation DESERT STORM by attacking a few key Iraqi power plants and distribution nodes. The effects of damage to critical infrastructure, such as the power grid will almost certainly cascade across other elements of the enemy system. Though effective at causing paralysis or system collapse, planners should be mindful of the potential cascading effects have for inducing unintended, second and third-order effects that may undermine achievement of objectives or the desired end state.

Effects of Sequential vs. Parallel Operations. In sequential operations, effects are imposed one after another. These operations are generally conducted when the effects generated through associated tasks and actions are required to enable follow on operations. Of the two, sequential operations are less likely to generate cumulative or cascading effects. To cause paralysis or enemy system failure, it is normally better to impose effects through parallel operations instead. Parallel operations aim to converge effects across an enemy system in a relatively short period, through various means and domains, to stress an enemy system in a manner that overwhelms its capacity to adapt. Functioning in this manner, parallel operations may take less time to achieve desired objectives.

However, parallel operations come at a cost. They require more resources (except time). Parallel operations are more complex to coordinate and should be planned more thoroughly, especially in terms of integration and synchronization of operations. Further, there may be reasons effects cannot or should not be imposed in parallel. In some cases, there may not be sufficient resources or capabilities. In other cases, a sequential approach is necessary because events need to happen in a certain order to enable other effects and ensure success. For example, in the opening minutes of Operation DESERT STORM, specific key early warning nodes were targeted to facilitate penetration of Baghdad's air defenses by other coalition aircraft. This one sequential strike helped ensure the success of the parallel efforts that followed.

APPENDIX D: AIR COMPONENT APPROACH TEMPLATE

Air component approaches are created using different templates to communicate the commander's vision. An example template is provided in this publication for reference.



APPENDIX E: EXAMPLE MODIFIED FIVE-PARAGRAPH ORDER

Subject: [Title or Serial] xxxxx

Originator: [Originator]

DTG: [date time group]

GENTEXT/SITUATION/

1. () SITUATION. <At a minimum, provide sufficient background information as to WHY the tasking is needed and timeline. Additional amplifying information, as necessary.>//

GENTEXT/MISSION/

2. () MISSION. <State the mission statement and address who, what, when, where, and why>//

GENTEXT/EXECUTION/

3. () EXECUTION.

3.A. () COMMANDER'S INTENT. <Sub-paragraphs include the purpose, end state, and risk>

3.B () CONCEPT OF OPERATIONS. <Sub-paragraphs may include a general description of type or scope of the requirement and establish times.>

3.C. () TASKS TO STAFF, ORGANIZATIONS, AND SUBORDINATE UNITS
<Sub-paragraphs provide specific tasks to staff, organizations, and subordinate units with sufficient information of task requirements>

3.D () COORDINATING INSTRUCTIONS.

3.D.1. C-DAY L-HOUR for deployments DDMMMYYYY____Z.

3.D.2. Target D-DAY L-HOUR DDMMMYYYY____Z.

3.D.3. <Estimated duration of the operation. Circumstance or date that automatically terminates operations.>

3.D.4. <Defense condition (DEFCON) or deployability posture.>

3.D.5. <Operational constraints, including any special ROE to include anti-terrorism and force protection issues applicable to this specific operation.>//

GENTEXT/FORCE SUSTAINMENT/

4. () FORCE SUSTAINMENT. <Sub-paragraphs may include areas such as logistics, concept of maintenance, quantities and class of supplies, resupply plan, health requirements, status protections, transportation, etc.>//

GENTEXT/COMMAND, SIGNAL, AND COMMUNICATIONS/

5. () COMMAND, SIGNAL, AND COMMUNICATIONS.

5.A () COMMAND RELATIONSHIPS: <State supported/supporting relationships>

<Sub-paragraphs provide type of control exercised by the supported commander (OPCON, TACON, direct liaison authorized [DIRLAUTH])>

5.B () COMMAND, CONTROL, COMMUNICATIONS, AND COMPUTER SYSTEMS (C4). <Sub-paragraphs may include requests for subordinate unit requirements, communications circuits to be used for reach-back, POCs, etc.> //

//SIGNED/JCD/DD Mmm YYYY//
JOHN C. DOE
Rank, USAF
Position

DECL/<source for classification>/<reason for classification>/<downgrade instructions or date>/<downgrading or declassification exemption code> //

(CLASSIFICATION)

APPENDIX F: AFPP KNEEBOARD

STEPS	INPUTS	ACTIVITIES	RESULTS/OUTPUTS
1: Planning Initiation	☐ HHQ Direction to start planning ☐ Commander's recognition of planning requirement	☐ Alert Staff ☐ Document relevant design elements ☐ Gather tools ☐ Make staff estimates ☐ Conduct initial assessment ☐ Issue Commander's guidance	☐ Planning timeline ☐ Planning team ☐ Planning Guidance ☐ Identification of necessary design elements ☐ Issue of preparatory order
2: Mission Analysis	☐ Strategic Guidance ☐ HHQ Planning directive ☐ Commander's initial planning guidance ☐ Description of the OE ☐ Definition of the Problem ☐ Operational or Air Component Guidance ☐ Commander's Intent	☐ Review HHQ Planning Activities, Strategic Guidance and identify Chain of Command ☐ Review Commander's initial planning guidance ☐ Identify design elements ☐ Determine Facts ☐ Analyze Most Likely and Most Dangerous threat to mission success (Enemy COAs for JPP & JPPA) ☐ Develop Initial assumptions ☐ Determine Constraints and Restraints ☐ Determine necessary elements of design to develop an operational approach or air component approach ☐ Analyze Friendly/Adversary COGs and Critical Factors (JPP/JPPA) ☐ Determine Root Cause of Problem that requires further planning ☐ Identify Specified, Implied and Essential tasks ☐ Determine Decision points ☐ Develop a Proposed Mission Statement ☐ Conduct Initial Resource Review ☐ Develop Mission success Criteria ☐ Develop Risk Assessment ☐ Determine Initial CCIRs ☐ Determine the problem, problem's route cause, and subordinate problem sets ☐ Develop Commander's Intent Statement ☐ Prepare/Update Staff Estimates ☐ Present a Commander's Mission Analysis Brief	☐ Publish an air component approach (JPP/JPPA) ☐ Publish Commander's Refined Planning Guidance ☐ Identified Specified, Implied and Essential Tasks ☐ Component(s) of the problem that need to be addressed ☐ Staff estimates ☐ Mission Statement ☐ Updated Commander's intent statement ☐ List of initial CCIRs
3: COA Development	☐ HHQ Guidance ☐ Operational Approaches ☐ Air Component Approaches ☐ Mission Analysis Results	☐ Brainstorm COAs ☐ Chart/Visualize COAs ☐ Prepare COA concept of operations statement, sketch, and task organization ☐ Brief Commander on COAs under consideration – COA Selection Brief ☐ Continue Staff Estimate process to enhance COAs ☐ Continue vertical and horizontal parallel planning	☐ CONOPS ☐ COMREL ☐ Required capabilities ☐ Deployment and Sustainment Concept ☐ COA statement (narrative description) ☐ Valid COAs
4: COA Analysis and Wargaming	☐ Valid COAs ☐ Mission Analysis results ☐ Operations approaches ☐ Air component approaches ☐ COA evaluation criteria ☐ Selected wargaming method ☐ Planning guidance	☐ Prepare for COA Analysis and Wargaming ☐ Conduct Analysis/Wargaming and record results ☐ Refine/Modify each COA ☐ Report results	☐ Analysis and Wargaming results Inform COA Comparison ☐ Advantages/disadvantages for each COA
5: COA Comparison	☐ Commander's evaluation criteria and definitions ☐ Weight considerations (if used) ☐ COA Analysis and Wargaming results ☐ Selected COA comparison method ☐ Associated guidance to the staff	☐ Compare COAs using selected COA comparison method	☐ Individual COA advantages and disadvantages ☐ COA discussion notes ☐ COA recommendation
6: COA Approval	☐ Individual COA Advantages and disadvantages ☐ COA Discussion Notes ☐ COA recommendation	☐ Prepare COA Decision Briefing ☐ Present COA Decision Briefing ☐ Commander selects and/or Modifies COA ☐ Refine Selected COA (as required) ☐ Prepare briefing, message, or Commander's Estimate for HHQ Approval	☐ Selected COA
7: Plans and Orders Development	☐ COA Approval Decision ☐ Valid COA ☐ COA Analysis and wargaming results ☐ Mission analysis results ☐ Air component design ☐ Air component approach ☐ Operational approach ☐ HHQ guidance	☐ Develop base paragraphs ☐ Publish base order/plan ☐ Develop appropriate annexes ☐ Publish annexes and supplemental guidance as FRAAGORDS	☐ An order or plan to disseminate to subordinate forces

REFERENCES

All websites accessed 15 January 2025.

Doctrine can be accessed through links provided at: <https://www.doctrine.af.mil/>

US AIR FORCE DOCTRINE: <https://www.doctrine.af.mil/>

- ★ AFDP 1, [The Air Force](#)
- ★ AFDP 1-1, [Mission Command](#)
- ★ AFDP 3-0, [Operations](#)
- ★ AFDP 3-0.1, [Command and Control](#)

JOINT DOCTRINE

Joint Electronic Library (JEL): <https://www.jcs.mil/Doctrine/>

JEL+: <https://jdeis.js.mil/jdeis/index.jsp?pindex=2>

- ★ JP 2-0, [Joint Intelligence](#)
- ★ JP 3-0, [Joint Campaigns and Operations](#)
- ★ JP 3-02, [Amphibious Operations](#)
- ★ JP 3-30, [Joint Air Operations](#)
- ★ JP 5-0, [Joint Planning](#)

MISCELLANEOUS PUBLICATIONS

DoDM 5200.01 Volume 2, [Marking of Information](#)

DoDI 5200.48, [Controlled Unclassified Information \(CUI\)](#)

[DoD Dictionary of Military and Associated Terms](#)

CJCSM 3130.03A, [Planning and Execution Formats and Guidance](#)

CJCSM 3105.01B, [Joint Risk Analysis Methodology](#)

[Air Force Glossary](#)

Dr Joseph Strange, [Centers of Gravity and Critical Vulnerabilities: Building on the Clausewitzian Foundation So That We Can All Speak the Same Language](#)

Maj (Ret) Jason B. Barlow, [Strategic Paralysis: An Airpower Theory for the Present](#)

Col (Ret) John A. Warden III., [Warden's Five-Ring System Theory](#)

Carl von Clausewitz, *On War*

Fact Sheet: 2023 DoD Cyber Strategy

This week, the Department of Defense (DoD) transmitted to Congress the classified *2023 DoD Cyber Strategy*. The *2023 DoD Cyber Strategy* establishes how the Department will operate in and through cyberspace to protect the American people and advance the defense priorities of the United States.

This strategy is subordinate to the *2022 National Security Strategy* and the *2022 National Defense Strategy*. It complements the *2023 National Cybersecurity Strategy* and builds upon and supersedes the *2018 DoD Cyber Strategy*. An unclassified summary is forthcoming.

The *2023 DoD Cyber Strategy* is grounded in real-world experience. Since 2018, the Department has conducted a number of significant cyberspace operations through its policy of defending forward, actively disrupting malicious cyber activity before it can affect the U.S. Homeland. This strategy is further informed by Russia's 2022 invasion of Ukraine, which has demonstrated how cyber capabilities may be used in large-scale conventional conflict.

These experiences have shaped the Department's approach to the cyber domain:

- The Department will maximize its cyber capabilities in support of **integrated deterrence**, employing cyberspace operations in concert with other instruments of national power.
- The Department will **campaign** in and through cyberspace below the level of armed conflict to reinforce deterrence and frustrate adversaries.
- Finally, the Department recognizes that the United States' **global network of Allies and partners** represents a foundational advantage in the cyber domain that must be protected and reinforced.

The Department confronts an increasingly contested cyberspace:

- **The People's Republic of China (PRC)** represents the Department's pacing challenge in the cyber domain. The PRC has made significant investments in military cyber capabilities and empowered a number of proxy organizations to pursue malicious cyber activities against the United States.
- **Russia** poses an acute threat in cyberspace, evidenced by its malign influence efforts against the United States and repeated cyber attacks against Ukrainian civilian critical infrastructure.
- **North Korea, Iran, and Violent Extremist Organizations** remain persistent cyber threats.
- **Transnational Criminal Organizations** represent a unique threat in cyberspace due to their technical aptitude and often close alignment with the foreign policy objectives of their host governments.

In order to address current and future cyber threats, the Department will pursue four complementary lines of effort:

- **Defend the Nation.** The Department will campaign in and through cyberspace to generate insights about malicious cyber actors, as well as defend forward to disrupt and degrade these actors' capabilities and supporting ecosystems. Additionally, DoD will work with its interagency partners to leverage all available authorities to enable the cyber resilience of U.S. critical infrastructure and to counter threats to military readiness.

- **Prepare to Fight and Win the Nation's Wars.** The Department will ensure the cybersecurity of the DoD Information Network and will further invest in the Joint Force's cyber resilience. Additionally, the Department will use cyberspace operations to generate asymmetric advantages in support of the Joint Force's plans and operations.
- **Protect the Cyber Domain with Allies and Partners.** The Department will assist U.S. Allies and partners in building their cyber capacity and capability, as well as expand avenues of potential cyber cooperation. DoD will continue to conduct hunt forward operations to build cyber resiliency and will reinforce responsible state behavior by encouraging adherence to international law and internationally recognized cyberspace norms.
- **Build Enduring Advantages in Cyberspace.** The Department will optimize the organizing, training, and equipping of the Cyber Operations Forces and Service-retained cyber forces. Furthermore, DoD will invest in the enablers of cyberspace operations, including intelligence, science and technology, cybersecurity, and culture.

With a robust and integrated cyber capability, the Department will work to deter conflict where it can and prevail where it must.

Air Force Future Operating Concept Executive Summary

The Air Force Future Operating Concept (AFFOC) articulates how Airmen will successfully fight in the future and deliver airpower to deter and, if necessary, stop aggression by a peer adversary. The AFFOC is a key component to accomplishing the needed “change” in General Brown’s *Accelerate, Change, or Lose* concept, guiding the U.S. Air Force to fly, fight, and win in a new era. Specifically, the AFFOC explains how Airmen offer unique contributions to the Joint Force – and the forces of our Allies and partners – in support of the National Security Strategy, the National Defense Strategy, and the Joint Warfighting Concept.

In a situation where a peer adversary initiates military aggression, the U.S. Air Force brings trained Airmen and advanced capabilities to accomplish its five core functions: Air Superiority; Global Strike; Rapid Global Mobility; Intelligence, Surveillance, and Reconnaissance; and Command and Control. In doing so, the U.S. Air Force creates opportunities for the rest of the Joint Force to apply their strengths and stop the aggression. No service or force element operates alone, but in the face of the advanced threat, it is the Airmen who open the door ... or kick it down.

The AFFOC presents the current military problem as six critical and concurrent fights that Airmen must win through “pulsed airpower,” which is the concentrating of airpower in time and space to create windows of opportunity for the rest of the force. The Department of the Air Force’s investments in seven Operational Imperatives will provide critical capabilities to make these pulses possible. The AFFOC states that winning the six fights depends on Airmen empowered to make decisions under a doctrine and culture of mission command.

The AFFOC’s primary audience is Airmen, including Headquarters Air Force (HAF) staff, the Air Force Major Commands (MAJCOMs), and Airmen across the Joint Force who lead strategy, planning, and mission execution in all geographic and functional combatant commands. When Airmen apply the principles of the AFFOC, it will enable the Joint Force to coalesce combat power in unstoppable, all-domain pulses.

To do this, Airmen must be able to prevail in six key “fights” as they apply airpower in the defense of Allies, partners, and national interests.

- The (1) **fight to compete with and deter** endures from competition through conflict. Even in combat, Airmen will compete to nurture partnerships and strengthen alliances and deter horizontal and vertical escalation throughout the spectrum of conflict. While *competing and deterring*, Airmen deploy according to the Air Force Force Generation model, fostering relationships with Allies and partners and sharing capabilities and concepts that are integrated by design. They pre-position key warfighting materiel, create resilient infrastructure, and practice Agile Combat Employment.
- When ready to act globally, Airmen will overcome adversary attacks and (2) **fight to get into theater**. Our adversaries have studied how we fight – Airmen should expect resistance and interference from the very beginning of any crisis. Airmen will execute Agile Combat Employment when *getting into theater*. This will require Advanced Battle Management Systems capabilities, agile control powered by the deployable Tactical Operations Center, and Adaptive Defense to protect Airmen and operating locations from multi-domain threats, both kinetic and non-kinetic.

- Whether supporting Allies and partners directly or generating combat power over long distances, Airmen must (3) **fight to get airborne**. No air base, even in our homeland, will be a sanctuary. Airmen will generate combat power, even in the face of enemy attempts to stop them. Key supporting concepts include an Advanced Battle Management System and an all-domain sensing grid that will be pivotal in the *fight to get airborne* as the Joint Force synchronizes efforts and effects in time and space.
- Tomorrow's adversaries will contest our control of the air in the attempt to prevent us from delivering effective support to the Joint Force, our Allies, and partners; therefore, Airmen must (4) **fight for air superiority** to create windows of opportunity for teamwork in all warfighting domains. Key concepts in the *fight for air superiority* are next-generation munitions, collaborative combat aircraft, the Advanced Battle Management System, agile electronic attack, and the sensing grid.
- To deliver devastating combat power against an aggressor, Airmen act as part of the Joint Force to (5) **fight to deny adversary objectives** by stopping an attack or invasion. Airmen must *deny adversary objectives* through next-generation and palletized munitions, as well as stand-off and stand-in strike capabilities powered by the Advanced Battle Management System that allows for distributed control. This fight will also require all-domain intelligence, surveillance, and reconnaissance, the all-domain sensing grid, electronic warfare, and cyber defense and attack teams.
- Airmen must (6) **fight to sustain** ourselves, our Allies, and partners in competition and conflict. The ability to conduct large-scale logistics has been a key strength of the U.S. military, and this will be a primary target in a future conflict. Airmen must be smarter, more agile, more responsive, and more flexible than our adversaries to survive and sustain the Joint Force. Airmen *sustain the fight* by conducting logistics under attack, adopting innovative ways to use Unmanned Aerial Systems for logistics, defending against all-domain effects, to include long-range missile threats, and recovering operating locations quickly after attacks, ensuring relentless combat pressure to stop aggression.

Airmen must be prepared to engage in several of these fights *at the same time*. For this reason, Airmen who are experts in a single core function will be unprepared to win against a peer adversary. Airmen will remain technical experts, but they will also be trained in additional skills that allow them to fight and win as agile teams. Tomorrow's Airmen must be multi-capable in order to win multiple fights.

The AFFOC will not work without Airmen skilled in mission command. Mission command is a philosophy of leadership that empowers Airmen to operate in uncertain, complex, and rapidly changing environments through trust, shared awareness, and understanding of the commander's intent. When empowered in this way, Airmen do not have to wait for orders from the higher headquarters to make bold decisions and take advantage of fleeting opportunities. This type of leadership does not just happen. It takes intentional development and practice.

The strength of our Nation's airpower is its Airmen. This AFFOC captures the value Airmen bring to the Joint Force by delivering unstoppable airpower to deter, confront, combat, and halt aggression.

Strategic Environment – Warfighting Concepts

SNCOA Warfighting Concepts Primer

01 May 2024

Joint All Domain Operations

(JADO): JADO involves the synchronized employment of military capabilities across all warfare domains (air, land, maritime, space, and cyberspace), plus the electromagnetic spectrum (EMS), to deter adversaries and ensure U.S. military superiority. As stated in AFDP 3-99, *The Department of*

the Air Force Role in Joint All Domain Operations, “**Actions by the joint force in multiple domains integrated in planning and synchronized in execution**, at speed and scale needed to gain advantage and accomplish the mission.”

Example: *The U.S. military's use of cyber capabilities to disrupt enemy communications while simultaneously deploying air and space assets for surveillance and precision strikes demonstrates JADO's approach to integrating efforts across domains for a strategic advantage.*

Real-World Example: The integration of cyber operations with kinetic actions in operations against ISIS, where cyber capabilities were used to disrupt the enemy's command and control while air strikes targeted critical infrastructure.

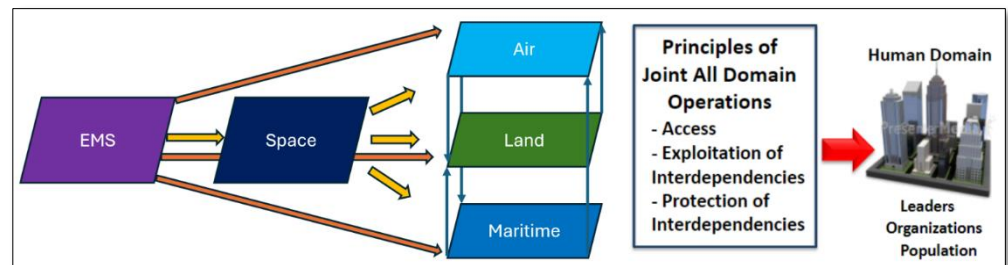


Figure 1 - Domain Interrelationships within Continuum of Domains, From: <https://www.nrdc-ita.nato.int/newsroom/insights/creating-competitive-space-through-a-framework-of-joint-all-domain-maneuver>

Joint All Domain Command and Control (JADC2): JADC2 is the Department of Defense's concept aimed at connecting sensors from all military services into a single network, enhancing decision-making and information-sharing across all domains. According to the *Department of Defense Joint All-Domain Command and Control (JADC2) Strategy*, JADC2, “provides a coherent approach for shaping future Joint Force C2 capabilities and is intended to produce the warfighting capability to sense, make sense, and act at all levels and phases of war, across all domains, and with partners, to deliver information advantage at the speed of relevance.” In this respect, **effective JADC2 accelerates our ability to Observe, Orient, Decide, Act (OODA) while simultaneously degrading our enemy's abilities.** **Example:** *The development and implementation of the Advanced Battle Management System*



Figure 2 - US Air Force Image, From: <https://news.usni.org/2022/04/06/panel-pentagon-needs-to-be-clearer-on-goals-for-jadc2>

Strategic Environment – Warfighting Concepts

(now the DAF Battle Network) by the U.S. Air Force exemplify JADC2 by integrating air, space, and cyber information to enable rapid decision-making and response to threats.

Note: As of May 2023, JADC2 is now referred to as CJADC2 or “combined” JADC2. “The ‘combined’ is to highlight that we’re going to start out with these capabilities by being interoperable from the beginning,” Lt. Gen. Mary O’Brien, director of C4 and cyber and chief information officer, J6 of the Joint Staff.

Real-World Example: The U.S. Air Force Program Executive Office for Command, Control, Communications and Battle Management is modernizing ABMS since it is not broad enough to completely address JADC2 command-and-control needs. The new system has been named the DAF Battle Network and will fulfill one of Secretary Kendall’s Operational Imperatives for the U.S. Air Force, “operationally-focused ABMS”.

Mission Command: Mission Command is a leadership philosophy that empowers subordinate decision-makers to make decisions based on their understanding of commander's intent, the mission, and context of the operation, promoting agility and decentralized execution. From AFDP 1-1, *Mission Command*, “The hallmark of mission command is decentralized execution through the delegation of authority to empower subordinate decision-making and enable flexibility, initiative, and responsiveness in the accomplishment of commander’s intent.” Mission Command principles are quite simple: ***provide clear commander’s intent, create shared understanding, exercise disciplined initiative, build teams through mutual trust, accept prudent risk, and use mission-type orders (MTOs) when appropriate.*** Note: MTOs tend to focus on purpose rather than details to delegate and empower decision-making at the lowest level possible. According to AFDP 1-1, *Mission Command*, “Together, the embodiment of the mission command culture, the practice of its principles, and guidance provided by commander’s intent enables the execution of operations through the framework of CC-DC-DE that allows the USAF to fly, fight, and win now and into the future.” **Example:** *The U.S. Army's use of mission command principles in multinational exercises enables leaders at all levels to make timely decisions, adapt to changing battlefield conditions, and exploit opportunities to achieve operational objectives.*

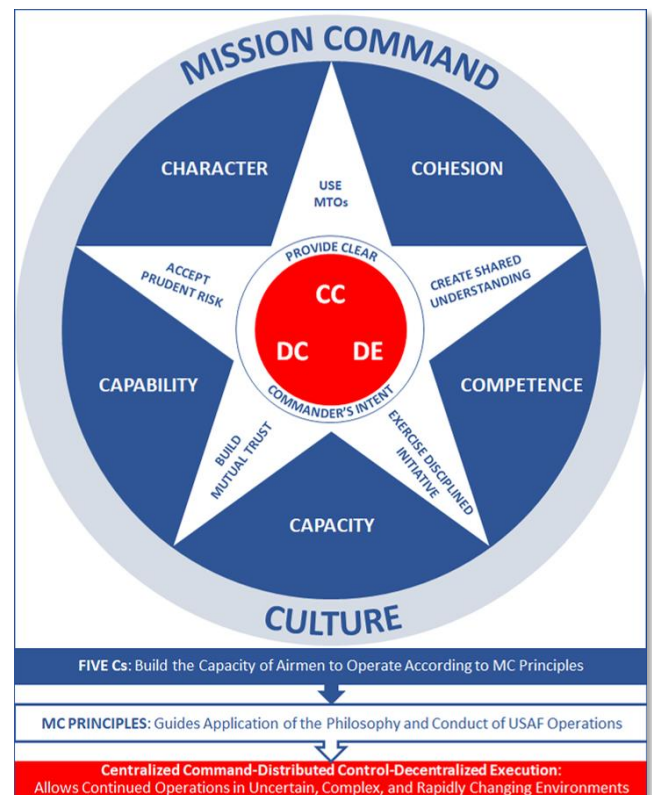


Figure 3 - From Air Force Doctrine Publication 1-1, *Mission Command*:

https://www.doctrine.af.mil/Portals/61/documents/AFDP_1-1/AFDP%201-1%20Mission%20Command.pdf

Real-World Example: On D-Day in 1944, Allied forces launched the largest amphibious invasion in history to begin the liberation of Nazi-occupied Europe. The operation was massive in scale, involving over 156,000 troops from the United States, Britain, Canada, and other Allied nations, landing on five beachheads along the Normandy coast of France. At Omaha Beach, American forces faced unexpected strong resistance from German fortifications. The pre-landing bombardments had failed to neutralize the German defenses, leading to heavy casualties among the first waves of landing troops. The situation seemed dire, and a textbook operation was not feasible due to the stiff resistance and chaotic conditions. In this context, junior officers and non-commissioned officers on the ground took the initiative, adapting their tactics on the fly. They organized ad hoc units from different companies and branches, directed fire support to suppress enemy positions, and led their troops to breach the defenses through gaps and improvised routes. This autonomy and adaptability, hallmarks of mission command, were critical in eventually overcoming the German defenses and securing

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the beachheads. The success of D-Day underscored the effectiveness of mission command in allowing units to respond with flexibility to unexpected challenges and opportunities, fostering innovation and resilience among troops facing the fog of war.

Agile Combat Employment (ACE):

ACE is a proactive and reactive operational scheme that enhances the survivability of forces and their ability to generate combat power by shifting operations from centralized physical infrastructures to a network of smaller, dispersed locations that can complicate adversary planning making it difficult for adversaries to target U.S. forces effectively. According to AFDN 1-21, *Agile Combat Employment*, “ACE is an operational concept that supports joint all-domain operations (JADO). Joint force operations are increasingly

interconnected, interdependent, and challenged. Anti-access and area denial threats, reduced freedom of maneuver, and rapid proliferation of advanced technologies challenge the Air Force’s ability to operate. The ***successful employment of ACE positions the force to observe, orient, decide, and act in concert across all domains. To achieve freedom of action, ACE enables convergence across domains, presenting an adversary with dilemmas at an operational tempo that complicates or negates adversary responses*** and enables the joint force to operate inside the adversary’s decision-making cycle.” **Example:** *The U.S. Air Force’s deployment of a multi-capable, rapidly deployable force package to austere locations in the Pacific demonstrates ACE by showing the ability to operate from multiple dispersed locations, enhancing resilience against enemy attacks.*

Real-World Example: The U.S. Air Force exercise “Bomber in a Box” in 2023 at Barksdale AFB and Fairchild AFB showcased ACE principles by demonstrating the capability to rapidly deploy Four B-52H Stratofortresses from the 2nd Bomb Wing. The bulk of the exercise took place at Fairchild AFB, but the maintenance personnel from the 2AMXS started their portion of ACE at Barksdale. They packed the necessary repair, maintenance and replacement equipment into an innovative cargo container called a BOCS (B-52 On-Board Cargo System) before flying with the aircrew to Fairchild. Each bomber transported their own mobile maintenance team and a BOCS to practice the capability of landing, rearming and repairing the aircraft anywhere that has enough runway.

ACE Operations, Activities, and Investments	
ACE operations present commanders with options to initiate, continue, or re-establish operations while balancing associated risk factors, as shown through a few examples below.	
Continuum event	ACE operations, activities, and investments
Cooperation	★ Create steady-state and contingency authorities with partner nations which allow for: overflight, direct coordination with host nation defense, staging of material/equipment, etc. ★ Organize, Train, and Equip forces for ACE constructs ★ Build interoperability with allies and partners
Competition	★ Exercise ACE operations and conduct capability demonstrations with joint forces and partner nations as an active deterrent to conflict ★ Maintain freedom of access and maneuver
Conflict	★ Conduct persistent mission generation and logistics under attack ★ Complicate enemy observe-orient-decide-act loop ★ Preserve combat capability ★ Generate combat power

Figure 4 - From AFDN 1-21: https://www.doctrine.af.mil/Portals/61/documents/AFDN_1-21/AFDN%201-21%20ACE.pdf

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Centralized Command-Distributed Control-Decentralized Execution (CC-DC-DE): CC-DC-DE is a foundational principle of U.S. military command and control (C2) strategy, evolving from centralized control and decentralized execution within the U.S. Air Force. AFDP 1-1, *Mission Command* further explains, “The delegation of authority to subordinate commanders who are capable and resourced to plan, coordinate, execute, and assess operations within an acceptable level of risk is the hallmark of successful distributed control. Likewise, the empowerment of subordinate decision-making guided by clear understanding of commander’s intent and guidance on risk tolerance is the hallmark of successful decentralized execution.” CC-DC-DE seeks to **balance the need for authoritative direction with the requirement for flexibility and rapid response at the tactical level** and is intended to provide a unifying framework for the development of new operating concepts, organizational approaches, and materiel solutions that address the challenges posed by the future operating environment.



Figure 5 - Image by AFSNCOA



Figure 6 - Image by AFSNCOA, Text From: AFDP 1-1, *Mission Command*

Real-World Examples:

Centralized Command (*Centralized Command ensures unity of effort and coherence in the overall mission and objectives, with strategic decisions made by senior leadership*): **Operation Desert Storm (1991)**

Overview: The coalition forces led by the United States against Iraq to liberate Kuwait.

Example of Centralized Command: General Norman Schwarzkopf, the commander of Coalition forces, maintained centralized command over the operation. He was responsible for the strategic decision-making and planning of the campaign, ensuring a unified effort and coordination among the coalition forces. This centralized command allowed for a cohesive strategy that effectively combined air, land, and sea operations.

Impact: The centralized command structure enabled efficient coordination of a large and diverse coalition force, leading to the successful liberation of Kuwait.

Distributed Control (*Distributed Control allows for dispersal of decision-making authority and control mechanisms across various echelons or platforms, enabling responsive and adaptive operations to dynamic conditions and threats*): **Counter-Insurgency Operations in Afghanistan (2001-2021)**

Strategic Environment – Warfighting Concepts

Overview: The U.S. and NATO forces' operations against insurgent groups in Afghanistan.

Example of Distributed Control: In Afghanistan, the challenging terrain and complex social landscape required a more flexible approach to command and control. U.S. and NATO forces adopted a distributed control model, where higher command levels provided the overall objectives and resources, but operational control was distributed among regional commanders. This allowed for more agile responses to insurgent activities and better adaptation to local conditions.

Impact: Distributed control facilitated better use of local intelligence and quicker adjustments to tactics, although the long-term success was mixed due to the complexity of the insurgency and other political factors.

Decentralized Execution (*Decentralized Execution allows for empowering subordinate decision-making to enable flexibility, initiative, and responsiveness in mission accomplishment*): **Operation Neptune Spear (2011)**

Overview: The U.S. Navy SEAL operation that resulted in the death of Osama bin Laden in Pakistan.

Example of Decentralized Execution: This operation exemplifies decentralized execution, where the strategic objectives were defined by high-level command (the President of the United States and top military leadership), but the execution was left to the SEAL team on the ground. The team had the autonomy to make real-time decisions based on the situation they encountered in the compound. This approach was critical due to the operation's sensitivity and the need for rapid, precise action.

Impact: The decentralized execution allowed the SEAL team to adapt to unforeseen challenges during the operation, leading to the successful elimination of the target with no casualties among the assault team.

Joint Warfighting Domains

The U.S. Department of Defense (DoD) recognizes multiple domains for joint warfighting, reflecting the diverse and interconnected battlefields of modern warfare. As stated in the *Cross-Domain Synergy in Joint Operations Planners Guide*, 2016, “Military operations are becoming more complex with the rise in the number and variety of options available to commanders. Today’s warriors must contend with computers and satellites in addition to bayonets and bullets. The expansion of military activity beyond the air, sea, and land domains to space and cyberspace has broadened the community of warfighters to include computer scientists and astrophysicists. Integrating this expertise to achieve operational effectiveness against an adaptive, complex enemy is the mission of Joint Force Commanders (JFCs) and their staffs.”

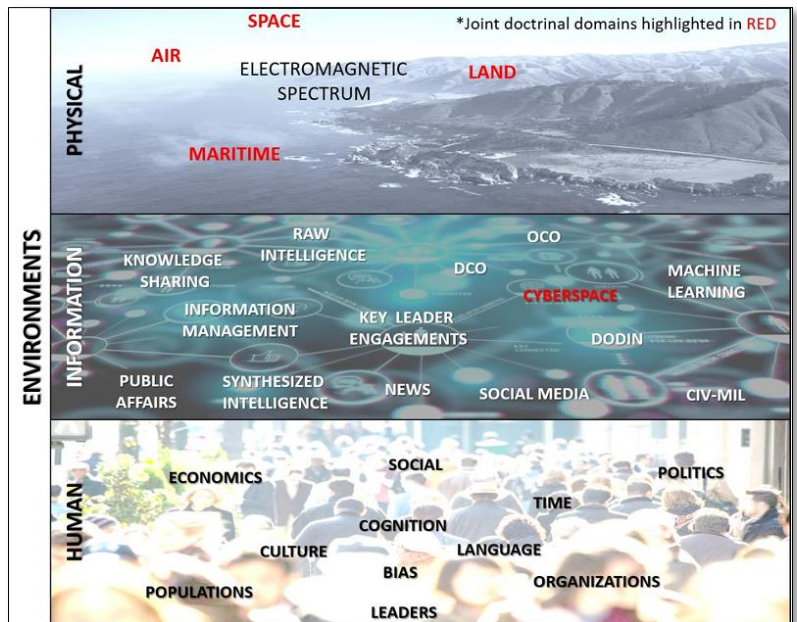


Figure 7 - Image sourced from AFDP 3-99, Appendix A

Land: This is the traditional domain of warfare involving ground troops, armored vehicles, and land-based military operations. It includes securing territory, engaging in direct combat, and supporting operations with logistics and infrastructure.

Strategic Environment – Warfighting Concepts

Air: Involves the use of military aircraft, including fighters, bombers, drones, and support aircraft for surveillance, air superiority, bombing, and transport. Airpower is crucial for providing support to ground forces, controlling airspace, and conducting strategic strikes.

Maritime (Sea): This domain covers operations on, under, and over the sea, involving naval ships, submarines, and maritime aircraft. It includes securing sea lines of communication, projecting power ashore, and defending against threats to maritime interests.

Space: Recognized increasingly as a vital warfighting domain, space operations involve satellites, space vehicles, and related technologies for communication, navigation, reconnaissance, and missile warning. Space capabilities enable global positioning systems (GPS), real-time communication, and surveillance.

Cyberspace: Encompasses the digital battlefield, including networks, systems, and cyber infrastructure. Operations in cyberspace are conducted to protect and exploit information systems, disrupt or degrade enemy capabilities, and defend against cyber-attacks.

Electromagnetic Spectrum (EMS): While sometimes considered part of cyberspace, EMS operations focus on exploiting, managing, and controlling electromagnetic signals for communication, radar, and electronic warfare. This includes jamming enemy communications, radar, and guidance systems, as well as protecting friendly use of the spectrum.

Information: Often integrated with cyberspace and EMS, the information domain focuses on the use, management, and denial of information to shape perceptions, manage knowledge, and influence decision-making processes. This can include psychological operations, propaganda, and counter-information campaigns.

Note: These domains are not isolated; they overlap and interconnect, requiring joint operations and coordination across all branches of the military to effectively conduct modern warfare. The integration of efforts across these domains is crucial for achieving strategic objectives and ensuring national security.

Warfighting Concepts Primer – Attachment 1 Reflection Questions

Question 1: Considering *Mission Command culture*, which of the 5 Cs (*Cohesion, Competence, Capacity, Capability & Character*) are lacking or need attention in your home unit? Why do you think that is? Please provide an example.

(Optional) Use the SEE format for the question above

State: *My home unit's Mission Command culture is most lacking _____.*

Elaborate: *This is because:* OR *In other words:*

Exemplify (examples): *For example, at work:*

Illustrate (OPTIONAL): *This is similar to:* OR *This is like:*

Question 2: How do you envision the role of Senior NCOs in your career field evolving within the *framework of ACE*. What steps could you take to prepare your team for these changes? (Optional)

Question 3: Discuss the importance of effective *communication in JADO* and share a personal experience where communication significantly impacted the outcome of an operation (preferably joint).

This image shows a full page of blank white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, providing a guide for writing. There are no margins, text, or other markings on the paper.

Warfighting Concepts Primer – Attachment 3 Resources

Student Preparation:

Watch: ACE Concepts (DVIDS video)

<https://www.dvidshub.net/video/872198/ace-concepts>

Read: Air Force Doctrine Publication 3-85, Electromagnetic Spectrum Operations

https://www.doctrine.af.mil/Portals/61/documents/AFDP_3-85/AFDP%203-85%20Electromagnetic%20Spectrum%20Ops.pdf

(Appendix A and B)

Read: Air Force Doctrine Publication 3-99, The Department of the Air Force Role in Joint All Domain Operations

https://www.doctrine.af.mil/Portals/61/documents/AFDP_3-99/AFDP%203-99%20DAF%20role%20in%20JADO.pdf

(Chapters 1 and 2; and page 12)

AFDP 1-1 https://www.doctrine.af.mil/Portals/61/documents/AFDP_1-1/AFDP%201-1%20Mission%20Command.pdf

(Pages 1-4 and 13-16)

Read: Warfighting Concepts Primer

Additional Resources (for increased understanding):

Air University Podcast, Joint All Domain Operations, Dr. Jeffery M. Reilly

<https://www.af.mil/News/Radio/mod/1781/player/527/audio/70524>

Air University Podcast, Agile Combat Employment, Lt Col Justin D.

<https://www.af.mil/News/Radio/mod/1781/player/527/audio/70525>

Air University Podcast, Information operations in ACE, space and cyberspace, Dr. Joshua Sipper

<https://www.af.mil/News/Radio/mod/1781/player/527/audio/72024/>

Air Force Doctrine Note 1-21, Agile Combat Employment

https://www.doctrine.af.mil/Portals/61/documents/AFDN_1-21/AFDN%201-21%20ACE.pdf

Air Force Doctrine Publication 3-99, The Department of the Air Force Role in Joint All Domain Operations

https://www.doctrine.af.mil/Portals/61/documents/AFDP_3-99/AFDP%203-99%20DAF%20role%20in%20JADO.pdf

Air Force Doctrine Publication 1-1, Mission Command

https://www.doctrine.af.mil/Portals/61/documents/AFDP_1-1/AFDP%201-1%20Mission%20Command.pdf

Air Force Doctrine Publication 3-85, Electromagnetic Spectrum Operations

https://www.doctrine.af.mil/Portals/61/documents/AFDP_3-85/AFDP%203-85%20Electromagnetic%20Spectrum%20Ops.pdf

Air Force Future Operating Concept

<https://www.af.mil/Portals/1/images/airpower/AFFOC.pdf>

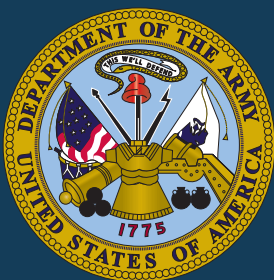
Summary of the Joint All-Domain Command and Control (JADC2) Strategy

<https://media.defense.gov/2022/Mar/17/2002958406/-1/-1/1/SUMMARY-OF-THE-JOINT-ALL-DOMAIN-COMMAND-AND-CONTROL-STRATEGY.PDF>

Cross-Domain Synergy in Joint Operations Planners Guide

https://www.jcs.mil/Portals/36/Documents/Doctrine/concepts/cross_domain_planning_guide.pdf?ver=2017-12-28-161956-230

Joint Publication 5-0



Joint Planning



01 July 2025



PREFACE

1. Scope

a. This publication provides the doctrinal foundation and fundamental principles that guide the Armed Forces of the United States in planning joint campaigns, contingency operations, and crisis response.

b. Joint planning is the process of identifying military ways and means (with associated risk) military leaders can recommend to the President to integrate with other instruments of national power (i.e., diplomatic, informational, and economic) to achieve strategic objectives. Joint planning provides the methodology the joint force commander (JFC) uses to develop and articulate solutions to achieve the objectives.

c. The crises and contingencies the joint force faces transcend multiple combatant commands, all domains, and all functions. Integrated planning addresses transregional, all-domain, and multifunctional challenges. JFCs unify efforts through global military integration, arranging cohesive military actions in time, space, and purpose. Additionally, JFCs work with allies, partners, and the interagency to advance a global, integrated, and all-domain approach to respond to crises and support national security objectives.

2. Purpose

This publication has been prepared under the direction of the Chairman of the Joint Chiefs of Staff (CJCS). It sets forth joint doctrine to govern the activities and performance of the Armed Forces of the United States in joint operations, and it provides considerations for military interaction with governmental and nongovernmental agencies, multinational forces, and other interorganizational partners. It provides military guidance for the exercise of authority by combatant commanders and other JFCs and prescribes joint doctrine for operations and training. It provides military guidance for use by the Armed Forces in preparing and executing their plans and orders. It is not the intent of this publication to restrict the authority of the JFC from organizing the force and executing the mission in a manner the JFC deems most appropriate to ensure unity of effort in the accomplishment of objectives.

3. Application

a. Joint doctrine established in this publication applies to the Joint Staff, combatant commands, subordinate unified commands, joint task forces, subordinate components of these commands, the Services, the National Guard Bureau, and combat support agencies.

b. This doctrine constitutes official advice concerning the enclosed subject matter; however, the judgment of the commander is paramount in all situations.

c. If conflicts arise between the contents of this publication and the contents of Service publications, this publication takes precedence unless the CJCS, normally in coordination

with the other members of the Joint Chiefs of Staff, has provided more current and specific guidance, or the Secretary of Defense has directed otherwise. Commanders of forces operating as part of a multinational (alliance or coalition) military command should follow multinational doctrine and procedures ratified by the United States unless they conflict with this guidance. For doctrine and procedures not ratified by the United States, commanders should evaluate and follow the multinational command's doctrine and procedures, where applicable and consistent with United States law, regulations, and doctrine.

For the Chairman of the Joint Chiefs of Staff:



DAGVIN R.M. ANDERSON
Lieutenant General, U.S. Air Force
Director for Joint Force Development

**SUMMARY OF CHANGES
REVISION OF JOINT PUBLICATION 5-0
DATED 01 DECEMBER 2020,
INCORPORATING CHANGE 1, 01 JULY 2024**

- **Improved usability and reduced length by reducing duplication and repetition from other doctrine.**
- **Ensured longevity by removing instances of repeated policy, which updates on a shorter cycle than doctrine, otherwise creating conflicts between doctrine/policy; maintains primacy of source documents, while capturing enduring doctrine principles.**
- **Recognized that joint force commanders (JFCs) follow law, policy, and strategic guidance, without the need for recapturing strategic direction**
- **Removed appendices reflecting joint operation plan example captured in the text, outdated global force management policy, and old theater distribution plans from United States Transportation Command.**
- **Moved “Operational Design” chapter ahead of the “Joint Planning Process” to reflect design preceding planning; refined operational design methodology in the context of understanding the operational environment.**
- **Renamed and refined Chapter V to reflect “Integrated Planning,” described planning across the competition continuum, and defined the relationship and interdependencies of campaign and contingency planning surrounding an adversary.**
- **Defined strategic planning frameworks and integrated plan sets.**
- **Refined how planners support JFCs throughout planning on what they account for, consider, and perform when organizing, planning, preparing, executing, and assessing joint operations across the competition continuum.**
- **Described how planners approach crisis within the context of a campaigns and contingency preparedness.**
- **Added Chapter VIII, “The Future of Joint Planning,” anticipating the future operating environment, while describing how technology can shape joint planning.**
- **Added Appendix C, “Planning Considerations for Joint All-domain Operations,” describing planning considerations for joint all-domain operations.**

- **Added Appendix J, “Strategy,” incorporating content from Joint Doctrine Note 2-19, *Strategy*, that will now be archived.**
- **Incorporated *Joint Warfighting Concept*-informed content throughout. Ensured consistency with doctrinal concepts, best practices, and terminology captured in Joint Publication (JP) 1, Volume 1, *Joint Warfighting*, and JP 3-0, *Joint Campaigns and Operations*.**
- **Removed all artifacts and obsolete terms in the text and glossary; updated terminology that evolved in practical application.**

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EXECUTIVE SUMMARY

1. Overview

a. Joint planning is the process of identifying military ways and means to integrate with other instruments of national power (i.e., diplomatic, informational, and economic) to achieve strategic objectives.

b. Strategic guidance provides the common thread that integrates the planning activities and operations of the Department of Defense. It provides purpose and focus to the planning for employment of military force.

c. This publication incorporates *Joint Warfighting Concept*-informed content and is consistent with doctrinal concepts, best practices, and terminology captured in Joint Publication (JP) 1, Volume 1, *Joint Warfighting*, and JP 3-0, *Joint Campaigns and Operations*.

2. Operational Approach and Joint Planning Process

a. Planners apply operational art to organize and understand the operational environment (OE) as a complex, interactive system. Understanding linkages across domains and instruments of power help planners identify an adversary center of gravity, as well as possible areas to exploit.

b. With an understanding of the OE, planners can distill strategic guidance to define a problem statement and analyze the mission. Through a process of developing, analyzing, and wargaming possible courses of action, planners develop a plan, ready for execution. The joint planning process (JPP) is an orderly, analytical set of logical steps to approach a problem and determine a solution. The seven-step process is a tool for planners but is not prescriptive. This includes planning for joint all-domain operations that enable joint force commanders to recognize and seize opportunities while mitigating risk.

3. Integrated Planning

a. Modern threats do not confine themselves by domain or function. They also do not fit neatly into divisions of separate campaign, crisis, and contingency planning and execution. Foreseeable contingencies will likely require the joint force to execute coordinated and synchronized operations across geographic regions, functions, domains, the electromagnetic spectrum, and time against adversaries. This requires integrated planning across the joint force.

b. The nation's adversaries continue to grow in capability and are increasingly interconnected and active. Adversaries advance their objectives asymmetrically while operating short of armed conflict, deliberately avoiding the activation of United States contingency plans. Siloed planning structures are insufficient in this environment and require an overhaul to achieve the analytical rigor, understanding, collaboration, and prioritized resourcing to operate with flexibility.

c. Strategic planning frameworks (SPFs) and integrated plan sets (IPSs) set the conditions for integrated planning. SPFs seek to provide a framework for senior leaders to understand the linkages between decisions and actions made across the competition continuum. Highlighting authorities and associated resources in competition is imperative to understanding the feasibility, risks, and opportunities of executing contingency plans. An IPS is a coordinating, authority-led planning effort composed of a core plan, concurrent plans, and conditional plans likely to execute contemporaneously, closing gaps in previously segregated plans.

d. Through SPFs, the joint force produces executable, resource-informed plans, while reducing siloing of campaign, crisis, and contingency planning activities executed across the competition continuum. This planning methodology drives integration across commands and domains by organizing multiple plans in a common framework oriented on a common threat scenario. Using a defined resource pool shared across the core plan and concurrent plans, the joint force develops IPSs to present resourcing decisions, tradeoffs, and risks early in the planning process which enable senior leader decision in feasible and executable plans. The resultant detailed planning informs global and regional campaign activities, crisis response options, and contingency plans for a global, all-domain planning effort that highlight the risks inherent in an environment producing multiple, simultaneous threats.

4. Operation Assessment

a. Operation assessments help the commander and staff determine progress toward mission accomplishment. Assessment results enhance the commander's decision making, enable more effective operations, and help the commander and the staff to keep pace with a constantly evolving OE. A secondary purpose is to inform senior civil-military leadership dialogue to support geopolitical and resource decision making throughout planning and execution.

b. The assessment process complements and is concurrent with the JPP in developing specific and measurable tasks, effects, and objectives during operational design. During execution, assessment provides information on progress toward creating effects and achieving objectives. The assessment process is continuous.

5. Transition to Execution

While SPFs help improve fluidity across the competition continuum, there remains a need to transition deliberate planning to execution. Disparities between planning assumptions and the actual OE conditions at execution drive refinement or adaption of the plan or order. Commanders and their staffs should accept that contingency plans, as written, will likely need changes on execution.

6. The Future of Joint Planning

a. The future OE demands a proactive approach where joint force commanders leverage emerging technologies to adapt their planning. Harnessed correctly, technology

can better integrate functions and collaboration to enable expedited decision timelines. Similarly, technology can process large volumes of data to help leaders anticipate and adjust to conditions in real time.

b. Preparing for this future goes beyond the static, linear planning of the past that created rigid products. While joint planning is a deliberative process, it should remain flexible enough for planners to anticipate future requirements, mitigate risk, and maximize opportunities for resource employment. Developing human capital to know “how to think” as opposed to “what to do” is increasingly important as technology plays an increased role in processing data and automating actions.

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CHAPTER I

JOINT PLANNING

*"I tell this story to illustrate the truth of the statement I heard long ago in the Army: **Plans are worthless, but planning is everything.** There is a very great distinction because when you are planning for an emergency you must start with this one thing: the very definition of 'emergency' is that it is unexpected; therefore, it is not going to happen the way you are planning."*

President Dwight D. Eisenhower
34th President of the United States, 1953-1961
Remarks at the National Defense Executive Reserve Conference,
November 14, 1957

1. Overview

a. Joint planning is the deliberate process of determining how to implement strategic guidance: to achieve objectives through the employment of military capabilities (i.e., ends, ways, means) within an acceptable level of risk at the operational level of warfare. Ideally, planning begins with specified strategic objectives that provide a unifying purpose around which to focus actions and resources. Leaders conduct joint planning to determine the best methods for employing the Department of Defense's (DoD's) capabilities to achieve objectives. Joint planning integrates the military ways and means with those of interagency partners, allies, and partner nations. Joint planning frames the issue, develops operational approaches, and gives leaders options with decision space and levels of risk. Combatant commanders (CCDRs) may propose objectives for the President and Secretary of Defense (SecDef). The Chairman of the Joint Chiefs of Staff (CJCS), as the principal military advisor to the President and SecDef, may offer military advice on proposed objectives and prioritization.

b. The strategic environment is uncertain, complex, and dynamic. For the first time since the end of the Cold War, joint force commanders (JFCs) face the prospect of planning for a protracted war, potentially on a global scale. JFCs increasingly face threats that are **transregional, all-domain** (land, air, maritime, space, and cyberspace), **and multifunctional**. Rapid advancements in space, cyberspace, and information capabilities, such as artificial intelligence (e.g., machine learning), digital editing, and Internet applications, enable sharing of information, including narratives, between areas of conflict, with national leaders, and to the global public. This shared information can include digitally manufactured events. An expanding range of adversaries, both state and non-state, can purchase, manufacture, and employ high-tech, homemade weapons that may create problems for nation-states. Although the character of conflict is evolving, the nature of war remains constant (fear, honor, interests). The act of imposing our will on another party to achieve a political aim through the sanctioned use of force is at its core, a human endeavor.

c. The Joint Staff (JS) implements global integration by aligning military actions in time, space, and purpose. The CJCS accomplishes global integration through a problem-

focused approach that connects campaign and contingency planning, balances resources, and enables risk-informed decision making. This enables the CJCS to provide timely military advice to the President and SecDef on campaigning and contingency planning and execution.

d. Joint planning serves three critical purposes (see Figure I-1):

(1) At the strategic level, joint planning provides the President, SecDef, and the CJCS with options and advice to achieve national security strategy (NSS) objectives through the employment of the joint force. Planning supports decision making by identifying alternative approaches at the operational level, along with probable outcomes, costs, and risks.

(2) At the operational level, joint planning translates strategic guidance into specific activities aimed at achieving strategic and operational objectives as directed in presidential, SecDef, and CJCS guidance. Joint planning ties the training, mobilization, deployment, employment, sustainment, redeployment, and demobilization activities of joint forces to achieve objectives in the service of enduring national interests.

(3) At the tactical level, joint planning enables commanders to conduct battles and engagements. Activities at this level focus on employing combat power to achieve

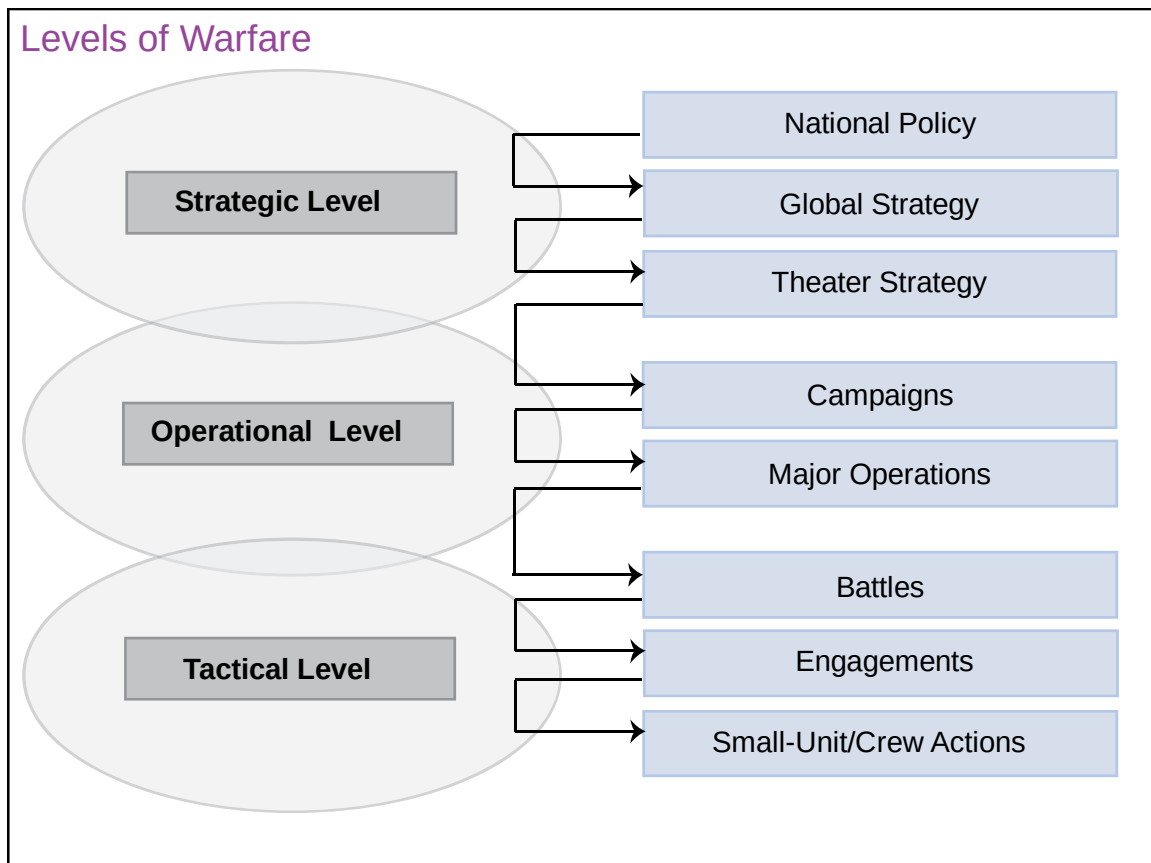


Figure I-1. Levels of Warfare

objectives. An engagement can include a wide variety of activities between opposing forces, normally occurring in a short period of time and limited physical or virtual space. A battle consists of a set of related engagements in time and space.

For more information on the levels of warfare, see Joint Publication (JP) 1, Volume 1, Joint Warfighting.

2. Strategy, Plans, Operations, and Assessments Cycle

a. Strategy, plans, operations, and assessments have an intertwined relationship. Plans translate the broad intent provided by a strategy into requirements and operations through the orders process; successful joint campaigns and operations achieve the strategy's objectives.

b. The effects of operations, successful or otherwise, change the operational environment (OE). Throughout planning and execution, commanders and staffs assess conditions and effects to identify whether changes to the OE support national interests and determine the impact their operations have on the OE. To maintain a competitive advantage, the joint force should constantly evaluate planned effects and objectives, align them with strategic objectives, and verify that they are still relevant and feasible. Assessments enable JFCs to evaluate the effects of friendly, adversarial, and third-party actions on the OE. Through their assessments, JFCs identify when conditions of the OE hinder the achievement of objectives, analyze ways and means to resolve those hindrances, and change their operations and activities to facilitate achieving the objectives.

c. As part of this assessment, the staff develops the commander's information requirements to identify key elements of the OE as indicators for success or failure of the strategy and then plans and works to continuously align strategic and operational approaches with national interests. The commander updates the command's strategy or operational approach to reflect the changed OE and ensures continued coherence with national policy. Simultaneously, the commander also updates operations to reflect the changed OE and updated strategy.

3. Planning

a. Joint planning is the art and science of interpreting national policy objectives, strategic direction, and guidance for plans, translating this guidance into executable operations and activities within imposed limitations and assessing the environment to place the joint force in the best position to achieve objectives. When approved, plans require a linked order that directs execution. Planning enables leaders to identify cost-benefit relationships, risks, and tradeoffs to determine the best concept of operations (CONOPS).

b. The four planning functions of strategic guidance analysis, concept development, plan development, and plan evaluation are generally sequential, but they often run simultaneously to accelerate the process. Leaders may direct staffs to refine or adapt a plan by entering the planning process through any of the functions. Planners adapt to changes in guidance and the OE during each planning function. The joint planning and execution

community (JPEC) synchronizes plans in the United States Government (USG) through ongoing civil-military dialogue.

c. **Strategy, Strategic Art, Operational Art, and Operational Planning**

(1) Strategy is an idea or set of ideas for employing the instruments of national power in a synchronized and integrated fashion to achieve theater, national, and multinational objectives. Strategy is also the art and science of determining a future state or condition (ends), determining the possible approaches (ways), and identifying the authorities and resources (e.g., time, forces, equipment, money) (means) to achieve the intended objective, all while managing the associated risk.

(2) Strategic art is the formulation, coordination, and application of ends, ways, and means to implement policy and promote national interests. Practitioners evaluate the competing interests and objectives of state and non-state actors and disruptive conditions of the operational area (OA), organize joint forces to implement policy, and sense when revision is prudent. Strategies should provide a coherent narrative to bridge the present to the desired future. Enduring, effective strategy provides the conceptual basis for an integrated military operation or campaign. Visualization and conceptualization of strategic success achieved or supported by military means is the foundation of operational art and operational design. The essence of strategic art is distillation—organizing and articulating the complex interrelationship between national interests, policy, strategic ends, and practice, in clear terms.

(3) Operational art is the cognitive approach by commanders and staffs—supported by their skill, knowledge, experience, creativity, and judgment—to develop campaigns and operations to organize and employ military forces. Operational art aids the joint force to link the tactical employment of forces to strategic objectives. This may be particularly important for protracted armed conflict consisting of multiple campaigns.

(4) Strategic art and operational art are mutually supporting. Strategic art provides policy context to objectives, while operational art demonstrates the feasibility and efficacy of a strategy. Operational planning informs and translates strategy into joint campaigns and operations, within resource and policy limitations.

4. Principles of Planning

a. **Focused on the Objective.** Joint planning orients around defining and achieving objectives. The commander and staff evaluate strategic guidance, analyze the OE, and coordinate with senior leadership to identify objectives. Joint planners align military plans with national priorities and direct them to achieve strategic objectives. The JS, CCDR, Service, and National Guard Bureau (NGB) staffs work with DoD leadership in this effort, and they often view problems from different perspectives. Examining and discussing the different perspectives is essential, because a directed military action may not necessarily result in the strategic objective envisioned by policymakers. Commanders, with their

staffs, identify gaps and risks between the objectives and the capabilities and limitations of employing the military and then discuss these gaps with their superiors.

b. Globally Integrated and Coordinated. Integrated planning addresses resources, timelines, decision points, and authorities across combatant commands (CCMDs) to achieve objectives. Additionally, integrated planning incorporates the JS, NGB, DoD agencies, interagency partners, multinational partners, and commercial sector entities. It produces a shared understanding of the OE, required decisions, resource prioritization, and risk.

(1) Integrated planning addresses complex strategic challenges that span multiple CCMD areas of responsibility (AORs) and affect CCDR transregional missions. The JS, NGB, DoD, USG departments and agencies, and multinational partners use integrated planning to ensure resources and timelines, decision matrices, and authorities are coordinated across the CCMDs. The CCDRs, using integrated plan development, in-progress reviews (IPRs), and assessments, inform the leadership how a conflict could develop, about options for response, and how operations by one CCMD could affect the OE.

(2) Military forces alone cannot achieve strategic objectives. Joint forces coordinate with USG departments and agencies, allied and partner nations, nongovernmental organizations (NGOs), international organizations, commercial entities (e.g., contractors), and local and regional stakeholders. Within an OA, JFCs and staffs also use network engagement, seeking to partner with friendly networks, engage neutral networks, and counter threat networks. Additionally, JFCs often leverage commercial solutions to complement military capabilities and address shortfalls. JFCs integrate commercial entity expertise into planning processes and establish mechanisms for rapid contracting, supply chain security, and technology integration. In most cases, these networks will have ties with diasporas or links with international elements of a particular network. Networks that will form, evolve, dissolve, and reform in different arrangements. Individuals will often belong to several types of networks at the same time. Planners coordinate and synchronize joint force actions with the operations of these networks and align military actions and resources with international organization, NGO, and commercial partner functions, consistent with legal authorities.

c. Resource Informed/Time Constrained. When translating strategic guidance into joint plans and orders, planners provide a realistic proposal for the application of forces, given current readiness, availability, location, posture, sustainment capabilities/limitations, available transportation, and speed of movement. Planning assumes operations employ forces and capabilities currently available—not future capabilities or capacities. Planners consider that available resources may change during plan execution.

d. Risk-Informed. Planners assess and articulate risks and identify potential mitigation. Identifying and communicating risk is fundamental in joint planning. In general, risk entails the probability and severity of loss linked to hazards. Effective risk management involves a thorough analysis of the OE, including the identification of

potential threats, vulnerabilities, and opportunities. Commanders and staffs consider both positive and negative risks, weighing the potential benefits of a particular course of action (COA) against the potential costs and consequences.

(1) Generally, risk and resources have an inverse relationship. At the operational level, as the resource commitment to a plan increases, the level of risk to that plan usually decreases. However, committing additional resources does not guarantee success or risk reduction. Planning can identify additional resources that reduce risk associated with the plan. However, national resources are inherently finite, and their use, or even planned use, may weaken overall United States (US) security by creating opportunity costs and elevated risk elsewhere.

(2) Assumptions are suppositions taken as true in the absence of proof. They are unavoidable in planning, but using assumptions incurs risk. Planners identify the role of assumptions in their plans and the impact if key assumptions are invalid. Planners also identify collection requirements to verify assumptions and identify the impact of constraints (things that must be done) and restraints (things that cannot be done) on the operation.

(3) There is no definitive formula for quantification of risk. Planners provide decision makers an assessment of the expected risks, costs, and benefits, as well as the potential consequences of proposed military actions and recognize that different leaders may assess or tolerate risk at different levels. This facilitates decisions to reduce, control, or accept risk through a shared knowledge of potential consequences.

e. Framed within the OE and Strategically Informed. Planning requires an understanding of the OE as it exists and as it changes. The starting position for any plan is the current OE. Planners do not assume away contentious issues or conditions. JFCs and planners expect adversaries to set the conditions for their advantage. Unlike concepts and future development, adaptive planning involves continuous monitoring and analysis of conditions affecting the OE (e.g., current friendly and threat postures, readiness, geopolitical conditions, perceptions of relevant state and non-state actors). Adaptive planning identifies changes that improve the probability of success or mitigate risks (e.g., additional forces; partner nation contributions; agreements, access, basing, and overflight permissions; preparation activities, including prepositioning of capabilities to include war reserve materiel). However, the starting position for any plan is the current OE. Planners should not overlook or dismiss contentious issues or conditions. Expect adversaries to take action for their advantage in the theater and, perhaps, globally. These actions can include their own version of intelligence preparation of the OE. Adversaries' actions, and the changes they cause to the OE, may challenge the assumptions of US plans.

f. Informs Decision Making. Senior leaders require joint planning that is agile and flexible enough to provide information to make critical decisions. It should identify the most significant contextual issues, key assumptions, likely resource requirements, cost-benefit trade-offs, and risks associated with different COAs, in a manner that enables key leaders to make decisions that best serve national interests.

g. **Adaptive and Flexible.** Planning is an adaptive process that occurs in a networked, collaborative environment. Joint planning requires dialogue among senior leaders, concurrent plan development, and collaboration across all echelons of organizations. Early planning guidance and frequent interaction between senior leaders and planners promotes an understanding of the operational problem, strategic and operational objectives, mission, assumptions, considerations, risks, and other factors. Clear strategic guidance ensures joint planners are in sync with senior leaders as they prepare, refine, and adapt plans to an ever-changing OE. This enables adaptive planning to produce and maintain up-to-date plans and to provide viable, flexible COAs for commanders and national decision makers to consider.

5. Planning Products

Joint planning encompasses the preparation of several planning and execution-related products. While the planning process is generally the same for campaign, contingency, or crisis planning, the output or products may differ. The two basic types of plans are campaign and contingency plans. Campaign and contingency planning encompass the preparation of plans that occur in non-crisis situations with a timeline generally not driven by external events. Crisis planning uses the same process, but external events typically drive it, and time constraints almost always limit it and do not permit the production of a plan. **Combatant command campaign plans (CCPs)** translate strategic guidance into operations and activities executable by CCMDs. CCP objectives are accomplished by forces when the plan transitions into orders and actions are directed. CCPs link current and future operations to strategic guidance and describe the conditions for successful achievement of contingency plan objectives. The planner needs to know the type of plan and the detail of planning required.

a. **Campaign Plans.** A campaign is a series of related military operations to achieve strategic and operational objectives in time and space. Campaign plans organize the enduring operations and activities to influence behaviors and the OE. While campaigning is an ongoing effort, JFCs' campaign plans do not simply seek to maintain the status quo. Rather, they include high-payoff efforts that give JFCs an advantage over adversaries. Through campaign plans, JFCs establish objectives, conditions, and tasks under which the CCMD and Service components build operations, activities, and investments (OAI) in support of strategic guidance. Campaigns are proactive and rarely feature a single measure of military success implying victory in the traditional sense.

(1) CCDRs document the full scope of their campaigns in plans that include the campaign plan and all its subordinate and supporting plans. JFCs direct execution through tasks to subordinates. CCDRs plan and conduct campaigns and operations, while Service and functional components conduct operations, activities, battles, and engagements but not independent campaigns. CCDRs can plan and conduct subordinate campaigns or operations in support of another CCDR's campaign. While intended primarily to guide the use of military power, discussions and decisions at the national level provide guidance for employing the different instruments of national power. CCDRs' CCPs account for the interdependencies of all instruments of national power.

(2) Campaign plans implement a CCDR's strategy by balancing all CCMD enduring activities, such as assuring allies and partners, deterring attack, competing below the level of armed conflict, and preparing for and responding to threats. A CCDR's strategy and resultant campaign plan seek to achieve prioritized operational objectives and coordinate all subordinate and supporting planning and operations. Campaign plans also help the CCDR identify resources required to execute tasks and achieve objectives directed by strategic guidance for input into budget and force allocation requests.

(3) Under this construct, plans developed to respond to contingencies exist as branches to the overarching campaign plan (global, functional, CCMD, or regional). They address scenarios that put one or more US strategic objectives in jeopardy and leave the United States no recourse other than to address the problem through military actions, not as part of an ongoing campaign or operation. Military actions can be in response to many scenarios, including armed aggression, regional instability, a humanitarian crisis, or a natural disaster.

(4) **Types of Campaign Plans**

(a) **Global Campaign Plans (GCPs).** GCPs are globally integrated campaign plans that integrate CCMD activities against priority adversaries to achieve unity of effort across enduring activities below the level of armed conflict. The GCPs have their own discrete objectives outside of armed conflict but account for integrated contingency planning efforts. Chairman of the Joint Chiefs of Staff Instruction (CJCSI) 3110.01, (*U*) *2023 Joint Strategic Campaign Plan* (short title: JSCP), identifies GCPs based on strategic guidance and the GCPs form the base of the JSCP. The CJCS manages GCPs on behalf of SecDef.

(b) **Regional Campaign Plans.** Regional planning guidance addresses regional threats or challenges that require coordination across multiple CCMDs. Generally, issues that require regional campaign plans are not as significant a threat to US interests as GCPs but require attention to ensure they do not devolve into a more significant crisis. If necessary, SecDef, through the CJCS, could direct a regional campaign plan and delegate coordinating authority to a CCDR.

(c) **Functional Campaign Plans.** Functional campaign plans address functional threats or challenges that are not geographically constrained and could require coordination across multiple CCMDs.

(d) **CCPs.** CCPs are the centerpiece of the CCMDs' planning construct and operationalize CCMD strategies. CCDRs orient their CCPs on objectives, tasks, and effects required for success. CCPs contain tasks to staff and subordinate units that drive actions and set the conditions during campaigning. They link support and contingency plans, set priorities, and identify risks in requirements placed on the CCMD. CCDRs operationalize their CCPs through orders and conduct periodic assessments.

b. **Contingency Plans.** JFCs prepare contingency plans in anticipation of a wartime scenario. JFCs expect to modify these plans during execution in response to actual conditions. If there is an approved contingency plan that closely resembles the emergent scenario, planners can adapt the plan as necessary.

(1) CCDRs prepare contingency plans in association with GCPs and CCPs for designated threats, catastrophic events, and contingent missions without a crisis at hand, pursuant to strategic guidance. Contingency plans account for the possibility that OAI for competition below the level of armed conflict could fail to deter aggression or preclude instability in a key state or region.

(2) Planning for contingencies is based on hypothetical situations and, therefore, relies heavily on assumptions regarding the circumstances that will exist when a crisis arises. Planning for a contingency encompasses the development of plans for the deployment, employment, sustainment, and redeployment of forces and resources in response to potential crises identified in joint strategic planning documents. Contingency planning enables the JPEC to deepen its understanding of the OE and sharpen its analytical expertise.

(3) Planners periodically review and update contingency plans to maintain their relevance. Contingency planning most often addresses situations in which military options focus on combat operations but can address contingencies in which the joint force is in support of another agency from the onset. These include defense support of civil authorities, support to stabilization efforts, and foreign humanitarian assistance.

c. **Planning Detail.** Commanders develop contingency plans to one of the four levels of planning detail or into an operation order (OPORD) (for execution).

(1) **Level 1 Planning Detail—Commander’s Estimate.** This level of planning has the least detail. It produces multiple COAs to address contingencies. The product for this level can be a COA briefing, command directive, commander’s estimate, or a memorandum with a proposed force list. The commander’s estimate provides SecDef with military COAs to meet a potential contingency. The estimate reflects the commander’s analysis of the various COAs and recommends a COA.

(2) **Level 2 Planning Detail—Base Plan.** A base plan describes the CONOPS, major forces, concepts of support, and anticipated requirements for completing the mission. It normally does not include annexes. A base plan may contain alternatives, including flexible deterrent options (FDOs) and flexible response options (FROs), to provide multiple options to address contingencies as they develop or to influence the developing situation.

(3) **Level 3 Planning Detail—Concept Plan (CONPLAN).** A CONPLAN is an operation plan (OPLAN) in an abbreviated format. It may require considerable expansion or alteration to convert into a complete and detailed level 4 OPLAN or an OPORD. It includes a plan summary; a base plan; and usually includes the following annexes: A (Task

Organization), B (Intelligence), C (Operations), D (Logistics), J (Command Relationships), K (Command, Control, Communications, Computer, and Cybersecurity Systems), S (Special Technical Operations), V (Interagency-Interorganizational Coordination), and Z (Distribution). The CCDR develops a level 3T plan when a time-phased force and deployment data (TPFDD) is the requirement. This requires consideration of the OE's potential impacts on deployment and distribution operations. In a level 3T contingency plan, a CCDR may direct the CCMD intelligence directorate of a joint staff (J-2) to collaborate with the Joint Staff J-2 [Intelligence Directorate] in the development of a national intelligence support plan (NISP). A NISP is a scalable intelligence support plan that addresses the CCDR's intelligence requirements (IRs), identifies intelligence gaps, outlines support strategies, and ensures effective interagency and non-DoD intelligence coordination to meet operational needs. A troop list and TPFDD also require an annex E (Personnel) and annex W (Operational Contract Support).

(4) **Level 4 Planning Detail—OPLAN.** An OPLAN is a complete and detailed plan. The OPLAN identifies the force requirements, functional support, and resources to execute the plan. It contains a full description of the CONOPS, all applicable annexes, a time-phased force and deployment list and a transportation-feasible notional TPFDD, as well as analysis of the impact of the OE on the joint deployment and distribution enterprise. A TPFDD enables JFCs to organize and move unit and individual requirements into the theater of operations to support the CONOPS and provide closure estimates. A CCMD may direct developing NISP for level 4 OPLANS. A CCDR may direct developing a NISP for level 4 OPLANS. An OPLAN is normally prepared when:

- (a) The contingency threatens national security and requires detailed prior planning.
- (b) The magnitude or timing of the contingency requires detailed planning.
- (c) The multinational forces (MNFs) involved require additional detailed planning.
- (d) A level of detailed planning is necessary to determine force deployment, employment, sustainment, and redeployment requirements; identify resources to fill requirements; and validate shortfalls.

Plan formats are available in Chairman of the Joint Chiefs of Staff Manual (CJCSM) 3130.03, Planning and Execution Formats and Guidance.

d. **Planning for a Crisis.** A crisis is an incident or situation that creates a condition of such national security importance that the President or SecDef may consider committing US military forces and resources to achieve or defend strategic objectives. A crisis may evolve over time (e.g., escalating civil war, humanitarian crisis) or develop quickly (e.g., hostage rescue, natural disaster) with little or no warning and require accelerated decision making. Planning in response to a crisis generally results in the publication of an order and the execution of an operation. Crisis planning should help senior leaders determine if a

military response can help achieve the desired objective at acceptable cost and risk levels. Normally, an OPORD and execute order (EXORD) result from crisis planning.

(1) Regardless if planning currently exists, in response to an emergent event or crisis, planners initiate crisis planning using the same construct as all other planning, albeit on a compressed timeline. Planners leverage previously prepared plans as a starting point in a crisis, modifying as required to meet the operational circumstances. Regardless of whether a plan exists, commanders can modify a similar plan, or planning begins from scratch; the basic tenets of integrated planning always apply. In some cases, commanders and staffs may need to develop and approve a feasible COA with a notional TPFDD and assess possible contested environments, publish the plan or order, prepare forces, verify high-demand forces or capabilities (e.g., communications systems; lift; precision munitions; intelligence, surveillance, reconnaissance) are sufficient, develop and execute an integrated intelligence and counterintelligence plan [annex B (Intelligence)], and arrange sustainment for the employment of US military forces. Figure I-2 provides a comparison of planning for a contingency and planning in a crisis.

(2) In a crisis or time-sensitive situation, the CCDR reviews previously prepared plans for suitability. The CCDR may adapt these plans into an executable OPORD or develop an OPORD from scratch when no useful contingency plan exists.

(a) When the President, SecDef, or CJCS decides to develop military options, the CJCS issues a planning directive to the JPEC initiating the development of COAs. This planning directive may request that the supported commander submit a commander's estimate of the situation with a recommended COA. Normally, commanders issue a warning order (WARNORD), but they may use a planning order (PLANORD) or alert order (ALERTORD) if the crisis warrants accelerated planning. In a quickly developing crisis, commanders may verbally communicate the initial WARNORD with follow-on documentation to inform the JPEC. SecDef approves directives that include a force deployment such as a prepare to deploy order (PTDO) or deployment order (DEPORD).

(b) The WARNORD should describe the situation, establish command relationships, and identify the mission and any planning constraints. It may identify forces and strategic mobility resources, or it may request that the supported commander develop these factors. It may establish tentative dates and times to commence mobilization, deployment, or employment, or it may solicit the recommendations of the supported commander regarding these dates and times. The WARNORD should also identify any planning assumptions, restraints, or constraints the President or SecDef have identified to guide the response. If the President, SecDef, or CJCS directs development of a specific option or COA, the WARNORD describes the COA and requests the supported commander's assessment. The amount of detail in the WARNORD depends on the known facts and time available. A WARNORD and commander's estimate examples are in the CJCSM 3130.03, *Planning and Execution Formats and Guidance*.

Planning Comparison

	Planning for a Contingency	Planning in a Crisis
Time Available	As defined in authoritative directives (normally 6+ months)	Situation dependent (hours, days, up to 12 months)
Environment	Distributed, collaborative planning	Distributed, collaborative planning and execution
Facts and Assumptions	Significant use of assumptions	Rely on facts and minimal use of assumption
JPEC Involvement	Full JPEC participation (Note: JPEC participation may be limited for security reasons.)	Full JPEC participation (Note: JPEC participation may be limited for security reasons.)
Operational Activities	Situational awareness Planning Evaluation	Situational awareness Planning Execution Assessment
Planning Functions	Strategic guidance analysis Concept development Plan development Plan evaluation	Strategic guidance analysis Concept development Plan development Plan evaluation
Document Assigning Planning Task	1. JSCP (CJCS)/CPG (SecDef) 2. Planning directive 3. WARNORD (for short suspense planning)	CJCS issues: 1. WARNORD 2. PLANORD 3. SecDef-approved ALERTORD
Forces for Planning	Directed readiness tables and apportionment tables.	Allocated in WARNORD, PLANORD, or ALERTORD.
Planning Guidance	CJCS issues JSCP or WARNORD. CCDR issues PLANDIR and TPFDD LOI.	CJCS issues WARNORD, PLANORD, or ALERTORD. CCDR issues WARNORD, PLANORD, or ALERTORD and TPFDD LOI to subordinates, supporting commands, and supporting agencies.
COA Selection	CCDR prepares COAs and submits to CJCS and SecDef for review. Specific COA may or may not be selected.	CCDR develops commander's estimate with recommended COA, and, as appropriate, alternative COAs to achieve different objectives.
CONOPS Approval	SecDef approves planning or directs additional planning or changes.	President/SecDef approve COA, disapproves or approves further planning.
Final Planning Product	Campaign plan. Level 1–4 contingency plan.	OPORD
Final Planning Product Approval	CCDR submits final plan to CJCS for review and SecDef for approval.	CCDR submits final plan to President/SecDef for approval.
Execution Document	Not applicable.	CJCS issues SecDef-approved EXORD. CCDR issues EXORD.
Output	Plan	Order/Execution

Legend

ALERTORD	alert order	LOI	letter of instruction
CCDR	combatant commander	OPORD	operations order
CJCS	Chairman of the Joint Chiefs of Staff	PLANDIR	planning directive
COA	course of action	PLANORD	planning order
CONOPS	concept of operations	SecDef	Secretary of Defense
CPG	Contingency Planning Guidance	TPFDD	time-phased force and deployment data
EXORD	execute order	WARNORD	warning order community
JPEC	joint planning and execution community		
JSCP	Joint Strategic Campaign Plan		

Figure I-2. Planning Comparison

(c) In response to the WARNORD, the supported commander, in collaboration with subordinate and supporting commanders and the rest of the JPEC,

reviews existing joint contingency plans for applicability; accounts for a complex or contested OE; then develops, analyzes, and compares COAs and prepares a commander's estimate. The commander's estimate provides recommendations and advice to the President, SecDef, or higher headquarters for COA selection. Based on the supported commander's guidance, supporting commanders begin their planning activities.

(d) Although an existing plan almost never completely aligns with an emerging crisis, planners can amend the plan to facilitate rapid COA development and adjustment to fit the specific situation.

(e) The CJCS, in consultation with other members of the Joint Chiefs of Staff and JPEC, reviews and evaluates the supported CCDR's estimate and provides recommendations and advice to the President and SecDef for COA selection. The Joint Chiefs of Staff review the supported CCDR's COAs and may accept, refine, or revise them, or develop a new COA(s) as needed. The President or SecDef selects a COA and then directs the initiation of detailed planning.

(f) Upon receiving directions from the President or SecDef, the CJCS issues a SecDef-approved ALERTORD to the JPEC. The order is a record communication stating the President or SecDef has approved the detailed development of a military plan to help resolve the crisis. The contents of an ALERTORD vary depending upon the crisis and amount of prior planning accomplished, but it should always describe the selected COA in sufficient detail to enable the supported commander, in collaboration with other members of the JPEC, to conduct detailed planning to deploy, employ, and sustain forces. However, the ALERTORD does not authorize execution of the approved COA.

(g) The supported commander then develops an OPORD using the approved COA. Planners determine the speed of OPORD development based on the amount of prior planning and the planning time available. The supported commander and subordinate commanders identify force requirements, commercial capability requirements, contracted support requirements, existing contracts and task orders, and mobility resources. Then JFCs describe the CONOPS in OPORD format. The supported commander assesses the available forces and determines which ones to use. If they identify a shortfall, the supported commander sends a request for forces to the JS.

For a detailed description of the global force management (GFM) allocation process refer to CJCSM 3130.06, Global Force Management Allocation Policies and Procedures.

(h) The supported CCDR submits the completed OPORD through the CJCS for SecDef or presidential approval. The President or SecDef may decide to begin deployment in anticipation of execution, execute the operation, place planning on hold, or cancel planning pending resolution by some other means. Detailed planning may transition to execution, as directed. The CCDR directs the effort, maintaining continuous situational awareness and adjusting planning products as necessary, unless otherwise directed.

(i) Operations for a selected plan do not commence until the President decides and SecDef signs an EXORD. Plan development continues after the President's or SecDef's execution decision. When the crisis does not lead to execution, the CJCS provides guidance regarding continued planning.

(j) **Abbreviated Procedures.** The preceding discussion describes planning activities sequentially. During a crisis, planners may conduct steps concurrently or compress them, depending on the conditions. The President or SecDef may decide to commit forces shortly after an event occurs, and as a result, planners compress planning activities. Planners manage the allocation process, which has standard timelines, but they can accelerate them if necessary. No specific length of time can be associated with any planning activity. **Severe time constraints may require planning participants to pass information verbally, including the decision to commit forces.** Commanders follow up verbal orders with written orders as soon as practical.

(3) **Orders.** Upon approval, CCDRs and Services issue orders directing action (see Figure I-3). Orders are authoritative and legally binding. By the CJCS's direction, the JS operations directorate of a joint staff (J-3) develops, coordinates, and prepares orders. The Joint Staff J-3 [Operations Directorate] prepares and coordinates the Secretary of Defense Orders Book to present recommendations to SecDef for decision.

(a) **WARNORD.** A WARNORD initiates the development and evaluation of military COAs by a supported commander and requests that the supported commander submit a commander's estimate. SecDef authorizes orders that contain the deployment of forces.

(b) **PLANORD.** A PLANORD provides essential planning guidance and directs the initiation of plan development before the directing authority approves a military COA.

(c) **ALERTORD.** An ALERTORD provides essential planning guidance and directs the initiation of plan development after the directing authority approves a military COA. An ALERTORD does not authorize execution of the approved COA.

(d) **PTDO.** A PTDO orders the force provider (FP) to have a unit ready and available to deploy within a specified response time. Unless otherwise stated, units placed on PTDO by the FP remain under the command and control (C2) of the FP. CJCSM 3130.06, *Global Force Management Allocation Policies and Procedures*, and the *Global Force Management Implementation Guidance* (GFMIG) discuss the PTDOs in more detail.

(e) **DEPORD.** SecDef directs the CJCS to issue a DEPORD, which empowers the CJCS to redistribute forces among CCMDs, Services, and DoD agencies. The DEPORD outlines the authorities that the gaining CCDR wields over the transferred forces. The CJCS uses the Global Force Management Allocation Plan as a comprehensive DEPORD, governing the allocation of all forces worldwide. FPs then deploy or ready forces for deployment in accordance with the Global Force Management Allocation Plan's

Joint Orders

	Order Type	Intended Action	Secretary of Defense Approval Required
Warning Order		Initiates development and evaluation of COAs by supported commander. Requests commander's estimate be submitted.	No. Required when WARNORD includes deployment or deployment preparation actions.
Planning Order	PLANORD	Begins planning for anticipated President or SecDef-selected COA. Directs preparation of OPORDs or contingency plan.	No. Conveys anticipated COA selection by the President or SecDef.
Alert Order	ALERTORD	Begins execution planning on President or SecDef-selected COA. Directs preparation of OPORD or contingency plan.	Yes. Conveys COA selection by the President or SecDef.
Operation Order	OPORD	Effect coordinated execution of an operation.	Specific to the OPORD.
Prepare To Deploy Order	PTDO	Increase/decrease deployability posture of units.	Yes (if allocates force). Refers to five levels of deployability posture.
Deployment/Redeployment Order	DEPORD	Deploy/redeploy forces. Establish C-day/L-hour. Increase deployability. Establish joint task force.	Yes (if allocates force). Required for movement of unit personnel and equipment into combatant commander's AOR.
Execute Order	EXORD	Implement President or SecDef decision directing execution of a COA or OPORD.	Yes.
Deployment/Redeployment Order	FRAGORD or FRAGO	Issued as needed after an OPORD to change or modify the OPORD execution.	No.

Legend

AOR	area of responsibility
C-day	unnamed day on which a deployment operation begins
COA	course of action
FRAGORD (or FRAGO)	fragmentary order
L-hour	specific hour on C-day at which deployment operation commences or is to commence
SecDef	Secretary of Defense

Figure I-3. Joint Orders

directed timeline. CJCSM 3130.06, *Global Force Management Allocation Policies and Procedures*, and the GFMIG discuss the DEPORD in more detail.

(f) **EXORD.** An EXORD is a directive to implement an approved CONOPS. Only the President and SecDef have the authority to approve and direct the initiation of military operations. The CJCS, by the authority of and at the direction of the President or SecDef, may subsequently issue an EXORD to initiate military operations. Supported and supporting commanders and subordinate JFCs use an EXORD to implement the approved CONOPS.

(g) **OPORD.** An OPORD is a directive issued by a commander to subordinate commanders that coordinates preparation to conduct the execution of an operation. The authority to conduct this operation is promulgated by an EXORD. Joint OPORDs are prepared under joint procedures in prescribed formats during a crisis.

(h) **Fragmentary Order (FRAGORD).** A FRAGORD is a modification to any previously issued order. It is issued to change an existing order or to execute a branch or sequel of an existing OPORD. It provides brief and specific directions that address only those parts of the original order that have changed.

For more information on plan and orders formats, see CJCSM 3130.03, Planning and Execution Formats and Guidance.

6. Risk Identification and Mitigation

a. Identifying Risk

(1) **Risk** is the probability and consequence of an event causing harm to something valued. Commanders and senior leaders evaluate risk as they assess the likelihood of mission success. They assess risk by considering the costs that threats or challenges impose on achieving the objective or the impact that these factors have on objective achievement. Military risk is usually described by two subsets, risk-to-mission or risk-to-force. Risk-to-mission is the estimated probability and consequence of the joint force's potential inability to achieve current or future objectives. Risk-to-force is the estimated probability and consequence of negative impacts on the joint force's ability to provide and sustain military resources.

(a) Operational risk (risk-to-mission) reflects the current force's ability to achieve objectives in the national military strategy (NMS) with acceptable human, material, and financial costs. Operational risk is a function of the probability and consequence of failure to achieve the mission while protecting the force from unacceptable losses.

(b) Force management risk (risk-to-force) reflects a Service or joint FP's ability to generate trained and ready forces within established rotation ratios and surge capacities to meet mission requirements. Force management risk is a function of the probability and consequence of not maintaining force generation balance ("breaking the force"). This risk subset considers the capacity to execute plans today (e.g., "fight tonight" on the Korean peninsula) compared to contingency requirements (e.g., potential conflict arising over an economic exclusion zone or a disputed territory) in the near-to mid-term. This may identify dilemmas for DoD's leadership, related to available forces for campaigns or contingencies, and their assessment of risk to other planned or potential missions.

(c) Risk management includes commercial support integration. For effective risk management in joint planning, JFCs include assessments of the availability, reliability, and resilience of commercial capabilities that support military operations.

For more information on joint risk; analysis methodology; and guidance for identifying, assessing, and managing risk, see CJCSM 3105.01, Joint Risk Analysis Methodology.

(2) Planners conduct an initial risk assessment during mission analysis and update it throughout the planning process. As they plan, they use logical, realistic, and essential assumptions in place of unknown facts. Planners continuously review these assumptions to determine if they remain valid. If an assumption proves critical, planners may develop a branch plan based on it. When new information invalidates an assumption, planners revise the plan, develop a new COA, or create a new plan.

(3) The CCDR develops an alternative approach when they expect resource shortages. CCDRs should solicit the advice of the Services, other CCDRs, joint FPs, Joint Staff J-35 [Joint Force Coordinator] joint functional managers as applicable, and other FPs in identifying forces.

b. **Mitigating Risk.** Planners and CCDRs identify methods to mitigate risk as part of the plan. Methods to mitigate risk include:

(1) **Reducing the Likelihood of Occurrence.** Reduce risk by decreasing the likelihood of events that can negatively affect joint force efforts. Mitigation measures may include protective measures (e.g., forward deploy theater ballistic missile defense systems), funding installation resiliency efforts (e.g., redundancy in critical infrastructure and systems at forward locations), informing an audience about the risk involved to discourage behavior in support of the event, developing assessments to clearly identify and understand risks (e.g., an assessment of potential risks to mission or risk-to-force or risks to civilians), and avoiding a potential hazard (e.g., use friendly secure seaports instead of unknown ports in potentially hostile areas).

(2) **Reducing the Cost of Occurrence.** Reduce risk by decreasing the potential negative effect of these events if they were to occur. Mitigation measures may include reactive measures (e.g., placing a reserve force in an area where they can quickly react to an event) and dispersion (e.g., placing capabilities at multiple locations so that an attack at one affects only capacity).

(3) **Increasing Nonorganic Support.** Reduce risk by use of capabilities that may fall outside of the JFC's organization or tasking authority (e.g., contracted support or host nation (HN) support. This type of increase may address limitations associated with strategic lift to enable the deployment of combat forces in lieu of combat service support forces. Alternatively, commercial capabilities may be a first option to help achieve the mission.

c. **Residual Risk.** Planners cannot eliminate risk, regardless of mitigation efforts. Identification of residual risk by senior leaders builds common understanding and clarifies decisions they make and their potential effects. Commanders should understand the impact of residual risk prior to mission execution.

d. **Risk Discussion.** Commanders discuss risk with senior leaders. Strategic-level risk is primarily diplomatic (e.g., impact on the US public, allies, adversaries, enemies, US objectives, and future US status). Consultations with interagency counterparts can supplement CCDRs' risk assessments with additional perspectives vital to senior leader decision making.

(1) The President and SecDef may identify options not previously available to the CCDR and planners to mitigate risk.

(2) Leaders conduct risk discussions using concrete terms and contextual accuracy to enable informed decision making. They acknowledge the limitations of quantifying all risk elements and the potential inaccuracies in analytic and modeling outputs. However, quantifying statements by using phrases such as "in our analysis, the mission will take six months versus two months," or "we expect casualties to increase from x to y," planners can better inform commanders and senior leaders of the differences between COAs.

(3) CCDRs provide their risk analysis and its implications to senior civilian leaders and stakeholders associated with joint operations. Although CCDRs focus on military operations, they often have broad exposure to the implications and impacts of military force employment, both at home and abroad. As a result, discussions with senior leaders may reveal issues that overlap with other instruments of national power (e.g., diplomatic, informational, and economic).

See CJCSM 3105.01, Joint Risk Analysis Methodology, for additional information on risk assessment and communicating risk.

7. Interorganizational Planning and Coordination

Interorganizational planning and coordination are the interaction among elements of DoD; participating USG departments and agencies; state, territorial, local, and tribal agencies; foreign military forces and government departments and agencies; international organizations; NGOs; and the private sector to achieve an objective. Unified action is the synchronization, coordination, and integration of the activities of governmental and nongovernmental entities with military operations to achieve unity of effort. Coordination of interorganizational and multinational plans facilitates unity of effort among multiple organizations by promoting common understanding of the capabilities, limitations, and consequences of military and nonmilitary actions. It also identifies common objectives and how military and civilian capabilities best complement each other to achieve these objectives.

a. **Interagency Coordination.** Interagency coordination links the US military and the other instruments of national power.

b. The nation achieves strategic objectives through unified action, built on unity of effort, by collaborating, synchronizing, and coordinating the diplomatic, informational,

military, and economic instruments of national power. Military power is most effective when used in conjunction with the other instruments of national power to advance and defend US values, interests, and objectives.

c. JFCs and their staffs actively collaborate with interagency partners to drive successful operations, campaigns, and transitions, particularly when supporting other USG departments and agencies. They assess and synchronize DoD capabilities with those of other USG departments and agencies. CCDRs coordinate directly with interagency representatives in their own command and with those in the National Capital Region. This cooperation addresses issues such as overflight rights and access agreements.

d. The Department of State (DOS) conducts planning at the department, bureau, and country levels. The following are key DOS planning documents that commanders and planners consult when planning.

(1) DOS develops the four-year **joint strategic plan** as their primary strategy, to set forth the direction and priorities to implement US foreign policy and development assistance for the coming years.

(2) **Regional Strategies.** The DOS regional bureaus develop a four-year regional strategy that articulates the priorities, goals, and areas of strategic focus within the region in alignment with US national interests. Regional strategies also provide a flexible framework within which regional bureaus and missions prioritize desired objectives, identify supporting resources, and respond to unanticipated events.

(3) **Integrated Country Strategies.** An integrated country strategy is a four-year, whole-of-government strategy developed by a US country team for a particular country. It articulates a common set of USG priorities and goals by setting the mission goals and objectives through a coordinated and collaborative planning effort. It provides the basis for the development of annual mission resource requests for DOS, as well as all USG security sector assistance. The chief of mission leads the development process and has final approval authority.

(4) **Country Development Cooperation Strategy.** A country development cooperation strategy, typically a five-year strategy, defines a chief of mission's chosen approach in a country, articulates the self-reliance trajectory, and details expected results. The country development cooperation strategy provides a road map for DOS to design and implement projects and activities. The strategy informs dialogue with Congress and engages HN partners and other stakeholders, including the private sector and civil society.

(5) **Functional Strategies.** DOS also produces functional strategies that address counterterrorism and countering weapons of mass destruction.

For additional information regarding chief of mission, see JP 1, Volume 1, Joint Warfighting, and JP 1, Volume 2, The Joint Force.

e. Interagency Considerations

(1) USG departments and agencies actively pursue their own policies, legal authorities, roles, and responsibilities, which can lead to complexity in interagency coordination. They also develop and implement their own procedures, decision-making processes, and security protocols, and may have distinct cultural norms. Meanwhile, nonmilitary organizations and NGOs take the lead in executing operations, and the military provides support as needed. In such instances, the understanding of military authorities and objectives may vary among the participants. The JFC should clearly articulate military capabilities, requirements, operational limitations, liaison, and legal considerations to interagency partners. Joint planners should understand the relationships and the types of support they can provide interagency partners. This includes learning the supported organization's processes, planning methodologies, planning products, policies, and operational limitations to identify areas where they can assist. Additionally, joint planners should understand federal interagency operational plans, or incident command systems for crisis planning, and how those processes align with the joint planning process (JPP). When other USG departments and agencies have provided institutional points of contact for defense planning, planners should coordinate directly with these individuals. The JFC's civil-military operations center, with oversight by the civil-military operations directorate of the command's staff, can facilitate these relationships. Products developed for network engagement will inform other annexes and provide additional local and regional information that planners might otherwise miss during the joint intelligence preparation of the operational environment (JIPOE) process. In the absence of a formal command structure, JFCs may need to build consensus to achieve unity of effort. Robust liaison efforts facilitate understanding, coordination, and mission accomplishment. Joint planners understand that most other agencies do not have the large planning staffs present in DoD and are cognizant that JFCs can quickly overwhelm interagency planning capabilities. Annex V (Interagency-Interorganizational Coordination) to the plan or order should address these considerations.

(2) During crisis planning and crisis management, DoD should coordinate with the crisis management and planning nodes across the interagency, and in particular with DOS, to agree on facts and assumptions prior to COA development.

For additional information on interagency considerations, see JP 3-08, Interorganizational Cooperation; CJCSM 3130.03, Planning and Execution Formats and Guidance, provides a planning outline for annex V (Interagency-Interorganizational Coordination).

8. Multinational Planning, Coordination, and Review

a. Multinational operations are military actions conducted by forces of two or more nations. Coalitions and alliances usually undertake such operations, and international organizations, such as the United Nations or Organization for Security and Cooperation in Europe, may provide supervision or oversight. In some cases, these international organizations may also take the lead or collaborate with coalitions and alliances to conduct

these operations. **Key to any multinational operation is unity of effort** among national and military leaders of participating nations emphasizing common objectives and shared interests, as well as mutual support and respect. Additionally, cultivating and maintaining personal relationships between counterparts in the participating nations is fundamental to achieving success. At times, US national interests may not be in complete agreement with those of the multinational organization or some of its individual nation states. These situations require additional coordination at the political and military levels to establish a common set of objectives to support unity of effort among nations. Additional complexities joint planners consider in multinational situations are additional time to receive plan approval from all participating international authorities, the degree and type of information and intelligence sharing with the MNF, unfamiliar chain of command, different delineations of command relationship between the US forces and the MNF, process and procedure for transferring authority between US control and multinational control, and degree of interoperability.

b. Joint planners frequently conduct planning within the context of multinational operations. The US military and its allies use JP 3-16, *Multinational Operations*, and North Atlantic Treaty Organization (NATO) Allied joint publication doctrine to guide their joint planning efforts for multinational operations. However, there is no single doctrine for multinational action, and each MNF develops its own protocols, OPLANs, CONPLANs, and OPORDs. US planning for multinational operations should accommodate and complement such protocols, plans, and orders. JFCs should also anticipate and incorporate planning factors such as applicable domestic, international, and HN laws, regulations, policies, and other international agreements, as well as operational limitations on the use of contributed forces, various weapons, and tactics.

(1) Commanders train and equip joint forces to operate in both permissive and hostile environments alongside forces from other nations, either under US command or foreign command, as part of an MNF.

(2) MNF commanders develop multinational strategies and plans in multinational channels. Supporting US JFCs perform planning for multinational operations in US national channels. At the national level, governments and multinational bodies actively coordinate in separate planning channels, while JFCs coordinate efforts at the theater strategic and operational levels, overseeing planning matters in both channels. US doctrine and procedures for joint planning are also applicable to multinational challenges. The general considerations for interaction with international organizations and partner-nation organizations are like those for interaction with USG departments and agencies.

c. Operational-Level Integration

(1) The commander of US forces dedicated to a multinational military organization integrates joint planning with multinational planning at the operational level. The CCCR or the subordinate JFC takes responsibility for planning and executing multinational operations within their AOR. These commanders always function within two chains of command during any multinational operation: the multinational chain of

command and the US national chain of command. Operating within multinational organizations, they command or support the designated MNF and plan, as appropriate, for multinational employment in accordance with the strategic guidance provided by multinational leadership. When in support of an MNF, a JFC may provide operational assistance (e.g., joint functions) to the MNF during combat operations to address a specific adversary, threat, or challenge. Operating within the US chain of command, they command US forces and prepare plans in response to strategic guidance from the President, SecDef, and the CJCS. These tasks include developing plans to support each multinational commitment within the AOR and planning for unilateral US contingencies within the same area. In this dual capacity, the US commander coordinates multinational planning with US planning.

(2) For NATO's operations, the United States and other NATO countries have developed and ratified an Allied joint doctrine hierarchy of publications outlining the doctrine and tactics, techniques, and procedures used during NATO operations. JFCs, their staffs, and subordinate forces should review and train with these publications prior to participating in NATO operations.

(3) Much of the information and guidance provided for unified action and joint operations remains applicable to multinational operations. However, commanders and staffs consider differences, including partner's institutional functions (e.g., governance, executive, generating, and operating functions), terminology, culture, politics, religion, language, caveats on authorized military action throughout the entire operation. CCDRs and JFCs develop plans to align US forces, actions, and resources, to include sustainment, in support of the multinational plan.

(4) When directed, US commanders participate with the armed forces of other nations in preparing bilateral or multilateral contingency plans. Commanders and their staffs assess the potential constraints, opportunities, security risks, and any additional vulnerabilities resulting from bilateral planning and how these plans impact the ability of the United States to achieve its objectives. Bilateral planning involves the preparation of combined, mutually developed, and approved plans governing the employment of the forces of two nations for a common contingency. Bilateral planning usually occurs within the framework of a treaty or alliance but may occur in the absence of such arrangements. JFCs conduct bilateral planning in accordance with US guidance or bilateral/multilateral strategic guidance signed by the leadership of all participating countries.

CHAPTER II

STRATEGIC GUIDANCE AND COORDINATION

1. Overview

This chapter introduces some of the major sources of planning guidance available to the commander and staff. It describes how commanders use strategic direction to develop military plans and orders. Finally, it discusses how to integrate other USG departments and agencies and multinational partners into overall joint planning efforts.

2. Joint Strategic Planning System

a. The Joint Strategic Planning System is the primary system the CJCS uses to execute statutory responsibilities assigned by Title 10, United States Code (USC), Section 153. The Joint Strategic Planning System enables the CJCS to conduct assessments; provide military advice to the President, SecDef, National Security Council, and Homeland Security Council; and assist the President and SecDef in providing strategic direction to the Armed Forces of the United States. The NMS and JSCP are the core strategic guidance documents the CJCS uses to provide direction and policy essential to implementation of the NSS. Other elements of Joint Strategic Planning System, such as the CJCS risk assessment, the Joint Force Readiness Review, and the annual joint assessment, inform decision making and identify new contingencies that may warrant planning and the commitment of resources.

CJCSI 3100.01, Joint Strategic Planning System, describes the system in detail.

b. **Strategic Direction.** The President, SecDef, and CJCS provide strategic direction by communicating broad objectives and issue-specific guidance to DoD. Strategic guidance provides the common thread that integrates the planning activities and operations of DoD. It provides purpose and focus to the planning for employment of military force. Strategic direction identifies the strategic objectives, national-level planning assumptions, and national-level limitations.

(1) **NSS.** The Executive branch prepares the NSS annually for Congress to outline US major national security concerns and how the administration plans to address them using instruments of national power. The document is often purposely general in content, and its implementation by DoD relies on elaborating direction provided in supporting documents (e.g., the national defense strategy [NDS] and NMS).

(2) **NDS.** SecDef promulgates the NDS, which outlines DoD's approach to implementing the President's NSS. The NDS establishes a set of overarching defense objectives that guide DoD's activities and provide direction for the NMS. The NDS objectives serve as links between military activities and those of other DoD agencies in pursuit of national goals.

(3) ***Unified Command Plan (UCP)***. The UCP, signed by the President, establishes CCMDs and responsibilities and missions of the CCDRs. The unified command structure identified in the UCP is flexible and changes as required to accommodate evolving US national security needs. The CJCS reviews the UCP not less often than every two years and submits recommended changes to the President through SecDef. This document provides broad guidance from which commanders and planners can derive tasks and missions during CCMD plan development and modification.

(4) ***Contingency Planning Guidance (CPG)***. The President signs the CPG, which provides written policy guidance on the preparation and review of campaign and contingency plans.

(5) ***Guidance for Employment of the Force (GEF)***. Issued by SecDef, the GEF provides guidance for JFCs' GCPs and CCPs.

c. **NMS**. The CJCS prepares a new or updated NMS for SecDef.

(1) The NMS provides the CJCS's amplifying guidance for planning, force employment, posture, and future force development. It provides the strategic framework to prioritize planning, allocate resources, and manage risk. As such, this classified military strategy serves as the starting point for all other Joint Strategic Planning System actions and constitutes the CJCS's military advice to SecDef and the President.

(2) The NMS defines the objectives (ends) and how to achieve these objectives (ways) and addresses the military capabilities (means) required to execute the strategy. The NMS objectives may often be the same as those in the NDS. The NMS provides focus for military activities by defining a set of interrelated objectives and joint operating concepts from which the Service chiefs and CCDRs identify required capabilities and against which the CJCS assesses risk. The NMS also looks beyond the near-term to identify long-range operational requirements and provides guidance in other areas within the CJCS's statutory requirements, such as joint force development and joint capability development.

d. **JSCP**. The JSCP fulfills the CJCS's statutory responsibilities in Title 10, USC, Section 153, to assist the President and SecDef in providing for strategic direction to the joint force and implementing the strategic guidance in the NSS, NDS, NMS, CPG, and GEF. The JSCP provides guidance for preparation of plans based on current military capabilities and strategic guidance, as well as planning guidance identified in the CPG. In addition to communicating to the CCMDs specific planning guidance, the JSCP operationalizes the strategic vision described in the NMS and nests with the strategic direction delineated by the NSS, NDS, and GFMIG. The JSCP also provides integrated planning guidance, strategic planning framework (SPF) information, and direction for planners to fulfill the CJCS's role as the global integrator.

(1) **Global Posture.** CCDRs submit posture plans every two years (with annual updates) to support campaign and contingency plans. Posture plans align basing and forces to ensure security, readiness to respond to contingency scenarios, and strategic flexibility.

For more information on posture plans, see Appendix F, “Posture Plans.” For detailed information and guidance on global defense posture (GDP) and posture plans, see the Global Posture Executive Council’s (GPEC’s) GPEC Posture Planning Guidance; supplemental guidance to the JSCP; and CJCSI 3110.01, (U) 2023 Joint Strategic Campaign Plan.

(2) **Global Distribution.** Strategic guidance provides broad themes for global distribution and posture coordinated through United States Transportation Command’s horizontal and vertical integration of global distribution planning. As a “plan of plans,” some CCPs include regional country plans, posture plans, and theater distribution plans that facilitate integration of resources, authorities, processes, and timelines to favorably affect conditions within the CCDRs’ AORs. Global distribution establishes the requirement for CCDRs with an AOR to submit theater distribution plans biennially to support campaign plans and inform contingency plans. Distribution plans support GCPs by interfacing with the relevant posture plans to support strategic lift, infrastructure, distribution enablers, agreements, policies, processes, and information systems.

For additional information on the JSCP, see CJCSI 3110.01, (U) 2023 Joint Strategic Campaign Plan.

(3) **GFMIG.** The GFMIG is the single, authoritative GFM document that provides procedures prescribed by SecDef on directed readiness, assignment, allocation, apportionment, and assessment. The processes apply within the force management and force planning constructs to support resource-informed planning and enable JFCs to dynamically employ forces, while allowing senior decision makers to assess the impact and risk of proposed changes.

e. **GFM.** GFM enables commanders to achieve strategic direction through the assessment, apportionment, assignment, allocation, and directed readiness of the joint force while proactively balancing risk. Detailed procedures for executing GFM are published by SecDef in the GFMIG. The CJCS provides implementing policy on allocation in the CJCSM 3130.06, *Global Force Management Allocation Policies and Procedures*.

3. Joint Planning and Execution Community

a. The JPEC includes the headquarters, commands, and agencies involved in joint planning or committed to a joint operation. Although not a standing or regularly meeting entity, the JPEC consists of the stakeholders shown in Figure II-1.

(1) The **supported CCDR** has primary responsibility for all assigned tasks. In the context of joint planning, **the supported commander can initiate planning at any time based on command authority** or in response to direction or orders from the

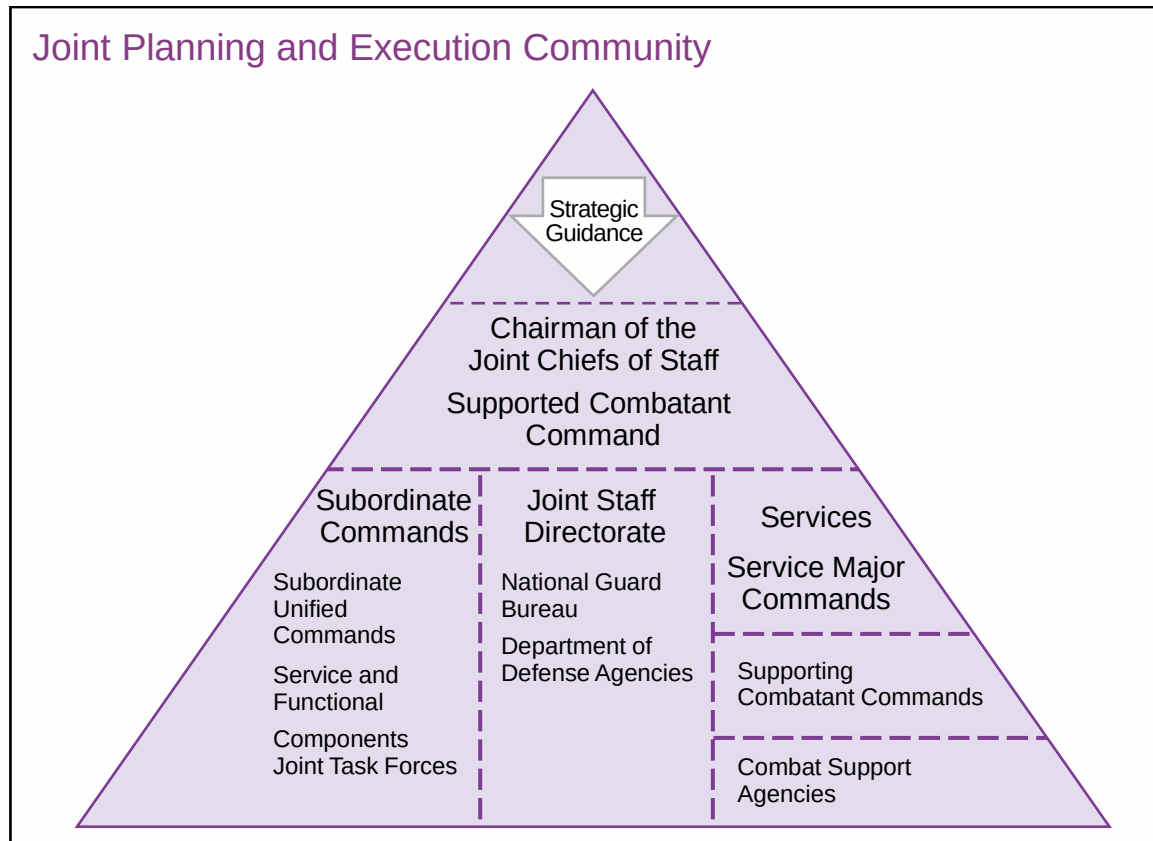


Figure II-1. Joint Planning and Execution Community

President, SecDef, or CJCS. The designated supporting commanders provide planning assistance, forces, or other resources to a supported commander, as directed.

(2) **Supporting Commanders.** Supporting commanders provide actions and other directed support to a supported commander in accordance with strategic guidance. Supporting commanders prepare supporting plans as required. A commander may be a supporting commander for one operation while being a supported commander for another.

(3) **Coordinating Authority.** Coordinating authority is the authority delegated to a commander or individual for coordinating specific functions and activities involving forces of two or more Military Departments, two or more joint force components, or two or more forces of the same Service (e.g., joint security coordinator exercises coordinating authority for joint security area operations among the component commanders) and may include USG departments and agencies and partner nations (e.g., as part of security cooperation planning). The CJCS assigns coordinating authority to a CDR for each GCP through the JSCP, with assigned collaborators to support planning. In this context, the CJCS delegates coordinating authority to a CDR with the preponderance of responsibility for developing plans in support of a GCP and associated contingencies but who does not receive additional command authority or authority to compel agreement. CDRs' authorized coordinating authorities convene collaborative forums to perform three functions: planning, assessing, and recommending changes to plans.

b. CCDRs use strategic guidance and direction to prepare command strategies focused on their command's specific capabilities and missions to link national strategic guidance to strategies and joint campaigns and operations. The command strategy identifies the command's broad, long-range objectives that contribute to national security. The command strategy provides the link between national strategic guidance and joint planning.

c. Global and Transregional Missions

(1) Since threats or opportunities presented by allies, partners, competitors, enemies, adversaries, and natural disasters do not restrict their operations by boundaries, CCDRs and their planners should integrate their plans with other CCDRs to ensure unified actions in support of objectives. Integrated planning also coordinates and addresses resources, timelines, decision points, and authorities across multiple CCMDs to achieve directed operational objectives.

(2) CCDRs tasked with global missions provide planning and assessment expertise to identify tasks and missions other CCMDs (supporting commands) perform to ensure success of global missions. Commands include supporting tasks as part of their campaign and contingency planning and coordinate to ensure assessments are complete. CCDRs with global responsibilities use the planning process to provide an assessment of risk from the global, cross-AOR perspective to ensure the military advice provided to the President and SecDef includes these considerations.

d. Planning Organization. CCMDs use joint planning groups, operational planning groups, or operational planning teams to direct planning efforts across the command, including implementation of plans and orders. In addition to members from the operations and planning directorates, these planning groups should also include members from the intelligence directorate, the logistics directorate, the communications directorate, and information planners, including public affairs. Some members of the commander's special staff (e.g., the chaplain, the judge advocate, the surgeon) should also be part of the planning group. The planning group leader should include others depending on the mission, (e.g., a political military advisor for a noncombatant evacuation). Initial consultation with these additional planners at the beginning of mission analyses aids the lead planner in determining whether their further involvement is warranted.

e. Strategic Estimate. The CCDR and staff, with input from subordinate commands and supporting commands and agencies, prepare a strategic estimate by analyzing and describing the political, military, economic, social, information, and infrastructure factors and trends; key relationships and links between relevant actors or networks; and the threats and opportunities that facilitate or hinder achievement of the objectives over the time frame of the strategy.

(1) The strategic estimate is a tool available to commanders as they develop plans. CCDRs use strategic estimates to facilitate the employment of military forces. The strategic estimate is more comprehensive than estimates of subordinate commanders,

encompasses all aspects of the OE, and is the basis for the development of the theater strategies.

(2) The CCCR, the CCCR's staff, and supporting commands and agencies evaluate the broad, strategic-level factors that influence the theater strategy.

(3) The estimate should include an analysis of strategic direction received from the President, SecDef, or the authoritative body of an MNF; an analysis of all states, groups, or organizations in the AOR that may threaten or challenge the CCMD's ability to advance and defend US interests; visualization of related geopolitical, geoeconomic, and cultural factors; an evaluation of major strategic and operational challenges facing the CCMD; an analysis of known or anticipated opportunities the CCMD can leverage; and an assessment of risks in the AOR.

(4) The result of the strategic estimate is a visualization and better understanding of the OE, to include allies, partners, neutrals, the civilian environment, adversaries, and enemy combatants. The strategic estimate process is continuous and provides input used to develop strategies and implement plans. The broad strategic estimate is also the starting point for conducting the commander's estimate of the situation for a specific operation.

(5) Supported and supporting CCRs and subordinate commanders all prepare strategic estimates based on assigned tasks. CCRs who support multiple JFCs prepare estimates for each supporting operation.

See Appendix A, "Strategic Estimate," for a notional strategic estimate format.

See CJCSI 3141.01, Management and Review of Campaign and Contingency Plans, for a more complete discussion of the JPEC and coordinating authority. See JP 3-0, Joint Campaigns and Operations, for a more complete discussion of command relationships.

CHAPTER III OPERATIONAL DESIGN

1. Overview

a. Operational art and operational design enable **understanding**. Understanding is more than just knowledge of the capabilities and capacities of the relevant actors or the scope and nature of the OE; it provides context for decision making and how the many facets of the problem are likely to interact, enabling commanders and planners to identify hazards, threats, consequences, opportunities, and risk.

b. **Operational art** is inherent in all aspects of operational design.

(1) Operational art relies on the ability of the commander and planners to identify the tools needed to address the planning problem presented. Execution of planning is based on the commander's and planners' experience and time available. Different commanders and planners use different tools to help them, as each person has inherent strengths, weaknesses, and bias. Similarly, every problem is different and may require different tools and skills for analysis. The tools chosen by the planner should be appropriate for the problem and should complement the planners' strengths and weaknesses.

(2) A JFC's ability to collect, organize, and use data across platforms and systems is essential to create advantage. JFCs prioritize the use of data in all operations, staffing actions, and decision making. Identification, collection, understanding, and integration of relevant data enables timely, accurate, and effective decision making.

(3) While data can enable military advantage, the amount of data readily available can overwhelm the planning process. A timely decision with incomplete information may present a better solution than waiting until all information is available. Integrating data as part of the planning process can help JFCs gain advantage through data while avoiding delays to decision making. Operational art supports a commander's ability to sort through the mass of information, identify key and critical information, and make decisions or provide guidance with incomplete information. Operational art aids the commander in identifying the point of diminishing returns in collection and analysis.

c. **Operational design** is the analytical framework that underpins planning. Operational design supports commanders and planners in organizing and understanding the OE as a complex interactive system. Commanders develop understanding of the audience and political environment to give the best military advice to civilian decision makers. Planners consider how to translate often-times confusing military jargon and concepts into a universally understood language; interagency partners are critical to this discussion. Operational design nests with the planning process to fill in gaps in guidance and information and provide a framework in which to plan, enabling planners to address the complexity of the OE, support mission analysis and COA development, and develop CONOPS with the highest likelihood of success.

(1) Operational design provides a framework for coordinating the operations and activities of the joint force within space and time. Operational design also aids in identifying trade-offs in the plan or operation. Planners can use operational design to look across factors of force, space, and time to identify the ability to exchange between those broad categories to best use the force available to achieve the objective at the lowest risk and cost. Planners employ the elements of operational design to identify the best method of identifying required forces and organizing the forces available, based on the OE. Operational design supports arranging and timing operations and organizing the OE to meet mission requirements. The output of operational design is problem definition and the operational approach.

(2) JFCs use operational design to organize their OAs, task-organize forces, and develop a C2 structure that supports the operation.

(3) While the JPP guides a planner from the receipt of guidance to issuance of the plan or order, operational design is a methodology planners can use to identify the problem, develop COAs in line with guidance, and identify the best methods for addressing the problem. Planners can modify the process described to support the specific operation or mission, based on the planner's analysis.

(4) Elements of Operational Design

(a) The commander and staff use the elements of operational design (see Figure III-1) to organize their thoughts, break down and identify the problem, understand the OE, develop their OAs, and orient the joint force on the objective.

(b) Use of these elements provides a common construct for planning and execution. Planners do not need to use all elements of operational design, only what is appropriate for the plans.

d. **Joint Functions.** Joint functions are related capabilities and activities integrated together to help JFCs synchronize and direct joint operations. Joint functions can facilitate

Elements of Operational Design

- Objective
- Center of Gravity
- Effects
- Decisive Points
- Lines of Operation
- Lines of Effort
- Direct and Indirect Approach
- Anticipation
- Transitions
- Operational Reach
- Culmination
- Arranging Operations
- Forces and Functions

Figure III-1. Elements of Operational Design

planning by providing a framework to ensure planners have addressed all aspects of an operation. Some functions, such as C2, information, and intelligence, apply to all operations. Others, such as fires, apply as the mission requires.

(1) Inadequate integration and balancing of these functions can undermine the cohesion, effectiveness, and adaptability of the force. For example, inattention to protection can deplete combat power, thereby undermining reserves and degrading the force's ability to capitalize on an opportunity or respond to an unforeseen problem. Likewise, inattention to intelligence can leave the force with inadequate information to support decision making or identify opportunities in time to exploit them.

(2) Most functions are mutually supporting. For example, movement and maneuver, such as freedom of navigation operations, can support the information joint function. For example, joint function activities that leverage information to affect behavior, such as deception operations or attacks on enemy information, information systems, and information networks, can enable movement and maneuver.

(3) Plans describe how the JFC uses military capabilities (i.e., organizations, people, and systems) to perform tasks associated with each joint function. Individual Service capabilities can often support multiple functions simultaneously or sequentially, while the joint force is executing a single task.

For more information on joint functions, see JP 3-0, Joint Campaigns and Operations.

2. Operational Design Methodology

Operational design requires recurring touch points between the commander and staff in developing an understanding of the OE, incorporating higher-level guidance, defining the problem, and developing an operational approach. The components have characteristics that exist independent of each other and are not necessarily sequential. As commanders and staffs develop their operational approach, they account for how information impacts the OE and the inherent informational aspects of activities. In doing so, joint force planners consider how information affects the behavior of friendly, neutral, and adversarial audiences across the competition continuum. Operational design is one of several tools available to help the JFC and staff understand the broad solutions for mission accomplishment and to understand the uncertainty in a complex OE. The process is continuous and cyclical, conducted prior to and during planning, and for follow-on joint operations. Additionally, it supports ongoing civil-military dialogue concerning the nature of the problem and an operational approach to achieve the desired objectives.

a. Understand the Strategic Directions and Guidance

(1) Direction from strategic guidance documents can be vague, incomplete, outdated, or conflicting. This is due to the different production timeframes, changes in personnel that result in differing opinions or policies, and the staffing process where compromises achieve agreement within the documents. During planning, commanders and staff read the directives and synthesize the contents into a concise statement. Since the

intent of some strategic guidance can be underdeveloped, the JFC and staff should obtain clear, updated direction through sustained civilian-military dialogue throughout the planning process. When clarification does not occur, planners and commanders identify those areas as elements of risk.

(2) Commanders maintain dialogue with leadership at all levels to resolve differences of interpretation of higher-level objectives and the ways and means to achieve these objectives. While CCDRs and national leaders may have a clear strategic perspective of the problem from their vantage point, operational-level commanders and subordinate leaders often have a better understanding of specific circumstances that comprise the operational situation and may have a completely different perspective on the causes and solutions. Both perspectives are essential to a plan. Subordinate commanders should be aggressive in sharing their perspective with their higher headquarters, and both should resolve differences at the earliest opportunity. While policy and strategic guidance clarify planning, it is equally true that planning informs policy formulation. An unrealistic strategy or impossible plan at the tactical level can be as detrimental to long-range US interests as tactical actions that accomplish a task but undermine the objectives.

b. Understand the OE

(1) The OE is the aggregate of the conditions, circumstances, and influences that affect the employment of capabilities and bear on the decisions of the commander. It encompasses all aspects of the air, land, maritime, space, and cyberspace domains, information, the civilian environment, and the electromagnetic spectrum (EMS). Understanding the OE helps the JFC identify the problem; anticipate potential outcomes; and understand the results of various friendly, adversary, enemy, and neutral actions and how these actions affect achieving the objective (see Figure III-2).

(2) To visualize an approach to solving the problem, the commander should be able to describe both the current state of the OE, the assessed OE throughout the operation or campaign, and the desired state of the OE when operations conclude, to include actor behavior. Planners should compare the current conditions of the OE with the desired conditions and delineate the tensions or frictions that prevent or hinder achieving the objectives. Identifying necessary objectives early in planning helps the commander and staff devise an operational approach with lines of effort (LOEs)/lines of operation (LOOs) that link decisive points and operations with the objectives.

(3) Describe the Current and Future OE

(4) Strategic-Level Considerations

(a) Strategic considerations include circumstances, and influences that affect national interests, and may impact the composition of alliances, establish competing requirements or priorities, or affect deployment and distribution operations. Some considerations are:

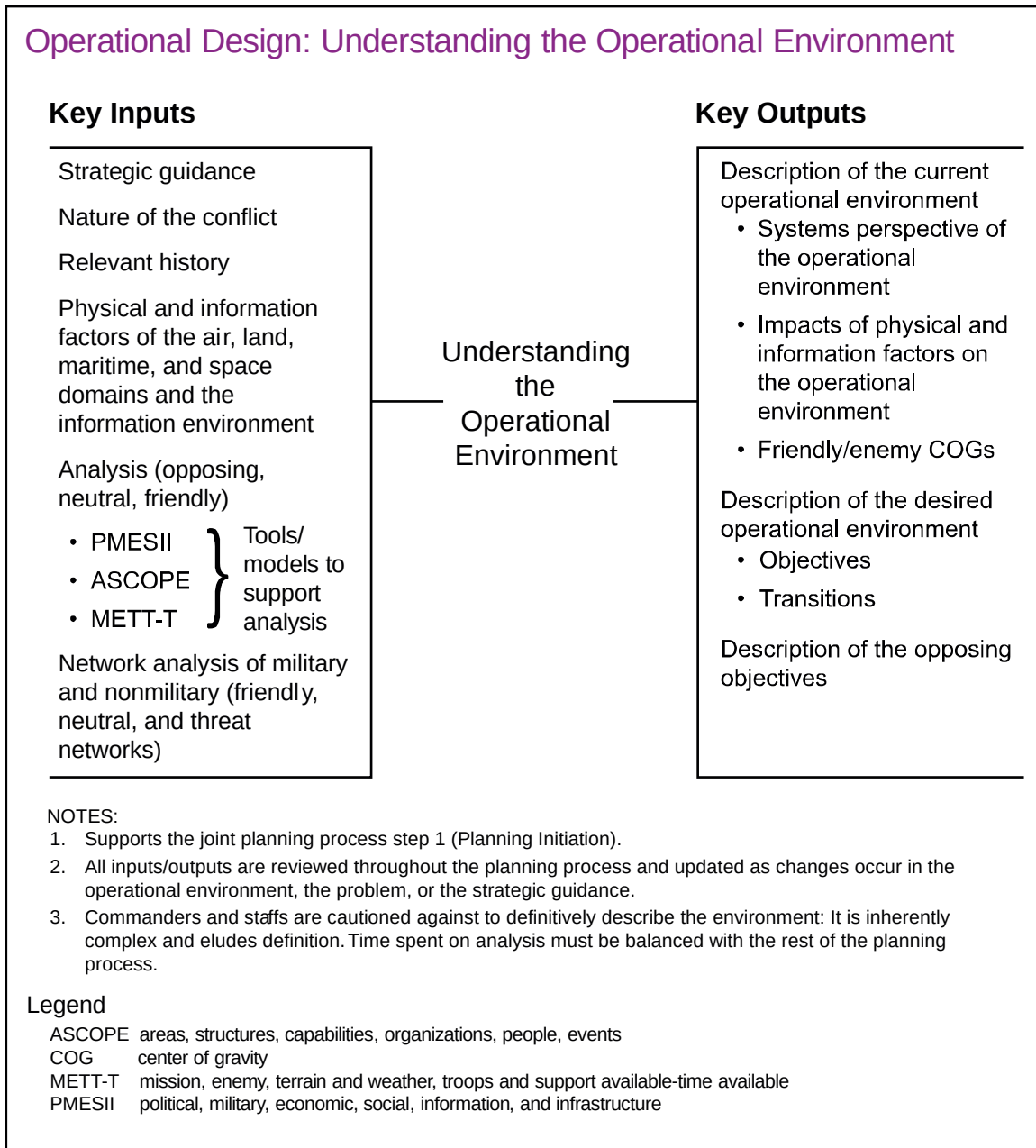


Figure III-2. Operational Design: Understanding the Operational Environment

1. What actions or planning assumptions will be acceptable given the current US policies and the diplomatic and political environment?

2. What impact will US activities have on third parties (focus on military impacts but identify possible diplomatic, economic, or commercial ramifications that may impact third-party willingness to support US activities including access, basing, and overflight decisions)?

3. What are the current US national strategic objectives? Are the objectives expected to be long-lasting or short-term only? What unintended consequences

could result (e.g., is there sufficient time to develop strong controls so that weapons provided to a nation do not serve unintended purposes)?

(b) Commanders undertake activities to support the achievement of objectives, which in turn advance or defend national interests. Commanders evaluate military activities against that strategic measure—do they contribute positively to the strategic objectives and advance or defend national interests? CCPs do this through reasoned sequencing of military operations; defining limits; and assessing benefits, costs, and risks for the use of military forces and capabilities.

1. Strategic-level considerations may include factors such as international law; diplomatic, information, military, and economic activities that influence other nations' policies and domestic and world opinion; adversary and friendly organizations and institutions; efforts to protect and restore elements within the civilian environment; and the capability and availability of logistics, transportation, space capabilities, and information technology. Strategic-level considerations of the OA analyze geopolitical regions and nations rather than local considerations. Additional considerations are commercial capabilities that provide access to and through all domains.

2. Nonmilitary aspects of the OE assume increased importance at the strategic level. For example, the industrial and technological capabilities of a nation or region influence the type of military force it fields and may influence the ability of a nation or region to endure a protracted conflict without outside assistance. In many situations, nonmilitary considerations may play a greater role than military factors in influencing adversary, partner, and neutral state decisions, behaviors, and approaches.

(c) JIPOE is a continuous, analytical process that joint intelligence organizations use to produce intelligence assessments, estimates, and other intelligence products in support of the JFC's decision making. JIPOE provides an understanding of the OE and assesses potential threat COAs in a wide variety of situations across the competition continuum. In addition to traditional threats, JIPOE should encompass other relevant actors, to include neutral and hostile networks that could delay, degrade, or prevent the joint force from accomplishing its mission, and deniable subversive activities, as well as those relevant actors that could help the joint force mission. The J-2 manages JIPOE with input from intelligence planners and other staff directorates or elements through the following four steps:

1. Define the OE.
2. Describe the impact of the OE on planned operations.
3. Evaluate the threat and other relevant actors.
4. Determine the COAs for the threat and other relevant actors, particularly the most likely COA and the most dangerous COA to friendly forces and mission accomplishment.

See the Joint Guide to Joint Intelligence Preparation of the Operational Environment and JP 2-0, Joint Intelligence.

(5) Operational-Level Considerations

(a) In analyzing the current and future OE, the staff can use a political, military, economic, social, information, and infrastructure analytical framework to determine relationships and interdependencies to the specific operation or campaign (see Figure III-3). This includes the civilian environment. The civilian environment includes the civilian population and the personnel, organizations, resources, infrastructure, essential services, and systems on which civilian life depends within an OE.

(b) The size and scope of the analysis varies depending on aspects of the OE and time available. For example, if a landlocked adversary has access to space-based services (organic or provided by third parties) or the capability to conduct cyberspace operations, then the portions of the contested environment extend worldwide, while maritime considerations might be minimal. While most joint operations at the operational level may encompass many or all political, military, economic, social, information, and infrastructure considerations and characteristics, the staff's balanced JIPOE efforts should vary according to the OE aspects of the operation or campaign. The CCDR's J-2 also coordinates with supporting CCDRs' J-2(s) to ensure the OE assessment identifies and

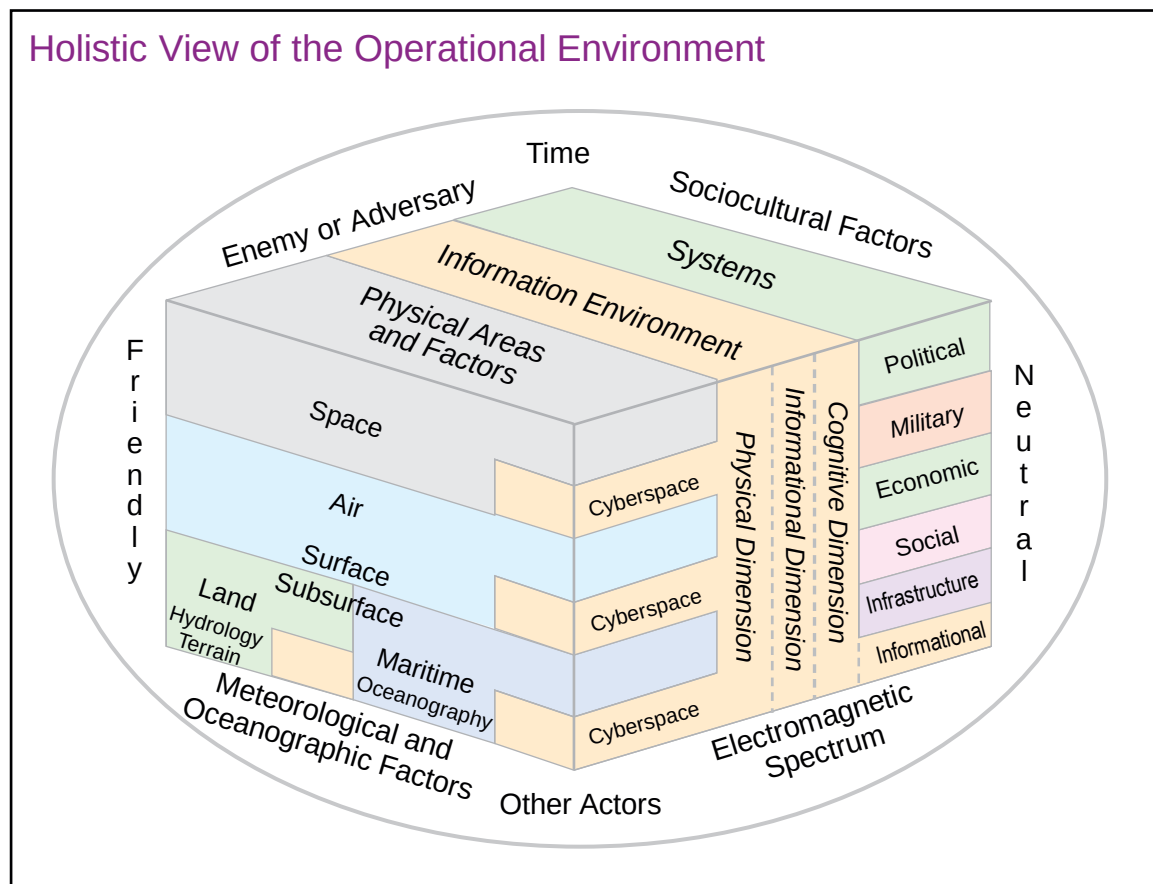


Figure III-3. Holistic View of the Operational Environment

characterizes threats that may degrade, disrupt, or destroy missions executed by supporting CCDRs. See JP 2-0, *Joint Intelligence*, and the *Joint Guide for Joint Intelligence Preparation of the Operational Environment* for additional information on analyzing and understanding the OE.

(c) When developing an understanding of the OE, planners consider how data enables that understanding, such as data availability, how JFCs can obtain data, how often does data update, the potential for collection automation, etc. JFCs consider data in the early stages of planning to enable understanding of the OE and inform subsequent decision making, assessments, and data collection plans (DCPs).

(d) Analysis of the OE should include an analysis of the information environment. This should include how information impacts the OE and is the basis for incorporating the use and application of information into the planning, execution, and assessment of joint operations. The JIPOE process should provide the analysis that answers the following questions related to describing the current OE in terms of its informational, physical, and human aspects:

1. How does information move in and through the OE; how is it received, processed, and used; by whom is it received, processed, and used; and for what purposes is that information used?

2. Who are the known or potentially relevant actors? Who are the decision makers and key influencers? Which individuals play multiple roles? With whom do relevant actors partner (e.g., protection, shared values)?

3. What key human and automated decision-making processes do the relevant actors use?

4. Who uses which information systems and networks, and how do the relevant actors interact with them?

5. What are the joint force activities that relevant actors are/will be observing?

6. How will relevant actors perceive, and what meaning will relevant actors assign to, joint force activities? What details and features of those activities (e.g., timing, duration, presence, posture, profile) will have the most impact on the meaning that relevant actors assign?

7. What relevant actor behaviors manifest can the JFC expect to because of those perceptions?

(e) Additional factors to consider, include:

1. Geographic, meteorological, and oceanographic features and characteristics.

2. Population demographics and dynamics.

a. Demographic concerns include ethnic groups, tribes, ideological factions, religious groups and sects, language dialects, sex, age distribution, income groups, public health issues, and population density.

b. Dynamics include patterns of life, cultural norms, the interconnected relationships between the civilian population, and other aspects of the civilian environment.

3. Social and cultural factors of enemies, adversaries, neutrals, and allies in the OA.

4. Political and socioeconomic factors (e.g., economic system, political factions, tribal factions, commercial influence).

5. Infrastructure, such as transportation, energy, and information systems, and its foreign or third-party ownership.

6. Rules of engagement (ROE)/rules for the use of force, or legal restrictions on military operations as specified in US law, international law, or HN agreements.

7. All friendly, adversary, and enemy conventional, irregular, and paramilitary forces and their general capabilities and objectives (including all known or suspected chemical, biological, radiological, and nuclear threats, and hazards).

8. Environmental conditions (e.g., earthquakes, volcanic activity, pollution, naturally occurring diseases).

9. Location of toxic industrial materials or facilities that may produce chemical, biological, radiological, or nuclear hazards.

10. All locations of foreign embassies, international organizations, and NGOs.

11. Friendly, adversary, and enemy space capabilities (military, civil, or commercial); their current or potential use; and critical vulnerabilities.

12. Knowledge of the capabilities and intent; centers of gravity (COGs); and critical vulnerabilities of forces, individuals, or organizations conducting cyberspace operations.

13. Financial networks that could impact an adversary's or enemy's ability to sustain operations.

14. Psychological characteristics of enemy or adversary decision making.

15. The presence, role, and objectives of friendly, neutral, and threat networks and their interrelationships.

16. Enemy and adversary capability to employ or develop improvised threat devices, weapons, or systems, especially unmanned aerial systems.

17. Friendly, adversary, and neutral EMS dependencies resulting in a complex electromagnetic environment. JFCs address adversary capabilities to contest friendly use of the EMS, spectrum congestion in the OA, and constraints put on friendly use of the EMS by HN policy.

For more information on EMS operations, see JP 3-85, Joint Electromagnetic Spectrum Operations.

(f) To produce a holistic view of the enemy, adversary, neutral, and friendly systems within a larger system that includes many external influences, analysis should define how these systems interrelate. Most important to this analysis is describing the **relevant relationships** within and between the various systems that directly or indirectly affect the problem at hand. Although the J-2 manages the JIPOE process, other directorates and agencies can contribute valuable expertise to develop and assess the complexities of the OE.

For more information on JIPOE, see JP 2-0, Joint Intelligence, and the Joint Guide for Joint Intelligence Preparation of the Operational Environment.

(g) Network engagement analysis can aid understanding of relationships. Network analysis also facilitates identification of information that might otherwise go unnoticed. This can include uncovering unrecognized positions of power within related groups, accounting for structure and organization, finding individuals or nodes who are critical, and facilitating measuring change over time. Once the JFC identifies the networks in the OA and understands their interrelationships, functions, motivations, and vulnerabilities, the commander can tailor the force to partner with friendly networks, engage neutral networks, and counter threat networks.

(h) **Tendencies and Potentials.** In developing an understanding of interactions and relationships in the OA, commanders and staffs consider observed tendencies and potentials in their analyses. Tendencies reflect the inclination to think or behave in a certain manner. Tendencies model thoughts or behaviors and help identify the range of possibilities that may develop with or without external influence. Once identified, commanders and staffs evaluate the potential of these tendencies to manifest within the OE. Potential is the inherent ability or capacity for the growth or development of a specific interaction or relationship. However, not all interactions and relationships support achieving the desired objective. The desired objective accounts for tendencies and potentials or other aspects of the OE. Early in the JPP, review pertinent lessons learned as part of the analysis to incorporate lessons into the plan. The Joint Lessons Learned Information System provides a database of lessons learned. However, seek out people experienced in the mission, OE, and lessons learned functions for their knowledge and experience.

(i) Describe the key conditions that have to exist in the future OE to achieve the objectives. Planners should put a time limit on this set of conditions to be able to conduct feasibility and acceptability analyses.

(j) Determine the relevant actor's objectives that affect the OE. Each participant has different sets of conditions for achieving their respective objectives. Opposing forces actively work to undermine the objectives of the US and its partner nations. Others, whether neutral or friendly, may not have an opposing mindset but may have desired conditions (including their unintended consequences) that jeopardize achievement of the JFC's objectives. In developing the plan, planners should consider if the COA addresses these conflicts. For example, a CDR's assessment (e.g., partnership assessment) while conducting security cooperation may identify conflicts of interest amongst parties within the partner nation.

For more information on network engagement and understanding friendly, neutral, and threat networks, see JP 3-25, Joint Countering Threat Networks.

c. Define the Problem

(1) Defining the problem is essential to addressing the problem. It involves understanding and isolating the root causes of the issue that are the essence of a complex, ill-defined problem. Defining the problem begins with a review of the tendencies and potentials of the relevant actors and identifying the relationships and interactions among their respective desired conditions and objectives. As plans have objectives that may address multiple problems, planners create a problem statement. The problem statement articulates how the operational variables can resist or facilitate transformation of current conditions. It also describes how to leverage inertia in the OA to enable the desired conditions to achieve the objectives.

(a) The problem statement identifies the areas that, when successfully acted upon, help transform the existing condition into the desired condition. Defining the problem extends beyond analyzing interactions and relationships in the OA (see Figure III-4). It identifies areas of tension, competition, and contested environments—as well as opportunities and challenges—that commanders address to transform current or anticipated conditions to achieve the desired objective. Tension is the resistance or friction among and between participants. The commander and staff identify the tension by analyzing tendencies and potentials of the OE.

(b) Critical to defining the problem is determining action to reconcile the differences between existing and desired conditions. Some of the conditions are critical to success, some are not. Some may change as a secondary or tertiary result of another condition. In identifying the problem, the planning team identifies the tensions among the desired conditions and identifies the areas of tension that merit further consideration as areas of possible intervention.

(c) The JFC and staff identify and articulate (see Figure III-5):

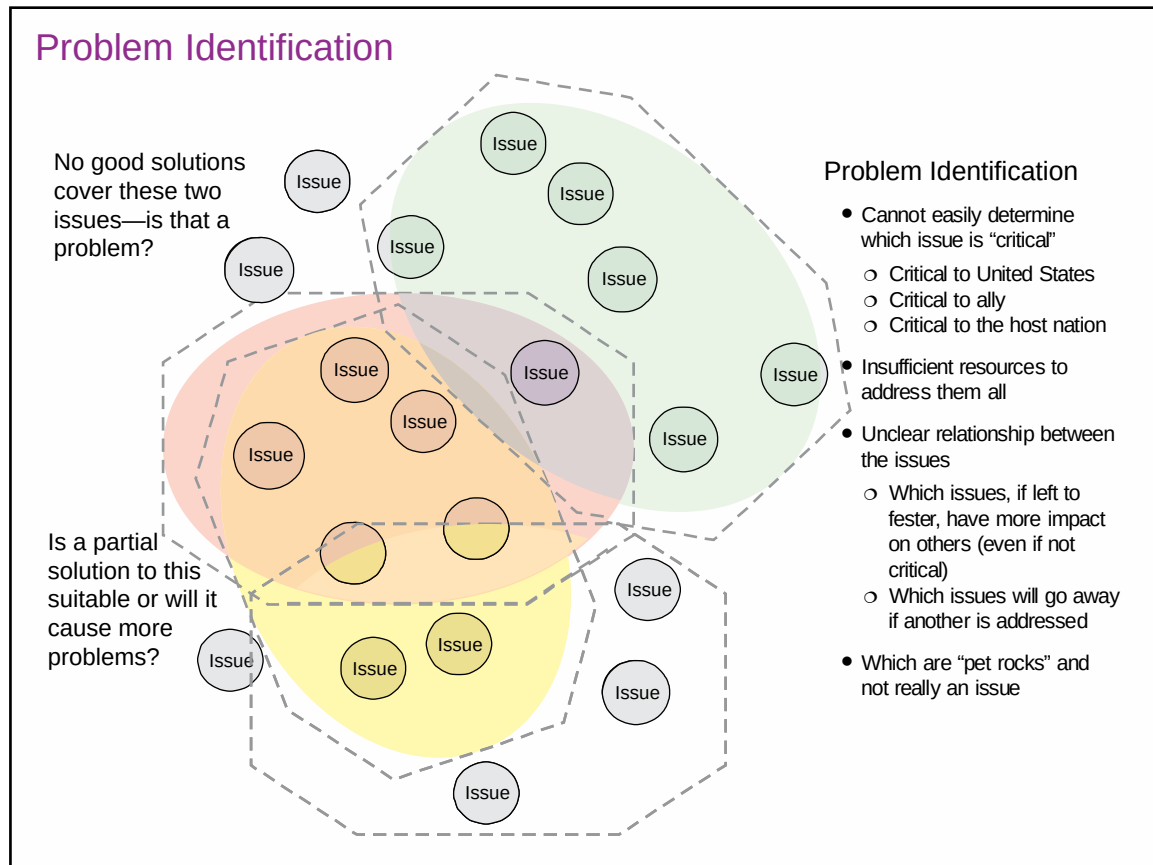


Figure III-4. Problem Identification

1. Tensions between current conditions and desired conditions at the objective.

2. Elements of the OE that JFCs have to influence to change or maintain to achieve the objective or create desired condition, including relevant actors’ behaviors, which can impede or support changing the current conditions in the OA to the desired conditions.

3. Opportunities to exploit or threats that impede the JFC from attaining the desired condition.

4. Effects of military operations on the civilian environment.

5. Operational limitations.

(d) A clear, concise, and precise problem statement is essential to provide definitive focus for development of a plan. The problem statement is the planner’s answer to the questions “what’s going on here?” and “why is this happening?” In other words, what situation or condition is threatening or presenting an opportunity, for which interests, and how. The problem statement considers how tension and competition affect the OE by identifying pathways to transform current conditions into new, more desirable conditions.

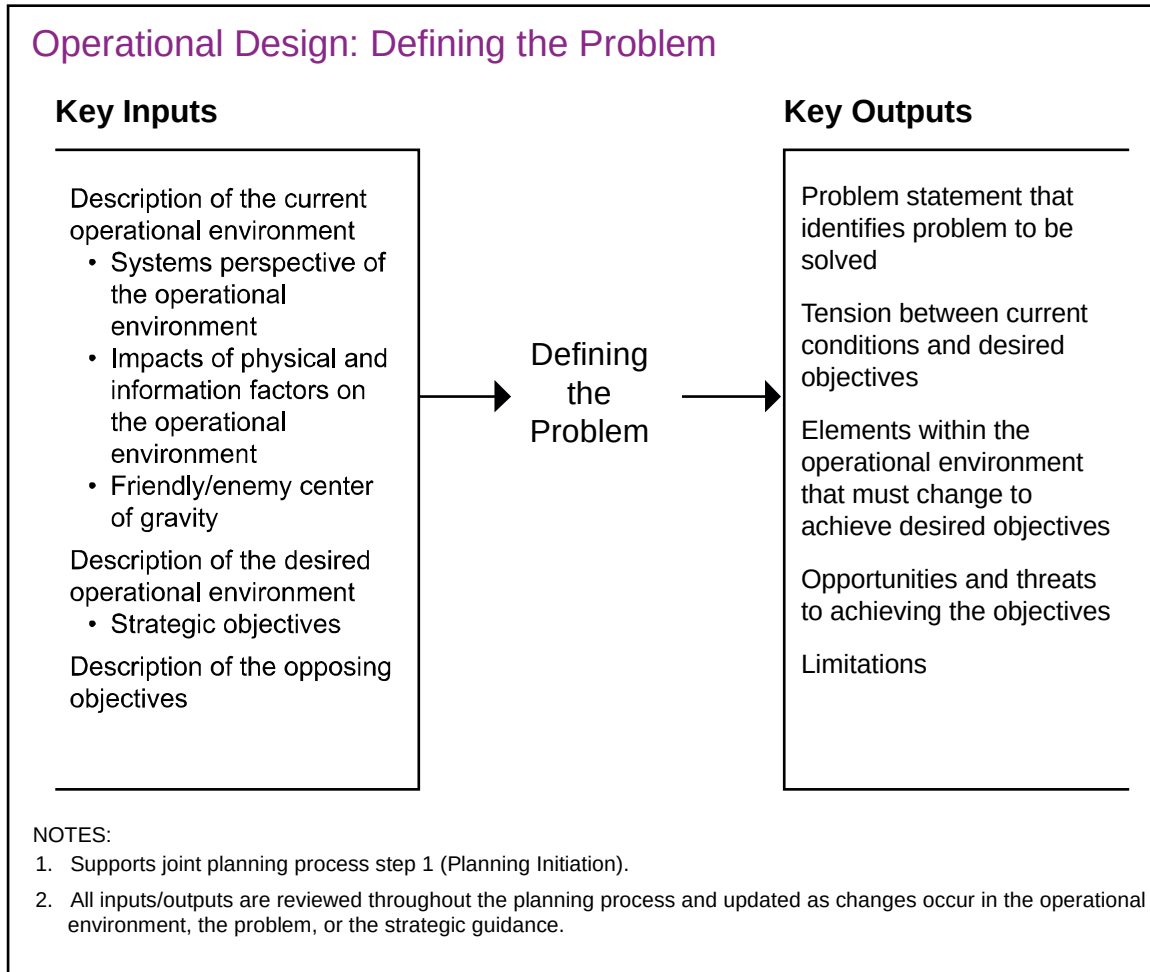


Figure III-5. Operational Design: Defining the Problem

d. Identify Assumptions

(1) Where there is insufficient information or guidance, the commander and staff identify assumptions to assist in framing solutions. At this stage, assumptions address strategic and operational gaps. These assumptions are necessary for the JFC to continue planning and develop the operational approach.

(a) Minimize assumptions, as each assumption adds to the probability of error in the plan and requires specific commander's critical information requirements (CCIRs) to continuously assess its validity and risk to the plan.

(b) Assumptions should only address gaps in information or guidance that are essential to the plan's success. If the plan can succeed if the assumption proves invalid, or planning can continue without the assumption, then the assumption is unnecessary.

(c) Assumptions address key decisions required by senior leaders to enable JFCs to continue planning.

(d) Assumptions should not overlook or dismiss a likely hostile COA or a friendly inability to execute or sustain the plan that could cause the plan to fail.

(2) Commanders and staff should review strategic guidance and direction to identify any assumptions from their higher headquarters that impose constraints or restraints on the planning process. They should also regularly discuss planning assumptions with supporting CCDRs, Office of the Secretary of Defense (OSD), and DoD leadership to account for changes to the OE, global requirements, policy, or guidance in the planning assumptions (e.g., basing or access permissions, allied or multinational contributions, alert and warning decision timelines, anticipated threat actions and reactions). Phrase assumptions in terms of will or will not (rather than using “should” or “may”) to establish specific conditions that enable planning to continue.

(3) During the JPP, the commander may develop additional assumptions to support detailed COA development as noted earlier in this chapter.

e. Develop Operational Approaches

(1) The operational approach is a commander’s broad description of the mission, operational concepts, tasks, and actions required to accomplish the mission. The operational approach is based largely on an understanding of the OE and the problem facing the JFC (see Figure III-6 and Figure III-7). A discussion of operational approaches within and between options forms the basis of the IPRs between the CCDR and SecDef. Once SecDef approves the approach, it provides the basis for beginning, continuing, or completing detailed planning. The JFC and staff should continually review, update, and modify the approach as policy, the OE, objectives, or the problem change. This requires frequent and continuing dialogue at all levels of command.

(2) There are three purposes for developing an operational approach:

(a) It provides the foundation for the commander’s planning guidance to the staff and other partners by providing the commander’s visualization of how the joint force’s operations will transform current conditions into the desired conditions—the way the commander envisions the OE at the conclusion of operations to support strategic objectives.

(b) It provides the model for execution of the campaign or operation and development of assessments for that campaign or operation.

(c) It enables a better understanding of the OE and of the problem.

(3) Commanders and their staffs can use operational design when planning any joint campaign or operation. Notwithstanding a commander’s judgment, education, and experience, the OE often presents situations so complex that just understanding them—let alone attempting to change them—exceeds the capability and capacity of the committed forces. This complexity complicates coherent planning. Bringing adequate order to

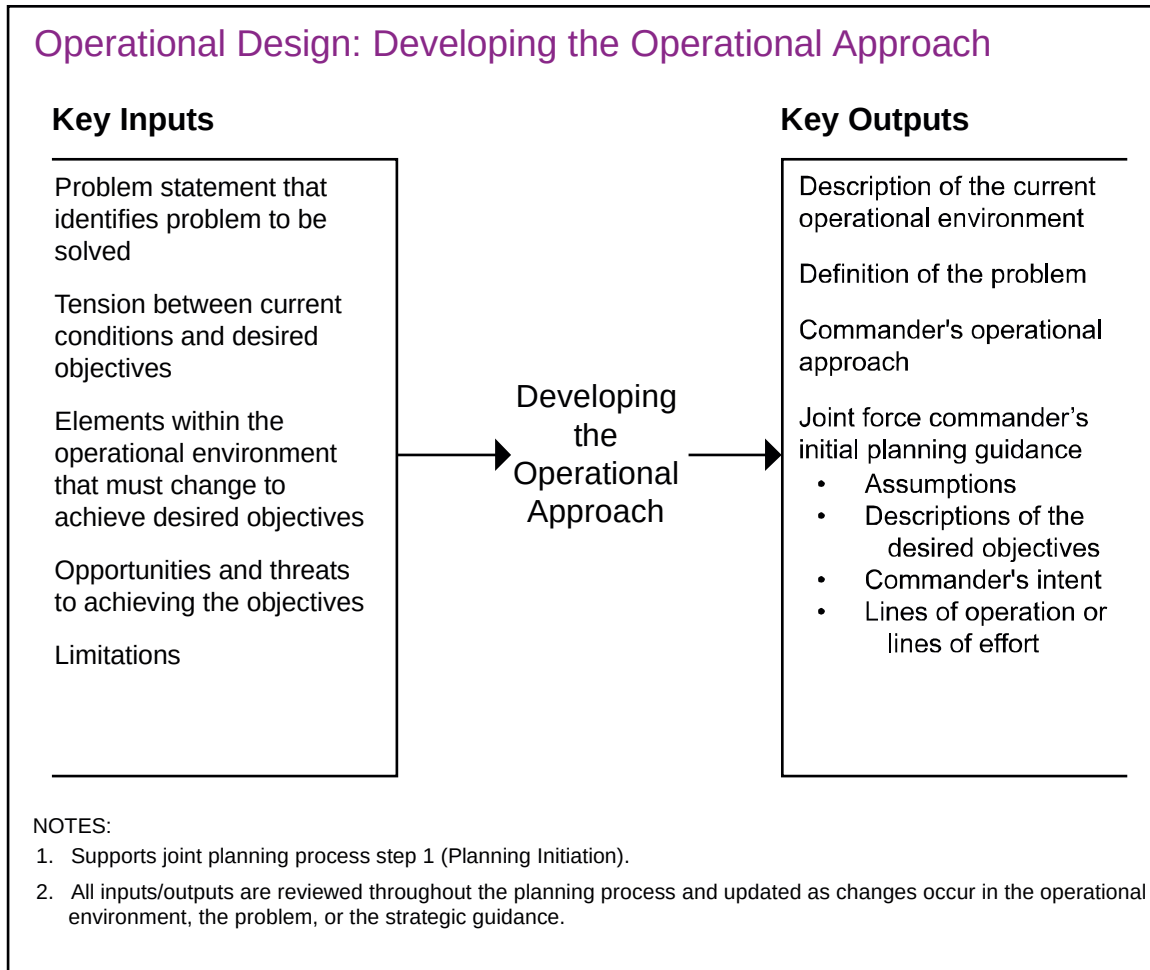


Figure III-6. Operational Design: Developing the Operational Approach

complex problems to facilitate further detailed planning requires an iterative dialogue between commander, supporting commanders, the planning staffs, and policy staffs. Rarely do members of both staffs recognize an implicit operational approach during their initial analysis and synthesis of the OE. Successful development of the approach requires continuous assessment of activities and changes to the OE, analysis, learning, dialogue, and collaboration between commander and staff, as well as other subject matter experts. The challenge is even greater when the joint operation involves other agencies, the private sector, and multinational partners (which is typically the case), whose unique considerations can complicate the process. Some friendly and neutral networks might also provide new capabilities, information, perspectives, or solutions to enable mission success.

(4) It is essential that commanders, through a dialogue with their staffs, planning teams, and initiative groups, first gain an initial understanding of the current and future OE. This includes understanding the US policy perspective and defining the problem facing the joint force prior to conducting detailed planning. The problem as presented and characterized in strategic guidance documents rarely includes all available information and may identify and describe symptoms rather than the actual problem. From this understanding of the OE and definition of the problem, commanders develop their broad,

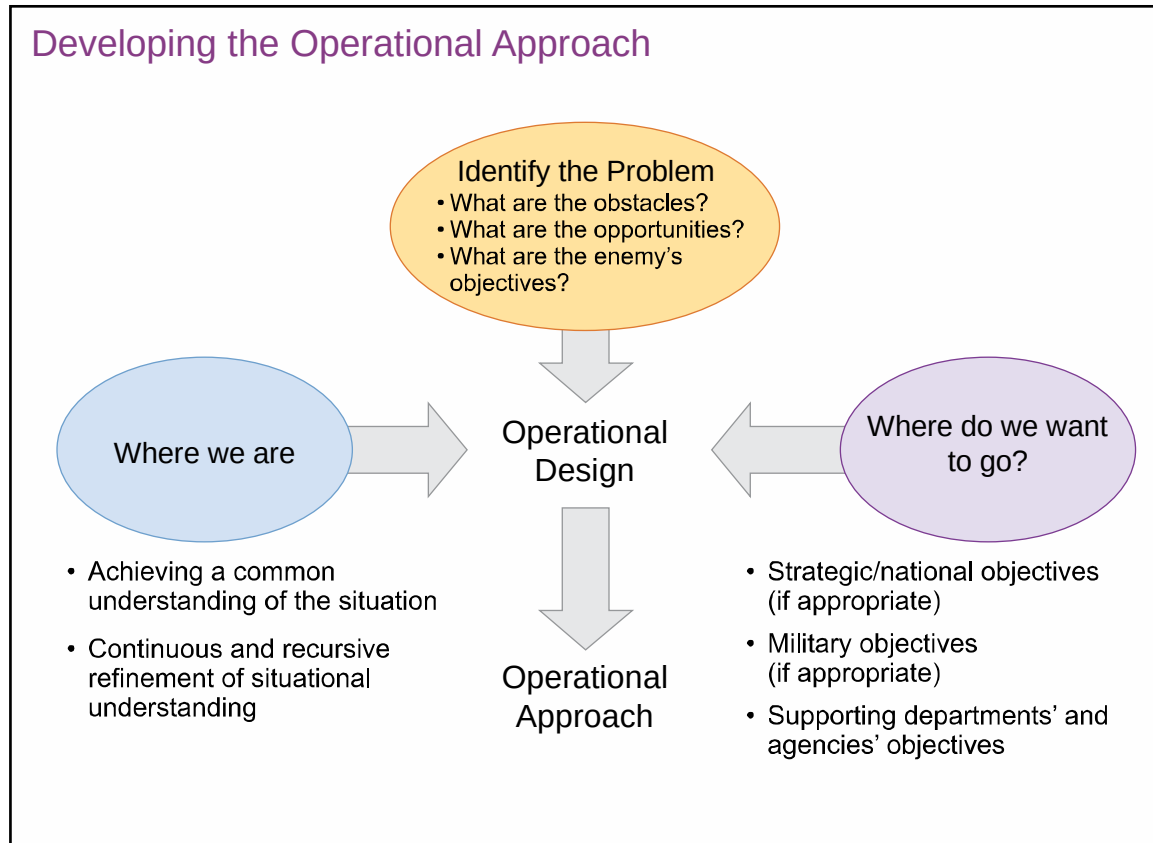


Figure III-7. Developing the Operational Approach

operational approach for transforming current conditions into desired conditions. The operational approach underpins the operation and the detailed planning that follows. The JFC and staff continually refine the operational approach as detailed planning occurs and as operations and joint force actions impact the OE.

(5) In developing an appropriate operational approach, the commander should address the following questions:

- (a) What are the strengths, weaknesses, intentions, and interrelationships of the various actors?
- (b) What vulnerabilities will the actor(s) likely seek to exploit?
- (c) What are the opportunities and threats?
- (d) How can the operational approach change the existing conditions to favorable conditions?
- (e) What will be the likely consequences of shaping the OA toward a desired set of conditions?

(f) How well understood are the consequences of actions or changes to the OE?

f. Identify Decisions and Decision Points

(1) During planning, the staff informs leadership ahead of key decisions and the uncertainty and risk accompanying decisions and any delay in making them. This provides military and civilian leaders a template and warning for decisions in advance and helps facilitate collaboration with interagency partners and allies to develop alternatives and exploit opportunities short of escalation. The decision matrix also identifies the expected indicators needed in support of operation assessment and IRs and collection plans.

(2) Decision points are the latest point in space and time when a commander can make a key decision concerning a specific COA. They account for the delay between deciding to implement the plan and the execution of the action. The space between the decision initiation and the decision point is the decision time available to the staff and commander. Early decision initiation may overwhelm the staff battle rhythm, prompt premature decisions, and telegraph operations to the enemy or adversary.

(3) Commanders ensure senior leaders understand the risk and timelines associated with the decision points and the possible impacts on the mission of delayed decisions.

g. Refine the Operational Approach

(1) Throughout the planning processes, commanders and their staffs conduct formal and informal discussions at all levels of the chain of command, supporting CCDRs, and subordinate commands. These discussions help refine assumptions, limitations, and decision points that could affect the operational approach and ensure the plan remains feasible, acceptable, and suitable.

(2) The commander adjusts the operational approach based on feedback from the formal and informal discussions at all levels of command and other information.

h. Prepare Planning Guidance

(1) **Develop Commander's Planning Guidance.** The commander approves a summary of the OE and the problem, along with a visualization of the operational approach created by the staff. Commanders share understanding through their planning guidance. As time permits, the commander may have been able to apply operational design to think through the campaign or operation before the staff begins the JPP. In this case, the commander provides initial planning guidance to help focus the staff during mission analysis. Commanders should continue the analysis to further understand and visualize the OE as the staff conducts mission analysis. Upon completing analysis of the OE, the commander issues planning guidance, as appropriate, to help focus the staff efforts. At a minimum, the commander issues planning guidance, either initial or refined, at the conclusion of mission analysis and provides refined planning guidance as understanding

of the OE, the problem, and visualization of the operational approach matures. It is critical for the commander to provide updated guidance as the campaign or operation develops, to adapt the operational approach to a changing OE or changed problem.

(2) The format for the commander's planning guidance varies based on the personality of the commander and the level of command but should adequately describe the logic to the commander's understanding of the OE, the methodology for reaching the understanding of the problem, and a coherent description of the operational approach. It may include the following elements:

(a) **Describe the Strategic Environment.** Some combination of graphics showing key relationships and tensions and a narrative describing the strategic environment help convey the commander's understanding to the staff and other partners. The description of the strategic environment should include assessed/anticipated enemy, adversary, or other relevant actor actions that extend beyond the OA, particularly those that may impact deployment, distribution, and other critical strategic capabilities.

(b) **Describe the Current and Future OE.** A combination of graphics showing key relationships and tensions, and a narrative describing the transition from the current OE to the future OE, helps convey the commander's understanding to the staff and other partners. The description of the OE includes the assessed and anticipated adversary, enemy, and other actor actions that could degrade, disrupt, or destroy successful accomplishment of the unit's mission and achievement of the conditions of the future OE.

(c) **Define the problem to solve.** A narrative problem statement that includes a time frame to solve the problem will best convey the commander's understanding of the problem.

(d) **Describe the operational approach.** A combination of a narrative describing objectives, decisive points, potential mission areas, LOEs, and LOOs, with a summary of limitations (constraints and restraints) and risk (what leaders can or cannot accept) will help describe the operational approach.

(e) **Provide the commander's intent.** The commander should also include intent in planning guidance. The commander's intent describes the purpose of the operations, objectives, and risks associated with joint campaigns and operations and describes the desired conditions in terms of behaviors needed to support enduring outcomes. It also includes where the commander will and will not accept risk during the operation. It organizes (prioritizes) desired conditions and the combinations of potential actions in time, space, and purpose. The JFC should envision and articulate how military power and joint operations, integrated with other applicable instruments of national power, will achieve strategic success and how the command intends to measure the progress and success of its military actions and activities. It should help staff and subordinate commanders understand the intent for unified action using interorganizational cooperation among all partners and participants. The commander's intent identifies the major unifying efforts during the campaign or operation, the points and events where operations must

succeed to control or establish conditions in the OA, and where other instruments of national power play a central role. The intent allows for decentralized execution. It provides focus to the staff and helps subordinate and supporting commanders take actions to achieve the objective without further orders, even when operations do not unfold or result as planned. While there is no specified joint format for the commander's intent, a generally accepted construct includes the purpose and risk. Chapter I, "Joint Planning," discusses purpose and risk in more detail.

1. The intent may also include objectives, method, and effects guidance.
2. The commander may provide additional planning guidance, such as information management, resources, or specific effects to create or avoid.

3. Primary Considerations for Operational Design

a. The primary considerations for operational design exist in four broad categories: overarching, space (OE), time, and forces.

(1) **Overarching** elements of operational design are those that drive the operation. Some, such as the objective, may come from higher-level guidance. Others, such as the COG, effects, and culmination, are determined from planners' analysis of the OE and other considerations such as available forces and time.

(2) The space of the **OE** requires planners to consider the physical characteristics, conditions, and geometry of the environment, to include how the commander should divide the operation for C2 purposes. The OE also accounts for the integration of the information environment including cyberspace and the EMS into planning, even though some aspects of the OE exist outside the physical terrain. Analysis of the OE also considers the distances required for deployment and physics of movement through environment.

(3) **Time** considerations lead planners to identify how long it could take to conduct operations from an initial decision to commit forces through planning, mobilization, deployment, and execution. When considering the factor of time, a commander needs to identify what information they require to make timely decisions, how operations might sequence, or how an operation might phase to ensure the force achieves its objective without culminating. Time can also drive other factors such as the resources required by a commander to successfully operate throughout the duration of an operation. Unlike space or forces, time cannot reconstitute once elapsed.

(4) **Force** requires JFCs to know the science of warfare, the capabilities and limitations of blue force weapon systems and tactics, and the capabilities and limitations of the enemy's weapon systems and tactics. Force accounts for the intangible factors of leadership and morale surrounding the planning process, as well as the capabilities of interagency, international, and commercial actors in the OA.

(5) The characterization of elements into categories is not meant to be exclusive. Many factors affect more than one condition. For example, a **decisive point** could be an enemy force, a key terrain feature, or ensuring sufficient food and medical supplies deliver on time (for humanitarian assistance).

b. **Integration.** Use operational design in conjunction with the principles of joint operations and the joint functions. Throughout the planning process, planners should continuously evaluate to ensure the plan addresses joint functions and the plan does not violate the principles of joint operations.

4. Elements of Operational Design

a. **Objective.** The objective is the single most important element of operational design. Objectives may be broad or narrow.

(1) Operational objectives are critical in plan development because they link to strategic objectives and provide the basis for determining necessary effects. This, in turn, drives the formulation of the missions to achieve those objectives.

(2) **Clearly define an achievable and decisive objective.** Joint planning integrates military actions and capabilities in time, space, and purpose to provide unity of effort to achieve the JFC's objectives. Objectives and their supporting effects provide the basis for identifying tasks to accomplish. **In JSCP-directed campaign plans, objectives (and their subordinate conditions or effects) define the path of the command's actions in achieving objectives.**

(3) Clear and concise objectives enable planners to better examine what the command needs to accomplish. They use military, diplomatic, economic, and informational terms and help define and clarify what planners do to support strategic guidance. Objectives developed at the national-strategic and theater-strategic levels are the defined, decisive, and attainable goals to direct all military OAI within the OA.

(4) There are four primary considerations for an objective:

(a) An objective establishes a single result.

(b) An objective (and its associated conditions/effects) should link directly or indirectly to higher-level objectives. Planners need to know the higher-level objective and should be able to identify how their objective supports the next higher-level objective.

(c) An objective is specific and unambiguous.

(d) An objective does not imply ways or means—it is not a task.

b. **Transitions**

(1) Planning transitions help JFCs visualize how to adapt to the OE relative to their on-going operations. JFCs develop and use transitions and relevant criteria for all types of joint planning - campaign, crisis, and contingency. Transitions apply to situations where JFCs contemplate changes in their forces' character of effort, posture, and composition to address an imminent or anticipated change to the OE. Aside from its obvious association with pursuing or maintaining objectives, anticipating transitions promotes unity of effort, facilitates synchronization, and helps clarify (and may reduce) the risk associated with the campaign or operation. Commanders account for and include transitions in their planning guidance.

(2) Transitions within CCPs

(a) JFCs are always campaigning, and the OE is perpetually changing. Therefore, JFCs continually assess their campaigns' effect on an OA and the OE's impact on their campaigns. As situations evolve, CCDRs diagnose the situation and modulate, revise, and change their OAI to address and confront these changes. Recognizing why and when to transition substantive portions of an AOR-wide campaign is a hallmark of maintaining the logic or validity of the approach. This recognition typically helps JFCs' planners to craft the specifics of transitions for what aspects of the force and its actions need to change.

(b) If an emerging situation exceeds the existing authorities and resources of one of the global campaigns, joint force senior leadership (Joint Chiefs of Staff and CCDRs) analyzes the problem and makes a diagnosis before recommending an appropriate response. The considerations include determining a range of possible outcomes and assessing the likely long-term consequences.

(3) Transitions for Crisis Planning

(a) For crisis planning, anticipating and recognizing the need (the why) to change because of the OE is an essential transition criterion and drives the specifics of options on how, what, and where. A crisis can develop rapidly and create a circumstance of such diplomatic, economic, or military importance that national leaders consider a commitment of military forces and resources. A crisis can occur anywhere across the competition continuum and the response can include almost any type of joint operation. For the United States, many responses to a crisis are cooperative efforts that seek to mitigate the consequences of natural disasters. The joint force's deployments and changes in posture are clear evidence of the US commitment, narrative, and strategic guidance. Successful response involves the ability to deploy the joint force rapidly with the required capabilities.

(b) In circumstances where the OE is rapidly deteriorating, armed conflict is imminent, or when adversaries threaten a nation's sovereignty or regional stability, US forces may intervene to bolster internal defense, conduct offensive or defensive operations, or signal a willingness and resolve to impose severe costs on the adversary. A prompt

response may preclude escalation of the situation. Effective early intervention can also deny an enemy time to set conditions in their favor and achieve their objectives.

(4) Transitions for Armed Conflict

(a) Planning the transitions for armed conflict can present significant challenges for the joint force. While JFCs do not assume they will receive abundant warning time and focus proactively on preparedness for armed conflict, they strive to recognize changes to the OE that warrant proper transitions. Defeating an enemy requires civilian leaders and commanders to transition the force optimized for global campaigns to a disposition for armed conflict. In addition to optimizing nuclear and conventional posture and readiness, JFCs plan and prepare for armed conflict by conducting operational preparation of the environment activities to develop knowledge of the OE; establish human, physical, or virtual infrastructure; and develop potential targets. Planning operational preparation of the environment includes active and passive observation, reconnaissance, and surveillance; area and network familiarization; site surveys; developing operational capability; prepositioning logistics; developing and using support networks; mapping the information environment; and mission rehearsals.

(b) Planning for transitions involving mobilization includes assembling and organizing resources to win a war. These mobilization transitions include alerting and deploying personnel and materiel for active duty military forces, activating the Reserve Component (including federalizing the National Guard), extending terms of service, surging and mobilizing the industrial base and training bases, and bringing the Armed Forces of the United States to a higher state of readiness. JFCs review the mobilization implications associated with their plans to identify, mobilize, and protect needed resources. Civilian and military leaders identify requirements for activating Reserve Component forces and applicable activation authorities, in a timely manner, as well as the need to expand the capability or capacity of other resource areas.

(5) Transitions Throughout Armed Conflict

(a) Joint warfighting is how all components of the joint force task organize together within an OA and unify in pursuit of a common military goal. The basis of joint warfighting is in the CCDRs' ability to integrate and, to the degree possible, synchronize mutually supporting and unified forces, campaigns, and operations with all the joint functions throughout all domains and multiple AORs. In these situations, JFCs adapt continually to evolving situations, opportunities presented by the enemy, lessons learned, and changes in available forces and capabilities.

(b) JFCs anticipate the many transitions necessary for armed conflict. A key component of joint warfighting is avoiding enemy strengths and surviving their attacks, while simultaneously creating and exploiting weaknesses and vulnerabilities through multiple cycles of offensives, counteroffensives, and transitions. Transitions during armed conflict primarily emphasize continuity above all but may include temporary defensive operations or operational pauses. Throughout the conduct of joint warfighting, JFCs

confront and endure surprise and failure while minimizing, where possible, inconclusive actions or periods of stalemate. JFCs adapt and transition throughout as armed conflict continues until one side begins to adapt and adjust to the OE more quickly, realizing increasing advantages. JFCs press and exploit these opportunities to maintain and expand these advantages. As JFCs' offensive operations begin defeating the enemy militarily through combinations of attrition, exhaustion, and annihilation, the United States can bring to bear other instruments of national power effectively.

(6) Transitions for New or Recharacterized Competitions

(a) Regardless of the reason for transition away from armed conflict, JFCs continue to orient on the enemy and anticipate new adversaries. JFCs plan for continuity of effort long after the cessation of hostilities to achieve and perpetuate the objectives. To successfully transition from armed conflict to the new competitions, JFCs avoid characterizing the continuing effort as merely a minimalist endeavor. Planning successful transitions requires a mindset, posture, and readiness to continue offensive operations – if necessary. As a function of the cessation of armed conflict with an enemy, the United States may impose terms, negotiate a settlement, or insist on regime change. Forcing an imposed settlement may occur through the selective destruction of critical functions or assets, such as C2 and infrastructure, or otherwise making the enemy unable to resist US will and resolve. Whether the United States imposes or negotiates a settlement, or the warring parties merely reach an armistice, the CCDRs' continuity of efforts have different characteristics depending on the situation.

(b) Even when the repercussions of armed conflict constitute a minor adjustment to the international order, success requires a great deal of effort, persistence, legitimacy, and restraint. In the case of regime change, only a comprehensive, long-term effort and committed follow-through can achieve strategic objectives. For each of these impositions, new competitions emerge and challenge all operational and strategic accomplishments; the joint force cannot simply walk away and expect any level of permanence. JFCs expect that the moment that the joint force ceases protecting the wartime gains, enemies and adversaries begin to probe, assault, and undo the hard-won achievements.

(7) After developing an understanding of the OE and the strategic guidance, JFCs and planners leverage the range of possible outcomes. Knowing the universe of possibilities helps JFCs identify specific considerations for transitions. The considerations enable JFCs in defining the long series of deliberate transitions leading to the point of an acceptable condition or a normalization of relations with the previous adversary. In most cases, this occurs years or even decades after the cessation of hostilities. The condition is when the United States formally recognizes the former enemy and re-establishes diplomatic relations. In the early stages of planning commanders and their staffs think through the conditions and behaviors that must exist to transition military-led operations on terms favorable to the United States and its partners. A hasty or ill-defined transition to the new or re-characterized competition may bring with it the possibility an enemy will renew hostilities, or third parties may interfere and potentially initiate new hostilities.

(8) Successful Transition Is Impossible Without Effective Preparation.

During planning, an integrated staff effort ensures the plan is a team effort and includes the knowledge gained across the staff. Integrated staff planning assists in identifying changes to the OE, developing guidance, and speeding the transition to execution. JFCs transition planning accounts for a wide variety of operational tasks the joint force needs to accomplish to ensure continuity. These tasks can include posture adjustments, increases in force protection, specific rehearsals, and appropriate changes to ROE. Throughout planning, JFCs present the possible transitions to SecDef as part of the IPR process to ensure the operational approach or CONOPS supports the strategic objectives. Once approved, however, the planned transitions may change. Any change could result in modifications to the operational-level objectives, as well as the commander's operational approach.

c. COG

(1) One of the most important tasks confronting the JFC's staff during planning is identifying and analyzing the COGs of friendly, enemy, and adversary forces. The COG is the source of power or strength that enables a military force to achieve its objective and is what an opposing force can orient its actions against that will lead to enemy failure. JFCs evaluate a COG on how effectively and efficiently targeting it will enable achieving the objective. Success requires protecting the friendly COG while defeating the enemy COG.

(a) COGs can exist at different levels. At the strategic level, a COG could be a military force, an alliance, political or military leaders, a set of critical capabilities or functions, or national will. At the operational and tactical level, a COG is often associated with the threat's military capabilities, such as a powerful element of the armed forces, but could include other capabilities in the OA.

(b) COGs may change over time as the OE changes due to shifts in friendly, neutral, or threat diplomatic, information, military (adversary or friendly), or economic/commercial conditions. A force's will to fight, for example, may increase or decrease throughout the course of an operation with dramatic impact on the joint force's success. Similarly, victories or defeats may cause adversaries or enemies to reassess objectives or strategies, which can alter the COG.

(2) COGs do not exist in a vacuum; they stem from the relationships between adversaries and enemies. JFCs define COGs through their view of the threats in the OA, as well as the requirements to develop and maintain power relative to their own objectives. Commanders, therefore, not only consider the adversaries' COGs but they also identify and protect their own.

(3) The COG construct is useful as an analytical tool to help commanders analyze friendly and adversary or enemy sources of strength as well as weaknesses and vulnerabilities. A faulty conclusion resulting from a poor or hasty analysis can have serious consequences, such as the inability to achieve objectives at an acceptable cost. The identification of COGs is not a static J-2 process conducted once during JIPOE. Planners continually analyze and

refine COGs due to actions taken by friendly forces and the threat's reactions to those actions. Figure III-8 shows several characteristics that may be associated with a COG.

(4) COG analysis requires knowledge of a threat's physical and psychological strengths and weaknesses and how the threat organizes, behaves, fights, and makes decisions. The determination and analysis of COGs (friendly and threat), as well as understanding the relationships between them, is a key step in operational design. Joint force intelligence analysts determine potential options for the threat COG by identifying what to weaken or destroy to defeat the enemy. The J-2, in conjunction with other operational planners, then determines if the proposed COGs are critical to the adversary's strategy or approach. This analysis is a linchpin in the planning effort. Planners in all sections on the joint force staff conduct similar analysis to identify friendly COGs and their strengths and weaknesses. JFCs and their staffs determine how to attack enemy COGs while protecting friendly COGs. Understanding the relationship among COGs facilitates greater precision in thought and expression in operational design.

(5) A CCP may have multiple COGs if it includes operations along multiple, independent LOEs. Without a well-defined threat, there is often no COG, as COGs are established by an adversarial or competitive relationship.

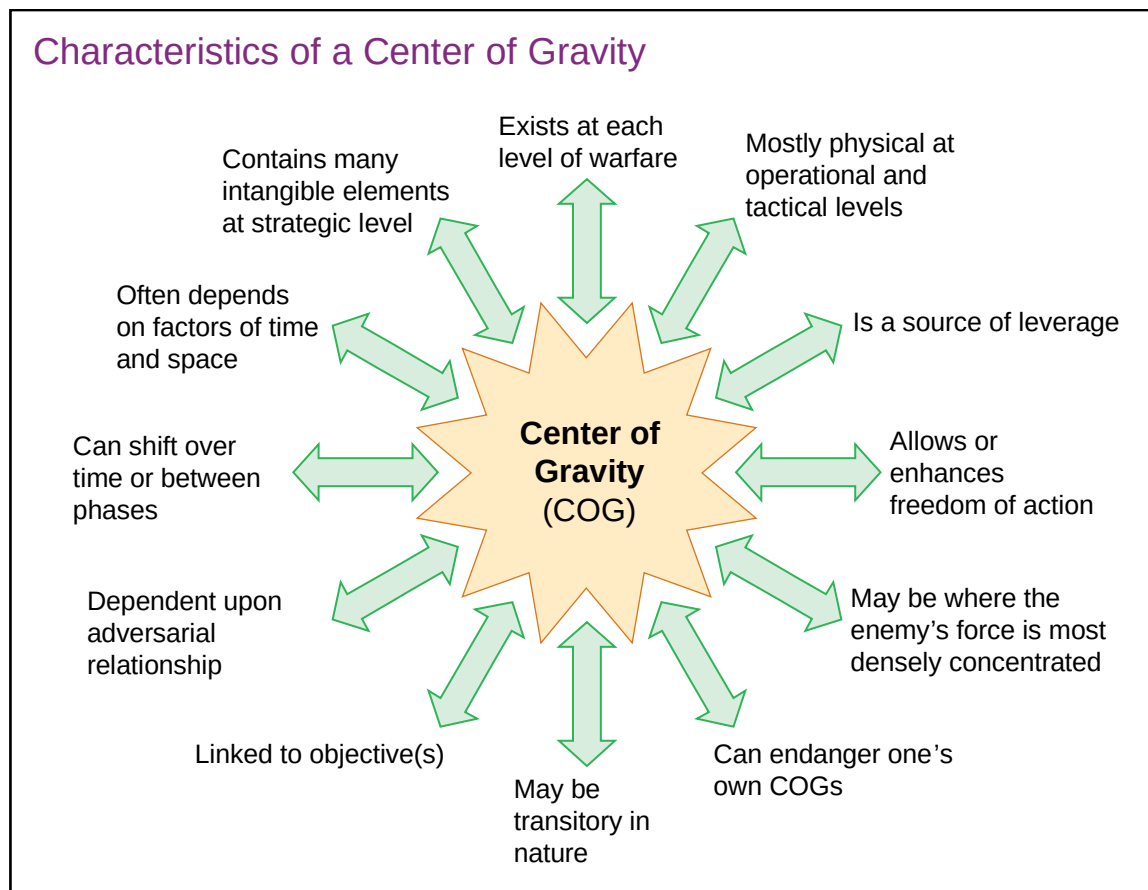


Figure III-8. Characteristics of a Center of Gravity

**ATTACKING ENEMY CENTER OF GRAVITY AT THE
BATTLE OF MIDWAY**

“To Nimitz it was clear that in Yamamoto’s disposition of force, the crucial elements were Nagumo’s four veteran carriers. Only they could provide the punch needed to knock out Midway’s ground and air defenses, and only they could provide a concentration of air power great enough to cover the other components of his fleet. They alone were essential to the attack. Therefore, Nimitz and his staff laid plans to eliminate the Nagumo carriers.”

—E.B. Potter

(6) Identifying a COG

(a) Critical factors analysis (CFA) is a framework to assist in analyzing and identifying a COG and to aid operational planning against the threat within the OA. During CFA, planners evaluate the operational design elements and identify those considered crucial for mission accomplishment. This analysis identifies those characteristics of a threat that present challenges to friendly forces and provide capabilities to that threat. CFA can be particularly useful for JFCs’ analysis of networks, including insurgent, resistance, terrorist, and criminal organizations that operate in the human aspects of the OE. Also, JFCs can apply CFA to friendly or neutral groups for vulnerability or gap analyses.

(b) A proper analysis of threat critical factors is based on the best available knowledge of how they organize, fight, think, and make decisions, along with their strengths and weaknesses. JFCs and their staffs develop an understanding of the threat’s capabilities and vulnerabilities, as well as the factors that would compel them to abandon their own objectives. They also envision how friendly forces and actions appear from the threat’s viewpoint. For more information on the CFA process, see JP 3-25, *Joint Countering Threat Networks*.

(7) Attacking a Threat’s COG. Once planners have identified the likely threat COG, they identify the best method to attack or weaken it. This process forms the core of COA development and assists with the identification of missions and tasks.

(a) Identify the missions and tasks by analyzing and deconstructing the COG within a framework of three components: critical capabilities, critical requirements, and critical vulnerabilities. In general, JFCs leverage their operational reach and combat power to take advantage of an enemy’s critical vulnerabilities while protecting friendly critical capabilities within the operational reach of an enemy. The JFC should seek opportunities to aggressively apply force against critical vulnerabilities

1. Critical capabilities are the primary ways essential to the achievement of their objective(s). JFCs express critical capabilities as verbs within an illustrative CONOPS or descriptive action.

2. Critical requirements are essential conditions, resources, and means the COG requires to employ the critical capability. JFCs express critical requirements as nouns that are essential to providing the action of the critical capability.

3. Critical vulnerabilities are those aspects of critical requirements vulnerable to an attack that significantly degrades the critical requirement. JFCs seek to create effects on the critical requirements by attacking these vulnerabilities.

(b) When identifying friendly and threat critical vulnerabilities, planners often want to focus their efforts against the critical vulnerabilities that will do the most decisive damage to a threat's COG. However, in selecting those critical vulnerabilities, planners also compare their criticality with their accessibility, redundancy, resiliency, and impact to balance those factors against friendly capabilities that can affect the critical vulnerabilities.

(c) Before solidifying COGs into the plan, planners should analyze and test the validity of the COGs. The defeat, destruction, neutralization, or substantial weakening of a valid COG should cause an adversary or enemy to change their approach or prevent them from achieving their objectives. If analysis, red teaming, or wargaming proves planners misidentified the COG, the CFA requires reexamination. The conclusions, while important to the planning process, must pair with continuous evaluation because COGs and critical vulnerabilities may change during the campaign due to the interactive nature of warfare and changes in the objectives of either combatant. JFCs and staffs should be prepared to adjust friendly plans and operations should the COGs change due to shifts in the threat or friendly objectives.

(8) On completion of the COG analysis, planners should consider:

(a) Will the joint force achieve its objectives after destroying the threat COG?

(b) Does accomplishment of this mission lead to the achievement of the objective? Does it support the next higher-level objective?

(c) If the joint force defeats the COG, what gaps, weaknesses, vulnerabilities, or vacuums does it create that may result in unforeseen consequences (second- and third-order effects)?

(d) Is a direct attack on the COG feasible or desirable? Identifying COGs is useful in understanding the system. Planners, however, should refrain from automatically assuming a strike on the COG is the solution to every operation.

1. The COG may be too difficult to attack or influence due to insufficient forces, complexity, or enemy or adversary defenses. In this case, an indirect approach may be more feasible than a direct attack.

2. Consider whether total collapse of the enemy or system supports the objectives and transitions. Striking a COG could lead to escalation or fracturing of the system, which might leave the commander and planning staffs with multiple unforeseen consequences that increase the mission's complexity and risk. Even if the commander and planning staffs identify a COG critical to an enemy, it may not be advantageous to strike it if the commander wants to avoid second- and third-order effects or the overall objective is to ensure stability within the system.

3. Planners may recommend affecting smaller elements of the whole enabling continued balance until reducing the entire problem to manageable parts or the COG changes. If using this approach, planners should remember that as the system changes, the COG may change in relation to the remaining whole.

(9) COG analysis may require operations to strengthen or protect the friendly COG, such as building interoperability with allies and partners.

For more information on COGs and the systems perspective, see the Joint Guide for Joint Intelligence Preparation of the Operational Environment.

d. **Effects.** An effect is a physical or behavioral state of a system that results from an action, a set of actions, or another effect. Desired effects are not tasks but are a result or condition that supports achieving an objective. JFCs assign tasks to subordinates that create desired effects. An undesired effect is a condition that can inhibit progress toward an objective.

(1) The CCDR plans joint operations based on analysis of senior-level guidance and development of objectives supported by measurable effects and assessment indicators. At the operational level, a subordinate JFC develops supporting plans that can include objectives supported by measurable operational-level desired effects and assessment indicators. This may increase operational- and tactical-level understanding of the purpose reflected in the higher-level commander's mission and intent. At the same time, commanders consider potential undesired effects and their impact on the tasks assigned to subordinate commands.

(2) There are five primary considerations for writing a desired effect statement.

(a) Each desired effect should link directly to one or more objectives.

(b) The effect should be measurable. However, cognitive effects are difficult to measure, and planners identify indicators that can enable assessment of these effects.

(c) The statement should not specify ways and means for accomplishment.

(d) The effect should be distinguishable from the objective it supports as a condition for success, and not appear as another objective or a task.

(e) Each JFCs' effect statement includes the observed system (who) and the intended behavior (what). Planners may also include the location of the desired behavior. Taken together, this equates to the effect.

(3) The proximate cause of effects in complex situations can be difficult to predict particularly when these effects relate to influencing the decision making of relevant actors. Further, there are always gaps in our understanding of the OE. Commanders and their staffs appreciate that unpredictable third-party actions, unintended consequences of friendly operations, subordinate initiative and creativity, and the fog and friction of conflict contribute to uncertainty of the OE. While gaps in understanding may be unavoidable, often the best strategy to mitigate them involves working with partners. Partners, particularly local relevant actors, can provide additional information and perspective that can help mitigate surprise from hard-to-predict effects or avoid unintended consequences.

(4) The use of effects in planning can help commanders determine the tasks required to achieve objectives. Effectiveness requires a logical association between objectives, targets, tasks, and effects. When used in this manner, there is a logical hierarchy within this aspect of operational design. Additionally, they facilitate the use of other elements of operational design by clarifying the relationships. Effects also enable a more intentional linking with higher-level objectives' required effects. After developing a systems perspective of the OE (with appropriate links and nodes identified), the linkage and relationship between COGs, LOOs, and decisive points is more obvious. This linkage enables efficient use of desired effects in planning. The JFC and planners continue to develop and refine desired effects throughout the JPP. Monitoring progress toward creating desired effects and avoiding undesired effects continues throughout execution. Identifying effects in a plan enables the development of a data collection method to facilitate assessment of whether operations are progressing as described in the plan.

(5) When JFCs determine tasks, they analyze their associated effects. Tasks initiate friendly actions, which create desired effects within the OA. The desired effects describe results or conditions to achieve objectives. JFCs link each of these elements of operational design together.

e. Culmination

(1) Culmination is that point when the operation can no longer maintain momentum.

(a) In the offense, the culminating point is when continuing the attack is no longer effective or possible and the force should revert to a defensive posture or attempt an operational pause. The attacker risks counterattack and defeat and continues attacking at great peril. Success in the attack is obtaining the objective before reaching culmination.

(b) In the defense, a defender reaches culmination when the defending force no longer has the capability and capacity to go on the counteroffensive or defend

successfully. Success in the defense is to draw the attacker to their offensive culmination, enabling the joint force to conduct a counteroffensive to exploit the enemy's culmination.

(2) The JFC ensures forces arrive at the appropriate times and places to support the campaign and that sufficient resources are available in the later stages of the campaign. Integration and synchronization of sustainment with combat operations can forestall culmination and help commanders control the tempo of their operations. At both tactical and operational levels, theater logistic planners forecast the drain on resources associated with conducting operations over extended distance and time. They respond by generating enough military resources at the right times and places to enable their commanders to achieve objectives before reaching their culminating points. If commanders cannot generate these resources, they should revise their CONOPS.

5. Organizing the Operation

Planners need to identify the method of organizing the plan and associated operation within a logical framework to achieve the objective. The two primary methods are LOOs and LOEs.

a. LOOs

(1) A LOO is geographic in nature and defines the interior or exterior orientation of the force in relation to the adversary. **LOOs organize, describe, and connect a series of decisive actions that lead to control of a geographic or force-oriented objective.** The objective associated with a LOO is physical terrain, and often defined as control of an area. Operations designed using LOOs generally consist of a series of actions executed according to a well-defined sequence, although multiple LOOs can exist at the same time (parallel operations). JFCs typically plan combat operations using LOOs. These doctrinal concepts relate offensive and defensive tasks and their transitions to the geographic and positional references in the OA. Commanders synchronize activities along complementary LOOs to achieve objectives.

(2) A force operates on **interior lines** when its operations diverge from a central point. Interior lines usually represent central position where a friendly force can reinforce or concentrate its elements faster than the enemy force can reposition. With interior lines, friendly forces are closer to separate enemy forces than the enemy forces are to one another. Interior lines enable an isolated force to mass combat power against a specific portion of an enemy force by shifting capabilities more rapidly than the enemy can react.

(3) A force operates on **exterior lines** when its operations converge on the enemy. Operations on exterior lines offer opportunities to encircle and annihilate an enemy force. However, these operations typically require a force stronger or more mobile than the enemy.

(4) The relevance of interior and exterior lines depends on the time and distance relationship between the opposing forces. Although an enemy force may have interior lines with respect to the friendly force, this advantage disappears if the friendly force is

more agile and operates at a higher tempo. Conversely, if a smaller friendly force maneuvers to a position between larger but less agile enemy forces, the friendly force may be able to defeat them in detail before they can react.

b. LOEs

(1) An LOE links multiple decisive points and tasks using the logic of purpose—cause and effect—to focus efforts. LOEs provide utility to operational design when positional references to an adversary or enemy have little relevance, such as in counterinsurgency operations. Similarly, mission areas link and focus multiple operational-level objectives and corresponding LOEs that lead to achieving strategic objectives. In operations involving many nonmilitary factors, they may be the only way to link tasks, effects, conditions, and the desired objectives (see Figure III-9). LOEs and mission areas are often essential to helping commanders visualize how military capabilities can support the other instruments of national power. LOEs are a valuable tool when used to achieve unity of effort in operations involving MNFs and civilian organizations, where unity of command is elusive.

(2) Commanders at all levels may use LOEs to develop tasks and to determine force capability requirements. Commanders synchronize and sequence related actions along multiple LOEs. Seeing these relationships helps commanders assess progress toward achieving the objective as forces accomplish missions.

(3) Commanders typically visualize enduring activities along mutually supporting LOEs. These enduring efforts tasks link military actions with the broader interagency efforts. A full array of LOEs might include offensive and defensive lines, a line for public affairs and other information activities, a line for counter-threat finance, and a line of for vendor threat mitigation, where each typically produces effects across multiple LOEs.

For more information, see Department of Defense Directive (DoDD) 3000.16, Vendor Threat Mitigation.

(4) Planners should focus LOEs for military plans on what the military does even though many LOEs require more than one instrument of national power to achieve the desired objective. LOEs should include awareness of, and support for, other instruments of national power when relevant, especially when those instruments are more likely to attain the strategic ends the military is supporting. Cross-cutting LOEs, such as establishing essential services or civil security, create a tendency toward more dynamic and open interagency coordination during execution because they require the synchronization of efforts of multiple USG departments and agencies. This type of construct brings to bear the capabilities and expertise of multiple elements of the USG, which makes it particularly effective toward achieving more complex objectives. Planners should remain aware that other departments and agencies lack the military's capacity and, therefore, need to actively seek participation from other organizations on overarching

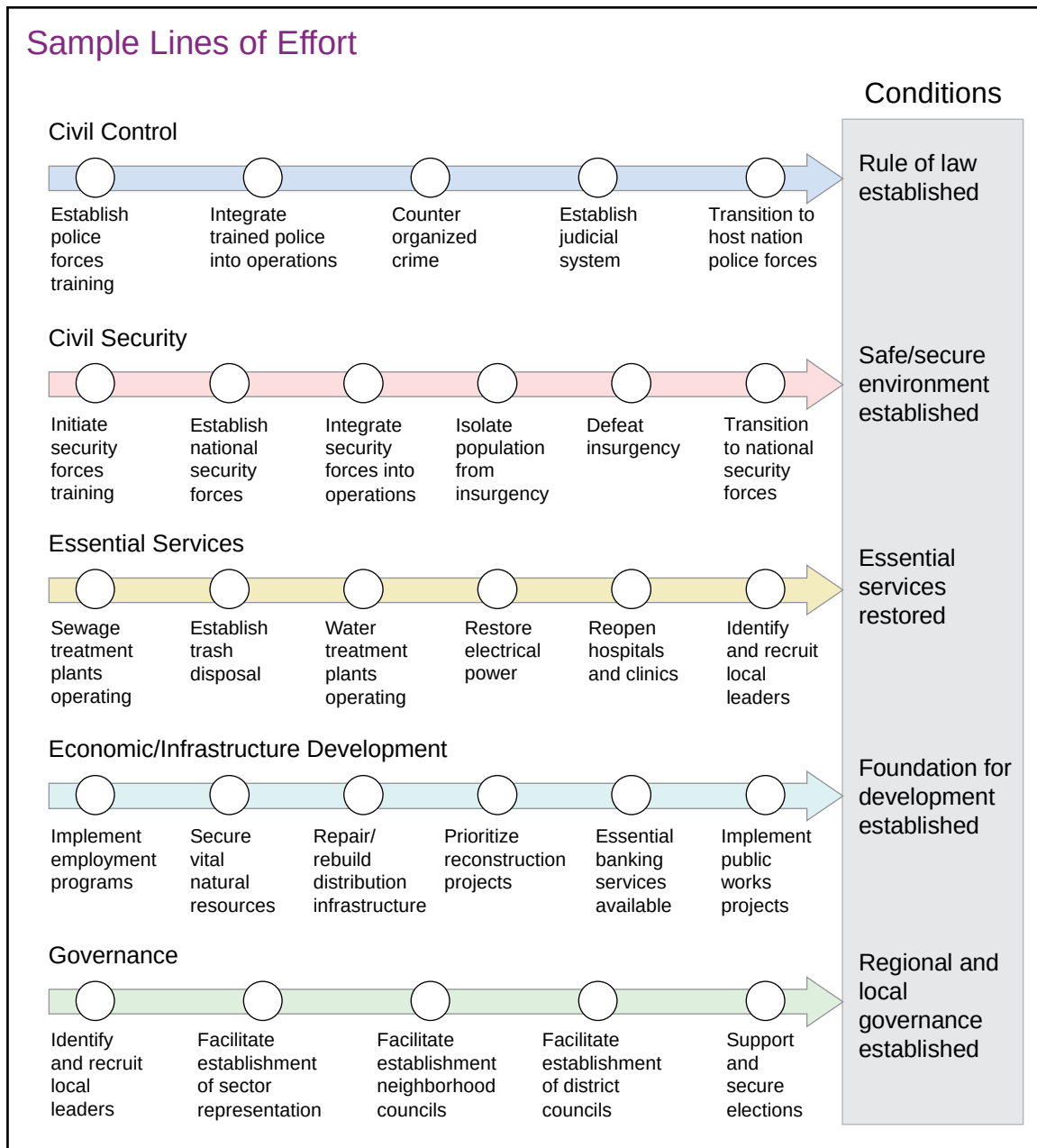


Figure III-9. Sample Lines of Effort

issues and critical specifics at the right time. Overwhelming partners with opportunities to interact will reduce partner input and buy in.

c. **Combining LOOs and LOEs.** Commanders may use both LOOs and LOEs to connect objectives to a central, unifying purpose. LOEs can also link objectives, decisive points, and COGs. Combining LOOs and LOEs helps commanders include nontraditional military activities and expected contributions from nonmilitary entities in the operational design. This combination helps commanders incorporate enduring tasks necessary to achieve the objective and enables commanders to consider the less tangible aspects of the OAs, where the other instruments of national power or nontraditional military activities

may be more effective. Nontraditional military activities occur in stability operations and conventional warfare—demonstrated by the military’s control of towns in World War II by graduates of the School of Military Government. Making these connections relates the tasks, effects, and objectives identified in the operation or campaign plan.

d. **Decisive Points**

(1) **A decisive point is key terrain, a key event, a critical factor, or a function that, when acted upon, enables a commander to gain a marked advantage over an enemy** or contributes materially to achieving success (e.g., creating a desired effect, achieving an objective). Decisive points can greatly influence the outcome of an action. Decisive points can be terrain features such as a constricted sea lane, a hill, or a geosynchronous orbit. Decisive points can also be specific things like a weapons of mass destruction material cache or facility, or an air base, but they could also include other elements, such as command posts, a satellite downlink station, or an undersea cable. In some cases, specific key events may also be decisive points, such as achievement of air, space, or maritime superiority; opening a supply route during humanitarian operations; or gaining the trust of a key leader. In other cases, decisive points may have a larger systemic impact and, when acted on, can substantially affect the threat’s information, financial, economic, or social systems. When dealing with an irregular threat, commanders and their staffs should consider how actions against decisive points affect not only the threat but also the relevant population’s perception of threat and friendly forces. Collateral effects on the local populace may impact stability in the area or region of interest.

(2) The most important decisive points stem from analysis of critical factors. Understanding the relationship between a COG’s critical capabilities, requirements, and vulnerabilities can illuminate direct and indirect approaches to the COG. It is likely most of these critical factors are decisive points to address further in the planning process.

(3) There may often be cases where the JFC’s combat power and other capabilities are insufficient to affect the enemy’s or adversary’s COGs rapidly with a single action. In this situation, the supported JFC selectively focuses a series of actions against the enemy’s or adversary’s critical vulnerabilities until the cumulative effects of these actions lead to mission success. Just as a combined arms approach is often the best way to attack an enemy’s field force in the military system, attacking several vulnerable points in other systems may degrade an enemy or adversary COG. The indirect approach may offer the most effective method to exploit enemy and adversary critical vulnerabilities through the identification of decisive points. **Although decisive points usually are not COGs, they are the keys to attacking or protecting them.**

e. **Direct and Indirect Approach.** The approach is the way a commander contends with a COG. **A direct approach attacks the enemy’s COG or principal strength by applying combat power directly against it.** However, COGs are generally well-protected and not vulnerable to a direct approach. Thus, commanders usually choose an indirect approach. **An indirect approach attacks the enemy’s COG by applying combat power**

against critical vulnerabilities that lead to the defeat of the COG while avoiding enemy strength.

(1) Direct attacks against adversary or enemy COGs resulting in their neutralization or destruction provide the most immediate path to victory. Since direct attacks against COGs mean attacking an opponent's strength, JFCs determine if friendly forces possess the power to attack with acceptable risk. Commanders normally attack COGs directly when they have superior forces, a qualitative advantage in leadership, or technological superiority over enemy weapon systems. Instead of attacking enemy strength, the goal is the application of our strength against selected enemy weakness to maximize advantage. In the event a direct attack is not a reasonable solution, JFCs should consider an indirect approach until conditions permit successful direct attacks (see Figure III-10). Whenever applicable, JFCs should consider developing simultaneous or synchronized action with both direct and indirect approaches. In this manner, the enemy or adversary's derived vulnerabilities can offer indirect pathways to gain leverage over its COGs.

(2) At the strategic level, indirect methods of defeating the enemy or adversary's COG could include depriving them of allies or friends, emplacing sanctions, weakening the national will to fight by undermining the public support, and breaking up cohesion of the threat alliances or coalitions.

(3) At the operational level, the most common indirect method of defeating an enemy's COG is to conduct a series of attacks against selected aspects of the enemy's combat power. For example, the JFC may sequence combat actions to force an enemy to

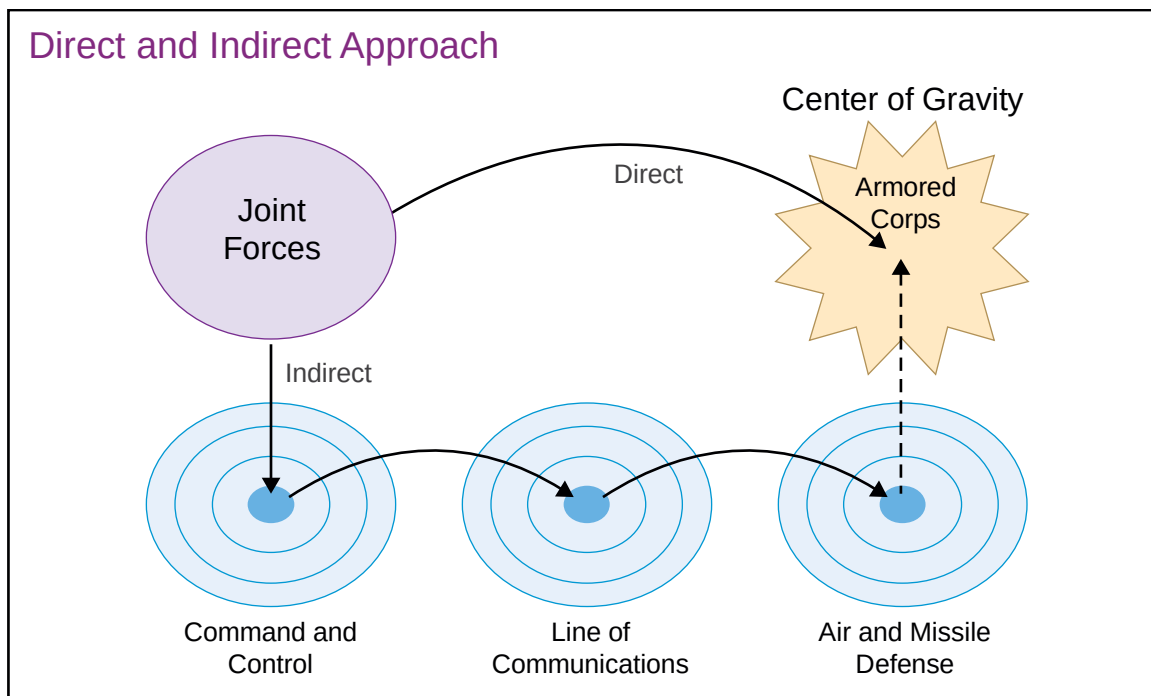


Figure III-10. Direct and Indirect Approach

divide its forces in theater, destroy the enemy's reserves or elements of the enemy's base of operations, or prevent or hinder the deployment of the enemy's major forces or reinforcements into the OA. Indirect methods of attacking the enemy's COG (through critical vulnerabilities) could entail reducing the enemy's operational reach, isolating the force from its C2, and destroying or suppressing key protection functions such as air defense. Additionally, in irregular warfare, a persistent indirect approach helps enable a legitimate and capable local partner to address the conflict's causes and to provide security, good governance, and economic development.

f. Operational Reach

(1) Operational reach is the distance and duration across which a joint force can successfully employ military capabilities. The geography, threats, and environmental conditions in and around the OA can constrain reach. Reach may extend through forward positioning of capabilities and resources, using information activities, increasing the range and effectiveness of weapon systems, leveraging HN support and contracted support (e.g., system support, external support, theater support), and maximizing the throughput efficiency of the distribution architecture. Joint force messages and images may reach outside of the OA to unintended audiences creating effects that are contrary to the JFC's objectives. Properly synchronized communication and proper execution of operations security procedures can maximize the impact and operational reach of information while mitigating the risk of this type of adverse consequence.

(2) Operational reach ties to the construct of LOOs. The geography surrounding and separating our threats influences operational reach. Locating forces, reserves, bases, pre-positioned equipment sets, and logistics forward extends operational reach. Operational reach is a factor of the increasing range of weapons; transportation availability; throughput of lines of communications; and wide use of multiple media sources, including social media and messaging applications. Given the appropriate level of superiority, some forces, such as air, space, and cyberspace, maintain a responsive global capability that extends operational reach. For any given joint campaign or operation, there is a finite range beyond which predominant elements of the joint force cannot operate.

(3) Basing, in the broadest sense, is an indispensable part of operational art tied to the construct of LOOs and operational reach. Basing directly affects the combat power and other capabilities a joint force can generate. In particular, the arrangement and positioning of advanced bases (often in austere, rapidly emplaced configurations) underwrites the ability of the joint force to shield its components from enemy and adversary action and deliver symmetric and asymmetric attacks. It also directly influences the combat power and other capabilities the joint force can generate because of its impact on critical factors such as sortie or resupply rates. Political and diplomatic considerations often affect basing decisions.

(4) US force basing options span the range from permanently based forces to temporary sea-basing. Bases enable the operational reach to affect enemies and adversaries. Analysis during planning determines if diplomatic support exists and whether

to obtain or use existing sufficient infrastructure, including ports and airfields, to support the operational and sustainment requirements of deployed forces. Support should afford some degree of security from attack. Determining where to locate infrastructure and bases poses critical challenges for planners since infrastructure and basing play a key role in enabling campaigns and operations. Enemies and adversaries will likely try to develop anti-access or area denial capabilities to prevent the buildup and sustainment of forces. One such approach could be a preemptive attack against US forces located outside the enemy or adversary's boundaries, so planners must also consider the risk of placing US combat capabilities within the enemy or adversary's operational reach. Planners must determine how to mitigate an enemy or adversary's efforts to deny access to the theater and its infrastructure and conduct operations as part of the campaign to set conditions for future operations.

See JP 3-0, Joint Campaigns and Operations, for additional considerations in organizing the OA for joint operations.

6. Time

a. Arranging Operations

(1) Commanders determine the best arrangement of joint force and component operations to conduct the assigned tasks and joint force mission. This arrangement is often a combination of simultaneous and sequential operations to attain the strategic conditions with the least cost in personnel and other resources. Commanders consider a variety of factors when determining this arrangement, such as geography of the OA, available strategic lift, changes in command structure, force protection, distribution and sustainment capabilities, enemy reinforcement capabilities, and public opinion. Thinking about the best arrangement helps determine the tempo of activities in time, space, and purpose. Planners should consider factors such as simultaneity, depth, timing, and tempo when arranging operations. Phases, branches and sequels, operational pauses, and the development of a notional TPFDD all improve the ability of the planner to arrange, manage, and execute complex operations.

(a) Simultaneity refers to the simultaneous application of integrated military and nonmilitary power against an enemy's or adversary's key capabilities and sources of strength. Simultaneity in joint force operations contributes directly to an enemy or adversary's collapse by placing more demands on their forces and functions than they can handle. This does not mean all elements of the joint force are employed or are employed with equal priority.

(b) Simultaneity also refers to the concurrent conduct of operations at the tactical, operational, and strategic levels. Tactical commanders fight engagements and battles understanding their relevance to the contingency plan. JFCs set the conditions for battles within joint operations or campaigns to achieve operational-level objectives. CDRs integrate theater strategy and operational art. At the same time, they remain acutely aware of the impact of tactical events. Because of the inherent interrelationships

between the various levels of warfare, commanders cannot be concerned only with events at their respective echelon. Commanders understand how their actions contribute to the objectives of all levels of warfare.

(c) The joint force should conduct operations at a tempo and time that maximizes friendly capabilities and inhibits enemies and adversaries. With proper timing, JFCs can dominate the action, remain unpredictable, and operate beyond the adversary's ability to react.

(d) The tempo of warfare increases because of technology, and innovative doctrines impact military operations. In many situations, JFCs may find it advantageous to maintain an operational tempo that stretches the capabilities of both friendly and enemy or adversary forces. On other occasions, JFCs may find it advantageous to conduct operations at a reduced pace. During selected phases of a campaign, JFCs could reduce the pace of operations, frustrating enemy or adversary commanders, while buying time to build a decisive force or tend to other priorities in the OA, such as relief to displaced persons. During other phases, JFCs could conduct high-tempo operations designed specifically to overwhelm enemy defensive capabilities. Assuring strategic mobility preserves the JFC's ability to control tempo by enabling freedom of theater access.

(2) Several tools are available to planners to assist with arranging operations. Phases, branches and sequels, and operational pauses all improve the ability of the planner to arrange, manage, and execute complex operations.

(a) **Phases.** Phasing is a way to organize a complex joint operation or campaign's CONOPS into manageable parts. The phases are unique to each operation or campaign as a tool to integrate and synchronize related activities, thereby enhancing C2 to improve flexibility and unity of effort during execution. Phasing should help the commander and staff understand the sequence of actions forces must execute to be successful. Achieving the objectives often requires JFCs to arrange an operation or campaign's CONOPS in several phases. Phases are sequential and do not overlap; however, there are often activities that continue and may overlap between phases. In a campaign, a phase can consist of one or more sequential or concurrent operations in varying scope, scale, and geographic location, while within an operation, a phase is normally defined by specific conditions and the completion of several subordinate tasks and related actions or activities.

1. Characterize a phase by the focus placed on it. Phases are distinct in time, terrain, or purpose and incorporate mutual support. They should represent a natural progression and subdivision of the campaign or operation. Each phase should have starting conditions and ending conditions. The ending conditions of one phase are the starting conditions for the next phase.

2. Link phases to gain significance in the larger context of the campaign. This requires that planners not break the campaign into numerous arbitrary components that may inhibit tempo and lead to a plodding, incremental approach.

3. Phases are distinct, but activities from one phase may continue into subsequent phases. The commander's vision of how a campaign or operation should unfold drives subsequent decisions regarding phasing. Phasing, in turn, assists with synchronizing the CONOPS and aids in organizing the assignment of tasks to subordinate commanders. By arranging operations and activities into phases, the JFC can better integrate capabilities and synchronize subordinate operations in time, space, and purpose. Each phase should orient the force on a specific objective. As such, a phase represents a definitive stage during which a large portion of the forces and joint/multinational capabilities are involved in similar or mutually supporting activities.

4. As a general rule, phasing should use condition-driven rather than time-driven terms. However, resource availability depends in large part on time-constrained activities and factors—such as sustainment or deployment rates—rather than the events associated with the operation. The challenge for planners is to reconcile the reality of time-oriented deployment of forces and sustainment with the condition-driven phasing of operations.

5. Phasing addresses how the joint force avoids reaching a culminating point. If resources are insufficient to sustain the force until achieving the objective, planners should consider phasing the campaign or operation to ensure continuity throughout and account for planned operational pauses and transitions. Such phasing enables the reconstitution of the joint force, but the JFC must understand this may provide the enemy an opportunity to reconstitute as well. In some cases, sustainment requirements and diplomatic factors may even dictate the purpose of certain phases, as well as the sequence of those phases. For example, phases may shift the main effort among Service and functional components to maintain momentum while one component is reconstituting.

6. Commanders determine the number and purpose of phases used during a campaign or operation. Within the context of these phases established by a higher-level JFC, subordinate JFCs and component commanders may establish additional phases that fit their CONOPS. For example, the joint force land component commander or a subordinate commander might have the following operations inside a single phase of a higher headquarters' operation: deploy, forcible entry, and defend the beachhead.

a. During planning, the JFC establishes conditions, objectives, or events for transitioning from one phase to another and plans sequels and branches for contingencies. Phases execute sequentially, but some activities from a phase may begin in a previous phase and continue into subsequent phases. For instance, the commander may transition to stabilization efforts in some areas while still conducting combat operations in other areas. The JFC adjusts the phases to exploit opportunities presented by the adversary, enemy, operational situation, or to react to unforeseen conditions. A joint campaign or operation may execute through multiple phases simultaneously if the OA has widely varying conditions.

b. Planners design transitions between phases as distinct shifts in focus by the joint force that may accompany changes in command or support relationships.

The activities that predominate during a given phase, however, rarely align with neatly definable breakpoints. Transitioning between phases is usually event or objective driven versus time or activity driven. The need to transition to another phase occurs when achieving objectives for the current phase or the adversary or enemy acts in a manner requiring a major change in focus for the joint force.

c. The phasing construct may not be practical for a CCP or GCP. At the theater level, campaigning is usually continuous and includes multiple mission areas and LOEs across the theater and the lifecycle of the CCP. CCPs are indefinite, without expectations for conclusion as circumstances that make for favorable conditions are always under pressure and changing. CCDRs continually prioritize and sequence OAI, anticipate changes in their AOR, identify necessary refinements through campaign assessments, and lead and drive adjustments to the operational objectives or to the campaign plan.

(b) **Branches and Sequels.** Many plans require adjustment beyond the initial stages of the operation. Consequently, JFCs build flexibility into plans by developing branches and sequels to preserve freedom of action in rapidly changing conditions. They include changing deployments or direction of movement and accepting or declining combat.

1. **Branches.** Branches represent expected contingencies that provide a range of alternatives often built into the basic plan. Branches add flexibility to plans by anticipating situations that could alter the basic plan.

2. **Sequels.** Sequels anticipate and plan for subsequent operations based on the possible outcomes of the current operation—victory, defeat, or stalemate.

3. Once the commander determines possible branches and sequels, they determine the what, when, and where of the decision points. Such decision points capture in space or time decisions a commander must make. To aid the commander, planners develop synchronization matrices, as well as a decision support matrix (DSM) to link those decision points with the earliest and latest timing of the decision, the appropriate priority intelligence requirement (PIR) (things the commander must know about the adversary, enemy, and the OE to make the decision), and friendly force information requirements (FFIRs) (things the commander must know about friendly forces to make the decision).

(c) **Operational Pause**

1. The supported JFC should aggressively conduct operations to obtain and maintain the initiative. However, there may be certain circumstances when this is not feasible due to logistic constraints or force shortfalls. Use operational pauses when a major operation is reaching the end of its sustainability. Operational pauses can provide protection and opportunity for sustainment and help avoid potential culmination of the force while the JFC retains the initiative in other ways. Execute an operational pause when the enemy or adversary will lack sufficient combat power to threaten the joint force or regain the initiative during the pause.

2. Operational pauses are useful tools for obtaining the proper synchronization of sustainment and operations. Normally, plan operational pauses to regenerate combat power or augment sustainment and forces for the next phase. When properly planned and sequenced, operational pauses ensure the JFC has sufficient forces to achieve the objectives. However, planners guard against cutting the margin of sustainment and combat effectiveness too thin. Executing a pause before it is necessary provides for flexibility in the timing of the pause and enables added agility under urgent conditions without unduly endangering the future effectiveness of the force.

3. Operational pauses can support strategic decisions such as opportunities for de-escalation or negotiation.

4. The primary drawback to operational pauses is the risk of forfeiting the initiative. It is therefore incumbent upon the JFC to plan for as few operational pauses as possible. If pauses are necessary, the JFC can alternate pauses among components to ensure continuous pressure on the enemy or adversary through offensive actions by some components while other components pause. Appropriately planned operational pauses provide opportunities for activities such as military deception.

b. Anticipation

(1) Anticipation is key to effective planning. JFCs consider what might happen and look for indicators of forecasted events. During execution, JFCs should remain alert for the unexpected and be prepared to exploit opportunities. JFCs continually gather information by personally observing and communicating with higher headquarters, subordinates, partner nations, and other organizations in the OA. JFCs may avoid surprise by gaining and maintaining the initiative at all levels of command, forcing the enemy or adversary to react rather than initiate, and by continuously wargaming to identify probable enemy or adversary reactions to joint force actions. Thorough wargaming assists JFCs in understanding and planning for the effects of operations, as well as the effects they have on the enemy, adversary, interagency and multinational partners, and civilian environment. JFCs should anticipate that impact on the civilian environment may be unavoidable and seek to mitigate undesirable effects.

(2) A shared, common understanding of the OE aids commanders and their staffs in anticipating opportunities and challenges. Knowledge of friendly capabilities; enemy and adversary capabilities, intentions, and likely COAs; and the location, activities, and status of dislocated civilians enables commanders to focus joint efforts where they can best, and most directly, contribute to achieving objectives.

(3) Anticipation is critical to the decision-making process. At times, reducing risk requires JFCs to make decisions well in advance. Decisions such as mobilization of Reserve Component forces and deploying or repositioning forces often require anticipation to ensure those capabilities are available when needed or when an opportunity arises.

(4) Anticipation is not without risk. If a commander plans for an anticipated action from the enemy or adversary, the commander could be susceptible to deception efforts or having forces out of position should opportunities or threats appear in other places. Therefore, commanders and their staffs should carefully consider all available information upon which decisions are based. Where possible, multiple or redundant sources of information should be employed to reduce risk in the decision-making process.

7. Forces and Functions

a. Forces

(1) Commanders and planners can plan campaigns and operations that focus on defeating either enemy forces, functions, or a combination of both. Typically, JFCs structure operations to attack both forces and functions concurrently to create the greatest possible impact on the enemy and chance for success. These types of operations are especially appropriate when friendly forces enjoy technological or numerical superiority.

(2) Commanders and planners need to know the technical capability of the enemy or adversary's forces as well as their own. During planning and wargaming, commanders and planners use this information to identify the best method to defeat an enemy or counter the adversary while minimizing risk to the joint force.

(3) Commanders should also use available resources to understand the intangible aspects of the threat, such as their doctrine, leadership, and morale. These factors contribute to the way a threat fights and supports the way the commander and planners should plan operations.

(4) JFCs can focus on destroying and disrupting critical enemy or adversary functions such as C2, sustainment, information, and protection. Attacking an enemy or adversary's functions normally intends to destroy their balance, thereby creating vulnerabilities for exploitation. The direct effect of destroying or disrupting critical enemy or adversary functions can create the indirect effects of uncertainty, confusion, and panic in their leadership and forces, which may contribute directly to the collapse of their capability and will. When determining whether functional attack should be the principal operational approach, JFCs should evaluate several variables within the context of anticipated events, such as time required to cripple the enemy or adversary's critical functions, time available to the JFC, the enemy or adversary's current actions, and likely responses to such actions.

b. Force Employment Mechanisms. Force employment mechanisms complement COG analysis. While COG analysis helps planners understand a problem, these mechanisms suggest ways to solve it. They provide a useful tool for describing how a JFC intends to achieve an operational objective and ensure understanding of the commander's intent by establishing common references for force employment.

(1) **Defeat Mechanisms.** Defeat mechanisms are the methods used in combat operations. Defeating an enemy is often temporary and centers on the enemy's ability and will to oppose or resist US aims. The three general philosophies or methods to prevail in armed conflict are exhaustion, attrition, and annihilation. Collectively, they guide or inform specific defeat mechanisms. Annihilation is the complete destruction or denial of the enemy's means. Attrition aims to gradually disrupt, degrade, or neutralize an enemy's armed forces or war-making capabilities by applying combat power with a cumulative operational or strategic impact, destroying the adversary's war-making capabilities over time. Exhaustion aims to impose unacceptable costs that erode the will of an enemy to continue fighting, even if that enemy is achieving tactical or operational military success. JFCs rarely employ one of these methods in isolation from the others. Collectively, all three reduce the effectiveness of an enemy through combinations of denial and loss of will, personnel, materiel, morale, or commitment. Planners should consider employing nonlethal force capabilities for both counter personnel and counter material applications. When waging war from a position of disadvantage, attrition may be the appropriate mechanism for gaining time and space to increase combat power and seize the initiative. In irregular warfare, the weaker force may choose to exhaust an enemy's will because it cannot win by annihilation or attrition of the enemy's armed forces or war-making capabilities. Specific defeat mechanisms include:

(a) **Destroy.** Denial by eliminating enemy forces and capabilities by applying combat power.

(b) **Dislocate.** Compel the enemy or adversary to expose forces by reacting to a specific action. This mechanism forces enemy or adversary commanders to either accept neutralization of part of their force or risk its destruction while repositioning.

(c) **Disintegrate.** Exploit the effects of dislocation and destruction to shatter the enemy's coherence. This mechanism typically follows destruction and dislocation, coupled with the loss of capabilities that enemy commanders use to develop and maintain situational understanding.

(d) **Isolate.** Limit the enemy or adversary's ability to conduct operations effectively by marginalizing critical capabilities or limiting the adversary's ability to influence events. This mechanism exposes the adversary to continued degradation through the massed effects of other defeat mechanisms.

(e) **Disrupt.** Denial by interrupting the effectiveness of an enemy or adversary's operations and activities without significantly degrading their ability to conduct future operations and activities. This mechanism is appropriate when policy, resource, risk, or other limitations prevent friendly forces from inflicting greater costs on an enemy or adversary.

(f) **Degrade.** Denial by reducing an enemy's will or ability to conduct future operations and activities. This mechanism imposes greater costs on the enemy than

disruption when policy, resource, risk or other limitations prevent friendly forces from defeating an enemy militarily.

(g) **Neutralize.** Render an enemy's ability to conduct operations or activities ineffective without necessarily destroying or degrading the enemy's capabilities. To achieve this, planners should consider employing nonlethal weapons as an intermediate force capability for both counter personnel and counter material applications.

(2) **Stabilization Mechanisms.** Stabilization is an inherently political endeavor requiring aligning USG efforts—diplomatic engagement, foreign assistance, and defense—to create conditions in which locally legitimate authorities and systems can peaceably manage conflict and prevent violence. To the extent authorized by law, DoD plans and conducts stabilization in support of mission partners to counter subversion, deter and mitigate conflict, and consolidate military gains. If directed, and consistent with available authorities, DoD leads USG stabilization efforts in extreme situations and less permissive environments until it is feasible to transition lead responsibility to other USG departments and agencies. Stabilization mechanisms may include enforce, control, influence, and support.

(a) **Enforce.** The threat or use of lethal or nonlethal force to establish dominance; affect behavioral change; enable USG or international stabilization efforts; or enforce cessation of hostilities, peace agreements, or other political arrangements. Legitimacy and compliance are interrelated. While legitimacy is vital to achieving and sustaining the compliance of local populations, compliance also depends on how the local populace perceives the joint or coalition force's ability to secure the OA and protect them from threats. The appropriate and discriminate use of force often forms a central component to success in stabilization activities; it closely ties to the perceived legitimacy of the joint force and supported local government. Depending on the circumstances, the threat or use of force can reinforce or complement efforts to stabilize a situation, gain consent, and ensure compliance with mandates and agreements. The misuse of force—or even the perceived threat of the misuse of force—can adversely affect the legitimacy of the mission or the joint force or MNF conducting the mission.

(b) **Control.** Establish public order and safety; secure borders, routes, sensitive sites, population centers, and individuals; physically occupy key terrain and facilities; and provide for the immediate needs of the population. DoD's core responsibility during stabilization is to support and reinforce the civilian efforts of the USG lead agencies consistent with available statutory authorities, primarily by providing forces in support of these missions. As a stabilization mechanism, control closely relates to the primary stabilization task: establish civil control. However, control is also fundamental to effective, enduring security. When combined with the stabilization mechanism compel, it is inherent to the activities that comprise disarmament, demobilization, and reintegration, as well as broader security sector reform programs. Without effective control, efforts to establish civil order—including efforts to establish both civil security and control over an area and its population—will not succeed. Establishing control requires time, patience, and coordinated, cooperative efforts across the OA.

(c) **Influence.** To alter the opinions and attitudes of targeted populations. DoD considers how US or partner military forces promulgate a coherent narrative consistent with USG objectives to counter adversaries and affirm effective and legitimate local governance. DoD uses civil-military teams to integrate key instruments of national power that complement local populations, international, allied, partner, civil society, and private entities to achieve stabilization objectives. Influence applies nonlethal capabilities to complement and reinforce enforcing and controlling effects. Influence also aims to effect behavioral change through nonlethal means. Results are more a product of public perception than a measure of operational success. Influence reflects the ability of forces to operate successfully among the people of the HN, interacting with them consistently and positively while accomplishing the mission in support of advancing integrated USG stabilization efforts. Here, consistency of actions, words, and deeds is vital. Influence requires legitimacy, cultural understanding, and assessment of conflict, to include intelligence collection and related activities to improve understanding of and ability to influence stability. Military forces earn the trust and confidence of the people through constructive activities. It contributes to success across the LOEs and engenders support among the people. Once achieved, maintain influence by consistently exhibiting respect for, and operating within, the cultural and societal norms of the local populace.

(d) **Support.** To establish, reinforce, or set the conditions necessary for the other instruments of national power to function. DOS leads the overall USG stabilization efforts. DoD provides support, including requisite security and reinforcing civilian efforts where appropriate and consistent with available statutory authorities. When required to achieve US objectives, and to the extent authorized by law and policy, DoD reinforces and complements civilian-led stabilization efforts, including providing logistical support, services, and other enabling capabilities to other USG departments and agencies. DoD solicits participation from mission-critical USG departments and agencies to plan, exercise, and wargame stabilization aspects of military plans, including transition from combat operations. As such, this mechanism requires coordinating and cooperating closely with HN civilian agencies and assisting aid organizations as necessary to secure humanitarian access to vulnerable populations. The joint force can quickly address the immediate needs of the HN and local populace by combining several stabilization activities in collaboration with the interagency partners such as establishing civil security, providing access to dispute resolution, delivering targeted basic services, and establishing a foundation for the return of displaced people and longer-term development. The joint force should avoid the improper use of support that has the potential to destabilize a situation by disrupting local power structures.

(e) **Competition Mechanisms.** When military forces are employed in operations within CCPs in either supporting or supported roles, planners should identify competition mechanisms for cooperation and competition below the level of armed conflict. These mechanisms are ways to apply or counter coercive measures and maintain or establish favorable conditions.

For more information on competition, see JP 1, Volume 1, Joint Warfighting, and JP 3-0, Joint Campaigns and Operations.

8. Balancing

a. Balancing Across and Tradeoff Between Factors

(1) The JFC seeks to understand the operational factors and their inter-relationships within the command. Commanders rarely have all the resources or time desired for an operation. By understanding the relationship between the elements of operational design, commanders and planners can balance different factors to maximize the likelihood of success in the most efficient manner.

(2) The operational commander decides which tradeoffs produce the best balance. For example, commanders may decide to use an indirect approach and several phases, due to insufficient forces, or a direct assault on a COG. Similarly, a commander may plan for an operational pause to use additional time to mobilize, deploy, or reconstitute forces.

(3) Commanders cannot regain lost time, which may lead them to commit forces before they are completely ready, to use forces which do not fit well, or to trade terrain for the time needed to deploy forces. This enables commanders to take advantage of opportunities as they appear or prevent an enemy or adversary from gaining the initiative.

(4) Alternatively, a commander may alter operational objectives to fit the balance of operational factors they can affect, ensuring the OAI continue to support the objectives.

b. **Balancing Risk.** During planning, commanders assess risk to ensure the plan has an acceptable level of risk across all elements of design regarding risk-to-mission and risk-to-force.

See Chapter I, “Joint Planning,” and CJCSM 3105.01, Joint Risk Analysis Methodology, for more information on assessing and presenting risk analysis as part of planning.

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CHAPTER IV JOINT PLANNING PROCESS

SECTION A. OVERVIEW OF PLANNING AND THE PLANNING FUNCTIONS

1. Overview

a. Although planning has an input (guidance) and an output (the plan or order), the planning process is a recursive, assessment-informed process and not linear. Issues discovered in later steps of the planning process can require repeating earlier steps. For example, COA wargaming may identify additional assumptions that could affect viability of options or COAs within an option.

b. The commander leads planning, using staff analysis and expertise to inform decisions.

c. Planning consists of four functions, the planning process, and an operational design methodology as depicted in Figure IV-1.

d. Operational design and the JPP are complementary tools of the overall planning process. Operational design provides an iterative process that enables the commander's

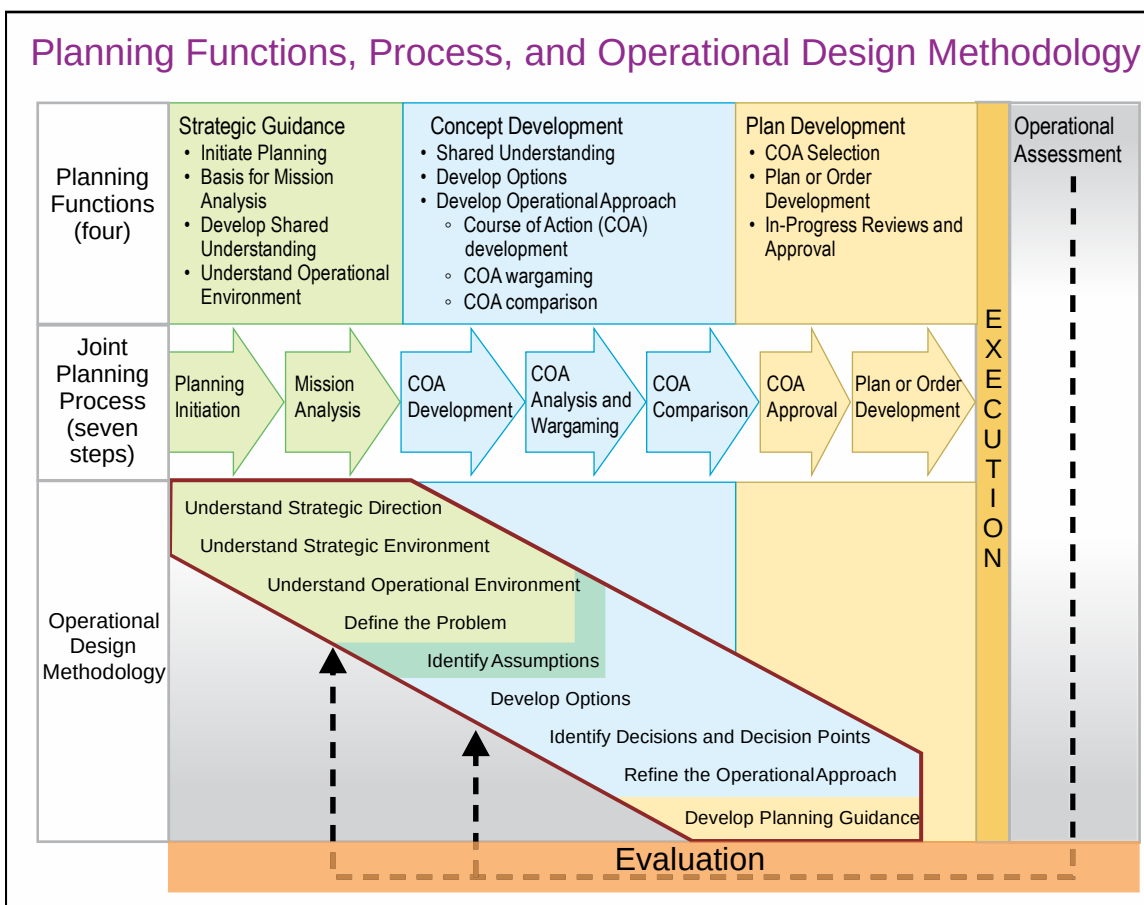


Figure IV-1. Planning Functions, Process, and Operational Design Methodology

vision and mastery of operational art to help planners answer ends—ways—means—risk questions and appropriately structure campaigns and operations in a dynamic OE. The commander, supported by the staff, gains an understanding of the OE, defines the problem, and develops an operational approach for the campaign or operation through the application of operational design during the initiation step of the JPP. Commanders communicate their operational approach to their staff, subordinates, supporting commands, agencies, and multinational/nongovernmental entities to translate their approach into executable plans. In applying the JPP, commanders may receive updated guidance, learn more about the OE and the problem, and refine their operational approach. Commanders provide their updated approach to the staff to guide detailed planning. This iterative process facilitates the continuing development and refinement of possible COAs into a selected COA with an associated initial CONOPS and eventually into a resource-informed executable plan or order.

e. The relationship between the application of operational art, operational design, and the JPP continues throughout the planning and execution of the plan or order. By applying the operational design methodology, in combination with the procedural rigor of the JPP, the command can monitor the dynamics of the mission and OE, while executing operations in accordance with the current approach and revising plans as needed. By combining these approaches, the friendly force can maintain the greatest possible flexibility and do so proactively (see Figure IV-2).

2. Planning Teams

a. As part of the plan initiation, planners identify expertise needed from within and outside their headquarters. Planners in a headquarters' plans branch or division rarely have the full expertise, manpower, or time necessary to conduct planning by themselves. Planners should establish a team of experts to support the planning process. These are often called joint planning groups, operational planning teams, operational planning groups, or cross-functional teams and are normally led by either the plans directorate of a joint staff (J-5) or the J-3.

b. Membership should include:

(1) Members of adjacent, supporting, or collaborating commands. At the CCMD level, this usually includes functional commands to address their specific capabilities; CCMDs that might provide staging, logistics, or transload locations; combat support agencies with significant support or coordination roles, such as the Defense Logistics Agency or NGB; and, if appropriate and authorized, ally and partner planners.

(2) **Staff Expertise.** Regular members include the J-2 to provide intelligence analysis, updates, and planning; the logistics directorate of a joint staff for sustainment planning; the communications system directorate of a joint staff for communications planning; information planners including public affairs; and the staff surgeon for medical planning.

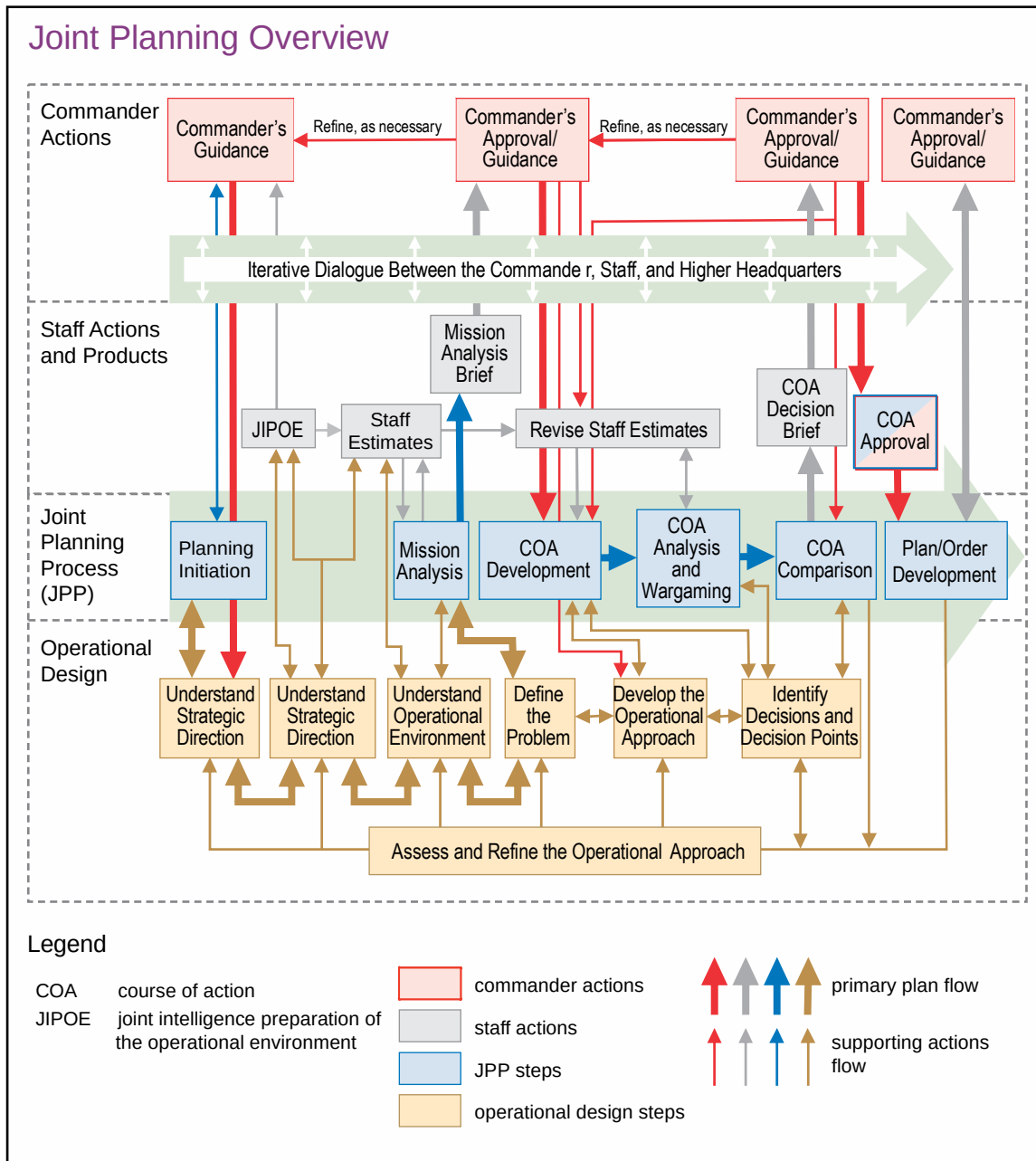


Figure IV-2. Joint Planning Overview

(3) **Service Component and Component Command Representation.** The CCMD Service representatives and component commands are essential in developing supporting plans, as well as identifying required capabilities and probable availability.

3. Planning Functions

The four planning functions of strategic guidance analysis, concept development, plan development, and plan evaluation are generally sequential, although often strategic guidance develops throughout the dialogue between civilian and military leaders, which

enables the overall planning process. The following subparagraphs should explain how the four planning functions run simultaneously to deepen dialogues and accelerate the planning processes. SecDef, the CJCS, or CCDR may direct the planning staff to refine or adapt a plan by reentering the planning process at any of the earlier functions. The time spent accomplishing each activity and function depends on the circumstances. In time-sensitive cases, commanders and planners collaborate in an open forum, compressing planning functions, making decisions, and issuing oral orders.

a. Strategic Guidance Analysis

(1) Strategic guidance analysis initiates planning, provides the basis for mission analysis, and enables the JPEC to develop a shared understanding of the issues, OE, objectives, and responsibilities.

(2) The CCDR provides input through sustained civilian-military dialogue that may include IPRs. The CCDR builds an approach that supports the guidance and consent of SecDef; if required, the CJCS offers advice to SecDef. This process begins with an analysis of existing strategic guidance. Planners conduct mission analysis, conduct threat assessment, and develop assumptions, and then brief SecDef on these during the IPR process.

(a) Some of the primary end products of strategic guidance analysis are assumptions, identification of available/acceptable resources, conclusions about the OE (nature of the problem), strategic and operational objectives, and the commander's mission.

(b) The CCDR maintains dialogue with DoD leadership to ensure they understand the above topics and the planning to date. This step can be iterative.

b. **Concept Development.** Concept development is the creative, problem-solving activity that planners conduct to devise a logical approach to the problem. The concept is the result of the design process and is the broad-brush overview of how the JFC intends to achieve the mission. The CONOPS is the centerpiece of the plan. During planning, the commander develops several COAs, each containing an initial CONOPS that the commander uses to identify threats to the joint deployment and distribution enterprise, determine major capabilities and authorities required, establish task organization, define major operational tasks for components to accomplish, conceptualize employment and sustainment, and conduct complete OE modeling and simulation to inform the commander's own assessment of risk. Each COA may contain multiple embedded alternatives to achieve designated objectives as conditions change (e.g., OE, problem, strategic direction). In time-sensitive situations, a WARNORD may not be issued, and a PLANORD, ALERTORD, or EXORD might be the first directive the supported commander receives with which to initiate planning. Using the guidance included in the directive and the CCDR's mission statement, planners solicit input from supporting and subordinate commands to develop COAs based upon the outputs of the strategic guidance analysis planning function.

(1) If an IPR occurs during concept development, the commander outlines the COA(s) and recommends them to higher authority for approval, and higher authority then approves and directs further development.

(a) The commander recommends a COA that achieves the directed objective(s) with the fewest trade-offs in forces and resources and incurs the lowest level of risk.

(b) Concept development should consider a range of COAs that integrate robust options to provide greater flexibility and to expedite transition during a crisis. CCDRs should be prepared to continue to develop multiple COAs to provide options to national-level leadership should the crisis develop.

(c) For CCPs, CCDRs should address resource requirements, expected changes in the OA, and how each COA supports achieving strategic objectives.

(d) Planners identify mission areas where contractor support is essential and ensure alignment with overall force structure and capability requirements. Early integration of commercial capabilities allows for a more resilient and adaptive operational design.

(e) The commander also requests SecDef guidance on interorganizational planning and coordination and makes appropriate recommendations, based on the interorganizational requirements identified during assessments, mission analysis, and COA development.

(2) One of the main products from the concept development planning function is approval for continued development of one or more COAs. Detailed planning begins upon COA approval in the concept development function.

c. **Plan Development.** Planners use this function to develop a feasible plan or order that they can transition into execution. This function fully integrates mobilization, deployment, employment, sustainment, transition, redeployment, and demobilization activities. The CCDR briefs the final plan to SecDef (or a designated representative) for approval.

d. **Execution.** Following plan approval, JFCs convert the plan into orders. In execution, JFCs turn contingency plans into orders if the anticipated threat scenario matches reality and strategic guidance.

e. **Assessment and Evaluation**

(1) Assessment is a continuous operation activity for execution and informs the commander's decision making. It determines whether current actions and conditions are creating the desired effects and changes toward the desired objectives and ensures operations remain appropriate for the changing conditions. For CCPs in execution,

assessments should enable the JFC to make a definitive, singular conclusion whether or not OAI are achieving the objectives.

(2) Operation Assessment

(a) Operation assessment identifies the progress of the joint force toward mission accomplishment. Operation assessment involves comparing desired conditions of the OE with actual conditions to determine the overall effectiveness of the campaign or operation. More specifically, operation assessment helps the JFC measure task performance, determine progress toward or regression from accomplishing a task, create an effect, achieve an objective, and issue the necessary guidance for change to guide forward momentum. Planners compare the follow-on operation assessments to the initial (ideally pre-execution) baseline and previous follow-on assessments; historical trends can aid the analysis and provide more definitive and reliable measures and indicators of change.

(b) During execution, operation assessment helps the command adapt and adjust operations as required to achieve the objectives. This analysis and adjustment of operations creates an organizational environment of learning and adaptation. Adaptation can range from minor operational adjustments to a radical change of approach, including transition of the operation. When fundamental changes challenge existing understanding or indicate a shift to the OE, commanders and staffs may develop a new operational approach that recognizes that the initial problem has changed, thus requiring a different approach toward the solution. The change to the OE could be so significant that it may require a review of the objectives, including discussions with higher authority to determine whether existing guidance is still viable.

(3) **Plan Evaluation.** (Refine, Adapt, Terminate Planning, Execute). Commanders continually review the plan to determine one of four possible outcomes—refine, adapt, terminate planning, or execute—and then act accordingly. Commanders and the JPEC continue to evaluate the situation for any changes that require updates to the plan. The CCCR briefs SecDef during plan update IPRs on modifications and updates to the plan, based on the CCCR's assessment of the situation, changes in resources or guidance, and the plan's feasibility and suitability.

(a) **Refine.** During all planning efforts, plan refinement is typically an orderly process that follows plan development and is part of the evaluation function. Planners refine the plan by continuously assessing operations to confirm changing OE conditions related to the plan or potential contingency. In a crisis, continuous operation assessment accommodates the fluidity of the crisis and facilitates continuous refinement throughout plans or OPOD development. Planners frequently adjust the plan or order based on evolving commander's guidance; results of force planning, support planning, or deployment planning; shortfall identification; adversary or MNF actions; changes to the OE; resolution or addition of assumptions; or changes to strategic guidance. Based on continuous operation assessment, refinement continues throughout execution, with changes typically transmitted in the form of FRAGORDs rather than revised copies of the plan or order.

(b) **Adapt.** Planners adapt plans when they determine the need for major modifications, driven by changes in strategic direction, the OE, or the problem facing the JFC. Planners continually monitor the situation for changes that necessitate adapting the plan, to include modifying the commander's operational approach and revising the CONOPS. When this occurs, commanders may need to initiate a plan revision.

(c) **Terminate.** For CPG- or JSCP-tasked plans, the CCDR requests permission to archive the plan through the Joint Staff J-5 [Strategy, Plans, and Policy Directorate], which staffs the request through the JPEC. Recommendations on CPG-tasked plans route through the appropriate OSD staff for approval by SecDef (with a CJCS recommendation attached). For plans tasked in the JSCP, planners will present transition recommendations to the CJCS for approval.

For more information on termination and archiving of plans, see CJCSM 3130.03, Planning and Execution Formats and Guidance.

(d) **Execute.** When ordered, commanders review the plan, validate assumptions and previously issued guidance, issue orders, and conduct operations as directed. ALERTORDs, WARNORDs, and EXORDs will identify any changes to plans and specific forces allocated for the operation. Additional information on transitioning to execution is in Chapter VII, "Transition to Execution."

SECTION B. THE JOINT PLANNING PROCESS

4. Overview

The commander is the central figure in planning due to authority and responsibility. The commander's judgment and decisions, informed by knowledge and expertise, guide the staff through the process. Generally, the more complex a situation, the more critical the role of the commander early in planning by leveraging their knowledge, experience, judgment, intuition, responsibility, and authority to generate a clearer understanding of the conditions needed to focus effort and achieve success.

a. Commanders distinguish the unique features of their current situations to enable development of innovative or adaptive solutions. They understand that each situation requires a solution tailored to the context of the problem. With operational design and the application of operational art, commanders develop innovative, adaptive alternatives to solve complex challenges.

b. Commanders use the knowledge and understanding gained from operational design, along with any additional guidance from higher headquarters, to provide guidance that directs and guides the staff through the JPP. Developing touchpoints throughout the planning process with the supported and supporting commanders and other stakeholders enables a shared understanding of the OE.

c. Operational design encourages discourse, leverages dialogue, and collaborates to identify complex, ill-defined problems. The commander empowers organizational learning and develop methods to determine whether modifying the operational approach is necessary during an operation or campaign. This requires assessment and reflection that challenge understanding of the existing problem and the relevance of actions addressing that problem. Partners can be sources of additional information on the problem or OE, and can provide differing perspectives, which can broaden understanding and challenge institutional biases. Red teaming can also be a means to improve understanding by challenging assumptions or preconceptions and proposing alternative views. Due to complexity and constant change, commanders should be comfortable in the recognition that they will never know everything about their OA and will never be able to fully define its problems. As such, many of the problems of the OE may not have solutions. Success may therefore require innovation and rapid, iterative adaptation based on assessment of results from activities and changes.

5. Joint Planning Process Steps

a. The JPP is an orderly, analytical set of logical steps to frame a problem; examine a mission; develop, analyze, and compare alternative COAs; select the best COA; and produce a plan or order. The application of operational design provides the conceptual basis for structuring campaigns and operations. The JPP provides a proven process which organizes the work of the commander, staff, subordinate commanders, and other partners to develop plans that appropriately address the problem. It focuses on defining the military mission and development and synchronization of detailed plans to accomplish that mission. It applies to both supported and supporting JFCs and to joint force component commanders when the components participate in joint planning. Together with operational design, the JPP facilitates interaction between and among the commander, staff, and subordinate and supporting headquarters throughout planning. The JPP helps the commander, and their staff organize their planning activities, share a common understanding of the mission and commander's intent, and develop effective plans and orders. Figure IV-3 shows the primary steps of the JPP.

(1) The JPP is applicable for joint planning. Like operational design, it is a logical process to approach a problem and determine a solution. It is a tool for planners but is not prescriptive. Based on the nature of the problem, planners can modify the process to account for other available tools and expertise. Similarly, planners can truncate, modify, or execute steps concurrently, depending upon the situation, subject, or time constraints of the planning effort. For example, force planning, as an element of plan development, is different for campaign planning and contingency planning.

(2) In a crisis, planners may execute the steps of the JPP simultaneously to speed the process. Supporting commands and organizations often conduct the JPP simultaneously and iteratively with the supported CCMD. In these cases, once mission analysis begins, it continues until the operation is complete. Moreover, steps 4-7 repeat as often as necessary to integrate new requirements (missions) into the development of the plan.

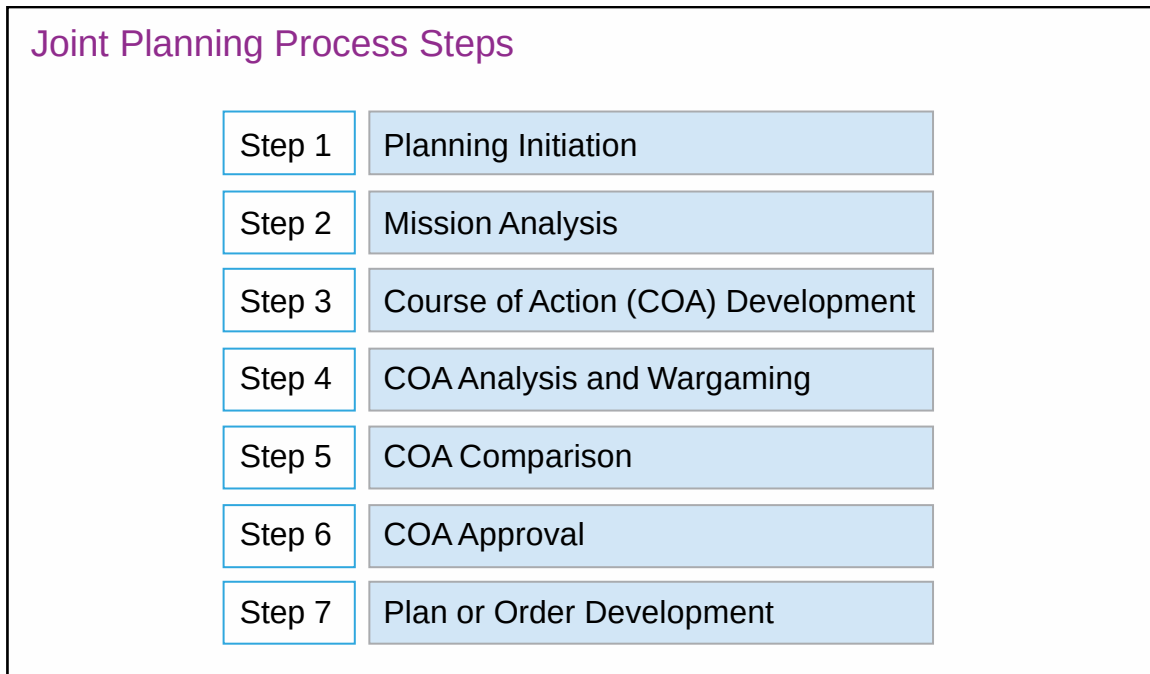


Figure IV-3. Joint Planning Process Steps

b. Planning Initiation (Step 1)

(1) Joint planning begins when an appropriate authority recognizes potential for military capability to be employed in support of enduring interests or in response to a potential or actual crisis.

(2) CCDRs, subordinate commanders, and supporting commanders also initiate planning on their own authority when they identify a planning requirement not directed by higher authority. Additionally, analyses of the OE or developing or immediate crises may result in the President, SecDef, or CJCS directing military planning through a planning directive. CCDRs normally develop military options in combination with other nonmilitary options so the President can direct a whole-of-government approach that involves all the appropriate instruments of national power. Whether or not planning begins as described here, the commander may act within approved authorities and ROE/rules for the use of force in an immediate crisis.

(3) The commander and staff will receive and analyze the planning guidance to determine the time available until mission execution; status of strategic and staff estimates; and intelligence products, to include JIPOE, and other factors related to the specific planning situation. The staff gathers and analyzes related information about friendly, neutral, and threat networks from both military and nonmilitary elements. The commander will typically provide initial planning guidance based upon current understanding of the OE, the problem, and the initial operational approach for the campaign or operation. Commander guidance could specify time constraints, outline initial coordination requirements, or authorize movement of key capabilities within the JFC's authority.

(4) Once execution begins, the planning focus changes due to new direction, significant changes to the current mission or planning assumptions, or the commander receiving a mission for follow-on operations.

(5) Planning for campaign plans is different from contingency plans in that contingency planning focuses on the anticipation of future events, while campaign planning assesses the current state of the OE and identifies how the command can influence the OE to deter adversaries and achieve the objectives.

(6) Operational design supports this step in the planning process by building understanding of **strategic direction, the strategic environment, and the OE**. This helps the commander and staff define the problem as they begin mission analysis.

(7) The Planning Directive

(a) The **planning directive** identifies planning responsibilities for developing joint force plans. It provides guidance and requirements to the staff and subordinate commands concerning coordinated planning actions for plan development. The JFC normally communicates initial planning guidance to the staff, subordinate commanders, and supporting commanders by publishing a planning directive to ensure everyone understands the commander's intent and to achieve unity of effort.

(b) Generally, the J-5 coordinates staff action for planning for the CCMD campaign and contingencies, and the J-3 coordinates staff action for a crisis. The J-5 staff receives the JFC's initial guidance and combines it with the information gained from the initial staff estimates. The JFC, through the J-5, may convene a preliminary planning conference for members of the JPEC who will be involved with the plan. This is the opportunity for representatives to meet face-to-face. At the conference, the JFC and selected members of the staff brief the attendees on important aspects of the plan and may solicit their initial reactions.

c. Mission Analysis (Step 2)

(1) The CCDR and staff develop a restated mission statement that enables subordinate and supporting commanders to begin their own estimates and planning efforts for higher headquarters' concurrence. **The joint force's mission is the task or set of tasks, together with the purpose, that clearly indicates action and the reason for the action.** Mission analysis studies the assigned tasks to identify all other tasks necessary to accomplish the mission. Mission analysis focuses the commander and the staff on the problem at hand and lays a foundation for planning. When the commander receives a mission tasking, analysis begins with the following questions:

(a) What is the purpose of the mission? (What problem is the commander asked to solve or what change to the OE is desired?)

(b) What tasks will accomplish the mission?

- (c) Will the mission achieve the desired objectives?
- (d) What limits my own forces' actions?
- (e) What forces do I need to support my operation?
- (f) How will I know when the mission is successful?
- (g) How will friendly, neutral, and threat networks affect the accomplishment of the mission?

(2) **The primary inputs to mission analysis** are strategic guidance; the higher headquarters' planning directive; and the commander's initial planning guidance, which may include a description of the OE, a definition of the problem, the operational approach, initial intent, and the JIPOE (see Figure IV-4). **The primary outputs of mission analysis** are the identified essential, specified, and implied tasks; friendly and threat COGs and their critical factors; staff estimates; the mission statement; a refined operational approach; the commander's intent statement; updated planning guidance; and initial CCIRs.

(3) Mission analysis helps the JFC understand the problem and purpose of the operation and issue appropriate guidance to drive the rest of the planning process. The JFC and staff can accomplish mission analysis through several logical activities, such as those shown in Figure IV-5. Include liaison officers from adjacent, supporting, and subordinate organizations in the planning process as soon as possible to ensure coherent planning across the command and with higher and adjacent units.

(a) Although some activities occur before others, mission analysis typically involves substantial concurrent processing of information by the commander and staff, particularly in a crisis.

(b) During mission analysis, planners clearly state the tasks (specified and implied) and their purposes to ensure planning encompasses all requirements, including any limitations on actions that the commander or subordinate forces may take. Mission analysis should also ensure the correlation between the commander's mission and intent and those of higher commanders. Evaluate resources and authorities to ensure there is not a mission-resource-authority mismatch and to enable the commander to prioritize missions and tasks against limited resources.

(c) Capture and track specific information to improve the end products. This includes requests for information regarding forces, capabilities, and other resources; questions for the commander or special assistant (e.g., legal); and proposed battle rhythm for planning and execution. Recording this information during the mission analysis process enables a more complete product and smoother mission analysis brief.

(d) Mission analysis should also assess the availability and capacities of commercial capabilities alongside military, interagency, and allied assets to enhance

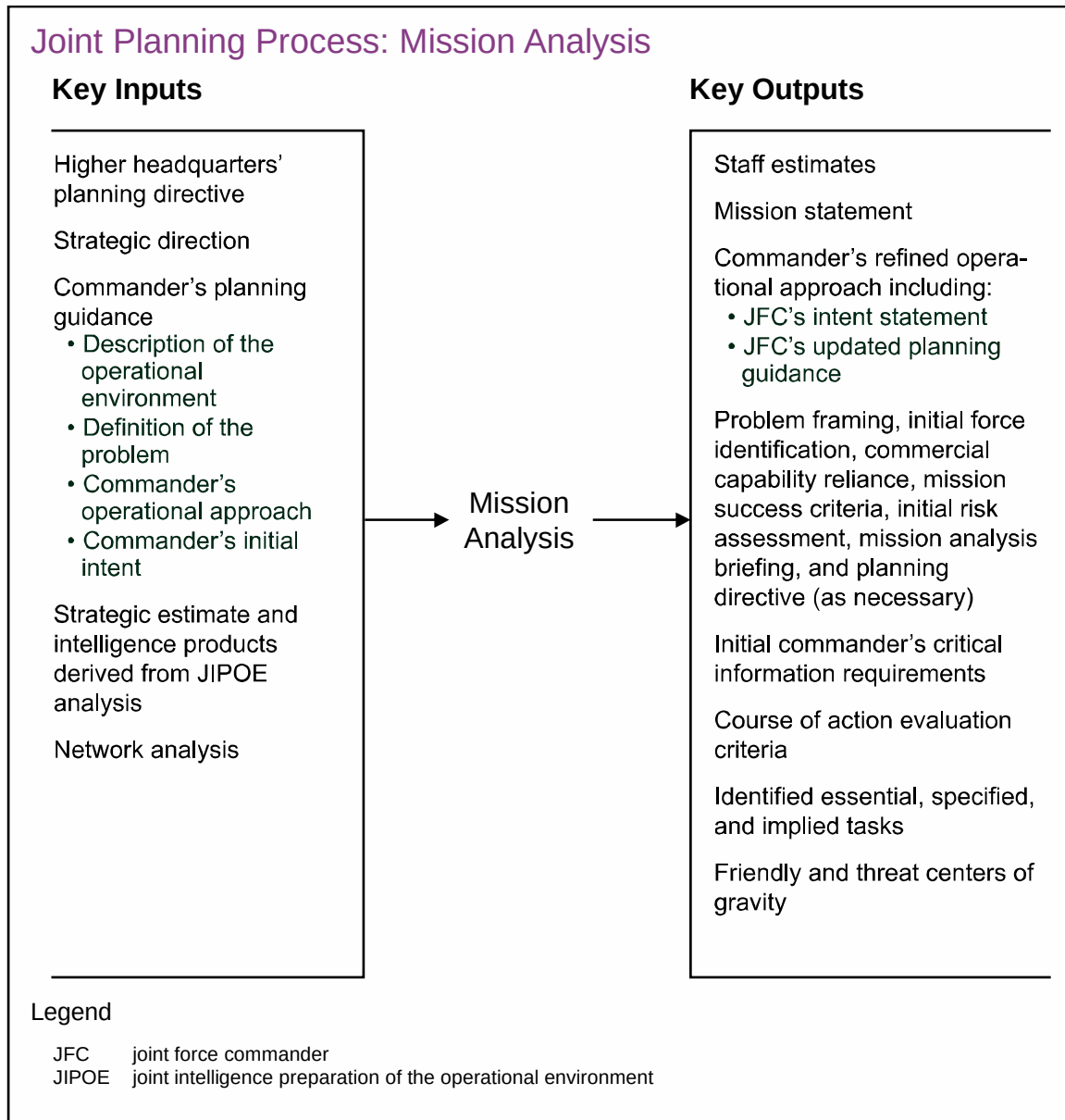


Figure IV-4. Joint Planning Process: Mission Analysis

operational effectiveness and sustainment. Integrating commercial capabilities and commercial support considerations enables a more adaptive, resilient, and sustainable force posture.

(4) Analyze Higher Headquarters' Planning Directives and Strategic Guidance

(a) Strategic guidance is essential to joint planning and operational design. The President, SecDef, and CJCS promulgate strategic guidance documents that cover a broad range of situations, and CCDRs provide guidance that covers a narrower range of situations. The CCDR's strategy provides the mid- to long-term (greater than three years)

Mission Analysis Activities

- Begin logistics supportability analysis.
- Analyze higher headquarters planning activities and strategic guidance.
- Review commander's initial planning guidance, including his initial understanding of the operational environment, of the problem, and description of the operational approach.
- Determine known facts and develop planning assumptions.
- Determine and analyze operational limitations.
- Determine specified, implied, and essential tasks.
- Develop mission statement.
- Conduct initial force identification.
- Develop risk assessment.
- Develop course of action evaluation criteria.
- Develop initial objectives.
- Develop commander's critical information requirements.
- Prepare staff estimates.
- Prepare and deliver mission analysis brief.
- Publish commander's updated planning guidance, intent statement, and refined operational approach.
- Conduct initial analysis to identify commercial capability reliance (e.g., commercial infrastructure, logistics and transportation, communications and information technology).

Steps are not necessarily sequential.

Figure IV-5. Mission Analysis Activities

CCMD vision for employment of capabilities prepared in the context of SecDef's priorities. CCCR strategy links strategic direction to joint planning.

(b) For a specific crisis, an order provides guidance, typically including a description of the situation, purpose of military operations, objectives, anticipated mission or tasks, and pertinent limitations. The GFMIG assignment and apportionment tables identify forces that planners can reasonably expect to be available. Generally, planners should not include future capabilities or posture because plans should be immediately executable, but there may be exceptions. Supported and supporting plans for the same military activity rely on the same resources. Additionally, there are likely to be operations and activities already ongoing at the onset of a crisis. Planners should remain aware that most plans do not execute individually and the finite resource pool for all operations and plans is the same.

(c) The CJCS may amplify apportionment guidance if appropriate. This planning can confirm or modify the guidance for an existing contingency plan or order.

This might simplify the analysis step, since consensus should already exist between the supported command and higher authority on the nature of the OE in the potential joint OA—such as the diplomatic, economic, social, and military circumstances—and potential US or multinational responses to various situations described in the existing plan. But even with a preexisting contingency plan, planners need to confirm the actual situation matches the hypothetical situation that the contingency plan was based on, as well as validating other assumptions. Significant changes may require refining or adapting the existing contingency plan. The dynamic nature of an emerging crisis can change many key aspects of the OE compared with earlier assumptions.

(d) In time-compressed situations, especially with no preexisting plan, the higher headquarters' assessment of the OE and objectives may be the only guidance available. However, this circumstance is one that can benefit the most from the commander and staff's independent assessment of circumstances to ensure a common understanding with higher headquarters assessment. This is why CCMD JIPOE efforts should be continuous; these efforts maintain the intelligence portions of the CCDR's strategic estimate. Keeping the strategic estimate up to date facilitates crisis planning, as well as the transition of contingency plans to execution for crisis situations.

(e) **Multinational Strategic Guidance.** Operational success requires that CCDRs, subordinate JFCs, component and supporting commanders, and their staffs understand both US and partner-nation objectives and the conditions set by national or multinational political leadership for the multinational military force to achieve, in terms of the internal and external balance of power, regional security, and geopolitics. To ensure unity of effort, planners should identify and attempt to resolve conflicts between participating nations' objectives and identify possible conflicts between different nations' national policy outcomes to ensure strategic planning accounts for these divergences. When multinational objectives are unclear, the senior US military commander should seek clarification and convey the positive or negative impact of continued ambiguity to the President and SecDef. For additional information on multinational operations, see JP 3-16, *Multinational Operations*. For specific information on NATO operations, see Allied Joint Publication-01, *Allied Joint Doctrine*; Allied Joint Publication-3, *Allied Joint Doctrine for the Conduct of Operations*; and Allied Joint Publication-5, *Allied Joint Doctrine for the Planning of Operations*.

(5) **Review Commander's Initial Planning Guidance.** Staff members and representatives from supporting organizations should maintain an open dialogue with the commander to better develop an appropriate solution to the problem and be able to adapt solutions to match the evolving OE and any potentially changing problems. Staffs should analyze the CCDR's initial planning guidance for the campaign or operation, which provides a basis for continued detailed analysis of the OE and of the tasks that may describe the mission and its parameters.

(6) **Determine Known Facts and Develop Planning Assumptions.** The staff assembles both facts and assumptions to support the planning process and planning guidance.

INFORMATION AND FACTS

Commanders and planners use caution in characterizing information as facts, as some items of information thought to be facts may be open to interpretation, based on the observer's perspective or incomplete information.

(a) Commanders and staffs gather key facts as they develop a plan. A fact is information used as evidence of something known to be true, exist, or have happened (such as verified locations of friendly and adversary force dispositions). Facts concerning operational variables serve as a basis for developing situational understanding during planning. When listing facts, planners ensure they are relevant to the plan or help commanders make decisions.

(b) An assumption provides a supposition about the current situation or future course of events, presumed to be true based on an assessment of available facts. Valid assumptions have three characteristics: logical, realistic, and essential for planning to continue. Commanders and staffs should never underestimate adversary capabilities or assume unrealistic friendly capabilities will be available. Assumptions address gaps in knowledge critical for the planning process to continue. Planners review all assumptions to ensure their validity and challenge if unrealistic, including those provided in strategic guidance or from higher headquarters. Planners acknowledge the higher headquarters' assumptions and assess the impact (risk) should they prove to be incorrect. JFCs may have information that invalidates a higher headquarters' assumption; however, it does not relieve subordinate headquarters from planning within the confines of the given assumption. Subordinate commanders do not develop assumptions that contradict valid higher headquarters assumptions.

1. Commanders and staffs should anticipate changes to the plan if an assumption proves to be incorrect. Because of assumptions' influence on planning, planners should either validate the assumptions (treat as facts) or invalidate the assumptions (alter the plan accordingly) as quickly as possible.

2. During wargaming or red teaming, planners should review the plan from both the perspective that the assumption will prove true and from the perspective that the assumption will prove false. Specifically, the red team should review the assumptions to identify influence from the cognitive bias of their own experience or point of view. Cognitive bias is a subconscious mental procedure for processing information that could adversely impact the result of analytic efforts and various processes such as joint planning, joint targeting, and JIPOE. The joint force can apply structured analytic techniques, analytic tradecraft practices, and red teaming to verify that planning, targeting, and intelligence processes/assessments are sound and minimize cognitive bias. This can aid in preventing biases or tunnel vision from impacting decision making.

3. Commanders should assess the impact higher headquarters assumptions have on the plan during wargaming. Although higher headquarters assumptions frame planning, they are assumptions validated for execution.

4. Plans may contain assumptions that are unresolvable until a crisis develops. As a crisis develops, replace assumptions with facts as soon as possible. The staff accomplishes this by identifying the information needed to validate assumptions and submitting an information request to an appropriate agency as an information requirement. If the commander needs the information to make a key decision, designate the information requirement a CCIR.

5. Planners work to limit assumptions to only those necessary for continued planning. Assumptions introduce possibility for error. If the assumption is not necessary to continue planning, its only effect is to introduce error and add the likelihood of creating a bias in the commander and planner's perspective. Since most plans require refinement, a plan with fewer assumptions allows the commander and staff to act and react with other elements of the OE (including enemies, adversaries, allies, and the physical element). However, assumptions are useful to identify those issues the commander and planners validate on execution, and strategic plans are heavily reliant on assumptions due to the ambiguity inherent in the strategic environment.

6. Identify all assumptions in the plan or decision matrix to ensure they review and validate prior to execution.

(7) **Determine and Analyze Operational Limitations.** Operational limitations are actions required or prohibited by higher authority and other restrictions that limit the commander's freedom of action, such as diplomatic agreements; access, basing, and overflight issues; and partner nation caveats.

(a) A **constraint** is a requirement placed on the command by a higher command that **dictates an action** ("must do"), thus restricting freedom of action. Constraints limit commanders by making them do things they might not otherwise be inclined to do. For example, to gain victory in World War II, national leadership directed General Dwight D. Eisenhower to enter the continent of Europe through the Italian peninsula first instead of entering directly through France.

(b) A **restraint** is a requirement placed on the command by a higher command that **prohibits an action** ("cannot do"), thus restricting freedom of action. Restraints prevent commanders from doing things they would otherwise like to do. For example, national leadership prohibited General Douglas MacArthur from striking Chinese targets north of the Yalu River during the Korean War.

(c) Many operational limitations transition to ROE/rules for the use of force. Operational limitations may restrict or bind COA selection or may even impede implementation of the chosen COA. Commanders examine the operational limitations imposed on them, understand their impacts (including risk posed by limitations), and develop options within these limitations to promote maximum freedom of action during execution.

(d) Other operational limitations may arise from laws or authorities, such as the use of specific types of funds or training events. Commanders ensure they have the authority to execute operations and activities.

(8) **Determine Specified, Implied, and Essential Tasks.** The commander and staff typically review the planning directive's specified tasks and discuss implied tasks during planning initiation to resolve unclear or incorrectly assigned tasks with higher headquarters. If there are no issues, the commander and staff will confirm the tasks in mission analysis and then develop the initial mission statement.

(a) **Specified tasks** are those assigned to a subordinate commander in a planning directive, usually because they are important to the higher command's mission or objectives. One or more specified tasks often become essential tasks for the subordinate commander.

EXAMPLES OF SPECIFIED TASKS

Ensure freedom of navigation for United States forces through the Strait of Gibraltar.

Defend Country Green against attack from Country Red.

(b) **Implied tasks** are additional tasks the commander has to accomplish, typically to accomplish the specified tasks, support another command, or otherwise accomplish activities related to the operation or achieving the objective. In addition to specified tasks within the higher headquarters' planning directive, the commander identifies implied tasks necessary to finalize the commander's operational approach. The commander can also deduce implied tasks from knowledge of the OE, such as the enemy situation and political conditions in the assigned OA. However, implied tasks do not include tasks or standard operating procedures inherent for most operations, such as conducting reconnaissance and protecting a flank.

EXAMPLES OF IMPLIED TASKS (RELATED TO ABOVE SPECIFIED TASKS)

Establish maritime superiority out to 50 miles from the Strait of Gibraltar. Be prepared to conduct foreign internal defense and security force assistance operations to enhance the capacity and capability of Country Green security forces to provide stability and security if a regime change occurs in Country Red.

(c) **Essential tasks** are those that the command must execute successfully to accomplish the mission. The commander and staff determine essential tasks from the lists of both specified and implied tasks. Depending on the scope of the operation and its purpose, the commander may synthesize certain specified and implied task statements into an essential task statement.

(9) **Develop Mission Statement.** The mission statement describes the mission in terms of the elements of who, what, when, where, and why. The “what” is typically the essential tasks determined by the commander. The “why” describes the purpose of the operation. The commander’s operational approach informs the mission statement and helps form the basis for planning. The commander includes the mission statement in the planning guidance, planning directive, staff estimates, commander’s estimate, CONOPS, and completed plan.

EXAMPLE MISSION STATEMENT

When directed [when], United States X Command, in concert with coalition partners [who], deters Country Y from coercing its neighbors and proliferating weapons of mass destruction [what] to maintain security [why] in the region [where].

(10) **Conduct Initial Force and Resource Analysis**

(a) **Initial Force Analysis.** During mission analysis, the planning team begins to develop a rough-order-of-magnitude list of required forces and capabilities necessary to accomplish the specified and implied tasks. Planners consider the responsiveness of assigned and currently allocated forces. While more deliberate force requirement identification efforts continue during concept and plan development, initial identification of readily available forces during mission analysis may constrain the scope of the proposed operational approach.

1. In some cases, joint force capacity limits the types of tasks and range of possible objectives a JFC can achieve. In these cases, JFCs should identify these limitations early and develop COA options that include objectives the joint force can achieve with less risk. However, planners also develop the best CONOPS possible for the objectives. Higher headquarters can make more informed cost-benefit-risk decisions when presented with multiple COAs that are distinguishable in how they achieve objectives. COA comparison that includes COAs with different objectives helps higher headquarters make informed cost-benefit-risk decisions.

2. Develop force requirements for a plan from forces assigned, allocated, or apportioned from the Directed Readiness Tables and Apportionment Tables and initially documented in a time-phased force and deployment list. The time-phased force and deployment list depicts general details of the CCDR’s force requirements (e.g., type of force, required delivery in theater) in appendix 1 (Time-Phased Force Deployment List) to annex A (Task Organization) of the OPLAN. Later in the planning process, planners enter these requirements into an information technology system (i.e., Joint Operation Planning and Execution System) as a baseline of forces with a more specific level of detail to support subsequent time phasing. From the start, planners should document forces in a format and system that enables GFM allocation to simplify plan execution.

3. In a crisis, assigned and allocated forces currently deployed to the CCMD's AOR may be the most responsive during the early stages of an emergent crisis. Planners may consider assigned forces as likely to be available to conduct activities unless allocated to a higher priority. Re-missioning previously allocated forces may require SecDef approval and require coordination through the JS using procedures outlined in CJCSM 3130.06, *(U) Global Force Management Allocation Policies and Procedures*.

4. Planners should also consider the time required for call-up and mobilization of reserve forces.

5. Planners should evaluate appropriate requirements against existing or potential contracts or task orders to determine if the contracted support solution could meet the requirements.

6. Force requirements for supported and supporting plans draw from the same inventory of DoD capabilities and compete with other global requirements for military activities and ongoing operations during plan execution. By using the Directed Readiness Tables and Apportionment Tables, planners can address the risk-associated shortfalls early in the planning process. Address the risk associated with reallocation or diversion of forces from one mission to another early in the planning process, to facilitate SecDef-directed GFM allocation decisions during execution.

7. Finally, planners compare the specified and implied tasks to the forces and resources available and identify shortfalls.

8. JFCs plan with extant forces/capabilities available in the joint force inventory.

(b) **Identify Non-Force Resources Available.** In many types of operations, the commander (and planners) may have access to non-force resources, such as commander's initiative funds, other funding sources (e.g., train and equip funding, support to foreign security forces funding), or can work with other security assistance programs (e.g., foreign military sales, excess defense article transfers). Planners and commanders can weave together resources and authorities from several different programs to create successful operations. Only currently available resources should be part of planning. For example, planners should not expect to have bases or facilities the military plans to build or acquire in the future.

See JP 3-20, Security Cooperation, for additional information on integrating multiple resources. See the GFMIG for more information on the GFM processes and CJCSM 3130.06, Global Force Management Allocation Policies and Procedures, for additional guidance on GFM allocation.

(11) **Develop Operational-Level Objectives**

(a) JFCs direct joint operations against **operational-level objectives**. During plan development, planners may use objectives to develop mission areas, LOOs or LOEs during formulation of the operational approach. Describing operational objectives

early helps commanders and staffs visualize and describe how effects in each LOO or LOE contribute to accomplishing the mission.

(b) Each operational objective guides the actions and effects of a LOO or LOE. Mission areas may include multiple operational-level objectives (e.g., GCPs/CCPs). The JFC may express objectives as a particular behavior the military operation will produce; a geographic area to seize, control, or isolate; or desired conditions in the OA, such as creating effects on enemy forces or increased freedom of action. Planners write operational-level objectives as short phrases in active voice that are clear, concise, specific, and prescriptive. Scope operational objectives to focus resources on points in time, space, or action.

(c) Objectives and effects, at any level, are **not** friendly tasks. Each objective should be broad enough to describe the net outcome of multiple subordinate actions. They also provide a foundation for effective operational assessment.

(12) **Develop COA Evaluation Criteria.** Evaluation criteria are standards the commander and staff use later to measure the relative effectiveness and efficiency of one COA relative to other COAs. Developing these criteria during mission analysis or as part of commander's planning guidance helps to eliminate a source of bias prior to COA analysis and comparison.

(a) Evaluation criteria address factors (e.g., hazards, threats, mitigation measures) that affect success and those that can cause failure. Commanders adjust criterion selection and weighting according to their own experience and vision. The staff member responsible for a functional area scores each COA using those criteria.

(b) The commander and staff use evaluation criteria during follow-on COA comparison (JPP step 5) for the purpose of selecting the best COA. The commander and staff consider various potential evaluation criteria during wargaming and select those that the staff use following the wargame during COA comparison to evaluate the effectiveness and efficiency of one COA relative to others. These evaluation criteria help focus the wargaming effort and provide the framework for data collection by the staff. These criteria are those aspects of the situation (or externally imposed factors) that the commander deems critical to mission accomplishment. Figure IV-6 shows examples of potential evaluation criteria.

(c) The staff presents the proposed evaluation criteria to the commander at the mission analysis brief for approval.

(13) **Develop Risk Assessment**

(a) Planners conducting a preliminary risk assessment should identify the obstacles or actions that may preclude mission accomplishment and then assess the impact of these impediments to the mission. Once planners identify the obstacles or actions, they assess the probability of achieving objectives and severity of loss linked to an obstacle or action and characterize the military risk. Military risk assessment is a function of judgment, probability, and consequences of an identified impediment.

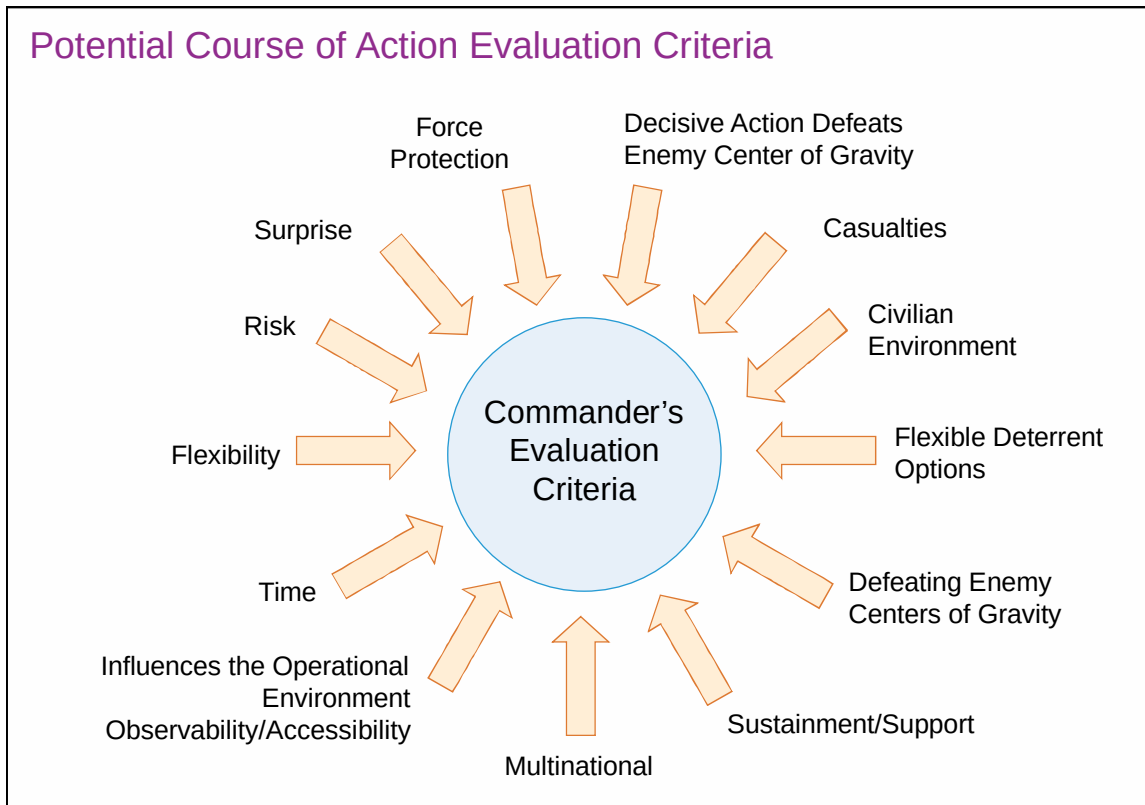


Figure IV-6. Potential Course of Action Evaluation Criteria

(b) Planners weigh the probability of the impediment occurring with the consequence of the action, should it occur. Planners refer to the current version of the CJCSM 3105.01, *Joint Risk Assessment Methodology*, for details.

(c) Determining military risk is more art than science. Planners use historical data, analysis, and judgment. Military risk characterization is a function of the assessed probability and consequence of an event causing harm to objectives or resources.

(d) Planners and commanders explain military risk to civilian leadership who may not be as familiar with military operations. Additionally, since military risk is often a matter of perspective and personal experience, commanders are able to help decision makers understand how they evaluated the probability of achieving objectives, how they characterized the resultant military risk, and the sources or causes of that risk.

(e) During decision briefs, planners explain risks using standard terms that support the decision-making process, such as **mission success** (accomplishing a mission), **time** (how much longer will a mission take to achieve success), **forces** (e.g., casualties, future readiness), and sourcing (e.g., solutions for critical requirements).

(14) **Determine Initial CCIRs**

(a) **CCIRs are elements of information the commander identifies as being critical to timely decision making.** CCIRs help focus information management

and help the commander assess the OE, validate (or refute) assumptions, identify achievement of objectives, and identify decision points during operations. **CCIRs belong exclusively to the commander. They are situation-dependent, focused on predictable events or activities, time-sensitive, and always established by an order or plan.** The CCIR list is normally short so the staff can focus its efforts and allocate resources. The CCIR list is not static; JFCs add, delete, adjust, and update CCIRs throughout plan development, plan evaluation, execution, and operation assessment, based on the information they need for decision making. PIRs and FFIRs constitute the total list of CCIRs (see Figure IV-7).

1. **PIRs** focus on the adversary and the OE and tie to commander's decision points. They drive the collection of information by all elements of a command, requests for national-level intelligence support, and requirements for additional intelligence capabilities. All staff sections can recommend potential PIRs they believe meet the commander's guidance. However, the joint force J-2 has overall staff responsibility for consolidating PIR nominations and for providing the staff recommendation to the commander. **Commander-approved PIRs are automatically CCIRs.**

For more information on PIRs, see JP 2-0, Joint Intelligence.

2. **FFIRs** focus on information the JFC requires for assessing the status of the friendly force and supporting capabilities. All staff sections can recommend potential FFIRs they believe meet the commander's guidance. **Commander-approved FFIRs are automatically CCIRs.**

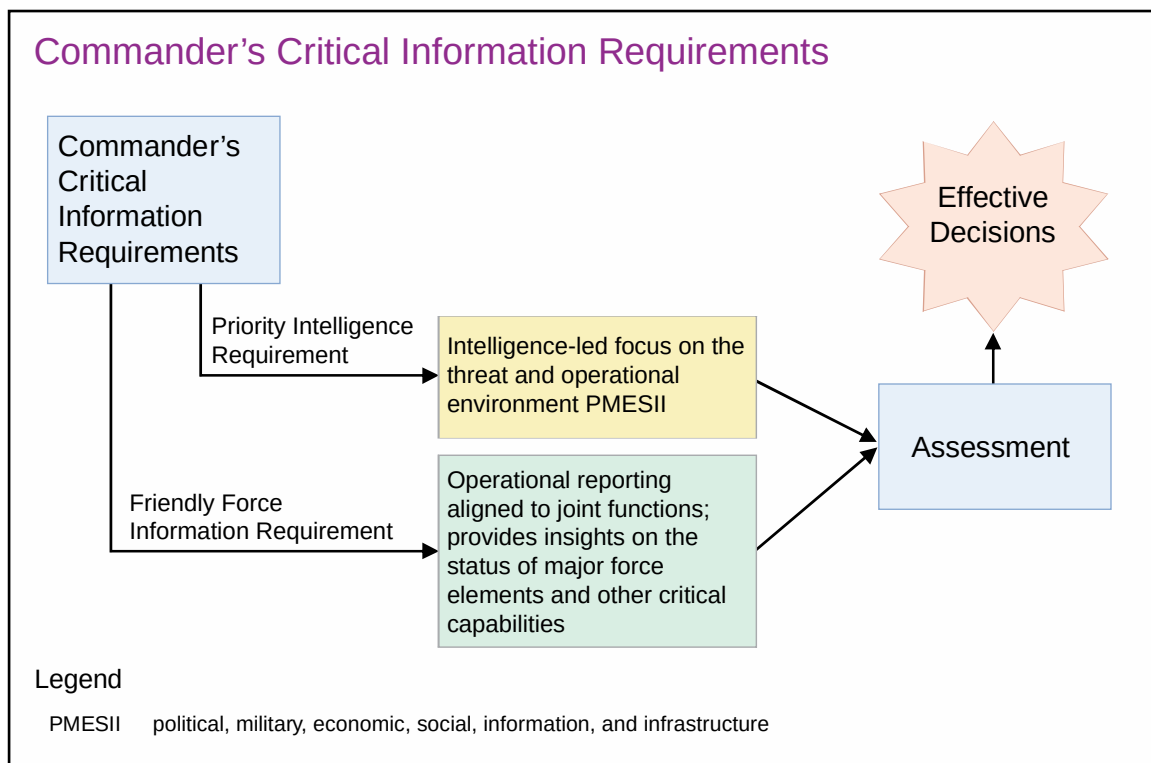


Figure IV-7. Commander's Critical Information Requirements

(b) **Decision Support.** CCIRs support the commander's future decision requirements and are often related to decision points, measures of effectiveness (MOEs), and measures of performance (MOPs). These should all support specific decisions the commander makes, to include operation assessment.

(15) **Prepare Staff Estimates**

(a) A staff estimate is an evaluation of how factors in a staff section's functional area support and impact the mission. The purpose of the staff estimate is to inform the commander, staff, and subordinate commands how the functional area supports mission accomplishment and to support COA development and selection.

(b) Initiate staff estimates during mission analysis, while functional planners focus on collecting information from their functional areas to help the commander and staff understand the situation and conduct mission analysis. Later, during COA development and selection, functional planners fully develop their estimates providing functional analysis of the COAs, as well as recommendations on which COAs are supportable. They should also identify critical shortfalls or obstacles that impact mission accomplishment. Continually update staff estimates based on changes in the situation. Operation assessment helps inform the staff estimates for each functional area. Staff estimates can expand to the annex or appendix section during plan development.

(c) Not every situation requires or permits a lengthy and formal staff estimate process. For a crisis, planners may deliver staff estimates orally to support the rapid development of plans. However, with sufficient time, planning demands a more formal and thorough process. Share staff estimates with subordinate and supporting commanders to help them prepare their supporting estimates, plans, and orders. This improves parallel planning and collaboration efforts of subordinate and supporting elements and helps reduce the planning times for the entire process.

(d) Intelligence support to joint planning includes Defense Intelligence Agency-produced dynamic threat assessments for each JSCP-directed integrated plan set (IPS) and select CCMD contingency plans and theater intelligence assessments with a two-to-five-year outlook to support CCP development and assessment. CCMD joint intelligence operations centers and subordinate JFC's joint intelligence support elements produce intelligence assessments and estimates resulting from the JIPOE process. The intelligence estimate constitutes the intelligence portion of the commander's estimate and resides in appendix 11 (Intelligence Estimate) to annex B (Intelligence) to a plan or an order. These are baseline information and finished intelligence products that inform the four continuous operational activities of situational awareness, planning, execution, and assessment.

For additional information on the intelligence estimate format and its relationship to the commander's estimates, see CJCSM 3122.01, Joint Operation Planning and Execution System (JOPES), Volume I, Planning Policies and Procedures, and CJCSM 3130.03, Planning and Execution Formats and Guidance.

(e) During mission analysis, intelligence planners lead the development of PIRs and collection plan(s) to close critical knowledge gaps in intelligence products, validate threat and OE-related planning assumptions, and provide indications of changes. Throughout the JPP, planners may nominate additional PIRs to support critical decisions needed throughout all phases of the operation. Intelligence planners then prepare a J-2 staff estimate, which is an appraisal of available capabilities within the intelligence joint function to satisfy PIRs. This estimate drives development of annex B (Intelligence) to a plan or an order. In annex B, the J-2 publishes the PIRs; describes the concept of intelligence and counterintelligence operations; specifies intelligence procedures; clarifies counterintelligence authorities, objectives, effects; and assigns intelligence tasks to subordinate and supporting agencies. Commanders delegated coordinating authority or a CCMD J-2 may also request a NISP for any level 3 or 4 plan. A NISP synchronizes and coordinates defense intelligence enterprise support to the CDR's plan. Combat support agencies and Service intelligence centers prepare annexes to the NISP called functional support plans. Functional support plans present supporting capabilities to CDRs for employment in either a deployed or reachback mode. Through the intelligence planning (IP) process, intelligence planners identify gaps and shortfalls in DoD intelligence capabilities. Left unmitigated, these gaps and shortfalls may present risks to the execution of the supported plan for consideration during plan assessment.

For additional information on the IP process, see JP 2-0, Joint Intelligence, and CJCSM 3314.01, Intelligence Planning.

(f) The commander's logistics staff and Service component logisticians develop a logistics overview, which includes, but is not restricted to, critical logistics facts, assumptions, and information required for the CCIRs; current or anticipated HN support and status; in coordination with applicable vendor threat mitigation activities, identification of existing contracts and task orders available for use; identifying aerial and sea ports of debarkation; any other distribution infrastructure and associated capacity; inventory (e.g., on-hand, prepositioned, theater reserve); combat support and combat service support capabilities; known or potential capability shortfalls; and contracted support required to replace or augment unavailable military capabilities. From this theater logistics overview, a logistics estimate can identify known and anticipated factors that may influence the logistics support.

(g) There are unique threats associated with the use of contracted support. The commander's logistics and intelligence staff plan and mitigate these risks of vendor threats. Vendor threat mitigation leverages all-source information and commander's guidance to identify threats to missions or national security.

For more information see DoDD 3000.16, Vendor Threat Mitigation.

(h) The commander's women, peace, and security advisor provides advice to JFCs and staff on incorporating differences between men and women in the planning and execution of OAI. The women, peace, and security advisor coordinates with staffs' other directorates and functions to incorporate their analysis and expertise into key planning products. For example, a women, peace, and security advisor analyzes patterns

of movement and friendly and adversarial behaviors. They help identify nontraditional COGs to inform joint planning.

For more information on women, peace, and security advising, see United States Strategy on Women, Peace, and Security.

(i) The commander's civilian harm mitigation and response officer coordinates and integrates civilian harm mitigation and response considerations into planning, operations, and assessments.

For more information on staff estimates and sample formats, see Appendix B, "Staff Estimates," and CJCSM 3130.03, Planning and Execution Formats and Guidance. For more information on operational contract support, see JP 4-10, Operational Contract Support; Department of Defense Instruction (DoDI) 3020.41, Operational Contract Support Outside the United States; and CJCSM 4301.01, Planning Operational Contract Support.

(16) **Prepare and Deliver Mission Analysis Brief**

(a) Upon conclusion of the mission analysis, the staff presents a mission analysis brief to the commander. This brief provides the commander with the results of the staff's analysis of the mission, offers a forum to discuss issues, and ensures the commander and staff share a common understanding of the mission. The results inform the commander's development of the mission statement. The commander provides refined planning guidance and intent to guide subsequent planning. Figure IV-8 shows an example mission analysis briefing.

(b) The mission analysis briefing may be the only time the entire staff is present and the only opportunity to make certain all staff members start from a common reference point. The briefing focuses on conclusions reached by mission analysis.

(c) Immediately after the mission analysis briefing, the commander approves a restated mission. This can be the staff's recommended mission statement, a modified version of the staff's recommendation, or one that the commander has developed personally. **Once approved, the restated mission becomes the unit mission.**

(d) At the mission analysis brief, the commander is likely to describe an updated understanding of the OE, the problem, and the vision of the operational approach to the entire assemblage, which should include representatives from supporting CCMDs, subordinate commands, and other partner organizations. This provides the ideal venue for facilitating unity of understanding and vision, which is essential to unity of effort.

(17) **Publish Commander's Refined Planning Guidance.** After approving the mission statement and issuing the intent, the commander may provide the staff (and supporting commands and subordinates in a collaborative environment) with additional guidance (including preliminary decisions) to focus the staff and subordinate planning

Example Mission Analysis Briefing

- Introduction
- Situation overview
 - Operational environment (i.e., operational area) including contested environments extending beyond the operational area, the civilian environment, and a threat overview
 - Political, military, economic, social, information, and infrastructure strengths and weaknesses
 - Enemy (including center[s] of gravity) and objectives
 - Neutral assessment (vulnerabilities and protection)
 - Nonmilitary threat networks (e.g., violent extremist organization, terrorist, criminal and insurgent) organization, composition, and disposition
- Friendly assessment
 - Facts and assumptions
 - Limitations—constraints/restraints
 - Vulnerabilities
 - Capabilities allocated
 - Legal considerations
 - Nonmilitary friendly networks (e.g., local police, local defense groups, local or regional civic groups) and their organization, composition, and disposition
 - Nonmilitary networks
 - Ability to help or hinder mission
- Communication synchronization
- Objectives, effects, and task analysis
 - United States Government interagency objectives
 - Higher commander's objectives/mission/guidance
 - Objectives and effects
 - Specified/implied/essential tasks
 - Centers of gravity
- Operational protection
 - Operational risk
 - Mitigation measures (existing/planned)
- Proposed initial commander's critical information requirements
- Mission
 - Proposed mission statement
 - Proposed commander's intent
- Command relationships analysis and options
- Conclusion—potential resource shortfalls
- Mission analysis approval and commander's course of action planning guidance

Figure IV-8. Example Mission Analysis Briefing

activities during COA development. This refined planning guidance could include new substantive threats (capabilities and intent), objectives, the conditions necessary for success, targeting guidance, levels of risk, and the JFCs visualization of the operational

approach. The commander should provide as much detail as appropriate to provide the right level of freedom to the staff in developing COAs. Planning guidance should also address the role of interorganizational and multinational partners in the pending operation and any related special considerations as required.

(a) Commanders describe their visualization of the forthcoming campaign or operations to help build understanding among the staff. Commanders provide enough guidance (preliminary decisions) to enable the subordinates to plan the action necessary to accomplish the mission consistent with commander's intent. The commander's guidance should focus on the essential tasks and associated operational-level objectives. It emphasizes in broad terms when, where, and how the commander intends to employ military capabilities integrated with other instruments of national power to accomplish the mission within the higher JFC's intent.

(b) The JFC may provide the planning guidance to the entire staff or subordinate JFCs, or meet each staff officer or subordinate unit individually, as the situation and information dictates. Commanders can pass guidance in written or oral form, without prescribed format. However, the guidance should provide a clear direction and detail to avoid unnecessary efforts by the staff or subordinate and supporting commands.

(c) Planning guidance can be very explicit and detailed, or it can be very broad, giving the staff or subordinate commands wide latitude in developing subsequent COAs. However, no matter its scope, arrange the content of planning guidance in a logical sequence to reduce the chances of misunderstanding and to enhance clarity. Moreover, planning guidance is only a starting point; planning enhances preparedness of the CCMD, and the staff must be flexible enough to leverage a given plan and adjust accordingly to address the actual OE. The JFC may issue successive planning guidance during the decision-making process, yet the focus of the JFC's staff should remain upon the framework provided in the initial planning guidance. The commander should continue to provide refined planning guidance during the rest of the planning process while understanding of the problem continues to develop.

(18) **Operational Design Implications.** The commander and staff continue to develop their understanding of the OE and refine their understanding of strategic direction during mission analysis. They continue to define the problem and begin to identify assumptions to support COA development. At this time, the commander should determine the need for multiple options. Options identify different ways, generally broadly defined in scope, to support differing objectives. COAs are subsets of options that identify specific military operations to achieve the objective. The purpose of options is to provide senior decision makers, usually SecDef or the President, the opportunity to better integrate the military within policy decisions. Mission analysis usually concludes with the commander providing refined planning guidance, to include selecting the option to develop for the COA.

d. COA Development (Step 3)

(1) Introduction

(a) **A COA is a potential way (solution, method) to accomplish the assigned mission.** Staffs develop multiple COAs to provide commanders with flexibility to achieve the objective. A good COA accomplishes the mission within the commander's guidance, provides flexibility to meet unforeseen events during execution, and positions the joint force for future operations. It also gives components the maximum latitude for initiative. All COAs are suitable, feasible, acceptable, distinguishable, and complete.

(b) Figure IV-9 shows the key inputs and outputs of COA development. The products of mission analysis drive COA development. Since the operational approach contains the JFC's broad approach to solve the problem at hand, each COA expands this concept with the additional details that describe **who** will take the action, **what type** of

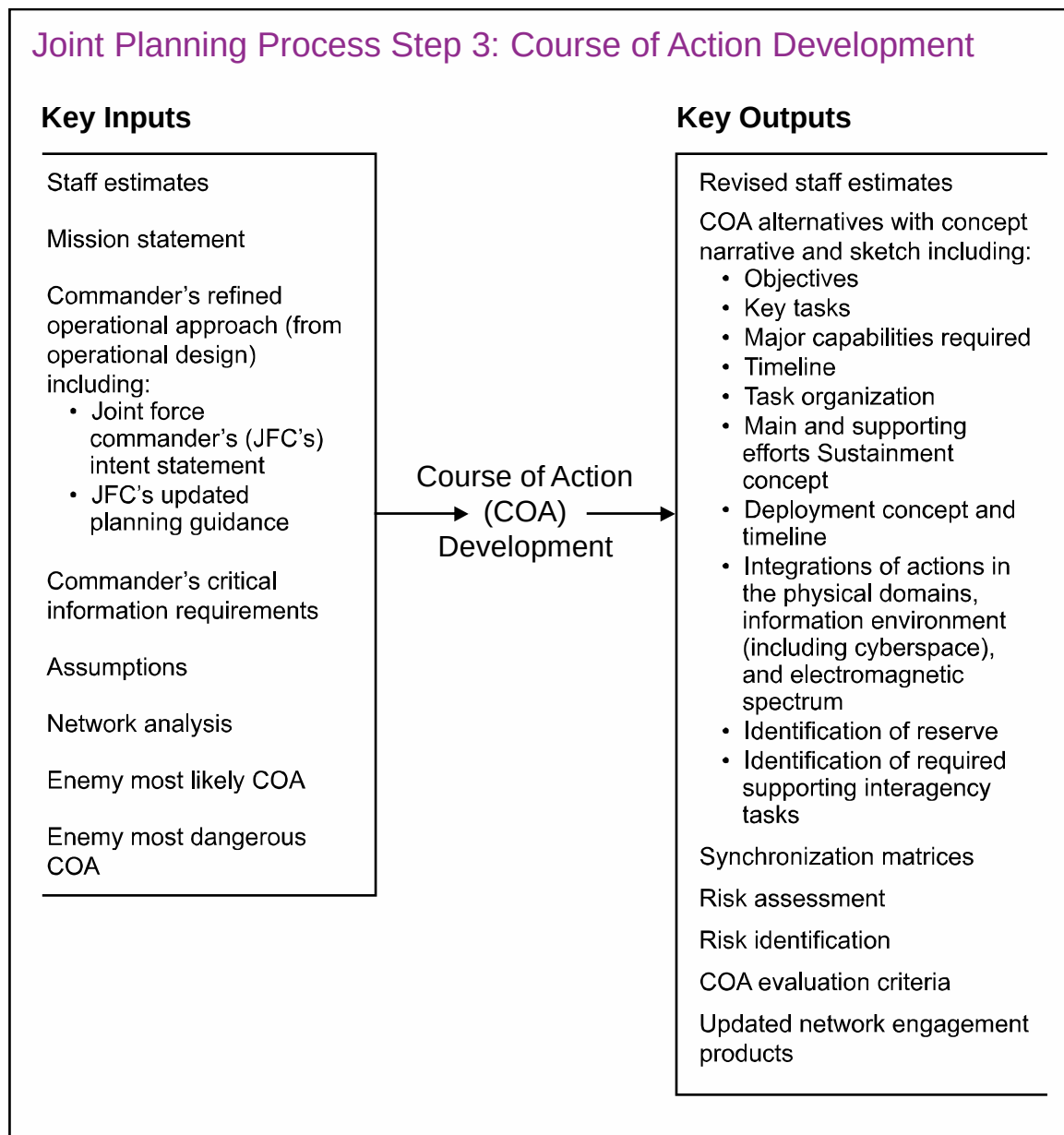


Figure IV-9. Joint Planning Process Step 3: Course of Action Development

military action will occur, **when** the action will begin, **where** the action will occur, **why** the action is required (purpose), and **how** the action will occur (method of employment of forces). Likewise, **the essential tasks identified during mission analysis (and embedded in the draft mission statement) are common to all potential COAs.**

(c) Planners can vary COAs by adjusting the use of joint force capabilities in the physical domains, information environment (including cyberspace), and EMS throughout the OE. Planners may recommend alternate or a range of objectives to distinguish COAs. COAs with different objectives offer senior military leaders and policy makers the opportunity to evaluate military options based on the broader objectives they can plausibly achieve at various levels of cost, risk, and force deployments.

(2) COA Development Considerations

(a) The inputs to COA development include the staff estimates (to include assessed most likely/most dangerous enemy COAs), mission statement, refined operational approach, and CCIRs.

(b) The products of COA development are potential **COAs**, with a sketch for each if possible. Each COA describes, in broad but clear terms, what is to be done throughout the campaign or operation, including consolidation, stabilization, and transition from combat operations; operations in and across the physical domains, the information environment (which includes cyberspace), and the EMS; the size of forces deemed necessary; the time in which joint force capabilities need to be brought to bear; and the risks associated with the COA. These COAs undergo additional validity testing, analysis, modeling/simulation, wargaming, comparison, and if necessary, elimination. These COAs provide conceptualization and broad descriptions of potential CONOPS for the conduct of operations that will achieve the objective.

(c) Developing valid COAs is critical, despite constraints on time and planning resources available. The JFC gives the staff additional considerations early in COA development to focus the effort, helping the staff concentrate on developing the most appropriate COAs. There are always multiple ways to accomplish the mission, but normally, the staff should strike a balance between the time desired for innovation and the time needed to build complete COAs that meet JFC guidance and the operational approach. Many staffs find they have time and resources to develop only two or three distinct COAs per operational approach. If time and personnel resources permit, different teams working in parallel could develop different COAs. This technique may ensure uniqueness while enabling greater output without sacrificing time for creative and critical thinking.

(d) Under some circumstances, COAs are a vehicle for a conversation between senior leaders, who may include civilians, about what a JFC can accomplish with available forces and acceptable levels of cost and risk. This requires the JFC to develop at least one COA to achieve the objective directed by the higher objective. Then the JFC builds one or more COAs to achieve the alternate objective. The COA briefing should compare the anticipated costs, benefits, and risks for all COAs. When presented to a higher headquarters, alternative COAs or CONOPS offer an opportunity to achieve a shared vision between the

JFC and senior leaders about acceptable cost-benefit-risk tradeoffs. This methodology can apply to both planning for contingencies and planning for a crisis at the discretion of the JFC.

(e) For each COA, the commander should envision the employment of all participants in the operation as a whole—US military forces, MNFs, and interagency and multinational partners—while accounting for input from senior leaders during plans reviews, operational limitations, political considerations, the OA, and existing FDOs. However, COA success should only depend on support from participants outside the JFC's control with the explicit approval of higher headquarters. Different levels of outside support often provide a good way to make alternate COAs unique.

(f) During COA development, the commander and staff consider all feasible enemy COAs. Other factors in the OA may also create difficult conditions to consider during COA development.

(g) Each COA typically has an associated initial CONOPS with a narrative and sketch and usually includes the following:

1. OA.
2. Objectives.
3. Essential tasks and purpose.
4. Forces and capabilities required, to include anticipated commercial capabilities, interagency roles, actions, and supporting tasks.
5. Integrated timeline.
6. Task organization.
7. Operational concept.
8. Sustainment concept.
9. Communication synchronization.
10. Risk.
11. Required decisions and decision timeline (e.g., mobilization, DEPOD).
12. Deployment concept.
13. Main and supporting efforts.

(3) COA Development Techniques and Procedures

(a) **Review information** contained in the mission analysis and commander's operational approach, planning guidance, and intent statement. All staff members understand the mission and the tasks to accomplish within the commander's intent to achieve mission success.

(b) Determine the COA Development Technique

1. The first decision in COA development is whether to conduct simultaneous or sequential development of the COAs. Each approach has distinct advantages and disadvantages. The advantage of simultaneous development of COAs is potential time savings. Separate groups are simultaneously working on different COAs. The disadvantage of this approach is that this disrupts the constructive collaboration of the joint planning group by breaking up the team. This approach is also manpower-intensive and requires component and directorate representation in each COA group, and there is an increased likelihood the COAs will lack distinctiveness. While there are potential time savings, experience has demonstrated that it is not an automatic result. The simultaneous COA development approach can work, but planners should address and accept risk of its inherent disadvantages.

2. There are several planning sequence techniques available to facilitate COA development. One option is the step-by-step approach (see Figure IV-10), which uses the backward-planning technique (also known as reverse planning).

(c) Review objectives and tasks and develop ways to accomplish tasks.

Planners review and refine objectives from the initial work done during the development of the operational approach. Tasks stem from CONOPS—intended sequencing and integration of air, land, maritime, special operations, cyberspace, and space forces. Prioritize tasks while considering the enemy's objectives and the need to gain advantage.

1. Regardless of the eventual COA, the staff should plan to accomplish the higher commander's intent by understanding its essential task(s) and purpose and the intended contribution to the higher commander's mission success.

2. The staff ensures all COAs developed fulfill the command mission and the purpose of the operation by conducting a review of all essential tasks developed during mission analysis. They should then consider ways to accomplish the other tasks.

(d) **Timing.** Once the staff has begun to visualize COA alternatives, they determine how best to arrange the actions of all the elements of the force. The staff estimates when key tasks need to occur over the duration of the operation. One method of arranging actions is the use of phasing, as discussed earlier. Planners should then **arrange and integrate** these requirements by using the joint functions of C2, intelligence, fires, movement and maneuver, protection, sustainment, and information. At a minimum, planners should make certain the arranged actions answer the following questions:

Step-by-Step Approach to Course of Action Development

Step Action

- 1 Within the limits of available forces, determine how much force will be needed in the theater at the end of the operation or campaign, what those forces will be doing, and how those forces will be postured geographically. Use troop-to-task analysis. Draw a sketch to help visualize the forces and their locations.
- 2 Looking at the sketch and working backwards, determine the best way to get the forces postured in Step 1 from their ultimate positions at the end of the operation or campaign to a base in friendly territory. This will help formulate the desired basing plan.
- 3 Using the mission statement as a guide, determine the tasks the force must accomplish in the physical domains, information environment (including cyberspace), and electromagnetic spectrum to achieve the desired objective. Draw a sketch of the maneuver plan. Make sure the force does everything the Secretary of Defense has directed the commander to do (refer to specified tasks from the mission analysis).
- 4 Determine the basing required to posture the force in friendly territory, and the tasks the force must accomplish to get to those bases. Sketch this as part of the deployment plan.
- 5 Determine if the planned force is enough for the joint force commander to accomplish the outcomes of the strategic guidance.
- 6 Given the required tasks, determine the order the forces should deploy into theater. This prioritization includes forces necessary for mobilization, deployment, force protection, joint reception staging, onward movement, and integration.
- 7 The information developed should now allow determination of force employment, major tasks and their sequencing, sustainment, and command relationships.

Figure IV-10. Step-by-Step Approach to Course of Action Development

1. How do land, maritime, air, space, cyberspace, and special operations forces integrate across the joint functions to accomplish their assigned tasks?

2. How does the joint force leverage the informational aspects of military activities to create relevant actor perceptions and drive relevant actors to behave in ways that support achieving the JFC's objectives?

3. What permissions and authorities are needed to execute the synchronized actions?

4. What timeline, directed dates, time limitations, or transitions are driving the establishment of D-Day or M-Day?

OPERATION MILESTONES

C-Day: The day deployment begins.

D-Day: The day operations began or are scheduled to begin.

H-Hour: The time (on D-Day) the operation is scheduled to begin.

L-Hour: The time (on C-Day) deployment operations begin.

M-Day: The day mobilization (partial or full) begins.

N-Day: The day a unit (active duty) is notified for deployment or redeployment.

(e) The COAs should focus on **COGs and decisive points**. The commander and the staff review and refine their COG analysis, begun during mission analysis, based on updated intelligence, JIPOE products, and initial staff estimates. They consider the refined enemy and friendly COG analysis, particularly the critical vulnerabilities, in the development of the initial COAs. The COG analysis helps orient the commander to the enemy and compare friendly strengths and weakness with those of the enemy. By looking at friendly COGs and vulnerabilities, the staff understands the capabilities of their own force and critical vulnerabilities that require protection. Protection resource limitations probably mean the staff cannot plan to protect every capability but, rather, look at prioritizing protection for critical capabilities and developing overlapping protection techniques. The strength of one force or capability may provide protection from the weakness of another.

(f) **Identify the sequencing** (simultaneous, sequential, or a combination) of the actions for each COA. Understand which resources become available, and when, during the operation or campaign. Resource availability significantly affects sequencing operations and activities.

(g) **Identify main and supporting efforts by phase**, the purposes of these efforts, and key supporting/supported relationships within phases.

(h) **Identify decision points with associated CCIRs**. The commander needs to know when to make a critical decision or when they achieve an objective. This requires integration of decision points and assessment criteria into the COA, as these processes anticipate a potential need for decisions from outside the command (SecDef, the President, or other command).

(i) **Identify component-level missions/tasks** (who, what, and where) that accomplish the stated purposes of main and supporting efforts. Think of component tasks in terms of the joint functions. Display them with graphic control measures as much as possible. A designated LOO or LOE helps identify these tasks.

(j) **Task Organization**

1. The staff should develop an outline task organization to execute the COA. The commander and staff determine appropriate command relationships and appropriate missions and tasks.

2. Determine command relationships and organizational options. Joint force organization and command relationships are based on the operation or campaign CONOPS, complexity, and degree of control required. Establishing **command relationships** includes determining the types of subordinate commands and the degree of authority delegated to each. Clear definition of command relationships further clarifies the intent of the commander and contributes to decentralized execution and unity of effort. The commander has the authority to determine the types of subordinate commands from several doctrinal options, including Service components, functional components, and subordinate joint commands. **Regardless of the command relationships selected, it is the JFC's responsibility to ensure these relationships are clear to all subordinate, adjacent, and supporting headquarters.** The following are considerations for establishing joint force organizations:

a. Joint forces normally organize with a combination of Service and functional components with operational responsibilities.

b. Functional component staffs should be joint, with Service representation in approximate proportion to the mix of subordinate forces. These staffs should organize and train prior to employment to be efficient and effective, which requires advanced planning.

c. Commanders may establish support relationships between peers and components to facilitate operations.

d. Commanders define the authority and responsibilities of functional component commanders, based on the strategic CONOPS, and may alter their authority and responsibility during an operation.

e. Commanders balance the need for centralized direction with decentralized execution.

f. Major changes in the joint force organization normally occur at phase changes.

(k) **Sustainment Concept. No COA is complete without a proper sustainment plan.** The sustainment concept is more than just gathering information on various logistic and personnel services. It entails identifying the requirements for all classes of supply (e.g., national stock number-level) and services and creating distribution, transportation, operational contract support, and disposition plans to support the commander's execution. Additionally, JFCs group joint medical capabilities across the task organization and position appropriately throughout the OA. Sustainment concepts also organize capabilities and resources into an overall campaign or operation sustainment

concept. It concentrates forces and material resources strategically, so the right force is available at the designated times and places to conduct decisive operations. It requires thinking through a cohesive sustainment for joint, single-Service, and supporting forces relationships in conjunction with combat support agencies, multinational, interagency, nongovernmental, private-sector, or international organizations.

(l) **Deployment Concept.** A COA includes a deployment concept to describe the general flow of forces (organic and nonorganic) into theater. There is no way to determine the feasibility of the COA without including the deployment concept and how the force will respond to a contested environment with enemy attacks on force flow. The concept should describe force buildup, sustainment requirements, and military-political considerations. The concept should account for how cohesive military actions in time, space, and purpose address transregional, all-domain, multifunctional challenges.

(m) **Nuclear Planning**

1. COA development includes nuclear planning, as required. Commanders assess the military and strategic impact of nuclear weapons employment on joint operations. This includes conventional-nuclear integration considerations, which consist of:

- a. The joint force's ability to survive the use of nuclear weapons.
- b. Planning and executing in a nuclear environment.
- c. Conventional support to nuclear operations.
- d. Nuclear support to conventional operations, such as when nuclear capabilities are required to deter strategic attack, reestablish deterrence, or manage escalation.
- e. The simultaneous employment of capabilities to generate effects within an option.

2. Presidential policy provides nuclear planning guidance and further clarified in other DoD documents, such as the nuclear supplement to the JSCP. Guidance issued to the CCDR is based on national-level considerations and supports the achievement of US objectives. United States Strategic Command is the lead organization for nuclear planning and coordination with appropriate allied commanders. United States Strategic Command, in collaboration with the other CCMDs, ensures optimal integration of US nuclear and conventional forces prior to, during, and after conflict. United States Strategic Command uses this framework to develop detailed mission plans for the appropriate nuclear forces. Due to the strategic and diplomatic consequences associated with nuclear operations and plans, only the President has the authority to direct the planning and employment of nuclear weapons.

For additional information, see JP 3-72, Joint Nuclear Operations.

(n) Define the OA

1. The OA is an overarching term that can encompass more descriptive terms for geographic areas. It provides flexibility/options or limitations to the commander. The specific geographic area impacts planning factors such as access, basing, overflight, and sustainment.

2. OAs include specific geographic areas such as AOR, theater of war, theater of operations, joint OA, amphibious objective area, joint special OA, and area of operations. CCDRs, with assigned AORs and their subordinate JFCs, designate smaller OAs on a temporary basis. OAs have physical dimensions composed of some combination of air, land, maritime, and space domains. JFCs define these areas with geographical boundaries, which facilitate the coordination, integration, and deconfliction of joint operations among joint force components and supporting commands. The size of these OAs and the types of forces employed within them depend on the scope and nature of the crisis and the projected duration of operations.

See JP 3-0, Joint Campaigns and Operations, for additional information on OAs.

(o) Integrate and synchronize capabilities that create effects across all domains within individual COAs (e.g., incorporating special access program-protected capabilities into COAs). This may include:

1. Conducting operations with authorized allies and other partners.
2. Supporting allies and other partners requests for use of US capabilities through integrated joint special technical operations.
3. Including special access program capabilities into CONPLANs and OPLANs and seeking early approval.
4. As appropriate, training and exercising all aspects of integrated special access program planning and execution.
5. Ensuring planners identify capabilities that may offer asymmetric advantages and matching those advantages to main or supporting efforts. This facilitates eventual development of a joint all-domain operations CONOPS and subordinate orders (e.g., integrated tasking order).
6. JFCs emphasizing the importance of avoiding siloed planning efforts by empowering planners to operate at a common security level through use of integrated planning and its inherent collaboration.

(p) **Develop Initial COA Sketches and Statements.** Each COA should answer the following questions:

1. Who (type of forces) will execute the tasks?
2. What are the tasks?
3. Where will the tasks occur? (Start adding graphic control measures, [e.g., areas of operation, amphibious objective areas].)
4. When will the tasks begin?
5. What are key/critical decision points?
6. How (but do not usurp the components' prerogatives) should the commander provide "operational direction" so the components can accomplish "tactical actions."
7. Why (for what purpose) will each force conduct its part of the operation?
8. How will the commander assess mission accomplishment?
9. What is the initial intelligence support concept?

(q) **Test the Validity of Each COA.** All COAs selected for analysis are valid, and the staff should reject COA alternatives that do not meet all five of the following validity criteria:

1. **Suitable**—Can accomplish the mission within the commander's guidance. Preliminary tests include:
 - a. Does it accomplish the mission?
 - b. Does it meet the commander's intent?
 - c. Does it accomplish all the essential tasks?
 - d. Does it meet the conditions for the objective?
 - e. Does it take into consideration the enemy and friendly COGs?
 - f. Where appropriate, are security objectives influenced by demographics of the local population?

2. Feasible—Can accomplish the mission within the established time, space, and resource limitations.

a. Does the commander have the force structure, posture, transportation, and logistics (e.g., munitions) (means) to execute it? The COA is feasible if executable with the forces, support, and technology available within the constraints of the OE and against expected enemy opposition.

b. Although this process occurs during COA analysis and the test at this time is preliminary, it may be possible to declare a COA infeasible (for example, resources are obviously insufficient). However, it may be possible to fill shortfalls by requesting support from the commander or other means.

3. Acceptable—Balances cost and risk with the advantage gained.

a. Does it contain unacceptable risks? (Is it worth the possible cost?) A COA is acceptable if the estimated results justify the risks. The basis of this test consists of an estimation of friendly losses in forces, time, position, and opportunity.

b. Does it consider the limitations placed on the commander (constraints, “must do;” restraints, “cannot do;” other physical or authority limitations)?

c. Are COAs reconciled with external limitations, including US and international law, USG policy, ROE, and acceptable risk? This requires visualization of execution of the COA against each enemy capability.

d. Although this process occurs during COA analysis and the test at this time is preliminary, it may be possible to declare a COA unacceptable if it violates the commander’s definition of acceptable risk.

4. Distinguishable—Are sufficiently different from other COAs in the following:

a. The focus or direction of main effort.

b. The scheme of maneuver, including integration and sequencing of actions in the physical domains, information environment (including cyberspace), and EMS.

c. Sequential versus simultaneous maneuvers.

d. The primary mechanism for mission accomplishment.

e. Task organization(s).

f. The use of reserves.

5. Complete—Does it answer the questions who, what, where, when, how, and why? The COA incorporates:

- a. Objectives, desired effects to create, and tasks to perform.
- b. Major forces and capabilities required, to include the forces and capabilities of international partners.
- c. Concepts for deployment, employment, and sustainment.
- d. Time estimates for achieving objectives.
- e. Mission success criteria (including the assessment: how the commander will know they have achieved success).

(r) Conduct COA Development Brief to Commander. Figure IV-11 provides suggested sequence and content.

(s) JFC Provides Guidance on COAs

- 1. Review and approve COA(s) for further analysis.
- 2. Direct revisions to a COA(s), combinations of COAs, or development of an additional COA(s).
- 3. Direct priority for which enemy COA(s) (most dangerous/most likely) to use during wargaming of a friendly COA(s).

(t) Continue the Staff Estimate Process. The staff continues to conduct their staff estimates of supportability for each COA.

(u) Conduct Vertical and Horizontal Concurrent Planning

- 1. Discuss the planning status of staff counterparts with both commander's and JFC components' staffs.
- 2. Coordinate planning with staff counterparts from other functional areas.
- 3. Permit adjustments in planning to accommodate additional details learned from higher and adjacent echelons, while allowing lower echelons to begin planning efforts to generate questions (e.g., requests for information).

e. COA Analysis and Wargaming (Step 4)

(1) Introduction

Example Course of Action Development Briefing

- Operations Directorate of a Joint Staff (J-3)/Plans Directorate of a Joint Staff (J-5)
 - Context/background (i.e., road to war)
 - Initiation—review guidance for initiation
 - Strategic guidance—planning tasks assigned to supported commander, forces/resources apportioned, planning guidance, updates, defense agreements, theater campaign plan(s), Contingency Planning Guidance/Joint Strategic Campaign Plan
 - Forces allocated/assigned
- J-2 [Intelligence Directorate of a Joint Staff]
 - Joint intelligence preparation of the operational environment
 - Enemy objectives
 - Enemy courses of action (COAs)—most dangerous, most likely; strengths and weaknesses
 - Nonmilitary threat networks effect on possible COAs
- J-3/J-5/J-9 [Civil-military Operations Directorate of a Joint Staff]
 - Update facts and assumptions
 - Mission statement
 - Commander's intent (purpose, method, objective)
 - Objective political/military transition criteria
 - Center of gravity analysis results: critical factors; strategic/operational
 - Joint operations area/theater of operations/communications zone sketch
 - Shaping activities recommended (for current theater campaign plan)
 - Flexible deterrent options with desired effect
 - For each COA, sketch and statement by phase
 - task organization
 - component tasking
 - timeline
 - recommended command and control by phase
 - lines of operation/lines of effort—logistics estimates and feasibility
 - COA risks
 - synchronization matrices
 - COA summarized distinctions
 - COA priority for analysis
 - Operations in the information environment
 - COA risks and opportunities from friendly nonmilitary networks and by friendly military forces.
 - COA risks and opportunities from neutral networks.
- Update COA Development Briefing to include:
 - Red objectives
- Commander's Guidance
- Reliance on commercial support after "Forces allocated/assigned" in the section describing J-3 and J-5

Figure IV-11. Example Course of Action Development Briefing

(a) COA analysis and wargaming are iterative and ensure outcomes inform COA development. COA analysis is the process of closely examining potential COAs to reveal details that enable the commander and staff to tentatively evaluate COA validity and identify the advantages and disadvantages of each proposed friendly COA. The commander and staff analyze each COA separately according to the commander's guidance. COA analysis is a valuable use of time that ensures COAs are valid.

(b) **Wargaming** is a primary means for COA analysis. Wargames are representations of conflict or competition in a synthetic environment, in which people make decisions and respond to the consequences of those decisions. COA wargaming is a conscious attempt to visualize the flow of the operation, given joint force strengths and dispositions, adversary capabilities and possible COAs, the OA, and other aspects of the OE. Each critical event within a proposed COA should be wargamed based upon time available using the action, reaction, and counteraction method of friendly or opposing force interaction. Modify the basic COA wargaming method to fit the specific mission and OE and apply to combat, noncombat, and CCP activities. Wargaming is most effective when it involves the following elements:

1. A well-developed, valid COA.
2. People making decisions.
3. A fair, competitive environment (i.e., the game should have no rules or procedures designed to tilt the playing field toward one side or another).
4. Adjudication.
5. Consequences of actions.
6. Iteration (i.e., new insights gained through iteration).

(c) COA wargaming involves a detailed evaluation of each COA as it pertains to the enemy and the OE. Each of the selected friendly COAs is then wargamed against selected enemy or OE COAs, as well as other relevant actor actions as applicable (e.g., wargaming CCMD campaign activities can provide insights about how an HN or others might react/respond). The commander selects the COAs for wargaming and provides guidance along with refined evaluation criteria.

(d) Wargaming stimulates thought about the operation so the commander can obtain ideas and insights that otherwise might not have emerged. An objective, comprehensive analysis of COA alternatives is difficult even without time constraints. Based upon time available, the commander should wargame each COA against the most likely and the most dangerous adversary COAs (or most difficult objectives in noncombat and campaign operations) identified through the JIPOE process. Figure IV-12 shows the key inputs and outputs associated with COA analysis.

(2) Analysis and Wargaming Process

(a) The analysis and wargaming process can be as simple as a detailed narrative effort that describes the action, probable reaction, counteraction, forces, and time used. A more comprehensive version is the “sketch-note” technique, which adds operational sketches and notes to the narrative process to gain a clearer picture. Sophisticated wargames employ more extensive means to depict a range of actions by competitors and the consequences of those actions. The most sophisticated form of wargaming is one where representatives emulate all competitors involved in armed conflict and fully explore the interactions within the OA. Modeling and simulation are distinct and separate analytic tools and different from wargames. Modeling and simulation can be complementary and assist wargaming through bookkeeping, visualization, and adjudication for well-understood actions. Planners use red teaming to support COA

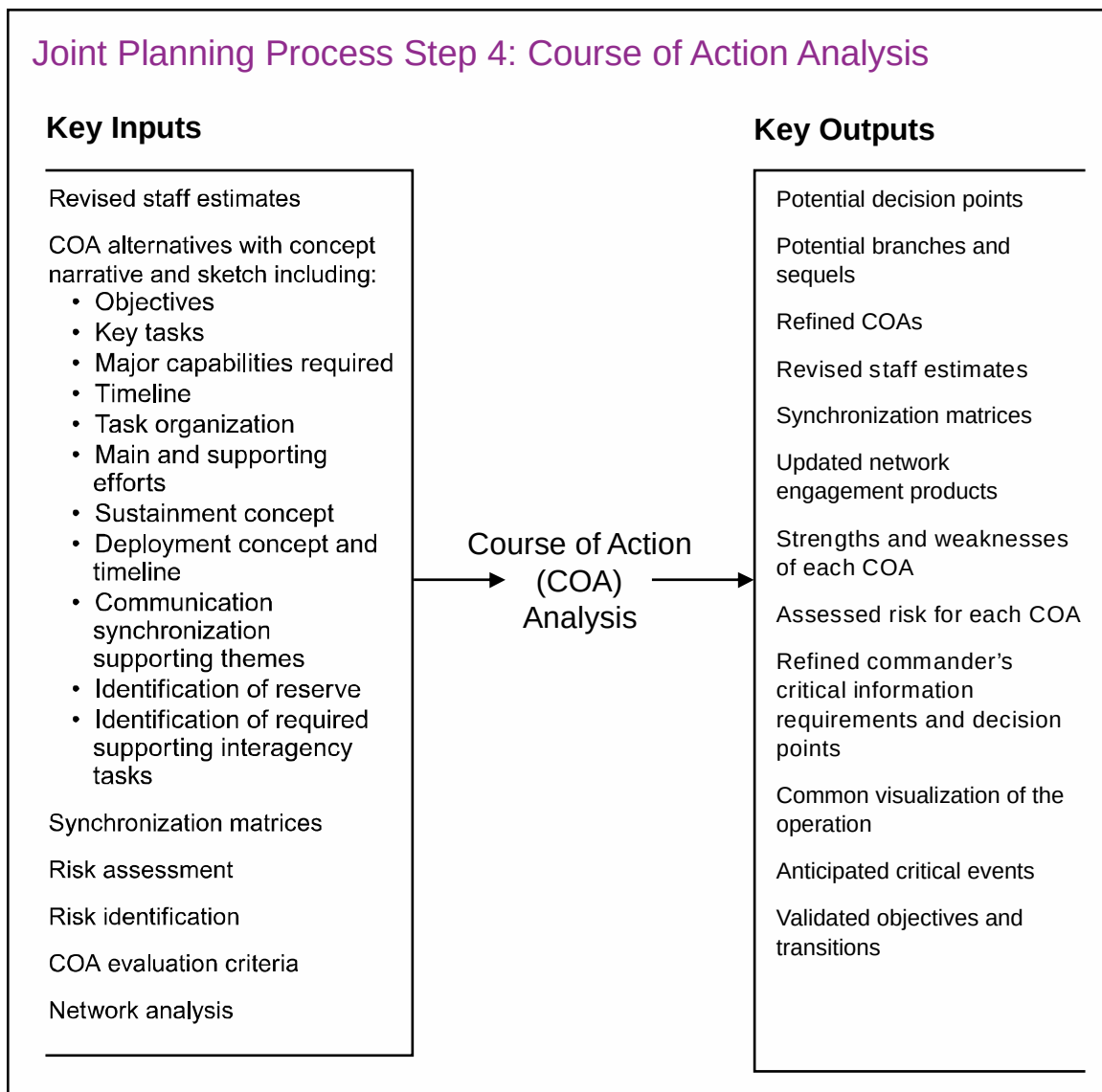


Figure IV-12. Joint Planning Process Step 4: Course of Action Analysis

analysis and wargaming. Red teaming enables understanding of implications, potential unintended consequences, and likely second- and third-order effects associated with each COA.

(b) Analysis of multiple COAs is vital to the commander's estimate process. The items selected for wargaming and COA comparison depend on the nature of the mission. For plans or orders involving contested deployment and combat operations, the staff considers opposing COAs based on enemy capabilities, objectives, an estimate of the enemy's intent, and activities of other relevant actors that are not favorable or that challenge achievement of the objective. For noncombat operations or CCPs, the staff may analyze COAs based on partner capabilities, partner and US planned activities and objectives, likely actions of other relevant actors, criticality, and risk.

(c) In the analysis and wargaming step, the staff analyzes the probable effect each opposing COA has on the chances of success of each friendly COA. The objective is to develop a basis for determining the feasibility and acceptability of the COAs. Analysis also provides the planning staff with a greatly improved understanding of their COAs and the relationship between them. COA analysis identifies which COA best accomplishes the mission while best positioning the force for future operations. It also helps the commander and staff to:

1. Determine how to maximize combat power against the enemy while protecting the friendly forces and minimizing collateral damage in combat or maximize the effect of available resources toward achieving JFCs' operational-level objectives.
2. Have as near an identical visualization of the operation as possible.
3. Anticipate adversary, enemy, and other relevant actor actions/events in the OA and potential reaction options.
4. Determine conditions and resources required for success, while also identifying gaps and seams.
5. Determine when and where to apply the force's capabilities.
6. Plan for and coordinate authorities to integrate information activities early.
7. Focus intelligence and operation assessment requirements.
8. Determine the most flexible COA.
9. Identify potential decision points.
10. Determine task organization options.

11. Develop data for use in a synchronization matrix or related tool.
12. Identify potential plan branches and sequels.
13. Identify high-value targets.
14. Assess risk to friendly forces, HN partners and infrastructure, and the civilian environment.
15. Determine COA advantages and disadvantages.
16. Recommend CCIRs.
17. Validate objectives.
18. Identify contradictions between friendly COAs and expected enemy objectives.
19. Identify the potential impact to the civilian environment.

(d) Wargaming is a disciplined process, with rules and steps that attempt to visualize the flow of the operation. The process considers friendly dispositions, strengths, and weaknesses; adversary, enemy, and other relevant actor capabilities and intent set forth as probable COAs; and characteristics of the political, information, commercial/economic, civilian, and physical environment. It relies heavily on joint doctrinal foundation, tactical judgment, and operational and regional/area experience. It focuses the staff's attention on each phase of the operation in a logical sequence. It is an iterative process of action, reaction, and counteraction. Wargaming stimulates ideas and provides insights otherwise not realized. It highlights critical tasks and provides familiarity with operational possibilities otherwise difficult to achieve. Wargaming is a critical portion of the planning process and should be allocated significant time. **Each retained COA should be wargamed against both the most likely and most dangerous enemy COAs and address contested environments across both the strategic environment and the OE.**

(e) During the wargame, the staff takes a COA statement and begins to add more detail to the concept, while determining the strengths or weaknesses of each COA, including the associated deployment and distribution requirements. Wargaming tests a COA and can provide insights to improve a developed COA. The commander and staff (and subordinate commanders and staffs) may change an existing COA or develop a new COA after identifying unforeseen critical events, tasks, requirements, or problems.

(f) For the wargame to be effective, the commander should indicate what aspects of the COA to examine. Wargaming guidance should include the list of friendly COAs to be wargamed against specific threat COAs (e.g., COAs 1, 2, and 3 against the enemy's most likely and most dangerous COAs), including contested environments impacting deployment and distribution operations, the timeline for the phase or stage of

the operations, a list of critical events, and level of detail (i.e., two levels down). For a valid COA comparison (JPP step 5), planners wargame each friendly COA against the same set of threat COAs.

(g) **COA Analysis Considerations.** Evaluation criteria and known critical events are two of the many important considerations as COA analysis begins.

For more information, see Appendix E, “Course of Action Comparison.”

1. Evaluation criteria change from mission to mission. It will be helpful during future wargaming steps for all participants to be familiar with the criteria to record any insights that influence a criterion for later comparison. The criteria may include anything the commander desires. The staff can derive criteria from the commander’s intent statement. Evaluation criteria do not stand alone and require clear definition. Precisely defining criteria reduces subjectivity and ensures consistent evaluation. The following sources provide a good starting point for developing a list of potential evaluation criteria:

- a. Commander’s guidance and commander’s intent.
- b. Mission accomplishment at an acceptable cost, which may include impacts to other global requirements.
- c. The principles of joint operations.
- d. Doctrinal fundamentals for the type of operation conducted.
- e. The level of residual risk to mission and force in the COA.
- f. Implicit significant factors relating to the operation (e.g., need for speed, security).
- g. Factors relating to specific staff functions.
- h. Elements of operational design.
- i. Other factors planners consider are diplomatic or political constraints, residual risks, the protection and restoration of the civilian environment, financial costs, flexibility, simplicity, surprise, speed, mass, sustainability, C2, and capability and infrastructure survivability.

2. **List Known Critical Events.** These are essential tasks, or a series of critical tasks, that require detailed analysis (such as the series of component tasks performed on D-day).

- a. This may expand to review component tasks over a phase(s) of an operation or over a period (C-day through D-day). The planning staff may wish at this

point to also identify decision points (those decisions in time and space that the commander makes to ensure timely execution and synchronization of resources). These decision points link to a critical event (e.g., commitment of the reserve force).

b. For CCMD campaigns, this includes identifying linked events and activities: the staff identifies if campaign activities are sensitive to the execution sequence and if subsequent activities are dependent on the success of earlier ones. If committing resources for an activity early in the campaign, the staff identifies to the commander the impact of the loss of that event (or if the results were different from those anticipated), a decision point to continue subsequent events, and alternates if the planned events were dependent on earlier ones.

(h) **There are three key decisions to make before COA analysis begins.** The first decision is to decide what type of wargame to use. This decision should stem from commander's guidance, time, resources available, staff expertise, and availability of simulation models. The second decision is to prioritize the enemy COAs or the friendly partner capabilities to analyze. Finally, planners choose the criteria for assessing the COAs before the wargame.

(3) **Conducting the Wargame**

(a) The primary steps are prepare for the wargame, conduct the wargame, evaluate the results, and prepare products. Figure IV-13 shows sample wargaming steps.

(b) **Prepare for the Wargame**

1. Wargames are either manual or digital. There are many forms of digital wargames; most require a significant amount of preparation to develop and load scenarios and then to train users. However, the potential to use the digital model for multiple scenarios or blended scenarios makes it valuable. For both types, consider how to organize the participants in a logical manner.

2. For manual wargaming, three distinct methods are available to run the event:

a. **Deliberate Timeline Analysis.** Consider actions day-by-day or in other discrete blocks of time. This is the most thorough method for detailed analysis when time permits.

b. **Phasing.** Used as a framework for COA analysis. Identify significant actions and requirements by functional area or joint task force (JTF) component.

c. **Critical Events/Sequence of Essential Tasks.** The sequence of essential tasks, also known as the critical events method, highlights actions necessary to establish the conditions for future operations, such as a sustainment capability and engage enemy units in the deep battle area. At the same time, the critical events method enables

Sample Wargaming Steps

Step Action

- 1 Prepare for the Wargame
 - Gather tools
 - List and review friendly forces and capabilities
 - List and review opposing forces and capabilities
 - List known critical events
 - Determine participants
 - Determine opposing course of action (COA) to wargame
 - Select wargaming method
 - manual or computer-assisted
 - Select a method to record and display wargaming results
 - narrative
 - sketch and note
 - war game worksheets
 - synchronization matrix
- 2 Conduct Wargame and Assess Results
 - Purpose of wargame (identify gaps, visualization)
 - Basic methodology (e.g., action, reaction, counteraction)
 - Record results
- 3 Prepare Products
 - Results of the wargame brief
 - potential decision points
 - evaluation criteria
 - potential branches and sequels
 - Revised staff estimates
 - Refined COAs
 - Time-phased force and deployment data refinement and transportation feasibility
 - Feedback through the COA decision brief

Figure IV-13. Sample Wargaming Steps

the planners to adapt if the enemy or others in the OA react in such a way that necessitates reordering of the essential tasks. This technique also enables wargamers to analyze concurrently the essential tasks required to execute the CONOPS. Focus on specific critical events that encompass the essence of the COA. If necessary, develop different measures for assessing different types of critical events (e.g., destruction, blockade, air control, neutralization, ensure defense). As with the focus on phasing, the critical events discussion identifies significant actions and requirements by functional area or by JTF component and enables a discussion of possible or expected reactions to execution of critical tasks.

3. Red Cell. The J-2 staff, augmented by supporting CCMD J-2 personnel provides a red cell to role-play and model the enemies and others in the OA during planning and specifically during wargaming.

a. A robust, well-trained, imaginative, and skilled red cell that aggressively pursues the enemy's point of view during wargaming is essential. By

accurately portraying the full range of realistic capabilities and options available to the enemy to include information activities, which include cyberspace operations and some electromagnetic and counter-space capabilities. For campaign and noncombat operation planning, the red cell provides expected responses to US actions, based on their knowledge and analysis of the OE. In some cases, the red cell is synonymous with the red team.

b. The red cell should be composed of personnel from the joint force J-2 staff and augmented by other subject matter experts.

c. The red cell develops critical decision points, projects enemy and other OE reactions to friendly actions, and estimates impacts and implications on the enemy forces and objectives. By trying to win the wargame, the red cell helps the staff identify weaknesses and vulnerabilities before a real enemy does.

d. Given time constraints, as a minimum, the most dangerous and most likely COAs should be wargamed and role-played by the red cell during the wargame and include plausible enemy contested environment actions targeting deployment and distribution operations.

4. White Cell. A small cell of arbitrators normally composed of senior individuals familiar with the plan is a smart investment to ensure the wargame does not get bogged down in unnecessary disagreement or arguing. The white cell provides overall oversight to the wargame and any adjudication required between participants. The white cell may also include the facilitator or highly qualified experts as required.

5. In addition to a red cell and a white cell, there should also be a blue cell that represents friendly forces and a green cell representing civilians, transnational groups, NGOs, and others in the OA.

(c) Conduct the Wargame and Evaluate the Results

1. The facilitator and the red cell chief get together to agree on the rules of the wargame. The wargame begins with an event designated by the facilitator. It could be an enemy offensive or defensive action, a friendly offensive or defensive action, or some other activity such as a request for support or campaign activity. They decide where (in the OA) and when (H-hour or L-hour) it will begin. They review the initial array of forces and the OE. Of note, they must come to an agreement on the effectiveness of capabilities and previous actions by both sides prior to the wargame. The facilitator must ensure all members of the wargame know what events to wargame and the techniques employed. This requires advanced coordination within the blue cell and between the blue cell and the red cell.

2. Each COA wargame has several turns, each consisting of three total moves: action, reaction, and counteraction. If necessary, each turn of the wargame may extend beyond the three basic moves. The facilitator, based on JFC guidance, decides how many total turns to conduct in the wargame.

3. During the wargame, the participants continually evaluate the COA's feasibility. Is it supportable? Can it accomplish the objective? Will it achieve the desired results? How will adversaries, enemies, and other relevant actors react? Are more forces, resources, intelligence collection capabilities, or time needed? Are necessary logistics and communications available? Is the OA large enough? Has the threat successfully impacted key enablers, like logistics or communications, or countered a certain phase or stage of a friendly COA? Based on the answers to the above questions, planners may need to revise the friendly COA. Do not revise a COA during a wargame. Instead, stop the wargame, make the revisions, and start over at the beginning.

4. The wargame is for comparing friendly COAs with the enemy COAs. Planners compare friendly COAs, and the associated outcomes and risks, with each other in JPP step 5—COA comparison. Planners avoid becoming emotionally attached to a friendly COA and avoid comparing one friendly COA with another friendly COA during the wargame so they can remain unbiased. The facilitator ensures adherence to the timeline. The facilitator allocates enough time to ensure the wargame thoroughly tests a COA.

5. A **synchronization matrix** is a decision-making tool and a method of recording the results of wargaming. Key results to record include decision points, potential evaluation criteria, CCIRs, COA adjustments, branches, and sequels. Using a synchronization matrix helps the staff, supporting commands, and component commands visually synchronize the COA across time and space in relation to the enemy's possible COAs or other activities that may impact the OE and broader strategic environment. The wargame and synchronization matrix efforts are useful in identifying cross-component support resource requirements.

6. The wargame considers friendly dispositions, strengths, and weaknesses; enemy forces and likely COAs; and characteristics of the OA, to include elements of the civilian environment. Through a logical sequence, it focuses the participants on essential tasks to accomplish. In addition to the synchronization matrix, the wargame captures tasks to specific units. The goal of the planning process, to include wargaming, is to produce an order or plan providing direction and tasks to subordinate and supporting organizations.

7. After completing the wargame, worksheet, and synchronization matrix, there should be enough detail to flesh out the bones of the COA and begin orders development (once the commander selects the COA in a later JPP step).

8. Additionally, the wargame will produce a refined event template and the initial decision support template (DST), decision points (and the CCIR related to them), or other decision support tools. Wargaming can help predict what adversaries, enemies, and other relevant actors will do and how partner nations or others will react to US actions or take advantage of the situation to achieve their respective objectives. Wargaming also provides the commander options for employing forces to counter an enemy action. Wargaming prepares the commander and the staff for a wide range of possibilities and a choice of immediate solutions.

9. The wargame relies heavily on doctrinal foundation, tactical and operational judgment, and experience. Wargaming generates new ideas and provides insights that otherwise may be overlooked. The dynamics of the wargame require the red cell to be aggressive, but realistic, in the execution of threat activities. The wargame:

a. Records advantages and disadvantages of each COA as they become evident.

b. Creates decision support tools (a game plan).

c. Focuses the planning team on the threat and commander's evaluation criteria.

d. Provides insights on what pre-conflict campaign activities could reduce risk and improve the chances of operational success.

(d) **Prepare Products.** Certain products should result from the wargame in addition to wargamed COAs. Planners enter the wargame with a rough event template and complete the wargame with a refined, more accurate event template. The event template with its named areas of interest and time-phase lines helps the J-2 focus the intelligence collection effort. An event matrix can serve as a “script” for intelligence reporting during the wargame. It can also tell planners if they are relying too much on one or two collection platforms and if their plan overextends assets.

1. A first draft of a DST should also come out of the COA wargame. As more information about friendly forces and threat forces becomes available, the DST may change.

2. The critical events are associated with the essential tasks identified in mission analysis. The decision points tie to points in time and space when and where the commander makes a critical decision. Decision points should tie to the CCIRs. CCIRs generate two types of information requirements: PIRs and FFIRs. The commander approves CCIRs. From a threat perspective, PIRs tied to a decision point require an intelligence collection plan that prioritizes and tasks collection assets to gather information about the threat. JIPOE ties PIRs to named areas of interests, which link to enemy COAs. The synchronization matrix is a tool that helps determine if adequate resources are available. Primary outputs are:

a. Wargamed COAs with graphic and narrative. Branches and sequels identified.

b. Information on commander's evaluation criteria.

c. Initial task organization.

d. Critical events and decision points.

- e. Newly identified resource shortfalls.
- f. Refined/new CCIRs and event template/matrix.
- g. Initial DST/DSM.
- h. Refined synchronization matrix.
- i. Refined staff estimates.
- j. Assessment plan and criteria.

3. The outputs of the COA wargame inform the JPP steps COA comparison, COA approval, and plan or order development. The results of the wargame are an understanding of the strengths and weaknesses of each friendly COA, the core of the back brief to the commander.

4. The commander and staff normally compare advantages and disadvantages of each COA during COA comparison. However, if the suitability, feasibility, or acceptability of any COA becomes questionable during the analysis step, the commander should modify or discard it and concentrate on other COAs. The commander may identify the need to create additional combinations of COAs.

f. COA Comparison (Step 5)

(1) Introduction

(a) COA comparison is both a subjective and objective process, whereby planners consider COAs independently and evaluate/compare them against a set of criteria established by the staff and commander during mission analysis. The objective is to identify and recommend the COA that has the highest probability of accomplishing the mission and is acceptable.

(b) Figure IV-14 depicts inputs and outputs for COA comparison. Other products not graphically shown in the chart include updated JIPOE products, updated CCIRs, staff estimates, and commander's identification of branches for further planning.

(c) COA comparison facilitates the commander's decision-making process by balancing the **ends, means, ways, and risk** of each COA. The product of this step is a recommended COA that is included in the COA decision brief in step 6. COA comparison helps the commander answer the following questions:

- 1. What are the differences between each COA?
- 2. What are the advantages and disadvantages?
- 3. What are the risks?

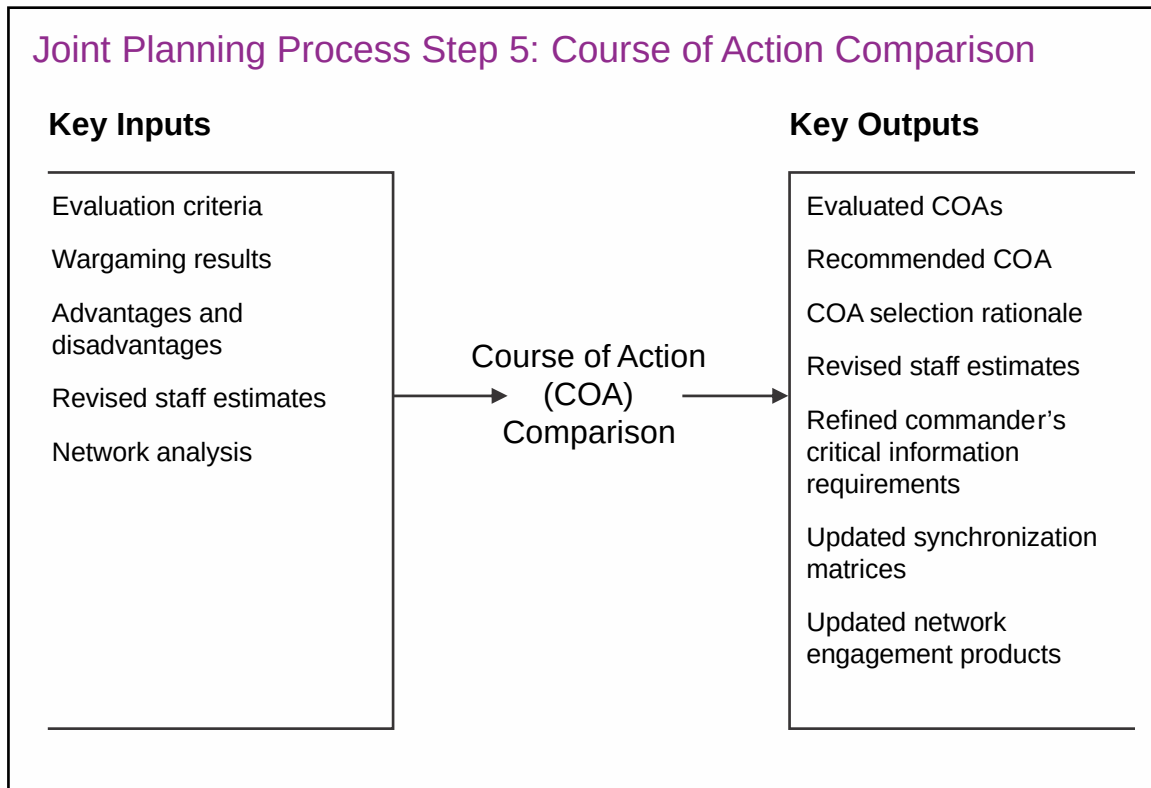


Figure IV-14. Joint Planning Process Step 5: Course of Action Comparison

(2) COA Comparison Process

(a) In COA comparison, the staff determines which COA performs best against the established evaluation criteria. The commander reviews the criteria list and adds or deletes, as required. The number of evaluation criteria varies, but there should be enough to differentiate COAs. Do not evaluate COAs against each other within any one criterion but, rather, evaluate COAs individually against the criteria established by the staff and commander. Compare their individual performances to enable the staff to recommend a preferred COA to the commander.

(b) Staff officers may each use their own matrix to compare COAs with respect to their functional areas. Matrices use the evaluation criteria developed before the wargame. Decision matrices alone cannot provide decision solutions. Their greatest value is providing a method to compare COAs against criteria that the commander and staff believe will produce mission success. They are analytical tools that staff officers use to prepare recommendations. Commanders provide the solution by applying their judgment to staff recommendations and deciding.

(c) The staff helps the commander identify and select the COA that best accomplishes the mission. The staff supports the commander's decision-making process by clearly portraying the commander's options and recording the results of the process. The staff evaluates feasible COAs to identify the one that performs best within the evaluation criteria against the enemy's most likely and most dangerous COAs.

(d) **Prepare for COA Comparison.** The commander and staff use the evaluation criteria developed during mission analysis to identify the advantages and disadvantages of each COA. Evaluating the strengths and weaknesses of one COA facilitates comparing its advantages and disadvantages with another COA.

1. Determine/define comparison/evaluation criteria if not done prior to wargaming. As discussed earlier, planners base criteria on the circumstances and situation. There is no standard list of criteria, although the commander may prescribe several core criteria for all staff directors to use. Individual staff sections, based on their estimate process, select the remainder of the criteria. Criteria are based on the circumstances and should be relative to the situation.

a. Review commander's guidance for relevant criteria.

b. Identify implicit significant factors relating to the operation.

c. Identify criteria relating to that staff function.

2. Define and determine the standard for each criterion.

a. Establish standard definitions for each evaluation criterion. Define the criteria in precise terms to reduce subjectivity and ensure the interpretation of each evaluation criterion remains constant between the various COAs.

b. Establish definitions prior to commencing COA comparison to avoid compromising the outcome.

c. Apply standards for each criterion to each COA.

3. The staff evaluates COAs with the commander's evaluation criteria to select the one most likely to succeed, considering both near-term and long-term (e.g., 10+ years) implications.

(3) **Determine the comparison method and record.** Actual comparison of COAs is critical. The staff may use any technique that facilitates reaching the best recommendation and the commander making the best decision. There are several techniques for comparing COAs. See Appendix E, "Course of Action Comparison."

(4) **COA comparison is subjective and should not be a strictly mathematical process.** Inform the commander on COA preference in terms of the evaluation criteria and the risk.

(5) If the COAs are significantly different, a side-by-side comparison for selection may not be appropriate, if they have differing objectives. However, this provides the commander the ability to show senior leaders the costs and risks of differing options rather than just different COAs within a single option to support strategic decision making.

g. COA Approval (Step 6)

(1) Introduction

(a) In this JPP step, the staff briefs the commander on the COA comparison and the analysis and wargaming results, including a review of important supporting information. The staff determines the preferred COA to recommend to the commander. Figure IV-15 depicts the COA approval inputs and outputs.

(b) The nature of the OE or contingency may make it difficult to determine the desired objective until the crisis occurs. In these cases, the JFC may choose to present two or more valid COAs for approval by higher authority. However, for a crisis, the desired objective should be based on the set of objectives approved by the President or SecDef.

(2) **Prepare and present the COA Decision Briefing.** The staff briefs the commander on the COA comparison, COA analysis, and wargaming results. The briefing includes a review of important supporting information such as forces available, the current JIPOE analysis, the operational approach, and assumptions used in COA development. All principal staff directors and the component commanders should attend this briefing (physically or virtually). Figure IV-16 shows a sample COA decision briefing guide.

(3) **Commander Selects/Modifies the COA.** COA selection is the result of the COA comparison process. Throughout the COA development process, the commander conducts an independent analysis of the mission, possible COAs, and relative merits and risks associated with each COA. The commander, upon receiving the staff's recommendation, combines personal analysis with the staff recommendation, resulting in a selected COA. It gives the staff a concise statement of how the commander intends to accomplish the mission and provides the necessary focus for planning and plan development. During this step, the commander should:

(a) Review staff recommendations.

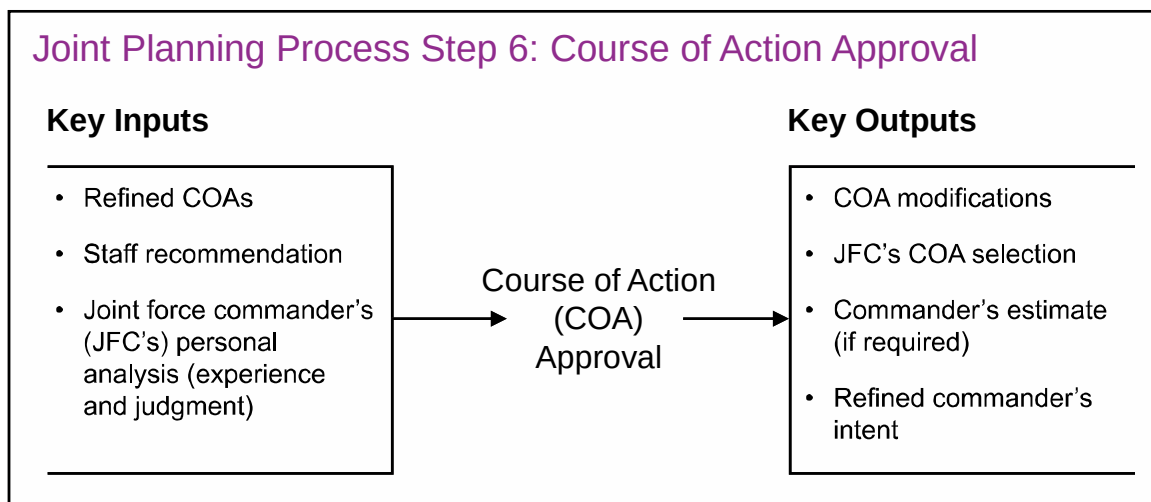


Figure IV-15. Joint Planning Process Step 6: Course of Action Approval

Sample Course of Action Decision Briefing Guide

- Purpose of the briefing
- Opposing situation
 - **Strength.** A review of opposing forces, both committed and available for reinforcement
 - **Composition.** Order of battle, major weapons systems, and operational characteristics (e.g., air, ground, and space-based capabilities, electronicmagnetic warfare capabilities, and cyberspace capabilities)
 - **Location and disposition.** Ground combat and fire support forces; air, naval, and missile forces; logistics forces and nodes; command and control facilities; and other combat power including forces contesting blue deployments
 - **Reinforcements.** Land; air; naval; missile; chemical, biological, radiological, and nuclear; other advanced weapons systems; capacity for movement of these forces
 - **Logistics.** Summary of opposing forces ability to support combat operations
 - Time and space factors. The capacity to move and reinforce positions
 - **Combat efficiency.** The state of training, readiness, battle experience, physical condition, morale, leadership, motivation, tactical doctrine, discipline, and significant strengths and weaknesses
- Nonmilitary threat networks (flexible format depending on situation and mission)
- Friendly situation (similar elements as opposing situation)
- Nonmilitary neutral networks (flexible format depending on situation and mission)
- Commander's intent statement
- Operational concepts and courses of action (COAs)
 - Any changes from the mission analysis briefing in the following areas:
 - assumptions
 - limitations
 - adversary and friendly centers of gravity
 - phasing of the operation (if phased)
 - Present COAs. As a minimum, discuss:
 - COA# _____ (short name, e.g., "Simultaneous Assault")
 - COA statement (brief concept of operations)
 - COA sketch
 - COA architecture
 - task organization
 - command relationships
 - organization of the operational area
 - major differences between each COA
 - summaries of COAs
 - COA analysis
 - review of the joint planning group's wargaming efforts
 - add considerations from own experiences
 - COA comparisons
 - description of comparison criteria (e.g., evaluation criteria) and comparison methodology
 - weigh strengths and weaknesses with respect to comparison criteria
 - COA recommendations
 - staff
 - components

Figure IV-16. Sample Course of Action Decision Briefing Guide

- (b) Apply results of own COA analysis and comparison.
- (c) Consider any separate recommendations from supporting and subordinate commanders.
- (d) Review guidance from the higher headquarters/strategic guidance.
- (e) The commander may:
 - 1. Concur with staff/component recommendations, as presented.
 - 2. Concur with recommended COAs, but with modifications.
 - 3. Select a different COA from the staff/component recommendation.
 - 4. Combine COAs to create a new COA.
 - 5. Reject all and start over with COA development or mission analysis.
 - 6. Defer the decision and consult with selected staff/commanders prior to making a final decision.

(4) **Refine Selected COA.** Once the commander selects a COA, the staff begin the refinement process of that COA into a clear decision statement for the commander's estimate. At the same time, the staff apply a final "acceptability" check.

- (a) Staff refines commander's COA selection into clear decision statement.
 - 1. Develop a brief statement that **clearly and concisely** sets forth the COA selected and provides whatever information is necessary to develop a plan for the operation (no defined format).
 - 2. Describe what the force is to do and as much of the elements of when, where, and how as may be appropriate.
 - 3. Express decision in terms of what to accomplish if possible.
 - 4. Use simple language so the meaning is unmistakable.
 - 5. Include statement of what is acceptable risk.
 - 6. Realize that many simulations are unable to capture qualitative data within the information environment.
- (b) Apply final "acceptability" check.

1. Apply experience and an understanding of situation.
2. Consider factors of acceptable risk versus desired objectives consistent with higher commander's intent and concept. Determine if gains are worth expenditures.

(5) **Prepare the Commander's Estimate**

(a) Once the commander selects the COA, provides guidance, and updates intent, the staff then completes the commander's estimate. The commander's estimate provides a **concise narrative statement** of how the commander intends to accomplish the mission and provides the necessary focus for campaign planning and contingency plan development. Further, it responds to the establishing authority's requirement to develop a plan for execution. The commander's estimate provides a continuously updated source of information from the perspective of the commander. Commanders at various levels use estimates during the JPP to support COA determination and plan or order development.

(b) A commander uses a commander's estimate as the situation dictates. The commander's initial intent statement and planning guidance to the staff can provide sufficient information to guide the planning process. The commander tailors the content of the commander's estimate based on the situation and ongoing analysis. A typical format for a commander's estimate is in CJCSM 3130.03, *Planning and Execution Formats and Guidance*.

1. Contents may vary, depending on the nature of the plan or contingency, time available, and the applicability of prior planning. In a rapidly developing situation, the formal commander's estimate may be impractical, and the entire estimate process may condense to a commanders' conference.

2. With appropriate horizontal and vertical coordination, the commander may brief their COA selection to SecDef. In the strategic context, where military operations are strategically significant, the President or SecDef approves a commander's selected COA. The commander's estimate then becomes a matter of formal record keeping and guidance for component and supporting forces.

(c) The supported commander may use simulation and analysis tools in the collaborative environment to evaluate a variety of options and may also choose to convene a concept development conference involving representatives of subordinate and supporting commands, the Services, JS, and other interested parties. Review of the resulting commander's estimate requires collaboration and coordination among all planning participants. The supported commander may highlight issues for future interagency consultation, review, or resolution to present to SecDef during the IPR.

(d) **CJCS Estimate Review.** The estimate review determines whether the scope and concept of planned operations will accomplish the mission, determines using available resources in the timeframes contemplated by the plan. It ensures the plan is proportional and worth the expected costs. As SecDef (or designated representative)

approves planning during an IPR, the commander's estimate informs the refinement of the initial CONOPS for the plan.

h. Plan or Order Development (Step 7)

(1) CONOPS

(a) The CONOPS clearly and concisely expresses what the JFC intends to accomplish and how it uses available resources. It describes the integration and synchronization of joint force component action and supporting organizations to accomplish the mission, including potential branches and sequels. The CONOPS:

1. States the commander's intent.
2. Describes the central approach the JFC intends to take to accomplish the mission.
3. Provides for the application, sequencing, synchronization, and integration of forces and capabilities in time, space, and purpose (including those of multinational and interagency organizations, as appropriate).
 - a. Describes a scheme of maneuver that articulates the application, arranging, sequencing, and integration of forces and capabilities in time and space.
 - b. Describes when, where, and under what conditions the supported commander intends to conduct operations.
4. Describes when, where, and under what conditions the supported commander intends to conduct operations and give or refuse battle, if required.
5. Focuses on friendly, allied, partner, and enemy COGs and their associated critical vulnerabilities.
6. Provides for controlling the tempo of the operation.
7. Visualizes the campaign in terms of the forces and functions involved.
8. Relates the joint force's objectives and desired effects to those of the next higher command and other organizations as necessary. This enables assignment of tasks to subordinate and supporting commanders.

(b) Planning results in a plan documented in the format of an order. During plan or order development, the staff, in collaboration with subordinate and supporting components and organizations, expand the approved COA into a detailed plan or OPORD by refining the initial CONOPS associated with the approved COA. The CONOPS is the centerpiece of the plan or OPORD.

(c) The staff writes (or graphically portrays) the CONOPS in sufficient detail so subordinate and supporting commanders understand their mission, tasks, and other requirements and can develop their supporting plans. During CONOPS development, the commander determines the best arrangement of simultaneous and sequential actions and activities to accomplish the assigned mission consistent with the approved COA, and resources and authorities available. This arrangement of actions dictates the sequencing of activities or forces into the OA, providing the link between the CONOPS and force planning. Planners perpetuate the link between the CONOPS and force planning through the sequencing of forces into the OA via a TPFDD. This structure ensures unit integrity, force mobility, and force visibility, as well as the ability to rapidly transition to branches or sequels as operational conditions dictate. Supported CCDRs determine the sequencing of forces in concert with supporting commands and capture this sequencing in the TPFDD as the commander's required delivery dates. Planners ensure the CONOPS, force plan, deployment plans, and supporting plans provide the flexibility to adapt to changing conditions and are consistent with the JFC's intent.

(d) If the scope, complexity, and duration of the military action contemplated to accomplish the assigned mission warrants execution via a series of related operations, then the staff outlines the CONOPS as a campaign. They develop the preliminary part of the operation in sufficient detail to impart a clear understanding of the commander's concept of how to accomplish the assigned mission.

(e) During CONOPS development, the JFC assimilates many variables under conditions of uncertainty to determine the essential military conditions, sequence of actions, and application of capabilities and associated forces to create effects and achieve objectives. **JFCs and their staffs are continually aware of the higher-level objectives and associated desired and undesired effects that influence planning at every juncture.** In execution, if the JFC's operational objectives do not link with the strategic objectives, eventually tactical actions can begin to drive the strategy toward cross-purposes.

CJCSM 3130.03, Planning and Execution Formats and Guidance, provides detailed guidance on content and format of orders and plans.

(2) Plan or Order Development

(a) For most plans and orders, the CJCS monitors planning activities; resolves shortfalls when required; and reviews the supported commander's plan for adequacy, feasibility, acceptability, completeness, and compliance with policy and joint doctrine. When required, the commander conducts one or more IPRs with SecDef (or designated representative) to confirm the plan's strategic guidance, assumptions (including timing and national-level decisions required), any limitations (restrictions and constraints), the mission statement, the operational approach, key capability shortfalls, areas of risk, acceptable levels of risk, and any further guidance required for plan refinement. During the IPRs, the CJCS and OSD separately address issues arising from, or resolved during, plan review (e.g., key risks, decision points). Commanders should show how the plan

supports the objectives identified in strategic guidance and identify the links to other plans, both within the AOR (or functional area) and with those of other CCMDs. The result of an IPR should include an endorsement of the planning to date or acknowledgement of friction points and guidance to shape continued planning. All four operational activities (situational awareness, planning, execution, and assessment) continue in a complementary and iterative process. CJCSI 3141.01, *Management and Review of Campaign and Contingency Plans*, provides further details on the IPR process.

(b) The JFC guides plan development by issuing a PLANORD or similar planning directive to coordinate the activities of the commands and agencies involved. The planning activities show in Figure IV-17 typically occur in a concurrent, collaborative, and iterative fashion rather than sequentially, depending largely on the planning time available. In a similar fashion as COA development, planners discover and eliminate shortfalls and conflicts within their command and with the other CCMDs.

(c) The CJCS planning family of documents referenced in CJCS Guide 3130, *Joint Planning and Execution Overview and Policy Framework*, provides policy, procedures, and guidance on these activities for organizations required to prepare a plan or order. These are typical types of activities that supported and supporting commands and Services accomplish collaboratively as they plan for joint operations.

Plan Development Activities

- Force planning
- Support planning
- Deployment and redeployment planning
- Shortfall identification
- Feasibility analysis
- Refinement
- Documentation
- Plan review and approval
- Supporting plan development

Figure IV-17. Plan Development Activities

CHAPTER V

INTEGRATED PLANNING

“The struggle to secure the adoption by all concerned of a common concept of strategical objectives is wearing me down. Everybody is too much engaged with small things of his own—or with some vague idea of larger political activity to realize what we are doing—rather not doing.”

Brigadier General Dwight D. Eisenhower, January 22, 1942

1. Overview

a. Threats do not confine themselves by domain or function. They also do not fit neatly into divisions of separate campaign, crisis, and contingency planning and execution. Foreseeable contingencies will likely require the joint force to execute coordinated and synchronized operations across geographic regions, functions, domains, the EMS, and time against adversaries. This requires integrated planning across the joint force.

b. The nation’s adversaries continue to grow in capability and are increasingly interconnected and active. Adversaries advance their objectives asymmetrically while operating short of armed conflict, deliberately avoiding the activation of US contingency plans. Siloed planning structures are insufficient in this environment and require an overhaul to achieve the analytical rigor, understanding, collaboration, and prioritized resourcing to operate with flexibility.

c. SPFs and IPSs set the conditions for integrated planning. SPFs seek to provide a framework for senior leaders to understand the linkages between decisions and actions made across the competition continuum. Highlighting authorities and associated resources in competition is imperative to understanding the feasibility, risks, and opportunities of executing contingency plans. An IPS is a coordinating authority-led planning effort composed of a core plan, concurrent plans, and conditional plans likely to execute contemporaneously, closing gaps in previously segregated plans. This chapter outlines the key principles of integrated planning to help commanders and planners align planning efforts to ensure preparedness to fight and win in the current environment.

d. Through SPFs, the joint force produces executable, resource-informed plans, while reducing siloing of campaign, crisis, and contingency planning activities executed across the competition continuum. This planning methodology drives integration across commands and domains by organizing multiple plans in a common framework oriented on a common threat scenario (CTS). Using a defined resource pool shared across the core plan and concurrent plans, the joint force develops IPSs to present resourcing decisions, tradeoffs, and risks early in the planning process, which enable senior leader decision in feasible and executable plans. The resultant detailed planning informs global and regional campaign activities, crisis response options, and contingency plans for a global, all-domain planning effort that highlight the risks inherent in an environment producing multiple, simultaneous threats.

2. Principles of Integrated Planning

a. To achieve planning integration, the joint force requires a global perspective and an understanding of threats, hazards, opportunities, and overall demand for forces, and sustainment. Integrated plans build upon the responsibilities directed by the UCP and enable the joint force to execute coordinated and synchronized operations across geographic regions, functions, domains, and time against adversaries. In developing integrated plans through an iterative process, commanders and their planners should consider the principles outlined in this section.

b. While specific instructions and timelines change, the principles of integrated planning remain the same.

c. **Campaign and Contingency Planning are Iterative and Interrelated Activities.** As depicted in Figure V-1, campaign and contingency plans inform each other through iterative, continuous planning cycles. The force posture and activities of the CCMDs for campaigning are a starting consideration for any contingency plan. Likewise, integrated contingency planning reveals risk that informs refinement to the campaign's objectives to establish favorable conditions that build relative advantage.

(1) Contingency planning ensures the joint force is prepared to militarily respond to major conflict with our adversaries. Concurrently, campaign plans develop the approach to assure allies and partners, deter attack, and counter antagonistic behavior to compete and set the conditions to enable transition and execution of contingency plans. Conceptually,

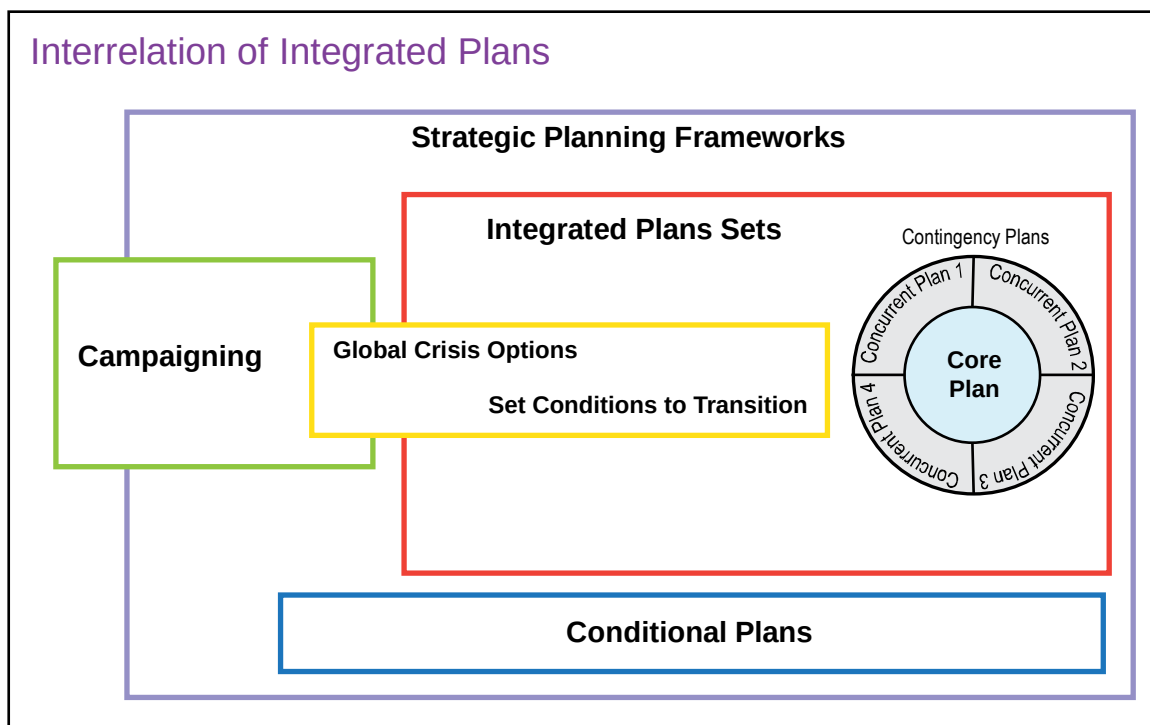


Figure V-1. Interrelation of Integrated Plans

campaign plans seek to influence and change the OE to set conditions to prevent or mitigate crises but also prepare the joint force to execute contingency plans should prevention fail.

(2) Both campaign and contingency plans seek similar strategic outcomes over different timeframes and commitment of resources. During execution of global and CCMD campaigns, current operations staffs oversee continuous refinements and adjustments. However, the weight of planner effort focuses on creating campaign and contingency plans and evaluating their viability, not execution. By creating and resourcing overlapping contingency plans in an integrated set under a global CTS, the product can efficiently apply planner resources to enable corresponding action across the competition continuum. The planning for common resources required for campaign and contingency plans are more complementary at the forefront.

(3) Upon transition from a crisis or contingency to the recharacterized OE, commands adjust their campaigning baseline by adapting to the new strategic environment. CCMDs assess the new environment following the cessation of hostilities and transition to the new or recharacterized competition. Such actions may include changes to force posture, assignment, and allocation of forces, force structure, and strategic priorities. These resultant changes then act as the basis for new iterations of campaign and contingency planning.

d. **CTS.** In coordination with CCMD J-2s and the intelligence community, the coordinating authority leads the development of the CTS to provide the joint force an overview of the objectives, approaches, decision points, and critical capabilities of a specific adversary. CCDRs and planners coalesce their objectives, activities, and initiatives around and against each adversary CTS. Planners leverage intelligence assessments about how adversary actions evolve across the competition continuum and incorporate these assessments into the development of the CTS. By developing the common scenario, informed by the joint strategic intelligence estimate, dynamic threat assessment, and JIPOE, planners can work from a global view of each threat, as opposed to a theater or region focus.

e. **Common Strategic Approach.** Planners should develop a common strategic approach that outlines the objectives the joint force needs to achieve. This approach describes how the joint force collectively would synchronize efforts to achieve these objectives. The approach should consider the role of other conditional plans that are not necessary to achieve objectives but provide optionality. However, conditional plans are not part of resourcing decision.

f. **Early and Iterative Resourcing Guidance.** The viability of multiple plans executed concurrently relies on clear resourcing and employment authority guidance that defines a shared resource pool for the integrated plan set. Integrating individual plans requires several iterations of SecDef guidance to provide critical resourcing direction early in the planning process, instead of identifying resource deficiencies after the aggregation of plans. By determining resourcing for a set of plans, instead of individual plans, planners can drive their efforts using a common understanding of what is available should the CTS materialize. Planners should consider alternative means to supplement plans or account

for resource shortfalls, such as other authorities or interagency and partner nation capabilities. Joint, multinational, and interagency partners are included in resource planning early to enable more effective resource allocation and coordination. This common understanding enables planners to build complementary plans, instead of plans that require adjustment upon resource adjudication in execution.

g. Common Terminology. Lead planners and commanders with coordinating authority should set common terms to provide uniformity of meaning across the plans in an integrated set. Terms like threshold, and transition require consistency across all plans in the set. Those same terms should also apply to the CTS and defined enemy COA or be specific to that individual plan. While doctrine defines these terms for individual plans, it is important that the definition match the perspective of the integrated set.

(1) In similar regard, it is important to delineate between plans and concepts. Often Services or staff elements develop concepts to illustrate a future capability to describe the need for resourcing. Separately, integrated plan concepts may describe an IPS CONOPS prior to detailed plan development that relies upon existing capabilities within the two-year time frame. IPS concept development and approval synchronizes integrated planning efforts across the CCMDs and enables for senior leader input prior to SecDef's approval of a plan.

(2) Establishing classification and access requirements for planners across the integrated planning sets a baseline for appropriate information sharing and prevents redundant or conflicting planning. Planners need to incorporate sensitive information and protected capabilities in a manner that enables their use throughout the entire plan set, not as an adjunct function planned without coordination. Integrating multiple plans require a common level of access, established early in planning to ensure appropriate use of sensitive capabilities.

h. Consolidating Global Crisis Options (GCOs) and Using Alongside Campaign OAI. FDOs and FROs developed within individual CCMD contingency plans form the backbone of GCOs that the joint force provides to decision makers for a crisis. By consolidating options from an integrated set, planners can access a variety of robust crisis options which tie to the detailed planning associated with contingency and campaign plans. This helps enable complementary action across the competition continuum. Campaign plans detail day-to-day activity aligned toward achieving objectives through a series of consolidated strategic opportunities consisting of OAI. Planners can employ GCOs in the same fashion, instead of reserving those options for contingency plan execution. Many GCOs apply across the competition continuum to further the campaign's objectives, while also complementing contingency preparedness. The consolidation and accessibility of GCOs with integrated planning increases their usefulness to planners, and optionality to commanders.

i. Partner and Ally Coordination. Incorporating allies and partners early in the planning process allows for coordination of planning assumptions before sensitive details limit sharing. Equally important, planners verify critical planning assumptions such as

access, basing, and overflight early in the integrated planning process to provide realistic options for execution. To the extent authorized by policy, integrated planning should extend to allies and partners. This ensures individual plans within the integrated planning set are complementary to allied action and support, when executing concurrently.

j. **Interagency Coordination.** The military does not go to war, the nation does. Failing to account for the contributions of the interagency inefficiently employs warfighting resources and limits the nation's effectiveness in armed conflict. However, the interagency does not have the resources available to contribute full-time planners to all efforts. The joint force should not consider the interagency as additional planners or subject matter advisors. Instead, joint force planners need an understanding of what the interagency can contribute to achieve the nation's objectives in a CTS. Oversharing is a distraction to interagency partners not trained in the application of military force. The joint force has the statutory responsibility for crafting military plans, but the professional responsibility to incorporate the capabilities of the broader government to ensure success.

k. **Integrated TPFDD.** An integrated TPFDD is one in which CCMDs, the Services, and combat support agencies share the same sequencing across multiple contingency plans. By planning for sets of contingency plans likely executed concurrently, logisticians, with input from planners and GFM specialists, can generate integrated TPFDDs that account for the entire set, instead of individual plans. This effort applied in conjunction with critical resource guidance (e.g., planning factors and logistics factor files) early in the planning process enables the joint logistics enterprise to conduct logistics supportability analysis. This analysis enables planners to identify global requirements, gaps, risk to plan execution, and mitigation strategies (e.g., including posture and war reserve materiel), to help prevent/reduce logistical shortcomings in execution. This also enables economy of effort in execution, where forces flow in a logical sense that enables sequenced execution of concurrent plans. By contrast, independent contingency TPFDDs become invalid as soon as the military activates another, concurrent plan while pulling from the same finite resources. While CCMDs will still execute their own plans' TPFDDs, integrated TPFDDs enable more accurate risk and feasibility assessments that highlight potential shortfalls to successfully resolving an adversarial threat scenario, as opposed to achieving objectives of an individual plan.

3. Planning Across the Competition Continuum

a. Adversaries and allies alike seek to avoid the commitment and costs of armed conflict, yet still desire the objectives a war-time strategy pursues. This makes thresholds for armed conflict more malleable and less defined. To increase effectiveness during the campaign, planners and CDRs should proactively use elements of IPSs, prior to a plan's execution.

b. By design, campaigns involve proactive action, whereas contingencies are traditionally reactive to an aggressor's action. Integrated planning identifies opportunities beyond response to influence an adversary in a contingency. An SPF drives deliberate, integrated planning and produces GCOs to provide options for CCMDs to employ across

the competition continuum. The detailed JIPOE assessments and the dynamic threat assessment define adversary objectives through analysis of their political, military, economic, social, information, and infrastructure linkages. By analyzing an adversary's system across the world, planners can identify opportunities to exploit while campaigning. This maximizes the results of the planner's efforts, while ensuring viability of contingency plans through complementary action during a campaign.

c. Contingency plans are a branch of a campaign but are the most resource intensive. GFM and force employment initiatives weigh current use against readiness and preparedness to execute armed conflict/war or contingency plans as written. This, in effect, prioritizes preparedness over utilization, at the expense of executing coherent campaign initiatives. The shared resourcing of integrated plan sets, as opposed to individual plans presents a more complete picture to decision makers on the cost of resourcing OAI prior to contingency. Doing the cost analysis upfront helps identify available resources to apply in campaign, while understanding where those resources flow from and to, should contingency plan execution be necessary. This informed resourcing helps instill discipline in requests for additional forces, while lowering the risk of inaction, which could weaken deterrence and lead to contingency.

4. Campaign, Contingency, and Crisis Planning

a. Successful integrated planning involves aligning campaign activity to complement IPSs. Campaign plans operationalize strategic guidance by organizing and aligning day-to-day OAI with resources to achieve the CCDRs' objectives and complement related USG efforts in the theaters or functional areas. Figure V-2 depicts the nesting relationship of the different planning guidance and products, and Figure V-3 depicts a notional campaign.

b. Campaign plans orient objectives in support of strategic goals directed by GCPs.

See JP 3-0, Joint Campaigns and Operations, for more information CCPs and functional campaign plans.

c. Campaign plans are composed of the CCMD-specific requirements established by GCPs, supporting plans for other CCMDs, posture plans, theater logistics and distributions plans, and other requirements based on the CCDR's assessment.

See CJCSM 3130.03, Planning and Execution Formats and Guidance, for additional information on how to develop campaign plans.

5. Leveraging Integrated Planning for a Crisis

a. Crises are independent events, declared and labeled as such by national leadership. A crisis could span a wide range of possible impetuses but includes a rapid onset, finite time frame, and a driving necessity to respond. The event may or may not correspond to an adversary strategy, but its required response should complement US objectives.

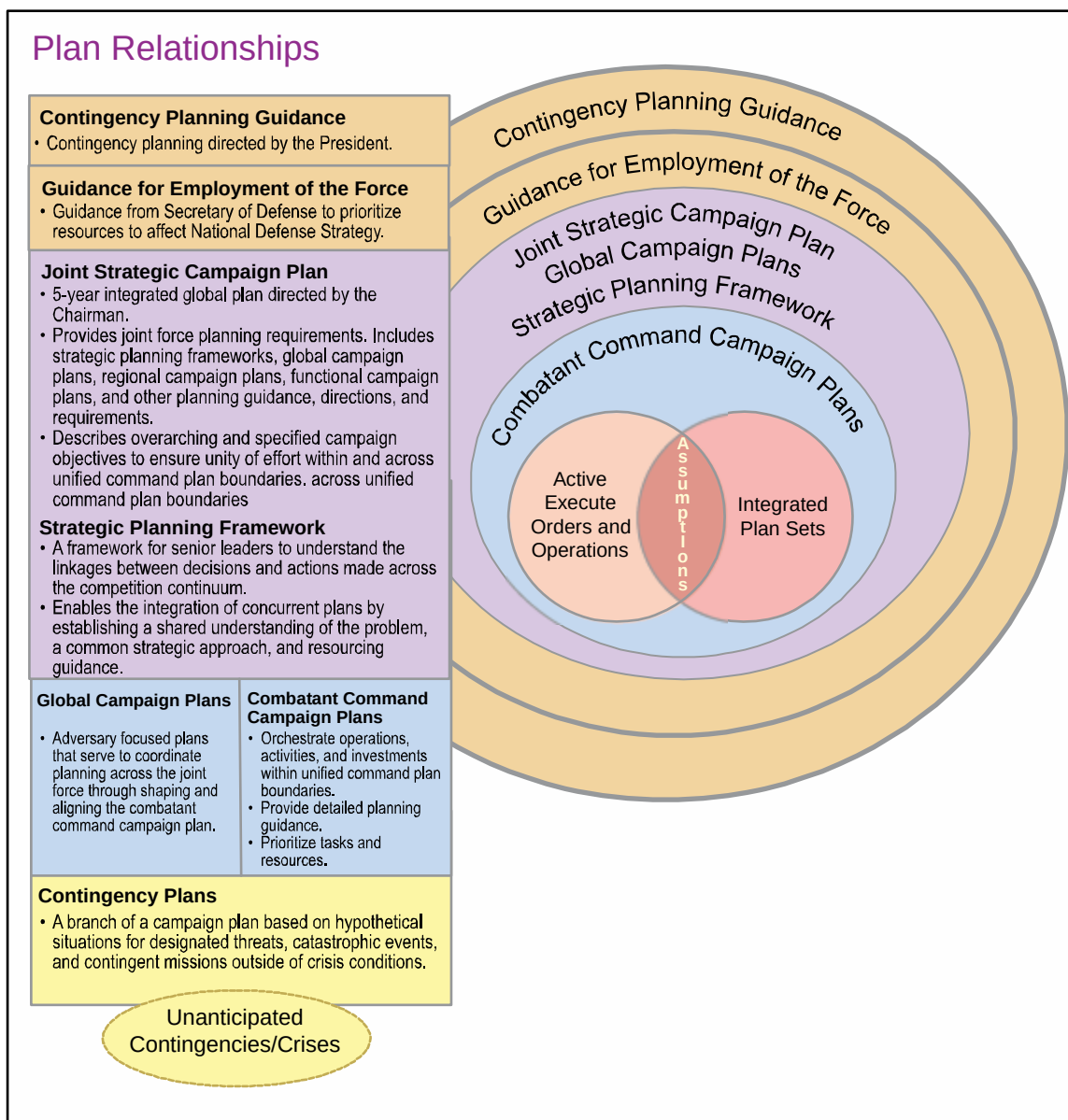


Figure V-2. Plan Relationships

b. In preparing to respond, commands recognize they do not enjoy the same time or resources afforded to deliberate planning efforts, due to the unanticipated and rapidly unfolding nature of a crisis. In contrast, contingency plans are the result of months of analysis, study, and assessment by multitudes of planners across the joint force. For this reason, it is imperative that the military incorporate the analytical and planning rigor done through contingency planning.

c. **Utilize pre-planned COAs and GCOs.** Options developed for a crisis provide flexibility but can lack the detailed planning and thorough coordination with existing joint force efforts. Contingency COAs have the analytical rigor and planning backbone to provide a measure of certainty to national leadership. IPSs incorporate COAs across

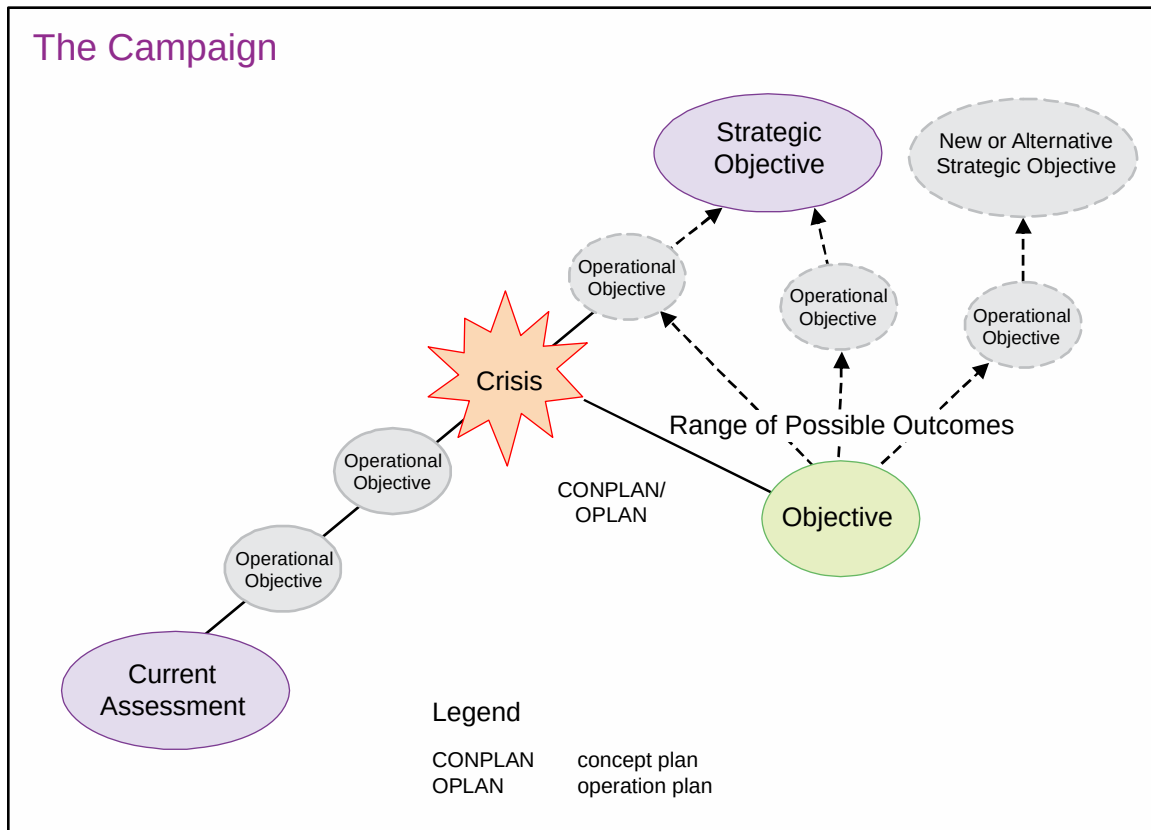


Figure V-3. The Campaign

multiple domains and geographic locations designed to achieve an objective. If not directly applicable to the crisis, the format and approach of the COA could create a similar effect.

(1) It is impossible to predict every possible crisis and a corresponding response. However, leveraging consolidated GCOs, to include those tangentially related to the primary subject of an integrated plan, enables the joint force to rapidly provide multiple options for a crisis.

(2) Planners incorporate logistics for the options and COAs associated with IPSs built into the deliberate planning effort. In turn, employing an option or COA for a crisis remains complementary to the readiness required to execute an IPS. By contrast, creating a new option independent of an IPS pulls from resources required for executing contingency plans, impacting readiness. While not always feasible to apply standing options from IPSs, using mature options saves significant time, while ensuring complementary action.

d. Articulate impacts to integrated plan sets. Military capacity is finite, and employment of forces is a zero-sum balance between readiness and current capacity. Utilizing forces today will impact readiness for future execution. Staffs present this tradeoff to military and DoD leadership for adjudication, but it is historically difficult to articulate and estimate the long-term impact of any one decision due to the large portfolio of disparate contingency plans. By integrating sets of contingency plans against a given adversary and

using a shared pool of resources commanders and planners can better articulate what they require to achieve an objective against an adversary across functions and geography. IPSs present a more complete picture of required forces than individual plans. In a crisis, commands articulate the impact and risk to an IPS focused on an adversary to best enable decision makers to ensure crisis response helps develop or aligns with IPS contingency plans.

e. Consider a proactive versus reactive approach. Planning for a crisis is inherently reactive in responding to an event. However, it could also serve as an opportunity to advance a campaign or better prepare for a contingency. To ensure crisis response is complementary to campaign plans and contingency preparedness, planners should assess how movement of military forces creates opportunity or initiative to deter conflict. A crisis, depending on the manner, size, and scope, may impact an adversary's framework and strategic approach. In this case, planners should proactively identify opportunities for advantage and ensure crisis response enables such a position, without inhibiting it. Every crisis response is also an opportunity to complement objectives and improve preparedness to execute integrated plan sets. GCOs should clearly show linkages to campaign and contingency plans.

6. Role of the Joint Staff in Integrated Planning

Inter-command planning necessitates an increased role by the joint staff to best enable the CCMDs. While not possible to unify or group all plans, integrating the plans likely executed concurrently establishes unity of effort. In producing the JSCP and informing the CPG and GEF, the JS sets the structure, timeline, and weight of effort given to specific IPSs, considering the bandwidth of planners across the CCMDs. The nature of integrated planning requires more frequent iteration and adjustment, driving an increased need for active management of timetables.

a. Delineate Methods to Conduct Integrated Planning. The JSCP drives the planning efforts of the CCMDs. In conjunction with the CPG and GEF, the document sets the top-down guidance for planning to align with national strategy. Through an SPF, the joint staff can increase cross-command standardization and unity of effort. An SPF may consist of a standard synchronization matrix with horizontal rows of plans to be conducted integrated with vertical rows indicating effort over time/phasing of missions. In coordination with the OSD staff, the JS can set the schedule, forums, and venues to conduct integrated planning, unifying the effort across CCMDs and staffs. Through this framework, the JS can fulfill the CJCS's role as global integrator, leveraging its convening authority to gather planners on a schedule that aligns with staff action to produce results. Similarly, the JS can set top-down direction for sensitive information access and provide the mechanisms and venues for inter-command planning. This sets a common security level for assessing the joint problem and deriving solutions.

b. Incorporate Existing Plans into Current Operations and Crisis Planning. Doctrine prescribes reviews of existing plans when planning for a crisis. However, the two largest inhibitors to accomplish this are time and access. When creating PLANORDs, the command J-3 should incorporate J-5 planners and instruct a review of applicable IPSs,

assessing pre-developed COAs and GCOs for applicability in the crisis scenario. The directive can also temporarily provide increased access to sensitive information to ensure planners tasked during a crisis can access the detailed information included in an IPS.

c. **Identify Real-Time Strategic Opportunity.** Operating across the competition continuum necessitates seizing opportunities to achieve positions of enduring advantage. One of the greatest strategic risks is inaction or failing to take advantage of a situation. Commands ensure options presented for decision are complementary, not detrimental to theater or global campaigns. Knowledge of IPSs and their ties to campaign plans enables deliberate, managed escalation in a manner that aligns with national objectives, while avoiding unintended consequences of unplanned action. This kind of integration requires a resident and persistent global perspective that can enable fast decision or action. Detailed knowledge of all the priority adversaries, global campaigns, and the posture to execute contingency plans associated with each challenge is resident in the J-5, while preserving the focus and detail within the CCMDs. Leveraging this breadth of knowledge is necessary to prevent third-party actors from taking advantage of instability or a strategic distraction while operating across the competition continuum.

d. **Adjudication Ownership.** In broadening the aperture to integrate planning efforts across the globe, the JS provides methods of adjudication for critical decisions or resourcing guidance, prior to SecDef approval of the plan. Building upon the CJCS's role as global integrator, the JS maximizes time and decision space of senior leaders in how it executes reviews of plans and resourcing guidance. This preserves bandwidth of CCMD planners, while obtaining the necessary decisions to further planning at a level that accounts for all CCMDs. In the absence of SecDef decisions, guidance, or approval of plans, the JS can provide recommendations in senior military forums.

7. Resource-Informed Planning and Global Force Management

a. JSCP-directed campaigns, unlike contingency plans, are not just plans, they are campaigns in execution, constrained by readiness, availability, and authorities of resources.

b. CCDRs plan, assess, and execute their JSCP-directed campaign plans. The CCMDs, however, receive limited funding and rely on the Services and their component commands to budget for and execute campaign activities. Therefore, the JFC involves components, JS, and FPs during the planning process to identify resources and tools that ensure the campaign plan is executable. The component commands can also identify options and activities of which the CCMD might not be aware.

c. Campaign planning requires planning across four resource timeframes (see Figure V-4).

(1) JFCs execute ongoing operations with the current budget with assigned or currently allocated resources. As the operations progress and the CCMD conducts its assessment, the commander may redesignate assigned and allocated resources, with the proper authorities, to other operations and activities or to address critical issues that may

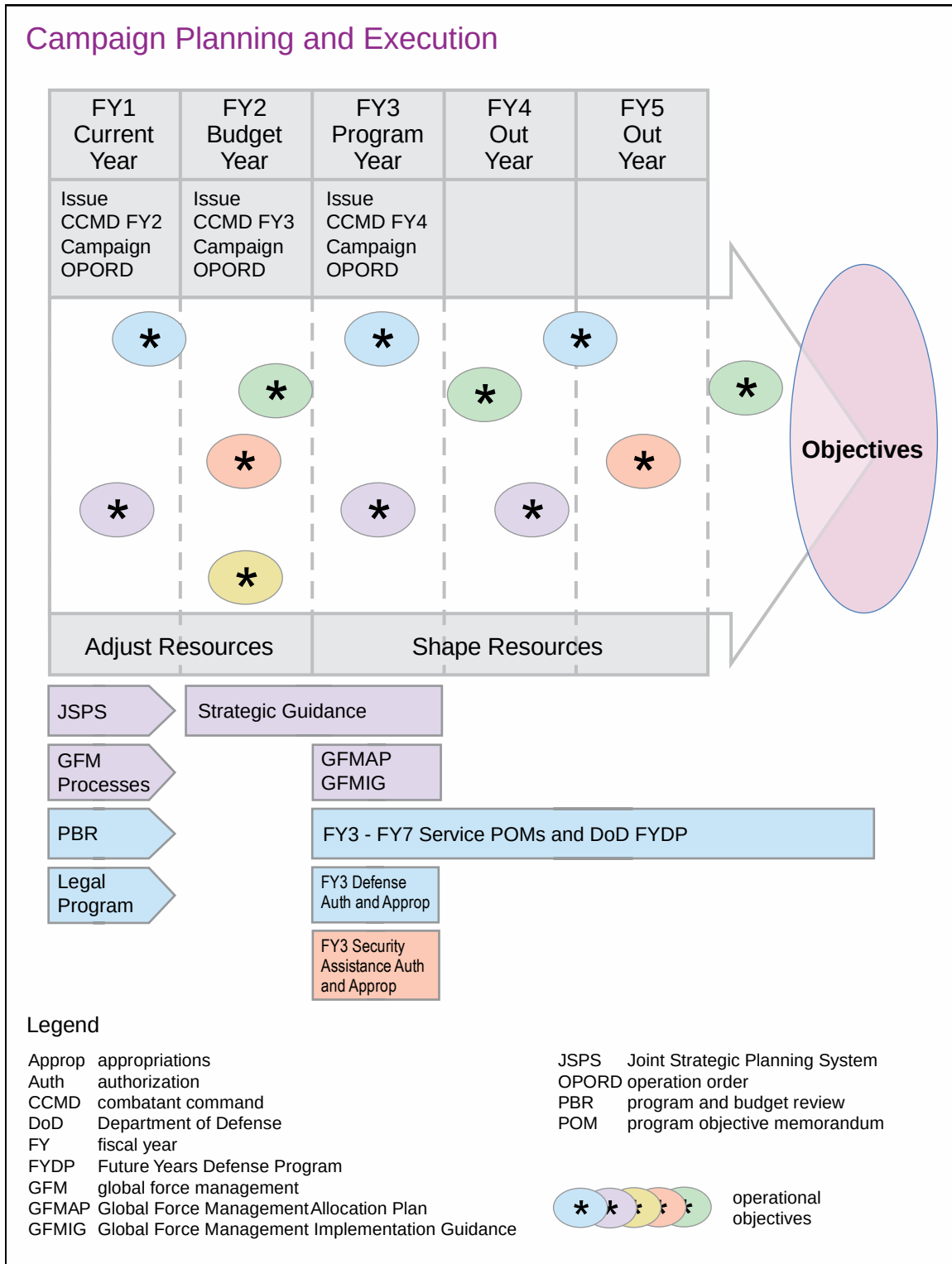


Figure V-4. Campaign Planning and Execution

arise. Simultaneously, the commander uses the ongoing assessment to project a resource requirement for two years in the future (the program year). The commander uses

assessment of the OE and the projection of the impact of activities in both the current and budget year.

(2) The commander develops and briefs the campaign plan for the upcoming year, considering the budget year forecast, assigned and allocated forces, and force apportionment tables. The commander updates objectives, develops new ones as appropriate, and prioritizes resources, based on the ongoing assessment of current year actions. The commander also identifies gaps and shortfalls in capabilities, along with associated risk, and includes them in assessments. These reports support the command's budget and force request for the budget and apportionment in development for the program year.

(3) The commander uses the current and budget year allocation, combined with the assessment, to develop a budget and resource request for the program years. Working with the JS, the command identifies opportunities for military engagement and exercises (joint and combined) and identifies future posture requirements that support the CCMD's campaign. Posture changes require long-lead times to implement, so the commander identifies these in time to conduct required diplomacy and stationing requirements (such as construction) to meet any posture changes.

(4) The commander can only execute those operations and activities for which there are resources and authorities. Authorization laws may further restrict the commander by limiting the use of the resources for specific programs or by requiring specific conditions before conducting the operation or activity.

8. Risk

a. CCDRs, Services, and DoD's senior leaders work together to reach a common understanding of risk (the strategic risk assessed at the CCMD level combined with the military risk), decide what risk is acceptable, and minimize the effects of accepted risk by establishing appropriate risk controls.

b. For strategic risk, CCDRs identify the probability and consequence of near (0-3 years) and mid-term (3-10 years) strategic events or crises that could harm US national interests, and they identify the impacts of long-term (10-20 years) trends and future adversary capabilities.

c. For military risk, CCDRs evaluate the impact of the difference between required and available capability, capacity, readiness, plans, and authorities on their ability to execute assigned missions. Assessments include:

(1) Future Years Defense Program budgetary priorities, tradeoffs, or fiscal constraints.

(2) Deficiencies and strengths in force capabilities identified during preparation and review of campaign and contingency plans.

- (3) Projected readiness of forces required to execute the campaign in future years.
- (4) Assumptions or plans, including risks, about the contributions or support of:
 - (a) Other USG departments and agencies.
 - (b) Alliances, coalitions, and other friendly nations.
 - (c) Commercial support to operations and contract support.
 - (d) Changes in adversary capabilities identified during the preparation of the strategic estimate and other intelligence products.
- d. Impact on or changes in the civilian population, governing organizations, and other related civilian infrastructure: Commanders are willing to try new activities to see if there are better or less risky methods to achieve objectives.

For additional information on risk, see CJCSM 3105.01, Joint Risk Analysis Methodology.

9. Opportunity

- a. Commanders and their staff identify opportunities they can exploit to influence the situation toward mission objectives. Limited windows of opportunity may open and the CCDR should be ready to exploit these to set the conditions that lead to successful transition. Execute in collaboration with interagency partners, international partners, and partner nations who are looking for opportunities to enhance resilience and mitigate conflict and who may have assessment tools.
- b. Commanders and their staff comprehend dynamics in the environment, such as evolving strategic guidance and mandates; the type of conflict; the strategic logic of adversaries; the impact of operations; and changing vulnerabilities and threats that relate to protection of civilians, resiliencies, and emerging opportunities, to change the OA evaluation or actors to support US objectives.
- c. Assessing the OE from the perspective of the root causes and immediate drivers of instability is essential to identify and create opportunities for longer-term processes to deal with the root causes.

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CHAPTER VI OPERATION ASSESSMENT

1. Overview

a. Operation assessments are an integral part of planning and execution of any campaign or operation, fulfilling the requirement to identify and analyze changes in the OA and to determine progress. Assessments involve the entire staff and other sources such as higher and subordinate headquarters, interagency and multinational partners, and other stakeholders. They provide perspective, insight, and the opportunity to correct, adapt, and refine planning and execution to make military operations more effective. Operation assessment applies to all levels of warfare and during all military operations.

b. Commanders maintain a personal sense of the progress of the operation or campaign, shaped by conversations with senior and subordinate commanders, key leader engagements, and battlefield circulation. Operation assessment complements the commander's awareness by methodically identifying changes in the OA by identifying and analyzing risks, opportunities, and decision points to formally provide recommendations to improve progress toward mission accomplishment. Assessment should integrate into the organization's planning, beginning in the plan initiation step, and operations battle rhythm to support the commander's decision cycle.

c. Integrating assessments into the planning cycle helps the commander ensure the operational approach remains feasible and acceptable in the context of higher policy, guidance, and orders. This integrated approach optimizes the feedback senior leadership needs to appropriately refine, adapt, or terminate planning to be effective in the OA.

KEY TERM

Assessment: Determination of the progress toward accomplishing a task, creating a condition, or achieving an objective. A continuous process that measures the overall effectiveness of employing capabilities during military operations.

d. CCMDs, Service components, joint functional components, and JTFs devote significant effort and resources to plan and execute operations. They apply appropriate rigor to determine effective planning and execution as needed to achieve objectives. Assessment complements that rigor by analyzing the OE objectively and comprehensively to estimate the effectiveness of planned tasks and measure the effectiveness of completed tasks with respect to desired conditions in the OA.

e. Background

(1) CCPs and country plans are continuously in some stage of implementation. Accordingly, CCMD planners should annually extend their planning horizon into the future year. The simultaneity of planning for the future while implementing a plan requires a

CCMD to continually assess its implementation to appropriately revise, adapt, or terminate elements of the evolving (future) plan. This synergism makes operation assessment a prerequisite to plan adaptation. Operation assessment is thus fundamental to revising implementation documents ahead of resource allocation processes.

(a) Events can arise external to the CCMD's control that affect both plan execution and future planning. Some of these events can impede achievement of one or more objectives, while others may present opportunities to advance the plan more rapidly than anticipated.

(b) External events generally fall into two categories. The first are those that change the JFC's OA (typically an intelligence focus). The second category involves those events that change the resource picture with respect to funding, forces, and time available (typically a force structure, resource, and assessment focus). Changes in strategic guidance can impact both categories. This document treats these two types of external events as separate considerations because they can influence plan implementation independent of each other.

(2) Throughout campaign planning and execution, the CCDR and staff continually observe the OE and assess the efficacy of the operations described (or directed by) the campaign plan. Because CCDRs conduct campaigns in a complex and dynamic environment, commands must be able to detect, analyze, and adapt to changes in the OA during execution. Local actors are typically the best suited to detect and understand changes in a complex OE, and their input can be instrumental for assessment. Planners refine or adapt the plan based on the guidance, their understanding of the OA, their understanding of the impact of actions or changes to the OE, the campaign's objectives, and the decisions that underpinned the original operational approach to refine or adapt the plan, or the approach.

(3) In addition to the command's internal assessment efforts, analysis and assessment of the strategic environment and OE by interagency partners is available to the CCMD. OSD and the JS can assist in obtaining these inputs. Promote cooperation events to enable sharing interagency partners' insights on environmental changes.

(4) The overall purpose of operation assessment is to provide recommendations to make operations more effective. As they relate to campaigns, where objectives frame the CCMD's mission, assessments help CCDRs and supporting organizations refine or adapt the campaign plan and supporting plans to achieve operational-level objectives or to changes within the AOR.

(5) The assessment process serves as part of the CCMD's feedback mechanism throughout campaign planning and execution. It also feeds external requirements such as the CCDR's inputs to the CJCS's annual joint assessment. Assessment analysis and products should identify where the CCMD's ways and means are sufficient to attain their ends, where they are not and why not, and support recommendations to adapt or modify

the campaign plan or its components. The analyses might provide insight into basic questions such as:

- (a) Are the objectives still valid and achievable given changes in the OA?
- (b) Is the current plan still suitable to achieve the objectives?
- (c) Do changes to the OE impose additional risks or provide additional opportunities to the command?
- (d) To what degree are the resources employed making a difference in the OA?

(6) Campaign assessment analyses and products should provide the CCDR and staff with sufficient information to make, or recommend, necessary adjustments to plans, policy, resources, or authorities in the next cycle of planning to make operations more effective. Assessments can inform OSD and CJCS reporting requirements as mandated by strategic planning documents.

(a) Campaign assessment activities should facilitate the CCDR's input to SecDef specifically, and DoD more broadly, regarding the capabilities and authorities required to accomplish the missions in the CCMD's contingency plans over the CCDR's strategic planning horizon. The campaign assessment should consider expected changes in threats and the strategic environment and OE.

(b) Campaign assessment analyses and products should also help the CCDR request additional resources or to recommend re-allocating available resources to desired priorities. Assessment analyses and products likewise inform SecDef and senior leaders' resourcing decisions across all CCMDs and DoD requests to Congress to add or reallocate resources through the Future Years Defense Program.

f. Campaign Assessments

(1) Campaign assessments evaluate progress toward achieving the objectives by determining progress of achieving the operational-level objectives. Essentially, a JFC may sequence operational-level objectives along the same LOO or part of the same LOE within the campaign.

(2) Strategic guidance provides CCMDs with strategic objectives. CCMDs translate and refine those into operational-level objectives that are behavioral, territorial, or conditional in character. In some cases, the CCMD's actions alone may not achieve strategic objectives, requiring other instruments of national power.

(3) One common method is for CCMDs to evaluate each LOE with progress toward an associated operational-level objective. In this method, the CCMD simultaneously assesses each LOE and then assesses the overall effort using products from

the LOE assessments. The following discussion uses several cross-functional staff organizations. The names merely provide context for the process and are not a requirement for organizations to follow (see Figure VI-1).

(4) The assessment needs to nest with and support the campaign and national objectives and cannot rely on accomplishment of specific tasks. Commanders and staffs seek to determine whether the ongoing OAI are effectively changing the OA.

(a) **LOE Assessment**

1. Leads. LOEs include the logical arrangement of objectives, tasks, effects (desired conditions), indicators, and associated metrics. A LOE may have a specific command relationship and task organization if a single organization cannot achieve the objective.

2. Output. The LOE assessment produces updated findings, insights, and recommendations by LOE. Staffs consolidate these for presentation and validation during the strategic assessment working group.

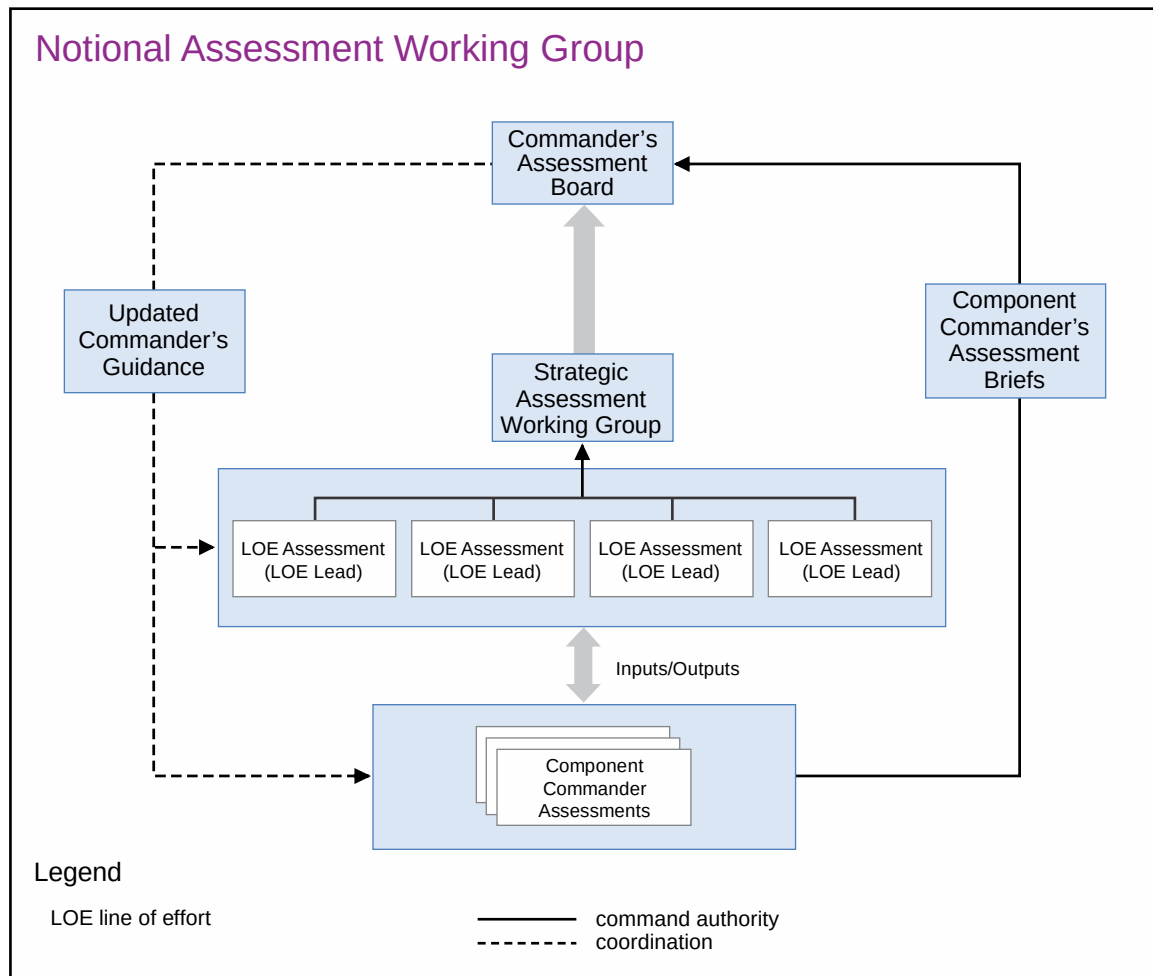


Figure VI-1. Notional Assessment Working Group

(b) **Strategic Assessment Working Group**

1. Leads. Designated lead (typically from a J-3; J-5; or force structure, resource, and assessment directorate of a joint staff element) chairs this O-6-level review working group. LOE assessors and leads brief their sub-campaign assessments, findings, insights, and recommendations to this group.

2. Output. The strategic assessment working group produces an assessment brief and recommendations for presentation and approval during the commander's assessment board.

(c) **Commander's Assessment Board**

1. Leads. CCDRs chair the commander's assessment board. Staffs and subordinates present a consolidated assessment brief with strategic assessment working group-validated, command-level recommendations for the commander's decision, centering on progress within the LOEs. As a note, this board may occur as part of the commander's council or the commander's update brief.

2. Outputs. The commander's assessment board validates recommendations for staff action and higher-level coordination and produces refined commander's guidance.

(5) **Component Command Assessment.** If required by the CCDR, component and subordinate commands provide an annual assessment to the CCDR detailing their progress toward key objectives and conduct of key operations and activities. CCDRs should account for all functions and activities that impact the information environment.

2. Purpose

a. Operation assessments help the commander and staff determine progress toward mission accomplishment. Assessment results enhance the commander's decision making, enable more effective operations, and help the commander and the staff to keep pace with a constantly evolving OE. A secondary purpose is to inform senior civil-military leadership dialogue to support geopolitical and resource decision making throughout planning and execution.

b. Integrating assessment during planning and execution can help commanders and staffs to:

- (1) Develop mission success criteria.
- (2) Compare observed OE conditions to desired conditions.
- (3) Determine validity of key planning facts and assumptions.

- (4) During execution, assess creation of the desired effects and objectives.
- (5) During execution, determine the effectiveness of allocated resources against specific task and mission performance and effects, and test the validity of objectives.
- (6) Determine whether to increase, decrease, or change resources.
- (7) Identify the risks and barriers to mission accomplishment.
- (8) Identify opportunities to accelerate mission accomplishment.

3. Tenets

The following tenets should guide the commander and the staff throughout assessment:

a. **Commander Centricity.** The commander's guidance in operation assessment is essential. The assessment plan should focus on the information and intelligence that directly support the commander's decision making.

b. **Subordinate Commander Involvement.** Assessments are more effective when used to support conversations between commanders at different echelons. Operation assessments link echelons of command by identifying the activities and impacts critical to success and sharing the assessment methods used to guide operational decisions. Understanding operational priorities enables subordinate commanders to directly communicate the most important information.

c. **Integration.** Staff integration is crucial to planning and executing assessments. Operation assessment is the responsibility of commanders, planners, and operators at every level and not the sole work of an individual advisor, committee, or assessment entity. It nests within the planning process and integrates roles across the staff. Properly structured, operation assessments enable the staff to examine and understand how actions are related. Integrating perspectives from across the staff should minimize errors that arise from limited focus (i.e., duplication of effort, incorrect identification of causes, or insufficient information to prioritize issues by level of impact).

d. **Integration into the Planning Process and Battle Rhythm.** To deliver information at the right time, the operation assessment should synchronize with the commander's decision cycle. The assessment planning steps occur concurrently with the steps of the JPP. The resulting assessment plan should support the command's battle rhythm.

e. **Integration of External Sources of Information.** Operation assessment enables the commander and staff to integrate information that updates the understanding of the OE to plan operations. To get a more complete understanding of the OE, it is important to share and solicit information with the HN, interagency, multinational, private-sector, and

nongovernmental partners. For aspects of the OPLAN for which nonmilitary influence has high impact, input from these sources is critical to refine understanding of the OE and to reduce risk. To this end, CCDRs and staff should ensure security classifications, foreign disclosure policy, and information sharing systems lend themselves to coordination with non-US military entities.

f. **Credibility and Transparency.** Assessment reports should cite all sources of information used to build the report. The staff should use methods that are appropriate to the environment and to the task of assessing a complex operation. As much as possible, sources and assessment results should be unbiased. Describe all methods and limitations in the collection of information and any assumptions used to link evidence to conclusions in the assessment report.

g. **Continuous Operation Assessment.** While an operation assessment product may follow a specific schedule, assessment is continuous in any operation. The information collected and analyzed can inform planning, execution, and assessment of operations.

4. Challenges to Operations Assessment

a. The commander's requirements for decision making should focus the assessment plan and activity. Assessment is a key component of the commander's decision cycle, helping to determine the results of OAI in the context of the overall mission objectives and providing recommendations for the refinement of plans and orders. If assessment products and analyses do not provide the commander with answers to specific questions pertaining to recommended actions to improve operational progress, acting on opportunities, or mitigating risks, they do not provide value.

b. Commanders establish priorities for assessment through their planning guidance, CCIRs, and decision points. Commanders tell their staff and subordinate commanders what they need, when (how often) they need it, and how they wish to receive it. Commanders also give their staffs guidance on where to focus limited collection and analytical resources.

(1) Commanders and staff balance collection and monitoring efforts between "what they can know" versus "what they need to know." The collecting and monitoring effort should reject the tendencies to measure things simply because they are measurable, demand measures where valid data does not exist, or ignore something pertinent because it is hard to measure. Understanding the difference may also help commanders and their staffs avoid burdening subordinates with overly detailed assessment and collection tasks.

(2) Commanders should leverage staff and subordinate commander assessments, personal observation of the OA, discussions with stakeholders, and experience and instincts to formulate their own assessment.

c. Assessment informs and strengthens the commander's understanding of the OE. Effective staffs leverage and integrate planning and operations processes and existing

reporting mechanisms whenever possible to enable synchronized assessments without adding significant additional requirements to personnel and subordinate units.

d. Significant challenges that staffs often have to overcome to enable an effective operation assessment activity include:

(1) **Integrating assessment into planning and execution from the outset.** The ongoing activities of situational awareness and assessment inform ongoing planning and execution and influence the overall decision cycle of the commander. The most successful staffs are those that routinely integrate and implement assessment activity at the onset of the planning process. Concurrently considering operational assessment during planning supports the development of well-written objectives and associated conditions or effects. Failing to consider how to assess an operation during planning can lead to tasks that do not tie to or support operational objectives.

(2) **Conducting adequate analysis before acting.** The assessment process, which includes detailed JIPOE products, improves the understanding of the OE, including understanding of the causal links between friendly OAI and changes within the OA; creating conditions favorable to mission accomplishment; and identifying actionable opportunities and risk mitigation measures to improve the likelihood of mission success. Close coordination between the assessment staff and intelligence personnel conducting JIPOE also supports brainstorming requests for information for baseline data. Use adequate analytic rigor to address complex issues to portray recommendations accurately. The staff should recognize the entire breadth of assessment contributing to the assessment operational activity of the command. The staff should consider leveraging already existing assessments and how the assessment of a specific plan contributes to the overall decision cycle of the command's higher headquarters. Recommendations provided by the assessment process provide insight to the commander and staff, enabling adjustments to current OAI and identifying planning refinement and adaptation efforts to enhance operational effectiveness.

(3) **Ensuring assessment keeps pace with a commander's appreciation of the OE.** Continual interaction with subordinate commanders, key leader engagement, stakeholders, and battle space circulation all inform the commander's understanding of the OE. Conversely, most staffs rely upon information provided by operational and intelligence reporting, usually within a set battle rhythm, and requiring consolidation, analysis, and some level of cross-staff vetting, often in the form of an assessment working group. Therefore, formal assessment reports and briefings are behind the pace of the operation. Further, senior staff and commanders will not wait for formal reports to act when necessary to adjust the operation. Staffs should leverage the collection management process, effectively calibrate assessment activity to the pace of operations, and recalibrate assessment requirements as the operation progresses to keep pace with and contribute meaningfully to the commander's understanding of the OE.

(4) **Ensuring recommendations facilitate the commander's decision making.** The staff considers what kinds of decisions the commander has to make to achieve

objective result in the intended outcomes. Decisions include both internal and external action. As such, recommendations developed during the assessment process should not be limited to only those resources and authorities over which the commander has control. Optimally, assessment recommendations should facilitate the commander's ability to provide guidance and directions to subordinates; request additional support from supporting organizations; and recommend additional diplomatic, informational, military, or economic actions to interagency and multinational partners.

(5) Resolving cross-organization resistance to assessment process requirements. All staff directorates should be aware of the importance of operations assessment to the commander as incomplete or missing data could lead to an inaccurate assessment and faulty decisions. Operation assessment is a cross-command process and developing ownership in the process and briefings (for example, where LOE working groups at the O-6 level present insights and recommendations to the commander) stimulates broader interest and quality.

(6) Integrating joint force component activities and efforts into the campaign assessment process. In most CCMDs, joint force components own most of the resources that operationalize the campaign plan. They focus on their own component support plan and Title 10, USC, activities, so it may require more effort by the CDR's campaign assessment process to make certain component operations and activities can focus to achieve the CDR's objectives.

(7) Lack of advocacy or commander disinterest. Senior staff needs to ensure the commander appreciates the value of assessment and strives to meet the assessment needs.

(8) Waiting for or requiring perfect knowledge. This concern is two-fold. First, with the ability to gather more information than the team is able to process, data can overwhelm the staff and commander leading to hesitation or inability to decide (decision/analysis paralysis). Second, a commander (or the staff) might continuously ask for additional information to confirm the assessment. Commanders and their staff need to determine when to make a decision and be comfortable making it with uncertainty.

5. Staff Organization

a. Cross-functional staff representation can effectively analyze progress toward achieving objectives. This provides the assessment activity with varied perspectives and broad expertise that are necessary for the assessment's credibility and rigor.

b. Roles and responsibilities for the assessment team are a key consideration. The ability to work across the staff impacts the quality and relevance of assessment efforts. The commander or chief of staff (COS) should identify the director or staff entity responsible for the collective assessment effort to synchronize activities, achieve unity of effort, avoid duplication of effort, and clarify assessment roles and responsibilities across the staff. The assessment activity should be routine and not ad hoc. The responsible director or staff

entity should have the authority to integrate and synchronize the staff when conducting the assessment process. The COS should play a pivotal role in staff synchronization for operation assessments, as the COS typically leads the command's operational cycle. Within typical staff organizations there are three basic locations where the responsible element could reside:

(1) **Special Staff Section.** In this approach, the assessment element reports directly to the commander, via the COS or deputy commander. Advantages of this approach may include increased access to the commander and visibility on decision making requirements, as well as an increased ability to make recommendations to the commander as part of the assessment process. Disadvantages may include isolation from the other staff sections and not having access to the information collected and monitored across the staff.

(2) **Separate Staff Section.** In this approach, the assessment element is its own staff section, akin to plans, operations, intelligence, logistics, and communications. Thus, assessment is a major staff function and enables the assessment team to participate in staff coordination and activities as co-equals with the other staff sections. A disadvantage to this approach is that it has the potential to create stove-piped assessment efforts without full collaboration for a whole-of-staff assessment.

(3) **Integrated in Another Staff Section.** In this approach, the assessment element integrates into the operations or plans sections, and the assessment chief reports to the plans chief or the operations chief. The advantage of this approach is that it tends to create close ties between the assessment team and either the plans or operations teams, but a significant disadvantage is that this approach limits the access of the assessment team to the commander and other elements of the staff and typically introduces another layer of review (and potential bias) of the assessment team's products.

6. Conducting Operation Assessment

The assessment process is continuous. Throughout the JPP, assessment supports operational design and operational art. The assessment process complements and is concurrent with the JPP in developing specific and measurable tasks effects, and objectives during operational design. These help the staff identify the information and intelligence requirements (including CCIRs). During execution, assessment provides information on progress toward creating effects and achieving objectives. Assessment reports are based on continuous situational awareness and OE analysis from internal and external sources and address changes in the OA and their proximate causes, opportunities to exploit and risks to mitigate, and recommendations to inform decision making throughout planning and execution.

See Appendix H, "Operation Assessment Plan," for details on the assessment process and examples.

CHAPTER VII

TRANSITION TO EXECUTION

“A good plan, violently executed now, is better than a perfect plan next week.”

General George S. Patton, United States Army

1. Overview

a. Plans rarely execute as written. Regardless of how much time and effort went into the planning process, commanders and their staffs should accept that the plan, as written, will likely need changes on execution. Often, the decision to employ military forces occurs in conditions significantly different from the assumptions. Planning provides a significant head start when called to deploy the military. Assessments and reframing the problem, if required, inform the applicability of, or necessary modifications to, the plan in response to changes in the OA.

(1) Effective planning enables transition. Integrated staff effort during planning ensures the plan is a team effort, sharing the knowledge gained across in the planning process. This staff work assists in identifying changes in the OA and guidance, speeding transition to execution.

(2) Detailed planning provides the analysis of the threat and the OE. The knowledge and understanding gained enables a well-trained staff to quickly identify what is different between their plan and current conditions and make recommendations based on their prior work.

(3) Detailed contingency level 3 CONPLANS and level 4 OPLANS may require more significant changes due to their specificity. Forces identified in the plan may not be available, assumptions may prove invalid, and policy and strategic decisions (and the decision timeline) may have changed or not support the original concept. However, the extra time spent on analysis provides a deeper understanding of the OE, threats, and the technical issues with projecting forces.

(4) Less-detailed plans (levels 1-2) may be more readily adaptable to execution due to their generality. However, they may require significantly more analysis (e.g., forces, transportation, logistics) to provide the detail required to enable decisions at the strategic level and ensure the plan’s executability and suitability for the problem at hand.

b. The decision to execute will often present as an examination of options in response to a developing crisis or action by a competitor state or threat (state or non-state) rather than a specific directive to execute a specific CONPLAN or OPLAN.

(1) If an existing plan is appropriate, the commander and staff should review and update the plan.

(2) If no existing plan meets the guidance, the commander and staff conduct crisis planning (planning in reduced timeline). Often, the commander and staff have conducted some previous analysis of the OE that will speed the planning process.

2. Types

a. There are three possible conditions for transitioning planning to execution:

(1) **Campaign Plan Execution.** Operations and activities within CCPs are in constant execution, as directed by orders.

(2) Contingency Plan Execution

(a) Contingency plans are typically prepared to address a specific scenario. If there is an approved contingency plan that aligns with the emergent situation, JFCs can refine or adapt that plan as necessary and execute.

(b) Members of the planning team may differ from the staff sections who oversee execution. The plans may move from J-5 contingency planning division to J-3 future operations division. They may have rotated out or be in the planning sections of the staff rather than the operations. This is the most likely situation where the conditions used in developing the plan will have changed, due to the time lag between plan development and execution. Staff from the planning team need to provide as much background information as possible to the operations team.

(c) The planning team should be a key participant, if not the lead, in updating the plan for the current (given) conditions. This enables the command to make use of the understanding gained by the staff during the planning process. The operations team should be the co-lead for the plan update to ensure they understand the decision processes and reasoning used in development of the operational approach and COAs. This speeds plan update, eases transition, and minimizes the time required to revisit the issues that arose during the initial plan development.

(3) **Crisis Planning to Execution.** Conduct crisis planning when an emergent situation arises. The planning team analyzes approved contingency plans with like scenarios to determine if an existing plan applies. If a contingency plan is appropriate to the situation, it may execute through an OPORD or FRAGORD. In a crisis, planning usually transitions rapidly to execution, so there is limited deviation between the plan and initial execution. Planners from the command J-5 can assist in the planning process through their planning expertise and knowledge gained of the OE during similar planning efforts.

b. Planning assumes forces and resources. Upon a decision to execute, replace these assumptions with the facts of actual available forces and resources. Disparities between planning assumptions and the actual OE conditions at execution drive refinement or adaption of the plan or order. Resource-informed planning during plan development

enables planners to make more realistic force and resource planning assumptions. Enabled by the common formats and collaborative systems, tools and processes, resource-informed planning intends to facilitate the transition of a plan or order by reducing the scope of required plan adjustments or refinements upon execution.

c. During execution, commanders may often consider updating objectives, effects, and tasks within a COA due to significant changes in the OA. This includes validation or invalidation of assumptions, identifying (through continuous assessment process) that the tactical actions are not resulting in the expected effects, addressing changes in the conditions of the OE, or progressing toward the objectives. The commander may determine one of three ways ahead:

(1) The current contingency plan is suitable, with either no change or minor change (such as execution of a branch)—the current operational approach remains feasible.

(2) The contingency plan's mission and objectives are sound, but the operational approach is no longer feasible or acceptable—requires a new operational approach.

(3) The mission or objectives are no longer valid, thus requiring a new COA—requires a new operational approach to support the further detailed planning.

d. Assessment could cause the JFC to shift the focus of the operation, which the JFC initiates with a new visualization manifested through new planning guidance for an adjusted operation or campaign plan.

3. Process

a. **Overview.** The transition process to contingency plan execution originates in the planning section with significant support from the intelligence staff. Planners synthesize strategic guidance from intelligence and existing plans. The output of this synthesis is a hand-off briefing to the crisis planning lead for the command (see Figure VII-1 and Figure VII-2). For crisis planning execution, plan transition follows similar steps but within the staff section responsible for crisis planning. The transition from plan to execution should consider the following points. These are not meant to be exclusive and execute simultaneously.

b. Transition Requirements

(1) Planning team validates and, if required, updates existing mission analysis.

(2) Evaluate environmental frame and intelligence analysis in the plan. Planners either validate or, if required, update existing products. Identify what has changed since plan development and assess potential impacts to planned execution.

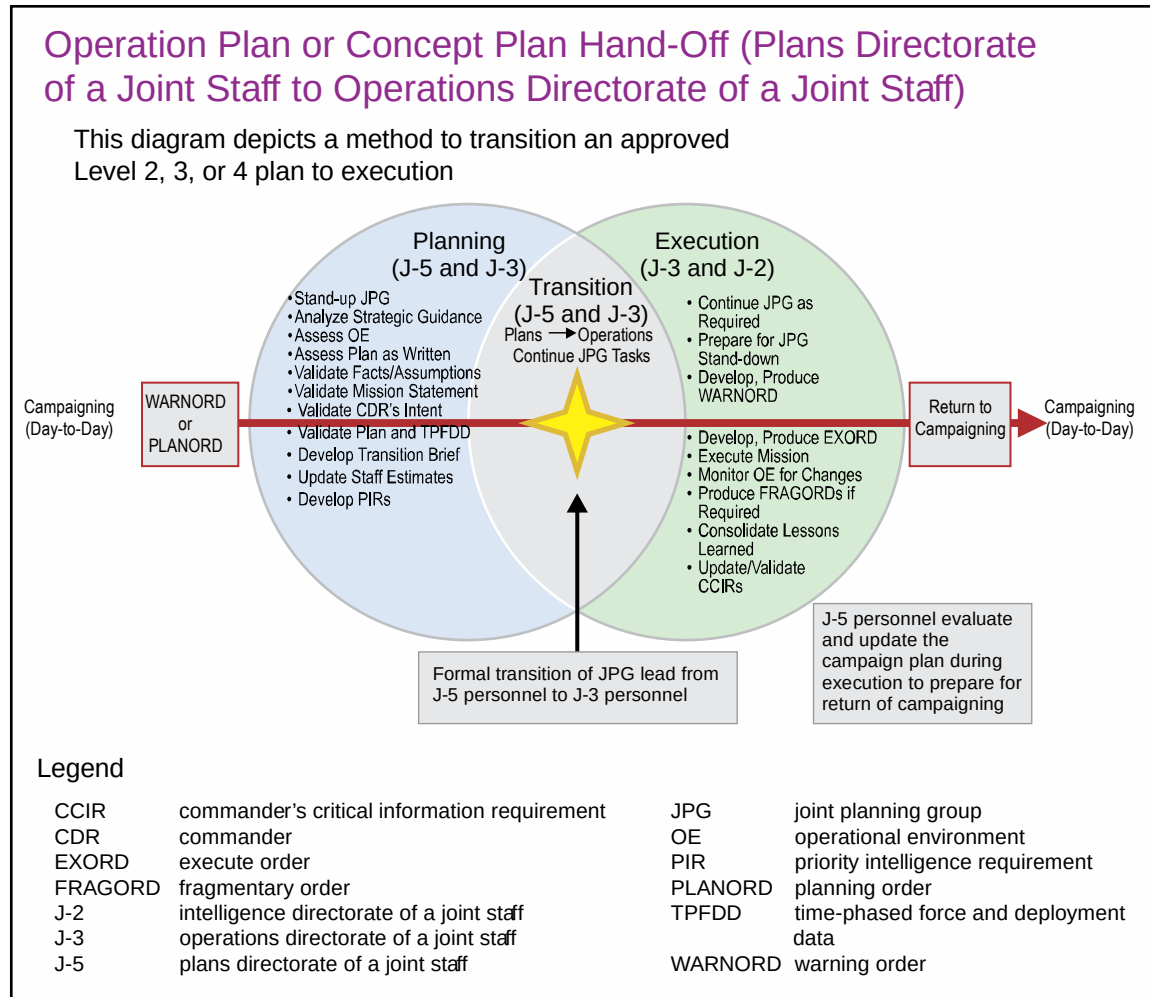


Figure VII-1. Operation Plan or Concept Plan Hand-Off (Plans Directorate of a Joint Staff to Operations Directorate of a Joint Staff)

(3) Identify any changes to strategic direction or guidance. This requires dialogue with senior civilian leadership to ensure the objectives remain synchronized with policy.

- (a) Confirm and update strategic objectives or ends.
- (b) Confirm and update operational limitations (constraints and restraints).
- (c) Validate assumptions.
- (d) Review and validate assessment criteria.

1. External (strategic) assumptions, especially those dealing with policy, diplomacy, and multinational partners, should be part of the plan review with senior civilian leadership. These are usually the assumptions dictated to the command through strategic directives (e.g., CPG, JSCP) or previous planning IPRs.

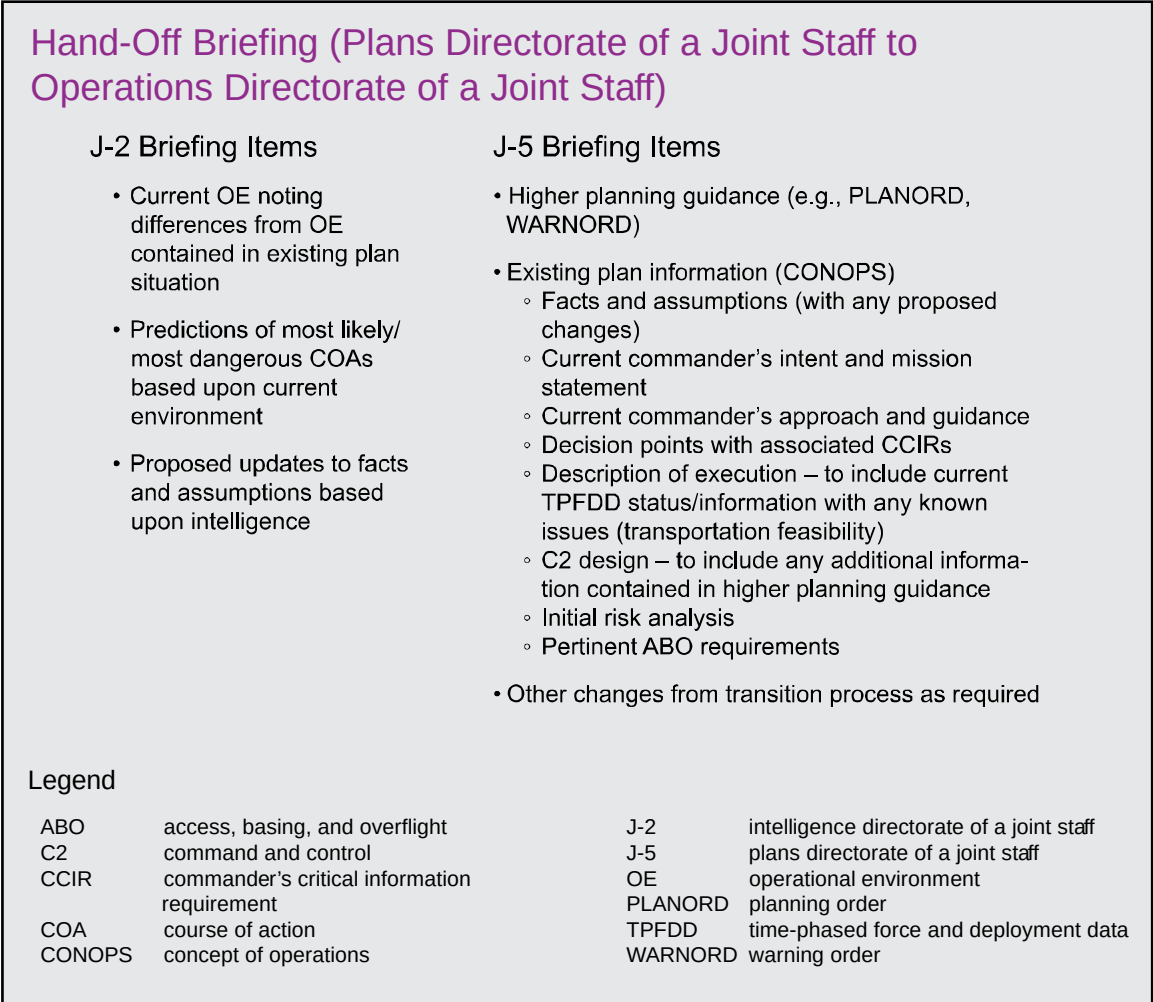


Figure VII-2. Hand-Off Briefing (Plans Directorate of a Joint Staff to Operations Directorate of a Joint Staff)

2. Internal (operational) assumptions validated by the staff through their update of the OE.

(e) Identify partners and allies.

(f) Identify interagency participation, actions, and responsibilities.

(4) Identify forces and resources, to include sustainment and transportation, with corresponding shortfalls. The forces assumed in planning are for planning purposes only; execution-sourced forces may or may not match those assumed in planning. Execution sourcing requires a dialogue between the supported CCDR, the JS, joint FPs, Services, the NGB, and United States Transportation Command.

(5) Identify decision points and CCIRs to aid in decision making. Ensure to take consideration of lead times, to include notification and mobilization for reserve forces; transportation timelines; procurement lead time (for contracts or commercial capabilities);

and joint reception, staging, onward movement, and integration requirements. These decision points are critical for senior DoD leadership to understand when to make decisions to enable operations and reduce risk. During this discussion, commanders and planners should identify alternative COAs, and the cost and risk associated with them should the commander delay or defer decisions. If decision points and CCIRs do not exist, develop and maintain them during crisis planning. Evaluate CCIRs and decision points in existing plans and update within the context of the new mission analysis and strategic guidance.

(a) **FDOs.** When and what FDOs should commanders deploy? The discussion should identify indicators that the FDOs are creating the desired effect.

(b) **FROs.** FROs can also be employed in response to coercion or aggression by a competitor or adversary. Like FDOs, the discussion should include indicators of their effectiveness and probability of consequences, desired and undesired.

(c) **De-Escalation.** During transition to execution, commanders should identify a means for de-escalation and steps to enable de-escalation without endangering US forces or interests.

(d) **Escalation.** Similarly, commanders need to identify decision points at which senior leaders make decisions to escalate to ensure strategic advantage, to include the expected risk associated should the adversary gain the advantage prior to US commitment.

(6) **Confirm Authorities for Execution.** Request and receive presidential or SecDef authority to conduct military operations. Authorities granted may be for execution of an approved plan or for limited execution of select phases of an approved plan.

(7) **Direct Execution.** The JS, on behalf of the CJCS, prepares orders for the President or SecDef to authorize the execution of an order. The authorities for execution, force allocation, and deployment are separate, vice in a comprehensive order. Upon approval, CCDRs and Services pass orders down the chain of command directing action ordered by higher headquarters. The following orders are some of those used in the process of transitioning from planning to execution: WARNORD, PLANORD, ALERTORD, OPORD, PTDO, DEPOD, EXORD, and FRAGORD.

(a) **Contingency Plans.** The authority to execute a contingency plan may come incrementally. Initial execution authority may limit to early phase activities (e.g., FDOs, FROs) and CCDRs should expect to request additional or modified execution authorities as an operation develops.

(b) **CCPs.** CCPs are in constant execution per order. The authorization to execute the campaign does not provide complete authority for the CCDR to execute all the individual operations and activities that comprise the plan. CCMDs coordinate to execute the discrete military activities within a campaign plan to include posture, force allocation, and country-team coordination.

See CJCSM 3130.03, Planning and Execution Formats and Guidance, for more information on the content and format of orders.

c. **Impact on Other Operations.** As the plan transitions to execution, the commander and staff synchronize that operation with the rest of the CCMD's campaign. Responses to contingencies will likely change the operational and strategic environments. This necessitates revisiting and updating the existing campaign plan based upon new conditions.

(1) The commander identifies how the additional operation will affect the campaign.

(a) **Resources.** Resources may divert from lower-priority operations and activities to support the new operation. This may require modifying the campaign or adjusting objectives.

(b) **Secondary Effects.** Adding new operations, especially combat operations, impacts the perception and effects of other operations within the AOR (and likely in other CCMD's AORs as well). Both the new operation and existing ones may need to adjust to reflect the symbiotic effect of simultaneous operations.

(2) The commander may require support from other CCMDs. In addition to support within the plan transitioning to execution, the CCDR may require external support to ensure continued progress toward the CCP's objectives. By using a pre-established capability (force) sharing agreement, a CCDR can gain the support needed without requiring additional JS or OSD coordination. Support from other CCMDs often requires shared battle rhythm activities. Balancing the benefit of improved awareness without overburdening commanders and their staffs remains a challenge. Informal cross-CCMD, directorate-level coordination has proven beneficial and can expand when security conditions necessitate deeper coordination and synchronization. However, identifying standardized staff organizations provides additional structure when planning and scheduling across organizational boundaries.

(3) Depending on the significance of the new operation, the CCDR may need to update the CCP's objectives. This requires a conversation with senior civilian leaders to see if the US national objectives should adjust given the change in the strategic landscape.

4. Tools

a. **Rehearsals and Exercises.** For most likely contingencies, the commander may hold rehearsals to ensure assessment and understanding of transition activities. When the situation allows, the commander could also conduct rehearsals of key events or use the scenario for an exercise (command post or field training exercise).

b. **Transition Book.** Commands may develop a condensed primer for the commander and staff that provides a rapid overview of options and requirements for a response. This could include checklists, a DST, FDOs, FROs, and location of key files and TPFDD. This

speeds the preparation of products to support the commander's discussion with higher headquarters or senior decision makers.

c. **Staff Checklists.** Staff sections should develop detailed checklists by both functional areas and timeline on actions required to support execution of crisis options.

CHAPTER VIII

THE FUTURE OF JOINT PLANNING

1. Anticipating the Future Operating Environment

In the future, JFCs and joint planners will face adversaries capable of functioning across the expanding OE; an environment that will reach from the bottom of the ocean to numerous deep-underground shelters, across the surface of the Earth, and into cislunar space. Within this vast operating area, the quantity of computers, processors, and sensors will be the densest on the Earth's surface and in urban areas. Humans will operate most of those devices in the physical environment, which will connect with one another through cyberspace and the EMS.

For more information on anticipating the future OE for operational design and joint planning, see Joint Operating Environment 2045.

a. Technological developments seek to consume and process large amounts of data to shorten the decision cycle by both ally and adversary alike. Combined with social media and the rapid flow of information, actions quickly play out on a global stage seeking to sway public opinion.

b. In a period where powerful nations compete, global powers will likely avoid escalating into major combat operations with each other and, instead, deliberately remain in a period of continual provocation and episodic armed conflict. Enduring conflicts will seek to stay beneath thresholds that trigger large combat operations but attrit capacity and erode readiness of standing militaries. Adversaries will seek to challenge and degrade the ability to govern and to provide services. Adversaries will also erode popular support through coordinated efforts in multiple domains. Nations will likely seek to contain vertical escalation of armed conflict intensity but leverage asymmetric escalations horizontally across regions to continue to weaken or coerce an adversary.

c. Developing human capital to know “how-to-think,” as opposed to “what-to-do” in the future is important to shape the development of future technology. The future OE demands a proactive approach where JFCs leverage emerging technologies to adapt their planning approaches.

2. Shaping Future Technology Development to Advance Planning Efforts

Technological advances in machine learning and artificial intelligence enable the processing of increased volumes of data. This technology is not exclusive to any one nation, and both allies and adversaries seek to harness these capabilities. Planners should consider how future technology can equip advisers and decision makers for better-informed action in a shorter time frame. Emerging, digital tools are intended to deliver integrated options and integrated commanders' estimate to senior leaders. This process is intended to expand senior leaders' decision space, optimize multi-tier parallel planning, and shorten the traditional joint planning timeline.

a. **Enable Expedited Decision Timelines.** For integrated planning in the future, planners will use digital tools that leverage advanced processing capabilities to expedite decision timelines for commanders and their planning staffs for all types of events. Planning teams will be able to leverage existing plans, risk assessments, and joint orders resident within the digital tools and input new considerations based on emerging requirements. Access to this type of information enables JFCs to create new integrated options and associated recommended LOEs, objectives, and activities. The accelerated digital tool-enabled planning process may significantly decrease initial planning timelines, but these tools do not replace the JPP—rather they enable and accelerate it.

b. **Integrated Commander's Estimate.** Planners can use these advanced software tools to create globally integrated commander's estimates and options along with their recommendation to facilitate senior leader decision making and unity of effort across CCMDs. Planner recommendations should reflect a synthesis of considerations, including the OE, the availability of forces, logistical sustainability, the expected effect on the enemy, and their anticipated response. The integration of multiple plans enabled by digital resources helps create a more integrated commander's estimate. The integrated estimate synthesizes multiple LOEs, objectives, and associated activities between and across CCMDs, interagency partners, and allies. This estimate provides senior leaders with a clear assessment of integrated options, framed in terms of risks, anticipated outcomes, and necessary resources that enables development of a truly integrated global CONOPS.

c. **Real-time, Conditions-Informed Decision Making.** Future technology could serve as a thermometer of a country's leadership, global partnerships, regional alliances, or a population's sentiment. Coercion and deterrence exist in the minds of individuals, but there are currently few ways to forecast and assess if an action will have a desired effect in the short or long terms. A thermometer can measure when an action pushes an entity too far or, conversely, when coercive actions have negligible or deleterious effect. It can also inform decision makers when events that may not be readily apparent may influence how another leader reacts.

(1) CCDRs staff their formations with expertise for the region under their span of control. However, they do not have the bandwidth to have the same understanding of other regions and how activity elsewhere impacts the campaign plans or preparedness for contingency in their AOR. Future technology's capacity to consume vast amounts of data while discerning patterns enables CCMDs to understand how action in other theaters furthers or detracts from their objectives. Similarly, CCMDs could leverage the data to identify adversary vulnerabilities in other theaters and coordinate across CCMDs to achieve results.

(2) A barometer could measure public information, weather events, economic data, and crisis, while merging with sensitive intelligence collection. While not all-inclusive, the volume of data would far exceed human capacity, driving a need for technology to compile and present potential impacts to decision makers. The resultant product could help the military and national leadership best navigate the volatile future environment, with less unintentional consequence from uninformed action or inaction.

d. **Digitally Aided Responsiveness for Armed Conflict.** The joint force dedicates the bulk of planning manpower and resources toward contingency planning and creating campaign plans. At the onset of a crisis, it is prudent to revisit campaign plans for an affected region and any related contingency plans. However, the short timeline and rapidly changing conditions associated with crises often do not allow planners to fully capitalize on previous planning work or products. Future technology may afford the means to quickly compile FDOs or FROs from across a series of contingency plans against an adversary or campaign plan activities in a region and generate a menu of options that achieve a desired outcome. Employing future technology to compile, categorize, and present the pre-planned actions could enable decision makers to quickly leverage and employ the deliberate planning efforts of a staff during crisis response. While a crisis is not necessarily a prelude to executing a contingency plan, much of the planning to deter or compel a behavior prior to activating a contingency plan is applicable to a stand-alone crisis. Using technology to make this planning, and the resultant options, more accessible increases responsiveness while decreasing crisis planning effort and response time. Cloud-based applications are being fielded and tested in various CCMD and Service component command headquarters. These applications allow planners to share planning information with other planners across the JPEC, which increases the speed of integrated planning. Some applications allow users to share common operating pictures in real time; rapidly produce orders, synch matrices, and command relationship diagrams; and track MOP and MOE changes in real time for faster assessments. Future cloud-based applications will be tied in with real-time TPFDD information and allow artificial intelligence to help validate planning assumptions and generate rapid wargames during crisis planning.

e. **Technology-Informed Resourcing.** Applying military force produces zero-sum outcomes that balance current capability against future readiness. The decision to apply force today degrades future readiness. However, measuring future readiness is difficult to quantify due to the wide variety and depth of plans.

(1) Future technologies that aggregate readiness information, future contingency requirements, and current capacity could impact readiness positively by providing decision makers with measurable employment options. Those technologies would provide a mechanism for continued assessment in periods of sustained or enduring crisis or conflict that erodes readiness and capacity over time.

(2) Data-informed decisions enable JFCs and planners to weigh probabilistic risk outcomes. Technology that merges OE data with intelligence assessments could highlight capability and capacity shortfalls, allowing Services and the defense industrial base to mitigate those shortfalls and risk outcomes rapidly.

(3) The range of challenges with TPFDDs residing on US systems and networks for contingency plans and operations range from data inaccuracies, mismatched data details between systems, and data sovereignty. Emerging technologies like generative artificial intelligence have also complicated the processing and dissemination of deployment data among users and planners. Moreover, to ensure data protection, sharing deployment data with allies and partners remains a challenge due to the ways each nation enforces its data

protocols. Mitigating these challenges will require technological improvements to deployment data development, storage, and transmitting practices, but mitigation also requires the adoption of flexible, adaptive strategies like increased collaboration between governments and the private sector.

3. Education and Development

Joint doctrine consists of the fundamental principles that guide the employment of joint forces to achieve unity of effort. It also provides the foundation for education and training. This foundation can encourage forward thinking by allowing JFCs to prepare for challenges of the future OE.

a. **Fostering Critical Thinking in Future Planners.** The next generation of joint planners needs critical thinking (careful, goal-oriented thinking) to adapt to the evolving character of war. Building upon the procedures and planning steps outlined in the JPP and operational design, planners should understand how nation states seek to achieve their strategic objectives short of war. This requires thinking beyond the rigidity of established planning cycles to present developed military options that fit within a campaign plan but still enable readiness to execute contingency plans.

b. **Expand Interagency Integration.** The US military combined with other instruments of power for national security purposes is an effective national strategy. Opportunities that increase crosstalk, partnerships, and cooperation with interagency partners promote unified action. However, interagency partners cannot match the size and scope of the military's planning apparatus. Military planners should, therefore, seek to better understand the authorities and capabilities of other USG departments and agencies and strive to incorporate them into campaign and contingency plans. Similar to the way joint forces work with partners and allies, incorporating interagency capabilities and perspectives enables more comprehensive options for decision makers.

c. **Expand Planning Beyond Resources.** Adversaries and nation states may avoid armed conflict by signaling their intentions and actions to avoid it because of the significant materiel and non-materiel costs they anticipate, including the disruptive externalities to global economies. Although armed conflict is one approach for adversaries or nations to achieve their objectives, they are more likely to pursue deliberate strategies of continuous confrontation and provocation to achieve the same ends. Future military planners should anticipate these strategies and understand how resource constraints prevent unintentional attrition of capacity that degrades joint force readiness and how expanded, or conditions-based authorities, provide options for JFCs.

d. **Combine Skillsets to Create Decision Support Cells.** Interoperability between different staff functions and action officers can assist in analyzing an adversary. A commander can create a decision support cell to drive cross-functional collaboration, across time horizons, to allow JFCs to better understand: the adversary's view of itself; the adversary's view of joint forces; and the joint force's view of the adversary.

e. **Understand and Integrate Commercial Capabilities.** The future is now with the commercial dimension, due to the inextricable link between commercial capabilities—infrastructures, services, and personnel—and joint force operations and joint force planners should be well-versed in commercial support, which extends beyond logistics, to include additional functionalities such as satellite systems, unmanned aircraft systems, exquisite system support, technical advisers, and other specialized capabilities not resident in our organic force. From competition to combat operations, DoD often employs commercial capabilities as an FDO in place of projecting force for myriad reasons. This trend may continue where well-planned integration of commercial capabilities displaces uniformed personnel deployments. The joint planner can leverage available training resources to understand, assess, and employ these options.

f. **Talent Development.** JFCs can use academic opportunities, inter-government fellowships, and private-industry forums to enhance education and training. Career talent management trajectories can nurture talent, grounded in a planning education that incorporates industry experience and other instruments of power, to build more breadth in planning professionals. Such a talent management system can deepen the bench of military professionals who are adept at generating military options across the competition continuum.

4. Conclusion

The future OE demands a proactive approach where JFCs leverage emerging technologies to adapt their planning approaches. Trained and well-informed planners remain critical to operational success in the future OE. JFCs should keep pace with emerging technologies while balancing limited resources to acquire and develop the human capital to use it. While joint planning is a deliberative process, it should remain flexible enough for planners to create opportunities to anticipate future requirements, mitigate risk, and maximize opportunities for resource employment.

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APPENDIX A STRATEGIC ESTIMATE

SECTION A. INTRODUCTION

1. Background

a. The strategic estimate is an analytical tool available to CCDRs before developing strategies; campaign plans; subordinate campaign plans; and deliberate plans, levels 1 to 4. Strategic estimates provide the commander's perspective of the strategic and operational levels of the OE, threats, and opportunities that could facilitate or hinder the achievement of NDS- and NMS-directed objectives, desired changes to meet objectives, and the commander's visualization of how to achieve those objectives. Developed annually and regularly updated, the strategic estimate is the basis for developing the CCDR's theater or functional strategy.

b. The CCDR, CCMD staff, supporting commands, and departments and agencies assess the factors that influence the OE, thus informing the ends, ways, means, and risks involved in achieving the prescribed objectives.

c. Both supported and supporting CCDRs prepare strategic estimates based on assigned tasks. CCDRs who support multiple commands may prepare strategic estimates for each supporting operation.

d. Section B, "Notional Strategic Estimate Format," presents a format a CCMD staff can use as a guide when developing a strategic estimate. The J-5 may provide the lead staff organization for the conduct of the strategic estimate with significant participation from the other staff directorates. The exact format and level of detail may vary somewhat among commands, based on theater-specific requirements and other factors.

e. The result of the strategic estimate is to better understand the OA. The strategic estimate process is dynamic and continuous and provides input for developing theater strategies and campaign plans. This strategic estimate is also the starting point for conducting more detailed staff estimates, as well as the commander's estimate of the situation for a potential contingency.

f. The CCDR's strategic estimate should identify potential for spillover, both from the AOR and functional area perspective, into other CCDRs' AORs or functional areas and into the CCDR's AOR or functional area, based on operations and activities outside the AOR.

SECTION B. NOTIONAL STRATEGIC ESTIMATE FORMAT

2. Strategic Direction

(This section analyzes broad policy, strategic guidance, and authoritative direction to the theater or global situation and identifies strategic requirements in global and regional dimensions.)

a. **US Policy Goals.** Identify the US strategic objectives and tasks assigned to the CCMDs.

b. **Non-US/Multinational Policy Goals.** Identify the multinational objectives and tasks assigned to or coordinated by the CCMD.

3. Operational Environment

a. **AOR.** (Provide a visualization of the relevant geographic, political, economic, social, demographic, historic, and cultural factors in the AOR assigned to the CDR.)

b. **Area of Interest.** (Describe the area of interest to the commander, including the area of influence and adjacent areas and extending into adversary territory. This area also includes areas occupied by enemy forces that could jeopardize the accomplishment of the mission.)

c. **Adversary Forces.** (Identify all states, groups, or organizations expected to be hostile to, or that may threaten, US and partner-nation interests and appraise their general objectives, motivations, and capabilities. Provide the information essential for a clear understanding of the magnitude of the potential threat, including threats to power projection activities.)

d. **Friendly Forces.** (Identify all relevant friendly states, forces, and organizations. These include assigned US forces, regional allies, anticipated multinational partners, and private and commercial partners. Describe the capabilities of the other instruments of national power [diplomatic, economic, and informational], US military supporting commands, and other USG departments and agencies that could have a direct and significant influence on the operations in this AOR.)

e. **Neutral Forces.** (Identify other states, groups, or organizations in the AOR and determine their general objectives, motivations, and capabilities. Provide the information essential for a clear understanding of their motivations and how they may impact US and friendly multinational operations.)

f. **The Civilian Environment.** (Identify factors to civilians and their communities, including the civilian population, personnel, organizations, resources, infrastructure, essential services, and other systems. Provide the information essential for a clear understanding of how US and multinational operations affect this environment.)

4. Assessment of the Major Strategic and Operational Challenges

- a. This is a continuous appreciation of the major challenges in the AOR.
- b. These may include a wide range of challenges, from direct military confrontation, peace enforcement, and security cooperation activities (that include security force assistance for building partner capacity and capability), to providing response to atrocities, and foreign humanitarian assistance.

5. Potential Opportunities

- a. This is an analysis of known or anticipated circumstances, as well as emerging situations, that the CCMD may use as leverage to improve the theater strategic situation and further US or partner-nation interests.
- b. Appraise each potential opportunity with respect to existing strategic guidance and operational limitations.

6. Assessment of Risks

- a. This assessment matches a list of the potential challenges with anticipated capabilities in the OA.
- b. Analyze risks associated with each major challenge separately and categorize according to significance or likelihood (e.g., most dangerous, most likely).
- c. The CCMD staff should develop a list of possible mitigation measures to these risks.

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APPENDIX B STAFF ESTIMATES

SECTION A. INTRODUCTION

1. Role of Estimates

a. Staff estimates are central to formulating and updating military action to meet the requirements of any situation. Staff estimates should start with the strategic estimate and be comprehensive and continuous and visualize the future, while optimizing the limited time available so as not to become overly time-consuming. Comprehensive estimates consider both the quantifiable and the intangible aspects of military operations. They translate friendly and enemy strengths, weapons systems, training, morale, and leadership into combat capabilities. The estimate process requires the ability to visualize the battle or crisis situations requiring military forces.

b. Estimates are an essential part of the operational design process. Through their estimates, the staff provides assessment of the OE and factors affecting planning and execution toward achievement of objectives.

c. Estimates are as thorough as time and circumstances permit. The JFC and staff, after collecting, processing, and evaluating information, update their estimates:

(1) When the commander and staff recognize new facts.

(2) When they replace assumptions with facts or find their assumptions invalid.

(3) When they receive changes to strategic direction based on high-level civilian-military dialogue or when assessment recommends to refine, adapt, or terminate planning.

d. Estimates for the plan in execution can often provide a basis for estimates for other plans, as well as changes to the plan in execution. Technological advances and near real time information estimates enable continuously updated estimates. Estimates must visualize the future and support the commander's visualization. They are the link between planning and execution and support continuous assessment. The commander's vision articulated in the strategic estimate guides the operations and activities. Each subordinate unit commander also requires the ability to envision the organization's objectives, as well as those desired by their opposition counterpart. Estimates contribute to this vision. Failure to make staff estimates can lead to errors and omissions when developing, analyzing, and comparing COAs.

e. Not every situation allows or requires an extensive and lengthy planning effort. It is conceivable that a commander could review the assigned task, receive oral briefings, make a quick decision, and direct writing of the plan to commence. This completes the process and is suitable if the task is simple and straightforward.

f. Most commanders, however, are more likely to demand a thorough, well-coordinated plan that requires a complex staff estimate process. Written staff estimates are carefully prepared, coordinated, and fully documented.

g. Because of the unique talents of each directorate, involvement of all is vital. Each staff estimate takes on a different focus that identifies certain assumptions, detailed aspects of the COAs, and potential deficiencies that are simply not known at any other level. Such a detailed study of the COAs involves the corresponding staffs of subordinate and supporting commands.

h. Each staff directorate:

(1) Reviews the OE, mission, and situation from its own staff functional perspective.

(2) Examines the factors and assumptions for which it is the responsible staff.

(3) Analyzes each COA from its staff functional perspective.

(4) Concludes whether the mission is supportable.

i. **This process produces revised, documented staff estimates.** These are extremely useful to the commander's J-5 staff, which extracts information from them for the commander's estimate. The estimates are also valuable to planners in subordinate and supporting commands as they prepare supporting plans. Although planners can delay documenting the staff estimates until after the preparation of the commander's estimate, they should distribute to subordinate and supporting commanders in time to help them prepare annexes for their supporting plans.

j. The principal elements of the staff estimate include mission, situation and considerations, analysis of opposing COAs, comparison of friendly COAs, and conclusions. The coordinating staff and each staff principal develop facts, assessments, and information that relate to their functional field. Types of estimates generally include operations, personnel, intelligence, logistics, communications, civil-military operations, military deception, and special staff. The details in each basic category vary with the staff performing the analysis. The principal staff directorates have a similar perspective—they focus on friendly COAs and their supportability. The J-2 staff estimate is separate from the intelligence estimate provided at the beginning of the planning process. Complete the staff estimate during the strategic guidance planning function and identify available CCMD intelligence collection and analytic capabilities or anticipated shortfalls that may limit the J-2's ability to support the proposed friendly COAs. Also, during the strategic guidance planning function, based on continuous JIPOE, the J-2 produces the intelligence estimate that serves as the baseline assessment of the OE, adversary capabilities (including requirements, vulnerabilities, and COGs), and an analysis of the various COAs available to the adversary according to its capabilities. The intelligence estimate conclusion indicates the adversary's most likely COA, identifies the effects of that COA on the accomplishment of

the assigned mission, and, where applicable, lists exploitable adversary vulnerabilities associated with that COA. The intelligence estimate informs the commander's estimate.

k. In many cases, the activities in the JPP COA development step are not separate and distinct, as the evolution of the refined COA illustrates. Staff estimates and assumptions used in the initial COA development may be based on limited staff support. But as concept development progresses, COAs evolve to include many of the following considerations:

- (1) What military operations did the COA consider?
- (2) Where will they act?
- (3) Who will conduct the operation?
- (4) When is the operation planned to occur?
- (5) How will the operation execute?

l. An iterative process of modifying, adding to, and deleting from the original tentative list refines COAs. The staff continually evaluates the situation as the planning process continues. Early staff estimates are frequently given as oral briefings to the rest of the staff. In the beginning, they tend to emphasize information collection more than analysis. It is only in the later stages of the process that the staff estimates indicate which COAs can be best supported.

m. **Sample Estimate Format.** The following is a sample format to guide developing an estimate. The exact format and level of detail may vary somewhat among joint commands and primary staff sections based on theater-specific requirements and other factors. Refer to CJCSM 3130.03, *Planning and Execution Formats and Guidance*, for the specific format when there is a requirement for the supported JFC to submit a commander's estimate.

SECTION B. SAMPLE ESTIMATE FORMAT

2. Mission

a. Mission Analysis

- (1) Determine the higher command's purpose. Analyze strategic direction, as well as appropriate guidance in partner nations' directions.
- (2) Determine specified, implied, and essential tasks and their priorities.
- (3) Determine objectives and consider desired and undesired effects.
- (4) Reassess if the strategic direction and guidance do not support the objectives.

b. Mission Statement

- (1) Express in terms of who, what (essential tasks), when, where, and why (purpose).
- (2) Frame as a clear, concise statement of the essential tasks to accomplish and the purpose—the action to take and the reason for doing so.

3. Situation and Courses of Action

a. Situation Analysis

(1) Geostrategic Context

(a) Domestic and international context: political or diplomatic long- and short-term causes of conflict; domestic influences, including public will, competing demands for resources and political, economic, legal, moral, and social constraints; and international interests (reinforcing or conflicting with US interests, including positions of parties neutral to the conflict), international law, positions of international organizations, and other competing or distracting international situations. Consider similar factors for theater and functional campaigns and noncombat operations.

(b) A systems perspective of the OE: political, military, economic, social, information, and infrastructure, and other related aspects.

(2) **Analysis of Enemy, Adversary, and Competitors.** Scrutiny of the situation, including capabilities and vulnerabilities (at the theater level, commanders normally have available a formal intelligence estimate), should include the following:

- (a) National and military intentions and objectives (to extent known).
- (b) Broad military COAs taken and available in the future.
- (c) Military strategic and operational advantages and limitations.
- (d) Possible external military support.
- (e) COGs (strategic and operational) and decisive points (including geographic places, key events, and critical factors).
- (f) Specific operational characteristics such as strength, composition, location, and disposition; reinforcements; logistics; time and space factors (including basing used and available); and combat/noncombat efficiency and proficiency in joint operations.
- (g) Reactions of third parties/competitors in theater and functional campaigns.

(3) **Friendly Situation.** Should follow the same pattern used for the analysis of the adversary. At the theater level, CCDRs normally have available specific supporting estimates, including personnel, logistics, and communications estimates. Multinational operations require specific analysis of partner nations' objectives, capabilities, and vulnerabilities. Consider interagency coordination required to achieve objectives.

(4) **Operational Limitations.** Actions either required or prohibited by higher authority, such as constraints or restraints, and other restrictions that limit the commander's freedom of action, such as diplomatic agreements, political or economic conditions in affected countries, and HN issues.

(5) **Assumptions.** Assumptions are intrinsically important factors upon which the conduct of the operation is based. Make assumptions only when necessary to continue planning.

(6) **Deductions.** Deductions from the analysis should yield estimates of relative combat power, including enemy capabilities that can affect mission accomplishment.

b. **COA Development and Analysis.** COAs are based on the above analysis and a creative determination of how to accomplish the mission. Each COA is suitable, feasible, and acceptable. State all practical COAs open to the commander that, if successful, will accomplish the mission. For a CCDR's strategic estimate, each COA typically constitutes an alternative theater strategic or operational concept and should outline the following:

- (1) Major strategic and operational tasks to accomplish in order of accomplishment.
- (2) Major forces or capabilities required (to include joint, interagency, and multinational).
- (3) C2 concept.
- (4) Sustainment concept.
- (5) Deployment concept.
- (6) Estimate of time required to achieve the objectives.
- (7) Concept for establishing and maintaining a theater reserve.

4. Analysis of Enemy and Adversary Capabilities and Intentions

a. Determine the probable effect of possible enemy or adversary capabilities and intentions on the success of each friendly COA from mobilization and projection of forces through achievement of the objectives.

b. Conduct this analysis in an orderly manner by time phasing, geographic location, and functional event. Consider:

- (1) The potential actions of subordinates two echelons down.
 - (2) Transition issues; think through own action, enemy reaction, and counteraction.
 - (3) The potential impact on friendly desired effects and likelihood that the enemy's or adversary's actions will cause specific undesired effects.
- c. Conclude with revalidation of friendly COAs. Determine additional requirements, make required modifications, and list advantages and disadvantages of each enemy or adversary capability.

5. Comparison of Own Courses of Action

- a. Evaluate the advantages and disadvantages of each COA.
- b. Compare with respect to evaluation criteria.
 - (1) Fixed values for joint operations (the principles of joint operations, the fundamentals of joint warfighting, and the elements of operational design).
 - (2) Other factors (for example, political constraints).
 - (3) Mission accomplishment.
- c. If appropriate, merge elements of different COAs into one.
- d. Identify risk specifically associated with the assumptions (i.e., what happens if one or more assumptions proves false).

ON WRITING SMART OBJECTIVES

Specific: Who is involved? What needs to be accomplished? Where will priorities be documented? What categories are important? Percentages and specific quantified metrics; Deadlines are specific metrics.

Measurable: On what observable objects, action and condition does achieving this objective rest? What terrain or resource is used across Operations, Activities, and Investments are related? What parts of the terrain are mission-relevant? Percentages are measurable quantities.

Achievable: What benchmarks/levels do we need to reach? What standards does our operations or tasks set?

Relevant: What is the directive that tells us to accomplish this objective? Who involved is also a matter of relevance?

Time-bound: What is our initial, aspirational deadline?

6. Recommendation

Provide an assessment of which COAs are supportable, an analysis of the risk for each, and a concise statement of the recommended COA with its requirements.

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APPENDIX C

PLANNING CONSIDERATIONS FOR JOINT ALL-DOMAIN OPERATIONS

a. Joint warfighting in the modern era requires an unprecedented level of integration across domains to achieve operational success. JFCs are tasked with the complex challenge of orchestrating diverse capabilities to create synchronized effects, ensuring that their operations are both adaptive and decisive. To meet these demands, commanders employ innovative planning frameworks that allow them to harness the full array of joint capabilities while responding to rapidly changing OEs.

b. Planning for joint all-domain operations (JADO) comes with several challenges that joint planners have to navigate. These challenges require innovative solutions, advanced technology, and strong collaboration among forces. Key obstacles include:

(1) **Interoperability Issues.** Integrating different military branches and allied forces with varying technologies and communication systems.

(2) **C2 Complexity.** Coordinating operations across all domains while maintaining flexibility and responsiveness.

(3) **Information Overload.** Managing vast amounts of data from intelligence, surveillance, and reconnaissance sources to make timely decisions.

(4) **Network Threats.** Protecting networks and systems from cyberspace attacks that could disrupt operations.

(5) **Adversary Adaptation.** Countering rapidly evolving threats and adversary strategies that challenge traditional military planning.

(6) **Logistical Constraints.** Ensuring movement of information and supply across air, land, maritime, space, and cyberspace.

c. JFCs start JADO planning with an understanding of the OE, the enemy's order of battle, and likely COAs. They base their CONOPS on enemy weaknesses rather than their strengths. They direct all-domain planning to leverage joint force advantages and target enemy vulnerabilities. Planning for JADO enables JFCs to recognize and seize opportunities while mitigating risk. In complex situations, joint operations go beyond deconflicted Service component actions. Joint warfighting requires coordinated operations with all-domain qualities and attributes working together to survive and succeed. The planning considerations include:

(1) **Intelligence.** Intelligence planners use battlespace intelligence to give JFCs insights and estimates. They develop IRs to address gaps and meet JFC intent.

(2) **Information.** Information planners provide JFCs with estimates on how to leverage information to support operations in the physical domains and cyberspace. This approach aims to use these operations to achieve an informational advantage.

(3) **Cyberspace.** Cyberspace plays an integral role in JADO. Cyberspace operations planners offer critical assessments to JFCs on the integration of cyberspace capabilities, countering adversarial activities within cyberspace, identifying and securing key digital terrain, and operating effectively in restricted environments.

(4) **Space.** Space planners provide JFCs evaluations on how to effectively integrate space capabilities. These assessments encompass operational needs, constraints, liaison requirements, legal limitations, and threats from directed energy, nuclear detonation, and electromagnetic sources.

(5) **Joint EMS Operations.** Joint EMS operations planners suggest ways to operate in the EMS. They create staff estimates to aid CONOPS development and selection. JFCs guide subordinate and component commanders on EMS activities and capabilities for planning. The JFC staff assigns tasks, identifies shortfalls, and proposes solutions during plan development.

(6) **Military Deception.** Military deception planners offer JFCs estimates on methods to analyze enemy beliefs and preferences, identify ways to exploit those preferences, influence enemy assumptions to generate exploitable perceptions, and orchestrate deception events. Military deception planners evaluate joint force vulnerabilities to enemy denial and deception across all domains.

(7) **Operations Security.** Operations security planners recommend to JFCs assessments on methods to manage detectable signatures and minimize exposure to enemy collection of vital information and indicators. JFCs utilize the operations security cycle to identify, manage, and safeguard critical information.

(8) **Protection.** Protection planners help JFCs by estimating ways to safeguard their forces through security measures, emergency management, and response actions. Protected assets range from physical objects and personnel to information and other valuable intangible resources.

(9) JADO targeting involves continuous analysis and integration to prioritize JFC objectives, targets, responses, requirements, capabilities, and ROE. JFCs synchronize various capabilities, information needs, and execution timing, considering complex enemies across all domains, EMS, and information environment.

(10) **Schemes of Maneuver, Fires, and Logistics.** The CONOPS's scheme of maneuver is based on an understanding of the enemy's COA and their capabilities relative to the JFC's mission. After identifying the challenges and opportunities for specific mission objectives, JFCs develop their schemes of maneuver, fires, and logistics. JFCs use CONOPS to guide the development of supporting plans or annexes and to explain how they plan to achieve the mission.

d. A JFC's JADO can vary in scope and scale from globally integrated operations across multiple CCMDs to a JTF with specific tasks within a defined joint OA inside a CCMD AOR. The interval of a specific joint all-domain operation may be as long as a week or as short as a few hours. JFCs' ability to refine, adapt, and transition JADO

throughout execution helps them maximize the potential of their forces. JFCs execute JADO using different approaches to task organizing, OAs, and control measures to converge capabilities and repeatedly create necessary effects.

e. Throughout planning, preparation, and execution, JFCs adjust OAs and associated fire support coordination measures to respond to changing circumstances. Additionally, they reprioritize, change, exchange, and add target groups or individual targets between adjacent or subordinate commanders. In planning an operation, JFCs evaluate available forces against enemy capabilities and expected actions. They determine how and when to integrate component forces across all domains, the information environment, and the EMS. Often, massing effects involves multiple commanders converging the right capabilities at specific decisive points. The JFC then assembles and rehearses the forces to achieve their intent.

f. Enemy capabilities, intentions, and methods of employing information, cyberspace, and the EMS are considered. An all-domain intelligence estimate identifies high-payoff targets; significant threat capabilities include space, cyberspace, the EMS, and the information environment. Disrupting, degrading, or destroying these targets increases an operation's likelihood of success. Thus, the intelligence staff's identification of high-payoff targets drives JFCs' targeting and informs CONOPS development.

g. During execution, JADO can create geographic, temporal, informational, and psychological effects that degrade enemy capabilities and their decision-making ability. Advantages from individual JADO are often temporary or limited in geographic scope, but successes from multiple JADO accumulate. Sequential actions enable follow-on JADO. JFCs' JADO objectives may involve seizing terrain, disrupting enemy key C2 nodes, attacking multiple decisive points, neutralizing enemy long-range fires, contesting enemy maneuver, and setting conditions for further offensive operations.

h. JFCs analyze CTSs involving enemies with global reach and develop integrated plan sets to inform eventual individual planning efforts. Armed conflict with a capable enemy may require coordination between multiple CDRs, focusing on unity of effort during planning. This planning accounts for JADO considerations to create specific effects at geographically separated decisive points. For example, a particular plan may require prosecuting a target outside of a joint OA, necessitating validation on a global integrated target list. Consequently, JFCs may conduct highly synchronized, globally integrated operations requiring some centralized control while allowing decentralized execution.

i. Commanders make resourcing and force presentation decisions based on integrated plan sets, impacting JFCs' ability to execute missions based on JADO CONOPS. Planners should consider JADO from the inception of integrated plan sets. JFCs who appreciate well-planned JADO may be willing to reallocate forces to other supporting or supported JFCs' AORs without incurring unacceptable levels of risk. Planning efforts account for JADO considerations and the potential need to execute multiple JADO in a short time, requiring rapid assessment of each operation to inform the next joint all-domain operation.

j. Planners adapt integrated plans rapidly to adjust to evolving OEs, such as discovering unexpected enemy vulnerabilities, ensuring JFCs have postured forces to react

and adjust schemes of maneuver, fires, and logistics across all domains. Through JADO, JFCs and planners combine diverse but mutually supporting joint capabilities to outperform and outmaneuver the enemy. Operating within coherent CONOPS, these capabilities enable faster, better decision making. JFCs respond to evolving situations by modifying targets, task organization, control measures, and OAs throughout execution. By integrating all capabilities, JFCs create synergy for effective fighting. They collect vital information, analyze it quickly and accurately, and disseminate it securely to relevant commanders through C2, ensuring similar situational awareness among subordinates.

k. A JFC's CONOPS drive, enable, and protect schemes of maneuver, fires, and logistics by integrating space, cyberspace, information-centric, and EMS capabilities. JFCs identify the most important information, make it actionable, avoid enemy strengths, and attack weaknesses. They prioritize targets and align tasks to generate desired effects, converging the right capabilities at decisive points. Planning results in coherent and executable integrated tasking orders. Subsequent sections characterize planning, targeting, organizations, C2, and assessment for JADO.

l. Previously, planners successfully integrated land, air, and maritime domain capabilities in joint operations. For JADO, JFCs expand traditional planning by emphasizing several domains, joint functions, and activities to inform planning and decision making. This guidance informs schemes of maneuver, fires, and logistics, driving targeting processes. Enhanced focus areas include space; cyberspace; the EMS; information; and protection, including operations security and military deception.

m. Guidance from targeting board outcomes refine target lists within joint integrated priority target lists. Planners provide considerations for targeting:

- (1) Influencing enemy leadership's decision making and behavior.
- (2) Optimizing effects across domains, the information environment, and the EMS.
- (3) Assessing previous mission effects, particularly fires.
- (4) Prioritizing and synchronizing fires on high-payoff targets across all domains.
- (5) Generating lethal and nonlethal effects, considering indirect, cumulative, cascading, and unintended consequences.
- (6) Disrupting, degrading, or destroying C2 nodes and sensors.
- (7) Identifying applicable and available assets.

n. JFCs prioritize capabilities over component or agency for tasking, focusing on deliberate and dynamic targeting and capability types. Joint targeting objectives stem from integrating enemy capabilities and COAs. Targets include known enemy strengths, critical capabilities, requirements, and vulnerabilities. JFCs consider global capabilities from all domains, authorities, and task organizations to achieve objectives.

APPENDIX D

FLEXIBLE DETERRENT OPTIONS AND FLEXIBLE RESPONSE OPTIONS

FDOs and FROs execute on order and provide scalable options to respond to a crisis. Commanders include FDOs and FROs as part of their plans to provide adaptive military options for SecDef or the President to deter or respond to a crisis. Both provide the ability to scale up (escalate) or de-escalate based on continuous assessment of an adversary's actions and reaction. While FDOs intend to prevent the crisis from worsening and enable de-escalation, FROs are generally punitive in nature. A planning outline for FDOs and FROs is in CJCSM 3130.03 series, *Planning and Execution Formats and Guidance*.

SECTION A. FLEXIBLE DETERRENT OPTIONS

1. General

a. FDOs are preplanned, deterrence-oriented actions tailored to signal to and influence an adversary's actions before or during a crisis. If necessary, FDOs may prepare for future operations, recognizing they may create a deterrent effect.

b. FDOs apply to each instrument of national power—diplomatic, informational, military, and economic—but they are most effective when combined across all instruments. FDOs facilitate early strategic decision making, rapid de-escalation, and crisis resolution by laying out a wide range of interrelated response paths.

c. FDOs enable a gradual increase in pressure to avoid unintentionally provoking combat and to enable leaders to analyze the situation and gain a better understanding of an adversary's capabilities and intentions. FDOs are elements of contingency plans executed to increase deterrence in addition to, but outside the scope of, the ongoing operations.

d. Examples of FDOs for each instrument of national power are in Figure D-1 through Figure D-4. Key objectives of FDOs are:

(1) Communicate the strength of US commitments to treaty obligations and regional peace and stability.

(2) Confront the adversary with the threat of unacceptable costs for their possible coercion or aggression.

(3) Isolate the threat from regional neighbors and attempt to split the adversary coalition.

(4) Rapidly improve the military balance of power in the AOR without precipitating armed response from the threat.

(5) Develop the situation, without provoking the threat, to better understand their capabilities and intentions.

Examples of Requested Diplomatic Flexible Deterrent Options

- Alert and introduce special teams (e.g., public diplomacy). Reduce international diplomatic ties
- Increase cultural group pressure
- Promote democratic elections
- Initiate noncombatant evacuation procedures
- Identify the steps to peaceful resolution
- Restrict activities of diplomatic missions
- Prepare to withdraw or withdraw United States (US) embassy personnel
- Take actions to gain support of allies and friends
- Restrict travel of US citizens
- Gain support through the United Nations. Demonstrate international resolve

Figure D-1. Examples of Requested Diplomatic Flexible Deterrent Options

e. Deterrence is threat-based and perception-focused. Since deterrence relies on perception, this requires JFCs to signal the threat effectively and ensure the intended target audiences receives the communication.

2. Description of Deterrent Actions

a. Deterrence seeks to influence an adversary's perceptions about an undesired action. Deterrence is a state of mind brought about by the threat's perception of three factors: the likelihood of denial of the expected benefits of action, the likelihood of having excessive costs imposed for taking the action, and the acceptability of restraint as an alternative. These effects are the results of a synchronized and coordinated use of all instruments of national power. FDOs are deterrent-oriented options that commanders may initiate based on evaluation of indicators of heightened regional tensions.

b. **FDOs serve two basic purposes.** First, they provide a visible and credible message to influence adversary perceptions about the costs and benefits of undesired activity. Second, they position US forces in a manner that facilitates implementation of CONPLANS/OPLANS or OPORDs if hostilities are unavoidable. They also facilitate an early decision by laying out a wide range of interrelated response paths tailored to avoid the classic response of "too much, too soon, or too little, too late." They initiate before and after unambiguous warning. Although not intended to place US forces in jeopardy if deterrence fails, risk analysis should be an inherent step in determining which FDO to use and how and when to use that FDO. FDOs have the advantage of rapid de-escalation if the situation precipitating the FDO changes.

Examples of Requested Informational Flexible Deterrent Options

- Impose sanctions on communications systems and intelligence, surveillance, and reconnaissance (ISR) technology transfer.
- Protect friendly communications systems and intelligence collection assets (defensive space control, cyberspace defense, operations security, cyberspace security).
- Increase public awareness of the problem and potential for conflict.
- Make public declarations of nonproliferation policy.
- Increase communication systems and ISR processing and transmission capability.
- Interrupt satellite downlink transmissions.
- Publicize violations of international law.
- Publicize increased force presence, joint exercises, military capability.
- Increase informational efforts:
 - Influence adversary decision makers (political, military, and social).
 - Promote mission awareness.
 - Increase measures directed at the opponent's military forces.
- Implement beaconing, interference, jamming, and intrusion of enemy informational assets.
- Maintain an open dialogue with the news media.
- Take steps to increase United States public support.
- Ensure consistency with strategic guidance.

Figure D-2. Examples of Requested Informational Flexible Deterrent Options

3. Flexible Deterrent Option Implementation

a. The President or SecDef directs FDO implementation, and the specific FDO or combination selected varies with each situation. Commanders can employ FDOs individually, in packages, sequentially, or concurrently. Employing FDOs in groups maximizes integrated results from all the instruments of national power. It is imperative that extensive, continuous coordination occurs with interagency and multinational partners to maximize the impact of FDOs.

b. Upon FDO implementation, the commander and staff conduct assessments to monitor changes in the OA. The commander, in consultation with senior leadership, prepares to review and adjust objectives of the plan as needed to accommodate new conditions to the OE.

Examples of Requested Economic Flexible Deterrent Options

- Freeze or seize real property in the United States where possible.
- Freeze monetary assets in the United States where possible.
- Freeze international assets where possible.
- Encourage United States (US) and international financial institutions to restrict or terminate financial transactions.
- Encourage US and international corporations to restrict transactions.
- Embargo goods and services.
- Enact trade sanctions.
- Enact restrictions on technology transfer.
- Cancel or restrict US-funded programs.
- Reduce security assistance programs.

Figure D-3. Examples of Requested Economic Flexible Deterrent Options

Examples of Requested Military Flexible Deterrent Options

- Increase readiness posture of in-place forces.
- Upgrade alert status.
- Increase intelligence, surveillance, and reconnaissance.
- Initiate or increase show-of-force actions.
- Increase training and exercise activities.
- Increase defense support to public diplomacy.
- Increase operations in the information environment.
- Deploy forces into or near the potential operational area.
- Increase active and passive protection measures.

Figure D-4. Examples of Requested Military Flexible Deterrent Options

SECTION B. FLEXIBLE RESPONSE OPTIONS

4. General

An FRO is a CONOPS that is easily scalable, provides a military option, and facilitates rapid decision making by national leaders in response to heightened threats or attacks against the US homeland or US interests. FROs also address terrorist actions or threats.

5. Description of Flexible Response Options

a. The basic purpose of FROs is to preempt or respond to attacks against the United States or its interests. FROs intend to facilitate early decision making by developing a wide range of prospective actions carefully tailored to produce desired effects, congruent with national security policy objectives. FROs present various military capabilities to the

President and SecDef, with actions appropriate and adaptable to existing circumstances, in reaction to any threat or attack.

b. FROs address both specifics and nonspecific threats. FROs are operations that are first and foremost designed to counter enemy attacks but also provide DoD the necessary planning framework to fast-track requisite authorities and approvals necessary to address dynamic and evolving threats.

c. CCMDs develop and maintain FROs to address multiple threats. FROs should support both long-term regional and national security policy objectives. Initially, FROs are pre-crisis tools used by CCMDs, based on intelligence collection and CFA, and then modified, refined, or developed real-time. FRO content guidelines are in Figure D-5.

d. Do not limit FROs to current authorities or approvals; rather, planning should be based on DoD's capabilities (overt, low visibility, clandestine, and covert) to achieve objectives, independent of risk. While unconstrained planning is not realistic or prudent, the intent of FROs is to provide national leaders multiple options, to include those prohibited in the OA. Planning must also identify expected effects (to include effects on third parties, partners, and allies), resources required, and risk associated with each option.

e. FROs have several key attributes. These attributes and the objectives determine the scope of FRO planning efforts:

- (1) Provides an effective signal of the US commitment and resolve.
- (2) Provides an effective counter to the adversary's actions.

Flexible Response Option Content Guidelines

- Identify critical enemy vulnerabilities and specific targets for each major vulnerability
- Operation objectives
- Desired effects
- Essential tasks
- Major forces and capabilities required
- Concept of deployment
- Concept of employment to include phasing, timing, major decision points, and essential interagency supporting actions
- Concept for sustainment
- Estimated time to achieve objectives
- Additional resources or shifts essential for execution
- Additional recommended changes in authority and approval required
- Additional risks associated with execution and mitigation approaches

Figure D-5. Flexible Response Option Content Guidelines

- (3) Provides a proportional and appropriate response.

f. **FRO Characteristics**

- (1) Provides military options to national leadership.
- (2) Military CONOPS at the operational or strategic levels.
- (3) Provides a start point for iterative planning.
- (4) Scalable based on situation and SecDef guidance.
- (5) Focused on enemy critical vulnerabilities.
- (6) Nested with the strategy.
- (7) Deliberate and synchronized expansion of the campaign rather than disparate actions.
- (8) A combination of direct and indirect actions.
- (9) Decisive action or set conditions for follow-on operations.
- (10) Include a discussion of risk and probability of escalation.

6. Flexible Response Option Implementation

a. The planning engine for FROs is the contingency planning process. In the event SecDef directs the execution of an FRO, the supported CCMD initiates planning to determine existing options or develop new ones for SecDef. Additionally, the supported CCMD coordinates the acquisition of authorities and approvals necessary to conduct appropriate military operations to disrupt threats or respond to attacks on the United States or US interests.

b. **Applications of FROs**

(1) **Disrupt** addresses both specific threats and heightened nonspecific threats. Disrupt options seek to preempt, interrupt, or reduce effectiveness of enemy attacks.

(a) **Specific Threats.** JFCs develop FROs to disrupt threats keyed by specific warning intelligence or identified attack plans spanning more than one AOR or otherwise requiring global integration, as determined by SecDef.

(b) **Nonspecific Threats.** JFCs develop FROs to disrupt threats keyed by general indications of increased threats. Periodically, intelligence assessments indicate enemy strength has increased despite current operations or attack preparations have

progressed to the point that national leadership is willing to consider additional operations, actions, and activities.

(2) **Response.** Response contingencies trigger because of a successful or unsuccessful attack against the United States or its interests. If efforts fail to preempt, disrupt, or defeat a major attack, response options rapidly provide flexible and scalable options to respond with global operations against the entire scope of the enemy (see Figure D-6). The following are examples of FRO scalability. Operations in each category can execute individually, concurrently, or sequentially.

(a) **Rapid Response.** Priority of effort is to demonstrate US resolve through speed of action. Rapid responses are most likely cyberspace operations, joint EMS operations, information activities, unilateral strikes, and raids against known targets with low collateral damage. Disadvantages of responding rapidly may include having limited strategic effects, being more likely to involve lethal engagement or unilateral actions, and a higher probability of negative international reactions.

(b) **Limited Response.** Priority of effort is to attack organizations directly attributed to the attack. The objective of this category is to maximize perceived legitimacy of US response. Limited response demonstrates restraint and is more likely to garner

Flexible Response Option Scalability		
Rapid Response Demonstrate Resolve	Limited Response Target Those Directly Responsible	Decisive Response Defeat Violent Extremist Organization
Priority of Effort: • Speed	Priority of Effort: • Legitimacy via attribution	Priority of Effort: • Direct attack on enemy center of gravity
Advantages: • Demonstrate resolve • Least impact of current operations	Advantages: • Response aimed directly at those responsible • Demonstrates restraint • International cooperation more likely	Advantages: • Proactive vice reactive • Targets critical enemy vulnerabilities • Greater impact on enemy
Disadvantages: • Limited strategic effect • More likely lethal in nature • Probable negative international reaction • More likely unilateral action	Disadvantages: • Uncertain timeline • Persistent operation may require reallocation of resources • United States remains vulnerable to other extremist organization elements	Disadvantages: • Potential to destabilize region of focus • Perception of United States overreaction • Higher risk • Unintended consequences

Figure D-6. Flexible Response Option Scalability

international cooperation. Disadvantages may include uncertain timeline due to requirement for attribution and continued vulnerability to networks not directly associated with the current attack.

(c) **Decisive Response.** Priority of effort is to attack the enemy operational COG to achieve a long-term disruption of its operational capability. This category is proactive vice reactive and seeks greater long-term impact on, or defeat of, the enemy. Disadvantages may include perception of US overreaction with possible negative public opinion consequences and the potential provocation of retaliatory responses.

APPENDIX E

COURSE OF ACTION COMPARISON

1. Overview

The most common technique for COA comparison is the weighted numerical comparison, which uses evaluation criteria to determine the preferred COA, based upon the wargame. Do not compare COAs to each other directly until considering each COA independently against the evaluation criteria. The CCDR may direct some of these criteria, but the joint planning groups develop most criteria, as detailed in Chapter IV, “Joint Planning Process.” Below are examples of common methods. JFCs can use these methods to aid their COA selection. They are not the only way to decide between COAs. Commanders apply logic, reason, their knowledge of the mission and the OE, and operational art to determine the best COA for the mission.

2. Weighted Numerical Comparison Technique

a. Use numerical methods with caution given the inherently subjective values and weighting assigned. The example in Figure E-1 and Figure E-2 provides a numerical method for differentiating COAs. Values reflect the relative preference of each COA within each criterion. All criteria use assigned weight criteria to reflect their relative preference to one another.

b. Recall the weight of each criterion determined in COA analysis. The staff leader responsible for a functional area scores each COA using those criteria. Multiplying the score by the weight yields the criterion’s value. The staff leader then totals all values but does not portray this simplified numeric method as the result of a rigorous mathematical analysis. Comparing COAs by criterion is more accurate than comparing total values.

(1) Evaluation criteria are those selected through the process described in Chapter IV, “Joint Planning Process.”

(2) Planners can weigh the evaluation criteria to reflect the most important criteria with the highest numbers and lesser criteria with progressively lower numbers.

(3) The highest number is best. The best criterion and the most advantageous COA ratings are those with the highest number. Values reflect the relative strengths and weaknesses of each COA.

(4) Each staff section does this separately, perhaps using different criteria on which to base the COA comparison. The staff then assembles and arrives at a consensus for the criterion and weights. The COS or deputy commander should approve the staff’s recommendations concerning the criteria and weights to ensure completeness and consistency throughout the staff sections.

Example Numerical Comparison

		Course of Action					
		COA 1		COA 2		COA 3	
Criteria	Weight	Rating	Product	Rating	Product	Rating	Product
Exploits maneuver	2	3	6	2	4	1	2
Attacks COGs	3	2	6	3	9	1	3
Integrates maneuver and interdiction	2	2	4	3	6	1	2
Exploits deception	2	1	2	2	4	3	6
Provides flexibility	2	1	2	3	6	2	4
CSS (best use of transportation)	1	3	3	2	2	1	1
Total		12		15		9	
Weighted total			23		31		18

NOTE: The higher the number, the better.

- The joint force commander's intent explained that the most important criterion was "attacking the enemy's COGs." Therefore, assign a value of 3 for that criterion and lower numbers for other criteria that the staff devises (this is the weighing criterion).
- For attacking the enemy COGs, COA 2 was rated the best (with a number of 3). Therefore, COA 2 = 9, COA 1 = 6, and COA 3 = 3.
- After the relative COA rating is multiplied by the weight given each criterion and the product columns are added, COA 2 (with a score of 31) is rated the most appropriate according to the criteria used to evaluate it.

Legend

COA course of action COG center of gravity CSS combat service support

Figure E-1. Example Numerical Comparison

Course of Action Comparison Matrix Format

Evaluation Criterion	Weight	COA 1		COA 2		COA 3	
		Score	Weighted	Score	Weighted	Score	Weighted
Surprise	2	3	6	1.5	3	1.5	3
Risk	2	3	6	1	2	2	4
Flexibility	1	3	3	1.5	1.5	1.5	1.5
Retaliation	1	1.5	1.5	3	3	1.5	1.5
Damage to alliance	1	3	3	1.5	1.5	1.5	1.5
Legal basis	1	2	2	3	3	1	1
External support	1	3	3	2	2	1	1
Force protection	1	2.5	2.5	2.5	2.5	1	1
OPSEC	1	3	3	1.5	1.5	1.5	1.5
Total			30		20		16

NOTE: The higher the number, the better.

Legend

COA course of action

OPSEC operations security

Figure E-2. Course of Action Comparison Matrix Format

3. Non-Weighted Numerical Comparison Technique

The same as the previous method except this does not have weighted criteria. In this case, raw numerical values are added for each COA. Again, the highest number is best for each of the criteria.

4. Narrative or Bulletized Descriptive Comparison of Strengths and Weaknesses or Advantages and Disadvantages

Summarize comparison of all COAs by analyzing strengths and weaknesses or advantages and disadvantages for each criterion. See Figures E-3 and E-4 for examples.

5. Plus/Minus/Neutral Comparison

Base this comparison on the broad degree to which selected criteria support or reflect in the COA. This organizes as a table showing (+) for a positive influence, (0) for a neutral influence, and (–) for a negative influence. Figure E-5 is an example.

Criteria for Strengths and Weaknesses Example

	Criteria 1		Criteria 2		Criteria 3	
COA 1	Strengths • •	Weaknesses • •	Strengths • •	Weaknesses • •	Strengths • •	Weaknesses • •
COA 2	Strengths • •	Weaknesses • •	Strengths • •	Weaknesses • •	Strengths • •	Weaknesses • •
COA 3	Strengths • •	Weaknesses • •	Strengths • •	Weaknesses • •	Strengths • •	Weaknesses • •

Legend

COA course of action

Figure E-3. Criteria for Strengths and Weaknesses Example

Descriptive Comparison Example

	Criteria 1		Criteria 2		Criteria 3	
COA 1	Advantages • •	Disadvantages • •	Advantages • •	Disadvantages • •	Advantages • •	Disadvantages • •
COA 2	Advantages • •	Disadvantages • •	Advantages • •	Disadvantages • •	Advantages • •	Disadvantages • •
COA 3	Advantages • •	Disadvantages • •	Advantages • •	Disadvantages • •	Advantages • •	Disadvantages • •

Legend

COA course of action

Figure E-4. Descriptive Comparison Example

Plus/Minus/Neutral Comparison Example

Criteria	COA 1	COA 2
Casualty estimate	+	-
Casualty evacuation routes	-	+
Suitable medical facilities	0	0
Flexibility	+	-

Legend

COA course of action

Figure E-5. Plus/Minus/Neutral Comparison Example

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APPENDIX F POSTURE PLANS

1. General

a. Posture plans are key elements of CCMD campaigns and strategies. They describe the forces, footprint, and agreements the commander needs to successfully execute the campaign and respond in a contingency. Commercial capabilities—infrastructure, services, and personnel—align to footprint, agreements, and forces, respectively. Commercial capabilities also provide an alternative to military posture, providing global agility for force projection, movement, and maneuver.

(1) CCMDs prepare posture plans, outlining their posture strategy, link to objectives with the means to achieve them, and identifying posture requirements and initiatives.

(2) The posture plan is the primary document used to advocate for changes to posture and to support resource decisions, the posture management processes, and departmental oversight responsibilities. It delineates the CCMD's posture status, with gaps, risks, required changes, and proposed initiatives substantiated by strategic direction provided in the NDS and NMS.

(3) Posture plans address the posture requirements of other DoD stakeholders in theater. Stakeholders include other CCMDs, Military Departments, DoD agencies, non-DoD agencies, and field activities. CCMDs coordinate their posture plans within an AOR with stakeholders to incorporate requirements.

(4) Posture plans enable CCMDs with specific responsibilities to identify their posture requirements to other CCMDs. United States Special Operations Command prepares the Special Operations Forces Posture Plan; United States Strategic Command prepares the Strategic Infrastructure Master Plan, which includes its En Route Infrastructure Master Plan; United States Transportation Command prepares its posture plan; and United States Cyber Command prepares the Functional Posture Plan for Cyberspace Operations. Each functional posture plan includes a strategic narrative that assesses posture gaps, associated risks, and posture initiatives recommended to address those gaps/risks. CDRs with functional responsibilities coordinate posture plans with their counterparts tasked with an AOR to facilitate incorporation of functional requirements into their posture plans.

b. Process

(1) CCMD posture plans outline the current posture and propose posture initiatives for a two-to-five-year time frame and beyond. Posture subject matter experts on strategic and operational access issues inform campaign planners. In turn, the strategy and the operational approach of the campaign plan inform the posture plan.

(2) The GPEC is DoD's senior posture governance body. The GPEC facilitates senior leader posture decision making; enables the CCMDs, Military Departments and Services, and DoD agencies to collaborate in DoD's GDP planning; and oversees the implementation and assessment of DoD's posture plans. The JS J-5, in coordination with Office of the Under Secretary of Defense for Policy, annually provides GPEC-endorsed posture guidance to guide the development of posture plans.

2. Elements of a Posture Plan

a. **Strategic Narrative.** The CCDR outlines the vision of posture linked to plans and objectives.

b. **Forces.** Forces are composed of assigned, allocated, and enabling units, personnel, and assets. It includes rotational and mobility forces.

c. **Footprint.** The footprint includes enduring locations, supporting infrastructure, and prepositioned equipment.

d. **Agreements.** Agreements provide access, basing, lawful mission execution, protection, and relationships that allow establishing a footprint and forces to execute their missions. Examples are access agreements, basic ordering agreements, transit agreements, status-of-forces agreements, and treaties.

e. **Appendices.** Four appendices convey the commander's posture:

- (1) Theater maps.
- (2) Summary chart of footprint by location.
- (3) Project summary sheets.
- (4) Enduring location change nominations.

3. Posture Terminology

a. **GDP Locations.** DoD uses an established lexicon for the types of overseas (in foreign countries or US territories overseas) locations from which it operates in its GDP framework.

(1) **Contingency Location.** A non-enduring location outside of the United States that supports and sustains operations during named and unnamed contingencies or other operations as directed by appropriate authority. Contingency locations categorize mission life-cycle requirements as initial, temporary, or semi-permanent.

(2) **Enduring Location.** A location is enduring when DoD intends to maintain access or use of that location for the foreseeable future. Enduring locations play a critical role in enabling DoD to deploy, or employ, US forces when and where necessary but need not have a continuous force presence or permanent structure sustained or constructed

through US appropriations. CCDRs nominate locations as enduring. The GPEC reviews and endorses the list, and OSD validates these nominations in consultation with DOS. There are three types of enduring site: cooperative security location (CSL), forward operating site (FOS), and main operating base (MOB). All three types of locations may include more than one distinct site designated for strategic access and support to US security interests for the foreseeable future.

(a) **CSL.** An enduring location characterized by the periodic presence of rotational US forces, with little or no permanent US military presence or US-owned infrastructure, used for a range of missions and capable of supporting requirements for contingencies. CSLs may feature a small, permanent presence of assigned support personnel (military or contractor). CSLs typically consist of mostly HN infrastructure, and CSL real property is often not US-owned (i.e., not part of the US real property inventory). However, CSLs may require US-funded infrastructure to meet operational requirements. CSLs are a focal point for security cooperation activities and provide contingency access, logistic support, and rotational use by operational forces and can support an increased force presence during contingencies of finite duration.

(b) **FOS.** An enduring location characterized by the sustained presence of rotational US forces, with infrastructure and quality-of-life amenities consistent with that presence, capable of providing forward staging for operational missions and support to regional contingencies. FOSs consist of US-owned real property, and they may feature a small, permanent presence of assigned support personnel (military or contractor). FOSs often support the stationing of pre-positioned force, equipment, or supplies, and they can serve as a regional hub in support of regional contingencies.

(c) **MOB.** An enduring location characterized by the presence of permanently assigned US forces and robust infrastructure that typically includes C2; highly developed force protection measures; and significant quality of life amenities, often including family support facilities. MOBs consist of US-owned real property and represent primary training and deployment locations for the United States overseas. MOBs can support both small- and large-scale operations and global contingencies.

(3) **En Route Location.** A term specific to global distribution; an intermediate node outside of the continental United States distribution network that supports refueling, maintenance, crew change/rest, or transload of cargo/passengers for onward movement to the destination. These nodes can be a fixed or a temporary location. En route locations are also MOBs, FOSs, or CSLs.

b. Other Terms

(1) **Posture Initiative.** A required change to US GDP that meets any one of the following criteria:

(a) Has policy significance.

(b) Comprises a significant change to an element of defense posture (forces, footprint, or agreements).

- (c) Comprises a significant change in capability.
- (d) Has significant impact on resourcing.

(2) **Policy Significance.** A change, or proposed change, expected to result in HN or US domestic political discussions at the national level or HN media coverage that may create political controversy at the national level. Examples include actions that directly affect foreign or defense relations between the United States and another government; require approval, negotiation, or signature at the OSD or diplomatic level of an international agreement or other document relating to US forces' presence in a foreign country; or generate a major increase or decrease in US operational or sustainment capability at one or more locations.

For more information on posture and posture planning, see the CJCSI 3110.01, (U) 2023 Joint Strategic Campaign Plan, and DoDI 3000.12, Management of US Global Defense Posture (GDP).

(3) **Theater Logistics Overview.** The theater logistics overview codifies the CDR's theater logistics analysis within the posture plan. The theater logistics overview provides a narrative overview, with supporting matrices of key findings and capabilities from the theater logistics analysis.

(4) **Theater Logistics Analysis.** The theater logistics analysis provides detailed country-by-country analysis of key infrastructure by location or installation (e.g., MOB, FOS, CSL), footprint projections, HN agreements, existing contracts, and task orders required to logistically support CCPs and their embedded contingency operations (e.g., contingency locations).

APPENDIX G

RED TEAMS

1. Background

The use of red teams contributes to reduce risk, avoid surprise, see opportunities, increase operational flexibility, broaden analysis, and enhance decision making. Red teams help organizations adapt to change and improve military planning and intelligence analysis by stimulating critical and creative thought. The JS has recommended the routine employment of red teams.

2. The Red Team Overview

a. All organizations, staff agencies, and work groups that study issues, draw conclusions, make plans, develop concepts, produce intelligence, create scenarios, conduct experiments, simulate adversaries, make recommendations, or decide issues can benefit from red teams. Red teams can help any staff element frame problems, challenge assumptions, counter institutional biases, stimulate critical and creative thought, and support decision making in every organization.

b. Joint forces should use red teams to enhance and complement regular processes and help understand all aspects of key problems to consider potential options.

c. Implicit tasks include countering the influence of institutional and individual bias and error; providing insight into the mindsets, perspectives, and cultural traits of adversaries and other relevant actors; and helping explore unintended consequences, follow-on effects, and unseen opportunities and threats. Red teams reduce risk by helping organizations anticipate, understand, prepare, and adapt to change.

KEY TERM

Red Team: An organizational element comprised of members that provide an independent capability to fully explore alternatives in plans and operations in the context of the operational environment and from the perspective of adversaries, the civilian environment, and other actors.

d. The red team is a specially trained, decision-support staff organization that can be employed throughout the joint force. The red team can complement all staff problem-solving and analytical efforts by serving as a “devil’s advocate” and generalized contrarian focused on supporting plans, operations, and intelligence. This element may be an ad-hoc element at tactical organizations (below CCMD and JTF) as needed.

(1) **Plans and Operations.** The red team supports planning and mission execution. This includes helping identify vulnerabilities, opportunities, and faulty or unstated assumptions; helping understand all aspects of the OE; and critically reviewing strategies, operational concepts, estimates, plans, and orders.

(2) **Intelligence.** The red team complements intelligence efforts by offering independent, alternative assessments and differing interpretations of information. This includes critical reviews of intelligence products, considering problem sets from alternative perspectives, and helping contribute informed speculation when reliable information is lacking.

3. Red Team Support to Joint Planning

a. **Support to joint planning is a core red team function.** Organizations that plan and execute operations may employ red teams to help them think critically and creatively and see planning issues and proposed COAs from alternative perspectives. The red team may also help the staff avoid common sources of error.

b. The red team could support virtually all aspects of the JPP. However, the team's capacity seldom matches the scale of requirements. Accordingly, this guidance focuses on those functions where red teams can have the greatest impact on planning. In most cases the red team has the greatest impact on planning during JPP step 2 (Mission Analysis) and Step 4 (COA Analysis and Wargaming).

c. Red team support to joint planning is through active participation in planning groups and the production of tailored papers and briefings that support the planning effort. When addressing key issues that may have wide-ranging effects on planning, it may be prudent to circulate comprehensive stand-alone red team products for the staff's review. While the red team may suggest alternatives for consideration, planners should weigh these inputs and either incorporate or set aside as appropriate before finalizing planning products. Critical red team observations may be developed into branch plans at the planner's discretion.

d. During multinational planning efforts, red teams should ensure foreign staff officers understand the red team's role as a devil's advocate, so they understand the purpose of the team's contrarian perspective. Some foreign services may have their own style of red teaming, and they might be able to make valuable contributions to the overall planning efforts once their red teaming efforts integrate with those of their US counterparts.

e. Planners use the strategic estimate to develop campaign plans. It encompasses all aspects of the commander's OE and is the basis for the development of the CCMD's theater or functional strategy. It provides the commander's perspective of the strategic and operational levels of the OA, the contested environment, desired changes required to meet objectives, and the commander's visualization of how to achieve those objectives. It addresses core issues which can benefit from red team scrutiny, such as an analysis of all states, groups, or organizations in the OA; a review of relevant geopolitical, economic, and cultural factors; an assessment of strategic and operational challenges facing the CCMD; an analysis of known or anticipated opportunities; and an assessment of risks inherent in the OA. Accordingly, the red team should carefully review the draft strategic estimate.

4. Joint Planning Activities, Functions, and Products

a. Intelligence assessments of current or projected situations and threats supports planning. It is generally not practical or advisable for the red team to offer independent, alternative assessments for each intelligence estimate, but within its capability, the red team should judiciously review key assessments and estimates and, when appropriate, suggest alternative assessments to alert the staff to previously unseen threats and opportunities that may require new planning initiatives.

b. The red team supports all planning functions through active participation in planning teams and critical reviews of draft planning materials. The red team should participate in the early stages of planning functions to ensure the staff has sufficient time to consider the red team's inputs before key decisions. The red team should not produce duplicative or competing planning materials but should instead seek to incorporate its inputs into the planning team's final products. In some cases, however, it may be useful to circulate comprehensive stand-alone "think pieces" to help the staff consider specific issues, but never criticisms of the planning team's products.

c. **Support to IPRs.** Periodic IPRs facilitate planning that provide up- and down-channel feedback, shaping and refining of the plan. Ideally, prior to an IPR, the red team should review their organizations' draft IPR briefings and papers and offer suggestions as appropriate. The red team's most critical contributions to any new plan will usually come during mission analysis, although preparations for later IPRs may involve more of the team's time and resources.

d. The red team should fully integrate into the planning process and assist in the initial development and revision of JPP products. When the red team is unable to support all aspects of a specific planning effort, the commander or J-5 should establish priorities for red team support.

e. **Potential red team roles in planning:**

(1) **Step 1 (Planning Initiation).** Commanders typically provide initial planning guidance to planning teams. The red team typically participates in the planning team's review of that guidance and recommending refinements back to the commander.

(2) **Step 2 (Mission Analysis)**

(a) One key input to mission analysis is JIPOE. If the red team has not participated in the JIPOE process, then it should conduct an independent, alternative assessment of the enemy or adversary's COG, critical capabilities, critical requirements, and critical vulnerabilities. The red team should then offer its alternative assessments for consideration by both the intelligence staff and the planners.

(b) One primary red team task during mission analysis is to help the planners frame the problem, define the objective, and assess known facts and key assumptions. The

red team should challenge weak assumptions or suspect facts and, as the situation evolves, consider whether the assumptions remain valid.

(c) If possible, the red team should help determine operational limitations, transitions, objectives, and mission success criteria, providing alternative perspectives and exploring how political will and cultural viewpoints might constrain operations and limit options. In addition, the team may also participate in developing specified, implied, and essential tasks, conducting risk analysis, writing the CCIRs, and drafting the mission statement. If possible, the team should participate in drafting the mission analysis brief and the commander's refined planning guidance.

(3) **Step 3 (COA Development).** The red team can often make useful contributions to COA development by helping the planners expand the range of COAs under consideration including those addressing contested friendly force deployment.

(4) **Step 4 (COA Analysis and Wargaming)**

(a) During COA analysis, the red team should advise planners of the potential contested environment and the cultural implications associated with each COA and should help explore the potential unintended consequences and likely second- and third-order effects associated with each COA. Red teaming can support COA analysis in situations where wargaming may not be suitable.

(b) During COA wargaming, the red team should help both the simulated friendly force and the opposition (enemy or adversary) force (red cell) consider a range of options during their respective moves. The red team should also advise both sides regarding how relevant actors may perceive their moves. Wargaming can include/integrate red teaming functions through their red cells.

(5) **Step 5 (COA Comparison).** COA comparison is an objective measurement of the relative merits of the COAs developed and analyzed in earlier steps. The red team should participate in the development of the COA comparison criteria and highlight those areas in which subjective and cultural issues might outweigh more tangible, more easily quantified factors.

(6) **Step 6 (COA Approval).** Planners should consider including a summary of wildcards, unintended consequences, and second- and third-order effects in the COA decision briefing to the commander.

(7) **Step 7 (Plan or Order Development)**

(a) If the decision is to develop an order, and the red team has been actively participating to this point, then order development may continue without additional red team input. However, the red team should support future revisions. If not involved in planning prior to this point, the red team should review assumptions and evaluate the potential impact of cultural factors. In addition, the red team should explore likely

unintended consequences, second- and third-order effects, and wildcards. Revising the order at this point can be extremely disruptive. Accordingly, the red team should advise the senior planner if the red team's review suggests serious shortcomings.

(b) The red team should participate in developing appendices, making certain themes, messages, and media are compatible with the mindsets of participants, understanding potential unintended consequences. The red team should also participate in developing the assessment annex, using its understanding of the OE to help ensure the relevance of the measurements.

(c) The OE and situation may evolve as planners draft the plan, and the red team should remain sensitive to developing threats and opportunities, wildcards, and other issues. In addition, plans typically address more issues, and in more detail, than the working groups, and these benefit from red team support.

(d) The red team should review key sections of the plan and offer recommendations while those sections are still in the draft stage. If the red team has not been involved in the planning effort before the decision brief, a red team review of these draft sections is critical.

(e) During plan development, it may be useful for the red team to circulate a document that consolidates its observations and concerns. This document can provide complete, fully reasoned descriptions of issues the red team might have previously raised in closed work groups, or it may propose new issues that planners should consider. Issues could include potential unanticipated and low probability/high impact events, likely unintended consequences and second- and third-order effects, and unseen threats and opportunities. In some cases, it may be prudent to include this document in the final plan as a reference for mission execution or to support future revisions, but in all cases, planners should understand that the primary purpose of the document is to support the development of the plan rather than serve as an after-the-fact critique.

(f) Completed plans are frequently refined or adjusted over time, and refinement continues even after execution. During refinement and adaptation, the red team helps assess the situation, develop new guidance, and support continued planning efforts.

5. Joint Planning During Execution

a. Red team support during mission rehearsal generally parallels that of wargaming. During rehearsal, the primary objective is to test the plan's CONOPS and COAs. In rehearsal, the red team focuses on helping the staff uncover previously unseen weaknesses, opportunities, and unintended effects. During rehearsal, the red team should attune to potential alternative COAs and assessments, which it may propose after rehearsal, when the staff may seek improvements or alternatives to the plan.

b. A crisis action team stands up during the initial stages of a crisis. While not part of its critical analysis function, the red team may support the crisis action team by providing

expertise in alternative interpretations of dynamic, uncertain situations; by helping frame problems; and by broadening the search for potential responses. A crisis action team normally uses streamlined decision-making procedures, and the primary red team mode of support often consists of active participation in work groups rather than formal written products.

c. Planning continues throughout execution in three venues, each focused on distinct but overlapping timeframes: future plans, future operations, and current operations. The red team plays distinct roles in each of these but should normally concentrate its efforts in future plans.

(1) Future plans address the next phase of operations or sequels to the current operation. Red team support to future plans generally follows that provided during JPP step 2 (Mission Analysis) but in an abbreviated form.

(2) The J-3 or, in some commands, an operations planning element addresses branches to current, on-going operations. Red team support to future operations often resembles that of future plans but with a more truncated time horizon and more streamlined processes.

(3) Current operations address immediate or very near-term issues associated with ongoing operations. The organization's joint operations center usually addresses current operations. Due to the compressed decision cycle, the red team provides limited alternative assessments of selected aspects of the ongoing situation.

d. In some commands, several working groups manage the flow of information to decision makers and to coordinate recurring decisions within the headquarters' battle rhythm. Planners often refer to these cross-functional staff organizations as boards, cells, centers, and working groups. The red team should support the following groups (or their equivalents), if formed:

- (1) Long-range planning group.
- (2) Operations planning element.
- (3) Commander's communication synchronization working group.
- (4) Commander's assessment board.

e. Assessment entails two distinct tasks: monitoring the situation and the progress of the operations and evaluating operations against established MOEs and MOPs to determine progress relative to established objectives. Dynamic interactions between friendly forces, adaptable adversaries and enemies, and populations can complicate assessment.

f. During assessment, the red team should analyze the situation from the perspective of the threat and other stakeholders. The most important measure of success may be how

the threat assesses their own situation, rather than whether friendly forces are maximizing MOP and MOE scores. Operation assessment especially during combat operations, should weigh against the enemy's perspective of their own condition, their own objectives, and their own unique mindset and world view. Even if all objective measurements and assessments portray the enemy as defeated, they may not believe it. For example, an enemy that has suffered extreme attrition, but can still conduct sporadic offensive operations, may see himself as heroic and undefeated, even when objective measures suggest otherwise. Overall, the red team should have access to the same information as the assessment elements. Whenever the red team's assessment of the threat's mindset portrays a different picture than that implied by assessment analyses, the red team input should present as a supplement to the assessment analyses.

g. As assessments and observations translate into lessons learned, the red team's external vantage point can be invaluable. The team's relative independence often helps it see issues and potential solutions that might not be apparent to those closer to the problem. The red team is also less inhibited in highlighting issues and proposing corrective measures than staff elements that might bear some responsibility for the problem or obligation to implement solutions.

6. Red Team Support to Intelligence Planning

a. US, ally, and partner-nation intelligence organizations conduct IP, as does the adversary. IP procedures fully integrate with joint planning. The IP process is a methodology to coordinate and integrate available intelligence capabilities to meet CCDR intelligence collection and production requirements. It ensures prioritized intelligence support aligns with CCDR objectives for each phase of an operation. The available US, ally, and partner-nation intelligence organizations develop intelligence products used by the joint force J-2 to provide the JFC and staff with situational understanding of the OE. Products developed by the CCMD J-2 during IP include intelligence estimates and annex B (Intelligence).

b. These intelligence products provide substantial support to senior leader decision making throughout planning and execution, should be free from analytical error and organizational bias, and consider all reasonable alternative interpretations. As such, red teams should help draft these products to ensure both accounting for adversary capabilities, as well as adversary IP.

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APPENDIX H OPERATION ASSESSMENT PLAN

SECTION A. THE ASSESSMENT PROCESS AND DEVELOPING AN ASSESSMENT PLAN

1. General

a. **There is no single way to conduct assessment.** Every mission and OE have their own unique challenges, making every assessment unique. The following steps can help guide the development of an assessment plan and assessment performance during execution. Assessment steps provide an orderly, analytical process to help organizations understand the underpinnings of desirable or undesirable action or behavior. Organizations should consider these steps as necessary to fit their needs. Figure H-1 provides an overview of the assessment process.

b. The assessment approach and plan nest within the operational design. The assessments team should determine the right measures and indicators to inform the collection effort on objectives, desired effects, decision points. The ever-changing OE requires continuous monitoring and adjustment of the plan. The operation assessment complements the planning process in answering three primary questions: “Where are we,” “How did we get here,” and “What’s next?” The complexity of an operation often makes it difficult to determine the criteria of success. Constructed properly, the assessment plan, nested with the CCIRs, enables appropriate monitoring and collection of necessary information and intelligence to inform critical decisions throughout planning and execution. The assessment does not replace but complements the commander’s intuition enabled through battlefield circulation and discussions with subordinate leaders.

c. Operation assessment enhances the effectiveness of planning and execution by identifying and measuring observable key indicators toward progress or regression, as well as providing recommendations to senior decision makers to correct deficiencies and exploit success. A major challenge for the assessment team is to understand “How much is enough” to not overstress finite collection assets.

d. The assessment approach should identify the information and intelligence needed to assess progress and inform decision points throughout planning and execution. This information and intelligence should nest completely with the CCIRs and should provide the basis for identifying changes in conditions within the OA related to specific objectives. The assessment approach should not create additional or duplicative IRs, especially in resource-constrained planning. CCIRs inherently seek to inform decision making and are tied to commander decision points, and thus, CCIRs should be sufficient to inform operation assessments. If operation assessment leads identify additional, new, or modified requirements, those should be staffed to the J-2 and J-3 via battle rhythm events for potential modifications to the commander-approved plan PIRs and FFIRs. Because success requires identifying specific conditions within the OA, assessment considerations should be part of their development. Integration, refinement, and adaptation of assessment

Operation Assessment Steps

Activity	Primarily in Planning or Execution	Personnel Involved	Input	Associated Staff Activity	Output
Develop Assessment Approach	Planning • Operational Design • JPP Steps 1-6	• Commander • Planners • Primary staff • Special staff • Assessment element	• Strategic guidance • Commander's Initial Planning Guidance • Description of OE • Problem to be solved • Operational approach • Commander's intent (purpose, risk)	• Conduct JIPOE • Develop operational approach • Support development and refinement of objectives, effects, and tasks • Conduct Joint Planning (JPP and operational design) • Determine and develop how to assess tasks, effects, objectives, and progress for each course of action • Identify indicators	Assessment approach which includes: • Assessment framework and construct • Specific outcomes (objectives, effects) • Commander's estimate/CONOPS (from JPP)
Develop Assessment Plan	Planning • JPP Steps 7	• Commander • Planners • Primary staff • Special staff • Assessment element • Operations planners • Intelligence planners • Subordinate commanders • Interagency and multinational partners • Others, as required	• Assessment approach which includes: assessment framework and construct • Specific outcomes (objectives, effects) • Commander's estimate/ CONOPS (from JPP)	• Document assessment framework and construct ◦ Finalize the data collection plan ◦ Coordinate and assign responsibilities for monitoring, collection, and analysis ◦ Identify how the assessment is integrated into battle rhythm/feedback mechanism • Vet and staff the draft assessment plan	• Approved assessment plan • Data collection plan • Approved contingency plan/operation order
Collect Information and Intelligence	Execution	• Intelligence analysts • Current operations • Assessment element • Subordinate commanders • Interagency and multinational partners • Others, as required	• Approved assessment plan • Data collection plan • Approved contingency plan/operation order	• JIPOE • Staff estimates • IR management • ISR planning optimization	• Data collected and organized, relevant to joint force actions, current and desired conditions
Analyze Information and Intelligence	Execution	• Primary staff • Special staff • Assessment element	• Data collected and organized, relevant to joint force actions, current and desired conditions	• Assessment working group • Staff estimates • Vet and validate recommendations	• Draft assessment products • Vetted and validated recommendations
Communicate Feedback and Recommendations	Execution	• Commander • Subordinate commanders (periodically) • Primary staff • Special staff • Assessment element	• Draft assessment products • Vetted and validated recommendations	• Provide timely recommendations to appropriate decision makers	• Approved assessment products, decisions, and recommendations to higher headquarters
Adapt Plans or Operations/ Campaigns	Execution Planning	• Commander • Planners • Primary staff • Special staff • Assessment element	• Approved assessment products, decisions, and recommendations to higher headquarters	• Develop ranches and sequels • Modify operational approach/plan • Modify objectives, effects, tasks • Modify assessment approach/plan	• Revised plans or fragmentary orders • Updated assessment plan • Updated data collection plan
Repeat Steps 36 until operation relieved/replaced/transitioned (Adjust using steps 1 and 2 as required during execution).					

Legend

CONOPS	concept of operations	JIPOE	joint intelligence preparation of the operational environment
IR	intelligence requirement	JPP	joint planning process
ISR	intelligence, surveillance, and reconnaissance	OE	operational environment

Figure H-1. Operation Assessment Steps

requirements throughout planning and execution help ensure decision points, objectives, and tasks enhance the effective allocation and employment of joint capabilities.

2. Operation Assessment Process

a. Operation assessment applies to both campaign plans and contingency plans and is continuous throughout planning and execution. For CCPs, an assessment plan is an annex or appendix of the campaign plan as the campaign's operational approach continues to refine. The decisive points accompanying metrics directly and measurably contribute to achieving objectives. Modify the campaign assessment plan should a contingency plan or crisis-generated order go into execution as a new campaign operation. For contingency plans, the supported CCDR determines whether an assessment plan should support the four planning functions and to enhance effectiveness and keep it up to date and ready for transition to execution. As the contingency plan adjusts, the assessment plan, if required, may likewise adapts.

b. Consider the impacts of friendly, threat, and neutral actions throughout the OA to a military plan and its execution. Operation assessment can help to identify significant actions and evaluate the results of these actions. This typically requires collaboration with other agencies and multinational partners—preferably within a common, accepted process—in the interest of unified action and facilitating the commander's understanding of the OE. The CCDR should receive reports on progress toward the campaign's objectives, as specified in the campaign plan evaluation annex or appendix, from the following: intelligence collection and analysis, components, JTFs, supporting commands, defense agencies, and country teams.

c. The prescribed format for an assessment plan is in accordance with CJCSM 3130.03, *Planning and Execution Formats and Guidance*. The CCP assessment plan is annex N (Assessments) within the campaign plan. Contingency plan evaluation plans, as required, should be annex N to the contingency plan.

3. Assessment Steps

A common method for an assessment uses six steps identified in Figure H-1.

a. **Step 1 – Develop the Assessment Approach.** In conducting an assessment, the first step is to determine what to assess and how to conduct the assessment (see Figure H-2).

(1) Operation assessment begins during the initiation step of the JPP when the command identifies possible operational approaches and their associated objectives, tasks, effects, and desired conditions of the OE. Concurrently, the staff begins to develop the operation assessment approach by identifying and integrating the appropriate assessment plan framework and structure needed to assess planning and execution effectiveness. The assessment approach identifies the specific information needed to monitor and analyze effects and conditions associated with achieving objectives. The assessment approach becomes the framework for the assessment plan and continues to mature through plan development, refinement, adaptation, and execution to understand the OE and measure whether anticipated and executed operations are having the desired impact on the OE. In

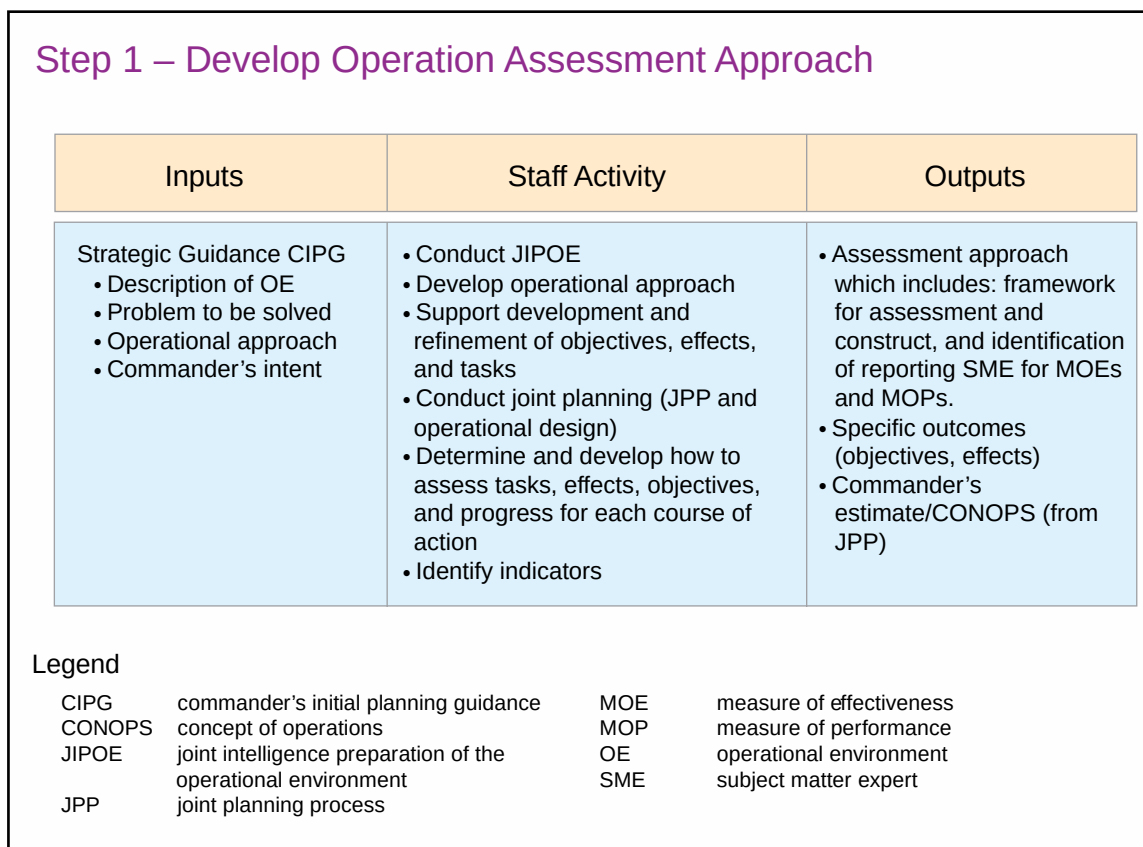


Figure H-2. Step 1 – Develop Operation Assessment Approach

short, the command tries to answer the following questions: “How will we know we are creating the desired effects,” “Are we achieving the objectives,” “What information do we need,” and “Who is best postured to provide that information?”

(2) The first step of the assessment approach aligns to all JPP steps, as assessment should complement and be concurrent with the planning effort. A common error is not considering assessment until completing the plan. Upon receipt of a new mission or revised strategic direction, identify assessment roles and responsibilities. Identifying an experienced assessment development lead can help planners throughout mission analysis, CCIR development, success criteria, COA selection, and eventual plan development. Forming an assessment team of the right subject matter experts across the key staff, as well as inclusion of interagency and multinational partners, encourages transparency, unity of effort, and avoids duplicative efforts. Additionally, focusing on future assessment requirements throughout planning and execution can ensure the anticipated and executed tasks and objectives are assessable and help establish a logical hierarchy from measured task completion status and the creation of effects toward achievement of objectives. Before any assessment development occurs, understand the purpose of the assessment and key decisions to inform throughout planning and execution.

(3) **Identify Information Requirements.** Strategic guidance documents serve as the primary guidance to begin planning at the CCMDs. CCDRs and other commanders

may also initiate planning on their own authority when they identify a planning requirement not directed by higher authority. Subordinate components and commands typically begin planning based on higher headquarters guidance but should be aware of strategic guidance to properly nest supporting plans within higher headquarters' plans. Military options employ in combination with other nonmilitary options so the President can respond with the appropriate instruments of national power. Staffs begin updating their estimates and gather the information necessary for mission analysis and continued planning. Specific information gathered regarding assessment includes:

- (a) The higher headquarters' plan or order, including the assessment annex if available.
- (b) If replacing a unit, any current assessment products.
- (c) Relevant assessment products (classified or open source) produced by civilian, government, military, and partner-nation organizations.
- (d) The draft objectives, and effects of the organization.

(4) Understand Current and Desired Conditions

(a) Fundamentally, operation assessment is about understanding current and desired conditions of the OE, observing changes in the OA, ascertaining the contribution of anticipated or completed tasks or missions to observed OE changes, and assessing progress or regression toward the desired OE conditions relative to the specified objective. Staffs compare current conditions in the OA against the desired conditions. During mission analysis, JIPOE and component-level intelligence preparation of the battlespace help develop an understanding of the current situation. The commander and staff identify the desired conditions and key underlying assumptions for an operation during joint planning. During execution, operational and intelligence reporting, the update of staff estimates, and any relevant information from external sources help them update and improve their understanding of the current conditions of the OE. Validate assumptions as soon as possible during execution. Likewise, desired conditions should be reevaluated as needed during execution.

(b) Understanding current and desired conditions requires acknowledging the underlying assumptions. Data analysis throughout the operation assessment challenges assumptions identified during planning. If disproven, planners should reframe the problem.

(c) During initiation and operational design, commanders and selected key personnel develop and issue planning guidance that includes initial intent. Following mission analysis, the commander issues CCIRs, approves the mission statement, and issues additional guidance to the planning team during COA development. The objective in the initial commander's intent describes the conditions the commander wants to change. The staff section reviews each assessment plan and desired condition mentioned in the

operational approach and commander's intent. These individual conditions provide focus for the overall planning, execution, and assessment of the operation. If the conditions that define the objective change during planning and execution, the staff updates these changes for the assessment plan.

(d) To assess progress, the staff identifies both the current situation and the desired objective. The staff identifies appropriate joint forces tasks, observable key indicators of task performance, and effect(s) of task completion on OE conditions and develops a plan to collect and analyze key indicator information, while continuously monitoring OE conditions relative to this desired objective. These indicators also identify the current and pre-hostility levels of essential services across the OA. By taking these actions, the staff establishes a mechanism to assess progress toward these required conditions so that operations planned to change them are most effective.

(5) Identify Assessment Indicators

(a) An assessment plan should have a structure that begins with the operation or campaign's implied, specified, and essential tasks that, if successfully accomplished, should create the desired effects that achieve objectives. These tasks support measurable, achievable objectives and through accompanying metrics based on carefully selected MOEs and MOPs from among available indicators. Combined with continuous JIPOE, the MOEs and MOPs facilitate staff observations and analysis of changes in the OA and their impact on planning and execution.

(b) The assessment plan should focus on identifying those CCIR indicators and associated information and intelligence that accurately reflect changes in the OA. Analyses should identify effectiveness and continually evaluate assumptions to validate or invalidate them. Indicators, associated information, and intelligence may require modification or replacement during planning and execution to respond to the dynamic conditions of the OE. Operation assessment leads may need to submit requests to periodically re-assess and adjust CCIRs as required.

(c) Establish collection requirements with identified data requirements, using CCIRs as the backbone. Essentially, the organization considers the rules and content of the DCP—the “who, where, when, how, and who (again)” (see Figure H-3). Develop collection requirements with functional subject matter experts, vetted throughout the staff, and included in the DCP. The DCP is not meant to counter or circumvent the command's joint collection management board. These broad actions typically fall under one of three categories: organize and collect, analyze, and communicate.

1. Organize and Collect

a. Organizing for assessment involves identifying the information needed to assess effectiveness throughout planning and execution. In the case of assessment, the required information should promote understanding of the OE. This understanding is critical to assess the difference between present and desired OE conditions

Information Collection Considerations

Commands should incorporate the following considerations when developing collection plans:

- Identify the information requirements that will indicate progress or regression regarding the objective.
- Identify the information requirements that will inform critical decisions throughout planning and execution.
- Identify the organization or individual responsible for collecting the data.
- Determine how often the data should be updated based on the scope of the campaign or operation and changes to the operational environment.

Figure H-3. Information Collection Considerations

toward achieving the objective, as well as to assess the performance and effects of completed tasks and missions.

b. Identifying the required information is an integral part of joint planning. Operational design should inform and influence assessment considerations. As success requires identifying specific conditions within the OA, assessment considerations should be part of their development. Integration, refinement, and adaptation of the required information throughout planning and execution help make certain that recommendations regarding the effectiveness of anticipated and completed tasks and missions support decision points.

c. After identifying information requirements, determine whether the required information exists or needs collection. Vet information requirements throughout the staff and include in the initial and subsequent CCIRs. This establishes an OE baseline to compare the effectiveness of anticipated and completed operations. While often considered part of execution, collection efforts for assessment should begin during planning. This process continues until planning or execution of an operation transitions to the next logical event.

2. Analyze. Analysis identifies operationally significant trends and changes to the OE; their impact on planning and execution (including risks and opportunities); and develops recommendations to refine, adapt, or terminate planning and execution. Include information considerations in the development of the assessment approach and assessment plan. Command internal analysis begins with plan initiation and continues throughout planning and execution. External analysis support typically follows plan or order approval, but informal collaboration with supporting CCMDs, Services, and DoD agencies should begin at plan initiation. The analysis should include the limitations of assessment plan indicators, such as degree of uncertainty, correlations versus causation, or gaps that may exist.

3. Communicate. The communicate action provides appropriate assessment products to all stakeholders and interested audiences. Internal communication

includes the commander (for decisions regarding the overall operation or areas where the commander has expressed interest), staff elements and subordinate commands requiring information related to the analyses (for additional analysis or internal functional decisions), and external audiences (whose products may require commander approval for release). Address communication considerations during development of the assessment approach and included in the assessment plan. However, the “communicate” action typically follows plan or order approval (e.g., CONPLAN, OPLAN) and analysis.

(d) During planning for an operation or campaign, a baseline understanding of the OE assists the commander and staff in setting objectives, if useful, for desired rates of change within the OA and thresholds for success and failure. This also focuses the assessment process on answering specific questions relating to the desired objectives of the plan.

1. Identifying the desired objective and the associated conditions is critical to determining progress in any operation or campaign. Poorly defined objectives typically result in ineffective planning and execution. Poorly defined objectives also increase the risks of wasting time, resources, and opportunities to successfully accomplish the mission. To avert this, the staff should identify clear objectives and tasks having observable performance and effects criteria to measure, refine, and adapt throughout planning and execution. In turn, analysis of anticipated and completed tasks should generate assessment recommendations to communicate.

2. Throughout planning and execution, the command defines the desired observable changes in the OA necessary to achieve the objective and may develop a DSM. The DSM is a written record of a wargamed COA that describes decision points and associated actions at those decision points. Among other information, the DSM lists decision points and the corresponding criteria to evaluate. It also lists the units that observe and report information affecting the criteria for decisions. Incorporate this information into the assessment plan and reflect the required information considerations described in Figure H-4.

3. Nonmilitary aspects of the OE may be critically important in some operations. Information derived from multiple external sources should contribute to tailored JIPOE products that address the relevant nonmilitary actors and relationships within the OA. Analysis addressing all relevant actors within the OA improves and informs the understanding of how conditions may change.

For more detailed discussion on developing indications, see Section B “Linking Effects and Objectives to Tasks through Indicators.”

b. Step 2 – Develop Operation Assessment Plan

(1) Developing, refining, and adapting the assessment plan is concurrent and complementary throughout joint planning and execution. This step overlaps with the previous step during identification of the objectives and effects. Developing the

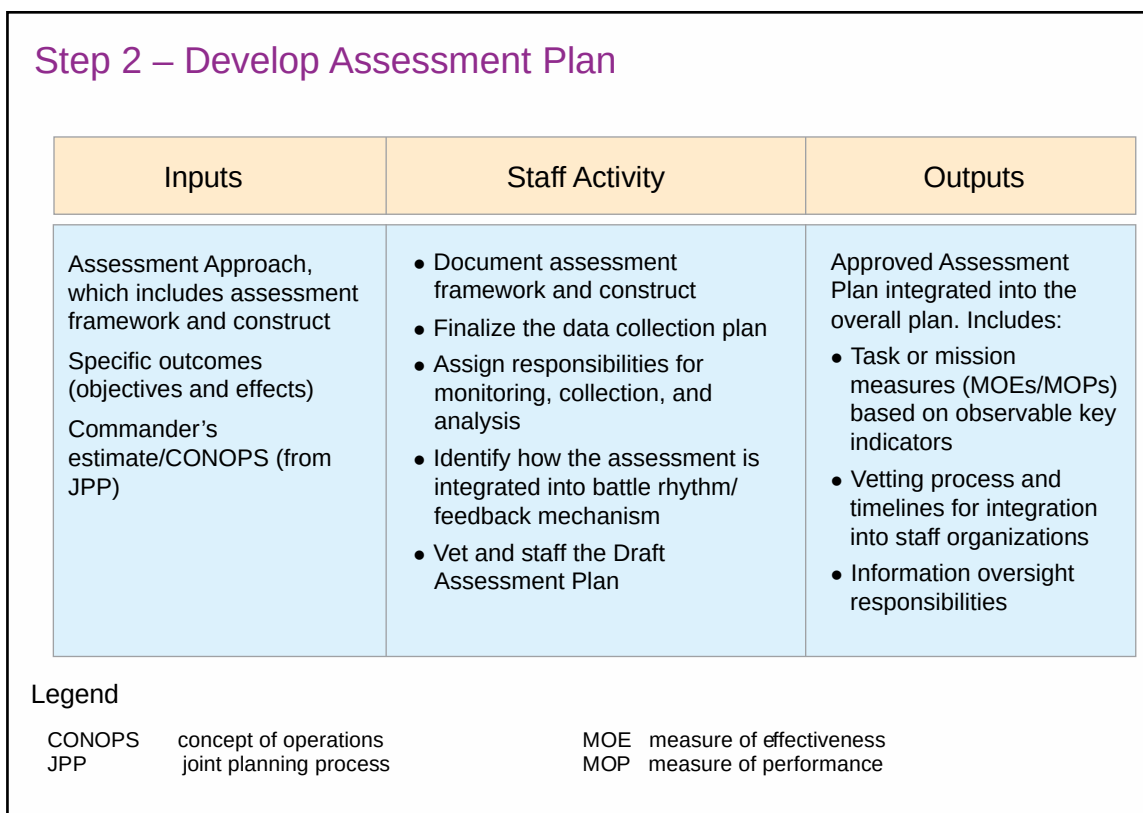


Figure H-4. Step 2 – Develop Assessment Plan

assessment plan is a whole-of-staff effort and should include other key stakeholders to better inform the assessment effort. The assessment plan should identify staff or subordinate organizations to monitor, collect, analyze information, and develop recommendations and assessment products as required. Requirements for staff coordination and presentation to the commander should nest in the plan and integrate into the command's battle rhythm to support the commander's decision cycle (Figure H-4).

(2) The plan framework (i.e., LOEs, objectives, effects, and supporting MOEs and MOPs) is the foundation for the assessment. As such, the assessment plan should link objectives to effects and finally to tasks (and supporting OAI, if appropriate or available). Within this framework, performance and effects can measure observable key indicators. The assessment plan should also include required information oversight responsibilities to gather, update, process, exploit, analyze, integrate, disseminate, classify, and archive the required information.

c. Step 3 – Collect Information and Intelligence (see Figure H-5). Throughout planning and execution, the joint force refines and adapts information collection requirements to gather information about the OE and the joint force's anticipated and completed actions as part of normal C2 activities. Typically, staffs and subordinate commands provide information about planning and execution on a regular cycle through specified battle rhythm events (e.g., joint collection management board or joint collection management board, which is typically co-led by a J-3 and J-2 general officer/flag officer).

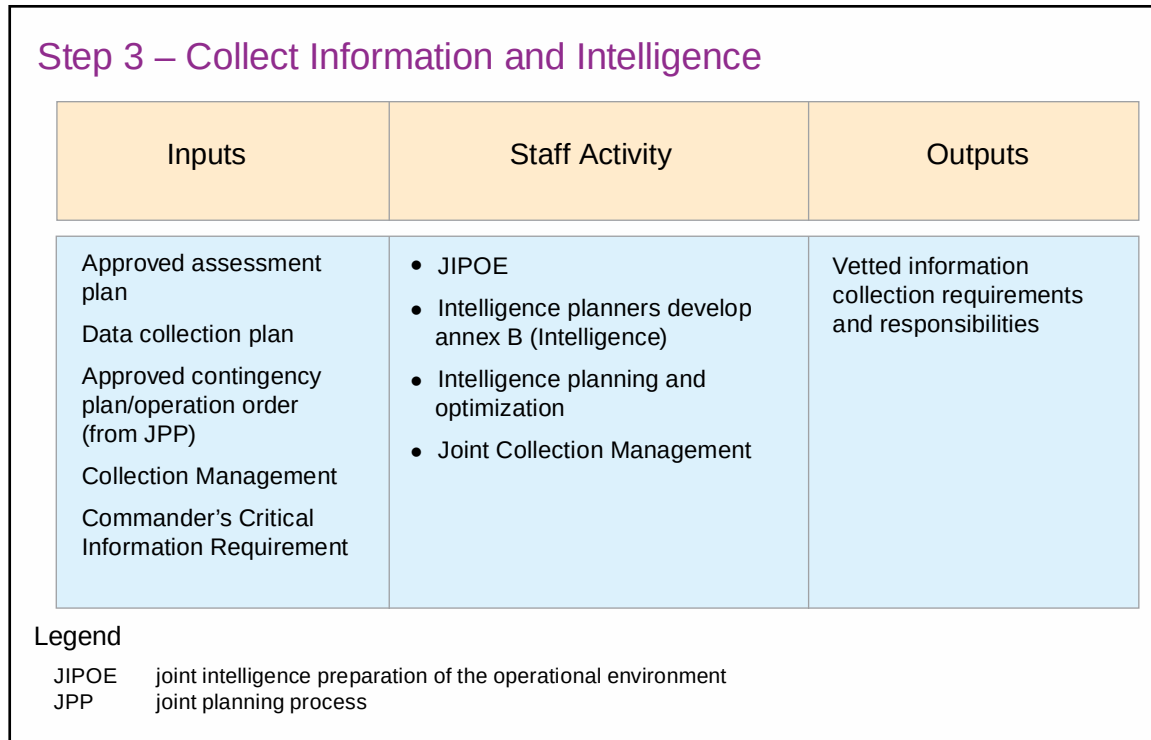


Figure H-5. Step 3 – Collect Information and Intelligence

Intelligence staffs continually provide intelligence about the OE and operational impact to support the collective staff assessment effort. In accordance with the assessment plan, assessment considerations may help the staff determine the presence of decision-point triggers and other mission impacts.

d. Step 4 – Analyze Information and Intelligence

(1) Accurate, unbiased analysis seeks to identify operationally significant trends and changes to the OE and their impact on the operation or campaign. To increase credibility and transparency, **conduct analysis with, and vet through, functional experts within the staff.** Some assessment elements may lack the expertise required to judge the impact to a particular functional area.

(2) Based on analysis, the staff can estimate the effects of force employment and resource allocation, determining decision points or objective success. Using these determinations, the staff also may identify additional risks and challenges to mission accomplishment or identify opportunities to accelerate mission accomplishment (see Figure H-6).

(3) To identify trends and changes, it is necessary to collect and analyze observable key indicators of those differences in conditions in the OA that are the result of completed tasks and missions rather than simply “noise” or normal variation. Analysis seeks to identify positive or negative movement toward creating desired effects or achieving objectives.

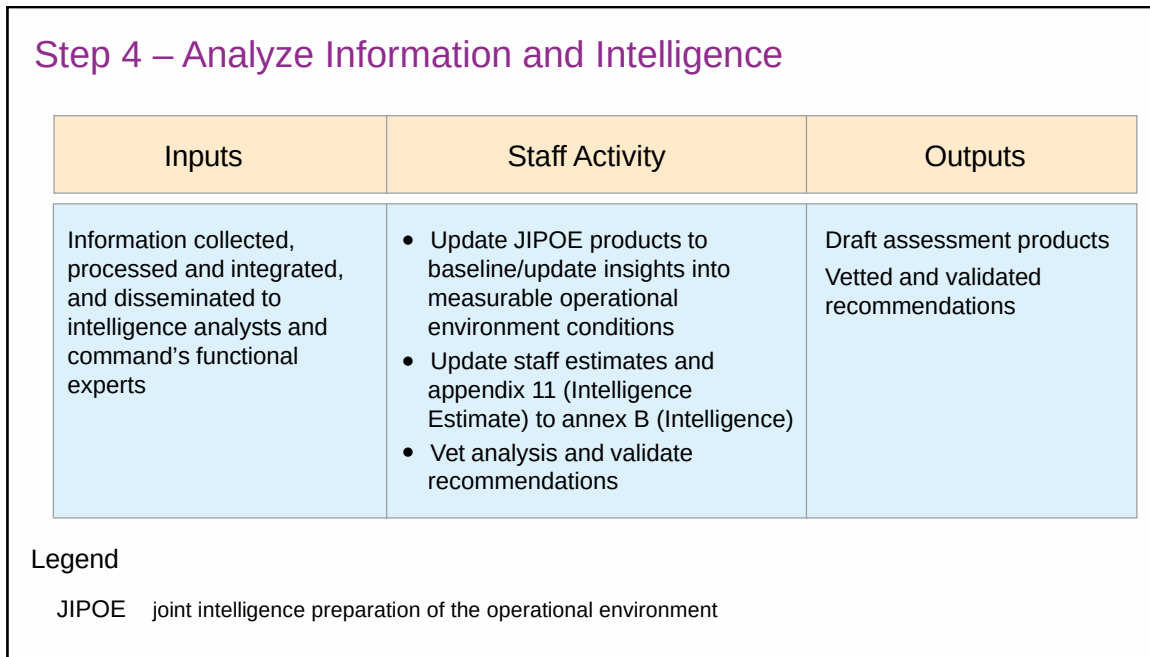


Figure H-6. Step 4 – Analyze Information and Intelligence

(a) While individual staff elements may be responsible for analysis within their functional area, vetting and validation across the staff should enable coherent, holistic assessment products that reflect and encompass numerous discrete analyses.

(b) Cautionary Notes

1. Military operations are inherently human endeavors. In contrast, models are abstract representations of the OE, limited by the perspectives of their developers and what their users are attempting to evaluate. Consequently, models may not include all the critical variables and relationships in the OA. The presence of numbers or mathematical formulae in an assessment does not imply deterministic certainty, rigor, or quality. However, despite the inherent presence of uncertainty in modeling outputs, with the appropriate supporting context, assessment models can assist in analyzing and understanding the OE. Models can assist planners and assessors in producing assessments by providing objective, rational, structured approaches toward complex systems and issues.

2. Military units often find stability activities the most challenging to assess accurately. Staff elements should use caution when seeking to quantify data related to social phenomena. They should ensure military and nonmilitary subject matter experts validate data quality and its appropriateness to the phenomena and answers sought. This type of data normally requires a statistical approach and expert interpretation to be in analysis.

3. While there can be seemingly predictable patterns, these may rest on foundations that change in ways not easily observed or fully understood. Given the challenges of gaining and maintaining understanding of complex environments and the need to maintain awareness of underlying changes, combining the perspectives of multiple groups

and organizations can be a key enabler. No single group or organization has all the information or the complete perspective on the complex environment. Analysis of complex environments should include subject matter experts on the range of issues addressed, as well as experts or local relevant actors knowledgeable about the local context and environment. Further, challenge organizational biases by developing alternate perspectives, either through inviting differing views from partners or through deliberate efforts such as red teaming.

(4) Using professional military judgment, the assessment describes progress toward or regression from achieving the objectives, decisive conditions, or creating effects by answering the assessment-essential questions:

- (a) Where are we?
- (b) What happened?
- (c) Why do we think it happened?
- (d) So what?
- (e) What are the likely future opportunities and risks?
- (f) What do we need to do?

(5) The conclusions generated by the staff analyses regarding achievement of the objective, force employment, resource allocation, validity of planning assumptions, and decision points should lead the staff to develop recommendations for consideration. Recommendations should highlight ways to improve the effectiveness of operations and plans by informing all decisions, including the following:

- (a) Update, change, add, or remove critical assumptions.
- (b) Transition between phases (as appropriate).
- (c) Execute branches and sequels.
- (d) Realign resources.
- (e) Adjust operations.
- (f) Adjust orders and objectives.
- (g) Adjust priorities.
- (h) Change priorities of effort.
- (i) Change support commands.

- (j) Adjust command relationships.
- (k) Adjust decision points.
- (l) Refine or adapt the assessment plan.

(6) Before presenting recommendations to the commander for action, the staff confirms that the assessment plan details the staff processes required to ensure assessment products are valid and any associated recommendations are achievable and improve the effectiveness of operations. Figure H-7 reflects a notional example of battle rhythm activities used to vet and validate assessment products. Many vetted and validated recommendations do not require commander involvement to implement. Present remaining contentious issues to the commander for approval and implementation guidance.

e. Step 5 – Communicate Feedback and Recommendations

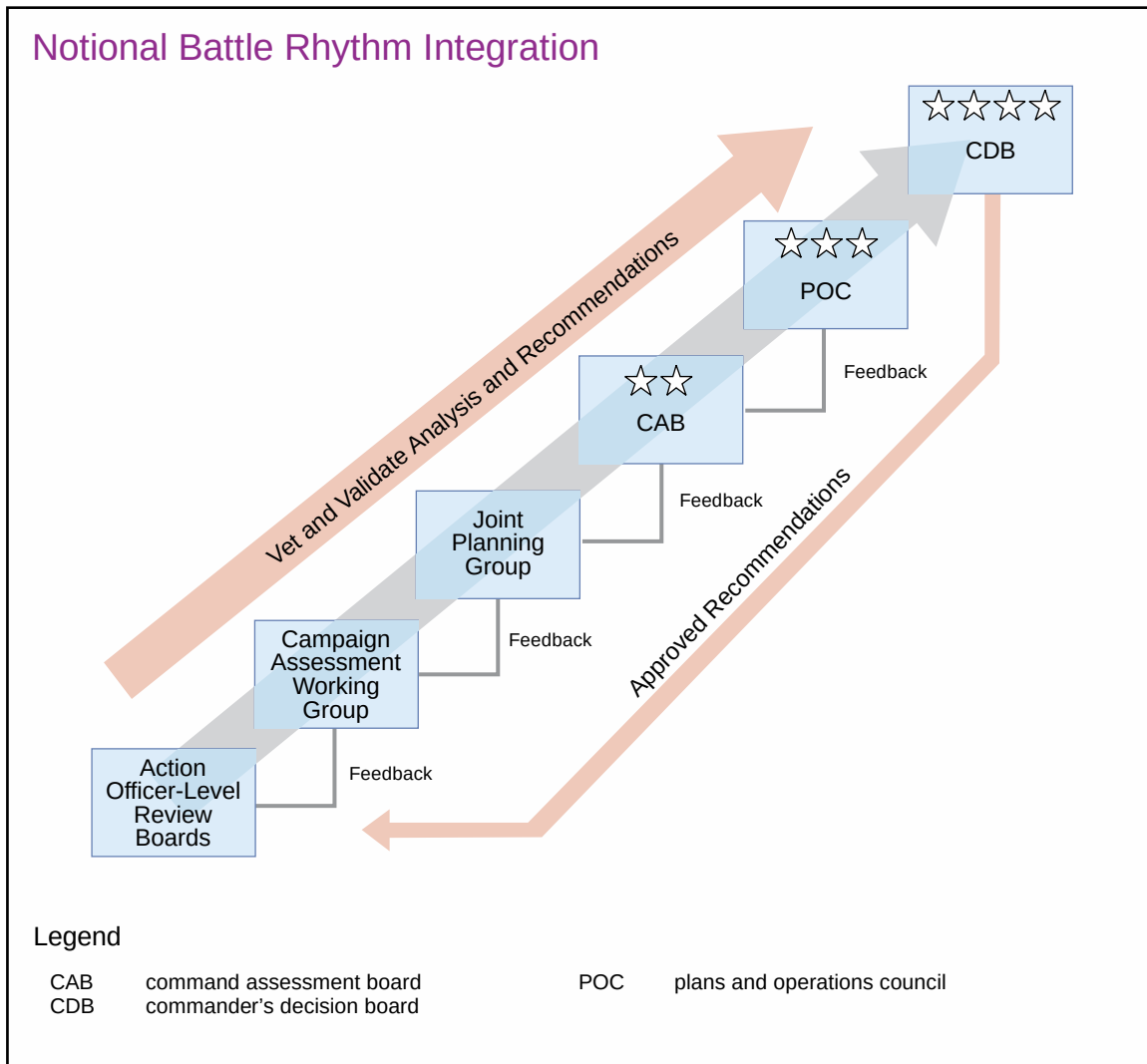


Figure H-7. Notional Battle Rhythm Integration

(1) The staff develops assessment products (which may include summary reports and briefings) containing recommendations for the commander based upon the guidelines set forth in the assessment plan. The commander's guidance is the most critical step in developing assessment products. Regardless of quality and effort, the assessment process is useless if the communication of its results is deficient or inconsistent with the commander's personal style of digesting information and making decisions.

(2) Assessment products are not the assessment itself. Neither are they the data collected for analysis. Assessment products serve the functions of informing the commander about current and anticipated conditions of the OE, determine the ability of the joint force to impact the OA, evaluate progress toward objectives, provide accountability to higher authority, and communicate progress to multinational and interagency partners.

(3) Staffs should strive to align their efforts when communicating assessment results and recommendations (see Figure H-8). Inclusion of various staff products may gain efficiencies by possibly eliminating duplicative briefings and decision boards. It also serves to convey proper context and assure staff-wide dialogue with the commander.

f. Step 6 – Adapt Plans or Operations/Campaigns

(1) Upon receiving feedback and recommendations, commanders typically direct changes or provide additional guidance that dictate updates or modifications to a plan. The commander's guidance may also induce modifications to the assessment plan (see Figure H-9). Even without significant changes to the plan or order, changes to the assessment plan may be necessary to reflect changes in the OA or adjustments to the information or intelligence requirements. Operation assessment leads submit recommended modifications to the plan CCIRs through battle rhythm events, such as the operational planning team, joint planning group, and joint collection management board.

(2) As the operation or campaign transitions between phases, if applied, the assessment plan will likely require updates to adjust to changes in objectives, effects, and tasks associated with the new phase. While anticipated during the original assessment plan

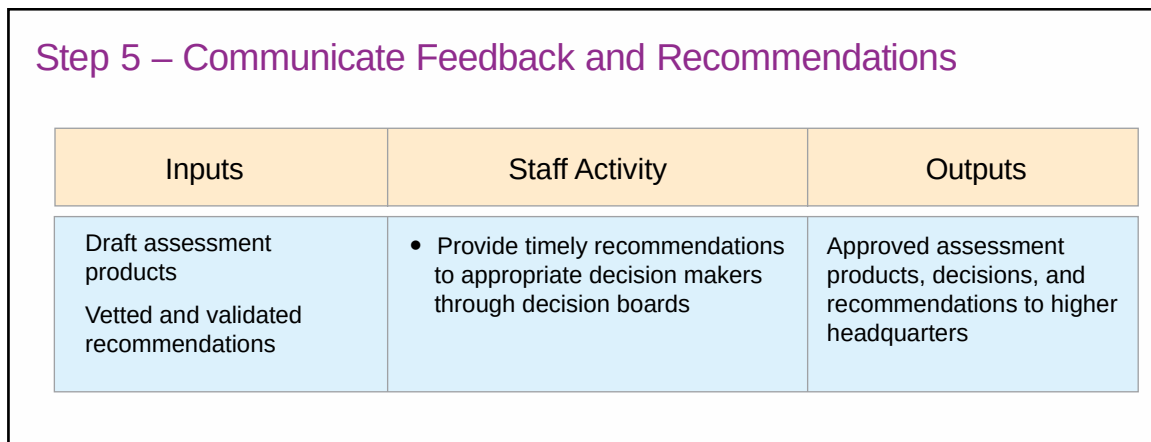


Figure H-8. Step 5 – Communicate Feedback and Recommendations

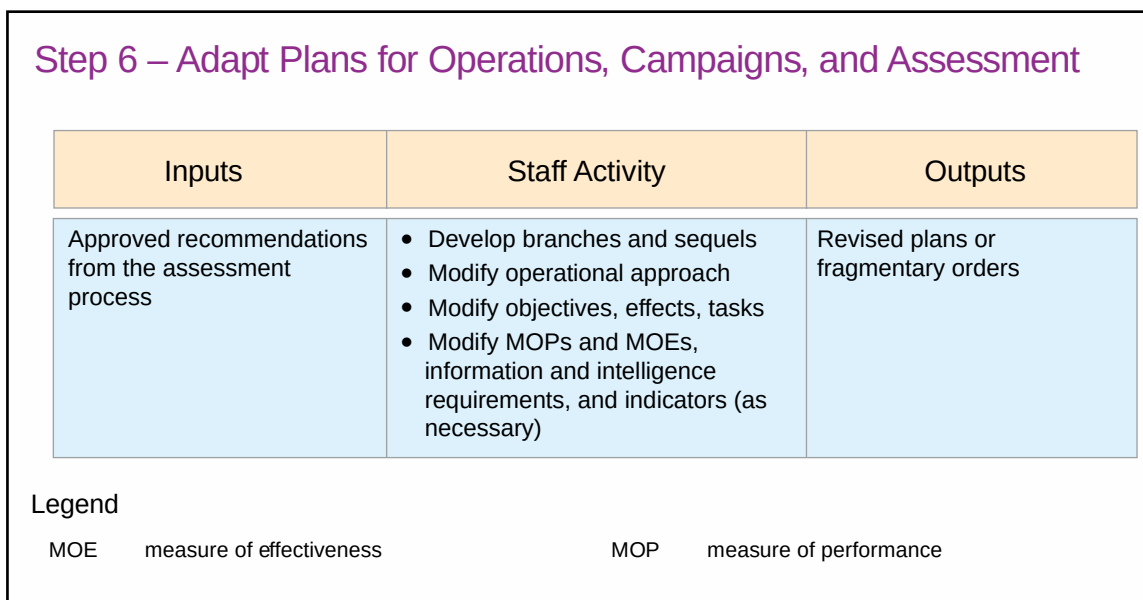


Figure H-9. Step 6 – Adapt Plans for Operations, Campaigns, and Assessment

development, revisions may be necessary to reflect actual conditions in the OA or changes to the plan or order.

(3) There should be organizational procedures associated with capturing the commander's decisions and guidance to ensure completion of necessary actions. The ongoing assessment process should account for these decisions and the actions taken.

4. Cyclical Nature of Assessment

a. Until achieving the objectives, or transitioning the operation, operational assessment remains an on-going process.

b. Adjustments to the plan or order based on commander's updated guidance or changes within the OA require similar updates or changes to the assessment and perhaps its DCP. Formalize updates to the plan or order with FRAGORDs for the widest possible dissemination. Each completed analysis identifies new baseline conditions for the OE and the new basis for analyses of progress.

c. As part of the command's campaign plan, appropriate intelligence and information requirements should nest with the CCIRs and the assessment plan (and perhaps the DCP) for the campaign plan.

d. Upon termination of planning, commands should document their assessment approach and assessment results as part of their lessons learned.

5. Assessment Plan Essentials

During the development of the assessment plan, the staff should ensure it nests with commander-approved CCIRs and:

- a. Document the MOEs and MOPs in terms of acceptable conditions, rates of change, thresholds of success/failure, and technical/tactical triggers.
- b. Document the selection of relevant aspects of the OE during mission analysis.
- c. Document the development of information and intelligence requirements and record the linkage to key MOE and MOP indicators.
- d. Document information and intelligence collection and analysis methods.
- e. Establish methods to estimate risk integrated with the command's risk management process.
- f. Establish methods to determine progress toward the objectives.
- g. Establish a method to evaluate triggers to the commander's decision points.
- h. Develop a terms of reference document.
- i. Establish the format for assessment products.
- j. Coordinate development of recommendations.

SECTION B. LINKING EFFECTS AND OBJECTIVES TO TASKS THROUGH INDICATORS

6. Introduction

a. An operation's desired effects and objectives should help focus the staff's assessment efforts by identifying and analyzing a subset of changes to the overall OA. As the staff develops the desired effects and objectives during planning, they should concurrently identify the specific pieces of information needed to infer changes in the OA supporting them (indicators).

KEY TERM

Indicator: In the context of assessment, a specific piece of information that infers the condition, state, or existence of something and provides a reliable means to ascertain performance or effectiveness.

b. Indicators share common characteristics with carefully selected MOPs and MOEs and link tasks to effects and objectives. Commanders and staffs should develop an approach that best fits their organization, operation, and requirements.

7. Guidelines for Indicator Development

a. Indicators should be relevant, observable, or collectable, responsive, and resourced.

(1) **Relevant.** A valid indicator bears a direct relationship to the desired effect or objective and accurately signifies the anticipated or actual status. This criterion helps avoid collecting and analyzing information that is of no value to a specific operation. It also helps ensure efficiency by eliminating redundant efforts.

(2) **Observable and Collectable.** Indicators should be observable (and therefore collectable), enabling detection of changes to measure and evaluate. The staff should make note of indicators that are relevant but not collectable and report them to the commander. Collection shortfalls can often put the analysis quality at risk. The commander decides whether to accept this risk, realign resources to collect required information, or modify the plan or order.

(3) **Responsive.** Indicators should signify changes in the OA timely enough to enable effective response by the staff and timely decisions by the commander. Assessors consider an indicator's responsiveness to stimulus in the OA. Should indicators react too slowly, the staff will likely miss opportunities for response, and if too quickly, it exposes the staff and commander to false alarms. The JFC and staff should consider the time required for a task or mission to produce desired results within the OA and develop indicators that can respond accordingly. Many actions directed by the JFC require time to implement and may take even longer to produce a measurable result.

(4) **Resourced.** CCMD and subordinate units should adequately resource collection related to indicators so they can obtain the required information without excessive effort or cost. Derived indicator information comes from other staff processes whenever possible. Assessors should avoid indicators that require development of an additional collection system. Staffs should include and monitor resource requirements for indicator collection efforts and analysis in plans. Incorporating threat data collection and analysis should align with the commander's PIRs with relevant tasks specified through annex B (Intelligence) to a plan or an order. Given the focus of PIRs, the collection and analysis they drive provides the commander with insights on changes associated with MOEs. On the other hand, FFIRs provide insights to the commander on the ability of major force elements and other critical capabilities to execute their assigned tasks. Thus, they are associated with MOPs and should reflect in annex C (Operations) with reporting requirements and procedures specified in annex R (Reports) and managed via the guidance in annex I (Knowledge, Information, and Data Management). Effective assessment planning can help avoid duplicating tasks and unnecessary actions, which in turn can help preserve combat power.

b. Assessment DCPs should clearly articulate why an indicator is necessary for the accurate assessment of an action. Collection may draw on subordinate unit operations, key leader engagements, joint functions and functional estimates, and battle damage assessment. Staffs need to understand the fidelity of the available information, choose appropriate information, and prioritize use of high-demand collection resources.

c. Some assessment indicators compete for prioritization and collection assets. Operation assessment leads leverage the commander's approved CCIRs and joint collection management board process to seek approval to add assessment collection requirements into the command's integrated collection plan.

8. Selecting Indicators

a. The two types of indicators commonly used by the joint forces are MOPs and MOEs.

(1) MOPs are indicators used to assess friendly (i.e., multinational) actions tied to measuring task accomplishment. MOPs commonly reside in task execution matrices and confirm or deny proper task performance. MOPs help answer the question: "Are we doing things right," "Was the action taken," or "Was the task completed to standard?"

(2) MOEs are indicators used to help measure a current system state with change indicated by comparing multiple observations over time to gauge the achievement of objectives. MOEs help answer the question, "Are we doing the right things to create the effects or changes in the conditions of the OE that we desire?"

b. Choose distinct indicators. Using indicators that are too like each other can result in the repetitious evaluation of change in a particular condition. In this way, similar indicators skew analyses by overestimating, or "double counting," change in one item in the OA.

c. Include indicators from different causal chains. When indicators have a cause-and-effect relationship with each other, either directly or indirectly, it decreases their value in measuring a particular condition. Measuring progress toward a desired condition by multiple means adds rigor to the analyses.

d. Use the same indicator for more than one objective, effect, task, condition, or mission when appropriate. This sort of duplication in organizing OE information does not introduce significant bias unless carried to an extreme.

e. Avoid or minimize additional reporting requirements for subordinate units. In many cases, commanders may use information generated by other staff elements as indicators in the assessment plan. Units collect many assessment indicators as part of routine operational and intelligence reporting. With careful consideration, commanders and staffs can often find viable alternative indicators without creating new reporting requirements. Excessive reporting requirements can render an otherwise valid assessment plan untenable.

f. **Maximize clarity.** An indicator describes the sought-after information, including specifics on time, information, geography, or unit, as necessary. Any staff member should be able to read the indicator and precisely understand the information it describes.

g. For complex situations or challenging-to-assess activities, developing multiple indicators may be necessary. In such cases, draw indicators from different types of data (i.e., qualitative, quantitative, subjective, objective) to provide a more comprehensive and robust understanding of trends and developments.

9. Understanding Information Categories and Data Types

a. **Information Categories.** The specific type of information expressed in indicators is quantitative or qualitative as well as subjective or objective.

(1) **Quantitative.** Numerical information relating to the quantity or amount of something.

(2) **Qualitative.** Information reflecting an observation of, relating to, or involving quality or kind, typically expressed as a word, sentence, description, or code that represents a category.

(3) **Subjective.** Information based on an individual interpretation of an observed item or condition.

(4) **Objective.** Information based on facts and the precise measurement of conditions or concepts that exist without distortion by personal feelings, prejudices, or interpretations.

(5) To ensure value and credibility, assessors understand and apply categorization considerations in their assessments and recommendations. Indicator information is usually a combination of the four information categories: quantitative-objective, quantitative-subjective, qualitative-objective, and qualitative-subjective (as shown in Figure H-10). As a standard of analytical rigor, categorize information when formulating analyses, reports, and recommendations.

b. **Information Types.** Commanders use assessment information to calculate, analyze, and recommend. Whenever possible, information should originate with observation or experience. Generally, there are four information types. Knowing the type is essential to understand the type of analysis to perform and whether one can interpret the information to draw conclusions, such as the quantity and speed of change in an OE condition over time. In increasing level of complexity and information content, the information types are:

(1) **Nominal.** Nominal information sorts into categories, with no difference in degree or amount between category and any ordering by category is arbitrary. For example, friendly forces categorized by the sending nation (e.g., from Albania, Belgium, Bulgaria).

Information Category Example		
	Quantitative	Qualitative
Objective	The number of no-fly zone violations that have occurred in the last week	The mandate to enforce a no-fly zone is approved
Subjective	The air component's assessment of the effectiveness of the no-fly zone, on a scale of 1 to 10	Enemy freedom of action is limited by the no-fly zone

Figure H-10. Information Category Example

(2) **Ordinal.** Ordinal information has an order but does not indicate the magnitude of discrete intervals within the information. A Likert Scale is a common application of ordinal information where “strongly agree” represents more agreement than “agree” but without specifying how much more. An example of ordinal data might include the rating of the capability of a unit from “able to perform independent operations” as the highest rating and “unable to perform operations without assistance” as the lowest rating.

(3) **Interval.** Interval information is ordinal data with the extra property of having the discrete intervals qualified. However, an interval scale has no value for zero and, therefore, ratios are meaningless. An example is temperature scales where 0° Celsius does not mean there is no temperature. To illustrate, the average daily temperature in Kabul, Afghanistan, in June may be 25° Celsius and in December 5° Celsius. While a difference of 20° Celsius between these months is meaningful, it's improper to state that June is five times as hot as December.

(4) **Ratio.** Ratio information has meaning in both intervals and ratios between measurements. Ratio information has a natural zero indicating the absence of whatever measured. For example, the number of personnel in the armed forces of NATO nations (1999 figures, in thousands) is United States, 1,372; Turkey, 639; Germany, 322; and so on. It is valid to say both that Turkey has 317,000 more military personnel than Germany and that the United States has more than twice as many military personnel as Turkey.

10. Linking Effects and Objectives to Tasks Through Indicators

a. Ensuring effects and objectives link to tasks through carefully selected MOPs and MOEs is essential to the analytical rigor of an assessment framework. Establishing strong, cogent links between tasks, effects, and objectives through MOPs and MOEs facilitates the transparency and clarity of the assessment approach. Additionally, links between tasks, effects, and objectives assist in mapping the plan's CONOPS to actual activities and conditions in the OA and subsequently to desired effects and objectives.

b. **Using Assessment Questions and Information and Intelligence Requirements.** This information guides the development of assessment questions and information and intelligence requirements to identify indicators.

(1) **Statements** about effects or objectives can refer to anything that specifies the change(s) in the OA sought. Within Figure H-11, the refinement of a statement into “smaller statements” refers to any statement or question that increases the specificity of the original statement. During this portion of the process, assessors help develop and refine specific indicators to measure effectiveness. These nested operational design elements may have one or more associated assessment questions.

(2) **Assessment questions** are those that, when answered, provide the commander and staff with direct answers to critical information pertaining to the OE and progress toward desired effects or objectives. Assessment questions take the general form of “How well are we creating our desired effects” and related questions such as, “How can we achieve our objectives more effectively—more quickly, qualitatively better, at less cost, or at less risk?” Answers to the CCIRs should ground the assessment process in the desired effects. They should be answerable with the information or data available to the command; relevant to the desired effects, objectives, and commander priorities; and **useful** to evaluate effects or performance in achieving objectives.

Comparison and Use of Information and Intelligence		
	Information	Intelligence
Perspective	Internal focus	External focus
Sources	Staff section and subordinate command reports, host-nation reports, nongovernmental organization information	All-source intelligence, intelligence agency reports, host-nation reports
Use in Plans	Friendly force information requirements assumptions linking force posture to operations	Priority intelligence requirements assumptions linking operations to effects
Use in Assessments	Identifies if planned actions are executed properly	Identifies if desired outcomes are achieved
Result of Assessment	Resource efficiency of the plan	Resource effectiveness of the plan
Example of Information or Intelligence Requirement	- Allocation of coalition trainers to train host-nation security forces within a specific region - Readiness assessment of host-nation security forces	Security assessment within a particular region

Figure H-11. Comparison and Use of Information and Intelligence

(3) **Information and intelligence requirements** should relate to the desired effects or objectives of the plan and should stem from the assessment questions. Information and intelligence requirements provide a foundation for the development of indicators and record the logical connection between indicators and assessment questions and the effects or objectives they support. Within the context of assessments, IRs seek to understand conditions within the OA while information requirements determine whether the joint force properly executed planned actions. By using both intelligence and information, assessment can provide more comprehensive analyses of the current OE and the joint force's impact on it. When developing information and intelligence requirements, here are some of the questions the staff may ask to determine the value of proposed requirements:

(a) **Use.** What aspect of the desired effects or objectives does this information or intelligence requirement inform?

(b) **Source.** How will the required information or intelligence become available? Who is collecting it? What is our confidence level in the reporting?

(c) **Measurability.** Is the information or intelligence requirement measurable? If the information or intelligence requirement is unavailable, are there other information or intelligence requirements that can serve as proxies?

(d) **Impact.** What is the impact of knowing the required information or intelligence? What is the impact of not knowing it? What is the risk if it is false?

(e) **Timeliness.** When is the required information or intelligence no longer valuable?

(f) **Cost.** What is the cost of data collection to answer the information and intelligence requirements (e.g., the risk to forces, resources, and or mission)?

(4) Indicators should answer information and intelligence requirements. Indicator characteristics are in paragraph 7, "Guidelines for Indicator Development."

SECTION C. DATA COLLECTION PLAN

11. Developing the Data Collection Plan

a. After establishing the assessment indicators, the staff develops a DCP in coordination with planners. This process should include members of staff who will collect data. Although there is no set format for a DCP, it should identify the following for each indicator:

(1) Data parameters, such as:

(a) Units of measurement.

- (b) Scale, if appropriate.
 - (c) Categorization for nominal or interval data.
 - (d) Upper and lower bounds (if required).
 - (e) Additional criteria.
- (2) Source of the data.
 - (3) Method of collection.
 - (4) Party responsible for collection.
 - (5) Record data formatting.
 - (6) Required frequency of collection.
 - (7) Data recipients (who needs the data), recognizing that recipients may change over time.
 - (8) Required frequency of reporting.
 - (9) Additional information.

b. The creation of the data collection matrix assists in clarifying the “measurability” of the selected indicator and may result in further refinement. The DCP should always synchronize and deconflict with established reports across the command.

(1) Military units and civilians provide indicators. While some of this information may be available prior to execution, most performance-related reporting occurs following execution.

(2) Other data for indicators associated with impacts on the OE require the designation of observers as part of the intelligence collection plan or the development of another mechanism for collection. Generally, it is helpful to establish a baseline as early as possible from which subsequent change can be determined. It may be possible to collect on some indicators prior to beginning an operation.

c. The DCP should publish with the final operation or campaign plan/order and incorporated into annex I (Knowledge, Information, and Data Management). Once the commander approves the campaign or operation, all levels of command should start the operation assessment collection process. Throughout planning and execution, the collection plan should adjust as required until terminating planning or the operation transitions.

12. Data Sources

a. Staff elements, in conjunction with commander's communication synchronization personnel, should identify expected and available sources of data. Data can originate from a variety of sources, including:

- (1) Local population (formal or informal surveys of both men and women).
- (2) HN officials (formal or informal surveys).
- (3) HN records.
- (4) Other USG departments and agencies (i.e., embassies, development departments).
- (5) International organizations working in area (e.g., United Nations, World Bank, International Monetary Fund, European Union, Organization for Economic Cooperation and Development).
- (6) NGOs working in area.
- (7) Friendly force observations (e.g., patrol reports, intelligence, training and exercise reports).
- (8) Media and other open sources (e.g., local, national, and home radio, Internet, social media, television, and print sources).
- (9) Commercial data sources.
- (10) Assessment products from superior, subordinate, and supporting commands.
- (11) Subject matter experts within the command.
- (12) Lessons learned and historical records.

b. Each data source requires appropriate scrutiny prior to and during use. When classification rules allow, the source should always link to the information collected to help provide full disclosure when reporting. Without this information, the credibility of any recommendation is disputable if the analysis and communication products appear to be overly positive or negative when compared with general perceptions and expectations.

c. Some data sources may be using that data for their own assessment purposes. Consider whether data sourced from other organizations is raw or processed. In the case of processed data, obtain knowledge about the raw data, assumptions, and processing methods.

d. The staff should specify the expected source of data for each indicator and, if available, identify back-up or corroborating sources for the following reasons:

(1) **Easier-to-Verify Data from Multiple Sources.** A data item from one source is not as valuable as corroborated data from multiple sources.

(2) **Mitigate Bias.** If the data item involves visual observations (e.g., number of open shops) or perceptive observations (e.g., sense of security in the town), the data may vary depending on the source chosen.

(3) **Keep Track of Data and Its Origin.** Whether taken from one or multiple sources, source identity is important for analysis purposes. An annex I (Knowledge, Information, and Data Management) should support manage and track data sources.

(4) **Data Archiving.** Historical records and data backups are essential. In addition to capturing lessons learned, analysts can work to improve measures by performing trend analysis of data over time. Improved historical data capture can also improve the ability to use predictive analytical techniques where opportunities arise.

13. Methods of Collection

a. Throughout planning and execution, planners and staff should identify assessment-related data collection methods. They should identify resources required for data collection, prepare data collection orders for subordinate and supporting commands, and identify appropriate liaison with mission partners to set up data exchange procedures. The command's mindset should be: "Everyone is a collector."

b. Figure H-12 provides some examples of data collection methods and associated advantages and disadvantages.

14. Assign Responsibility for Data Collection

a. The staff should normally assign individual units or organizations with responsibility for each data collection item in the data collection matrix. In some situations, an individual could be responsible for one or more indicators.

b. Codifying responsibility is important for the following reasons:

(1) Assigning responsibility increases the likelihood of the task accomplishment.

(2) Identify and communicate a reporting chain.

(3) In the event of a data query, the analyst can direct data item questions to the person, unit, or organization responsible.

(4) Facilitates data archiving and data analysis.

Data Collection Methods

Data Collection Method	Description	Advantages	Disadvantages
Military Survey	A selective and planned questioning by military forces	• Ease of tasking military forces. • Superior mobility. • Ability to access difficult environments. • Good for gathering raw qualitative data.	• Response of subjects may be biased by negative perception of military. • Military forces may not have specific skills in surveying.
Survey	A selective and planned questioning of subjects by nonmilitary parties (e.g., charities, nongovernmental organizations, or specialists)	• Survey by independent bodies can be more impartial. • Reduced burden on military • May use survey specialists.	• Difficult to task nonmilitary organizations and extra financial cost may be involved. • Nonmilitary organizations may have reduced ability to access difficult environments.
Focus Group	A group of people are asked about their perceptions, opinions, beliefs towards a subject of interest. Questions are asked in an interactive group setting where participants are free to talk with other participants.	• Enables collection of in depth attitudes, belief and anecdotal data. • Group dynamics facilitate idea generation. • Participants not required to read or write, relies on oral communication.	• Requires strong, trained facilitator. • Difficult to make conclusions that represent a population view. • Limited ability to repeat data collection over time.
Structured Interview	A planned, targeted discussion with a subject where the objectives of the discussion are pre-determined and noted in the data collection matrix.	• Often captures richer information than a survey. • Many methods may be used: face to face, electronic mail, telephone, video conference, etc. • Provides opportunity to probe and explore ideas in depth.	• Interviews are very time consuming. • May require trained interviewer or transcriber.
Structured Observation/ Debrief	A set of specific observations collected during routine work, followed by formal question, or asked to report observations at a specific time.	• Good approach to discover behaviours.	• Structured observations can be time consuming. • Human bias of difference in perceptions.
Military Situation Report	A formatted report intended to convey a pre-defined set of information in relation to a specific event or activity, or a routine (time dependant) report.	• A standardized set of information which helps in consistency of reporting. • A normal part of military business rather than an extra burden.	• Limited opportunities for reporting nonstandard data, or for changing report formats for mission specific data.
Automatic Media Collection	An automated collection and analysis of open source media (e.g., Rich Site Summary feeds, online market data, social media, country watch reports) or closed-source media.	• Resources and time are saved in the efforts required for the data collection.	• More analyst time is required to sort through data. • Automatic methods may either collect too much or too little, or miss vital data.
Manual Media Collection	Manual observation (e.g., reading documents, logging events, photocopying of open-source media or closed-source media.	• The data collection can be more thorough, with a certain amount of analysis being done simultaneously.	• Manual data collection is time consuming.

Figure H-12. Data Collection Methods

(5) If data collection tasks fall to persons or organizations outside the military, establish links with those persons or organizations.

(6) If data originates from an organization not likely to be compliant, then seek an alternative source of data or even an alternative indicator.

15. Collection and Reporting Timelines

Planners and staff, with guidance from the commander, should determine the frequencies for data collection and reporting.

a. **Data Collection Frequency.** For each indicator, identify the number of times (e.g., per day, week, month) to collect the data should.

(1) Is the requested frequency commensurate with the possible observable change? It makes no sense to record the number of enemy aircraft sorties per day when only a few occur each month. Conversely, if monitoring the incidence of a highly contagious disease in a refugee camp, daily figures are more appropriate than the number of new infections each month.

(2) Is the data collection influenced by important events? For example, while “attacks per month” is sufficient for most cases, in the month leading up to the regional elections it may be prudent to capture “attacks per day.”

b. **Data Reporting Frequency.** The number of times (e.g., per day, week, or month) to report the data to identified users. Different users may require different frequencies.

c. **Data collection frequency and data reporting frequency may not be the same.** Typically, the requested collection frequency supports the most rapid reporting frequency. For example, the requirement to collect the “number of attacks” daily can support a tactical commander who requires reporting on the attacks per day but also support a higher command’s requirement for attacks per week or attacks per month.

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APPENDIX J STRATEGY

1. Strategic Overview

a. Introduction

(1) The objective of strategy is to serve policy—the positions of governments and others cooperating, competing, or waging war in a complex environment. The goal of strategy is to achieve policy objectives by maintaining or modifying elements of the strategic environment to serve those interests.

(2) **NSS.** The NSS is a presidential document for developing, applying, and coordinating the instruments of national power to achieve strategic objectives. The character of war is changing at an accelerating pace—national strategy secures and advances a nation’s long-term, enduring, core interests over time. The most common expression of national strategy for JFCs is the President’s NSS and policy guidance issued through the National Security Council.

(3) **NDS.** The NDS is SecDef’s framework to prioritize DoD strategic guidance and activities, and it structures assessments and deliberations.

2. Military Strategy

a. Military strategy is the creation, employment, and articulation of the military instrument of national power to achieve policy objectives. The ends in military strategy are the defense strategy’s objectives, while the ways and means represent how the joint force will execute the defense strategy. The frameworks in a military strategy provide a lens for subsequent campaign planning and contingency planning.

b. **The Logic of Military Strategy.** While military strategy is principally a function of creative art, the logic or science behind every strategy should be rigorous and founded upon the evidence of history; the arithmetic of available resources; a clear acknowledgment of time horizons and distances; and an astute analysis of friendly, neutral, adversary, and enemy interests and will. Developing military strategy requires an understanding of facts and assumptions to inform strategic decision making. Its logic is both inductive and deductive, guiding purposeful action toward its end.

c. **Strategic Uses of Military Force.** The United States leverages all instruments of national power to pursue its national interests. Reinforcing America’s traditional tools of diplomacy, DoD provides military options to ensure the President and our diplomats negotiate from positions of strength. DoD is in a supporting role when the military instrument of national power is not the predominant instrument for the strategy. When directed or if the other instruments of national power prove insufficient, the military becomes the nation’s primary instrument. In either case, the military facilitates and supports the application of the other instruments.

3. Types of Military Strategy

a. **NMS.** The CJCS submits an NMS biennially to Congress. It describes how the Armed Forces of the United States will support the objectives of the United States. The broad scope of the CJCS's responsibilities suggests a continuum of strategic direction spanning three overlapping time horizons corresponding to how the joint force employs, adapts, and innovates the force to meet the requirements of law, policy, and defense strategy. This provides the final piece of the ends, ways, and means construct.

b. **CCMD Strategy.** These strategies describe the CDR's vision to position forces, designate necessary campaigns, and prepare to execute contingency plans. The strategy takes into consideration policy, available resources, Service force development plans, and budget forecasts. CCMD staffs identify mismatches between ends, ways, and means during strategy development, reconcile them, and articulate residual risk that the CDR accepts.

c. **Service or Institutional Strategy.** Unlike a strategy of force employment, strategies for the military Services and other institutions tend to look internally for their implementation. Such strategies translate a senior leader's vision for their organization into direction for the future force while meeting today's commitments, consistent with their responsibilities and authorities.

4. Military Strategy in Joint Campaigning

a. **Operational Art from Strategy.** Operational art is the cognitive approach by commanders and staffs—supported by their skill, knowledge, experience, creativity, and judgment—to develop strategies, campaigns, and operations to organize and employ military forces by integrating ends, ways, and means.

b. **Global Campaigning.** Implementing national strategic guidance through operational art with a global perspective has multiple aspects. The first aspect is global campaigns, which direct day-to-day operations and are the purview of global campaign planning that occurs at the JS and CCMD levels. Day-to-day campaigns span the competition continuum.

c. **Joint Warfighting with Multiple Supported CCMDs and Multiple AORs.** The second aspect of implementing strategic guidance is contingency campaigning, operations executed in response to changes in the strategic environment that require a branch from the GCPs or CCPs. National, defense, and military strategies, through the presidentially approved CPG, direct contingency plans to address designated threats, potential catastrophic events, and contingent missions without a crisis that put one or more national interests at risk in ways that warrant military response options.

d. **GFM and Posture.** The GFM process aligns force apportionment, assignment, allocation, and readiness methodologies in support of the NDS.

5. Institutional Strategy

a. **Implementing Institutional Strategies.** Institutional strategies translate military strategy to an organization's internal tasks to shape investments for the future force and ensure force resilience. Those investments are adaptive in the case of force development or innovative in the case of force design.

b. **Military Strategy in Force Development.** Implementing military strategy in force development occurs within the years of the Future Years Defense Program. It assesses and identifies required capabilities for the future joint force. Those capabilities may come from strategy, campaign plans, and contingency plans at multiple levels. Collectively, readiness reporting for contingency planning and assessments of strategy and campaign planning inform the Joint Forces Readiness Review, which is the joint force's capstone assessment product.

c. **Military Strategy in Force Design.** The joint force's primary document for force design is the *Capstone Concept for Joint Operations: Joint Force 2030*. Vetted through war games and experimentation, the capstone concept, and the other documents in the family of joint concepts provide the intellectual basis for Service concepts that describe each of the Services' contributions to the future joint force. The required capabilities articulated in those concepts also bridge force design to force development by highlighting capability gaps that require long lead times to mitigate.

6. Risk and Strategic Assessment

a. Assessing risk, the probability and consequence of an event causing harm to something valued, is a key element of decision making. Accurately appraising risk enables commanders and staffs to manage and communicate risk, inform decisions, and provide information across disparate processes. A formal methodology to assess strategic risk to national interests, as well as military risks to missions and to forces, appears in CJCSM 3105.01, *Joint Risk Analysis Methodology*.

b. Strategy assessments often address the entire continuum of strategic direction spanning force employment, force development, and force design, and are often more focused on identifying longer-term trends. In contrast, campaign and contingency assessments and readiness reporting focus principally on near-term force employment using the current force structure and posture.

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APPENDIX K REFERENCES

The development of JP 5-0 is based on the following primary references:

1. General

- a. Title 10, USC.
- b. Title 22, USC.
- c. *2022 National Security Strategy of the United States of America.*
- d. National Security Presidential Memorandum, *Organization of the National Security Council, the Homeland Security Council, and Subcommittees.*
- e. *3D Planning Guide: Diplomacy, Development, Defense.*

2. Department of Defense Publications

- a. (U) *2022 National Defense Strategy of the United States of America.*
- b. *Contingency Planning Guidance (CPG).*
- c. *Defense Strategy Review.*
- d. (U) *The National Military Strategy of the United States of America 2022.*
- e. *Fiscal Years 2026 to 2028 Global Force Management Implementation Guidance.*
- f. DoDD 3000.05, *Stabilization.*
- g. DoDD 3000.06, *Combat Support Agencies (CSAs).*
- h. DoDD 3000.07, *Irregular Warfare (IW).*
- i. DoDD 3000.10, *Contingency Basing Outside the United States.*
- j. DoDD 3000.16, *Vendor Threat Mitigation.*
- k. DoDD 4500.09, *Transportation and Traffic Management.*
- l. DoDD 5100.01, *Functions of the Department of Defense and Its Major Components.*
- m. DoDD 5100.03, *Support of the Headquarters of Combatant and Subordinate Unified Commands.*
- n. DoDD 5205.14, *DoD Counter Threat Finance (CTF) Policy.*

- o. DoDI 1100.22, *Policy and Procedures for Determining Workforce Mix*.
- p. DoDI 3000.12, *Management of US Global Defense Posture (GDP)*.
- q. DoDI 3000.17, *Civilian Harm Mitigation and Response*.
- r. DoDI 3020.41, *Operational Contract Support Outside the United States*.
- s. DoDI 5000.68, *Security Force Assistance (SFA)*.
- t. DoDI 5025.12, *Standardization of Military and Associated Terminology*.
- u. *Department of Defense Strategy for Countering Weapons of Mass Destruction*.
- v. *Department of Defense Civilian Harm Mitigation and Response Action Plan (CHMR-AP)*, August 25, 2022.

3. Chairman of the Joint Chiefs of Staff Publications

- a. CJCSI 2300.01D, *International Agreements*.
- b. CJCSI 2420.01F, *(U) United States Freedom of Navigation Program*.
- c. CJCSI 3100.01F, *Joint Strategic Planning System*.
- d. CJCSI 3110.01L, *(U) 2023 Joint Strategic Campaign Plan*.
- e. CJCSI 3110.02J, *(U) Intelligence Supplement to the Joint Strategic Campaign Plan*.
- f. CJCSI 3141.01F, *Management and Review of Campaign and Contingency Plans*.
- g. CJCSI 3150.25H, *Joint Lessons Learned Program*.
- h. CJCSI 3210.06A, *Irregular Warfare*.
- i. CJCSI 3401.01E, *Joint Combat Capability Assessment*.
- j. CJCSI 3401.02B, *Force Readiness Reporting*.
- k. CJCSI 3500.01J, *Joint Training Policy for the Armed Forces of the United States*.
- l. CJCSI 5705.01H, *Standardization of Military and Associated Terminology*.
- m. CJCSI 5715.01D, *Joint Staff Participation in Interagency Affairs*.
- n. CJCSI 5840.01, *Civilian Harm Mitigation and Response*.
- o. CJCSI 8501.01C, *Chairman of the Joint Chiefs of Staff, Combatant Commanders, Chief, National Guard Bureau, and Joint Staff Participation in the Planning, Programming, Budgeting, and Execution Process*.

- p. CJCSM 3105.01B, *Joint Risk Analysis Methodology*.
- q. CJCSM 3122.02F, *Joint Operation Planning and Execution System (JOPES), Volume III, Time-Phased Force and Deployment Data Development and Deployment Execution*.
- r. CJCSM 3130.03B, *Planning and Execution Formats and Guidance*.
- s. CJCSM 3130.06D, *Global Force Management Allocation Policies and Procedures*.
- t. CJCSM 3150.01E, *Joint Reporting Structure General Instructions*.
- u. CJCSM 3314.01A, *Intelligence Planning*.
- v. CJCSM 3500.03E, *Joint Training Manual for the Armed Forces of the United States*.
- w. CJCSM 4301.01, *Planning Operational Contract Support*.
- x. JP 1, Volume 1, *Joint Warfighting*.
- y. JP 1, Volume 2, *The Joint Force*.
- z. JP 1-0, *Joint Personnel Support*.
- aa. JP 2-0, *Joint Intelligence*.
- bb. JP 3-0, *Joint Campaigns and Operations*.
- cc. JP 3-04, *Information in Joint Operations*.
- dd. JP 3-05, *Joint Doctrine for Special Operations*.
- ee. JP 3-07, *Joint Stabilization Activities*.
- ff. JP 3-08, *Interorganizational Cooperation*.
- gg. JP 3-09, *Joint Fires*.
- hh. JP 3-11, *Operations in Chemical, Biological, Radiological, and Nuclear Environments*.
- ii. JP 3-12, *Joint Cyberspace Operations*.
- jj. JP 3-14, *Joint Space Operations*.
- kk. JP 3-16, *Multinational Operations*.
- ll. JP 3-20, *Security Cooperation*.
- mm. JP 3-22, *Foreign Internal Defense*.
- nn. JP 3-24, *Counterinsurgency*.

- oo. JP 3-25, *Joint Countering Threat Networks*.
- pp. JP 3-28, *Defense Support of Civil Authorities*.
- qq. JP 3-30, *Joint Air Operations*.
- rr. JP 3-33, *Joint Force Headquarters*.
- ss. JP 3-34, *Joint Engineer Operations*.
- tt. JP 3-35, *Joint Deployment and Redeployment Operations*.
- uu. JP 3-40, *Joint Countering Weapons of Mass Destruction*.
- vv. JP 3-41, *Chemical, Biological, Radiological, and Nuclear Response*.
- ww. JP 3-57, *Joint Civil-Military Operations*.
- xx. JP 3-60, *Joint Targeting*.
- yy. JP 3-61, *Joint Public Affairs*.
- zz. JP 3-85, *Joint Electromagnetic Spectrum Operations*.
- aaa. JP 4-0, *Joint Logistics*.
- bbb. JP 4-01, *The Defense Transportation System*.
- ccc. JP 4-02, *Joint Health Services*.
- ddd. JP 4-05, *Joint Mobilization Planning*.
- eee. JP 4-10, *Operational Contract Support*.
- fff. JP 6-0, *Joint Communications*.
- ggg. *Joint Guide for Joint Intelligence Preparation of the Operational Environment*.

4. Service Publications

- a. Air Force Doctrine Publication 1-1, *Mission Command*.
- b. Air Force Doctrine Publication 3-0, *Operations*.
- c. Marine Corps Doctrinal Publication 1-2, *Campaigning*.
- d. Marine Corps Doctrinal Publication 5, *Planning*.
- e. Army Doctrine Publication (ADP) 3-0, *Operations*.

- f. ADP 3-05, *Army Special Operations*.
- g. ADP 3-07, *Stability*.
- h. ADP 3-90, *Offense and Defense*.
- i. ADP 5-0, *The Operations Process*.
- j. ADP 6-0, *Mission Command: Command and Control of Army Forces*.
- k. *Field Manual (FM) 3-14, Army Space Operations*.
- l. FM 3-24, *Insurgencies and Countering Insurgencies*.
- m. FM 5-0, *Planning and Orders Production*.
- n. FM 6-0, *Commander and Staff Organization and Operations*.
- o. Naval Doctrine Publication 1, *Naval Warfare*.
- p. Navy Warfare Publication 5-0, *Navy Planning*.

5. Multiservice Publications

a. Army Techniques Publication 5-0.3/Marine Corps Reference Publication 5-10.1/Navy Tactics, Techniques, and Procedures 5-01.3/Air Force Tactics, Techniques, and Procedures 3-2.87, *Multi-Service Tactics Techniques and Procedures for Operation Assessment*.

b. FM 6-05/Marine Corps Reference Publication 3-30.4/Navy Tactics, Techniques, and Procedures 3-05.19/Air Force Tactics, Techniques, and Procedures 3-2.73/United States Special Operations Command-Pub 3-33, *Multi-Service Tactics, Techniques, and Procedures for Conventional Forces and Special Operations Forces Integration, Interoperability, and Interdependence*.

6. Other Publication

North Atlantic Treaty Organization Assessment Handbook.

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APPENDIX L

ADMINISTRATIVE INSTRUCTIONS

1. User Comments

Users in the field are highly encouraged to submit comments on this publication using the Joint Doctrine Feedback Form located at https://jdeis.js.mil/jdeis/jel/jp_feedback_form.pdf and e-mail it to js.pentagon.j7.mbx.jedd-support@mail.mil. These comments should address content (accuracy, usefulness, consistency, and organization), writing, and appearance.

2. Authorship

The lead agent and Joint Staff doctrine sponsor for this publication is the Director for Strategy, Plans, and Policy (J-5).

3. Supersession (if required)

This publication supersedes JP 5-0, *Joint Planning*, 01 December 2020, Incorporating Change 1, 01 July 2024. In addition, relevant material from Joint Doctrine Note (JDN) 2-19, *Strategy*, is incorporated into the publication. Accordingly, JDN 2-19, *Strategy*, will be removed from the joint doctrine hierarchy.

4. Change Recommendations

a. To provide recommendations for urgent or routine changes to this publication, please complete the Joint Doctrine Feedback Form located at https://jdeis.js.mil/jdeis/jel/jp_feedback_form.pdf and e-mail it to js.pentagon.j7.mbx.jedd-support@mail.mil.

b. When a Joint Staff directorate submits a proposal to the CJCS that would change source document information reflected in this publication, that directorate will include a proposed change to this publication as an enclosure to its proposal. The Services and other organizations are requested to notify the Joint Staff J-7 when changes to source documents reflected in this publication are initiated.

5. Lessons Learned

The Joint Lessons Learned Program's (JLLP's) primary objective is to enhance joint force readiness and effectiveness by contributing to improvements in doctrine, organization, training, materiel, leadership and education, personnel, facilities, and policy. The Joint Lessons Learned Information System (JLLIS) is the DoD system of record for lessons learned and facilitates the collection, tracking, management, sharing, collaborative resolution, and dissemination of lessons learned to improve the development and readiness of the joint force. The JLLP integrates with joint doctrine through the joint doctrine development process by providing insights and lessons learned derived from operations, events, and exercises. As these inputs are incorporated into joint doctrine, they become

institutionalized for future use, a major goal of the JLLP. Insights and lessons learned are routinely sought and incorporated into draft JPs throughout formal staffing of the development process. The JLLIS Website can be found at <https://www.jllis.mil> (NIPRNET) or <http://www.jllis.smil.mil> (SIPRNET).

6. Releasability

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b. The Joint Staff does not print hard copies of JPs for distribution. An electronic version of this JP is available on:

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(2) SIPRNET JEL+ at <https://jdeis.js.smil.mil/jdeis/index.jsp>.

GLOSSARY
PART I—SHORTENED WORD FORMS (ABBREVIATIONS, ACRONYMS,
AND INITIALISMS)

ADP	Army doctrine publication
ALERTORD	alert order
AOR	area of responsibility
C2	command and control
CCDR	combatant commander
CCIR	commander's critical information requirement
CCMD	combatant command
CCP	combatant command campaign plan
CFA	critical factors analysis
CJCS	Chairman of the Joint Chiefs of Staff
CJCSI	Chairman of the Joint Chiefs of Staff instruction
CJCSM	Chairman of the Joint Chiefs of Staff manual
COA	course of action
COG	center of gravity
CONOPS	concept of operations
CONPLAN	concept plan
COS	chief of staff
CPG	contingency planning guidance
CSL	cooperative security location
CTS	common threat scenario
DCP	data collection plan
DEPORD	deployment order
DoD	Department of Defense
DoDD	Department of Defense directive
DoDI	Department of Defense instruction
DOS	Department of State
DSM	decision support matrix
DST	decision support template
EMS	electromagnetic spectrum
EXORD	execute order
FDO	flexible deterrent option
FFIR	friendly force information requirement
FM	field manual (USA)
FOS	forward operating site
FP	force provider
FRAGORD	fragmentary order
FRO	flexible response option

GCO	global crisis option
GCP	global campaign plan
GDP	global defense posture
GEF	Guidance for Employment of the Force
GFM	global force management
GFMIG	Global Force Management Implementation Guidance
GPEC	Global Posture Executive Council
HN	host nation
IP	intelligence planning
IPR	in-progress review
IPS	integrated plan set
IR	intelligence requirement
J-2	intelligence directorate of a joint staff
J-3	operations directorate of a joint staff
J-5	plans directorate of a joint staff
JADO	joint all-domain operations
JFC	joint force commander
JIPOE	joint intelligence preparation of the operational environment
JP	joint publication
JPEC	joint planning and execution community
JPP	joint planning process
JS	Joint Staff
JTF	joint task force
LOE	line of effort
LOO	line of operation
MNF	multinational force
MOB	main operating base
MOE	measure of effectiveness
MOP	measure of performance
NATO	North Atlantic Treaty Organization
NDS	national defense strategy
NGB	National Guard Bureau
NGO	nongovernmental organization
NISP	national intelligence support plan
NMS	national military strategy
NSS	national security strategy
OA	operational area
OAI	operations, activities, and investments

OE	operational environment
OPLAN	operation plan
OPORD	operation order
OSD	Office of the Secretary of Defense
PIR	priority intelligence requirement
PLANORD	planning order
PTDO	prepare to deploy order
ROE	rules of engagement
SecDef	Secretary of Defense
SPF	strategic planning framework
TPFDD	time-phased force and deployment data
UCP	Unified Command Plan
US	United States
USC	United States Code
USG	United States Government
WARNORD	warning order

PART II—TERMS AND DEFINITIONS

1. JP 5-0, *Joint Planning*, 01 July 2025, Active Terms and Definitions

acceptability. The plan review criterion for assessing whether the contemplated course of action is proportional, worth the cost, consistent with the law of war, and militarily and politically supportable. (Approved for incorporation into the DoD Dictionary.)

adequacy. The plan review criterion for assessing whether the scope and concept of planned operations can accomplish the assigned mission and comply with the planning guidance. (Approved for incorporation into the DoD Dictionary.)

alert order. A planning directive that provides essential planning guidance and directs the initiation of planning but does not authorize execution. Also called **ALERTORD**. (Approved for incorporation into the DoD Dictionary.)

allocation. 1. Distribution of forces and resources for employment among competing requirements. (JP 3-30) 2. The temporary transfer of forces between combatant commanders including rotational requirements and requests for capabilities or forces (unit or individual) in response to crisis or emergent contingencies. (JP 5-0) (Definition #2 sourced to JP 5-0 approved for incorporation into the DoD Dictionary.)

apportionment. The forces provided for planning purposes but not necessarily an identification of the actual forces that may be allocated for use when a plan transitions to execution. (Approved for incorporation into the DoD Dictionary.)

base plan. An operation plan that describes the concept of operations, major forces, sustainment concept, and anticipated timelines for completing the mission, without annexes or time-phased force and deployment data. Also called **BPLAN**. (Approved for incorporation into the DoD Dictionary.)

branch. The contingency options built into a base plan used for changing objective, orientation, or direction of movement of a force based on anticipated events, opportunities, or disruptions. (Approved for incorporation into the DoD Dictionary.)

campaign. A series of related operations aimed at achieving strategic and operational objectives. (Approved for incorporation into the DoD Dictionary.)

campaign plan. A plan for a series of related operations aimed at achieving strategic or operational objectives. (Approved for incorporation into the DoD Dictionary.)

C-day. The unnamed day on which a deployment operation is to commence. (Approved for incorporation into the DoD Dictionary.)

center of gravity. The source of power that provides moral or physical strength, freedom of action, or will to act. Also called **COG**. (DoD Dictionary. Source: JP 5-0)

civilian environment. The factors within the operational environment that relate to civilians and their communities, including the civilian population and the personnel, organizations, resources, infrastructure, essential services, and systems on which civilian life depends. (DoD Dictionary. Source: JP 5-0)

combat support agency. A Department of Defense agency so designated by Congress or the Secretary of Defense that supports military combat operations. Also called **CSA**. (DoD Dictionary. Source: JP 5-0)

commander's estimate. The commander's initial assessment that provides options in a concise statement that defines who, what, when, where, why, and how to implement the course of action. (Approved for incorporation into the DoD Dictionary.)

commander's required delivery date. The original date, relative to C-day, specified by the combatant commander for arrival of forces or cargo at the destination. (Approved for incorporation into the DoD Dictionary.)

concept of operations. A description that expresses how the commander intends to employ available resources to achieve the objectives. Also called **CONOPS**. (Approved for incorporation into the DoD Dictionary.)

concept plan. An abbreviated plan, in the format of an operation plan, that expands on the concept of operations. Also called **CONPLAN**. (Approved for incorporation into the DoD Dictionary.)

concurrent plan. A problem set plan, likely to be executed concurrently to achieve essential objectives, which is aligned and collectively resource informed to the core plan and other concurrent plans through shared mission analysis. (Approved for inclusion in the DoD Dictionary.)

conditional plan. An identified problem set plan that presents different options for the achievement of integrated plan set objectives that may go beyond those objectives to address less likely or alternative adversary scenarios. (Approved for inclusion in the DoD Dictionary.)

constraint. In the context of planning, a requirement placed on the command by a higher command that requires an action. (Approved for incorporation into the DoD Dictionary.)

contingency. A situation requiring military operations in response to natural disasters, terrorists, subversives, or as otherwise directed by appropriate authority to protect United States interests. (DoD Dictionary. Source: JP 5-0)

contingency plan. A branch of a campaign plan planned based on hypothetical situations. (Approved for incorporation into the DoD Dictionary.)

core plan. The plan at the center of an integrated plan set around which concurrent and conditional plans are framed. (Approved for inclusion in the DoD Dictionary.)

culminating point. The point at which a force can no longer maintain momentum in its operations. (Approved for incorporation into the DoD Dictionary.)

decision point. The latest time when the commander anticipates making a key decision concerning a specific course of action. (Approved for incorporation into the DoD Dictionary.)

decisive point. Key terrain, key event, critical factor, or function that enables commanders to gain a marked advantage or to achieve success. (Approved for incorporation into the DoD Dictionary.)

deployment order. A directive for the deployment of forces for operations or exercises. Also called **DEPORD**. (Approved for incorporation into the DoD Dictionary.)

deterrence. Discouraging an actor from taking unwanted action by signaling and demonstrating the existence of a credible threat of unacceptable counteraction or the adversary's belief that the costs of a future action outweigh the perceived benefits. (Approved for incorporation into the DoD Dictionary.)

essential task. A specified or implied task an organization must perform to accomplish the mission. (DoD Dictionary. Source: JP 5-0)

estimate. An appraisal of the friendly and threat forces, vulnerabilities, and potential courses of action. (Approved for incorporation into the DoD Dictionary.)

execute order. An order to initiate military operations as directed by the Chairman of the Joint Chiefs of Staff. Also called **EXORD**. (Approved for incorporation into the DoD Dictionary.)

feasibility. The plan review criterion for assessing whether the plan could accomplish the assigned mission using available resources within the time contemplated by the plan. (Approved for incorporation into the DoD Dictionary.)

force planning. In the context of joint planning, an element of plan development where the supported combatant command, in coordination with its supporting and subordinate commands, determines force requirements to accomplish an assigned mission. (Approved for incorporation into the DoD Dictionary.)

force sourcing. The identification of the actual units, their origins, ports of embarkation, and movement characteristics to satisfy the time-phased force requirements of a supported commander. (DoD Dictionary. Source: JP 5-0)

fragmentary order. An abbreviated operation order issued as needed to change or modify an order or to execute a branch or sequel. Also called **FRAGORD**. (DoD Dictionary. Source: JP 5-0)

global campaign plan. Primary means by which the Chairman of the Joint Chiefs of Staff or designated combatant commander arranges for unity of effort and purpose and

through which they guide the planning, integration, and coordination of joint operations across combatant command areas of responsibility and functional responsibilities. Also called **GCP**. (DoD Dictionary. Source: JP 5-0)

H-hour. 1. The specific scheduled hour on D-day at which a particular operation is to commence. (JP 5-0) 2. In amphibious operations, the time the first surface wave is scheduled to land on the beach. (JP 3-02) (Definition #1 approved for incorporation into the DoD Dictionary.)

implied task. In the context of planning, a task derived during mission analysis that an organization must perform to accomplish a specified task or the mission but which the higher headquarters order does not state. (Approved for incorporation into the DoD Dictionary.)

integrated planning. Planning that incorporates strategic planning frameworks and integrated plans sets and builds upon the responsibilities and missions directed by the Unified Command Plan and enables the joint force to execute coordinated and synchronized operations across geographic regions, joint functions, all domains, and time. (Approved for inclusion in the DoD Dictionary.)

integrated plan set. Identifies and links specific core, concurrent, and conditional contingency plans likely to execute contemporaneously, closing gaps in segregated plans. Also called **IPS**. (Approved for inclusion in the DoD Dictionary.)

joint planning. Planning activities by combatant commanders and their subordinate commanders. (Approved for incorporation into the DoD Dictionary.)

joint planning group. A planning organization consisting of representatives of the joint force headquarters staff sections, joint force components, and other supporting organizations or agencies as deemed necessary by the joint force commander. Also called **JPG**. (Approved for incorporation into the DoD Dictionary.)

joint planning process. A process to analyze a situation, select the best course of action, and produce a campaign or joint operation plan or order. Also called **JPP**. (Approved for incorporation into the DoD Dictionary.)

L-hour. 1. The specific hour on C-day at which a deployment operation is scheduled to commence. (JP 5-0) 2. In amphibious operations, the time at which the first airwave is scheduled to touch down in a landing zone. (JP 3-02) (Definition #1 sourced to JP 5-0 approved for incorporation into the DoD Dictionary.)

limitation. An action required or prohibited by higher authority, such as a constraint or a restraint, and other restrictions that limit the commander's freedom of action. (Approved for incorporation into the DoD Dictionary.)

line of effort. Links, tasks, and effects to achieve objectives. Also called **LOE**. (Approved for incorporation into the DoD Dictionary.)

line of operation. A line that defines the orientation of the force in relation to the enemy or that connects actions, nodes, or decisive points related in time and space to objectives. Also called **LOO**. (Approved for incorporation into the DoD Dictionary.)

measure of effectiveness. An indicator used to measure change in a current system state, by comparing multiple observations over time. Also called **MOE**. (Approved for incorporation into the DoD Dictionary.)

measure of performance. An indicator to gauge how well an action is executed. Also called **MOP**. (Approved for incorporation into the DoD Dictionary.)

mission statement. A short sentence or paragraph that describes the organization's essential task(s), purpose, and action containing the elements of who, what, when, where, and why. (DoD Dictionary. Source: JP 5-0)

multinational. Between forces or agencies of two or more nations or coalition partners. (Approved for incorporation into the DoD Dictionary.)

objective. A clearly defined and achievable aim toward which an operation is directed. (Approved for incorporation into the DoD Dictionary.)

operational approach. A broad description of the mission, operational concepts, tasks, and actions required to accomplish the mission. (DoD Dictionary. Source: JP 5-0)

operational design. The conception and construction of the framework that underpins planning. (Approved for incorporation into the DoD Dictionary.)

operation assessment. A continuous process that measures the effectiveness of military operations to achieve stated objectives and to determine the progress toward accomplishing a task, creating an effect, changing a condition, or achieving an objective. (Approved for incorporation into the DoD Dictionary.)

operation order. A commander's directive that defines the mission and the subordinate commanders' roles and responsibilities for the operation. Also called **OPORD**. (Approved for incorporation into the DoD Dictionary.)

operation plan. A detailed plan implementing the concept of operations, including all annexes and a time-phased force and deployment list. Also called **OPLAN**. (Approved for incorporation into the DoD Dictionary.)

phase. A stage of a campaign or operation during which a large portion of the forces is involved in similar or mutually supporting activities for a common purpose. (Approved for incorporation into the DoD Dictionary.)

planning order. A directive that provides essential planning guidance and directs the development, adaptation, or refinement of a plan or order. Also called **PLANORD**. (Approved for incorporation into the DoD Dictionary.)

preferred forces. Specific units that are identified to provide assumptions essential for continued planning and assessing the feasibility of a plan. (DoD Dictionary. Source: JP 5-0)

prepare to deploy order. An order to a force provider to have a unit ready and to deploy within a specified response time. Also called **PTDO**. (DoD Dictionary. Source: JP 5-0)

priority integrated plan set. An integrated plan set consisting of a core plan, concurrent plans, and conditional plans that is prioritized by strategic guidance and is of a higher priority than other integrated plan sets. (Approved for inclusion in the DoD Dictionary.)

red team. Personnel who challenge planning assumptions from an adversary's perspective. (Approved for incorporation into the DoD Dictionary.)

restraint. In the context of planning, a requirement placed on the command by a higher command that prohibits an action. (Approved for incorporation into the DoD Dictionary.)

scheme of maneuver. An aspect of the commander's concept for operations covering how arrayed forces will accomplish the mission. (Approved for incorporation into the DoD Dictionary.)

sequel. The subsequent operation or phase based on the possible outcomes of the current operation or phase. (DoD Dictionary. Source: JP 5-0)

shortfall. A lack of forces, reflected as the difference between the resources identified as a plan requirement and those quantities identified as apportioned for planning. (Approved for incorporation into the DoD Dictionary.)

specified task. In the context of planning, a task that is specifically assigned to an organization by its higher headquarters. (DoD Dictionary. Source: JP 5-0)

staff estimate. A continual evaluation of how factors in a staff section's functional area support and impact the planning and execution of the mission. (DoD Dictionary. Source: JP 5-0)

strategic planning framework. A description developed by the Joint Staff for senior leaders to explain the linkages between decisions and actions made across the competition continuum. Also called **SPF**. (Approved for inclusion in the DoD Dictionary.)

time-phased force and deployment data. The time-phased force, non-unit cargo, and personnel data combined with movement data for the operation plan, operation order, or ongoing rotation of forces. Also called **TPFDD**. (DoD Dictionary. Source: JP 5-0)

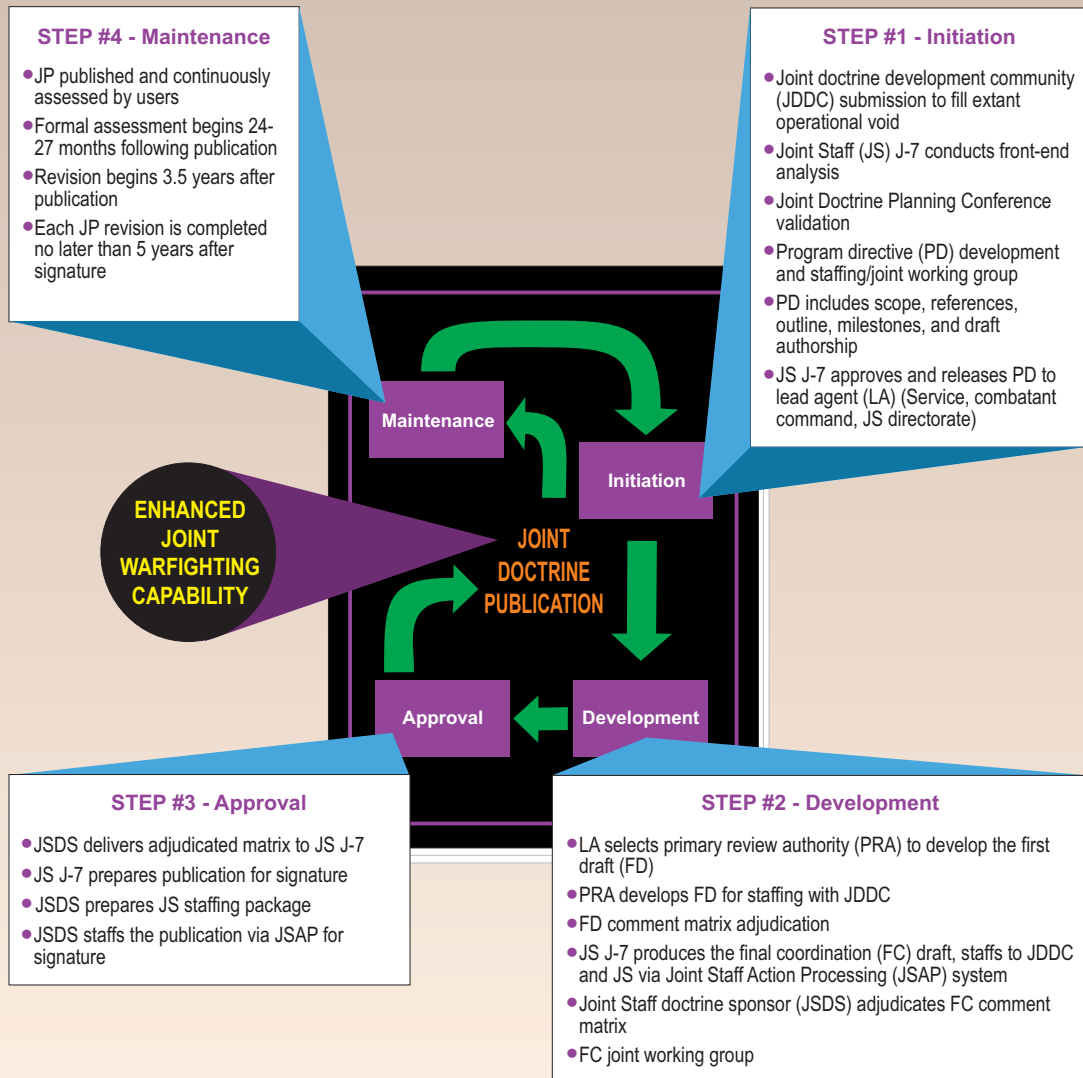
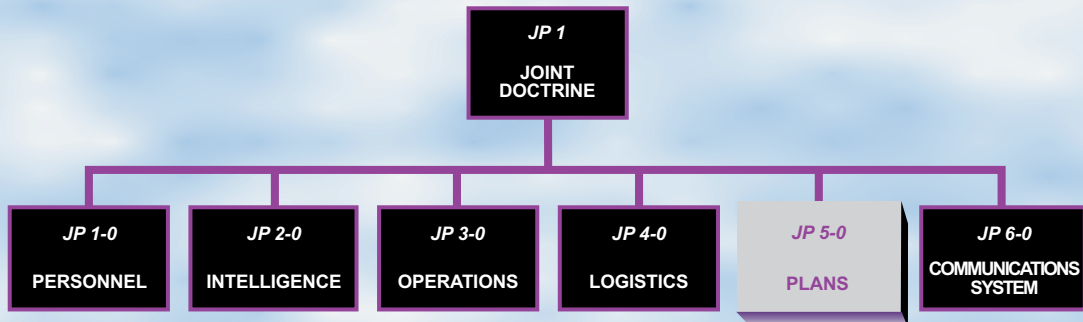
transportation feasible. A determination made by the supported commander that a draft operation plan can be supported with the identified or assumed transportation capabilities. (Approved for incorporation into the DoD Dictionary.)

warning order. A preliminary notice of an order or action, or a planning directive that initiates the development and evaluation of courses of action by a commander. Also called **WARNORD**. (Approved for incorporation into the DoD Dictionary.)

2. Terms Removed from the DoD Dictionary

- **Supersession of JP 5-0, *Joint Planning*, 01 December 2020, Incorporating Change 1, 01 July 2024:** assumption; completeness; course of action; critical capability; critical requirement; critical vulnerability; current force; decision; deployment planning; dispersion; employment; global integration framework; implementation; indicator; joint planning and execution community; Joint Strategic Planning System; leverage; limiting factor; major force; operational characteristics; operational pause; planning factor; strategic direction; strategic estimate; strategic guidance; subordinate campaign plan; supporting plan; times; Universal Time; validate

JOINT DOCTRINE PUBLICATIONS HIERARCHY





Joint Doctrine Note 1-19



Competition Continuum



03 June 2019



Unclassified

PREFACE

1. Scope

This joint doctrine note (JDN) describes the competition continuum; discusses some of its implications for how the joint force campaigns; and describes aspects of campaigning through cooperation, adversarial competition below armed conflict, and armed conflict.

2. Purpose

A JDN is a publication intended to facilitate information sharing on problems and potential solutions as a supporting effort of formal joint doctrine development and revision. This publication addresses potential gaps in joint doctrine identified by the *Joint Concept for Integrated Campaigning* (JCIC). Many JCIC terms and ideas have already been incorporated into joint and Service concepts, training, education, and processes. This publication is the first step in the further refinement of those ideas from the JCIC into doctrine.

3. Application

The guidance in this JDN is not authoritative. If conflicts arise between the contents of this JDN and a joint publication (JP), the JP will take precedence for the activities of joint forces, unless the Chairman of the Joint Chiefs of Staff, in coordination with the other members of the Joint Chiefs of Staff, has provided more current and specific guidance.



DANIEL J. O'DONOHUE
Lieutenant General, USMC
Director, Joint Force Development

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EXECUTIVE SUMMARY

- Introduces the competition continuum.
 - Discusses the competition continuum and integrated campaigning.
 - Outlines campaigning through cooperation, competition below armed conflict, and armed conflict.
 - Discusses the competition continuum and deterrence.
-

The Competition Continuum

Competition is a fundamental aspect of international relations. As states and non-state actors seek to protect and advance their own interests, they continually compete for diplomatic, economic, and strategic advantage.

The joint force, however, employs many constructs and procedures that reflect an artificial distinction between an environment of armed conflict and peace without significant military competition.

For the joint force to play its role in advancing national interests, it must adopt a better framework for understanding, describing, and participating within a competitive operational environment.

Rather than a world either at peace or at war, the **competition continuum** describes a world of enduring competition conducted through a mixture of cooperation, competition below armed conflict, and armed conflict.

Competition Continuum and Integrated Campaigning

Integrated campaigning requires the skillful combination of cooperation, competition below armed conflict, and, when appropriate, armed conflict in conjunction with diplomatic, informational, military, and economic efforts to achieve and sustain strategic objectives.

To accommodate the inevitable shifts in policy over any extended period, joint force commanders

and planners adopt a mindset of **campaigning** rather than of **campaigns**. This is a flexible approach that recognizes joint force activities of all kinds—not just armed conflict—should be continually adapted in response to evolving strategic conditions and policy objectives.

Campaigning Through Cooperation

Campaigning through cooperation is usually an enduring activity with no discrete start or end point; the relationship with the ally or partner is in place and will continue for the foreseeable future. In some cases, however, cooperation in specific areas with a partner whose overall relationship with the US is neutral, or even adversarial, may be necessary. Cooperative activities can take many forms, from security force assistance with a partner in a quiet region to multinational operations and activities in an armed conflict.

Campaigning Through Competition Below Armed Conflict

Competition below armed conflict tends to occur over extended periods of time. In comparison to armed conflict, actions are often more indirect and the expenditure of resources less intense, thus allowing for a more protracted effort. As an inherently constrained and measured approach, it is not generally used by competitors requiring quick results. For the joint force to successfully campaign through competition below armed conflict, it should adopt a similar long-term approach but one supple enough to react to rapid changes in the political, diplomatic, and strategic environment.

Campaigning Through Armed Conflict

One implication of the competition continuum is that it is not solely sufficient to excel in campaigning through armed conflict. Even in an international armed conflict, success requires the skillful application of both cooperation and competition below armed conflict. If these are ignored or treated as strictly ancillary to the armed

conflict effort, then the joint force is at increased risk for failure to meet some or all of the desired objectives. Commanders and staffs should be aware of the interrelated nature of these various elements.

The joint force should also campaign in armed conflict with a long-term view toward the transition period following the end of the main period of armed conflict. Rarely do wars end with a complete end of armed conflict. Wars disrupt political, social, and economic structures, networks, and institutions to the point it is often impossible to simply return to a pre-conflict state.

The Competition Continuum and Deterrence

Deterrence applies across the competition continuum, though in different fashions according to the situation.

Within armed conflict, there are many aspects to deterrence. The joint force seeks to deter a conventional attack against a partner or ally, but even if that occurs, it is still possible to deter the attacker from expanding the war geographically or from using certain forms of weapons.

Deterrence in competition below armed conflict is similarly nuanced and perhaps harder to judge. As with armed conflict, the joint force can deter future actions in competition below armed conflict by effectively responding to current challenges.

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COMPETITION CONTINUUM

"Our traditional way that we differentiate between peace and war is insufficient to [the dynamic of competition below armed conflict]."

"We think of being at peace or war...our adversaries don't think that way."

General Joseph F. Dunford, Jr.
Chairman of the Joint Chiefs of Staff
21 September and 5 October 2016

1. Introduction

a. Competition is a fundamental aspect of international relations. As states and non-state actors seek to protect and advance their own interests, they continually compete for diplomatic, economic, and strategic advantage. The Cold War was a clear example of the many facets of international competition. The US and the Soviet Union competed with each other in numerous ways (e.g., for greater influence in international organizations, for the next great achievement in space, and for more medals won at the Olympics). Rather than engage in direct armed conflict with the other, each state fought through and with proxies as an indirect means to gain advantage. Yet, the two superpowers also cooperated, such as when both backed actions in the United Nations Security Council. And, just as competitors can cooperate, friendly states can compete. Within an alliance, individual countries naturally seek to tilt policy in the direction most advantageous for their long-term interests. Diplomats, trade representatives, and other members of the US Government who routinely interact with foreign actors intuitively recognize that any strategic relationship mixes elements of both competition and cooperation.

b. The joint force, however, employs many constructs and procedures that reflect an artificial distinction between an environment of armed conflict and peace without significant military competition. There are some reasons for this. The Constitution, domestic law, and international law all use the peace/war distinction as a means to regulate the act of employing violence on behalf of the state. Maintaining a sharp distinction between peace and war also has purposes within the Department of Defense. The designation of a state of war (or armed conflict) justifies the expenditure of resources, the imposition of hardship on Service members, and the acceptance of risks far beyond what would normally be acceptable. For an institution as large as the Department of Defense, it is far easier to treat these matters in a few distinct categories rather than a complex sliding scale. Though, the joint force has long been conducting competitive activities outside of armed conflict, the previous operational environment was not so competitive that, in most cases, the legal and institutional advantages of the peace/war binary model outweighed the disadvantages created by its artificial simplicity.

c. The current operational environment requires a more nuanced model. Geopolitical rivals such as Russia and China employ a mixture of instruments of national power to achieve significant strategic advantages in a manner calculated not to trigger our legal or institutional thresholds for armed conflict. For the joint force to play its role in advancing national interests,

it must adopt a better framework for understanding, describing, and participating within a competitive operational environment.

2. The Competition Continuum

a. Rather than a world either at peace or at war, the **competition continuum** describes a world of enduring competition conducted through a mixture of cooperation, competition below armed conflict, and armed conflict. These descriptors refer to the relationship between the US and another strategic actor (state or non-state) in relation to a set of specific policy objectives. This description allows for simultaneous interaction with the same strategic actor at different points along the competition continuum. For instance, the US might be in a state of competition below armed conflict with China in regard to some interests, such as freedom of navigation in disputed areas, and cooperation in others, such as counter-piracy. By providing a lexicon to describe this complexity, the competition continuum facilitates shared understanding, both within the Department of Defense and also with the interagency partners who often have a leading role. This will enable better and more precise communications, planning, and decision making.

b. The **competition continuum** describes, from the joint force perspective, the environment in which we apply the instruments of national power (diplomatic, informational, military, economic) to achieve objectives. In practice, all instruments of

ELEMENTS OF THE COMPETITION CONTINUUM

Armed Conflict. Situations in which joint forces take actions against a strategic actor in pursuit of policy objectives in which law and policy permit the employment of military force in ways commonly employed in declared war or hostilities.

Competition Below Armed Conflict. Situations in which joint forces take actions outside of armed conflict against a strategic actor in pursuit of policy objectives. These actions are typically nonviolent and conducted under greater legal or policy constraints than in armed conflict but can include violent action by the joint force or sponsorship of surrogates or proxies. Competition below armed conflict does not preclude some cooperation in other areas. Competition below armed conflict may include diplomatic and economic activities; political subversion; intelligence and counterintelligence activities; operations in cyberspace; and the information environment, military engagement activities, and other nonviolent activities to achieve mutually incompatible objectives, while seeking to avoid armed conflict. Within competition below armed conflict, joint force actions may include security cooperation activities, military information support activities, freedom of navigation exercises, and other nonviolent military engagement activities. Competition below armed conflict does not preclude armed conflict or cooperation in other areas. Concurrent with competition below armed conflict, potentially hostile actors may engage in forms of indirect armed conflict (e.g., external support of an

indigenous insurgency, counterinsurgency, or resistance movement) through proxies or surrogates that engage each other or the sponsor's adversaries in direct armed conflict.

Cooperation. Situations in which joint forces take actions with another strategic actor in pursuit of policy objectives. Cooperation does not preclude some element of competition or even armed conflict when their objectives are not in complete alignment. Within cooperation, joint force actions may include security cooperation activities, multinational training and exercises, information sharing, personnel exchange programs, and other peaceful military engagement activities. Military cooperation may also occur in the form of multinational operations and activities during an armed conflict or adversarial competition (or competition short of international armed conflict).

Various Sources

national power should function together as an interrelated and integrated whole. For instance, competition below armed conflict might be addressed through a mixture of diplomatic, informational, military, and economic activities.

c. The competition continuum is not a three-part model substitute for the two-part peace/war model. As the vignette demonstrates, cooperation, competition below armed conflict, and armed conflict can occur simultaneously. Because the joint force rarely

DEPARTMENT OF DEFENSE LAW OF WAR MANUAL

- International armed conflict occurs between two or more states.
- Non-international armed conflict occurs between states and non-state armed groups, or between two or more non-state armed groups, even if the conflict is transnational or transregional. Examples include civil wars, rebellions, insurrections, insurgencies, and resistance to military occupation.
- Mixed armed conflict occurs when parts of the conflict are international in character, while other parts are non-international in character. An example would be an armed conflict where a state is simultaneously engaged in hostilities with a rebel or resistance movement and with another state supporting that movement.
- Internal armed conflict occurs within the borders of a single state. It does not include internal disturbances and tensions such as riots, isolated and sporadic acts of violence, and other acts of a similar nature. The intensity of the conflict and the organization of the parties are criteria that distinguish internal armed conflict from internal disturbances and tensions.

HISTORICAL EXAMPLE OF THE COMPETITION CONTINUUM

World War II illustrates that even in an international conflict all three elements of the competition continuum are present. The US was in conflict with the Axis powers. In Europe, the US cooperated with the United Kingdom, France, and the Soviet Union, coordinating operations and providing large amounts of arms and equipment to each. Yet, at the same time, there was also a degree of competition with the Soviet Union. The US and United Kingdom had competed with the Soviet Union to contain the spread of communism prior to the war, and many anticipated a return to competition afterward. In the Pacific theater, the Soviets offered minimal cooperation until the very end of the war, and their late entry was arguably motivated more by a desire to compete with the US in the post-war order than to cooperate in the combined effort to defeat Japan. Even with the United Kingdom and France, differences in visions for the disposition of former colonies led to low-key competition among the allies that influenced the conduct of war. The role of both civilian policymakers and military leaders was to understand these many strategic relationships, weigh their relative importance, tailor military campaigns in accordance with those priorities, and then continually reassess and adjust as the strategic and political situation evolved.

Various Sources

operates unilaterally in any significant campaign, cooperation is a feature of nearly every significant military action. In an interconnected world, there are few circumstances in which a major joint force activity does not have some ramification for competition below armed conflict with at least one of the US's global or regional rivals. For instance, competition below armed conflict with Iran to shape regional security alignments accompanied armed conflict against the Islamic State in Syria and Iraq. Cooperation and competition below armed conflict are always occurring and so the presence or absence of armed conflict is the only variable element. Therefore, the notions of the US being either "in competition" or "in conflict" actually refer to whether the joint force is or is not using armed force as a means to achieve policy objectives. In either case, the joint force will be conducting cooperative activities with partners and competitive activities below armed conflict to counter adversaries who are seeking to turn the competition or conflict to their advantage. The joint force is never solely **in** cooperation (or **in** competition below armed conflict or **in** armed conflict) but instead campaigns **through** a mixture of cooperation, competition below armed conflict, and armed conflict calculated to achieve the desired strategic objectives.

3. Competition Continuum and Integrated Campaigning

a. Integrated campaigning requires the skillful combination of cooperation, competition below armed conflict, and, when appropriate, armed conflict in conjunction with diplomatic, informational, military, and economic efforts to achieve and sustain strategic objectives. This requires a different framework than that employed for tactical operations, particularly in situations where the US is in a long-term competition with a

global or regional power. For instance, it would have been meaningless to try to impose operational phases to the strategic competition during the Cold War, stretching from the end of World War II through the collapse of the Soviet Union. Describing the Berlin Airlift, Greek Civil War, or Cuban Missile Crisis as shaping phase applies an inappropriate operational model to a strategic situation, as if the Cold War were a single major operation. Similarly, it would be misleading to describe Korea, Vietnam, or the Soviet invasion of Afghanistan as a dominate phase. Just as rigid phases associated with definite military end states would not have captured the complexity of the Cold War, such constructs are equally unhelpful in the context of enduring competitions for strategic advantage, which are likely to extend decades into the future. In enduring competition, even when policy objectives are definitively stated at one point, they will evolve with the strategic and political situation. This is also true for shorter periods of a specific armed conflict. The experience of past wars demonstrates that initial policy objectives shift as the conflict rearranges the political and strategic context. To accommodate the inevitable shifts in policy over any extended period, joint force commanders and planners adopt a mindset of **campaigning** rather than of **campaigns**. This is a flexible approach that recognizes joint force activities of all kinds—not just armed conflict—should be continually adapted in response to evolving strategic conditions and policy objectives.

DESCRIPTION OF KEY TERMS

Armed Conflict

Defeat. Create conditions to impose desired strategic objectives upon the adversary.

Deny. Frustrate the strategic objectives of the adversary.

Degrade. Reduce the adversary's ability and will to the greatest extent possible within resource constraints and acceptable risk.

Disrupt. Temporarily interrupt the enemy's activities or the effectiveness of enemy organizations by interdiction, subversion, or coercion.

Competition Below Armed Conflict

Enhance. Achieve strategic objectives, prevent the competitor from achieving incompatible objectives, and improve relative strategic or military advantage without causing an escalation to armed conflict.

Manage. Maintain relative strategic or military advantage to ensure the competitor achieves no further gains; only seek to improve the US advantage when possible with existing resources and in a manner that does not jeopardize interests elsewhere.

Delay. Achieve the best possible strategic objective within given resources or policy constraints, recognizing that this lesser objective entails risk that the competitor will achieve further gains.

Cooperation

Engage Selectively. Cooperate transactionally with a partner (who is often a competitor elsewhere) to achieve a specific objective.

Maintain. Sustain an open-ended cooperative relationship with an ally or partner and secure bilateral advantage but without significant increase in resources or commitment.

Advance. Establish and improve an open-ended cooperative relationship with an ally or partner by significantly increasing resources or commitment.

b. To facilitate shared understanding as strategic objectives change, the competition continuum includes further descriptors to direct the relative intensity with which the joint force competes in a given situation.

c. These expanded elements provide a fuller lexicon that can capture the nuances of strategic prioritization and change. This benefit is of particular use in relation to cooperation and competition below armed conflict but pertains to armed conflict as well. For instance, as shown in the example of World War II (see vignette), the initial Allied approach was to **defeat** Germany while **denying** Japan further gains in the Pacific until more favorable conditions would allow its **defeat**. Such distinctions are particularly important in competition below armed conflict, which, by its very nature, is a mode of campaigning within policy restraints; competition below armed conflict exists when policy dictates that military forces not employ the full array of military capabilities. Thus, identifying the desired scale, intensity, and tempo of the competition below armed conflict is crucial to prioritization of resources and granting of authorities and permissions in globally integrated operations.

d. The competition continuum facilitates dialogue between civilian policymakers and military leaders by providing means to more precisely convey degrees of strategic initiative or restraint. In turn, military advice should clearly articulate when the Armed Forces of the United States cannot provide effective or efficient means to achieve the desired policy objective with given resources and authorities. Just as in armed conflict, something less than defeat of the enemy is more realistic if policymakers do not allocate sufficient resources, authorities, and permissions; so too in competition below armed conflict, the decision to restrain the intensity of competition might cause continued degradation of the US's strategic position. Restraint might be directed to reduce the risk of escalation, to use resources elsewhere, or to preserve limited resources for a major armed conflict. What balance policymakers choose to strike among competing demands will vary, but the competition continuum lexicon facilitates the civil-military dialogue, whatever balance is struck.

e. The competition continuum complements and supports a campaigning mindset. Though policymakers might limit the intensity with which the joint force competes in relation to a specific policy objective, this does not alleviate the responsibility to still

compete to achieve the best possible strategic objective under the circumstances. Indeed, commanders and staffs must be particularly resourceful and imaginative when confronted with severely limited resources and authorities. Inherent within the campaigning mindset is a long-term perspective. In enduring competitions, the joint force does not **win** or **lose** but is in the process of **winning** or **losing**. Every setback contains the possibility of future success, and every victory, the seeds of potential failure. For instance, if a competitor achieves some gain through competition below armed conflict, the joint force must seek ways to turn that to the US's advantage by taking the opportunity to deepen ties with a partner or by exacting some cost so that the short-term success for the competitor actually becomes a long-term loss.

4. Campaigning Through Cooperation

a. Campaigning through cooperation is usually an enduring activity with no discrete start or end point; the relationship with the ally or partner is in place and will continue for the foreseeable future. In some cases, however, cooperation in specific areas with a partner whose overall relationship with the US is neutral, or even adversarial, may be necessary. Cooperative activities can take many forms, from security force assistance with a partner in a quiet region to multinational operations and activities in an armed conflict. The common thread is that campaigning through cooperation is a purposeful activity to achieve or maintain policy objectives.

b. Campaigning through cooperation requires patience, consistency, and empathy. The most productive relationships take time to build. A partnership is unlikely to reach its potential if the joint force approaches military engagement as discrete events rather than as part of a deliberate, continuous process. Commanders and staffs must have an understanding of the environment, a realistic appraisal of the partner's objectives, and the nature of the partner's relationship with the US to derive a range of feasible and productive military options that lead to sustainable and acceptable outcomes for the US and its partner. If done well, the resulting relationships can yield not only immediate tactical or operational benefits but also enduring benefits such as increased commitment of a foreign military to the rule of law or greater willingness to assist US efforts in a crisis. Though the immediate benefits of cooperative relationships are not always apparent, history demonstrates that long-term relationships can pay dividends in unanticipated ways. For instance, the defense relationship with Saudi Arabia paid dividends by creating a foundation for cooperation during Operations DESERT SHIELD and DESERT STORM.

c. Campaigning through cooperation aids competition below armed conflict and armed conflict by integrating the contributions of partners and allies, but cooperation should not be regarded as merely a subordinate activity to those other forms of competition. Particularly in an enduring competition with an adversary seeking to overturn the existing security order, the consolidation of strategic gains can be achieved through cooperation with partners. If, for instance, an adversary is using aggressive air or maritime incursions to intimidate US partners or allies, the joint force might increase its activities and challenge the competitor. If credible, these tactical actions reassure the partner of US capability and intentions, making the partner more likely to deepen cooperation through actions such as

increased intelligence sharing, allowing greater US presence, or more closely integrating its air defenses into a regional network. If campaigning through cooperation creates a more favorable strategic balance in the region, then it has countered the adversary's use of competition below armed conflict. The joint force can combine competition below armed conflict against a competitor and cooperation with partners to turn an adversary's attempted intimidation into an overall gain for the US. Such actions also help deter future efforts by adversaries to destabilize regional security through competition below armed conflict.

5. Campaigning Through Competition Below Armed Conflict

a. Competition below armed conflict tends to occur over extended periods of time. In comparison to armed conflict, actions are often more indirect and the expenditure of resources less intense, thus allowing for a more protracted effort. As an inherently constrained and measured approach, it is not generally used by competitors requiring quick results. For the joint force to successfully campaign through competition below armed conflict, it should adopt a similar long-term approach but one supple enough to react to rapid changes in the political, diplomatic, and strategic environment. These competing demands create challenges for commanders and staffs who work through processes to employ military forces that operate on different, and often inflexible, time horizons—administrative and logistical preparations, the request and receipt of authorities and permissions, the granting of access, coordination with interorganizational partners, and the request and expenditure of funds. Nonetheless, the joint force should set conditions to enable the maximum range of options to accommodate and respond to changing political and diplomatic situations. When properly executed, competition below armed conflict creates strategic opportunities for the US and its partners.

b. The methods employed in competition below armed conflict will vary with the situation, but successful action will feature several characteristics. First, the joint force should begin with the best possible understanding of how relevant actors will perceive action. Second, the joint force and its partners should conduct a broad array of activities: establish access to critical areas, forward position units, establish appropriate and timely presence, organize exercises, share intelligence, prepare the environment for crisis response, and conduct operations in the information environment, to include efforts to counter and undermine the competitor's narrative. Third, the joint force and its partners should ensure the creative and flexible conduct of these activities within a fluid, strategic situation and pervasive information environment. Fourth, the joint force should conduct continual reassessment of the competitor's intentions and capabilities, which will change over time. For these actions to have tangible effects on a competitor's behavior, it is essential to have a deep understanding of competitor perceptions and decision making, as well as the close integration of diplomatic, informational, military, and economic efforts. Finally, in comparison to armed conflict, competition below armed conflict makes use of latent, rather than direct military power.

THE GREAT GAME

The Great Game was a political and diplomatic confrontation that existed for most of the nineteenth century between the British Empire and the Russian Empire over Afghanistan and neighboring territories in Central and Southern Asia. Russia was fearful of British commercial and military inroads into Central Asia, and Britain was fearful of Russia adding "the jewel in the crown," India, to the vast empire that Russia was building in Asia. This resulted in an atmosphere of distrust and the constant threat of war between the two empires.

The Great Game began in 1830 as Britain intended to gain control over the Emirate of Afghanistan and make it a protectorate, and to use the Ottoman Empire, the Persian Empire, the Khanate of Khiva, and the Emirate of Bukhara as buffer states between both empires. This would protect India and also key British sea trade routes by stopping Russia from gaining a port on the Persian Gulf or the Indian Ocean. Russia proposed Afghanistan as the neutral zone. The results included the failed First Anglo-Afghan War of 1838, the First Anglo-Sikh War of 1845, the Second Anglo-Sikh War of 1848, the Second Anglo-Afghan War of 1878, and the annexation of Khiva, Bukhara, and Kokand by Russia. The Great Game ended in 1895 when the border between Afghanistan and the Russian empire was defined.

Various Sources

c. The enduring nature of competition below armed conflict poses unique challenges for consolidation of strategic objectives. Local successes rarely mean the end of the larger competition and few gains are reliably permanent. Instead, the situation is reminiscent of the "Great Game" of the nineteenth century between the British Empire and Russian Empire over control of Afghanistan and neighboring territories, in the sense that each party continually sought to improve its position and guard against the competitor undermining the desired order. In this context, consolidation is an ongoing effort to protect and advance national interests and position the joint force for the next evolution of competition.

d. Competition below armed conflict reflects a choice by the US Government to pursue policy objectives while constraining military actions so they remain short of armed conflict. Though this competition will typically be directed against a strategic competitor that has also resolved to compete below armed conflict, the two antagonists will rarely be equal in willingness to commit resources and accept risk. This policy asymmetry is a defining feature of competition below armed conflict. When the US is more committed than the competitor, the joint force seeks to translate the operational advantage of greater resources and risk acceptance to achieve strategic objectives. When the competitor is willing to exert greater effort than the US, the joint force seeks the best possible outcome under the circumstances.

6. Campaigning Through Armed Conflict

a. Naturally, joint and Service doctrine already address many aspects of campaigning through armed conflict. Yet, one implication of the competition continuum is that it is not

solely sufficient to excel in campaigning through armed conflict. Even in an international armed conflict, success requires the skillful application of both cooperation and competition below armed conflict. If these are ignored or treated as strictly ancillary to the armed conflict effort, then the joint force is at increased risk for failure to meet some or all of the desired objectives. Commanders and staffs should be aware of the interrelated nature of these various elements. For instance, in World War II, even as the Allies were defeating Germany, the conditions for what eventually became the competition with the Soviet Union during the Cold War were being established. At the same time, the depth of the US's post-war cooperative relationships with France, the United Kingdom, and other Western European allies were influenced by the conduct of operations. Operation IRAQI FREEDOM offers another example. While Multinational Force-Iraq was in armed conflict with several groups of insurgents, the manner in which it cooperated with the Iraqi Security Forces, Sons of Iraq, and Kurdish Peshmerga had implications for longer-term political transitions. Moreover, cooperation and armed conflict took place against a background of competition below armed conflict with Iran, which was trying to extend its influence at the expense of the US. In sum, it is neither realistic nor helpful for the joint force to concentrate all of its efforts on armed conflict until "peace," when it can then devote any remaining resources to cooperation and competition below armed conflict. These elements should be regarded as an integrated campaigning effort. In some instances, this might mean adopting what would normally be considered a sub-optimal military course of action in relation to the armed conflict to bolster the US's position within a cooperative relationship with an ally or in relation to a third-party competitor.

b. The joint force should also campaign in armed conflict with a long-term view toward the transition period following the end of the main period of armed conflict. Rarely do wars end with a complete end of armed conflict. Wars disrupt political, social, and economic structures, networks, and institutions to the point it is often impossible to simply return to a pre-conflict state. Destruction of government and societal institutions can create conditions for intense competition among internal, regional, and global actors seeking to retain or gain power, status, or strategic advantage within a new order. Thus, the joint force might be directed to fight in an armed conflict against enemy combatants who become guerrillas, warlords, or criminal militias. Global or regional competitors can then exploit these conditions by supporting these groups as proxies or surrogates to continue the armed conflict. Therefore, the immediate "post-war" period still requires the joint force to campaign through a mixture of cooperation, competition below armed conflict, and armed conflict. Violent, chaotic transitions are made more difficult when significant resources are withdrawn and redeployed because "peace" has returned. Another significant complication is that ongoing military activities may be essential to maintaining an acceptable security situation but may be conducted in support of the activities of other governmental department or interorganizational partners.

7. The Competition Continuum and Deterrence

a. Deterrence applies across the competition continuum, though in different fashions according to the situation. Within armed conflict, there are many aspects to deterrence. The most absolute form is nuclear deterrence in defense of the homeland; if such an attack

occurs, deterrence failed. In other instances, deterrence must be thought of over extended periods of time. The joint force seeks to deter a conventional attack against a partner or ally, but even if that occurs, it is still possible to deter the attacker from expanding the war geographically (such as by attacking US installations elsewhere) or from using certain forms of weapons (such as weapons of mass destruction). Moreover, if deterrence fails because an adversary miscalculates US will or capability, but is then decisively defeated, the one “failure” of deterrence might deter future adversaries from taking similar risks.

b. Deterrence in competition below armed conflict is similarly nuanced and perhaps harder to judge. For instance, if an adversary supports proxy “separatists” in a neighboring country, this cannot be taken as proof that deterrence has failed. The adversary might have preferred to make an overt incursion but concluded the risks were too great. In that case, successful deterrence of armed conflict led to competition below armed conflict. Perhaps the adversary might have employed proxies, but successful intelligence and security cooperation between the US and an ally caused them to conclude such operations would be fruitless, so they confined their actions to propaganda to discredit the US. Once again, the act of competition may in fact be the result of successful deterrence rather than its failure. And, as with armed conflict, the joint force can deter future actions in competition below armed conflict by effectively responding to current challenges. If the use of proxies fails to achieve the desired objectives, the same or other competitors might not resort to similar methods in the future.

c. Deterrence also applies to cooperation, though this is not a desirable outcome. Nonetheless, if partners conclude that working with the US was diplomatically/politically or militarily counterproductive, then securing cooperation from them or others will be more difficult.

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GLOSSARY

(There is no glossary for this joint doctrine note.)

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Foreword

In retrospect, it seems clear that the new era of Great Power competition that is the subject of the chapters in this volume began to take shape almost as soon as the last era had drawn to a close. With the collapse of the Soviet Union and the sudden end of the Cold War, the United States found itself in a position of unchallenged (and seemingly unchallengeable) global preponderance. Surveying the scene in the early 1990s, American policymakers quickly decided to put aside their previous grand strategy of containment in favor of what the George H.W. Bush administration initially described as “collective near-term engagement” aimed at the global “promotion of peace and democracy,” and what William Clinton administration national security adviser Anthony Lake subsequently labeled a policy of “engagement and enlargement.”¹ Henceforth, Lake declared, the goal of U.S. policy would be to encourage the spread of “democracy and market economics” to places where these had not yet taken firm root, most notably across the vast expanse of Eurasia, an area that included China, Russia, the newly independent nations of the former Soviet Union, and the former members of its erstwhile empire in Eastern and Central Europe.²

What Lake and his colleagues had in mind was nothing less than the fulfilment of Woodrow Wilson’s vision for an all-encompassing liberal world order, an international system made up of states bound together by free trade, international rules and institutions, and a shared commitment to the principles of democratic governance and universal human rights. This was the third time in the course of the 20th century that American policymakers had sought to remake the world along liberal lines. Wilson’s first attempt, at the close of World War I, had ended in failure. Twenty-five years later his successors would try again, only to find their path blocked by the descent of the Iron Curtain and the start of the Cold War. In place of a truly global liberal order, in the wake of World War II, U.S. policymakers had to settle for a partial, geographically limited subsystem that ultimately came to include the advanced industrial democracies of Western Europe, East Asia, and the Western Hemisphere. Despite its constrained scope, this collection of nations (often referred to loosely and somewhat inaccurately as the “West”) proved to be enormously successful in generating both wealth and power. Over a 40-year period of intense and sometimes dangerous rivalry, its members were able to out innovate, outproduce, and ultimately outlast their Communist competitors.

With the Cold War over, American policymakers hoped, in effect, to extend the boundaries of the Western system, expanding a partial order operating on liberal principles to encompass the entire globe, including nations that had previously chosen to remain outside its limits. Over the course of the 1990s, the United States and its European allies were able to induce the smaller and weaker nations of the former Soviet empire to reshape their economic and political systems along liberal lines by making reform a requirement for full

membership in the Western system. Lacking equivalent leverage with Russia and China, the democracies chose to take the opposite tack: incorporating these major powers as fully as possible into existing institutions (and, in particular, into the international economy) in the hopes that doing so would, in itself, promote liberalizing reforms. It is the failure of this approach, evidence of which has been accumulating for at least the past decade, that set the stage for the new era of Great Power competition.

Traveling by different routes, in the 30 years since the end of the Cold War, Russia and China have ended up in broadly similar positions. The regimes that govern these two powers have managed to integrate into the global economy, enjoying the benefits of international trade and investment without evolving into true market-based economies or surrendering their grip on domestic political power. Indeed, to the contrary, since the turn of the century both the Russian and Chinese states have become more repressive and more militantly nationalistic, tightening their grip over society and the economy at home, while engaging in increasingly aggressive behavior on the international stage.

Vladimir Putin's Russia and Xi Jinping's China are driven by a mix of resentment, insecurity, and ambition.³ Both seek redress for what they regard as past wrongs and humiliations; both feel themselves threatened by the physical proximity of the United States and its democratic allies and by the persistence of an international system built on principles that challenge the legitimacy of their own illiberal regimes. Albeit to varying degrees, both Russia and China are revisionist powers; each aims to alter the status quo in its immediate neighborhood, pushing back what Chinese propagandists describe as "hostile foreign forces" and establishing a zone of effective control along its periphery. As its strength and self-confidence have grown, Beijing has also begun to reveal broader ambitions. In ways that are not yet fully specified, it intends, in Xi's words, to "move closer to the center of the world stage," reshaping existing international rules, norms, and institutions to better reflect its power and serve its interests.⁴

As the authors of several of the chapters in this volume point out, in terms of their relative national power China and Russia appear to be following very different trajectories. In the long run, the former will likely be able to mount a far more serious challenge to the United States and its allies than the latter. Notwithstanding this divergence, there are still some notable similarities in the tactics and techniques that the two nations are currently employing in pursuit of their objectives. Both are using surveillance, censorship, and nationalist propaganda to harden their societies against what they see as the deadly subversive threat of liberal ideas and influences. Both have developed forces and concepts of operation suitable for ambiguous, low-level aggression in the geographic and strategic "gray zone." And both are strengthening their cyber warfare, antisatellite, nuclear, and conventional antiaccess/area-denial capabilities in hopes of deterring the United States from intervening in future large-scale conflicts along their peripheries.

Developments in communications technology, and the vestiges of the failed policies of "enlargement" and "engagement," have left the economies, societies, and political systems of the Western nations open to penetration and manipulation, vulnerabilities that their authoritarian rivals have been quick to discover and exploit. Thanks to the lingering after-effects of the 2008 financial crisis and now the COVID-19 pandemic, the democracies are also weaker and more divided at present than they were throughout most of the Cold War.

For these reasons, at least in its opening stages, the new era of Great Power competition could prove to be even more challenging than the one that preceded it. Success will require creative and wide-ranging thinking of the sort that the Institute for National Strategic Studies was designed to promote, and which the following pages contain.

—AARON L. FRIEDBERG

Princeton University

July 2020

Notes

¹ For a review of the post–Cold War evolution of U.S. grand strategy, see the series of National Security Strategy documents published between August 1991 and July 1994, available at <<https://history.defense.gov/Portals/70/Documents/nss/nss1991.pdf?ver=2014-06-25-121158-580>>; <<https://history.defense.gov/Portals/70/Documents/nss/nss1991.pdf?ver=2014-06-25-121158-580>>; <<https://history.defense.gov/Portals/70/Documents/nss/nss1993.pdf?ver=2014-06-25-121210-297>>; and <<https://history.defense.gov/Portals/70/Documents/nss/nss1994.pdf?ver=2014-06-25-121219-500>>.

² See Anthony Lake, “From Containment to Enlargement,” Johns Hopkins University, School of Advanced International Studies, September 21, 1993, available at <<https://www.mtholyoke.edu/acad/intrel/lakedoc.html>>.

³ This section draws on the analysis in Aaron L. Friedberg, *The Authoritarian Challenge: China, Russia and the Threat to the Liberal International Order* (Tokyo: Sasakawa Peace Foundation, 2017), available at <https://www.spf.org/_jpus-j_media/img/investigation/The_Authoritarian_Challenge.pdf>.

⁴ Simon Denyer, “Move over, America. China Is ‘Blazing a New Trail’ for the World,” *Washington Post*, October 19, 2017. For more on what China’s leaders may have in mind, see Nadège Rolland, *China’s Vision for a New World Order*, NBR Special Report #83 (Seattle: National Bureau of Asian Research, January 2020), available at <https://www.nbr.org/wp-content/uploads/pdfs/publications/sr83_chinasvision_jan2020.pdf>.

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Major Findings on Contemporary Great Power Competition

This strategic assessment is both firmly focused on the dynamics of contemporary Great Power competition (GPC) and respectful of past strategic assessments generated by the Institute for National Strategic Studies (INSS) over the course of almost 40 years. As an homage to the format of several historical INSS strategic assessments, this one begins with a summary of major findings within the current volume.

The chapters that follow provide multiple insights and analytical conclusions about contemporary GPC. This prelude provides many of the most significant and substantive findings and conclusions found within them. The findings are provided with an explicit reference to the book chapters where they are found. Readers are encouraged to consult the referenced chapters for deeper analysis and insight into these major research conclusions about, and considerations for, a new era of GPC.

Before perusing these major findings, readers may be entertained by three key findings extracted from the INSS *Strategic Assessment 1998: Engaging Power for Peace*. These findings underscore the degree to which the world of a mere 20 years ago was breathtakingly different from the one today featuring an emerging new era of GPC:

- The United States now enjoys a secure and promising position in the world. . . . The other most successful nations are its closest friends; its few enemies are comparatively weak, isolated, and swimming against the current of . . . globalization [which] is both integrating and extending the core of free-market democracies, thus favoring U.S. interests and winning converts to the norms of state behavior.
- Great uncertainties still exist: the future of China and other large transition states. . . . Because of its capabilities, the United States has considerable influence, and a crucial stake, in how these uncertainties are resolved.
- In the best plausible case, an expanded core or commonwealth of peaceful democracies could encompass most of the planet—with U.S. partners shouldering an increased share of the burden of defending common interests and norms. China would reform and integrate into the core. . . . In the worst case, U.S. friends could be free riders instead of responsible partners, China's reforms would founder . . . leaving the United States superior but beleaguered.¹

The major findings from *Strategic Assessment 2020: Into a New Era of Great Power Competition* follow:

- A Great Power displays three conspicuous attributes: *capabilities*, *behavior*, and *status attribution* by other states in the international system. It has unusual capabilities in comparison with other states. It uses those unusual capabilities to pursue broad foreign policy interests beyond its immediate neighborhood. It is perceived by other states as powerful, having influence, and is thus treated accordingly. In the dawning era of new GPC, the United States, China, and Russia fit this description (chapter 1).
- Competition is not synonymous with conflict; competition exists on a continuum of interactions between and among states. On one end of the spectrum is cooperation, and on the other is direct armed conflict. In between, states compete in varying states of collaboration and confrontation. They edge toward cooperation and collaboration when geopolitical goals are aligned. They drift toward confrontation and armed conflict when main geopolitical aims are perceived as divergent and mutually unattainable (chapter 1).
- Power has absolute, relative, and transitional properties. State power exists in two major dimensions: hard power (or the *coercive* use of military power and leveraging economic power as a *payoff*) and soft power (which includes cooperative and collaborative interactions that attain influence by *attraction*: cooperative economic arrangements, ideological appeal, cultural and social engagements, diplomatic acumen, and reciprocal information exchanges). *Smart power* is sometimes today used to describe policy choices that effectively mix coercive hard power and the attractive features of soft power. *Sharp power* has become a vogue phrase to describe state actions that twist soft power attributes in a manipulative or confrontational manner to undermine or severely distort the political system or social order of a competitor state (chapter 1).

Part I. Conceptualizing the New Era of Great Power Competition

- All states, especially Great Powers, compete to gain relative advantage in the classic objectives of power, prosperity, status, and influence. More critically, Great Powers contend for these relative advantages in five distinct categories of interaction: political and diplomatic, ideological, informational, military, and economic. States apply power capabilities (for example, foreign policy tools) in these categories. Effective state use of power capabilities establishes the degree to which they attain relative influence and secure strategic advantage. A complete analysis of GPC status requires a review of the comparative aims and relative power capabilities of each Great Power in these five competitive categories (chapter 2).
- As Great Powers compete, these particular states inevitably confront the dilemma of transition in relative power status. Great Power transition challenges rising states with the dilemma of how to assert their relative power gains without provoking outright clash with the dominant state(s). Transition also confronts the dominant but relatively declining state with a vexing choice of how to accommodate its rising challenger(s) in a manner that avoids both destructive military clash and an unacceptable change in its preferred status quo (chapter 2).

- Great Power transitions play out over decades or centuries, not years. Three-quarters of Great Power transitions since 1500 have culminated with—or featured during—a destructive period of direct violent clash. The inevitability of war between Great Powers during times of transition is not foreordained; Great Powers may channel or expend their worst animus in nonviolent categories of competition: politico-diplomatic, economic, ideological, and informational (chapter 2).
- Once a Great Power competition is under way, the most reliable indicator of when a war will erupt is when one or both sides recognizes a shift in the relative alignment of economic and military power that is perceived as immutable and untenable. As states view the relative power alignment moving decisively against them, they are much more inclined to risk a preemptive conflict than when they perceive a stable power status quo (chapter 2).
- Although incompatible ideologies and caustic informational exchanges about a Great Power rival's people are not a lone determinant of when Great Power rivalry will devolve into direct violent clash, they are strong yet lagging indicators of insurmountable contentiousness. Great Power leaders should appreciate the degree to which blanket invective of a rival's entire population differs from criticism of a competitor's political leadership. The latter, circumspect approach to official criticism—a feature of the peacefully resolved dyadic competitions between Great Britain and the United States and later the United States and the Soviet Union—was correlated with avoidance of war between those Great Power competitors (chapter 2).
- During power transition periods, Great Power competitors may not perceive their own various forms of power accurately. Too often, misperceptions of relative power, rather than detailed and empirical assessments of power, inform and then drive policymakers. Even when accurate assessments of relative decline or vulnerability are made, domestic or bureaucratic interests may retard agile adaptation necessary to mitigate risks. Thus, success in GPC requires extraordinary political leadership in both international statecraft and in generating domestic renewal and adaptation (chapter 2).
- During periods of dynamic technological change, the likelihood of strategic surprise or operational obsolescence is greater in the military dimension of GPC. States may overestimate or underestimate the potential combat power of new innovations, whether they are technological or conceptual. The dawning era of GPC is in just such an era—one featuring a fourth industrial revolution. The convergence of new technologies, including artificial intelligence, quantum computing, robotics, 3D printing, nanotechnology, biotechnology, energy storage, and autonomous vehicles, among other breakthroughs, increases the risks of strategic surprise (chapters 2, 4).
- The United States enters the emerging era of GPC as the dominant of the three rivals. Its preferred norms, rules, and institutions for interstate interactions today set the patterns for all major categories of global activity. The emerging strategic aims of China and Russia are incompatible with those established by United States, and this conflict has produced the return of a historically dominant pattern of GPC in the international system. But the strategic challenges posed by China and Russia diverge significantly, portending a long-term Sino-American strategic competition,

while the U.S.-Russian rivalry is more likely to be a more regional set of contestations (chapter 3a).

- China is the most important—albeit presently less threatening—Great Power challenger to American power and policy interests. China is the lone contemporary rising Great Power with the combination of a positivist strategic vision for the future and the ambition to push for changes in the international system on near- and long-term bases. Moreover, gross power indicators in 2020, and projections for the next 5 to 10 years, clearly indicate that China is the Great Power best poised to displace America from its long-dominant position. While a net power comparison between the United States and China indicates that their power transition timelines are longer than some now fear, the Sino-American competitive dyad is likely to be the dominant Great Power rivalry into the future (chapters 3a, 3b).
- Vladimir Putin's Russia is an urgent but transient security risk for the United States and China, with the potential to do enormous military damage to the world if miscalculation leads to military clash. But Russia is a Great Power competitor without any positivist, global strategy or discernable norms, institutions, and procedures for establishing an alternative international order. Instead, it practices a reactive, disruptive strategy aimed to pacify its immediate borders and question contemporary institutions and processes it perceives as a threat. Putin's Russia has generated limited power factors that align well with the short-term, geographically limited strategy it is pursuing, thus making its long-term status as a Great Power questionable (chapters 3a, 3b).
- China and Russia may continue a tactical entente over the coming 5 to 10 years, working together on common near-term strategic interests to erode U.S. power, frustrate U.S. actions, challenge U.S.-dominated institutions, and question U.S.-underwritten norms and rules these states deem threatening. However, divergent long-term Sino-Russian strategic interests make it unlikely they will form a long-term alliance. The United States should remain careful not to misunderstand tactical coordination between Beijing and Moscow that balances U.S. power as evidence of some deeper strategic cooperation (chapter 3a).
- All three contemporary Great Powers are dissatisfied with some aspects of international order and are growing less willing to make compromises and sacrifices to keep the order working. Thus, there is heightened potential for Great Power rivalry to reduce the effectiveness of global institutions in managing complex regional and global problems. The absence of Great Power cooperation to confront the 2019–2020 novel coronavirus (COVID-19) pandemic was symptomatic of this breakdown (chapter 3a).
- Over the next 5 to 10 years, U.S. economic and strategic interests in the Indo-Pacific region and in Eastern Europe most conflict with Chinese and Russian regional interests. Thus, the competition will be fiercest, and risks of misperception and violent confrontation greatest, there (chapter 3a).
- Space and cyberspace are interrelated contested domains where GPC is increasingly on display, and inclining toward direct confrontation and clash. These relatively unregulated areas also present an opportunity for Great Power dialogue and a

chance to craft norms and rules emphasizing greater deconflation of interests and reduction in the risks from unbridled competition (chapter 3a).

- The United States, China, and Russia face major internal structural, economic, and demographic challenges. Political leadership's decisions in each state about how to address these internal dynamics as well as international challenges will determine the future power each will possess and the future policy options each might pursue. Russia appears most likely to confront these challenges first, then China, and then the United States, although national leadership choices will greatly impact timing (chapter 3b).
- The crucible of emerging technologies that make up the fourth industrial revolution is today changing the manner in which products have been made, distributed, and used internationally over the past 40 years. The fourth industrial revolution is fueling deglobalization by eroding many aspects of global markets and supply chains, most notably moving product manufacturing closer to natural markets. This movement will mean less global economic integration and greater supply chain regionalization (and even localization) as the world moves into this era of GPC (chapter 4).
- The United States has distinct advantages over both China and Russia as the fourth industrial revolution begins to reshape the world. The one key American weakness is the gridlock in our current political systems. Failure to adjust American laws and regulations to fourth industrial revolution realities risks squandering nascent American advantages in higher education, innate innovation, entrepreneurial spirit, and the largest natural market in the world (chapter 4).
- China also could benefit greatly from the fourth industrial revolution. It is heavily subsidizing priority high-technology manufacturing sectors as part of its Made in China 2025 plan. Simultaneously, it is shifting its economy from export-based growth to domestic consumption as an economic engine. However, China must also deal with a looming dramatic reduction in labor demand and associated unemployment caused by the fourth industrial revolution while addressing the social and economic impact from a rapidly aging and less productive work force. The Chinese Communist Party (CCP) must find a way to manage these challenges to continue its decades-long economic rise into the era of GPC (chapter 4).
- Russia, in contrast, is not well positioned for the fourth industrial revolution. It suffers from a combination of low investor confidence, a poor innovative culture, a low-quality workforce and education system, and a rapidly aging population. As a kleptocracy, Putin's Russia does a poor job in allocating capital to the industries and properties most likely to benefit from the convergence of critical fourth industrial revolution technologies. These multifaceted economic challenges reinforce the tenuous position of Russia as a durable Great Power (chapter 4).

Part II. Warfighting, Innovation, and Technology in a New Era of Great Power Competition

- The technologies of the fourth industrial revolution will not change the fundamental nature of war or bring clarity about its imminence or brevity to its conduct. War will remain the domain of fog, friction, and uncertainty. Each society will use emerging technologies in unique ways that are best suited to it, and any conflict will evolve based on the reciprocal and dynamic interaction of all societies involved (chapter 5).
- The fourth industrial revolution is reducing the price of precision and advanced manufacturing, creating a new generation of smaller, smarter, and cheaper weapons. Yet the United States, China, and Russia continue to pursue exquisite, high-end systems such as fifth-generation fighters, heavy bombers, and aircraft carriers. As manufacturing continues to rapidly change in the era of GPC, a key question appears: Which nation can most rapidly and effectively adapt to this revolution (chapter 5)?
- In the conventional military arena, the revolution of small, smart, and cheap favors the United States over China or Russia. Operationally and tactically, the United States is on the defensive in both Eastern Europe and Asia. In the Indo-Pacific region, the United States could move from easily targeted bases and platforms to multiple locations and mobile systems that could disperse through the First Island Chain, denying China tactical military advantage for at least some time. In Europe, if the United States and its Allies in the North Atlantic Treaty Organization (NATO) are willing to equip frontline states with a mix of inexpensive drones and improvised explosive devices, and invest in autonomous drones and cruise missiles, they could deter and defend in depth against the most challenging Russian security threats (chapter 5).
- The foundation of modern Great Power wealth and competitive advantage has essentially changed from one dominated by industrial era technology to one in which information technology (IT) has become the source of geopolitical power. This change has been affecting the balance of global power in favor of China for over a decade, and is about to enter a dramatic new phase (chapter 6).
- Where China and Russia are concerned, information power is more likely than industrial power to determine the outcomes of long-term geopolitical contests. Indeed, no amount of American investment in industrial era technology could do much to defend against the damage being done by autocratic states' political-psychological operations or help the United States respond with information operations that uphold American values and characteristics (chapter 6).
- To compete in the critical IT arena, the United States must work with other developed nations, and in public-private partnerships, to reprioritize resources into key information technologies and capabilities. Simultaneously, America and its partners must effectively counter China's ability to steal intellectual property and Beijing's quest to control global information flows (chapter 6).
- Russia and China have been increasingly waging foreign propaganda campaigns on social media platforms, including Facebook and Twitter. These campaigns are enticing because they are cheap, easy to execute, allow targeting of specifically

refined audiences, and provide anonymity that limits the risks of attribution or reprisal (chapter 7).

- Russia has proved adept at using social media and other online channels of international influence and global propaganda in the new era of GPC. Multiple Russian campaigns since 2013 have been skillful and disruptive in many other states, but it remains unclear that they have met particular strategic goals or changed audience attitudes, behaviors, or beliefs in ways that Moscow intended. Nonetheless, the confusion sewn in these repetitive campaigns makes it likely Russia will value them in the future. Moscow appears prepared to adapt and persist in the face of any emerging countermeasures (chapter 7).
- China has gained a strong reputation for effectively stifling and influencing online debate within its borders, effectively censoring illicit content on the Web, and shaping online conversations. But it has struggled to weaponize social media or online tools to influence international policy and popular opinion abroad. There can be no doubt that Beijing will learn lessons and generate new projects to build international social media presence, leveraging its large state-generated IT investments to generate better scale and scope for international social media influence (chapter 7).
- To succeed in the GPC for influence through propaganda in social media, the United States must pursue broad, agile approaches to limiting the threat—featuring public and private domestic cooperation and close international collaboration. First, it must track, highlight, and block adversarial social media content. Then it must build resilience in at-risk populations at home and in allied states targeted by adversaries. Finally, it must better organize government to counter adversary propaganda (chapter 7).
- Weapons of mass destruction (WMD) remain a critical feature and potentially dynamic factor in GPC. For Russia and China, WMD contribute to multiple goals: conflict deterrence at the strategic and regional levels, regime survival, coercion of rival states, and, potentially, as an adjunct to conventional forces to support operations. The erosion of longstanding arms control treaties and nonproliferation norms increases the risks of arms races and the use of WMD in conflict. Ongoing advances in the technologies that underpin WMD could lead to the emergence of novel threats with uncertain consequences for GPC (chapter 8).
- The risk of an arms race in nuclear weapons, delivery systems, and missile defenses is growing as Great Power relations become more competitive and even confrontational. Systems of arms control treaties, which for decades limited U.S. and Russian nuclear forces, are under great strain and could even collapse. China is investing more in nuclear capabilities, modernizing and expanding strategic systems, and developing dual-capable theater-range platforms that would heighten nuclear risks in Indo-Pacific conflicts. And while the introduction of hypersonic vehicles by the Great Powers is unlikely to affect the balance of nuclear power in the next few years, as these capabilities are deployed in larger numbers, the risks to nuclear stability at both the strategic and theater levels are likely to grow (chapter 8).
- China and Russia may perceive chemical and biological warfare agents, including agents developed through new scientific and manufacturing techniques, as important

capabilities for a range of operations against the United States and its allies. Chemical or biological attacks could be difficult to attribute and may be well suited to support Russian and Chinese objectives in operations below the threshold of open armed conflict (chapter 8).

Part III. Geostrategic Interactions in a New Era of Great Power Competition

- U.S. and Chinese strategic interests are less aligned in the Indo-Pacific region than anywhere else in the world, and the importance of those interests to both countries makes the region a central venue for GPC. The U.S. Free and Open Indo-Pacific vision is not compatible with China's aspirations for increasing control within its First Island Chain and wider Chinese regional aims sometimes espoused as a "community of common destiny" (chapters 3a, 9).
- China has economic dominance in markets and investment across most of the region. It also has eroded the U.S. military advantage in potential locations of confrontation near its shores and inside the First Island Chain. The United States retains an overall advantage in military technology and power projection across the wider Indo-Pacific region, a resonant ideology and ability to communicate it, along with a regional political and military alliance structure unmatched by China (chapter 9).
- China's superior ability to use market access and other economic tools to provide benefits means that Indo-Pacific countries will not give up their economic ties with Beijing, even if Washington attempts to decouple from the Chinese economy. But Washington needs to be actively involved in regional economic affairs, both to advance specific U.S. economic interests and to shape rules and norms in the most dynamic region in the world. A policy that combines engagement with China and attention to nurturing a balance of economic power around Beijing as a hedge would best serve U.S. interests (chapter 9).
- The United States should build on its relative political-military advantages to sustain and strengthen its overall Indo-Pacific security position. Reinforcing present alliances, building military partnerships, extending cooperative training, and expanding interoperability are techniques that states in the region would embrace and which would work against unilateral Chinese efforts to intimidate or erode the U.S. alliance system (chapter 9).
- As long as American society models and promotes open, transparent, and democratic institutions, the United States likely will appear as an ideological and even existential threat to CCP leaders. But strong and consistent messaging with Indo-Pacific allies and partners could send a positive signal to the Chinese people about the value of good, representational governance and provide other states around the region a positive alternative framework that contrasts with China's authoritarian model (chapter 9).
- Although best understood as a contemporary Great Power, Russia could be alternatively considered to be a rogue, disruptor, or spoiler state, such as Iran and the Democratic Republic of Korea. These are countries that lack the military and

long-term economic power and/or transnational cultural appeal to match U.S. power globally or stabilize an alternative international political order. They are motivated by a combination of regime survival, aspirations for regional dominance and sometimes global relevance, as well as an inclination to confront the United States, which they believe is the main obstacle to their own aspirations (chapter 10).

- Rogue states tend to confront the United States below the threshold of active armed conflict and across multiple domains in the contemporary era of GPC. As they do, these states' actions divert American attention and resources away from longer term objectives, thus impeding the United States and benefiting China. However, Russian, Iranian, and North Korean provocative behavior is not uniformly beneficial to China (chapter 10).
- The prospect of a robust and fully cooperative anti-U.S. rogue state axis in the early 2020s remains remote. While U.S.-Chinese competition will yield limited prospects for burden-sharing between Beijing and Washington to confront Russian, Iranian, or North Korean conduct harmful to the United States, China also must fear negative spillover from such conduct onto its own economic interests and strategic aims. The United States thus can expect a mixture of cooperative and obstructive responses from China when addressing these actors on a case-by-case basis in the new era of GPC (chapter 10).
- Terrorism is far from eradicated and will not go away in the emerging era of GPC. Instead, American counterterrorism efforts will confront a set of new realities. Recent American counterterrorism operations in Syria likely will be the model of the future. The U.S. Government should reconsider counterterrorism authorities, new technologies, and other tools that could help manage the risks from small-footprint deployments—especially those with active proxies—and that hold sponsor states accountable for actions by proxies against U.S. counterterrorism forces (chapter 11).
- Russia must be expected to undermine U.S. counterterrorism objectives, either directly or indirectly. Moscow will likely try to destabilize U.S. objectives by fomenting right-wing and other homegrown violent extremists indirectly through media campaigns. Russia also will confront U.S. forces, especially in the Middle East and North Africa, as the forces attempt to mitigate threats to the U.S. homeland. As in Syria, Russia will combine diplomatic initiatives, proxy warfare, and electronic warfare to foil U.S. military dominance (chapter 11).
- Regional states will continue to pursue their own counterterrorism objectives. Sometimes they will deploy their forces in a manner that the United States finds unacceptable. Sometimes they will utilize proxy forces in a destabilizing manner. In other instances, important regional states will have objectives, ways, and means that align with U.S. goals for countering violent extremist organizations. The best way to mitigate the risk of regional states acting in an unruly manner is to be involved—even to a minimal degree if necessary—and then leverage American influence with them (chapter 11).
- In the dawning age of GPC, Europe is a resilient but troubled region. Europe has shown considerable hardiness while overcoming the 2008 Great Recession and saving the euro in the face of a serious sovereign debt crisis. But its cohesion and

solidarity also have been severely tested by terrorism, uncontrolled migration, Brexit, and most recently, the still-evolving complications from the COVID-19 pandemic. These problems have generated extremist populist movements across the continent that challenge liberal democracy and inhibit cohesive European policy positions or security activities (chapter 12).

- Despite a share of global economic output comparable to that of the United States in 2020, Europe is not a Great Power. Europe finds itself an object of Great Power rivalry on the continent rather than a subject competitor itself. In 2020, it confronts a more aggressive Russia, growing Chinese power, and reduced trust in the longstanding U.S. commitment to Europe's security and the wider construct of transatlanticism (chapter 12).
- European cohesion and stability has long been a function of both American support and a collaborative Franco-German core. In 2020, that core is weak, as Germany and France lack common positions on many critical issues, including on European defense. As Brexit moves the United Kingdom out of the European Union, much about Europe's way forward will be decided in Washington and Moscow. Europeans worry that the United States may detach itself from Europe—particularly NATO—even as Europe today remains unable to create an autonomous system of security and defense. While Europeans mistrust Russia generally, their perception of Russia as a security threat varies greatly. All know that Europe cannot alone defend member states from Russia. Should America move to detach from NATO, Europe may intensify accommodation with Russia—and even with China—believing this move to be the least-worst path to the evolving competition among the Great Powers (chapter 12).
- The era of GPC will confront U.S. policymakers both with the challenge of how to shift greater resources and attention toward Russian and Chinese traditional spheres of influence—in the Indo-Pacific region and Europe—as well as the challenge of whether and how to compete with Moscow and Beijing on a global scale. Washington will require distinct strategies for competing with Russia and China, a recalibration of U.S. interests across the world, and a discerning approach that reduces the prospects of pulling U.S. regional partners into an unrestricted, zero-sum competition (chapter 13).
- Russia and China present distinct competitive threats to the United States around the globe. China's behavior is grounded in its global investment strategy and desire to shape an international political order more conducive to Chinese interests, even if not fully Sino-centric. Russia's desire to be a global Great Power is not grounded in a proactive vision for a new global geopolitical order (chapters 3a, 13).
- In many regions, Russia often poses the more immediate challenge, whereas the repercussions of Chinese economic investments manifest themselves subtly and will likely undermine U.S. strategic interests more gradually. Both are only nominally united in their desires to compete with and displace U.S. influence across Latin America, the Middle East, Africa, and the Arctic (chapter 13).
- States in Latin America, the Middle East, Africa, and the Arctic are often eager recipients of Russian and Chinese attention and resources due to convenience rather

than ideological commitment. With few exceptions, these countries accept support without any allegiance to China's global vision or Russia's cynicism vis-à-vis Western norms and institutions. The limitations to Russian and Chinese approaches—as well as the transactional hedging strategies of many of these smaller states—should induce caution and prudence in Washington (chapter 13).

- Few, if any, smaller states wish to be pulled into a zero-sum U.S.-China or U.S.-Russia GPC. Thus, the United States should avoid imposing regional strategies that view Russian or Chinese activities as uniformly harmful to U.S. interests and detrimental to the stability of recipient states across these regions. Instead, Washington should emphasize American strengths as an economic partner, the quality and quantity of its military assistance, and the positive and benign nature of its military forward presence (chapter 13).

Part IV. Preparing to Compete

- Great Power competitions and accompanying power transitions are rarely resolved without a holistic approach, managed within an appropriate strategic framework. They require leadership involvement, disciplined priorities, sustainable resourcing, and adaptive oversight. In 2020, Russia is a dangerous competitor in the near term, but the U.S.-China competitive dyad is the one that will determine the prospects for continued global stability and the contours of any geopolitical Great Power transition (chapters 3a, 3b, 14).
- After more than two decades of geopolitical dominance featuring mainly cooperative interactions and relative comfort, the United States must acquire a competitive mindset. A competitive American mindset must understand that while interstate cooperation remains feasible today in areas of shared mutual interests, competitive tensions can occur in formerly cooperative political and economic categories so long as they are contested within established bounds. The challenge is to expand on the potential for cooperation while carefully managing the competition to keep it short of armed conflict, all without compromising vital national interests (chapter 14).
- In the Sino-American competition, the United States cannot merely accommodate China's rising power by acquiescing to its ambitions. An effective U.S. strategy must create leverage and accept risk. It must create leverage by working with allies and partners to strengthen rules and norms, set standards, collaborate on industrial policy, cooperate on critical information technologies, rejuvenate research and development, enhance innovation, invest in higher education, and share best practices. It must accept risk, while standing with these partners, to counter aggressive Chinese behaviors, violations of standing rules and norms, or the suborning of human rights. These principles frame a strategy of Strategic Balancing for America's role vis-à-vis China in the new era of GPC (chapter 14).
- The COVID-19 pandemic is far less likely to change the basic trajectory of emerging GPC than to accelerate it and expose underlying dynamics. The three contemporary Great Powers are likely to remain dominant rivals through at least the middle of

the dawning decade. If any of the three might falter, Russia seems the most likely candidate, but the odds of that are long. China and the United States may joust over whose model best dealt with COVID-19 crisis, but the basic outlines of their strategic disagreements will remain. Moreover, the phenomena of deglobalization and partial economic decoupling seems most likely to continue (chapter 15).

- As the COVID-19 pandemic recedes, Washington will face the same choice that it confronted before: whether, where, and how to compete. As the dominant Great Power in a multipolar competition, America could contest or confront its Great Power rivals today with a resolve to sustain its global position and the standing rules, norms, institutions, and alliances of the current international order. Alternatively, it could abdicate leadership of the global order and allow Russia to trample it and then an increasingly powerful China to extend its own version of global norms, rules, and institutions. The former course entails risks of expanding confrontation and potential direct military clash, but the latter course would not necessarily avoid a military fight, especially if the United States comes to view an increasingly Chinese-ordered world to be unacceptable (chapters 3b, 15).
- The history of rivalrous dyads played out in periods of multilateral GPC offers several informative principles for competing effectively while minimizing the prospect of Great Power transition collapsing into Great Power war. Four stand out.
 - » Firmness with Flexibility. The dominant Great Power must demonstrate firmness with flexibility. It must clearly signal to the fullest extent possible the strategic aims it will defend at all costs and then offer the prospect for dialogue on those aims it may be willing to negotiate. While firm on its nonnegotiable goals, it should be flexible in finding issues and venues where win-win outcomes are possible. For example, the United Kingdom accepted U.S. primacy in the western Atlantic as a better path to sustaining high seas primacy on vital routes for its Middle East and Asian colonies—and preferable to naval confrontation in recognition of growing American power. At the same time, the rising United States came to accept the once-abhorrent British monarchy in recognition of growing political enfranchisement for a great number of British citizens. Is there such a trading room today for the United States and China to agree on rules for collaboration in space and cyberspace while at the same time negotiating over reduced CCP domestic economic and human rights constraints (chapters 2, 15)?
 - » Durable Partnerships and Alliances. The dominant Great Power must build and maintain durable interstate alliances and provide would-be partners with alternatives to the either-or choices posed by a hard-charging rival. Great Britain was right to seek strategic partnerships and allies in its rivalry with Napoleonic France, parlaying these alliances into first containment of the threat and then its defeat. Napoleon had no such partnerships, relying instead on conquest of allies. Today, the United States has a far greater base for building economic and military partnerships than any Great Power in modern history, and it confronts a rising Great Power in China with little experience or inclination in this area. Washington has an enormous opportunity to construct alternative economic,

diplomatic, and political geometries with an array of partners giving them alternatives to Chinese enticements and blandishments (chapters 2, 15).

- » The Peril of Reciprocal Societal Denigration. Successful GPC short of direct military confrontation is unlikely if the rivals give into a poisonous, open, and reciprocal denigration of each other's people. The choice to criticize the government or policies of a rival state while distinguishing it from the people is not as fraught with peril—although a tightrope must be walked to maintain the difference. Once the British and German press went after the character of the other's society, the march toward World War I accelerated. So, too, World War II in the Pacific loomed ominously once the United States and Tojo's Japan devolved to mutual societal recrimination. But the U.S. Government's conscious Cold War effort to distinguish the Soviet Union's Communist Party from the Russian people, reserving greatest criticism toward the Party and offering outreach to its people, generated a far different result. American leaders are likely to compete best with China while clearly distinguishing between its criticism of the CCP and its feelings for the Chinese people (chapters 2, 15).
- » Play for Time. Some argue that time works in favor of the rising Great Power in a competitive dyad, putting the dominant Great Power at more risk should it not take confrontational and decisive action. But this thesis rests on a false assumption that the rising power will continue to ascend in mainly a linear fashion and not confront problems or challenges on the way. The United States has its own domestic inconsistencies and challenges, but these pale in comparison to those certain to play out in China. The CCP faces multifaceted challenges to safeguarding both its political position and an economic rise, including environmental degradation; rising income inequalities; a rapidly aging and less productive population; chronic worry about abuses of political power; widespread corruption; restive domestic regions, including Tibet, Xinjiang, and Mongolia; and a poor record on human rights. As China's economy shifts and its economic growth decelerates, these issues are likely to move to the fore. Thus, a U.S. strategy that plays for time as China's challenges grow seems best suited for successful contemporary GPC (chapter 15).

Note

¹ Hans Binnendijk and David C. Gompert, eds., *Strategic Assessment 1998: Engaging Power for Peace* (Washington, DC: NDU Press, 1998), xiii, available at <<https://apps.dtic.mil/dtic/tr/fulltext/u2/a354594.pdf>>.

Part I
Conceptualizing the New Era of
Great Power Competition

Chapter 1

Introduction

By Thomas F. Lynch III

This chapter establishes the return of Great Power competition (GPC) as the fully acknowledged, dominant paradigm of interstate relations in 2017 after a 25-year absence from mainstream thinking. It establishes that competition is not synonymous with confrontation and clash and that GPC features a continuum of friendly-to-confrontational interactions between the competitors. The chapter notes the important linkage between GPC and Great Power transitions, observing that power transitions do portend greater instability and possible military clash (war). It establishes that Great Powers compete for an array of interests with a mixture of hard and soft power tools. It also defines a Great Power as one with three major characteristics in comparison to other states: unusual capabilities, use of those capabilities to pursue broad foreign policy interests beyond its immediate neighborhood, and a perception by other states that it is a major player. This makes the United States, China, and Russia today's Great Powers. After a brief introduction of the volume's 15 chapters, this chapter provides a short analytical evaluation of 4 relevant topics to contemporary GPC that cannot be addressed fully herein: space, cyberspace, homeland security, and climate change.

Great Power competition (GPC) is a framework for understanding global interstate relations that dominated global political affairs for centuries prior to World War II. Many past GPC eras have featured multiple powerful states jockeying for relative status and position. During the Cold War (1945–1991), GPC played out as a two-state competitive dyad between the United States and the Soviet Union. After lying dormant during a relatively short two-decade period of post–Cold War globalization and American international ascendancy, the construct of GPC returned to the vocabulary of international relations and security studies in earnest during the late 2010s.¹

The National Security Strategy of 2017 openly advanced the idea that America and fellow Great Powers, Russia and China, had transitioned formally from a more than 20-year period of collaboration and cooperation into one of competition.² In Washington, DC, 2017 was the year of fully acknowledged Great Power rivalry.³ The National Security Strategy simultaneously identified three additional threats to U.S. security: North Korea, Iran, and transnational terrorist and criminal organizations, but it clearly focused American security

"China and Russia challenge American power, influence, and interests, attempting to erode American security and prosperity. They are determined to make economies less free and less fair, to grow their militaries, and to control information and data to repress their societies and expand their influence. . . . These competitions require the United States to rethink the policies of the past two decades—policies based on the assumption that engagement with rivals and their inclusion in international institutions and global commerce would turn them into benign actors and trustworthy partners. For the most part, this premise turned out to be false."

—National Security Strategy of the United States of America, December 2017

and future prosperity on the ability to compete with the two emerging Great Powers headquartered in Moscow and Beijing.⁴

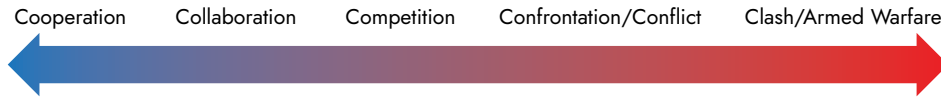
The United States National Defense Strategy of 2018 defined the central challenge to U.S. and Western prosperity and security as the reemergence of long-term strategic competition with those it classified as "revisionist powers," particularly Russia and China.⁵ It also observed that the emerging security environment would now be characterized by rapid technological advancements and a changing character of war. Specifically, technological changes would both broaden and unhelpfully blur the lines of competition and conflict.⁶

Debates about the meaning and relevance of GPC have been prominently featured in strategic discussions from 2017

to 2020. Beneath the surface of these discussions lurks another historic feature of GPC: *Great Power transition*. Transition is concerned with differential state growth rates and the impact on relative power between and among the states. A transition of differentiated power levels generates new relationships and the formation of new political and economic entities. One byproduct of differential state growth is a high potential for conflict when a challenger (or challengers) to a dominant country approaches the stage of relative equivalence of power, and specifically when the challenger is dissatisfied with the status quo.⁷

Strategic analysts and political scientists have haggled over the precise definition and detail of *greatness*, *power*, and *competition* in a raft of literature.⁸ A complete analysis of these debates and disputes is beyond the scope of this volume, but a few elements of these debates are important to set the scene for the chapters that follow.

First, competition is not synonymous with conflict. To a worrisome degree, some Western pundits have begun to conflate competition with clash, asserting that most if not all interactions between and among the three contemporary Great Powers must now be confrontational or even more extreme.⁹ Students and policy practitioners of this new era need to be mindful that competition exists on a continuum of interactions among states, nonstate actors, and some super-empowered individuals (see figure 1.1). On one end of the spectrum is cooperation. *Cooperating states* are exceptionally aligned in geopolitical goals and means of achieving them, thus able to pursue means of attaining them harmoniously. *Collaborating states* have similar goals and a general agreement on the means of achieving them. *Competing states* recognize that some, but not most, of their major goals are compatible and simultaneously disagree on the best means for attaining mutual gains in their remaining aligned goals. *Confrontational states* are characterized by incompatible aims in almost all major goals and in general conflict about what constitutes legitimate means for attainment of national goals. On the far end of the interaction continuum is the undesirable

Figure 1.1. Continuum of Major State Interaction Postures

environment where state goals are perceived as so incompatible, and even threatening, that the dominant form of state-to-state interaction devolves into armed conflict.¹⁰

As it exists on a continuum of interactions, competition features elements of collaboration and some aspects of cooperation. At the same time, competition is not conflict, and it is definitely not a clash between states (or armed warfare). And yet the ongoing move toward a competitive-dominant interactive framework among the three most mighty states and several others in the new era of GPC interweaves more elements of conflict and confrontation into competition and more preparations for clash than witnessed in recent history. Recent history was dominated by a preference for cooperation and collaboration among the major states of the post–Cold War era—an era that came to be known in some circles as an ascent of a liberal international order.¹¹ This volume proceeds from the perspective that the emerging era of GPC, while not exactly the same as previous eras with two or more competitive powers, now features three dominant states with robust capabilities in the major areas of power interactions. These states have moved from a phase of generally cooperative and collaborative interactions to those now dominated by competitive and confrontational dynamics.¹²

Second, power is a multifaceted construct having to do with a state's ability to attain its aims vis-à-vis another. Max Weber defined the essence of *power* as the “probability that one actor within a social relationship will be in a position to carry out his own will despite resistance.”¹³ Power has absolute, relative, and transitional properties. Scholars have long referred to power as relative to the type of actor, the goals the actor seeks, and the kind of relationships at play. Assessing relative power between states is an inherently challenging task.¹⁴ Power exists in two major dimensions: hard power (or the *coercive* use of military power and leveraging economic power as a *payoff*) and soft power (which includes cooperative and collaborative interactions that attain influence by *attraction*: partnered economic arrangements, ideological appeal, cultural and social engagements, diplomatic acumen, and reciprocal information exchanges).¹⁵ The term *smart power* is sometimes today used for policy choices that display an effective mix of coercive hard power and the persuasion and attraction of soft power in the pursuit of national interests.¹⁶ More recently, the term *sharp power* has become vogue as a phrase to describe state actions that twist soft power attributes in a manipulative or confrontational manner, especially the co-option of culture, educational institutions, media, and entertainment interactions by one state in a manner aimed to undermine or severely distort the political system or social order of another.¹⁷ The contributors in this volume provide thoughts on these different kinds of power and their relevance to the competition between the Great Powers today. Contributors also indicate the role of these kinds of Great Power relationships with other countries and global institutions.

Third, the notions of what constitutes “greatness” as an actor on the international stage are addressed in an operational manner throughout the majority of this volume. The framework of GPC tacitly acknowledges that not all players in the international arena of today are

as equally important as the others. At the same time, Great Powers act on and interact with international entities, states, nongovernmental actors, and critical individuals (such as Apple's Tim Cook, Tesla's Elon Musk, or Virgin's Richard Branson) in a manner that critically impacts the prospects for relative gain and relative loss in the varying areas of competition between them. An operationalized definition for a Great Power state has three substantive features: capabilities, behavior, and status attribution by other states in the system. First, a Great Power is a state that has unusual capabilities—in comparison to other states—with which to pursue its interests and to influence interstate relations. Second, a Great Power uses those capabilities to pursue broad foreign policy interests beyond its immediate neighborhood. Finally, that state's relative status is perceived by other states to be major in nature, and the other states act toward that state accordingly.¹⁸ Using this operational definition, this volume—for the most part—considers there to be three Great Powers in the contemporary era: the United States, Russia, and China.¹⁹ Only in chapter 10 is there analysis of Russia as though it is not a contemporary Great Power.

Intent, Audience, and Contributors

This volume provides a succinct, expert, and nuanced understanding of important emerging dimensions of GPC today. It primarily focuses on the critical interactions and activities among the United States, China, and Russia. It simultaneously develops many of the major implications of these interactions for other state and for nonstate actors and processes. Therefore, its analyses and recommendations are generally framed for the years from 2020 to 2025. The speed and pace of change in global power relationships and activities requires an update of credible analysis by mid-decade. Indeed, impactful events such as the 2019–2020 novel coronavirus (COVID-19) pandemic will generate important changes that are beginning to take shape only as this volume is published.

As begun in this chapter, contributors provide readers with text, analysis, and notes that reference primary documents and sources to the fullest extent possible. Where such sources are not available, the contributors cite and then note for reference the most prominent and influential secondary sources and analytical pieces available in the field. The volume is designed to enable each reader to gain access to the best and most relevant writings on the topics of contemporary GPC.

The volume is written with chapters that are short, are self-contained for standalone use, and tie back to the central themes of GPC developed in early chapters (chapters 2–4). Each chapter and the overall elements of the volume's key findings and conclusions are for graduate-level students within professional military institutions, graduate students in civilian political science and national strategy programs, and mid- to upper-level career civil servants in the U.S. interagency community and other security establishments. Volume editors and authors hope that individual chapters and combinations of chapters will be of use to students and policy practitioners both in an academic setting and for personal study and understanding.

The contributors include many who have been directly engaged as thought leaders and policymaking pioneers grappling with the emerging contours of GPC. One contributor helped write the 2018 National Defense Strategy, working on that project directly for then-Secretary of Defense James Mattis. Another contributor has served as a subject matter

expert and consultant to U.S. Cyber Command. Two others are regular consultants and advisers for military and civilian U.S. Government organizations focusing on strategic deterrence and weapons of mass destruction. Others have served across the U.S. military and wider interagency community and intelligence organizations as subject matter consultants and as red-teaming leaders on China—providing uniquely informed insights on the key dimensions of China’s rise and interaction on the world stage. One contributor has been instrumental in developing U.S. and allied military leaders’ understanding about the future of terrorism and counterterrorism in an era of GPC. Two of the main chapter contributors have provided direct analysis and recommendation about the implications to the U.S. military from the emerging GPC dynamics across South Asia, the Middle East, and the Indo-Pacific region. Last, one contributor has been a leading public voice for understanding how the competitive space of social media has become and will continue to evolve as a critical nexus between Great Powers and their surrogates—aiming to generate power and influence in the emerging era.

Each chapter includes original author research, analysis, and insights—much of it generated from direct contact with senior U.S. Government policymakers and other global security leaders. Some chapters include text drawn from pieces the contributors have published elsewhere in recent years. All chapters feature original, updated analysis and insights for understanding GPC in the specific period from 2020 to 2025 and beyond.

Sections and Overviews

This strategic assessment of the new era of GPC is organized into 4 discrete sections and a total of 15 numbered chapters that include a couplet chapter, chapters 3a and 3b.

The first section focuses on conceptualizing this new era of GPC. Its four chapters undertake a focused assessment of historic cases of Great Power rivalry among three or more parties and generates important lessons for the current era. It then provides an overview of the major geostrategic dynamics and technological competitive aspects of the present Great Power rivalry among China, Russia, and the United States.

Chapter 2 takes a short, focused historical look at past eras of GPC. Thomas Lynch and Frank Hoffman survey the major understandings of Great Power identities and transitions in historic eras of GPC. From this survey, they conclude that past eras featuring major power transitions trended toward direct military clash (war) between the rising and declining state absent an atypical exercise in sage Great Power leadership. They also develop five universally applicable aspects/categories of interstate competition and the competitive elements in each: political and diplomatic, informational, ideological, military, and economic. The authors then apply these categories to an analysis of the dynamics of major state competition in four GPC case studies, focusing on the most critical dyadic rivalries in each: France–the United Kingdom (UK) (late 18th and 19th centuries), UK–Germany (late 19th and 20th centuries), the United States–UK (late 19th and 20th centuries), and the United States–Japan (early 20th century). All were contested in eras of multiple Great Power rivals with one ascendant Great Power worried about relative power decline in the face of at least one rising challenger and under conditions of emergent, disruptive technological change. Thus, they are cases with strong parallels to current GPC. The chapter concludes with a

summary of major insights and a framework for analysis of contemporary GPC tethered to relevant historical vignettes.

Chapter 3a looks at contemporary Great Power geostrategic dynamics in light of declared strategies and revealed strategic preference forces. It focuses on the main strategies and relationship dynamics among the United States, China, and Russia in 2020; how these dynamics developed; and where they should be anticipated to evolve by 2025. Phillip Saunders and Thomas Lynch utilize the five key categories of interstate activities important in past GPC eras developed in chapter 2 to assess the most critical aspects of emerging competitive postures and strategies of the three Great Powers. The authors make an analytic evaluation of relative strategic interest force by the Great Powers in various regions of the world, indicating how these could impact forthcoming GPC dynamics.

Chapter 3b extends beyond chapter 3a by analyzing contemporary GPC dynamics from the perspective of state power assets and the tool sets available to the three Great Power protagonists—evaluating their objective means to achieve the strategic preferences established in chapter 3a. Utilizing the main competitive elements defined in the five key categories of past GPC eras developed in chapter 2, the authors provide an array of quantitative and qualitative measures that evaluate the main power dynamics at play between the contemporary Great Powers. They assess Great Power relative present strengths and future trajectories. Chapters establish that—for the foreseeable future—Russia’s tool kit makes it an urgent but transient security challenger to the United States, while China’s growing power tools make it the long-term challenger to American national interests and global policy preferences.

Chapter 4 takes a focused look at contemporary Great Power competition through the lens of technological competitive factors central to the ongoing fourth industrial revolution. This revolution describes the blurring of boundaries among the physical, digital, and biological worlds. It is a fusion of advances in artificial intelligence (AI), robotics, the Internet of Things, 3D printing (additive manufacturing), genetic engineering, quantum computing, energy, and biotechnology. Authors T.X. Hammes and Diane DiEuliis trace these technologies that most impact GPC in the short term and explain the importance of each individually. They also describe how together these technologies will revolutionize the global economy. The authors then indicate how these factors impact the economic growth potential and the relative strategic interests and power positioning of the three Great Powers.

The second section of the book takes up from chapter 4 with a featured focus on the critical dynamics of technology, innovation, and the evolving character of war in a new era of Great Power competition.

In chapter 5, T.X. Hammes begins this four-chapter part with specific attention to key technologies that are leading to a revolution of small, smart, and cheap in emerging warfare. He focuses on the important role of autonomous weapons. Hammes also tackles some dimensions and implications of hypersonic weapons for GPC and conflict. He concludes with thoughts about how small, smart, cheap, and super-fast weaponry will impact the dynamics of defense competition in the coming 5 years and, perhaps, the possibility of Great Power conflict in the distant future.

Chapter 6 looks directly at the emerging impact of AI, quantum computing, and 5G wireless technologies for GPC. Richard Andres explains that where control of industrial resources was once key to geopolitical power, today control of information resources is the

most important factor. The chapter examines the Sino-American competition in these critical technologies and explains why China's state-led ascent in them is a serious challenge to American power and a key element of the ongoing GPC.

Chapter 7 addresses the contemporary dynamics and strategic implications of social media that influence operations technologies for Great Power competition and conflict. Todd Helmus of RAND explains why foreign propaganda campaigns on social media platforms have become prolific. The chapter reviews how three key U.S. adversaries—Russia, China, and the so-called Islamic State—have exploited modern technologies to attain political objectives. It evaluates the aims, capabilities, and limitations of online propaganda practiced by each of these American adversaries, concluding with recommendations for the United States to counter their use of online propaganda in the new era of GPC.

Chapter 8 focuses on weapons of mass destruction (WMD) and strategic deterrence in the emerging era of GPC. Paul Bernstein and his co-authors assess the prospects for nuclear competition between the Great Powers and the potential role of biological and chemical weapons in Russian and Chinese strategies for various forms of competition and conflict. They identify both geopolitical and technology drivers of future competition in WMD and the steps the United States should take to manage strategic and operational risk in this important and potentially volatile area of GPC.

The third section of this strategic assessment examines selected geostrategic interactions in the new era of GPC.

Chapter 9 examines the Indo-Pacific competitive space—perhaps the most contentious geopolitical region in the emerging GPC era. Thomas Lynch, James Przystup, and Phillip Saunders develop American and Chinese strategic visions for the region, formerly described as the Asia-Pacific. They highlight the divergence of strategic interests between America's "Free and Open Indo-Pacific" vision and China's "community of common interest" framework. Then the chapter conducts an analytical comparison of relative Chinese and American strengths in the competitive categories of political and diplomatic, ideological, informational, and military and economic. The analysis reveals a mixture of relative power advantages, indicating that despite growing regional tensions between the two Great Powers, there are opportunities to both secure stability and pursue selected collaboration if both parties identify and accept their relative power limitations.

Chapter 10 takes a direct and comparative look at Russia, North Korea, and Iran as a grouping of "rogue, disruptive, and spoiler states." In doing so, the chapter treats Russia in a manner that diverges from its treatment in other chapters, and more like those contemporary scholars who view Moscow as more of a nuclear weapons-led, muscle-bound declining state with a disruptive rather than a constructive global worldview.²⁰ The chapter develops key motivations and activities for rogue state activities and the new era of GPC. It explains why Russian, Iranian, and North Korean provocative behaviors are not uniformly harmful to the United States or beneficial for China. It further elucidates that the prospect of a robust and fully cooperative anti-U.S. rogue axis remains remote. It recommends appropriate U.S. strategic principles to meet these rogue state realities in a new era of GPC.

Chapter 11 addresses the future of counterterrorism missions by the U.S. military in the era of GPC. Kim Cragin and her co-authors assess that, over the next 3 to 5 years, Great Power competition likely will constrain the ability of U.S. forces to achieve even limited

counterterrorism objectives. They explore the insights from America's recent counterterrorism experiences in Syria—where Russia was extensively involved—demonstrating that lessons from that experience inform the future of U.S. counterterrorism operations. The chapter then explores the long-lasting regional rivalry between Iran and Saudi Arabia, explaining how this contest will exacerbate future threats to the United States from violent extremist organizations (VEOs). Then it demonstrates how Australia and similar countries might help reduce VEO risks to the United States, even as Washington shifts strategic priorities toward GPC. The chapter finally summarizes the VEO risks that America's military should prioritize and the new authorities and technologies it should pursue for counterterrorism success in GPC.

Chapter 12 asks the question, “Whither Europe in the era of GPC?” Steven Kramer and Irene Kyriakopoulos trace Europe's recent history: its lost promise, its major troubles, its relations with the three contemporary Great Powers, and its potential for evolution and regeneration. The authors explain how Europe in 2020 is a region troubled by recent financial, migration, and pandemic crises and one questioning the future of the transatlantic security alliance. They document the resilience of Europe in the face of these troubles and provide an understanding of how the experiment of post–World War II Europe should be expected to evolve between now and 2025.

Chapter 13 concludes the third section with a look at the competing visions and activities for the Great Powers in several critical regions of the world: the Middle East, Africa, Latin America, and the Arctic. Bryce Loidolt and his co-authors take an analytical look at the strategies and impacts of Chinese and Russian competitive activities across these regions. They trace regional receptivity to and potential repercussions from them. The chapter finds that the challenges posed to the United States are rarely grounded in an ideological commitment to Beijing's global vision or Moscow's cynicism. This points to the need for American regional strategies that appreciate the diverse challenges that China and Russia pose and American approaches that avoid pulling important U.S. regional partners into an unrestricted zero-sum competition.

The fourth section of the book features two chapters that ask “for what” and “how best” can the United States prepare to compete successfully in the era of GPC.

In chapter 14, Frank Hoffman provides a framework for thinking about U.S. competitive alternatives for the emerging era of GPC. He sketches out the elements of five possible strategy alternatives for the United States with its primary competitor, China. He reviews the key features of one collaborative alternative (bilateral bargain), two mainly competitive alternatives (managed competition and enhanced balancing), and two predominantly confrontational ones (compression and contested primacy), focusing on each one's suitability, feasibility, and sustainability. The chapter details how each alternative leverages relative American and Chinese strengths and weaknesses and the international and domestic support likely for each. The author concludes that an American strategy of “enhanced balancing” best prepares America for successful competition with China.

Chapter 15 concludes the volume with a short summary and substantive extension of major insights about GPC. Thomas Lynch evaluates the main features of evolving GPC. He then situates major contemporary GPC dynamics in the context of past periods. The chapter offers an interim assessment about what the COVID-19 pandemic will mean for

dominant GPC trends, assessing that these are likely to accelerate rather than be supplanted by new ones. The chapter flags three main imperatives for U.S. competitiveness in the Sino-American dyadic rivalry, especially the wise choice of strategic allies and partners. It concludes with a historically framed assessment of four important dynamics for American success in a long-haul competition with China: firmness with flexibility, prioritization of partnerships and alliances, confronting China's leaders rather than its people, and playing for time.

Finally, this volume features two appendixes. The first appendix is a selected bibliography of critical documents, books, and articles featured within book chapters. The second is an original, Web-only selected database compiled in the research for this volume titled "Contemporary Great Power Dynamics." This modest catalogue of focused data provides readers with a ready reference for nine of the most significant quantitative indicators of relative state status (gross domestic product [GDP], GDP per capita, composite index of national capability, population, birthrate, level of industrialization, percentage of global financial markets, and innovation rank). Each indicator is provided for the three contemporary Great Powers and five other states. This data is measured for 7 different years between 1980 and 2025. Some of this data is assimilated and assessed in chapter 3b. Mainly, this appendix is intended as a one-stop reader resource for follow-on investigations and as a living document that will be updated periodically by the research team in the Institute for National Strategic Studies at the National Defense University. The Contemporary Great Power Dynamics Matrix can be found at the following URL: <https://ndupress.ndu.edu/Contemporary-GPC-Dynamics-Matrix/>.

Important GPC Topics Without Individual Chapters

There is no way to cover all the rich and varied topics that might be addressed in a full assessment of the new era of GPC. Reader attention could not be expected to withstand a barrage of separate chapters on every potential topic of relevance in a single volume. As a consequence, there are some interesting topics involving GPC that could not fit into these pages as standalone contributions. Many such topics are mentioned in context within several chapters, including the emerging implications of the COVID-19 pandemic. All also are covered in an authoritative manner in other publications. Among these, four topics stand out for specific mention and reader attention: space, cyberspace, homeland security, and climate change. It is likely that these topics will evolve significantly over the next half-decade—generating worthy chapter topics for the final years of the 2020s. For now, this chapter flags some key elements in these areas for contemporary Great Powers. It also offers readers some detailed reference sources that provide more insight into how these areas now impact emerging GPC.

Space

The United States and the Soviet Union competed in space for decades during the Cold War. After a two-decade period of U.S.-Russian space cooperation and relative absence of Chinese space activity, outer space has once again become a key location for measuring relative power and conducting GPC.²¹ In early 2019, echoing calls for a U.S. Space Force, then-Chairman of the U.S. Joint Chiefs of Staff General Joseph Dunford reiterated the need

for the Nation to be intensely engaged on all five interactive domains (land, sea, air, space, and cyberspace) to stay competitive with Russia and China.²² Currently, NASA is working on a Moon-to-Mars mission that, in theory, will give the United States a more competitive edge.²³ Despite an evolving competition, the United States remains tethered to a cooperative and codependent framework of working with Russia in space. As of 2020, Washington still uses the Russian Soyuz system to bring astronauts to the International Space Station, and the Lunar Gateway program remains a joint U.S.-Russian effort to ease access to the Moon.²⁴

Meanwhile, during a reorganization of its military in 2015, China created a “space force” aimed at operating satellites and running counterspace missions, including a home-grown GPS system named the Beidou Navigation Satellite System, which has grown from 1 to 54 satellites from December 2011 to March 2020.²⁵ In 2019, former U.S. Acting Defense Secretary Patrick Shanahan stressed that Russia and China have weaponized space and that the United States now must join suit.²⁶

A growing number of space-capable nations are filling Earth’s orbits with a high volume of satellites contesting a shrinking number of orbital spaces.²⁷ In addition to the three Great Powers, seven other states in 2020 are reported as developing or considering the development of one or more types of counterspace systems.²⁸ Intensifying GPC is a driver of antisatellite (ASAT) development and use, including high-profile tests by India in 2019 and a series of them by Russia including one in April 2020.²⁹ Recognized norms, standards, and treaties do not exist to divvy up the limited volume of space or to regulate the risk from ASAT proliferation. The absence of a multilateral cooperative framework for Great Power interactions in space makes it more likely that competition may beget confrontation in this medium.³⁰

Cyberspace

The pursuit of competitive advantage and confrontational dominance extends far outside the arena of conventional warfare. Cyberspace has emerged as a new Great Power battlespace and has motivated the United States, Russia, and China to develop their own cyber attack and defense capabilities.³¹ The current U.S. Defense Cyber Strategy is aimed at preventing aggressive actors, specifically Russia and China, from conducting campaigns that impact the United States and its allies.³² U.S. Cyber Command has focused efforts on thwarting clandestine Russian intelligence and civilian proxy agent interference like that which occurred during the 2016 U.S. national elections.³³ Chinese cyber espionage historically has been aimed at the U.S. commercial sector, but Beijing’s 2015 military strategy placed increasing cyber emphasis on domestic security—protecting its infrastructure from foreign interference and allowing its military to integrate further with the technological scene.³⁴ Russian cyber operations are more aggressive and aimed at laying the groundwork for future major military and infrastructure disruptions.³⁵

The cyber medium for competition will become increasingly important to defend as Great Power dynamics drive greater and more sophisticated cyber innovations.³⁶ Chapter 6 provides some important analysis about the wide-ranging dynamics of GPC in cyberspace. But chapter 6 is limited by the need to address additional critical factors involving computing automation, AI, and big data analytics.³⁷

Homeland Security

The dominant construct for homeland security during the past 30 years has been that of defense from catastrophic terrorist attack. From the 1990s through the early 2010s, only the very latent risk of unforetold nuclear exchange between the United States and Russia stood outside the dominant paradigm of homeland protection from the threat of terrorism. With the dawn of openly announced GPC, the straitjacket on homeland security thinking has been removed. As North Korea launched nuclear-capable missiles in 2017 with a range to threaten the United States (and interestingly also Moscow and Beijing), the prospects for other nuclear weapons-capable states to attain global reach came astride as a major feature of the new global order.

Simultaneously, targeted cyber and social media activities against military, national, and civilian infrastructure capacity became—during the 2010s—a more substantive threat with ongoing homeland security implications. Finally, the emergence of unmanned platforms capable of operations against targets inside a national sovereign space launched from outside platforms on the ground, at sea, in the air, and from space are becoming far more significant than even a year or two ago. The 2018 National Defense Strategy states that “the homeland is no longer a sanctuary,” and “[d]efending the homeland from attack” is the number one defense objective listed for the United States as new threats develop on both private and public fronts.³⁸ Russian and Chinese views on this also are changing, with Beijing looking inward toward bolstering social and economic stability and Chinese Communist Party dominance.

While chapters in this volume address many of the GPC competitive dynamics with homeland security implications—including those involving nuclear weapons, biochemical weapons, unmanned platforms, and social media—there are several other new dynamics at play in the rapidly evolving construct of homeland security in an era of GPC.³⁹

Climate Change

Over the past decade, the multifaceted implications of climate change and humankind’s role in that change have grown in salience while remaining contentious. Fossil fuels, textiles, plastics, and meat production industries have been placed under scrutiny for generating high levels of carbon pollutants damaging to water sources, the air, ambient temperatures, and the wider ecosystems they touch. While some countries and leaders resist the science attributing human activity for climate change, others contend that our world is in a climate emergency.⁴⁰ Younger generations have become increasingly animated, with some ascribing “climate change anxiety” to the idea that many of the world’s youth under 30 are anxious about the impending doom of wider Earth ecosystems during their lifetimes.

Almost 5 years ago, in December 2015, 175 of the world’s governments adopted the Paris Climate Agreement to arrest increasing man-made stressors on the Earth’s ecosystems. The Paris Agreement focused on keeping global warming below 2 degrees Celsius by capping and apportioning greenhouse emission percentages among the signatory nations. That agreement resulted from two decades of international negotiations, but unanimous consensus about the urgency of the problem or the imperative for significant restraint of human activities remained elusive.

China and the United States are the world's biggest economies and also the biggest consumers of global natural resources and polluters.⁴¹ China is the world's largest polluter, producing 30 percent of global carbon emissions.⁴² It signed and ratified the Paris Agreement in 2016 but remains the country most obviously afflicted by growing climate challenges. More than a half billion of its citizens live near oceans and face rising sea levels. Major Chinese cities are overwhelmed by smog and other air pollution requiring face mask protection and threatening pulmonary health. Chinese rivers have been ruined by decades of unregulated toxic waste.⁴³ The United States is currently the second worst polluter, producing 15 percent of global carbon emissions.⁴⁴ In 2017, the Trump administration announced its intention to quit the Paris Agreement and began the year-long process of withdrawal in late 2019. Absent a change in policy, the United States will exit the agreement in November 2020. America's carbon emissions percentage is projected to increase after its planned Paris Agreement exit.⁴⁵ Russia is the world's fourth largest producer of global carbon emissions with nearly 5 percent of the total.⁴⁶ In 2019, Russia finally ratified the Paris Agreement after a 3-year delay. But as of early 2020, Russia has not taken any actual measures to reduce greenhouse emissions. The government initially attempted to take small measures by imposing emissions quotas, but the Russian Union of Industrialists and Entrepreneurs lobbied against these measures and prevented their implementation.⁴⁷

The World Health Organization states that climate change is an evolving human health crisis, slower moving but no less ominous than deadly pandemics for the future of global prosperity and stability.⁴⁸ The economic and security implications of accelerating climate change on land, at sea, and in the air are only now coming into view. Over the coming years, the changing climate will shape Great Power natural endowments and national security in at least three important ways.⁴⁹ First, rising global temperatures will affect resource availability. The changes will shift productive agriculture toward the Earth's poles while making crop growth more difficult near the equator. Russia and Canada may be relative winners in new arable land, but uncertainties remain. Second, as the world transitions from fossil fuels to renewable energy sources, the relative importance of carbon resources will shift. Here, the United States has an inherent advantage over China, but less so with Russia. Third, as the Great Powers—and especially the United States and China—jockey for the competitive edge in high-tech industries and renewable energy technology, competition over critical mineral resources in major producing countries such as Australia, Brazil, Chile, Congo, and South Africa will intensify.

It remains difficult to predict precisely how climate change will impact Great Power competition into the future. Although natural endowment factors altered by climate change could become a point of contention between the United States and China, increasing risk of confrontation, these factors may also become a source of collaboration and cooperation. Washington and Beijing could work together on critical minerals, including research and development on less destructive ways to mine and refine these materials. Moreover, as climate change challenges agricultural productivity worldwide, international cooperation and trade could play an important part in adapting to changing conditions.⁵⁰ Only one thing seems certain: Climate change will matter to new patterns of Great Power competition. Greater clarity about the dominant patterns will be present by mid-decade. Moreover, there are other major climate change factors influencing the dynamics of GPC in the emerging era.⁵¹

Transitions

This volume now turns to framing the backdrop for a new era of Great Power competition. Its first section sketches the key areas of historic GPC and assesses lessons from four representative past eras. Then in two “couplet” chapters, the section takes a look at contemporary GPC geostrategic dynamics, major power strategies, and available resources for competition. The section finishes with a chapter about the impact of emerging, revolutionary technological factors influencing the dynamics of emerging GPC.

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Notes

¹ For examples, see Robert Kagan, *The Return of History and the End of Dreams* (New York: Vintage Books, 2009); Aaron L. Friedberg, *A Contest for Supremacy: China, America, and the Struggle for Mastery in Asia* (New York: Norton, 2011); John Mearsheimer, *The Tragedy of Great Power Politics* (New York: Norton, 2014); Shivshankar Menon, “How Great Power Competition Has Changed,” *Brookings*, May 4, 2015, available at <<https://www.brookings.edu/blog/order-from-chaos/2015/05/04/how-great-power-competition-has-changed/>>; Max Fisher, “The New Era of Great Power Competition,” *Vox*, April 13, 2016, available at <www.vox.com/2016/4/13/11421352/ash-carter-deterrence-power-competition>; Uri Freidman, “The New Concept Everyone in Washington Is Talking About,” *The Atlantic*, August 6, 2019, available at <www.theatlantic.com/politics/archive/2019/08/what-genesis-great-power-competition/595405/>.

² *National Security Strategy of the United States of America* (Washington, DC: The White House, December 2017), 2–3, available at <<https://www.whitehouse.gov/wp-content/uploads/2017/12/NSS-Final-12-18-2017-0905.pdf>>.

³ *Rivalry* is the relationship between two or more actors (states) who regularly compete, while *competition* is the action taken to contest ascendancy in a particular field. Rivalry describes the reciprocal competition between two or more states for ascendancy.

⁴ *National Security Strategy of the United States of America*, 2–4.

⁵ *Summary of the 2018 National Defense Strategy of the United States of America: Sharpening the American Military's Competitive Edge* (Washington, DC: Department of Defense, 2018), 2, available at <<https://dod.defense.gov/Portals/1/Documents/pubs/2018-National-Defense-Strategy-Summary.pdf>>.

⁶ *National Security Strategy of the United States of America*, 4–5.

⁷ Ronald L. Tammen, Jacek Kugler, and Doug Lemke, “Power Transition Theory,” Trans-Research Consortium Work Paper #1, December 2011, available at <<https://static1.squarespace.com/static/576ef1a0be65941edd80fcf7/t/578d56e2c69cfbb-1192c4cf/1468880611440/Power+Transition+Theory.pdf>>. For a detailed look at power transition theory, see A.F.K. Organski, *World Politics*, 2nd ed. (New York: Knopf, 1968); A.F.K. Organski and Jacek Kugler, *The War Ledger* (Chicago: University of Chicago Press, 1980); Jacek Kugler and Douglas Lemke, eds., *Parity and War: Evaluations and Extensions of the War Ledger* (Ann Arbor: University of Michigan Press, 1996). Related transition theories of hegemonic decline and global cycles are found in Robert Gilpin, *War and Change in World Politics* (Cambridge: Cambridge University Press, 1981); George Modelski, *Long Cycles in World Politics* (New York: Palgrave Macmillan, 1987), respectively.

⁸ Thomas J. Wright, *All Measures Short of War: The Context for the 21st Century and the Future of American Power* (New Haven:

Yale University Press, 2017); Ali Wyne, “America’s Blind Ambition Could Make It a Victim of Global Competition,” *The National Interest*, February 11, 2019, available at <<https://nationalinterest.org/feature/americas-blind-ambition-could-make-it-victim-global-competition-44227>>.

⁹ For examples of this tendency in 2018–2020, see Geoff Hertenstein, “DIME Without the ‘M’ is DIE: A Case for Conventional Military Power in Modern Strategy Discourse,” *Strategy Bridge*, September 22, 2019, available at <<https://thestrategybridge.org/the-bridge/2019/9/22/dime-without-the-m-is-die-a-case-for-conventional-military-power-in-modern-strategy-discourse>>; Nikki Haley, “How to Confront an Advancing Threat from China,” *Foreign Affairs*, July 18, 2019, available at <www.foreignaffairs.com/articles/china/2019-07-18/how-confront-advancing-threat-china>; Matthew Kroenig, *The Return of Great Power Rivalry* (Oxford: Oxford University Press, 2020).

¹⁰ This continuum is an original construct for this volume. It was developed from several major writings on the range of major state-to-state interactions, especially those found in Charles Glaser, *Rational Theory of International Politics: The Logic of Competition and Cooperation* (Princeton: Princeton University Press, 1986); Robert Jervis, “Cooperation Under the Security Dilemma,” *World Politics* 30, no. 2 (January 1978), 167–214; Robert Koehane, *After Hegemony* (Princeton: Princeton University Press, 1984), 51–52; Kenneth N. Waltz, “Structural Realism After the Cold War,” *International Security* 25, no.1 (Summer 2000), 5–41.

¹¹ See G. John Ikenberry, *Liberal Leviathan: The Origins, Crisis, and Transformation of the American World Order* (Princeton: Princeton University Press, 2011), especially 35–75, 132–145. For a discussion of how the liberal international order has been used as a shorthand for three separate but related phenomena—the creation of a rules-based order enshrined in institutions, the establishment of alliance relationships with the United States as security guarantor, and the propagation of liberal values—see “Is There a Liberal International Order?” *Strategic Survey 2018: The Annual Assessment of Geopolitics* (London: International Institute for Strategic Studies, 2018), 23–32.

¹² This approach differs from the one advanced by RAND analyst Michael Mazarr and his colleagues that focuses on separate Great Powers or major players. The major analytical difference is established in the paragraph after next with an operational definition of what constitutes a Great Power. For the RAND take on why contemporary global interactions do not qualify as a period of Great Power competition, see Michael J. Mazarr et al., *Understanding the Emerging Era of International Competition: Theoretical and Historical Perspectives*, RR2726 (Santa Monica, CA: RAND, 2018).

- ¹³ Max Weber, cited in Isidor Wallimann, Nicholas C. Tatsis, and George V. Zito, "On Max Weber's Definition of Power," *The Australian and New Zealand Journal of Sociology* 13, no. 3 (December 1977), 231–235, available at <<https://doi.org/10.1177/144078337701300308>>.
- ¹⁴ This challenge dates to the philosophical writings of Sir Francis Bacon and his construct of scientific knowledge as the necessary essence of political power. See Timothy Paterson, "The Secular Control of Scientific Power in the Political Philosophy of Francis Bacon," *Polity* 21, no. 3 (Spring 1989), 457–480. Estimating state power—particularly relative military power—continues to be a subject of deep discussion and debate. See A.W. Marshall, *Problems of Estimating Military Power* (Santa Monica, CA: RAND, 1966), available at <<https://www.rand.org/pubs/papers/P3417.html>>.
- ¹⁵ The most resonant operational definitions of *hard power* and *soft power* are best developed in two books by Harvard political scientist Joseph Nye, Jr. See *Bound to Lead: The Changing Nature of American Power* (New York: Basic Books, 1990); *The Paradox of American Power* (London: Oxford University Press, 2002).
- ¹⁶ The term *smart power* was coined by Nye in the early 2000s to respond to what he identified as a growing misperception that his writings asserted that soft power alone could produce effective foreign policy, as he explained in "Get Smart: Combining Hard and Soft Power," *Foreign Affairs* 88, no. 4 (July/August 2009).
- ¹⁷ *Sharp power* has been developed and promulgated by U.S. National Endowment for Democracy employees Christopher Walker and Jessica Ludwig and focused mainly on what they view as the deliberate efforts of China and Russia to craft this new category of power. See Christopher Walker, "What Is 'Sharp Power'?" *Journal of Democracy* 29, no. 3 (July 2018), 9–13; Christopher Walker, Shanthi Kalathil, and Jessica Ludwig, "Forget Hearts and Minds," *Foreign Policy*, September 14, 2018, available at <<https://foreignpolicy.com/2018/09/14/forget-hearts-and-minds-sharp-power/>>. Sharp power is controversial and not fully accepted. Joseph Nye rejects it as but a longstanding form of hard power. He writes, sharp power is the "deceptive use of information for hostile purposes" and one that has been practiced by the intelligence and information services of democracies as well as those of authoritarian regimes. See Joseph S. Nye, Jr., "How Sharp Power Threatens Soft Power," *Foreign Affairs*, January 24, 2018, available at <www.foreignaffairs.com/articles/china/2018-01-24/how-sharp-power-threatens-soft-power>. China has labeled the expression as one concocted by the West to vilify China and without merit. See Xin Liu, "What Is Sharp Power? It's Nothing But 'Unsmart' Power," *USC Center on Public Diplomacy CPD*, November 15, 2018, available at <<https://www.uscpublicdiplomacy.org/blog/what-sharp-power-it%E2%80%99s-nothing-%E2%80%99Cunsmart%E2%80%9D-power>>.
- ¹⁸ This operational definition is taken from Thomas J. Volgy et al., "Major Power Status in International Politics," in *Major Powers and the Quest for Status in International Politics: Global and Regional Perspectives*, ed. Thomas J. Volgy et al. (New York: Palgrave Macmillan, 2011), 1–26. Their definition derives from an amalgam of several other classic efforts at defining Great Power status, including the following historic references: J. David Singer and Melvin Small, "Formal Alliances, 1815–1939: A Quantitative Description," *Journal of Peace Research* 3, no. 1 (March 1966); Jack Levy, *War in the Modern Great Power System: 1495–1975* (Lexington: University Press of Kentucky, 1983); John J. Mearsheimer, *The Tragedy of Great Power Politics* (New York: Norton, 2001); and Benjamin O. Fordham, "Who Wants to Be a Major Power? Explaining the Expansion of Foreign Policy Ambition," *Journal of Peace Research* 48, no. 5 (2011), 587–603.
- ¹⁹ As noted, this formal definition precludes inclusion of the other eight states listed in the RAND study of 2018. The other states in that study, including India, Japan, Turkey, and Brazil, do not meet the operational definition. They have no truly unusual capabilities. They do not use unique capabilities globally to pursue broad policy interests. Finally, they are not perceived as "Great" by other states. They do matter in consideration of GPC interactions, but in 2020 are not contemporary Great Powers. See Mazarr et al., *Understanding the Emerging Era of International Competition*.
- ²⁰ For contemporary scholars who view Russia as less of a Great Power and more of a disruptor state, see Wyne, "America's Blind Ambition Could Make It a Victim of Global Competition"; David J. Kramer, "Russia Is No Great Power Competitor," *The American Interest*, April 24, 2019, available at <www.the-american-interest.com/2019/04/24/russia-is-no-great-power-competitor/>; Timothy Frye, "Putin Touts Russia as a Great Power. But He's Made It a Weak One," *Washington Post*, June 6, 2019, available at <www.washingtonpost.com/opinions/2019/06/06/putin-touts-russia-great-power-hes-made-it-weak-one/>.
- ²¹ Christian Trotti and Mark Massa, "Lessons from Apollo: Industry and Great-Power Competition," Atlantic Council, July 26, 2019, available at <<https://www.atlanticcouncil.org/blogs/new-atlanticist/lessons-from-apollo-industry-and-great-power-competition/>>.
- ²² C. Todd Lopez, "Great Power Competition's Resurgence," *Defense.gov*, November 19, 2019, available at <<https://www.defense.gov/explore/story/Article/1792014/great-power-competitions-resurgence/>>; "War in the Fifth Domain," *The Economist*, November 19, 2019, available at <www.economist.com/briefing/2010/07/01/war-in-the-fifth-domain>.
- ²³ Trotti and Massa, "Lessons from Apollo."
- ²⁴ "Why Does America Still Use Soyuz Rockets to Put Its Astronauts in Space?" *The Economist*, October 16, 2018, available at <www.economist.com/the-economist-explains/2018/10/16/why-does-america-still-use-soyuz-rockets-to-put-its-astronauts-in-space>; Trotti and Massa, "Lessons from Apollo."
- ²⁵ Kevin L. Pollpeter, Michael S. Chase, and Eric Heginbotham, *The Creation of the PLA Strategic Support Force and Its Implications for Chinese Military Space Operations*, RR2058 (Santa Monica, CA: RAND, 2017), available at <https://www.rand.org/pubs/research_reports/RR2058.html>.
- ²⁶ Lara Seligman, "The New Space Race," *Foreign Policy*, May 14, 2019, available at <<https://foreignpolicy.com/2019/05/14/the-new-space-race-china-russia-nasa/>>.
- ²⁷ As of late 2018, there were reportedly 6,000 satellites among 24,000 Earth-orbiting space objects (satellites, space objects, space debris) being tracked by the U.S. Space Surveillance Network (SSN). Another 100 satellites per year—mainly commercial but also governmental—are forecast to join an increasingly crowded Earth orbit during the 2020s. SSN is a component of the U.S. Space Command and is the only agency in the world tracking and forecasting collision risks for space objects. SSN has no authority for space orbit management and can only alert commercial companies about potential risk of collision in the event that a model forecasts such risks. See "Space Surveillance," *Science Direct*, available at <www.sciencedirect.com/topics/engineering/space-surveillance>; Alexandra Witze, "The Quest to Conquer Earth's Space Junk Problem," *Nature*, September 5, 2018, available at <www.nature.com/articles/d41586-018-06170-1>.
- ²⁸ Brian Weeden and Victoria Samson, eds., "Global Counterspace Capabilities: An Open Source Assessment," *Security World Foundation*, April 2020, available at <https://swfound.org/media/206970/swf_counterspace2020_electronic_final.pdf>; Todd Harrison et al., *Space Threat Assessment 2020* (Washington, DC: Center for Strategic and International Studies, March 2020), available at <https://aerospace.csis.org/wp-content/uploads/2020/03/Harrison_SpaceThreatAssessment20_WEB_FINAL-min.pdf>.
- ²⁹ Nathan Strout and Aaron Mehta, "Russia Conducted Anti-Satellite Missile Test, Says U.S. Space Command," *C4ISR*, April 16, 2020, available at <www.c4isrnet.com/battlefield-tech/space/2020/04/15/russia-conducted-anti-satellite-missile-test-says-us-space-command/>;

Aaron Mehta, "Indian Anti-Satellite Test Shows Importance of Space Force, Shanahan Says," *Defense News*, March 27, 2019, available at <www.defensenews.com/global/asia-pacific/2019/03/27/in-wake-of-indian-anti-satellite-test-shanahan-urges-thoughtfulness-in-space/>.

³⁰ "President Donald J. Trump Is Unveiling an America First National Space Strategy," The White House Briefings and Statements, March 23, 2018, available at <<https://www.whitehouse.gov/briefings-statements/president-donald-j-trump-unveiling-america-first-national-space-strategy/>>; Alexander Bowe, *China's Pursuit of Space Power Status and Implications for the United States* (Washington, DC: U.S.-China Economic and Security Review Commission, April 11, 2019), available at <https://www.uscc.gov/sites/default/files/Research/USCC_China%27s%20Space%20Power%20Goals.pdf>; Anatoly Zak, *Russian Military and Dual-Purpose Spacecraft: Latest Status and Operational Overview* (Arlington, VA: CNA, June 2019), available at <https://www.cna.org/CNA_files/PDF/IOP-2019-U-020191-Final.pdf>; Omar Lamrani, "Great Power Competition Feeds the Threat Posed by Anti-Satellite Technology," *Real Clear Defense*, April 19, 2019, available at <www.realcleardefense.com/articles/2019/04/19/great_power_competition_feeds_the_threat_posed_by_anti_satellite_technology_114350.html>.

³¹ Francis C. Domingo, "Conquering a New Domain: Explaining Great Power Competition in Cyberspace," *Comparative Strategy* 35, no. 2 (March 2016), 154–168, available at <<https://doi.org/10.1080/01495933.2016.1176467>>.

³² "Fact Sheet: 2018 DoD Cyber Strategy and Cyber Posture Review," Department of Defense, available at <https://dodcio.defense.gov/Portals/0/Documents/Factsheet_for_Strategy_and_CPR_FINAL.pdf>.

³³ Ellen Nakashima, "Pentagon Launches First Cyber Operation to Deter Russian Interference in Midterm Elections," *Washington Post*, October 23, 2018, available at <www.washingtonpost.com/world/national-security/pentagon-launches-first-cyber-operation-to-deter-russian-interference-in-midterm-elections/2018/10/23/12ec6e7e-d6df-11e8-83a2-d1c3da28d6b6_story.html>.

³⁴ Lyu Jinghua, "What Are China's Cyber Capabilities and Intentions?" Carnegie Endowment for International Peace, April 1, 2019, available at <<https://carnegieendowment.org/2019/04/01/what-are-china-s-cyber-capabilities-and-intentions-pub-78734>>; Matthew Bey, "Great Powers in Cyberspace: The Strategic Drivers Behind U.S., Chinese, and Russian Competition," *Cyber Defense Review* (Fall 2018), 5, available at <https://cyberdefensereview.army.mil/Portals/6/Documents/CDR%20Journal%20Articles/Fall%202018/BEY_CDR_V3N3.pdf?ver=2018-12-18-101630-063>.

³⁵ Michael Kofman, "Russia's Great Power Raiding Strategy," *War on the Rocks*, June 14, 2018, available at <<https://warontherocks.com/2018/06/raiding-and-international-brigandry-russias-strategy-for-great-power-competition/>>.

³⁶ Bey, "Great Powers in Cyberspace," 5.

³⁷ *National Cyber Strategy of the United States of America* (Washington, DC: The White House, September 2018), available at <<https://www.whitehouse.gov/wp-content/uploads/2018/09/National-Cyber-Strategy.pdf>>; Dmitry Adamsky, "From Moscow with Coercion: Russian Deterrence Theory and Strategic Culture," *Journal of Strategic Studies* 41, nos. 1–2 (2018), 33–60, available at <<https://doi.org/10.1080/01402390.2017.1347872>>; Nicu Popescu and Stanislav Secieru, eds., *Hacks, Leaks, and Disruptions: Russian Cyber Strategies*, Challi Paper No. 148 (Paris: European Union Institute for Security Studies, October 2018), available at <https://www.iss.europa.eu/sites/default/files/EUISSFiles/CP_148.pdf>; Jinghua, "What Are China's Cyber Capabilities and Intentions?"

³⁸ *Summary of the 2018 National Defense Strategy*, 14.

³⁹ *National Strategy for Counterterrorism of the United States* (Washington, DC: The White House, October 2018), available at <https://www.dni.gov/files/NCTC/documents/news_documents/NSCT.pdf>; *The DHS Strategic Plan: Fiscal Years 2020–2024* (Washington, DC: Department of Homeland Security, 2019), available at <https://www.dhs.gov/sites/default/files/publications/19_0702_plcy_dhs-strategic-plan-fy20-24.pdf>; Anthony H. Cordesman, *China's New 2019 Defense White Paper: An Open Strategic Challenge to the United States, But One Which Does Not Have to Lead to Conflict* (Washington, DC: Center for Strategic and International Studies, July 2019), 3–5, available at <<https://www.csis.org/analysis/chinas-new-2019-defense-white-paper>>; Olga Oliker, "Unpacking Russia's New National Security Strategy," Center for Strategic and International Studies, January 7, 2016, available at <<https://www.csis.org/analysis/unpacking-russias-new-national-security-strategy>>.

⁴⁰ "The Stern Review on the Economic Effects of Climate Change," *Population and Development Review* 32, no. 4 (2006), 793–798.

⁴¹ Wyatt Scott, Francis Gassert, and Sharon Burke, "Great Power Resource Competition in a Changing Climate," *New America's Natural Security Index*, October 23, 2019, available at <<https://www.newamerica.org/resource-security/reports/great-power-resource-competition-changing-climate/>>.

⁴² "Global Greenhouse Gas Emissions Data," Greenhouse Gas Emissions, Environmental Protection Agency, last modified September 13, 2019, available at <www.epa.gov/ghgemissions/global-greenhouse-gas-emissions-data>.

⁴³ Avinash Godhole, "Paris Accord and China's Climate Change Strategy: Drivers and Outcomes," *India Quarterly* 72, no. 4 (2016), 361–374.

⁴⁴ "Global Greenhouse Gas Emissions Data."

⁴⁵ Godhole, "Paris Accord and China's Climate Change Strategy."

⁴⁶ "Global Greenhouse Gas Emissions Data."

⁴⁷ Godhole, "Paris Accord and China's Climate Change Strategy."

⁴⁸ Ellen Drage O'Reilly, "WHO Warns of 13 Emerging Health Threats Including Possible Pandemics," *Axios*, January 20, 2020, available at <www.axios.com/who-emerging-health-challenges-pandemic-climate-change-inequalities-f83e0dd1-6d00-40c4-aaca-11523270648c.html>.

⁴⁹ For a more detailed review of these, see Scott, Gassert, and Burke, "Great Power Resource Competition in a Changing Climate."

⁵⁰ *Ibid.*

⁵¹ Roger A. Pielke, "What Is Climate Change?" *Issues in Science and Technology* 20, no. 4 (2004), 31–34; Godhole, "Paris Accord and China's Climate Change Strategy"; Barry Rabe, *Statehouse and Greenhouse: The Emerging Politics of American Climate Change Policy* (Washington, DC: Brookings Institution Press, 2014); Nate Hultman, "Climate Change and the Next Administration," in *Brookings Big Ideas for America*, ed. Michael E. O'Hanlon (Washington, DC: Brookings Institution Press, 2017), 167–175; David Sandalow, *Guide to Chinese Climate Policy* (New York: Columbia SIPA Center on Global Energy Policy, 2019); Bo Wang, "Exploring China's Climate Change Policy from Both International and Domestic Perspectives," *American Journal of Chinese Studies* 16, no. 2 (2009), 87–104; Laura A. Henry and Lisa McIntosh Sundstrom, "Russia's Climate Policy: International Bargaining and Domestic Modernization," *Europe-Asia Studies* 64, no. 7 (2012), 1297–1322.

Past Eras of Great Power Competition

Historical Insights and Implications

By Thomas F. Lynch III and Frank Hoffman

The chapter reviews the major contemporary theories about interstate power competition and state power transitions. It surveys many of the recent major studies about Great Power transitions since 1500, establishing that the vast majority of such transitions include some form of direct Great Power clash (war). The chapter develops a framework for evaluating the main competitive categories of Great Power competition (GPC): political and diplomatic, ideological, informational, military, and economic. It then applies these categories in analysis of four distinct dyadic rivalries contested in three post-1780 eras of GPC: the United Kingdom (UK) and France; UK and Imperial Germany; UK and the United States, and the United States and Imperial Japan. These eras were chosen due to several important parallels with the emerging era of GPC. It concludes with 10 major insights that hub around the broad conclusion that although periods of Great Power rivalry that involve major power transitions generally lead to direct clash (war) between them, adept statesmanship can arrest this tendency if properly attentive to both the geopolitical and domestic drivers of Great Power war.

This chapter provides a short overview of the historical context and construct for understanding the emerging era of Great Power competition (GPC). It begins with a discussion of the theoretical bases for understanding power: hard power and soft power, as well as the manner in which scholars of international affairs have understood the nature of Great Power competition and the transition between Great Powers. The chapter then operationalizes the main historic dimensions of interstate competition, focusing on five major categories of Great Power interaction: political and diplomatic, ideological, informational, military, and economic. It provides a compact overview of the major research literature findings about GPC dynamics and outcomes over the past 500 years. The chapter then features an analytical review of four cases of rivalry dyads contested during eras of GPC, evaluated across the five categories of state-to-state interaction. It concludes with delineation of 10 major insights and implications about Great Power competitions and transitions that seem germane for the dawning era of GPC.

Understanding Great Power Identities and Transitions

For the past several decades, political scientists and international relations theorists have written about global interactions while referencing the relative difference in power between the major protagonists. In realist theories, states are the primary actors, and they are locked in a constant struggle for security (a security dilemma) in a system that features anarchy (the absence of any supreme authority or sovereign). For realists, the anarchic international system gravitates toward confrontation and war absent a dominant power, or a balance of power among several states.¹ Realist musings focus on the relative disparity between post-Westphalian states' military power (or hard power) differences. In neorealist theory, military power remains the basis of the relative power measurement—but the disparity is understood with reference to the “polarity” of the wider international system, or the number of states with “top drawer” hard power capabilities.² Alternatively, liberal and institutionalist theories dispute the deterministic nature of international system anarchy. They believe that durable interstate cooperation is feasible with the right international framework—like some constellation of cooperative rules, norms, and institutions. They compare international actor power with a wider array of measures, or soft power aspects, including economic size, diplomatic capability, cultural and social attractiveness, and informal relationships built on rules, norms, and protocols.³ Finally, constructivist international relations theory disagrees with the notion that anarchy or any permanent tendency conditions the international system at all. For constructivists, the individual actors together make the international system anything they want it to be.⁴

International relations scholars also pay some attention to the rise and fall in the relative power positions of international actors—with greatest attention on the transitions in power positions among and between those actors on the high end of the power spectrum, that is, the Great Powers.⁵ The realist framework for understanding Great Power transitions focuses on military capabilities and the manner in which these rise and fall based on individual states' choices about developing and projecting military power. The neorealist modification of realism agrees that military power capabilities establish the essential framework for relative status transitions, but neorealists argue that it is the distribution of military power across the international system (the relative polarity of the system)—not individual state choices—that sets in motion power shifts among the states. Neorealism foretells unipolarity (a circumstance where one state is militarily dominant over all other states) is not only least likely to result in catastrophic violence but also inherently untenable. The multitude of smaller states will combine in balance against that dominant power, reducing it with a thousand tiny cuts until its military capabilities are exhausted. The once-dominant state then is caught and passed by one or more rising states. Neorealists contend that while bipolarity is inherently most stable, all other forms of multipolarity across the international system will encourage greater degrees of instability, direct violence among and between the Great Powers, and drive turbulent, hard power-dominated transitions among the major powers.

Hegemonic stability theory (HST) advances an alternative framework for understanding power transitions among the major actors, that is, the states. While not a realist theory, HST agrees with realism that relative military power is the bedrock for understanding status relations among the states at any given time. HST contends that a Great Power's temporal dominance in military attributes (its relative position as a hegemon) enables the framing of

system-wide rules, norms, and procedures in interstate relations that favor that dominant state's ideology and policy preferences. These rules also reduce the costs to police that system. HST is an international political economy-based, game theory aligned understanding of the conditions for interstate cooperation rather than competition. Yet most HST proponents align with realists in one critical point: it is the hard power capabilities and the will to use of the dominant state that underwrite the rules-based relationships among the actors. Thus, without a dominant state (hegemon) willing to use its hard power for enforcement, cooperative and collaborative arrangements inevitably will give way to the inherent chaos and violence of the anarchical world system. Then a new state will seek the dominance necessary to establish its own norms and institutions as ascendant across all elements of international interaction.⁶

Liberal and institutional constructs contend that Great Power state transitions need not become violent contests. As noted earlier, they do not believe that anarchy is the default setting for international relations. Instead, they view durable interstate cooperation as feasible with proper arrangements. Thus, in limited agreement with HST, liberal institutionalists believe that a hegemon (or even a combination of Great Powers) can establish norms, rules, and procedures for interstate relations that are fundamentally cooperative and collaborative. Unlike HST advocates (and also realists and neorealists), liberal institutionalists believe that well-constructed cooperative and collaborative norms, rules, and institutional arrangements can take on a life of their own, long outlasting the military power and dominance of the state(s) that created them. In liberal institutionalism, it is the cumulative cost of challenging cooperative arrangements with more competitive and conflictual ones that dissuades a break from well-established, peace-sustaining norms.⁷ For constructivists, the transitions in global order—from cooperative to conflictual or from clash to collaboration—are dependent on the state-to-state interactions. It is about the choices made by the states themselves—and in particular the role of individual decisionmakers in each state and how leadership decisions shape the understandings of their people and other leaders in the global system—that establish the basic conditions of the system itself.⁸

The history of Great Power transitions over the past 500 years does not provide optimism for those hoping to see modern U.S. dominance culminate with a peaceful ascent of a successor state or combination. American political scientist Graham Allison's study of 16 major cases of rising power(s) versus an established Great Power found that only 4 of those cases—or 25 percent—ended without war.⁹

Allison's work—and that of many other international relations analysts and historians over the past decade—captures the dominant worry of the moment: Must the transition of relative power dominance between the United States and its successor become violent? During the generation from 1992 to 2008, this worry was mooted. The United States stood alone—atop the power structure at a unipolar moment in history.¹⁰ Its post-World War II bipolar rival, the Soviet Union, was vanquished, and with it Moscow's vision of universal communism and the superiority of command-directed economics.¹¹ The next most feasible rising power, China, had declared itself in 1978 an aspirant to the international capitalist economy and found itself over the ensuing 30 years to be the benefactor of an American orthodoxy. In this orthodoxy, the rise of a Chinese entrepreneurial middle class would inevitably demand political liberalization and individual freedoms that would swamp the

"In the 1990s and 2000s, American leaders believed that Russia and China were converging with the West on basic questions of world order. Countries would work together on common challenges while old geopolitical rivalries would matter much less. The 'era of convergence' came to an end because Russian and Chinese leaders concluded that if the liberal order succeeded globally, it would pose an existential threat to their regimes."

—Thomas J. Wright, *"The Return to Great-Power Rivalry Was Inevitable,"*
The Atlantic, September 12, 2018

Chinese Communist Party and see Beijing become a co-stakeholder in the web of liberal rules, norms, and institutions nurtured by American power from 1945 to 1990.¹²

After 2008, the promise of a reliably cooperative set of relations among these states was shaken, but not fully jettisoned. Almost a decade into his increasingly illiberal domestic political regime, Russian President Vladimir Putin oversaw a short, sharp invasion of neighboring Georgia and declared Russia a badly treated country that deserved Great Power recognition that the United States was unwilling to grant. That same year, Beijing—long wary of America's insistence that Western values of liberal democracy and human freedoms were lagging in China—witnessed American capitalism narrowly avoid a complete, catastrophic financial

meltdown. China took away a lesson that there can and must be a Chinese alternative to American global dominance. By 2014–2015, the dawn of a new competition-dominant era came fully into view—as a de facto if not fully acknowledged era of Great Power rivalry. Russia conducted a covert military invasion of Crimea and annexed it from the Ukraine. U.S.-led Western countries slapped Russia with various forms of economic sanctions and expelled Moscow from diplomatic and economic organizations that it had belonged to for a couple of decades. During those same years, Chinese territorial assertiveness in the South China Sea and its increasingly restrictive business and communications practices at home catalyzed competitive-to-confrontational American responses: insistence of unfettered freedom of navigation, open condemnation of Chinese industrial espionage practices, and exclusion of China from planning for a broad new trans-Pacific trading partnership.

It took another couple of years, but the passing of America's "unipolar moment" begat the U.S. National Security Strategy of 2017 and National Defense Strategy (NDS) of 2018. Both declared that a two-decade-long dominant paradigm of cooperation in international relations was over, and competition would be the hallmark of the way forward among the three dominant states. A fully acknowledged era of GPC had begun. As noted in chapter 1, the dominance of a competitive framework in the emerging era does not exclude many parallel residual and dynamic elements of cooperation and collaboration from the past. Liberal institutionalist theories are under duress, but far from disproved. At the same time, growing competition among the Great Powers does not make realists or neorealists correct; it remains uncertain that the only potential outcome from this new era of GPC is overt confrontation and violent conflict.

Instead, the reemergence of GPC in this new era puts the world in a phase that has been the norm over the past 500 years, just not in recent memory. It is thus informative to extract a set of representative historical cases in which competition among the Great Powers provides useful lessons about the nature and evolution of such competitive eras and the transformation of relative power that often ensues. First, we must establish the key

Table 2.1. Frameworks for Thinking About Categories/Dimensions of Competition

	Howard (1974)	Friedberg (1988)	Kennedy (1987)	NDS (2018)
Social and Cultural	X			
Military		X	X	X
Political and Diplomatic				X
Science and Technological	X		X	X
Logistical	X			
Ideological		X		
Economic	X	X	X	X
Financial		X		
Informational				X

dimensions of GPC: How do they compete? Then how have some critical cases of historic competition been conducted in the contested categories?

How Do States Compete? The Dimensions of Interstate Competition

It is important to establish the main historic dimensions of interstate competition. Since the Treaty of Westphalia in 1648 codified statehood as the recognized form of geopolitical organization, state-to-state competition has been multifaceted, incorporating an array of state interactions. The international relations and security studies communities have used a variety of different dimensions of strategy or national power to capture the complexity of competition.

In the post-Vietnam era, British historian Sir Michael Howard identified four dimensions of national strategy and competition reflective of successful wartime circumstance.¹³ A decade later, American political scientist Aaron Friedberg defined four different categories of state-to-state power competition resonant during a Great Power transition in his assessment of Great Britain's relative power decline over the late 1800s and early 1900s.¹⁴ About the same time, Yale historian Paul Kennedy in *The Rise and Fall of the Great Powers* stressed three major areas focusing on a state's economic production base and productivity as key dimensions of Great Power competition while highlighting the importance of the military dimension.¹⁵ Leveraging these three authors and more, the NDS defined five dimensions of GPC in its conception of the expanded competitive space relevant to the future security environment. The various concepts found in these major works about GPC are summarized in table 2.1.

"The competition is likely to be multilayered and interactive. No single theme or model will capture the complex mosaic of global competition, and the intersections among diverse types of competition—how success or failure in one area exacerbates or mitigates others—will be a crucial determinant of relative success."

—Michael J. Mazarr et al., *Understanding the Emerging Era of International Competition*, RAND (2018)

Subsequent to the publication of the NDS, RAND political scientist Michael Mazarr and his colleagues conducted a historical review and identified seven objectives of traditional interstate competition:

- power and security
- status, standing, and prestige
- material and economic prosperity
- resources
- territory and sovereign claims
- values and ideology
- rules, norms, and institutions of the system.¹⁶

The seven objectives added texture but did not fundamentally reshape the construct of the five competitive categories (or aspects of competition) delineated in the NDS. Thus, for purposes of this chapter and the wider framework found in the volume, the competitive categories listed in table 2.2 will be applied to analyses of Great Power interactions.

Understanding Prototypical GPCs: Four Cases

In the past 5 centuries, Great Power transitions have played out over decades, not years. As noted earlier in this chapter, in 16 historical cases of GPC from the late 1400s to the present studied by a Graham Allison–led team at Harvard in the 2017 version of the *Thucydides's Trap Case File*, 12 of them (75 percent) resulted in Great Power war.¹⁷ In a review of great strategic rivalries from the classical world to the Cold War, U.S. Marine Corps War College military historian James Lacey's 2016 edited volume *Great Strategic Rivalries* demonstrated that rarely do strategic competitions during eras of major state power shifts end without at least one direct major military clash. The U.S.-Soviet Cold War denouement and the United Kingdom (UK) accommodation of U.S. power at the end of World War II stand out as just 2 of 15 Great Power transition cases where a major direct military clash did not occur.¹⁸ The University of Michigan's decades-old *Correlates of War Project* chronicles major and minor wars since the Napoleonic era and provides significant evidence that the dawn of Great Power rivalries inevitably introduces a heightened risk of major war into the international system.¹⁹ Moreover, since the fall of Napoleon in 1815, over half of all wars have been between enduring Great Power rivals. If one adds early conflicts among proto-rivals, that number climbs to over 80 percent.²⁰ Thus, the most persuasive research done on past eras of GPC demonstrates that a majority of them involved power transitions among Great Powers and that a full three-quarters of them culminated with—or featured within the competitive transition period—a destructive period of violent Great Power clash (war).

And yet the inevitability of direct military clash among Great Powers during times of relative power transition is not foreordained. As neorealists observe, the structure of the international system can mitigate competition so that it culminates without direct combat—with a bipolar system being most likely to remain stable and great state competition settling into patterns of rivalry short of war.²¹ Other systemic factors may help Great Powers channel (or expend) their worst animus in one of the other four nonviolent categories of interstate competition: politico-diplomatic, economic, ideological, or informational. Liberal

internationalists offer a “liberal peace” theory, arguing that states with shared ideological values—including liberal democratic political institutions and deference to individual human rights—are resistant to major war with each other.²² An alternative institutionalist thesis asserts that states with a high degree of economic interdependence are more likely to compete in categories and a manner short of war.²³ Constructivists assert that national leadership can condition interactions and cultures to refrain from confrontation and clash with attention to social norms and cultural symmetries that build identities and communities of peace.²⁴ Great Powers also may vent violent tendencies in surrogate conflicts and proxy wars. They may seek out networked partnerships or alliance systems with combinations of lesser powers in order to increase their prospects for successful competition in nonmilitary categories and/or to deter any move toward armed clash. Or they may seek common arrangements to restrain confrontation and inhibit armed clash within agreements such as the informal Concert of Europe (1815–1854), the League of Nations (1919–1930), and the United Nations (1945–present).²⁵ Finally, the enormous destructive power and generational damage wrought by nuclear weapons may—as it has seemingly done since 1945 with rare exception—inhibit a resort to major violence and warfare between Great Powers.²⁶

With the five primary dimensions of interstate competition established and the general understandings of research into past GPCs and power transitions highlighted, this chapter now turns to a short analytical review of several representative historical cases.²⁷ Four dyadic cases within broader periods of GPC have been selected:

- France and the UK, from the late 18th to the early 19th centuries
- the UK and Imperial Germany, from the late 19th to the mid-20th centuries
- the UK and the United States, from the late 19th to mid-20th centuries
- the United States and Imperial Japan during the early 20th century.²⁸

These four rivalries were contested in a multipolar global system, like the era emerging in 2020. All four featured an ascendant Great Power worried about relative power decline and at least one rising power seeking recognition and status—also evident in 2020. Finally, all four were contested during periods of emergent, disruptive technologies driving global economic dynamics from one paradigm to another. In the late 1700s, agrarian economies and mercantilist trade preferences were giving way to industrialization and free trade networks across Western Europe. By the late 1800s and early 1900s, maturing industrial economies acted out globally across a landscape featuring contested colonial empires and a growing number of protectionist trading networks. In 2020, the world is moving beyond industrial economies and even digitized ones to those featuring the hallmarks of a fourth industrial revolution: the blurring of boundaries between the physical, digital, and biological worlds.²⁹ At the same time, powerful political forces are questioning the wisdom of global free trade, but without any clear alternative framework.

Three of these rivalries resulted in war—a percentage consistent with the broader historic percentage of Great Power competitions culminating in war discussed earlier. The wars fought out of these competitive transition eras were three of the five most deadly conflicts in human history: World War II, World War I, and the Napoleonic Wars, respectively.³⁰ Only in the case of the rise of the United States in an era of relative UK power decline did a

Table 2.2. A Framework for Assessing the Aspects/Categories of Competition	
Competitive Aspect/Category	Main Competitive Elements
Political and Diplomatic	Levels of influence in multilateral institutions, key posts held that control multilateral institutions, number and strength of political alliances.
Ideological	Values and political systems' appeal.
Informational	The manner and degree of transnational communications: open and transparent vs. closed and restrictive. Extent of denigration of "the other" in mass communications.
Military	Size, posture, technological edge of armed forces. Cohesion and capacity of military alliances.
Economic	Size, technological breadth, diversity, and resource base of national economy. Innovation ecosystem of national economy, including access to and management of financial capital.

Table 2.3. Major Dyadic Dynamics During Four Eras of Great Power Competition				
	France/UK	UK/Germany	United States/UK	United States/Japan
Political and Diplomatic	UK ascendancy challenged	Competition over global colonies; later continental hegemony by Germany	UK global colonies preeminence and UK dominance of western Atlantic Ocean challenged	U.S. Asia-Pacific hegemony and rules for China
Ideological	Aristocracy vs. Populist, charismatic rule	Limited ethnic prestige issues; later clash over human rights and liberties	Limited ethnic tensions, only lingering with Irish-Americans	Liberal democracy vs. divine rule and messianism
Informational	Exchange among elites vs. empowerment of popular passions; negative popular imagery	Press-fueled nationalism/xenophobia; later clash over state-dominated propaganda and expression limitations	Limited and linked to economic concerns; favorable reciprocal popular press	Open, individual communications vs. hierarchical information; xenophobic press eventually
Economic (with science and technology factors)	Trade and commercial dominance across Europe (level of industrialization)	Lines of communication and colonial preferences (chemical and industrial manufacturing industries)	UK colonial trade order vs. U.S. commerce preferences (manufacturing innovation mainly in electromagnetic spectrum)	"Open Door Policy" vs. Co-Prosperity Sphere (access to metals, oil products, rubber)
Military	Royal Navy vs. France's Conscripted, Massed Army	Naval arms race	Limited to naval parity	Naval supremacy (emerging aviation)

century of chilly interstate contestations resolve short of war. The U.S.-UK dyad ultimately produced an alliance and partnership during two global wars against other Great Power competitors. Their victorious alliance in the Second World War made way for the final accommodation of ascendant U.S. power into an alliance oriented toward a wider period of system stability and international peace.

A summary of the major dynamics of these four GPC dyads across the five dimensions of interaction listed in table 2.2 is provided in table 2.3. A short analytical review of each of



Figure 2.1. Britain (William Pitt) and Napoleon Carving Up the World. Source: James Gillray, *The Plumb-Pudding [sic] in Danger* (London: H. Humphrey, 1805).

these episodes follows that table. The reviews illuminate wider lessons about GPC histories, transition dynamics, and patterns of competitive interaction with relevance for 2020.

France and the United Kingdom, Late 18th to Early 19th Centuries

Between 1790 and 1820, longstanding European powers and rivals France and the United Kingdom entered a period of intensified competition and confrontation that resulted in two major wars—the French Civil War (and its continental spillover from 1792 to 1802) and the Napoleonic Wars (1803–1815). The international system at that time was multipolar, with other Great Power contestants including Russia, Austria-Hungary, Prussia, and the Ottoman Empire.

The UK and France had been competitors for more than a century. From the 1680s through the 1780s, the two jostled globally over colonies and international resource access and on the continent of Europe over economic and religious ascendancy. For the 100-year period from 1648 to 1763, France was the dominant power and the UK its rising challenger.³¹ By the end of the 1700s—after four decade-long periods of continental war and interrelated colonial proxy war between the two conducted over a century—the power relationship had flipped. The UK commenced the 1790s as the dominant global and European power in command of the seas with its Royal Navy, flourishing as the most rapidly industrializing state astride Europe.³²

From the 1500s to the mid-1700s, Great Britain and France contested economic supremacy in a mercantilist system. Mercantilist economics was based on the premise that

a nation's relative wealth and power were best secured by the accumulation of land, gold, and silver. The accumulation of overseas colonies also generated national wealth and relative power by assuring the export of home country goods to loyal colonists while limiting the number of foreign products available for purchase in those colonies. In the mid-1700s, France squandered its mercantilist advantages in continental land mass and in its overseas colonial presence. It lagged in modern techniques of crop rotation and fertilizer use, and its industrial advances in textiles, mining, and metallurgy came mainly from innovation by British entrepreneurs. The Seven Years' War (1756–1763) increased royal debt and then lost France nearly all its North American colonial possessions. At the same time, the British industrial revolution began. Its innovations in steam engines, textile machines, tool-making, and railroads vaulted the UK into position as the world's most prosperous economy. In turn, UK businessmen became leaders in international commerce, trade, shipping, and banking—with London quickly becoming the financial capital and focus of the world economy. The British military, and especially its Royal Navy, grew from the wealth of the nation. London leveraged this military advantage to assure preferential trade and exchange between growing British colonies and the home country.

Early British public empathy for the 1789 French Revolution waned by 1793. Never amused by the revolution, the British government first supported a failed Austro-Prussian military effort to march on Paris and end the revolution in the fall of 1792. Then the vulgar beheading of King Louis XVI and the bloody excesses of the Reign of Terror compromised all but the rashest British common folk sympathies for the rebels. Britain passed an Aliens Act that prohibited French radicals from travel into the UK, and London kept a wary eye on France's activities. While the early years of revolutionary chaos in France helped Britain's overall power status, persistent French-inspired turbulence on the continent left London worried about roiling instability there. The populist interim French government and its successor Napoleonic Empire both promised to topple or replace the Divine Right and standing of the monarchies across Europe. With this persistent ideological threat, the French Revolution introduced outright political and ideological conflict into what had been a serious but often constrained competitive rivalry among fellow monarchies. The rules of competition that dominated (and often moderated) the normally bitter French-English competition from the late 17th to mid-18th centuries vanished.

Populist France also posed a threat to the UK and fellow continental monarchies' preferred means of communication and information exchange among royal elites—threatening to directly agitate anti-monarchy massed unrest in London and across the continent.³³ The UK's economic access to the continent through the Low Countries and Spain was put in jeopardy by revolutionary France's crusade-like activism, sending mass armies into Austria, Belgium, Italy, the Netherlands, and Spain. England's lead in industrial productivity also seemed in jeopardy should France conquer England's favored European economic partners.³⁴ France's turn to universal military conscription for its army (*levée en masse*) and a new preference for total war instead of limited, seasonal campaigns—begun during the Directorate and perfected by Napoleon—disadvantaged traditional monarchic armies and threatened unconstrained French continental dominance. Fearful that France might establish absolute continental power and turn that power outward into serious confrontation against the UK's overseas colonial holdings—a fear crystalized during the 1798–1801

French campaign into Egypt and Syria—British resolve steeled. London forged continental military alliances against France and commenced armed military conflict geared to contain and then collapse France's expansionist aims and ambitions.³⁵

This UK-France period of dyadic rivalry was contested during a 26-year period following a dramatic upheaval in French politics and against a backdrop of multiple Great Powers jockeying for position in Europe—a relatively unstable multipolar system. It peaked into full-out warfare when London's ascendancy on the seas and largely unfettered access to preferred overseas colonies, once contested by Spain and France, seemed to be jeopardized by a radicalized France. The period of French-British competition turned exceptionally violent when the prospects for collaboration and/or accommodation between them in the five major areas of state-to-state interactions became seemingly impossible. The UK turned to anti-French military ground alliances across continental Europe and leveraged its superior navy to destroy the French fleet at Trafalgar in 1805. These direct military clashes confined revolutionary France into a landlocked country with limited economic activities and constrained ideological reach.³⁶ The UK's clever exploitation of allied and partner continental armies to converge against French military forces from 1813 to 1814 and again in 1815 eliminated the main French threat and reset the norms of monarchy-to-monarchy power competition on the European continent in political, ideological, informational, and economic areas of interaction. In a display that realist theorists would label an astute balance of power maneuver, Great Britain reestablished—for a time—its self-perceived relative power advantage against all European states, its general freedom of economic action on the continent, and its unrivaled ascendancy in overseas commerce. It is important to note that the main domain of commerce and communications/information of that time—the high seas—was the arena of competition where the UK worked most vigorously and independently to sustain its dominance and to secure future ascendancy over its Great Power rival, revolutionary France.

Great Britain and Imperial Germany, Late 19th to Early 20th Centuries

From 1870 to 1945, the United Kingdom again found itself challenged by a new and rising Great Power, Imperial Germany. Germany rose to challenge UK dominance in an era marked by a multipolar distribution of global power. The UK was the globally ascendant power; France was an established but constrained Great Power; Russia, Austria-Hungary, and the Ottoman Empire were established but declining Great Powers; and the United States, Germany, and Japan were economically rising states and aspiring Great Powers.

Berlin threatened London by direct competition for overseas colonies enabled by an ascent to be a global sea power and with potential dominance on the European continent. This dyadic rivalry played out within a wider era of GPC and culminated in two major global conflagrations: World War I and World War II. Initially, Imperial Germany's challenge to an ascendant United Kingdom was an economic one and without sharp competition in the other four categories of major state interaction (see table 2.2). As they consolidated control of a majority of the European continent via wars in 1864, 1866, and 1870–1871, Imperial Germany's founders effectively leveraged expanding German access to natural resources and labor. They forged a state with industrial prowess and manufacturing capacity that surpassed that of the long-dominant United Kingdom.³⁷ In 1870, the UK produced almost

Table 2.4. Great Power Iron and Steel Production in Selected Years (in Tons)

	1870	1910	1935	1950
UK	5,819,492	12,050,361	12,905,243	18,800,000
Germany	1,560,000	25,500,000	6,498,873	10,600,000
United States	375,000	25,643,871	51,100,000	80,100,000
France	1,417,073	11,200,000	33,301,000	10,600,000
Russia	2,336,000	11,900,000	27,918,000	53,200,000
Japan	0	180,000	4,703,000	7,800,000

Sources: Tibor Fabian, "A Linear Programming Model of Integrated Iron and Steel Production," *Management Science* 4, no. 4 (July 1958), 415–449; M.S. Birkett, "The British Iron and Steel Industry," *Economica*, no. 5 (June 1922), 149–161; John B. Parrish, "Iron and Steel in the Balance of World Power," *Journal of Political Economy* 64, no. 5 (October 1956), 369–388; Muzaffer Erselcuk, "Iron and Steel Industry in Japan," *Economic Geography* 23, no. 2 (April 1947), 105–129; Abraham Berglund, "The Iron and Steel Industry of Japan and Japanese Continental Policies," *Journal of Political Economy* 30, no. 5 (October 1922), 623–654; Charles Will Wright, *The Iron and Steel Industries of Europe*, Economic Paper 19 (Washington, DC: Government Printing Office, 1939); S.L., "Iron and Steel in the Soviet Union," *The World Today* 8, no. 5 (May 1952), 210–222; Robert C. Allen, "International Competition in Iron and Steel, 1850–1913," *The Journal of Economic History* 39, no. 4 (December 1979), 911–937; James E. Rowe, "The Development of the Russian Iron and Steel Industry," *The Geographical Bulletin*, vol. 10 (May 1975), 24–30; Daniel J. Morrell, "Iron and Steel," in *Reports of the United States Commissioners to the Paris Universal Exposition, 1878*, vol. 3, *Iron and Steel, Ceramics and Glass, Forestry, Cotton* (Washington, DC: Government Printing Office, 1880), 1–112; William P. Blake, "Ceramics," in *Reports of the United States Commissioners to the Paris Universal Exposition, 1878*, vol. 3, 113–224; John B. Parrish, "Iron and Steel in the Balance of World Power," *Journal of Political Economy* 64, no. 5 (October 1956), 369–388; Henry Harrison, "Iron Ore, Pig-Iron, and Steel Production," *Scientific American* 110, no. 22 (May 1914), 444.

four times as much iron and steel as Imperial Germany and by 1910 the Germans twice as much as the British (see table 2.4). In the subset of steel—the product of a more advanced industrial process and of greater use in modern machine tools, construction, and advancing implements—the UK produced twice as much steel as Imperial Germany in 1870, but by 1910 the Germans twice as much as the British.³⁸

Over the same 40-year period, Germany increased coal production by 800 percent; the British increase was only 200 percent.³⁹ As it grew, Imperial Germany protected its industrial manufacturing and its agricultural production from outside competition with steep import tariffs.⁴⁰ The UK had quit protectionism and embraced free trade in 1840 so had no similar import tariffs to counter the German taxes (or, for that matter, similar American tariffs). Despite recurring British government efforts to establish countervailing protectionist tariffs, domestic British export merchant and financial interests prevailed in preventing any new UK tariff regime that could hurt their strong export businesses.⁴¹ In 1910, the UK had twice as much in credit bank assets as Germany.⁴²

Beginning in the 1890s, Berlin turned its growing manufacturing advantage—especially in steel—into the quest for a world-class navy. The Kaiser and key German interest groups (or leagues) wanted a navy that could rival that of the UK for prestige and that could compete with imperial London for acquisition and maintenance of global colonies.⁴³ In 1897, Imperial Germany began emphasis of a so-called world policy (*Weltpolitik*) that shifted popular attention from growing domestic social issues to foreign policy by focusing on overseas colonial expansion and the construction of a high seas fleet.⁴⁴

Kaiser Wilhelm II perceived Germany's quest for a dominant navy as one to gain London's respect as a peer Great Power to be accommodated. But this fateful policy choice had an opposite effect in London.⁴⁵ Badly spooked, the UK made diplomatic moves during the 1890s to curtail Anglo-French enmity that had lasted over 800 years. London's diplomats also took steps to moderate a 60-year, often intense competition with Russia over colonies and boundaries in the Near East and Far East: the "Great Game."⁴⁶ London took these steps in order to seek an agreement with Paris and later Moscow to constrain Germany's rise on the European continent.⁴⁷ British diplomacy also undertook a "Great Rapprochement" with another putative Great Power rival of the era—the United States.⁴⁸



Figure 2.2. Germany, Britain, and Russia in Pursuit of Colonies. Source: Thomas Nast, "The World's Plunderers: Germany, England, and Russia Grab What They Can of Africa and Asia," *Harper's Weekly*, 1885 (Sarin Images/Granger).

By 1905, the UK turned the full British Empire's attention—including London's notorious press—toward the German threat. Germany returned the favor. The popular press in both Germany and the UK played a substantial role in accelerating Anglo-German hostilities. From about the mid-1890s, the British mass press featured a steady diet of invective against a militarized and menacing Germany. At the same time, the German press took delight in selling copy by accusing Britain of standing in the way of Germany's rightful place as a global leader. German papers also made their mark by publicizing British troops in South Africa as cravenly bayonetting to death innocent Boer babies in their conduct of the Boer War.⁴⁹ German publications fanned the flames of extreme nationalism advocated by the Pan-German League, founded in 1891 by Ernst Hasse.⁵⁰ This xenophobic, right-wing league had a notable influence on German public opinion against Anglo-Saxons, Yankees, and Russians before World War I and set the stage for the interwar rise of Nazism's hateful intolerance of all things insufficiently German.⁵¹

Of note, Imperial Germany and the UK had relatively similar political systems during their pre-World War I era of rivalry. Both were constitutional monarchies, and their aristocratic classes were heavily intermarried and related by blood. Both were growing more democratic during the period, each enfranchising more and more voters into its political systems. But unlike the cases of the UK and the United States addressed later, this political "sameness" did not limit descent into toxic rivalry.⁵²

UK fears of decline vis-à-vis Germany and German grievances against British haughtiness contributed to the development of a European military alliance structure steeped in complex animosities and tethered to hair-trigger war plans in the event of crisis. Germany's

major alliance partner, Austria-Hungary, was wrapped into a mesh of self-determination breakaway wars across Eastern Europe where Russian meddling fueled military grievance and a seemingly inevitable Russo-Austrian armed clash. In turn, the UK's 1907 Entente with Russia did not fully appreciate St. Petersburg's risky adventurism in the Balkans or the degree to which the Russian army was no match for the German war machine. Thus, each rival tethered itself to allies with great incentives to fight, and these incentives produced extremely risky war plans that set the conditions for the miscalculations of 1914 and the calamity of World War I.⁵³

The aftermath of World War I featured a "victor's peace," with stifling economic war reparations against Germany, insufficient British economic power to moderate global economic shocks, and a postwar economic titan—the United States—unwilling to undertake an economic or diplomatic leadership role.⁵⁴ These conditions accelerated global economic instability and military rivalries that fueled the rise of fascist and communist ideologies. The struggle over power transition and ascendance that generated World War I also set the conditions of fragmentation and confrontation between the UK, its allies, and Nazi Germany that followed the Great War. By the early 1930s, Nazi Germany and the UK again had major, intractable grievances across the political, economic, ideological, and informational dimensions of state-to-state interaction. These combined to produce a second, even more all-consuming global conflagration: World War II in Europe and the North Atlantic.⁵⁵

The Anglo-German competitive dyad was contested over a 75-year period and in a multipolar world—one that neorealism views as the least stable and most prone to interstate war. It evolved from a period of wary competition between the UK and Germany that lasted from 1870 to 1895, when political, ideological, informational, and economic forms of interaction were reasonably collaborative.⁵⁶ By 1900, this GPC dyad devolved into stark competition, confrontation, and major war when Germany moved to turn its economic advantage toward creation of a globally relevant navy and sought to become a colonial power on equal footing with the UK and France. After World War I, Germany's grievances, Britain's inability to fully recognize its accelerating decline of economic and political power as an outcome of the Great War, and the very wide differences between London's and Berlin's narratives about proper political, economic, human rights, and institutional norms set the stage for another violent and even more vicious clash of arms, World War II.

It is again worth noting that the UK became most worried that it would be eclipsed once Germany asserted a clear challenge for dominance over the principal medium of commerce and communications in that era—the high seas. After London's early 1900s determination of the primacy of Germany's challenge, the UK shifted into diplomatic, informational, and then military confrontation and clash with Imperial Germany.

The United Kingdom and the United States, Late 19th to Mid-20th Centuries

The United States and the UK participated in a dyad of Great Power competition from the 1880s through 1940. Their competition nested in the multiple rivalries between the many Great Power states of the era. The UK was the ascendant but declining power and the United States the rising one. London and Washington mainly contested naval dominance across the Western Hemisphere and about the rules and ascendant order in global economic exchange.⁵⁷ In 1870, U.S. gross domestic product became equal to that of the UK. By 1890, American industrialization rapidly eclipsed that of the UK, and by the turn of the



Figure 2.3. U.S. and UK Rapprochement—Uncle Sam and John Bull Shake Hands. Source: Victor Gillam, "Hands across the Sea," *Judge*, June 11, 1898.

century, American economic industrialization and manufacturing prowess was on the path to eclipse the size of the entire economy of the British Empire.⁵⁸ While the UK pursued an economic policy of openness, the United States established and maintained post-Civil War protectionist tariffs with special preference to industrial manufacturing and agriculture.⁵⁹ Yet U.S.-UK trade expanded throughout the period as London tolerated American protectionism while pursuing the consumer and financial services economic benefits of liberal international trade.

A century of Anglo-American political and ideological animus began a slow decline in the 1880s. From before the American Revolution, the United States demonstrated a strong distaste for the British colonial empire. So, too, British politicians had a deep disdain for American views about liberal democracy that denigrated constitutional monarchies and often called for toppling old European dynasties and the demise of colonial empires.⁶⁰ But during the 1880s and 1890s the UK became more democratic by enfranchising more voters, and the United States gained its own form of territorial empire.⁶¹ This growing sense of sameness along political and ideological lines allowed London and Washington to view each other's power in less threatening ways. Between 1887 and 1901, London concluded a number of treaties and agreements with the United States that settled residual questions about Canadian boundaries, Alaskan fisheries, and the future of the Panama Canal Zone, among others.⁶²

By the late 1890s, Americans began to convert economic might into diplomatic initiatives and naval military power. Where the Royal Navy had been ascendant over the Atlantic Ocean from the late 1790s, a small but increasingly capable and active U.S. Navy now began to assert itself. Rumors of possible war between the United States and the UK during 1895

Table 2.5. Percentage of International Exports: Selected Years

	1900	1910	1920	1935	1950
United States	14	11.1	22.4	11.6	16.7
Germany	10.9	11.2	3.8	8.9	3.2
UK	14.6	13.5	14	12.1	10
France	7.9	7.6	7.5	5.3	5
Russia	N/A	N/A	N/A	N/A	N/A
Japan	1	1.4	3.1	4	1.3

Source: United Nations (UN), *International Trade Statistics 1900–1960* (New York: UN, 1962).

and 1896 involving a territorial row between London and Venezuela caused panic in New York and worry in London.⁶³ Yet cooler heads prevailed in both countries. Diplomats navigated Anglo-American frictions, so they were viewed less moralistically and with greater realism, inhibiting the potential for violent clash.

Around the same time, dynamics in the multipolar international system came to reinforce the nascent U.S.-UK rapprochement. As previously noted, the UK perceived greater risks to its colonial holdings and to a hostile takeover of the seas from Germany.⁶⁴ British admiralty worries about the German naval buildup closer to home caused the War Office of the early 1900s to de-prioritize planning for possible war with the United States. Gradually, British politicians joined Royal Navy analysts in determining that America's economic might meant that Washington could afford a navy that Britain could never hope to match. Moreover, without any powerful UK allies in the Western Hemisphere, it would be folly to engage in a military quarrel with the United States.⁶⁵ The UK's decision to seek accommodation and collaboration with the United States at the turn of the 19th century (the Great Rapprochement) paid future dividends.⁶⁶ A historically isolationist United States eventually joined the UK in its fight against Imperial Germany during World War I and later against the consolidated European fascist powers led by Adolf Hitler's Germany in World War II. The United States also became the global standard bearer for liberal democratic norms and institutions after World War II when the UK's power and prospects for global influence sharply declined. Yet the British decision to accommodate rather than confront U.S. power negatively impacted long-term UK interests in one significant area—economics. Washington's distaste for the British colonial empire generated a post-World War II set of global commercial and financial rules and norms that transferred fiscal and trade dominance from the UK to the United States and that saw the rapid disintegration of the once mighty British colonial empire and an attendant decline in British trade as a percentage of global trade (see table 2.5).⁶⁷

The Great Rapprochement between the UK and the United States during 1895 and 1915 was far from the certain thing that is often assumed today.⁶⁸ In many ways, the Anglo-American competition prior to the 1880s featured more areas of confrontation (political, economic, and informational) than areas of collaboration (ideological and military). However, Britain made a proper assessment of its inherent economic and military-industrial power limitations compared to those held by a late 19th-century America. It also came

to see the United States as a Great Power rival with tolerable political, ideological, and information preferences. Realists contend that it was London's balance of power calculus against Germany that led to a strategic decision to contest the rise of Berlin and convert tensions with its former American colonists into a long-term special relationship and geostrategic partnership. Liberal institutionalists argue this as a case where common norms, values, and political systems facilitated a peaceful Great Power transition. HST proponents view British acquiescence to the ascent of American power as a logical outcome between a dominant, declining hegemon and a rising state that viewed the future of global norms, orders, and procedures in a similar way.

The United States and Imperial Japan, Early 20th Century

Japan and the United States engaged in a 45-year period of rivalry over economic influence across the Asia-Pacific and for control of the Pacific Ocean sea lines of communication. Although both were largely rising powers in the Asia-Pacific region when compared to established Great Powers such as Britain and France, the United States perceived itself as the dominant commercial and maritime power in the region and the Japanese as the rising challenger to its regional hegemony. This contest culminated in a 5-year, deeply destructive war across Asia-Pacific fought as part of World War II.

American interest in commerce and free enterprise in the Asia-Pacific traced back to Commodore Matthew Perry's maritime engagement with the Japanese in the early 1850s. U.S.-Japan relations were relatively positive and featured modest economic trade and cooperation during the late 1800s. After the Spanish American War of 1898, the United States took a great and growing interest in setting the rules for trade and commerce across the Asia-Pacific. Rapidly increasing American industrial might resulted in high-volume exports that needed import partners across Asia. American industrial prowess also was invested into a modern recast: iron- and steel-hulled U.S. Navy capable of enforcing commercial exchange in the Pacific Ocean. The 1899 U.S. declaration of an Open Door Policy calling for equal trading rights for all nations in China and recognition of Chinese territorial integrity set the stage for increasing friction with Imperial Japan. Japan already had territorial interests in China and on the Korean Peninsula, secured during the First Sino-Japanese War (1894–1895). Tokyo believed that it had to control selected tracts of territory on the Asian land mass to access natural resources and control markets necessary to be a regional Great Power. Japan reaffirmed its intent and its capability to assert imperial control of continental Asian territories with victory in the Russo-Japanese War (1904–1905) where Japanese victory confirmed its control of land in Korea and Manchuria. In 1915, Japan announced its Twenty-One Demands for the Republic of China, asserting a far greater Japanese economic and territorial remit there.⁶⁹

During late World War I and its immediate aftermath, the United States sought modest accommodation with Japan in a manner that enhanced Japan's belief that it had the right to special territorial interests in China due to its geographic proximity but that maintained the public perception that America's Open Door Policy remained intact.⁷⁰ The United States and the UK also sought to constrain the burgeoning post-World War I naval arms race among them, Japan, France, and Italy with a Five-Power Treaty (1921–1922) that locked warship ratios among the UK, United States, and Japan at a 5:5:3 ratio.⁷¹ Initially acceptable to Japanese



Figure 2.4. U.S. Army Anti-Japanese Propaganda Poster. Source: "Our Next Boss?" World War II propaganda poster (U.S. Army/University of Minnesota Libraries, Upper Midwest Literary Archives).

politicians, this treaty became a growing aggravation with Japanese military leadership during the 1920s and 1930s. It also became a poster child codifying Tokyo's grievance that, led by the United States, Western powers were treating Japan unfairly as a second-class power.⁷² At the same time, longstanding ideological and racial tensions between the United States and Japan hardened when President Woodrow Wilson refused a Japanese request for a racial equality clause or an admission of the equality of the nation in the League of Nation's charter. U.S. isolationism in the 1920s and early 1930s meant that competition between Washington and Tokyo festered beneath the surface in the United States. But Japanese frustration with the United States grew increasingly palpable as its 1920s economy suffered from a post-

World War I recession and an early 1930s depression jolted by the U.S. stock market collapse and subsequent draconian U.S. tariffs on all imports.⁷³

Jarred by severe economic depression, Japan's military leaders gradually muscled aside its political leaders from 1931 to 1937, pressing the government into ever greater military adventures throughout eastern Asia. Japanese military and strident nationalist politicians became increasingly assertive in claims for colonial ascendance across Asia and in China. Japan invaded Manchuria in 1931, exciting international condemnation. It withdrew from the League of Nations in 1933 over this criticism and increasingly turned away from international diplomatic and economic fora. In 1937, Japan launched an open war against China in Beijing, seeking expanded control of critical infrastructure and resources. From 1937 to 1940, the United States and Japan coexisted in an uneasy truce—with the United States refusing formal entry into the war on the side of China and continuing to abide by the terms of the 1911 U.S.-Japan Treaty of Commerce and Navigation. But by 1940, Japan's continually brutal war in China, coupled with its diplomatic and economic overtures toward the Axis powers of Germany and Italy and its formal declaration of a "Greater East Asia Co-Prosperity Sphere" against all Western interests in the Pacific, triggered a much sharper U.S. reaction.

As a strong signal of growing worry about Japan's unrelenting war in Asia and its growing naval fleet, America repositioned its Pacific Fleet Headquarters from San Diego to the territorial port of Pearl Harbor, Hawaii, during the summer of 1940. In late 1940, Washington levied restrictive sanctions against Japan's economy and stepped up overt aid and assistance to China. America sought to contain Japanese imperial expansion in the Asia-Pacific with an embargo of Japan's access to critical materials, such as oil, rubber, and scrap iron. Japan viewed this policy with increasing alarm and perceived that it was being placed under existential threat. After the United States placed a full oil embargo on Japan in the middle of 1941, Japan's military leadership resolved to preemptive war—hoping to strike a decisive blow against the U.S. Pacific Fleet early in a clash that would cause Washington to sue for peace in the Asia-Pacific. Japan properly recognized that U.S. industrial might and latent military capability would inevitably dwarf that of Japan, but its military junta would not auger long-term accommodation with Washington, instead gambling on success in a short, sharp 6-month war.⁷⁴ This military gambit failed at Pearl Harbor, and 4½ years later, Japan's imperial ambitions and its country lay in ruins.

During the early 1900s, U.S.-Japan GPC in the Asia-Pacific was contested in a multipolar world where the major powers of Europe along with Japan and the United States sought political and economic influence on the continent of Asia. A relatively dominant United States—led by its modern growing navy and strong economic influence—managed the palpable tensions with Japan through restraint and a number of frequently secret bilateral and multilateral arrangements involving Tokyo's continental aims. Economic interests on the continent of Asia greatly diverged, but Japan and the United States continued direct trade with each other in a number of key categories, including those of energy and mineral resources. The two countries also found space for mutual financial benefit in secret deals impacting China. Ideological differences were obvious as U.S. aversion to Japanese and Chinese immigrants played out in public, but American and Japanese diplomacy sidestepped the issue until the outbreak of war. Japan's growing military might was addressed in a series of temporal arms control treaties that weathered poorly when Japanese military leadership took center stage in Tokyo during the 1930s. Disarmament and arms control efforts at collaborative competition also fell short because they did not address emerging technologies with critical warfighting impact such as naval aviation and aircraft carriers, submarines, and electromagnetic sensing devices. Political and diplomatic niceties remained ascendant until the 1930s, but then sharply eroded once Japan's government took a militaristic and fascist tone.

A destructive U.S.-Japan military clash was not foreordained. But the deep divisions in all the major dimensions of state-to-state interaction between them by the 1930s made peaceful competition exceptionally fraught. Economic accommodations made from the early 1900s to the late 1930s waned as Japan's government turned militaristic, and long-festered cultural and ideological differences were thrown into stark relief as Japan pursued empire and America pushed back. Japan's military became Tokyo's preferred means of interaction as its quest for self-contained regional economic hegemony in the Asia-Pacific confronted America with a choice to abandon the region or engage in confrontation. Japan's initial advantage in naval aviation and amphibious military operations did not hold up well in the face of America's enormous industrial and manpower advantages. Its quest for ascendant power in the Asia-Pacific lay in ruins by late 1945.

Insights from Selected GPC History

There are a number of insights about GPC that can be drawn from analysis done in this chapter—including the selected analysis made of our four historic rivalry dyads. Among these insights, 10 stand out.

First, for all but liberal-institutionalist theorists, international relations scholars and modern strategic analysts identify the historical relations among the most powerful states (the Great Powers) as critical to determining the levels of peace and stability across the international system. All agree that GPC involves more than just military power, and many view military power as the most critical attribute. Geopolitical scholars agree that soft power matters to GPC and includes political, ideological, informational, economic, and emerging technological dimensions. Most of these strategic dimensions or instruments are interdependent to some degree.⁷⁵

Second, over the past 500 years, interactions among Great Powers in the international system normally are dominated by competition in one or more of the major interactive categories (see table 2.2) with simultaneous elements of collaboration and conflict. Pure cooperation (unfettered peace and stability) and direct violent clash (war) among them have been the rarest forms of interaction.⁷⁶ Thus, the unipolar moment featuring unchallenged American military ascendance, absence of war among major states, and the primacy of cooperation and collaboration in the international order from 1992 to 2008 was atypical.

Third, Great Power transitions play out over decades or centuries, not years. Three-quarters of transitions since 1500 have culminated with—or featured during—a destructive period of war. The inevitability of war among states during times of transition is not foreordained; Great Powers may channel or expend their worst animus in one of several other nonviolent categories of competition: politico-diplomatic, economic, ideological, and informational. Great Powers also may vent matters of confrontation or clash through surrogate agents, covert activities, and proxy forces short of war. In addition, GPC among nuclear armed actors may decrease the likelihood of rivalry moving into direct violent conflict due to the swift, comprehensive destruction threatened should nuclear weapons be used.⁷⁷

“The history of relations among Great Powers is a story of persistent rivalry and recurrent warfare, punctuated by occasional, usually brief, periods of peace. . . . In the absence of a [global] higher power, states are always, in some measure, insecure. . . . The mistrust, military buildups, and diplomatic maneuvering . . . can sometimes result in periods of dynamic balance and tenuous stability . . . however, these have always broken down eventually, giving way to major wars.”

Aaron L. Friedberg,
A Contest for Supremacy (2012)

Fourth, an alignment of conflictual demands or grievances across the five major areas of interstate competition (political and diplomatic, ideological, informational, economic, and military) sets the conditions for GPC to drift toward direct military clash. This fateful five-layer negative alignment occurred between the United Kingdom and Revolutionary France, and again between the United States and Imperial Japan. Such alignment did not occur between the United States and the UK. Late 19th-century leaders in Washington and especially in London found enough common ground in political, ideological, and economic interactions that they moved beyond residual diplomatic challenges and military

power realignments. Here, enlightened leadership mattered. Like those in late Victorian England and the United States, leaders can inhibit the alignment of intractable demands across all five areas, looking for avenues to undertake collaborative or cooperative activities where such opportunities exist.

Fifth, once a GPC rivalry is under way, the most reliable indicator of when a clash will erupt is when one side or both recognizes a shift in the relative alignment of economic and military power that is perceived as immutable and untenable. As states view the relative power alignment moving against them in an unacceptable way, they are much more inclined to risk a preemptive conflict than when they perceive a stable power status quo.⁷⁸ The UK most acutely feared loss of naval ascendancy in its rivalry with Revolutionary/Napoleonic France and later Imperial/Nazi Germany. Also, Imperial Japan perceived its imperative for naval dominance in the Pacific untenable if it did not strike first against encroaching U.S. forces. Applied to 2020, longstanding U.S. ascendancy at sea, in space, and in cyberspace points to areas where a perceived shift in relative power among the Great Powers in these areas could foretell growing risk of direct armed clash.

Sixth, although incompatible ideologies and caustic informational exchanges about the rival's people are not a lone determinant of when Great Power rivalry will devolve into direct violent clash (war), they are strong lagging indicators of insurmountable contentiousness. Limited but noteworthy UK popular support for the French Revolution turned irredeemably hostile when an increasingly bloody insurrection generated popular press revulsion for everything French across Great Britain. The UK press and that of Imperial Germany turned to ad homonym attacks on each other's national character, and this crescendoed as the contest over colonies and naval ascendancy peaked. U.S. antipathy for Japanese militarism and governance choices played out in crass press attacks against Japanese society during the late 1930s, and Tokyo propaganda returned the favor. Conversely, U.S. and UK similarities in culture, governance, and general worldview found positive press in America and Britain during the dawn of their late 1800s rapprochement. Modern analysts must beware when U.S. and/or Chinese press caricatures of "others" become uniformly negative—a historically bad sign for peaceful resolution of GPC transitions.

Seventh, during power transition periods, competitors may not perceive their own various forms of power accurately. Moreover, even when accurate assessments of relative decline or vulnerability are made, domestic or bureaucratic interests may retard agile adaptation necessary to mitigate risks. The UK of the late 1800s was afflicted by this challenge. It was aware of relative economic decline and made sensible foreign policy changes regarding the United States. But it found itself unable to persuade domestic constituencies favoring unfettered free trade to consider some targeted tariffs as a means to generate manufacturing innovation into new critical technologies. This inability to adapt domestic resource priorities left the UK disadvantaged versus Imperial Germany in emerging technologies such as industrial chemicals, machine tools, and military-grade steel. London required Swedish, Swiss, and U.S. assistance to compensate during World War I. Like the UK a century ago, American politicians in 2020 already may be finding themselves confronting similar resistance from exporters and financial business leaders to any adaption of national production priorities and trade activities toward Great Power competition.

Eighth, perceptions of relative power, rather than detailed and empirical assessments of power, are likely to inform and then drive policymakers.⁷⁹ Late 19th-century America and Germany might have benefited more domestically and seemed less threatening abroad with reduced economic tariffs as they grew, but politicians could not sway powerful agrarian and export political constituencies demanding such immutable protections. Here again, leadership matters. Enlightened state leaders can seek and promulgate factual assessments of national strengths and weaknesses. They can guard against the pitfall of blaming others and then acting out violently based on false perceptions, instead turning to and educating their people about empirical assessment of relative economic or military strengths.⁸⁰ As the siren's song of rapacious protectionism plays out in 2020, the United States and China might benefit from contemplation of this lesson from past GPC.

Ninth, during periods of dynamic technological change, the likelihood of strategic surprise or operational obsolescence is greater in the military dimension of GPC. States may overestimate or underestimate the potential combat power of new innovations, whether they are technological or conceptual. Napoleon's tactical genius and his innovative use of massed artillery made him a formidable land opponent but did not translate into strategically vital sea power in his rivalry with the UK. Similarly, Imperial Germany's superior battlefield use of railways and Nazi Germany's perfection of the tactical use of airpower in blitzkrieg did not translate into an effective strategic challenge to the Royal Navy. Imperial Japan had a temporary advantage in strategic naval airpower, but insufficient national economic power to survive a U.S. industrial onslaught when its Pearl Harbor gambit fell short. In 2020, the rapid rise of new and novel forms of military and protomilitary technologies may contribute to improper estimates of relative power.

Finally, Great Power success in geostrategic competitions requires extraordinary political leadership, in both the conduct of statecraft and generating requisite forms of domestic renewal and institutional adaptation.⁸¹ The UK exhibited such statecraft in its competitions with post-revolutionary France and with a rising United States. The UK proved less adept during competition with Imperial and then Nazi Germany. Japan did not succeed in its contest with the United States over ascendancy in Asia-Pacific. The challenge for leaders in 2020 is at least as great as it was for those in past eras of GPC.

Conclusion

The history of Great Power competition and related debates over power rise and decline offer numerous insights. The most important conclusion is that while GPC is the historical norm, relative decline and violent clash among rivals are not predestined in any way. Instead, these outcomes reflect choices that leaders make and their capacity to assess and adapt. In 2020, contemporary debates about the evolving era of GPC among the United States, China, and Russia often betray a degree of fatalism about American power and where Washington stands in economic, military, or geostrategic influence terms.

As Aaron Friedberg noted in the afterword to his 1988 study of Great Britain's policies vis-à-vis the United States at the dawn of the 20th century, "it would be unwise to bet against the resilience and adaptability of the American system."⁸² But it would be equally unwise to allow complacency or inertia to drive national security strategy. In a world defined by disruptive political and socioeconomic change, and a potentially revolutionary

altered character of warfare, the United States finds itself at the dawn of an era where it can no longer assume that its power advantages will adapt. Leadership matters. Understanding the emerging dynamics in an era of interstate relations dominated by three Great Powers across the five major arenas of interaction—politico-diplomatic, ideological, informational, economic, and military—cannot be underappreciated. When it comes to the proper development and utilization of U.S. power from 2020 to 2025, a competitive mindset in Washington is necessary. In turn, a solid historical understanding of GPC is important. Finally, a proper assessment of the strategic imperatives and relational dynamics as well as the competitive elements and power toolsets available to Washington, Beijing, and Russia must be established. The next two chapters provide an overview of these critical geostrategic elements for our new era of GPC.

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Notes

¹ Hans Morgenthau, *Politics Among Nations* (New York: Knopf, 1948).

² Kenneth N. Waltz, *Theory of International Politics* (New York: Random House, 1979).

³ Robert O. Keohane and Joseph S. Nye, Jr., *Power and Interdependence* (Boston: Little, Brown, 1977). In this seminal work on this topic, Keohane and Nye identified three pillars of soft power: political values, culture, and foreign policy. Each extends these constructs in subsequent works, including Joseph S. Nye, Jr., *Bound to Lead: The Changing Nature of American Power* (New York: Basic Books, 1990); Joseph S. Nye, Jr., *The Paradox of American Power* (London: Oxford University Press, 2002); Joseph S. Nye, Jr., *The Future of Power* (New York: Perseus Books, 2011); Robert O. Keohane, *International Institutions and State Power* (New York: Westview Press, 1989); Robert O. Keohane, *Power and Governance in a Partially Globalized World* (New York: Routledge, 2002).

⁴ Alexander Wendt, "Anarchy Is What States Make of It: The Social Construction of Power Politics," *International Organization* 46, no. 2 (Spring 1992), 391–425; E. Adley, "Seizing the Middle Ground: Constructivism in World Politics," *European Journal of International Relations* 3, no. 3 (1998), 319–363; Steve Walt, "International Relations: One World, Many Theories," *Foreign Policy*, no. 110 (1998), 29–46.

⁵ Robert Gilpin, *U.S. Power and the Multinational Corporation: The Political Economy of Foreign Direct Investment* (New York: Basic Books, 1975); Robert Gilpin, *War and Change in World Politics* (Cambridge: Cambridge University Press, 1981); Jack S. Levy, "Theories of General War," *World Politics* 37, no. 3 (April 1985), 344–374.

⁶ Gilpin, *War and Change in World Politics*; Robert Gilpin, *The Political Economy of International Relations* (Princeton: Princeton University Press, 1987); Duncan Snidal, "The Limits of Hegemonic Stability Theory," *International Organization* 39, no. 4 (Autumn 1985), 579–614. Economist Charles Kindleberger also is considered a proponent of hegemonic stability theory, but with a focus on trade and finance tools (soft power), rather than military ones, as the primary means for the hegemon to establish and sustain a liberal, free trade order. See Charles P. Kindleberger, *The World in Depression, 1929–1939* (Berkeley: University of California Press, 1973), 288–305.

⁷ Robert O. Keohane, *After Hegemony: International Change and Financial Reform* (Princeton: Princeton University Press, 1984); John Ikenberry, *After Victory: Institutions, Strategic Restraint, and the Rebuilding of Order After Major Wars* (Princeton: Princeton University Press, 2001); John Ikenberry, *Liberal Leviathan: The Origins, Crisis and Transformation of the American World Order* (Princeton: Princeton University Press, 2011).

⁸ E. Adler, "Seizing the Middle Ground: Constructivism in World Politics," *European Journal of International Relations* 3, no. 3 (1998), 319–363.

⁹ Graham Allison, *Destined for War: Can America and China Escape Thucydides's Trap?* (New York: Houghton Mifflin, 2017), 244–286. Even the conclusion of 25 percent is generous; one of the four cases studied was that of Germany versus the United Kingdom and France for political influence in Europe from 1990 onward. Arguably, this case is indeterminate because the overwhelming influence of American military power and political preferences across Europe remains both dominant and an inhibitor of political tensions migrating toward violent confrontation, and because the case is not yet concluded, so no final outcome is known.

¹⁰ Charles Krauthammer, "The Unipolar Moment," *Foreign Affairs* 70, no. 1 (1991), 23–33; Fareed Zakaria, *The Post-American World* (New York: Norton, 2008); Hal Brands, *Making the Unipolar Moment: U.S. Foreign Policy and the Rise of the Post-Cold War Order* (Cornell, NY: Cornell University Press, 2016).

¹¹ As the Soviet Union was wavering, American political scientist Francis Fukuyama questioned whether this moment of ascendant Western liberal democracy represented an unabashed and irreversible victory of economic and political liberalism. His pregnant question was premature. Liberalism and free trade orthodoxy may prove rather fragile achievements. Instead, the "desire for recognition and respect" may not be satisfied by prosperity or political participation. See Francis Fukuyama, "The End of History?" *The National Interest*, no. 16 (Summer 1989), 3–18; Francis Fukuyama, *Identity: The Demand for Dignity and Politics of Resentment* (New York: Farrar, Straus and Giroux, 2018).

¹² Kurt M. Campbell and Ely Ratner, "The China Reckoning: How Beijing Defied American Expectations," *Foreign Affairs* (March/April 2018).

¹³ Michael Howard, "Military Science in an Age of Peace," *The RUSI Journal* 119, no. 1 (March 1974), 4–23.

¹⁴ Aaron L. Friedberg, *The Weary Titan: Britain and the Experience of Relative Decline, 1895–1905* (Princeton: Princeton University Press, 1988), 3–20.

¹⁵ Paul Kennedy, *The Rise and Fall of the Great Powers* (New York: Random House, 1987).

¹⁶ Michael J. Mazarr et al., *Understanding the Emerging Era of International Competition: Theoretical and Historical Perspectives*, RR2726 (Santa Monica, CA: RAND, 2018), available at <www.rand.org/pubs/research_reports/RR2726.html>. These seven objectives meld classic realist definitions of state competitive objectives and those later defined by liberal institutionalists. See Morgenthau, *Politics Among Nations*, 29–114; Keohane and Nye, *Power and Interdependence*, 11–19.

¹⁷ Allison, *Destined for War*, 244–286.

¹⁸ James Lacey, ed., *Great Strategic Rivalries: From the Classical World to the Cold War* (New York: Oxford University Press, 2016).

¹⁹ See "COW Typology of War: Defining and Categorizing Wars" and "COW War List" at the *Correlates of War Project*, available at <www.correlatesofwar.org/>.

²⁰ Gary Goertz and Paul F. Diehl, "Enduring Rivalries: Theoretical Constructs and Empirical Patterns," *International Studies Quarterly* 37, no. 2 (June 1993).

²¹ Waltz, *Theory of International Politics*.

²² But that same research into this "democratic peace theory" indicates that war is more likely between liberal democratic states and those that are not and that the propensity for war between liberal democracies and those perceived as illiberal democracies also is greater than between two similar liberal democratic states. See Michael W. Doyle, "Kant, Liberal Legacies, and Foreign Affairs," *Philosophy & Public Affairs* 12, no. 3 (Summer 1983), 222; David Lake, "Powerful Pacifists: Democratic States and War," *American Political Science Review* 86, no. 1 (March 1992), 24–37.

²³ The economic interdependence argument is made in Keohane and Nye, *Power and Interdependence*. The counterargument regarding the origins of World War I and the intense economic interdependence among European monarchies that did not inhibit major Great Power war is found in Barbara Tuchman, *The Proud Tower: A Portrait of the World Before the War, 1890–1914* (New York: Macmillan, 1966).

²⁴ Walt, *Theory of International Politics*, 40–42.

²⁵ Mazarr et al., *Understanding the Emerging Era of International Competition*, 6–11.

²⁶ This controversial "nuclear peace theory" is established in such major works as Kenneth N. Waltz, "The Spread of Nuclear Weapons: More May Be Better," *Adelphi Papers* 21, no. 171 (1981), available at <<https://fliphtml5.com/lhec/zexq/>>; John Mearsheimer, "The Case for a Ukrainian Nuclear Deterrent," *Foreign Affairs* 73, no. 2 (Summer 1993), 50–66; Robert Rauchhaus, "Evaluating the Nuclear Peace Hypothesis: A Quantitative Approach," *The Journal of Conflict Resolution* 53, no. 2 (2009), 258–277, available at <www.jstor.org/stable/20684584>. The rare but important exception is the war between nuclear weapons states India and Pakistan in July 1999, known as the Kargil War. While neither state was a Great Power by the definition used in this volume, the fact that each state possessed large and relatively modern militaries and engaged them in a month-long major war despite the specter of nuclear escalation stands as a caution against any

orthodoxy that an absence of Great Power war in the nuclear weapons era is valid. See Neil Joeck, "The Kargil War and Nuclear Deterrence," in *Nuclear Proliferation in South Asia Crisis Behavior and the Bomb*, ed. Sumit Ganguly and S. Paul Kapur (New York: Routledge, 2008).

²⁷ We chose a short list of historical cases that represents several important dimensions of modern Great Power competition (GPC). This case study exercise is intended as illustrative rather than comprehensive. In that context, the U.S.–Soviet Union Cold War competition from 1945 to 1991 was considered but not chosen for review. It was omitted due to the expansive trove of analytical literature that already exists to describe the critical dynamics and outcomes from the 46-year rivalry. Cold War case study literature establishes in great detail the kinds of analytical observations featured in this chapter about the four less well-understood cases chosen. Under Mikhail Gorbachev, the Soviet Union (temporarily) ended the Cold War rivalry by quitting the competition in its three key dimensions: ideology, economic organization, and political identity. In this chapter and those that follow, the authors refer to many of the authoritative texts on the Cold War, with special consideration to the following seminal works: John Lewis Gaddis, *We Now Know: Rethinking Cold War History* (Oxford: Oxford University Press, 1997); Mikhail Gorbachev, *Memoirs* (New York: Doubleday, 1995); John Lewis Gaddis, *The Cold War: A New History* (New York: Penguin, 2005); Sergey Radchenko, *Two Suns in the Heavens: The Sino-Soviet Struggle for Supremacy, 1962–1967* (Stanford: Stanford University Press, 2009); Vladislav M. Zubok, *A Failed Empire: The Soviet Union in the Cold War from Stalin to Gorbachev* (Chapel Hill: University of North Carolina Press, 2007).

²⁸ Some of our reviewers specifically questioned why the rise of Japan from the 1970s to 1990s was not chosen for this chapter. Although briefly considered, the Japan–U.S. trade competition of the Cold War was not selected as a critical case for understanding modern GPC for a number of reasons. Three stood out. First, Japan never met the operational definition of a Great Power provided in chapter 1. Specifically, Japan never had unusual capabilities in terms of the other states during the Cold War and was never perceived by the other states in the system as a state able to leverage its commercial capabilities for a change in the international order. Second, Japan at that time was a veritable client of the United States in its military, political, and diplomatic activities, thus not truly an independent actor (much less a Great Power) in four of the five aspects of international competition defined in table 2.2. Third, the "competition" between the United States and Japan over economic practices had no parallel ideological, informational, communications, or military competitive dimensions. Also, the limited economic dimensions contested in this power dyad never infected other critical bilateral interactions. For these and additional conclusions rendering the U.S.–Japan trade competition during the Cold War relatively unhelpful for understanding GPC historically or today, see Keyu Jin Keyu, "No, China Is Not the 'Next Japan,'" *World Economic Forum*, October 13, 2016, available at <www.weforum.org/agenda/2016/10/no-china-is-not-the-next-japan/>; and Wendy Wu, "Why China's U.S. Trade Stand-Off Is Not a Replay of Japan's in the 1980s," *South China Morning Post* (Hong Kong), April 16, 2018, available at <www.scmp.com/news/china/diplomacy-defence/article/2141824/why-chinas-us-trade-stand-not-replay-japans-1980s>.

²⁹ For details, see Klaus Schwab, *The Fourth Industrial Revolution* (New York: Crown Publishing Group, 2016).

³⁰ World War II took an estimated 70 million lives over the course of 15 years. World War I cost about 17 million dead in a 4½-year period. The Napoleonic Wars ended an estimated 6.5 million lives over a 14-year series of military clashes. Only the 13th-century Mongol conquest of Europe (about 60 million dead) and the 17th-century, 60-year Qing Dynasty conquest

of China (about 25 million dead) rank in the same category of deadliness for Great Power warfare. This accounting omits incorporation of civil war periods such as the multiple instances in China and that in 19th-century Russia—many of which are estimated to have killed more than 10 million people. See Matthew White, *Atrocities: The 100 Deadliest Episodes in Human History* (New York: Norton, 2011).

³¹ In the 1500s, Portugal and Spain were ascendant Global Powers with international trade reach, naval prominence, and growing colonies. But in 1578, King Sebastian I of Portugal, who had no heir, was killed in the Battle of Ksar El Kebir. This led to the Portuguese succession crisis of 1580, which allowed Philip II of Spain to unite the two kingdoms under the rule of the Spanish kings for the next 60 years. Philip II mismanaged Portuguese colonies, declared war on England, and suffered the disastrous destruction of a large part of the Spanish Armada in 1588, which marked the end of Spanish colonial supremacy. During the 1600s, France, England, and the Netherlands overtook Spain and Portugal in commercial and naval supremacy. The Dutch gained an early trade advantage in South Asia, the Far East, and North America. England and France reduced the Dutch advantage with a century of shifting alliance combinations, continental battles, and proxy fights over disputed colonies. Beginning with the Treaty of Westphalia (1648) and culminating during the collapse of the Spanish monarchy and the War of Spanish Succession (1701–1714), France attained an ascendant power position on the European continent and a relative ascendancy in global colonies that it retained during the early 18th century. See Kennedy, *The Rise and Fall of the Great Powers*, 99; John A. Lynn, *The Wars of Louis XIV, 1667–1714* (Harlow, UK: Longman, 1999), 17; and Francois Crouzet, *Britain Ascendant: Comparative Issues in Franco-British Economic History* (Cambridge: Cambridge University Press, 1990), 12.

³² Many historians mark the UK rise and ascendancy over France to the erosion of France's colonial reach and its naval prowess, highlighted most dramatically in Paris's loss of the Seven Years' War to the UK and its allies in 1763, and from the attendant erosion of its colonial possessions worldwide—especially in North America—that ensued from the 1763 Treaty of Paris. See Kennedy, *The Rise and Fall of the Great Powers*, 120.

³³ William Doyle, *The Oxford History of the French Revolution* (Oxford: Oxford University Press, 2002), 197; Michael V. Leggiere, “Enduring Strategic Rivalries: Great Britain vs. France During the French Wars (1792–1815),” in *Great Strategic Rivalries: From the Classical World to the Cold War*, ed. James Lacey (New York: Oxford University Press, 2016), 289–390.

³⁴ Ironically, British economic concerns with Revolutionary France lagged in the finance markets. In 1803, England's Barings Bank financed two-thirds of the U.S. Louisiana Purchase price of \$15 million by giving the French \$10 million in cash for American bonds used in the purchase, then selling the bonds on international markets. Since Napoleon sold the Louisiana territories to fund France's military war efforts, a major British bank basically enabled the growth of its Great Power military rival. See Paul Strathern, *Empire: A New History of the World* (New York: Pegasus Books, 2020), 225.

³⁵ Clive Emsley, *Britain and the French Revolution* (London: Routledge, 2014), especially 53–76; Michael V. Leggiere, *The Fall of Napoleon* (Cambridge: Cambridge University Press, 2007), 2–5; and Kennedy, *The Rise and Fall of the Great Powers*, 124.

³⁶ Paul Kennedy, *The Rise and Fall of British Naval Mastery*, 2nd ed. (New York: Humanity Books, 1983), 123–148.

³⁷ Kennedy, *The Rise and Fall of the Great Powers*, 202.

³⁸ Joachim Remak, *The Origins of World War I, 1871–1914* (Hinsdale, IL: Dryden Press, 1967), 77–78.

³⁹ *Ibid.*, 77.

⁴⁰ These German import tariffs, forged from the so-called Marriage of Iron and Rye and implemented in 1879, became sacrosanct domestic political features that limited Germany's politicians from adjusting to British calls for tariff reductions as Germany's manufacturing moved from developing to dominant in the early 1900s. See Cheryl Schonhardt-Bailey, “Parties and Interests in the ‘Marriage of Iron and Rye,’” *British Journal of Political Science* 28, no. 2 (April 1998), 291–332; Edgar Crammond, “The Economic Relations of the British and German Empires,” *Journal of the Royal Statistical Society* 77, no. 8 (July 1914), 777–824; Friedberg, *The Weary Titan*, 39–40.

⁴¹ Friedberg, *The Weary Titan*, 33–36.

⁴² Crammond, “The Economic Relations of the British and German Empires,” 784–785.

⁴³ Paul Kennedy, *The Rise of the Anglo-German Antagonism, 1860–1914* (New York: Humanity Books, 1980), 410–432; Remak, *The Origins of World War I, 1871–1914*, 35. For more on the role of Germany's influential web of “interest leagues” and their important role in colonialism and a growing navy, see Gordon Craig, *Germany, 1866–1945* (Oxford: Oxford University Press, 1981), 250–297.

⁴⁴ *Weltpolitik* took over as Germany's foreign policy at the same time as the Kaiser's adoption of the Tirpitz Plan (named for its author, Admiral Alfred von Tirpitz, head of the German navy). The plan was for Germany to achieve world power status through buildup of its navy while at the same time co-opting and minimizing growing economic and social issues by elevating a domestic policy of social imperialism. See James Joll, *The Origins of the First World War* (London: Pearson Education Limited, 1984), 131–134.

⁴⁵ Remak, *The Origins of World War I, 1871–1914*, 78–80.

⁴⁶ After almost 40 years of intense rivalry over influence in Central Asia and South Asia, British and Russian conflicting interests in 1900 centered on Afghanistan, Iran, and Tibet. Each of these states comprised competitive buffers between Britain's and Russia's colonial possessions in Asia. Germany's rise as a competitive empire with global aspirations, along with the Russian navy's embarrassing defeat by the Japanese in the Russo-Japanese War of 1904–1905, persuaded key leaders in London and Moscow to resolve their respective differences across Asia. By 1914, Russia and Britain agreed on terms to divide Iran, to recognize the standing Afghanistan border, and to conclude an agreement on Tibet that established the McMahon Line. See George P. Gooch and Harold Temperley, eds., *British Documents on the Origins of the War, 1898–1914*, vol. 4, *The Anglo-Russian Rapprochement, 1903–07* (London: Her Majesty's Stationery Office, 1929), 618–621.

⁴⁷ London's diplomacy resulted in the Triple Entente of Britain, France, and Russia. See George P. Gooch, “Chapter V: Continental Agreements, 1902–1907,” in *The Cambridge History of British Foreign Policy, 1793–1919*, ed. A.W. Ward and George P. Gooch, vol. 3, 1866–1919 (Cambridge: Cambridge University Press, 1923), 305–365; Remak, *The Origins of World War I, 1871–1914*, 44–46.

⁴⁸ Charles S. Campbell, *From Revolution to Rapprochement: The United States and Great Britain, 1783–1900* (New York: John Wiley & Sons, 1974), especially 194–204.

⁴⁹ Remak, *The Origins of World War I, 1871–1914*, 82–85.

⁵⁰ Mildred S. Wertheimer, *The Pan-German League, 1890–1914* (New York: Octagon, 1924), 101–102.

⁵¹ *Ibid.*, 198.

⁵² Some argue that Imperial Germany's democracy was too new to withstand the rapid rise of its industrial era and social interest groups, thus making it less resilient than the UK. See Kennedy, *The Rise of the Anglo-German Antagonism, 1860–1914*, 361. A Chinese author on Germany's rise even goes so far as to intimate that it was too much democracy in Germany that set a course for German adventurism and eventual clash with the UK. This interpretation is far from persuasive given other factors but is a window into a potential particular psyche in modern China regarding the inherent dangers of too much democracy in a rapidly expanding economic state. See Xu Qiyu, *Fragile Rise: Grand Strategy and the Fate of Imperial Germany, 1871–1914* (Cambridge: MIT Press, 2017), 150–153.

⁵³ Kennedy, *The Rise of the Anglo-German Antagonism, 1860–1914*, 167–222; James Lacey, “Great Strategic Rivalries and the Return of Great Power Politics,” *The Strategy Bridge*, August 22, 2018, available at <<https://thestrategybridge.org/the-bridge/2018/8/22/great-strategic-rivalries-the-return-of-geopolitics>>; H.C. Allen, *Great Britain and the United States: A History of Anglo-American Relations, 1783–1952* (New York: Saint Martin's Press, 1955), 518–610; Gordon Martel, *Origins of the First World War*, 4th ed. (New York: Routledge, 2017).

⁵⁴ Kindleberger, *The World in Depression, 1929–1939*, 288–300. For evidence of U.S. economic ascendance during the interwar period, see table 4.2, where steel and iron production is a good proxy for relative economic size and strength.

⁵⁵ Williamson Murray, “Britain and Germany: 1914–1945,” in Lacey, *Great Strategic Rivalries* (New York: Oxford University Press, 2016), 371–416; William Shirer, *The Rise and Fall of the Third Reich: A History of Nazi Germany* (New York: Simon and Schuster, 1960).

⁵⁶ Kennedy, *The Rise of the Anglo-German Antagonism, 1860–1914*, 291–410; William Harbutt Dawson, “Chapter III: Imperial Policy in the Old and the New World, 1885–1899,” in Ward and Gooch, *The Cambridge History of British Foreign Policy*, vol. 3, 125–130, 200–222.

⁵⁷ Kennedy, *The Rise and Fall of British Naval Mastery*, 149–176.

⁵⁸ Kennedy, *The Rise and Fall of the Great Powers*, 200–201; Adam Tooze, *The Deluge: The Great War, America, and the Remaking of the Global Order, 1916–1931* (New York: Viking, 2014), 13. Also see table 4.2 for relative steel and iron production.

⁵⁹ Douglas A. Irwin, *Historical Aspects of U.S. Trade Policy* (Cambridge, MA: The National Bureau of Economic Research, 2006), available at <<https://data.nber.org/reporter/summer06/irwin.html>>.

⁶⁰ Allen, *Great Britain and the United States*, 30–52, 518–548, 941–948; Charles Kindleberger, *World Economic Primacy, 1500–1990* (New York: Oxford University Press, 1996), 35–36, 125–148, 172–189.

⁶¹ While London and many other world capitals viewed American expansion and subsequent acquisition of former Spanish colonies in the Caribbean and the Pacific as evidence of Washington's own imperialism, American lore rejects any notion of American empire in favor of manifest destiny and expulsion of Spanish empire. It was the British perception that helped advance UK–U.S. rapprochement irrespective of divergent viewpoints about empire. See Kori Schake, *Safe Passage: The Transition from British to American Hegemony* (Boston: Harvard University Press, 2017), 280–281; Strathern, *Empire*, 10–11, 224–225.

⁶² Dawson, “Chapter III,” in Ward and Gooch, *The Cambridge History of British Foreign Policy*, 223–227.

⁶³ Campbell, *From Revolution to Rapprochement*, 174–189; Ernest R. May, *Imperial Democracy: The Emergence of America*

as a Great Power (New York: Harcourt, Brace and World, Inc., 1961), 57–59; Kenneth Bourne, *Britain and the Balance of Power in North America, 1815–1908* (Berkeley: University of California Press, 1967), 351; Walter A. McDougall, *Promised Land, Crusader State* (Boston: Houghton Mifflin Company, 1997), 101–121; Kennedy, *The Rise and Fall of British Naval Mastery*, 185–210.

⁶⁴ Kaiser Wilhelm II's 1896 telegram of congratulations to the Boer president after the capture of British citizens in the Transvaal of Africa was effective as well. The Kaiser's missive led many Americans to take the British side and reminded the British that they could use friends offshore from Europe. Campbell, *From Revolution to Rapprochement*, 183–184; Valentine Chirol, “Chapter IV: The Boer War and the International Situation, 1899–1902,” in Ward and Gooch, *The Cambridge History of British Foreign Policy, 1783–1919*, vol. 3, 263–266.

⁶⁵ Bourne, *Britain and the Balance of Power in North America, 1815–1908*, 351; Kennedy, *The Rise and Fall of the Great Powers*, 203; Friedberg, *The Weary Titan*, 97.

⁶⁶ Campbell, *From Revolution to Rapprochement*, 91–204.

⁶⁷ Allen, *Great Britain and the United States*, 944–948; Kindleberger, *The World in Depression, 1929–1939*, 302–305.

⁶⁸ For a similar conclusion, see Schake, *Safe Passage*.

⁶⁹ Herbert Feis, *Road to Pearl Harbor: The Coming of the War Between the United States and Japan* (Princeton: Princeton University Press, 1950), 3–7, available at <www.jstor.org/stable/j.ctt13x0wpg.4>; S.C.M. Paine, “China, Russia, and Japan and the Competition to Create a New World Order,” in Lacey, *Great Strategic Rivalries*, 417–448; William M. Morgan, “Enduring Strategic Rivalries: U.S.–Japan Strategic Rivalry to 1941,” in Lacey, *Great Strategic Rivalries*, 479–510.

⁷⁰ Of particular note in this respect was the Lansing–Ishii Agreement of 1917, where the United States and Japan agreed in public that the Open Door Policy would remain intact but established in a secret annex that Japan had “special rights” in territories across China and that the two would share financial spoils from major rail and infrastructure investments into China in a special banking consortium. Lansing–Ishii set the stage for numerous other secret post–World War I treaties that promised colonial enclaves in China be passed from Germany, Russia, and Austria–Hungary to Japan and the victorious World War I Allies. See Barbara Tuchman, *Stillwell and the American Experience in China, 1911–1945* (New York: Grove Press, 2001); J. Chal Vinson, “The Annulment of the Lansing–Ishii Agreement,” *Pacific Historical Review* 27, no. 1 (February 1958), 57–69.

⁷¹ Michael Graham Fry, “The Pacific Dominions and the Washington Conference, 1921–22,” in *The Washington Naval Conference, 1921–22*, ed. Erik Goldstein and John Maurer (London: Frank Cass & Company, 1994), 60–101.

⁷² Sadao Asada, “From Washington to London: The Imperial Japanese Navy and the Politics of Naval Limitation, 1921–1930,” in Goldstein and Maurer, *The Washington Naval Conference, 1921–22*, 147–191.

⁷³ “Japanese–American Relations at the Turn of the Century, 1900–1922,” *Milestones in the History of U.S. Foreign Relations* (Washington, DC: Department of State Office of the Historian, n.d.), available at <<https://history.state.gov/milestones/1899-1913/japanese-relations>>; Masato Shizume, *The Japanese Economy During the Interwar Period: Instability in the Financial System and the Impact of the World Depression*, Bank of Japan Review 2009–E–2 (Tokyo: Institute for Monetary and Economic Studies, May 2009), available at <www.boj.or.jp/en/research/wps_rev/rev_2009/data/rev09e02.pdf>.

⁷⁴ Hiroyuki Agawa, *The Reluctant Admiral: Yamamoto and the Imperial Navy*, trans. John Bester (Tokyo: Kodansha International, Ltd., 1979), 202–223. The 6-month quote is to Fumimaro Konoe, *Konoye Ayamaro Ko Shuki: Memoirs of Prince Ayamaro Konoye* (Tokyo: Asahi Shimbun-sha, 1946), 3.

⁷⁵ Mazarr et al., *Understanding the Emerging Era of International Competition*; Robert D. Blackwill and Jennifer M. Harris, *War by Other Means: Geoeconomics and Statecraft* (Boston: Harvard University Press, 2017).

⁷⁶ Kennedy, *The Rise and Fall of the Great Powers*; Mazarr et al., *Understanding the Emerging Era of International Competition*.

⁷⁷ On proxy wars, see Andrew Mumford, “The New Era of the Proliferated Proxy War,” *The Strategy Bridge*, November 16, 2017, available at <<https://thestrategybridge.org/the-bridge/2017/11/16/the-new-era-of-the-proliferated-proxy-war>>; Zack Gold and William Rosenau, “The Future of Conflict Is Proxy Warfare, Again,” *Defense One*, July 25, 2019, available at <www.defenseone.com/ideas/2019/07/future-conflict-proxy-warfare-again/158697/>; Geraint Hughes, *My Enemy's Enemy: Proxy Warfare in International Politics* (Chicago: Independent Publishers Group, 2012).

⁷⁸ Lacey, “Great Strategic Rivalries.”

⁷⁹ Friedberg, *The Weary Titan*, 14–17, 286.

⁸⁰ Ibid., 282–283, 286.

⁸¹ Ibid., 10, 15–16; Brian D. Blankenship and Benjamin Denison, “Is America Prepared for Great Power Competition?” *Survival* 61, no. 5 (October–November 2019), 43–64.

⁸² Friedberg, *The Weary Titan*, 318.

Chapter 3a

Contemporary Great Power Geostrategic Dynamics

Relations and Strategies

By Thomas F. Lynch III and Phillip C. Saunders

This chapter provides a comparative assessment of the strategic objectives for the three contemporary Great Powers: the United States, China, and Russia. It first traces the evolution of each power's strategic interests from 2000 to 2017, indicating where important milestones transitioned the powers' relations from relative cooperation and collaboration into de facto rivalry (by 2014 to 2015) and then a formally acknowledged rivalry (in 2017). The chapter next outlines the Great Powers' current strategic viewpoints and how they contrast across the five major categories of state interaction: political and diplomatic, ideological, informational, military, and economic. It demonstrates that each power has many divergent strategic interests, making rivalry inevitable. The chapter indicates where varying strategic interest intensity combines to make risks of Great Power clashes most worrisome in the coming 5 years: the Indo-Pacific, cyberspace, outer space, and, to a receding degree, the Middle East. It concludes that Russian strategic aims make Moscow a transient security risk to U.S. geopolitical dominance, while China's ideological vision and aspirations make it the most important, albeit presently less threatening, rival to the U.S. status as the head of the global liberal international order.

This chapter focuses on the three modern Great Powers—the United States, China, and Russia—and the broad framework of their contemporary interactions. It provides an overview of the ongoing major debates about the nature and degree of challenges posed by these three major states. It traces the recent trajectory of their strategic interaction narratives from 2000 to 2014–2015, establishing the dominant evolutionary themes for each over that time. The chapter next sketches the national strategy and major strategic aims for each country in 2020 and what these mean for the upcoming 5 years of Great Power competition (GPC). It then makes an analytical evaluation of what each of these national strategies implies about the Great Power's aims for international norms, rules, and institutions, followed by a discussion of what these strategies suggest for Great Power relations in major geographic regions. It concludes with 10 major insights and implications for GPC from 2020 to

2025, setting the table for a more detailed assessment of specific Great Power capabilities to achieve their strategic aims found in the next chapter.

Great Power Relevance and Geostrategic Dynamics in the Early 21st Century

As described in chapter 1, this volume defines *Great Power state* as one that has three main characteristics: unusual capabilities in comparison with those of other states, behavior that indicates a willingness to use those capabilities in and beyond the state's immediate neighborhood, and the perception by other actors in the system that the state has both unusual capabilities and the will to use them, making it an actor that must be treated as a major power.¹ From the late 20th century and into 2020, three states have satisfied these criteria: the United States, China, and Russia.

Yet these three Great Powers are far from uniform in status. Each differs substantively in terms of the strategic outcomes it wishes to assure, the relative capabilities it possesses, and the specific cases and places where it seems ready to pursue its main strategic goals. In turn, these differences color the perceptions held by other global actors—states and non-state entities—about where and how to treat each major power.

“First, who is America’s principal competitor? The Trump administration’s national security strategy and the aforementioned national defense strategy often refer to China and Russia jointly. That two countries of such significantly different economic proportions can both be considered Great Powers—China’s gross domestic product in 2017 was nearly eight times as large as Russia’s (\$12.2 trillion versus \$1.6 trillion)—suggests that analysts should delineate more clearly the criteria for earning that designation.”

—Ali Wyne, “America’s Blind Ambition Could Make It a Victim of Global Competition,”
The National Interest, February 2019

Although most global analysts and international relations scholars generally agree that the United States remains the most powerful of the three contemporary Great Power states in 2020, pundits diverge widely in terms of how great the challenge from Russia or China is to current U.S. dominance.² They also diverge in practical terms about how the U.S. Government should conceive of competition between these two rivals.³ Some observers even question the pairing of Russia and China as Great Power competitors to the United States, noting that Beijing is a resurgent, selective revisionist power, while Moscow is a faltering, disruptive, and opportunist one.⁴

Among those pundits worried about con-

flation of Russia with China, most agree that China will pose a greater long-term challenge to Washington. History reminds us, though, that declining powers can enact more acute short-term disturbances.⁵

These debates make it important to carefully compare the commonalities and differences among the three modern Great Powers. This chapter does so first with a direct comparison of two dimensions of current global Great Power relations: their recent geostrategic trajectories and their contemporary national strategies for current and future geostrategic relations. Once established, these comparisons are applied to an overview of the major aspects of their interactions, with the current global institutions and norms and then in the major geographic locations of interaction. The following chapter then compares

and evaluates the Great Power national capabilities (tool sets) and their capacity and willingness to use these tools in pursuit of geostrategic ambitions as developed in this chapter.

In keeping with the framework developed in chapter 2 (table 2.2), this chapter and the next address Great Power strategies, capabilities, and willingness to use their tool sets in a framework focusing on five competitive categories: political and diplomatic, ideological, informational, military, and economic. A credible assessment of GPC in the emergent era must begin with a brief summary of the recent trajectory of geostrategic relations from the perspectives of each major power.

U.S. Perspectives from 2000 to 2015

From 1992 to 2008, the United States stood alone—atop all aspects of the international power structure—at a unique unipolar moment.⁶ Its post–World War II rival, the Soviet Union, was vanquished in 1991 with the end of the Cold War, and so was Moscow’s vision of universal communism and the superiority of command-directed economies. Another latent potential rising power, China, began its own “opening up and reform” in late 1978, when the Chinese Communist Party (CCP) incorporated market principles into “socialism with Chinese characteristics.”⁷ With a pragmatic foreign policy focused on external stability, China began a gradual but accelerated integration into global economic and governance structures; this effort was accompanied by an end to CCP efforts to control thought and micromanage peoples’ daily lives. Relative to life during the Mao era, Chinese citizens gained increasing control over their life choices and enjoyed greater political and ideological freedom as long as they did not challenge CCP political control.⁸ Policymakers in Washington welcomed these trends toward internal liberalization and external moderation. They were cautiously optimistic that a liberalizing China would eventually become integrated into the web of global economic, informational, political, and ideological norms established by the United States after World War II. Washington pursued “strategic engagement” in its official relations with both Moscow and Beijing.⁹ Thus, U.S. strategy and policies toward China and Russia from 1992 through 2008 were dominated by cooperative interactions and collaborative programs. The goal of this collaboration was supporting domestic reforms and integrating both countries as responsible members of the global community defined by U.S. norms and preferences. As documented in chapter 2, this dominant cooperative paradigm coincided with a period of U.S. military ascendance rarely seen in the past 500 years.

From a U.S. perspective, the era of stable cooperative/collaborative relations wavered and then collapsed between 2008 and 2014–2015 due to a series of Russian and Chinese military and paramilitary actions in Georgia, Ukraine, the South China Sea, and the East China Sea and in commercial interactions that demonstrated clear disagreement with key aspects of the U.S.-led liberal international order.¹⁰ Russia conducted a covert military invasion of Crimea in 2014 and then annexed it from Ukraine. U.S.-led Western countries slapped Russia with various economic sanctions and expelled Moscow from diplomatic and economic organizations it had joined in the immediate post–Cold War world. From 2013 through 2015, Chinese assertiveness in maritime territorial disputes, increasing state intervention to support Chinese businesses at the expense of foreign competitors, and Xi Jinping’s centralization of power and tightening of political and information controls catalyzed U.S. responses. In 2014 and into 2015, the Obama administration asserted freedom

of navigation rights and challenged Chinese Pacific maritime claims. It openly condemned Chinese industrial espionage and intellectual property practices, and it reimaged a broad new Trans-Pacific Partnership as a lever to reshape Chinese economic policies. The U.S. policy toward its former geopolitical “strategic engagement” partners chilled gradually during the second term of the Obama administration, with a public hardening toward both nations during 2014 and 2015.¹¹ Thus, 2014 and 2015 were the years that a de facto, three-party Great Power rivalry became obvious—although not yet fully acknowledged.¹² The formal declaration of the Great Power rivalry was acknowledged and conveyed in U.S. strategy documents published in 2017 and 2018: the 2017 National Security Strategy (NSS) and the 2018 National Defense Strategy (NDS).¹³

China’s Perspectives from 2000 to 2014–2015

The central challenge facing Chinese leaders in the post–Cold War era was to take advantage of the opportunities provided by a globalizing world economy and rebuild domestic legitimacy while managing China’s vulnerability to outside pressure, especially that of ideological and military pressure from what it viewed as a dominant and unrestrained United States. After the United States imposed sanctions following the 1989 Tiananmen domestic political crackdown, Chinese leaders concluded that Washington was pursuing a strategy of “peaceful evolution” to end CCP rule and seeking to Westernize (*xihua*) and split up (*fenhua*) China. They adopted a strategy of trying to resist U.S. ideological subversion and limit pressure while maintaining a cooperative relationship with Washington by compromising on less important interests and deferring goals, such as unification with Taiwan and China’s maritime territorial claims.¹⁴ Chinese leaders sought to extend a post–Cold War “period of strategic opportunity” to build China’s comprehensive national power relative to that of the United States and to allow an inevitable global trend toward multipolarity to erode U.S. dominance and constrain its unilateral behaviors. This restrained policy was consistent with Deng Xiaoping’s dictum after the collapse of the Soviet Union that China should “bide its time and hide its capabilities” and avoid premature efforts to play an international leadership role.¹⁵

The U.S. response to the September 11, 2001, terrorist attacks aided China’s strategic approach. Washington’s plunge into long-term commitments in Afghanistan and Iraq diverted U.S. attention to the Middle East and ensnared the U.S. military. Then the 2008 global financial crisis—which produced a prolonged U.S. recession even as China’s economy returned to its rapid growth trajectory—led many Chinese analysts to see an acceleration of U.S. relative economic decline as a sign of growing multipolarity that created new opportunities for China. Although Chinese leaders sought to avoid a direct clash with Washington, they hastened efforts to expand China’s regional and global economic presence and influence, sometimes at the expense of the United States. Rapid and sustained economic growth, especially following China’s accession to the World Trade Organization (WTO) in 2001, helped the CCP increase domestic legitimacy based on its ability to build China’s power and raise living standards. Growth also gave Chinese leaders more resources and new channels of influence as additional countries became dependent on access to China’s market and sought loans and economic assistance from Beijing.

China's economic success was based largely on orthodox development economic advice about maintaining a stable financial system while giving markets the dominant role in reallocating labor and resources to their most productive uses.¹⁶ China's openness to foreign direct investment brought an infusion of Western capital, technology, and management practices that contributed to rapid growth and turned China into an export powerhouse, as Western and Asian multinational corporations incorporated inexpensive Chinese labor into their production networks. At the same time, China rejected advice from the International Monetary Fund and World Bank to fully liberalize its capital account, preferring to manage its currency in order to create competitive advantage and avoid the risk of the destabilizing capital flight that brought down multiple governments in the 1997–1998 Asian financial crisis. China also drew from the post–World War II experiences of Japan, South Korea, Singapore, and Taiwan, all of which adopted economic policies that involved a higher degree of state intervention to accelerate export-oriented development and build domestic companies into globally competitive national champions.

China's model involved a large role for state-owned enterprises in which the CCP directly controlled management and provided support in the form of subsidies; preferential access to capital; protected domestic markets; and favorable laws, regulations, and courts. A series of reforms allowed Chinese state-owned enterprises to become more competitive by rationalizing their operations and jettisoning older workers as well as pension and social welfare obligations.¹⁷ China's rapid and sustained growth, while maintaining a communist political system, yielded a growing sense of self-confidence among CCP leaders and theorists. Although CCP leaders initially downplayed praise of a "Beijing consensus" by Western analysts, in recent years Chinese leaders have argued that China's development experience with authoritarian capitalism is a valid alternative model that deserves respect and has useful lessons for other developing countries.¹⁸ Some even argue that China's performance in responding to the 2020 novel coronavirus demonstrates the superiority of China's model over Western approaches.

At the same time, Chinese leaders worried about a range of potentially serious domestic threats to sustained CCP rule. These included separatist threats in Tibet and Xinjiang, where ethnic minorities mounted violent protests in 2008; the political impact of rising inequality and worries about what might happen if economic growth slowed; and a growing number of mass incidents in which citizens protested local CCP corruption and governance. CCP leaders also worried about U.S. subversion. Beijing shared Russia's view that the United States had fomented a series of "colored revolutions" in the Middle East, Georgia, Kyrgyzstan, and Ukraine to overthrow authoritarian governments, and it worried that China was a future target. China had conditionally tolerated U.S. alliances in Asia, as long as they were not aimed against China and contributed to regional stability. As Washington sought to modernize its alliances for the post–Cold War era and to increase security cooperation with Taiwan, Chinese leaders and strategists began to view the United States as encircling and constraining China in order to slow its growth and obstruct its regional ambitions. These concerns increased with the Obama administration's rebalance to Asia, which was formally announced in 2011.¹⁹

By the time of Xi's accession to the CCP Secretary General position in November 2012, China's political leadership saw a new strategic moment, with both opportunity and threat.

The strategic opportunity came from China's dramatic economic and technological advances, increased influence and military power, and the perception that the United States and the West were entering a state of irrevocable economic and moral decline. The strategic threat came from internal pressures that could jeopardize stability and the potential for the United States to step up ideological efforts to subvert the Chinese socialist system and economic and military efforts to encircle and contain China. Although internal scholarly debates about whether its "moment had come" continued into Xi's premiership, China's self-concept and CCP leaders' vision of a proper future world order dramatically changed between 2000 and 2014.²⁰

Russian Perspectives from 2000 to 2014

Vladimir Putin ascended to political leadership in Russia at the end of 1999. Under his predecessor Boris Yeltsin—Russia's first post–Cold War elected leader—the country emerged from an unraveling Soviet Union as a weakened international power with an agenda to adapt to a global world order valuing individual freedoms, liberal democracy, capitalism, openness, and transparency. Yeltsin's Russia aimed for accommodation and assimilation into the U.S.-led world order. The 1990s saw Moscow enter such economic and financial institutions as the G7/G8 and the WTO and witnessed Russia grudgingly accede to expansion of the North Atlantic Treaty Organization (NATO) military alliance into Eastern Europe while gaining an observer status in NATO as a form of compensation.²¹

Russian cooperation with and convergence into a Western set of norms and institutions wavered as early as the mid- to late 1990s, with Washington and Moscow disagreeing over U.S./NATO intervention in the Balkan wars and aggressive U.S. expansion of NATO.²² Russian cooperation with the West reversed fully under Putin, who swiftly generated a Russian national security concept that decried the post–Cold War world as fundamentally unjust and untenably dismissive of Russia's proper role as a Great Power:

*The world situation is characterized by a system of international relations . . . [with] attempts to create an international relations structure based on domination by developed Western countries in the international community, under U.S. leadership and designed for unilateral solutions (primarily by the use of military force) to key issues in world politics in circumvention of the fundamental rules of international law. . . . Russia is one of the world's major countries, with a centuries-old history and rich cultural traditions. Despite the complicated international situation and difficulties of a domestic nature, Russia objectively continues to play an important role in global processes by virtue of its great economic, science-technological, and military potential and its unique strategic location on the Eurasian continent.*²³

Putin immediately singled out NATO as an entity hostile to Russian security interests and insisted that the Alliance's eastward expansion represented a dire threat to Moscow.²⁴ Later, in a 2007 speech, he explicitly warned NATO to cease eastward expansion.²⁵ Putin also tapped into historic Russian nationalism to contest U.S. and Western "disrespect" for Russia's rightful role as a Great Power. Many Russian elites joined Putin, amplifying an already extant narrative that the United States and its Western allies had taken advantage

of Russia at its moment of greatest weakness—building a sense of national victimhood.²⁶ Putin began a two-decade pushback on U.S. military and political power, tactically and operationally challenging the military and its allies in a series of global activities that included an invasion of the state of Georgia in 2008 and an annexation of its province, Abkhazia.²⁷ Six years later, Putin orchestrated a paramilitary invasion and annexation of Crimea, followed by the initiation of a proxy war in eastern Ukraine that continues into 2020. Tough Western financial sanctions followed the Crimea/Ukraine adventures of 2014, and Russia withdrew its observers from NATO, found itself expelled from the G7/G8, and became a key player in the dissolution or abrogation of multiple post-Cold War arms control regimes.²⁸

Russian domestic politics regressed from the early 1990s, with Putin crafting rules that moved Russia toward illiberal democracy and authoritarian rule, including abrogation of term limits for the Russian president. Economics under Putin devolved into what many Western analysts decried as a “military-industrial-political-criminal complex” designed to launder high volumes of Russian money captured by Putin-friendly oligarchs. Russian economic growth became increasingly tethered to global oil prices—its main export. Rising prices underwrote a period of heady prosperity in Russia from 2000 to 2006, but the majority of the country’s economy entered into long-term stagnation before and especially after the global financial crisis of 2008–2009. Putin’s team failed to diversify or modernize Russian industries. Massive state-backed construction projects became a gold mine for crooked officials in Russia, and a vast amount of Russian wealth got transferred out of the country by a cohort of Putin-connected oligarchs into Western banks, real estate ventures, and other personal investments.

Putin and his oligarch consorts welcomed those aspects of the international financial system that circulated their personal expropriation of Russian wealth. At the same time, they chafed against economic and financial institutions that joined in various ever-increasing sanctions against Russian individuals and entities indicted for illegal transactions, participation in lethal Russian espionage against “disloyal” expatriates, and involvement in Russian paramilitary interventions and encounters in Ukraine, Syria, and elsewhere. For most of the 2010s, Russia invested heavily in traditional and modern media agencies around the world—leveraging military intelligence expertise as well as commercial technologies—to develop and deliver an array of anti-Western propaganda, conspiracy theories, and disinformation. This novel “information management” undertaking has generated enormous global impact by questioning the legitimacy of longstanding Western political institutions, societal norms, and leader legitimacy—enhancing the Russian narrative that U.S. and Western values and frameworks are illegitimate (except for those that the Putin kleptocracy deems useful).

At the end of 2019—20 years into the era of Putin—Russia stood as a Great Power state with pride in its recent past. It is a military Great Power. It has a reorganized military on which Putin has spent a disproportionate amount of Russian gross domestic product—an average of 4 percent per year since 2010—to ensure that its nuclear weapons remain a viable deterrent and its ground and air forces can protect its borders and act decisively against threatening states in its near abroad, and it continues to demonstrate limited but effective global projection abilities while leveraging Russian airlift and some sealift to enable a mixture of military and paramilitary forces.²⁹ At the same time, Putin’s Russia displays

characteristics in its economic, political, and ideological elements that make it seem less than a Great Power. To some extent, Putin has offset these serious liabilities masterfully while leveraging modern communications and information messaging to “question everything” about the Western-led world order.

With this short analytical overview of the past 20 years of U.S.-China-Russia relationship trajectories established, this chapter now turns to delineation of the current strategies of each of the Great Powers and what these strategies indicate for specific activities and policies from 2020 to 2025.

Great Power Strategies

U.S. National Strategy and Geostrategic Trajectory

The National Security Strategy (NSS) of December 2017 asserted that the United States and fellow Great Powers Russia and China had transitioned formally from an almost 20-year period of cooperation and collaboration into a new era of competition.³⁰ The 2017 NSS identified three additional threats to U.S. security—North Korea, Iran, and transnational terrorist and criminal organizations—but clearly premised U.S. security and future prosperity on the ability to compete with Moscow and Beijing.³¹ Although the advancement of NSS and NDS premises into action during 2017–2020 generated tensions between U.S. economic and security aims as well as between U.S. administration political aspirations and ideological norms, these documents clearly build on trends present prior to 2016.

First, the United States continued its halting but longstanding efforts to rebalance economic, military, and informational priorities toward the Asia-Pacific region. The Trump administration formalized a new term, *the Indo-Pacific region*, to highlight the increasing priority placed on India and South Asia by successive U.S. administrations.³² The George W. Bush administration had declared the 21st century to be the “Asia-Pacific century” but then got enmeshed in the South Asian and Middle Eastern wars against terrorist organizations.³³ Beginning in 2011, the Obama administration announced a long-anticipated “Rebalance to Asia” but, like its predecessor, found itself shackled to counterterrorism activities in other parts of the world that reduced the salience of a rebalance to Asia.³⁴

Second, the 2017 NSS embedded several U.S. economic, military, and ideological norms for the Indo-Pacific region that evolved during the Bush and Obama administrations and adhered closely to longstanding post–World War II U.S. preferences for global order and interaction. Prior to the 2017 NSS, they were most clearly articulated in the U.S.-India Joint Strategic Vision for the Indo-Pacific Region of January 2015: commitment to growth and broad-based prosperity in the region; assurance of free and unfettered trade; assurance of freedom of navigation and flight; commitment to the peaceful resolution of disputes among states; adherence to international norms and protocols (especially those regarding sovereignty); collective action against terrorism, piracy, and the proliferation of weapons of mass destruction; and the commitment to universal human rights.³⁵

Third, growing U.S. energy independence in the early 21st century began to erode American interests in the provision of stability in and commercial access to areas of the world important primarily for their fossil fuel exports, particularly in the Middle East. This

trend was encouraged by growing domestic wariness of overseas military activities and a move toward insular nationalism that emerged before 2017.³⁶

Conversely, the 2017 NSS exhibited a new and skeptical attitude toward post–World War II U.S. alliance structures and partnerships. NSS language focusing on the autonomous, unilateral imperatives of U.S. national strategy threw into stark relief the longstanding U.S. valuation of international institutions, multilateral alliances, and partner nations. An unwavering U.S. commitment to politico-military institutions such as NATO and its bilateral Pacific alliances with Japan and South Korea could no longer be taken for granted.³⁷ This change signaled reduced U.S. interest in bearing the costs of maintaining global norms, rules, and procedures in economic, military, diplomatic, and informational domains. In 2020, it remains to be seen how this devaluation of post–World War II institutions and alliances can be reconciled with consistent and continuing U.S. preferences for global rules and norms.

At the same time, U.S. foreign policy ended the 2010s in an ambiguous place. Strategic writings champion the importance of alliances, economic norms, and multilateral institutions, but U.S. actions between 2017 and 2019 veered between strong support and the view that these institutions are “fundamentally unfair.” U.S. foreign policy has focused on renegotiating agreements to redress the U.S. trade deficit but has found itself unable to “easily win” bilateral trade wars with an array of states, including China. Its efforts revealed that many domestic economic constituencies value multilateral free trade and fear American decoupling from both the Chinese economy and the wider global one. These tensions in early 2020 suggest that current foreign and domestic policies are impediments to the U.S. ability to successfully engage in the kind of strategic Great Power competition envisioned by the NSS and NDS.³⁸

China’s National Strategy and Geostrategic Trajectory

Avoiding a hostile relationship with Washington has been a consistent element of China’s post–Cold War national strategy. Simultaneously, Beijing has sought to reduce its vulnerability to U.S. power by building its own comprehensive national power (a Chinese construct that includes all elements of power) and improving relations with other major powers, countries on its periphery, and developing countries in other regions. Chinese leaders have relied primarily on economic, diplomatic, and informational instruments to advance foreign policy goals. China has pursued strategic partnerships with other Great Powers and major regional powers around the world to strengthen political and economic relations without the binding commitments inherent in alliances. China also has begun establishing and supporting new regional and global institutions that can expand its influence and provide a counterweight to the United States. They include the Shanghai Cooperation Organization, focused on Central Asia; the Brazil, Russia, India, China, South Africa (BRICS) grouping; the Conference on Interaction and Confidence-Building Measures in Asia; and the Asian Infrastructure Investment Bank, which provides an alternative to the World Bank and the Asian Development Bank.³⁹ To dampen concerns about rising Chinese military power, Beijing launched a so-called charm offensive that articulated “win-win” policies based on economic cooperation and sought to downplay growing Chinese military capabilities and economic leverage over smaller states.⁴⁰

Beijing's efforts have been most complicated in the Indo-Pacific region, where economic and military power from China poses the greatest direct threat to its regional neighbors (see chapter 9). For countries outside the region, China successfully portrayed itself as an economic partner of opportunity with a vast supply of cheap labor that could reduce production costs for multinational companies; a large market with 1.3 billion potential consumers; and, by the mid-2000s, an important source of foreign direct investment, loans, technology, and foreign aid that could help other developing countries.

China's rapid growth was achieved through increased integration into the global economy, a course that not only took advantage of opportunities in a globalizing world but also created new vulnerabilities. As China became the "workshop of the world" following its 2001 entry into the WTO, its large trade surpluses meant that employment of many Chinese workers became dependent on continued access to developed country markets in North America and Europe. These critical markets were subject to unpredictable and unpleasant external economic developments, such as the 2008 global financial crisis. The production and consumption needs of China's booming economy made Beijing increasingly reliant on imported oil and natural gas, natural resources, and foodstuffs. In response, the Chinese government urged Chinese companies to "go global" in search of markets, natural resources, and technology.⁴¹

China's strategic successes produced an expanded overseas footprint of investments, loans, companies, and workers—many in unstable parts of the world—that had to be protected. Success also increased China's dependence on sea lines of communications that passed through maritime chokepoints such as the Malacca Strait and the Gulf of Aden. These areas were vulnerable to disruption by nonstate actors (such as Somali pirates) and to interdiction by major naval powers such as the United States and India. President Hu Jintao articulated the "new historic missions" for the People's Liberation Army (PLA) in 2004, tasking the Chinese military to expand beyond its traditional missions of defending China's sovereignty and territorial integrity and ensuring CCP rule to also protect China's overseas interests and right to development. This shift led the PLA to begin antipiracy patrols in the Gulf of Aden in December 2008 and to conduct evacuations of Chinese citizens from Libya (2011) and Yemen (2015).

Under President Xi, these various economic, diplomatic, and military initiatives have been drawn together into a more coherent strategy. In 2013, Xi launched the Belt and Road Initiative (BRI), which gathers many of China's overseas investments, loans, and infrastructure projects under one grand banner.⁴² BRI seeks to strengthen China's land and maritime connections to Eurasia with new ports, roads, and railroad infrastructure, funded by Chinese loans and built by Chinese companies. The vision is of a future in which China stands at the center of a vast Eurasian regional system, integrated economically and tied together by road, rail, and pipeline infrastructures. The economic benefits to China are obvious, but the BRI also has strategic implications in terms of expanding Chinese economic influence over participating countries and constructing alternative trade routes that bypass maritime chokepoints.⁴³ Western critics of BRI fear that it will not only enhance China's access to material resources and markets but also enable future capabilities for surveillance and social control and boost Chinese influence to push for broader changes in global governance and

international norms.⁴⁴ BRI has expanded geographically to include Africa, the Middle East, and Latin America and functionally to encompass a “digital silk road.”

During its reform era, China initially focused on exploiting its comparative advantage in cheap labor, but Chinese leaders also sought to help state-owned enterprises and private companies innovate and move up the global “value chain.” This effort initially involved relatively decentralized joint ventures and efforts to acquire foreign technology, quality control, and management skills. But increasingly this strategic endeavor involved national-level industrial policy tools to promote indigenous innovation and help Chinese companies become globally competitive producers and exporters. Promulgated in 2015, Beijing’s Made in China 2025 is a 10-year economic development plan that leverages a range of government subsidies to make China dominant in global high-tech manufacturing. It includes a range of practices that skirt global investment, intellectual property rights, and technology transfer norms.⁴⁵

China’s economic development has benefited greatly from access to an open, globalized world economy supported by relatively liberal rules, norms, and institutions and undergirded by U.S. power. As China’s power has grown, Beijing has expanded efforts to strengthen China’s influence in various international institutions and to seek changes in international rules and norms to better accommodate its national interests. In early 2020, China does not seek to challenge the United States for global leadership; China’s domestic fragility would make it difficult for Beijing to take on many of the responsibilities and burdens that such a role would entail.⁴⁶ At the same time, China has no interest in shoring up the foundations of U.S. global leadership and is working with other countries such as Russia to promote a multipolar world where the United States is less dominant. Chinese scholars and officials have articulated a range of areas where China seeks modifications in international rules and norms: ensuring that China and other developing countries have more influence in global institutions, increasing the degree to which the United States is actually constrained by global rules and norms, and reducing the role of U.S. alliances and military deployments that might constrain China.⁴⁷

China is broadly comfortable with the existing United Nations (UN) system and its formal emphasis on sovereign equality of states, not least because Beijing possesses a veto on the Security Council and can block actions against its interests. Chinese complaints are usually couched in terms of the need for fairness for developing countries and increased “democracy” in international relations. But the underlying demand is for a Chinese seat at the table and a greater role for Beijing in shaping international rules and norms. That said, China opposes many of the liberal principles embedded in the U.S.-led system, such as the emphasis on individual political rights rather than collective economic rights. It favors stronger norms of nonintervention and the rights of sovereign states to choose their political systems and control what happens inside their borders. After years of downplaying Western claims that China has a unique development model, the so-called Beijing consensus, China has begun to argue that its economic success showcases strengths of its CCP-controlled political system and presents a new development model that may have useful lessons for other countries.⁴⁸ This message has appeal for authoritarian governments that hope to replicate Chinese economic success without liberalizing their political systems, although the extent to which China has a coherent model that other countries could emulate is debatable.⁴⁹

Articulated after Xi became secretary general in 2012, his China Dream of national rejuvenation includes the aspiration to build a powerful and prosperous China by 2049. Subsequent CCP statements indicate that unification with Taiwan is considered part of “national rejuvenation.” This broad goal is accompanied by the “two centenary goals” of building both a moderately prosperous society by 2021 and a “prosperous, strong, democratic, civilized, harmonious, and beautiful modernized socialist strong country” by 2049.⁵⁰ Xi’s 2017 report to the 19th Party Congress also articulated a three-step goal for military modernization: The PLA should achieve mechanization and make strides in applying information technology and developing strategic capabilities by 2020; national defense modernization should be basically completed by 2035; and the PLA should become a “world class military” midcentury, on its 100th anniversary in 2049.⁵¹ China has also published a range of narrower national plans and strategies in fields such as high-tech manufacturing, space, and artificial intelligence.

As the preceding discussions suggest, Chinese leaders proclaim national objectives formulated in terms of power and employ a range of diplomatic, economic, and military tools to advance those objectives; however, China’s public articulations of such regional and global goals consistently emphasize vague principles—“mutual respect for each other’s territorial integrity and sovereignty” and “peaceful resolution of disputes through dialogue”—while downplaying conflicts of interest between states and the central role that power plays in international relations. For example, China’s 2017 white paper on Asia-Pacific security cooperation stresses “common development” and “political and security partnerships” but expresses a negative view of alliances.

The document draws a distinction between “major powers” and “small and medium powers,” which “need not and should not take sides among big countries.”⁵² However, the white paper does not mention relative power or balance of power, even though the perceived vulnerability of other Asian countries to China’s rising power is the central dynamic in the Indo-Pacific. This diplomatic effort to obscure relative power and discourage regional efforts to balance against Chinese power is inconsistent with Chinese internal, military, and academic analyses, which regularly stress the importance of relative power and power tra-

jectories. It is also inconsistent with Chinese diplomatic practice, which reflects an acute awareness of power relations and a willingness to use power to reward and punish.⁵³ China’s efforts to articulate a global vision for a “harmonious world” and a “community of common destiny” suffer from similar shortcomings—the implication being that less powerful countries facing a more powerful China must rely on China’s uniquely peaceful nature. The real message is that China will not compromise in pursuing its interests and that less powerful countries must be prepared to give way.

—Aaron L. Friedberg, *A Contest for Supremacy* (2012)

Beijing's lack of candor about its regional and global objectives leaves space for competing assessments of China's international ambitions. There is an increasing consensus that China seeks to dominate the Indo-Pacific region; the analytic differences lie mostly in whether this authority will involve an intrusive effort to micromanage the region or a looser sphere of influence where China seeks a veto on actions that might damage its security interests.⁵⁴ There is agreement that Beijing seeks greater influence over global rules, norms, and institutions, but disagreement on the scope and urgency of China's ambitions and likelihood of realizing them.

Some analysts view China as a moderate revisionist power that seeks to change aspects of global rules and norms that affect its specific national interests, but that has limited ideological ambitions and remains willing to deal with countries regardless of their regime type.⁵⁵ Those with this view see China's articulation of the goal of building a "community of common destiny for mankind" as a continuation of past Chinese efforts to express normative principles that should govern international relations rather than a coherent program for systematic changes in the current rules, norms, and institutions of the post-World War II order. From this perspective, China's efforts to use its economic achievements to win international respect for its governance model are aimed more at building domestic legitimacy than at forcing others to follow China's example. Others view China's authoritarian regime as requiring external validation and an accommodating international environment in order to keep the CCP in power, and thus see "a community of common destiny" as a far more ambitious effort to transform international rules and norms and propagate authoritarian values in order to maintain domestic stability.⁵⁶ Both groups agree that China prefers authoritarian norms in areas such as cyber sovereignty and prioritizing collective economic rights over individual political rights, but the latter faction suggests that liberal norms pose an existential political threat to the CCP and thus require its active efforts to transform the international system rather than simply adapt to it.

In early 2020, China seems to conceive a national security strategy in concentric circles. Its primary focus is on internal security, both by maintaining political support from the Han majority and by managing a host of separatist and nontraditional security challenges. For China, this includes preventing Taiwan independence, which it considers a domestic issue. The second ring involves countries on its periphery in the Asia-Pacific region (the Indo-Pacific). China will work to maintain regional stability, achieve a satisfactory resolution of its territorial disputes, and reconcile the region to a dominant Chinese role. The United States stands as an obstacle to these ambitions, so China will work to erode U.S. power and influence in the region while seeking to avoid direct confrontation. The third ring lies outside Asia, where China will seek to maintain access to resources and markets, protect its expanding overseas interests, and expand its influence in regional and global institutions while introducing alternative institutions where feasible. China's approach to relations with its fellow Great Powers will be to seek recognition of its status as a global player and deference to its interests in Asia and beyond. Beijing will avoid direct military challenges or confrontations where possible and will compete in the economic, technological, military, and diplomatic spheres to improve its regional and global position.⁵⁷

"The competition for global influence between the United States and Russia has been exaggerated and fueled in large measure by a combination of inflated assessments of Russian capabilities and ambitions, the United States' expansive definition of its interests, and a pervasive consensus within the foreign policy establishment that the exercise of U.S. leadership demands a central role for Washington in engineering American solutions to global problems. . . . The only realistic and sustainable solution to these challenges is to focus on their root causes and adopt a more realistic view of Russia and a more disciplined, restrained, and judicious approach to defining U.S. interests around the globe."

—Eugene Rumer and Richard Sokolsky, *"Thirty Years of U.S. Policy Toward Russia,"* Carnegie Endowment for International Peace, June 20, 2019

Russia's National Strategy and Geostrategic Trajectory

In Vladimir Putin's Russia in 2020, the perception of existential threats drives its strategy of "aggressively defensive" policies aimed at disrupting the Western world.⁵⁸ Russian global strategy is loosely tethered to a nostalgic view of imperialism and the bygone era of the Russian Empire.⁵⁹ Unlike the Soviet Union, with its positivist strategic aim of promulgating global communism, Putin's Russia pursues reactive and limited strategic outcomes. It asserts rights of control in the historic lands of the Russian and Slavic peoples of its "near abroad" and domestic borders, sometimes referred to as the construct of neo-Eurasianism.⁶⁰ Simultaneously, Russia seeks to despoil U.S.-led rules, norms, and institutions around the globe. It pursues a strategy of reactive resistance to U.S. leadership in-

ternationally and proactive assertion of the right to historic imperial dominance over the states of its near abroad. Moscow seeks a multicentric world that impedes and resists U.S.-led Western institutions. Simultaneously, it asserts regional power and authority based on bilateral, transactional military and economic relations advantages.⁶¹ Both elements of this strategic approach (versus a formal, lucid strategy) represent Putin's desire that Russia be viewed as a global power despite Moscow's demonstrably weak position across all but the military and informational dimensions of strategic interaction.⁶²

Russia seeks to manage its relationship with the United States, the European Union, and NATO to deter "supposed" hostile action by weakening the cohesion of these alliances.⁶³ Moscow also leverages a tactical relationship with China, the Chinese-led Shanghai Cooperation Organization, and the international BRICS consortium countries to tarnish U.S. influence abroad.⁶⁴ Although the Russian government cooperates with China on a handful of political issues, it remains wary of its Asian Great Power competitor and seeks to limit Chinese influence in the former Soviet states.⁶⁵ Russia's regional priorities closely focus on consolidating Russian ascendancy in former Soviet space and Eurasia and on projecting relevance in the Middle East and the Arctic.⁶⁶

Putin understands that his control in Russia is not limitless, and his policy actions directly reflect his wishes to maintain personal power and legitimacy.⁶⁷ He has become a master of deflection, shifting the dialogue from the problems Russia faces at home to managing conflicts abroad through diplomatic and military force.⁶⁸ By flexing Russian specialized military and diplomatic power in the Middle East, Putin cultivates solidarity from "other dictators threatened by revolution," but he does not demonstrate any capacity to bring nations together or work toward common goals or mutual betterment.⁶⁹ Simulta-

neously, Putin seeks Russia's status as a Great Power, but his reliance on the narrative of neo-Eurasianism (sometimes referred to as *confessional imperialism*) is a far cry from the comprehensive ideological framework of the former Soviet Union or the modern, cohesive ideology of China.⁷⁰ Unlike the Soviet Union, which oversaw the First and Second Communist International forums that proselytized a formal and structured vision of world rules, norms, and institutions featuring Marxist/Leninist sociopolitical and command-driven economies, Putin's Russia does not have any form of a positivist vision for reordering the world political space.⁷¹

Thus, Russia is a contemporary Great Power with a peculiar mix of short-term power capabilities and long-term challenges. It leverages a history of martial prowess with unique global capabilities in several areas of military might, especially nuclear weapons, space, and aerospace, and specialized but limited precision weapons and elite forces power projection through recently modernized air and sea platforms. Modern Russia also generates considerable disruptive capacity and will do so over cyberspace and social media, undermining Western political, ideological, and informational narratives with a limited-aspiration, maximum-confusion campaign. As chronicled in the following chapter, Russia's current capabilities match well with Putin's limited strategic aspirations and transactional aims; however, the future of Russia's Great Power status is in doubt, as its major power indicators are receding today and promise an even greater downward turn into the future.

Evolving GPC Bilateral and Trilateral Geostrategic Dynamics: Norms, Institutions, and Geographic Regions

Based on the national strategies and geostrategic trajectories of the three Great Powers in early 2020, one can discern in and among them today broad philosophical and specific regional dynamics that are likely to remain salient over the coming 5 years. Some have to do with U.S.-Russia relations; several involve U.S.-China relations; and others engage Chinese and Russian dynamics.

First, all three Great Power states have unique perspectives and attitudes about the *established rules, norms, and institutions of the international system*, which produce consequential strategic contrasts and policy imperatives.

The United States was the dominant architect of post-World War II norms, rules, and institutions for international interactions and exchange. In general, these American preferences continue to dominate the contemporary global system, which bears the hallmarks of a U.S./Western desire for multilateralism, the peaceful resolution of disputes, cultural pluralism, free and open global trade and finance, open and transparent communications, and individual human rights.⁷²

However, Washington increasingly has become sensitive to the fiscal and human costs of maintaining and enforcing the existing order. Since 2017, the Trump administration has amplified existing American concerns that preferred U.S. rules and norms are too costly and other states are unwilling to pay a fair share of maintenance costs (the free-rider problem). The Trump administration has chafed more openly than its predecessors at the fact that multilateral organizations and regimes constrain unilateral U.S. bargaining power. Growing insularity and nationalism have masked U.S. self-awareness that it derives enormous benefits from current rules that are unlikely to last should the standing order erode.

Simultaneously, the United States remains oblivious to the ways in which it has acted outside of global rules when convenient. It also has been unable and/or unwilling to generate new rules about international issues, such as currency valuation, the use of space, the policing of cyberspace, and others.⁷³

Although invited into the global order during 1945–1946, the Soviet Union spent the Cold War ideologically opposed to the U.S./Western “first-world” order, limiting its participation to parts of the United Nations (UN) system. After a brief period of attempted assimilation during the 1990s, Putin’s Russia entered 2020 with a mixed attitude toward that order: working to erode many aspects of the system while leveraging the selective benefits of the order where Russian national interests are met, such as at the UN and in the international banking and finance system. Working with China and other states, Russia desires to reshape some international rules and norms that constrain its power. At the same time, Russia is unlikely to accept integration into institutions that it did not design, as Putin believes Great Powers do not dissolve into other integration projects but forge their own.⁷⁴

Contemporary Russia can be expected to support rules that allow for authoritarian regimes, resisting those that assert a “duty to intervene” against totalitarian or abusive governments. Russian political and diplomatic interests remain aligned with tethering friendships and transactional state-to-state engagements with all states willing to entertain Moscow’s presence—especially when those friendly states join Russia in opposing longstanding Western norms. Putin’s Russia will work against norms of nonintervention and military restraint—actively cultivating paramilitary and proxy forces that violate fellow-state sovereignty—in the pursuit of dominance in its near abroad and when responding in support of a friendly state anywhere in the world.⁷⁵ Moscow will exploit today’s trade and finance systems to its advantage but will resist and subvert economic system norms when they mandate too much transparency or exact too much fiscal pain. Its approach to cyberspace, social media, and other forms of mass communication will continue to sow confusion and derision in the activities of competitor states, resisting the creation of new rules and norms mandating reciprocal freedom and openness in this informational realm.

The People’s Republic of China was not involved directly in the establishment of the post–World War II global order.⁷⁶ As mentioned, Beijing began engaging global institutions beyond the UN system as part of its reform and opening up policy in 1978, with a focus on those areas with direct benefit to China’s growth and development. China claims to act as a representative of developing countries in global institutions, even though its own status and interests have diverged as its power has grown. China has benefited from many economic institutions that support trade and commerce, such as the WTO and the World Bank; however, China has sought to use its status as a developing country to resist or evade some commitments and has taken advantage of gaps in international rule and norms in areas such as currency valuation. China has sought a greater voting share and increased influence in institutions such as the World Bank, the International Monetary Fund, and the Asia Development Bank, but it has also begun to develop parallel institutions such as the Asian Infrastructure Investment Bank as vehicles for its interests.⁷⁷

Great Power national strategies and geopolitical aims combine in 2020 in a manner that highlights some areas where collaboration may remain feasible. They also indicate

Table 3a.1. Basic Postures and Compatibility of Strategic Aims, 2020–2025

	United States	China	Russia	Remarks
Political and Diplomatic	Liberal democratic governance	Authoritarian, one party state rule	Authoritarian rule with illiberal democratic facade	U.S.-China = incompatible; China-Russia = short-term compatible; U.S.-Russia = short-term incompatible
Ideological	"Free and open societies" with individual freedoms, universal human rights, and pluralism	"A community of common destiny" (ambiguous); state sovereignty and collective order, limited human rights	Loose "neo-Eurasianism" and multipolarity; state ascendancy, Russian sovereignty and disruption of global norms	U.S.-China = incompatible; U.S.-Russia = short-term clash/long term? China-Russia = short-term compatible, long term?
Informational	Free and open exchange, only limited restrictions	State control and cyber sovereignty restrictions; increasing external propaganda	State overwatch with selective closure; complementary disinformation	U.S. incompatible with China and Russia
Military	High-tech, robust quantity mix of regular and irregular forces, great deployment reach	Improving tech, high-quantity, regular forces; gradually expanding regional deployability	Some critical high-tech, limited quantity regular with irregular forces; global deployability in selected areas/missions	(details provided in chapter 3b)
Economic (Trade and Finance)	Free and open trade; transparent, free-flowing finances	Open trade with strong state role; managed financial system; barriers to trade	State-monopolized trade; exploitation of international finances for oligarch gains	U.S.-China = compatible in near term if U.S. underpins; China-Russia = compatible and largely complementary; U.S.-Russia = compatible if U.S. accepts Russia free riding and stops use of trade sanctions and financial listing to achieve political aims

areas where strategic aims and values are likely to be competitive or even confrontational. Table 3a.1 helps paint a picture of the prospects for strategic compatibility in the five areas of state interactions that underpin global rules, norms, and institutions.

Political and Diplomatic

The current system aligns with U.S. goals and strategic aims, supporting the role of the UN, multinationalism in diplomatic relations, and peaceful and collective resolution of disputes. It also is biased toward a U.S./Western preference for liberal democratic governance. Yet Washington demonstrates increasing ambivalence about many of these norms and institutions; it remains stolid in declared support but far less certain in its policy actions to sustain them. China values the UN and its protection of state sovereignty but is increasingly using its economic and military power to pursue its interests in the Indo-Pacific region at the expense of other countries. With its domestic CCP dominance and preference for a state-centric global order, China is opposed to liberal democratic governance norms and is

seeking to expand its influence in regional and global institutions. Russia, with its Security Council veto power, also favors the status quo institution of the UN. Moscow has recently demonstrated a desire to take charge of multilateral political institutions such as Interpol and global counterterrorism organizations, presumably from a desire to bend these toward Russian interests.⁷⁸ Moscow is less accepting of peaceful, collective resolution of disputes. Under Putin, Russia practices single-man authoritarianism with illiberal democratic institutions. After Putin, Russia's political culture may change, but that outcome seems unlikely in the coming decade.

Ideological

In 2020, the U.S./Western ideology valuing free and open societies, commercial markets, and protection of political rights clashes directly with Beijing's desire to leverage Chinese power to obtain regional deference, preference for authoritarian norms, and pursuit of the ambiguous goal of a *community of common destiny*. There is a growing awareness of this clash, but the impacts from it moving forward remain uncertain. Moscow has no overarching ideology save that of sustaining historic Russian pride and prominence, maintaining ascendance in its territorial near abroad, and exercising global access and avoiding cooperation by a U.S./Western order.

Informational

U.S. preferences are for the free and open exchange of ideas with little restriction and a global communications architecture that features consensus-based cooperation. Russia and China find this construct threatening and prefer closed and restrictive communications and exchange, with the state having the right to control the flow of information within and across its borders. Both states have well-developed propaganda and censorship apparatuses for both online and traditional media and seek to use these operations to shape foreign perceptions. China denies Western accusations that it engages in extensive commercial and cyber espionage. Russia also has been willing to leverage the current openness in the system to flood it with disinformation and discordant themes, confronting its regional opponents and the United States with uncomfortable dissonance in open communications systems.

Military

The current international system espouses the peaceful resolution of disputes and multilateral cooperation to deal with aggressor states. The UN was founded on this premise, and many subsequent regional and functional organizations and norms have grown up around sustainment of the practice. But this promise has not been met in practice, and in 2020, there are differing views by the Great Powers on where and how to use the military instrument. In general terms, the United States has a dominant military across most of the use-of-force spectrum and an unparalleled ability to project military power. Washington continues to emphasize a high-tech, large-quantity force with a dominant set of deployment and sustainment resources. China has a military that is rapidly improving its technological capabilities and has a limited but growing reach. Chinese military reforms are improving naval, air, and missile capabilities and beginning to build a joint force focused on fighting and winning informationalized wars. Over the past decade Russia has recapitalized its

military in a manner that has sustained its nuclear ascendance, incorporated some critical advanced technologies, and moved toward utilization of a mixture of regular military forces with irregular and commercial forces. Russia also has modernized to sustain global deployability in limited numbers for much of its limited-sized military force. (The following chapter addresses in much greater detail the capabilities of the Great Power military forces.)

Economic (Trade and Finance)

Never perfect in the post–World War II era, modern trade and finance systems were built around the norms of freedom and openness. The General Agreement on Trade and Tariffs established relative norms for free and open trade and was succeeded by the WTO. The World Bank provided pooled funds from developed countries to foster economic growth in underdeveloped ones. The International Monetary Fund offered donor funding to offset short-term imbalances in international payments between countries. As world currencies became convertible, tariffs were reduced and private international investment again became robust.⁷⁹ The United States continues to underwrite the norms of freedom and openness represented in these institutions—but with increasingly obvious “donor fatigue.” China has greatly benefited from these norms and institutions and continues to support many of them. Beijing values free-flowing trade and finance, but with a model that emphasizes a large state role in economic decisionmaking. China has been slow to fundamentally lower domestic barriers to overseas ownership and has chafed at making its state-led economic decisions transparent. As noted, China has recently begun to establish parallel institutions and programs to compete with Western institutions in trade and finance, such as the BRI and the Asian Infrastructure Investment Bank. Russia, too, prefers state-monopolized trade. Moscow openly accepts and welcomes the elements of international trade and financial flows that sustain Putin and his oligarch constituency’s financial gains. Russia does not support or adhere to norms of freedom or openness in its general commercial activities.

As of 2020, trade and financial disputes have become primary flashpoints among the three global powers, and especially between the United States and China. Russia has bridled under U.S. and Western trade and financial sanctions in response to Moscow’s military adventures in its near abroad. More substantively, the United States and China entered a trade war in mid-2018 that continues into 2020. This emergent economic rivalry between the United States and China seems unlikely to just ease into a “normal business struggle,” similar to that between Japan and the United States in the 1980s, which waned with a decade of Japanese stagnation in the 2000s. Many in Washington are growing more and more convinced that an authoritarian, CCP-led China seeks to use unfair competition to challenge U.S. economic leadership and displace U.S. military dominance in the western Indo-Pacific in the near term. Beijing also may wish to supplant Washington’s preferred

“From the outset [June 2018], it was clear that the friction between the world’s two largest economies was about far more than just trade. At issue were long-simmering differences of ideology and values and a context for global geostrategic influence that was increasingly being played out in the economic and technological spheres.”

—“China’s Concept of World Order: Theory and Practice,” IISS Strategic Survey 2019

Table 3a.2. Geographic Regions and Great Power Strategic Interest Intensity, 2020–2025

	United States	China	Russia	Remarks
Indo-Pacific	Primary	Primary	Secondary	Elevated risk of confrontation or clash between the United States and China
Europe	Secondary	Tertiary	Primary	Competition unless Russian near-abroad security interests challenged
Middle East	Secondary	Secondary	Secondary	Mainly competition over resources, but increasingly for prestige
Africa	Tertiary	Secondary	Tertiary	China resource interests—mainly competition
Latin/South America	Primary	Tertiary	Tertiary	U.S. pride and prestige interests
The Arctic	Secondary	Tertiary	Secondary	Limited risk of confrontation short of miscalculation
Space	Primary	Primary	Primary	Very high risk of confrontation and clash in this unregulated competitive space
Cyberspace	Primary	Primary	Primary	Already evident confrontation and concern about greater clash without new norms and rules

international rules, norms, and processes with a more state-centric model of economic activity in the more distant future.⁸⁰

All three Great Power states also have different aims and attitudes toward different *geographic regions* in the competitive space of GPC.

As demonstrated in table 3a.2, there are eight major global regions and environments where the three Great Powers will compete. The strategies of each Great Power reveal that each nation has differing levels of interest intensity in these regions—all do not have primary interests in all spaces. A primary strategic interest intensity is defined as one in which the Great Power believes a significant risk to national security is found in that region, and where it could risk military conflict with a Great Power rival to defend that interest in the next decade. A secondary strategic interest intensity is one in which the state believes only a modest risk to its national security is in play, and where its preferred means of interaction with the other Great Powers will remain competitive and avoid confrontation or direct clash in other than accidental circumstances. A tertiary strategic interest involves a limited to no perceived risk to state security, and where Great Power interactions might be focused on activities that feature at least some collaboration, as well as subdued or proxy-level competition and (very rarely) confrontation.

As noted in this chapter's review of the main Great Power strategic interests, the United States and China have primary interests in the Indo-Pacific region that conflict. Here, their competition could turn toward confrontation or a military clash if careful diplomacy is not exercised. Russia has a primary interest in Europe, with special sensitivity to its near abroad—the former Soviet Union provinces. American and European diplomacy will remain challenged to stanch Russian misadventures without generating overt confrontation or clash. The United States retains a historic interest in primacy across the

Western Hemisphere, and this appears unlikely to be similarly important in the near term by the other two and likely to remain an area of less direct competition absent unforeseen miscalculation. Conversely, the Middle East promises to be an area of dynamic competition and occasional nonmilitary confrontation in the coming decade—with access to resources as the principal focal point of interaction. However, as U.S. and Russian interests in energy sources wane, the competitive focus in the Middle East seems destined to be that of prestige and resonance of ideological narratives.

Interestingly, two nontraditional competitive venues, space and cyberspace, are those where all three Great Powers have primary interests engaged now and into the foreseeable future. There is high risk that intensifying competition in space could lead to greater confrontation there between the modern Great Powers. Agreement on some viable rules and norms for collaborative use and cooperative actions in space would seem a vital undertaking to reduce the growing risks of confrontation and miscalculation leading to clash. Likewise, the absence of cooperative rules and norms in cyberspace has already witnessed this medium for state interaction take a dark turn toward confrontational dynamics; in the absence of new norms and standards for cooperation, this medium of Great Power interaction risks an even greater set of malevolent and confrontational activities in coming years.

Major GPC Comparative Insights and Implications

This chapter's analytical review of Great Power strategic postures and geostrategic preferences provides several important insights about the new era of GPC. Among them, 10 stand out.

First, the United States enters this new era as the dominant Great Power of the three rivals—its preferred norms, rules, and institutions for interstate interactions color all major categories of global activity. But the strategic aims of the three modern Great Powers are incompatible and thus assure the return to a historically dominant pattern of competition between Great Powers in the international system.

Second, China is the one rising Great Power with the combination of a positivist vision for the future and the ambition to push for changes in the international system on a near-term and long-run basis. China may not intend to do so, but the United States finds this combination disruptive to the standing—and Washington's preferred—international order. China's lack of a meaningful boundary between public and private ventures and its wide-ranging and intrusive efforts to gain competitive advantage and coercive leverage over states within its region carry significant risk of escalating confrontation with the United States. Thus, the primary competition in the emerging era—the only truly global, comprehensive national-level competition—likely will be between the United States and China.⁸¹

Third, Russia's Great Power aims are not grounded in a positivist global strategy with discernible alternative norms, institutions, and procedures for an international order. Instead, Moscow in 2020 practices a reactive and often disruptive strategy oriented toward questioning contemporary institutions and processes by which it feels threatened and simultaneously keeping subservient the smaller states on its geographic boundaries. Putin's Russia, therefore, has the potential to cause difficulties on specific issues but does not have the global aspiration to reshape the international system.⁸²

Fourth, Chinese and Russian challenges to the existing U.S.-established global order of 2020 should be sharply distinguished. It is incorrect and unhelpful to lump China and

Russia together as the same kind of comprehensive U.S. rival now or into the future. The divergent degree of their global ambitions means that, for the United States, the future portends an overarching political, ideological, information, and economic competition with China, with secondary, largely regional contestations with Russia.⁸³

“China and Russia refer to their relationship as a ‘comprehensive strategic partnership,’ in which Russia supplies oil to China and the two countries hold joint military exercises. And, officially, their relationship has rarely been better. But trade is lopsided in China’s favor; the fall in energy prices has made China considerably less dependent on Russia. Russia sells arms to China’s adversaries, India and Vietnam. And China has copied Russian weapons designs. These deeper geopolitical realities mean China and Russia will be only allies of convenience.”

—Robert Kaplan, *The Return of Marco Polo’s World* (2018)

Fifth, despite China’s long-term strategic vision and ambition, both China and Russia have regional interests in 2020 that appear more important and urgent than their global interests. Each desires *de facto* spheres of influence free from outside interference—on their borders and in nearby regions.⁸⁴ This is especially true in Central Asia and in the Russian Far East.

Sixth, China and Russia may continue their tactical entente over the coming 5 to 10 years, working together to erode U.S. power, frustrate U.S. actions, challenge U.S.-dominated institutions, and question U.S.-underwritten norms and rules they deem threatening. However, divergent long-term Sino-Russian strategic interests make it unlikely they will form an enduring

alliance.⁸⁵ The United States should remain careful not to misunderstand, as evidence of some deeper strategic cooperation, tactical coordination between Beijing and Moscow that balances U.S. power.⁸⁶

Seventh, all three contemporary Great Powers are dissatisfied with some aspects of international order and are growing less willing to make compromises and sacrifices to keep it running. Thus, there is heightened potential for GPC rivalrous activities to reduce effectiveness of global institutions in managing complex regional and global problems. Some observers argued that the absence of Great Power cooperation or collaboration during the height of the early 2020 coronavirus pandemic was symptomatic of this breakdown.⁸⁷

Eighth, U.S. economic and strategic interests in the Indo-Pacific and in Europe challenge Chinese and Russian regional interests most. These are competitive regions with the greatest near-term salience. The Middle East is evolving as a secondary area of GPC where less important Great Power interests collide in various ways. Other regional areas are of tertiary importance, where GPC interests exist but vary greatly and where competitive dynamics are less clear.

Ninth, space is an arena where longstanding cooperation and a relative absence of strategic competition is giving way to Great Power rivalry and the potential for confrontation. Great Power geopolitical competition is increasingly observed from and managed by space-based platforms. Thus, as global Great Power rivalries heat up, more and more states will develop technologies, including antisatellite weapons, that put human access to or effective use of space at risk. The risk of confrontation and clash in this unregulated competitive

arena is great, but so too is an opportunity to craft norms and rules of cooperation and deconfliction capable of reducing the risk of confrontation in space.⁸⁸

Tenth, cyberspace is already a major medium impacting relative state power and an important element of the emerging era of Great Power competition. The risk of cyber-clash will grow without concerted effort to effect agreed-to global standards and norms that reconcile cyber-openness with security and safety concerns.⁸⁹

From Aspirations to Actions: From “What to Do” to “How to Do It”

The end of the 2010s heralded an indisputable shift of the major dynamics of international affairs from cooperation and collaboration and into a far more competitive set of international relations, especially between the United States, Russia, and China.

In a nutshell, the modern Great Powers have divergent strategic interests, meaning they will compete across five major interaction categories in the next 5 years. Cooperation and collaboration remain possible, but episodes of confrontation and clash are likely, even over nonmilitary issues. The United States must proceed with a clear-minded strategic approach that understands that Putin’s strategic aims make Russia a transient security risk, while Beijing’s ideological vision and aspirations make China the more important, albeit presently less threatening, security threat. A U.S. strategy that cooperates when possible, competes smartly, confronts only when necessary, and concurrently builds out unique U.S. strategic tools and power capabilities across all of its economic, diplomatic, ideological, informational, and military categories appears to be one best suited to the beginning of a new era of GPC.

The next chapter moves beyond an assessment of the strategic aims and intentions of the modern Great Powers. It will explore whether the United States, China, or Russia can achieve their aims. Do they have now, or will they soon possess, the capabilities necessary to achieve their strategic goals? It will itemize and evaluate many of the most important power capabilities and the relative abilities of the United States, China, and Russia now and into the coming decade—with special attention to the next 5 years.

“[I]f China’s power continues to grow, and if it continues to be ruled by a one-party authoritarian regime, its relations with the United States are going to become increasingly tense and competitive. That is the path along which events are presently proceeding, and like it or not, it is the future for which America and its allies have no choice but to prepare.”

—Aaron L. Friedberg, *A Contest for Supremacy* (2012)

Notes

¹ This operational definition is derived from Thomas J. Volgy et al., “Major Power Status in International Politics,” in *Major Powers and the Quest for Status in International Politics: Global and Regional Perspectives*, ed. Thomas J. Volgy et al. (New York: Palgrave Macmillan, 2011), 1–26. Their definition derives from an amalgam of several other classic efforts at defining Great Power status, including the following historic references: Melvin Small and J. David Singer, “Formal Alliances, 1815–1939: A Quantitative Description,” *Journal of Peace Research* 3, no. 1 (1966), 1–32; Jack Levy, *War in the Modern Great Power System,*

1495–1975 (Louisville: University Press of Kentucky, 1983); John Mearsheimer, *The Tragedy of Great Power Politics* (New York: Norton, 2001); Benjamin O. Fordham, “Who Wants to Be a Major Power? Explaining the Expansion of Foreign Policy Ambition,” *Journal of Peace Research* 48, no. 5 (2011), 587–603.

² For examples of the range of these perspectives on China, see Aaron L. Friedberg, *A Contest for Supremacy: China, America, and the Struggle for Mastery in Asia* (New York: W.W. Martin and Company, 2011); Charles Edel and Hal Brands, “The Real Origins of the U.S.-China Cold War,” *Foreign Policy*, June 2, 2019,

available at <<https://foreignpolicy.com/2019/06/02/the-real-origins-of-the-u-s-china-cold-war-big-think-communism/>>; Xiaoye Pu and Chengli Wang, "Rethinking China's Rise: Chinese Scholars Debate Strategic Overstretch," *Chatham House Papers* 94, no. 5 (September 2018). The range of views about the challenge posed by Russia to the United States and the international order is discussed in "How Big a Challenge Is Russia? Foreign Affairs Asks the Experts," *Foreign Affairs*, November 13, 2017, available at <www.foreignaffairs.com/ask-the-experts/2017-11-13/how-big-challenge-russia>.

³ Michael J. Mazarr et al., *Understanding the Emerging Era of International Competition: Theoretical and Historical Perspectives* (Santa Monica, CA: RAND, 2018), available at <www.rand.org/pubs/research_reports/RR2726.html>; Ashley J. Tellis and Robert D. Blackwill, *Revising U.S. Grand Strategy Toward China* (New York: Council on Foreign Relations Press, 2015), available at <www.cfr.org/report/revising-us-grand-strategy-toward-china>; Ian Bremmer, "The End of the American International Order: What Comes Next?" *Time*, November 18, 2019, available at <<https://time.com/5730849/end-american-order-what-next/>>.

⁴ Chapter 10 of this volume addresses the perspective that Russia is not really a Great Power but instead a disruptive (or spoiler) state. Those looking for an alternative view of Russia's role in an era of Great Power competition are encouraged to turn to that chapter.

⁵ Ali Wyne, "America's Blind Ambition Could Make It a Victim of Global Competition," *The National Interest*, February 11, 2019, available at <<https://nationalinterest.org/feature/americas-blind-ambition-could-make-it-victim-global-competition-44227>>. Among those concerned with the short-term Russian risks, Soviet-expat-turned-Norwegian research fellow Pavel Baev's concern is most concise: "Russian revisionism from a position of weakness is both ambitious and desperate." See "How Big a Challenge Is Russia? Foreign Affairs Asks the Experts."

⁶ Charles Krauthammer, "The Unipolar Moment," *Foreign Affairs* 70, no. 1 (1990/1991), 23–33; Fareed Zakaria, *The Post-American World* (New York: Norton, 2008); Hal Brands, *Making the Unipolar Moment: U.S. Foreign Policy and the Rise of the Post-Cold War Order* (Ithaca: Cornell University Press, 2016).

⁷ Susan L. Shirk, *How China Opened Its Door: The Political Success of the PRC's Foreign Trade and Investment Reforms* (Washington, DC: Brookings Institution Press, October 1, 1994); Barry Naughton, *Growing Out of the Plan: Chinese Economic Reform, 1978–1993* (Cambridge, MA: Cambridge University Press, 1995).

⁸ The 1989 crackdown on Tiananmen protestors and the Chinese Communist Party's 1999 campaign to eradicate the Falun Gong movement stand out as exceptions to these general trends.

⁹ The belief that China's economic liberalization would most likely lead to middle-class demands for greater national political participation and a weakening-to-collapse of Communist Party control became somewhat bipartisan political orthodoxy in Washington. See Henry Kissinger, *On China* (New York: Penguin, 2011), 356–367, 393; "Asia," *Strategic Survey* 119, no. 1 (January 1, 2019), 68–133, available at <<https://doi.org/10.1080/04597230.2019.1676992>>.

¹⁰ Russian experts observe that Moscow saw the United States as a Great Power rival much earlier. This is documented in subsequent paragraphs. China experts suggest that Beijing would argue that its rivalry with the United States did not become public until 2015. China's perspective is documented in the following section.

¹¹ Among the tell-tale U.S.-China events signaling the arrival of their de facto Great Power rivalry in 2014 and 2015, several stood out. First, the U.S. Department of Justice made a first-ever public indictment of Chinese military hackers for cyber espionage against U.S. corporations in May 2014. See Department of Justice, Office of Public Affairs, "U.S. Charges Five Chinese Military Hackers for Cyber Espionage Against

U.S. Corporations and a Labor Organization for Commercial Advantage," May 19, 2014, available at <<https://www.justice.gov/opa/pr/us-charges-five-chinese-military-hackers-cyber-espionage-against-us-corporations-and-labor>>. Second, the U.S. Department of State published a paper formally challenging China's assertion of sovereignty over maritime claims related to its "dashed line" encircling islands and waters in the South China Sea. This paper set the legal predicate for freedom of navigation operations conducted in the South China Sea by U.S. warships that began in 2015. See *Limits in the Seas: Maritime Claims in the South China Sea*, Paper no. 143 (Washington, DC: Bureau of Oceans and International Environmental and Scientific Affairs, December 5, 2014), available at <www.documentcloud.org/documents/1376464-us-limits-in-the-seas-dos-no143-china-in-scs-12.html>. Finally, President Barack Obama's November 2014 summit in China did realize agreement by both states to sign the Paris Accord, but it did not find formal compromise for simmering grievances on the two critical issues of Chinese cyber espionage or maritime disputes. See Jane Perlez, "China's 'New Type' of Ties Fails to Sway Obama," *New York Times*, November 9, 2014, available at <www.nytimes.com/2014/11/10/world/asia/chinas-new-type-of-ties-fails-to-sway-obama.html>.

¹² See the similar conclusion found in International Institute for Strategic Studies (IISS), "A New U.S. Consensus on China?" in *Strategic Survey: The Annual Assessment of Geopolitics* (London: Routledge, 2019), 390–391.

¹³ *National Security Strategy of the United States* (Washington, DC: The White House, December 2017), available at <<https://www.whitehouse.gov/wp-content/uploads/2017/12/NSS-Final-12-18-2017-0905.pdf>>; *Summary of the 2018 National Defense Strategy of the United States of America: Sharpening the American Military's Competitive Edge* (Washington, DC: Department of Defense, 2018), available at <<https://dod.defense.gov/Portals/1/Documents/pubs/2018-National-Defense-Strategy-Summary.pdf>>. The 2018 National Defense Strategy (NDS) is a classified document and thus not in public circulation; this NDS summary is the published, public record of that strategy.

¹⁴ Phillip C. Saunders, "Managing Strategic Competition with China," in *Global Strategic Assessment 2009: America's Security Role in a Changing World*, ed. Patrick M. Cronin (Washington, DC: NDU Press, 2009), 260–264.

¹⁵ Deng Xiaoping's statement came in the context of whether China should seek leadership of the international communist movement but was applied more widely as a guide to Chinese policy vis-à-vis the United States.

¹⁶ Naughton, *Growing Out of the Plan*; Nicholas Lardy, *Sustaining China's Economic Growth After the Global Financial Crisis* (Washington, DC: Peterson Institute for International Economics, December 2011).

¹⁷ For an example from the Chinese steel industry, see Edward S. Steinfeld, *Forging Reform in China: The Fate of State-Owned Industry* (Cambridge, MA: Cambridge University Press, 1998).

¹⁸ The "Beijing Consensus," also known as the Chinese Economic Model, is a phrase coined by Joshua Cooper Ramo to distinguish China's economic growth experience featuring authoritarian politics and state-managed capitalist economics from the "Washington Consensus," which blends privatized, deregulated capitalism and liberal democratic politics. See Joshua Cooper Ramo, *The Beijing Consensus: Notes on the New Physics of Chinese Power* (London: Foreign Policy Centre, 2004); Joshua Kurlantzick, "Why the 'China Model' Isn't Going Away," *The Atlantic*, March 21, 2013, available at <www.theatlantic.com/china/archive/2013/03/why-the-china-model-isnt-going-away/274237/>. Some Western observers see the Chinese model as emphasizing the need for a strong, centralized state to constrain rampant economic inequality and social injustice associated with rapacious capitalism and liberal democracy.

See IISS, "China's Concept of the World Order: Theory and Practice," in *Strategic Survey 2019*, 27–28; Friedberg, *A Contest for Supremacy*, 245–263; Aaron Friedberg, "Getting the China Challenge Right," *The American Interest*, January 10, 2019, available at <www.the-american-interest.com/2019/01/10/getting-the-china-challenge-right/>.

¹⁹ See Yawei Liu and Justine Zheng Ren, "An Emerging Consensus on the U.S. Threat: The United States According to PLA Officers," *Journal of Contemporary China* 23, no. 86 (2014), 255–274; "Why Is the U.S. So Keen on 'Color Revolutions'?" *People's Daily Online*, October 11, 2014, available at <<http://en.people.cn/n/2014/1011/c98649-8793283.html>>; Phillip C. Saunders, *The Rebalance to Asia: U.S.-China Relations and Regional Security*, INSS Strategic Forum 281 (Washington, DC: NDU Press, August 2013).

²⁰ For evidence of a continuing debate, see Pu and Wang, "Rethinking China's Rise." On the certainty of China's change in strategic approach, see Kissinger, *On China*, 478–513.

²¹ Eugene Rumer and Richard Sokolsky, *Thirty Years of U.S. Policy Toward Russia: Can the Vicious Circle Be Broken?* (Washington, DC: Carnegie Endowment for International Peace, 2019), available at <<https://carnegieendowment.org/2019/06/20/thirty-years-of-u.s.-policy-toward-russia-can-vicious-circle-be-broken-pub-79323>>.

²² For a review of the trajectory of Russian grievances against its post-Cold War treatment by the United States and the West, see Angela Stent, *The Limits of Partnership: U.S.-Russian Relations in the Twenty-First Century* (Princeton: Princeton University Press, 2014).

²³ "Section I: Russia in the World Community," in *National Security Concept of the Russian Federation* (Moscow: The Ministry of Foreign Affairs of the Russian Federation, January 1, 2000), available at <www.mid.ru/en/foreign_policy/official_documents/-/asset_publisher/CptICk6B6Z29/content/id/589768>.

²⁴ "Section III: Threats to the Russian Federation's National Security," in *National Security Concept of the Russian Federation*.

²⁵ The 2008 promise of membership to Georgia and Ukraine—a U.S.-led North Atlantic Treaty Organization (NATO) announcement—pushed across an invisible but distinct Putin line in the sand. Russia viewed it as another sign of NATO's boundless ambitions, and it raised the prospect that Moscow might lose control, or at least immediate influence, over two neighbors that it saw as critical to its security, well-being, and prestige as a major power. Given the Alliance's commitment to spreading democracy, Russia also perceived its neighbors' intended accession as a threat to its domestic stability. See Rumer and Sokolsky, *Thirty Years of U.S. Policy Toward Russia*.

²⁶ Ibid.

²⁷ The Russo-Georgian war of 2008 was a turning point in U.S.-Russia relations and in Russian relations with the West and proved a harbinger of competitive relations into the future, despite brief efforts by the Obama administration to "reset" them to a more cooperative footing. See Michael Kofman, "Raiding and International Brigandry: Russia's Strategy for Great Power Competition," *War on the Rocks*, June 14, 2018, available at <<https://warontherocks.com/2018/06/raiding-and-international-brigandry-russias-strategy-for-great-power-competition/>>.

²⁸ Rumer and Sokolsky, *Thirty Years of U.S. Policy Toward Russia*. Of course, the United States took the lead in abrogating some Cold War arms-control treaties with Russia as well, most notably the Anti-Ballistic Missile Treaty.

²⁹ Ibid. Also see John W. Parker, *Putin's Syria Gambit: Sharper Elbows, Bigger Footprint, Stickier Wicket, Strategic Perspectives* 25 (Washington, DC: NDU Press, July 2017), 6–20, available at <<https://inss.ndu.edu/Portals/68/Documents/stratperspective/>

inss/Strategic-Perspectives-25.pdf>. For a review of Russian military spending as a percentage of gross domestic product—greater than that by the United States, China, or any Western country since 2010—see SIPRI Military Expenditure Database, available at <<https://www.sipri.org/databases/milex>>.

³⁰ *National Security Strategy of the United States*, 2–3.

³¹ Ibid., 2–4.

³² Rex Tillerson, "Defining Our Relationship with India for the Next Century: An Address by U.S. Secretary of State Rex Tillerson," Center for Strategic and International Studies, 2017, available at <<https://www.csis.org/events/defining-our-relationship-india-next-century-address-us-secretary-state-rex-tillerson>>; "President Trump's Administration Is Advancing a Free and Open Indo-Pacific Through Investments and Partnerships in Economics, Security, and Governance," The White House, November 18, 2018, available at <<https://www.whitehouse.gov/briefings-statements/president-trumps-administration-advancing-free-open-indo-pacific-investments-partnerships-economics-security-governance/>>.

³³ For a review of the George W. Bush administration efforts to increase the priority on Asia, see Nina Silove, "The Pivot Before the Pivot: U.S. Strategy to Preserve the Power Balance in Asia," *International Security* 40, no. 4 (Spring 2016), 45–88.

³⁴ "Fact Sheet: Advancing the Rebalance to Asia and the Pacific," The White House, November 16, 2015, available at <<https://obamawhitehouse.archives.gov/the-press-office/2015/11/16/fact-sheet-advancing-rebalance-asia-and-pacific>>; Brian Michael Jenkins, "President Obama's Controversial Legacy as Counterterrorism-in-Chief," RAND, August 22, 2016, available at <<https://www.rand.org/blog/2016/08/president-obamas-controversial-legacy-as-counterterrorism.html>>. For a perspective suggesting that the Obama administration did more toward realizing an Asia rebalance before 2011 than is accepted in other accounts, see Jeffrey A. Bader, *Obama and China's Rise: An Insider's Account of America's Asia Strategy* (Washington, DC: Brookings Institution Press, 2012).

³⁵ "U.S.-India Joint Strategic Vision for the Asia-Pacific and Indian Ocean Region," The White House, January 25, 2015, available at <<https://obamawhitehouse.archives.gov/the-press-office/2015/01/25/us-india-joint-strategic-vision-asia-pacific-and-indian-ocean-region>>.

³⁶ Alec Tyson, "Should U.S. Strike First If Threatened? Americans Are Divided," Pew Research Center, November 28, 2017, available at <<https://www.pewresearch.org/fact-tank/2017/11/28/americans-are-split-on-the-principle-of-pre-emptive-military-force/>>; "Public Opinion on Foreign Affairs, Terrorism, and Privacy," Pew Research Center, June 26, 2014, available at <<https://www.people-press.org/2014/06/26/section-6-foreign-affairs-terrorism-and-privacy/>>.

³⁷ "New Troublemakers Emerge," *The Economist*, December 7, 2019, available at <www.economist.com/leaders/2019/12/07/new-troublemakers-emerge>; Maria Bartiromo, "President Trump Touts Tariffs as 'Very Powerful' Tool," video, Fox News, June 26, 2019, available at <www.youtube.com/watch?v=citAb7fdq44>; @realDonaldTrump, "South Korea Has Agreed to Pay Substantially More Money to the United States in Order to Defend Itself from North Korea. Over the Past Many Decades, the U.S. Has Been Paid Very Little by South Korea, but Last Year, at the Request of President Trump, South Korea Paid \$990,000,000," Twitter, August 7, 2019.

³⁸ IISS, "China's Concept of the World Order," 390–398; Brian D. Blankenship and Benjamin Denison, "Is America Prepared for Great-Power Competition?" *Survival* 61, no. 5 (October–November 2019), 43–64, available at <www.iiss.org/publications/survival/2019/survival-global-politics-and-strategy-october-november-2019/615-04-blankenship-and-denison>.

- ³⁹ See Marc Lantaigne, *China and International Institutions: Alternative Paths to Global Power* (New York: Routledge, 2005); Scott L. Kastner, Margaret M. Pearson, and Chad Rector, *China's Strategic Multilateralism: Investing in Global Governance* (New York: Cambridge University Press, 2018).
- ⁴⁰ Joshua Kurlantzick, *Charm Offensive: How China's Soft Power Is Transforming the World* (New Haven: Yale University Press, 2008).
- ⁴¹ Phillip C. Saunders, *China's Global Activism: Strategy, Drivers, and Tools*, Occasional Paper 4 (Washington, DC: NDU Press, 2006); David Shambaugh, *China Goes Global: The Partial Power* (New York: Oxford University Press, 2013); Elizabeth Economy and Michael Levi, *By All Means Necessary: How China's Resource Quest Is Changing the World* (New York: Oxford University Press, 2013).
- ⁴² Nadege Rolland, "A Concise Guide to the Belt and Road Initiative," National Bureau of Asian Research, April 11, 2019, available at <<https://www.nbr.org/publication/a-guide-to-the-belt-and-road-initiative/>>.
- ⁴³ Joel Wuthnow, *Chinese Perspectives on the Belt and Road Initiative: Strategic Rationales, Risks and Implications*, China Strategic Perspectives 12 (Washington, DC: NDU Press, 2017).
- ⁴⁴ Daniel Kliman and Abigail Grace, *Power Play: Addressing China's Belt and Road Strategy* (Washington, DC: Center for a New American Security, September 2018), available at <<https://www.cnas.org/publications/reports/power-play>>; IISS, "China's Concept of the World Order," 35.
- ⁴⁵ Jost Wübbeke et al., *Made in China 2025: The Making of a High-Tech Superpower and Consequences for Industrial Countries* (Berlin: MERICS, 2016), available at <<https://www.merics.org/en/papers-on-china/made-china-2025>>; Max J. Zenglein and Anna Holzmänn, *Evolving Made in China 2025: China's Industrial Policy in the Quest for Global Tech Leadership* (Berlin: MERICS, 2019), available at <<https://www.merics.org/en/papers-on-china/evolving-made-in-china-2025>>. By early 2019, China had begun to downplay "Made in China 2025," not mentioning it in public since the beginning of a very public trade war launched against Beijing by the United States in summer 2018. Nonetheless, most observers believe Made in China 2025 remains a major Chinese economic and foreign policy underpinning. See Emily Crawford, "Made in China 2025: The Industrial Plan that China Doesn't Want Anyone Talking About," *Frontline*, May 7, 2019, available at <<https://www.pbs.org/wgbh/frontline/article/made-in-china-2025-the-industrial-plan-that-china-doesnt-want-anyone-talking-about/>>.
- ⁴⁶ For example, the United States kept its goods and financial markets open after the 2008 global financial crisis at the cost of a deeper recession and increased domestic unemployment. The Chinese system would have difficulty bearing that strain.
- ⁴⁷ Thomas Fingar, "China's Vision of World Order," in *Strategic Asia 2012–13: China's Military Challenge*, ed. Ashley J. Tellis and Travis Tanner (Washington, DC: National Bureau of Asian Research, 2012), 342–373.
- ⁴⁸ Ramo, *The Beijing Consensus*; "The China Model: The Beijing Consensus Is to Keep Quiet," *The Economist*, May 6, 2010, available at <<https://www.economist.com/asia/2010/05/06/the-beijing-consensus-is-to-keep-quiet>>.
- ⁴⁹ For an affirmative view, see Stefan Halper, *The Beijing Consensus: How China's Authoritarian Model Will Dominate the Twenty-First Century* (New York: Basic Books, 2010); for a skeptical view, see Scott Kennedy, "The Myth of the Beijing Consensus," *Journal of Contemporary China* 19, no. 65 (June 2010), 461–477.
- ⁵⁰ These dates are the 100th anniversary of the establishment of the Chinese Community Party and the People's Republic of China, respectively.
- ⁵¹ Xi Jinping, "Secure a Decisive Victory in Building a Moderately Prosperous Society in All Respects and Strive for the Great Success of Socialism with Chinese Characteristics for a New Era," Report to the 19th National Party Congress of the Communist Party of China, Xinhua, October 18, 2017, available at <www.xinhuanet.com/english/download/Xi_Jinping's_report_at_19th_CPC_National_Congress.pdf>.
- ⁵² *China's Policies on Asia-Pacific Security Cooperation* (Beijing: State Council Information Office of the People's Republic of China, January 2017), available at <http://english.www.gov.cn/archive/white_paper/2017/01/11/content_281475539078636.htm>.
- ⁵³ For a useful assessment of the Chinese white paper that illustrates these points, see Daljit Singh, "China's White Paper on Security Cooperation in the Asia-Pacific Region and Chinese Grand Strategy," ISEAS Yusof Ishak Institute Perspective 22, April 7, 2017, available at <https://www.iseas.edu.sg/images/pdf/ISEAS_Perspective_2017_22.pdf>.
- ⁵⁴ Graham Allison, "The New Spheres of Influence: Sharing the Globe with Other Great Powers," *Foreign Affairs* (March–April 2020), available at <<https://www.foreignaffairs.com/articles/united-states/2020-02-10/new-spheres-of-influence>>.
- ⁵⁵ Fingar, "China's Vision of World Order"; Phillip C. Saunders, "Implications: China in the International System," *The PLA in 2025* (Carlisle, PA: U.S. Army War College Press, 2017); Alastair Iain Johnston, "China in a World of Orders: Rethinking Compliance and Challenge in Beijing's International Relations," *International Security* 44, no. 2 (Fall 2019), 9–60.
- ⁵⁶ In the less benign interpretation, "a community of common destiny" is the core of a Chinese revisionist strategy that is incompatible with the current international order; it is based on the ideology that a strong centralized state is the only way to arrest the inherent inequalities and unacceptable aspects of Western-model electoral democracy, pluralistic civil society, and laws that emphasize protection of individual rights. In this framework, the Belt and Road Initiative and "Made in China 2025" are viewed as strategic enablers aimed toward the Chinese global economic dominance and integration necessary for a new international order where political and social norms of noninterference within state borders will supplant the freedom and openness norms predominant today. See IISS, "China's Concept of the World Order," 27, 390–398; Blankenship and Denison, "Is America Prepared for Great-Power Competition?"; Rolland, *A Concise Guide to the Belt and Road Initiative*; Liza Tobin, "Xi's Vision for Transforming Global Governance: A Strategic Challenge for Washington and Its Allies," *Texas National Security Review* 2, no. 1 (November 2018), 155–166; François Godement, "Global Values: China's Promotion of New Global Values," in *Strategic Asia 2019: China's Expanding Strategic Ambitions*, ed. Ashley J. Tellis, Alison Szalwinski, and Michael Wills (Washington, DC: National Bureau of Asian Research, 2019); Melanie Hart and Blaine Johnson, *Mapping China's Global Governance Ambitions* (Washington, DC: Center for American Progress, 2019); Friedberg, *A Contest for Supremacy*.
- ⁵⁷ IISS, "China's Concept of the World Order," 35–36.
- ⁵⁸ Peter B. Zwack and Marie-Charlotte Pierre, *Russian Challenges from Now into the Next Generation: A Geostrategic Primer*, INSS Strategic Perspectives 29 (Washington, DC: NDU Press, 2019), available at <<https://inss.ndu.edu/Portals/82/Documents/Strategic%20Perspectives/SP%2029%20Final%20for%20Web.pdf?ver=2019-03-25-100758-543>>.
- ⁵⁹ Jeffrey Mankoff, "A Century after the Armistice, the World Is Still Coping with the End of Empires," Center for Strategic and International Studies, November 13, 2018, available at <<https://www.csis.org/analysis/century-after-armistice-world-still-coping-end-empires>>.
- ⁶⁰ The oft-cited declaration of this construct of neo-Eurasianism is found in Vladimir Putin's October 2011 article in *Izvestia*,

"A New Integration Project for Eurasia: The Future in the Making." See Christopher S. Chivvis, *Understanding Russian Hybrid Warfare and What Can Be Done About It* (Santa Monica, CA: RAND Corporation, 2017), available at <https://www.rand.org/content/dam/rand/pubs/testimonies/CT400/CT468/RAND_CT468.pdf>; Nadezda Arbatova, "Three Faces of Russia's Neo-Eurasianism," *Survival* 61, no. 6 (December 2019/January 2020), 7–24.

⁶¹ Each of these core elements of Russian strategy is detailed in "Section I: General Provisions," *Foreign Policy Concept of the Russian Federation* (Moscow: Ministry of Foreign Affairs of the Russian Federation, February 18, 2013), available at <https://www.mid.ru/en/foreign_policy/official_documents/-/asset_publisher/CptICk6BZ29/content/id/2542248>.

⁶² Rumer and Sokolsky, *Thirty Years of U.S. Policy Toward Russia*; Chivvis, *Understanding Russian Hybrid Warfare*, 2.

⁶³ Karl-Heinz Kamp, "From Wales to Warsaw: NATO's Future Beyond the Ukraine Crisis," *American Foreign Policy Interests* 36, no. 6 (November 2014), 361–365, available at <<https://doi.org/10.1080/10803920.2014.995544>>.

⁶⁴ Rachel S. Salzman, *Russia, BRICS, and the Disruption of Global Order* (Washington, DC: Georgetown University Press, 2019), available at <www.jstor.org/stable/j.ctvcj2sb2>.

⁶⁵ "Russia, China Agree to Natural Gas Deal," *Stratfor*, May 21, 2014, available at <<https://worldview.stratfor.com/article/russia-china-agree-natural-gas-deal>>; "China, Russia: An End to an Island Dispute," *Stratfor*, July 17, 2008, available at <<https://worldview.stratfor.com/article/china-russia-end-island-dispute>>; and, Zwack and Pierre, *Russian Challenges from Now into the Next Generation*.

⁶⁶ Zwack and Pierre, *Russian Challenges from Now into the Next Generation*.

⁶⁷ Michael Crowley and Julia Ioffe, "Why Putin Hates Hillary," *Politico*, July 25, 2016, available at <<http://politi.co/2EwrOVM>>. Putin's grab for permanent control took a new twist in March 2020 when a Putin-inspired initiative passed the Russian Duma, which allows him to serve another two 6-year terms beyond 2024—effectively allowing him to be president for life. See "Russia's President Reluctantly Agrees to 16 More Years in Power," *The Economist*, March 12, 2020, available at <<https://www.economist.com/leaders/2020/03/12/russias-president-reluctantly-agrees-to-16-more-years-in-power>>.

⁶⁸ Tom Kutsch and Michael Pizzi, "Russia May Be Wading into a Quagmire in Syria," *Al Jazeera America*, October 2, 2015, available at <<http://america.aljazeera.com/articles/2015/10/2/the-risks-of-russias-intervention-in-syria.html>>.

⁶⁹ Anne Applebaum, "Putin's Grand Strategy," *South Central Review* 35, no. 1 (2018), 22–34, available at <<https://doi.org/10.1353/scr.2018.0001>>; Jeffrey Mankoff, *Russian Foreign Policy: The Return of Great Power Politics* (Lanham, MD: Rowman and Littlefield, 2011), 197; Stephen M. Walt, "I Knew the Cold War. This Is No Cold War," *Foreign Policy*, March 12, 2018, available at <<https://foreignpolicy.com/2018/03/12/i-knew-the-cold-war-this-is-no-cold-war/>>.

⁷⁰ Chris Buckley, "Xi Jinping Thought Explained: A New Ideology for a New Era," *New York Times*, February 26, 2018, available at <<https://www.nytimes.com/2018/02/26/world/asia/xi-jinping-thought-explained-a-new-ideology-for-a-new-era.html>>; James H. Anderson, "A New Global Paradigm: The United States versus Russia," in *Great Strategic Rivalries: From the Classical World to the Cold War*, ed. James Lacey (New York: Oxford University Press, 2016), 479–510.

⁷¹ Applebaum, "Putin's Grand Strategy"; David Gibbs, "Does the USSR Have a 'Grand Strategy'? Reinterpreting the Invasion of Afghanistan," *Journal of Peace Research* 24, no. 4 (December 1987), 365–379. For a review of the Communist International

framework for a global world and the Soviet Union's domination of that entity throughout its history, see Edward B. Richards, "The Shaping of the Comintern," *Slavic and East European Review* 18, no. 2 (April 1959), 197–204, available at <<https://www.jstor.org/stable/3001362>>; Robert H. McNeal, "The Legacy of the Comintern," *International Journal* 201, no. 2 (Spring 1966), 199–204, available at <<https://www.jstor.org/stable/40200330>>. For a discussion of how Putin's Russia overrules no such cohesive, globalist positive vision, instead preferring a strategic approach that sustains Russian oligarch kleptocracy and the destabilization of global norms and institutions along with disruption of rival state political cohesion, see Fiona Hill and Clifford G. Gaddy, *Mr. Putin: Operative in the Kremlin* (Washington, DC: Brookings Institution Press, 2015), especially 312–384; Walt, "I Knew the Cold War. This Is No Cold War."

⁷² Robert O. Keohane and Joseph S. Nye, Jr., *Power and Interdependence*, 2nd ed. (Glenview, IL: Scott, Foresman and Company, 1989), 279–282; Charles P. Kindleberger, *Manias, Panics, and Crashes: A History of Financial Crises*, 3rd ed. (New York: John Wiley and Sons, Inc., 1996), 179–186.

⁷³ Ivo H. Daalder and James M. Lindsay, "The Committee to Save the World Order," *Foreign Affairs* 97, no. 6 (November–December 2018), available at <<https://www.foreignaffairs.com/articles/2018-09-30/committee-save-world-order>>; Matthew Bey, "Trump and the WTO's Uncertain Future," *Stratfor*, November 1, 2018, available at <<https://worldview.stratfor.com/article/trump-and-wtos-uncertain-future>>.

⁷⁴ Timofie Bordachev, "Russia and the Eurasian Economic Union: The View from Moscow," European Council on Foreign Relations, January 21, 2015, available at <https://www.ecfr.eu/article/commentary_russia_and_the_eurasian_economic_union_the_view_from_moscow403>.

⁷⁵ Simultaneously, Putin's Russia will both obfuscate its interventionism and assert that multiple shameless U.S. interventions into sovereign states—Iraq, Libya, Syria, etc.—during the post-Cold War era make the United States ill-positioned to offer criticism.

⁷⁶ It was Chiang Kai-Shek's China—and Mao Zedong's civil war rival—that was awarded a seat on the United Nations Security Council that the People's Republic of China finally inherited in 1971.

⁷⁷ The Asian Infrastructure Investment Bank (AIIB) "is a multilateral development bank with a mission to improve social and economic outcomes in Asia. Headquartered in Beijing." See "About AIIB Overview—AIIB," available at <<https://www.aiib.org/en/about-aiib/index.html>>; David Dollar, "The AIIB and the 'One Belt, One Road,'" *Brookings*, June 21, 2015, available at <<https://www.brookings.edu/opinions/the-aiib-and-the-one-belt-one-road/>>.

⁷⁸ "Russia," *Interpol*, available at <<https://www.interpol.int/en/Who-we-are/Member-countries/Europe/RUSSIA>>; Matt Apuzzo, "Interpol Rejects Russian as President, Electing South Korean Instead," *New York Times*, November 21, 2018, available at <<https://www.nytimes.com/2018/11/21/world/europe/interpol-russia-south-korea.html>>; Amy Mackinnon, "The Scourge of the Red Notice," *Foreign Policy*, December 3, 2018, available at <<https://foreignpolicy.com/2018/12/03/the-scurge-of-the-red-notice-interpol-uae-russia-china/>>; Robbie Gramer, "China and Russia Take the Helm of Interpol," *Foreign Policy*, November 10, 2016, available at <<https://foreignpolicy.com/2016/11/10/china-and-russia-take-the-helm-of-interpol/>>.

⁷⁹ For a review of these economic system traits, see Charles P. Kindleberger, *World Economic Primacy: 1500–1990* (Oxford: Oxford University Press, 1996), 225.

⁸⁰ Joel Gahrke, "State Department Preparing for Clash of Civilizations with China," *Washington Examiner*, April 30, 2019,

available at <<https://www.washingtonexaminer.com/policy/defense-national-security/state-department-preparing-for-clash-of-civilizations-with-china>>; Edward Wong, “U.S. Versus China: A New Era of Great Power Competition, but Without Boundaries,” *New York Times*, June 26, 2019, available at <<https://www.nytimes.com/2019/06/26/world/asia/united-states-china-conflict.html>>.

⁸¹ Mazarr et al., *Understanding the Emerging Era of International Competition*, 32–33.

⁸² *Ibid.*, 32.

⁸³ *Ibid.*, 18.

⁸⁴ See Graham Allison, “The New Spheres of Influence: Sharing the Globe with Other Great Powers,” *Foreign Affairs* (March/April 2020).

⁸⁵ For more details on the limits to greater Sino-Russian strategic cooperation, see Robert Sutter, *China-Russia Relations: Strategic Implications and U.S. Policy Options*, NBR Special Report #73 (Washington, DC: National Bureau of Asian Research, September 2018), available at <https://www.nbr.org/wp-content/uploads/pdfs/publications/special_report_73_china-russia_cooperation_sep2018.pdf>.

⁸⁶ Of particular note, Beijing and Moscow have been careful not to use the word *ally* in regard to each other until relatively recently. Russia began using it quite casually, but China has not gone along. Beijing continues to avoid the term at an official level, preferring official wording about an “all-encompassing partnership and strategic interaction.” See Vasily Kashin, “Tacit Alliance: Russia and China Take Military Partnership to a New Level,” Carnegie Moscow Center, October 22, 2019, available at <<https://carnegie.ru/commentary/80136>>.

⁸⁷ Colum Lynch and Robbie Gramer, “U.S. and China Turn Coronavirus into a Geopolitical Football,” *Foreign Policy*, March 11, 2020, available at <<https://foreignpolicy.com/2020/03/11/coronavirus-geopolitics-china-united-states-trump-administration-competing-global-health-response/>>.

⁸⁸ For a brief review of contemporary Great Power competition in space—a topic not addressed in great detail in this volume—see the paragraph in chapter 1.

⁸⁹ For greater detail, see Justin Sherman, “How the Internet Is Taking Center Stage in Great Power Competition,” *New America*, February 7, 2019, available at <<https://www.newamerica.org/weekly/edition-235/internet-great-power-world-order-china/>>. Also see chapter 6 in this volume.

Chapter 3b

Contemporary Great Power Geostrategic Dynamics

Competitive Elements and Tool Sets

By Thomas F. Lynch III and Phillip C. Saunders

The chapter assesses the hard and soft power tools of the three contemporary Great Powers. It focuses on the tools that each has today and is likely to attain in the coming 5 to 7 years, analyzing how each might use these tools to advance its major interests and strategic aims in the five major categories of state interaction: political and diplomatic, ideological, informational, military, and economic. The chapter observes that the tools of competition traditionally associated with one category of interaction in less rivalrous eras will be used more often in other categories in this era of Great Power competition. It assesses that for the foreseeable future, Russia's tool kit makes it an urgent but transient security challenger to the United States, while China's growing power tools make it the true challenger to American national interests and global policy preferences. An assessment of both gross and net power indicators between the United States and China indicates that Beijing's ongoing power transition timeline is longer than some now fear. This allows American and Chinese leaders time to negotiate mutually acceptable changes to contemporary international norms, rules, and institutions in order to prevent what would be a truly unwelcome and destructive direct military clash, should such accommodation be elusive.

Chapter 3a establishes the geostrategic trajectories and primary strategic aims of the three contemporary Great Powers: the United States, China, and Russia. It provides analysis of where their major strategic narratives align and diverge. The chapter also provides an assessment of each state's national interest intensity in specific locations and domains—indicating where its most critical strategic interests come into conflict.

This chapter turns to an evaluation of the tools and main capabilities possessed by each Great Power to advance its general strategic aims and specific strategic goals. It considers the assets the Great Powers bring to their competitive interactions, with a focus on the tools each power now possesses, those likely to be attained in the next 5 to 7 years, and each nation's ability to employ these tools to advance its interests and attain strategic aims. The chapter

first evaluates power as the physical resources that states can draw on to attain strategic interests—their *means*. It then assesses each power's potential to employ these means to attain strategic aims—their *ends*.¹ The chapter briefly addresses the difference between gross power indicators and net power indicators, using net indicators to demonstrate that a power transition between the United States and China may be less imminent than most now imagine. Although the details anticipated from contemporary Great Power competition across many specific regions of the world are provided in chapters that follow, this chapter concludes with an assessment of how current and forecast future power tools of the major Great Powers should be anticipated to interact in major competitive areas from 2020 to 2025.

From Aspirations to Actions: Measuring State Ability to Do What Is to Be Done

As operationalized in the chapter 1 definition, a *Great Power state* has three substantive features: capabilities, behavior, and status attribution by other states in the system.² In chapter 3a, we establish the major goals and strategic interests of the three Great Powers and demonstrate that their interests display broad, global foreign policy aims and activities, with China's strategic aims more ambitious and global in reach than Russia's, and that other states in the international system view all three as major players and treat them accordingly. Chapter 3b now turns to an analytical review of the first aspect of our Great Power definition: the power capabilities (tool kits) they possess with which to pursue these strategic interests.

As observed in chapter 2, measuring power is a fraught enterprise and the subject of extensive scholarly debate. Determining what to measure as state power is contentious. So too is determining how to measure even agreed-on categories of state power.

The historic challenge of power assessment is manifested today in the fact that many scholars argue that the United States is in relative *overall decline* given China's rapid economic growth. But this conclusion is far from certain. First, the baseline premise is debatable. Gross domestic product (GDP), the most frequently utilized state power metric, is a narrow basis for making an assessment. GDP was originally designed to measure mid-20th-century manufacturing economies, not those of the 21st century.³ The more knowledge-based and globalized a country's production, the more GDP underestimates its true size. Thus, GDP likely overvalues China's economic prowess and undervalues American advantage on the cutting edge of economic modernization. A significant amount of China's economic success can be explained by the fact that it started from a low base and took good advantage of "catchup" opportunities, although the country is now trying to move up the value chain and pursue indigenous innovation. While China has been catching up in manufacturing, the United States has been expanding advantage in key industries, in nonindustrial processes, in financial services growth and rent extraction, and (generally) in the quality of higher education.⁴ At the same time, the GDP gap between the United States and Russia is growing, and the United States continues to outperform Europe and Japan in economic growth since the end of the late 2000s financial crisis.⁵

Based on these economic considerations and a review of major military factors, some authors argue not only that the conventional wisdom of relative U.S. decline is off base but also that America is actually still a Great Power on the rise.⁶ They admit that America's military advantage in 2020 has eroded in locations proximate to Russia and China, such as in

Crimea and the South China Sea, compared to 2000 and that these are now intense strategic interest areas and likely to be contested sternly in Moscow and Beijing. But they assess that with enough at stake and enough political will, the United States still has an unmatched capacity to concentrate overwhelming military in any area of armed hostilities.⁷ They argue that the advantage is greater to the extent that America's allies in a given armed conflict zone are willing participants. Their judgment is that America's advantage in this area of hard power deployment/concentration ability will remain robust through 2030.

This chapter cannot grapple with all the details involved in measuring state power. Most scholars measure power in terms of resources, specifically wealth and military assets. They assume these gross power indicators are good enough, serving as "rough but reliable" measures of power, and they are the "best comparable indicators available given data constraints."⁸ In general, the chapter acknowledges inherent measurement limitations but mainly utilizes canonical measures of Great Power capabilities. Yet it also moves beyond canonical economic and military factors to consider some other nonstandard power tools important to Great Power strategic influence in the 21st century, such as levels of economic innovation and engagement with private global financial markets; resonance and popular appeal of state ideology, language, and culture; and a consideration of a relatively new approach to understanding national power and comparison of relative state power with an index that better considers net economic and military resources (assets less costs) than historic national capability indicators. The chapter thus focuses on the power factors most germane to the five categories of major state interaction developed in table 2.2, table 3a.1, and reprised in table 3b.1.

"Since the 1990s, and especially since the 2008 financial crisis, hundreds of books and thousands of articles and reports have asserted that the United States' economic and military edge over other nations is eroding and that the world will soon become multipolar. . . . The main evidence cited for these trends is China's rising GDP and military spending and various statistics that are essentially subcomponents of GDP—most notably, China's massive manufacturing output; volume of exports; trade surplus with the United States; infrastructure spending; consumer spending and large government bureaucracy and scientific establishment. . . . The problem, however, is that these are the same gross indicators that made China look like a superpower during its century of humiliation: in the mid-1800s. . . . China may (today) have the world's biggest economy and military, but it also leads the world in debt; resource consumption; pollution; useless infrastructure and wasted industrial capacity; scientific fraud; internal security spending; border disputes; and populations of invalids, geriatrics, and pensioners. China also uses seven times the input to generate a given level of economic output as the United States and is surrounded by nineteen countries, most of which are hostile toward China, politically unstable, or both."

—Michael Beckley, *"The Power of Nations,"* International Security 43, no. 2 (Fall 2018)

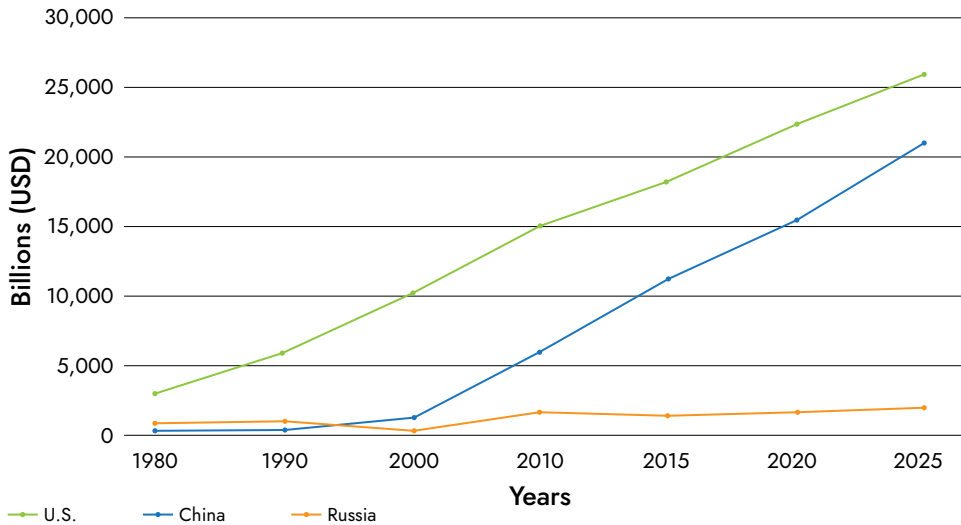
Major Political and Diplomatic Tools

These tools include objective measures of the state's presence in multilateral political institutions and qualitative assessment of its influence in intergovernmental organizations. They

Table 3b.1. A Framework for Assessing the Aspects/Categories of Competition	
Competitive Aspect/Category	Main Competitive Elements
Political and Diplomatic	Levels of influence in multilateral institutions, key posts held that control multilateral institutions, and number and strength of political alliances.
Ideological	Values and political systems' appeal.
Informational	The manner and degree of transnational communications: open and transparent vs. closed and restrictive. Extent of denigration of "the other" in mass communications.
Military	Size, posture, and technological edge of armed forces. Cohesion and capacity of military alliances.
Economic	Size, technological breadth, diversity, and resource base of national economy. Innovation ecosystem of national economy, including access to and management of financial capital.

also include a quantitative overview of the state's geopolitical diplomatic presence and its political tools for securing key state partners and multistate alliances.

- **Major Ideological Tools:** These power factors consist of qualitative assessments regarding the attractiveness of the Great Power's values, narratives, and political system in other states. The trajectory of this soft power attractiveness is best evaluated in geopolitical polling results.
- **Major Informational and Communications Capabilities:** The qualitative dimensions of this competitive category are more important than the quantitative ones. State power tools include the degree of penetration by key communications technologies in vital geostrategic regions and around the globe. In addition, the state tool kit in this area incorporates the manner in which competitive visions of information pathways and system openness play with Great Power states and those lesser states integrated/integrating in the communications networks.
- **Major Military Capabilities:** These capabilities include the classical quantitative hard power comparative assessments of available major weapons systems, which are the easiest to measure. This chapter also focuses on military systems and capabilities harnessing emerging commercial technologies from leading-edge commercial areas. Military tools include those with the potential for influence and suasion, such as the cohesion and capacity of military alliances and the manner in which military technology sales and military personnel exchanges work to enhance state aims and potential adversary perspectives.
- **Major Economic Tools—Commercial and Financial:** Economic power is often understood as the ultimate foundation of military power and a strong influence on the other forms of state power. Here, the canonical economic growth dynamics are measured in nominal GDP, nominal GDP per capita, and level of industrialization. The chapter also considers the amount and impact of outbound direct foreign investment—governmental and private. To properly understand the influence of modern and future economic power factors, financial linkages and innovation potential are described and analyzed with reference to the percentage of global

Figure 3b.1. Comparison of Great Power GDP

Source: Chart generated by authors. Data sources are at the online appendix B of this volume, available at <https://ndupress.ndu.edu/Contemporary-GPC-Dynamics-Matrix/>.

private investment transactions. Additionally, the level and trajectory of economic innovation potential in the state is discussed.

The chapter ends with the use of a relatively new index of net overall power that purports to better capture economic and military capabilities less cost by combining GDP with GDP per capita as well as military capabilities and internal costs—yielding an indicator that better accounts for overall net national size and efficiency.⁹

Great Power Competitive Postures and Tool Sets

U.S. Competitive Posture and Tool Sets

General American Power Factors and Approaches. Nominal U.S. GDP in 2019 was an estimated \$22.32 trillion, ahead of China's \$15.3 trillion and Russia's \$1.7 trillion (see figure 3b.1). America remained the world's top and most dynamic national economy, generating 23.9 percent of global GDP in 2019, far ahead of its nearest competitor, China, with 15.9 percent that year.¹⁰ The U.S. share of the global GDP remains in slow relative decline, but it is anticipated to generate about 21 percent of global GDP in 2035. Given canonical projections in 2020, China's nominal GDP would be about 25 percent of the global total in 2035, surpassing that of the United States in about 2030.¹¹ The United States is not an export-dependent economy, but about 10 percent of its nominal GDP in 2018 was goods exports (\$2.5 trillion). America also is the world's leader in services exports, with \$828 billion in 2018 nominal value led by audiovisual technology, banking services, energy, express delivery, information technology, insurance, and telecommunications service industries.¹² Top U.S. export customers in the 2010s were China, Canada, and Mexico, all of which were

Table 3b.2. Percentage Disparity Between Gross and Net Overall Power Factors Using Beckley's Net Power Index

Great Power Rivalries				
	1990	2000	2010	2015
U.S. vs. Russia	10	10	4	5
U.S. vs. China	24	30	35	23
Russia vs. China	32	45	49	37

Note: Values calculated with the formula provided by Michael Beckley and from the data found at tab 2 of the online appendix B of this volume, available at <<https://ndupress.ndu.edu/Contemporary-GPC-Dynamics-Matrix/>>. The noteworthy “blip” in these comparative gross-to-net power index values beginning in 2010 comes from the revaluation of one of the input Correlates of War (COW) Composite Index of National Capability index values significantly reevaluated in the late 2000s by COW to better account for economic factors.

linked into complex regional and global supply chains accounting for a high percentage of U.S. manufactured exports.¹³ Forty-seven percent of the U.S. economy is industrialized (see table 3b.2). Almost 19 percent of American manufactured exports were in the high-technology category in 2018, lower than the 31 percent of China's exports in 2017 but near the global average of 18 percent for 2018.¹⁴ In mid-2018, the United States began a trade war with China, which over its first 18 months had become the most serious disruption in global commerce in nine decades.¹⁵ Analysis of the Sino-U.S. trade war through September 2019 revealed the effort to be a double-edged sword. China's lost export revenue was triple that of the United States (\$53 billion to \$14.5 billion), but it had not achieved any substantive movement in the Chinese economic behaviors it was seeking to change. Moreover, key

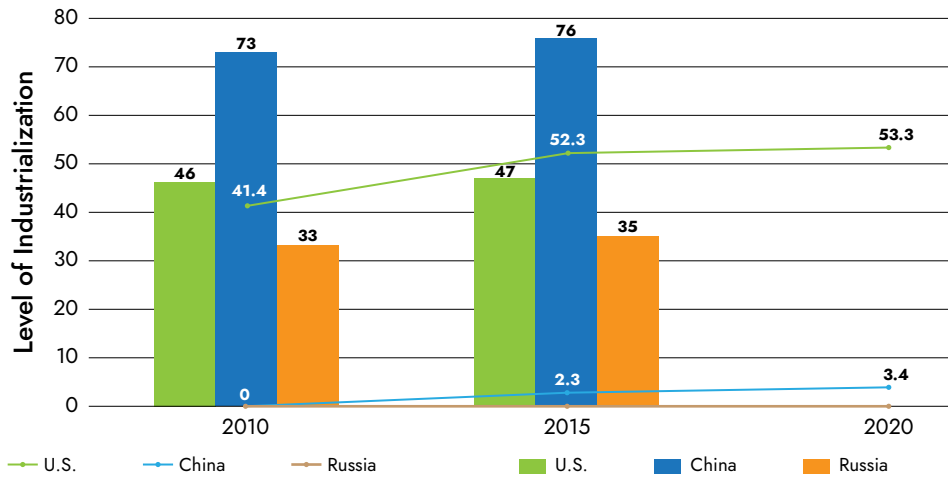
“Those [states] that have the resources to do so will generally try to increase their military capabilities so as to reduce their vulnerability to coercion and attack. . . . Both strong and weak states may also enter into alliances intended to fend off potential enemies, or to overwhelm opposing powers or coalitions. . . . As a state's capabilities grow, . . . rising powers typically attempt not only to secure their borders but also to reach beyond them, taking steps to ensure access to markets, materials, and transportation routes; to protect their citizens far from home and defend their foreign friends and allies; to promulgate their religious or ideological beliefs; and, in general, to have what they consider to be their rightful say in the affairs of their region and of the wider world.”

—Aaron L. Friedberg, *A Contest for Supremacy* (2012)

sectors of the U.S. economy—exporters of minerals and ores, forestry products, agribusiness, and transportation systems—lost substantial amounts of revenue and were disturbed that China has found alternative global suppliers, meaning potential lasting damage to their export revenues.¹⁶ The “phase one” U.S. trade deal with China announced in January 2020 involved Chinese agreement to lift some retaliatory tariffs and commit to substantial increases in imports from the United States.¹⁷ But trade disruptions due to the novel coronavirus (COVID-19) pandemic of 2020 make it unlikely that China will fulfill those commitments.

The United States has been ranked in the top dozen innovative countries over the past decade and is expected to remain a global leader as an incubator for innovative manufacturing and services delivery for at

Figure 3b.2. Correlation Between Level of Industrialization and Percentage of Commercial Global Financial Markets



Source: Chart generated by authors. Data sources are at the online appendix B of this volume, available at <https://ndupress.ndu.edu/Contemporary-GPC-Dynamics-Matrix/>.

least the next decade. The United States also has been the dominant nation in private global financial markets for decades, accounting for 53 percent of activity in 2018.¹⁸ China and Russia barely register on this measure of financial activity and corresponding revenue from global commodity and services exchange (see table 3b.2). The U.S. share of commercial financial markets has continued to grow over the past decade, and the anticipated continuation of this trend is a significant future U.S. economic growth advantage not easily accessed by other governments, so long as U.S. financial entities remain attractive compared to other options.¹⁹

At the same time, U.S. Government management of foreign access—governments and individuals—to America's dominant financial markets is both a power opportunity and a challenge. It gives the United States a unique coercive economic power tool to modify foreign entity behaviors by denying or restricting foreign access to preferred U.S. financial institutions should their behaviors counter American national interests. Sharper than diplomatic censure while gentler than military confrontation, financial sanctions can be appealing as an asymmetric U.S. power tool. The United States has used sanctions with far greater frequency since 2000, especially against known and suspected terrorists and their financing agents during the war on terror.²⁰ But overuse of financial sanctions can backfire, encouraging U.S.-sanctioned countries and groups to turn to alternative financial arrangements, reducing the effectiveness of the sanctions and straining relations with longstanding U.S. economic partners affected by the secondary impact. Over time, sanctioning can create incentives for friends and foes to develop alternative financial structures and arrangements that can work around U.S. entities and erode American current dominance in commercial financial services.²¹

Another risk to U.S. economic strength and future power standing lies in its level of national debt. As of late 2018, the United States had a nominal total national debt of \$21.8

trillion, almost 100 percent of its nominal GDP. About 28 percent of that debt (\$6.3 trillion) was held by foreign entities, with the top three holders being China (\$1.1 trillion), Japan (\$1 trillion), and Brazil (\$0.3 trillion).²² Thus, 72 percent of U.S. total national debt was held by U.S.-based investors, making it more of a domestic than a foreign relations challenge. The risks to U.S. economic standing and military investments would come in the out years if debt continues to grow. Interest on the national debt and spending for mandatory government programs, such as Medicare, could eat an ever-increasing piece of the Federal budget, leaving fewer dollars for the application of U.S. military, diplomatic, and economic power and influence activities around the world. Financing the debt also could drive up interest rates and reduce private investment in economic activity and technological innovation.²³

The 2020 COVID-19 pandemic will have significant impact on the global economy as a whole, but its impact will not be evenly distributed. The United States could conceivably come out of the pandemic with economic scars that would see atrophy in these significant advantages. However, unless it makes some truly intemperate health or economic policy decisions, it seems more likely that post-COVID-19 America will remain in a relatively similar economic posture with respect to its Great Power rivals.

America's three-decade advantage of military-technological superiority—especially in precision-guided weapons and their sensors, information technology, and space-based networks—can no longer be assumed in this dawning era of Great Power competition. Yet the United States remains demonstrably superior across most major measures of military power. Absent some crushingly ill-considered decisions, America should remain the dominant military state for at least the coming decade. After several years of low-level increases, American military spending grew from \$716 billion during fiscal year 2019 to \$738 billion for fiscal year 2020. These figures triple the Chinese official defense budget and are 16 times that of Russia. This spending is forecast to slow again into the 2020s as domestic priorities and concerns about U.S. national debt return to political prominence.²⁴

The U.S. nuclear arsenal remains below its treaty-limited number of 1,600 warheads, with 1,350 strategic nuclear warheads deployed on 650 land-based intercontinental missiles, 158 strategic bombers, and 18 nuclear submarines. In 2018, the United States launched a program to modernize its nuclear weapons arsenal and delivery systems—especially the overworked, dual-capable strategic bomber fleet—to assure its strategic deterrent value in the face of fast-emerging delivery technologies. The all-volunteer U.S. military consists of approximately 1.29 million Active-duty forces as of 2017: 470,000 active in the Army, 183,000 in the Marine Corps, 320,000 in the Navy, and 313,000 in the Air Force. The United States maintains forces with global reach and operational capability across a full spectrum of military operations.

The Navy is optimized for global presence with 12 top-end aircraft carriers and 12 smaller ones, compared to 1 operational carrier for China (with 2 more on the way) and 1 inoperable carrier in Russia. Its surface fleet features almost 90 frigates and destroyers, most with anti-air, antiballistic missile, and antisubmarine capable systems. The Navy has 68 modern submarines, 18 of which are ballistic missile carriers and all of which are nuclear powered and thus capable of extended range and duration operations. Sixty percent of Navy assets and operations took place in the Pacific Ocean during 2018. The combined U.S. aviation fleet constitutes a force with global reach, featuring 2,300 modern fighter aircraft,

2,800 attack aircraft, 1,150 transport aircraft, and over 5,700 total helicopters, including 970 attack helicopters.²⁵ As of 2017, the United States also has a fleet of 7,500 unmanned aerial systems (UAS), 10 percent of which (786) were strategic-range, mid- and high-altitude, long-endurance intelligence or strike platforms. The United States will spend about \$12 billion per year in 2020 and 2021 to retain its lower end UAS fleet while investing in research and development for new strategic UAS technologies to modernize a strategic fleet that has become increasingly vulnerable to the growing number of air defense and electronic warfare capabilities of the Great Powers and other international actors.²⁶ America's number of troop transports, helicopters, and fighter jets dwarfs those of any other nation, and more than doubles those found in the next largest force, the Chinese military. The Army has engaged in two decades of low-intensity conflict and special operations forces counterinsurgency and counterterrorism operations but retains a fleet of 6,000 technologically sophisticated main battle tanks, 39,000 armored fighting vehicles, and 3,000 artillery and rocket forces—most of which are operational and deployable worldwide.

America's military reach and power capacity are significantly augmented by formal military alliances in Europe and Asia and strategic partnerships with individual nations in the Middle East, South Asia, Africa, and the Americas. These alliances have come under increasing duress in the late 2010s, but their structures remain intact. U.S. military forces have military interoperability with the 28 North Atlantic Treaty Organization (NATO) nations and with two dozen other key allies and strategic partners across the Middle East and the Pacific. Perhaps more significantly, in a 2019 Pew International Survey, a median of 27 percent of respondents in 17 countries named the United States as their state's most dependable partner, while only 6 percent cited China and 4 percent Russia.²⁷ The United States remains the world's largest arms exporter, with 36 percent of the global total in 2018. Of the 25 countries buying the most weapons from the United States, 10 are either NATO member nations or part of other alliances formed with the United States since the Cold War.²⁸ With many of its military partners, the United States conducts security assistance and arms sales programs. Annual government-to-government U.S. defense assistance to foreign allies and partners averaged approximately \$43 billion per year over the decade of the 2010s, with conspicuous spikes above that in 2012 and 2018.²⁹ Major partner states for U.S. Government military foreign sales support and security assistance are the states of the Middle East, Israel, Egypt, and Pakistan.

American diplomatic and political assets have been under strain for a decade, but they still provide the United States with unmatched structural power and opportunities to favorably compete with would-be rivals. The United States is a full member of 22 formal international government organizations, including the United Nations Security Council Permanent Five, and the largest voting member of the International Monetary Fund and World Bank, and has been a leading member of the World Trade Organization and World Health Organization.³⁰ For the past two decades, and especially since 2017, American policymakers have confronted a persistent question: Are the benefits of leadership of post-World War II, cooperative international organizations that help lock in predictable, U.S.-friendly policy orientations and minimize the need for Washington's use of coercive power worth the price of the institutional maintenance costs and the sacrifice of a degree of political autonomy?³¹

Well before the administration of Donald Trump, American skeptics of global institution participation tapped insular and nationalist tendencies in the U.S. public to erode support for participation and maintenance, but national polling in 2018 indicated that a majority of Americans continue to view U.S. international engagement and leadership of global institutions and alliances worth the cost.³² U.S. leaders, if they choose to do so, have the public support and material resources to reverse the recent accelerated skepticism of U.S. international engagement. The still-dominant U.S. position in many intergovernmental and nongovernmental organizations means it can recover from the domestic political exploitation of ordinary citizens' grievances with "globalism." American influence in longstanding international institutions can be reinvigorated with the political will to sponsor and fund programs that enhance transparency into global institutions and, at the same time, generate domestic policies that do a much better job of compensating aggrieved U.S. national constituencies that consider themselves "losers" from the economic and social adjustments resultant from the international trade, financial flows, migration, technological change, and international legal decisions identified with globalization.³³

Measures of American ideological resonance and cultural identity are mixed in 2020, but generally remain stronger than those for its main competitors. The years 2017 and 2018 witnessed a sharp decline in international views of the United States and its Presidential leadership. In 2016, a median of 64 percent of international respondents in the Pew Global Attitudes Survey held a favorable view of the United States and had confidence in President Barack Obama to direct America's role in the world. By 2017, only 49 percent viewed the Nation positively and just 22 percent felt confident in President Trump's leadership.³⁴ Yet a subsequent 2019 Pew survey revealed that despite this decline, many countries still view the United States as the nation their country can most rely on as a dependable ally into the future. The allure of the American Dream—the interdependence of prosperity, individual freedom, and liberal democracy—remains a positive attribute in most corners of the world.³⁵ While not without backlash, American products, popular culture, and basic individual rights are resonant around the globe. They are well marketed and do not require Americans to be physically present to exert this influence.³⁶

Despite some notable erosion in confidence about the American ideal, its global appeal dwarfs that of any other country's narrative and provides a fungible power attribute if applied wisely. The COVID-19 pandemic provides another opportunity for Washington to market this appeal wisely. Equally important, English is the dominant global language for business, industry, and cultural exchange—1.8 billion people speak English, and an increasing number of multinational corporations require English as their common corporate language.³⁷ Fifty-five of the top 100 universities in the world are in the United States, and despite a decline in international student applications that began in 2012 and accelerated in 2017, revenue generated by American higher education in 2018 was nearly \$41 billion—double the value of soybean exports and only modestly below U.S. automobile exports (\$53 billion).³⁸

In the information dissemination and communications arenas, longstanding American preferences are under duress in 2020. Yet here again, the United States possesses the economic strength and technological know-how to compete favorably, if not perfectly, across the information domain. Global acceptance of the U.S. preference for free and open

communications, including over the Internet and social media networks, has been badly bruised in the decade since 2010. The once dominant U.S.-led Internet Corporation for Assigned Names and Numbers oversight framework and its preference for openness and multi-stakeholder cooperation have encountered political challenges and then technological ones. China, Russia, and a number of other authoritarian states around the world believe that free and open communications on a mass scale present a threat to their national sovereignty and maintenance of domestic political order.³⁹ They each have developed technologies and protocols to constrain the free flow of information across and within their borders; some, mainly China and Iran, have completely blocked Internet and global telecommunications flows during periods of public unrest or government worry.

The United States today relies primarily on the private sector to project external information about American values and ideals—the core of the liberal capitalist brand.⁴⁰ During the Cold War, the U.S. Government invested substantial resources in *public diplomacy*, a term that covered a host of overseas activities—from libraries to lecture tours, art exhibits to world's-fair-style expositions, international visitor programs to radio and television broadcasts meant to undermine Soviet censorship.⁴¹ As conducted by the United States Information Agency and the Department of State's Division of Cultural Relations, these activities sought to convey “a full and fair picture” of America. The Voice of America, Radio Free Europe, Radio Liberty, and Radio Free Asia were all publicly funded efforts to counter propaganda and provide alternative sources of information to oppressed populations.⁴² Today, these U.S. public information activities are mostly gone, repurposed for other diplomatic duties, or cash-starved enterprises. The State Department does maintain a Bureau of Educational and Cultural Affairs and a Bureau of Global Public Affairs that perform some similar functions (including Internet and social media outreach), but with significantly less funding and impact than their Cold War-era equivalents.⁴³

Instead, America's most successful export, commercial entertainment, has stepped into the void and become far more influential than the remaining public diplomacy activities. Recently, a U.S. political scientist investigated the implications of this privatization of public diplomacy and found it often to be counterproductive: “Instead of showing the interdependence of prosperity, democracy, and freedom, contemporary [American] popular culture tends to single out freedom and portray it in ways that are very entertaining, but often also very alien to the concerns of most people in the world.”⁴⁴ There is a case to be made for far greater U.S. Government attention to its external information messaging in the era of Great Power competition. America's projection of information during and after the COVID-19 pandemic will provide an opportunity to generate such additional attention. One recent positive step that might be built on was the establishment of the interagency Global Engagement Center in 2016 to coordinate U.S. Government efforts to “recognize, understand, expose, and counter foreign state and nonstate propaganda and disinformation efforts aimed at undermining or influencing the policies, security, or stability of the United States, its allies, and partner nations.”⁴⁵

But even if the U.S. Government devoted more resources and attention to external messaging and counter-propaganda activities, it would face a major challenge given Russian and Chinese robust propaganda and domestic censorship activities—including those that block or distort the global Internet. Technology may provide a bit of an answer to this

challenge, for satellites do not currently play a major role in global Internet infrastructure. As addressed in chapter 5, within the next decade, a new generation of satellites could lay the foundations for an “Internet from space.” That fast-emerging concept would use thousands of satellites instead of the tens of satellites used in contemporary telecommunications systems today. Laser light could then link these satellites to each other to form a network that would be able to reach remote, isolated, or even blocked regions of the world that today have no or limited access to the Internet.⁴⁶ Should America choose leadership in developing these technologies, and those associated with land-based relay stations and handheld phone receptors, Washington may again leverage the advantages of innovation and technology to counter the ongoing balkanization of the Internet and compete to see its norms for telecommunications and information exchange dominate this domain of state-to-state interaction. In turn, this could provide a rejuvenated U.S. public diplomacy with a conduit to project a more realistic, cohesive, and attractive American brand.

Key American Power Tools and Their Strategic Utility. For the period 2020 to 2025 and beyond, the United States remains relatively strong in military hard power and most of the soft power tools necessary to compete favorably in the five major areas of Great Power competition. Washington’s military might—while not as dominant as in the early 2000s—is still unmatched in global power projection capacity. Although the relative size of the U.S. economy and its manufacturing base is in decline compared to China, American financial dominance is unchallenged, its innovation dynamism far more robust, and its demographic profile more conducive to long-term economic adaptation and expansion. Combined, its economic power tools—unless self-limited—remain most formidable in modern economic competition.

Core U.S. ideological messages featuring freedom, openness, transparency, and universal human rights resonate in many parts of the world, providing America with an ability to attract other states and individuals to act favorably toward American objectives and interests. Nevertheless, Washington’s internal information management has been under duress from outside interference, and its external messaging remains largely a private commercial enterprise without a mandate or the means to compete with other Great Power narrative projections. Moreover, the government’s recent uneven policy support for these influential soft power attributes has begun to reduce the decades-long U.S. role as the most influential Great Power in the information and ideological domains. The U.S. response to the 2020 COVID-19 pandemic, and how it is publicized domestically and internationally, will determine if America’s advantages in these vital areas are eroded or reinforced.

American political and diplomatic power tools are now and will remain under duress in many specific locations around the globe in the decade of the 2020s. But even these are unlikely to be displaced by Russia or China in the near term.

China’s Competitive Posture and Tool Sets

General Chinese Power Factors and Approaches. China’s emergence as a global power is the product of its three-decade ascent to economic superpower status. From 1979 to 2018, China’s economy grew at an annual rate of 9.4 percent. China became the world’s second largest economy, largest manufacturer, largest trader in goods, second largest consumer of commodities, second largest recipient of foreign direct investment (FDI), and

largest holder of foreign exchange reserves.⁴⁷ In 2019, China's GDP was second only to that of the United States, at an estimated \$15.27 trillion in exchange rate terms (see figure 3b.1). China's percentage of global GDP almost tripled in the decade from 2007 to 2017, moving from 6.18 percent to 15.25 percent.⁴⁸ Should China continue annual growth between 5 and 6 percent per year and the U.S. growth per year remain around 2.2 percent, then China's GDP will pass that of the United States between 2030 and 2034.⁴⁹ Less optimistic estimates of China's growth rate and the negative implications to productivity from a rapidly aging Chinese population versus that of the United States pushes this crossover point out to beyond 2040, or later—if ever.⁵⁰

China's economic rise has been significantly fueled by export growth, as Western and Asian companies relocated production to tap low-cost Chinese labor. Chinese exports as a percentage of its GDP fell 8 percentage points in the decade from 2007 to 2017, down to 9 percent by 2017. This decline reflects China's increasing reliance on domestic consumption for growth. At the same time, Asian economies tightly coupled to Chinese supply chains, resource-rich countries that export to China, and developed countries selling to China's consumer market are growing more dependent on China.⁵¹

Manufacturing plays a critical role in the Chinese economy. By 2018, China accounted for 35 percent of global manufacturing output.⁵² China's manufacturing base is largely industrialized, with a 73 percent industrialization rate in 2010 and a 76 percent rate in 2015 (see figure 3b.2). Thirty-one percent of China's manufactured exports were high-tech ones in 2017.⁵³ Most of these exports were originally produced by Western and Asian multinationals using imported components assembled in China, but Chinese firms are moving up the technology chain to produce more of these goods on their own. Sustaining strong economic growth and accelerating China's advantages in high-end manufacturing and cutting-edge technologies are critical objectives of the Chinese Communist Party (CCP). As noted in chapter 3a, these objectives were codified in the 2015 Made in China 2025 10-year economic development plan. China's trade surplus has been as high as 10 percent of GDP but declined to 2.2 percent of GDP (about \$336 billion) in 2018. China's trade profile has been put under duress from a trade war with the United States that began in mid-2018, when the Trump administration implemented multiple rounds of tariffs on goods imported from China to force changes in Chinese industrial policies.

This trade war is the most serious disruption in global commerce in nine decades. Although China's losses from a year of this bilateral trade war totaled \$53 billion, China found alternative partners for its exports, shifted some imports away from U.S. suppliers in retaliation, and has refused to make the policy changes the United States demanded.⁵⁴ As addressed, the phase one bilateral trade deal aimed at easing the trade war signed in January 2020 consisted primarily of Chinese pledges to increase imports of U.S. goods and some modest measures to address U.S. concerns about forced intellectual property transfers.⁵⁵ But the economic impacts of COVID-19 make it unclear if China will fulfill those commitments.

Despite policies aimed at stimulating "indigenous innovation," China ranks as the 29th nation in the world for economic innovation, much closer to Russia's 48th-place ranking than America's 5th-place ranking. However, some analysts see Chinese progress. One 2019 study observed that over the past decade, China narrowed the gap with the United States

on 36 indicators of scientific and technological progress and led the United States on some indicators.⁵⁶ As of 2015, China accounts for only 2.3 percent of global commercial financial transactions, massively below the American 52.3 percent share and well below Japan and Western European countries (see figure 3b.2). China's trajectory in the innovation and financial markets categories suggest that it will not be immune to challenges over the coming decade in an increasingly service-based and high-tech global economy.⁵⁷

Long a net recipient of foreign direct investment, China became a net exporter of investment capital in 2015. China's Belt and Road Initiative (BRI) investments have driven its outward FDI flows to above \$150 billion per year since 2015, while its FDI inflows have remained near \$140 billion annually.⁵⁸ The China Global Investment Tracker estimates total Chinese outbound FDI at \$1.2 trillion from 2005 to 2019 and total Chinese overseas construction projects (often funded by loans from Chinese state banks) at \$800 billion. Chinese overseas investment is focused on access to resources (especially oil and natural gas), factories, and infrastructure projects that piggyback on Chinese trade and efforts to acquire advanced technology that will support China's innovation and industrial upgrading from the United States, Europe, and Asia. Overseas construction largely focuses on the energy, telecommunications, and transportation sectors.⁵⁹ In addition to direct investment, China has about \$3.1 trillion in foreign currency reserves and is one of the two (with Japan) largest holders of U.S. Government securities, with about \$1.1 trillion as of January 2020.⁶⁰ In addition to economic value, China's state-managed investments generate influence with foreign elites by contributing to economic development in other countries, while at the same time potentially enabling coercion if countries cannot service their loans and become overly indebted to Beijing.⁶¹

In 2020, China has an increasingly capable military with many instruments of power but does not yet match the United States and Russia. China is competitive in many areas of conventional force and weapons numbers but lags U.S. and Russian forces in several notable areas: overall level of technology, capabilities of individual systems, and power projection capacity. However, the modernization of the People's Liberation Army (PLA) has made impressive progress over the last 15 years toward the goal of being able to "fight and win informationized wars," including a major reorganization in late 2015 that will greatly improve its ability to conduct joint operations.⁶²

The PLA Army is the largest of the services, making up about half of the PLA's 2 million soldiers (China also has about 510,000 reservists). The postreform army is organized in a standardized group army-brigade-battalion structure, with each of the 13 group armies equipped with 6 combined-arms operational brigades and 6 specialized support brigades, including artillery, air defense, special operations forces, and army aviation. PLA combined-arms brigades have operational (armored, mechanized infantry, or light infantry) and support battalions. The result is modular, relatively flexible units that can perform multiple functions and deploy by rail or air to fight away from their home garrison.⁶³ As part of efforts to build a fully mechanized force by 2020, the PLA operates 5,850 main battle tanks, although only about half of these are modern, frontline systems, along with 5,800 infantry fighting vehicles and 3,950 armored personnel carriers.⁶⁴ The army also has six amphibious brigades that could be used in an invasion of Taiwan.

The PLA Navy has been upgrading and developing new major combat platforms (surface ships, submarines, and aircraft) that incorporate modern technology, including advanced antiship cruise missiles and advanced surface-to-air missiles. Its best surface platforms, such as the new Type-055 cruiser currently in sea trials, approach U.S. and Russian capability levels, and it is outbuilding the U.S. and Russian navies as it replaces older ships with much more capable modern replacements. The navy currently operates one rebuilt Ukrainian aircraft carrier, is conducting sea trials on an indigenously built carrier, and is building a third flat-deck carrier that can launch aircraft capable of offensive operations. The navy's aircraft carriers, advanced destroyers (28) and frigates (52), replenishment vessels, and amphibious assault ships give it an increasing ability to operate further from China's coast, including into the Western Pacific and Indian Ocean. This capability is necessary to protect China's overseas interests and is reflected in China's new naval doctrine of "near seas defense and far seas protection."⁶⁵

Over the last 20 years, purchases of advanced aircraft from Russia and improvements in the ability of China's aviation industry to produce modern aircraft have significantly enhanced the PLA Air Force's combat capabilities. Although not on technical par with the most advanced U.S. and Russian aircraft, the Chinese air force now operates more than 900 modern fourth-generation fighters such as the J-10 and J-11, has deployed its first squadron of J-20 stealth fighters, and is developing new medium- and long-range stealth bombers to augment its existing force of about 176 H-6 bombers. The air force also controls China's paratrooper corps and transport aircraft, which provide the PLA a degree of strategic mobility. Air force doctrine has shifted from territorial air defense to conducting both offensive and defensive missions, including a growing emphasis on long-range strategic attack and bombing operations over water.⁶⁶ China has a growing UAS program that features robust low-altitude, low-endurance systems; three known variants of mid-altitude, long-endurance surveillance or strike drones; and at least one high-altitude, long-endurance UAS that has been observed in the South China Sea and near the disputed Sino-Indian border.⁶⁷

China's Rocket Force, formerly known as the Second Artillery Corps, operates China's intercontinental (about 100), intermediate-range (about 72), and medium-range (about 80) nuclear ballistic missiles and a large conventional force with ballistic and cruise missiles of various ranges that can target Taiwan and U.S. bases throughout the region. It has primary responsibility for deterring a nuclear attack and being prepared to retaliate if deterrence fails. China's nuclear policy calls for a "lean and effective" nuclear force focused on deterring nuclear attack in accordance with China's "no first use" nuclear policy. Accordingly, China has been satisfied with a much smaller nuclear arsenal than the United States and Russia (which both have about 1,600 deployed strategic warheads), although the size has expanded to about 300 deployed warheads as the Rocket Force has increased the number of missiles (including some with multiple warheads) and the navy has deployed four ballistic missile submarines.⁶⁸ The Rocket Force probably also controls other strategic capabilities based on missiles, such as China's direct-ascent antisatellite weapons and the antiship ballistic missile versions of the DF-21 and DF-26 missiles.

The PLA performs a range of tasks, including domestic missions such as maintaining political security and social stability, traditional military missions such as nuclear deterrence and protecting China's sovereignty and security, new missions such as protecting

China's economic development and China's interests in space and cyberspace, and non-traditional security missions such as emergency rescue, disaster relief, and international security cooperation. The PLA also supports China's foreign policy and broader strategic objectives by engaging in military diplomacy, with a focus in 2020 on the United States and Russia, and by engaging with the countries on China's periphery in the Pacific region.⁶⁹ PLA diplomacy places special emphasis on senior-level visits, exercises with foreign militaries, and naval port calls. In 2018, the PLA conducted more than 60 bilateral and multilateral exercises with foreign militaries.⁷⁰

China has the second largest defense budget. Its estimated \$250 billion in expenditures in 2018 was roughly 40 percent of the U.S. base defense expenditure budget of \$650 billion, but 4 times higher than Russia's \$61.4 billion.⁷¹ China remains a major importer of weapons and military technology, depending on Russia for jet engines, advanced missiles, sensors, and other military systems—although this dependence has declined significantly over time and will probably end in the next decade.⁷² China's improved military industrial base makes it a major arms exporter to developing states, particularly in Asia. Between 2008 and 2018, China exported some \$15.7 billion worth of conventional weapons across the globe, making it the fifth largest arms supplier in the world—behind the United States, Russia, Germany, and France. The lion's share of these exports—about 75 percent—went to Asia. An additional 20 percent flowed into Africa.⁷³ China's arms exports niche has historically been medium-cost, medium-capability systems, and its export potential is also limited by the fact that many countries will not procure Chinese arms for political reasons. From 2014 to 2018, China delivered major arms to 53 countries, compared with 32 from 2004 to 2008. Pakistan was the main recipient (37 percent) from 2014 to 2018, as it has been for all 5-year periods since 1991. From 2014 to 2018, China became the largest exporter in the niche market of unmanned combat aerial vehicles, partly because the United States has restrictions on exports of these systems, and Russia has lagged in UAS development.⁷⁴

China's principal military weakness relative to the United States and Russia is its limited power projection capability. China has invested in antiaccess/area-denial capabilities such as advanced diesel submarines, advanced surface-to-air missiles, antiship cruise missiles, and an innovative antiship ballistic missile designed to attack U.S. aircraft carriers. These capabilities raise the costs and risks for U.S. forces operating near China. The PLA's current limitations are partially offset by its geographic location and priority area for strategic focus in the Pacific. The United States, Russia, and other potential military contestants face challenges in projecting power and influence into the Western Pacific and Asia, where the Chinese are most obviously optimizing military capabilities for the coming decade.

However, the PLA's power projection capabilities fall off rapidly with distance, and China lacks allies or a network of overseas bases that could extend its range into other regions.⁷⁵ Nevertheless, the PLA is gradually expanding its global reach.⁷⁶ China has invested in a range of antisatellite capabilities that could degrade, interfere with, or directly attack U.S. satellites and has extensive cyber capabilities to collect intelligence and attack U.S. military computer networks. The PLA is developing a range of hypersonic weapons (and has deployed the DF-17 medium-range ballistic missile with a hypersonic glide vehicle) and is investing heavily in military applications of artificial intelligence.⁷⁷

China is using its political and diplomatic clout to advance its influence within existing global institutions and by creating alternative ones.⁷⁸ In the late 2010s, China was a full member of 15 major intergovernmental organizations and an observer in two dozen others. These numbers were unremarkable and actually only two-thirds the number (22) of formal intergovernmental organization (IGO) memberships held by the United States and Russia at the same time.⁷⁹ However, China has been using its influence within the United Nations (UN) system and with other intergovernmental organizations to pick up diplomatic and political “distressed assets” abandoned by the United States and its allies and repurpose them to serve its strategic goals. China is now the second largest funder of the UN (behind the United States) and provides more troops to UN peacekeeping missions than any of the other permanent members of the UN Security Council (UNSC). Over the past decade, Chinese candidates have taken on senior leadership positions at the World Bank, Interpol, the United Nations Industrial Development Organization, the International Telecommunication Union, and the Montreal-based International Civil Aviation Organization. China has also sent military officers to lead UN peacekeeping missions in Western Sahara and Cyprus.

China’s pursuit of crucial international organization posts has raised alarm among human rights and free speech advocates who fear Beijing will set back progress on these issues. After a former Chinese official was appointed head of Interpol in 2016, Beijing successfully used Interpol’s “red notice” system to pursue critics living abroad. Beijing has also pressed to cut funding for human rights investigators in UN peacekeeping operations. In Geneva, the UN has stifled Chinese human rights advocates from making their case before the world. China’s play for leadership of UN Educational, Scientific, and Cultural Organization (UNESCO) revealed that it views the Paris-based organization as more than just an overseer of world heritage sites and educational programs. Beijing also sees UNESCO as a vehicle to regulate the global Internet.⁸⁰ In addition, China practiced “lawfare” by leveraging its positions in treaties and regimes to ignore or reinterpret canonical provisions of international agreements when these undercut Beijing’s preferences, as with UN Convention on the Law of the Sea provisions involving the right of all ships to innocent passage through its territorial seas.⁸¹

A second line of effort involves China creating alternative organizations that compete with existing international arrangements. Its BRI and Asian Infrastructure Investment Bank (AIIB) stand out as state-led infrastructure development programs that provide alternatives to multilateral UN development organizations such as the World Bank, International Monetary Fund, and Asian Development Bank. India and the United States rejected the state-led BRI model, and the United States shunned participation in the AIIB for similar reasons. Some argue that China’s effort to build a parallel alternative framework for global infrastructure development does not pose a major challenge due to resource limitations and the AIIB’s status as a multinational entity relying on standing commercial markets.⁸² However, the existence of alternatives undercuts World Bank efforts to incorporate anti-corruption, labor, and environmental standards in lending to developing countries. Others note that as BRI reached its 5-year anniversary in late 2018, as many as 14 percent of its projects (accounting for 32 percent of global BRI project value) had run into some kind of trouble. Many BRI projects confront local pushback from performance delays, lack of local workforce participation, and predatory project loan terms—including in states friendly

"We assess that China's intelligence services will exploit the openness of American society, especially academia and the scientific community, using a variety of means."

—Daniel R. Coats, *Worldwide Threat Assessment of the U.S. Intelligence Community*, ODNI, January 29, 2019

to China such as Pakistan, Sri Lanka, and Malaysia.⁸³ The extent to which China's alternative global institutional framework will gain traction and produce significant leverage for Beijing in the coming decade remains unclear.

China has extensive propaganda and communications tools to get its message out to international audiences, but the content

of that message limits its effectiveness. Beijing's ability to craft and disseminate its preferred ideology in a resonant and positive message has been improving over the past decade but still exhibits significant liabilities and shortcomings. China's ideological framework of "a community of common destiny" glosses over conflicts of interest between nations and instead places emphasis on state sovereignty at the expense of human rights and freedoms, which inherently limits appeal. These values resonate with autocratic elites but not so much with ordinary citizens.

China has historically maintained an extensive censorship and propaganda apparatus to get the party's message out and to control and censor competing messages within China. The ruling Communist Party has adapted this apparatus to the Internet age, investing heavily in modern technologies (sometimes called the Great Firewall of China) to ban unwanted information from public view on the Internet and on social media. China has also increasingly pursued an assertive ideological strategy aimed at international audiences. With massive infusions of money—funding advertorials, state-owned newspapers and television networks, sponsored journalistic coverage, and positive messages from co-opted boosters—China has been trying to reshape global views of itself by exploiting the vulnerabilities of the international free press and higher education establishments.

Beijing's main means of international influence has been through print, television, and radio. Its pays for Chinese "information supplements" to appear in respected international newspapers like the *Washington Post*. It oversees Xinhua, a state-run global media service that produces CCP-friendly stories for worldwide dissemination in multiple languages and boasts an 11.5-million-follower Twitter account (despite the fact that Twitter access is banned in China). It endorsed the acquisition of Hong Kong's *South China Morning Post* in 2015 by the CEO of the Alibaba Group (e-commerce), who inserted a management team that promised to provide a positive view of China. It generates content from its state-run China Radio International for use by broadcast networks from Norway to Turkey to Australia. It has generously funded a globally positioned China Global Television Network—rebranded in 2016 as the international arm of China Central Television—promising local journalists across the world excellent money and opportunity so long as they tell China's story well. The content emphasizes the generosity of the Chinese people and the benign nature of the Chinese government while amplifying the chaotic and unpredictable nature of Western politics and liberal democracies. Finally, China invested extensively in several hundred Confucius Institutes at universities around the world to promote Chinese language and culture and to promulgate CCP perspectives on an array of international and Chinese-related issues. These have drawn scrutiny in recent years for stipulating that the Chinese government must

approve teachers, events, and speakers at events on Western university campuses, while the Chinese government has refused U.S. State Department efforts to set up American Cultural Centers on Chinese college campuses.⁸⁴

These efforts at information dissemination and image-making have met with uninspiring results. Across 34 countries surveyed, the 2019 Pew Global Attitudes Survey found a median of 40 percent had a favorable opinion of China, compared with a median of 41 percent who had an unfavorable opinion. Asia-Pacific, North America, and Western Europe saw a decline in favorable views of China compared to 2018.⁸⁵ But it will be interesting to see how Western European perspectives trend after China's very public assistance to states hard hit by the 2020 COVID-19 pandemic there. Conversely, African views of China are generally positive, averaging 62 percent favorable ratings.⁸⁶ Polling results indicate that China's image-making tools are having limited success in the regions of greatest concern—the Asia-Pacific and South Asia. Just one state in the region considers Beijing to be a friend: Pakistan. Even North Korea and Iran have transactional relations with Beijing. The Chinese language also limits the effectiveness of Chinese propaganda efforts. Very few people speak Chinese fluently as a second language around the world, while English is spoken fluently by almost 1.8 billion people, including hundreds of millions across the Indo-Pacific. The language barrier and the heavy role of state censorship has limited China's ability to use music, film, and entertainment as global soft power tools. These also are impediments to China's ability to use education as a source of cultural influence. Although China hosted 492,000 foreign students in 2019 (third most in the world), the quality of Chinese higher education institutions varies widely, and Xi Jinping's efforts to tighten the CCP's ideological grip over college lesson plans are likely to leave a negative impression on foreign visitors.⁸⁷

Key Chinese Power Tools and Their Strategic Utility. China's contemporary overall power rests largely on its status as an economic global giant with growing resources and a steadily improving technology base. Beijing's global economic influence already exceeds that of the United States in some important categories that will continue to expand over the coming 5 to 7 years. Its capital reserves, level of industrialization, and attention toward high-tech innovation and military modernization underpin ongoing efforts both to assume leadership in current international economic and political institutions and to develop new ones more conducive to Beijing's interests. Yet China's projected economic power advantages may not be sustainable in the out years, unless it finds a way to redress weaknesses that may constrain growth, including a fast-aging population, an educational and intellectual culture that constrains innovation, and an undersized presence in financial markets that limits the revenue potential and influence of Chinese financial services.

Beijing is spending far more on military forces than Russia and more than any other country except the United States. If its military spending and investment trends continue, China has the capacity to equal or surpass the U.S. economy and U.S. military forces at some point during the next two decades but not in the coming one. China's emergence as a full superpower is uncertain, and the timing is impossible to predict.⁸⁸

China's military capabilities have grown over the last decade to the point where it can compete with the U.S. military in East Asia and the Western Pacific. Its forces can deny U.S. naval and air forces uncontested access to areas near the Chinese coast, and it can hold

major U.S. air and naval weapons platforms at risk. However, China will be pressed to project power outside the second island chain over the next 5 to 10 years.

China's diplomatic power tools are important but not impressive. Beijing has filled leadership vacuums left by recent American and European withdrawals from international organizations and attempted to use these to advance Chinese national interests and/or change the institutional rules to suit Beijing's preferences.

China displays clear deficiencies in its ideological, cultural, and communications power dynamics. Beijing has no real multilateral political or military alliances, and only one true long-term strategic relationship. It pursues transactional interactions with economic and investment partners often wary of Chinese interests and financial terms. China gets poor ratings and survey responses regarding levels of international respect and trust. Despite an intense effort to improve global messaging, its national narrative focused on state control and social order over individual liberties resonates poorly outside of authoritarian circles. Finally, China continues to demonstrate limited language, cultural, or academic appeal. Beijing's proactive global response to the 2020 COVID-19 pandemic may help overcome these deficiencies and offset criticism that its repressive internal politics hid the problem from the rest of the world for far too long. How this plays out remains to be seen.⁸⁹

China's power tools are skewed toward the economic today but have long-term potential to develop more broadly. China's trade and investment prowess make it a major force in the economic competitive space, and its long-term plan to leverage this economic advantage to develop military, political, informational, and ideological capability is palpable. China's contemporary power factors do not present a current urgent military threat, but in the long term, China's growth and global aspirations make it the most important potential Great Power challenger to the current U.S. global position and to longstanding American/Western values, norms, and institutions.

Russia's Competitive Posture and Tool Sets

General Russian Power Factors and Approaches. In 2020, Russia's application of its power resources to international competition remains as it has been for the prior decade: tactically successful despite severe structural shortcomings. Some analysts assert that Russia has a viable long-term strategy for use of its limited power base, a "raiding" strategic framework.⁹⁰ They see an underlying strategic logic behind a decade of Russian activism that includes Moscow's ongoing interventions into Georgia and Ukraine; its ventures into Syria and Libya; its norms-busting interactions with Iran, North Korea, and Venezuela; its ongoing use of cyber tools to disrupt and discredit elections in Europe and the United States; and its tactical rapprochement with China. Yet most strategic observers do not believe Russia has a true international strategy and credit Prime Minister Vladimir Putin with masterfully playing a weak and eroding power base to maximum short-term effect.⁹¹

Russia's major power factors are not generally positive but do include critical military capabilities. Russian military power tools are a mixed bag. Moscow retains a vast nuclear arsenal, one equal in size to that of the United States, with an estimated, treaty-authorized 1,600 active deployed strategic nuclear warheads.⁹² Russian nuclear weapons underpin Moscow's claim to Great Power status and are distributed between an estimated 860 land-based delivery missiles, 10 ballistic missile submarines, and 50 bomber aircraft.⁹³ Russia is actively

modernizing this nuclear weapons force with new single-warhead and multiple-warhead missiles, a hypersonic glide missile delivery vehicle, new cruise missiles for its bombers, and reportedly new intermediate-range missiles and a rail-mobile missile—both long banned by Cold War-era arms control treaties that have either expired or are not likely to survive in the future.⁹⁴ Despite many reported delays and frequent testing challenges, Russian modernization efforts convince many military analysts that Moscow will be able to sustain a secure, second-strike nuclear deterrent at a price far more affordable than the prohibitive costs of developing robust antiballistic missile systems.⁹⁵

Since 2011, Putin's Russia also has modernized its conventional military forces. The modernization has not generated a globally relevant conventional force. In 2019, Russia had 1 million members in its active-duty, conscript-based military, with more than 800,000 of these in ground and aviation units with home-country defense missions rather than deployment-capable ratings. Its navy featured 1 inoperable aircraft carrier, 56 aging submarines in varying states of repair, and a surface fleet heavy on Corvettes and shore patrol craft compared with an American fleet featuring 12 capital aircraft carriers, 68 fully operable submarines, and a surface fleet dominated by more than 90 frigates and destroyers. Russia's air force possessed fewer than 900 fighters compared to the 2,400 in the U.S. Air Force; an attack aircraft fleet of 1,500 that was half that of the United States; helicopter units with only 25 percent of the U.S. military's 5,800; and a transport aircraft fleet of 400, barely more than one-third the size of the U.S. force.⁹⁶ In addition, Russia's recent conventional military track record features multiple mishaps and embarrassments that call into question its ability to sustain global reach: Its only aircraft carrier, the geriatric *Admiral Kuznetsov*, suffered debilitating mechanical and safety incidents from 2016 to 2018 that have placed it in dry dock through at least 2021; its vaunted and extremely expensive T-14 Armata main battle tank reportedly failed many operational tests before its prototype broke during rehearsal for the May 2015 Victory Parade in Moscow; and its military aircraft—fighters and transports—began crashing at a regular and alarming rate from 2015 to 2019.⁹⁷

However, since the 2008 Russo-Georgian war, strategically targeted military investments have underwritten a significant and meaningful upgrade of conventional and gray zone Russian military capabilities.⁹⁸ Generating a small annual defense budget compared to China and especially the United States, Russia spent 50 percent of its military budget over the decade—a disproportionately large share—on procurement of precision-guided and enhanced conventional strike weapons.⁹⁹ It capitalized a new generation of precision-guided munitions, modernized almost 1,000 of its current helicopter fleet of 1,485, and generated 1,000 new or modernized combat aircraft out of a force of 1,500.¹⁰⁰ The Russian military has increased its operational UAS fleet to over 2,000 systems, most of which are tactical and all of which are intelligence and surveillance models, not strike variants. Moscow has budgeted over \$10 billion to develop combat UAS programs by 2020, and it has been aggressively developing counter-UAS capabilities featuring electronic warfare, counter-GPS spoofing, and kinetic detect-and-kill systems.¹⁰¹ These enhanced and modernized systems mesh well with a smaller and more professional and deployable Army and Special Forces military cadre, often intermixed with civilian Russian private military company or mercenary forces.¹⁰² Russia also has built flotillas of small surface ships and diesel-powered submarines in the Black Sea and Caspian Sea—both equipped with long-range, sea-launched cruise missiles—

and a number of cruise missile-carrying frigates. These weapons expand Russia's military strike reach to waters around most of Eurasia.¹⁰³ Russia aims to build a similar fleet in the Baltic Sea over the coming decade.

As of 2020, Russia's nominal GDP is only \$1.7 trillion compared with \$15 trillion for China and \$22 trillion for the United States (see figure 3b.1). Russia is a "one crop economy" with a heavy dependence on energy exports (mainly oil and natural gas) that accounted for almost 60 percent of Russian exports and almost all of Moscow's \$120 billion trade surplus in 2017.¹⁰⁴ Russia's dependence on energy export revenue is high, but its share of global exports remains low for its size—only 2.6 percent in 2018. Almost 55 percent of Russian exports went to Europe, with another 37 percent going to Asian trading partners, mainly China and South Korea.¹⁰⁵ Russia's level of industrialization hovers in the 30 percent range, well below other modern economies, and has remained relatively unchanged for more than two decades (see figure 3b.2). Russian high-tech manufacturing is subpar and declining, with only 11 percent of its 2018 manufactured exports consisting of high-tech products—just over one-half of America's 19 percent, one-third of China's 31 percent, and well below the global average of 18 percent.¹⁰⁶

These numbers correspond with Russia's relatively low rankings on two other measures of modern economic performance: level of business innovation and share of global financial market transactions. In 2018, Russia's global innovation ranking was 46 out of 118 countries, relatively unchanged in the last decade, far below other modern economies, and a factor that restrains Moscow's ability to modernize its economy for a fast-changing future. Russia's share of global financial market transactions in 2018 was less than 1/10 of 1 percent—far below all other modern economies and a statistic indicating the inability of Russia—save for a handful of Putin-linked autocrats—to derive profit from the dynamic and expanding elements of the broader global services economy. The political institutions for an effective market economy in Russia are largely missing, its currency (the ruble) is an untrustworthy investment instrument, and robber-baron state capitalism lacks the kind of effective regulation and predictability that generates the trust necessary for economic investment, exchange, or growth.¹⁰⁷ Combined, Russia's trade as well as its financial and innovation limitations bode ill for its ability to generate sustainable tools for successful Great Power competition today or into the future.¹⁰⁸

Russia fares just as poorly in measures of ideological resonance and cultural identity. It has demonstrated some success in promulgating a message of mistrust for Western institutions and values around the world, modestly increasing Russia's relative stature as an influential state.¹⁰⁹ Russia also has obvious linguistic and cultural affinity in former Soviet states and its "near abroad," especially in Belarus, Central Asia, and Mongolia, but the Russia brand and narrative do not resonate more widely. Only 34 percent of global respondents in a 2018 Pew International Survey had a favorable view of Russia, and 63 percent had no confidence in Vladimir Putin.¹¹⁰ Few outside Russia speak Russian or watch Russian films or Internet programming in Russian. No Russian universities are ranked in the global top 100. Putin's Russian lifestyle lacks global appeal. Its public health system is weak; average Russian life expectancy is 5 years shorter for men and women than in Europe and a dozen years lower for men than in the United States. Russia's current population of 145 million

is forecast to decline to 121 million by mid-century, calling into question the ability of the Russian social system to support itself.¹¹¹

Despite a negative global image and constrained ideological appeal, Russia does possess and wield several diplomatic and communications power tools to good effect. Putin's Russia has been a member of 22 major intergovernmental organizations for a decade or more (it was expelled from the G7/8 in 2015 after its annexation of Crimea and invasion of eastern Ukraine). Its political IGO affiliations include permanent member status on the UNSC, International Atomic Energy Agency (IAEA), and Interpol.¹¹² Moscow applies this diplomatic power to advance Russian policy interests and to make its case for recognition of controversial policies. For example, in 2015 Moscow protested listing a Sevastopol nuclear facility as belonging to Ukraine in an annual IAEA report annex, insisting that the facility in the Crimean city be listed as Russian.¹¹³ Putin's Russia also advances its preferred standards of international policing and criminal accountability via Interpol, where in late 2018 the Russian nominee for the president was rejected by Western delegates based on Moscow's history of using Interpol to target Putin's political foes.¹¹⁴

In the information and communications space, Putin's Russia has funded and managed three substantive agencies for overt and covert dissemination of its global viewpoints: Russian Television (RT), Sputnik radio, and the Internet Research Agency (IRA). All are funded by the Russian government and work to disseminate propaganda and put out disinformation intended to polarize and confuse non-Russian audiences in a manner aimed to sow mistrust of Western media and institutions.¹¹⁵ These tools modernize and update techniques used by Soviet Union intelligence agencies during the Cold War, today making the viewpoints of Putin's Russia available in 24 languages, especially English. They also amplify manifestly fake but disturbing stories that are difficult to disprove and create the feeling that no one knows quite what is real.

Relying heavily on the multiplier effect of high-volume retweeting and forwarding of its specious stories, the daily deluge of disinformation produced by RT and Sputnik has a nontrivial societal impact across much of the West and is a tactic that other authoritarian regimes are seeking to replicate.¹¹⁶ The IRA is the covert social media influence and operation funded by the Russian government that works with Russian military intelligence hacking units to promulgate targeted disinformation and propaganda designed to distort voter perceptions and manipulate participation in democratic elections across Europe and in the United States.¹¹⁷ At the same time, Putin's Russia has been developing the tools to isolate Russia from the global Internet, passing a 2019 law that allows such a cutoff and testing the technology necessary to operate a Russia-only Intranet.¹¹⁸

Russia has attempted to counter a longstanding negative humanitarian image in reply to the COVID-19 pandemic. Moscow sent hard-hit Italy a shipment of pandemic assistance materials in late March 2020, in an apparent effort to contrast itself with the European Union that reportedly sent Italy nothing. It remains to be seen if Putin's humanitarian gesture is remembered as genuine or a publicity ploy, especially since reports from Italy were that 80 percent of what Russia sent was of little use to Italy.¹¹⁹

Key Russian Power Tools and Their Strategic Utility. Even though Putin's Russia is in unambiguous relative economic decline compared to the United States and China, Russia in 2020 possesses a geographic expanse, a skilled workforce, and the vast natural resources

to balance against U.S. hegemony and China's rise for at least the coming decade. Its most important power tools are its nuclear weapons, its skills in cyber technology and the promulgation of information/disinformation, and its vast stores of oil and gas. Russia also has a diplomatic gravitas and a limited, modernized military and paramilitary capability necessary for projection of force in selected areas where its most significant strategic interests are engaged.

Russia's limited economic and ideological power attributes and potent but declining military, diplomatic, and communications tools make Moscow most capable of achieving foreign policy outcomes in its near abroad: Eurasia.

Outside Eurasia, the region where Russian diplomatic, military, and communications capability appears to be the most relevant is the Middle East. There, Moscow can use military bases in Syria and Iran to selectively employ its new conventional strike assets—conventional military and contractor ground forces equipped with precision-guided conventional weapons including missiles, rotary and fixed-wing attack aircraft, and conventional long-range cruise missiles launched from land and sea.

Russia's power limitations require it to avoid direct military confrontation with the United States and to seek tactical accommodation with China in areas where Sino-Russian key interests align for the coming decade. For at least the next 5 years, the two will continue to cooperate closely in the UNSC, take similar positions on cyber sovereignty and Internet governance, and use various diplomatic frameworks such as the BRICS (Brazil, Russia, India, China, and South Africa) grouping and the Shanghai Cooperation Organization to coordinate joint security and policy positions. They also should be anticipated to continue arrangements that share nonnuclear military technology and to conduct joint military exercises on a symbolic, limited basis. Despite the Sino-Russian entente, Russia continues to sell arms and provide advanced military technology to countries that have territorial disputes with China, such as Vietnam and India.

Russia's relative power capabilities are heavily concentrated in the military arena, with tools ranging from a formidable nuclear weapons arsenal, to significant conventional military power projection capabilities, and to a successfully employed set of gray zone armed actors. Moscow's information operations potential is equally impressive and unscrupulous, adding to the short-term capacity of Russia to pursue a strategy of disruption against Western institutions and organizations. Yet Russia's economic, ideological, and political power tools are substandard for a durable global power now and are likely to atrophy further over the next 5 years. Its severe limitations in many critical areas of power development and projection make Russia an urgent but not a grave threat to many immediate American/Western competitive interests. They also render it a dubious long-term challenger for Great Power ascendance.

Net Power vs. Gross Power Indicators: A Less Imminent Great Power Transition?

The conclusions reached above about the relative status of American, Chinese, and Russian power attributes today and into the future—a future featuring the disruptive technologies of the fourth industrial revolution (addressed in chapter 4) and increasingly service-based economies—were made from comparison of their gross factors of power—that is, factors of power available for use before any internal “costs” or “taxes” on them from domestic needs

and constraints are factored in. Some political scientists, most notably Michael Beckley of Tufts University, contend that gross power indicators misrepresent actual state power. Beckley believes that one must move beyond gross power factors and calculate the net index of Great Power factors to get a realistic feel for relative state power.

Beckley has generated such a net power index. It focuses on *net power resources*, which he defines as the resources available to a country after subtracting production costs, welfare costs, and security costs. His net power approach captures the fact that countries with large populations and potential domestic challenges will spend most of their gross power resources supporting their people and maintaining domestic stability, leaving fewer net resources available for external use in Great Power competition. Beckley argues that for populous countries such as China, gross power often significantly overstates actual capabilities and net power provides a more accurate assessment. His determination of net power factors can be applied by a comparative index to the U.S.-China, U.S.-Russia, and China-Russia competitive dyads over the past 25 years. Table 3b.2 makes this application and demonstrates the discrepancy between gross power balances between two states and net power balances, measured as a percentage difference in the two calculations.

Low numbers indicate a small gap between gross power factor comparisons (from the Correlates of War Composite Index of National Capability) and net power comparisons, while higher numbers indicate a greater disparity.¹²⁰ The calculated numbers for 2015 indicate that the net power disparity between Russia and the United States closely matches the gross power disparity, which validates a large gap in overall power attributes. Conversely, the figures for the United States vs. China and Russia vs. China show large disparities, suggesting that gross power calculations significantly exaggerate China's actual power. The 23 percent gap between U.S. and China net and gross power comparison reflects the high costs China faces in maintaining domestic stability and generating military forces that is not captured in gross power calculations. This difference is similar to the number Beckley calculated in net-vs.-gross power between Germany and Russia in the 1890 to 1917 period, where gross power calculations severely overestimated Russian power. The even higher 37 percent gap for Russia and China in 2015 is similar to the disparity Beckley calculated between Britain and China between 1840 and 1910, a period where Britain's small size misassessed the huge power deficiency that came from China's enormous internal security and societal costs. The implication is that China's contemporary internal challenges and costs again make its gross power indicators exaggerate its actual overall power potential.¹²¹ In turn, the use of a net power comparison indicates that a Great Power transition between the United States and China is far from imminent.

Comparative Insights and Implications: 2020–2025 and Beyond

This detailed review of contemporary Great Power factors and their strategic utility reveals eight major insights.

First, the modern Great Powers—the United States, Russia, and China—will compete across the five categories listed in table 3b.1 in a manner featuring some cooperation and collaboration but with increasing episodes of confrontation, especially over nonmilitary issues. The tools of competition traditionally associated with one category of interaction in a less rivalrous era will be used more and more often to achieve strategic effects in another

"The competition is likely to be multilayered and interactive. No single theme or model will capture the complex mosaic of global competition, and the intersections among diverse types of competition—how success or failure in one area exacerbates or mitigates others—will be a crucial determinant of relative success."

—Michael J. Mazarr et al., *Understanding the Emerging Era of International Competition*, RAND (2018)

category during Great Power competition. This is both a return to historic dynamics of interstate rivalry and a reason that concepts such as sharp power, gray zone operations, and geoeconomics now appear frequently in the writing and thinking of those today grappling with old geopolitical concepts now made new.

Second, the most important gross power indicators available today and their projections for the next 5 to 10 years clearly indicate that for Washington, Russia is an

urgent but transient security risk, while China is the most important—albeit presently less threatening—Great Power challenger to U.S. national interests and global policy preferences. A net power comparison between the United States and China indicates that their power transition timeline is longer than some now fear.

Third, America's military advantage—albeit not what it was in the two-decade period from 1992 to 2014—remains robust. America remains largely unrivaled in the one area that matters most to its military power potential: its ability to deploy effective forces anywhere in the world in the event of a crisis.¹²² However, this advantage could be less definitive if China or Russia is able to pick favorable political and geographic ground for a short but decisive military conflict and limit America's ability to bring its full power advantages to bear in a particular setting. This is especially true in the Indo-Pacific region and is addressed in detail in chapter 9.

Fourth—and in alignment with the third insight—neither Russia nor China possesses the power to prevail in a protracted military clash with the United States today or for the foreseeable future. Each knows this and, unless guilty of a serious miscalculation, will seek to avoid a direct military clash with the United States if at all possible, between now and 2025. Washington can leverage this to its advantage while pursuing a strategy that collaborates when possible, competes smartly, and confronts unacceptable behaviors and policy challenges adroitly. American military dominance is an asset in this new era of Great Power competition, but the United States needs to develop new competitive tools in nonmilitary areas. Its current strength gives it the opportunity to wean itself from its post-Cold War addiction to military instruments and develop more fungible capabilities across the diplomatic, ideological, informational, and economic categories of Great Power interaction. Improving America's capacity and ability to wield nonmilitary instruments effectively is necessary to compete effectively in a new era of Great Power rivalry.

Fifth, China today has the economic and communications/information power necessary to compete with the United States (and Russia) for access and influence around the world. Its use of foreign investment through the BRI and the AIIB demonstrates China's ability to win access and influence, at least in the short run. The longer run strategic impact remains uncertain as the downsides of Beijing's ideological message and its often-criticized predatory economics model may not provide sustainable influence in the future. China has the ability to seriously constrain U.S. (or Russian) military activities in East Asia, in

the Western Pacific, in cyberspace, and increasingly in space. These factors suggest that the United States must calibrate a competitive strategy for these regions that leverages U.S. ideological and soft power advantages and seeks to undercut China's economic strengths by highlighting the downsides for its partners.

Sixth, Russia's power factors align well with the short-term, geographically limited strategy it has been pursuing. Moscow has clear military, economic, and communications advantages in its near abroad (Eurasia) and a limited but nontrivial ability to project these tools for influence in the Middle East, the Arctic, and cyberspace. However, Russia's economic, ideological, and political challenges are likely to erode its power tools for influence beyond its near abroad as the decade of the 2020s progresses. A worthy U.S. approach to competition with Russia might optimize soft power and deterrent postures along Russia's immediate periphery combined with more assertive competition (and, where necessary, nonmilitary confrontation) against Moscow's use of global institutions and communications structures to delegitimize openness, transparency, and truth in the rest of the world.

Seventh, the combination of Great Power strategic interests and their current and future power potentials makes it clear that Russia is a dangerous near-term strategic competitor to the United States with the potential to do enormous military damage to America and the world if miscalculation leads to a military clash. China is a less insidious short-term challenger, but it is the Great Power with the strategic interests and the growing power potential to dramatically alter current norms, rules, and procedures preferred for international interactions by the United States and its Western allies. Washington must treat each Great Power accordingly. In the 2020 to 2025 window, Washington must choose whether, where, and how to compete. Put starkly, the United States can contest or confront its Great Power rivals today in accordance with a resolve to sustain its global position and the standing rules, norms, institutions, and alliances of the current international order, or it can abdicate leadership of the global order and allow a much more powerful China to extended its own version of global norms, rules, and institutions. The former course entails risks, but the latter course would not necessarily avoid a military confrontation, especially if the United States comes to view an increasingly Chinese ordered world to be unacceptable.¹²³

Finally, past performance is not a guarantee of future results. The United States, China, and Russia each face major internal structural, economic, and demographic challenges.¹²⁴ The choices each state's political leadership make about how to address these domestic dynamics as well as their international challenges will determine the future power they will possess and the future policy options they might pursue.

Technology, Innovation, and GPC Considered

The next chapter rounds out this volume's first section dedicated to providing a conceptual framework for understanding a new era of Great Power competition. Major advances in technology have been demonstrated to empower the rise of new states into international preeminence. Emerging powers become dominant because they develop new economic spheres that become leading sectors of the global economy, underpinning the economic vitality and military power of that state in a reordered global hierarchy.¹²⁵ At the same time, the diffusion of key technological know-how or inventiveness to other countries has been correlated with Great Power decline.¹²⁶ Chapter 4 looks at the Great Powers in context with

the ongoing fourth industrial revolution—one being driven by the rapid emergence and convergence of multiple technologies, including robotics, artificial intelligence, 3D printing, energy, biotechnology, and food production. Chapter 4 considers critical technologies and their meaning for a new era of Great Power competition.

Notes

¹ This distinction from power literature, in particular, can be found in David Baldwin, *Paradoxes of Power* (New York: Bail Blackwell, 1989). The priority focus of this chapter on power-as-material-resources, before consideration of power as the ability-to-realize-ends, aligns with the approach in Stephen G. Brooks and William C. Wohlforth, *America Abroad: The United States' Global Role in the 21st Century* (Oxford: Oxford University Press, 2016).

² This operational definition was derived from Thomas J. Volgy et al., "Major Power Status in International Politics," in *Major Powers and the Quest for Status in International Politics: Global and Regional Perspectives*, ed. Thomas J. Volgy et al. (New York: Palgrave Macmillan, 2011), 1–26. Their definition derives from an amalgam of several other classic efforts at defining Great Power status, including the following historic references: J. David Singer and Melvin Small, "Formal Alliances, 1815–1939: A Quantitative Description," *Journal of Peace Research* 3, no. 1 (March 1966); Jack Levy, *War in the Modern Great Power System: 1495–1975* (Lexington: University Press of Kentucky, 1983); John J. Mearsheimer, *The Tragedy of Great Power Politics* (New York: Norton, 2001); and Benjamin O. Fordham, "Who Wants to Be a Major Power? Explaining the Expansion of Foreign Policy Ambition," *Journal of Peace Research* 48, no. 5 (2011), 587–603.

³ On the basic point of the misleading nature of gross domestic product (GDP) to capture overall national power decline, see Brooks and Wohlforth, *America Abroad*, 14–47.

⁴ As an example, the United States realized \$127 billion in receipts for the use of intellectual property in 2017, while China only netted \$4.8 billion. On the number of America's nonmanufacturing economic advantages underrepresented in GDP and other economic measurements, see Thomas J. Wright, *All Measures Short of War: The Contest for the Twenty-First Century and the Future of American Power* (New Haven: Yale University Press, 2017), 165–168; Brooks and Wohlforth, *America Abroad*, 14–15. For data on comparative revenue from the global use of intellectual property in 2017, see "Charges for the Use of Intellectual Property, Receipts (BOP, Current US\$)—China, World, United States, East Asia & Pacific," World Bank, available at <<https://data.worldbank.org/indicator/BX.GSR.ROYL.CD?end=2017&locations=CN-1W-US-Z4&start=2000>>.

⁵ Wright, *All Measures Short of War*, 165.

⁶ *Ibid.*, 165–168.

⁷ *Ibid.*

⁸ Insight from Michael Beckley, "The Power of Nations: Measuring What Matters," *International Security* 43, no. 2 (Fall 2018), 8–9, who argues for a use of net power indicators (power assets minus power liabilities). For insights on the historic use and defense of gross measures of state power in theory and in practice, see Paul M. Kennedy, *The Rise and Fall of the Great Powers: Economic Change and Military Conflict from 1500 to 2000* (New York: Random House, 1987); John Mearsheimer, *The Tragedy of Great Power Politics* (New York: Norton, 2014), 55–138; Joseph Nye, Jr., *The Future of Power* (New York: PublicAffairs, 2011), 25–81; Ashley J. Tellis et al., *Measuring Power in the Postindustrial Age* (Santa Monica, CA: RAND, 2000), 1–33; Beckley, "The Power of Nations," 8–12.

⁹ This new index is based on the premise that while GDP systematically exaggerates the wealth and military capabilities of poor and populous countries—because they tally countries' resources without deducting the costs countries pay to police, protect, and serve their people—thoughtful military studies show that the higher a country's GDP per capita, the more efficiently its military fights in battle. The reason is that a vibrant civilian economy helps a country produce advanced weapons, train skillful military personnel, and manage complex military systems. GDP per capita provides a rough but reliable measure of economic and military efficiency. The new index reduces the distortions described by multiplying GDP and GDP per capita before making power comparisons. See Beckley, "The Power of Nations," 14–19; Michael Beckley, "Economic Development and Military Effectiveness," *Journal of Strategic Studies* 33, no. 1 (February 2010), 43–79.

¹⁰ Jeff Desgardins, "The \$86 Trillion World Economy—In One Chart," *World Economic Forum*, September 10, 2019, available at <www.weforum.org/agenda/2019/09/fifteen-countries-represent-three-quarters-total-gdp/>; Rob Smith, "The World's Biggest Economies in 2018," *World Economic Forum*, April 18, 2018, available at <www.weforum.org/agenda/2018/04/the-worlds-biggest-economies-in-2018/>.

¹¹ Desgardins, "The \$86 Trillion World Economy."

¹² "The Benefits of International Trade," U.S. Chamber of Commerce, available at <www.uschamber.com/international/international-policy/benefits-international-trade>.

¹³ Ken Roberts, "In Top 10 U.S. Exports for 2017, 3 Countries Keep Popping Up: China, Canada and Mexico," *Forbes*, February 28, 2018, available at <www.forbes.com/sites/kenroberts/2018/02/28/in-top-10-u-s-exports-for-2017-three-countries-keep-popping-up-china-canada-and-mexico/#3965f39860d3>.

¹⁴ World Bank, "High-Technology Exports (% of Manufactured Exports)," available at <<https://data.worldbank.org/indicator/TX.VAL.TECH.MF.ZS?view=chart>>.

¹⁵ Ken Roberts, "China Trade War Losses at \$84 Billion and Counting—Does New Data Suggest Tide Turning Toward U.S.?" *Forbes*, December 16, 2019, available at <www.forbes.com/sites/kenroberts/2019/12/16/china-trade-war-losses-at-84-billion-and-counting--does-new-data-suggest-tide-turning-toward-us/#35c3b543bd40>. Clearly, the global economic disruption from the 2019–2020 novel coronavirus will eclipse this trade war by quite some margin.

¹⁶ Steve Liesman, "Trade War Losses for the U.S. and China Grow into the Tens of Billions of Dollars," CNBC, November 5, 2019, available at <www.cnbc.com/2019/11/05/trade-losses-for-the-us-china-mount-into-tens-of-billions-of-dollars.html>.

¹⁷ See *Economic and Trade Agreement Between the Government of the United States of America and the Government of the People's Republic of China* (Washington, DC: Office of the U.S. Trade Representative, 2020), available at <https://ustr.gov/sites/default/files/files/agreements/phase%20one%20agreement/Economic_And_Trade_Agreement_Between_The_United_States_And_China_Text.pdf>.

¹⁸ In 2018, U.S. financial markets contributed 7.4 percent (\$1.5 trillion) to nominal U.S. GDP. In context, this was equal to the total size of the Russian economy that year. See "Financial

Services Spotlight: The Financial Services Industry in the United States,” *Select USA*, n.d., available at <<https://www.selectusa.gov/financial-services-industry-united-states/>>.

¹⁹ Unless otherwise specified by individual note, the economic data cited in this paragraph and the prior one are found in the online appendix B of this volume, available at <<https://ndupress.ndu.edu/Contemporary-GPC-Dynamics-Matrix/>>.

²⁰ In 2014, the U.S. Treasury had some 6,000 financial sanctions in place; by late 2018, that number had reached 7,967. See Cathy Gilsinan, “A Boom Time for U.S. Sanctions,” *The Atlantic*, May 3, 2019, available at <www.theatlantic.com/politics/archive/2019/05/why-united-states-uses-sanctions-so-much/588625/>.

²¹ Ibid.

²² *Foreign Holdings of Federal Debt*, RS22331 (Washington, DC: Congressional Research Service, July 26, 2019), available at <<https://fas.org/sgp/crs/misc/RS22331.pdf>>.

²³ Although not without economic pain and political risk, the United States has generated viable plans to reduce national debt and attendant risk to future power. For the most prominent recent one, the so-called Simpson-Bowles Plan, see Kevin Robillard, “Report: New Simpson-Bowles Plan,” *Politico*, February 19, 2013, available at <www.politico.com/story/2013/02/report-new-simpson-bowles-plan-087769>.

²⁴ Mark E. Cancian, “U.S. Military Forces in FY 2020: The Strategic and Budget Context,” Center for Strategic and International Studies (CSIS), September 30, 2019, available at <www.csis.org/analysis/us-military-forces-fy-2020-strategic-and-budget-context?gclid=CjwKCAiAuqHwBRAQEiwAD-zr3W5VXp25g3B_08uUJYHBc5_q9z58pywaOs_SZD5HGS-xRlobyECggoC2dEQAvD_BwE>.

²⁵ All U.S. military data derived from “Chapter 2: Comparative Defence Statistics,” in *The Military Balance* (London: Institute of International Strategic Studies [IISS], 2019).

²⁶ Zachary Morris, “U.S. Drones: Smaller, Less Capable Drones for the Near Future,” *Military Review* (May–June 2018), available at <<https://www.armyupress.army.mil/Journals/Military-Review/English-Edition-Archives/May-June-2018/US-Drones-Smaller-Less-Capable-Drones-for-the-Near-Future/>>.

²⁷ Laura Silver, “U.S. Is Seen as Top Ally in Many Countries—But Others View It as a Threat,” Pew Research Center, December 5, 2019, available at <www.pewresearch.org/fact-tank/2019/12/05/u-s-is-seen-as-a-top-ally-in-many-countries-but-others-view-it-as-a-threat/>.

²⁸ Arms transfers from the United States bring security guarantees, which basically entail diplomatic and military assistance in case of troubles. See *SIPRI Yearbook 2019: Armaments, Disarmament and International Security—Summary* (Stockholm: Stockholm International Peace Research Institute [SIPRI], March 2019), 6–9, available at <www.sipri.org/sites/default/files/2019-06/yb19_summary_eng.pdf>.

²⁹ “U.S. Arms Sales and Defense Trade,” fact sheet, Department of State, May 21, 2019, available at <<https://www.state.gov/u-s-arms-sales-and-defense-trade/>>. Many analysts question the strategic value of military assistance and arms sales programs. For example, see Mara Karlin, “Why Military Assistance Programs Disappoint,” *Foreign Affairs* (November/December 2017), <www.brookings.edu/articles/why-military-assistance-programs-disappoint/>.

³⁰ “Diplomatic Dashboard,” available at <<http://diplodash.pardee.edu/>>.

³¹ For this conceptualization, see John Ikenberry, “State Power and the Institutional Bargain: America’s Ambivalent Economic and Security Multilateralism,” in *U.S. Hegemony and International Organizations: The United States and Multilateral Institutions*, ed. Rosemary Foot, S. Neil MacFarlane, and Michael Mastanduno (Oxford: Oxford University Press, 2003).

³² Dina Smeltz et al., *America Engaged: American Public Opinion and U.S. Foreign Policy* (Chicago: The Chicago Council on Global Affairs, 2018), available at <www.thechicagocouncil.org/sites/default/files/report_ccs18_america-engaged_181002.pdf>.

³³ Stewart M. Patrick, “President Trump and the Future of Global Governance,” Council on Foreign Relations, January 31, 2017, available at <www.cfr.org/blog/president-trump-and-future-global-governance>.

³⁴ Jacob Poushter and Kristen Bialik, “Around the World, Favorability of the U.S. and Confidence in Its President Decline,” Pew Research Center, June 26, 2017, available at <www.pewresearch.org/fact-tank/2017/06/26/around-the-world-favorability-of-u-s-and-confidence-in-its-president-decline/>.

³⁵ See Martha Bayles, “How the World Perceives the New American Dream,” *The Atlantic*, October 10, 2015, available at <www.theatlantic.com/international/archive/2015/10/american-dream-world-diplomacy/410080/>. Of note, the international projection of this “dream” has not been pushed by the U.S. Government since the Cold War. See Thomas G. Mahnken, *Forging the Tools of 21st Century Great Power Competition* (Washington, DC: Center for Strategic and Budgetary Assessments, 2020), 30–31, 37.

³⁶ Lane Crothers, *Globalization and American Popular Culture*, Vol. 4 (New York: Rowman and Littlefield, 2017).

³⁷ Tsedal Neely, “Global Business Speaks English,” *Harvard Business Review* (May 2012), available at <<https://hbr.org/2012/05/global-business-speaks-english>>.

³⁸ “One of America’s Most Vital Exports, Education, Never Goes Abroad, but It Still Faces Threats,” *New York Times*, January 3, 2019, available at <www.nytimes.com/2019/01/03/magazine/one-of-americas-most-vital-exports-education-never-goes-abroad-but-it-still-faces-threats.html>.

³⁹ *China’s National Defense in the New Era* (Beijing: The State Council Information Office of the People’s Republic of China, July 2019), 5–7; Joel Wuthnow, *Chinese Perspectives on the Belt and Road Initiative: Strategic Rationales, Risks and Implications*, China Strategic Perspectives 12 (Washington, DC: NDU Press, October 2017), 9.

⁴⁰ Bayles, “How the World Perceives the New American Dream.”

⁴¹ Ibid.; Mahnken, *Forging the Tools of 21st Century Great Power Competition*, 30–31.

⁴² Mahnken, *Forging the Tools of 21st Century Great Power Competition*, 31.

⁴³ As of early 2020, the leadership of this State Department Bureau had no political appointee, indicating its relative stature in American diplomacy. See Bureau of Global Public Affairs (Web site), Under Secretary for Public Diplomacy and Public Affairs, U.S. Department of State, available at <<https://www.state.gov/bureaus-offices/under-secretary-for-public-diplomacy-and-public-affairs/bureau-of-global-public-affairs/>>.

⁴⁴ Bayles, “How the World Perceives the New American Dream.”

⁴⁵ See Global Engagement Center (Web site), Department of State, available at <<https://www.state.gov/bureaus-offices/under-secretary-for-public-diplomacy-and-public-affairs/global-engagement-center/>>.

⁴⁶ Brandi Vincent, “Laser-Linked Satellites Could Deliver ‘Internet from Space,’” *Nextgov*, December 20, 2019, available at <www.nextgov.com/emerging-tech/2019/12/laser-linked-satellites-could-deliver-internet-space/162009/>.

⁴⁷ *China and the World in the New Era* (Beijing: The State Council Information Office of the People’s Republic of China, September 2019), 4.

⁴⁸ Desgardins, “The \$86 Trillion World Economy”; Rob Smith, “The World’s Biggest Economies in 2018.”

- ⁴⁹ Central Intelligence Agency, "China," *World Factbook*, available at <<https://www.cia.gov/library/publications/the-world-factbook/geos/ch.html>>; Malcolm Scott and Cedric Sam, "Here's How Fast China's Economy Is Catching Up to the U.S.," *Bloomberg*, May 21, 2019, available at <www.bloomberg.com/graphics/2016-us-vs-china-economy/>.
- ⁵⁰ Li Fuxian, "Why Ageing China Won't Overtake the U.S. Economy as the World's Biggest—Now or in the Future," *South China Morning Post* (Hong Kong), March 29, 2019, available at <www.scmp.com/comment/insight-opinion/article/3003524/ageing-china-simply-cannot-overtake-us-economy-worlds>.
- ⁵¹ Yen Nee Lee, "McKinsey Research Finds the World Becoming More Exposed to China—But Not the Reverse," *CNBC*, July 21, 2019, available at <www.cnbc.com/2019/07/15/mckinsey-world-has-become-more-exposed-to-china-but-not-the-reverse.html>.
- ⁵² Jonathan Woetzel et al., *China and the World: Inside the Dynamics of a Changing Relationship* (Washington, DC: McKinsey Global Institute, July 2019), i–ii.
- ⁵³ World Bank, "High-Technology Exports (% of Manufactured Exports)."
- ⁵⁴ Liesman, "Trade War Losses for the U.S. and China Grow into the Tens of Billions of Dollars."
- ⁵⁵ "What's in the U.S.-China Phase 1 Trade Deal," *Reuters*, January 15, 2020, available at <www.reuters.com/article/us-usa-trade-china-details-factbox/whats-in-the-us-china-phase-1-trade-deal-idUSKBN1ZE2IF>.
- ⁵⁶ Robert D. Atkinson and Caleb Foote, *Is China Catching Up to the United States in Innovation?* (Washington, DC: Information Technology and Innovation Foundation, April 2019), available at <<http://www2.itif.org/2019-china-catching-up-innovation.pdf>>.
- ⁵⁷ Unless otherwise specified by individual endnote, the economic data cited in this paragraph is found in the online appendix B of this volume, available at <<https://ndupress.ndu.edu/Contemporary-GPC-Dynamics-Matrix/>>.
- ⁵⁸ Wayne M. Morrison, *China's Economic Rise: History, Trends, Challenges, and Implications for the United States*, RL33534 (Washington, DC: Congressional Research Service, updated June 25, 2019), 16, available at <<https://fas.org/sgp/crs/row/RL33534.pdf>>.
- ⁵⁹ Derek Scissors, *China's Global Investing in 2019: Going Out Goes Small* (Washington, DC: American Enterprise Institute, January 2020), available at <www.aei.org/wp-content/uploads/2020/01/Chinas-global-investment-in-2019-1.pdf>.
- ⁶⁰ U.S. Department of the Treasury, "Major Foreign Holders of Treasury Securities," January 2020, available at <<https://ticdata.treasury.gov/Publish/mfh.txt>>.
- ⁶¹ In developing countries, China buys political influence through development finance; in emerging and medium-sized economies, China uses state-owned enterprises and investment funds to buy what is on sale; and in large, advanced economies, China uses state-backed funds and Chinese private investors to buy shares in large companies with hopes of both realizing economic gain and reducing skepticism regarding the goals and effects of Chinese investments and global influence. See Frank Mouritz, "China's Economic Coercion," in *China's Global Influence: Perspectives and Recommendations*, ed. Scott D. McDonald and Michael C. Burgoyne (Honolulu: Asia-Pacific Center for Security Studies, 2019), 174–189.
- ⁶² See Phillip C. Saunders et al., eds., *Chairman Xi Remakes the PLA: Assessing Chinese Military Reforms* (Washington, DC: NDU Press, 2019).
- ⁶³ See Denis Blasko, "The Biggest Loser in Chinese Military Reforms: The PLA Army," in Saunders et al., *Chairman Xi Remakes the PLA*, 357–365.
- ⁶⁴ Reserve, tank, infantry fighting vehicle, and armored personnel carrier numbers are from *The Military Balance 2020* (London: IISS, 2020), 260.
- ⁶⁵ Ship numbers are from *The Military Balance 2020*, 262–263. Also see Andrew Erickson, "Power vs. Distance: China's Global Maritime Interests and Investments in the Far Seas," in *Strategic Asia 2019: China's Expanding Strategic Ambitions*, ed. Ashley J. Tellis, Alison Szalwinski, and Michael Wills (Washington, DC: National Bureau of Asian Research, 2019), 247–277; and Defense Intelligence Agency (DIA), *China Military Power 2019* (Washington, DC: DIA, 2019), appendix B.
- ⁶⁶ Aircraft numbers are from *The Military Balance 2020*, 265; DIA, *China Military Power 2019*, appendix C; also see Richard P. Hallion, Roger Cliff, and Phillip C. Saunders, eds., *The PLA Air Force: Evolving Concepts, Roles, and Capabilities* (Washington, DC: NDU Press, 2012).
- ⁶⁷ Rick Joe, "China's Growing High-End Military Drone Force," *The Diplomat*, November 27, 2019, available at <<https://thediplomat.com/2019/11/chinas-growing-high-end-military-drone-force/>>.
- ⁶⁸ Intercontinental ballistic missile, intermediate-range ballistic missile, and medium-range ballistic missile numbers are from *The Military Balance 2020*, 259; DIA, *China Military Power 2019*, appendix D; also see David C. Logan, "Making Sense of China's Missile Forces," in Saunders et al., *Chairman Xi Remakes the PLA*, 393–436.
- ⁶⁹ Phillip C. Saunders and Jiunwei Shyy, "China's Military Diplomacy," in McDonald and Burgoyne, *China's Global Influence*, 209–210.
- ⁷⁰ Ibid., 214–216; also see Phillip C. Saunders, "China's Global Military-Security Interactions," in *China and the World*, ed. David Shambaugh (New York: Oxford University Press, 2020), 181–207.
- ⁷¹ "Military Expenditure by Country, in Constant (2017) US\$ m., 1988–2018," SIPRI Military Expenditure Database, available at <www.sipri.org/sites/default/files/Data%20for%20all%20countries%20from%201988%E2%80%932018%20in%20constant%20282017%29%20USD%20%28pdf%29.pdf>.
- ⁷² Pieter D. Wezeman et al., "Trends in International Arms Transfers," SIPRI fact sheet, March 2019, available at <www.sipri.org/publications/2019/sipri-fact-sheets/trends-international-arms-transfers-2018>.
- ⁷³ A combined 61.3 percent of China's conventional weapons sales since 2008 have found their way to Pakistan, Bangladesh, and Myanmar. Other Asian countries have purchased an additional 14 percent of Chinese arms. See "How Dominant Is China in the Global Arms Trade?" *China Power*, available at <<https://chinapower.csis.org/china-global-arms-trade/>>.
- ⁷⁴ *Annual Report to Congress: Military and Security Developments Involving the People's Republic of China 2019* (Washington, DC: Office of the Secretary of Defense, 2019), 27–28, available at <https://media.defense.gov/2019/May/02/2002127082/-1/-1/1/2019_CHINA_MILITARY_POWER_REPORT.pdf>.
- ⁷⁵ China has established one overseas logistics support facility in Djibouti but is unlikely to be able to use it to support combat operations.
- ⁷⁶ *Assessment on U.S. Defense Implications of China's Expanding Global Access* (Washington, DC: Department of Defense, December 2018), available at <<https://media.defense.gov/2019/Jan/14/2002079292/-1/-1/1/EXPANDING-GLOBAL-ACCESS-REPORT-FINAL.PDF>>; see Joel Wuthnow, Phillip C. Saunders, and Ian Burns McCaslin, "PLA Joint Operations in the Far Seas," in *Going Global? The People's Navy in a Time of Strategic Transformation*, ed. Ryan Martinson and Andrew Erickson (Newport, RI: Naval War College Press, forthcoming).
- ⁷⁷ *The Military Balance 2020*, 234.

⁷⁸ See Katherine Morton, "China's Global Governance Interactions," in Shambaugh, *China and the World*, 156–180.

⁷⁹ "Diplomatic Dashboard."

⁸⁰ Since 2017, China has sought to portray itself as a responsible alternative, highlighting its support for international agreements, including the Paris Agreement and the Joint Comprehensive Plan of Action (Iran nuclear deal), and affirming its commitment to the United Nations (UN). President Xi Jinping's January 2017 speech at the UN's European headquarters in Geneva rejected isolationism and trade protectionism and placed Beijing at the center of UN-overseen international affairs. Of note, human rights advocates and other nongovernmental organizations were barred from attending Xi's speech. See Colum Lynch and Elias Groll, "As U.S. Retreats from World Organizations, China Steps In to Fill the Void," *Foreign Policy*, October 6, 2017, available at <<https://foreignpolicy.com/2017/10/06/as-u-s-retreats-from-world-organizations-china-steps-in-the-fill-the-void/>>.

⁸¹ Jonathan G. Odom, "Understanding China's Legal Gamesmanship in the Rules-Based Global Order," in McDonald and Burgoyne, *China's Global Influence*, 190–206.

⁸² See John Ikenberry and Darren J. Lim, *China's Emerging Institutional Statecraft* (Washington, DC: The Brookings Institution, April 2017), available at <www.brookings.edu/wp-content/uploads/2017/04/chinas-emerging-institutional-statecraft.pdf>.

⁸³ Ritika Passi, "China's BRI in Doldrums: Multilateralism to the Rescue?" *ORF Online*, October 10, 2018, available at <www.orfonline.org/expert-speak/bri-china-doldrums-multilateralism-rescue-44893/>.

⁸⁴ Benjamin Wermund, "Chinese-Funded Institutes on U.S. College Campuses Condemned in Senate Report," *Politico*, February 27, 2019, available at <www.politico.com/story/2019/02/27/china-college-confucius-institutes-1221768>.

⁸⁵ Laura Silver, Kat Devlin, and Christine Huang, "People Around the Globe Are Divided in Their Opinions of China," Pew Research Center, December 5, 2019, available at <www.pewresearch.org/fact-tank/2019/12/05/people-around-the-globe-are-divided-in-their-opinions-of-china/>.

⁸⁶ "How Are Global Views on China Trending?" CSIS, December 20, 2019, available at <<https://chinapower.csis.org/global-views/>>.

⁸⁷ Shaun Breslin, "China's Global Cultural Interactions," in Shambaugh, *China and the World*, 137–155; Institute of International Education, "Project Atlas: China," available at <www.iie.org/en/Research-and-Insights/Project-Atlas/Explore-Data/China>; George Gao, "Why Is China So . . . Uncool?" *Foreign Policy*, March 8, 2017, available at <<https://foreignpolicy.com/2017/03/08/why-is-china-so-uncool-soft-power-beijing-censorship-generation-gap/>>.

⁸⁸ Anthony H. Cordesman, *China and the United States: Cooperation, Competition, and/or Conflict* (Washington, DC: CSIS, October 2019), available at <www.csis.org/analysis/china-and-united-states-cooperation-competition-andor-conflict>.

⁸⁹ On whether or how the novel coronavirus could play to China's advantage, see Kurt M. Campbell and Rush Doshi, "The Coronavirus Could Reshape Global Order," *Foreign Affairs*, March 18, 2020, available at <www.foreignaffairs.com/articles/china/2020-03-18/coronavirus-could-reshape-global-order>; Mira Rapp-Hooper, "China, America, and the International Order After the Pandemic," *War on the Rocks*, March 24, 2020, available at <<https://warontherocks.com/2020/03/china-america-and-the-international-order-after-the-pandemic/>>.

⁹⁰ Michael Kofman, "Raiding and International Brigandry: Russia's Strategy for Great Power Competition," *War on the Rocks*, June 14, 2018, available at <<https://warontherocks.com/2018/06/raiding-and-international-brigandry-russias-strategy-for-great-power-competition/>>.

⁹¹ For variations on this conclusion of Russian overachievement relative to limited strategy and a strapped resource base, see Christopher S. Chivvis, *Understanding Russian 'Hybrid Warfare' and What Can Be Done About It* (Santa Monica, CA: RAND, 2017), available at <www.rand.org/content/dam/rand/pubs/testimonies/CT400/CT468/RAND_CT468.pdf>; Nadezda Arbatova, "Three Faces of Russia's Neo-Eurasianism," *Survival* 61, no. 6 (December 2019–January 2020), 7–24; Eugene Rumer and Richard Sokolsky, *Thirty Years of U.S. Policy Toward Russia: Can the Vicious Circle Be Broken?* (Washington, DC: Carnegie Endowment for International Peace, June 20, 2019), available at <<https://carnegieendowment.org/2019/06/20/thirty-years-of-u-s-policy-toward-russia-can-vicious-circle-be-broken-pub-79323>>.

⁹² Revisions to the U.S.–Russia Strategic Arms Reduction Treaty limited the number of strategically deployed U.S. and Russian nuclear warheads to no more than 1,600 by February 2018. In April 2020, Russia had an estimated 1,572 strategically deployed weapons and the United States had 1,600. See Hans M. Kristensen and Matt Korda, *Status of World Nuclear Forces* (Washington, DC: Federation of American Scientists, April 2020), available at <<https://fas.org/issues/nuclear-weapons/status-world-nuclear-forces/>>.

⁹³ "Strategic Rocket Forces," *Russian Strategic Nuclear Forces*, June 2017, available at <<http://russianforces.org/missiles/>>; and DIA, *Russia Military Power: Building a Military to Support Great Power Aspirations* (Washington, DC: DIA, 2016), 47.

⁹⁴ Hans M. Kristensen and Matt Korda, "Russian Nuclear Forces, 2019," *Bulletin of the Atomic Scientists* 75, no. 2 (March 2019), available at <www.tandfonline.com/doi/full/10.1080/00963340.2.2019.1580891>.

⁹⁵ Michael Kofman, "Emerging Russian Weapons: Welcome to the 2020s (Part 1—Kinzhalt, Sarmat, 4202)," *Russian Military Analysis*, March 4, 2018, available at <<https://russianmilitaryanalysis.wordpress.com/2018/03/04/emerging-russian-weapons-welcome-to-the-2020s-part-1-kinzhalt-sarmat-4202/>>; Matthew Bodner, "Russia Releases Video of Its Modernized Ballistic Missile Defense System," *Defense News*, February 20, 2018, available at <<https://www.defensenews.com/land/2018/02/20/russia-releases-video-of-its-modernized-ballistic-missile-defense-system/>>.

⁹⁶ Military numbers derived from "Chapter 2: Comparative Defence Statistics," in *The Military Balance*.

⁹⁷ Mark Epizkopos, "Is Russia's Only Aircraft Carrier Doomed?" *The National Interest*, November 16, 2019, available at <<https://nationalinterest.org/blog/buzz/russias-only-aircraft-carrier-doomed-97246>>; "Russia's T-14 Armata Tank Is Amazing (But There Is a Big Problem)," *The National Interest*, April 6, 2019, available at <<https://nationalinterest.org/blog/buzz/russias-t-14-armata-tank-amazing-there-big-problem-51022>>; "Russia's Most Advanced SU-57 Fighter Jet Suffers First Crash," Reuters, December 24, 2019, available at <www.reuters.com/article/us-russia-airplane-crash/russias-most-advanced-su-57-fighter-jet-suffers-first-crash-idUSKBN1YS164>.

⁹⁸ Kofman, "Emerging Russian Weapons—Part 1."

⁹⁹ Russia's military expenditure in 2018 was \$44 billion (the same size as France, despite an economy the size of Spain), only 20 percent of that spent by China and just 6 percent of that spent by the United States. See Richard Connolly, *Russian Military Expenditure in Comparative Perspective: A Purchasing Power Parity Estimate*, CNA Occasional Paper (Alexandria, VA: CNA, October 2019), 7–8.

¹⁰⁰ Richard Connolly and Michael Kofman, "Russian Defense Expenditure and Military Modernization: How Much Does the Military Spend?" video, CSIS Discussion Event, December 4, 2019, available at <www.csis.org/events/russian-defense-expenditure-and-military-modernization>.

¹⁰¹ David Oliver, "Russia's Rapid UAV Expansion," *Armada International*, March 22, 2019, available at <<https://armadainternational.com/2019/03/russias-rapid-uav-expansion/>>.

Nikolai Novichkov, "Russia Develops Multilayered C-UAS Systems," *Janes*, November 29, 2019, available at <www.janes.com/article/92922/russia-develops-multilayered-c-uas-system>.

¹⁰² The most well-known of the Russian mercenary units is the Wagner Group, but in recent years, competitors to it have sprung up in Russia, including those known as "Patriot" and "Shield." Detailed analyses emphasize that these companies are not monolithic paramilitary entities but highly personalized, riven with factionalism, and subservient to the personal relationships and transactional allowances by their leaders with Vladimir Putin. See Neil Hauer, "The Rise and Fall of a Russian Mercenary Army," *Foreign Policy*, October 6, 2019, available at <<https://foreignpolicy.com/2019/10/06/rise-fall-russian-private-army-wagner-syrian-civil-war/>>; Nathaniel Reynolds, *Putin's Not-So-Secret Mercenaries: Patronage, Geopolitics, and the Wagner Group* (Washington, DC: Carnegie Endowment for International Peace, July 8, 2019), available at <<https://carnegieendowment.org/2019/07/08/putin-s-not-so-secret-mercenaries-patronage-geopolitics-and-wagner-group-pub-79442>>.

¹⁰³ Nikiolai Sokov, *Russia's New Conventional Capability: Implications for Eurasia and Beyond*, PONARS Eurasia Policy Memorandum No. 472 (Washington, DC: Elliott School of International Affairs, May 2017), available at <www.ponarseurasia.org/memo/russias-new-conventional-capability-implications-eurasia-and-beyond>.

¹⁰⁴ "Russia," *OECD World Profile*, 2018, available at <<https://oec.world/en/profile/country/rus/>>.

¹⁰⁵ See Daniel Workman, "Russia's Top Trading Partners," *World's Top Exports*, November 30, 2019, available at <www.worldstopexports.com/russias-top-import-partners/>.

¹⁰⁶ The World Bank, "High-Technology Exports (% of Manufactured Exports)—United States, Russian Federation," available at <<https://data.worldbank.org/indicator/TX.VAL.TECH.MF.ZS?locations=US-RU>>.

¹⁰⁷ Joseph S. Nye, Jr., "How to Deal with a Declining Russia," *The Strategist*, November 6, 2019, available at <www.aspistrategist.org.au/how-to-deal-with-a-declining-russia/>.

¹⁰⁸ Unless otherwise specified by individual note, the economic data cited in this paragraph are found in the online appendix B of this volume, available at <<https://ndupress.ndu.edu/Contemporary-GPC-Dynamics-Matrix/>>.

¹⁰⁹ Clark Letterman, "Image of Putin, Russia Suffers Internationally," Pew Research Center, December 6, 2018, available at <www.pewresearch.org/global/2018/12/06/image-of-putin-russia-suffers-internationally/>.

¹¹⁰ *Ibid.*

¹¹¹ Angela Dewan, "Russia vs. U.S.: Where Is Life Better?" CNN, March 14, 2018, available at <www.cnn.com/2018/03/14/europe/russia-us-election-compare-intl/index.html>.

¹¹² "Diplomatic Dashboard."

¹¹³ "Russia Wants IAEA to Recognize Crimea's Legal Status," *World Nuclear News*, June 10, 2015, available at <www.world-nuclear-news.org/NP-Russia-wants-IAEA-to-recognise-Crimeas-legal-status-10061501.html>.

¹¹⁴ Ann M. Simmons, "Russian Is Nominated to Lead Interpol, Alarming Western Officials," *Wall Street Journal*, November 20, 2018, available at <www.wsj.com/articles/russian-is-nominated-to-lead-interpol-alarming-western-officials-1542723109>.

¹¹⁵ Elizabeth Flock, "After a Week of Russian Propaganda, I Was Questioning Everything," PBS, May 2, 2018, available at <www.pbs.org/newshour/arts/after-a-week-of-russian-propaganda-i-was-questioning-everything>.

¹¹⁶ Robert Elliott, "How Russia Spreads Disinformation via RT Is More Nuanced Than We Realize," *The Guardian*, July 26, 2019,

available at <www.theguardian.com/commentisfree/2019/jul/26/russia-disinformation-rt-nuanced-online-ofcom-fine>.

¹¹⁷ Alina Polyakova, "What the Mueller Report Tells Us About Russian Influence Operations," *Lawfare*, April 19, 2019, available at <www.lawfareblog.com/what-mueller-report-tells-us-about-russian-influence-operations>; Robert S. Mueller III, *Report on the Investigation into Russian Interference in the 2016 Presidential Election—Volume I* (Washington, DC: Department of Justice, March 2019), available at <www.justice.gov/storage/report.pdf>.

¹¹⁸ Patrick Tucker, "Russia Will Test Its Ability to Disconnect from the Internet," *Defense One*, October 24, 2019, available at <www.defenseone.com/technology/2019/10/russia-will-test-its-ability-disconnect-internet/160861/>.

¹¹⁹ Isabel Togoh, "From Russia with Love? Putin's Medical Supplies Gift to Coronavirus-Hit Italy Raises Questions," *Forbes*, March 26, 2020, available at <www.forbes.com/sites/isabeltogoh/2020/03/26/from-russia-with-love-putins-medical-supplies-gift-to-coronavirus-hit-italy-raises-questions/#313feeed4a47>.

¹²⁰ The Composite Index of National Capability (CINC) is a statistical measure of gross (not net) national power used in the University of Michigan Correlates of War (COW) Project that began in 1963. CINC uses an average of percentages in six different components that represent economic, demographic, and military strength to generate a "CINC number" for relative power. Its six components are total population of a country ratio, urban population of a country ratio, iron and steel production of a country ratio, primary energy consumption ratio, military expenditure ratio, and military personnel ratio. The COW Project is available at <<http://www.correlatesofwar.org/data-sets/national-material-capabilities>>.

¹²¹ For the historical comparative numbers, see Beckley, "The Power of Nations," 22–25.

¹²² This conclusion aligns with that of Wright, *All Measures Short of War*, 166. However, it is modified to focus on U.S. predominance in a "protracted" military clash—the kind that neither Russia nor China would willingly fight today with the United States. The third insight indicates that China or Russia might today prevail in a military clash proximate to their own territory and where America's full power advantages might never be brought to bear.

¹²³ Here, "unacceptable" relates to the phrase "immutable and untenable" in the fifth major insight of chapter 2's historic review of Great Power competition: "The most reliable indicator of when a clash (war) will erupt is when one or both sides recognize a shift in the relative alignment of economic and military power that is perceived to be immutable and untenable."

¹²⁴ Anthony H. Cordesman, *China and the U.S.: Cooperation and/or Conflict: An Experimental Assessment—Part Two: China's Emerging Economic Power*, with Max Molog (Washington, DC: CSIS, October 2019), 3, available at <https://csis-prod.s3.amazonaws.com/s3fs-public/publication/191001_China_Grand_Strategy_Part%202_.pdf>.

¹²⁵ See especially, Robert Gilpin, *U.S. Power and the Multinational Corporation: The Political Economy of Foreign Direct Investment* (New York: Basic Books, 1975), 65–68; Robert Gilpin, *War and Change in World Politics* (Cambridge, UK: Cambridge University Press, 1981), 156–185.

¹²⁶ William R. Thompson, "Long Waves, Technological Innovation, and Relative Decline," *International Organization* 44, no. 2 (Spring 1990), 201–233; Daniel Drezner, "State Structure, Technological Leadership, and the Maintenance of Hegemony," *Review of International Studies* 27, no. 1 (January 2001), 3–25.

Chapter 4

Contemporary Great Power Technological Competitive Factors in the Fourth Industrial Revolution

By T.X. Hammes and Diane DiEuliis

The convergence of new technologies is creating a fourth industrial revolution that will transform almost every aspect of 21st-century life. Even as the new technologies generate much greater wealth, the revolution will reshape trade patterns as it returns both manufacturing and services to home markets. The United States is particularly well positioned to take advantage of these changes—but only if it revises its immigration policies to attract and retain the best minds from around the world. China is also well positioned, but it must overcome increasing distrust of its government. Russia is dealing with an ongoing demographic crisis even as foreign and domestic investors have lost trust in its potential for growth.

As with earlier iterations, the fourth industrial revolution is developing from the convergence of multiple technologies. Despite the shorthand sometimes used to identify the three previous revolutions—steam, electricity, digital—none was driven by a single technology.¹ Each needed a merging of numerous new technologies alongside relevant economic, social, and political change before it could evolve.

In *The Fourth Industrial Revolution*, Klaus Schwab states that the unifying of new technologies—“artificial intelligence [AI], robotics, internet of things, autonomous vehicles, 3D printing, nanotechnology, biotechnology, material science, energy storage, and quantum computing, to name a few”²—is going to revolutionize almost every aspect of life, mostly in a positive way. But when discussing the fourth industrial revolution’s impact on international security, he was concerned that the technologies will provide much greater power to nonstate actors and create instability in many regions. Therefore, the fourth industrial revolution is a critical factor in the emerging era of Great Power competition (GPC) headlined in 2020 by the United States, China, and Russia.

In this chapter, we do not attempt to deal with all the technologies driving the fourth industrial revolution; instead, we focus only on those that will most directly impact GPC, economic potential, and international trade in the short term: robotics, AI, 3D printing, energy, and biotechnology. The chapter also considers two important factors essential to

U.S. efforts to exploit new technologies: current immigration policy and research and development (R&D) investments.

Although we discuss each of these technologies individually, it is the combination of them in new ways that will revolutionize the global economy. For instance, the first industrial revolution is often referred to as steam driven, but steam had to be combined with improved steel production, new manufacturing techniques, the telegraph, and other technologies both new and old to create the railroads that revolutionized trade and production. While space limitations forced us to select only a few of the technologies driving today's revolution, readers should keep in mind that the others are essential to enabling the chosen technologies (for instance, advances in material science are essential to advances in the chosen technologies).

Robotics

Robotics covers a vast field that will fundamentally alter how humans do things—from in the deep sea to outer space. We start by examining industrial robots and then move on to collaborative robots.

General Motors purchased its first industrial robots in 1961. Since then, robots have steadily evolved with improvements in degrees of freedom, range of motion, strength, speed, reliability, accuracy, and repeatability. Industrial robots' increasing flexibility and effectiveness are resulting in rapid and steady growth of sales. Sales averaged 115,000 per year from 2005 to 2008 but increased to 422,000 by 2018.³

Sales keep growing because robots dramatically improve productivity. The U.S. steel industry offers a prime example. From 1962 to 2005, it shed 75 percent of its workforce, but its shipment of steel products in 2005 equaled that of the early 1960s. Robots increased output per worker by a factor of five. Despite the massive shedding of jobs in the industry, steel manufacturing has been one of the fastest growing industries in the past three decades, behind only computer software and equipment.⁴

New sensors and improved mechanics mean industrial robots are becoming cost effective even in high-tech industries. China's Changying Precision Technology Company has automated its mobile phone production lines and cut factory personnel from 650 to just 60 while increasing productivity by 250 percent.⁵ Although perhaps an extreme example, this type of streamlining is driving chief executive officers to explore how industrial robots can improve their companies' competitiveness. Robots tasked with routine computer electronics assembly cost about \$7.25 per hour to operate, and the purchase cost of robots is expected to come down 22 percent by 2025, even as these machines become easier to integrate into current operations.⁶ In 2020, Chinese labor costs \$6.50 per hour.⁷

Products with potential for even greater growth—and hence greater impact on many aspects of life—are collaborative robots, or *cobots*. Unlike industrial robots, which must be separated from humans for safety, cobots are specifically designed to work in collaboration with people. And unlike industrial robots, which are expensive and have limited flexibility, cobots are cheap and flexible. Designed to be mobile, they are easy to be moved to different locations and assigned new tasks. They are already working in homes, laboratories, hospitals, nursing homes, warehouses, farms, and distributions centers to tend, test, carry, assemble, package, pick, place, count, secure, and inspect.⁸ They are even being used as

exoskeletons to provide strength and protection to people. And unlike most large industrial robots, cobots are relatively easy to upgrade.⁹

In 2019, basic cobots cost about \$24,000.¹⁰ Assuming a 40-hour workweek and 3-year lifespan, this works out to approximately \$4 an hour—well below U.S. labor costs and competitive with wages in emerging economies. And if a plant is running with three shifts, the hourly cost is about \$1.35—well below even most emerging economy wages. Of course, robots need no medical, retirement, or leave benefits. Just as important, cobots are easily programmable. In fact, a “non-technical person can teach [a cobot] what to do through arm movement and simple button presses, and [a cobot] can master a new task in half an hour or so. There is also little assembly or setup required.”¹¹ Moreover, the low prices, minimal technical support required, and flexibility mean that many of the 6 million small and medium enterprises worldwide will buy cobots. Goldman Sachs notes that today’s versions have a payback period as short as 6 months.¹² ABI Research predicts global revenue from cobots will “grow at an annual rate of 49.9 percent between 2016 and 2025 compared to 12.1 percent for industrial robots.”¹³

Even as costs of cobots come down quickly, their capabilities are growing at an exceptional rate. In 2012, the Defense Advanced Research Projects Agency awarded a \$1.3-million grant to develop a robot that could sew. Inevitably nicknamed “sewbots,” these systems are mastering the complex task of sewing—and thus are threatening to disrupt the global clothing industry.

The field of robotics seems to have reached the knee of its exponential growth curve—the point at which the curve turns vertical. From heavy-duty industrial robots to small personal robots, the range of capabilities is expanding rapidly even as cost drops and ease of use improves. By eliminating any labor-cost advantage to production in low-cost regions, robots are, and will continue to be, central to the return of production to home markets. Even as industry masters the mechanical aspects of robotics, key advances are being made in the field of AI, which will further enhance the advantages robots provide in production.

Artificial Intelligence

AI will play the role electricity did in the second industrial revolution, when the world moved from steam to electrical power; it will be an integral part of every new technology. From task-specific AI that autonomously executes a job such as controlling a mining truck in an open pit to more powerful AI that assists with complex planning, analysis, and decisionmaking, this technology will be essential. Unfortunately, the term *artificial intelligence* causes considerable confusion. Much of the current discussion concerns artificial general intelligence (AGI), which itself boasts a range of different definitions. In fact, there is a great deal of disagreement in the AI research community as to when or even if AGI is achievable. However, this chapter focuses on how limited, or task-specific, AI is rapidly improving productivity. Task-specific AI is essentially a machine operating with a set of guidelines to accomplish chores. Although such a system will provide great practical capabilities in its specific field, it will not be capable of fully independent operation.

Task-specific AI-driven robots are executing manual tasks in virtually every field of human endeavor, but AI is not limited to physical work. In January 2017, Japanese insurance company Fukoku Mutual Life Insurance replaced 34 insurance claim workers with

software from IBM Watson Explorer. The software scans hospital records and other documents to determine insurance payouts, factoring in the specific injuries, each patient's medical history, and the procedures administered. Fukoku Mutual reportedly spent \$1.7 million (200 million yen) to install the AI system, and now pays \$128,000 per year for maintenance. By using the software, the firm saves roughly \$1.1 million per year on employee salaries—meaning it hopes to see a return on the investment in less than 2 years.¹⁴

This kind of breakthrough acts as a major incentive for other firms to follow suit—simply to remain competitive in their industries. The result has been a steady return of service industries to their home countries as AI takes over many of the back-office tasks, such as computer programming, bookkeeping, handling insurance adjustments, and manning call centers, that used to be contracted to firms in India or the Philippines.

Another form of AI can develop optimal designs for a wide variety of structures. Autodesk's "Dreamcatcher system allows designers to input specific design objectives, including functional requirements, material type, manufacturing method, performance criteria, and cost restrictions."¹⁵ But the improved designs often cannot be produced using conventional manufacturing techniques,¹⁶ thus the need for another emerging technology: 3D printing.

3D Printing

The first 3D-printing patent was granted in 1986. Slow speeds and uneven finishes meant that 3D printing was initially used mainly to produce prototypes and a limited number of unique low-volume products. But in the past decade, it has transformed from an industry focused on prototyping to one engaged in creating a wide range of products. In addition to increasing the speed of printing and refining the finishes, a great deal of effort has gone into expanding the number of materials that can be used. More exciting, 3D manufacturing is rapidly developing entirely new materials: "up to 140 different digital materials can be realized from combining the existing primary materials in different ways."¹⁷ 3D printing is quite literally changing what can be made.

Meanwhile, researchers and 3D-printing companies continue to pursue both versatility and speed. Multimaterial printers were one of the big steps in this effort. Instead of printing a series of pieces that then must be assembled, the multimaterial machine forms the assembly in one go. As businesses learn to use these multimaterial printers, the range of products they will be able to print will expand exponentially. Furthermore, 3D printing's efficiency is unmatched: Its material wastage is near zero.

The range of products—from medical devices and aircraft parts to buildings and bridges—and the order of magnitude increase in the speed of printing are already challenging traditional manufacturing.¹⁸ Better printing speeds mean that 3D printing has moved beyond prototyping and high-value parts. In April 2016, Carbon3D released a commercial printer that was 100 times faster than existing printers. Such improvements allowed 3D printing to capture 20 percent of the global plastics manufacturing market in 2016.¹⁹ Not to be outdone, metal printers have combined high speed and low cost to make them a system of choice even for mass production of small parts. The fact that key patents are expiring soon will further accelerate enhancements in printer capabilities and capacities.²⁰

3D printing opens up the possibility of a totally different supply chain, one that runs with lower costs and a smaller carbon footprint. The materials and energy required for manufacturers to create new parts will be harnessed into electronic design files that can be printed on demand anywhere in the world. In these new transformative supply chains, many spare parts may not even need to exist, which could translate into huge savings on warehousing costs.²¹

3D printing is revolutionizing manufacturing in many industries. The ability to change each product by changing the software means the era of mass customization and local production is on us. No longer will parts have to be shipped across oceans and then trucked to the user; they will be printed on site. 3D printing is clearly on the path to causing major disruptions in global supply chains.

Energy

Energy, in the form of petroleum, natural gas, and coal, has been a key component of global trade for the past half-century, but the influence of petroleum on both global trade and national security may be waning. Rapid advances in energy technology are changing the world's energy markets—and in many cases moving energy sources from overseas companies to locally produced oil, gas, and renewables. In its *World Energy Outlook 2017*, the International Energy Agency stated, “Four large-scale shifts in the global energy system set the scene for the World Energy Outlook 2017: the rapid deployment and falling costs of clean energy technologies, the growing electrification of energy, the shift to a more services-oriented economy and a cleaner energy mix in China, and the resilience of shale gas and tight oil in the United States.”²²

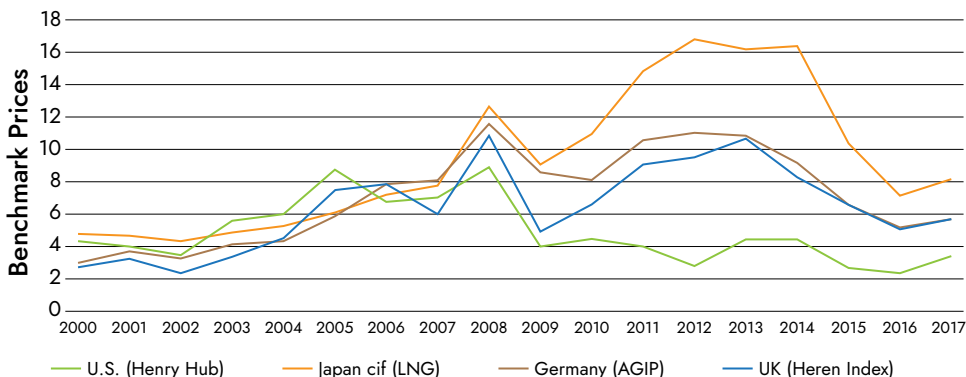
Gas and Oil

The U.S. fracking revolution is driving a global increase in demand for natural gas, even as oil use may have peaked. In November 2016, the *Wall Street Journal* ran a front-page article reporting that global oil producers such as Royal Dutch Shell and even state-owned Saudi Aramco anticipate that the world has reached peak oil usage and are preparing for a future decline in demand.²³

U.S. shale oil production (fracking) has been the fastest growing source of oil globally. In 2012, the U.S. Energy Information Administration (EIA) estimated that shale oil production could reach 2.8 million barrels per day (bpd) by 2035; it did so by 2013.²⁴ In late November 2014, the EIA predicted the United States would become the world's largest oil producer by 2020.²⁵ This milestone was reached by June 2014.²⁶

Perhaps the most powerful aspect of U.S. shale oil operations is the speed with which they can be closed or reopened in response to demand. From October 2016 to January 2017, U.S. crude production increased 500,000 bpd in response to the West Texas Intermediate (WTI) price increasing from \$45 to \$50 a barrel.²⁷

The combination of the novel coronavirus pandemic and the short-term Saudi-Russian “oil wars” resulted in a collapse in the price of oil to less than \$20 a barrel. Even the April 2020 tentative pact between Russia and Saudi Arabia did not result in major price increases. In mid-April 2020, *The Economist* predicted that oil prices will remain low as the oil industry restructures post-pandemic.²⁸ Although price spikes will still occur during times of crisis, U.S.

Figure. Global Natural Gas Benchmark Prices in USD per MMBtu

shale oil supplies may well ensure that the WTI price will not exceed \$60 a barrel (in 2018 dollars). Even the September 14, 2019, attack on Saudi oil facilities resulted in only a few days' increase to the mid-\$60s before WTI returned to the mid-\$50s.

Natural Gas

Fracking has also created a natural gas boom in the United States. As recently as 2007, U.S. companies were racing to build liquid natural gas (LNG) import facilities. The demand for natural gas was growing quickly, and U.S. production was falling. In October 2005, the Henry Hub (U.S.) spot price rose to \$13.42 per million British thermal units (MBTU). Then the fracking revolution occurred, which forced a massive drop in price to \$1.93 per MBTU by December 2015—a reduction of 85 percent (see figure). Suddenly companies were applying for permits to turn their LNG import facilities into LNG export facilities.

As late as 2005, those U.S. industries that made heavy use of natural gas for their products (for instance, the petrochemical industry) or to generate energy for production were at a disadvantage in global competition. U.S. companies paid significantly more for this vital input. By 2008, fracking had completely reversed the situation, and the cost advantage to U.S. manufacturers has only increased since then.²⁹ The result has been heavy investment in new U.S. petrochemical plants that use natural gas as a feedstock, with 310 new projects under way that will satisfy most U.S. demand and increase U.S. exports from \$17 billion in 2016 to \$110 billion by 2027.³⁰ This unplanned advantage for America's chemical subsector will have spillover benefits as the expansion stimulates local development of directly related service and manufacturing businesses.

Renewables and New Transmission Lines

In its 2017 report, the International Energy Agency (IEA) noted new renewables installation increased to almost two-thirds of all newly installed electrical energy production.³¹ By 2021, global generation from renewables should be “equivalent to the total electricity generation of the United States and the European Union put together today.”³² In its 2019 *Global Energy Perspective*, McKinsey & Company predicted that, even as energy consumption doubles by 2050, renewables will generate over 50 percent of the world's electricity by 2035.³³

In 2006, the IEA predicted that by 2013 global installed photovoltaic capacity would be 20 gigawatts; it was actually 7 times larger—140 gigawatts—and increased to 227 gigawatts by 2015. These statistics reflect only utility-scale generating capacity and thus do not account for the growing private installation of renewable energy. Both developing and developed countries are seeing huge growth in local production and grids.³⁴ In fact, Bangladesh is the world's largest market for home-based solar power systems.

During the first three quarters of 2016, 15 percent of additional residential and nonresidential solar generating capacity was privately installed in the United States.³⁵ Business is also growing in providing renewable energy to major corporations.³⁶ However, two major problems with renewables persist. First, they are inherently intermittent. Wind intensity varies in unpredictable ways, and solar fails every night and during bad weather. Even hydropower is subject to reduction during periods of drought. Second, the best solar exposure or steadiest winds are often far from the places where people live. Thus, renewable energy and power transmission must be thought of as an integrated problem. Around the world, nations and private business are installing renewables along with long-line high-voltage transmission systems.

Like Europe and China, the United States needs to invest in transmission lines if it is to maximize the use of renewables. Fortunately, it has a successful model—Texas. Because its power grid is contained within the state, Texas overcame the various forms of political resistance and built transmission lines from its windy western plains to its energy-hungry eastern cities. At times, wind provides 40 percent of the state's power needs. Other regions have great potential for renewables—for example, the Great Plains and offshore for wind and the Southwest for solar. In March 2017, Xcel announced plans to install 800 megawatts of new wind generation capability in the Dakotas and Minnesota.³⁷ Delayed by regulators since 2005, a 3,000-megawatt line is finally being built to take Wyoming's wind energy to southern California.³⁸ Other investors are seeking to link wind and solar energy to the southeastern and eastern United States.

Impact of Batteries on Energy Sources

Whether users are major power companies looking for a way to store power to feed back into the grid or individual homeowners seeking to get off of it, batteries can offer them an alternative to fossil fuel backups. This is another field in which many researchers—commercial, government, and academic—are pursuing a variety of possibilities. Major battery technology breakthroughs in 2019 include a battery that can fully charge in 10 minutes, thermal-energy devices that can store 1.2 megawatt-hours and can be hooked in series to create almost unlimited storage, and new processes that could double the storage capacity of lithium-ion batteries.³⁹

Commercial power companies need massive storage capability to take over the “peaker” function now performed by natural gas-burning plants. These plants come online only during periods of peak load to prevent brownouts or even blackouts. Driven partly by California's Public Utilities Commission, Southern California Edison plans to install a 100-megawatt storage battery by 2020. Moving much more quickly, Elon Musk combined Tesla Motors and SolarCity to create a new way of supplying power. In July 2017, he signed a contract to provide 100 megawatts of storage in Australia and had the system running

and highly profitable by January 2018.⁴⁰ By October, it was on “track to make back a third of its construction costs in its first year of operation.”⁴¹ Tesla has contracted to increase the capacity of this system by 50 percent in 2020.⁴²

The convergence of fracking, renewables, energy grids, and batteries means that more and more energy production will be local or regional. Renewables can be moved vast distances via transmission lines, so they can tie a region together. But there will not be a global market for renewables. Unlike oil, propane, coal, and LNG, it is impractical and unnecessary to move renewable energy across oceans. Thus, unlike the gas/oil market, which contributes to globalization, the renewables market will contribute to regionalization—and even to localization—as more and more businesses and homes take advantage of better battery capacity to move off-grid.

Emerging Biotechnology

Just as information technology and the Internet have transformed society, business, government, and warfare since the late 20th century, emerging biotechnology will shape the global landscape for the next several decades. The world is entering a new era of biotechnology, highlighted by the advancing ease with which genomes can be engineered for specific purposes. Synthetic biology and associated genome-editing tools will be essential for addressing the global challenge of resource scarcity and environmental sustainability, while providing unprecedented advances in public health and medicine. The expanding U.S. biotechnology industry, including a wide range of startup companies, along with larger scale corporations, is already exploring capabilities for manufacturing high-value products, creating what is now referred to as the bioeconomy. Products of the bioeconomy include the creation of biology-based commodities, fuels, textiles, and consumer goods—all of which are proposed to be produced on innovative biomanufacturing platforms. Because the locus of this economic innovation is in industry, and particularly startups, government funding does not primarily drive the bioeconomic trajectory, making government just one of many actors shaping the field. In order to establish and maintain global leadership in biotechnology, the United States requires a holistic national approach that supports innovation and growth in the bioeconomy, establishes strategic priorities, and ensures responsible use.

Emerging biotechnology will have important implications for the Department of Defense (DOD), and internal DOD leadership has acknowledged this by including biotechnology as one of its 11 modernization priorities.⁴³ Moreover, DOD intends to create a community of interest in biotechnology to coordinate biotechnology R&D across the Armed Forces. Just some of the many promising advances for defense could include innovative body armor designed from spider silk, jet fuels or runway material produced from algae, living plant-based sensors, and flame-resistant coatings. Many other advances, such as those related to skin or gut microbiotics, could benefit the health or performance of warfighters directly; however, DOD is not driving biotechnology innovation, and many challenges exist to the most beneficial incorporation of biotechnology into DOD requirements. Outside of traditional force health protections and the development of medical countermeasures, what advantages over adversaries could biotechnology provide? What unique DOD challenges and problems are best met with biotechnological solutions versus

other emerging technologies? These questions have yet to be addressed as industry continues to innovate.

At the same time, new dual-use technologies for defense will present significant challenges to biodefense, in addition to the ethical and moral dilemmas they have already created; the capabilities that drive the U.S. bioeconomy are the same tools that could allow for the creation of bioweapons (see also chapter 8). Most recently, the National Academy of Sciences published a biodefense consensus report, providing a framework for assessing those capabilities that are the most concerning to the DOD warfighter—namely, the creation of viruses from scratch, the modification of harmful bacteriological pathogens, and the development of harmful chemicals through biomanufacture.⁴⁴ DOD will need to address these issues in ways that do not stifle the technology's advancement or America's competitiveness in the global bioeconomy.

Enabling Factors

As noted in the introduction, two factors should enable U.S. technological advances. The first, investment, is on a positive trend; the second, immigration policy, is having a powerful negative effect.

Obviously, in a period of swift technological change, robust investment in R&D is essential to leveraging those shifts. Fortunately, despite all the hype about the Made in China 2025 plan to dominate 10 key emerging technologies, the United States still invests significantly more in R&D than does China. Unfortunately, from 2003 to 2016, U.S. Government spending in nondefense R&D was essentially flat. Since then, government investment has increased sharply.⁴⁵ In 2018, the United States invested 2.84 percent of its gross domestic product (GDP) in R&D. China invested only 1.97 percent of a significantly smaller GDP.⁴⁶ The key question is whether the United States will sustain this investment in the face of rapidly increasing debt-servicing costs and continued deficit spending.

The bad news is that current U.S. immigration policy is having a major negative impact on America's progress toward a fourth industrial revolution economy. While immigration policy is a hot political issue, one element that is not usually associated with immigration is the intellectual nature of the fourth industrial revolution. Exploiting the revolution requires large numbers of smart, skilled, and educated people. Innovation at the top of the scale requires advanced education in science, technology, engineering, and math (STEM) skills. The 330 million people of the United States cannot hope to stay ahead of the 7 billion people in the rest of the world. In the past, the United States has had great success by encouraging the best STEM students from around the world to come to America for an education and then stay to work.

In 2017, foreign nationals in the United States accounted for 81 percent of the full-time graduate students in electrical engineering and petroleum engineering; 79 percent in computer science; 75 percent in industrial engineering; 69 percent in statistics; 63 percent in mechanical engineering, and economics; 59 percent in civil engineering; and 57 percent in chemical engineering.⁴⁷

The United States is generally recognized as having the finest university system in the world, and it attracts large numbers of the best foreign students. This is a major advantage, but what really counted was the fact that the number of foreign STEM graduates who chose

to remain in the United States to work increased by 400 percent from 2008 to 2016. Essentially, the United States was attracting and keeping some of the finest minds in the world.

The policy of encouraging immigrants to stay paid off. Despite representing only 13 percent of the U.S. population, immigrants start more than 25 percent of the new businesses in the United States. More than 20 percent of the chief executive officers of the 2014 Inc. top 500 business are immigrants.⁴⁸ And 55 percent of the new companies worth more than \$1 billion have at least 1 immigrant as a founding member.⁴⁹

Then, in 2017, two things happened that dramatically reversed the flow of foreign students into U.S. universities. First, the U.S. Government made it much harder for students to obtain visas or to be certain they could renew them year to year. At the same time, the Trump administration restricted the number of graduates who can remain in the country to work. The result has been a major downturn in the number of foreign students enrolled in U.S. universities. Other countries, having seen the success that the United States was having, have started their own aggressive recruiting programs to attract foreign students—and current U.S. policy is only assisting those foreign programs in attracting top students. It makes no sense for a foreign student to invest heavily in a U.S. education when the visa might not be renewed and thus the investment will not pay off in a degree; even if he or she succeeds in getting a degree, the U.S. job market, which needs STEM graduates, will be closed. It makes much more sense for these students to go to school in Canada, Australia, or the United Kingdom. Canada and Australia, in particular, are encouraging overseas students.

Not only do current U.S. policies deprive the United States of intelligent, productive students and potential citizens, they also hurt America universities. Overseas students pay much higher tuitions than do U.S. citizens and so, in effect, subsidize their education. Universities across the country are having to cut STEM programs due to the reduction in funds flowing in from overseas.⁵⁰ In short, U.S. immigration policies in 2020 are directly responsible for reducing the flow of the people America needs to thrive in the fourth industrial revolution. By failing to effectively distinguish between types of potential immigrants, the United States is excluding a great deal of talent. In sharp contrast, forward-thinking governments elsewhere are enticing the best and brightest from around the world to move to their nations. Current immigration policies are damaging American prospects for success in a rapidly changing global economy.

Conclusion

The convergence of the aforementioned technologies will change societies in ways that are hard to imagine. It is already clear these technologies will change what, how, and where we manufacture commodities. Most important, manufacturing will be located near the market, which means the trend of onshoring manufacturing to America will continue. New technologies are also returning service-industry jobs to the markets they aid, and renewable energy is inherently regional rather than global. Environmental movements will reinforce these trends by pushing to reduce the impact of manufacturing and agriculture on the environment.

The fourth industrial revolution is shifting trade networks from the global to regional and even local levels. The United States may well be the nation best positioned to benefit from this shift. The U.S. economy already derives 84 percent of its GDP from the United

States, Canada, and Mexico. Although the supply chains are deeply embedded in Asia, U.S. manufacturing and services have been in the process of moving production back to the United States for years. The United States benefits from effective rule of law, strong protection of intellectual property, the largest market in the world, an exceptional university system, heavy investment in R&D, and a pervasive entrepreneurial spirit. These attributes will allow the country to thrive in this era of rapid technological change.

Until 2017, foreign investors thought so too. Foreign direct investment (FDI), particularly in manufacturing, was running at record levels in 2015 and 2016. Unfortunately, the uncertainty introduced by tariffs has reduced FDI into the United States by over 50 percent. However, the long-term advantages are still present, and FDI flows should recover once the United States establishes and sticks to stable trade agreements. America must also ensure its immigration and R&D policies do not hold it back. Economically, the United States has distinct advantages over both China and Russia as the fourth industrial revolution begins to reshape our world. The one key weakness is the gridlock in America's current political systems. Failure to adjust our laws and regulations to the new reality risks squandering those advantages.

China could also benefit greatly from the fourth industrial revolution. The Chinese Communist Party specifically developed its Made in China 2025 plan to take advantage of the new technologies. The party selected 10 priority sectors to subsidize, which included robotics, green energy, artificial intelligence, biosciences, and materials. Essentially, China is investing heavily in each of the technologies discussed in this chapter; it is also working hard to shift its economy from export based to consumption based, to decrease its reliance on exports. It needs to do so because it is already suffering from businesses leaving to protect intellectual property rights and increasing labor costs. However, even as these efforts progress, China will have to deal with the dramatic reduction in labor required in its industries and its rapidly aging, and hence less productive, population. It must do so even as its economy slows significantly and pushback to its Belt and Road Initiative grows. While the Chinese government has once again turned to stimulus spending, a key question is whether the cumulative public and private debt is manageable.

On the positive side, China's working population has been declining for the past couple of years and will continue to do so. Thus, it will need fewer jobs. At the same time, China has an enormous talent pool due to the sheer size of its population. Furthermore, Chinese universities have steadily improved in global rankings. Most important for China's future is the fact that Asian trade is also regionalizing and China is central to the fastest growing region in the world. If the Chinese Communist Party's use of centralized management can mitigate these challenges, China too can be a big winner in the fourth industrial revolution.

In contrast, Russia is not well positioned to benefit. Transitioning to advanced manufacturing requires major investment, and Russia currently suffers from a lack of investor confidence. The Institute of International Finance ranked Russia "last among 23 emerging economies in terms of 'real' FDI."⁵¹ Russian businesses have not been particularly innovative, nor has the country created an environment that encourages foreign innovators to establish businesses there. Compounding its problems, Russia's economy remains dependent on exporting energy, which generated 60 percent of its GDP.⁵² Today, technological improvements in various fields as well as global warming concerns are driving a worldwide

shift to alternative energy. Russia also faces major human capital issues due to its demography and low-quality university system. No Russian institution rated in the top 250 of *U.S. News & World Report's* university rankings, and only 2 made it into the top 400.⁵³ Finally, as a kleptocracy, Russia does a poor job of allocating capital to the industries that benefit most from the convergence of these new technologies.

Notes

- ¹ Klaus Schwab, *The Fourth Industrial Revolution* (Cologne, Switzerland: World Economic Forum, 2016), 6–7.
- ² *Ibid.*, 1.
- ³ “Global Industrial Robot Sales Reach Record Level in 2018,” Xinhua, September 19, 2019, available at <www.china.org.cn/business/2019-09/19/content_75222098.htm>.
- ⁴ Allan Collard-Wexler and Jan De Loecker, “Reallocation and Technology: Evidence from the U.S. Steel Industry,” *American Economic Review* 105, no. 1 (January 2015), 131–171.
- ⁵ Mihai Andrei, “Chinese Factory Replaces 90% of Human Workers with Robots: Production Rises by 250%, Defects Drop by 80%,” *ZME Science*, February 3, 2017, available at <www.zmescience.com/other/economics/china-factory-robots-03022017>.
- ⁶ “Industrial Robots in United States to Cut Labor Costs,” *Technavio* (blog), June 1, 2016, available at <<https://blog.technavio.com/blog/industrial-robots-united-states-cut-labor-costs-21-2020>>.
- ⁷ Erin Duffin, “Manufacturing Labor Costs per Hour for China, Vietnam, Mexico from 2016 to 2020 (in U.S. Dollars),” *Statista*, August 9, 2019, available at <www.statista.com/statistics/744071/manufacturing-labor-costs-per-hour-china-vietnam-mexico/>.
- ⁸ Frank Tobe, “Why Co-Bots Will be a Huge Innovation and Growth Driver for Robotics Industry,” *IEEE Spectrum*, December 30, 2016, available at <<http://spectrum.ieee.org/automation/robotics/industrial-robots/collaborative-robots-innovation-growth-driver>>.
- ⁹ Bruce Geiselman, “Upgrades to Collaborative Robots Bring Cost Savings,” *Plastics Machinery*, May 31, 2017, available at <www.plasticsmachinerymagazine.com/blow-molding/article/13002309>.
- ¹⁰ Ben Halder, “How China’s ‘Cobot’ Revolution Could Transform Automation,” *OZY*, March 24, 2019, available at <www.ozy.com/fast-forward/how-chinas-cobot-revolution-could-transform-automation/93044>.
- ¹¹ “Lightweight Cobots Take on Heavy-Duty Jobs—Even Welding,” *Universal Robots*, November 9, 2017, available at <<https://blog.universal-robots.com/lightweight-cobots-take-on-heavy-duty-jobs-even-welding>>.
- ¹² “Goldman Sachs Summary on Cobots,” *Robotonomics*, July 26, 2016, available at <<https://robotonomics.wordpress.com/tag/cobots/>>.
- ¹³ Halder, “How China’s ‘Cobot’ Revolution Could Transform Automation.”
- ¹⁴ David Gershgorin, “Japanese White-Collar Workers Are Already Being Replaced by Artificial Intelligence,” *Quartz*, January 2, 2017, available at <<http://qz.com/875491/japanese-white-collar-workers-are-already-being-replaced-by-artificial-intelligence>>.
- ¹⁵ Carlos E. Perez, “The Alien Style of Deep Learning Generative Design,” *Intuition Machine*, December 24, 2016, available at <<https://medium.com/intuitionmachine/the-alien-look-of-deep-learning-generative-design-5c5f871f7d10#hp9oywmsv>>.
- ¹⁶ *Ibid.*
- ¹⁷ “The Free Beginners Guide,” *3D Printing Industry*, available at <<https://3dprintingindustry.com/3d-printing-basics-free-beginners-guide/materials>>.
- ¹⁸ Bridget O’Neal, “ORNL Researchers Improve 3D Printing at the Nanoscale with Simulation Guided Process,” *3D Print*, August 16, 2016, available at <<https://3dprint.com/146101/ornl-3d-printing-nanoscale>>.
- ¹⁹ Olivier Scalabre, “The Next Manufacturing Revolution Is Here,” *TED Talks*, video, 12:19, May 2016, available at <www.ted.com/talks/olivier_scalabre_the_next_manufacturing_revolution_is_here?language=en>.
- ²⁰ Jelor Gallego, “A Host of Soon-to-Be-Expired Patents Are Set to Revolutionize 3D Printing,” *Futurism*, May 17, 2016, available at <<https://futurism.com/expiring-patents-set-to-improve-3d-world/>>.
- ²¹ Mary Ann Yule, “A Manufacturing Renaissance: How 3D Printing Can Bring Back Local,” *Digital*, December 5, 2016, available at <<https://techvibes.com/2016/12/05/d-3d-printing-mary-ann-yule-president-of-hp-canada>>.
- ²² *World Energy Outlook 2017* (Paris: International Energy Agency, 2017).
- ²³ Sarah Kent and Brian Spegele, “Oil Firms Anticipate Day of Reckoning,” *Wall Street Journal*, November 27, 2016, 1.
- ²⁴ Catherine Ngai, “Shale Boom Confounds Forecasts as U.S. Set to Pass Russia, Saudi Arabia,” *Reuters*, July 9, 2014, available at <www.reuters.com/article/us-usa-oil-shale-forecasts-analysis-idUSKBN0FE0CJ20140709>.
- ²⁵ Isaac Arnsdorf, Wael Mahdi, and Grant Smith, “Saudis Go Back to the Future to Take on U.S. Shale Rivals,” *Bloomberg*, November 4, 2014, available at <www.bloomberg.com/news/articles/2014-11-04/saudis-go-back-to-the-future-to-take-on-u-s-shale-rivals>.
- ²⁶ U.S. Energy Information Administration, “Hydraulic Fracturing Accounts for About Half of Current U.S. Crude Oil Production,” March 15, 2016, available at <www.eia.gov/todayinenergy/detail.php?id=25372>.
- ²⁷ Francine Lacqua and Javier Blas, “IEA Sees Significant Gains in U.S. Shale Oil as Prices Rise,” *Bloomberg*, January 18, 2017, available at <www.bloomberg.com/news/articles/2017-01-18/iea-sees-significant-increase-in-u-s-shale-oil-as-prices-rise>.
- ²⁸ “A View of the Future,” *The Economist*, April 18–24, 2020, 8.
- ²⁹ “Natural Gas Prices,” *BP*, available at <www.bp.com/en/global/corporate/energy-economics/statistical-review-of-world-energy/natural-gas.html#natural-gas-prices>.
- ³⁰ Christopher Matthews, “Abundant Shale Oil Byproducts Reshape the Plastics Industry,” *Australian* (New South Wales), June 27, 2017, 28.
- ³¹ “Renewables 2017: A New Era for Solar Power,” International Energy Agency, video, 1:38, October 4, 2017, available at <www.youtube.com/watch?v=mfNE_6ae6B4>.
- ³² *Medium-Term Renewable Energy Market Report 2016* (Paris: International Energy Agency, 2016).

- ³³ *Global Energy Perspective 2019: Reference Case* (New York: McKinsey & Company, January 2019), 3.
- ³⁴ *Renewables 2016: Global Status Report* (Paris: REN21 Secretariat, 2016), available at <www.ren21.net/wp-content/uploads/2019/05/REN21_GSR2016_FullReport_en_11.pdf>.
- ³⁵ Solar Energy Industries Association, “Solar Market Insight Report 2016 Q4,” available at <www.seia.org/research-resources/solar-market-insight-report-2016-q4>.
- ³⁶ Michael Holder, “Solar Panel Researchers Investigate Powering Trains by Bypassing Grid,” *The Guardian*, January 10, 2017, available at <www.theguardian.com/environment/2017/jan/10/solar-panel-research-power-trains-imperial-college-london-1010>.
- ³⁷ Mike Hughlett, “Xcel Unveils New Phase of Wind Power Construction, with Huge Plant in South Dakota,” *Star Tribune* (Minneapolis), March 16, 2017, available at <www.startribune.com/xcel-unveils-new-phase-of-wind-power-construction-with-huge-plant-in-south-dakota/416340794>.
- ³⁸ James Temple, “How to Get Wyoming Wind to California, and Cut 80% of U.S. Carbon Emissions,” *MIT Technology Review*, December 28, 2017, available at <www.technologyreview.com/s/609766/how-to-get-wyoming-wind-to-california-and-cut-80-of-us-carbon-emissions/>.
- ³⁹ Nick Lavars, “The Biggest Battery Breakthroughs of 2019,” *New Atlas*, December 8, 2019, available at <<https://newatlas.com/science/battery-breakthroughs-2019/>>.
- ⁴⁰ Fred Lambert, “Tesla’s Giant Battery in Australia Made Around \$1 Million in Just a Few Days,” *Electrek*, January 23, 2018, available at <<https://electrek.co/2018/01/23/tesla-giant-battery-australia-1-million/>>.
- ⁴¹ Calla Wahlquist, “South Australia’s Tesla Battery on Track to Make Back a Third of Cost in a Year,” *The Guardian*, September 27, 2018, available at <www.theguardian.com/technology/2018/sep/27/south-australias-tesla-battery-on-track-to-make-back-a-third-of-cost-in-a-year>.
- ⁴² Lavars, “The Biggest Battery Breakthroughs of 2019.”
- ⁴³ Under Secretary of Defense for Research and Engineering, “Modernization Priorities,” Department of Defense, available at <<https://www.cto.mil/modernization-priorities/>>.
- ⁴⁴ National Academy of Sciences, *Biodefense in an Age of Synthetic Biology* (Washington, DC: The National Academies Press, 2018).
- ⁴⁵ Science News Staff, “Trump, Congress Approve Largest U.S. Research Spending Increase in a Decade,” *Science*, March 23, 2018, available at <www.sciencemag.org/news/2018/03/updated-us-spending-deal-contains-largest-research-spending-increase-decade>.
- ⁴⁶ Sintia Radu, “The Big Spenders in Research and Development,” *U.S. News & World Report*, November 9, 2018, available at <www.usnews.com/news/best-countries/articles/2018-11-09/these-countries-are-the-top-spenders-on-research-and-development>.
- ⁴⁷ National Foundation for American Policy, “The Importance of International Students to American Science and Engineering,” October 2017, 3, available at <<http://nfap.com/wp-content/uploads/2017/10/The-Importance-of-International-Students.NFAP-Policy-Brief.October-20171.pdf>>.
- ⁴⁸ Adam Bluestein, “The Most Entrepreneurial Group in America Wasn’t Born in America,” *Inc.*, February 2015, available at <www.inc.com/magazine/201502/adam-bluestein/the-most-entrepreneurial-group-in-america-wasnt-born-in-america.html>.
- ⁴⁹ Stuart Anderson, “55% of America’s Billion Dollar Startups Have an Immigrant Founder,” *Forbes*, October 25, 2018, available at <www.forbes.com/sites/stuartanderson/2018/10/25/55-of-americas-billion-dollar-startups-have-immigrant-founder/#766218e548ee>.
- ⁵⁰ Nick Wingfield, “The Disappearing American Grad Student,” *New York Times*, November 3, 2017, available at <www.nytimes.com/2017/11/03/education/edlife/american-graduate-student-stem.html>.
- ⁵¹ Lloyds Bank, “Foreign Direct Investment (FDI) in Russia,” available at <www.lloydsbanktrade.com/en/market-potential/russia/investment>.
- ⁵² “Russia’s Natural Resources Valued at 60% of GDP,” *Moscow Times*, March 14, 2019, available at <www.themoscowtimes.com/2019/03/14/russias-natural-resources-valued-at-60-of-gdp-a64800>.
- ⁵³ “Best Global Universities in Russia,” *U.S. News & World Report*, available at <www.usnews.com/education/best-global-universities/russia>.

Part II
Warfighting, Innovation, and
Technology in a New Era of
Great Power Competition

Chapter 5

Key Technologies and the Revolution of Small, Smart, and Cheap in the Future of Warfare

By T.X. Hammes

The convergence of fourth industrial revolution technologies is making possible smaller, smarter, and cheaper weapons systems that will challenge the few and exquisite systems of today's militaries. Based on land, sea, and air, these small, smart, and cheap weapons will fundamentally change the character of war and may come to dominate Great Power conflicts.

The U.S. Civil War demonstrated the impact of the Industrial Revolution on warfare. The advent of mass production, railroads, telegraphs, and steam warships meant mass became a dominant, if not *the* dominant, element on the battlefield. World War I reinforced this fact. Even with the advent of maneuver warfare, World War II remained largely a competition of mass. In the 1980s, American leaders turned to precision weapons to defeat the Warsaw Pact's mass.¹ Thankfully, we never found out if precision weapons could defeat the Warsaw Pact, but the technologies developed to do so proved devastating in the 1991 Gulf War and Operation *Iraqi Freedom*. Unfortunately, these weapons and the platforms to deliver them have become progressively more expensive. The result is the United States entered the 21st century building fewer but ever-more-capable weapons systems.

Twenty years into the new century, the United States, China, and Russia continue to pursue high-end systems such as fifth-generation fighters, heavy bombers, and aircraft carriers. Yet by reducing the price of precision and advanced manufacturing, the fourth industrial revolution is creating a new generation of smaller, smarter, and cheaper weapons that challenges these weapons systems. In short, we are moving to an era of mass *precision*. A key question in the unfolding era of Great Power competition is which nation can most rapidly and effectively adapt to this revolution.

Technological Convergence

Like all previous industrial revolutions, this one is not based on a single technology but the convergence of an array of technologies that are rapidly maturing. This chapter studies only those most directly affecting military capabilities in the next 5 years (2020–2025).

Nanotechnology has made long-range precision systems affordable. Hypersonic weapons, although expensive, are creating yet another challenge. Furthermore, since each of these technologies is new, improvements in capability are happening fast while, with the exception of hypersonic weapons, costing a fraction of what it costs to improve current weapons systems technology.

Nanotechnology

Nanotechnology is science, engineering, and technology conducted at the nanoscale, which is about 1 to 100 nanometers. To put this scale into perspective, a sheet of newsprint is about 100,000 nanos thick.² Nanotechnology is applicable to chemistry, biology, physics, and materials science to dramatically increase a range of properties such as strength, combustibility, and conductivity.

The most immediate military application of nanotechnology is nano-explosives. Open-source reporting indicates nano-explosives have demonstrated an explosive power 10 times that of conventional explosives.³ Nano-research continues, which means much greater conventional explosive power is available to states and, inevitably, nonstate actors as well. Increased explosive capacity can give new, smaller weapons the same effectiveness of much larger older ones.

The second area of interest is nanomaterials. Carbon nanotubes (created from graphene) are over 400 times stronger than steel and are exceptional conductors.⁴ They are being used to reduce the weight needed for structural strength in many products—electronics, vehicles, medical devices, and even water purification. When techniques under development eventually make graphene available cheaply in large quantities, they will free designers to improve performance across the spectrum, from bicycles to high-end military applications.

Nanomaterials also dramatically increase battery storage capacity, and graphene-based supercapacitors can recharge almost instantly.⁵ Nanomaterials are making smart, adaptive fabric that changes to match the background and nano-based paints, making the acquisition of electronically or visually coated objects challenging.⁶ Potentially, some of the most valuable nanoproducts are radiation-hardened circuits to protect electronics from microwave weapons. In short, improvements in materials, circuitry, energy storage, and explosives based on nanotechnology will lead to major increases in range, payload, and stealth for various vehicles.

Drones

Exponential improvements in drone capabilities are being driven by commercial interests. Autonomous air, ground, and maritime drones are available to anyone who wishes to purchase them for surveillance/reconnaissance, communications, logistics, and even strike.

Surveillance/Reconnaissance Drones. Long-endurance (30+ hours), military, remotely piloted unmanned aerial vehicles such as the Global Hawk are expensive (\$130 million) and require extensive support structures to maximize their effectiveness—up to 171 personnel for each orbit.⁷ Today, commercial demand for long-endurance surveillance systems is resulting in solutions that are much less expensive and require only a few personnel. Defiant Labs' DX-3 autonomous vertical takeoff and landing (VTOL) drone has a mission range of 900 miles and carries light detection and ranging technology or integrated optical and ther-

mal cameras with high-resolution zoom.⁸ Aeroovel's autonomous VTOL Flexrotor provides long-term (32 hours) and long-range (1,800 miles) observation. It carries visual and infrared video and still cameras as well as synthetic aperture radar.⁹ Each costs about \$200,000.

Clearly, the rapid advances in commercial, autonomous surveillance drones and machine interpretation is lowering the cost of operating long-range, multispectral, and radar surveillance missions.

Communications Relay Drones. Long-endurance or tethered drones can provide temporary nodes either to extend the range of existing systems or to provide rapid replacement for a node that has sustained damage.

Strike Drones. Predator-type remotely piloted military strike drones have received extensive coverage over the past two decades. Like a traditional airplane, it is a runway takeoff and runway landing platform. But the important advances are taking place in autonomous, VTOL drones. At the high end, for \$2 million or 1/45th the cost of an F-35, the Kratos XQ-222 drone features a 3,000-mile range (over two times the F-35) with a 600-pound payload.¹⁰ The cost advantage actually increases with time. With its vertical takeoff and recovery system, the XQ-222 has minimal monthly training costs, needs no airfield, and requires no training pipeline for pilots and maintainers, no pilot bonuses, no retirement costs, no health care, and so forth. The U.S. Air Force is testing a version it has designated the XQ-58A. Unfortunately, the XQ-222's presence at the Paris Air Show in 2017 indicates America will not be the only nation with this capability.

Strike drone technology is not limited to million-dollar systems, however. The United States has shipped over 4,000 of its \$15,000 Switchblade drones to Afghanistan since 2011. Measuring 2 feet long and 3 inches wide, and weighing only 6 pounds, it can fly for approximately 6.2 miles and provide live video to the operator. Because it is armed, operators can use it to "suicide" into any targets they find.¹¹ Moreover, major states do not have a monopoly on strike missions. Iran demonstrated its capabilities in the strike on the Saudi Arabian Abqaiq oil facility in September 2019. Tehran also provided weapons for Yemeni cruise missile attacks on U.S. warships. Even before Iran's activities, the so-called Islamic State conducted over 80 drone attacks between October 2016 and January 2017 that resulted in about a dozen dead and 50 injured enemy soldiers.¹²

Commercial firms are testing autonomous VTOL drones that are GPS-independent, hardened against electronic interference, range hundreds of miles, deliver with an accuracy of 1 meter, and are inexpensive enough to be disposable.¹³

Maritime Drones. The commercial shipping industry is working on an array of unmanned surface vessels (USVs), from North Sea ferries to electric coastal cargo vessels to full-size container ships and tankers. By 2018, the U.S. Navy was developing its own USVs to detect and strike enemy combatants, clear a mined strait within hours using a swarm of vessels, trail an adversary's submarines, and guard a critical infrastructure.¹⁴

Imaginatively employed, unmanned underwater vehicles (UUVs) could be a relatively inexpensive substitute/augmentation for a submarine force. Both the U.S. and Chinese navies are pursuing numerous autonomous maritime platforms.

Ground Drones. In Iraq, Shia militia employed four armed robots to fight the Islamic State.¹⁵ Russia deployed its Uran-9 remotely operated minitank to Syria with mixed results.¹⁶ This is not surprising. All revolutionary systems encounter major problems during

initial deployments. Despite early setbacks, many nations continue to develop unmanned ground vehicles.

In short, both military and commercial drone usage—on the ground, at sea, in the air—is exploding. The global smart commercial drone market is projected to reach \$179 billion by 2025,¹⁷ and there appear to be few limits to the planned uses.

Counter-Drone Efforts. Since 2002, Defense Advanced Research Projects Agency (DARPA) has sponsored the annual Black Dart live-fire exercise to explore ideas for defeating drones. Since then, anti-drone technology has taken many different paths: kinetic kill, directed energy (lasers and electromagnetic pulse), software attacks, and electronic and GPS jamming.

Lasers, software attacks, and electromagnetic pulse initially appeared to have the most potential impact on swarms. Unfortunately, each approach has significant limitations. Certain atmospheric conditions (dust, haze, water vapor) as well as reflective paints and ablative coatings can significantly reduce the effectiveness of lasers. Software attacks require the enemy to cooperate by leaving a path into his system to insert the software. Electromagnetic weapons can be defeated by hardening a drone's electronics. During the Cold War, mission-critical electronics were protected from the potential electromagnetic pulse of a nuclear detonation by enclosing the electronics within Faraday cages.¹⁸ Today, radiation-hardened electronics can be placed into a system during its manufacture.¹⁹

In a new approach, DARPA is exploring the use of drone swarms to *defeat* drone swarms. It sponsored a 2017 competition among teams from the U.S. Service academies to see which could develop the most effective software and tactics for one swarm to defeat another. The teams then flew mixed swarms of 25 drones against each other with remarkable success.²⁰ This concept has not been tested against larger swarms, and no attempt has been made to operationalize the concept. The fact remains, however, that current civil and military systems cannot defeat swarms of autonomous drones.

Artificial Intelligence

Artificial intelligence (AI) is a wide-ranging, dynamic field. Chapter 6 of this volume addresses a number of AI implications for wider defense and security. In this chapter, we address two AI areas of particular importance in the rapid evolution of small, smart, and cheap weapons: GPS-independent navigation and target identification. At the cheap end, a University of Pennsylvania quadcopter “uses a smartphone for autonomous flight, employing only onboard hardware and vision algorithms—no GPS is involved.”²¹ At the expensive end, DARPA is developing precise navigation using variations in Earth's magnetic field.²² Both systems are immune to GPS jamming or spoofing—a growing challenge from Russia and inevitably China.²³ These systems can get a drone to the target area but they cannot ensure it can hit a specific target.

Thus, the second key element for truly autonomous drone strike operations is accurate target identification. Many researchers are working on limited AI that will provide accurate identification from onboard sensors.²⁴ At the low end, an early 2019 online journal rated seven commercial drones that can identify and follow a person.²⁵ These drones can certainly identify an aircraft on a parking apron. At the high end, the Air Force Research Lab is using a neuromorphic chip to identify military and civilian vehicles in radar-generated aerial imagery.²⁶

Swarms

Truly autonomous drones will require no human intervention to execute missions, thus they will be able to operate in large numbers. This requires they be inexpensive. In 2014, a University of Virginia team 3D printed a drone in a single day, then added a small electric motor, 2 batteries, and an Android phone for guidance to produce an \$800 autonomous drone with a range of approximately 30 miles.²⁷ Today, a factory with 1,000 Carbon3D printers could print 100,000 such drone bodies daily. The limitation is now the assembly and shipment of the finished products. Both processes can be automated with robots.

Launching thousands of drones will be challenging, but by using standard shipping containers set up like multiple-launch rocket systems, numerous drones can be moved and employed quickly. Sheer numbers will overwhelm any current anti-air system.

Small states and insurgents are not only operating but also manufacturing military drones. Azerbaijan builds a licensed version of Israel's Orbiter 1K.²⁸ In 2016, the Polish army contracted to buy 1,000 Polish-manufactured combat drones annually at a price of about \$7,000 each.²⁹ They have high-explosive, high-explosive anti-tank, and fuel-air explosive (also known as thermobaric) warheads.

Small Warheads Technology

As of 2020, design is clearly mastering the first two challenges to drone swarms: autonomous navigation and target identification. The last challenge—the payload limitation of small drones—can be overcome by three separate approaches. The first and least technically challenging approach is “bringing the detonator.” The second approach, the use of explosively formed penetrators (EFPs) as warheads, requires more technical expertise. The third approach uses swarms and counts on the cumulative damage of dozens of small warheads to accomplish the mission.

“Bringing the detonator” uses the drone to deliver a small initiating charge to the much larger supply of explosive material provided by thin-skinned targets such as ammunition dumps, parked aircraft, fuel trucks, or rocket launchers. A few ounces of explosives delivered to the right point on these targets will initiate secondary explosions that completely destroy them. Russian separatists have repeatedly used small drones to drop simple thermite grenades on Ukrainian ammunition dumps. The resulting secondary explosions have destroyed hundreds of thousands of *tons* of ammunition.³⁰

To penetrate nonexplosive targets such as supply trucks or light armored vehicles, one can use the second approach, EFPs. A properly built thumb-sized EFP weighing only 3 ounces can penetrate up to one-half inch of steel. Increasing the size of the EFP to only a few pounds allows it to destroy even well-armored vehicles.

The combination of drones, AI, and advanced manufacturing means there could be thousands of mobile, smart, active hunters spread widely over tomorrow's battlespaces.

Space

Even as governments and large firms squeeze more capability out of expensive satellites, cube satellites—or CubeSats—measuring about 4 inches on a side are creating cheaper alternatives.³¹ Costing under \$125,000, CubeSats are key to a new industry that sells space as a service. Using a network of CubeSats, Planet, a commercial venture, has achieved the

elusive goal of taking a medium-resolution photo of every place on Earth every day.³² Viasat is building a network that will allow customers to view images and video from the ever-increasing network of observation satellites in near real time.³³

Major powers have lost their monopoly on the use of space. “Cheap space” means that in an era of disruptive technologies and Great Power competition, all military planners must assume almost any enemy can see their forces whenever they are moving.

Hypersonic Weapons

Hypersonic weapons, also known as hypervelocity weapons, travel between 5 and 25 times the speed of sound.³⁴ There are three distinct approaches to achieving hypersonic weapons—boost-glide rockets, cruise missiles, and artillery projectiles. Each approach is rapidly maturing and will dramatically change the character of conflict. Most are not projected to be operational in large numbers within the next 5 years.

Boost-glide rocket systems require a ballistic missile to attain speed and altitude before they maneuver in the manner that gives them their name. When operational, they will provide maneuverable, long-range precision with devastating destructive power.³⁵ Both Russia and China seem to be developing boost-glide as strategic weapons with a focus on carrying nuclear warheads. The United States hopes to have a boost-glide system operational by 2028.³⁶

In contrast, hypersonic cruise missiles have the potential to be much cheaper and launched by a variety of ground, sea, and air platforms but are much smaller and have limited range. China’s DF-17 hypersonic glide vehicle will have a range of about 1,200 miles and is nearing operational readiness.³⁷ Russia’s Tsirkon hypersonic cruise missile, with a range of 250 to 600 miles against land and sea targets, will be operational by 2023. It can be fired by surface ships and submarines.³⁸ The U.S. Air Force is working on the AGM-183 air-launched rapid-response weapon, and the U.S. Army is developing a long-range hypersonic weapon that it plans to transition to a program of record in 2024.³⁹ Both provide specific threats to fixed facilities such as airfields and seaports.

The final technology, hypersonic artillery shells, is cheaper but with much less range—currently only about 50 miles.⁴⁰ The U.S. Army has set a goal of an artillery piece with a 1,000-mile range but has no projected date for fielding such a system.⁴¹

In sum, these advances in nanotechnology—drones, AI, swarms, space, and hypersonics—will revolutionize combat in all domains of warfare in the emerging era of Great Power competition and change the tactical and operational context of future conflicts.

Tactical Impact

The ascent of small, smart, cheap technologies will have major tactical impacts on each individual domain of combat: ground, sea, air, space, cyber, and electromagnetic—and more importantly on cross-domain operations.

Ground

When we combine simple drones, additive manufacturing, autonomous navigation, and target identification, ground forces may face thousands or even tens of thousands of autonomous aerial drones. Furthermore, if packaged in standard commercial containers or trucks for movement within the battlespace, they will be difficult to identify or preempt.

Combined with mines, GRAMM (guided rockets, artillery, mortars, and missiles) fires, and other precision weapons, the emerging family of drones can create a deep zone denied to movement by either side.

Today, ground forces are in the first stage of the historically demonstrated process where a new technology starts out assisting the old. Then, as the technology improves, it becomes a partner until it finally replaces the old system (assistant-to-partner-to-replacement process). Ground forces are working out how to employ weapons to assist their current forces in combat. However, with the rapid improvements in cruise missiles and drones, division commanders may soon control weapons with a 1,000-mile range and be full partners with manned aircraft. Moreover, joint doctrine will have to adjust if ground forces must soon control weapons that exceed the range of fighter bombers.

The final step—replacement—will require careful consideration of which combat functions can be allocated to autonomous drones. The AI limitations of such autonomous systems indicate they will be better at conventional conflict than unconventional, where determining human intention is much more challenging. This area requires extensive wargaming, computer simulation, field experiments, and air-ground coordination.

Sea

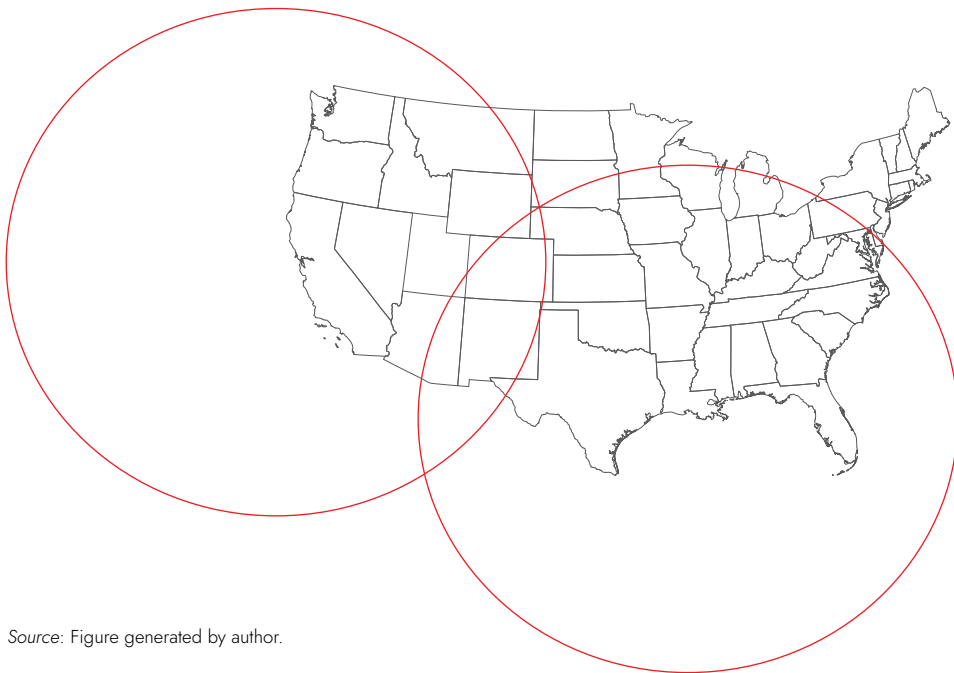
At sea, it will soon be possible to pit numerous relatively cheap drones as well as cruise and ballistic missiles against the few but exquisite platforms of the U.S. Navy. Studies by the Navy's assessment division indicate that in naval combat the first salvo wins.⁴² Although winning the first salvo wins the battle against another conventional fleet, it is less applicable for a fleet fighting a land-based force. The ground force can hide, creating no signature until it fires. In contrast, ships and aircraft create signatures merely by moving.

Russia, China, and Israel have already produced containerized cruise missiles, which can provide various platforms with ship-killing weapons. Container ships could be converted to warships by the addition of 40 to 50 containerized weapons and a simple command and control system for about \$125 million, or a fraction of the purchase price of a Navy littoral combat ship (LCS). Unlike the LCS, container ships are well built due to the requirement to resist hogging and sagging of their long hulls in heavy seas. In short, they could be a survivable and cheaper class of ship and could add numerous missiles to a fleet.

In addition to increasingly capable cruise missiles, VTOL drones such as the XQ-58A described earlier in this chapter mean almost any seagoing vessel can be a small aircraft carrier. Drones do not have to carry weapons large enough to sink a ship to achieve a mission kill. A drone detonating against an aircraft on the deck of a carrier or firing a fragmentation charge against a phased array radar will significantly degrade that platform's capabilities. In addition, many warships carry weapons in box launchers topside. These provide easily identifiable target points and great potential for secondary explosions. Aircraft carriers may well be the most vulnerable. The USS *Oriskany*, USS *Forrestal*, and USS *Enterprise* fires all demonstrate that a small explosion on a deck full of armed, fueled aircraft can result in a carrier being put out of action for weeks to months.⁴³

Undersea weapons pose an even greater challenge to navies. The wide array of commercial unmanned underwater vehicles may provide even small nations with a submarine force. If developed as weapons systems, commercial UUVs could dramatically change naval

Figure 5.1. Kalibr Cruise Missile Ranges if Launched from West or Gulf Coasts



Source: Figure generated by author.

combat; the skills and organization needed to build and employ a short-range UUV force are orders of magnitude less than those needed for a submarine force.

While currently unable to intercept ships under way, UUVs can serve as self-deploying smart mines. With the right fuze, they could wait for a specific class of ship—commercial or military—before attacking. Although these systems cannot stop trade, damaging a few ships will cause major increases in maritime insurance rates. To date, no nation is capable of rapidly clearing smart mines with a high degree of confidence. Clearly, mines provide another major advantage to the defense.

The arrival of numerous relatively cheap, autonomous drones and cruise missiles will drastically change war at sea in the emerging era of Great Power competition. To adapt, navies must conduct rigorous and ruthless wargames to examine how a near-peer competitor will use many small, smart, and cheap weapons as well as a limited number of hypersonic missiles to destroy the few and exquisite platforms most navies currently deploy.

Air

Small, smart, cheap new technologies present two enormous challenges to aviators: how to protect critical high-demand, low-density assets and how to protect aircraft at their bed-down sites.

Clearly, China sees U.S. enablers—Airborne Warning and Control Systems aircraft and aerial tankers—as a critical vulnerability and have dedicated major resources to crippling American airpower by defeating enablers with long-range, air-to-air missiles. Beijing is

working hard to extend its cruise and ballistic missile ranges to force the United States to base its enablers and fighters so far from China's shore that they are no longer a threat. In 2018, China announced the activation of a brigade of 22 DF-26 missile launchers that can range U.S. bases on Guam.⁴⁴ Prudent planners must also assume China will experiment with anti-aircraft missiles in shipping containers. Mounted on various commercial ships, these could ambush tankers en route to the fight. Another way China seeks to reduce the effectiveness of and perhaps even defeat America's advanced aircraft like the B-21 is to destroy them on the ground. Or, as figure 5.1 shows, Chinese ship-based *Kalibr*-class cruise missiles can range almost all U.S. tanker and bomber bases from launch positions off the coasts of the United States. Without tanker support, manned aircraft are rendered impotent in the Pacific.

Fortunately, destroying air bases will not affect mobile, vertical launch drones, which can hide easily in urban or most rural terrain. These drones can also almost turn any ocean-going vessel into an aircraft carrier—particularly if they are employed as suicide drones and the ship does not have to be around for recovery. Small ships will carry only a few containers, but it is important to remember that 18 Harpy drones fit in a single 20-foot container. Each Harpy can carry a 55-pound payload to a range of 600 miles—and hunt autonomously in visual, infrared, and electromagnetic spectrums. From mainland China, these drones can reach Kadena Air Force Base in Okinawa, Japan. A Chinese battery could launch 54 Harpies while a battalion could launch 162. Thus, the sheer number of drones could overwhelm defensive systems.

The combination of cheap drones and much more capable cruise missiles offers opportunities to overcome legacy, manned airpower's key vulnerabilities. But it also may offer even small- and medium-sized states antiaccess/area-denial and precision, long-range strike capabilities. Airpower is facing enormous change. Relatively cheap, autonomous drones and somewhat more expensive but more capable cruise and ballistic missiles and potentially hypersonic missiles are rendering manned aircraft obsolete. The United States does not currently have a way to defend airbases against this new generation of weapons. Aircraft carriers, although tougher targets, are also under increasing threat. Nor can current or envisioned systems find and attack these mobile systems to preempt them. Airmen must embrace drones and work rapidly to transition them from their current status as partners for manned aircraft to that of replacements.

Space

In space, the advent of micro- and cube satellites, paired with commercial launch platforms, will provide space surveillance and even make attack from space a possibility. In April 2020, the Northrup Grumman Mission Extension Vehicle successfully docked with Intelsat 901 and serviced it in orbit.⁴⁵ The system could be used to destroy as well as repair. Whether space assets will be used to attack targets on Earth remains an open question.

Because China has demonstrated the ability to use both kinetic and soft kill on satellites, we should assume the United States is developing similar capabilities.⁴⁶ This evinces a clear understanding that any major conflict will involve fighting in space to defend one's own assets. Resilience is also required. A combination of small satellites, long-endurance solar-powered drones, and perhaps even balloons can mitigate the loss of current space

assets and is being actively explored by the United States. It is good news that the U.S. Air Force is shifting the emphasis of U.S. Space Command from that of a service provider to that of a warfighting command.⁴⁷

Cyber

Cyber operators have enjoyed enormous success in stealing data and money but much more varied results in attempts at physical damage. China has been at the forefront of cyber espionage and focused its teams “on the satellite, aerospace, and communications sectors.”⁴⁸ The Snowden files confirmed that China had penetrated Lockheed Martin’s top-secret data on the F-35.⁴⁹ There is no question that many nations continue to pursue cyber espionage. Even the most advanced cyber actors that have attempted it (Russia, Israel, and the United States) have achieved relatively modest physical destruction outcomes despite often months or even years of intensive efforts.⁵⁰ Perhaps the only reassuring aspect of the increasing state-sponsored cyber activity is that it may be reducing the chance of a major surprise. Historically, devastating surprise attacks have usually taken place after a long period of peace when new operational concepts clash for the first time.

In contrast, cyber warfare is constant, global, ongoing, and evolving. The many sides in these cyber contests are constantly sharpening both offensive and defensive skills and testing them against each other. In addition, cyber attacks focused on societies are, by definition, attacking complex adaptive systems. These systems show remarkable resilience and power of recovery. The ongoing conflict in cyber space will accelerate as the fourth industrial revolution continues. However, it is virtually impossible to predict if a specific attack will succeed or how long it will remain effective.

One aspect not normally discussed is the potential for physical attacks on an enemy’s cyber capability. All networks have nodes in the real world. For instance, satellite downlinks, servers, and fiber-optic networks are vulnerable. Furthermore, unlike cyber attacks that have great uncertainty, physical attacks on known locations can cause more predictable major damage to an opponent’s cyber networks. It is also easier to coordinate real-world attacks than to mix cyber and real-world attacks. This is an area ripe for cross-domain exploitation.

Electromagnetic

Use of the electromagnetic (EM) spectrum is essential to conduct warfare in every other domain. If a force can dominate the spectrum, it can severely limit or even defeat an enemy’s efforts in land, sea, air, space, and cyber. With proper and constant training, naval and air forces can operate in emission-controlled environments, but such a posture severely degrades their capabilities. Land forces can fight without communications, but unless well trained and practiced in maneuver warfare, the result is a series of disconnected local engagements. If an enemy truly dominates the EM spectrum, it can make it difficult for an adversary to communicate with space assets. Also, cyber systems often rely on EM communications links that could be subject to jamming, degradation, or deception.

Although the United States has not yet officially recognized the EM spectrum as a domain of war, China has made it a key element of its newly organized Strategic Support Force (SSF). The SSF not only integrates space, cyber, information, and electronic warfare

(EW) in support of Chinese operations, but also directs advanced research in EW.⁵¹ For its part, Russia has dedicated millions of dollars to upgrading its EW capabilities. As a result, Moscow has been able to use EW to damage or destroy Ukrainian command networks; jam radios, radars, and GPS signals; and control fires.⁵² Clearly both China and Russia have assigned high priority to operating in this environment.

Miniaturization of electronics, massive increases in computer power, creation of electromagnetic pulse systems, and improved sensors are driving the EM spectrum to be a key area of high-tech competition—one that may well determine the outcome of future engagements. The vulnerability of current communications systems provides great impetus to develop truly autonomous, sealed weapons systems.

Operational Implications

Technological convergence is driving the “democratization” of military power by providing small states—and even groups—capabilities that used to be the preserve of major powers. Five factors will have a direct impact on the operational level of war: range obsolescence, the loss of immunity to attack, the tactical dominance of defense, the return of mass, and the requirement to mobilize.

Range Obsolescence

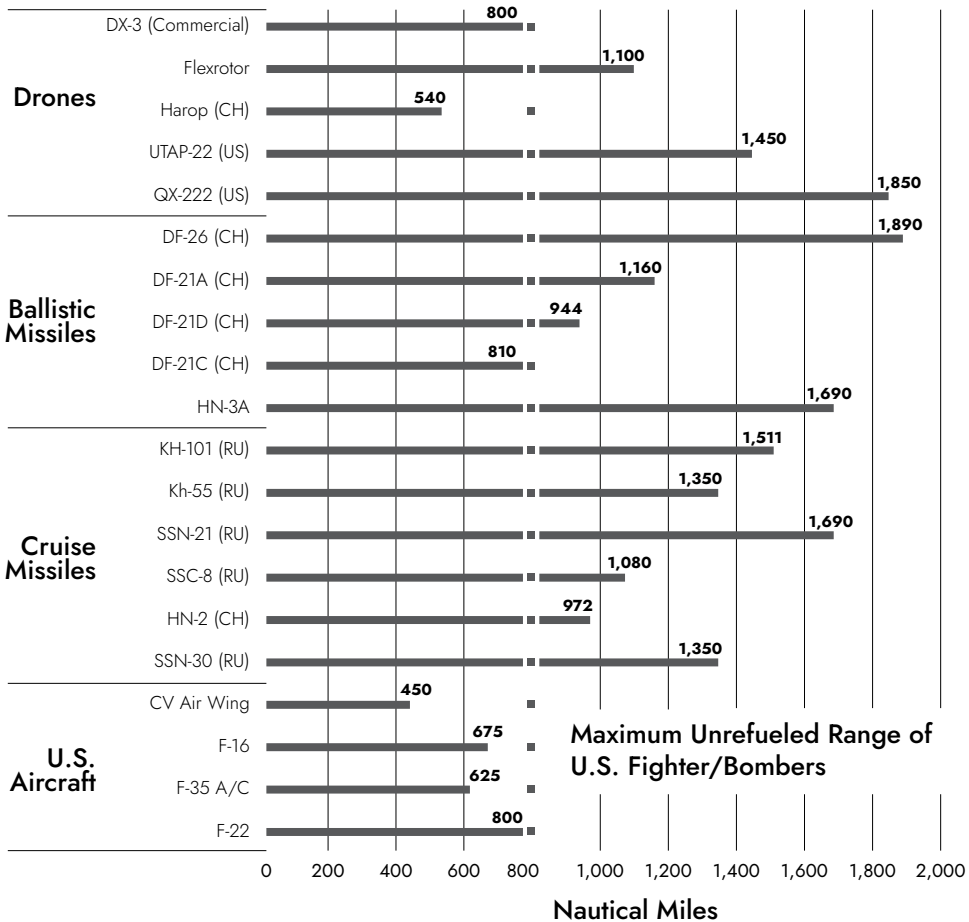
As discussed, missiles and drones have three major advantages over manned aircraft—range, basing flexibility, and cost. An increasing number of drones and ballistic and cruise missiles outrange all fighter bombers. Because many are truck mobile, they are extremely difficult to suppress, much less destroy. Moreover, they are forcing U.S. warfighters to be based so far from the frontlines that the fighters are effectively neutralized. Finally, they are relatively cheap (see figure 5.2).

Loss of Immunity to Attack

Long-range drones, containerized weapons on commercial ships, sea mines, and submersible drones will provide small states and even nonstate actors the capability to strike air and sea ports globally. The United States will no longer project power anywhere in the world with impunity. Future enemies will be able to impose real costs that directly affect U.S. citizens, and they may not be shy about employing these weapons. Small states and nonstate actors have shown cyber capabilities that have distinctly reduced U.S. immunity to counter-attack when America gets involved in a conflict overseas.

Internationally, attacks on Saudi oil facilities and U.S. facilities on Iraqi bases have demonstrated major increases in opponents’ abilities to threaten intermediate bases. If threatened, would friendly nations allow U.S. forces to use their transit facilities? Can the United States protect commercial airfields, key government facilities, and key economic assets in host countries to ensure continued access?

Small precision weapons also provide opportunities for state sponsors to vastly improve the capabilities of proxies, as demonstrated by the Yemini firing cruise missiles at U.S. warships.⁵³ Of more immediate concern will be the far larger number of weapons that can hit critical operational support bases such as Bagram, Afghanistan. Even if the United States can protect major fixed bases, can it defend all patrol bases, gatherings of local leaders, and

Figure 5.2. Range of Missiles, Drones, and Aircraft in Nautical Miles

Source: Figure generated by author.

host-nation facilities essential to current U.S. operations? Even more challenging than protecting fixed facilities will be defending the convoys that move personnel, equipment, and supplies within the theater.

Tactical Dominance of Defense

Emerging autonomous systems will provide an inherent advantage to defenders because they do not have to generate any active signature until they choose to fire. Finding a defender's system hidden in urban clutter, underground facilities, or even the complex littoral environment is much harder than finding the aggressor's system as it moves to attack—particularly if it moves in the air or on the sea.

This is a major advantage of the United States and its allies because we are on the operational defensive in both Asia and Europe. If the United States chooses to create a new generation of inexpensive, smart weapons, it can share them with its allies. Japan, South Korea, Australia, and Europe all have defense industries that could produce these weapons

to provide their nations with affordable, effective defensive capabilities against either China or Russia. In addition, the United States could provide these systems to smaller allies like those on the First Island Chain and the Nordic and Baltic countries. With allies adopting these systems, U.S. forces could integrate more effectively into the defense of these nations. This ability could significantly improve the capabilities of U.S. contact and blunt layers (as described in the National Defense Strategy) while making it easier for the surge forces to enter the theater.⁵⁴

The proliferation of precision weapons may create a situation similar to World War I, where any person in range above the ground could be cheaply killed. The result was static, trench warfare. Able to reach out 1,000 miles in the surface, subsurface, and air domains, these drones and cruise and ballistic missiles may again make defense the tactically dominant form of warfare. Power projection may be limited to strike operations using a family of drones and cruise and hypervelocity missiles.

Proponents of directed energy weapons (DEW)—lasers and microwave systems—suggest their systems will defeat such drone and missile swarms and thus return offense to the tactical battlefield. Unfortunately, these systems are still expensive and power hungry, ineffective under some environmental conditions, and may be subject to defeat by relatively inexpensive countermeasures. At the same time, DEW will favor defenders who have the huge advantage of fixed-power facilities and the ability to blend into complex terrain.

If DEW systems ever become capable of defeating thousands of drones, they may also be able to defeat the much fewer conventional aircraft, guided bombs, and missiles the United States can deploy. This ability would reinforce the dominance of the defense. A key question that must be explored is whether land power—by taking advantage of complex terrain, unlimited magazines, massive power networks, and the ever-increasing range and speed of land-based weapons—will come to dominate the sea, air, and space domains.

Return of Mass

As noted at the beginning of this chapter, technological convergence means that mass with precision has arrived. Additive manufacturing will make numerous cheap drones and advanced GRAMM systems available. Sheer numbers of cheap, mass-produced, unmanned precision weapons may simply overwhelm the exceptionally capable but manned U.S. legacy weapons systems. Large numbers of relatively inexpensive precision weapons will defeat the current U.S. inventory of exquisite but few systems. In response, the Pentagon must figure out how to produce precision weapons in mass.

Requirement to Mobilize

Mobilization in World War II was possible because civilian industry could rapidly convert to military production. By 1990, the complexity of modern military weapons systems made rapid mobilization difficult if not impossible.⁵⁵ Advanced manufacturing—particularly 3D printing and robotics guided by task-specific AI—may radically change this situation. 3D printing is inherently flexible because the product produced depends only on the materials the printer can use, the design of the printer, and the software that is loaded. With a change of software, 3D printers can go from producing commercial products to producing weapons. Thus, as 3D printing assumes a greater role in industry, the possibility

of industrial mobilization will reemerge. However, successful mobilization cannot be accomplished quickly without a significant planning effort. The Pentagon must be prepared to provide the computer files for the 3D printers as well as produce the required government-furnished equipment. The Pentagon must also be prepared to enlist and train new personnel, build them into coherent units, and then move those units and the weapons to an overseas battlefield. Unfortunately, the Pentagon has only recently begun to think about these issues.⁵⁶

Cumulative Impact on Military Operations

The convergence of technology over the next 10 to 20 years means everyone—even nonstate actors—will have precision strike capabilities. Nations with access to as few as 100 3D printers could have access to thousands of precision drones. By pairing these drones with passive sensors, defenders will not have to emit any signal until they choose to fire. In contrast, attackers must create a signal when moving to the target area. Thus, defenders will choose when to initiate the fight and will be able to do so with a massive barrage of precision weapons. In short, a prepared defense will be able to create tactical dominance.

Unfortunately, because of the huge physical and electronic signature of major military bases, they will prove the exception to the dominance of the defense. Rapidly improving inexpensive drones can reach 500 miles today. Relatively inexpensive cruise missiles reach 1,000 miles or more. Both can be deployed on platforms that will offer new opportunities for surprise mass attacks on U.S. bases at home and overseas.

The combination of these factors means that states and nonstates will be able to impose significant casualties and even economic damage to opponents who try to land ground forces in their home territories. In human and financial terms, intervention will be much more costly in the future than it is today.

Impact on Great Power Competition and Conflict

The technologies of the fourth industrial revolution will not change the fundamental nature of war. It will still be driven by Carl von Clausewitz's primary trinity of passion, chance, and reason. It will remain the domain of fog, friction, and uncertainty. Technology will not bring clarity or brevity. For millennia, political and military leaders have embarked on wars where they thought they understood the situation and could win a short and decisive war—and subsequently paid the price for ignoring the true nature of war.⁵⁷

In the conventional arena, the revolution of small, smart, and cheap favors the United States over China or Russia. Operationally and tactically, the United States is on the defensive in both Eastern Europe and Asia.

China is obviously the most serious military challenge. Its rapid buildup of conventional forces, fundamental reorganization to improve jointness, and intensive development of "system attack and destruction warfare" directly challenges U.S. doctrine for force projection.⁵⁸ In response, the Department of Defense is working on a joint warfighting concept to meet these new challenges. While still a work in progress, the emergence of small, smart, and cheap technologies can do much to neutralize China's ongoing investments in system attack.

By adopting mobile missile systems, VTOL drones, missile merchants, and smart mines, U.S. forces could move from dependence on large, fixed, easily targeted bases and platforms to systems that can disperse and hide throughout the First Island Chain as part of both the contact and blunt layers noted in the National Defense Strategy.⁵⁹ This would greatly strengthen the American strategy that is based on defending the First Island Chain, denying China use of the seas inside the First Island Chain, and dominating the seas outside that chain.

As noted, the relatively low cost and simplicity of these systems mean the United States can share production of these weapons with Australia, Japan, Singapore, and South Korea. The United States can also share them with the Philippines and Indonesia to provide these nations with weapons that can challenge Chinese forces. Forward basing or rotating similarly equipped U.S. forces to train with these allies will reinforce political aspects of the alliances even as it strengthens their military power.

In Europe, the fundamental challenge for the North Atlantic Treaty Organization (NATO) is to prevent Russia from seizing territory before the Alliance can mobilize and respond. The concern is that once Russia seizes territory, it will “escalate to deescalate” by threatening to use tactical nuclear weapons against any forces moving to eject Russian forces. Given the reluctance of the population of NATO member states to engage in conflict with Russia, it is uncertain if many states will fight even if not threatened by nuclear weapons.⁶⁰ If Russia threatens the use of nuclear weapons, the probability of NATO action will decrease significantly. Given the long mobilization and deployment times of Alliance forces, the only way to prevent this outcome is to prevent Russia from seizing the territory in the first place.

By equipping front-line states with a mix of inexpensive drones and improvised explosive devices, NATO could create a tough, quickly mobilized, and deep defense on its borders. If NATO states invest heavily in autonomous drones and cruise missiles, they could provide supporting fires in a matter of hours rather than the weeks or months currently envisioned.⁶¹

The Changing Character of War

In contrast to the unchanging nature of war, the character of war—how it is fought—has changed continually. Despite America’s love of technology, how people fight wars will remain based more on the political, economic, and social aspects of their societies than the technology involved. Historically, each society’s employment of new technology has been heavily shaped by these same aspects. For over 200 years, Swiss pikemen dominated the battlefields of Europe. There was no technological mystery to making or even using pikes, but only the Swiss had the necessary social cohesiveness and trust to fight in tight formations under elected leaders. The hierarchical political entities of Europe could not.

Each society will use the emerging technologies in ways best suited to its unique needs. Furthermore, conflict will not be based solely on those aspects of one society but the interactions of all the societies in the conflict. In the dawning era of Great Power competition, wars are likely to be bloodier, longer, and more financially ruinous. Fortunately, the emergence and convergence of fourth industrial revolution technologies provide real advantages for the United States—if it can seize them.

Notes

- ¹ Robert Work, "The Third U.S. Offset Strategy and Its Implications for Partners and Allies," Department of Defense, January 28, 2015, available at <<http://www.defense.gov/News/Speeches/Speech-View/Article/606641/the-third-us-offset-strategy-and-its-implications-for-partners-and-allies>>.
- ² "What Is Nanotechnology?" National Nanotechnology Initiative, available at <<https://www.nano.gov/nanotech-101/what/definition>>.
- ³ Louis A. Del Monte, *Nanoweapons: A Growing Threat to Humanity* (Lincoln, NE: Potomac Books, 2017), 47–48.
- ⁴ Michael Berger, "Carbon Nanotubes—What They Are, How They Are Made, What They Are Used For," *Nanowerk*, available at <www.nanowerk.com/nanotechnology/introduction/introduction_to_nanotechnology_22.php>.
- ⁵ Li Yi Hsu, Thomas Lepetit, and Boubacar Kante, "Extremely Thin Dielectric Metasurface for Carpet Cloaking," *Progress in Electromagnetic Research* 152 (2015), 33–40.
- ⁶ S. R. Vadera and Narendra Kumar, "Nanotechnology and Nanomaterials for Camouflage and Stealth Applications," *Nanowerk*, January 30, 2015, available at <www.nanowerk.com/spotlight/spotid=38899_3.php>.
- ⁷ Winslow Wheeler, "The MQ-9's Cost and Performance," *Time*, February 28, 2012, available at <<http://nation.time.com/2012/02/28/2-the-mq-9s-cost-and-performance>>.
- ⁸ "New 24-Hour Endurance Hybrid Drone Developed for Monitoring and Inspection," *Unmanned Systems News*, December 12, 2016, available at <www.unmannedsystemstechnology.com/2016/12/defiant-labs-announces-new-dx-3-drone-for-monitoring-inspection>.
- ⁹ "Payload Options," *Serovel.com*, available at <<http://aerovel.com/flexrotor/>>.
- ¹⁰ Tyler Rogoway, "More Details Emerge on Kratos' Optionally Expendable Air Combat Drones," *The Warzone*, February 7, 2017, available at <www.thedrive.com/the-war-zone/7449/more-details-on-kratos-optionally-expendable-air-combat-drones-emerge>.
- ¹¹ Sofia Bledsoe, "PEO Workers Receive Army Acquisition Awards," *Army*, May 11, 2017, available at <https://www.army.mil/article/148323/PEO_workers_receive_army_acquisition_awards>.
- ¹² Eric Schmitt, "Papers Offer a Peek at ISIS's Drones, Lethal and Largely Off-the-Shelf," *New York Times*, January 31, 2017, available at <www.nytimes.com/2017/01/31/world/middleeast/isis-drone-documents.html?_r=1>.
- ¹³ Lora Kolodny and Darren Weaver, "These Drones Can Haul a 20-Pound Load for 500 Miles and Land on a Moving Target," *CNBC*, May 26, 2018, available at <www.cnbc.com/2018/05/26/volans-i-drones-can-haul-cargo-for-500-miles-and-land-on-a-moving-ship.html>.
- ¹⁴ Rob Wittman, "U.S. Navy's Unmanned Vehicle Efforts Are the Answer to Detering Adversaries," *Defense News*, April 26, 2018, available at <www.defensenews.com/unmanned/2018/04/26/us-navys-unmanned-vehicle-efforts-are-the-answer-to-detering-adversaries/>.
- ¹⁵ Adam Rawnsley and Austin Bodetti, "The Warbot Builders of the Middle East Spill Their Secrets," *Wired*, February 2, 2017, available at <www.wired.com/2017/02/warbot-builders-middle-east-spill-secrets>.
- ¹⁶ Daniel Brown, "Russia's Uran-9 Robot Tank Reportedly Performed Horribly in Syria," *Business Insider*, July 9, 2018, available at <www.businessinsider.com/russias-uran-9-robot-tank-performed-horribly-in-syria-2018-7>.
- ¹⁷ "Global Smart Commercial Drones Market Is Expected to Reach US\$ 179600 Million by 2025 with 83.3% CAGR," *Benzinga*, June 10, 2019.
- ¹⁸ "What Is a Faraday Cage?" *Faraday Cage*, available at <www.faradaycage.org/>.
- ¹⁹ John Keller, "The Evolving World of Radiation-Hardened Electronics," *Military & Aerospace Electronics*, June 1, 2018, available at <www.militaryaerospace.com/computers/article/16707204/the-evolving-world-of-radiationhardened-electronics>.
- ²⁰ "Service Academies Swarm Challenge Pushes the Boundaries of Autonomous Swarm Capabilities," Defense Advanced Research Projects Agency, May 11, 2017, available at <<https://www.darpa.mil/news-events/2017-05-11>>.
- ²¹ Evan Ackerman and Celia Gorman, "This Drone Uses a Smartphone for Eyes and a Brain," *IEEE Spectrum*, April 9, 2015, available at <<http://spectrum.ieee.org/video/robotics/aerial-robots/this-drone-uses-a-smartphone-for-a-brain>>.
- ²² Brian Wang, "Quantum Assisted GPS Would Be One Thousand Times More Accurate Than Any Existing GPS and Other High Potential DARPA Projects," *Next Big Future*, May 23, 2014, available at <www.nextbigfuture.com/2014/05/quantum-assisted-gps-would-be-one.html>.
- ²³ Dana A. Goward, "Russia, China Alliance on Navigation Satellites Threatens GPS," *National Defense Magazine*, August 23, 2019, available at <<https://www.nationaldefensemagazine.org/articles/2019/8/23/viewpoint-russia-china-alliance-on-navigation-satellites-threatens-gps>>; *Above Us Only Stars: Exposing GPS Spoofing in Russia and Syria* (Austin, TX: C4Ads, 2019), available at <<https://static1.squarespace.com/static/566ef8b4d8af107232d5358a/t/5c99488beb39314c45e782da/155354942554/Above+Us+Only+Stars.pdf>>.
- ²⁴ "Cameras and Sensors," HSE-UAV, available at <www.agricultureuavs.com/photos_multispectral_camera.htm>.
- ²⁵ "The 7 Best Follow You Drones—[Updated 2020] Follow Me Drone Review," *Drone Enthusiast*, April 9, 2020, available at <www.dronethusiast.com/drones-that-follow-you/>.
- ²⁶ Andrew Rosenblum, "Air Force Tests IBM's Brain-Inspired Chip as an Aerial Tank Spotter," *MIT Technology Review*, January 11, 2017, available at <www.technologyreview.com/2017/01/11/5848/air-force-tests-ibms-brain-inspired-chip-as-an-aerial-tank-spotter/>.
- ²⁷ Jordan Golson, "A Military-Grade Drone That Can Be Printed Anywhere," *Wired*, September 16, 2014, available at <www.wired.com/2014/09/military-grade-drone-can-printed-anywhere>.
- ²⁸ Arie Egozi, "Azerbaijan Starts Producing Zarba Armed UAV," *Flight Global*, January 10, 2017, available at <www.flightglobal.com/news/articles/azerbaijan-starts-producing-zarba-armed-uav-433014>.
- ²⁹ "Misiewicz: 1,000 Attack Drones for the Polish Army in 2017," *Defence* 24, December 5, 2016, available at <www.defence24.com/503694/misiewicz-one-hundred0-attack-drones-for-the-polish-army-in-2017>.
- ³⁰ Tony Wesolowsky, "Ukraine's Exploding Munition Depots Give Ammunition to Security Concerns," *Radio Free Europe*, October 6, 2017, available at <<https://www.rferl.org/a/ukraine-exploding-munitions-security-concerns-russia/28777991.html>>.
- ³¹ "What Are SmallSats and CubeSats?" *NASA*, February 26, 2015, available at <<https://www.nasa.gov/content/what-are-smallsats-and-cubesats>>.
- ³² Robinson Meyer, "Google Remakes the Satellite Business, by Leaving It," *The Atlantic*, February 7, 2017, available at <www.theatlantic.com/technology/archive/2017/02/google-gets-out-of-the-satellite-business/515841>.

- ³³ Kent Leka, "Viasat Will Enable Near-Instant Satellite Images and Video Delivered Anywhere," Viasat, May 15, 2018, available at <<https://corpblog.viasat.com/viasat-will-enable-near-instant-satellite-images-and-video-delivered-anywhere/>>.
- ³⁴ Kris Osborn, "Hypersonic Weapons: Everything You Need to Know About the Ultimate Weapon," *The National Interest*, July 22, 2017, available at <<https://nationalinterest.org/blog/the-buzz/hypersonic-weapons-everything-you-need-know-about-the-21637>>.
- ³⁵ "Gliding Missiles That Fly Faster Than Mach 5 Are Coming," *The Economist*, April 6, 2019.
- ³⁶ Kelley M. Saylor, *Hypersonic Weapons: Background and Issues for Congress*, R45811 (Washington, DC: Congressional Research Service, March 4, 2020), 5, available at <<https://fas.org/sgp/crs/weapons/R45811.pdf>>.
- ³⁷ "Hypersonic Weapons and Strategic Stability," *International Institute of Strategic Studies Strategic Comments* 26, no. 4 (March 2020), 2, available at <https://www.iiss.org/publications/strategic-comments/2020/hypersonic-weapons-and-strategic-stability?mc_cid=f55b265518&mc_eid=0a46c72ebe>.
- ³⁸ Saylor, "Hypersonic Weapons," 11.
- ³⁹ *Ibid.*, 6.
- ⁴⁰ David Axe, "Mach 7.3 'Bullets': The U.S. Navy Tested a 'Hypervelocity Cannon Round,'" *The National Interest*, January 9, 2019, available at <<https://nationalinterest.org/blog/buzz/mach-73-bullets-us-navy-tested-hypervelocity-cannon-round-41022>>.
- ⁴¹ Joseph Trevithick, "Photos Surface of the Army's Crazy 1,000 Mile Range Supergun Concept: Here's What We Know," *The Warzone*, February 25, 2020, available at <www.thedrive.com/the-war-zone/32352/photos-surface-of-the-armys-crazy-1000-mile-range-supergun-concept-heres-what-we-know>.
- ⁴² Megan Eckstein, "Panel: Navy Must Invest in Counter-ISR, Unmanned Boats, Railgun to Prepare for Future Fight," *USNI News*, March 9, 2017, available at <<https://news.usni.org/2017/03/09/panel-navy-must-invest-counter-c4isr-unmanned-boats-railgun-prepare-future-fight>>.
- ⁴³ For the history of these U.S. aircraft carrier deck incidents from 1966 to 1969, see Sebastian Roblin, "A Rocket Explosion Almost Destroyed a U.S. Navy Nuclear-Powered Aircraft Carrier," *National Interest*, December 17, 2019, available at <<https://nationalinterest.org/blog/buzz/rocket-explosion-almost-destroyed-us-navy-nuclear-powered-aircraft-carrier-105891>>.
- ⁴⁴ Zachary Keck, "China's DF-26 'Carrier-Killer' Missile Could Stop the Navy in Its Track (Without Firing a Shot)," *The National Interest*, April 20, 2018, available at <<http://nationalinterest.org/blog/the-buzz/chinas-df-26-carrier-killer-missile-could-stop-the-navy-its-25493>>.
- ⁴⁵ Christian Davenport, "In Historic First, an Aging Satellite Is Resurrected by Another in a Technology That Could Reduce Junk in Space," *Washington Post*, April 20, 2020, available at <www.washingtonpost.com/technology/2020/04/20/new-technology-creates-fountain-youth-aging-satellites-potentially-reducing-space-junk/>.
- ⁴⁶ For example, Jeffrey G. Lewis, "They Shoot Satellites, Don't They?" *Foreign Policy*, August 8, 2014; George Kulacki and Jeffrey G. Lewis, "Understanding China's Antisatellite Test," *The Nonproliferation Review* 15, no. 2 (2008), 335–347.
- ⁴⁷ Matthew Bienart, "Air Force Space Command Shifting to Warfighting Operational Mindset," *Defense Daily Network*, March 2, 2018, available at <www.defensedaily.com/air-force-space-command-shifting-warfighting-operational-mindset/>.
- ⁴⁸ Adam Segal, "Chinese Cyber Espionage: We Know the Who, How, Why, and Why It Matters—We're Missing the What to Do," Council on Foreign Relations, June 11, 2014, available at <<https://www.cfr.org/blog/chinese-cyber-espionage-we-know-who-how-why-and-why-it-matters-were-missing-what-to-do>>.
- ⁴⁹ Franz Stefan-Gady, "New Snowden Documents Reveal Chinese Behind F-35 Hack," *The Diplomat*, January 27, 2015, available at <<https://thediplomat.com/2015/01/new-snowden-documents-reveal-chinese-behind-f-35-hack/>>.
- ⁵⁰ Dustin Volz and Timothy Gardner, "In a First, U.S. Blames Russia for Cyber Attacks on Energy Grid," Reuters, March 15, 2018, available at <www.reuters.com/article/us-usa-russia-sanctions-energygrid/in-a-first-u-s-blames-russia-for-cyber-attacks-on-energy-grid-idUSKCN1GR2G3>.
- ⁵¹ John Costello, "The Strategic Support Force: Update and Overview," *China Brief* 16, no. 19 (December 2016), available at <<https://jamestown.org/program/strategic-support-force-update-overview/>>.
- ⁵² Sergey Sukhankin, "Russian Electronic Warfare in Ukraine: Between Real and Imaginable," *Eurasia Daily Monitor* 14, no. 71 (May 24, 2017), available at <<https://jamestown.org/program/russian-electronic-warfare-ukraine-real-imaginable/>>.
- ⁵³ Idrees Ali and Matt Spetalnick, "U.S. Warship Targeted in Failed Missile Attack from Yemen: Official," Reuters, October 15, 2016, available at <www.reuters.com/article/us-yemen-security-usa/u-s-warship-targeted-in-failed-missile-attack-from-yemen-official-idUSKBN12G004>.
- ⁵⁴ *Summary of the 2018 National Defense Strategy of the United States of America: Sharpening the American Military's Competitive Edge* (Washington, DC: Department of Defense, 2018), 7, available at <<https://dod.defense.gov/Portals/1/Documents/pubs/2018-National-Defense-Strategy-Summary.pdf>>.
- ⁵⁵ Matthew Richardson and Evan Hooper, "Lockheed Martin Lands \$5.5B Contract for Dozens of F-35s—More to Come," *Dallas Business Journal*, July 11, 2017, available at <www.bizjournals.com/dallas/news/2017/07/11/lockheed-martin-lands-5-5b-contract-for-dozens-of.html>.
- ⁵⁶ Daniel Goure, "The First Signs of a National Mobilization for War Are Appearing," *Real Clear Defense*, May 12, 2017, available at <www.realcleardefense.com/articles/2017/05/12/the_first_signs_of_a_national_mobilization_for_war_are_appearing_111371.html>.
- ⁵⁷ Cathal J. Nolan, *The Allure of Battle: A History of How Wars Have Been Won and Lost* (Oxford: Oxford University Press, 2017).
- ⁵⁸ Matt Devost, "What You Should Know About China's 'Destruction Warfare' Doctrine," *OODA Loop*, July 23, 2019, available at <www.oodaloop.com/archive/2019/07/23/what-you-should-know-about-chinas-destruction-warfare-doctrine/>.
- ⁵⁹ *Summary of the 2018 National Defense Strategy of the United States of America*, 7.
- ⁶⁰ Kate Simmons, Bruce Stokes, and Jacob Poushter, "NATO Publics Blame Russia for Ukrainian Crisis, but Reluctant to Provide Military Aid," Pew Research Center, June 10, 2015, available at <<http://www.pewglobal.org/2015/06/10/nato-publics-blame-russia-for-ukrainian-crisis-but-reluctant-to-provide-military-aid/>>.
- ⁶¹ For a detailed description of how such a defensive posture in Europe could work, see T. X. Hammes, *The Melians' Revenge: How Small, Frontline, European States Can Employ Emerging Technology to Defend Against Russia*, Issue Brief (Washington, DC: Atlantic Council, June 2019), available at <https://www.atlanticcouncil.org/wp-content/uploads/2019/09/The_Melians_Revenge.pdf>.

Chapter 6

Emerging Critical Information Technology and Great Power Competition

By Richard Andres

Over the past few decades, the foundation of Great Power competition has changed. Where control of industrial resources was once the key to geopolitical power, today control of information resources is most important. China is currently investing heavily in three critical new information technologies—5G wireless, quantum computing, and artificial intelligence—that, as part of its information strategy, will vastly increase its control of the global information flow. The United States has a short window to contest China’s state-led ascent in these technologies, as well as in the underlying conditions that are allowing China to outpace the United States in this wider field. If the United States does not prevent China from dominating global flows of information, China will attain a clear advantage in its rise to replace the United States as the world’s leading Great Power.

Over the past few decades, and as observed in the chapter 4 discussion of the fourth industrial revolution, the foundation of Great Power wealth and competitive advantage has fundamentally changed from one dominated by industrial era technology to one in which information technology (IT) has become the primary source of geopolitical power. U.S. businesses were quick to recognize and act on this change. Today, the top three U.S. IT companies are worth 70 times as much as the top three U.S. car manufacturers. Apple Incorporated alone could buy all five major U.S. defense contractors with its cash on hand. Like U.S. private businesses, China’s government has seen and acted on this new IT reality. It has poured billions of dollars into key information technologies and bankrolled its so-called private company Huawei’s schemes to dominate global information infrastructure. China has focused vast military and commercial resources on stealing its adversaries’ intellectual property, infesting their critical infrastructure with malware, and conducting social media-based influence campaigns at home and abroad. Unfortunately, unlike U.S. commercial interests and the Chinese government, U.S. Government defense policy has been slow to respond to the changing foundations of global power. Today, U.S. defense resources generally go toward industrial era capabilities, and America’s strategy remains fixed on winning industrial era battles.

This chapter focuses on the critical role of IT in current geopolitics. It argues that the foundation of geopolitical power has shifted from industrial output to information control. This change has been affecting the global balance of power in favor of China for over a decade but is about to enter a dramatic new phase as China pours state-managed resources into new technologies—most critically, 5G wireless communication, quantum computing, and artificial intelligence (AI)—with the goal of increasing its control over the global flow of information. Meanwhile, although recognizing the problem on paper, U.S. defense policy remains wedded to the quixotic Cold War–era notion that U.S. entrepreneurialism and technology will eventually overcome China’s aggressive information policy—something that cannot happen so long as China continues to steal technology as fast as U.S. entrepreneurs and laboratories develop it.

It is imperative that U.S. policymakers recognize and act on the new reality. Information, not physical resources, is now the foundation of geopolitical power. Just as IT has reversed the relative value of physical- and information-based businesses in the past decade, in the next decade, IT will invert the effectiveness of physical- and information-based Great Power policies and politics. The United States has only a short window to come to grips with and act on the new foundation of global power. To do this, America must prioritize countering China’s current ability to steal intellectual property and otherwise control the flow of information within the United States and other developed nations while reprioritizing resources into key technologies and capabilities that will allow it to contest China’s ability to increase its future power and act in cyberspace.

This chapter is presented in six sections. The first describes the relationship between technology and geopolitical power. The second describes the difference between the way industrial and information era technologies affect global power. The third describes the way autocratic countries are currently using information technology against the United States and its allies. The fourth discusses the race for the three information technologies that will determine which country leads in the emerging geopolitical contest over control of the global flow of information. The fifth describes the problem with current U.S. defense policy, focusing on the futility of racing to develop technology when China can quickly steal it. The chapter concludes with a call to prioritize information over physical conflict as the key to success in Great Power competition.

Technology and Geopolitical Power

Throughout history, states have pursued their political interests using various instruments of national power. The types of issues that states compete over vary. In the second half of the 20th century, the two main actors—the United States and the Soviet Union—vied over which superpower would control which regions of the globe and whether weaker states would be governed by communist or capitalist economic systems. Today’s main players are the United States and China, and the main issues in contention center on whether the U.S.-led liberal global order will persist or be supplanted by one based on China’s autocratic system and preferences. While the United States still dominates in industrial age military power, China has acted aggressively using information-based power. As a result, China’s system is beginning to hold sway over important issues in a number of countries in Asia and, increasingly, in Europe. On some key issues, such as its right to steal intellectual prop-

erty without penalty or to force foreign business such as the National Basketball Association to conform to Chinese Communist Party *dictat*, its system could be said to prevail even inside the United States.¹

In the current system, as in previous eras, geopolitical power is mainly determined by the amount of economic and military capabilities major players can project abroad, and these abilities in turn are shaped by the era's dominant technology. Between the 15th and 19th centuries, both economic and military power were principally shaped by ocean-borne trade and sea power. During this period, the nations best able to control the flow of commerce on the world's seas tended to make the world's rules. During the 20th century, the major powers best able to harness industry tended to dominate global politics; the countries able to bring the most men and materiel to bear tended to make the rules. In the current era, information technology is the key to both economic and military power. In this era, the countries best able to control the flow of information across the world's networks tend to make the system's rules.²

Industrial vs. Information and Global Power

To understand how states have been using information technology to shape geopolitics and how they are likely to utilize emerging IT over the next decade, it helps to consider how the dynamics and incentives connected with IT differ from those associated with traditional industrial power.

In the past century, a state's power was closely linked with its industrial capacity. To increase their power, major players worked to bolster their manufacturing base at home and often attempted to seize other states' resources through military action. Both world wars were caused by nations acting on the belief that they could increase their power by seizing territory, and throughout the Cold War, the United States and Western Europe based much of their defense policies on the fear that the Soviet Union would invade Western Europe in order to seize its industrial resources.

To counter the industrial era incentive for invasion during the Cold War, nations built alliances to bolster their military capability; they expanded their conventional military power and developed large, often hair-trigger nuclear forces. By the end of the war, the Earth was encircled in competing military alliances and ringed in bases, bombers, and aircraft carriers. The United States and the Soviet Union, with the largest industrial capacities, dominated geopolitics, generally set the rules within their respective spheres of influence, and used their economic and military power to shape world politics.

Yet, even during the Cold War, there were signs that industrial power was beginning to lose its place to information power as the dominant technology in geopolitics. In a well-known story, the Soviets were among the first to recognize this change. At the height of the Cold War, Soviet analysts noted that, while the Soviet Union could produce far more steel and mobilize a far larger army than the United States (the two traditional indicators of industrial-military capability), the West's information economy allowed it to more than overcome its industrial disadvantages with superior information-based military technology.³ Even worse, from the Soviet perspective, was the West's ability to use what the Soviets saw as psychological-information operations to foment insurrection within the Eastern

bloc. The first fear was proved accurate by the success of U.S. information age weapons in the 1990 Gulf War, the second by the fall of Soviet communism to internal insurrection.

While the Soviet Union did not survive long into the information age, the People's Republic of China did. Writing in the early 2000s, holding up the fall of the Soviet Union as evidence, Chinese geopolitical strategists, such as Major General Xu Hezhen, spoke of the threat posed by the United States to the Chinese Communist Party.⁴ As a counter to this, China proposed a national strategy based on information rather than industrial era methods. According to this argument, the most effective route to geopolitical power in the current century involved information. This led China to an overall strategy that sought control over the flow of knowledge, secrets, and beliefs rather than simply raw materials and industrial output.⁵ This included, for instance, its Three Warfares doctrine and an industrial policy aimed at controlling key industries that produced, among other things, software, undersea cables, microchips, and telecommunications rather than an approach aimed at merely controlling territory.⁶

With the advantage of hindsight, a U.S. strategist might characterize the idea as attempting to encircle the world in a virtual network rather than trying to compete directly with America's physical network of bases, bombers, and ships. The new information age methods do not make conventional and nuclear war obsolete any more than nuclear forces made industrial era technology irrelevant or railroad and other industrial era technology made navies irrelevant. Rather, information technology has been layered on top of and throughout the older system. This overall philosophy has infused much of China's geopolitical strategy for the past two decades.⁷

Geopolitical Targets of Autocratic Countries

The Chinese strategy is effective. To benefit economically from emerging IT, the United States and other developed nations have connected essentially everything in their territories to computer networks. This includes businesses, critical infrastructure, and social media networks. Once connected, all these institutions are potentially vulnerable to exploitation or destruction by anyone connected to the global telecommunications grid.⁸ Meanwhile, as an autocratic state, China's government has been able to control information domestically by fencing off many of its own information vulnerabilities from outside penetration.⁹

After two decades of experimentation by states attempting to use information as an instrument of geopolitics, three main types of targets have emerged. The first is economic. In traditional industrial era economies, wealth is generated and stored in physical assets, and this pattern continues in many less developed countries, including China and Russia. In the developed world, over the past few decades, firms have increasingly generated and stored wealth in nonphysical assets. By one estimate, as early as 2010, about 80 percent of U.S. corporate value was stored in intellectual property and trade secrets vulnerable to cyber theft.¹⁰ That percentage is almost certainly considerably higher today. Beyond this, in the United States, large amounts of wealth—much of which can be stolen—are stored in the knowledge and research held within state-funded universities.

In the industrial age, if an ambitious Great Power hoped to plunder the most valuable resources of the United States or Europe, it would have had to physically defeat North Atlantic Treaty Organization armed forces. Today, to steal the West's assets, it is necessary only

to penetrate the computer networks where companies and nonprofit organizations store their wealth. According to the independent U.S. Intellectual Property Commission, China uses these methods to steal hundreds of billions of dollars of intellectual property from U.S. and European firms every year.¹¹ The overall effect is to stunt the economies of plundered nations while vastly accelerating China's economic growth. Over the past two decades, this approach has contributed to China's rapid economic growth. Information theft is not the only cause of China's economic miracle, but it is a necessary one. If it stops, China's swift economic progress would slow considerably, but if it continues over the next decade, all things equal, it will likely lead to China's economy eclipsing that of the United States. No state-sponsored physical piracy campaign in history has had anything like the geopolitical impact of China's virtual piracy campaign. Given the costs of occupying conquered nations, it is unlikely that China could have gained as much wealth from even a Soviet-style conquest of the small countries on its borders.

The second main target for information operations involves civilian and military critical infrastructure. As early as 2010, when Stuxnet malware was found in Iranian nuclear centrifuges at the country's Natanz facility, firms around the world began to realize they were vulnerable to software-based attacks on their hardware. Moreover, on inspection, thousands of critical infrastructure-providing companies discovered that their computers were infected with malware, and many more learned that unknown entities had developed methods of accessing their equipment. In 2015, the commander of U.S. Cyber Command informed the Senate that China and other countries had the means in place to take down U.S. critical infrastructure.¹² On one end of the spectrum, such malware could be used to cause individual civilian or military systems to temporarily stop functioning; on the other, it could be used to create continent-wide, months-long infrastructure failures that could cause millions of deaths.¹³ As China works to increase its ability to attack civilian infrastructure, it also has integrated information operations into all aspects of its military capability and posture.¹⁴

Methods that use information technology to take down an adversary's critical infrastructure and military systems are generally not as dependable as methods that use conventional or nuclear weapons. They are, however, superior to kinetic industrial era techniques in at least two ways. First, they are less expensive. Developing the bases, navies, and air forces necessary to project power globally costs trillions of dollars and is, at least currently, something only the United States can accomplish. Malware is less expensive and allows poor countries to project power cheaply. Second, information-based attacks on critical infrastructure can be calibrated and conducted in low-intensity situations. Thus, a country that might fear attacking the United States with conventional or nuclear weapons might be willing to conduct cyber attacks on critical infrastructure that it believes could work below the threshold that would elicit a violent response from America. Such capabilities could plausibly be used to coerce or deter the United States.¹⁵ The Department of Homeland Security has repeatedly warned that China and Russia have infected U.S. critical infrastructures with malware that could be used in this manner.¹⁶

The third target for information operations involves populations. In recent decades, firms have acquired access to most individuals in most countries (through social media) and developed AI technologies to manipulate targets' purchasing behaviors. More recently,

nations have begun using similar methods to manipulate targets' political preferences and passions. A growing body of literature suggests that humans are susceptible to these techniques.

At present, China is the leader in the technology associated with state-centric manipulation of political behavior. Using its close relationship with Chinese Internet service providers, the Chinese government has developed methods to manipulate information, thought, and political activity within its borders and, increasingly, in other nations.¹⁷ Like China, Russia has experimented with these methods at home and abroad. Over time, both countries are likely to increase and refine their technologies and techniques. While it is unclear how effective these techniques will be in the long term as a tool of political control used against Western populations, the success of Chinese computer-based political-psychological operations against Chinese citizens and of Russian actions in Europe and North America suggest a clear danger. The 20th-century Soviet information operations that led to the global rise of communism represent a possible low-technology precedent that bears consideration. Just as the Soviet Union once persuaded over half the planet's population to abandon that era's dominant political systems in favor of communism, today's autocratic information operations have the potential to do enormous harm to the current liberal world order.

Race for Three Information Technologies

Given the ways IT can be used in geopolitical competition, if an autocratic Great Power was to gain complete control over cyberspace, eventually it would gain the wealth and military power to set the rules for the international system in much the same way the United States now does. Analogously, a major power gaining unfettered access to the world's computer networks today would be similar to the Soviet Union gaining control of Western Europe's industrial capacity during the Cold War.

To date, despite ongoing efforts, neither China nor Russia has gained unfettered access to Western networks. They have failed to do so largely because of the immense resources that governments and private industry dedicate to computer defenses. The battle for access is a constant struggle to control the domain that is played out every time a new piece of software or hardware is added to a network. It involves attempts to find holes in operating systems, commercial software, phone apps, hardware, and nearly everything associated with the growing Internet of Things. This contest is reminiscent of trench warfare during World War I in that it is a persistent whole-of-nation contest played out daily in millions of individual duels among individuals, firms, and agencies for small advantages, and what is won one day is often lost the next.

While battles in cyberspace are often for small and fleeting advantage, several contests occurring today are likely to have large and enduring results that will determine which Great Power dominates cyberspace for future decades.

The First Contest: 5G

The first major contest for control of cyberspace involves the fifth-generation wireless communications technologies supporting cellular data networks, generally known as 5G.¹⁸ For much of the coming decade at least, this technology will provide the backbone for future cell phone communication and for the Internet of Things.¹⁹ This includes technologies

such as self-driving cars, supervisory control and data acquisition infrastructure, and the networks that military systems depend on. A country able to dominate 5G systems potentially will have access to and a good deal of control over most information flowing through cyberspace.²⁰

The nearly unfettered access that 5G dominance provides to user information goes beyond simple data collection. A country with this type of access can use it to map critical systems throughout an adversary's territory. It gains real-time intelligence about the physical and network location of individuals and systems.²¹ It could deny or corrupt information received by human or machine users, civilian or military (once such systems are installed nationally, a military would be hard pressed not to use them). These capabilities would vastly increase the ability of the controlling nation to conduct espionage, sabotage machines that are connected to digital networks, and perform a range of operations involved with conducting social-psychological operations against adversaries' populations.

Traditionally, the United States has been able to purchase IT equipment from autocratic adversaries without much fear that they will be able to exploit the hardware; the end users have retained physical control of most of the software and hardware. But 5G is different in that it pushes far more functionality from the user to the supplier. For instance, in a 5G environment, the apps on a phone might reside entirely on servers in Beijing. Moreover, in so far as a 5G provider uses proprietary software to perform a range of controls conducted by hardware in earlier wireless technology, the provider now largely controls access and control of information passing through its system. In certain places in China where 5G is well established, this ability sometimes reaches the point where customers no longer need handsets. Distributed cameras and microphones in places such as hotel lobbies or sidewalks identify users and respond directly to commands, completely removing any possibility of user-supplied defense or control. In general, the more bandwidth becomes available, the more control can be pushed from the user to the company supplying the service.²²

Over the past two decades, China has enacted a national industrial policy that, among other things, seeks to make China the global leader in communications technology in general and 5G in particular. China's industrial strategy aims to make the nation the world leader in 5G by supporting national champions, Huawei in particular, via direct funding from the state (mainly via subsidized loans) and by a campaign to steal cutting-edge telecommunications technology from Western firms and provide that technology to its champions for free. The overall effect has been to cut Huawei's cost of producing goods far below that of its competition, thereby allowing it to undercut the prices of Western competitors. As a result, Huawei is able to sell 5G technology at rates far below those of other nations—in some cases, as much as 60 percent below market rates.²³ With the exception of countries such as the United States that eschew Huawei's high-quality/low-cost services out of concerns about national security, most nations in the world installing 5G are installing Huawei systems.

Huawei's 5G dominance of global networks will not result in China instantly dominating cyberspace. If Huawei acts too aggressively on China's interests, countries are likely to spend the money necessary to replace Huawei's systems with hardware built in other countries. More than that, although China is currently using 5G within its own borders to control the flow of information, it will take time to develop and adapt software, tactics,

techniques, and procedures that work abroad. Yet, over time, China will be able to become ever more aggressive as countries become increasingly dependent on Huawei's goods and services. At some scale, China will be able to get away with virtually any use of 5G networks because the price of replacing the network will become too high. Two years ago, Congress was barely able to force a handful of small U.S. telecoms to divest themselves of Huawei's hardware. In the future, China will be able to extract significant geopolitical information rents from countries that are dependent on its systems.²⁴

The Second Contest: Quantum

The second major contest for control of cyberspace involves the race for quantum technology.²⁵ Because quantum computers are able to perform a range of operations at rates that greatly exceed older processing technology, they hold out the possibility of being able to break existing encryption.²⁶ This ability would not only render current encryption obsolete but also most likely allow its holder to decrypt decades of older encoded messages. Because encryption is generally the single point of failure in all computerized defenses, having this capability would vastly increase its holder's power.²⁷

For decades, quantum computing has been the Holy Grail of computer technology but has always been out of reach. Over the past 2 years, however, several companies have introduced computers with limited quantum capabilities.²⁸ While none of these companies so far has advertised the ability to break codes, this technology is within reach. While it might seem that quantum technology would equally aid both encryption and decryption, this is not the case. For practical reasons, the technology will do much more to aid decryption than encryption, and it is likely that the country to win the race to produce and utilize the technology will gain a significant advantage in the contest to control the cyber domain.²⁹ Beyond this scenario, while both China and the West have many state secrets, the West simply has immensely more secrets than China—both Western industry and militaries depend on encryption in ways China's industry and military do not.

Over the past few years, the United States has invested considerably in quantum technology, but recently China has begun to spend vastly more. While the United States is probably ahead in this race—at least as far as industry is concerned, with IBM and Google fielding computers with limited quantum capability—that lead is unlikely to last.³⁰ Given forecasted spending trends, China is likely to finish the race to utilize quantum technology in support of national security goals well ahead of the United States.³¹ When it does, its ability to steal industrial secrets and infect and sabotage critical civilian and military systems will vastly increase.

The Third Contest: Artificial Intelligence

The third major contest for control of cyberspace involves AI and has two main implications for national security.³² The first involves the potential for AI use in computer defenses. In recent years, AI has proved capable of vastly increasing both the offensive and defensive cyber capabilities by finding and exploiting or plugging gaps in defenses. The second involves social-psychological applications. Currently, firms are developing AI to create psychological profiles of individuals to market goods and services to them.³³ AI applications are capable of processing vast amounts of information on individuals and conducting psychological

experiments on large populations in order to discover how to predict and manipulate their actions and beliefs. Experiments have demonstrated that this type of AI is often vastly better at predicting individual subjects' decisions than even subjects' friends and spouses.³⁴

Over the past half-decade, both Russia and China have taken advantage of AI for political purposes. The full extent of these operations is not public, but China is generally believed to spend vast resources on experimenting with AI to monitor and control its own population, and Russia has made ample use of marketing AI supplied by Facebook and other companies to target and manipulate foreign populations' political beliefs and passions. As AI advances, its potential for this type of social-political manipulation is likely to increase considerably.³⁵

At present, both the United States and China are investing heavily in AI research. However, from the perspective of national security, China has two significant advantages. The first is that China's political and economic systems allow it to provide much more data to government and corporate users than is the case in the United States.³⁶ This data is a critical asset in creating practical applications for AI. Second, for the purposes of national security, the U.S. Government invests far less in this technology than does the Chinese government.³⁷ Thus, while U.S. and Chinese companies are able to compete on a somewhat even basis, China's government has a nearly unassailable lead over the U.S. Government in terms of how each uses AI in geopolitical information.³⁸

The Big Picture and the Red Queen: The Problem with Current U.S. Defense Policy

The outcomes of the three technology contests will be critical to the larger contest for cyberspace. Looked at individually, each does not tell the whole story about China's overall strategy to gain control of information networks or how it will use them in geopolitics. China promulgated a broad manufacturing plan, Made in China 2025, that focuses on specific Chinese government support for these critical technologies for future Chinese dominance, but that aims for much more.³⁹

Regarding China's overall plan to gain control of cyberspace and dominate the information contest of the future, the three technologies are best seen as the tip of a much larger iceberg. According to the notion propounded by many Soviet and Chinese geopolitical strategists and also behind the U.S. third offset strategy, the United States defeated the Soviet Union in the Cold War because it was able to make up for its shortcomings in manpower and industrial capacity with advantages in science and technology.⁴⁰

To stay ahead in technologies such as the three described, China has dedicated enormous resources to education in science, technology, engineering, and mathematics (STEM). This policy has both a domestic and foreign component. At home, China produces around eight times as many STEM graduates as does the United States. Abroad, in 2016, 43 percent of students in U.S. science and engineering schools were Chinese. In computer-related U.S. college programs, only 21 percent were American.⁴¹

The skills these students bring home, combined with the technology Chinese businesses and intelligence agencies steal from U.S. and allied countries, create a Red Queen problem for U.S. technology.⁴² The faster the United States runs, the faster it must run to stay ahead. Since new U.S. technology is immediately taught to Chinese citizens by U.S.

professors and stolen by Chinese hackers, U.S. technological ingenuity and investment are not likely to overcome China's lead in manpower and industrial capacity the way it overcame that of the Soviet Union three decades ago. Because China has invested heavily in controlling the global flow of information, China freely garners the benefits of U.S. public and private investment in science and technology. As a result, it is free to focus its own investments into 5G, quantum capabilities, AI, and other technologies that will expand its ability to control the global information flow.⁴³ As a result, the United States cannot win the geopolitical information age contest simply by spending more on research. The Cold War paradigm will not work in the current era.

The Way Ahead: Prioritizing Information Over Physical Conflict

There have been several attempts to address the technology problems described in this chapter. The Department of Defense's third offset strategy attempted to address these issues by investing in new technology faster than China.⁴⁴ The National Security Strategy defines specific technologies the United States will pursue to keep its lead, particularly singling out those related to information technology.⁴⁵ The National Cyber Strategy further focuses on key information technologies the United States will pursue in its competition with Russia and China.

The problem with these approaches is not their specific proposals; it is more that they are written from an industrial era perspective. They tend to portray the contest for control of information and information technology as one more aspect of *national security policy* rather than the emerging foundation of future *geopolitical power*. While it is true that traditional industrial power projection capability will play a central role in small state competition for the foreseeable future, where China and Russia are concerned, information power is more likely than industrial power to determine the outcomes of long-term geopolitical contests.⁴⁶

A basic idea expressed—and then for the most part ignored—in each of the U.S. security policy documents referenced above is that the United States could do more to reduce threats to its interests and increase its geopolitical influence by solving information-based problems than by increasing its kinetic military power. Stopping Chinese intellectual property theft must be first on the agenda because little else will work until this problem is solved. This problem has been with us for a long time.⁴⁷ Securing U.S. networks against industrial espionage would considerably bolster the U.S. economy and decrease Chinese piracy-based economic growth. Further increasing the U.S. military lead over Russia or China does neither. Securing U.S. critical infrastructure against cyber attacks from Russia, China, Iran, and North Korea would significantly increase U.S. defenses and decrease each state's ability to deter or compel the United States in a crisis. More kinetic power would increase defenses and crisis bargaining power only marginally, if at all. Perhaps most important, no amount of investment in industrial era technology would do much to defend against the damage being done by autocratic states' political-psychological operations or help the United States respond with information operations with American characteristics.

America's long-time policy of paying lip service to the transition from the industrial to the information era—while mainly resourcing industrial era technologies and methods—was merely a curiosity when the United States was at its unipolar apex and did not face Great Power

competitors. That period is over. Today, the United States faces two autocratic rivals that are actively pioneering ways of using information age technologies and strategies to undermine the U.S.-led liberal international order. If the United States hopes to win this competition, it will need to change its approach. It will need to rethink the importance of defending its citizens, its firms, and its military from being quietly exploited by foreign militaries using steadily advancing cyber methods. It will need to resource its pursuit of key technologies that it has long prioritized on paper while paying whatever it takes to prevent adversaries from stealing that technology. It will have to decide how it will prevent adversaries from building and dominating the world's information networks and supply chains. Most important, it will have to change its mindset in such a way as to understand that it must defend virtual property and territory as it currently does to protect physical space. The industrial age is over, and it is time U.S. defense policy comes to terms with the emerging reality.

Notes

¹ Jonah Blank, "China Bends Another American Institution to Its Will," *The Atlantic*, October 10, 2019, available at <www.theatlantic.com/international/archive/2019/10/nba-victim-china-economic-might/599773/>.

² Richard B. Andres, "Cyber Conflict and Geopolitics by Richard B. Andres," *Great Decisions 2019*, Foreign Policy Association, 2019.

³ Andrew F. Krepinevich, *The Military-Technical Revolution: A Preliminary Assessment* (Washington, DC: Center for Strategic and Budgetary Assessments, 1992), 6.

⁴ Timothy L. Thomas, *Decoding the Virtual Dragon: Critical Evolutions in the Science and Philosophy of China's Information Operations and Military Strategy—The Art of War and IW* (Fort Leavenworth, KS: Foreign Military Studies Office, 2007).

⁵ For an overview of early Chinese thinking on this topic, see Timothy L. Thomas, "Nation-State Cyber Strategies: Examples from China and Russia," in *Cyberpower and National Security*, ed. Franklin D. Kramer, Stuart H. Starr, and Larry K. Wentz (Washington, DC: NDU Press, 2009).

⁶ On the Three Warfares doctrine, see Sangkuk Lee, "China's 'Three Warfares': Origins, Applications, and Organizations," *Journal of Strategic Studies* 37, no. 2 (2014), 198–221. For an overview of China's information-industrial policy, see William C. Hannas, James Mulvenon, and Anna B. Puglisi, *Chinese Industrial Espionage: Technology Acquisition and Military Modernization* (Abingdon, UK: Routledge, 2013). For an overview of underlying Chinese information and industrial strategy, see Dean Cheng, *Cyber Dragon: Inside China's Information Warfare and Cyber Operations* (Santa Barbara, CA: Praeger, 2016); Gregory C. Allen, *Understanding China's AI Strategy: Clues to Chinese Strategic Thinking on Artificial Intelligence and National Security* (Washington, DC: Center for New American Security, February 2019), available at <www.cnas.org/publications/reports/understanding-chinas-ai-strategy>.

⁷ Cheng, *Cyber Dragon*. See also Jacqueline Newmyer, "The Revolution in Military Affairs with Chinese Characteristics," *Journal of Strategic Studies* 33, no. 4 (August 2010), 483–504.

⁸ Richard Andres, "Air Power and Cyber," in *Routledge Handbook of Air Power*, ed. John Andreas Olsen (Abingdon, UK: Routledge, 2018), 203–214.

⁹ Such defenses include censorship, data localization, and cross-border data flow limits. See "Digital Trade," Congressional Research Service, March 29, 2019, available at <<https://>

crsreports.congress.gov/product/pdf/IF/IF10770>; Elsa Kania, "China: Active Defense in the Cyber Domain," *The Diplomat*, June 12, 2015, available at <<https://thediplomat.com/2015/06/china-active-defense-in-the-cyber-domain/>>.

¹⁰ IP Commission, *The IP Commission Report: The Report of the Commission on the Theft of American Intellectual Property* (Seattle: National Bureau of Asian Research, 2013), available at <http://ipcommission.org/report/IP_Commission_Report_052213.pdf>.

¹¹ IP Commission, *Update to the IP Commission Report: The Theft of American Intellectual Property: Reassessments of the Challenge and United States Policy* (Seattle: National Bureau of Asian Research, 2017), available at <http://ipcommission.org/report/IP_Commission_Report_Update_2017.pdf>.

¹² Jamie Crawford, "The U.S. Government Thinks China Could Take Down the Power Grid," CNN, November 21, 2014, available at <www.cnn.com/2014/11/20/politics/nsa-china-power-grid/index.html>.

¹³ "An Interview with Paul M. Nakasone," *Joint Force Quarterly* 92 (1st Quarter 2019), 4–9, available at <https://ndupress.ndu.edu/Portals/68/Documents/jfq/jfq-92/jfq-92_4-9_Nakasone-Interview.pdf>.

¹⁴ Jeffrey Engstrom, *Systems Confrontation and System Destruction Warfare: How the Chinese People's Liberation Army Seeks to Wage Modern Warfare* (Santa Monica, CA: RAND, 2018), available at <https://www.rand.org/pubs/research_reports/RR1708.html>.

¹⁵ "An Interview with Paul M. Nakasone."

¹⁶ Richard A. Clarke and Robert K. Knake, *Cyber War: The Next Threat to National Security and What to Do About It* (New York: Penguin, 2010), 158.

¹⁷ Chris Buckley and Paul Mozur, "How China Uses High-Tech Surveillance to Subdue Minorities," *New York Times*, May 22, 2019, available at <www.nytimes.com/2019/05/22/world/asia/china-surveillance-xinjiang.html>.

¹⁸ *National Strategy to Secure 5G of the United States of America* (Washington, DC: The White House, March 2020), available at <www.whitehouse.gov/wp-content/uploads/2020/03/National-Strategy-5G-Final.pdf>.

¹⁹ Of note—and an example of Moore's law of technological progress—while most consumers still have yet to upgrade to 5G wireless service around the world and its deployment is

still only in roll-out stages, Beijing is already signaling plans to develop 6G technology. See Peter Suci, "5G Is Old News: China Wants 6G for Its Military," *The National Interest*, April 28, 2020, available at <<https://nationalinterest.org/blog/buzz/5g-old-news-china-wants-6g-its-military-148806>>.

²⁰ Tom Wheeler, "5G in Five Not So Easy Pieces," Brookings, July 9, 2019, available at <<https://www.brookings.edu/research/5g-in-five-not-so-easy-pieces/>>.

²¹ Huawei argues that it would not share information with the Chinese Communist Party; however, China's Counter-Espionage Law of 2014 and National Intelligence Law in 2017 require companies to comply with requests from intelligence services. See Daniel Harsha, "Huawei, a Self-Made World-Class Company or Agent of China's Global Strategy?" Ash Center for Democratic Governance and Innovation at Harvard University, available at <<https://ash.harvard.edu/huawei-self-made-world-class-company-or-agent-chinas-global-strategy>>.

²² For a discussion of how China uses these capabilities domestically, see Kai-Fu Lee, *AI Superpowers: China, Silicon Valley, and the New World Order* (Boston: Houghton Mifflin Harcourt, 2018).

²³ On state funding, see Henry Tugendhat, "Banning Huawei's 5G Won't Halt China's Tech Revolution," *The Guardian*, January 30, 2020, available at <www.theguardian.com/commentisfree/2020/jan/30/banning-huawei-5g-china-tech-revolution-free-market>. On Huawei undercutting competitors, see Alex Capri, "Beijing's Global 5G Ambitions Threaten to Disrupt Telecoms," *Nikkei Asian Review*, August 31, 2018, available at <<https://asia.nikkei.com/Opinion/Beijing-s-global-5G-ambitions-threaten-to-disrupt-telecoms>>.

²⁴ The Chinese Communist Party has a history of using economic threats to censor U.S. companies' and individuals' speech. See Jennifer Pan and Margaret Roberts, "These 3 Factors Explain Why the NBA and Other Companies Struggle to Push Back Against Chinese Censorship," *Washington Post*, October 16, 2019, available at <www.washingtonpost.com/politics/2019/10/16/these-factors-explain-why-nba-other-companies-struggle-push-back-against-chinese-censorship/>.

²⁵ Quantum technology has numerous national security implications beyond those tied to cyberspace and information technology.

²⁶ See Princeton University's Center for Information Technology Policy, *Implications of Quantum Computing for Encryption Policy*, Encryption Working Group Paper (Washington, DC: Carnegie Endowment for International Peace, April 25, 2019), available at <<https://carnegieendowment.org/2019/04/25/implications-of-quantum-computing-for-encryption-policy-pub-78985>>. For a more detailed discussion, see Richard A. Clarke and Robert K. Knake, *The Fifth Domain: Defending Our Country, Our Companies, and Ourselves in the Age of Cyber Threats* (New York: Penguin Press, 2019), chapter 16.

²⁷ E.B. Kania and J.K. Costello, "Quantum Technologies, U.S.-China Strategic Competition, and Future Dynamics of Cyber Stability," *2017 International Conference on Cyber Conflict* (Washington, DC: Institute of Electrical and Electronics Engineers, 2017), 89–96.

²⁸ David Grossman, "Did Google Just Achieve Quantum Supremacy?" *Popular Mechanics*, September 23, 2019, available at <www.popularmechanics.com/technology/a29190975/google-quantum-supremacy/>.

²⁹ The usual argument is that encryption is used by billions of users, whereas decryption is used by only a handful of intelligence organizations. As a result, it will be easier for a small number of intelligence agencies working on decryption to obtain expensive new quantum devices than for the general public to obtain use of quantum computers.

³⁰ Paul Smith-Goodson, "Quantum USA vs. Quantum China: The World's Most Important Technology Race," *Forbes*, October 11, 2019, available at <www.forbes.com/sites/moorinsights/2019/10/10/quantum-usa-vs-quantum-china-the-worlds-most-important-technology-race/>.

³¹ Tim Johnson, "China Speeds Ahead of U.S. as Quantum Race Escalates, Worrying Scientists," *McClatchy DC*, October 23, 2017, available at <www.mcclatchydc.com/news/nation-world/national/national-security/article179971861.html>; Smith-Goodson, "Quantum USA vs. Quantum China."

³² For a general discussion of these security and related threats posed by artificial intelligence (AI), see U.S. National Security Commission on Artificial Intelligence (NSCAI), *Interim Report* (Washington, DC: NSCAI, November 2019), 11–13.

³³ For an overview of the national security implications of AI, see Clarke and Knake, *The Fifth Domain*.

³⁴ Frank Luerweg, "The Internet Knows You Better than Your Spouse Does," *Scientific American*, March 14, 2019, available at <www.scientificamerican.com/article/the-internet-knows-you-better-than-your-spouse-does/>.

³⁵ Regarding China's resourcing, see Tiffany Lo, "Big Brother Is Watching You! China Installs 'The Worlds Most Advanced Video Surveillance System' with over 20 Million AI-Equipped Street Cameras," *Daily Mail*, September 25, 2017, available at <www.dailymail.co.uk/news/article-4918342/China-installs-20-million-AI-equipped-street-cameras.html>. On the Russian use of Facebook and other commercial AI, see Alina Polyakova, "Weapons of the Weak: Russia and AI-Driven Asymmetric Warfare," Brookings, November 15, 2018, available at <<https://www.brookings.edu/research/weapons-of-the-weak-russia-and-ai-driven-asymmetric-warfare/>>.

³⁶ See NSCAI, *Interim Report*, 17–20.

³⁷ See Martijn Rasser et al., *The American AI Century: A Blueprint for Action* (Washington, DC: Center for New American Security, December 2019), 9, available at <https://s3.amazonaws.com/files.cnas.org/documents/CNAS-Tech-American-AI-Century_updated.pdf?mtime=20200103081822>; NSCAI, *Interim Report*, 19.

³⁸ NSCAI, *Interim Report*, 1, 17–20.

³⁹ Released in 2015, but subsequently deemphasized in 2018 and 2019 by the Chinese government when "Made in China 2025" sparked international controversy, the document's goal was to comprehensively upgrade Chinese industry, making it more efficient and integrated so that it can occupy the highest parts of global production chains. The plan identified the goal of raising domestic content of core components and materials to 40 percent by 2020 and 70 percent by 2025. The focus for attaining dominance is aimed at 10 priority sectors: new advanced information technology, automated machine tools and robotics, aerospace and aeronautical equipment, maritime equipment and high-tech shipping, modern rail transport equipment, new energy vehicles and equipment, power equipment, agricultural equipment, new materials, and biopharma and advanced medical products. Quite clearly, dominance in new information technology was "aim #1," but within a wider construct of aims for modern technology dominance. See Scott Kennedy, "Made in China 2025," Center for Strategic and International Studies, June 1, 2015, available at <<https://www.csis.org/analysis/made-in-china-2025>>; Melissa Cyrill, "What Is Made in China 2025 and Why Has It Made the World So Nervous?" *China Briefing*, December 28, 2018, available at <www.china-briefing.com/news/made-in-china-2025-explained/>.

⁴⁰ Paul McLeary, "The Pentagon's Third Offset May Be Dead, But No One Knows What Comes Next," *Foreign Policy*, December 18, 2017, available at <foreignpolicy.com/2017/12/18/the-pentagons-third-offset-may-be-dead-but-no-one-knows-what-comes-next/>.

⁴¹ Arthur Herman, "America's High-Tech STEM Crisis," *Forbes*, September 10, 2018, available at <www.forbes.com/sites/arthurherman/2018/09/10/americas-high-tech-stem-crisis/#f6730d8f0a25>.

⁴² The hypothesis is mainly used in evolutionary biology and is based on Lewis Carroll's *Through the Looking-Glass*. It is termed *Red Queen* because populations have to "run" or evolve to stay in the same place, or else go extinct. See Leigh Van Valen, "A New Evolutionary Law," University of Chicago, 1973, available at <<https://www.mn.uio.no/cees/english/services/van-valen/evolutionary-theory/volume-1/vol-1-no-1-pages-1-30-1-van-valen-a-new-evolutionary-law.pdf>>.

⁴³ China outlines this strategy in its Medium- and Long-Term Plan for Science and Technology Development, 2006–2020. The plan calls for copying foreign innovation and focusing resources to achieve breakthroughs in targeted strategic areas of technological development and basic research. For an assessment of the plan and its outcome, see Hannas, Mulvenon, and Puglisi, *Chinese Industrial Espionage*. Private sources estimate the annual value of intellectual property stolen by China from the United States at \$225 billion to \$600 billion. See IP Commission, *The IP Commission Report*.

⁴⁴ McLeary, "The Pentagon's Third Offset May Be Dead, But No One Knows What Comes Next."

⁴⁵ *National Security Strategy of the United States of America* (Washington, DC: The White House, December 2017).

⁴⁶ For a similar perspective, see Michael C. Horowitz, "Artificial Intelligence, International Competition, and the Balance of Power," *Texas National Security Review* 1, no. 3 (May 2018).

⁴⁷ For example, see James Andrew Lewis, *Intellectual Property Protection* (Washington, DC: Center for Strategic and International Studies, January 2008), available at <https://csis-prod.s3.amazonaws.com/s3fs-public/legacy_files/files/publication/080802_LewisIntellectualProperty_Web.pdf>.

Chapter 7

Social Media and Influence Operations Technologies

Implications for Great Power Competition

By Todd C. Helmus

Nation-states have increasingly been waging foreign propaganda campaigns on social media platforms such as Facebook and Twitter. Such campaigns are enticing because they are cheap and easy to execute; they allow planners to identify, target, and reach specific audiences; and the campaign's anonymity limits the associated political and foreign policy risks. Russia, China, and the so-called Islamic State are three key U.S. adversaries that have exploited online technologies for propaganda. This chapter reviews the aims, capabilities, and limitations of online propaganda for each of these entities. The chapter also highlights key recommendations that the United States should adopt in order to counter adversary use of online propaganda.

As the world has entered a new era of Great Power competition over the past decade, nation-states have been increasingly waging foreign propaganda campaigns on social media platforms such as Facebook and Twitter, effectively turning such platforms into influence operations technologies.¹ A study from the University of Oxford documented that some 70 countries around the world are engaged in manipulating social media to serve domestic and foreign policy ends. This is up from 48 countries in 2018 and 28 countries in 2017. In particular, the study documented foreign propaganda campaigns conducted by Russia, China, India, Iran, Pakistan, Saudi Arabia, and Venezuela.²

Why are states increasingly relying on social media as a tool of foreign propaganda? It is cheap and easy to operate and allows campaign planners to identify, target, and reach specific overseas audiences, such as individuals, voter demographics, and ethnic groups. Governments also seek to engage in such campaigns anonymously, thereby limiting the associated political and foreign policy risks. The campaigns can also be conducted at scale, and they can be informed by a wealth of easy-to-access big data on target audiences.

States conduct online propaganda campaigns in a number of ways, including using “bot” and “troll” accounts. Bots are automated social media accounts, often on Twitter, that employ code to replicate human activity to promote a particular message. To enable more

sophisticated interactions with other users, bot campaigns employ real people or trolls to monitor and control fake social media accounts. Campaigns employ these types of accounts in numerous ways. They can spread progovernment content, attack adversary positions, distract or divert conversations or criticism away from an issue, promote divisions and polarization, and suppress participation through attacks or harassment.³

This chapter offers a look at how three contemporary U.S. adversaries have used and are using online content and platforms to engage in foreign influence campaigns. It focuses on Russia, China, and the so-called Islamic State (IS). The first two are, as defined earlier in this volume, America's modern Great Power rivals. As noted in the 2018 U.S. National Defense Strategy, and discussed in detail in chapter 11, IS remains, despite major recent setbacks, a modern violent extremist organization with global reach and sustained influence that will challenge America into this Great Power era. For each case study, the chapter identifies the adversary's aims and objectives in using online technologies for influence operations and propaganda. It also identifies the capabilities and limitations for using online propaganda. In addition, because online platforms are not the only means of disseminating propaganda, the chapter briefly describes relevant offline mechanisms for influence. The chapter does not explicitly list U.S. Government aims, objectives, capabilities, or limitations for using social media and online technologies for external propaganda, instead offering an implicit assessment of these technologies in the final section on recommendations. This approach for the chapter has been selected in part because of the limited information available in open-source reporting.

Russia

Aims and Objectives

In a recent report on hostile social manipulation, RAND political scientist Mike Mazarr and his colleagues identified several key strategic aims for Russian online operations. First, they noted that Russia has long believed it is a target of adverse information operations, and Moscow may use its social manipulation efforts to counter these "disinformation" programs. Second, they noted that Russia uses social manipulation to pursue what it calls "discrete policy objectives" to influence a particular policy debate or foreign policy in order to suit its interests.⁴ In addition, they noted that some analysts see an animus to push foreign societies to a "posttruth" environment—one in which the distinction between fact and falsehood is immaterial, objectivity is unattainable, and reality is malleable."⁵

Diego Martin and Jacob Shapiro offered a glimpse of Russian influence objectives by analyzing a specially created database of influence campaigns waged between 2013 and 2018. The campaigns were waged by Russia, China, Saudi Arabia, and Iran and targeted 14 countries, including the United States, the United Kingdom, Australia, Germany, the Netherlands, and Ukraine.⁶ Russia accounted for some 38 of 53 identified influence efforts, highlighting the obvious importance Russia places on this aspect of its foreign policy.

Capabilities

First, we should note that Russia benefits from a broad set of capabilities for influencing overseas governments and publics. Elizabeth Bodine-Baron and colleagues specifically identified four key categories of capabilities.⁷ The first category includes actors that are part of the Russian state, such as the Main Intelligence Unit (*Glavnoye razvedyvatel'noye upravleniye*) or Sputnik online media. Second is the RT international news network, which is a nonprofit news organization that is visibly supported by the Russian state. Third are those actors who knowingly work on behalf of the Russian government but whose connections to the state are concealed. This includes the Internet Research Agency (IRA; also known as the St. Petersburg troll factory) and patriotic Russian hackers and networks run by criminal oligarchs. Last, in the fourth category, are the various proxies and potential proxies. These include those actors who may or may not hold pro-Russian views but are nonetheless motivated to spread messages in line with Russian campaign objectives.

The media and messages produced by these different arms often work together in a seemingly systematic manner. Following the poisoning of former Russian spy Sergei Skripal, various channels, combined with official press statements and Russian bots and trolls, produced “a blizzard of falsehoods” designed to muddy the waters of international investigations and opinion on Russian blame for the assassination.⁸

The most famous example of Russian propaganda stems from its systematic campaign to target the U.S. 2016 election. Russia employed a high-volume array of social media content to include 10.4 million tweets, 1,000 YouTube videos posted on 17 accounts, 116,000 Instagram posts from 133 accounts, and 61,500 unique Facebook posts across 81 pages. These postings yielded 77 million engagements on Facebook, 187 million engagements on Instagram, and 73 million engagements on original Twitter content.

The content sought to promote wide-ranging themes specially targeted at different U.S. demographic groups. Targets included African-American communities to promote black separatism, inflame opinions toward police, and undermine confidence in the electoral system. Content engaged in voter suppression tactics included promoting third-party candidates, encouraging voters to stay home on election day, and creating confusion about voting rules. Beginning in the primaries and continuing through the election, the content promoted pro-Trump operations and countered Hillary Clinton.⁹

Many of these competing themes appeared to promote disunity among the American electorate. Campaigns sought to promote both Black Lives Matter and Blue Lives Matter themes to inflame opinions. Skillful manipulation of social media even proved successful at promoting dual street protests.¹⁰ Other campaigns sought to promote secession by the state of Texas, anti-immigrant causes, gun rights, patriotism, Tea Party culture, and so forth.¹¹

Russia has since engaged in a number of other online influence campaigns. For example, Russia supported the Brexit referendum in 2016, heavily promoted Catalonia's independence referendum in 2017, and sought to undermine the presidential election of Emmanuel Macron in France. Statistics tabulated by Martin and Shapiro suggest that Russia engaged in at least 28 campaigns in 2017 and 21 in 2018.¹²

Overall, Russia appears to demonstrate a relatively skilled approach to these information operations. First, we should note that Russia was one of the first countries to recognize the potential value of using social media for Great Power competition, and Russia's efforts to

implement social media-based information operations have unfortunately paved the way for other nations, such as Iran and China, to follow suit. Second, by some accounts, Russia implemented the campaign with skill. IRA staffers visited the United States in order to conduct “market research.” IRA social media accounts clustered around identity- or issue-based online communities and built sizable audiences—all without tipping off audiences (at least so far) to the Russian origin of the campaigns. Russian accounts were retweeted by political figures and news media. As Tom Uren and colleagues noted, the “IRA campaign displayed a clear understanding of audience segmentation, colloquial language, and the ways in which online communities framed their identities and political stances.”¹³

Why is Russia so adept at using social media and other online channels of influence? Some analysts point to a 2013 article published on “ambiguous warfare” by the chief of the Russian general staff and general of the army, Valery Gerasimov, which led to some scholars believing that the Russian information campaigns are the result of an “elaborate strategy” developed and executed by Russian planners.¹⁴ Sergey Sanovich, a researcher at Princeton’s Center for Information Technology Policy, alternatively suggests that Russia’s online propaganda tools were “conceived and perfected” in the liberal Russian economy and politics of the 1990s. After the 1990s, Russia proved unsuccessful at using its network of bots and trolls to curb domestic online discussions, and the government was unwilling to “ban the platforms outright.” Consequently, it then adapted and stepped up its online international influence game.¹⁵

Limitations

The Russian campaign may have been skilled, but was it effective? Clearly, audiences have seen and interacted with Russian content. However, clear scientific evidence is lacking on whether such social media campaigns helped Russia meet any particular campaign goals or change audience attitudes, behaviors, or beliefs in the ways that the Russian planners intended.¹⁶

Interestingly, one study assessed the IRA’s impact on political attitudes and behaviors of American Twitter users in late 2017. The study identified users who interacted with IRA content on Twitter and analyzed whether outcomes of six distinctive measures of political attitudes and behavior changed over a 1-month period. The study found no such evidence of change and suggested that the Russian trolls might have failed because they directly targeted those who were already highly polarized in their attitudes and beliefs.¹⁷

Still, the campaign’s ingenuity and scope have garnered it significant attention from politicians, journalists, researchers, and the American public. The social media campaign, combined with other elements of a broader Russian interference campaign—including hacking the Democratic National Committee (DNC) server and the release of DNC emails—has led at least some audiences to question the legitimacy of the 2016 election.¹⁸ Such an outcome may well lead Russian planners and politicians to conclude that the campaign was a success.

One key limitation or challenge for Russia and other disinformation actors is that Facebook, Twitter, and other such platforms are now on the lookout for Russian content. Bots will need to be more sophisticated to overcome the various bot detectors that exist in the

market. Russia will no longer be able to pay for advertisements with rubles or post content directly from IRA-affiliated computers.

An arms race will certainly ensue. Russians will likely adapt to any countermeasures Western governments or online platforms put in place. Evidence suggests this is already taking place. Russia has sought to test new disinformation tactics on the African continent. Instead of creating fake Facebook groups from the IRA offices in St. Petersburg, it has rented or purchased accounts already created and cultivated by locals. Russia also has created local media organizations in select African countries that post the content on behalf of Russia. The seemingly authentic nature of the content will make detection more difficult.¹⁹

China

Aims and Objectives

By documenting the expansion of the Communist Party media influence since 2017, analyst Sarah Cook identified several broad and overarching aims of the Chinese Communist Party (CCP) communication and influence strategy. First, she noted the government seeks to “promote a positive view of China and the CCP’s authoritarian regime.” By the same token, it also seeks to “marginalize, demonize, or entirely suppress anti-CCP voices” and other information that might cast a negative light on its government or leaders. China also seeks to promote nationalistic sentiment at home and abroad, promote the reunification of Taiwan with the mainland, and quell the protests in Hong Kong.²⁰

Capabilities

China has gained a strong reputation for effectively stifling and influencing online debate within its borders, and it has fostered a number of seemingly effective offline tools for international influence. However, China’s ability to use online tools to influence international policy and opinion remains in a relatively nascent state.

Censorship and Influence at Home

China uses both the “Great Firewall” and its “Golden Shield” to stifle dissent at home. The Great Firewall blocks access to restricted foreign Web sites. If a China-based user tries to access a restricted site, it will not load and the user will receive a time-out message.²¹ Additionally, China uses its Golden Shield to regulate information on domestic sites. According to Gillian Bolsover at the University of Oxford, social media sites in China actively monitor user-generated content to ensure that posted information is not deemed illegal by the state. Examples of content often censored include information related to political scandals, political leaders, and efforts to organize protests.²²

China also actively seeks to shape the social media-based conversations and discussions of its citizens. Starting in 2011, the Chinese government saw a need to do more than just censor online content; it engaged in political communication. The Central Party, state institutions, state-run media, and individual party cadres soon began setting up government social media accounts. By 2014, the government had created more than 100,000 official social media accounts on WeChat and 180,000 profiles on Sina Weibo.²³

One potential tool for this internal information control is the use of 50-cent accounts. Academics and policy experts have written about an army of volunteers who are paid 50

cents per post to “attack critics and support the state” online and to do so in a way that appears these attacks come from ordinary people.²⁴ A 2017 Harvard paper studied a data leak associated with the 50-cent accounts. The authors estimate that the government fabricates and posts about 448 million social media comments a year. Instead of engaging in direct arguments with potential skeptics of the government, these bogus users often try to change the subject and engage in cheerleading for China and the CCP.²⁵ Much of this work is done by government employees who post part time outside their regular day jobs.

Influence Abroad: “Offline” Capabilities

Although the focus of this chapter is online influence operations and propaganda technologies, China has built a robust and multimodel offline approach to influence overseas populations and governments. This approach was addressed earlier in chapter 3b but merits brief repetition here. First, China has built an expanding capacity for global media reach. China’s most prominent state-owned media outlets offer an international presence.²⁶ The China Global Television Network (CGTN), for example, broadcasts in English, Spanish, French, Arabic, and Russian to every region in the world via satellite and cable. In addition, Chinese state media have distributed content associated with the *China Daily* and the *Washington Post* in newsstands in New York City and congressional offices in Washington. Diplomats and other key influencers draft various op-ed articles and assiduously work to build relationships with foreign journalists. China also often threatens to withhold access to its markets if representatives of various business interests do not toe the party line.²⁷

In addition, China’s state media organs have broadened their reach with the use of social media. The English-language Facebook pages for *China Daily*, the official Xinhua News Agency, and CGTN, according to disinformation researcher Renee DiResta, have amassed more than 75 million followers each, a sum two to three times greater than CNN or Fox News. DiResta argues that China’s heavy use of paid social media advertisements played a key role in cultivating this large following.²⁸

Influence Abroad: Chinese Online Capabilities

China has attempted to use online tools, including fake social media accounts, to advance its Taiwan unification campaign and its efforts to counter the Hong Kong protests. China has engaged in several small-scale efforts at using social media to promote Taiwanese unification. In one interesting case, following a typhoon that disabled a bridge to Osaka’s Kansai International Airport, a post on the Professional Technology Temple (PTT), a Taiwanese-focused bulletin board, falsely suggested that the Chinese consulate could evacuate Taiwan citizens from Osaka only if they identified themselves as Chinese citizens. Researchers traced the story on PTT to an account on the Chinese microblogging site Weibo and a “content farm” that posted on mainland media sites, where it was then picked up by PTT messaging.²⁹ The consequences were harmful: Taiwan’s foreign ministry representative in Osaka committed suicide due to pressure stemming from his inability to aid Taiwanese citizens.

In another case, it was discovered that a large number of PTT accounts, some of which were deemed “influential,” were purchased on an online auction site active in Taiwan and Southeast Asia. Many of these accounts switched their content from being pro-democratic

to leaning pro-Chinese. The accounts posted at a time to allow the Taiwanese public to see the posts first thing in the morning.³⁰

China has most recently been caught using sham Facebook and Twitter accounts in an attempt to counter the Hong Kong protests. On August 19, 2019, both Twitter and Facebook announced the discovery of the Chinese campaign. Twitter identified 936 accounts that originated within the People's Republic of China that were "deliberately and specifically attempting to sow political discord in Hong Kong, including undermining the legitimacy and political positions of the protest movement on the ground."³¹ Fake accounts were also detected on Facebook and YouTube.³²

Subsequently, the Australian Strategic Policy Institute (ASPI) released a detailed analysis of the archive of terminated Twitter accounts. The researchers discovered that the 940 false accounts had disseminated 3.6 million tweets and identified three key narrative themes of the accounts: condemnation of the protesters, support for the Hong Kong police and the "rule of law," and conspiracy theories about Western involvement in the protests.³³ The authors described the campaign as relatively small and "hastily assembled" and lacking in sophisticated advance planning. They observed that the accounts were cheaply acquired repurposed spam or marketing accounts. Prior account owners, for example, tweeted in Arabic, English, Korean, Japanese, and Russian on topics that ranged from British football to pornography.³⁴

The researchers also observed that there was "little attempt to target online communities with any degree of psychological sophistication."³⁵ In contrast, they noted that carefully crafted and long-running influence operations on social media, such as those conducted by the Russian state, are often characterized by tight network clusters associated with key target audiences. Analysis of the Chinese database revealed no such network characteristics. They finally observed that the Chinese dialect in some of the tweets was a dead giveaway for Chinese mainland authors.³⁶

Beyond the anti-Hong Kong protest campaign, the researchers at ASPI found evidence of relatively small campaigns targeting China's political opponents. The largest campaign targeted Guo Wengui, a Chinese businessman and bookseller now residing in the United States who has publicly levied allegations of corruption against senior members of the Chinese government. Over 38,000 tweets from 618 accounts targeted Wengui with vitriolic attacks on his character.³⁷ Two other smaller campaigns targeted two dissidents who had already been arrested in China.³⁸

With the 2019–2020 outbreak of the novel coronavirus (COVID-19), China launched a new propaganda campaign partially aimed at avoiding blame for the virus and promoting its own relief efforts. First, China has pushed social media content arguing that the virus may not have originated from China. On March 7, for example, the Chinese embassy in South Africa tweeted, "Although the epidemic first broke out in China, it did not necessarily mean that the virus is originated from China, let alone 'made in China.'"³⁹ The tweet further speculated that the virus originated in the United States. Chinese media promoted a conspiracy theory that a U.S. military cyclist may have brought the disease to Wuhan from Fort Detrick, the location of the U.S. Army's premier biological laboratory. The spokesman and deputy director general of the Information Department of China's Foreign Ministry

also speculated on Twitter that the United States secretly concealed COVID-19 deaths in its count of flu fatalities.⁴⁰

Second, China has promoted its domestic and international response to COVID-19. In February, Chinese state-run media began running social media advertisements that praised Secretary General Xi Jinping for his leadership in containing the virus.⁴¹ On March 9, the official account of China's Ministry of Foreign Affairs tweeted, "China's endeavor to combating the epidemic has bought time for [international] preparedness."⁴² China also promoted its international relief efforts. State-run media pushed advertisements that promoted stories of countries such as Italy and Serbia expressing gratitude to China for supporting them with medical supplies.⁴³ Pro-China bots pushed out tweets promoting Chinese medical relief efforts in Italy with two hashtags: #forzaCinaeItalia, which means "Come on China and Italy," and #grazieCina, which means "Thank you, China." Chinese diplomatic Twitter accounts also used these hashtags.⁴⁴

Limitations

The discovered Chinese online propaganda campaigns targeting Taiwan and protesters in Hong Kong suggest that China has struggled to weaponize social media to influence audiences abroad. China clearly has had success at home in terms of effectively censoring illicit content on the Web and shaping online conversations. The control it enjoys domestically over the Internet is not so easily replicated abroad, where it must contend with competing narratives that cannot be suppressed.⁴⁵

China will likely learn its lesson. On the same August 2019 day that Twitter announced China's suspensions, China's Internet regulator put out notice for a contract to help it "operate and grow" overseas social media accounts on platforms such as Facebook. The project sought a team of experts who could "tell China's stories with multiple angles, express China's voice, and get overseas audience recognition and support for Jinping Thought." The state news agency, China News Services, also announced that it has started a new project to build its social media presence overseas. It specifically seeks to increase Twitter followers on its 2 accounts by 580,000 within 6 months. It wants at least 8 percent of the accounts to come from North America, Australia, and New Zealand. Overall, China is spending more than \$1 million on both accounts.⁴⁶

Further aiding China will be its investment in artificial intelligence, which, if effectively integrated in highly scaled social media campaigns, could prove a serious social media threat.⁴⁷ China's stake in the rapidly growing TikTok social media application could also help further the country's message. Suspicions have already arisen about China using the application to promote censorship and manipulation. Growing access to a widening public could be an information advantage for the Chinese and give them a new platform for influence.

The So-Called Islamic State and Social Media

Aims and Objectives

According to a leaked strategy document, IS had three main aims for its information operations campaign: recruitment, governance, and media.⁴⁸ First, IS sought to increase the

recruitment of local and foreign fighters into the organization. Second, IS developed a plan that included a sophisticated information and intelligence apparatus designed to help IS expand and maintain control over its territory. Finally, IS's well-crafted media system sought to empower its recruitment and governance efforts while strengthening its embrace of atrocities intended to deter opponents and energize supporters with maximum psychological impact.

Other studies have sought to understand IS information objectives and aims by examining the content of its propaganda campaigns. Researchers in a RAND study isolated a community of ardent IS members and supporters on Twitter and lexically analyzed the content. They found that IS appeared to demonstrate "a more self-aware social media strategy" than any of the other groups, with disproportionately high usage of social media terms such as *spread*, *link*, *breaking news*, and *now released*.⁴⁹ In addition, themes of religion and belonging resonated strongly. In reference to violent IS activities, users employed noble phrases such as *lions of the Islamic State* and *mujahideen* and coopted the trappings of real states by using terms such as *army* and *soldiers of the Caliphate*.⁵⁰

Capabilities

To understand IS capabilities for using online tools in radicalization and recruitment, it is important to first understand the evolution of extremist radicalization and recruitment. Al Qaeda and other militants in places such as Bosnia, Chechnya, and the Palestinian territories used to disseminate propaganda content via DVD and cassette videos. Many of these videos featured depictions of atrocities that were meant to inflame opinions and sermons of religious leaders who sought to lay the intellectual and spiritual groundwork for terrorist actions. Recruitment into militant groups was often done through small social groups. For example, gatherings at private homes or mosques targeted Saudi recruits for Iraq and served as a venue where a "harmless discussion about Islam" turned to the U.S. war in Iraq and U.S.-committed atrocities.⁵¹

This approach to extremist propaganda and recruitment began to evolve slowly as al Qaeda turned to the Internet to aid in recruitment efforts. After initially disseminating speeches on al Jazeera, Osama bin Laden and his deputy, Ayman al-Zawahiri, began posting long and dry speeches. Anwar al-Awlaki, a key al Qaeda ideologue, gained particular fame through sermons posted to YouTube. His success as a propagandist led to him being targeted by U.S. forces in Yemen.⁵² During the Iraq War, al Qaeda gained notoriety by posting videos of improvised explosive device attacks on American troops.⁵³ Throughout this process, al Qaeda and other groups used the Internet primarily as a broadcast tool, but limited effort was placed into harnessing social media platforms or the social aspects of online life.

IS helped revolutionize how extremist groups used the Internet and its social media platforms. IS decentralized its propaganda production and dissemination, used a multilayered set of media centers to produce its official media publications, and then leveraged a network of supporters in order to distribute its message. It also learned how to use direct messaging capabilities to reach out and connect to prospective recruits. A range of outlets helped IS produce its official propaganda. Official outlets Al-Furqan Media and Al-Hayat Media produced and released top-level messaging, such as the sermons of Abu Bakr al-Baghdadi and issues of the

sleek English-language magazine *Dabiq*. Various “provincial” media outlets also produced an assortment of content and propaganda videos.

IS sought to disseminate this content not only on various Internet sites but also through a variety of social networking applications, including Facebook, Instagram, Tumblr, Ask.fm, and, most famously, Twitter. These channels allowed IS to enlist a worldwide army of supporters in both propaganda creation and dissemination. In addition to disseminating its own content on the Internet, IS welcomed supporters and members taking the initiative to create their own unofficial propaganda. In her 2014 study of tweets posted by 29,000 accounts belonging to Western IS foreign fighters, Jytte Klausen found that disseminators outside the conflict zone, including a handful of particularly influential women, played a key role in this effort to “build redundancy by spreading the material, often posting and reposting material provided by the feeder accounts belonging to organizations and fighters based in Syria.”⁵⁴

IS also worked with individual fighters who had their own Twitter, Instagram, or Tumblr accounts. IS worked to coordinate and synchronize the postings of fighters. These tales describing life and success on the battlefield gained a wide following.⁵⁵ In addition, a report by the International Centre for the Study of Radicalisation and Political Violence emphasized the role of disseminators based primarily in the West, arguing that many foreign fighters informed themselves about the conflict by following unofficial disseminators rather than official IS channels.⁵⁶

IS also used social media to support its recruitment efforts. IS enlisted “units of specialized recruiters operating around the clock from Internet cafes in Iraq and Syria, interacting on an individual level with prospective recruits.”⁵⁷ Individuals who liked, retweeted, or positively commented on IS social media accounts self-identified themselves as potentially suitable targets for recruitment. Recruiters could then make contact using private communications such as direct messaging on Twitter or Facebook. The two-way dialogue allowed the recruiter to groom the target and promote action (be it conducting attacks in the West or emigrating to IS territory). As J.M. Berger notes, such recruitment efforts became increasingly important as IS sought to recruit from lands farther than Iraq and Syria.⁵⁸

Overall, these efforts appeared successful. IS gained significant media exposure and notoriety for the conduct of its social media campaign. Ultimately, it recruited over 40,000 people (32,809 men, 4,761 women, and 4,640 children) to join its ranks. Over 5,900 foreign fighters joined from Western Europe and 753 joined from the Americas.⁵⁹ IS operatives also launched various deadly attacks, most notably the Brussels airport and subway bombings of March 22, 2016, which killed 32 individuals, and the Paris attacks of November 13, 2015, which resulted in 130 fatalities. IS-inspired attacks also took place in the United States, Canada, Australia, Tunisia, Turkey, and Egypt.⁶⁰

Limitations

IS success with social media did not last. In 2015, Twitter and other social media firms initiated an effort to suspend IS social media accounts based on terms of service violations. IS in turn worked diligently to overcome these suspensions. As soon as Twitter suspended one IS supporter’s account, the supporter was supposed to move to a backup account.⁶¹ However,

even under the best of circumstances, such remediation efforts could not maintain the previous breakneck pace of propaganda dissemination.⁶² Soon the platforms created new artificial intelligence tools that could detect and terminate accounts in near real time. In the first 6 months of 2017, Twitter took down nearly 300,000 terrorist accounts.⁶³ As a result, Maura Conway and colleagues wrote in their research on Twitter takedowns that “the costs for most pro-IS users of engaging on Twitter (in terms of deflated morale, diffused messages, and persistent effort needed to maintain a public presence) now largely outweigh the benefits. This means that the IS Twitter community is now almost nonexistent.”⁶⁴

IS has since sought new online territory from which to recruit and radicalize. For some time, IS turned to Telegram, an application that, while lacking the unique broadcast capability of Twitter, allowed IS a secure way to “communicate with likeminded supporters across the world, disseminate official and unofficial [IS] media, and provide instructional material for operations.”⁶⁵ However, a series of account suspension efforts by Telegram, mostly recently conducted in collaboration with Interpol in late 2019, has left IS once again looking for more friendly territory.

Recommendations

The growing use of social media as a technology tool for strategic influence operations and nation-state propaganda represents a significant threat to U.S. interests as America moves into a new era of Great Power competition. While Russian social media may not have decisively impacted the U.S. election of 2016, it is clear the campaign likely negatively impacted American trust in that election. Additionally, it seems clear that such attacks will continue against the United States, its allies, and other nations’ democratic elections. As demonstrated by its assertive social media activities during 2020 to shape a factually suspect COVID-19 narrative, Beijing has targeted the United States and its allies with online propaganda. Clearly, the United States needs to safeguard the authority, legitimacy, and respect of American norms, values, and institutions from such adversaries.

Numerous documents and reports lay out an array of recommendations for how the United States can best counter this threat.⁶⁶ Reviewing the long lists of recommendations is beyond the scope of this chapter; however, it is useful to briefly consider some broad approaches that the United States could undertake to limit the threat.

Track, Highlight, and Block Adversarial Content

The U.S.-led and Western campaign against IS propaganda proved successful largely because social media platforms were able and willing to identify IS content and terminate IS social media accounts. Overall, the platforms seem uniquely able to identify and target such content in part because extremists clearly branded their campaigns for driving terrorist recruitment. The U.S. Government should continue to work with Twitter, Facebook, and other platforms to ensure suspensions of extremist accounts. The challenge is much greater for nation-state campaigns, which wage stealth propaganda using fake accounts and targeting a litany of causes. An arms race is already taking place between those charged with planning and executing online propaganda and those seeking to detect and remove such campaigns. The U.S. Government will need to work closely with technology firms and the

academic community to support this arms race and enable new approaches for detecting campaigns.

Build Resilience of At-Risk Populations

It will be crucial to help audiences be more critical consumers of social media. Doing so means giving audiences the skills and capabilities to identify fake news, consider the credibility of sources on social media, and recognize their role in countering such content or limiting its propagation. Media literacy campaigns represent one potential avenue for this resilience-building, and efforts are under way to develop and implement relevant educational curriculums in the United States and elsewhere. Governments should also warn citizens when they detect or otherwise suspect that adversaries are targeting such citizens with online influence campaigns.

Support Allies Targeted by U.S. Adversaries

The United States should help its allies stand up against online propaganda. For example, Russia has been engaged in a near-persistent propaganda campaign against the government of Ukraine as well as other Eastern European countries. The United States should work with the targeted countries to give them the necessary capabilities to withstand and counter these campaigns. The specific policy prescriptions will vary, but efforts may include training local governments in better communication strategies, improving training for journalists, providing funds to support outing adversary propaganda, and establishing media literacy campaigns.

Better Organize to Counter Adversary Propaganda

The U.S. Government must ensure it is properly organized to fight online disinformation. The Intelligence Community will need the necessary capabilities and funding to help detect foreign influence campaigns before or as they occur. Interagency coordination will be critical as the Department of Homeland Security, Department of State, and Department of Defense will each be responsible for countering a particular component of adversary campaigns.⁶⁷ Coordination should continue down to the state level to help states prepare for elections and address locally targeted campaigns. It is critical that the United States and social media firms work closely together in an information-sharing capacity to ensure communication of threats and adversary campaigns and to coordinate on counterpropaganda activities.

Notes

¹ This chapter uses the terms *propaganda* and *disinformation*. Merriam-Webster defines *propaganda* as “Ideas, facts, or allegations spread deliberately to further one’s cause or to damage an opposing cause.” *Disinformation* is considered a type of propaganda. Christina Nemr and William Gangware describe disinformation as including authentic material used in a deliberately wrong context to make a false connection as well as fake news sites, manipulated sites, or outright false information shared through graphics, images, and videos. See Christina Nemr and William Gangware, *Weapons of Mass Distraction: Foreign State-Sponsored Disinformation in the Digital Age* (Washington, DC: Park Advisors, 2019).

² Samantha Bradshaw and Philip N. Howard, *The Global Disinformation Order: 2019 Global Inventory of Organized Social Media Manipulation*, Working Paper No. 2019.3 (Oxford, UK: Project on Computational Propaganda, 2019).

³ *Ibid.*, 1, 13.

⁴ Michael J. Mazarr et al., *Hostile Social Manipulation: Present Realities and Emerging Trends* (Santa Monica, CA: RAND, 2019).

⁵ *Ibid.*, 61.

⁶ Diego A. Martin and Jacob N. Shapiro, *Trends in Online Foreign Influence Efforts*, Working Paper Version 1.2 (Princeton: ESOC Publications, 2019).

- ⁷ Elizabeth Bodine-Baron et al., *Countering Russian Social Media Influence* (Santa Monica, CA: RAND, 2018).
- ⁸ Joby Warrick and Anton Troianovski, "Agents of Doubt: How a Powerful Russian Propaganda Machine Chips Away at Western Notions of Truth," *Washington Post*, December 10, 2019, available at <www.washingtonpost.com/graphics/2018/world/national-security/russian-propaganda-skripal-salisbury/>.
- ⁹ Robert S. Mueller III, *Report on the Investigation into Russian Interference in the 2016 Presidential Election*, vols. 1 and 2 (Washington, DC: Department of Justice, 2019).
- ¹⁰ Claire Allbright, "A Russian Facebook Page Organized a Protest in Texas. A Different Russian Page Launched the Counterprotest," *Texas Tribune* (Austin), November 1, 2017, available at <www.texastribune.org/2017/11/01/russian-facebook-page-organized-protest-texas-different-russian-page-l/>.
- ¹¹ Renee DiResta et al., *The Tactics and Tropes of the Internet Research Agency* (New York: New Knowledge, 2018), available at <<https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1003&context=senatedocs>>.
- ¹² Martin and Shapiro, *Trends in Online Foreign Influence Efforts*.
- ¹³ Tom Uren, Elise Thomas, and Jacob Wallis, *Tweeting Through the Great Firewall: Preliminary Analysis of PRC-Linked Information Operations Against the Hong Kong Protests*, Issues Paper Report No. 25/2019 (Canberra: Australian Strategic Policy Institute, 2019), 5, available at <<https://www.aspi.org.au/report/tweeting-through-great-firewall>>.
- ¹⁴ Sergey Sanovich, *Computational Propaganda in Russia: The Origins of Digital Misinformation*, Working Paper No. 2017.3 (Oxford, UK: Computational Propaganda Research Project, 2017), available at <<http://compprop.oii.ox.ac.uk/wp-content/uploads/sites/89/2017/06/Compprop-Russia.pdf>>.
- ¹⁵ Ibid., 15.
- ¹⁶ Looking at the Russian campaign targeted at the U.S. 2016 elections, it appears that Russia's ad-targeting efforts were not focused on the battleground states critical to Presidential victory. Russia spent less than \$2,000 on advertisements in Wisconsin, Pennsylvania, and Michigan, suggesting the campaign was not decisive in changing votes associated with the Electoral College. See Nemr and Gangware, *Weapons of Mass Distraction*.
- ¹⁷ Christopher A. Bail et al., "Assessing the Russian Internet Research Agency's Impact on the Political Attitudes and Behaviors of American Twitter Users in Late 2017," *Proceedings of the National Academy of Sciences* 117, no. 1 (January 2020), 243–250.
- ¹⁸ Matthew Nussbaum, "Worries About Trump's Legitimacy Resurface with Russia Indictment," *Politico*, February 16, 2018, available at <www.politico.com/story/2018/02/16/muller-indictment-trump-election-legitimacy-416163>.
- ¹⁹ Davey Alba and Sheera Frenkel, "Russia Tests New Disinformation Tactics in Africa to Expand Influence," *New York Times*, October 30, 2019, available at <www.nytimes.com/2019/10/30/technology/russia-facebook-disinformation-africa.html>.
- ²⁰ Sarah Cook, *Beijing's Global Megaphone: The Expansion of Chinese Communist Party Media Influence Since 2017*, Special Report (Washington, DC: Freedom House, 2020), available at <https://freedomhouse.org/sites/default/files/2020-02/01152020_SR_China_Global_Megaphone_with_Recommendations_PDF.pdf>.
- ²¹ Gillian Bolsover, *Computational Propaganda in China: An Alternative Model of a Widespread Practice*, Working Paper No. 2017.4 (Oxford, UK: Computational Propaganda Research Project, 2017), available at <<https://blogs.oii.ox.ac.uk/politicalbots/wp-content/uploads/sites/89/2017/06/Compprop-China.pdf>>.
- ²² Ibid.
- ²³ Hauke Gierow, Karsten Luc, and Kristin Shi-Kupfer, "Governance Through Information Control: China's Leadership Struggles with Credibility in Social Media Staged Propaganda, Everyday Tips, Lack of Interactivity," *China Monitor* 26 (January 19, 2016), available at <https://www.merics.org/sites/default/files/2019-08/China_Monitor_No_26_Social_Media_EN.pdf>.
- ²⁴ Bolsover, *Computational Propaganda in China*.
- ²⁵ Gary King, Jennifer Pan, and Margaret E. Roberts, "How the Chinese Government Fabricates Social Media Posts for Strategic Distraction, not Engaged Argument," *American Political Science Review* 111, no. 3 (2017), 484–501.
- ²⁶ Cook, *Beijing's Global Megaphone*. China has also exerted significant influence on Hollywood by granting select films access to Chinese markets. The films *Martian* and *Gravity* received significant praise in the Chinese market for giving Chinese astronauts key roles in saving beleaguered American astronauts. See Gus Lubin, "18 Hollywood Films That Pandered to China's Giant Box Office," *Business Insider*, October 14, 2016, available at <www.businessinsider.com/hollywood-movies-in-china-2016-10#cloud-atlas-removed-nearly-30-minutes-from-its-chinese-cut-largely-plotlines-and-scenes-with-controversial-sexual-relations-3>.
- ²⁷ Such was learned when Daryl Morey, the general manager of the Houston Rockets, published his now-deleted tweet: "Fight for freedom, Stand with Hong Kong." He quickly deleted the tweet, apologized, and the National Basketball Association worked overtime to avoid alienating itself from the lucrative Chinese basketball market. See Theodore Yu, "One Tweet, a Week of Turmoil: NBA Steps Out of Bounds with China and Pays the Price," *Sacramento Bee*, October 13, 2019, available at <www.sacbee.com/sports/nba/sacramento-kings/article235933242.html>.
- ²⁸ Renee DiResta, "For China, the 'USA Virus' Is a Geopolitical Ploy," *The Atlantic*, April 11, 2020, available at <www.theatlantic.com/ideas/archive/2020/04/chinas-covid-19-conspiracy-theories/609772/>.
- ²⁹ Gary Schmitt and Michael Mazza, *Blinding the Enemy: CCP Interference in Taiwan's Democracy* (Washington, DC: Global Taiwan Institute, October 2019).
- ³⁰ Ibid., 8–9.
- ³¹ Twitter Safety, "Information Operations Directed at Hong Kong," *Twitter*, August 19, 2019, available at <https://blog.twitter.com/en_us/topics/company/2019/information_operations_directed_at_hong_kong.html>.
- ³² Nathaniel Gleicher, "Removing Coordinated Inauthentic Behavior from China," *Facebook*, August 19, 2019, available at <<https://about.fb.com/news/2019/08/removing-cib-china/>>; Chris Welch, "YouTube Disabled 210 Accounts for Spreading Disinformation About Hong Kong Protests," *The Verge*, August 22, 2019, available at <www.theverge.com/2019/8/22/20828808/youtube-hong-kong-protests-china-disabled-accounts-suspension-disinformation>.
- ³³ Uren, Thomas, and Wallis, *Tweeting Through the Great Firewall*.
- ³⁴ Ibid., 4.
- ³⁵ Ibid.
- ³⁶ Ibid., 11.
- ³⁷ Ibid., 14.
- ³⁸ Ibid., 22.
- ³⁹ Bethany Allen-Ebrahimian, "Beijing's Coronavirus Propaganda Blitz Goes Global," *Axios*, March 11, 2020, available at <www.axios.com/beijings-coronavirus-propaganda-blitz-goes-global-f2bc610c-e83f-4890-9ff8-f49521ad6a14.html>.
- ⁴⁰ DiResta, "For China, the 'USA Virus' Is a Geopolitical Ploy."
- ⁴¹ Ibid.

- ⁴² Allen-Ebrahimian, "Beijing's Coronavirus Propaganda Blitz Goes Global."
- ⁴³ DiResta, "For China, the 'USA Virus' Is a Geopolitical Ploy."
- ⁴⁴ Bethany Allen-Ebrahimian, "Bots Boost Chinese Propaganda Hashtags in Italy," *Axios*, April 1, 2020, available at <www.axios.com/bots-chinese-propaganda-hashtags-italy-cf92c5a3-cdcb-4a08-b8c1-2061ca4254e2.html>.
- ⁴⁵ Echo Huang, "No Little Red Guidebook: Why China Isn't as Skillful at Disinformation as Russia," *Quartz*, September 19, 2019, available at <https://qz.com/1699144/why-chinas-social-media-propaganda-isnt-as-good-as-russias/>.
- ⁴⁶ "China's Propaganda Machine Is Spending Over \$1 Million to Buy Influence on Foreign Social Media," *Quartz*, August 21, 2019, available at <https://finance.yahoo.com/news/china-propaganda-machine-spending-over-150453611.html>.
- ⁴⁷ Christopher Paul and Marek Posard, "Artificial Intelligence and the Manufacturing of Reality," *RAND Blog*, January 20, 2020, available at <www.rand.org/blog/2020/01/artificial-intelligence-and-the-manufacturing-of-reality.html>.
- ⁴⁸ See Linda Robinson et al., *Modern Political Warfare: Current Practices and Possible Responses* (Santa Monica, CA: RAND, 2018); Shiv Malik, "IS Papers: Leaked Documents Show How IS Is Building Its State," *Guardian*, December 7, 2015, available at <www.theguardian.com/world/2015/dec/07/leaked-isis-document-reveals-plan-building-state-syria>.
- ⁴⁹ Elizabeth Bodine-Baron et al., *Examining IS Support and Opposition Networks on Twitter* (Santa Monica, CA: RAND, 2016), available at <https://www.rand.org/content/dam/rand/pubs/research_reports/RR1300/RR1328/RAND_RR1328.pdf>.
- ⁵⁰ Another study analyzed the content of IS propaganda magazines *Dabiq* and *Rumiyah*. Common themes associated with the articles in the magazines included IS theological justification and inspiration for violence; descriptions of community, belonging, and meaning; stories of progress or heroism; establishment of a common enemy, that is, the West and Muslim "apostates"; and instructional and inspirational articles empowering individual violent action. See Tyler Welch, "Theology, Heroism, Justice, and Fear: An Analysis of IS Propaganda Magazines *Dabiq* and *Rumiyah*," *Dynamics of Asymmetric Conflict* 11, no. 3 (2018), 186–198.
- ⁵¹ Thomas Hegghammer, *Saudi Militants in Iraq: Backgrounds and Recruitment Patterns* (Kjeller: Norwegian Defence Research Establishment, February 5, 2007), available at <https://www.fhi.no/en/publications-archive/saudi-militants-in-iraq-backgrounds-and-recruitment-patterns>; Alexandria Zavis, "Foreign Fighters in Iraq Seek Recognition, U.S. Says," *Los Angeles Times*, March 17, 2008, available at <www.latimes.com/world/la-fig-iraqi7mar17-story.html>.
- ⁵² "The Propaganda Wars Since 9/11," *Washington Post*, May 8, 2015, available at <www.washingtonpost.com/graphics/national/propaganda/>; Scott Shane, "The Lessons of Anwar al-Awlaki," *New York Times*, August 27, 2015, available at <www.nytimes.com/2015/08/30/magazine/the-lessons-of-anwar-al-awlaki.html>.
- ⁵³ Edward Wyatt, "Anti-U.S. Attack Videos Spread on Web," *New York Times*, October 6, 2006, available at <www.nytimes.com/2006/10/06/technology/06tube.html>.
- ⁵⁴ Jytte Klausen, "Tweeting the Jihad: Social Media Networks of Western Foreign Fighters in Syria and Iraq," *Studies in Conflict & Terrorism* 38, no. 1 (October 2014), 1–22.
- ⁵⁵ Ibid.
- ⁵⁶ Joseph Carter, Shiraz Maher, and Peter Neumann, *#Greenbirds: Measuring Importance and Influence in Syrian Foreign Fighter Networks* (London: International Centre for the Study of Radicalisation and Political Violence, 2014), 5.
- ⁵⁷ Landon Shroder, "The Islamic State's Propaganda War: Advertisers and Marketers Weigh in on the World's Angriest Ad Campaign," *VICE News*, July 14, 2015, available at <www.vice.com/en_us/article/8x3nnv/the-islamic-states-propaganda-war-advertisers-and-marketers-weigh-in-on-the-worlds-angriest-ad-campaign>.
- ⁵⁸ J.M. Berger, "Tailored Online Interventions: The Islamic State's Recruitment Strategy," *Combating Terrorism Center Sentinel* 8, no. 10 (October 2015), available at <https://www.ctc.usma.edu/posts/tailored-online-interventions-the-islamic-states-recruitment-strategy>.
- ⁵⁹ Joana Cook and Gina Vale, *From Daesh to Diaspora: Tracing the Women and Minors of the Islamic State* (London: International Centre for the Study of Radicalisation, 2018).
- ⁶⁰ Karen Yourish, Derek Watkins, and Tom Giratikanon, "Where IS Has Directed and Inspired Attacks Around the World," *New York Times*, March 22, 2016, available at <www.nytimes.com/interactive/2015/06/17/world/middleeast/map-isis-attacks-around-the-world.html>.
- ⁶¹ As one IS supporter urged in a blog, "To counter suspensions, you must support each other. Don't make the brothers and sisters ask for a shout out. If someone you recognize has followed you after a suspension, tell your followers to follow them." See "Blog Post Suggests Ways for IS Supporters to Boost Social Media Effectiveness," SITE Intelligence Group, July 17, 2015.
- ⁶² J.M. Berger and Heather Perez, *The Islamic State's Diminishing Returns on Twitter: How Suspensions Are Limiting the Social Networks of English-Speaking IS Supporters*, Occasional Paper (Washington, DC: Program on Extremism at George Washington University, 2016).
- ⁶³ Madhumita Murgia, "Twitter Takes Down 300,000 Terror Accounts as AI Tools Improve," *Financial Times*, September 19, 2017, available at <www.ft.com/content/198b5258-9d3e-11e7-8cd4-932067fb946>.
- ⁶⁴ Maura Conway et al., "Disrupting Daesh: Measuring Takedown of Online Terrorist Material and Its Impacts," *Studies in Conflict & Terrorism* 42, nos. 1–2 (2019), 141–160.
- ⁶⁵ Bennett Clifford and Helen Powell, *Encrypted Extremism: Inside the English-Speaking Islamic State Ecosystem on Telegram* (Washington, DC: Program on Extremism at George Washington University, 2019), available at <https://extremism.gwu.edu/sites/g/files/zaxdzs2191/f/EncryptedExtremism.pdf>.
- ⁶⁶ See, for example, Todd C. Helmus et al., *Russian Social Media Influence: Understanding Russian Propaganda in Eastern Europe* (Santa Monica, CA: RAND, 2018), available at <https://www.rand.org/pubs/research_reports/RR2237.html>; Bodine-Baron et al., *Countering Russian Social Media Influence*; Alina Polyakova and Daniel Fried, *Democratic Defense Against Disinformation 2.0* (Washington, DC: Atlantic Council, 2019), available at <https://www.atlanticcouncil.org/wp-content/uploads/2019/06/Democratic_Defense_Against_Disinformation_2.0.pdf>; Paul Barret, Tara Wadhwa, and Dorothee Baumann-Pauly, *Combating Russian Disinformation: The Case for Stepping Up the Fight Online* (New York: NYU Stern Center for Human Rights, 2018), available at <https://issuu.com/nyusterncenterforbusinessandhumanri/docs/nyu_stern_cbhr_combating_russian_di?e=31640827/63115656>.
- ⁶⁷ There is no single U.S. agency or government entity leading the work to counter foreign propaganda and influence. See *Homeland Security Advisory Council Interim Report of the Countering Foreign Influence Subcommittee* (Washington, DC: Department of Homeland Security, June 2019), 10, available at <https://www.dhs.gov/sites/default/files/publications/ope/hsac/19_0521_final-interim-report-of-countering-foreign-influence-subcommittee.pdf>; Melissa Dalton et al., *By Other Means—Part II: U.S. Priorities in the Gray Zone* (Lanham, MD: Rowman & Littlefield, 2019), 6–11, available at <https://csis-prod.s3.amazonaws.com/s3fs-public/publication/Hicks_GrayZone_II_full_WEB_0.pdf>.

Chapter 8

Weapons of Mass Destruction, Strategic Deterrence, and Great Power Competition

By Paul Bernstein, Justin Anderson, Diane DiEuliis, Gerald Epstein, and Amanda Moodie

Weapons of mass destruction (WMD)—nuclear, biological, and chemical weapons and the means to deliver them—are an important feature of the global security environment and a key element of Great Power competition. For Russia and China, WMD contribute to multiple goals: conflict deterrence at the strategic and regional levels; regime survival; coercion of rival states; and, potentially, as an adjunct to conventional forces to support operations. U.S.-Russia competition in nuclear weapons has been constrained in recent decades by various arms control agreements, but the erosion of this regulatory regime in the context of deteriorating bilateral relations could create new competitive pressures. China has elevated the importance of its nuclear forces, modernized and expanded its strategic nuclear capabilities, and fielded a growing number of dual-capable theater-range missile systems whose role (whether conventional or nuclear) in a future crisis or conflict could complicate deterrence and heighten escalation risks. China and Russia may perceive chemical and biological warfare agents, including agents developed through new scientific and manufacturing techniques, as important capabilities for a range of operations against the United States and its allies. Chemical or biological attacks could be difficult to attribute and may be well suited to support Russian and Chinese objectives in operations below the threshold of open armed conflict.

Weapons of mass destruction (WMD) remain a significant concern for U.S. defense planning. Core strategy and policy documents such as the National Security Strategy, National Defense Strategy (NDS), Nuclear Posture Review, and Department of Defense Strategy for Countering Weapons of Mass Destruction highlight these weapons as an enduring feature of the security environment.¹ They are also a potentially dynamic factor in Great Power relations. Understanding the role of nuclear, biological, and chemical weapons in competition—from strategic nuclear deterrence to regional conflict (traditional, hybrid, irregular) to operations below the level of armed conflict—is essential to manage geopolitical risk, limit the possibility of surprise, prepare the joint force for future operations, and

inform the American people about the dangers associated with potential crises and conflicts. Advances in the technologies that shape WMD could make these tasks more difficult.

Nuclear Weapons in Great Power Competition

New pressures for competition in nuclear capabilities among the Great Powers are likely to emerge in the next few years. In the U.S.-Russia relationship, a weakened arms control regime and the introduction of new technologies could catalyze a period of competition and arms-racing in both offensive and defensive systems, which could affect the nuclear balance in uncertain ways. China's continued expansion and improvement of its nuclear forces may create the basis for a more competitive stance vis-à-vis the United States, should Beijing decide this is necessary to advance its global and regional aspirations. Emerging technologies enabling improved prompt strike (for example, hypersonic vehicles) and active defense against missile attacks could be a factor, as various nuclear competitions unfold at both strategic and nonstrategic levels. Additionally, Great Power nuclear competition could have important effects on U.S. extended deterrence relationships, prospects for further nuclear proliferation, and the future of the global nonproliferation regime.

The United States and Russia

Deterrence and arms control have been central to managing nuclear competition between Washington and Moscow for decades. Both sides have adhered to a concept of mutual deterrence that has ensured neither government saw advantage in mounting a surprise nuclear attack or using nuclear weapons in a crisis. A series of arms control agreements dating back to the 1970s has sought to reinforce deterrence by first capping and then reducing or eliminating nuclear delivery systems, including those considered to have destabilizing effects. This framework of deterrence and force reduction has been successful in limiting pressures for nuclear arms-racing and in mitigating (though not fully eliminating) other dangers of the nuclear age, including crisis instability and accidental launch.

Today, stress on this framework is growing, as bilateral relations have deteriorated and the network of treaties designed to ensure nuclear stability continues to erode. To be sure, neither country has been standing still regarding strategic nuclear forces. Russia has nearly completed modernizing its entire strategic nuclear arsenal and has also introduced or stated its intent to develop several nontraditional nuclear systems (so-called exotic weapons) that are important, from Moscow's vantage, to pose a credible retaliatory threat to the United States.² The United States is in the early stages of executing a program to replace all three legs of its strategic nuclear triad by the 2030s.³ These respective strategic force upgrades have long been planned; their origins predate the downturn in bilateral relations and adoption of a Great Power competition framework by the 2018 National Defense Strategy. For both Washington and Moscow, the fundamental purpose of these programs is to ensure parity in strategic forces going forward—and thereby sustain a status quo that has long delivered mutual and global security benefits.

The question today is whether either side might see the need or the opportunity in the near term to move toward a more open and unconstrained rivalry in strategic nuclear forces, in order to achieve strategic competitive advantage. A new nuclear arms race is hardly inevitable—neither is it clear that this would be in America's interest. Some have

argued that robust and evidently superior nuclear forces would yield meaningful geopolitical advantage for the United States and a clear edge in competitions with other nuclear-armed states.⁴ Others suggest that the costs and risks of such a posture are likely to outweigh any benefits. Nothing in current U.S. strategy and policy documents

states or suggests a need to pursue nuclear superiority over Russia, or that achieving the goals of the NDS requires nuclear capabilities and policies markedly different from those that define U.S. planning.

The United States is not well postured for more open nuclear competition with Russia, as compared with earlier historical periods. Although today there is in Congress a bipartisan political consensus favoring the triad replacement program, it is unlikely this consensus would survive an effort to pursue a more expansive—and expensive—nuclear development program in the name of outcompeting Russia. Many of those who support modernization also question the cost of the triad replacement program in relation to that of the other capabilities required to fulfill the main objectives of the NDS. Accordingly, as a practical matter, it makes sense for the United States to avoid inviting an unconstrained competition in nuclear capabilities with an adversary that seems willing (to a point) to bear high costs in prioritizing nuclear forces.

This possibility points to the need to maintain a focus on stability characterized by parity in strategic nuclear forces, regulated through an arms control regime.⁵ The U.S. goal should be to extend and adapt the New Strategic Arms Reduction Treaty (New START) and thereby create for strategic nuclear weapons a type of “competition sanctuary” that will limit strategic risk generally and avoid diverting resources from areas of competition that are arguably more important to fully implement the NDS.⁶

If the New START, currently set to expire in February 2021, is not extended or updated, some degree of heightened nuclear competition seems likely, as either side could then act without legally binding constraints.⁷ How significant a competition this would be is not entirely clear. Russia might see an advantage to rapidly building up the number of warheads it can deliver on its strategic systems or expanding its strategic nuclear capabilities in other ways. Moscow could conclude that this was a relatively easy and cost-effective way to establish a degree of benefit and impose additional risk on the United States. Russia might feel a stronger incentive to move in this direction if it was experiencing the weight of other competitive pressures in nonnuclear domains, such as global nonnuclear strike, outer space, or cyber. After all, nuclear weaponry is one of the few strategic technology areas in which Russia is capable of competing effectively with the United States. But Moscow likely will not want to trigger a strong U.S. counter-response that creates new risk and prospective high additional costs for Russia, so it can be expected to exercise caution in moving too quickly or too aggressively toward a larger deployed force.⁸

The United States might or might not respond with similar steps. The political and psychological importance of maintaining the perception and reality of numerical parity would be an important consideration that could lead Washington to reverse the reductions

“I would like to tell those who have been trying to escalate the arms race for the past 15 years, to gain unilateral advantages over Russia . . . the attempt at curbing Russia has failed.”

—Russian President Vladimir Putin, March 1, 2018

taken under the New START. But as recent studies have demonstrated, the United States could remain within the treaty limits even if Russia did not—and it could do so without undermining its nuclear deterrent, as long as the resilience inherent in the U.S. nuclear triad is sustained.⁹ Furthermore, neither side has the capacity to upend the strategic nuclear balance by exceeding the treaty limits if the other chooses to do so too.¹⁰ In other words, one highly plausible outcome of renewed nuclear competition is a modified form of parity at higher levels of strategic forces.¹¹

Competitive pressures in strategic nuclear weapons could also be shaped by Russia's assessment of U.S. missile defense capabilities. Moscow has demonstrated that it will go to great lengths and bear considerable costs to ensure that its strategic nuclear forces can reliably overcome U.S. missile defenses if it needs to deliver a retaliatory strike in the event of a nuclear exchange. Russia fears that the United States will significantly expand its missile defense capabilities, and new nuclear systems being introduced by Russia, such as the Avangard nuclear-armed hypersonic glide vehicle (HGV), are intended mainly as a hedge against a U.S. breakout in missile defenses. While some analysts see the introduction of this and similar capabilities as signaling a new arms race or posing a qualitatively novel strategic threat, the number of such systems that Russia fields is likely for reasons of strategy and cost to be calibrated against U.S. missile defense deployments and the requirements for a secure second strike. That said, significant departures from current U.S. missile defense policy (for example, development of space-based interceptors) or an open effort to develop defenses tailored to HGVs could lead Russia to take more dramatic steps to ensure the survivability of its strategic nuclear forces.¹² This effort, in turn, could unleash new competitive pressures.

At the regional level, a somewhat different competitive landscape has taken shape in recent years. Russia continues to field modern land-, sea-, and air-based nonstrategic nuclear weapons (NSNW)—a category of weaponry in which it has long enjoyed uncontested advantage in relation to the United States and its Allies in the North Atlantic Treaty Organization (NATO). In the post-Cold War period, this imbalance in NSNW has been a source of concern precisely because of the fear that it could contribute to deterrence instability in Europe. Rather than compete with Russia in theater nuclear systems, the United States and NATO have sought repeatedly to extend the bilateral arms control framework to capture these capabilities. Moscow consistently has refused.

Still, during a period in which a nuclear crisis seemed a remote possibility, the United States judged the risk posed by this persistent asymmetry in NSNW to be manageable. Today, in light of Russia's conduct and its continued investment in these capabilities, this possibility is less remote and the risk therefore higher. Of particular concern is the threat that, in a regional conflict, Russia might see an advantage in escalating to the limited use of NSNW in the belief that the United States or NATO lacks the means to respond proportionately. Accordingly, mitigating this danger is now a priority for the United States. This strategy does not require matching Russia's large, diverse NSNW capabilities or its doctrine, but it does require a more tailored form of competition to narrow the imbalance in forces and convey resolve to strengthen the U.S. regional deterrence posture. To accomplish this, the United States will develop and field two nonstrategic nuclear capabilities: a low-yield option for existing submarine-launched ballistic missile warheads and a nuclear-armed sea-launched cruise missile.¹³ Additionally, combatant commands and the Services have been

directed to strengthen the ability of the joint force to operate effectively in a nuclear environment following an adversary's limited use of nuclear weapons. More openly competitive measures designed to achieve parity or advantage in this category of nuclear weapons are not necessary for the United States to meet the requirements of regional deterrence.

A second issue in considering regional nuclear competition with Russia is the demise in 2019 of the Intermediate-Range Nuclear Forces (INF) Treaty. It remains to be seen what enduring effect this loss will have.¹⁴ To date, it does not appear that Russia's deployment of previously prohibited INF systems alters the balance of power in Europe. The United States currently has no plans to develop a nuclear-capable INF system for deployment in Europe or elsewhere. The United States could choose to develop a land-based nonnuclear INF missile in support of NATO, which could help narrow the gap with Russia in such systems. In fact, Washington is much more focused on developing capabilities to close the conventional missile imbalance in East Asia that threatens to disadvantage the joint force in a future conflict with China. In the immediate period ahead, the most salient post-INF competitions will likely feature conventionally armed theater missiles.

The United States and China

Compared with the near-term dynamics that could shape U.S.-Russia nuclear competition, the prospects for U.S.-China nuclear competition need to be viewed over a longer time horizon. It is difficult to anticipate dramatic changes in the next 5 years, though trends in Chinese and U.S. capabilities should be monitored carefully; they could contribute to conditions that lead to a more competitive bilateral nuclear relationship in the future.

China has moved definitively away from its small, static strategic nuclear force of the past. As part of its broad-based modernization of its armed forces, China now fields a modern strategic dyad composed of intercontinental ballistic missiles (ICBMs) based on land and on submarines.¹⁵ Structural reform of the People's Liberation Army has elevated the Rocket Force to coequal status with land, sea, and air forces, making more resources available for nuclear force development. Modern ICBMs (a growing number of them mobile) and submarine-launched ballistic missiles constitute an increasingly dynamic force designed to give Beijing high confidence that it possesses a survivable deterrent against U.S. nuclear forces and missile defenses—one that would allow it to resist nuclear coercion in a crisis and press its advantage in a local or regional military conflict with the United States. Clearly, China's leadership sees such a capability as a critical component of long-term competitive strategy toward the United States.

But China historically has not sought to engage in nuclear competition with the United States, choosing instead to maintain only those capabilities needed to deter and respond to nuclear threats. China's political leadership continues to have a generally conservative view of the role of nuclear weapons and has long accepted a large disparity in capabilities—a posture embodied in slogans such as “nuclear strategy of self-defense” and “lean and effective nuclear force.”¹⁶ While Beijing views a credible strategic nuclear deterrent as indispensable to a stable relationship with Washington, achieving equal status in numbers and types of nuclear weapons has not been its goal. China does not wish to be seen as an arms-racing global power.

“While China’s declaratory policy and doctrine have not changed, its lack of transparency regarding the scope and scale of its modernization program raises questions regarding its future intent.”

— Nuclear Posture Review, *DOD* (2018)

As the strategic environment changes and as the technological impediments to fielding larger and more advanced forces continue to fall, it is reasonable to ask whether and under what conditions China could adopt a more competitive approach to its nuclear forces. One possibility is that leadership will decide that parity (or some-

thing close to it) in strategic nuclear weapons is necessary to enhance China’s status as a coequal global power and a dominant force in East Asia. Especially if U.S. force levels remain relatively static, this goal may become increasingly attractive; it likely would be even more attractive should the United States decide in 2020 or beyond to reduce current levels of operationally deployed forces, either unilaterally or through a renewed commitment to arms control with Russia. Should U.S. (and Russian) forces fall to, say, two-thirds of New START limits, Beijing’s task in moving toward parity would be much more manageable—assuming it continues to rebuff U.S. entreaties to join the process of making negotiated reductions. If China were to achieve parity or equivalence in deployed or deliverable warheads, it might then be expected to explore ways to translate this status into political and military advantage.

Another possibility is that, as with Russia, competitive pressures for China could be driven by changes to U.S. missile defense and defeat capabilities. Should the United States expand its regional missile defense network, move toward a larger or more sophisticated homeland defense capability (for example, boost-phase kill, space-based interceptors), and field advanced theater-range missiles capable of precision strike against Chinese nuclear sites (fixed and mobile), concerns about the survivability of its nuclear forces could lead China to consider any number of steps to ensure the credibility of its deterrent. These types of offense-defense dynamics might have little to do with a decision by China to pursue nuclear parity as an explicit policy goal, but they nonetheless point to the possibility that, under certain circumstances, Beijing could feel compelled to undertake a significant expansion of its strategic nuclear forces.

The United States must also consider the possibility of nuclear competition with China at the regional level. Beijing historically has eschewed theater or tactical nuclear weapons that would support more expansive deterrence concepts and more operational scenarios that envision the limited use of nuclear weapons. But there are indications that this stance is changing as China considers how to strengthen its options for coercing and deterring the United States (and its allies) at different stages of conflict in an increasingly complex operating environment. Should China move decisively in the direction of limited nuclear options, it will need to consider how the United States might respond.¹⁷ The United States does not station land-based or air-delivered nuclear weapons in the Far East and has no current plans to do so. If China’s theater nuclear footprint expands, then U.S. allies in the region could press Washington to take countervailing steps. This potential competitive dynamic bears watching.

The converse could happen as well. The United States currently extends nuclear deterrence to regional allies through over-the-horizon nuclear capabilities; however, if the

nuclear crisis with North Korea is not resolved and indeed worsens, Washington could face pressures from the Republic of Korea and Japan to provide a more visible nuclear presence in the theater. This could lead the United States to deploy nuclear weapons to the region as a means to deepen extended deterrence relationships with these allies and perhaps other regional security partners. Should the U.S. nuclear presence in the region grow significantly and in a visible way—even for the purpose of countering a North Korean threat—Beijing might see the need to respond in kind by building up its own theater nuclear presence.

Finally, other regional developments could shape Chinese thinking about nuclear forces. Notably, China is closely watching India's efforts to develop a nuclear triad. While China was an important factor in India's decision to acquire nuclear weapons, India's nuclear force is not optimized for use against China or any other state; rather, India's nuclear deterrent historically has been more existential in nature, with the goal of maintaining India's strategic independence and keeping it free from intimidation or coercion by other nuclear powers. For its part, Beijing is reticent to acknowledge that India's nuclear arsenal could impact China's security. Still, the two nations have divergent and potentially competing interests and ambitions in South Asia and the Indian Ocean. As Beijing assesses future nuclear risks in the region, it cannot ignore India's progress in developing new and improved nuclear-capable delivery systems or India's efforts to build missile defense capabilities.

The Impact of Competition in Hypersonic Systems

The introduction of hypersonic vehicles by the Great Powers is not likely to have a major impact on the global balance of nuclear power in the next few years. To date, only Russia, as noted, is fielding a nuclear-armed hypersonic missile as part of its strategic forces. This capability will not in itself alter the U.S.-Russia nuclear balance in a significant way. Over time, if the Great Powers deploy intercontinental-range hypersonic missiles in growing numbers, it will be necessary to consider the strategic implications regardless of whether these systems are nuclear or conventionally armed. Could the conventional hypersonic systems of one Great Power pose a credible threat to the strategic nuclear deterrent of another? How would strategic stability be affected if the United States fielded a new generation of missile defenses capable of defeating Russian and Chinese hypersonic platforms?

Such questions will become more important in the period ahead, but strategists and defense planners must also closely examine the potential impact of hypersonic weapons on nuclear stability at the theater level. One concern is whether the widespread use of hypersonic missiles in a regional conflict would undermine stability by creating pressures for early nuclear use. If, as might be anticipated, theater-range hypersonic weapons give both sides to a conflict the capability to inflict more decisive nonnuclear damage (for example, against power projection forces, air defenses, or missile arsenals) at an earlier stage, then these reciprocal vulnerabilities could not only reinforce deterrence and restraint but also generate pressures to consider limited use of nuclear weapons to avoid or redress major operational setbacks.¹⁸ How much more dangerous would such a scenario be if each side faced uncertainty about the payload of hypersonic missiles that were, in fact, dual capable?

Extended Deterrence and Proliferation Implications of Great Power Nuclear Competition

More open nuclear competition among the Great Powers likely would reverberate in the security environment in several ways. The impact on U.S. allies and security partners is one area of concern. The prospect of nuclear arms racing and heightened nuclear tension would make many partners anxious. This anxiety could lead some to demand a return to arms control and risk-reduction measures, while others are likely to demand stronger nuclear security guarantees from the United States. Either way, Washington would face new challenges in alliance management.

Indeed, the United States cannot dismiss the possibility that one or more of its allies, in the face of Great Power nuclear competition and a weakened arms control regime, could decide to pursue an independent nuclear weapons capability; other, less friendly states could make the same calculation as nuclear dangers rise. This is one way that Great Power nuclear competition could fuel proliferation. Additionally, sharper nuclear competition among the Great Powers is certain to be viewed by many nonnuclear states party to the Treaty on the Non-Proliferation of Nuclear Weapons (NPT) as additional evidence that the “nuclear haves” remain unprepared to make progress toward nuclear disarmament. This would add to growing questions about the utility of the NPT in the 21st century and to pressures facing allies under the U.S. nuclear umbrella and others to embrace the Treaty on the Prohibition of Nuclear Weapons (Nuclear Weapon Ban Treaty) as an alternative to the NPT—a development that could undermine the legitimacy of nuclear deterrence as a security strategy.

Biological Weapons in Great Power Competition

There has been a broad consensus for decades that biological weapons are not useful as instruments of warfare because their effects are too difficult to control and too dependent on conditions such as weather, which cannot be predicted long in advance. Even in a strategic role beyond the battlefield, use of a contagious biological agent on a large scale could threaten to spread back to the attacker. Although the Soviet Union was not deterred by such considerations, or by the Biological Weapons Convention (BWC), from pursuing an extensive offensive biological weapons program, that threat receded with the end of the Cold War and the exposure and dismantling of many Soviet biological weapons facilities. Thereafter the biological weapons threat came to be seen by the United States as tied principally to terrorists rather than to nation-states, especially after 9/11 and the anthrax attacks that followed.

The BWC, which entered into force in 1975, embodies this global consensus on the lack of military utility of biological weapons and the importance of establishing a norm prohibiting their use.¹⁹ The question today is whether, 45 years later, advances in technology and the revival of Great Power competition could challenge these assumptions. For example, progress in the life sciences could lead to the emergence of disease agents that are more easily controlled than their natural counterparts. Certain manipulations, such as conferring antibiotic resistance or hardening agents to environmental conditions, are likely to be within the capacity of scientists working for national defense establishments. Should U.S. competitors become interested in a new generation of biological weapons, defense planners would have to anticipate a threat of greater sophistication than in the past—one that could be employed in a set of contingencies that extends well beyond those associated with terrorists or violent extremist groups.²⁰

Table 8.1. Biological Weapons and Gray Zone Operations

Ambiguity	Pathogens naturally present in the affected area could be selected to disguise the deliberate origins of an attack.
Nonlethality	Pathogens could be disseminated to cause nonlethal rather than fatal diseases.
Delay	Effects from the release of a biological weapon would be delayed from the time of release, with the length of the delay depending on the dose received and the health of the victim.
Psychological Amplification	The invisible, time-delayed, and potentially lethal consequences of exposure to disease agents—whether deliberate or natural—could cause great anxiety among civilian or military personnel in the vicinity of any disease outbreak, greatly amplifying the disruptive effects.
Behavioral Consequences	Advanced biological agents might be developed that could influence the mood and behavior of those infected, rather than causing illness or death, with uncertain consequences.

Thus, an important task is to understand how the “new biology” could help determined competitors such as Russia and China overcome the traditional obstacles to effective battlefield or even strategic employment of disease agents—and to what potential effect. Should Russia or China be willing to violate its BWC commitments, either country could seek to develop more effective capabilities designed to target specific U.S. military activities and facilities important to the prosecution of a regional conflict. Examples could include attacks on U.S. power projection or logistical supply activities critical to flowing U.S. forces and equipment, such as operations at ports of embarkation or debarkation. Such attacks, if successful, would interrupt key supply chains by disabling or killing unprotected civilian port workers. Military personnel could also be affected by a no-notice attack that infects them before physical or medical protections could be put into place. Panic in the general populace could further complicate military activities.

Aside from the question of effects on military activities is the challenge of attribution. It is already difficult to determine whether a disease outbreak is natural or deliberate and to identify with high confidence the source of an intentional attack. A new generation of biological agents could make it even more challenging and thus encourage other powers to consider militarizing such agents as a tool of asymmetric warfare for various nontraditional battlefields. Here, the Department of Defense needs to look beyond the possibility of major war. In the types of irregular or hybrid operations exemplified by recent Russian activities, or even in so-called gray zone operations, innovations in the life sciences could allow peer competitors to exploit the ambiguity associated with biological agents; these actors could pursue important goals while avoiding accountability. Table 8.1 notes the attributes of biological weapons that could make them attractive in such an effort.

Emerging biological applications have the potential to shape Great Power competition more broadly. Bioinspired innovations—such as advanced materials, “living” sensors, engineered medicines, and new forms of energy production—could confer advantage on those militaries best able to develop, field, and exploit them. State competitors could also invest in biotechnology to directly enhance warfighter capabilities—for example, through machine interfaces or more direct physical enhancements or protections. These kinds of military applications may be legitimate in the sense that they do not violate international

legal prohibitions against the use of biological weapons, but they could also mask illicit programs to develop such weapons.

Additionally, technological advances in the life sciences could lead to the exploitation of genetic vulnerabilities and the genetic targeting of populations. Knowledge of the genetic makeup of key individuals might indicate health conditions to which they are especially vulnerable; more speculatively, in the event that relationships may be gleaned between genetic traits and behavior, this knowledge could indicate propensities to act in certain ways. Advances in genetics and biotechnology raise related questions about the possibility that genetic weapons might be developed that selectively target individuals or groups based on specific genetic signatures. A 2018 report from the National Academies of Sciences, Engineering, and Medicine finds that developing such a weapon confronts a number of technical challenges; however, the rate of technical progress in relevant fields suggests that this issue should continue to be monitored.²¹ To the extent that these possibilities exist, genetic databases containing information on a nation's population or leaders assume national security sensitivity.

Looking ahead, the United States will need to develop intelligence capabilities that can anticipate, monitor, and assess the range of advanced agents that modern biotechnologies may make possible and adversary efforts to militarize these agents. This is a formidable challenge, not least because of the dual-use nature of work in the life sciences. The potential for adversary use of biological weapons with deliberate ambiguity requires the United States to develop techniques and practices that can reliably identify disease outbreaks and differentiate natural from deliberate attacks, attribute the source of a possible attack, and accelerate the execution of medical management strategies. Given the importance of allies and coalitions to U.S. defense strategy and the prosecution of any regional conflict, the United States must work with partners to ensure an adequate level of preparedness for plausible biological weapons attacks. The United States cannot anticipate all possible weapons applications of the life sciences and biotechnology, but it must be postured to respond effectively to biological warfare threats so as to deny any meaningful advantage to adversaries seeking to exploit these technologies for military gain.²²

The coronavirus pandemic underscores this point. Although the novel coronavirus is clearly not a biological weapon (despite persistent efforts by those hostile to U.S. interests to assert otherwise), the pandemic nonetheless could hold lessons for adversaries inclined to see utility in asymmetric or unconventional means of conflict—and who have invested in modern biology. Future threat assessments will need to consider how potential adversaries view the economic and social disruption caused by the virus in the United States and for some of its partners, the challenges in mounting an effective response, and possible effects on the readiness of the joint force.

Chemical Weapons in Great Power Competition

As with biology, advances in chemical science and technology could result in novel military threats, new proliferation risks, and further challenges to the Chemical Weapons Convention (CWC). For example, the convergence of chemistry and biology—an important aspect of what has been called the fourth industrial revolution (as described in chapter 4)—has led some analysts to ask whether bad actors could use processes such as peptide

synthesis or metabolic engineering to create toxins or other chemical agents in order to exploit loopholes in the CWC or avoid detection through its routine verification measures.²³ Likewise, the convergence of chemistry and information technology means that many chemical production processes could become automated or capable of being performed remotely, which would make it more difficult to detect the existence of a covert or illicit chemical weapons program.²⁴

Another potential challenge is the interest of several countries, including Russia and China, in developing incapacitating chemical agents or central nervous system (CNS)-acting chemicals for domestic law enforcement purposes.²⁵ The CWC permits member states to use chemical agents for this purpose and does not rule out the use of incapacitants or CNS-acting chemicals under this exemption. Although the treaty prohibits the use of such agents as weapons, it is possible that Russia or China would consider this prohibition a useful tactic in military operations below the level of open armed conflict, while claiming that the use was both legal and acceptable under a broad interpretation of the law enforcement provision of the CWC.

Russia has already demonstrated its willingness to use chemical agents in operations other than war. In 2002, Russian security forces employed a fentanyl derivative to incapacitate Chechen separatists in a Moscow theater, leading to the death of 117 hostages. In 2018, a failed assassination attempt against a former Russian intelligence agent using a new variant of nerve agent developed by the Soviet Union in the 1970s resulted in the accidental death of a British citizen.²⁶ Similar incidents, such as the 2017 assassination of the half-brother of North Korean leader Kim Jong-un using the nerve agent VX, suggest that chemical agents may be an attractive option for governments seeking plausibly deniable means to conduct tailored operations short of war.²⁷

Far more troubling is the use of chemical agents to support major military operations. The government of Bashar al-Asad has repeatedly employed chemical weapons in the Syrian civil war, ranging from a massive attack using sarin gas in August 2013²⁸ to numerous smaller scale chlorine attacks in the following years,²⁹ even after Syria acceded to the CWC in September 2013. The apparent effectiveness of these attacks and the lack of a forceful, sustained international response in their aftermath could lead other autocratic governments to conclude that chemical weapons have utility in ensuring internal security and regime survival. Russia's shielding of the Syrian government's attacks further suggests that Moscow itself does not view the longstanding taboo against chemical weapons—or the international censure that might result from their use—as a constraint on its behavior, especially in the gray zone and when not confronting the United States directly.

The United States has been investing for decades in protection for deployed forces that might be exposed to chemical weapons. Whether the Department of Defense in a new era of Great Power competition now needs to consider additional threat possibilities is a reasonable question. Russia might not contemplate the use of chemical weapons in a major conflict with NATO, but such employment cannot be ruled out. Facing significant operational challenges or setbacks from a NATO counterattack, Moscow might well consider asymmetric responses to regain the initiative, including chemical attacks against NATO ground formations, air bases, and forward logistics sites. It is possible that such operations' effectiveness would be limited if Russia were to choose to constrain its

preparations (material and nonmaterial) for fear of signaling its intent to violate the CWC. Nonetheless, the possibility of such attacks and their impact on Alliance operations should be incorporated into future plans and exercises, in order to strengthen deterrence and battlefield preparations.

While China also engages in a wide range of gray zone activities in an attempt to assert and defend controversial maritime and land border claims and shape the political environment, it has not yet violated the norm against the use of chemical weapons and does not appear prepared to do so.³⁰ In the past 3 years, the Department of State's annual arms control report addressing CWC compliance has made no mention of China.³¹ Chinese military leaders may not perceive a need for chemical weapons on current or future battlefields; this may dovetail with a broader political strategy under which China seeks to counter the negative effects of its regional policies by maintaining a cooperative posture with the United States on other issues, including WMD proliferation.

The world may have entered an era in which the norm against chemical warfare continues to weaken while the incentives to resort to unconventional weapons could grow. Ongoing technology developments could further undermine traditional constraints against the use of chemical weapons—while enhancing their appeal as a tool to achieve specific political or operational goals.³² Varied uses of chemical agents could become a more common occurrence across the spectrum of competition and conflict. This possibility is an important feature of the new era of Great Power competition and should inform U.S. strategic thinking and defense planning.

Notes

¹ *Summary of the 2018 National Defense Strategy of the United States of America: Sharpening the American Military's Competitive Edge* (Washington, DC: Department of Defense [DOD], January 2018), 2–3; *2018 Nuclear Posture Review* (Washington, DC: DOD, February 2018), V–VI, available at <<https://media.defense.gov/2018/feb/02/2001872886/-1/-1/1/2018-nuclear-posture-review-final-report.pdf>>; *Department of Defense Strategy for Countering Weapons of Mass Destruction* (Washington, DC: DOD, June 2014), 3.

² See Matthew Kroenig, Mark Massa, and Christian Trotti, *Russia's Exotic Nuclear Weapons and Implications for the United States and NATO* (Washington, DC: Atlantic Council, March 2020), available at <www.atlanticcouncil.org/in-depth-research-reports/issue-brief/russias-exotic-nuclear-weapons-and-implications-for-the-united-states-and-nato/>.

³ Land-based intercontinental ballistic missiles (ICBM), *Columbia* class nuclear submarines carrying submarine-launched ballistic missiles (SLBM), and long-range bombers carrying air-launched cruise missiles and gravity bombs.

⁴ See, for instance, Matthew Kroenig, *The Logic of American Nuclear Strategy: Why Strategic Superiority Matters* (New York: Oxford University Press, 2018).

⁵ See Rose Gottemoeller, "Don't Let the New START Treaty Lapse," *New York Times*, November 8, 2019.

⁶ The New Strategic Arms Reduction Treaty (New START) aggregate limits are 700 deployed ICBM, SLBM, and nuclear-equipped heavy bombers equipped for nuclear armament, and 1,550 operationally deployed strategic nuclear warheads associated with these systems. See Department of State, "New START Treaty," available at <www.state.gov/new-start/>.

⁷ With the U.S. withdrawal from the Intermediate-Range Nuclear Forces (INF) Treaty in August 2019, following years of Russia violating the accord, New START is the only nuclear arms control treaty currently in force between the two countries. New START can be extended for up to 5 years, if both the United States and Russia agree to an extension.

⁸ Vince Manzo, *Nuclear Arms Control Without a Treaty? Risks and Options After New START* (Arlington, VA: CNA, March 2019), 59. The author notes, "Russia has prioritized funding for nuclear forces, but it probably does not have the economic wherewithal for a massive arms buildup."

⁹ *Ibid.*, 59–60.

¹⁰ *Ibid.*, 2.

¹¹ An additional consideration is the ability of the Department of Energy to produce warheads in numbers beyond those required for the triad replacement program of record. Constraints on the department's production capacity could be a limiting factor.

¹² The *2019 Missile Defense Review* discusses the potential benefits of space-based interceptors and states that DOD will examine concepts and technologies to support such a capability. See *2019 Missile Defense Review* (Washington, DC: DOD, 2019), 36–37, available at <<https://media.defense.gov/2019/Jan/17/2002080666/-1/-1/1/2019-MISSILE-DEFENSE-REVIEW.PDF>>.

¹³ The *2018 Nuclear Posture Review* fully explains the strategic logic behind the decision to field limited new nonstrategic nuclear capabilities. See *2018 Nuclear Posture Review*, 52–55.

¹⁴ For an overview of the INF Treaty, see Bureau of Arms Control, Verification and Compliance, Department of State,

"The Intermediate-Range Nuclear Forces Treaty," available at <www.state.gov/inf>. For the U.S. statement withdrawing from the treaty, see Michael R. Pompeo, "U.S. Withdrawal from the INF Treaty on August 2, 2019," Press Statement, Department of State, available at <www.state.gov/u-s-withdrawal-from-the-inf-treaty-on-august-2-2019/>. For a discussion of implications of the treaty's demise, see Amy Woolf, *U.S. Withdrawal from the INF Treaty: What's Next*, IF11051 (Washington, DC: Congressional Research Service, updated January 2, 2020), available at <<https://crsreports.congress.gov/product/pdf/IF/IF11051>>.

¹⁵ China may be moving toward a nuclear triad through the fielding of upgraded aircraft that can carry a nuclear-armed air-launched ballistic missile. See *Annual Report to Congress: Military and Security Developments Involving the People's Republic of China 2019* (Washington, DC: Office of the Secretary of Defense, May 2019), 67, available at <[https://media.defense.gov/2019/May/02/2002127082/-1/-1/1/2019%20CHINA%20MILITARY%20POWER%20REPORT%20\(1\).PDF](https://media.defense.gov/2019/May/02/2002127082/-1/-1/1/2019%20CHINA%20MILITARY%20POWER%20REPORT%20(1).PDF)>.

¹⁶ *China's National Defense in the New Era*, 2019 Defense White Paper (Beijing: State Council Information Office of the People's Republic of China, July 2019); *China's National Defense in 2006*, 2006 Defense White Paper (Beijing: State Council Information Office of the People's Republic of China, December 2006).

¹⁷ *2018 Nuclear Posture Review*, 32.

¹⁸ See Paul Bernstein and Harrison Menke, "Russia's Hypersonic Weapons," *Georgetown Journal of International Affairs*, December 12, 2019, available at <<https://gjia.georgetown.edu/2019/12/12/russias-hypersonic-weapons/>>.

¹⁹ The Biological Weapons Convention prohibits signing parties from developing, producing, stockpiling, or otherwise acquiring or retaining biological agents or toxins "of types and in quantities that have no justification for prophylactic, protective, or other peaceful purpose." These prohibitions also include "weapons, equipment, or means of delivery designed to use such agents or toxins for hostile purposes or in armed conflict." See "Convention on the Prohibition of the Development, Production, and Stockpiling of Bacteriological (Biological) and Toxin Weapons and on Their Destruction," Article I, available at <<http://disarmament.un.org/treaties/t/bwc/text>>.

²⁰ See Gerald L. Epstein, "Biodefense and Great Power Competition," *Nonproliferation Review*, Spring/Summer 2020 (forthcoming).

²¹ See *Biodefense in the Age of Synthetic Biology* (Washington, DC: National Academies Press, 2018), 71–84, available at <<http://nap.edu/24890>>.

²² Kim Riley, "Blue Ribbon Study Panel on Biodefense Focuses on DOD Role in Protecting U.S. Against Bio Attacks," *Homeland Preparedness News*, February 6, 2019, available at <<https://homelandprepnews.com/countermeasures/32417-blue-ribbon-study-panel-on-biodefense-focuses-on-dod-role-in-protecting-u-s-against-bio-attacks/>>.

²³ Jonathan B. Tucker, "The Future of Chemical Weapons," *The New Atlantis*, no. 26 (Fall 2009/Winter 2010), 3–29.

²⁴ *Spiez Convergence: Report on the Third Workshop 11–14 September 2018* (Zurich: Spiez Laboratory, November 2018), available at <www.labor-spiez.ch/pdf/en/rue/Spiez_CONVERGENCE_Report_on_the_3rd_workshop_2018.pdf>.

²⁵ Michael Crowley and Malcolm Dando, "The Use of Incapacitating Chemical Agent Weapons in Law Enforcement," *The International Journal of Human Rights* 19, no. 4 (2015), 465–487.

²⁶ Government of the United Kingdom, "Novichok Nerve Agent Use in Salisbury: UK Government Response, March to April 2018," April 18, 2018, available at <www.gov.uk/government/news/novichok-nerve-agent-use-in-salisbury-uk-government-response>.

²⁷ Oliver Holmes and Tom Phillips, "Kim Jong-nam Killed by VX Nerve Agent, Say Malaysian Police," *The Guardian*, February 24, 2017, available at <www.theguardian.com/world/2017/feb/24/kim-jong-nam-north-korea-killed-chemical-weapon-nerve-agent-mass-destruction-malaysian-police>.

²⁸ United Nations General Assembly Security Council, *Report of the United Nations Mission to Investigate Allegations of the Use of Chemical Weapons in the Syrian Arab Republic on the Alleged Use of Chemical Weapons in the Ghouta Area of Damascus on 21 August 2013*, A/67/997-S/2013/553, September 16, 2013, available at <https://digitallibrary.un.org/record/756814/files/A_67_997%26S_2013_553-EN.pdf>.

²⁹ *Third Report of the Organization for the Prohibition of Chemical Weapons–United Nations Joint Investigative Mechanism*, S/2016/738, August 24, 2016, available at <<https://undocs.org/S/2016/738>>.

³⁰ John A. Stevenson, Belinda Bragg, and Sabrina Pagano, *Violating Normal: How International Norms Transgressions Magnify Gray Zone Challenges*, NSI Concept Paper (Arlington, VA: NSI, February 2017), available at <<https://nsiteam.com/how-international-norms-transgressions-magnify-gray-zone-challenges-2/>>.

³¹ *Compliance with the Convention on the Prohibition of the Development, Production, Stockpiling and Use of Chemical Weapons and on Their Destruction*, Condition, (10)(c) Report (Washington, DC: Department of State, April 15, 2019), available at <www.state.gov/wp-content/uploads/2019/05/AVC-Condition-10c-2019.pdf>. See also 2018 and 2017 reports, both available at <www.state.gov/adherence-to-and-compliance-with-arms-control-nonproliferation-and-disarmament-agreements-and-commitments-compliance-report/>.

³² Peter Beaumont, "The Taboo on Chemical Weapons Has Lasted a Century—It Must Be Preserved," *The Guardian*, April 18, 2018, available at <www.theguardian.com/commentisfree/2018/apr/18/chemical-weapons-taboo-syria>; Filipa Lentzos, "Strengthen the Taboo Against Biological and Chemical Weapons," *Bulletin of the Atomic Scientists*, July 26, 2018, available at <<https://thebulletin.org/2018/07/strengthen-the-taboo-against-biological-and-chemical-weapons/>>.

Part III
**Geostrategic Interactions in a New
Era of Great Power Competition**

Chapter 9

The Indo-Pacific Competitive Space

China's Vision and the Post—World War II American Order

By Thomas F. Lynch III, James Przystup, and Phillip C. Saunders

This chapter examines the major strategic goals, interests, and policies being pursued by Washington and Beijing—the two major Great Power rivals in the Indo-Pacific region. It highlights the divergence of strategic interests between America's "Free and Open Indo-Pacific" vision and China's "community of common interest" framework. This divergence and the strategic importance of each country's regional interests make the Indo-Pacific region the most hotly contested geopolitical space at the dawn of the 2020s. An analysis of U.S. and Chinese critical power tools for attaining strategic outcomes finds a mix of relative advantages. China has clear advantage in economic leverage across the region and has developed some meaningful advantage in military tools necessary for success in conflict within the First Island Chain. On the other hand, the United States continues to possess demonstrable advantages in alliance diplomacy, ideological resonance, informational appeal, and broad military capabilities. Despite great and growing regional tensions, there are opportunities for collaboration between the Great Power competitors so long as both accept relative power limitations and rejuvenated American regional leadership provides a clear signal to Beijing that accommodating a continuing U.S. presence is a better choice than stoking conflict.

This chapter, like the volume itself, continues to analyze the emerging era of Great Power competition (GPC) with a framework focused on the three contemporary Great Powers: the United States, China, and Russia. In the Indo-Pacific region, however, Russian strategic interests and power capabilities pale in comparison with those of China and the United States. Thus, while in 2020 Moscow signals that it is interested in a greater role in the Indo-Pacific, significant doubts about its abilities to influence that region remain.¹ This chapter thus discusses Russia only to consider its role as an object rather than a subject in the dominant regional Great Power competition.

This chapter focuses on the Sino-American Great Power competitive dyad in the Indo-Pacific. It first summarizes the decade and a half of relations in Asia and the Pacific

that immediately followed the Cold War, examining how these relations set the conditions for significant strategic changes that began after 2008. The chapter then evaluates Chinese strategy and interests in the Indo-Pacific region since 2009, followed by the evolution of American strategic aims over the 2010s. The Sino-American competition for relative power across the Indo-Pacific region is then evaluated in the five categories of interstate competition established in table 2.2: political and diplomatic, ideological, informational, military, and economic. The differential power resources held by Washington and Beijing in these respective categories are assessed for 2020 and projected for at least the next half-decade. The unique case of Taiwan is considered in light of its special resonance to the regional rivalry.

The chapter establishes that each side pursues strategic goals that display historic continuity: The United States pursues unfettered economic and diplomatic access to the region with a preference toward open communications and human liberties, while China seeks domestic stability and to assert its sovereignty over long-contested geographic spaces with state-led management. While major aims have remained consistent, the power differential between Washington and Beijing has changed over the past 20 years. It has moved in Beijing's favor in terms of economic influence and selected measures of conventional military power, most notably in areas near China. At the same time, American relative power advantages remain strong in the diplomatic, ideological, and informational categories and the political-military aspects of defense competition. For America to compete effectively in the Indo-Pacific region now and into the near-term future, it must better leverage its advantages by strengthening existing alliances and security partnerships while actively promoting an attractive alternative vision for regional development. At the same time, it must parry Chinese efforts at economic diplomacy and military coercion to undercut the political foundations of U.S. alliances and American regional presence.

U.S. Regional Vision, Alliances, and Activities: 1992–2008

America's engagement with Asia began before the United States existed. In February 1784, the ship *Empress of China* departed New York harbor, arriving in Macau in August of that year. During the 20th century, Washington pursued Open Door trading rights in China, fought a war with Japan to sustain maritime access and commercial rights, and then developed a robust economic and security framework toward the region consistent with an array of American national interests. The common principle underlying various American policy approaches to the Western Pacific and Asia has been the concept of "access"—that is, economic access to the markets of the region to pursue U.S. commercial interests, strategic and physical access to allies to ensure confidence in U.S. security commitments, and political access to allow for the promotion of democracy and human rights.

Since the end of World War II, the United States has not only championed the evolution of a postwar liberal, open, rules-based international economic order allowing for the free flow of commerce and capital but also promoted efforts to support international stability and the peaceful resolution of disputes.² These global commitments applied firmly in America's post-Cold War approach to the Indo-Pacific and contributed significantly to the stability and prosperity of the region. To do so, the United States relied on military primacy expressed through an informal "hub and spoke" alliance structure. With Washington as the hub, American security treaties with Japan, the Republic of Korea (ROK), Australia,

the Philippines, and Thailand connected as spokes and served as the arrangement that Washington used to protect and advance its Asian and Pacific security interests.³

At the end of the Cold War in 1992, the United States was poised to enter what many pundits had dubbed the “Asian century.” Although post–Cold War America had urgent imperatives to consolidate the gains from newly liberated former Soviet bloc states and assist with the safe denuclearization of thousands of Soviet strategic weapons, Washington took strong steps to expand its economic competitiveness and influence in the dynamic Far East. Globally, and especially in Asia and the Pacific, the United States pursued a strategy of engagement (remaining active and connected globally) and enlargement (expanding the reach and strength of liberal political and economic rules and norms).⁴ In a September 1993 address, President Bill Clinton explained that the “successor to a doctrine of containment must be a strategy of . . . enlargement of the world’s free community of market democracies.”⁵ China was a key part of America’s approach there.

As detailed in chapter 3a, Beijing started its own economic metamorphosis from a command economy into a market-based one in 1978. A reforming China was a key beneficiary of American policies. Despite a lingering wariness after the 1989 Tiananmen Square crackdown by Chinese Communist Party (CCP) leadership security forces against human rights and democracy protestors, American leadership across the political spectrum converged during the mid-1990s around the view that including China in global institutions and supporting Chinese market reforms offered the best chance that China would rise peacefully and become a responsible global economic power and stakeholder within the existing order. Many Americans were optimistic that a rising Chinese middle class would demand a direct political voice and challenge CCP authoritarian rule.⁶ Washington opened American markets to Chinese goods, encouraged China’s introduction into regional supply chains, allowed the transfer of advanced civilian technologies, paved Beijing’s way into the World Trade Organization in 2001, and encouraged Beijing to become more engaged and influential in both regional and global diplomacy.⁷ Washington believed its support helped produce explosive growth of Chinese foreign trade from about \$20 billion in the late 1970s to \$475 billion in 2000.⁸

Between 1992 and 2008, American companies turned toward China to access its rapidly growing market and to use cheap Chinese labor to lower production costs.⁹ In 1993, the World Bank and International Monetary Fund published projections that China was on the verge of replacing Japan as the world’s second largest economy.¹⁰

At the same time, Washington’s alliance with Japan evolved into a genuine strategic partnership. Tensions over the trade imbalances that bedeviled Tokyo and Washington during the 1980s subsided as Japan entered a lengthy economic stagnation, bilateral voluntary trade restraints took hold, and Japanese production shifted to the Asian mainland. Despite a period of worry in Tokyo, numerous shared interests and similar democratic values enhanced the political and military cohesion of the alliance. Japan provided the military

“I would argue that both the domestic dynamics and each country’s increasingly gloomy assessment of the other’s true intentions against the backdrop of China’s rise help explain the current state of affairs.”

—James B. Steinberg, “What Went Wrong,”
Texas National Security Review 3, no. 1
(Winter 2019/2020)

bases and other logistical support that undergirded America's regional military dominance and helped maintain regional stability conducive to U.S. strategic interests. Common democratic values reinforced U.S.-Japan relations, making the bilateral commitment more than a mere strategic expediency. Public opinion polls throughout the 1990s and 2000s demonstrated an American-Japanese popular consensus that the alliance and common values mattered a great deal to their bilateral relationship and were elements missing in the two countries' relations with China.¹¹

During the early post-Cold War period, the United States tried to manage a sullen, stagnant North Korea (simultaneously pursuing deterrence and diplomacy to try to eliminate the North's nuclear program); maintain stability in the tense relations between China and Taiwan; foster greater Asia-Pacific multilateral economic, political, and security cooperation; and integrate China into regional and global economic and security regimes. These ambitions progressed in fits and starts. While regional economic integration got a boost from China's rapid growth and openness to foreign investment, Beijing's assimilation into the World Trade Organization fell short of obligations and Western expectations. The Clinton administration responded to the 1995–1996 Taiwan Strait security crisis with an effort to build a partnership with China, including limited military-to-military cooperation. The incoming Bush administration took a more skeptical view of China's military potential, and the accidental collision of a Chinese navy fighter and a U.S. reconnaissance plane on April 1, 2001, produced a tense diplomatic standoff and a freeze in Sino-American military contacts.

The terrorist attacks on the United States in September 2001 detoured U.S. attention from the “Asian century” to the Middle East for the better part of a decade. After 9/11, the U.S. Government engaged Asian states for support in the war on terror and instability in the Middle East and South Asia.¹² It also leveraged its strategic relationships with Japan and South Korea to move equipment and materiel into American-led counterterrorism activities in Afghanistan and Iraq. Both contributed money and personnel to the counterterrorism missions. When America did engage with China in the aftermath of 9/11, it was to appeal for China to help deal with North Korea's nuclear weapons program and to serve as a “responsible stakeholder” in the U.S.-led international system. Chinese responsiveness remained tepid. While China was restrained in employing force, its military modernization accelerated, supported by large increases in defense budgets that raised concerns across the region. The 2007 announcement of an 18 percent increase in military spending led Vice President Dick Cheney to state, “A China military buildup is not consistent with the country's stated goal of a ‘peaceful rise.’”¹³

Months earlier, America's main regional ally, Japan, publicly announced its worries over China's growing strategic challenge. In a major speech, “The Arc of Freedom and Prosperity,” Japanese Foreign Minister Taro Aso established a framework of policies to structure East Asia marked by “value oriented diplomacy,” based on “universal values” such as democracy, freedom, human rights, the rule of law, and a true market economy.¹⁴ Then Japanese Prime Minister Shinzo Abe in August 2007 addressed the Indian parliament and called for a “Confluence of the Two Seas” running from Japan to India where shared fundamental values such as freedom, democracy, and respect for basic human rights were honored.¹⁵ These Japanese statements—and the effort to engage India—were seen as efforts to alert a

distracted America and the rest of the world that China's rise had the potential to undercut liberal values at the regional and international levels.

China's Regional Vision and Activities: 1992–2008

For China, the Indo-Pacific is the most important region of the world in economic, security, and political terms. This was true during the Cold War and especially after, when China became more actively and deeply engaged with neighboring countries.

In the economic realm, the region serves as a source of raw materials; as a supplier of components, technology, and management expertise for production networks operating in China; and as a market for finished Chinese products. During the 1990s, China's increasing role in world trade and expectations of future growth made it an attractive market and gave Beijing leverage in dealing with nearby Asian and Pacific trading partners and enabled negotiation of regional and bilateral free trade agreements (FTAs). China worked to persuade Asian countries that they would share in the benefits of its rapid growth, while simultaneously advancing its own interests through commercial diplomacy. *Win-win* and *mutual benefit* became the watchwords of China's economic diplomacy.

Geography makes the Indo-Pacific region critically important to China from a security perspective. China shares land borders with 14 East Asian, South Asian, and Central Asian countries. Chinese leaders worry that neighboring countries could serve as bases for subversion or for military efforts to contain China. This is of particular concern because much of China's ethnic minority population, which Chinese leaders view as a potential separatist threat, lives in sparsely populated border regions such as Xinjiang and Tibet. Chinese concerns about threats posed by "terrorism, separatism, and religious extremism" have prompted increased efforts at security cooperation with its Central and South Asian neighbors.

China's unresolved territorial claims all are in Asia, including claims to the Spratly and Paracel islands in the South China Sea, the Diaoyu/Senkaku Islands, a disputed maritime boundary with Japan in the East China Sea, a 1,600-mile-long disputed land border with India, and China's self-described "core interest" in unification with Taiwan. China also worries about the possibility of encirclement and threats from conventional military forces based on its periphery. Chinese strategists have been highly sensitive to U.S. regional alliances and partnerships that might someday be turned against China.

Beijing also views Asia as politically critical. Its preference is for a stable environment that permits rapid Chinese economic growth to continue and supports increased Chinese regional influence. Chinese officials and analysts acknowledge that the U.S. role in supporting regional stability and protecting sea lines of communication has made a significant contribution to regional stability and supported Chinese interests. Beijing opposes alliances in principle but has tolerated them so long as they are not aimed toward China and help constrain Japanese militarization. The potential for U.S. power and alliances to be turned against China makes Chinese analysts uneasy, especially as changes in the U.S.-Japan alliance now appear to be empowering Japan rather than restraining it. China disclaims any desire to dominate Asia, declaring that it will never seek hegemony and talking about cooperation on the basis of equality, mutual respect, and noninterference in the internal affairs of other nations. But Chinese elites also appear to expect that

Figure 9.1. China's Nine-Dash Line (in green) in the South China Sea



Source: South China Sea (Islands), Perry-Castañeda Map Collection (Central Intelligence Agency, 1988)

weaker countries will defer to Chinese wishes as the country grows more powerful.¹⁶

Aggressive Chinese behavior toward Taiwan and in the South China Sea from 1994 to 1996 created regional alarm about a “China threat.” In late 1994, Beijing seized and then fortified a small shoal, Mischief Reef, claimed by the Philippines. This event highlighted China’s controversial “nine dash line” claims to sovereignty over the land features and most of the waters in the South China Sea, including a number of features claimed and occupied by other countries (see figure 9.1).

In late 1995 and in March 1996, China used military exercises (including firing ballistic missiles into waters near Taiwan) to express its concerns about the Taiwan independence movement and its displeasure at

a U.S. decision to permit Taiwan president Lee Teng-hui to visit the United States. Washington responded to Chinese attempts to intimidate Taiwan before its elections by deploying two carrier strike groups to nearby waters. These actions prompted numerous articles and books highlighting China’s military modernization and growing nationalism and asking whether China posed a threat to the Asia-Pacific region.¹⁷ More restrained Chinese behavior and assurance measures adopted from 1997 to 2008 helped ease some regional concerns.¹⁸ During this period, many Asian views shifted from regarding China as a potential threat to regarding China as an opportunity.¹⁹ However, as noted earlier, Japan did not join in this view and instead introduced a policy vision between 2006 and 2007 for a future in Asia that challenged preferred Chinese outcomes.

The growth of Chinese military power in the mid-2000s was driven partly by the military’s desire to convert China’s economic strength into military power and partly by CCP leadership concerns about vulnerability to unconstrained U.S. power. China observed and then participated in bilateral and multilateral military exercises with neighboring countries as a confidence-building measure and an opportunity for Asian militaries to interact with the People’s Liberation Army (PLA). As observed in chapter 3a, the 2008 global financial crisis—which produced a prolonged U.S. recession even as China’s economy returned to its rapid growth trajectory—led many Chinese analysts to see an acceleration of U.S. rel-

ative economic decline as a sign of growing multipolarity that created new opportunities for China. Although Chinese leaders sought to avoid a direct clash with Washington, they accelerated efforts to expand China's regional presence and influence. China also began to adopt a more assertive approach to its maritime territorial claims in the South China and East China seas. These developments set the stage for increased regional tensions and a negative turn in U.S.-China regional relations.

China and America in the Indo-Pacific Region: 2009 and Beyond

America's counterterrorism entanglements in the Middle East and South Asia did not change the underlying view in Washington that the center of gravity for American interests was in Asia. As the world crawled out from under the Great Recession of 2008–2009, the Obama administration began with a series of Indo-Pacific speeches and policy initiatives to extend cooperation with China, India, and longtime U.S. regional allies and partners in a manner that would “uphold international norms and [respect] universal human rights.”²⁰ As noted in chapter 3a, Beijing's increased influence and military power reinforced a belief that the United States had entered an accelerated period of decline that presented China with an opportunity to set the agenda in U.S.-China relations and regional affairs. At the same time, Chinese leaders continued to worry about domestic vulnerabilities (as evidenced by ethnic unrest and violence in Tibet and Xinjiang in 2008) and concerns about possible U.S.-led subversion à la the colored revolutions that had overthrown Middle Eastern dictators and pro-Russian authoritarian leaders in the Ukraine and Kyrgyzstan. Xi Jinping was part of the CCP collective leadership that charted a more assertive regional policy, a trend that would intensify once he became the CCP general secretary in November 2012. The conditions for a dramatic change in the tone of U.S.-Chinese relations had been set.

China's Geostrategic Aims and Trajectory

Beginning in 2009, a more assertive Chinese posture emerged on a wide range of bilateral, regional, and global issues.²¹ Within the space of 18 months, Chinese diplomatic bullying, assertive military and paramilitary actions, and disregard for foreign reactions undid many of the gains from Beijing's decade-long charm offensive in the Indo-Pacific region. In particular, the means used to advance Chinese maritime sovereignty claims in the South China Sea and East China Sea—including harassment of U.S. military ships and aircraft operating legally in international waters or within China's exclusive economic zone—did considerable damage to Beijing's efforts to persuade others that China's rise would be peaceful.²²

The shift in tone and substance of Chinese policy had both international and domestic causes. As noted in chapter 3a, when Chinese growth resumed, and the United States and Europe remained mired in the 2008–2009 recession, Chinese officials and analysts appear to have exaggerated the negative impact of financial problems on U.S. global leadership and mistakenly concluded that a fundamental shift in the global balance of power was under way. Chinese officials also appear to have misinterpreted Obama administration efforts to increase bilateral cooperation and expand China's role in global institutions as a sign of U.S. weakness and an opportunity to press Washington for concessions.²³ This assessment played into a nationalist mood in China, where many commentators argued that a more

powerful China should take a hard line on challenges to Chinese territorial claims and use its economic leverage to punish the United States for arms sales to Taiwan.²⁴

Chinese officials and scholars denied that Beijing changed its foreign policy goals, expanded its territorial claims, or adopted a more assertive attitude toward maritime disputes. They argued that other countries, emboldened by U.S. support, had stepped up their challenges to China's long-established territorial claims. The May 2009 deadline for submissions to the United Nations Convention on the Law of the Sea (UNCLOS) did spur many Asian countries (including China) to reinforce their claims to disputed islands and waters. Chinese officials and military officers argued that restraint in response to provocations was misinterpreted as weakness.²⁵ Beijing employed economic coercion in some of the sovereignty disputes, including a temporary ban on exports of rare earth elements to Japan following the 2010 arrest of a Chinese fishing boat captain and import restrictions on Philippine bananas in 2012. China also took a tough line on military activities in its exclusive economic zone, acting to interfere with U.S. ships (including a March 2009 incident off Hainan Island when Chinese paramilitary vessels attempted to snag the towed sonar array of the USNS *Impeccable*).²⁶

During this period, Chinese policymakers talked about the need to maintain the proper balance between the competing goals of defending Chinese sovereignty [*weiquan*] and maintaining regional stability [*weiwen*]. But under President Xi, China began placing more emphasis on pursuing territorial claims and exhibiting less concern about the negative impact on relations with its neighbors and with the United States. Tactics to assert sovereignty include patrols by Chinese coast guard and naval forces, occupying land features, enforcement of fishing regulations in disputed waters, oil and natural gas exploration, harassment of military ships and aircraft operating in disputed areas, and use of legal means to press tendentious Chinese interpretations of international law.

In 2013, the year after Xi's political ascension, China began land reclamation projects in the South China Sea on several low-tide elevations, geologic features that do not extend above water at high tide. China's efforts at land reclamation were not unprecedented: Malaysia, the Philippines, Taiwan, and Vietnam have also engaged in such projects since the 1980s. In May 2014, China deployed an oil rig into waters near the Paracel Islands claimed by Hanoi, raising tensions and setting off collisions between Chinese and Vietnamese coast guard ships and virulent anti-Chinese demonstrations in Vietnam. By June 2015, China's land reclamation projects totaled "more than 2,900 acres, or 17 times more land in 20 months than the other claimants combined over the past 40 years, accounting for approximately 95 percent of all reclaimed land in the Spratly Islands" (see figure 9.1).²⁷

In September 2015, President Xi pledged that China would not "militarize" the artificial islands that it had constructed, but the commitment was vague.²⁸ Soon China began to use the airfields and port facilities for both military and civilian purposes. China has never precisely specified the exact nature or the legal basis for its South China Sea maritime claims under international law. Beijing's position is that "China has indisputable sovereignty over the Nansha islands and their adjacent waters," with "sovereignty and relevant rights . . . formed over the long course of history and upheld by successive Chinese governments."²⁹ However, on July 12, 2016, the Permanent Court of Arbitration, in a case brought by the Philippines contesting Chinese claims in the South China Sea, ruled in favor of Manila

and found that most of Beijing's claims—including to historic waters inside “the nine dash line”—had no basis under UNCLOS. China denied that the court had jurisdiction, did not participate in the hearings, and refused to accept the court's judgment.

While benefiting from a stable regional order underpinned by U.S. power and alliances, China gradually moved to form new regional institutions to advance its interests that mostly excluded the United States. Initial steps involved the Shanghai Cooperation Organization in 2005 and Chinese efforts to exclude the United States from the nascent plans for an East Asian Summit. Since 2013, new initiatives include the Asian Infrastructure Investment Bank, Belt and Road Initiative (BRI), Regional Comprehensive Economic Partnership, and reinvigoration of the Conference on Interaction and Confidence-Building Measures in Asia (CICA) as vehicles for Xi's “Asia for Asians” security concept. In announcing the “Asia for Asians” concept at a 2014 CICA summit, Xi argued,

*Asia has come to a crucial stage in security cooperation where we need to . . . strive for new progress . . . to move from the 20th century with the outdated thinking from the age of the cold war and zero-sum game . . . to innovate our security concept to establish a new regional security cooperation architecture . . . that is shared by and win-win to all.*³⁰

An accompanying Xinhua article characterized U.S. alliances as the “Achilles' heel” of and a major impediment to ‘a peaceful Asia.’”³¹ Themes blaming “outside powers” for stirring up trouble in the region have become a staple of Chinese propaganda and diplomatic messaging.

America's Geostrategic Aims and Trajectory

As observed in chapter 3a, the American narrative on Sino-U.S. interactions is that the era of cooperative relations with China stumbled beginning in 2008, with efforts at cooperation gradually faltering and competitive aspects of relations coming to the fore with a de facto shift toward strategic competition in 2014/2015—one formalized by the United States in 2017/2018.

After taking office in January 2009 and despite an enormous American military footprint straddling South Asia and the Middle East, Obama administration officials proclaimed a U.S. “return to Asia.” In formally announcing the rebalance in a November 17, 2011, address to the Australian parliament, President Barack Obama argued that “Our new focus on this region reflects a fundamental truth—the United States has been, and always will be a Pacific nation. . . . Here we see the future.” The President noted that Asia is “the world's fastest growing region,” “home to more than half of the global economy,” and critical to “creating jobs and opportunity for the American people.” He described the rebalance as “a deliberate and strategic decision” to increase the priority placed on Asia in U.S. policy.³² Secretary of State Hillary Clinton elaborated on the rationale for the rebalance, arguing that “harnessing Asia's growth and dynamism is central to American economic and strategic interests” and that the United States had an opportunity to help build “a more mature security and economic architecture to promote stability and prosperity.”³³

While the main objective of the rebalance was to bring American foreign policy commitments in line with the global distribution of U.S. strategic interests, it also responded to China's increasingly assertive regional policies, especially in maritime territorial disputes. Countries across the Asia-Pacific region urged Washington to play a more active role in regional economic, diplomatic, and security affairs in order to demonstrate U.S. commitment and help maintain regional stability in the face of a more powerful and active China. One early U.S. response was at the May 2010 meeting of the Association of Southeast Asian Nations (ASEAN) Regional Forum, where the United States offered to assist countries in the peaceful resolution of concerns with China's assertive maritime policies, noting that these posed a threat to freedom of navigation. China urged regional states to keep silent, and, when they spoke up, Chinese Foreign Minister Yang Jiechi walked out in protest, only to return the following day to remind ASEAN states that "China is a big country and other countries are small countries and that is just a fact."³⁴

Obama administration officials stressed that the rebalance included diplomatic, economic, and military elements, coupled with efforts to build a more cooperative and stable Sino-U.S. relationship. The broad U.S. strategy of seeking to integrate China more fully within the existing global order, while discouraging any efforts to reshape that order by force or intimidation, remained in place. Washington sought to make the rebalance robust enough to reassure U.S. allies and partners of its capability and will to maintain a presence in Asia over the long term while not alarming Chinese leaders to the point where they abandoned bilateral cooperation.³⁵ Nevertheless, the rebalance was widely viewed as evidence of increasing U.S.-China competition for regional influence.

From 2013 through 2015, Chinese assertiveness in maritime territorial disputes, increasing state intervention to support Chinese businesses at the expense of foreign competitors, and Xi's centralization of power and tightening of political and information controls catalyzed American responses. In 2014 and into 2015, the Obama administration publicly stated that Article V of the Japan Security Treaty extended to the Senkaku Islands in the East China Sea, asserted freedom of navigation rights in the South China Sea, and conducted more frequent freedom of navigation operations to challenge illegitimate Chinese maritime claims. It openly condemned Chinese industrial espionage and intellectual property practices, and it reimagined a broad new Trans-Pacific Partnership (TPP) as a lever to reshape Chinese economic policies. U.S. policy toward both China and Russia chilled gradually during the second term of the Obama administration, with a public hardening increasingly evident during 2014 and 2015.³⁶ This gradual hardening did not precipitate a formal rupture in U.S.-Sino relations, but it set the conditions for a new U.S. administration. The administration of President Donald Trump, elected in 2016 and inaugurated in January 2017, took the bilateral hardening to a new and official level.

The December 2017 U.S. National Security Strategy argued that a "geopolitical competition between free and repressive visions of world order is taking place in the Indo-Pacific region," labeling China a "revisionist power."³⁷ That report was especially critical of China's use of "economic inducements and penalties, influence operations, and military threats" to alter the regional order.³⁸ A principal architect of the 2017 strategy, H.R. McMaster, later wrote that a careful study of history and experience teaches that the CCP will not liberalize internally and will not act abroad according to U.S.-led international rules. Instead, McMaster wrote,

China's goal is to replace the current international order with one led by the CCP. China will continue to engage in "economic aggression" and seek to exert control of "strategic geographic locations and establish exclusionary areas of primacy." In other words, China's goal is to reduce, then eliminate, U.S. influence in the Indo-Pacific region.³⁹ This assertion of hostile Chinese aims contrasted with the more equivocal tone on China's rise in Bush 43- and Obama administration-era strategic documents.⁴⁰ In May 2018, the U.S. military renamed its U.S. Pacific Command to U.S. Indo-Pacific Command (USINDOPACOM), symbolizing the growing importance of India in intensifying U.S. competition with China.⁴¹ The Defense Department's June 2019 Indo-Pacific report likewise noted China's "campaign of low-level coercion to assert control of disputed spaces in the region, particularly in the maritime domain."⁴² In testimony before the Senate Armed Services Committee in February 2019, USINDOPACOM's inaugural commander was even more direct in his dire assessment of the future threats posed by China to the United States and its partners in the Indo-Pacific region. He testified:

China represents our greatest long-term strategic threat to a Free and Open Indo-Pacific and to the United States. . . . Those who believe this is reflective of an intensifying competition between an established power in the United States, and a rising power in China, are not seeing the whole picture. . . . Rather, I believe we are facing something even more serious—a fundamental divergence in values that leads to two incompatible visions of the future. . . . Through fear and coercion, Beijing is working to expand its form of ideology in order to bend, break, and replace the existing rules-based international order. . . . In its place, Beijing seeks to create a new international order, one with "Chinese characteristics" and led by China—an outcome that displaces the stability and peace of the Indo-Pacific that has endured for over 70 years.⁴³

In November 2017, shortly before his administration released the 2017 National Security Strategy, President Trump announced a "free and open" Indo-Pacific (FOIP) vision at a summit of Asia-Pacific Economic Cooperation business leaders in Da Nang, Vietnam.⁴⁴ In July 2018, U.S. Secretary of State Mike Pompeo explained that the core principles of U.S. FOIP are freedom from coercion, good governance, open access to seas and airways, and free and fair trade.⁴⁵ The State Department's November 2019 report on the U.S. Indo-Pacific FOIP vision stated that the Trump administration was implementing a "whole of government strategy" to defend these principles. The document noted that U.S. trade in the region topped \$1.9 trillion in 2019, supporting more than 3 million American jobs.⁴⁶

Trump's emerging Indo-Pacific economic policies placed firm emphasis on bilateral FTAs, in contrast to the Obama administration's pursuit of the multilateral TPP, which was an immediate casualty of domestic politics and the new administration. In April 2018, a State Department official explained that this shift reflected the view that a multilateral deal would provide fewer benefits to "U.S. workers and U.S. businesses" than bilateral agreements.⁴⁷ Another change has been greater emphasis on infrastructure development, as exemplified by the October 2018 Better Utilization of Investments Leading to Development (BUILD) Act, which raised U.S. financing from \$29 billion to \$60 billion.⁴⁸ Of note, these funds, while funneled through a U.S. Government entity, were to combine some government dollars with a lot of

Figure 9.2. The First and Second Island Chains

Source: *Annual Report to Congress: Military Power of the People's Republic of China 2009* (Washington, DC: Department of Defense, 2009), 18.

loan guarantees for anticipated private equity, unlike the fully state-backed loans that support the Chinese BRI and state-owned enterprises that are China's biggest overseas investors.

From 2017 to 2019, the Trump administration successfully strengthened key alliances in the Indo-Pacific region, namely with Japan and Australia. It expanded military cooperation with traditional allies such as Australia, Japan, and South Korea, while using exercises and dialogues to reach out to nontraditional partners such as India, Malaysia, and Vietnam.⁴⁹ The Defense Department's *Indo-Pacific Strategy Report*, released in June 2019, illuminated how the acquisition and deployment of advanced capabilities, new operational concepts, and initiatives to strengthen security partnerships (highlighting Taiwan, New Zealand, and Mongolia) would contribute to the preservation of a "free and open" region and dissuade Chinese adventurism.⁵⁰ That report featured plans for the strengthening of America's five treaty alliances; expanded partnerships with Taiwan, New Zealand, and Mongolia; and emerging partnerships with other nations from South Asia to the Pacific islands.

The American strategy also prioritized greater development of a security partnership framework known as the "Quad," featuring the United States, Japan, Australia, and India. It encouraged greater trilateral regional security partnerships, greater American engagement with the ASEAN Regional Forum (ARF), and support for intra-Asian partnerships (most notably between Australia and Japan and Japan and the Philippines; trilateral cooperation among Australia, Japan, and the United States; and among Japan, the ROK, and the United

States). The document also prioritized the American purchase of fifth-generation aircraft, long-range antiship missiles, offensive cyber capabilities, and the development of new operating concepts. It highlighted American arms sales to Taiwan (\$10 billion) and India (\$16 billion), funding for a Southeast Asia Maritime Security Initiative (\$356 million), the Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation, greater U.S. Coast Guard engagement, and an explosives removal package for Vietnam (\$340 million).

Thus, the American strategic design—its policy interplay—across the Indo-Pacific is to seek partners that support and extend FOIP. This is a contrast to what the United States views as China's coercive efforts to advance its sovereignty claims inside the First Island Chain and increasingly to assert military dominance through the Second Island Chain (see figure 9.2). Washington sees China's desire to build a "community of common destiny" as part of efforts to exclude the United States from the Indo-Pacific and build a new regional order emphasizing values of state sovereignty, collective order, and limited human rights and freedoms.

American and Chinese Competitive Power Tools and Prospects for Indo-Pacific Success

As described in chapter 3a, several factors are driving the U.S.-China relationship into Great Power competition. Although this competition has global and extra-regional dimensions, the Indo-Pacific region is the most important venue, especially for diplomatic and military aspects of the competition. This leaves countries in the region in an uncomfortable position. Most share U.S. concerns about the risks of Chinese hegemony and have no desire to be left alone in a Chinese sphere of influence, forced to subordinate their interests to those of Beijing.

At the same time, countries across the Indo-Pacific region do not want Washington to drag them into a confrontation with China that could damage their economies (all of which depend heavily on trade with China), destabilize the region, and potentially lead to a devastating war. They seek to maintain a balance that allows them to cooperate economically with both the United States and China and limits Beijing's opportunities for coercion for fear of driving them into Washington's arms. Maintaining this balance is the difficult challenge for regional leaders as U.S.-China Great Power competition broadens and intensifies.

This chapter analyzes the Indo-Pacific competitive arena in terms of the five major categories first described in table 2.2:

political and diplomatic, ideological, informational, military, and economic. Below, U.S. and Chinese competitive advantages in these categories are briefly compared and assessed. This evaluation establishes that, while China's power tools have grown over the past several years, the United States retains formidable assets and capabilities if properly marshaled in cooperation with regional allies and partners.

"What has changed in recent years are not the CCP's goals, but rather the means available to achieve them, as well as Beijing's willingness to exercise its growing power in order to do so. Since the mid-1990s, China's rapid economic growth has enabled it to fund a wide-ranging and sustained modernization of its armed forces."

—Aaron L. Freidberg, "Getting the China Challenge Right," *The American Interest* 14, no. 4 (January 2019)

Political and Diplomatic Tools and Prospects

The United States. America's longstanding network of alliances and friendships across the Indo-Pacific region has traditionally been a huge advantage over China. Habits of co-operation have been institutionalized between Washington and a number of regional allies: Japan, South Korea, Australia, the Philippines, and Thailand. The United States also has strategic partnerships with other important Indo-Pacific nations—India, Singapore, Vietnam, Indonesia, and Malaysia—that view Washington as a preferable partner to Beijing so long as not forced to make an either-or choice.

America's diplomatic approach to the region was undergoing change before the Presidential election of 2016. It has undergone even more since. Since 2017, Washington has placed less emphasis on some multilateral regional forums that it encouraged and later joined during the post-Cold War years.⁵¹ President Trump appears more interested in bilateral engagements with major countries than in multilateral forums, although his Cabinet officials have continued to participate regularly in multilateral meetings such as the ARF and ARF Defense Ministers meetings.⁵² The Trump administration has pursued a series of bilateral initiatives—economic, security, and diplomatic—that have tested allies and partners in ways that many analysts have found worrisome. Interestingly, the diplomatic balance sheet of 2020 finds that longstanding American allies have weathered rather well the Trump administration's frequent questioning and testing, coming through as strong and resilient. Among these, America's alliances with Japan and Australia have been updated and enhanced.⁵³ Even longstanding partnerships with challenging allies—South Korea and the Philippines—have endured despite public spats over American military bases and cost-sharing.

The November 2019 State Department document titled *A Free and Open Indo-Pacific: Advancing a Shared Vision* pledged a wide range of American diplomatic, economic, and security programs to engage and sustain American interests and partnerships across the Indo-Pacific region.⁵⁴ The document emphasized continuing American diplomatic engagement with regional partners and institutions via programs including the Pacific Pledge (\$100 million) and the Lower Mekong Initiative (\$3.8 billion). It also championed continuing good governance with a Transparency Initiative (\$600 million) and a Myanmar Humanitarian Assistance Program (\$669 million), among others. The United States also invested in regional human capital development with a number of programs, including enhanced Fulbright Fellowships, a Young Southeast Asian Leaders Initiative, and a Food for Peace Program. These and other American diplomatic programs are carried out under many different labels, banners, and names, which may undermine their collective impact. American leaders might consider a common branding for these and related economic and security assistance initiatives.

At the same time, American diplomatic efforts remain under-resourced for their critical Indo-Pacific role. As of 2019, China surpassed the United States with the largest number of diplomatic posts in the world and also outmanned Washington in number and staffing of embassies and consulates across the Indo-Pacific region.⁵⁵ This relative decline in U.S. diplomatic presence risks undoing the programs Washington has crafted for political outreach and could turn a longstanding American regional diplomatic advantage into weakness.⁵⁶

China. China's political and diplomatic framework for activity in the Indo-Pacific region is based on looser and more complicated relationships with major regional neighbors and regular participation in multilateral regional forums such as Asia-Pacific Economic Cooperation, the East Asian Summit, ASEAN + 3 (China, Japan, and South Korea), and the China-ASEAN Forum. Its regional policy often seeks to serve multiple objectives and avoid embarrassing setbacks to its domestic interests and public image.⁵⁷ The priority Beijing gives to domestic concerns and its territorial disputes with countries such as India, Japan, Indonesia, Vietnam, and Malaysia can leave China isolated and awkwardly positioned, giving the United States a potential advantage in coalition-building based on longstanding strategic relationships.⁵⁸ China's preference for nonbinding strategic partnerships allows it to cooperate on a selective basis with most countries in the region but does not translate readily into coalitions for deeper strategic goals. China's longstanding "all-weather friendship" with Pakistan is a conspicuous exception to the paucity of alliances, partnership, and friends for Beijing in the Indo-Pacific region.

China's competitive approach to regional diplomacy is oriented on a three-pronged framework. First, China contrasts its five principles of peaceful coexistence, a new model of international relations centered on mutually beneficial cooperation, and the vague notion of collective security embodied in the notion of community of common destiny against Washington's supposed "Cold War mentality" and alliance-based approach to regional security.⁵⁹ Beijing increasingly portrays U.S. alliances and the U.S. military presence as stirring up trouble for the region and unfairly aiming to choke off China's legitimate rise. Second, China seeks to use access to its market and preferential benefits from its BRI infrastructure projects and other investment programs to increase its influence in the region and to dissuade countries from taking actions against its interests. This is a potential source of diplomatic leverage, although China's record of using economic sanctions and pressure to alter partner behaviors has been mixed at best, often producing blowback from states it is seeking to influence. Third, China has sought to undercut U.S. diplomatic initiatives through strategically targeted high-level visits and improved relations with traditional U.S. allies and would-be American partners. Xi Jinping's October 2019 visit to New Delhi and his planned visit to Tokyo in 2020 stand as cases in point.⁶⁰ China's public response to the Trump administration FOIP vision criticizes U.S. partnership activities as out of touch with regional needs, complains that American initiatives are intended to encircle China, and critiques the initiative as insultingly under-resourced for success. Chinese analysts and strategists are particularly focused on limiting the salience and effectiveness of the Quad security arrangement involving the United States, Japan, Australia, and India—with special attention to India as both the most worrisome strategic partner and the weakest link in the structure.⁶¹

During 2019 and into early 2020, a much more assertive Chinese diplomatic approach became evident globally, especially in the Indo-Pacific region. Critics complain that Chinese diplomats have become more strident and combative, including in aggressive efforts to deflect Chinese responsibility for mistakes in managing the novel coronavirus (COVID-19) outbreak, to threaten economic retaliation against states calling for an investigation of the origins of the virus, and to cite China's eventual success in taming the virus as evidence of the superiority of the Chinese political system.⁶² For example, Australia explicitly rejected what it called economic coercion by China in April 2020 when China's Australian

ambassador stated that the Chinese public could avoid Australian products and universities should Australia continue to press for an independent investigation into the origins and early actions in China surrounding COVID-19.⁶³ Chinese officials argue that more active diplomacy simply reflects Beijing's more prominent role on the world center stage and the West's relative decline.⁶⁴

Although China's diplomatic presence and activity continues to grow, its overall political influence across the region remains low compared with that of the United States, especially in terms of its ability to mobilize countries for costly actions. But America's advantages are tenuous. Should it abandon efforts to exert leadership in regional organizations, understaff and under-resource its diplomatic presence, or take its regional allies and partners for granted, America's palpable advantage in Indo-Pacific diplomatic competition could suffer a noteworthy downturn.

Ideological Tools and Prospects

The United States. The FOIP vision captures critical elements of historic American aims and interests in the Indo-Pacific region. It rests on the bedrock of American ideals of liberal democracy and a free trade system: respect for the rule of law; individual rights; freedom of navigation and overflight, including open shipping lanes; peaceful resolution of disputes; and transparency in the free flow of information. Some contend that these aims and values are "Western" and not inherently "Asian," claiming Asian culture and history prioritize strong central governments, collective responsibilities, and social harmony over economic liberty and political rights. But Asia's history since the late 1980s has challenged the notion of any sort of "Asian exceptionalism." Progress has been mixed, but countries such as Japan and South Korea remain impressive democracies even as states such as Thailand and the Philippines have wobbled in recent years. Taiwan also stands as a success story.⁶⁵ At the same time, the United States has been pragmatically applying its preferred values of liberal democracy and individual rights in the Indo-Pacific region. Singapore's authoritarian governance with a democratic veneer has been acceptable to Washington, and Thailand's rough-hewn, frequently illiberal democracy has not resulted in its termination as an American treaty ally.

FOIP vision themes have wide regional appeal, showing up in bilateral accords and in prominent regional bilateral and multilateral documents. Even before the Trump administration, these ideas appeared as cornerstones of two major regional vision documents: the U.S.-India Strategic Vision of January 2015 and the India-Japan Vision 2025 signed in December 2015.⁶⁶ Since 2017, FOIP's ideological foundations have been included in other major regional declarations, including American Two-Plus-Two Joint Statements with Japan and Australia and in other key partnership diplomatic documents.

American culture and social engagement remain robust in the Indo-Pacific region. Despite some downturn since 2017, the United States is an enormously attractive location for regional pursuit of graduate and tertiary education. In 2018, more than one-third of Chinese students who studied overseas did so in the United States (321,625), as well as half of Japan's overseas tertiary students (14,787), almost one-third of those from Australia (4,286), and almost half of those from India (142,618).⁶⁷ In 2019, some 730,000 students from the Indo-Pacific region were at graduate or fellowship programs in American colleges

and universities, representing more than two-thirds of all foreign students in American higher education programs.⁶⁸

Americans remain the predominant and highest spending tourists across the region and were—before COVID-19 complications—anticipated to spend \$257 billion in the region (28 percent of the market) by 2020, compared with a projected \$60 billion by China (8 percent).⁶⁹ English is the language of business in the Indo-Pacific region, and Australia, New Zealand, Singapore, the Philippines, India, Pakistan, and Nepal are either English-speaking or recognize English as an official language.⁷⁰ The proliferation of Americans and Americanisms across the region gives the United States a decided advantage in promulgating its ideological viewpoint and ideas.

This plays out in Indo-Pacific popular culture, including within China itself. There, American-imported entertainment offers an attractive alternative to state television's tightly controlled lineup dominated by historical costume dramas and anti-Japanese war films. As an example, when Hollywood actor Alan Thicke died in December 2016, there was an outpouring of sympathy on Chinese social media by a generation of Chinese that had come of age watching Thicke's character on *Growing Pains* during the 1980s and 1990s, one of the first American shows to air there. In late 2013, China established a national security committee to focus on "unconventional security threats" to thinking in Chinese youth, including Hollywood movies. By 2015, China's then-Minister of Education Yuan Guiren reportedly ordered university officials to disallow teaching materials that "disseminate Western values."⁷¹

China. Xi Jinping's work report at the CCP's 19th Party Congress in October 2017 introduced some new themes with specific resonance in the Indo-Pacific region.⁷² Xi called for CCP members to focus on governance, politics, and ideology with an emphasis that "Ideology determines the direction a culture should take and the path it should follow as it develops." He called on China's writers and artists to produce work that not only is thought-provoking but also extols "our party, our country, our people and our heroes." Chinese state media openly declared China's socialist system to be an alternative ideological model for the developing world and a clear competitor with Western liberal democracy.⁷³ Ideological competition now stands as a significant feature in China's efforts to build support across the Indo-Pacific region, especially with illiberal regimes.⁷⁴

As noted in chapter 3b, Beijing's ability to craft and disseminate its preferred ideology in a resonant and positive message has improved, but still exhibits significant liabilities and shortcomings. China's ideological framework of a community of common destiny is a vague slogan that glosses over conflicts of interest between nations, including China's territorial disputes with many of its neighbors.⁷⁵ China's emphasis on state sovereignty at the expense of human rights and freedoms is inherently limited in appeal, resonating with autocratic elites but not so much with ordinary citizens, even in the Indo-Pacific region.⁷⁶ Moreover, leaders and people in the region judge China's lofty principles against the reality of an increasingly authoritarian China whose growth is slowing and as a big country that increasingly uses coercive means to get its way with smaller countries.

China continues to expand efforts to generate soft power to persuade others in the region to pursue its goals and values or to emulate its behavior. Flows of tourists and students between China and other Asian countries continue at record highs, with about 47.8

million Chinese citizens visiting other East Asian countries in 2018.⁷⁷ China hosted more than 295,000 students from the region in 2018, with South Korea, Thailand, Pakistan, and India sending the most.⁷⁸ The Chinese government has supplemented student exchanges by establishing Confucius Institutes in foreign countries to teach Chinese language and promote Chinese culture. As of 2020, 19 Indo-Pacific countries hosted some 97 Confucius Institutes, with South Korea, Japan, and Thailand hosting at least 10 apiece.⁷⁹

Appeals to cultural and linguistic affinities have been important in dealing with countries that have significant ethnic Chinese minorities. Malaysia and Indonesia, which have historically viewed their ethnic Chinese populations with considerable suspicion, came to regard them as an asset in building economic relations with China. However, Beijing's recent efforts to strengthen outreach to the ethnic Chinese diaspora in Asia are renewing these concerns. Beijing found some sympathy in Southeast Asia for appeals to "Asian values" in the 1990s, but this has been tempered by the deepening of democracy in Japan, South Korea, Taiwan, and some Southeast Asian countries. Cultural and linguistic diversity in Asia is likely to limit China's ability to harness purported common "Confucian values" as a diplomatic tool.⁸⁰

In the cultural sphere, some Chinese products reflect traditional Chinese culture in ways that resonate within Asia, but most have limited appeal due to their focus on Chinese domestic concerns, derivative nature, political constraints on content, and language barriers. Films have arguably been China's most successful cultural exports. Some of these constraints may ease as China becomes richer, but for now other Asian countries are producing work with more regional impact and influence. It is worth noting that many of the most successful Chinese artists achieved their fame with work done outside China, including Nobel Prize-winning novelist Gao Xingjian.⁸¹

The American FOIP vision and China's community of common destiny are competing regional visions for a diverse Indo-Pacific region. As of 2019, regional views suggest that America and its global vision remains most appealing, although with some recent relative decline. A December 2019 Pew Survey reported China receiving unfavorable reviews from all but Pakistan in the region. In Japan, 85 percent have an unfavorable opinion of China, with 63 percent of South Koreans, 57 percent of Australians, and 54 percent of Filipinos sharing this view. Indonesian opinion of China plunged 17 percent between 2018 and 2019, the most negative drop in regional countries over the past decade.⁸² For now, American ideas and ideology, and its projection of them, continue to resonate in the Indo-Pacific region. It is unclear how China will close this gap, but American complacency might still give Beijing a chance.

Informational Tools and Prospects

The United States. As noted in the section on ideology, American public diplomacy and information dissemination in the Indo-Pacific region remains reliant on its post-Cold War medium of private sector journalism and entertainment. It also relies heavily on the penetration of social media images and interpersonal interactions where those are not blocked outright by the host nation government, as in China. The Indo-Pacific region accounts for more than half of all social media users worldwide, with 426 million active Facebook users and one-third of all global Twitter users. Taiwan, Hong Kong, Singapore, Australia, New

Zealand, and Malaysia all rate above 50 percent in national social media penetration—and have access to universal content. China ranks at only 46 percent penetration—but without global access.⁸³ These numbers give American cultural, social, and ideological themes high resonance—for better or for worse—in a broad and deep messaging arena.

U.S. public information activities prominent during the Cold War are largely gone, but the State Department does maintain a Bureau of Educational and Cultural Affairs and a Bureau of Global Public Affairs that engage in some Internet and social media outreach. In 2016, the State Department stood up an interagency Global Engagement Center to coordinate U.S. Government efforts to expose and counter foreign state and nonstate propaganda and disinformation.⁸⁴ Radio Free Asia is a U.S.-based, private, nonprofit corporation funded by the U.S. Agency for Global Media that broadcasts news and information to Asian countries whose governments prohibit access to a free press.⁸⁵

China. One of China's strengths in the information domain is a well-developed propaganda and influence apparatus that delivers consistent and coordinated messaging through a range of official and semiofficial channels. In 2018, China conducted a major reorganization of CCP and state bureaucracies that consolidated organs engaged in international propaganda and influence projection under Party control.⁸⁶ As noted in chapter 3b, China oversees Xinhua, a state-run global media service that produces CCP-friendly stories for worldwide dissemination in multiple languages and boasts an 11.5-million-user Twitter account (despite the fact that Twitter access is banned in China). It endorsed the acquisition of Hong Kong's *South China Morning Post* in 2015 by the chief executive officer of the Alibaba e-commerce group, which inserted a management team that promised to provide a positive view of China. It generates content from its state-run China Radio International for use by broadcast networks from Norway to Turkey to Australia. It has lavishly funded the China Global Television Network—rebranded in 2016 as the international arm of China Central Television—recruiting local journalists across the world with excellent pay and airtime to contribute stories acceptable to the Chinese propaganda apparatus. The content seeks to fulfill Xi Jinping's charge to "tell China's story well," emphasizing the generosity of the Chinese people and the benign nature of the Chinese government while amplifying the chaotic and unpredictable nature of Western politics and liberal democracy. However, the consistency and coordinated nature of the CCP messaging apparatus may actually work against the effectiveness of the message as received by foreign audiences. Public opinion of China is very low across the Indo-Pacific region, and China is deemed to be relatively untrustworthy.⁸⁷ Moreover, the extensive media and Internet censorship and message control that China practices at home undercuts Beijing's credibility in projecting a positive image overseas.

The COVID-19 pandemic provides a good illustration of China's ability to promulgate its narrative domestically and internationally. Confronted with a negative image as the source of the coronavirus outbreak, the CCP began a concerted effort to reshape the adverse narrative of China as an authoritarian power slow to sound the alarm and reluctant to share information to one of China as a global leader that stepped up when others did not. Interestingly, it sometimes did so in coordination with Iran and Russia. The Chinese government went from letting Russian disinformation claiming the United States was the source of the virus proliferate in Chinese social media, to raising questions on state media

about the virus's origin, to promoting disinformation that the United States was the source of the virus.⁸⁸ Simultaneously, it orchestrated heavy media coverage of Chinese provision of tests kits and face masks around the globe, with Chinese diplomats browbeating host governments for positive statements praising Chinese generosity. The full fruits of these efforts remain to be seen, but initial returns have not been favorable. Reports of faulty test kits, defective masks, and Chinese imperiousness combined to sour popular opinion of Beijing across the wider Indo-Pacific region, especially in countries such as Australia, India, and the Philippines.⁸⁹ China expert Bates Gill notes that, although the Chinese propaganda apparatus seeks to promote a positive image of China internationally, the most important audience for that message is domestic.⁹⁰

Despite noteworthy disorganization and atrophy of official information channels, American informational tools in the Indo-Pacific region remain substantial and gain credibility from the diversity of viewpoints expressed. China's external messaging is hindered by its unified propaganda message and hypersensitivity to criticism, which leads Chinese diplomats to complain about any foreign government statements or media coverage that paint China in an unattractive light. The quality of a government's information apparatus is important, and the U.S. Government needs to devote more resources and attention to its public diplomacy and informational tools. But ultimately it is the content of the message—both in terms of the performance of competing governance models and an attractive regional vision that others want to follow—that makes informational tools effective.

Military Tools and Prospects and the “Hard Case” of Taiwan

Since the end of World War II, the United States has been the dominant maritime and air-power in Asia. As noted, America has relied on a network of alliances and arrangements with allies and friendly partners in the Indo-Pacific region to support naval and air access and freedom of maneuver. The regional military balance in terms of relative U.S. and Chinese capabilities is important, but the real U.S. strategic center of gravity is the political-military relationships that underpin its alliances and the forward-deployed military presence that they support. Some American military advantages have eroded over the past 2 decades as PLA capabilities have grown, but the U.S. military is welcome in the region in a way that the Chinese military is not.

In 2020, the U.S. military enjoys significant quantitative and qualitative advantages over the PLA, especially in a long conflict that would allow it to bring all its assets to bear. USINDOPACOM oversees a Pacific Fleet with a complement of about 50 capital ships, 2 or 3 aircraft carriers, and approximately 30 advanced U.S. submarines operating in the region. USINDOPACOM manages 3 numbered U.S. Air Forces with an unrivaled mixture of some 2,000 fighter, bomber, and mobility aircraft. It also oversees 80,000 U.S. Army and Marine Forces stationed throughout the region and has access to another 100,000 deployable troops on command from the continental United States, if required. The United States also has advantages in its proven ability to employ space-based intelligence, surveillance, and reconnaissance and cyber capabilities to support its ground, naval, and air assets.⁹¹

Conversely, the PLA Navy (PLAN) has 3 fleets with about 140 capital ships, but lacks long-range, blue water warfare capabilities. Its two aircraft carriers use a ski-ramp design that limits the payload of their aircraft; the PLAN will not field a modern carrier until 2023. Its air

forces are large and composed primarily of multirole fighters, with a limited number of stealth fighters coming into the force. Its current bombers are based on a late 1950s Soviet design, although they are equipped with modern engines and capable land-attack and antiship cruise missiles. PLA Army and marine forces have been reorganized in a corps-brigade-battalion structure to improve their abilities to conduct combined arms and joint operations with other services. The PLA has a limited number of army and marine amphibious units but lacks the sea lift capability to deploy and sustain them too far from the Chinese mainland. The PLA is optimized for fighting conventional land conflicts along its borders, but for the last 15 years, its modernization efforts have emphasized the need to develop its naval, air, and missile forces and to develop the ability to conduct joint operations employing the full range of PLA capabilities.⁹² These efforts have significantly improved the PLA's ability to project power within and beyond the First Island Chain (see figure 9.2).⁹³

One aspect of the U.S.-China military competition in the Indo-Pacific region involves Chinese efforts to use increasing military and paramilitary presence and coercion to enhance its effective control of the maritime territories it claims in the South China and East China seas and U.S. military efforts to operate in these disputed waters to maintain the principles of freedom of navigation and international law. The United States does not take a position on the merits of the competing claims to sovereignty over land features, but insists on the principles of peaceful resolution of disputes and compliance with international law.⁹⁴ Chinese aggressive tactics to enforce its claims—which the United States regards as incompatible with UNCLOS—have involved the creative use of civilian fishing vessels and coast guard ships on the front line, backed by naval capabilities.⁹⁵ China has practiced gray zone tactics that seek to avoid the use of lethal force while employing a range of military, paramilitary, economic, diplomatic, legal, and informational tactics to reinforce its maritime claims.

These actions have increased the willingness of countries in the region to spend more on their militaries and their interest in enhanced security cooperation with the United States and other major powers. Absent U.S. intervention, China now has the military capability to seize and hold the disputed land features in the South China Sea, but this would be a bloody affair that would severely damage China's relations with claimant and nonclaimant states alike and stimulate military balancing against China. To date, Beijing has judged the costs of a military solution to be too high. This low-level war of nerves on the high seas is likely to continue without a definitive resolution for some time to come.

In addition to continuing military presence missions such as freedom of navigation operations, the United States remains well postured to help regional militaries—prioritizing Taiwan, Japan, Vietnam, and the Philippines—develop the ability to challenge China's power-projection capabilities.⁹⁶ Formal American alliances and partnerships, while under some recent duress, remain robust and growing. From 2017 to 2019, the Trump administration successfully strengthened key alliances and expanded military cooperation with traditional allies such as Australia, Japan, and South Korea, while using exercises and dialogues to reach out to nontraditional partners such as India, Malaysia, and Vietnam.⁹⁷ The Defense Department's *Indo-Pacific Strategy Report*, released in June 2019, illuminated the contributions made by the U.S. acquisition and deployment of advanced capabilities, new operational concepts, and initiatives to strengthen security partnerships.⁹⁸ The United

States also could support India's efforts to pose military dilemmas for China, a relatively low-cost means of complicating the PLA's ability to concentrate attention and resources on U.S. forward locations across East Asia and the Western Pacific.⁹⁹

U.S.-China strategic competition in the Indo-Pacific region will also have a high-end conventional military aspect where U.S. qualitative advantages in military hardware, ability to project power globally, and proven ability to conduct effective joint combat operations are partially offset by China's geographic advantages when operating from its own home territory. Since the mid-1990s, the paradigmatic PLA planning and modernization scenario has been an invasion of Taiwan in response to a *de jure* declaration of independence, with the United States intervening on Taipei's behalf. This scenario would require air and sea lift capabilities to get a PLA invasion force onto the island, but the ranges required would be relatively limited since the island is less than 100 miles away.

The PLA has invested in an array of antiaccess/area-denial (A2/AD) capabilities intended to raise the costs and risks for U.S. forces operating near China, with the goal of deterring or delaying U.S. intervention. These include advanced diesel submarines, which could attack U.S. naval forces deploying into the Western Pacific; surface-to-air missiles such as the Russian S-300, which could target U.S. fighters and bombers; and antiship cruise missiles and antiship ballistic missiles optimized to attack U.S. aircraft carrier battle groups. China has invested in a range of accurate conventional missiles that can target the bases and ports the U.S. military would use in a conflict. China has also sought to exploit U.S. military dependence on space systems by developing a range of antisatellite capabilities that could degrade, interfere with, or directly attack U.S. satellites and their associated ground stations. It has invested in cyber capabilities to collect intelligence and degrade the U.S. military's ability to employ computer networks in a crisis or conflict. In a conflict, the PLA would attempt to use multidomain attacks to paralyze U.S. intelligence, communications, and command and control systems and force individual units to fight in isolation, at a huge disadvantage.¹⁰⁰ This would make American defense of allies and national interests inside the First Island Chain difficult.

A potential U.S.-China conflict over Taiwan represents a "hard case" where China might hope to mount a successful surprise attack and force Taiwan's capitulation before the U.S. military could bring its forces to bear. This could present the United States with a hard-to-reverse *fait accompli*. China considers Taiwan part of its historic territory and is committed to eventually achieving unification as part of the "great rejuvenation of the Chinese nation." The United States abrogated its security treaty with the Republic of China (Taiwan's formal name) in 1979 and agreed to have only unofficial relations with it as part of the process of normalizing relations with the People's Republic of China. The United States does not have a formal security commitment to Taiwan, but the 1979 Taiwan Relations Act requires providing Taiwan with defensive arms and states that U.S. policy is to retain the capability to resist the use of force or coercion to undermine Taiwan's security.¹⁰¹

This task has become more difficult given the tyranny of distance and successful PLA modernization. RAND's 2015 evaluation of U.S.-China military force capability trends found that the United States had "major advantages" in 7 of 10 critical capability areas in a Taiwan scenario in 1996, but that by 2017, the United States would have clear "advantages" in only 3 categories, and the PLA would enjoy advantages in 2: its ability to attack

U.S. airbases and its ability to attack U.S. carriers. China's advances in ballistic missiles, cruise missiles, and modern diesel attack submarines now give it advantages it did not have during the 1990s Taiwan standoff.¹⁰² The U.S. Air Force ended its 16-year bomber forward presence on Guam in late April 2020 in recognition of China's enhanced missile capabilities, especially its DF-26 "Guam killer."¹⁰³

In a South China Sea scenario, where PLA weaknesses in power projection would matter more, RAND assessed that the U.S. military would retain an edge in 2017, but even there the PLA would have made up considerable ground.¹⁰⁴

The implications for the U.S. ability to defend Taiwan are significant in the context of U.S.-China Great Power rivalry in the Indo-Pacific region. As noted in chapter 3b and above, while China is not close to catching up to the U.S. military in terms of aggregate military capabilities (quality and quantity), it does not need parity to frustrate U.S. intervention in a short conflict on its immediate periphery. Despite ongoing U.S. military efforts to develop new military capabilities and operational concepts to operate in an A2/AD environment, China has made significant improvements in its ability to attain a decisive military advantage in confrontation scenarios near China's coast—such as with Taiwan. These

emerging realities suggest that American and Taiwanese militaries should consider an active denial strategy that reduces the risk from preemptive attack and makes Taiwan a more resilient target. Chapter 5 discusses some of the small, smart, and cheap alternatives that Taiwan and the United States might consider in this vein. Washington and other Western governments should continue to make it clear to China that aggression against Taiwan would carry immense costs and risks, but they must be more circumspect in predicting any absolute ability to prevail in armed conflict. Taiwan's allies and friends should simultaneously engage China on issues of strategic stability and escalation to reduce the prospects for miscalculation.¹⁰⁵

"The United States' ambiguous commitment to Taiwan's defense is making the actual defense of Taiwan less tenable. Two decades ago, when the threat of a Chinese invasion could be deterred by sending a U.S. aircraft carrier through the Taiwan Strait, a mostly symbolic military relationship between the two countries was a sufficient way to keep the peace. This is no longer true. The PLA has grown strong enough that neither Taiwan nor the United States can afford to have the Taiwanese military devote another decade to suboptimal arm purchases. If the United States wants to increase the defensive power of Taiwan's armed forces, then Washington must find other ways to give the Taiwanese leaders the symbolic victories they seek from arms packages."

—Tanner Greer, "Taiwan's Defense Strategy Doesn't Make Military Sense,"
Foreign Affairs, September 17, 2019

Economic Power and Influence Prospects

The United States. Almost every government in the Indo-Pacific region is focused on increasing economic growth and raising living standards for its people, which gives economic instruments of power great salience. U.S. leadership and support for the open global economic order has underpinned the Asian economic miracle that saw first Japan, then the four tigers (South Korea, Singapore, Taiwan, and Hong Kong), and finally China enjoy

rapid economic development. Access to the U.S. market and U.S. technology continues to play a major role in sustaining regional growth, and reciprocal access to growing Indo-Pacific markets and technology is important for U.S. growth and prosperity. While free trade produces mutual benefits, every government seeks to maximize the benefits for its country and companies by capturing an increased share of high-value-added products and rapidly growing markets.

UN trade data shows that China was the number one export market for other Indo-Pacific countries in 2017, taking \$413 billion in their exports (plus an additional \$82 billion routed through Hong Kong), compared with \$343 billion for the United States.¹⁰⁶ China is the number one export market for most countries in the Indo-Pacific region, including most U.S. allies. Yet these figures greatly understate the U.S. role in the regional economy because many exports to China are raw materials or components that are processed or assembled into final products and subsequently exported to the United States or other developed country markets. Such goods show up in the data as part of Chinese exports to the United States, which totaled \$526 billion in 2017.¹⁰⁷ U.S. companies play an important role in these regional production networks (including by owning and operating factories in China) and the U.S. market is the final destination for many of these products. There is extensive trade integration within and across the Indo-Pacific region, and countries want to be included in regional production networks and to access both U.S. and Chinese markets.

The centerpiece of the Obama administration's Indo-Pacific economic policy was the Trans-Pacific Partnership, an "ambitious, next-generation Asia-Pacific trade agreement" including 12 regional countries and extensive environmental, labor, and intellectual property standards.¹⁰⁸ The TPP did not include China, but advocates hoped that the prospect of eventual membership would provide incentives for China to modify its economic practices to comply with TPP rules. The TPP agreement was signed on February 4, 2016, but the Obama administration did not submit the agreement to Congress for approval in the face of bipartisan opposition. One of the Trump administration's first acts was to withdraw from participation in the TPP.

Since 2017, the Trump administration has focused on trying to change the terms of U.S. trade with foreign partners, including U.S. allies in the Indo-Pacific region. This has included tariffs on steel and threatened tariffs on automobiles and auto parts based on "national security grounds," renegotiating the Korea-U.S. FTA, and a bilateral agreement to increase access to the Japanese market for U.S. agricultural goods. These bilateral deals continued a long-term U.S. approach to trade. But abandoning the TPP was a major deviation in U.S. regional economic strategy, which places the United States outside the rules-setting role of the successor Comprehensive Progressive Agreement for Trans-Pacific Partnership (CPTPP) established in 2018 by the 11 other signatories of the TPP.¹⁰⁹ Preferring to resolve trade disputes through bilateral negotiations, the Trump administration also has blocked the appointment of judges to the World Trade Organization's appellate body, rendering this multilateral institution unable to rule on trade disputes.¹¹⁰

In the absence of a multilateral trade agenda, the U.S. FOIP vision has focused on developing alternative forms of regional infrastructure assistance to compete with China's BRI regional investment infrastructure initiatives. In July 2018, Secretary of State Mike Pompeo announced a \$113 million "down payment" on U.S. investments in the digital economy,

energy, and infrastructure sectors.¹¹¹ In October 2018, Trump signed the BUILD Act, which raised the ceiling on U.S. global development financing to \$60 billion. In November 2018, the U.S. Government signed a memorandum with its Japanese and Australian counterparts to create a new partnership designed to provide financing for projects that “adhere to international standards and principles for development,” an implicit critique of Chinese projects viewed as falling short of those standards.¹¹² Other elements included funding for the Millennium Challenge Corporation (\$2.3 billion) and establishment of a Blue Dot Network to set financially sustainable standards for infrastructure development.

In mid-2018, the United States began a trade war with China by imposing a series of tariffs covering most Chinese exports to the United States. China responded with tariffs targeting U.S. exports, including agricultural exports from farm states crucial to American electoral politics. The trade war was the most serious disruption in global commerce in the post–World War II era.¹¹³ Analysis through September 2019 revealed the effort to be a double-edged sword. China’s lost export revenue was triple that of the United States (\$53 billion to \$14.5 billion, respectively), but the United States had not achieved any substantive movement in the Chinese economic behaviors it was seeking to change. Key sectors of the U.S. economy—exporters of minerals and ores, forestry products, agribusiness, and transportation systems—lost substantial revenue and were disturbed that China has found alternative suppliers, meaning potential lasting damage to export revenues.¹¹⁴ The “phase one” U.S. trade deal with China announced in January 2020 involved Chinese agreement to lift some retaliatory tariffs and to substantially increase imports from the United States, but making no major concessions on the issue of government industrial policies.¹¹⁵ Many economists were skeptical that the targets were realistic, and trade disruptions due to the COVID-19 pandemic make it unlikely that China will fulfill those commitments.¹¹⁶

The uncertainties arising from the U.S.-China trade relationship, and particularly U.S. policy discussions about trying to “decouple” the U.S. economy from China, are moving alliance partners to consider alternative trade structures.¹¹⁷ These include the Japan-European Union FTA; ongoing negotiations for a Regional Comprehensive Economic Partnership that will include China (but not the United States); and continuing negotiations between Japan, China, and South Korea for a trilateral FTA. This region-wide hedging may have long-term negative political and strategic consequences. A serious U.S. effort to use tariffs to dismantle regional production networks and force countries to move production out of China is likely to meet with significant resistance given the importance of trade and investment ties with China to virtually every country in the Indo-Pacific region.

China. The growing economic dependence of other Indo-Pacific countries on the China market is a potential source of influence for Chinese leaders, but a tricky one to use. The desire to maintain market access makes Indo-Pacific countries reluctant to take actions that might offend China, but Chinese efforts to use restrictions on trade and tourism as an active coercive tool have had mixed results.¹¹⁸ In many cases, such as China’s efforts to punish South Korea for agreeing to host U.S. Terminal High Altitude Area Defense antiballistic missile systems, these measures have signaled Beijing’s unhappiness and imposed costs on the target country but have not succeeded in forcing it to make the policy changes China wanted.

China has had more success using economic incentives such as FTAs, outbound investment, foreign aid, and infrastructure loans as carrots that provide concrete benefits to

recipient countries and give substance to China's vision of a stable, prosperous region with extensive economic integration. China is building on its existing FTAs with ASEAN, New Zealand, and Singapore via ongoing negotiations for a Regional Comprehensive Economic Partnership and a trilateral Japan, China, and South Korea FTA, both of which would exclude the United States and expand China's preferential access to regional markets. China has emerged as a huge overseas investor over the past two decades. Although Indo-Pacific countries account for only about 20 percent of Chinese overall outbound investment, as of 2019, this totaled almost \$250 billion, along with an additional \$242 billion in construction projects by Chinese companies.¹¹⁹

China does not publish a detailed breakout of its foreign aid programs, but poorer countries in Southeast Asia and Oceania have been significant recipients of Chinese development assistance, receiving at least \$38 billion from 2000 to 2016.¹²⁰ Much of this assistance goes to improve transportation infrastructure connecting South and Southeast Asia with China; many of these projects are now included as part of the BRI. These infrastructure investments, some of which are funded by the Asian Infrastructure Investment Bank established by China in 2016, not only contribute to economic development but also link these countries more closely to the Chinese economy and will produce greater trade dependence in the future. Expanding BRI financing is an important tool for advancing China's regional influence, but there is growing skepticism in some Chinese quarters about spending massive sums on foreign development.¹²¹

In summary, the United States and China bring different strengths to Great Power competition in the Indo-Pacific region. In terms of political and diplomatic tools, China has raised its diplomatic game but is unlikely to find much support for efforts to limit U.S. presence in the region because most countries in the Indo-Pacific want the United States involved to help balance against Chinese power. In terms of ideological tools, the U.S. regional vision resonates with a number of countries and is likely to have more appeal than China's vague call for a community of common destiny. The soft power of both countries is likely to be damaged by poor performance in governance and the disjuncture between their stated regional visions and actual policies. China has a clear informational advantage in its ability to articulate and reinforce a consistent message, but the fact that this message is usually parroting CCP talking points that are inconsistent with Chinese behavior undercuts the effectiveness of its informational efforts. The United States has a more appealing message, but American government tools to express that message to Indo-Pacific countries have atrophied and need more resources.

U.S. military dominance has eroded as Chinese military capabilities have improved, but the fact that the Chinese military is viewed as a threat and the U.S. military is viewed as a partner is a huge political-military advantage. The United States needs to improve its ability to operate in an A2/AD environment, including its willingness to accept operational risk in peacetime settings. Neither side is likely to attain decisive military advantage in the region. The question is whether the high costs and escalation risks of a major war could continue to maintain a cold peace. In terms of economic tools, China has significant advantages in its ability to mobilize and direct resources and to provide countries with valued opportunities to increase their economic growth. The U.S. Government must rely primarily on creating incentives and opportunities for private-sector actors. The lack of a multidi-

mensional regional trade strategy and recent efforts to strong-arm U.S. allies and partners have reduced American economic influence in the region. If the United States tries to force countries to participate in an aggressive effort to decouple their economies from China's, Washington seems likely to meet significant regional resistance.

Conclusions

U.S. and Chinese strategic interests are less aligned and more important to both countries in the Indo-Pacific region than in any other area of the world, making it a central venue for Great Power competition. Over the past decade, Beijing has become more critical of the U.S. military presence and U.S. alliance system, arguing that it reflects Cold War thinking and emboldens U.S. allies to challenge Chinese interests. The U.S. Free and Open Indo-Pacific vision and increased U.S. regional security cooperation in activities such as the Quad have stoked Chinese fears of U.S. encirclement or containment. Beijing has resisted making any binding commitments that might restrict its military capabilities or ability to employ military power to defend its core interests. Its increasing military capabilities and more assertive approach to maritime territorial disputes have heightened regional concerns about how a strong China could behave, leading most countries to improve their security ties with the United States.

To protect and advance its interests, the United States will need to acknowledge that the appeal of access to China's superior market and the weight of Beijing's other economic tools make Indo-Pacific countries unlikely to give up economic ties with Beijing, even if Washington attempts to decouple from the Chinese economy. Washington's relative disadvantage in economic power limits its ability to persuade countries to fully align with it economically against China now, and for some time. Thus, it must develop flexible policies that allow China's neighbors to avoid an explicit choice of aligning completely with Washington or Beijing in the regional Great Power competition, unless they are compelled by Chinese behavior to do so. But Washington still has a full agenda to pursue, both to advance specific U.S. economic interests and to shape rules and norms in the most dynamic region in the world. A U.S. Indo-Pacific strategy that combines some degree of engagement with China and attention to nurturing a balance of economic and military power around Beijing as a hedge would best serve U.S. interests. In that spirit, the United States needs to find a pragmatic basis for bilateral economic relations with China that protects what is working and helps adjust what is not. A trade war or full economic decoupling is unlikely to achieve that end.

At present, the Trump administration is attempting to address trade and market access issues unilaterally through tariffs and other administrative measures. The United States should reconsider participation in the TPP in order to promote the integration with the Indo-Pacific economies that would be needed to form a truly viable counterweight to China. Admittedly, many U.S. interest groups and citizens have grown skeptical of FTAs, but the shortcomings of the past should not be allowed to hamper what is needed economically and strategically for the country's future. The successful negotiation and implementation of a new TPP-like agreement—including accession into the CPTPP—could be a powerful collective lever to reshape objectionable Chinese economic practices so that Beijing can eventually participate.

In the security domain, the United States is at a relative military power disadvantage if a clash over Taiwan or another sovereignty issue breaks out within the First Island Chain, where China could use its A2/AD capabilities to full effect. Thus, Washington should review military dispositions in these areas and adjust strategies, capabilities, and operational concepts with an eye toward making better use of emerging technologies and increasing the resilience of its military forces in theater. Chinese A2/AD capabilities will necessitate some tough military planning choices in these special cases. U.S.-China military-to-military relations are unlikely to overcome these competitive dynamics in areas where China's core interests lie, but they could have value enhancing deterrence, increasing transparency, and dispelling unfounded worst-case suspicions. They also can help improve communication mechanisms and understandings about how military ships and aircraft would behave when they encounter each other, which would help avoid incidents and provide more effective crisis management tools.¹²²

At the same time, the United States should build on its relative political-military advantages to sustain and strengthen its regional security position. Reinforcing present alliances, building military partnerships, extending cooperative training, and expanding interoperability are techniques that regional states will embrace and will work against unilateral Chinese efforts to intimidate. If the United States emphasizes its alliances, expands security cooperation with other partners, and actively engages in regional multilateral institutions, it can deal with Chinese regional security initiatives and actions from a position of strength and resist Chinese efforts to erode the U.S. alliance system. Conversely, if Washington appears disengaged, it will become less relevant and less able to shape the evolving regional security environment.

America's advantages in alliance diplomacy, relative trustworthiness, resonance of ideological vision, and (for the time being) approach toward open information and communications should be highly valued and enhanced. At present under Xi Jinping, the CCP is moving China in the direction of increased authoritarian control and a greater state role in the economy, policies that prioritize stability over growth. These are likely to have adverse side effects within China that undercut the appeal of China's model.¹²³ If Washington can prioritize its many concerns with China and partner with like-minded allies and partners to develop a practical agenda, there eventually may be renewed support within China for past reform proposals that are currently on hold due to resistance from Chinese special interests. Washington should work with regional and extra-regional partners to provide outside pressure that might help reenergize these reforms.¹²⁴

At the same time, American interests will be strongly advanced by working with Indo-Pacific partners to articulate and build regional support for the vision and values that underpin the FOIP. If China violates international law and regional norms, the United States should say so consistently. As long as American society models and promotes open, transparent, and democratic institutions, the United States likely will appear as an ideological and even existential threat to CCP leaders.¹²⁵ But strong and consistent messaging with allies and partners could send a positive signal to the Chinese people about the value of good, representational governance and provide other states around the region a positive alternative framework that contrasts with China's authoritarian model.

Put a different way, support for human rights and democracy in the Indo-Pacific region today makes sense given Washington's relative power advantage in the competitive categories of ideology, informational openness, and diplomacy. Standing publicly with supporters of human rights and political reform in China could be a key part of any U.S. strategy for a Great Power competition that is about values as much as it is about relative economic or military power.¹²⁶ Rallying support of this type today requires that U.S. officials act realistically about the nature of the challenge and spell out clearly what meeting it requires. It also requires articulating how addressing the Chinese challenge would be central to preserving the relatively stable, open, and democratic Indo-Pacific region that has taken hold over the past seven decades. In turn, this demands a level of sober but resolved political leadership in Washington. The size and scope of China's economic presence across the Indo-Pacific region means that the United States will need a strategy that is as broad and enduring as the threat it is meant to counter.¹²⁷ Chapter 14 considers several American strategies and evaluates which might best rise to this test.

"China surely has an extended time horizon, [but] its theory of influence actually betrays a short-term approach, favoring immediate, coercive results over shared values and extended cooperation. Power is likely to be more sustainable when exercised within and reinforced by a shared system of mutual benefit. . . . The U.S. helped to build precisely such a system in the form of a post-war, rules-based order, one that *endures* as a profound U.S. competitive advantage."

—Michael J. Mazarr and Ali Wyne,
"The Real U.S.-China Competition,"
The Interpreter, January 29, 2020

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Notes

¹ Russia rates this lesser status for three major reasons. First, Vladimir Putin's Russia has an unrelenting strategic focus on its troubled relations with the West. Second, Russia lacks capacity in its Far East and will take a long time to generate capabilities even if it determined to generate them. Third, the Sino-Russian relationship is highly uneven despite the tactical value each sees in specific cooperation against the United States. Beijing possesses a vision, resources, and a game plan for the Pacific region, while Russia has no fully articulated or resourced Asia policy. Thus, despite some noteworthy short-term common interests and concerns, Moscow and Beijing priorities diverge significantly. While Moscow wishes to derail the present U.S./Western-dominated international system, China sees much residual value in the current international framework, so long as Beijing can expand its influence and work to address its perception of major flaws. Russia and China can today agree to constrain American hegemonic power and liberal interventionism, but they do not have a convergent strategic vision and are unlikely to cooperate on grand strategy or the framework for post-Western international rules, norms, and institutions globally or in the Indo-Pacific region. For more detail, see the review of Russian strategy in chapter 3a. Also see Bobo Lo, *Once More with Feeling: Russia and the Asia-Pacific* (Sydney, Australia: Lowy Institute, 2019), available at <<https://www.lowyinstitute.org/publications/once-more-feeling-russia-and-asia-pacific>>; Malin Ostevik and Natasha Kuhr, "The Russian Far East and Russian Security Policy

in the Asia-Pacific Region," in *Russia's Turn to the East*, ed. Helge Blakkisrud and Elana Wilson Rowe (London: Springer, 2017), 75–94; Nivedita Kapoor, "The Long Road Ahead: Russia and Its Ambitions in the Far East," Observer Research Foundation, September 11, 2019, available at <<https://www.orfonline.org/expert-speak/the-long-road-ahead-russia-and-its-ambitions-in-the-far-east-55378/>>; Liz Bagot and Josh Wilson, "The Russian Far East, Gateway to Asia," *GeoHistory Today*, October 1, 2011, available at <<https://geohistory.today/russian-far-east/>>; James Brown, "Japanese Investment in Russia Floundering Despite Arctic Energy Deal," *The Moscow Times*, October 16, 2019, available at <www.themoscowtimes.com/2019/10/16/japanese-investment-floundering-despite-arctic-energy-deal-a67754>.

² These economic norms and principles were underwritten by international organizations established and supported by the United States, including the Bretton Woods institutions of the World Bank, International Monetary Fund, General Agreement on Tariffs and Trade, and its successor, the World Trade Organization.

³ The Southeast Asia Treaty Organization (SEATO), a multilateral treaty established in 1954 to resist the expansion of communism in Southeast Asia, is a partial exception. SEATO was dissolved in 1977.

⁴ For a review of successive American national security strategies from 1993 through 1998 that articulated the objectives of

"engagement and enlargement," see "National Security Strategy Archive," available at <<http://nssarchive.us/>>.

⁵ William J. Clinton, "Address by President Bill Clinton to the UN General Assembly," September 27, 1993, available at <<https://2009-2017.state.gov/p/io/potusunga/207375.htm>>; Anthony Lake, "From Containment to Enlargement" speech at The Johns Hopkins University School of Advanced International Studies, Washington, DC, September 21, 1993, available at <<https://www.mtholyoke.edu/acad/intrel/lakedoc.html>>. Also see Richard N. Haass, "Fatal Distraction: Bill Clinton's Foreign Policy," *Foreign Affairs* No. 108 (Autumn 1997), 113.

⁶ As noted in chapter 3b of this volume, this assertion is contested. For the contrasting arguments, see Kurt M. Campbell and Ely Ratner, "The China Reckoning: How Beijing Defied American Expectations," *Foreign Affairs*, March/April 2018, available at <www.foreignaffairs.com/articles/china/2018-02-13/china-reckoning>; Aaron Friedberg, "Getting the China Challenge Right," *The American Interest*, January 10, 2019, available at <www.the-american-interest.com/2019/01/10/getting-the-china-challenge-right/>; Alastair Iain Johnston, "The Failures of the 'Failure of Engagement' with China," *Washington Quarterly* 42, no. 2 (Summer 2019), 99–114, available at <<https://doi.org/10.1080/0163660X.2019.1626688>>.

⁷ Jim Mann, "Clinton's Asia Policy Puts China First," *The Los Angeles Times*, March 18, 1998, available at <www.latimes.com/archives/la-xpm-1998-mar-18-mn-30092-story.html>.

⁸ Nicholas R. Lardy, "Issues in China's WTO Accession," Brookings Institution, May 9, 2001, available at <<https://www.brookings.edu/testimonies/issues-in-chinas-wto-accession/>>.

⁹ Since the late 1970s, U.S. multinational corporations were key proponents of normalizing the relationship with China, and their operations in China have been central to ties between the two countries. Over the past quarter century, these firms have transferred technology, created jobs, and helped reshape the Chinese economy. See *Two-Way Street: 25 Years of U.S.-China Direct Investment* (New York: National Committee on U.S.-China Relations, 2016), 1, available at <https://www.ncusr.org/sites/default/files/page_attachments/Two-Way-Street-2016_Exec-Summary.pdf>.

¹⁰ *World Development Report 1995: Workers in an Integrating World* (London: Oxford University Press, 1995); *World Economic Outlook* (Washington, DC: International Monetary Fund, 1995).

¹¹ Neil E. Silver, *The United States, Japan, and China: Setting the Course* (New York: Council on Foreign Relations, 1999).

¹² Kurt M. Campbell, *The Pivot: The Future of American Statecraft* (New York: Hatchett Book Group, 2016), 3.

¹³ "Cheney Concerned About China and North Korea," *New York Times*, February 23, 2007, available at <www.nytimes.com/2007/02/23/world/asia/23iht-cheney.4698458.html>.

¹⁴ Taro Aso, "Arc of Freedom and Prosperity: Japan's Expanding Diplomatic Horizons," November 30, 2006, available at <<https://www.mofa.go.jp/announce/fm/aso/speech0611.html>>.

¹⁵ Shinzo Abe, "Confluence of the Two Seas," Ministry of Foreign Affairs–Japan, August 22, 2007, available at <<http://www.mofa.go.jp/region/asia-paci/pmv0708/speech-2.html>>.

¹⁶ Denny Roy, *More Security for Rising China, Less for Others?* AsiaPacific Analysis No. 106 (Honolulu: East-West Center, January 2013); Odd Arne Westad, "Legacies of the Past," in *China and the World*, ed. David Shambaugh (New York: Oxford University Press, 2020), 25–36.

¹⁷ Denny Roy, "Hegemon on the Horizon? China's Threat to East Asian Security," *International Security* 19, no. 1 (Summer 1994), 149–168; Richard Bernstein and Ross H. Munro, *The Coming Conflict with China* (New York: Knopf, 1997). For a survey of regional views and strategies, see Alastair Iain Johnston and

Robert S. Ross, eds., *Engaging China: The Management of an Emerging Power* (New York: Routledge, 1999).

¹⁸ David M. Finkelstein, "China's New Security Concept: Reading Between the Lines," *Washington Journal of Modern China* 5, no. 1 (1999), 37–50.

¹⁹ David Shambaugh, ed., *Power Shift: China and Asia's New Dynamics* (Berkeley: University of California Press, 2006); Evelyn Goh and Sheldon W. Simon, eds., *China, the United States, and Southeast Asia: Contending Perspectives on Politics, Security, and Economics* (New York: Routledge, 2008).

²⁰ Barack Obama, "Remarks by President Obama to the Australian Parliament," Canberra, November 17, 2011, available at <<https://obamawhitehouse.archives.gov/the-press-office/2011/11/17/remarks-president-obama-australian-parliament>>.

²¹ See Jeffrey A. Bader, *Obama and China's Rise: An Insider's Account of America's Asia Strategy* (Washington, DC: Brookings Institution Press, 2012), chapter 7. Also see the discussion in Michael D. Swaine, "Perceptions of an Assertive China," *China Leadership Monitor* 32 (2010).

²² Michael D. Swaine and M. Taylor Fravel, "China's Assertive Behavior—Part Two: The Maritime Periphery," *China Leadership Monitor* 35 (2011).

²³ For an insider's perspective on Obama administration thinking about building a partnership with China, see Bader, *Obama and China's Rise*, chapter 1. Chinese overconfidence in Western decline and the increasing power of the developing world is evident in the shifting language used to describe the trend toward a multipolar world in the 2008, 2011, and 2013 Chinese defense white papers. Also see Andrew Scobell and Scott W. Harold, "An 'Assertive' China? Insights from Interviews," *Asian Security* 9, no. 2 (2013), 111–131.

²⁴ Suisheng Zhao, "Foreign Policy Implications of Chinese Nationalism Revisited: The Strident Turn," *Journal of Contemporary China* 22, no. 82 (2013), 535–553.

²⁵ Authors' interactions with Chinese officials, military officers, and scholars, 2009–2013.

²⁶ Mark E. Redden and Phillip C. Saunders, *Managing Sino-U.S. Air and Naval Interactions: Cold War Lessons and New Avenues of Approach*, China Strategic Perspectives 5 (Washington, DC: NDU Press, 2012).

²⁷ Ibid.

²⁸ Jeremy Page, Carol E. Lee, and Gordon Lubold, "China's President Pledges No Militarization in Disputed Islands," *Wall Street Journal*, September 25, 2015, available at <www.wsj.com/articles/china-completes-runway-on-artificial-island-in-south-china-sea-1443184818>.

²⁹ "Foreign Ministry Spokesperson Lu Kang's Regular Press Conference on October 27, 2015," Ministry of Foreign Affairs of the People's Republic of China, October 27, 2015, available at <www.fmprc.gov.cn/mfa_eng/xwfw_665399/s2510_665401/t1309625.shtml>.

³⁰ Xi Jinping, "New Asian Security Concept for Progress in Security Cooperation," remarks at the Fourth Summit of the Conference on Interaction and Confidence Building Measures in Asia, delivered at the Shanghai Expo Center, May 21, 2014, available at <[https://www.fmprc.gov.cn/mfa_eng/zxxx_662805/t1159951.shtml](http://www.fmprc.gov.cn/mfa_eng/zxxx_662805/t1159951.shtml)>.

³¹ Adam P. Liff, "China and the U.S. Alliance System," *China Quarterly* No. 233 (March 2018), available at <<https://www.cambridge.org/core/journals/china-quarterly/article/china-and-the-us-alliance-system/1FF369905B4A8110DC8693A3C8A7857B/core-reader>>.

³² Obama, "Remarks by President Obama to the Australian Parliament."

³³ Hillary Clinton, “America’s Pacific Century,” *Foreign Policy* No. 189 (November–December 2011), 56–63.

³⁴ Ian Storey, “China’s Missteps in Southeast Asia: Less Charm, More Offensive,” *Jamestown China Brief* 10, no. 25 (December 17, 2010), available at <<https://jamestown.org/program/chinas-missteps-in-southeast-asia-less-charm-more-offensive/>>.

³⁵ China also made efforts to stabilize bilateral relations, proposing that the United States and China build a “new type of major country relations.” The Obama administration was prepared to consider the framework but did not want to accept obligations to respect China’s self-defined “core interests” (including claims to Taiwan) or suggest that Washington would cut deals with Beijing at the expense of its regional allies. See Jane Perlez, “China’s ‘New Type’ of Ties Fails to Sway Obama,” *New York Times*, November 9, 2014, available at <www.nytimes.com/2014/11/10/world/asia/chinas-new-type-of-ties-fails-to-sway-obama.html>.

³⁶ Among the tell-tale U.S.–China events signaling the arrival of their de facto Great Power rivalry in 2014 and 2015, a few stood out. First, the U.S. Justice Department made a first-ever public indictment of Chinese military hackers for cyber espionage against U.S. corporations in May 2014. See Department of Justice, Office of Public Affairs, “U.S. Charges Five Chinese Military Hackers for Cyber Espionage Against U.S. Corporations and a Labor Organization for Commercial Advantage,” May 19, 2014, available at <<https://www.justice.gov/opa/pr/us-charges-five-chinese-military-hackers-cyber-espionage-against-us-corporations-and-labor>>. Second, the U.S. State Department published a paper formally challenging China’s assertion of sovereignty over maritime claims related to its “dashed-line” encircling islands and waters in the South China Sea. This paper set the legal predicate for freedom of navigation operations conducted in the South China Sea by U.S. warships that began in 2015. See Bureau of Oceans and International Environmental and Scientific Affairs, U.S. Department of State, “Maritime Claims in the South China Sea,” *Limits in the Seas* No. 143 (December 5, 2014), available at <<http://www.documentcloud.org/documents/1376464-us-limits-in-the-seas-dos-no143-china-in-scs-12.html>>. Finally, President Obama’s November 2014 summit in China did realize agreement by both states to sign the Paris Climate Accord, but the administration did not find formal compromise for simmering grievances on the two critical issues of Chinese cyber espionage or maritime disputes. See Perlez, “China’s ‘New Type’ of Ties Fails to Sway Obama.”

³⁷ *The National Security Strategy of the United States of America* (Washington, DC: The White House, December 2017), 45.

³⁸ *Ibid.*, 46.

³⁹ H.R. McMaster, “How China Sees the World: And How We Should See China,” *The Atlantic*, May 2020, available at <www.theatlantic.com/magazine/archive/2020/05/mcmaster-china-strategy/609088/>.

⁴⁰ For comments on U.S.–China cooperation, see *The National Security Strategy of the United States of America*, 27; *National Security Strategy* (Washington, DC: The White House, February 2015), 24.

⁴¹ Idress Ali, “In Symbolic Nod to India, U.S. Pacific Command Changes Name,” Reuters, May 30, 2018, available at <www.reuters.com/article/us-usa-defense-india/in-symbolic-nod-to-india-us-pacific-command-changes-name-idUSKCN1IV2Q2>.

⁴² *Indo-Pacific Strategy Report: Preparedness, Partnerships, and Promoting a Networked Region* (Washington, DC: Department of Defense, June 1, 2019), 8.

⁴³ Admiral Phil Davidson, Testimony Before the Senate Armed Services Committee: Opening Remarks, February 12, 2019, available at <<https://www.pacom.mil/Media/Speeches-Testimony/Article/1755445/senate-armed-services-committee-sasc-opening-statement/>>.

⁴⁴ “Remarks by President Trump at APEC CEO Summit,” Da Nang, Vietnam, November 10, 2017, available at <<https://vn.usembassy.gov/20171110-remarks-president-trump-apec-ceo-summit/>>. Henceforth, this chapter uses “FOIP vision” to describe the Free and Open Indo-Pacific framework first articulated by President Donald Trump at the 2017 APEC CEO Summit. While FOIP is more of a *concept* than a *vision*, it is certainly not a *strategy*, for it lacks a clear linkage of ends, ways, and means. It is, as described in the June 2019 Department of Defense *Indo-Pacific Strategy Report*, “our nation’s vision . . . a commitment to a safe, secure, prosperous, and free region that benefits all nations . . . [and] flows from common principles that underpin the current international order . . . principles we have a shared responsibility to uphold.” See *Indo-Pacific Strategy Report*, 3. Also see reference to FOIP vision in Ely Ratner et al., *Rising to the China Challenge: Renewing American Competitiveness in the Indo-Pacific* (Washington, DC: Center for New American Security, December 2019), available at <<https://s3.amazonaws.com/files.cnas.org/documents/CNAS-Report-NDAA-final-6.pdf?mtime=20200116130752>>; *Assessing the Global Environment: Asia* (Washington, DC: Heritage Foundation, October 30, 2019). The authors acknowledge that the November 2019 State Department document talks about a “FOIP strategy” in its cover letter and main text but does so interchangeably with *FOIP vision* and its actual subtitle uses the term *vision*. See *A Free and Open Indo-Pacific: Advancing a Shared Vision* (Washington, DC: Department of State, November 4, 2019), available at <<https://www.state.gov/wp-content/uploads/2019/11/Free-and-Open-Indo-Pacific-4Nov2019.pdf>>.

⁴⁵ See “Sec. Pompeo Remarks on ‘America’s Indo-Pacific Economic Vision,’” Inaugural Indo-Pacific Business Forum, U.S. Chamber of Commerce, Washington, DC, July 30, 2018, available at <<https://asean.usmission.gov/sec-pompeo-remarks-on-americas-indo-pacific-economic-vision/>>.

⁴⁶ *A Free and Open Indo-Pacific*, 6.

⁴⁷ “Briefing on the Indo-Pacific Strategy,” Department of State, April 2, 2018, available at <<https://www.state.gov/briefing-on-the-indo-pacific-strategy/>>.

⁴⁸ For details, see Shayerah Ilias Akhtar and Marian L. Lawson, *BUILD Act: Frequently Asked Questions About the New U.S. International Development Finance Corporation*, R45461 (Washington, DC: Congressional Research Service, January 15, 2019).

⁴⁹ For a recent overview of these activities, see Admiral Harry B. Harris, Jr., *Statement Before the Senate Armed Services Committee Hearing on U.S. Pacific Command Posture*, February 23, 2016, available at <www.armed-services.senate.gov/imo/media/doc/Harris_02-23-16.pdf>.

⁵⁰ *Indo-Pacific Strategy Report*.

⁵¹ For a review of U.S. engagement with these multilateral Indo-Pacific organizations including Asia-Pacific Economic Cooperation, the East Asia Summit, the U.S.–ASEAN Summit, the ASEAN Regional Forum, and others, see *U.S.–East Asia Relations: A Strategy for Multilateral Engagement* (New York: Asia Society Task Force Report, November 2011), 9–20. Also see James Steinberg, “By Abandoning Asia’s Multilateral Organizations, U.S. Empowers China,” *Nikkei Asian Review*, November 21, 2019, available at <<https://asia.nikkei.com/Opinion/By-abandoning-Asia-s-multilateral-organizations-US-empowers-China>>.

⁵² Trump has held bilateral summits in India, China, and Japan, but has attended only two of the six East Asian, Asia-Pacific Economic Cooperation, and U.S.–ASEAN summit meetings while in office. See Jonah Langan-Marmur and Phillip C. Saunders, “Absent Without Leave? Gauging U.S. Commitment to the Indo-Pacific,” *The Diplomat*, May 6, 2020, available at <<https://thediplomat.com/2020/05/absent-without-leave-gauging-us-commitment-to-the-indo-pacific/>>.

⁵³ 2017 *Foreign Policy White Paper* (Canberra: Government of Australia, November 2017); “Trump Marks U.S.-Japan Security Pact with Call for Stranger, Deeper Alliance,” Reuters, January 19, 2020, available at <www.reuters.com/article/us-usa-japan-security/trump-marks-u-s-japan-security-pact-with-call-for-stronger-deeper-alliance-idUSKBN1ZI050>.

⁵⁴ *A Free and Open Indo-Pacific*.

⁵⁵ Bonnie Bley, “The New Geography of Global Diplomacy: China Advances as the United States Retreats,” *Foreign Affairs*, November 27, 2019, available at <www.foreignaffairs.com/articles/china/2019-11-27/new-geography-global-diplomacy>.

⁵⁶ Ivo H. Daalder and James M. Lindsay, *The Empty Throne: America's Abdication of Global Leadership* (New York: PublicAffairs, 2018), 164–170.

⁵⁷ Douglas H. Paal, *America's Future in a Dynamic Asia* (Washington, DC: Carnegie Endowment for International Peace, January 31, 2019), available at <<https://carnegieendowment.org/2019/01/31/america-s-future-in-dynamic-asia-pub-78222>>.

⁵⁸ *Ibid*.

⁵⁹ China's five principles of peaceful coexistence are mutual respect for sovereignty and territorial integrity, mutual nonaggression, noninterference in the internal affairs of others, equality and mutual benefit, and peaceful coexistence. These have been essential to Chinese foreign policy for more than 60 years and were amplified in “China's Policies on Asia-Pacific Security Cooperation,” State Council Information Office of the People's Republic of China, January 2017, available at <http://english.www.gov.cn/archive/white_paper/2017/01/11/content_281475539078636.htm>. Also see Ankit Panda, “Reflecting on China's Five Principles, 60 Years Later,” *The Diplomat*, June 2014, available at <<https://thediplomat.com/2014/06/reflecting-on-chinas-five-principles-60-years-later/>>.

⁶⁰ Feng Liu, “The Recalibration of Chinese Assertiveness: China's Responses to the Indo-Pacific Challenge,” *International Affairs* 96, no. 1 (January 2020), 23–24.

⁶¹ Joel Wuthnow, *Just Another Paper Tiger? Chinese Perspectives on the U.S. Indo-Pacific Strategy*, INSS Strategic Forum 305 (Washington, DC: NDU Press, June 2020), 13–14.

⁶² See Lily Kuo, “Australia Called ‘Gum Stuck to China's Shoe’ by State Media in Coronavirus Investigation Stoush,” *The Guardian*, April 28, 2020, available at <www.theguardian.com/world/2020/apr/28/australia-called-gum-stuck-to-chinas-shoe-by-state-media-in-coronavirus-investigation-stoush>.

⁶³ Kristy Needham, “Australia Rejects Chinese ‘Economic Coercion’ Thread amid Planned Coronavirus Probe,” Reuters, April 27, 2020, available at <www.reuters.com/article/us-health-coronavirus-australia-china-idUSKCN2290Z6>.

⁶⁴ See “West Feels Challenged by China's New ‘Wolf Warrior’ Diplomacy,” *Global Times* (Beijing), April 15, 2020, available at <<https://www.globaltimes.cn/content/1185776.shtml>>.

⁶⁵ Ambassador Derek Mitchell, *American Leadership in the Asia Pacific, Part 3: Promoting Democracy, Human Rights, and the Rule of Law*, Testimony Before the Senate Foreign Relations Subcommittee on East Asia, the Pacific and International Cybersecurity Policy, July 12, 2017, available at <<https://www.usip.org/publications/2017/07/american-leadership-asia-pacific-part-3-promoting-democracy-human-rights-and->>.

⁶⁶ Thomas F. Lynch III and James J. Przystup, *India-Japan Strategic Cooperation and Implications for U.S. Strategy in the Indo-Pacific Region*, INSS Strategic Perspectives 24 (Washington, DC: NDU Press, 2017).

⁶⁷ “Global Flow of Tertiary-Level Students,” United Nations Educational, Scientific and Cultural Organization, available at <<http://uis.unesco.org/en/uis-student-flow>>. At the same time, growing American fear of systematic, state-led spying by

Chinese students at American higher learning institutions is poised to reduce Chinese exchange student numbers. In July 2019, U.S. Federal Bureau of Investigation Director Christopher Wray testified to a Senate Judiciary Committee that China poses the greatest threat of intellectual and economic property theft in the United States and that much of this theft flows through Chinese students at American universities. See Ken Dilanian, “U.S. Universities Are a Soft Target for China's Spies, Say Officials,” NBC News, February 2, 2020, available at <www.nbcnews.com/news/china/american-universities-are-soft-target-china-s-spies-say-u-n1104291>.

⁶⁸ *A Free and Open Indo-Pacific*.

⁶⁹ “U.S. Travelers Will Dominate Tourism Spending in Asia-Pacific Through 2020,” *Skift*, May 18, 2016, available at <<https://skift.com/2016/05/18/u-s-visitor-spending-in-asia-pacific/>>.

⁷⁰ Victor Kiprop, “English Speaking Countries in Asia,” *World Atlas*, December 12, 2018, available at <www.worldatlas.com/articles/english-speaking-countries-in-asia.html>.

⁷¹ Eric Fish, “China's Youth Admire America Far More than We Knew,” *Foreign Policy*, February 9, 2017, available at <<https://foreignpolicy.com/2017/02/09/chinas-youth-admire-america-far-more-than-we-knew-surprising-survey-results-ideological-university-crackdown/>>. As observed in this article, it is important to note that Western researchers do not believe that Chinese youth—or Chinese in general—affinities for America and the West are about wholesale adoption of Western liberal democracy. Instead, decades of research consistently suggests that many Chinese yearn to adopt certain elements of Western governance, such as freedom of the press, personal liberties, and official transparency and accountability.

⁷² Abraham Denmark, “Ideological Competition in the Indo-Pacific,” Woodrow Wilson Center for Scholars, March 27, 2018, available at <<https://www.wilsoncenter.org/blog-post/ideological-competition-the-indo-pacific>>.

⁷³ “Milestone Congress Points to a New Era for China, the World,” Xinhua, October 24, 2017, available at <http://www.chinadaily.com.cn/china/19thpcnationalcongress/2017-10/24/content_33648051.htm>.

⁷⁴ Denmark, “Ideological Competition in the Indo-Pacific.”

⁷⁵ Denghua Zheng, “The Concept of ‘Community of Common Destiny’ in China's Diplomacy: Meaning, Motives and Implications,” *Asia and the Pacific Policy Studies* 5, no. 2 (March 2018), 196–207.

⁷⁶ As Aaron Friedberg observed, China's ideological appeal is constrained by the Chinese Communist Party's actions at home. The expanded resources available to the CCP regime have given it a widening array of options for crushing domestic dissent. It has strengthened the “Great Firewall” to block unwanted internet content. It also is moving toward a nationwide “social credit” system that will use facial recognition software and big data analytics to monitor the activities, track the movements, and assess the political reliability of virtually every person in China. This is a capability of which the 20th-century's totalitarian dictators could only dream. See Friedberg, “Getting the China Challenge Right.”

⁷⁷ Authors' calculations from United Nations World Tourism Organization (UNWTO) 2018 data. See “Country Profile-Outbound Tourism,” UNWTO, available at <<https://www.unwto.org/country-profile-outbound-tourism>>. An additional 45 million visited Hong Kong and Macao.

⁷⁸ Figures for 2010 outbound Chinese students are from the UNESCO Institute for Statistics. See “Data for the Sustainable Development Goals,” UNESCO, available at <<http://www.uis.unesco.org/Education/Pages/international-student-flow-viz.aspx>>. Figures for Indo-Pacific students in China are from

"Statistical Report on International Students in China for 2018," Ministry of Education, April 17, 2019, available at <http://en.moe.gov.cn/documents/reports/201904/t20190418_378692.html>.

⁷⁹ "Confucius Institutes: Asia," Office of Chinese Language Council International, available at <http://english.hanban.org/node_10971.htm>.

⁸⁰ Jonathan Stromseth, *The Testing Ground: China's Rising Influence in Southeast Asia and Regional Responses* (Washington, DC: Brookings Institution, November 2019), available at <<https://www.brookings.edu/research/the-testing-ground-chinas-rising-influence-in-southeast-asia-and-regional-responses/>>.

⁸¹ See Shaun Breslin, "China's Global Cultural Interactions," in *China and the World*, 137–155.

⁸² Laura Silver, Kat Devlin, and Christine Huang, "People Around the Globe Are Divided in Their Opinions of China," Pew Research Center, December 5, 2019, available at <<https://www.pewresearch.org/fact-tank/2019/12/05/people-around-the-globe-are-divided-in-their-opinions-of-china/>>.

⁸³ "Social Media Usage in the Asia Pacific," *Pixium Digital*, available at <<https://pixiumdigital.com/social-media-usage-asia-pacific/>>.

⁸⁴ "Global Engagement Center," Department of State, available at <<https://www.state.gov/bureaus-offices/under-secretary-for-public-diplomacy-and-public-affairs/global-engagement-center/>>.

⁸⁵ "About Radio Free Asia," RFA, available at <<https://www.rfa.org/about/>>; "U.S. Agency for Global Media," U.S. Broadcasting Board of Governors, available at <<https://www.usagm.gov/>>.

⁸⁶ See Larry Diamond and Orville Schell, eds., *China's Influence and American Interests: Promoting Constructive Vigilance* (Stanford: Hoover Institution Press, 2019), appendix 1.

⁸⁷ Silver, Devlin, and Huang, "People Around the Globe Are Divided in Their Opinions of China."

⁸⁸ This claim attributed to U.S. Department of State Global Engagement Center Director Lea Gabrielle, reported in Betsy Woodruff Swan, "State Report: Russia, Chinese, and Iranian Disinformation Narratives Echoing Each Other," *Politico*, April 21, 2010, available at <www.politico.com/news/2020/04/21/russia-china-iran-disinformation-coronavirus-state-department-193107>.

⁸⁹ Needham, "Australia Rejects Chinese 'Economic Coercion' Threat amid Planned Coronavirus Probe"; Yasmeen Serhan, "Indians Aren't Buying China's Narrative," *The Atlantic*, April 21, 2020, available at <www.theatlantic.com/international/archive/2020/04/india-china-pandemic-coronavirus-distrust/610273>; Philip Citowicki, "COVID-19 Escalates the China-Australia Contest in the Pacific," *The Diplomat*, April 10, 2020, available at <<https://thediplomat.com/2020/04/covid-19-escalates-the-china-australia-contest-in-the-pacific/>>.

⁹⁰ Bates Gill, "China's Global Influence: Post-COVID Prospects for Soft Power," *The Washington Quarterly* 43, no. 2 (Summer 2020), 97–115, available at <https://twq.elliott.gwu.edu/files/2020/06/Gill_TWQ_43-2.pdf>.

⁹¹ These numbers are extracted from "Asian-Pacific Military Strength (2020)," *Global Firepower*, available at <www.globalfirepower.com/countries-listing-asia-pacific.asp>; "Asia," Heritage Foundation, October 30, 2019, available at <<https://www.heritage.org/military-strength/assessing-the-global-operating-environment/asia>>. U.S. Indo-Pacific Command claims far more robust numbers for its regional forces on its public-facing information sites, contending that it has 5 carrier groups, 200 ships, and more than 300,000 military personnel. These numbers have understandable deterrent value, but overcount actual in-theater assets by claiming credit for all U.S. military assets that could be deployed in the event of a major surge requirement.

See "About USINDOPACOM," available at <<https://www.pacom.mil/About-USINDOPACOM/>>.

⁹² See Joel Wuthnow and Phillip C. Saunders, *Chinese Military Reforms in the Age of Xi Jinping: Drivers, Challenges, and Implications*, China Strategic Perspectives 10 (Washington, DC: NDU Press, March 2017); *China Military Power: Modernizing a Force to Fight and Win* (Washington, DC: Defense Intelligence Agency, 2019).

⁹³ While the notion of "island chains" is a construct that the Chinese military borrowed from American World War II military and diplomatic talk, and there is no authoritative Chinese military definition of each of the chains, most Chinese sources include Taiwan, Japan/the Ryukyu Islands, and the Philippines as being in the First Island Chain, and all include the Northern Mariana Islands in the Second Island Chain. See Andrew S. Erikson and Joel Wuthnow, "Barriers, Springboards, and Benchmarks: China Conceptualizes the Pacific 'Island Chains,'" *The China Quarterly* No. 225 (January 2016), 6–9.

⁹⁴ The United States recognizes Japanese administrative control over the Diaoyu/Senkaku Islands and therefore regards them as covered under the U.S.-Japan security treaty but does not take a position on the underlying sovereignty dispute.

⁹⁵ See Lyle J. Goldstein, *Five Dragons Stirring Up the Sea: Challenge and Opportunity in China's Improving Maritime Enforcement Capabilities* No. 5 (Newport, RI: U.S. Naval War College China Maritime Studies Institute, April 2010).

⁹⁶ Ratner et al., *Rising to the China Challenge*.

⁹⁷ For a recent overview of these activities, see Harris, *Statement Before the Senate Armed Services Committee Hearing on U.S. Pacific Command Posture*.

⁹⁸ *Indo-Pacific Strategy Report*.

⁹⁹ Ratner et al., *Rising to the China Challenge*.

¹⁰⁰ The People's Liberation Army calls this "systems attack" or "systems confrontation." See Jeff Engstrom, *Systems Confrontation and System Destruction Warfare: How the Chinese People's Liberation Army Seeks to Wage Modern Warfare* (Arlington, VA: RAND, 2018).

¹⁰¹ For a concise overview of U.S. policy, see Richard C. Bush, *A One-China Policy Primer*, East Asia Policy Paper, No. 10 (Washington, DC: Brookings Institution, March 2017), available at <<https://www.brookings.edu/research/a-one-china-policy-primer/>>.

¹⁰² "An Interactive Look at the U.S.-China Military Scorecard," RAND–Project Air Force, available at <<https://www.rand.org/paf/projects/us-china-scorecard.html>>. RAND defined *advantage* to mean that one side is able to achieve its primary objectives in an operationally relevant time period, while the other side would have trouble doing so.

¹⁰³ Stephen Bryen, "Why the U.S. Withdrew Its Bombers from Guam," *Asia Times* (Hong Kong), April 28, 2020, available at <<https://asiatimes.com/2020/04/why-the-us-withdrew-its-bombers-from-guam/>>; Lee Jeong-ho, "China Releases Footage of 'Guam Killer' DF-26 Ballistic Missile in 'Clear Message to the U.S.,"" *South China Morning Post* (Hong Kong), January 28, 2019, available at <www.scmp.com/news/china/military/article/2183972/china-releases-footage-guam-killer-df-26-ballistic-missile-clear>.

¹⁰⁴ "An Interactive Look at the U.S.-China Military Scorecard."

¹⁰⁵ Similar conclusions are found in Eric Heginbotham et al., *The U.S.-China Military Scorecard Forces, Geography, and the Evolving Balance of Power, 1996–2017* (Santa Monica, CA: RAND, 2015), available at <https://www.rand.org/pubs/research_reports/RR392.html>.

¹⁰⁶ Figures are 2017 data from the United Nations Comtrade Database, using the trade dashboard to count exports by Japan, North Korea, South Korea, Australia, New Zealand, India,

Bangladesh, Sri Lanka, Bhutan, and Nepal. Hong Kong re-exports to China are calculated at the 2018 rate of 55 percent.

¹⁰⁷ This figure is U.S. imports from China, which captures Chinese exports routed to the United States through Hong Kong.

¹⁰⁸ Members of the Trans-Pacific Partnership included Australia, Brunei Darussalam, Canada, Chile, Japan, Malaysia, Mexico, New Zealand, Peru, Singapore, the United States, and Vietnam.

¹⁰⁹ The Comprehensive Progressive Agreement for Trans-Pacific Partnership established an 11-country trading bloc representing 495 million people and a combined gross domestic product of \$13.5 trillion. It was written with an eye to future expansion. If the United States chooses to join, it would add nearly 330 million people and make the bloc the largest free-trading single market in the world, at 40 percent of world trade. See Keith Johnson, "The U.S. Wants Back in the TPP? Good Luck with That," *Foreign Policy*, March 29, 2018, available at <<https://foreignpolicy.com/2018/03/29/united-states-wants-back-in-tpp-good-luck-with-that-trans-pacific-partnership-asia/>>; "Overview and Benefits of the CPTPP," Government of Canada, available at <<https://www.international.gc.ca/trade-commerce/trade-agreements-accords-commerciaux/agr-acc/cptpp-ptpgp/overview-apercu.aspx?lang=eng>>.

¹¹⁰ Keith Johnson, "How Trump May Finally Kill the WTO," *Foreign Policy*, December 9, 2019, available at <<https://foreignpolicy.com/2019/12/09/trump-may-kill-wto-finally-appellate-body-world-trade-organization/>>.

¹¹¹ "Sec. Pompeo Remarks on America's Indo-Pacific Economic Vision."

¹¹² "Joint Statement of the Governments of the United States of America, Australia, and Japan," The White House, November 17, 2018, available at <<https://www.whitehouse.gov/briefings-statements/joint-statement-governments-united-states-america-australia-japan/>>.

¹¹³ Subsequently, this distinction was superseded by the economic impact of the COVID-19 pandemic. See Ken Roberts, "China Trade War Losses at \$84 Billion and Counting—Does New Data Suggest Tide Turning Toward U.S.?" *Forbes*, December 16, 2019, available at <www.forbes.com/sites/kenroberts/2019/12/16/china-trade-war-losses-at-84-billion-and-counting—does-new-data-suggest-tide-turning-toward-us/#35c3b543bd40>. Clearly, the global economic disruption from COVID-19 will eclipse this trade war by quite some margin.

¹¹⁴ Steve Liesman, "Trade War Losses for the U.S. and China Grow into the Tens of Billions of Dollars," CNBC, November 5, 2019, available at <www.cnbc.com/2019/11/05/trade-losses-for-the-us-china-mount-into-tens-of-billions-of-dollars.html>.

¹¹⁵ See *Economic and Trade Agreement Between the Government of the United States of America and the Government of the People's Republic of China* (Washington, DC: Office of the U.S. Trade Representative, 2020), available at <https://ustr.gov/sites/default/files/files/agreements/phase%20one%20agreement/Economic_And_Trade_Agreement_Between_The_United_States_And_China_Text.pdf>.

¹¹⁶ See Scott Kennedy, "China's Poor Purchasing Performance: How Should the United States Respond?" Center for Strategic and International Studies, May 8, 2020, available at <<https://www.csis.org/analysis/chinas-poor-purchasing-performance-how-should-united-states-respond>>.

¹¹⁷ See Sam Kim, "Korea Warns of Trade Pain That Will Shift Supply Chains," *Bloomberg*, May 10, 2020, available at <www.bloomberg.com/news/articles/2020-05-10/korea-warns-of-sharper-trade-pain-that-will-shift-supply-chains>; Carmen Reinicke, "These 8 Asian Countries Will Get the Biggest Boost If the Trade War Forces U.S. Companies to Leave China," *Markets Insider*, August 28, 2019, available at <<https://markets>>.

businessinsider.com/news/stocks/8-asian-countries-that-could-benefit-us-china-trade-war-2019-8-1028481289>.

¹¹⁸ Peter Harrell, Elizabeth Rosenberg, and Edoardo Saravalle, *China's Use of Coercive Economic Measures* (Washington, DC: Center for a New American Security, June 11, 2018), available at <<https://www.cnas.org/publications/reports/chinas-use-of-coercive-economic-measures>>.

¹¹⁹ "China Global Investment Tracker," American Enterprise Institute, April 30, 2020, available at <<https://www.aei.org/china-global-investment-tracker/>>. Figures do not include Chinese investment or construction projects in Hong Kong or Macao.

¹²⁰ Figures are calculated from "China's Public Diplomacy Dashboard Dataset (Version 1.0)," *AidData*, December 9, 2019, available at <<https://www.aiddata.org/china-public-diplomacy>>; also see Cheng Cheng, *The Logic Behind China's Foreign Aid Agency* (Washington, DC: Carnegie Endowment for International Peace, May 2019), available at <<https://carnegieendowment.org/2019/05/21/logic-behind-china-s-foreign-aid-agency-pub-79154>>.

¹²¹ Wuthnow, *Just Another Paper Tiger?* 15.

¹²² See Joel Wuthnow, "Managing U.S.-China Military Relations in an Era of Strategic Competition," paper presented at a University of Pennsylvania colloquium on U.S.-China relations, June 4, 2020.

¹²³ For a discussion of these challenges, see Thomas Fingar and Jean C. Oi, "China's Challenges: Now It Gets Harder," *Washington Quarterly* 43, no. 1 (Spring 2020), 65–82.

¹²⁴ Paal, *America's Future in a Dynamic Asia*.

¹²⁵ Charles Edel and Hal Brands, "The Real Origins of the U.S.-China Cold War," *Foreign Policy*, June 2, 2019, available at <<https://foreignpolicy.com/2019/06/02/the-real-origins-of-the-u-s-china-cold-war-big-think-communism/>>.

¹²⁶ *Ibid.*

¹²⁷ *Ibid.*

Chapter 10

Rogues, Disrupters, and Spoilers in an Era of Great Power Competition

By Bryce Loidolt, Mariya Omelicheva, and James Przystup

This chapter reviews the interests and behavior of Russia, Iran, and North Korea, so-called rogue, disrupter, and spoiler states. Motivated by goals ranging from a desire for regime survival to aspirations for regional dominance and even global relevance, these countries threaten to divert U.S. attention and resources away from the imperatives of Great Power competition and draw the United States into escalating and destructive crises. At first glance, then, there might appear to be strong incentives for China to form enduring, fully cooperative relationships with each of these countries. Yet this chapter also finds that Russian, Iranian, and North Korean provocative behavior is not uniformly beneficial for China, and the prospect of a robust and fully cooperative anti-U.S. axis in 2020 remains remote. U.S. policy-makers should anticipate the threat from each of these states to persist, but not necessarily to become more pronounced, as U.S.-Chinese competition intensifies.

As the United States continues to move into an era of Great Power competition featuring long-term rivalry and the prospect of Great Power transition with China, it will continue to encounter destabilizing activities from so-called rogue, disrupter, and spoiler states and regimes. We define this group of countries as those that lack the military and long-term economic power and/or transnational cultural appeal to match U.S. power globally or stabilize an alternative international political order. These states also tend to confront the United States below the threshold of active armed conflict and across multiple domains.¹ In defining this category of states based both on material and nonmaterial attributes as well as on conduct, this chapter reaches across what are often subjective, analytically blurry, and historically contingent concepts and definitions.² Although many countries meet the criteria identified above, this chapter focuses specifically on the challenges posed by the Russian Federation, the Islamic Republic of Iran, and the Democratic People's Republic of Korea (DPRK).

Notably, the grouping of these three countries within the same analytic category is a departure from the National Defense Strategy and the National Security Strategy, both of which describe the DPRK and Iran as rogue regimes and Russia as a near-peer competitor

on a par with China.³ This categorizing also deviates from the other chapters in this volume that treat Russia as a near-peer Great Power competitor. This change is intentional. The inclusion of Russia and exclusion of China—whose behavior is sometimes consistent with the definitional criteria offered above—is not intended to argue that Russian foreign policy lacks a desire for global relevance or that Russia is currently unable to compete with the United States. It is also not intended to sidestep the dimensions of Chinese behavior consistent with rogue, disrupter, or spoiler states. Instead, we posit that Russia and China's divergent economic trajectories imply a divergence in their future military strength. If Russian material power continues to decline and China's increases in the years to come, then Moscow's somewhat debatable status as a contemporary Great Power will recede even further.⁴ Thus, the alternative view of Moscow offered in this chapter may be an increasingly useful lens through which to analyze and understand Russian behavior and its implications for an era that becomes increasingly dominated by a U.S.-China Great Power dyad.⁵

The DPRK, Iran, and Russia are motivated by a combination of regime survival, aspirations for regional dominance and sometimes global relevance, as well as an inclination to confront the United States, which they all see as the main obstacle to their own aspirations. And lacking a proactive vision for or means to stabilize an alternative international order, these countries employ a variety of coercive instruments—ranging from proxy warfare to direct military threats—to pursue their interests.⁶ Moreover, their development of nuclear weapons and, in some cases, proliferation of ballistic missiles poses a serious threat to regional stability.

Through these activities, each of these countries threatens to undermine the security of U.S. allies and partners, erode U.S. credibility and influence abroad, and mire the United States and its allies in a labyrinth of internal challenges by impairing the legitimacy of their democratic political processes. Although such coercive activities tend to fall below a threshold that would prompt a conventional military response from the United States, they also risk escalating into potentially lethal crises. These states could not only divert U.S. attention and resources away from longer term objectives but also draw the United States into more distracting and costly confrontations.

Costs for the United States imply benefits for China, naturally raising the troubling specter of a more robust strategic alignment among these states against the United States. Indeed, China's security and economic relationships with Russia, Iran, and the DPRK often serve to constrain U.S. power. Welcoming the diversionary and constraining benefits of these countries' activities, Beijing could seek to strategically instigate these states' destabilizing behavior to the detriment of the United States.

This chapter finds, however, that Russian, Iranian, and North Korean provocative behavior is not uniformly beneficial for China, and the prospect of a robust and fully cooperative anti-U.S. axis in 2020 remains remote. U.S.-Chinese competition will yield limited prospects for burden-sharing between Beijing and Washington in comprehensively addressing Russian, Iranian, or DPRK conduct that is harmful to the United States. Given the negative externalities, the behavior of each country likely limits the depth of its relationship with Beijing.

China will need to balance the costs that these states can inflict on the United States and its allies with the potential spillover effects rogue and spoiler activities can have on

Beijing's economic interests and strategic partnerships. Ultimately, U.S. policymakers can anticipate neither fully cooperative nor obstructive responses from China to address the challenges each of these countries poses to regional stability.

The remainder of this chapter begins with an overview of the interests and behavior of the Russian Federation, the Islamic Republic of Iran, and the DPRK—paying specific attention to how these states' activities affect U.S. security interests. The second section turns to a discussion of China's relationship with these countries as well as the positive and negative consequences that their behaviors have for Chinese interests. It concludes with a projection for how these security challenges might evolve over the next 5 years and the implications for U.S.-China strategic competition.

Russian Federation

As detailed in chapter 3a of this volume, Russia's foreign policy is motivated by Great Power aspirations, a desire to reconstitute a sphere of influence in a multipolar world, and the desire for a buffer zone along its western, southern, and eastern borders to protect the country from potential security threats.⁷ Russian leadership views the unipolar world dominated by the United States as the gravest threat to its national interests. Russia perceives the United States as a dangerous meddler in the domestic affairs of sovereign states and as a rogue disrupter of stability in the Middle East and other parts of the world. This shift in Russian foreign policy from cooperation to competition with the United States did not occur all at once in 2014 but, rather, in the mid-1990s, when Russia's foreign policy establishment, disillusioned with the lack of reciprocity to its accommodation and integration with the West, began conceiving of a unipolar world as inimical to its own national interests.

As delineated in chapter 3b of this book, while Russia tends to pursue its interests through unconventional means of coercive influence, its conventional military capabilities loom large.⁸ Beginning in Ukraine in 2014, Russia's disinformation campaign and use of "little green men" were deemed effective substitutes for the direct application of military power, and in Syria, Russia's military proxies have allowed the Kremlin to plan and direct military actions under the cover of plausible deniability. Yet it is improved conventional military capabilities—rapidly deployable force; air defense; command, control, intelligence, surveillance, and reconnaissance; long-range strikes—that have been equally, if not more, decisive for Moscow in both theaters.⁹ Conventional military power has helped Moscow sustain pressure on Kyiv; in Syria, Russia's expeditionary capabilities have forged it into a key powerbroker in the conflict. Russia's gray zone tactics, which encompass psychological, cyber, computer network, proxy warfare, and electronic operations, are thus complementary to its conventional military capabilities.

Together with its information warfare in Ukraine, Moscow's interference in U.S. domestic politics represents a blueprint for a so-called strategy of active defense. Discussed by a chief architect of Russia's military doctrine, General Valery Gerasimov, this preemptive strategy encompasses a range of information tactics applied to destabilize potential threats to Russia's interests.¹⁰ These approaches include the use of Internet trolls (government-funded individuals who exploit social cleavages through fake blogs, offensive and inflammatory comments, and false information for sowing discord or swaying public opinion); leaking adverse, sensitive, or misleading information on foreign government officials

and institutions; and using government-funded mass media to disseminate propaganda favoring Russia. Releasing a significant amount of sensitive information or disinformation ahead of foreign elections or at the height of an international crisis involving Russia, using trolls and bots to amplify it, and publicizing these findings on Russia-sponsored outlets are used synergistically to orchestrate Russia's information strategy.¹¹

The spread of Russia's cyber intrusions and operations—ranging from the distributed denial-of-service (DDoS) and Structured Query Language injection attacks to phishing and eavesdropping—speaks to the sophistication of the country's cyber tools and institutional architecture for implementing them. The latter combines multiple security agencies with vast expertise in foreign and domestic intelligence-gathering with proxy-cyber activists, the so-called patriotic hackers, cyber criminals, and even legitimate cyber tech firms. Outsourcing cyber attacks allows Russia to create plausible deniability and lower the risks and costs associated with controversial foreign information campaigns. These attacks can also be used to sabotage critical physical infrastructure—banks, state institutions, and power plants—on a massive scale (as they did in Estonia, Georgia, Ukraine, and Montenegro). Multiple international organizations, including the World Anti-Doping Agency and the Organisation for the Prohibition of Chemical Weapons (OPCW), have also been targets of Russia's cyber attacks.¹² Russia's hacking attempt at the chemical weapons watchdog took place against the backdrop of the OPCW's ongoing investigation into the 2018 use of a military-grade nerve agent attack against Sergei Skripal in the United Kingdom.¹³

Proxies and mercenaries are other assets that allow Russia to accomplish its objectives without resorting to conventional military means. Russian mercenaries have fought alongside regular forces in complex battlespaces in Syria and eastern Ukraine and have operated in various capacities in the Central African Republic (CAR), Libya, Sudan, and even Venezuela. In all these contexts, the presence of Russian private military companies (PMCs) on the ground has allowed the Kremlin to play a critical role in security policies of these states. Because Russia's PMCs, in particular the Wagner Group, rely on the profits from natural resources seized on behalf of regimes in Syria, Sudan, or CAR for reimburse for their military service, their involvement in conflicts redirect the supported governments' operational priorities. Furthermore, in Sudan and CAR, Wagner has not only operated in a combat role but also provided these regimes with training, site defense, and security provision for top-level officials.¹⁴

Of course, Russia also employs more traditional means, using diplomacy, foreign aid, and arms transfers to pursue its strategic objectives. For example, in the United Nations Security Council, Russia has continued to veto resolutions aimed at investigating or sanctioning Bashar al-Asad's use of chemical weapons and wider brutality during the Syrian civil war.¹⁵ In Latin America, Russia has extended critical economic support to the embattled regime of Nicolás Maduro in Venezuela, stymying hopes for a democratic transition and undermining regional stability.¹⁶ Additionally, through its arms transfers, Russia has sought to exploit or otherwise create strategic daylight between the United States and its allies.¹⁷ Moscow has upped the ante through coercive messaging to Western audiences, stressing Russian resolve for nuclear retaliation and touting its purported superiority in hypersonic and other weapons systems.¹⁸ Russia often accompanies these announcements

with displays of force in massive wargames and provocative air force maneuvers near North Atlantic Treaty Organization (NATO) territory.¹⁹

These activities from Russia have far-reaching consequences for political stability, societal harmony, and continuous functioning of democratic institutions in its immediate neighborhood and around the world. Cyber attacks on physical infrastructure, particularly when they are combined with political trolling, not only have immediate grim consequences but also stimulate public fear, disengagement, and mistrust in the ability of the government to protect citizens. Information operations and the use of proxies have allowed Russia to maintain or expand its political and military influence (and supplant that of the United States) in many theaters around the world at relatively low cost. Turkey's status and NATO's continued viability and integrity are also in question, as Ankara is being pulled away from the West by the conflict. The unresolved war in Ukraine threatens to transform into a frozen conflict that would preclude Kyiv's integration into an array of liberal, democratic, and open European states. By offering security to embattled autocratic leaders in Africa, Russia also threatens to undermine U.S. interests by weakening local governance; this elevates Moscow's geopolitical posture and its material gains derived from weapons sales and access to natural resources.

The Islamic Republic of Iran

A mix of ideational and material factors motivates Iranian foreign policy behavior and underpins Iran's pursuit of regional interests. Iranian foreign policy revolves around the survival of the Islamic Republic in the face of perceived internal and external threats. Although the revolutionary zeal that characterized Iranian foreign policy throughout the 1980s has withered, Iran seeks to displace U.S. and Israeli regional dominance. Iran has often framed its policies in defensive terms. The Chief of Staff of the Iranian Armed Forces, Major General Mohammad Baqeri, declared in early 2019, "Regional enemies should know that in addition to [a] doctrine of peace, Iran has a strong military presence."²⁰ Nevertheless, Iran has been able to exploit the instability generated by the Arab Spring to counter what it views as U.S. and Israeli imperialist hegemony in the region, while also shoring up traditional allies and creating new ones in the Levant.²¹

Hampered by economic sanctions, particularly after the U.S. May 2018 withdrawal from the Joint Comprehensive Plan of Action (JCPOA), and facing a hostile geopolitical environment, Iran has been unable to address the threats to its national security through conventional military power. Since the Iran-Iraq War, the Islamic Republic has not had access to the foreign inputs required to rebuild and modernize its conventional military forces; it also lacks the requisite indigenous defense industrial base to do so unilaterally.²² Moreover, the devastating Iran-Iraq War has been seminal in shaping Iran's force structure decisions, creating a strong inclination among Iranian decisionmakers to avoid conventional warfare altogether.²³ As a result, Iran pursues its goals through a more asymmetric approach that aims to coerce and deter the United States and Israel, as well as rival Arab Gulf states, without prompting a conventional military response from its better equipped and more militarily proficient adversaries.

Iran's regional defense strategy relies on several instruments of coercion, deterrence, and defense, each of which can be calibrated to meet an array of threats. First, Iran has

developed a standoff strike capability that allows it to credibly threaten military, economic, and civilian targets within its rival's borders.²⁴ These weapons, which include ballistic and cruise missiles and unmanned aerial vehicles (UAVs), afford Iran not only a lower cost but also a plausibly deniable alternative to manned aircraft. In many ways, this obviates Iran's need for an air force capable of long-range precision strikes.²⁵ For example, on September 14, 2019, an attack using a combined 25 missiles and drones allowed Iran to jeopardize roughly half of Saudi Arabia's oil output. Moreover, in response to the U.S. killing of Islamic Revolutionary Guard Corps (IRGC) Qods Force commander Qassem Suleimani and Kata'ib Hezbollah commander Abu Mahdi al-Mohandis, Iran fired more than a dozen ballistic missiles at two Iraqi bases hosting U.S. and coalition personnel in January 2020.²⁶

Iran maintains several capabilities that have allowed it to disrupt global shipping in the Strait of Hormuz, a key maritime chokepoint. Although Iran has continued to threaten to close down the strait, the technical requirements and subsequent countermeasures that such an attempt would instigate make this prospect unlikely.²⁷ Nevertheless, Iran has several options that it could employ in various combinations to affect the calculus of the shipping industry, raising the costs of transiting the Strait of Hormuz and threatening to constrain the global energy supply; they include shore-based antiship cruise missiles, naval mines, armed IRGC navy speed boats, and UAVs.²⁸ The continued, albeit recently less prevalent, incidence of provocative behavior by the Iranian navy demonstrated a willingness to obstruct maritime traffic by attacking civilian tankers in the gulf with limpet mines in June 2019.²⁹

Next, Iran maintains an array of militant clients that, although varying in the degree of their responsiveness to Tehran, allows Iran to extend its influence to neighboring countries, forming what Iranian officials commonly refer to as the "Axis of Resistance."³⁰ This includes longstanding clients such as Lebanese Hezbollah and Hamas, as well as the Ansar Allah or Houthi rebels in Yemen, al-Ashtar Brigades in Bahrain, and the Afghan Fatimiyun and Pakistani Zaynabiyun Brigades that fight in Syria.³¹ Iran's diverse network of militant partners also contains several Shia groups in Iraq, encompassing long-term Iranian partners such as Asaib Ahl al-Haq, Kataib Hezbollah, and the Badr Organization, as well as recent iterations of smaller so-called special groups, that, although lesser in terms of pure numbers, are suspected to have closer ties to Tehran.³² These clients allow Iran to extend its political influence while offering it additional platforms from which it can attack U.S. and allied interests.

Iran has also sponsored terrorist attacks farther abroad, beyond areas where Iran lends support to militant proxies in the Middle East. For example, throughout the 1990s, Iranian operatives supported or directly executed a series of assassinations of Iranian dissidents in Europe.³³ In 2018, law enforcement authorities in Belgium, France, and Germany arrested Iranian operatives—including a government official—who had planned to bomb a political rally in France.³⁴ IRGC-led plots to attack Western and Israeli targets have also been disrupted in Nigeria, Kenya, Uruguay, and the United States.³⁵

Iran has recently expanded its cyber activities to offensive intrusions and attacks on foreign companies.³⁶ From September 2012 through January 2013, a hacker group known as the Izz al-Din al-Qassam Cyber Fighters carried out several DDoS attacks against U.S. financial institutions. Moreover, Iran is suspected to be behind the August 2012 cyber attack on Saudi Aramco as well as a 2016 attack that affected the Saudi General Authority for Civil Aviation and the Central Bank.³⁷

Finally, since the U.S. withdrawal from the JCPOA, Iran has gradually resumed its nuclear enrichment activities. In 2018, Iran prepared to expand its manufacturing and enrichment capacity, remaining within the JCPOA-prescribed limits.³⁸ In the wake of the U.S. killing of Qassem Suleimani, Iran also announced that it would be abandoning operational limits imposed by the nuclear deal.³⁹

Though not by any means an existential threat to the United States, Iran's set of coercive options and activities presents hazards to the stability of U.S. partners, allies, and regional interests. Beyond the recent U.S. strikes against Suleimani and Katib Hizballah facilities, U.S. allies and partners in the Middle East have come into conflict with Iranian-backed clients. This includes the Saudi-led coalition in Yemen as well as Israel, which has taken a more forward-leaning role in striking Iranian-backed proxies in Iraq and Syria.⁴⁰ Iran's activities in the Strait of Hormuz, including the June 2019 downing of a U.S. UAV and the seizing of civilian tankers, further risk instigating tit-for-tat escalations that could significantly disrupt maritime traffic through a key strategic chokepoint—just as a resumption in Iran's nuclear activities could spark a dangerous regional conflict.⁴¹

Democratic People's Republic of Korea

The DPRK operates more as a quasi-criminal enterprise than a legitimate nation-state.⁴² At its core, the Kim Jong-un government is most concerned with sustaining its family enterprise and ensuring the survival of the regime at all costs. It also prioritizes the reunification of the Korean Peninsula as a means to the end of regime survival and an endstate of itself.⁴³ This apparent lack of international ambition on the part of the Kim regime might suggest an attenuated threat to global stability, but its activities have far-reaching consequences for U.S. security interests in the new era of Great Power competition.

Through the development of weapons of mass destruction, use of chemical weapons, and aggressive posturing of its conventional forces, the DPRK threatens regional stability and global norms. North Korea is estimated to have somewhere between 15 and 60 nuclear warheads, as well as approximately 650 ballistic missiles that could threaten cities in South Korea, Japan, and eastern China.⁴⁴ It has also tested intercontinental ballistic missiles that could be capable of striking the United States.⁴⁵ At the same time, North Korea continues to pose a conventional threat to South Korea and Japan. The People's Army, an estimated 1.2 million in strength, is overwhelmingly forward-deployed toward the Demilitarized Zone in an offensive posture.⁴⁶ Kim has also pursued more advanced conventional capabilities, including more precise artillery and ballistic missile capabilities as well as UAVs.⁴⁷ In this environment, North Korea's nuclear arsenal provides Pyongyang with the potential for nuclear blackmail, allowing it to engage in lower level conventional provocations and, at the same time, affect South Korean and U.S. decisions on kinetic responses or induce economic concessions.

Underscoring the criminal nature of the regime, in February 2017, the DPRK carried out the assassination of Kim's half-brother using the nerve agent VX in Malaysia.⁴⁸ North Korea's malign behavior has historically extended beyond Asia and included weapons transfers to hostile states and armed groups in the Middle East. Iran has been accused of being "one of the two most lucrative markets for DPRK military-related cooperation."⁴⁹ Pyongyang has indeed engaged in an ongoing relationship with Iran featuring sales and

the transfer of military technology that has served to advance the development of Tehran's ballistic missile programs.⁵⁰ The DPRK has also pursued military cooperation and technology transfers in the Sudan and offered small arms and ballistic missiles to the Houthi rebels in Yemen through a Syrian intermediary.⁵¹ North Korea has also exported the SCUD-D, a newly tested advanced short-range ballistic missile, to Syria.⁵²

To sustain the regime in the face of international sanctions and condemnation, North Korea has resorted to a wide range of illegal activities that violate global norms. Pyongyang has employed its cyber capabilities to hack banks across the globe, reportedly carrying out successful cyber heists against banks in Bangladesh, Chile, Guatemala, India, Kuwait, Mexico, Pakistan, the Philippines, South Korea, Taiwan, Turkey, and Vietnam.⁵³ These attacks can be quite lucrative; it has been reported that in one such attack against the Central Bank of Bangladesh in 2016, North Korea netted \$81 million.⁵⁴ A United Nations Security Council report estimates that in total the DPRK may have acquired as much as \$2 billion through its cyber operations.⁵⁵ North Korean cyber operations have also targeted media outlets that it deems critical of its policies or of Kim in particular.⁵⁶

Currency counterfeiting and narcotics trafficking have helped the regime generate funds and offset the effects of sanctions. In the late 1970s, Pyongyang began to put counterfeit U.S. currency into circulation, featuring “supernotes”—phony bills of remarkably high quality—of \$50 and \$100 denominations. There is a high degree of uncertainty regarding the value of this activity; estimates range from \$1.25 million to \$250 million per annum.⁵⁷ Since the 1970s, when Pyongyang began to sponsor opium cultivation and the production of opiates, North Korean diplomats have been arrested in antinarcotics operations across the globe. As of 2007, North Korea has been linked to drug seizures in at least 20 countries.⁵⁸ In the 1990s, North Korea reportedly began manufacturing crystal methamphetamine for exports using Chinese triads, the Japanese Yakuza, and the Russian mafia as distribution channels.⁵⁹ To weaken the effect of sanctions on North Korea's exports, Pyongyang has moved to step up production of illicit drugs to earn the hard currency needed to fund its nuclear and missile development programs.⁶⁰

North Korea thus represents a multidimensional threat to the prosperity and security of the United States and its allies in the Indo-Pacific region, as its nuclear weapons, ballistic missiles, and conventional posture place several U.S. allies at risk. The DPRK's willingness to use chemical weapons and its involvement in transnational criminal activity similarly violate international norms, and the aforementioned cyber operations have important consequences for the security of the global financial sector.

Rogue, Disrupter, and Spoiler State Behavior and U.S.-China Competition

Beijing currently maintains collaborative, if sometimes distant, relations with Russia, Iran, and the DPRK. China shares with all of these countries a general displeasure with U.S. hegemony and dominance of international institutions. Iran is considered a “comprehensive strategic partner” by Beijing, and Chinese-Iranian cooperation spans the economic and security spheres.⁶¹ With respect to the former, since 2005, Chinese investments in and contracts with Iran have topped \$27 billion.⁶² And China has reportedly assisted in developing Iran's ballistic missiles, antiship mines, fast-attack boats, and other weapons technology.⁶³ In

2018, China also imported \$15 billion worth of oil from Iran.⁶⁴ China and Russia maintain a “comprehensive cooperative” strategic partnership, which may be emblematic of deepening Sino-Russian cooperation.⁶⁵ In addition to regular diplomatic and military exchanges under the auspices of regional and international organizations, Russia agreed to assist China in building a strategic missile early-warning system and may view a relationship with China as a valuable avenue through which it can challenge the United States.⁶⁶ China’s share of Russia’s trade and investments has also grown.⁶⁷ While Beijing officially continues to support United Nations sanctions on North Korea, cross-border trade with North Korea has bolstered regime stability. In July 2019, the *South China Morning Post* reported a 14.3 percent increase in China’s trade with North Korea in the first half of 2019, amounting to \$1.25 billion.⁶⁸ President Xi Jinping’s visit to Pyongyang in June 2019 stands as a clear indication of China’s political and economic support for the DPRK.

Although Chinese interests in maintaining relationships with each state are distinct, ranging from shared hostility toward the United States to China’s energy needs and desire to maintain a peaceful neighborhood, China’s behavior has the consequence of insulating Russia, Iran, and the DPRK from the costs of their provocative behavior. In the case of Iran and the DPRK, economic relations with Beijing offer relief from international sanctions. Additionally, Iran has relied on China for advanced conventional capabilities. Chinese trade and largesse similarly offer Russia an economic lifeline. China has also been willing to purchase Russian combat aircraft and surface-to-air missile systems despite U.S. sanctions.⁶⁹

In turn, each state’s provocative activities offer some important perks for Beijing. First, China benefits from having additional voices questioning the value and wisdom of U.S. hegemony and international norms. This benefit is perhaps most apparent in the cyber domain, where both Russia and China have advocated for a different set of norms on cyber and information security that emphasizes state sovereignty and prioritizes constraints on the free flow of information over the safeguarding of critical cyber infrastructure and networks.⁷⁰

The diversionary benefits of each country’s behavior are also considerable. Iranian provocations tie U.S. resources down in a volatile and often hostile region rather than the Indo-Pacific.⁷¹ Indeed, since May 2019, the United States has deployed 14,000 additional troops to the Middle East, coinciding with a rise in tensions between Iran and the United States.⁷² China similarly benefits from Russia’s propensity for distracting the United States from China’s potentially destabilizing and convention-breaking activities around the world.⁷³

These countries’ behavior also poses important risks for Beijing, however: for one, the prospect of crisis escalation between the United States and any of these countries would be enormously costly for China. China’s objectives toward the Korean Peninsula and North Korea in particular have remained consistent since the beginning of the nuclear crisis in the early 1990s. China seeks to avoid war on the peninsula and inhibit the collapse of the North Korean regime, while also pursuing the peaceful denuclearization of the DPRK. As a result, China likely views U.S.-DPRK sabre rattling with a degree of alarm. Thus, while seeking the survival of the Kim regime as a major priority, Beijing has continued to encourage diplomatic engagement between Washington and Pyongyang even as it offsets the effects of U.S. sanctions.⁷⁴ An active conflict in the Persian Gulf that could come about as a result

of U.S.-Iranian tensions would similarly be devastating to China, which relies on the Gulf states for roughly 45 percent of its energy imports.⁷⁵

Next, Iranian and Russian behavior could frustrate China's other strategic partnerships. China maintains a diverse and somewhat contradictory alliance portfolio in the Middle East, including not only Iran but of some its regional adversaries as well. China relies heavily on Saudi Arabia for its energy needs and has pursued a relationship with Israel in part to acquire advanced technologies. Any increase in destabilizing Russian behavior in Africa could similarly complicate Beijing's relationships with states in that region.⁷⁶

This balance of risks and benefits has likely motivated China's policies of cautious enablement—rather than complete endorsement—of North Korean, Russian, and Iranian activities. Even as it maintains rather friendly relations with Moscow, Beijing has refused to recognize the independence of Abkhazia and South Ossetia after the Russo-Georgian war of 2008, and abstained from, rather than vetoing, the 2014 United Nations resolution condemning Moscow's seizure of Crimea.⁷⁷ The pursuit of a diplomatic path toward the denuclearization of the Korean Peninsula remains a priority for China, even as it alleviates the effects of economic pressure. Beijing has likewise continued to engage with Tehran, while also responding to Iran's calls for more confrontational policies toward Washington and proposed accession into the Shanghai Cooperation Organization with little enthusiasm.⁷⁸

Conclusion

Russia, Iran, and the DPRK pose a threat within and beyond their respective regions. Russia is the most materially capable of these states and has deftly employed a mix of information, cyber, and proxy warfare to foment instability and erode the legitimacy of democratic political processes across the globe. The threats of Iran and the DPRK are more pronounced in their immediate regions but still undertake activities and behaviors that are global in scope. Iran has demonstrated a willingness to employ its precision-strike capabilities against civilian targets within Saudi borders and is located in a geopolitically sensitive region. Additionally, DPRK missile tests pose a danger to proximate states. Through their cyber intrusions, chemical weapons use, and other coercive and convention-breaking behavior, all of these states threaten to erode international norms.

Each country's provocative behavior can tie down U.S. resources while undermining Washington's global standing. This naturally produces a strong set of incentives for Beijing to build and maintain partnerships with all three of these states, and by establishing bilateral relationships that often span the economic and security domains, China can shield these states from some of the costs of what the United States perceives to be malign behaviors.

Nevertheless, the negative repercussions—real and potential—that each state's behavior poses for China inhibit the extent of these relationships. Iranian escalatory actions threaten China's energy supplies and regional partners, just as destabilizing behavior from Russia might complicate its relations with African states. DPRK recalcitrance and coercive threats also have the potential to yield a miscalculation and a devastating confrontation on the Korean Peninsula. Somewhat paradoxically, the more China insulates these countries from the costs of their activities through more in-depth cooperation, the more Beijing risks increasing the audacious behaviors that can threaten China's key interests.

China is likely well aware of these risks. Beijing will thus be unlikely to deepen its cooperation with these countries solely as a means of confronting and frustrating the United States; instead, U.S. policymakers can expect Beijing to adopt a more delicate approach that seeks to limit, but not completely eliminate, malign behavior. In turn, this means that U.S. policymakers should be cautiously optimistic that the prospects of an in-depth, fully cooperative bi- or multilateral anti-U.S. strategic alliance taking hold across these states will remain remote.

U.S. policymakers may even be able to identify cooperative space with Beijing in addressing some of the more detrimental dimensions of Iranian, Russian, and DPRK actions. With respect to Iran, the United States could find common ground with China in limiting Tehran's destabilizing activities in the Persian Gulf and its anti-Israel proxies, many of which threaten not only global energy supplies but also important Chinese partners.⁷⁹ The United States might similarly be able to leverage Chinese support for containing the destabilizing effects of Russian activities in Africa, just as it can rely to some degree on Chinese diplomatic support in reigning in Kim Jong-un.

Even so, U.S. policymakers should harbor no illusions regarding the potential for more robust cooperation from Beijing in implementing comprehensive punitive measures against any one of these countries. China will instead seek to keep Russian, Iran, and DPRK behaviors below a tolerable threshold. The United States can thus anticipate the threat of these states to persist, but not necessarily to become more pronounced, as it moves forward into a new era of Great Power competition marked by increasing rivalry with China.

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Notes

¹ On measures short of armed conflict, commonly referred to as gray zone tactics or political warfare, see, for example, Michael Mazarr, *Mastering the Gray Zone: Understanding a Changing Era of Conflict* (Carlisle Barracks, PA: Army War College Press, December 2015); Van Jackson, "Tactics of Strategic Competition: Gray Zones, Redlines, and Conflicts Before War," *Naval War College Review* 70, no. 3 (Summer 2017), 39–62; Linda Robinson et al., *Modern Political Warfare: Current Practices and Possible Responses* (Santa Monica, CA: RAND, 2018).

² Since the end of the Cold War, states have often been referred to by the term *rogue* if they pursue or proliferate weapons of mass destruction; provide material support for terrorist, mercenary, or other nonstate proxies; or habitually violate the principles, norms, and practices of the post–Cold War international order. See, for example, Kelly P. O'Reilly, "Perceiving Rogue States: The Use of the 'Rogue State' Concept by U.S. Foreign Policy Elites," *Foreign Policy Analysis* 3, no. 4 (October 2007), 295–315; Michael Rubin, *Dancing with the Devil: The Perils of Engaging Rogue Regimes* (New York: Encounter Books, 2014), 2–10; Mary Caprioli and Peter F. Trumbore, "Rhetoric Versus Reality: Rogue States in Interstate Conflict," *Journal of Conflict Resolution* 49, no. 5 (October 2005), 770–791; Thomas H. Henriksen, "The Rise and Decline of Rogue States," *Journal of International Affairs* 54, no. 2 (Spring 2001), 349–373; Maria Gabrielsen Jumbert, "How Sudan's 'Rogue' State Label Shaped U.S. Responses to the Darfur Conflict: What's the Problem and Who's in Charge?" *Third World Quarterly* 35, no. 2 (March 2014), 284–299. The phrase *spoiler state* has historically been reserved for countries

with enough economic clout to "do damage to the [international financial] system, but not substantial enough to stabilize it." See, for example, Charles P. Kindleberger, "Dominance and Leadership in the International Economy: Exploitation, Public Goods, and Free Rides," *International Studies Quarterly* 25, no. 2 (June 1981), 242–254. The term *spoiler* to describe this category of country appears in David A. Lake, "International Economic Structures and American Foreign Economic Policy, 1887–1934," *World Politics* 35, no. 4 (July 1983), 517–543. And *disrupter* describes states that employ "short-of-war tactics and techniques" in an effort to "disrupt the established international system as well as delegitimize the United States as the leading power of the current world order." See, for example, Isaiah Wilson III and Scott Smitson, "Solving America's Gray-Zone Puzzle," *Parameters* 46, no. 4 (Winter–Spring 2016), 65. For other terms policymakers have used to describe these states, see Gerry J. Simpson, *Great Powers and Outlaw States: Unequal Sovereigns in the International Legal Order* (Cambridge, UK: Cambridge University Press, 2004); Anthony Lake, "Confronting Backlash States," *Foreign Affairs* 73, no. 2 (March–April 1994), 45–55; Ronald Reagan, "The New Network of Terrorist States," *Studies in Conflict and Terrorism* 9, no. 2 (1987), 101–112.

³ *Summary of the National Defense Strategy of the United States of America: Sharpening the American Military's Competitive Edge* (Washington, DC: Department of Defense, 2018), 2; *National Security Strategy of the United States of America* (Washington, DC: The White House, December 2017), 26.

⁴ See chapter 1 for this volume's three-attribute operational definition of a Great Power, which consists of unusual capabilities, truly global behavior, and Great Power status attribution by other states in the system. See chapter 3a for a review of how Russia's power tools make its long-term retention of Great Power status uncertain.

⁵ On the material and nonmaterial limitations of Russian power and influence, see James Dobbins, Howard J. Shatz, and Ali Wyne, *Russia Is a Rogue, Not a Peer; China Is a Rogue, Not a Peer* (Santa Monica, CA: RAND, 2018); Peter B. Zwack and Marie-Charlotte Pierre, *Russian Challenges from Now into the Next Generation: A Geostrategic Primer*, INSS Strategic Perspectives 29 (Washington, DC: NDU Press, March 2019); Daniel R. Coats, *Worldwide Threat Assessment of the U.S. Intelligence Community* (Washington, DC: Office of the Director of National Intelligence [ODNI], February 13, 2018), 36–37, available at <www.dni.gov/files/documents/Newsroom/Testimonies/2018-ATA---Unclassified-SSCI.pdf>.

⁶ For more on Russia's lack of a proactive ideology for crafting an alternative world order—especially in comparison to the United States and China—see chapter 3a of this volume.

⁷ Thomas Graham, “Let Russia Be Russia: The Case for a More Pragmatic Approach to Moscow,” *Foreign Affairs*, November/December 2019, available at <www.foreignaffairs.com/articles/russia-fsu/2019-10-15/let-russia-be-russia>; Julia Gurganus and Eugene Rumer, *Russia's Global Ambitions in Perspective* (Washington, DC: Carnegie Endowment for International Peace, 2019); Mariya Y. Omelicheva, “Critical Geopolitics on Russian Foreign Policy: Uncovering the Imagery of Moscow's International Relations,” *International Politics* 53, no. 6 (2016), 708–726.

⁸ For an overview of Russian military capabilities, see *Russia Military Power: Building a Military to Support Great Power Aspirations*, DIA-11-1704-161 (Washington, DC: Defense Intelligence Agency [DIA], 2017), available at <www.dia.mil/portals/27/documents/news/military%20power%20publications/russia%20military%20power%20report%202017.pdf>.

⁹ Nicole Ng and Eugene Rumer, *The West Fears Russian Hybrid Warfare: They're Missing the Bigger Picture* (Washington, DC: Carnegie Endowment for International Peace, July 2019), available at <<https://carnegieendowment.org/2019/07/03/west-fears-russia-s-hybrid-warfare-they-re-missing-bigger-picture-pub-79412>>.

¹⁰ Valery Gerasimov, “Vektory razvitiya voennoi strategii,” *Krasnaya Zvezda* [Red Star] (Moscow), April 3, 2019, available at <<http://redstar.ru/vektory-razvitiya-voennoj-strategii/>>; Dara Massicot, “Anticipating a New Russian Military Doctrine in 2020: What It Might Contain and Why It Matters,” *War on the Rocks*, September 9, 2019, available at <<https://warontherocks.com/2019/09/anticipating-a-new-russian-military-doctrine-in-2020-what-it-might-contain-and-why-it-matters/>>. See also *Assessing Russian Activities and Intentions in Recent U.S. Elections*, Intelligence Community Assessment 2017-01D (Washington, DC: ODNI, January 6, 2017), available at <www.dni.gov/files/documents/ICA_2017_01.pdf>.

¹¹ Andrei Soldakov and Irina Borogan, “Russia's Approach to Cyber: The Best Defense Is a Good Offense,” in *Hacks, Leaks, and Disruptions: Russian Cyber Strategies*, ed. Nicu Popescu and Stanislav Secrieru (Paris: European Union Institute for Security Studies, 2018), 15–24. Additional detail on Russian social media and information operations is provided in chapter 7 of this volume.

¹² For further information, see Soldakov and Borogan, “Russia's Approach to Cyber,” 15–24.

¹³ Coats, *Worldwide Threat Assessment of the U.S. Intelligence Community*, 8.

¹⁴ Andrew Linder, “Russian Private Military Companies in Syria and Beyond,” *New Perspectives in Foreign Policy*, no. 16 (Fall 2018), 17–21; Nathaniel Reynolds, *Putin's Not-So-Secret Mercenaries:*

Patronage, Geopolitics, and the Wagner Group (Washington, DC: Carnegie Endowment for International Peace, 2019).

¹⁵ “Russia's 12 UN Vetoes on Syria,” *RTE* (Dublin), April 11, 2018, available at <www.rte.ie/news/world/2018/0411/953637-russia-syria-un-veto/>.

¹⁶ John E. Herbst and Jason Marczak, *Russia's Intervention in Venezuela: What's at Stake?* Policy Brief (Washington, DC: Atlantic Council, September 12, 2019), available at <www.atlanticcouncil.org/in-depth-research-reports/report/russias-intervention-in-venezuela-whats-at-stake/>.

¹⁷ Henry Meyer and Ilya Arkhipov, “Russia Wants to Sell Its Missiles to U.S. Allies,” *Bloomberg*, June 22, 2018, available at <www.bloomberg.com/news/articles/2018-06-22/russia-wants-to-sell-its-missiles-to-u-s-allies>.

¹⁸ David Brennan, “Russia Deploys First of ‘Invulnerable’ Hypersonic Missiles Putin Claims Can Defeat Any American Defenses,” *Newsweek*, December 27, 2019, available at <www.newsweek.com/russia-deploys-first-invulnerable-hypersonic-missiles-putin-defeat-american-defenses-1479362>.

¹⁹ Elias Groll, “Russian Military Escalates Provocation Campaign in Europe,” *Foreign Policy*, October 30, 2014, available at <<https://foreignpolicy.com/2014/10/30/russian-military-escalates-provocation-campaign-in-europe/>>.

²⁰ “Besides Doctrine of Peace, Iran Has Strong Military Presence: Commander,” Islamic Republic News Agency (IRNA), January 1, 2019, available at <<https://en.irna.ir/news/83154430/Besides-doctrine-of-peace-Iran-has-strong-military-presence>>. For an overview of Iranian strategic culture, see Michael Eisenstadt, *The Strategic Culture of the Islamic Republic of Iran: Operational and Policy Implications* (Quantico, VA: Marine Corps University Press, 2011).

²¹ For an overview, see, for example, Michael Eisenstadt, *Operating in the Gray Zone: Countering Iran's Asymmetric Way of War* (Washington, DC: Washington Institute for Near East Policy, 2020); DIA, *Iran Military Power: Ensuring Regime Survival and Securing Regional Dominance* (Washington, DC: DIA, 2019), available at <www.dia.mil/Portals/27/Documents/News/Military%20Power%20Publications/Iran_Military_Power_LR.pdf>.

²² Anthony H. Cordesman et al., *U.S.-Iranian Competition: The Gulf Military Balance—I* (Washington, DC: Center for Strategic and International Studies [CSIS], 2013); Michael Wahid Hanna and Dalia Dassa Kaye, “The Limits of Iranian Power,” *Survival* 57, no. 5 (September 2015), 173–198; DIA, *Iran Military Power*.

²³ Michael Eisenstadt, *Iran After Sanctions: Military Procurement and Force-Structure Decisions* (London: International Institute for Strategic Studies, December 2017), available at <www.washingtoninstitute.org/uploads/Documents/opeds/Eisenstadt20171219-IISS-chapter.pdf>.

²⁴ For an overview of several of these, see Seth Jones et al., *Iran's Threat to Saudi Critical Infrastructure: The Implications of U.S.-Iranian Escalation* (Washington, DC: CSIS, 2019).

²⁵ Shahryar Pasandideh, “Under the Radar, Iran's Cruise Missile Capabilities Advance,” *War on the Rocks*, September 25, 2019, available at <<https://warontherocks.com/2019/09/under-the-radar-irans-cruise-missile-capabilities-advance/>>; Sitakanta Mishra, “Cruise Missiles: Evolution, Proliferation and Future,” *Journal of Defence Studies* 6, no. 3 (July 2012), 139–142; Dennis M. Gormley, *Dealing with the Threat of Cruise Missiles* (New York: Routledge, 2013).

²⁶ Mike R. Pompeo, “Department of State Remarks to Traveling Press,” *The Iran Primer*, September 19, 2019, available at <<http://iranprimer.usip.org/index.php/blog/2019/sep/18/pompeo-briefing-saudi-attacks>>; Natasha Turak, “Drone and Missile Debris Proves Iranian Role in Aramco Attack, Saudi Defense Ministry Claims,” *CNBC*, September 18, 2019, available at <

saudi-arabia-drone-and-missile-debris-proves-iranian-role-in-attack.html>; Jon Gambrell, Aya Batrawy, and Fay Abuelgasim, "Mike Pompeo Calls Drone Attacks on Saudi Oilfields 'An Act of War' by Iran," Associated Press, September 18, 2019, available at <www.businessinsider.com/pompeo-drone-attacks-saudi-oilfields-act-of-war-by-iran-2019-9>; "DOD Statement on Iranian Ballistic Missile Attacks in Iraq," Department of Defense, January 7, 2010, available at <www.defense.gov/Newsroom/Releases/Release/Article/2052103/dod-statement-on-iranian-ballistic-missile-attacks-in-iraq>.

²⁷ Sidharth Kaushal, "Shutdown Showdown: How the Strait of Hormuz Factors into the U.S.-Iran Crisis," *The National Interest*, July 7, 2019, available at <https://nationalinterest.org/feature/shutdown-showdown-how-strait-hormuz-factors-us-iran-crisis-65596?page=0%2C1>; Caitlin Talmadge, "Closing Time: Assessing the Iranian Threat to the Strait of Hormuz," *International Security* 33, no. 1 (Summer 2008), 82–117; Abhijit Singh, "Dark Chill in the Persian Gulf—Iran's Conventional and Unconventional Naval Forces," *Maritime Affairs* 6, no. 2 (February 2011), 108–132; Ilan Goldenberg, Jessica Schwed, and Kaleigh Thomas, *In Dire Straits? Implications of U.S.-Iran Tensions for the Global Oil Market* (Washington, DC: Center for a New American Security, 2019).

²⁸ "Iran Says Has Built 'Smart' Remote-Controlled Submarine," *Eurasia Review*, January 22, 2012, available at <www.eurasiareview.com/22012012-iran-says-has-built-smart-remote-controlled-submarine>; DIA, *Iran Military Power*, 51–56.

²⁹ *Outlaw Regime: A Chronicle of Iran's Destructive Activities* (Washington, DC: Department of State, 2018), 27–30, available at <www.state.gov/outlaw-regime-a-chronicle-of-irans-destructive-activities/>; Jim Garamone, "Selva Details Iranian Oil Attacks, Need for International Efforts," Department of Defense, June 19, 2019, available at <www.defense.gov/explore/story/Article/1879849/selva-details-iranian-oil-attacks-need-for-international-efforts/>; Robert Burns, "U.S. Military Official: Iran Naval Forces Halt 'Provocations,'" Associated Press, March 15, 2018, available at <https://apnews.com/a36e23a8d549464caaea7dc1932babae/US-military-official:-Iran-naval-forces-halt-%27provocation>.

³⁰ "Iran Making Progress on Basis of Rationality, Resistance in Int'l Arena," IRNA, August 2, 2019, available at <https://en.irna.ir/news/83421453/Iran-making-progress-on-basis-of-rationality-resistance-in-int-l>.

³¹ For overviews, see Afshon Ostovar, "The Grand Strategy of Militant Clients: Iran's Way of War," *Security Studies* 28, no. 1 (October 2018), 159–188; *Outlaw Regime*, 9–17; Seth G. Jones, *War by Proxy: Iran's Growing Footprint in the Middle East* (Washington, DC: CSIS, 2019).

³² Joseph Felter and Brian Fishman, *Iranian Strategy in Iraq: Politics and "Other Means"* (West Point, NY: Combating Terrorism Center, 2008); Michael Knights, "Iran's Expanding Militia Army in Iraq: The New Special Groups," *CTC Sentinel* 12, no. 7 (August 2019).

³³ Daniel Byman, "Iran, Terrorism, and Weapons of Mass Destruction," *Studies in Conflict & Terrorism* 31, no. 3 (February 2008), 169–181; Bruce Hoffman, *Recent Trends and Future Prospects of Iranian Sponsored International Terrorism* (Santa Monica, CA: RAND, 1990); Michael Rubin, "Iran's Revolutionary Guards—A Rogue Outfit?" *Middle East Quarterly* 15, no. 4 (Fall 2008), 37–48.

³⁴ "France Blames Iran for Foiled Paris Bomb Plot," BBC News, October 2, 2018, available at <www.bbc.com/news/world-europe-45722523>.

³⁵ *Outlaw Regime*, 15–17.

³⁶ Michael Eisenstadt, *Iran's Lengthening Cyber Shadow* (Washington, DC: Washington Institute for Near East Policy, 2016).

³⁷ "Iranian Regime Malign Activities During Negotiations with Iran and During JCPOA," Department of State, October 17, 2019, available at <www.state.gov/iranian-regime-malign-activities-during-negotiations-with-iran-and-during-jcpoa/>.

³⁸ Coats, *Worldwide Threat Assessment of the U.S. Intelligence Community*, 10.

³⁹ "Iran Takes Final Step by Abandoning JCPOA Restrictions," IRNA, January 5, 2020, available at <https://en.irna.ir/news/83622509/Iran-takes-final-step-by-abandoning-JCPOA-restrictions>.

⁴⁰ David M. Halbfinger, Ben Hubbard, and Ronen Bergman, "The Israel-Iran Shadow War Escalates and Breaks into the Open," *New York Times*, August 28, 2019, available at <www.nytimes.com/2019/08/28/world/middleeast/israel-iran-shadow-war.html>; Nadav Ben Hour and Michael Eisenstadt, "The Great Middle Eastern War of 2019," Washington Institute for Near East Policy, August 20, 2018, available at <www.washingtoninstitute.org/policy-analysis/view/the-great-middle-eastern-war-of-2019>.

⁴¹ "Iran Seizes British Tanker in Strait of Hormuz," BBC News, July 20, 2019, available at <www.bbc.com/news/uk-49053383>.

⁴² *National Security Strategy of the United States of America*, 46. The nature of the North Korean regime is exemplified by the sanctions imposed on Pyongyang by the international community. United Nations Security Council (UNSC) resolutions now operative include UNSC 825, May 11, 1993; UNSC 1695, June 15, 2006; UNSC 1718, October 14, 2006; UNSC 1874, June 12, 2009; UNSC 2087, January 22, 2013; UNSC 2094, March 7, 2013; UNSC 2270, March 2, 2016; UNSC 2321, November 30, 2016; UNSC 2356, June 2, 2017; UNSC 2371, August 5, 2017; UNSC 2375, September 3, 2017; and UNSC 2397, December 22, 2017. In addition, the United States, South Korea, Japan, Australia, Russia, and the European Union have adopted individual country sanctions on North Korea. See Eleanor Albert, "What to Know About Sanctions on North Korea," *Council on Foreign Relations*, July 16, 2019, available at <www.cfr.org/backgrounder/what-know-about-sanctions-north-korea>; James Schoff, "Making Sense of UN Sanctions on North Korea," Carnegie Endowment for International Peace, n.d., available at <https://carnegieendowment.org/publications/interactive/north-korea-sanctions>.

⁴³ *Summary of the National Defense Strategy of the United States of America*, 2.

⁴⁴ Gian Gentile, *Four Problems on the Korean Peninsula: North Korea's Expanding Nuclear Capabilities Drive a Complex Set of Problems* (Santa Monica, CA: RAND, 2019).

⁴⁵ Nuclear Threat Initiative, "North Korea," August 2019, available at <www.nti.org/learn/countries/north-korea/>.

⁴⁶ Andrew Scobell and John M. Sanford, *North Korea's Military Threat: Pyongyang's Conventional Forces, Weapons of Mass Destruction, and Ballistic Missiles* (Carlisle Barracks, PA: Strategic Studies Institute, 2007).

⁴⁷ Coats, *Worldwide Threat Assessment of the U.S. Intelligence Community*, 28.

⁴⁸ *Ibid.*, 8.

⁴⁹ UNSC, "Report of the Panel of Experts Established Pursuant to Resolution 1874 (2009)," S/2019/171, February 21, 2019, 34, available at <www.securitycouncilreport.org/atf/cf/%7B65BFCF9B-6D27-4E9C-8CD3-CF6E4FF96FF9%7D/s_2019_171.pdf>.

⁵⁰ Coats, *Worldwide Threat Assessment of the U.S. Intelligence Community*, 8.

⁵¹ UNSC, "Report of the Panel of Experts Established Pursuant to Resolution 1874 (2009)," February 21, 2019, 44–45.

⁵² UNSC, "Report of the Panel of Experts Established Pursuant to Resolution 1874 (2009)," S/2019/691, July 31, 2019, 135, available at <www.securitycouncilreport.org>.

org/atf/cf/%7B65BFCF9B-6D27-4E9C-8CD3-CF6E4FF96FF9%7D/S_2019_691.pdf>.

- ⁵³ "Treasury Sanctions North Korean State-Sponsored Malicious Cyber Groups," Department of the Treasury, September 13, 2019, available at <<https://home.treasury.gov/index.php/news/press-releases/sm774>>; "North Korean Hackers Blamed for Wave of Cyberattacks on Banks," *South China Morning Post* (Hong Kong), October 3, 2018, available at <www.scmp.com/news/asia/east-asia/article/2166869/north-korean-hackers-blamed-wave-cyberattacks-banks>; UNSC, "Report of the Panel of Experts Established Pursuant to Resolution 1874 (2009)," July 31, 2019, 109–112.
- ⁵⁴ Jim Finkle, "Cyber Security Firm: More Evidence North Korea Linked to Bangladesh Heist," Reuters, April 3, 2017, available at <www.reuters.com/article/us-cyber-heist-bangladesh-northkorea-idUSKBN1752I4>.
- ⁵⁵ UNSC, "Report of the Panel of Experts Established Pursuant to Resolution 1874 (2009)," July 31, 2019, 4.
- ⁵⁶ "Update on Sony Investigation," Federal Bureau of Investigation, December 19, 2014, available at <www.fbi.gov/news/pressrel/press-releases/update-on-sony-investigation?utm_campaign=emailImmediate&utm_medium=email&utm_source=national-press-releases&utm_content=386194>; David Sanger, David Kirkpatrick, and Nicole Perlroth, "The World Once Laughed at North Korean Cyberpower. No More," *New York Times*, October 15, 2017, available at <www.nytimes.com/2017/10/15/world/asia/north-korea-hacking-cyber-sony.html>.
- ⁵⁷ Greg Walters, "North Korea's Counterfeit Benjamins Have Vanished," *Vice News*, March 16, 2016, available at <www.vice.com/en_us/article/vb8pk9/north-korea-s-counterfet-benjamins-have-vanished>. Also see Sheena Chestnut Greitens, *Illicit: North Korea's Evolving Operations to Earn Hard Currency* (Washington, DC: Committee for Human Rights in North Korea, 2014).
- ⁵⁸ Raphael F. Perl, *Drug Trafficking and North Korea: Issues for U.S. Policy*, RL32167 (Washington, DC: Congressional Research Service, January 25, 2007).
- ⁵⁹ Mike Ives, "Crystal Meth Is North Korea's Trendiest Lunar New Year's Gift," *New York Times*, February 12, 2019; Kim Young Il, "North Korea and Narcotics Trafficking: A View from the Inside," *North Korea Review* 1, no. 1 (March 2004).
- ⁶⁰ Julian Ryall, "North Korea 'Ramps Up Manufacture of Illegal Drugs' amid Sanctions," *Deutsche Welle*, August 21, 2017, available at <www.dw.com/en/north-korea-ramps-up-manufacture-of-illegal-drugs-amid-sanctions/a-40169753>.
- ⁶¹ For an overview of Chinese-Iranian relations, see Ariane Tabatabai and Dina Esfandiary, *Triple-Axis: Iran's Relations with Russia and China* (New York: Bloomsbury Publishing, 2018); Scott W. Harold and Alireza Nader, *China and Iran: Economic, Political, and Military Relations* (Santa Monica, CA: RAND, 2012).
- ⁶² Derek Scissors, "China Global Investment Tracker," American Enterprise Institute, available at <www.aei.org/china-global-investment-tracker/>.
- ⁶³ Harold and Nader, *China and Iran*, 6–7.
- ⁶⁴ "Bilateral Trade Between China and Islamic Republic of Iran in 2018, Product: 2709 Petroleum Oils and Oils Obtained from Bituminous Minerals, Crude," International Trade Commission. Most of these imports took place prior to the reimposition of U.S. sanctions and the expiration of U.S. waivers. See Dan Katz, "Despite Sanctions, China Is Still Doing (Some) Business with Iran," Atlantic Council, October 1, 2019, available at <www.atlanticcouncil.org/blogs/iransource/despite-sanctions-china-is-still-doing-some-business-with-iran/>.
- ⁶⁵ Vassily Kashin, "Tacit Alliance: Russia and China Take Military Partnership to New Level," Carnegie Moscow Center, October 22, 2019.
- ⁶⁶ Ibid.; Dmitri Trenin, "U.S. Obsession with Containment Driving China and Russia Closer," Carnegie Moscow Center, July 31, 2019.
- ⁶⁷ Rajeswari Pillai Rajagopalan, "Russia-China Strategic Alliance Gets a New Boost with Missile Early Warning System," *The Diplomat*, October 25, 2019, available at <[https://thediplomat.com/2019/10/russia-china-strategic-alliance-gets-a-new-boost-with-missile-early-warning-system/](http://thediplomat.com/2019/10/russia-china-strategic-alliance-gets-a-new-boost-with-missile-early-warning-system/)>.
- ⁶⁸ Le Jeong-ho, "China-North Korea Trade Up 14.3 Percent in First Half to US\$1.25 Billion," *South China Morning Post* (Hong Kong), July 24, 2019, available at <www.scmp.com/news/china/diplomacy/article/3019940/chinas-first-half-trade-north-korea-recovers-old-allies>.
- ⁶⁹ Marshall Billingslea, *Statement Before the U.S. House Financial Services Subcommittee on Monetary Policy and Trade*, September 26, 2018, available at <<https://home.treasury.gov/news/press-releases/sm496>>.
- ⁷⁰ See, for example, James Carden, "Time to Pursue an International Cyber Treaty?" *The Nation*, April 30, 2019, available at <www.thenation.com/article/international-cyber-treaty-russia-china-dnc/>; United Nations General Assembly, "Developments in the Field of Information and Telecommunications in the Context of International Security," A/C.1/73/L.27, October 22, 2018, available at <<https://undocs.org/A/C.1/73/L.27>>; United Nations General Assembly, "Developments in the Field of Information and Telecommunications in the Context of International Security," A/C.1/73/L.37, October 18, 2018, available at <<https://undocs.org/A/C.1/73/L.37>>.
- ⁷¹ John W. Garver, "Is China Playing a Dual Game in Iran?" *The Washington Quarterly* 34, no. 1 (Winter 2011), 75–88.
- ⁷² "DOD Statement on Deployment of Additional U.S. Forces and Equipment to the Kingdom of Saudi Arabia," Department of Defense, October 11, 2019, available at <www.defense.gov/Newsroom/Releases/Release/Article/1987575/dod-statement-on-deployment-of-additional-us-forces-and-equipment-to-the-kingdom/>.
- ⁷³ Andrea Kendall-Taylor and David Shullman, "A Russian-Chinese Partnership Is a Threat to U.S. Interests," *Foreign Affairs*, May 14, 2019, available at <www.foreignaffairs.com/articles/china/2019-05-14/russian-chinese-partnership-threat-us-interests>.
- ⁷⁴ Shannon Tiezzi, "Why China Isn't Mourning the Collapse of the Trump-Kim Summit," *The Diplomat*, March 1, 2019, available at <[https://thediplomat.com/2019/03/why-china-isnt-mourning-the-collapse-of-the-trump-kim-summit/](http://thediplomat.com/2019/03/why-china-isnt-mourning-the-collapse-of-the-trump-kim-summit/)>.
- ⁷⁵ Alexander Simoes and César Hidalgo, *The Economic Complexity Observatory: An Analytical Tool for Understanding the Dynamics of Economic Development*, Workshops at the 25th AAAI Conference on Artificial Intelligence, August 2011, available at <<https://oec.world>>.
- ⁷⁶ Vita Spivak, "Russia and China in Africa: Allies or Rivals?" Carnegie Moscow Center, October 25, 2019, available at <<https://carnegie.ru/commentary/80181>>. For an overview of Russian activities in Africa, see Paul Stronski, *Late to the Party: Russia's Return to Africa* (Washington, DC: Carnegie Endowment for International Peace, 2019).
- ⁷⁷ Leon Aron, "Are Russia and China Really Forming an Alliance?" *Foreign Affairs*, April 4, 2019, available at <www.foreignaffairs.com/articles/china/2019-04-04/are-russia-and-china-really-forming-alliance>. For an alternative view, see Kendall-Taylor and Shullman, "A Russian-Chinese Partnership Is a Threat to U.S. Interests."
- ⁷⁸ Alex Vatanka, "China's Great Game in Iran," *Foreign Policy*, September 5, 2019, available at <<https://foreignpolicy.com/2019/09/05/chinas-great-game-in-iran/>>.
- ⁷⁹ Brett McGurk, "China's Risky Middle East Bet," *The Atlantic*, April 29, 2019, available at <www.theatlantic.com/ideas/archive/2019/04/chinas-risky-middle-east-bet/588166/>.

Chapter 11

Counterterrorism and the United States in a New Era of Great Power Competition

By R. Kim Cragin, Hassan Abbas, Zachary M. Abuza, and Mariya Omelicheva

This chapter addresses the likely impact of Great Power competition on future counterterrorism missions by the U.S. military; it argues that the military should prioritize preventing external operations, directed or virtually planned by foreign violent extremist organizations (VEOs), against the U.S. homeland and minimizing the ability of foreign VEOs to inspire attacks by sympathizers in the West, commonly referred to as homegrown violent extremists. Yet the chapter also observes that, over the next 3 to 5 years, Great Power competition will likely constrain the ability of U.S. military forces to achieve even these more limited counterterrorism objectives. The U.S. Government, therefore, will need to cooperate closely with allies and partners to manage global terrorist threats. The military also will need to preserve its ability to conduct unilateral operations to protect the U.S. homeland. Given these requirements, this chapter recommends that the U.S. military revisit its risk threshold for small-footprint deployments, especially force protection requirements. It also should reconsider counterterrorism authorities, technologies, and other tools in light of the new realities created by Great Power competition. And, in this context, the U.S. Government should explore more ways to deter actions by surrogates and proxies against U.S. forces engaged in counterterrorism and to hold sponsors accountable.

The September 11, 2001, attacks by al Qaeda focused the attention of the U.S. national security community on the threat posed by nonstate adversaries. Since then, the United States and its allies have conducted multiple military operations to mitigate this threat: Operation *Enduring Freedom*, Operation *Iraqi Freedom*, Operation *Resolute Support*, and Operation *Inherent Resolve*, to name a few. Yet the emphasis on countering violent extremist organizations (VEOs) could not last indefinitely. Nation-state adversaries, such as China, Russia, Iran, and North Korea, also present immediate and future, and arguably greater, challenges to U.S. national security interests. A reprioritization was inevitable and, as chronicled in chapter 3a, took place with the publication of two new U.S. security documents in

2017 and 2018: the 2017 National Security Strategy (NSS) and the 2018 National Defense Strategy (NDS).¹ The 2017 NSS retained counterterrorism as an important component of protecting the U.S. homeland, rebranding this effort as “pursuing threats at their source.”² The NDS acknowledged the need to counter VEOs, such as al Qaeda’s network and the so-called Islamic State (IS), but it prioritized threats from near-peer competitors and other nation-state adversaries above counter-VEO operations.³ Together, these new strategic documents indicated that the United States would assume some increased risk from terrorism in the emerging era of Great Power competition.

These documents adjusted the U.S. strategic framework for countering VEOs. The 2018 National Strategy for Counterterrorism captured these adjustments while emphasizing the importance of future counter-VEO activities in a context of prioritization and partnerships. Two passages in this document stand out:

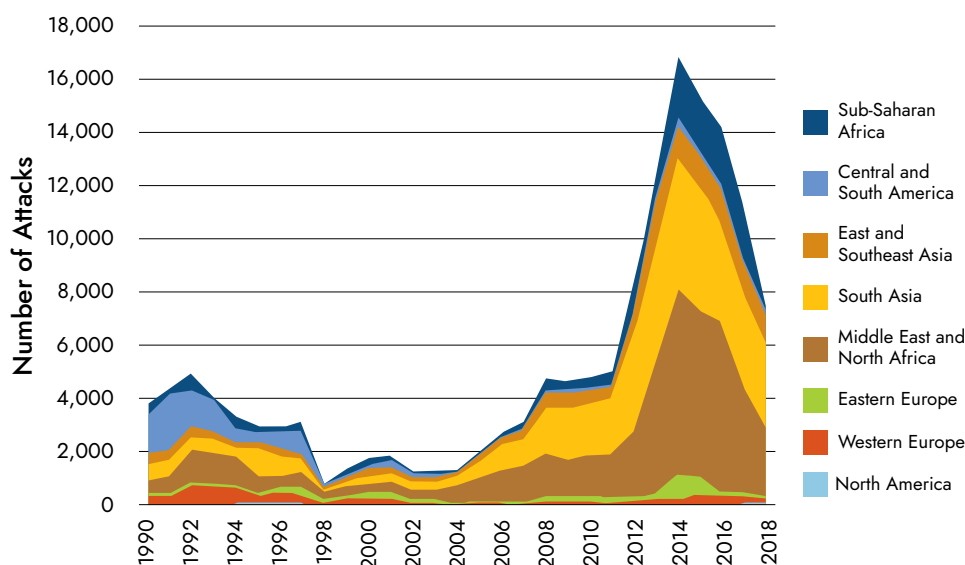
*Experience has . . . highlighted the importance of strong partnerships in sustaining our counterterrorism efforts. Whenever possible, the United States must develop more efficient approaches to achieve our security objectives, relying on our allies to degrade and maintain persistent pressure against terrorists. This means collaborating so that foreign governments take the lead wherever possible and working with others so that they can assume responsibility in the fight against terrorists.*⁴

*We will not dilute our counterterrorism efforts by attempting to be everywhere all the time, trying to eradicate all threats. We can and will, however, optimize and focus our resources to effectively prevent and counter those terrorists who pose a direct threat to the United States homeland and vital national interests.*⁵

Since 2018, the NDS has become the baseline document for the U.S. military as it plans, trains, and organizes the joint force. Thus, this chapter proceeds from the perspective of the NDS that the U.S. military will prioritize its near-peer competitors as well as rogue states in defense planning. This chapter also aligns with the definition of *Great Power competition* found in chapter 1, which validates three contemporary near-peer rivals: Russia, China, and the United States. The chapter affirms that VEOs will continue to be a major security issue for the U.S. military and its allies; it focuses on the fact that over the next 3 to 5 years, al Qaeda, the IS, and their associates will continue to pursue local insurgencies and external operations against the West. The U.S. military will be presented with the choice of engaging these VEOs unilaterally, as part of a coalition, indirectly through local partners, or not at all. Whatever the U.S. counterterrorist approach, Russia and China also will weigh the threats by VEOs to their own national security and make their own choices among a range of similar options. Other countries will do the same. The United States may benefit directly from the actions taken by Russia, China, or other nation-states against VEOs. Alternatively, these interventions may exacerbate the VEO threat to the U.S. homeland. In any case, these risks will need to be managed, and counterterrorism in a new era of Great Power competition just got a lot more complicated.

Subsequent paragraphs address this complication. The chapter provides an overview of the evolving terrorist threat, not only emphasizing the United States and its allies but

Figure 11.1. Number of Successful Attacks and Failed Plots Conducted by VEOs by Region, 1990–2018



Source: National Consortium for the Study of Terrorism and Responses to Terrorism, Global Terrorism Database, 2018, available at <<https://start.umd.edu/data-tools/global-terrorism-database-gtd>>.

also touching on threats to Russia and China. It then delves more deeply into the U.S. military's Operation *Inherent Resolve* in Syria. Syria represents the most concrete example of how U.S. counterterrorism objectives can be undermined by the presence of other Great Powers (in this instance, Russian military forces and their proxies). Next, the chapter explores the long-lasting regional rivalry between Iran and Saudi Arabia. Iran has been defined as a “disruptor” state in this book (see chapter 10); nevertheless, it remains a U.S. adversary, and its competition with Saudi Arabia for regional hegemony exacerbates the threat posed by VEOs. Finally, this chapter provides a discussion of Australia's role in Southeast Asia. Australia represents an important contrast to Russia and Iran: It illustrates how allies can assist the United States in its fight against VEOs, allowing the U.S. military to shift resources away from counterterrorism missions with reduced risk. The chapter concludes with thoughts on the implications of Great Power competition for U.S. counterterrorism in this new era.

Understanding the Threat Posed by Violent Extremists

The United States faces threats from a wide range of extremist groups, including domestic right-wing terrorists, left-wing terrorists, and foreign operatives linked to al Qaeda or IS. The U.S. military retains responsibility for countering VEOs that originate abroad but not for domestic terrorist groups. Among the foreign VEOs, the greatest priorities for the United States fall under the subcategory of Salafi-jihadists.⁶ This section, therefore, provides a broad overview of global trends in terrorism over the next 3 to 5 years, but it focuses more on the evolving threat from Salafi-jihadists to the United States and its allies.

Finally, to provide greater context to the threat from Salafi-jihadists, the section concludes with a short discussion of domestic left-wing and right-wing terrorism.

Global Trends in Terrorism

Any chart depicting overarching trends in terrorist attacks worldwide tends to be alarming; figure 11.1 is no different. It displays the total number of successful attacks and failed plots by VEOs globally since 1990. The trend line reveals several fluctuations in global terrorism: a steady decline in attacks between 1992 and 2000; a gradual increase in the number of attacks, starting in 2001, until the numbers flatline between 2007 and 2011; and a dramatic spike in the number of attacks between 2012 and 2014, followed by a decline again, beginning in mid-2015 until the present. While the overarching trend since 2015 is encouraging—for example, the numbers are decreasing—it still has not reached the relatively low levels of the late 1990s and early 2000s.

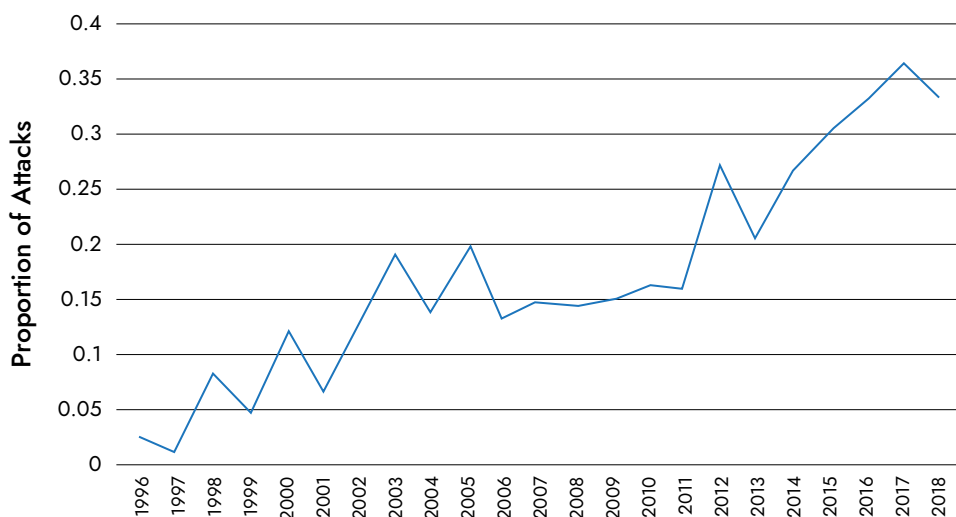
A closer examination of figure 11.1 offers a slightly more nuanced assessment. Although it is easy to assume, for example, that most attacks have taken place in the Middle East and North Africa over the past three decades, the data does not support this conclusion. Instead, the relative number of attacks by VEOs in the Middle East and North Africa, as compared with those in other regions, ranges from a low of 13 percent in 1991 to a high of 45 percent in 2016. Other regions, such as Central and South America in the early 1990s or sub-Saharan Africa at present, also experience relatively high numbers of attacks on their populations.

Interestingly, none of the regions with Great Powers—North America, Eastern Europe, East and Southeast Asia—have experienced high levels of terrorism since 1990. Beyond these overarching numbers, it is noteworthy that foreign VEOs have not systematically targeted Russia or China in recent years.⁷ A few exceptions exist. In Egypt, IS fighters placed an improvised explosive device on a Russian airline flying from Sinai to Saint Petersburg in October 2015.⁸ The Baluchistan Liberation Army attacked the Chinese consulate in Karachi, Pakistan, in November 2018.⁹ Nevertheless, the limited number of attacks by foreign VEOs on Russian and Chinese interests explains, in part, why Moscow and Beijing tend to emphasize domestic terrorist threats over those posed by foreign VEOs.¹⁰ These realities also limit the potential areas of cooperation between the Great Powers on countering foreign VEOs.¹¹ Now and into the future, the U.S. Government is more likely to find itself hampered in its counterterrorism operations due to the expanding military, political, and economic influence of these other two Great Powers than it is to find substantial areas for cooperation.¹²

The Salafi-jihadist Threat

Salafism offers literalist, rigid, and puritanical approaches to Islam. It emerged from political developments in the second half of the 19th century, when Muslim-majority regions were confronted with the spread of European ideas. Salafists often extol the first three generations of Muslims as well as the most geographically expansive caliphate of the 1200s.¹³ Most Salafists pursue their fundamentalist beliefs peacefully. Others advocate for a violent political revolution in the Muslim world; these individuals are referred to interchangeably as either Salafi-jihadists or *takfiris*, which is a derogatory term that means Muslims who declare others apostates and kill them. Between 1996 and 2014, al Qaeda was at the vanguard

Figure 11.2. Proportion of Attacks Conducted by VEOs in Salafi Jihadist Movement, 1996–2018



Source: National Consortium for the Study of Terrorism and Responses to Terrorism, Global Terrorism Database, 2018, available at <<https://start.umd.edu/data-tools/global-terrorism-database-gtd>>.

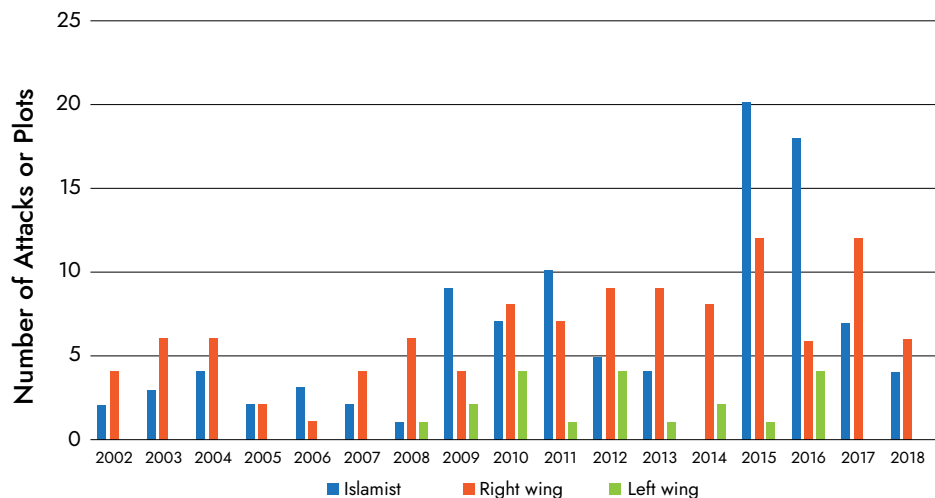
of the Salafi-jihadi movement. Al Qaeda leaders argued that their political revolution was failing because the United States propped up corrupt Arab regimes.¹⁴

The so-called Islamic State emerged in 2014—in many ways as an alternative to al Qaeda. Led by Abu Bakr al-Baghdadi until his death in November 2019, IS rejected al Qaeda’s emphasis on jihad against the West and instead aimed to establish an Islamic caliphate within the territories under its control.¹⁵ That said, IS also pursued external operations, or terrorist attacks, that took place outside its territorial control, in Syria, Iraq, and other countries. In fact, IS leaders met in Tabqah, Syria, in November 2015 to plan a way forward for the IS external operations campaign.¹⁶ Since then, al Qaeda and IS leaders have fought for preeminence among Salafi-jihadists, with most VEOs taking sides.¹⁷

Figure 11.2 depicts the overall trajectory of attacks by Salafi-jihadists, beginning after al Qaeda leaders announced their war against “Americans Occupying the Lands of the Two Holy Mosques” in 1996.¹⁸ It was derived from the Global Terrorism Database maintained by the START Consortium at the University of Maryland. The numbers report attacks by Salafi-jihadists as a proportion or ratio of all attacks by VEOs worldwide.¹⁹ Figure 11.2 also shows that the overarching trajectory of attacks by Salafi-jihadists remains upward, despite the recent territorial defeat of IS.²⁰

Significantly, most of the attacks shown in figure 11.2 were conducted against local targets or targets within easy reach—within the same countries—of IS territorial control. They were not external operations. Only 2 percent of all attacks by Salafi-jihadists since 1996 can be considered “external operations,” and the IS has been the most aggressive VEO in this regard. Indeed, while the IS recently lost both its caliphate and its caliph (al-Baghdadi), its decisive defeat remains a remote prospect. The new IS leader, Abdullah Qardash, has vowed

Figure 11.3. Number of Successful Attacks and Failed Plots by U.S. Homegrown Extremists, 2002–2018



Source: Anti-Defamation League, H.E.A.T. Map, 2018, available at <https://www.adl.org/education-and-resources/resource-knowledge-base/adl-heat-map>.

to avenge al-Baghdadi's death and has embarked on rebuilding the group's strengths and re-energizing IS supporters and sympathizers.²¹ Most experts agree that IS retains the capacity to direct an external attack against the U.S. and allies' homelands; it also continues to target U.S. forces and citizens abroad.²²

That said, among Salafi-jihadists, homegrown violent extremists—individuals inspired by foreign VEOs, such as the IS, to conduct attacks locally—represent the most persistent threat to the U.S. homeland over the next 3 to 5 years. IS has sustained a sophisticated media campaign propagating a “long war” against its enemies.²³ This campaign includes outreach to local sympathizers in the West, urging them to execute attacks. According to the Federal Bureau of Investigation, the number of active investigations related to international terrorism in the United States has stayed about the same since 2014, hovering around 4,000. The percentage of IS-related arrests for domestic attacks, however, has continued to increase over the past 2 years, even after the IS's territorial defeat.²⁴ As of August 2019, 194 individuals in the United States had been charged with offenses related to IS. Forty percent of these cases involved traveling or attempts to travel abroad; 32 percent of the cases were plots of terrorist attacks within the U.S. homeland.²⁵ These data suggest that the IS's message continues to reach and resonate with U.S. audiences.

Other Terrorist Threats to the United States

There are terrorist risks to the United States beyond Salafi-jihadists.²⁶ Over the past decade, domestic terrorism motivated by a range of far-right and white supremacist ideologies also has increased (see figure 11.3). According to a report by the New America Foundation, between 2002 and 2018, American Salafi-jihadists killed an estimated 104 people in the U.S. homeland, while the death toll from far-right, white supremacist, and other nonreli-

gious extremist ideologies stood at 125 people.²⁷ Left-wing terrorists also have historical roots inside the United States. According to the Global Terrorism Database, the number of attacks by left-wing extremists inside the United States has dropped from more than 20 per year in 2003 to less than 5 per year in 2018.²⁸ And, for the past several years, attacks by left-wing and anarchist groups have held steady at about 3 percent of all domestic terrorism inside the United States, according to both the Global Terrorism Database and the Anti-Defamation League (figure 11.3).²⁹ Much like their jihadist counterparts, domestic terrorists are empowered by the Internet and social media platforms, where they share their ideas and resources.³⁰ Although these individuals are largely decentralized and operate in small independent cells, many take ideological cues and inspiration from broader, more global movements.³¹ Importantly, at least for the purposes of this chapter, investigative journalism and academic research indicates that Russia's disinformation campaign, directed against the West, has amplified right-wing extremist movements.³²

In summary, this threat assessment suggests that the U.S. military should prioritize preventing external operations, directed or virtually planned by foreign VEOs, against the U.S. homeland, and minimize the ability of foreign VEOs, or even foreign nation-state adversaries, to inspire attacks by sympathizers in the West, commonly referred to as *homegrown violent extremists*. This way forward accepts risk, including the possibility of terrorist attacks on our allies' and partners' homelands.³³ It also includes risks from greater local or regional instability, which may in turn weaken some of our allies or partners.³⁴ Still further, some terrorism scholars have argued that too much U.S. counter-VEO retrenchment will allow for a third generation of global jihadists and homegrown violent extremists to emerge, setting us back to where we were before September 2001.³⁵ As the United States transitions into the 2020s, its defense strategy now accepts these risks when weighed against the threats from near-peer competitors and rogue states. The next sections review how Great Power rivals are likely to constrain U.S. counter-VEO endeavors in the future and how regional state actors and partners might exacerbate or mitigate VEO risks to the U.S. homeland, its vital interests, and its allies.

Counterterrorism and the Impacts from Great Power Competition: Syria

Syria represents the best, most concrete example of counterterrorism in the new era of Great Power competition. Recent experiences in Syria underscore the likelihood that the U.S. military will no longer be able to execute counterterrorism operations with little to no interference in the near future. It will face direct or indirect opposition from other nation-states. To be successful, the U.S. military will need to adjust its authorities, technologies, and other resources accordingly.

Operation *Inherent Resolve* is a counterterrorism mission begun in late 2014 against the Islamic State and led by the United States. Between 2015 and 2018, IS leaders used Syrian cities such as Tabqah, Raqqah, and Ayn al-Arab (Kobani) to orchestrate external terrorist operations against the West. IS media campaigns and other outreach to local sympathizers further exacerbated this threat. The U.S. national security community determined that the only way to halt the IS external operations campaign was for the U.S. military to reduce IS control over territory in Syria. It initially chose to accomplish this mission through a

combination of precision strikes and working by, with, and through local Kurdish militias, renamed the Syrian Democratic Forces (SDF). This approach allowed the United States to pursue its military objectives with a limited footprint, approximately 2,000 U.S. Soldiers, Sailors, and Airmen. Beginning in late 2019, the Trump administration reduced the U.S. force level in Syria even more, although counterterrorism operations continue at present.

But U.S. interests in Syria—narrowly focused on counterterrorism—clash with those of Russia, Iran, and Turkey, all of which view the conflict more broadly through the lens of their own geopolitical influence. Since 2011, Syrian president Bashar al-Asad and his security forces have struggled greatly against a wide range of opposition forces, even going so far as to deploy chemical weapons in this campaign.³⁶ During both the Obama and Trump administrations, the U.S. Government has articulated its opposition to the Syrian regime often and repeatedly, thereby assuring that Asad has no interest in U.S. military forces on the ground in Syria. Asad did, however, welcome—and count on—the presence of Iranian, and later Russian, military and paramilitary assistance in the fight. Turkey also was a critical party to the Syrian civil war, with its own unique mix of strategic interests and objectives. Initially an opponent of Asad and a supporter of anti-Asad insurgents, Turkey's involvement has proved fluid and its motives far from fully aligned with those of the United States. Ankara's longstanding battle against domestic Kurdish antigovernment militias made Turkey incredibly wary of Kurdish ethnic forces operating in Syria. Turkey's direct proximity to Syria meant that Turkish forces and their proxies had a serious impact on the U.S. counterterrorism mission against IS. The presence of Russian, Iranian, and Turkish forces and their proxies has complicated the U.S. counterterrorism mission. Indeed, the intermixing of cross-cutting Russian, Iranian, Turkish, and U.S. strategic aims during a time of reemerging Great Power competition has made *Inherent Resolve* a poster child for the complexities that U.S. counterterrorism missions will likely face in the future. The following paragraphs explore these experiences in greater detail.

President Vladimir Putin's Russia has three primary interests in the Syrian conflict: It prefers to see President Asad retain power in the country; it wants to reduce the threat posed by returning foreign fighters, primarily from Chechnya and the Caucasus; and it wishes to establish Putin as an alternative "broker" to the United States in the Middle East and North Africa.³⁷ Putin's Russia has sent as many as 4,000 soldiers and military advisors to Syria—along with rotary-wing aircraft, heavy weapons, and artillery—to bolster the Asad regime.³⁸ The Russian air force (RuAF) also has conducted airstrikes in support of regime forces and sent airborne intelligence, surveillance, and reconnaissance platforms to Syria: unmanned aerial vehicles, Il-20 Coots, and Tu-214R.³⁹ As noted in chapter 3b, Syria became a testing ground for some of the new Russian military capabilities heavily invested in by Putin since the mid-2000s. The RuAF has used the conflict in Syria to test its stealth bombers (T-50s) and fighters (Su-57s) against U.S. forces.⁴⁰ In a pattern begun with Russian operations in Chechnya in 2014, at least 2,500 Russian mercenaries have fought on behalf of the Syrian regime, demonstrating an often-overlooked capability critical to the strategic reach of Russia in its rivalry with the United States.⁴¹

Likewise, Iran—a regional hegemon with both animus toward the United States and hatred of IS—has played a key role in the Syrian conflict. Syria and Iran have a longstanding alliance. Iran has provided financial and military aid to Asad. It also has sent military advi-

sors from the Islamic Revolutionary Guards Corps–Qods Forces (IRGC-QF) to work with regime security forces. Specifically, IRGC-QF advisors oversee approximately 25,000 fighters from Lebanon (primarily Hizballah), Iraq, and Afghanistan, who have reinforced Asad's forces in his fight against opposition troops.⁴² To make the situation even more complicated, Russian military advisors reportedly have trained Hizballah and IRGC-QF personnel to call in RuAF airstrikes, in an effort to reduce fratricide for its close air support.⁴³

Unlike Russia and Iran, Turkey opposes Asad and his regime. President Recep Tayyip Erdogan has provided approximately 15,000 opposition forces within Syria with weapons and other forms of support.⁴⁴ Turkey also has deployed its own security forces into Syria as part of two military operations: Euphrates Shield (August 2016) and Olive Branch (March 2018). Turkish leaders did this to eliminate Syrian Kurds' control over territory close to the Turkish border. Turkey views Syrian Kurds as closely tied to a Turkish VEO, referred to as the Kurdistan Workers Party. In addition, Erdogan has accused Syrian Kurds of small-scale cross-border attacks with mortars.⁴⁵ Unfortunately, these Syrian Kurds are the same militias that form the backbone of the SDF and have partnered with the U.S. military to fight against IS. U.S. counterterrorist operations against IS had to factor in not only Russian and Iranian motivations and presence but also complex Turkish grievances against a major U.S.-allied, anti-IS militia.

In the midst of these crosswinds, U.S. forces have had to execute missions against IS in the thick of a highly complex operational environment, one not seen in the post–Cold War world prior to the return of Great Power rivalry. Despite efforts to deconflict ongoing operations, the presence of Russian, Iranian, and Turkish security forces, along with their proxies, has reduced U.S. military effectiveness. The following are some concrete examples of what happens to U.S. forces when they attempt to execute counterterrorism missions in the midst of Great Power competition:

- In October 2015, reports emerged of Russia jamming U.S. military communications equipment, navigation systems, and aircraft in Syria. The former commander of U.S. Special Operations Command subsequently described Syria as the “most aggressive [electronic warfare] environment on the planet.”⁴⁶
- In May 2017, Iranian-backed Shia militias stationed near the outskirts of At-Tanf garrison, Syria, entered the deconfliction zone, threatening U.S. forces and their partners (an Arab militia called Maghaweir al-Thowra). Russian jets provided the Shia militias with close air support.⁴⁷
- In November 2017, a Russian jet flew an unsafe flight profile—dangerously low—over U.S. forces in the Euphrates Valley. Two U.S. Air Force F-22s reportedly warned off the Russian jets by releasing infrared flares.⁴⁸
- In February 2018, approximately 300 Russian mercenaries attacked U.S. forces and their SDF partners near Dar el-Zour.⁴⁹ U.S. forces called for close air support, and F-15E Strike Eagles responded, killing approximately 200 of the mercenaries.⁵⁰
- In October 2019, Turkish-based opposition forces threatened U.S. Soldiers as they withdrew from Ain Issa, Syria. F-15E Strike Eagles responded with a show of force.⁵¹

In summary, American experiences in Syria illustrate how Great Power competition could complicate a narrowly defined U.S. counterterrorism mission. Over the next 3 to 5 years, the U.S. military should expect Russia and China to continue to position themselves as alternatives to the United States—not only in the Middle East and North Africa but also in other regions.⁵² Thus far, China appears less likely to use military forces outside East and Southeast Asia. But the presence of Russian ground forces and their proxies, combined with close air support provided by RuAF to these forces, will indelibly complicate the U.S. military's ability to conduct counterterrorism operations. Great Power rival use of electronic or cyber warfare also will likely constrain U.S. military options.

Regional States and VEO Risk Exacerbation or Mitigation

Salafi-jihadists have directed most of their violence against local and regional targets, not the United States or the West. Thus, it makes sense that other Great Powers and regional states will have their own calculations about the threat posed by local, regional, and global VEOs; they also will decide whether to intervene directly against VEOs or counter them with proxy forces. This dynamic exists in many areas, including the Middle East, North Africa, Central Asia, and South and Southeast Asia. In some places, the interventions by regional powers will likely reduce the threat of VEOs to the U.S. homeland, but in other areas, regional states will likely intervene in such a way that they exacerbate the threat of foreign VEOs to the United States and its interests abroad. These risks will need to be managed, even as the U.S. military prioritizes Great Power competition. The following are two distinct and contrasting examples of this dynamic, underscoring the complex requirements of countering VEOs in this new era of Great Power competition.

Saudi Arabia vs. Iran in the Middle East as VEO Risk Exacerbators

The escalating proxy war between Saudi Arabia and Iran represents the best example of how regional powers can strengthen VEOs—either directly or indirectly—in their pursuit of dominance. The competition between Saudi Arabia and Iran has played out in various theaters across the Middle East, North Africa, and into South Asia. For the United States, Saudi Arabia is a valued ally and Iran is an adversary. But their proxy war elevates the risks of unleashing empowered VEOs with international reach as well as adding to the volatility of many nation-states around them, many of which host either IS or al Qaeda fighters. This Saudi-Iranian confrontation has been stoking conflicts in Iraq and Yemen; they offer two instructive examples of how regional power competition can create environments hospitable to VEOs.

The rivalry between Saudi Arabia and Iran is not new. It started off as a cold war between regional competitors, both surviving on petrodollars while adhering to two different shades of Islam that have historically been at odds. Both of these world views—Shiism in Iran and Sunni-Wahhabism in Saudi Arabia—have been revolutionary in essence, highly political in orientation, and particularly zealous in proselytization. Over time, the competition between Saudi Arabia and Iran has turned into a balance-of-power game.⁵³ As of 2020, this competition has not yet led to a direct military confrontation.⁵⁴ However, Saudi Arabia and Iran have targeted each other's core national security interests, and these efforts have included the use of proxies and all actions short of war.⁵⁵

Within the past decade, Iraq has been at the center of this rivalry. In Iraq, the Saudi footprint has been less visible than Iran's. Since the United States toppled Saddam Hussein from power in Iraq in 2003, Iran has used its relationship with Iraq's Shia population to garner influence within its neighbor. To do this, Iran has focused attention on solidifying its influence in the Iraqi south, where pilgrimage routes in and around Shia-dominated Najaf and Karbala have religious significance. Iran also was able to take advantage of IS expansion and control over territory within Iraq between 2014 and 2019. Iran overtly sponsored several of the Hashd Al-Shaabi, or Popular Mobilization Forces (PMF) in Iraq, which emerged after Iraqi Shia Ayatollah Ali al-Seestani's 2014 fatwa encouraging Iraqis to assist government forces as they attempted to push back the Salafi-jihadist IS. The PMF played an important role in confronting IS fighters in Iraq between 2014 and 2016, in a parallel effort to *Inherent Resolve* in Iraq, led by U.S. forces. PMF success, in turn, reinforced Iranian influence in Iraq.

For Saudi Arabia, Iran's expanding influence in Iraq has been a grave concern. More sympathetic to Iraq's minority Sunni population, Riyadh has been trapped between IS expansion into Sunni-dominated areas of Iraq on one side and Iranian influence on the other. While Riyadh has been somewhat concerned about the regional threat posed by IS fighters and operatives, it has been equally worried about its waning influence in Iraq. Iraq's political leadership routinely has sympathized with Iranian interests against Saudi Arabia over the past decade. For Saudi Arabia and Iran, everything in the region is a zero-sum game, and so their competition for influence in Iraq only feeds into brewing instability in the wider Middle East.⁵⁶

Yemen offers an even more severe example of the consequences of this poisonous rivalry. Yemen has been in the midst of a civil war since early 2015. Its main factions are those loyal to the more formalized government of exiled President Abd-Rabbu Mansour Hadi, supported by Saudi Arabia, and the minority Shia-affiliated Huthi rebel movement, supported by Iran. Beyond these factions, al Qaeda in the Arabian Peninsula (AQAP) has retained control over some territory in Yemen, as has its competitor Salafi-jihadist outfit, IS in Yemen. AQAP remains the most threatening terrorist outfit to the United States and its allies, plotting multiple attacks over the years, including a foiled attack against Northwest Airlines 253 in Detroit (December 2009), a series of plots that prompted the U.S. Government to close 22 Embassies worldwide (August 2013), and a successful attack against the *Charlie Hebdo* newspaper in Paris (January 2015). AQAP also claimed responsibility for a December 2019 attack by a Saudi military officer at a U.S. base in Pensacola, Florida.⁵⁷

For Iran, Yemen represents a launching pad to target the Saudi military through proxies. The IRGC-QF has provided weapons, ammunition, communications equipment, and other support to Huthi rebels in Yemen's ongoing civil war. In December 2019, for example, the U.S. Government sanctioned the ESAIL Shipping Company and Mahar Air for smuggling weapons to Yemen on behalf of the IRGC-QF.⁵⁸ Many other materials have made their way from Iran to Yemen via well-established criminal smuggling networks.⁵⁹ Saudi Arabia and its close ally, the United Arab Emirates, have responded with airstrikes against Huthi targets. The resulting war has displaced more than 3.3 million people and has exacted an estimated death toll ranging from approximately 7,000 to more than 65,000 people.⁶⁰ Yemen stands destroyed, and it will take decades to rebuild its infrastructure.

The Yemen conflict also has posed direct challenges for the U.S. military. Huthi rebels have used weapons provided by IRGC-QF to attack U.S. military targets and personnel in the region, some of which are there for the explicit purpose of countering the threat posed by AQAP to the United States and its allies.⁶¹ For example, in June 2019, Huthi fighters shot down a U.S. MQ-9 Reaper drone as it flew over western Yemen, an area with local popular support and freedom of movement for AQAP fighters. U.S. Central Command subsequently stated the MQ-9 was shot down by a SA-6 and accused Iran of enabling the attack.⁶² The Huthis responded by threatening to attack any future drones that fly outside of AQAP territory, viewing them as a direct threat.⁶³

The Saudi-Iranian struggle will likely continue to motivate both sides to invest in destabilizing each other—and search for new proxies to serve their interests.⁶⁴ These proxies, as we have seen in Yemen, are inextricably linked to weapons-smuggling networks in the region. Furthermore, Salafi-jihadists have taken advantage of this instability to threaten the United States and its allies.⁶⁵ Granted, from a contemporary U.S. policy perspective, Iranian-backed proxies may be worse than those spawned by Sunni-led governments, but the net result of their mutual hostility continues to make space for VEOs that threaten the U.S. homeland, U.S. and allied military units, and vital U.S. interests. Washington must continue to manage the Saudi-Iranian rivalry in a manner that offsets that contest's ongoing potential to exacerbate the risks from truly dangerous VEOs in the era of Great Power competition.

Australia in Southeast Asia as a VEO Risk Mitigator

Australia represents an important contrast to the proxy warfare risks generated by the struggle between Saudi Arabia and Iran. Counterterrorism in Southeast Asia is a distant priority for the United States in an era of Great Power rivalry, well behind coping with a rising and more assertive China, freedom of navigation operations, alliance management, and the denuclearization of North Korea. Given these U.S. priorities, it is reasonable to expect that within the Indo-Pacific region, U.S. counterterrorism efforts will remain focused on the Philippines, where a number of pro-IS, al Qaeda, and other Salafi-jihadist groups continue to spread the poorly resourced Philippine military thin. For the rest of the region, including Indonesia, Malaysia, and Singapore, the United States will likely maintain a smaller intelligence and law enforcement engagement.

Australia, therefore, represents a critical partner in combating terrorism in the Indo-Pacific region during the dawning era of Great Power competition. Australia has paid far closer attention to Indonesia than the United States and has legitimate concerns about what it considers the “arc of instability” to its immediate north. Australian Federal Police have worked actively with their Indonesian counterparts, and they have been at the forefront of the Jakarta Centre for Law Enforcement Cooperation. The United States should be confident in both Australia's leadership and approach. For no other reason than proximity, Australia has taken the threat of terrorism in Southeast Asia more seriously than the United States.

Australia also has increased its bi- and multilateral engagement with the Armed Forces of the Philippines (AFP). During the 5-month-long siege of the Philippine city of Marawi by pro-IS militants in mid-2017, Australia deployed two AP-3C Orion aircraft to provide intelligence, surveillance, and reconnaissance support to the Philippine

military.⁶⁶ The Australian Defence Forces have conducted urban warfare training with the AFP. Australia also has assisted the Philippine police with forensic instruction, an improvised explosive device database, and other training. In 2019, Australian aid to the conflict-plagued Philippines was \$85 million, with a large portion of that money going to support the peace process between the government and the largest Islamist secessionist movement, the Moro Islamic Liberation Front.⁶⁷

Moreover, Australia has been an important backer of the trilateral maritime patrols established in 2017 among the Philippines, Indonesia, and Malaysia. In some ways, Australia is better poised to assist in the maritime component of counterterrorism, as Malaysia and Indonesia have largely resisted the involvement of the U.S. Navy. Australia is seen as a more acceptable partner, as U.S. Naval presence—and in particular its freedom of navigation operations—is often viewed as provocative. While the trilateral patrols have been going on for more than 2 years already, there is still no fusion center, and intelligence is shared on an ad hoc basis. Australia has been reluctant to join any freedom of navigation operations in the South China Sea, yet the United States wants Canberra to play a greater maritime role in the region. Australia also has been able to leverage its multilateral task force experience from the Persian Gulf and the Gulf of Aden.

Finally, while Australia has every reason to take a leadership role in countering VEOs in Southeast Asia, it has its own domestic IS challenge. On a per capita basis, Australia had one of the highest rates of foreign fighters in Iraq and Syria.⁶⁸ Additionally, there have been a number of IS-inspired attacks in Australia, including the 2014 cafe siege and a 2017 shootout in Melbourne. A plot to import weapons and explosives for a campaign against churches and embassies was thwarted in July 2019; it was the 16th alleged mass casualty attack to have been foiled in Australia since 2014.⁶⁹ So Canberra feels the need to play defense overseas as well. These actions reinforce Australia as a willing and able partner for countering VEOs, with longstanding commitment and credibility in Southeast Asia. In this sense, Australia exemplifies the opportunities presented by partner regional states becoming more engaged in countering VEOs, even as the United States devotes more of its resources to Great Power competition.

Implications

The cases reviewed in this chapter imply several realities for U.S. counterterrorism in the new era of Great Power competition. First, U.S. national security officials should assume that Putin's Russia will continue to undermine U.S. counterterrorism objectives, either directly or indirectly. Moscow will likely do this by fomenting right-wing and other home-grown violent extremists indirectly through a media campaign. Russia also will confront U.S. forces, especially in the Middle East and North Africa, as they attempt to mitigate VEO threats to the U.S. homeland. For the former, Syria will likely be the model, with Russia combining diplomatic initiatives, proxy warfare, and electronic warfare to foil U.S. military dominance.⁷⁰

The U.S. Government, therefore, should reconsider its counterterrorism authorities, technologies, and other tools in light of the evolving realities created by Great Power competition. The U.S. military also should revisit its risk threshold for small-footprint deployments, especially force protection requirements in areas with active proxies. In this

context, the U.S. Government should explore more ways to deter actions by proxies against U.S. forces and hold all sponsors accountable.

Finally, U.S. national security officials should assume that regional states will continue to pursue their own counterterrorism objectives. Sometimes they will deploy their forces in a manner that the U.S. Government will find unacceptable. Sometimes they will utilize proxy forces in a destabilizing manner. In other instances, important regional states will have objectives, ways, and means that align with U.S. aims for countering VEOs. The best way to mitigate the risk of regional states acting in an unruly manner is to be involved—even to a minimal degree if necessary—and then truly leverage U.S. influence with them. That said, U.S. military defense priorities outlined in the 2018 NDS make regional power struggles, including their potential to exacerbate some VEO threats, an area where the U.S. Government will accept some additional risk.

Notes

¹ *Summary of the 2018 National Defense Strategy: Sharpening the American Military's Competitive Edge* (Washington, DC: Department of Defense, 2018).

² "A New National Security Strategy for a New Era," The White House, December 18, 2017, available at <www.whitehouse.gov/articles/new-national-security-strategy-new-era/>.

³ *Summary of the 2018 National Defense Strategy*.

⁴ See *National Strategy for Counterterrorism of the United States (NSCT)* (Washington, DC: The White House, October 2018), available at <www.whitehouse.gov/wp-content/uploads/2018/10/NSCT.pdf>; "White House Releases National Strategy for Counterterrorism," October 4, 2018, available at <www.dni.gov/index.php/nctc-newsroom/item/1911-white-house-releases-national-strategy-for-counterterrorism>.

⁵ NSCT, 11.

⁶ Ibid.

⁷ Mariya Omelicheva and Lawrence Markowitz, *The Nature and Sources of Terrorist Threat in Russia: An "Armed Underground" or ISIL?* Eurasia Policy Memo 550 (Washington, DC: Institute for European, Russian and Eurasian Studies, November 2018), available at <www.ponarseurasia.org/memo/armed-underground-or-isis-nature-sources-terrorist-threat-russia>; Mariya Omelicheva and Lawrence Markowitz, "When Does the Crime-Terror Nexus Fail to Materialize? Drug Trafficking, Militants, and the State in Russia," *Journal of Global Security Studies*, March 17, 2020, 1–19, available at <<https://doi.org/10.1093/jogss/ogz078>>.

⁸ "Russia Plane Crash: 'Terror Act' Downed A321 over Egypt's Sinai," BBC News, November 17, 2015, available at <www.bbc.com/news/world-europe-34840943>.

⁹ Maher Ahmad and Salman Masood, "Chinese Presence in Pakistan Is Targeted in Attack on Consulate in Karachi," *New York Times*, November 23, 2018, available at <www.nytimes.com/2018/11/23/world/asia/pakistan-karachi-attack-chinese-consulate.html>.

¹⁰ Omelicheva and Markowitz, *The Nature and Sources of Terrorist Threat in Russia*; Ekaterina Stepanova and R. Kim Cragin, "Countering Terrorism and Violent Extremism," in *A Roadmap for U.S.-Russia Relations*, ed. Andrey Kortunov and Olya Oliker (Washington, DC: Center for Strategic and International Studies, 2017), available at <www.csis.org/analysis/roadmap-us-russia-relations>.

¹¹ Stepanova and Cragin, "Countering Terrorism and Violent Extremism." This report was the result of 2 years of Track II diplomacy between American and Russian academics on areas for cooperation in national security, including topics related to counterterrorism.

¹² Ibid.

¹³ For more information, see Shiraz Maher, *Salafi-Jihadism: The History of an Idea* (New York: Oxford University Press, 2016); Gilles Kepel, *Jihad: The Trail of Political Islam* (London: I.B. Tauris, 2002); Majid Fakhry, *A History of Islamic Philosophy* (New York: Columbia University Press, 2004); Sayyid Qutb, *Milestones* (Location unknown: International Islamic Federation of Student Organizations, n.d.).

¹⁴ Fawaz A. Gerges, *The Far Enemy: Why Jihad Went Global* (New York: Cambridge University Press, 2005); R. Kim Cragin, "An Early History of al-Qaeda," *The Historical Journal* 51, no. 4 (December 2008), available at <www.cambridge.org/core/journals/historical-journal/article/div-classtitleearly-history-of-al-qaadaposidativ/712B4E593C762040899BB4A78E62AEA1>.

¹⁵ Michael Weiss and Hassan Hassan, *ISIS: Inside the Army of Terror* (New York: Regan Arts, 2015); Abu Bakr Naji, *Management of Savagery: The Most Critical Stage Through Which the Ummah Will Pass*, trans. William McCants (Cambridge, MA: John M. Olin Institute for Strategic Studies, 2006).

¹⁶ Rukmini Callimachi, "How a Secretive Branch of ISIS Built a Global Network of Killers," *New York Times*, August 3, 2016. See also Martin Chulov, "How ISIS Laid Out Its Plans to Export Chaos to Europe," *The Guardian*, March 25, 2016.

¹⁷ R. Kim Cragin, "The November 2015 Paris Attacks: The Impact of Foreign Terrorist Fighters," *Orbis*, February 2017; Louise Callaghan, "Clandestine ISIS Unit Training Britons to Launch Attacks on UK," *The Times* (London), August 6, 2017, available at <www.thetimes.co.uk/edition/news/ clandestine-isis-unit-training-britons-to-launch-attacks-on-uk-9hjm3b6gr>.

¹⁸ Cragin, "An Early History of al-Qaeda."

¹⁹ Beginning in 2012, the START Consortium at the University of Maryland changed its data-collection methodology, which caused a spike in the overall number of terrorist attacks in the Global Terrorism Database. The proportional numbers allow accounting for this data-collection bias, revealing a more accurate trend in attacks.

²⁰ Readers should note that there is a slight downward turn after 2018. It is too soon to determine if this downturn will continue and represent a true change in trajectory, or if it simply represents a small bounce.

²¹ *Operation Inherent Resolve and Other Overseas Contingency Operations: Lead Inspector General Report to the United States Congress, October 1, 2018–December 31, 2018* (Washington, DC: Government Publishing Office, 2019), available at <https://media.defense.gov/2019/Feb/05/2002086500/-1/-1/1/FY2019_LIG_OIRREPORT.PDF>.

²² Christopher Wray, *Global Terrorism: Threats to the Homeland: Statement of the Federal Bureau of Investigation Director Before the House Homeland Security Committee*, October 30, 2019, available at <<https://www.fbi.gov/news/testimony/global-terrorism-threats-to-the-homeland-103019>>.

²³ Michael Munoz, “Selling the Long War: Islamic State Propaganda After the Caliphate,” *CTC Sentinel* 11, no. 10 (November 2018), available at <<https://ctc.usma.edu/selling-long-war-islamic-state-propaganda-caliphate/>>.

²⁴ Seamus Hughes and Deborah Margolin, “The Fractured Terrorism Threat to America,” *Lawfare*, November 19, 2019, available at <www.lawfareblog.com/fractured-terrorism-threat-america>.

²⁵ Lorenzo Vidino and Seamus Hughes, *ISIS in America: From Retweets to Raqqa* (Washington, DC: George Washington Program on Extremism, 2015), available at <<https://extremism.gwu.edu/sites/g/files/zaxdzs2191/f/downloads/ISIS%20in%20America%20-%20Full%20Report.pdf>>.

²⁶ For additional detail, see NSCT, 1, 10.

²⁷ Peter Bergen et al., “Part IV: What Is the Threat to the United States Today?” in *Terrorism in America After 9/11* (Washington, DC: New America, n.d.), available at <www.newamerica.org/in-depth/terrorism-in-america/what-threat-united-states-today/>.

²⁸ *Trends in Global Terrorism: Islamic State’s Decline in Iraq and Expanding Global Impact; Fewer Mass Casualty Attacks in Western Europe; Number of Attacks in the United States Highest Since 1980s* (College Park, MD: START Consortium, 2019), available at <https://www.start.umd.edu/pubs/START_GTD_TerrorismIn2018_Oct2018.pdf>.

²⁹ Ibid. See also Anti-Defamation League, “Murder and Extremism in the United States in 2019,” February 2020, available at <<https://www.adl.org/murder-and-extremism-2019/key-findings>>.

³⁰ Maura Conway, Ryan Scrivens, and Logan Macnair, *Right-Wing Extremists’ Persistent Online Presence: History and Contemporary Trends*, Policy Brief (The Hague: International Centre for Counterterrorism, October 2019), available at <<https://icct.nl/wp-content/uploads/2019/11/Right-Wing-Extremists-Persistent-Online-Presence.pdf>>.

³¹ Ibid.

³² See, for example, Clint Watts, “America Has a White Nationalist Terrorism Problem. What Should We Do?” Foreign Policy Research Institute, May 1, 2019, available at <www.fpri.org/article/2019/05/america-has-a-white-nationalist-terrorism-problem-what-should-we-do/>.

³³ Bruce Hoffman, “Why the Fall of Raqqa Doesn’t Signify the End of the Islamic State,” NPR Radio, interview with Scott Simon, October 21, 2017, available at <www.npr.org/2017/10/21/559215236/why-the-fall-of-raqqa-doesnt-signify-the-end-of-the-islamic-state>.

³⁴ Seth G. Jones et al., *The Evolution of the Salafi-Jihadist Threat* (Washington, DC: Center for Strategic and International Studies, 2018), available at <https://csis-prod.s3.amazonaws.com/s3fs-public/publication/181221_EvolvingTerroristThreat.pdf?_y6E2kgbXecCEcHFqTayRVST5k6KbTB>; see also Charles Lister, “Al-Qaeda and the Decentralization of Terrorism,”

Asharaq al-Wasat, August 8, 2019, available at <<https://aawsat.com/english/home/article/1849491/charles-lister/al-qaeda-and-decentralization-terrorism>>.

³⁵ Charles Lister, “Leaving Syria Won’t Make Things Better—for the U.S. or for Syria,” Middle East Institute, August 14, 2019, available at <www.mei.edu/publications/leaving-syria-wont-make-things-any-better-us-or-syria>; Lara Seligman, “In Overflowing Refugee Camps, Extremism Takes Root,” *Foreign Policy*, July 29, 2019, available at <foreignpolicy.com/2019/07/29/in-overflowing-syrian-refugee-camps-extremism-takes-root-syria-bashar-assad-islamic-state-isis/>.

³⁶ Jared Malsin, “Assad’s Regime Is Still Using Chemical Weapons in Syria,” *Time*, September 14, 2016, available at <time.com/4492670/syria-chemical-weapon-aleppo-assad-regime/>; Charles Lister, *The Syrian Jihad: al-Qaeda, the Islamic State, and the Evolution of an Insurgency* (New York: Oxford University Press, 2015).

³⁷ Since Soviet times, Russia has used facilities in the small Syrian port of Tartus, the only Russian naval outpost beyond the Bosphorus. While the utility of this naval base has been a matter of debate, it allows Russia to firm up its naval presence in the Mediterranean region. This port represents one of many reasons why Russia prefers that President Bashar al-Assad remain in power.

³⁸ Mark Mazzetti, Anne Barnard, and Eric Schmitt, “Putin Gains Leverage as Army Wins Syria Battles: By Bludgeoning Rebels, Russia Gets Upper Hand in Proxy War Against U.S.,” *New York Times*, August 8, 2016.

³⁹ Ralph Shield, “Russian Airpower’s Success in Syria: Assessing Evolution in Kinetic Counterinsurgency,” *Journal of Slavic Military Studies* 31, no. 2 (April 2018), 214–239.

⁴⁰ Nina Khrushcheva, “Mercenaries Cloud Russia’s Real Casualty Figures in Syrian War,” *The Australian* (New South Wales), March 2, 2018; David Axe, “Putin Sends His Stealth Jets into Air War over Syria,” *The Daily Beast*, February 23, 2018.

⁴¹ Ben Brimelow, “Russia Is Using Mercenaries to Make It Look Like It’s Losing Fewer Troops in Syria,” *Business Insider*, February 14, 2018; Maria Tsvetkova and Anton Zverev, “Russians ‘Backed by the Kremlin’ Are Secretly Dying in Syria Fighting for Bashar al-Assad,” *The Independent* (London), November 4, 2016.

⁴² Liz Sly, “Syria’s War Could Be Entering Its Last and Most Dangerous Phase,” *Washington Post*, August 10, 2018.

⁴³ Shield, “Russian Airpower’s Success in Syria.”

⁴⁴ Jonathan Spyer, “Fresh Sparring by Old Rivals,” *The Australian* (New South Wales), November 28, 2015, 17; Sarah el Deeb, “Factors Realign as U.S. Pulls Out of Syria,” *The Hamilton Spectator* (Ontario), January 7, 2019.

⁴⁵ “Erdogan Warns Kurds and Syrian Ceasefire Gets Off to a Rocky Start,” Al Jazeera, October 18, 2019, available at <www.aljazeera.com/news/2019/10/erdogan-warns-kurds-syria-ceasefire-rocky-start-191018183950579.html>.

⁴⁶ Lara Seligman, “Russian Jamming Poses a Growing Threat to U.S. Troops in Syria,” *Foreign Policy*, July 30, 2018; Alan Dupont, “Putin’s Gambit Changes the Game Strategy,” *The Australian* (New South Wales), October 10, 2015.

⁴⁷ Nicholas Blanford, “At Remote Desert Garrison in Syria, a U.S.-Iran Confrontation Is Brewing,” *Christian Science Monitor*, June 6, 2017, available at <www.csmonitor.com/World/Middle-East/2017/0606/At-remote-desert-garrison-in-syria-a-US-Iran-confrontation-is-brewing>.

⁴⁸ Axe, “Putin Sends His Stealth Jets into Air War over Syria.”

⁴⁹ Eric Schmitt, Ivan Nichepurechenko, and C.J. Chivers, “The Truth About the Brutal Four-Hour Battle Between Russian Mercenaries and U.S. Commandos,” *New York Times*, May 26, 2018; Neil Hauer, “Russia’s Mercenary Debacle in Syria,” *Foreign*

Affairs, February 26, 2018, available at <www.foreignaffairs.com/articles/syria/2018-02-26/russias-mercenary-debacle-syria>.

⁵⁰ This kind of fatality-producing incident between the U.S. and Russian military or paramilitary forces had not occurred since accidental encounters during the Balkan wars of the late 1990s. For further information, see Oriana Pawlyk, "Fewer Russian Mercenaries Entering Syria Since Airstrike: Mattis," *Military.com*, March 27, 2018, available at <www.military.com/daily-news/2018/03/27/fewer-russian-mercenaries-entering-syria-airstrike-mattis.html>; Brimelow, "Russia Is Using Mercenaries"; Tsvetkova and Zverev, "Russians 'Backed by the Kremlin' Are Secretly Dying in Syria"; and Seligman, "Russian Jamming Poses a Growing Threat to U.S. Troops in Syria."

⁵¹ Taylor Heyman and Jack Dutton, "Putin and Erdogan Speak as Syrian Regime Takes Town of Manbij," *The National* (Qatar), October 15, 2019.

⁵² For more information on Russia's strategy of "active defense," see a March 2019 speech by Russian Chief of General Staff Valery Gerasimov, "The Development of Military Strategy Under Contemporary Conditions," trans. Harold Orenstein and Timothy Thomas, *Military Review*, November 2019, available at <www.armyupress.army.mil/Journals/Military-Review/Online-Exclusive/2019-OLE/November/Orenstein-Gerasimov/>.

⁵³ F. Gregory Gause III, *Beyond Sectarianism: The New Middle East Cold War*, Doha Centre Analysis Paper 11 (Washington, DC: Brookings Institution, July 2014), available at <www.brookings.edu/wp-content/uploads/2016/06/English-PDF-1.pdf>.

⁵⁴ Banafsheh Keynoush, "Saudi Arabia and Iran Are Adapting to Perpetual Conflict," *Atlantic Council*, May 23, 2019, available at <www.atlanticcouncil.org/blogs/iransource/saudi-arabia-and-iran-are-adapting-to-perpetual-conflict/>.

⁵⁵ For how they view each other, see Mohammad Javad Zarif, "Saudi Arabia's Reckless Extremism," *New York Times*, January 10, 2016, available at <www.nytimes.com/2016/01/11/opinion/mohammad-javad-zarif-saudi-arabias-reckless-extremism.html>; Adel Bin Ahmed Al-Jubeir, "Can Iran Change?" *New York Times*, January 19, 2016, available at <www.nytimes.com/2016/01/19/opinion/saudi-arabia-can-iran-change.html>.

⁵⁶ Jeffrey Goldberg, "The Obama Doctrine, R.I.P.," *The Atlantic*, April 7, 2017, available at <www.theatlantic.com/international/archive/2017/04/the-obama-doctrine-rip/522276/>.

⁵⁷ "Al-Qaeda in Yemen Claims Charlie Hebdo Attack," *Al Jazeera*, January 14, 2015, available at <www.aljazeera.com/news/middleeast/2015/01/al-qaeda-yemen-charlie-hebdo-paris-attacks-201511410323361511.html>; Mark Sappenfield, "Al Qaeda Embassy Threats: Sign of Shrinking Ambition?" *Christian Science Monitor*, August 4, 2013, available at <www.csmonitor.com/USA/Foreign-Policy/2013/0804/Al-Qaeda-embassy-threats-Sign-of-shrinking-ambition>; Thomas Joscelyn, "AQAP Claims 'Full Responsibility' for Shooting at Naval Air Station Pensacola," *The Long War Journal*, February 2, 2020, available at <<https://www.longwarjournal.org/archives/2020/02/aqap-claims-full-responsibility-for-shooting-at-naval-air-station-pensacola.php>>.

⁵⁸ Darryl Coote, "Iran Shuns New U.S. Sanctions for Smuggling Weapons into Yemen," *UPI*, December 12, 2019, available at <www.upi.com/Top_News/US/2019/12/12/Iran-shuns-new-US-sanctions-for-smuggling-weapons-into-Yemen/7161576141059/>.

⁵⁹ Ahmed Nagi, "Oman's Boiling Yemeni Border," *Carnegie Middle East Center*, March 22, 2019, available at <carnegie-mec.org/2019/03/22/oman-s-boiling-yemeni-border-pub-78668>.

⁶⁰ "UN and Partners to Hold Conference Seeking Urgently Needed Funds to Save Millions in Yemen from 'Horrific' Plight," *UN News*, February 24, 2019, available at <news.un.org/en/story/2019/02/1033401>; "Yemen War Deaths on Overall Decline as UN Works to Salvage Truce," *Armed Conflict Location and*

Event Data Project, March 20, 2019, available at <www.acleddata.com/2019/03/20/press-release-yemen-war-deaths-on-overall-decline-as-un-works-to-salvage-truce/>.

⁶¹ In December 2009, al Qaeda in the Arabian Peninsula (AQAP) attempted to bomb Northwest Airlines 253, flying from Amsterdam to Detroit, utilizing a new recruit from Nigeria (Umar Farouk Abdulmutallab) in an effort to foil airport security. He successfully boarded the flight with an explosive device, but it ultimately failed to detonate. This event sparked a concerted effort by the U.S. military against AQAP fighters in Yemen. See Mark Sappenfield, "Detroit Attack: Terrorist Once Again Confounds Airport Security," *Christian Science Monitor*, December 27, 2009, available at <www.csmonitor.com/USA/2009/1227/Detroit-attack-terrorist-once-again-confounds-airport-security>.

⁶² "CENTCOM: MQ-9 Reaper Shot Down Last Week," *Military Times*, June 16, 2019, available at <www.militarytimes.com/news/your-military/2019/06/16/centcom-mq-9-reaper-shot-down-over-yemen-last-week/>.

⁶³ Ammar al-Ashwal, "Houthi Tout Missiles that Down Advanced U.S. Drone," *al-Monitor*, August 30, 2019, available at <www.al-monitor.com/pulse/originals/2019/08/yemen-houthi-down-us-drone-al-qaeda.html>.

⁶⁴ See Patrice Taddonio, "As Oil Attacks Raise Questions, Revisit Saudi Arabia and Iran's Bitter Rivalry," *PBS Frontline*, September 16, 2019, available at <www.pbs.org/wgbh/frontline/article/as-oil-attacks-raise-questions-revisit-saudi-arabia-and-irans-bitter-rivalry/>.

⁶⁵ In August 2013, the U.S. Government closed more than 20 Embassies worldwide due to an intercepted conversation between AQAP operatives indicating an imminent attack. Additionally, in January 2015, AQAP orchestrated an attack on the offices of the *Charlie Hebdo* newspaper in Paris, killing 12 people, and, in December 2019, AQAP claimed responsibility for the attack by a Saudi military officer in Pensacola, Florida. See "Al-Qaeda in Yemen Claims Charlie Hebdo Attack"; Sappenfield, "Al Qaeda Embassy Threats"; Joscelyn, "AQAP Claims 'Full Responsibility' for Shooting at Naval Air Station Pensacola."

⁶⁶ Manuel Mogato and Tom Westbrook, "Australia to Send Spy Planes to Help Philippines Fight Militants," *Reuters*, June 22, 2017, available at <www.reuters.com/article/us-philippines-militants-australia-idUSKBN19D2TU>.

⁶⁷ "Overview of Australia's Aid Program to the Philippines," Australian Government Department of Foreign Affairs and Trade, available at <<https://www.dfat.gov.au/geo/philippines/development-assistance/Pages/development-assistance-in-philippines>>.

⁶⁸ Nash Jenkins, "You'll Never Guess Which Country Is the Biggest Per Capita Contributor of Foreign Jihadists to ISIS," *Time*, June 23, 2014, available at <time.com/2911040/australia-isis-syria-iraq-terrorism/>; Andrew Zammit, *Australian Foreign Fighters: Risks and Responses* (Sydney: Lowy Institute for International Policy, April 2015), 30.

⁶⁹ "Australia Police Arrest Three Men over Terrorist Attack Plan," *Al Jazeera*, November 19, 2018, available at <www.aljazeera.com/news/2018/11/australia-police-arrest-men-terrorist-attack-plan-181120003913960.html>; Sally Rawsthorne and Matt Bungard, "'Sworn Allegiance to ISIS': Sydney Men Arrested over Foiled Terror Plot," *The Sydney Morning Herald*, July 2, 2019, available at <www.smh.com.au/national/nsw/three-alleged-isis-supporters-arrested-in-sydney-accused-of-terror-plot-20190702-p523az.html>; Rob Taylor, "Australia Foils Alleged Beheading Plot Linked to Islamic State," *Wall Street Journal*, September 18, 2014, available at <www.wsj.com/articles/australia-detains-suspects-in-terrorism-probe-1411008044>; Ashley Fantz and Chelsea J. Carter, "Australian PM: Public Execution Terror Plot Foiled," *CNN*, September 18, 2014,

available at <www.cnn.com/2014/09/17/world/asia/australia-terror-arrests/index.html>.

⁷⁰ Over the past several years, within the United Nations (UN), Russia has begun to seek greater diplomatic influence as it relates to the global counterterrorism agenda. Most notably, Vladimir Voronkov of the Russian Federation was appointed as the first Under Secretary General of the United Nations' Counterterrorism Office in June 2017. Voronkov received a broad mandate to provide strategic leadership for the complex counterterrorism architecture within the UN with the aim of strengthening coordination and improving efficiency of the UN counterterrorism system. While the UN Counterterrorism Office and the Under Secretary General are supposed to be politically neutral, the critics of the new agency and its head have warned about Russia's possible influence on the institution.

Chapter 12

Whither Europe in a New Era of Great Power Competition? Resilient but Troubled

By Steven Philip Kramer and Irene Kyriakopoulos

This chapter examines Europe's role and prospects as an important player in the emerging era of Great Power competition. Although the European Union's (EU's) share of global economic output is comparable to that of the United States, it is not a Great Power, never realizing some forecasts that it would become a superpower in the post-Cold War world. Instead, the European integration experience has been fraught with challenges. Europe has shown considerable resilience while overcoming the 2008 Great Recession and saving the euro in the face of a serious sovereign debt crisis. However, the EU's cohesion and solidarity have been severely tested by terrorism, uncontrolled migration, Brexit, and, most recently, the still-evolving complications from the 2019–2020 coronavirus pandemic. Across the continent, these problems have generated extremist populist movements that challenge liberal democracy and inhibit cohesive EU policy positions or security activities. Europe faces a more aggressive Russia, growing Chinese economic and political power, and a lack of trust in the U.S. commitment to the transatlantic union and European common defense. Despite its enormous latent power potential, Europe has become an object of Great Power rivalry on the continent rather than a subject competitor itself. Europe's responses to its ongoing and potential future challenges will shape its role in the new era of Great Power competition.

Early 20th-century French poet and political philosopher Charles Péguy's comment that "everything begins with mystique and ends in politics" could be aptly applied to European integration.¹ Embarked on just a few years after World War II, with origins in the 1949 Schuman Plan—then the 1951 European Coal and Steel Community, and later the European Economic Community of 1957—European integration was based on the romantic vision of a continent transcending conflict and forming an ever-closer union. Over the years, European integration evolved into a complex and opaque mix of federalism and intergovernmentalism. The crowning organizational jewel of the European

integration experiment—the European Union (EU), established by the 1992 Treaty of Maastricht and entered into force in 1993—has struggled yet endured.

The EU never has been greatly beloved by its citizens, who felt they had too little direct say in its governance. Even though it gradually expanded to include parts of the former Soviet bloc in a deepening economic and political union with security and defense cooperation, the EU has rarely preempted crises, has often been slow in reacting to crises, and has struggled to sustain the consent of those it governs. In addition, the EU has never realized the aspirations held out for it by integration optimists: It has not risen—as a whole or in any subgroup—to the status of a Great Power. The EU has never cleaved to the degree necessary to meet the three-feature definition of a Great Power in the post-Cold War world, and thus finds itself an important but lower tier player in the emerging new era of Great Power competition being shaped by the United States, China, and Russia.² There is also the matter of population size. If demography is destiny, then the prognosis for Europe's future as a potential Great Power is not good; beneath the lack of dynamism lurk the problems of low birth rates, an aging society, and internal migration. Europe's population today is about 10 percent of world population; it was about 20 percent in the 18th century. Europe's place in the world is shrinking.³

Yet the EU—as the experiment of European integration as a whole—has managed to survive with surprising resilience. Despite its shortcomings and problems, the EU has transformed Europe from a region prone to and deeply damaged by war to an area of peaceful cooperation.

The EU enters the 2020s grappling with its challenges and seeking a way forward. In the past 10 years, it has been assailed by a series of internal and external problems: a debt crisis, economic stagnation, Islamist terrorism, uncontrolled migration, and the 2019–2020 coronavirus (COVID-19) pandemic. The first four challenges, together with Brexit, the United Kingdom's rocky departure from the union, have destabilized liberal democracy across Europe. Geopolitically, the EU is facing a darkened international situation with an assertive Russia, the cohesion of its longstanding transatlantic alliance in doubt, and the problems posed by China's ascent as a significant economic competitor with growing political and military power.

This chapter examines the challenges facing the EU and traces their implications for the future of Europe. The debt crisis, economic stagnation, terrorism, migration, and the changing security environment are the hallmarks of Europe's status in 2020 and its path forward to 2025. So, too, is the impact of COVID-19, but that is a legacy yet to unfold.

Legacy of the Financial Crisis

The greatest test to EU resilience since its founding was posed by the effects of the financial turmoil that erupted in the United States in 2008 and precipitated the Great Recession. In the United States, the crisis originated mainly in the private financial sector of the economy. Its transmission to Europe led to a sovereign debt crisis. The EU managed to survive the Great Recession, but the crisis threatened the very existence of the common currency, the euro.

Europe's leaders were determined to preserve monetary union, even though they lacked traditional macroeconomic policy instruments.⁴ The Eurozone has a monetary

arm—the European Central Bank (ECB)—but lacks a fiscal arm or European treasury; EU treaties do not allow for that. Aware of these limitations, European leaders devised ad hoc solutions to deal with the immediate challenge of the possible default of member states on their sovereign debt. The term *Grexist*, coined by Citigroup economists, signified the serious risks to the Eurozone if Greece, the weakest member of the Eurozone, defaulted and left the common currency. There was grave concern that such an event would trigger fiscal contagion and loss of confidence in the ECB. To save the euro, the EU took drastic measures that tested all of Europe's institutions to the limit. Faced with an existential crisis, the EU devised novel policy measures to avert default by Ireland, Portugal, Greece, and later, Cyprus, and to rescue Spain's banking sector.⁵

The course the EU pursued from 2010 to 2018, in partnership with the International Monetary Fund, generated controversy that continues to this day. The approach consisted of two types of measures: rescue loans and austerity policies requiring cuts in public spending, tax increases, drastic structural reforms to liberalize labor and product markets; and the creation of a rescue fund of €750 billion. Key ingredients of the program were packaged so as to avoid violations of the EU's treaty obligations. Specifically, rescue loans for the states were structured to sidestep the no-bailout clause of the Maastricht Treaty.⁶ A special debt-relief measure was applied to the case of Greece, calling for a 50 percent writeoff of Greek sovereign debt held by banks, hedge funds, and private investors. Another debt-relief measure was a "bail-in" used in the case of Cyprus; it involved special levies on bank deposits of over €100,000 as a condition for Cyprus's rescue loan.⁷

The rescue loans imposed high costs on recipient EU states. As a condition for loan disbursement, the affected countries had to legislate and pass a broad set of fiscal and structural austerity measures. These macroeconomic stipulations brought major cuts in public spending and steep tax increases to the people of the recipient states. Recessionary forces exacerbated preexisting economic imbalances, leading to sharp increases in national unemployment and budget deficits. Unemployment rates in the worst affected countries of the EU periphery skyrocketed, reaching over 15 percent in Spain and nearly 25 percent in Greece. The level of indebtedness of Eurozone countries rose from an average of 85 percent of gross domestic product (GDP) in 2010 to nearly 93 percent of GDP by mid-2014. The ECB committed to do "whatever it takes" to save the euro.⁸ The ECB has persisted in providing monetary stimulus to the Eurozone economy through 2020. Ten years into the crisis, the euro remains intact.

But Europe's economy never fully recovered. In 2020, the EU is still dealing with anemic economic growth, and popular dissatisfaction with economic conditions is high. Structural economic reforms in various bailout recipient states, mandated to promote efficiency and competitiveness, brought dramatic transformations to Europe's economic and social landscape. Rescued states had to impose reductions in welfare benefits, expand part-time work, cut the minimum wage, freeze pay, eliminate bonuses, cut pensions, increase the retirement age, reduce severance and holiday pay, and ease restrictions of firings and layoffs, among other measures. Each of these actions contributed to a steep erosion in how Europe's social market economy functioned. In combination with fiscal retrenchment, structural reforms reduced living standards across the board and hit the least well-off across Europe especially hard. Increases in poverty, homelessness, and social

exclusion, affecting as much as a quarter of the population, were recorded throughout Europe in the wake of the wider financial crisis.

Popular resistance to the effects of strict macroeconomic discipline grew across Europe during the 2010s, with widespread political instability testing the EU's resilience. Governments that resorted to fiscal contraction and structural economic reforms were voted out of office amid rising social discontent. But political events that were deemed consequential a decade ago seem less relevant in 2020. Victories such as the 2012 election of François Hollande, France's first socialist president in 17 years, or Greece's Syriza, a radical left-wing party that became the governing party in 2015, were ephemeral. Ten years after the onset of the debt crisis, the anti-austerity rhetoric has subsided. The return to power of mainstream parties seems to affirm the electorate's assessment that saving the euro was of paramount importance. Europe's common currency has maintained its support by a large majority of Eurozone citizens throughout its first two decades, reaching the highest level in 2019, the euro's 20th birthday.⁹

Europe's ability to weather the financial crisis and the Great Recession demonstrated the resilience of its institutions and their adaptability. By 2019, the EU's economy had recovered to its 2010 level. EU member states' budget deficits have been nearly eliminated. Unemployment rates have come down; the EU average, nearly 10 percent in 2011, was about 6 percent in 2019.

Even as wages and real per capita income recovered from the Great Recession, the EU debt crisis exacerbated wealth and income inequalities.¹⁰ The distribution of the burden of austerity was borne disproportionately by those least able to afford it: the unemployed and the poor across the board and the EU's member states with the weakest economies. Greece was the worst hit of all; it suffered a recession greater in length and severity than the United States did after the 1929 stock market crash. Yet sovereign default was averted. Rescue programs for Ireland, Portugal, Spain, and Cyprus ended in 2017. Even Greece's bailout program ended in August 2018, while the economy was transitioning to the recovery phase.¹¹ In early 2020, the temporary respite from the worst of the debt crisis seemed likely to pass. There will be serious economic consequences from the COVID-19 pandemic playing across Europe, and it remains to be seen if these consequences will pose another episode of serious trauma for Europe's economic union.

At the same time, Europe's recovery from the debt crisis involved a new role for China on the continent—setting the conditions for new triangular relations among Europe, China, and the United States. While most international investors—including those from the United States—fled Europe in the worst days of its debt crisis, China-based companies surged in. Chinese cash flowed directly to some of the hardest hit countries in the EU, including Greece, Spain, Portugal, and Italy. But Chinese investors also moved into reduced-value assets across Europe during the debt challenge, investing heavily in developed European countries such as Germany, France, and especially Great Britain to acquire commercial brands and important technologies. From 2010 to 2012, the stock of Chinese direct investment in the EU quadrupled from a level that was smaller than holdings by Nigeria before 2010 to a main stakeholder by 2013.¹² Research into the implication of this prolific investment demonstrates that China has tried to use its economic strength to gain political concessions in Europe but has had limited success.¹³ Many eyes are now on the

issue of 5G Internet in Europe to see if China's ability to parlay economic presence into political influence holds, or if China's post-debt crisis economic prowess will lead Europe to a decision that goes against specific U.S. strategic interests to limit Chinese information technology penetration there.¹⁴

The Chinese investment surge notwithstanding, the EU's ability to manage the debt crisis was due to its application of a multiplicity of measures requiring more extensive and deeper fiscal and financial integration, or in the words of German Chancellor Angela Merkel, "More Europe."¹⁵ The European monetary union remains incomplete—lacking a common banking system, a common treasury, and a central fiscal policy, among other features. Still, the actions taken by European decisionmakers to sustain the EU's most compelling symbol, the euro as a common currency, held its financial status in the international community. The euro emerged from the debt crisis still the second most important currency in the international financial system. In 2020, the euro remains second only to the dollar in the size of its share in global foreign reserves, at 20 percent. The euro's share of global payments in 2019 was relatively even more impressive, at 33 percent versus the dollar's share at 45 percent.¹⁶ By contrast, the role of China's and Russia's currencies in the international financial system remains minimal. The Chinese renminbi has limited convertibility; its share of global foreign exchange reserves is less than 2 percent.¹⁷ The Russian ruble is not a reserve currency.

Thus, the most optimistic dream of Europe becoming "the first postmodern superpower" built on the prowess of the largest common market and common currency in the world—a dream with traction during the heady days of the early 21st century—was dealt a serious blow by the debt crisis. The EU did weather the storm with an intact common currency and an enviable global economic standing.¹⁸ Europe's profound investments in salvaging its monetary union from collapse have secured the euro's key role and Europe's important position at the international financial "high table."

The Challenge of Migration

Migration was instrumental in rebuilding Europe in the post-World War II era. To meet the demand for workers, governments of richer and growing northwestern European countries (Belgium, France, Germany, Luxembourg, the Netherlands, Sweden, and Switzerland) systematically recruited labor from Europe's periphery, mostly in the south. Inter-European migration was skills based, legal, and highly controlled, and migrant workers were treated as temporary guests.¹⁹

The ongoing migration challenge to the EU's common market for population flows paralleled the debt crisis but differs from it. In recent years, Europe has become a destination of choice for millions of people trying to escape war, political turmoil, and poverty in the Middle East, Africa, and Asia. From 2015 to 2016, nearly 1 million migrants and refugees passed through Greece's Aegean Islands and mainland toward central and northern Europe. Since 2015, nearly 4 million people have applied for asylum in the EU's 28 member states. After a temporary pause in 2018, the number of asylum seekers and unlawful entrants has continued to increase. Almost a quarter-million illegal crossings, mainly through sea routes, and illegal stays were detected in 2019.²⁰ The EU has been slow and ineffective

in dealing with a new challenge to its border security: uncontrolled population movements through European territory.

The migration crisis has not yet presented an overt threat to the EU, home to more than 500 million inhabitants, but it has the potential to pose a fundamental challenge by sowing discord and weakening solidarity and cohesion, key pillars of European integration. The influx of uninvited people from distant lands and foreign cultures—which peaked in 2015—has generated social, political, and economic pressures in a continent still recuperating from the financial shocks of the Great Recession.

The composition of population inflows changed significantly between 2015 and 2019. Early on, most arrivals were war refugees; by 2019, the majority of arrivals were economic migrants. Thus, the political economy of EU countries near or at its external borders has been destabilized. Regions and communities in frontline states, such as Greece and Italy, were overwhelmed by the asymmetric burden of the migration crisis. In these areas, it felt then and still feels in 2020 as if Europe is indeed under siege and without effective defenses. The rise of nationalist parties across Europe is a symptom of mounting dissatisfaction with the status quo in the prevention, deterrence, and management of illegal migration.²¹

Politics and geography have contributed to these developments. EU decisionmakers were caught unprepared to deal with chaotic migrant flows in a concerted fashion. Policy paralysis was overtaken by Chancellor Merkel's open-door policy announcement in 2015, which proved to be a major pull factor to migrants and refugees.²² During the peak of the migration crisis in 2015, hundreds of thousands of migrants and refugees arriving at the Greek islands were transported by boat to the mainland, northern Greece, and beyond. Neighboring states sought protection from sudden and massive migrant flows by temporarily blocking entry or closing their borders altogether. More than 80,000 migrants and refugees thus became stuck in Greece, unable to move to central and northern Europe, their original destinations. The Dublin Regulation, which came into force in 1997 and requires that all asylum seekers register at their first point of entry into the EU, virtually assured that many of those who escaped Greece would be returned there and to other frontline states.²³

Newly restrictive national migration policies adopted in haste often clashed with aspects of EU law, such as the common visa requirement and free movement of the Schengen area, which consists of 26 European states that officially abolished all passport and other types of control at their mutual borders in 1995. Repeated attempts by the European Commission to establish quotas for asylum seekers and a more equitable distribution of the migration burden among member states ultimately failed. Finally, an EU-Turkey statement issued in April 2016 promised billions of euros as compensation to non-EU member Turkey for stemming the exodus of migrants and refugees crossing over to EU member Greece.²⁴ This arrangement was deemed vital, aiming to make Turkey the stopping point of migrant flows with a money-for-refugee-camp swap.²⁵

The EU-Turkey migration deal has been fraught with significant capacity constraints and delays. Fragile from its inception, this arrangement began unraveling in early 2020. In February, Turkey announced that it would no longer keep refugees on its territory, threatened to let thousands cross into Europe, and accused the EU of falling short on commitments of financial support. Some 35,000 migrants massed on Turkey's border with Greece, and many were thrust back by Greek border forces. In March 2020, EU leaders

made their highest level visit ever to Greece's border with Turkey in an effort to defuse the crisis. There, they pledged solidarity with Greece and promised human and financial resources to enhance border security of the country that is "Europe's shield in the migrant crisis."²⁶ As of spring 2020, the future of the EU-Turkey deal remains in doubt, further complicated by geopolitical considerations.

The relevance of the North Atlantic Treaty Organization (NATO) in Europe's migration crisis has been mostly symbolic. In November 2016, NATO launched a non-Article 5 maritime security operation called Sea Guardian.²⁷ Its goals were similar to those of the EU-Turkey agreement and included reduction in the flows of migrants and refugees from the Asia Minor coast to the Greek islands and interdiction of traffickers and smugglers. Sea Guardian involved limited operations (during certain hours and days of the week). Its effectiveness has not been assessed, but it has not been perceived to be much of a success.

A leader in welcoming migrants to Europe during 2015, Angela Merkel paid a political price at home for this stance and announced in 2018 that she is not seeking reelection.²⁸ Backlash to migrant transit across Europe via the so-called Balkan Corridor, stretching from the northern Greek border to points north, became severe. That migrant pathway was sealed in 2016, as Austria, Hungary, and other European states closed their borders. Buffeted by the anger about migrant encroachment in multiple European states and a growing backlash at home, Germany's policy stance changed. In 2018, Germany announced it would set up border camps for migrants.²⁹ By 2019, German Interior Minister Horst Seehofer announced the *Geordnete-Rückkehr-Gesetz*—or "Orderly Return Law"—which facilitates the deportation of failed asylum seekers.³⁰

In response to these developments, EU decisionmakers also have adapted their strategic goals, assigning higher priority to stopping irregular migration flows at their source, before they reach Europe's borders. In 2019, new policy ideas, including French President Emmanuel Macron's idea for a "European Renaissance," emphasized the need for a unified asylum policy and stricter external border controls.³¹ The European Commission's "European Agenda on Migration Four Years On," introduced on March 6, 2019, considers it essential to pursue a comprehensive approach to restrict migration flows and strengthen external border protection.³² But these goals may undermine the essence of the border-free, visa-free Schengen area, the backbone of Europe's single labor market. Nor is it clear that the EU can allocate resources for a reinforced security infrastructure at its external borders. Border protection will remain a divisive issue, due to the social, political, and resource burdens that it would impose on the most vulnerable member states. A related factor is the size and effectiveness of the European Border and Coast Guard Agency (FRONTEX), which coordinates contributions of border guards from EU member states.³³ A 2019 announcement by the EU Commission to increase the size of FRONTEX to 10,000 by 2027 is viewed as highly unrealistic. There is no political consensus on the matter, and no resources have been allocated to such an enterprise.³⁴

Then there is the issue of a common asylum policy; the EU has been grappling with it since the beginning of the migration crisis. Absent a common policy, each state has been granting asylum based on its own laws and interpretations of the United Nations charter. According to the EU commissioner for migration, "The current system is broken."³⁵ To fix it, the commissioner submitted in 2019 seven proposals requiring agreement on

all components of a Common European Asylum System, such as stepping up returns of individuals not qualified to receive asylum, ensuring that return decisions are enforced, making plans to address the root causes of migration, and more tightly enforcing the EU-Turkey deal. Aiming for consensus to stem irregular migration at the source in northern Africa, the commission has set up a European Emergency Trust Fund for Africa, with planned contributions of nearly €3 billion from the EU budget.³⁶ But that fund is unlikely to be fully funded anytime soon.

The EU's newly developed strategy of containment of the migration crisis through a unified common asylum policy, tighter border controls, and stricter management of migrant flows at their source is an acknowledgment that the status quo is unsustainable and must be reformed. The implementation of policy reforms proposed by the EU Commission requires political consensus from all heads of state. Such a consensus cannot be formed as long as the migration burden is placed unevenly on European border states.

The migration crisis reinforced preexisting centrifugal forces unleashed by serious macroeconomic and resource imbalances across the EU. As noted, several EU member states have renationalization policies dealing with border controls and other measures to protect or project national sovereignty. These unilateral moves strongly indicate that the migration challenge poses a greater danger to European integration than the debt crisis. The EU was determined to devise and impose radical new policies to save the euro. By contrast, neither the EU's institutions nor its member states have invoked a comparable common European symbol to justify imposition of EU-wide policies to resolve the migration crisis. Repeated references to solidarity as the basis for the formulation of new EU-wide migration and people movement measures have not advanced common policies.

In fact, popular sentiment points in the opposite direction. Most Europeans believe that the migration crisis has been mismanaged, and the continued influx of illegal migrants remains their top concern. This is tethered to popular fears that illegal migration increases the incidence of terrorism in their countries.³⁷ The results of elections for the European parliament in May 2019 have indeed led to changes in the European Commission's policy portfolios. New policy appointments may foretell further changes to the EU's stance on future refugee inflows and unlawful migration. It remains to be seen whether the EU might begin to overcome the divide between border and nonborder states on the migration issue, achieve a more unified common asylum policy, enhance border security, and move toward a more equitable distribution of migration burden. The EU has everything to gain in meeting the challenge of migration—and much to lose if it does not. As it struggles to find a common way forward on migration, an internally divided Europe risks losing agency on the world stage in a new era of Great Power competition.³⁸

European Politics in Transition

In 2020, the major challenge of the EU is its own survival. In the past 10 years, the EU has faced daunting problems and disruptive crises. It has surmounted immediate perils but has not eliminated their root causes. Nor has it fundamentally reformed its system of governance. To do so requires unanimity among member states and, in some states, referenda as well—extremely difficult to achieve. The financial crisis was temporarily contained, but the economic fallout from the COVID-19 pandemic could portend another severe round

of recession and severe Euro-trauma. The challenge of migration and migrant crises threatened and can yet endanger EU survival. Brexit has destabilized the United Kingdom and introduced new tensions in Britain's relationship with Europe. In the short term, whoever "wins" the Brexit conflict, Britain is likely to be the loser.³⁹

The financial crisis and the demography challenge of migration have had major political consequences. They exacerbated longstanding political tensions within member European states, weakened centrist parties, strengthened right-wing extremism, and contributed to the development of illiberal democracy in Eastern Europe. New and unprecedented forms of social protest such as the *gilets jaunes* in France arose. France and Germany are less able to exercise leadership in Europe, and the Franco-German partnership, the engine of progress in Europe, has been immobilized. Among other Franco-German coordination challenges, Germany is leery about responding to President Macron's ideas for reforming the EU.⁴⁰ A vicious circle of weak leadership has ensued, and Russia has tried to take advantage of this EU weakness.

The postwar European model of a mixed economy and extensive social safety net has been eroded. This model was at the heart of European integration; its purpose was to prevent the kind of economic and social insecurity of the 1930s that spawned fascism and Nazism and resulted in World War II. But economic security is hard to guarantee in a postindustrial, fourth industrial revolution society, and the cost of the old European welfare state is becoming prohibitive.⁴¹ The impact of rapid economic and technological change today is only equaled by the disruptions caused by the first industrial revolution. The proposed remedy to Europe's serious economic and political challenges of the 2000s and 2010s was "reform," but reform was tantamount to an exercise in austerity. The remedy was imposed by Germany and execrated by large elements of the European population, who lost benefits and whose standard of living declined. Those affected were unlikely to welcome "More Europe." Blame for economic crises was attributed to the governing parties.

To the fear that tomorrow might be harsher than today was added the fear of uncontrolled migration in Europe. Many Europeans were not especially comfortable with Muslim immigrants; thus, political parties like the French National Front thrived on xenophobia. Muslim immigrants often came to the continent just before the process of deindustrialization set in across Europe during the 1970s and 1980s. As a result, they suffered economic, social, and residential marginalization and racism, which impeded true integration. While many Muslims ascended to the middle class, disaffected Muslims, especially in the underclass, were susceptible to Islamism.⁴² The migration crisis catalyzed the immigrant problem, just as the financial crisis catalyzed the erosion of the European welfare state. The problem of Muslim minorities was exacerbated by jihadist terrorism, which reinforced the impression that the state (and the EU) are unable to guarantee basic citizen safety, which in turn threatens the fragile social situation of Muslims. Deadly Islamist terrorist attacks took place in France (*Charlie Hebdo*, *Hyper Cacher*, and Bataclan in 2015; Nice in 2016).⁴³ In 2019, a police officer in the Paris prefecture killed four of his colleagues, leading to fears that terrorists could act as a fifth column within state institutions. In 2020, there is also concern about the return to Europe of jihadists who fought with the so-called Islamic State.

The apparent haplessness of the EU and its member states regarding uncontrolled migration shook confidence in the EU and in establishment political parties, including

Christian Democrats and Social Democrats, which had been the centrist mainstays of the EU. Socialist parties that had pioneered the concept of the “people’s home” now found that the far right was trying to appropriate the concept. The far right’s claim was that only by leaving the EU, erecting barriers to globalization, and keeping out immigrants could the people’s home be saved for the “real people”—that is, the native European population. The political system in Western Europe has generally managed to absorb, contain, or neuter the nativist extremists, but with difficulty. At the same time, some centrist political parties have taken on the rhetoric of the far right to protect their political bases.

In Europe’s East, however, the forces of illiberal democracy have been far more successful, especially in Hungary, under Prime Minister Viktor Orban, and in Poland, where the conservative regime attempted to undo separation of powers and insisted on acceptance of a nationalist version of Polish history.⁴⁴ The same crisis of establishment parties occurred in Germany and France. In the early 2000s, before the crisis, the German Social Democratic Party (SPD), under then–Prime Minister Gerhard Schröder, had shown its “sense of national economic responsibility” by enacting the Hartz labor and market reforms. This made German labor costs more competitive globally but made German workers less financially secure. The SPD lost much of its working-class base and declined from around 40 percent of the electorate to probably less than 20 percent.⁴⁵ Conversely, the left-wing Greens in Europe have experienced significant growth in both Germany and Austria (where they are now part of a coalition government). As noted, the migrant crisis undermined Merkel’s standing and that of her long-governing, right-leaning Christian Democratic Union. The very right-wing and hyper-nationalist Alternative für Deutschland (AfD) became the official opposition in the German parliament (Reichstag) in 2017, providing a party known as a political pariah with a kind of legitimacy.

The old East Germany may be moving toward less, rather than more, integration with the rest of the country. Its major parties are anti-system. The Left Party evolved from the old Communist Socialist Unity Party of Germany. A vote for it is a slap in the face to the West German establishment. The AfD in the East is far to the right of the AfD in the West. In Germany’s Thuringia state, where the AfD won 23.5 percent of the vote in the October 2019 lander election, its leader does not hesitate to use rhetoric recalling Nazism.⁴⁶ Antisemitism and antisemitic violence reemerged, and for first time since World War II, a significant political figure was assassinated. Beleaguered, Merkel entered 2020 in no position to lead Europe or to be a voice for change in Europe, especially if European reform might require financial sacrifice by German taxpayers.

Britain’s longstanding reluctance to commit to membership in an integrated Europe resulted in its joining late, in 1973, and remaining ambivalent. Dissension in the Conservative Party led then–Prime Minister David Cameron to call a referendum in 2016 which, to his surprise, rejected EU membership. The ensuing debate about what kind of British exit from the EU to pursue and whether to hold a second referendum divided Parliament and the country and paralyzed the conservative government of Prime Minister Theresa May from 2017 through 2019. The victory of her ultraconservative Tory Party rival, Boris Johnson, in the elections of December 2019 meant that there would be no second referendum and indicated that the United Kingdom would pursue a “hard” Brexit. Initially, Brexit raised fears about the survival of the EU, but the process so destabilized Britain that it seems likely

to serve as a deterrent (at least in the short term) to other member states considering a similar maneuver. Brexit will complicate British-U.S. relations, adding to preexisting trade tensions.⁴⁷

In France, both establishment parties, Les Républicains and Socialists, collapsed in the 2017 presidential and parliamentary elections. Emmanuel Macron took advantage of the vacuum to become president, and his brand-new political party gained an absolute majority in the National Assembly. As president, Macron is attempting to modernize France through neoliberal reforms, the need for which has long been proclaimed but which previous presidents have failed to accomplish because of popular resistance. The French political class has repeatedly stepped back from labor reform for fear of bringing on another period of popular uprisings like those in May 1968 and because of the ability of special interest groups and unions to bring France to its knees. Ironically, it was not the unions that successfully challenged Macron but rather something new and unexpected: the *gilets jaunes*.⁴⁸ The *gilets jaunes* crisis was an unprecedented social revolt different from anything seen before in France, an anti-system action of “the people” with no organization and leadership, based on a rural and small town population. It reflected opposition to the domestic policies of Macron and his elitist style. At the revolt’s high point, it was not obvious that the president would survive. There is no guarantee that the effort cannot be renewed.

As of late 2019, Macron had managed to nudge French unemployment down to its lowest figure in years. In 2020, prior to the COVID-19 pandemic, he took on a potentially explosive issue: reform of France’s 42 special pension systems, which provide privileged groups benefits far beyond the norm. But the people will not give them up without a fight. Macron’s only electoral rival is the Rassemblement National (RN), the new name of the National Front Party, a virulently anti-immigrant, anti-Muslim, racist party. Longtime RN leader Marine Le Pen abjured antisemitism, but few are convinced that the party’s base has truly experienced a change of heart. The RN in power would be a game changer for France and for Europe.

Domestic politics have weakened the Franco-German partnership, which has always been the sine qua non of effective EU leadership. There was widespread hope that the 2017 election of Macron in France would mark the beginning of a renewal of the partnership, but that positive scenario has not transpired. Given her position of weakness, Merkel has been especially cautious and tends to hew to the status quo on almost all issues related to Europe. There has been much talk about the emergence of Macron as the potential leader of Europe. He certainly seems eager to assume that role but has generated resentment by not seeking consensus. The European parliament showed this resentment by rejecting France’s candidate for the economics portfolio on the European Commission, Sylvie Goulard—a humiliating blow to Macron’s European reputation. Macron’s status in Europe is more a reflection of Merkel’s weakness than his own strength.⁴⁹ The weakness of Franco-German leadership has an adverse effect on Europe’s ability to assert its interests in an age of Great Power competition.

Europe’s political challenges are important internally and in the context of the new era of Great Power competition. While China’s role in exploiting Europe’s political fracturing remains limited, Russia has been capitalizing on European vulnerability. Russia is the foreign state that tries to influence European politics and decisionmaking most.

Discussed in chapters 3b and 7 of this volume, Moscow's influence activities around the globe and especially in Europe support three main Russian strategic objectives: regime security, predominance in Russia's near abroad, and world-power status for Russia. The long-term objective of Russian influence activities is to weaken NATO and the EU. In the shorter term, it is to lift the sanctions imposed after the Russian intervention in Ukraine in 2014. Russia has exploited Europe's challenges with minorities, refugees, and extremists, using them to weaken EU cohesion. Russia also uses the energy sector, business, and corruption as venues for influence on the European polity. It cultivates and exploits an extensive network of allies and front organizations and reconstructs reality and rewrites history to legitimize itself and undermine European leaders.⁵⁰ Among the modern Great Powers, Russia has the most to gain should political cohesion in Europe remain wobbly or further erode.

New Parameters of European Security

The 1991 end of Cold War bipolarity and the reunification of Germany transformed the security landscape of Europe and led the EU and its governance institutions to gradually adapt their own strategic perspectives. The foundations of the EU's Common Foreign and Security Policy, laid out in the 1992 Maastricht Treaty, placed security and defense policy in a common EU-wide context. By 2016, almost a quarter-century later, the EU had formulated a more comprehensive European Agenda for Security (EAS), which prioritized "terrorism, organized crime, and cybercrime as interlinked areas with a strong cross-border dimension."⁵¹ The EU's global strategy, adopted in June 2016, called for greater coherence in EU external action and identified five security and defense priorities: "the security of the Union; state and social resilience to our east and south; an integrated approach to conflicts and crises; cooperative regional orders; and global governance for the 21st century."⁵²

In summer 2019, the EU's 3-year review of its global strategy validated it as a vital component of a "path toward a stronger Europe."⁵³ That 2019 strategy review endorsed continuation of several activities and policies. First, it encouraged the EU's role as a global maritime security provider in activities such as operations *Atalanta* (fighting piracy off the Horn of Africa) and *Sophia* (chasing smugglers and human traffickers in the Mediterranean). It also endorsed an expanded European Border Security Force, FRONTEX, to deter illegal migration, unlawful crossings in the Mediterranean, and illegal stays in the EU.⁵⁴ The 2019 review also advocated enhancement of its defense initiative, the Permanent Structured Cooperation, which has for a while aimed to provide a new framework for joint European defense investments and cooperation and operational readiness among 25 member states, covering 34 new defense cooperation projects from cyber security to military mobility. The EU budgeted €590 million from 2017 to 2020 for these programs and entered 2020 planning to expend more funds to incentivize such European defense cooperation.⁵⁵ The expanding scope of security and defense initiatives reflects the EU's intensifying quest for greater strategic autonomy. In addition to the European defense initiatives listed above, the EU EAS announced an expansion of its security assistance in the Western Balkans; a desire to preserve the nuclear deal with Iran (Joint Comprehensive Plan of Action) despite the 2018 U.S. withdrawal from it; and partnership with African countries in multiple policy and security fields, including the field of migration.

Whether the EU's global strategy truly represents a break with the past remains to be seen. The EU's budget to address global security challenges remains miniscule. Furthermore, EU's governance rules mean that it cannot usurp the powers of national leaders on matters of security and defense. But a significant impetus for change in Europe's security strategy is the eroding cohesion of the transatlantic alliance between Europe and the United States. Over the past decade, the United States has been pressuring European members of NATO to increase their defense spending to 2 percent of GDP from an average of 1.3 percent in 2017. The response of several European countries, including Germany, France, Italy, and Sweden, has been to increase defense expenditures; total defense spending by the EU is planned to surpass \$300 billion a year by 2021.⁵⁶ But the U.S. administration also has objected to the EU's global strategy on grounds that it signals a "Europe First" approach to industrial policy, limiting the access of U.S. producers to Europe's domestic manufacturing base.⁵⁷ Complicating matters further, the December 2019 NATO summit revealed chasms among Allies' perspectives on their respective roles in countering China and Russia.⁵⁸

As the future of the transatlantic alliance is open to question, the EU's global strategy reflects the shifting balance of challenges in EU-China relations. The EU has become China's biggest trading partner, while China is the EU's second largest, with trade between the two entities worth €1.5 billion per day. But China's expanding foreign direct investments, its "new Silk Road" (also known as the Belt and Road Initiative [BRI]) stretching extensively into multiple European states, and Beijing's competition with the United States and the EU for preeminence in advanced technology have led to a reassessment of EU-China relations. While EU foreign direct investment in China was approximately €6.8 billion in 2014, China's investment in the EU had grown to around €35 billion in 2016.⁵⁹

A key area of concern for the EU is the increasing influence of China's European investments in support of the BRI. China's access to the EU's economy and market—growing during the debt crisis—greatly expanded in 2016 when the China Ocean Shipping Company acquired a majority stake in Greece's port of Piraeus, the main entry point for Chinese exports to Europe. Chinese investments of over \$8 billion to upgrade rail transportation between Belgrade and Budapest are part of a larger initiative to improve infrastructure in Central and Eastern European countries. Similar investments are planned in Europe's south. In 2019, Italy became the largest European country to participate in China's BRI, through cooperation in the development of Italy's infrastructure, civil aviation, ports, energy, and telecommunications.⁶⁰ These investments are well linked to those in the port of Piraeus, further enhancing the transportation, logistics, and warehousing infrastructure required for BRI.

EU concerns center on China's growing economic presence, potential political influence on EU policymaking, and Chinese funds displacing Russian energy as sources of dependency for Europe. The European Commission has promoted legislation requiring screening of Chinese investments, more controls over potential Chinese product "dumping," and greater scrutiny of China's offers to facilitate finance infrastructure spending.⁶¹ But Europe is not unified in how to approach China, allowing individual European states to pursue their own policies. Across Europe, China's global technological presence has already accentuated competition among Great Powers Russia and especially the United States. In many ways, this competition has developed with Europe as the object of rivalries. For example, the EU

has yet to develop a unified stance on China's role in Europe's technology base, especially its 5G networks, connecting billions of sensitive information and communications technology systems in crucial sectors. At present, France and Germany seem to diverge on what to do about allowing Chinese telecom companies Huawei and ZTE to bid for the next-generation 5G networks. While Paris remains silent on the topic, it appears that Berlin is about to allow Huawei to take part in the 5G competition in Germany.⁶² In its new EU-China Strategic Outlook in March 2019, the European Commission described its approach to China as a cooperating partner in some areas and a negotiating partner in others.⁶³ But it also named China as "an economic competitor in pursuit of technological leadership and a systemic rival promoting alternative models of governance."⁶⁴

Shifts may also be evidenced in the EU's approach to Russia. While Russia remains a security threat as a nuclear superpower, its geopolitical position in Eurasia is altogether different compared with the Cold War era. EU-Russia relations reflect a long history of complex interdependencies based on the Partnership and Cooperation Agreement of 1997.⁶⁵ These interconnections involve an array of policy areas, including trade, science and technology, and transport. Russia is now the fourth largest trading partner of the EU and the fourth largest export destination of EU goods (€85.3 billion in 2018). Russia—the largest oil, gas, uranium, and coal exporter to the EU—is the third largest source of goods imports (€168.3 billion in 2018); energy products account for nearly 70 percent of the EU's imports from Russia. For its part, the EU remains a key trading partner for Russia, representing €253.6 billion, or 43 percent of Russia's trade in 2018. And the EU is by far the largest investor in Russia, with a total stock of foreign direct investment there that approached €235.2 billion in 2018.⁶⁶

Financial interdependencies as well as geopolitical considerations will shape the future of EU-Russia relations. The energy sector will continue to dominate the EU-Russia trade and investment relationship, despite the controversies surrounding the Nord Stream, a 1,225-kilometer gas pipeline through Europe. Denmark's recent approval of a permit for Nord Stream 2 virtually guarantees that more Russian gas will be pumped into Europe.⁶⁷

There is widespread concern that Russia interferes in European political life, in elections, and possibly in the Brexit referendums; that Moscow subsidizes extremist parties; and that Moscow seeks to undermine the political cohesion of the EU and member states. But the perception of Russia as a security threat varies greatly among EU member states, depending on geography: hardly at all in the Mediterranean or Iberia, but seriously in the North and especially in the Baltics. While there is no fear that Russia could or would invade Europe as a whole, the possibility of hybrid operations in the Baltics is real.⁶⁸ Yet Europe alone cannot defend its member states against Russia, and the EU does not have a legal mandate to do so. European defense remains predicated on NATO, and U.S. guarantees are indispensable for NATO to be meaningful. Thus, Europeans worry that the United States might detach itself from the continent.⁶⁹ Some EU leaders (including Germany) try to avoid debate over the future of NATO and hope that future U.S. administrations will return to traditional transatlantic policies. To the extent that they feel less certain about U.S. commitment to NATO Article V (the mutual self-defense clause), Europeans may pursue and intensify mutually advantageous relations with Russia in as many sectors as possible. The EU could seek continued accommodation and deeper cooperation with China and Russia

as the optimal path to survive and adapt to the evolving competition among the world's Great Powers.

Another approach, not necessarily mutually exclusive, advocated by President Macron, is one of European defense autonomy. In the past, French support for European defense autonomy was based on neo-Gaullist opposition to perceived U.S. domination. In late 2019, Macron, who stated that NATO is “brain dead,” expressed far more concern that “America is turning its back on us so quickly on strategic issues.” He added that if Europe “can’t think of itself as a global power [it] will disappear.”⁷⁰ Macron took note of how the United States broke with its Kurdish allies in Syria with a controversial military departure that left Kurdish adversary Turkey holding the cards on what to do there. Macron’s comments were taken by some as recognition that Europe needs to be able to defend itself—a legitimate call to action to create genuine European defense cooperation. If the United States follows its historic policy of close cooperation with Europe on security, with a focus on the primacy of NATO, this is unlikely to happen. If, however, U.S. policy shifts even more deliberately to a more unilateral approach in the period of 2020 to 2025—and if Europe feels that there is an existential threat to its sovereignty and that the United States could not be counted on to come to its aid—the EU has the means to create an autonomous defense. In that case, Europe, very much against its inclinations, could move from a post-World War II experiment relying on U.S. security guarantees to a more ambitious system of integrated European security and defense with a single military structure.

Europe in 2025

When European integration began with the Schuman Plan in 1949, it was a superb strategy for transcending generations of conflict and creating new union. When the Cold War ended, the EU dared to expand toward the very limits of the continent. Operating in a moment of dominant U.S. power and in the absence of Great Power rivalry, Europeans thought large and operated boldly. That dynamism is lacking today.

At a time when other parts of the world, such as China, are exhibiting unprecedented economic growth, Europe is stagnating; its percentage of global GDP is in decline. The international context of 2020 is presenting Europe with a more aggressive Russia, growing Chinese power, and lack of trust in the U.S. commitment to European defense and to transatlanticism as a whole. Concurrently, the EU seems to be facing serious instability as a result of the weakening of its Franco-German core. Germany and France have not found common positions on many significant issues, including European defense.

Europe will remain vulnerable to internal and external events. Among such outside events, the COVID-19 pandemic that is sweeping the world and hitting much of Europe especially hard is precisely the kind of exogenous shock that might thrust the EU into an entirely new paradigm. Europe did not have enough resources or time to provide robust internal assistance to member states, such as Italy, hit hardest by the virus early on. The United States did not have a policy focus or the independent resources to fill that humanitarian assistance role either. Thus, it was China—and, to a lesser extent, Russia—that provided direct assistance with forms of anti-virus expertise to Italy and then to other European states during the crisis.⁷¹ By assuming a traditional U.S.-led role in international humanitarian health assistance for Europe’s own states, China in particular demonstrated the interest and

the capability to grow cultural goodwill and greater potential political influence with Europeans in an era of Great Power competition.

As with the pandemic, several trends now visible in 2020 could easily develop into major problems for EU cohesion over the course of the coming few years:

- A significant policy change toward NATO in a second Trump administration that could weaken European faith in the U.S. commitment toward Article V guarantees to member states. This could trigger aggressive Russian behavior toward a NATO state or intensification of the war in Ukraine. It could also begin a decoupling of European and U.S. defense.
- A renewed migration crisis or massive terrorist attacks. Even smaller attacks, like the murder of four policemen in the prefecture of police in Paris by a policeman-turned-jihadist, can have outsized consequences on the state and public opinion. This might further fragment Europe's right and center, exacerbating tensions over common population movement policies.
- A new financial crisis, requiring more radical interventions to save the euro and EU banks. A lengthy coronavirus-generated recession or a similar serious economic shock could lead to increased political friction and might further undermine the solidarity of the EU.
- A social or political explosion that threatens the survival of an elected government or the integrity of a European state. The *gilets jaunes* crisis indicates high levels of social discontent in France and, by extension, possibly elsewhere. The worker strikes of December 2019 through January 2020 emphasized that risk. The Catalan separatist crisis in Spain has not become violent, but nothing prevents it from deepening.
- Although the December 2019 British elections resolved the immediate question of whether Brexit would actually take place, there remains the daunting task of deciding on its terms in negotiations that are supposed to terminate within only 1 year. Brexit could still degenerate. It has bitterly divided the United Kingdom. It could lead to the fragmentation of the two major parties there and a shakeup of the parliamentary system itself. Britain may be threatened with dissolution; the likelihood of Scotland's independence and even Irish unification has increased. British prestige has suffered. The British role in European defense may also decline. Britain's problems are Europe's problems, even if Britain leaves the EU.

At the dawn of a new era of Great Power competition, Europe finds itself at a far different place than it did a mere decade ago. Europe did not become a "superpower" in the post-Cold War and post-Maastricht world, as predicted by some. In the past decade, Europe has weathered a serious array of stress-inducing challenges: a debt crisis, a migration crisis, increasing pressure on its security and political institutions by a resurgent Russia and by the rise an extreme right wing, the encroachment of Chinese economic strength, and a decline in transatlantic solidarity. In spite of it all, European integration remains a reality. Europe has been resilient but troubled. The coming years will continue to test that resilience.

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Notes

¹ In *Notre Jeunesse, Notre Jeunesse, Oeuvres Complètes de Charles Péguy 1873–1914, Oeuvres de Prose, Notre Jeunesse, Victor-Marie, Comte Hugo* (Paris: Nouvelle Revue Française, 1916), 59.

² See chapter 1 of this volume for the three-feature operational definition of a *Great Power* as derived from Thomas J. Volgy et al., “Major Power Status in International Politics,” in *Major Powers and the Quest for Status in International Politics: Global and Regional Perspectives*, ed. Thomas J. Volgy et al. (New York: Palgrave Macmillan, 2011), 1–26.

³ Jack Goldstone, “The New Population Bomb,” *Foreign Affairs* (January/February 2010).

⁴ Discussed in detail in Irene Kyriakopoulos, “Economic and Monetary Union Governance: A Post-Crisis Assessment,” *World Economics Journal* 12, no. 4 (October–December 2011), available at <www.world-economics-journal.com/Journal/Papers/Economic%20and%20Monetary%20Union%20Governance.details?ID=496>.

⁵ For a detailed analysis, see Irene Kyriakopoulos, “In the Name of the Euro: What Have the EU’s Policies Achieved in Greece?” *Intereconomics* 49, no. 6 (2014), available at <https://archive.intereconomics.eu/year/2014/6/in-the-name-of-the-euro-what-have-the-eus-policies-achieved-in-greece/>.

⁶ This is a provision of the Maastricht Treaty, which established Europe’s monetary union. The provision establishes that member states are not liable and will not assume the debts of any member state. Thus, if a member state defaults on its sovereign debt, other member states and/or the Eurozone as a whole are not liable. The member state is responsible for the consequences of a default. For further reading, see Council on Foreign Relations, “The Eurozone Crisis as Historical Legacy,” September 24, 2010, available at <www.cfr.org/report/eurozone-crisis-historical-legacy>.

⁷ The Eurogroup, which manages financial affairs for the Eurozone, decided to finance a €10 billion rescue program through a levy imposed on uninsured deposits over €100,000 (\$130,000 USD) in Cyprus’s banks.

⁸ Verbatim of the remarks by Mario Draghi, European Central Bank, July 26, 2012, available at <https://www.ecb.europa.eu/press/key/date/2012/html/sp120726.en.html>.

⁹ Felix Roth and Lars Jonung, “Public Support for the Euro and Trust in the ECB: The First Two Decades,” *VOX EU*, December 13, 2019, available at <https://voxeu.org/article/public-support-euro-and-trust-ecb>.

¹⁰ *Understanding the Socio-Economic Divide in Europe* (Paris: Organisation for Economic Co-operation and Development, January 26, 2017), available at <www.oecd.org/els/soc/cope-divide-europe-2017-background-report.pdf>.

¹¹ Andrew Walker, “Eurozone Bailout Program Is Finally Over,” BBC News, August 19, 2018, available at <www.bbc.com/news/business-45186511>.

¹² Jamil Aderlini, “Chinese Investors Surged into EU at Height of Debt Crisis,” *Financial Times*, October 6, 2014, available at <www.ft.com/content/53b7a268-44a6-11e4-ab0c-00144feabdc0>.

¹³ Po-Kuan Wu, “Examining the EU-China Relationship in the Aftermath of the Economic Crisis,” *International Journal of Public Administration* 40, no. 14 (April 2017), available at <www.tandfonline.com/doi/full/10.1080/01900692.2017.1295268?src=recsys>.

¹⁴ Kelvin Chan, “Europe Resists Mounting U.S. Pressure on Huawei 5G Technology,” *Associated Press*, February 18, 2020, available at <www.c4isrnet.com/battlefield-tech/it-networks/5g/2020/02/18/europe-resists-mounting-us-pressure-on-huawei-5g-technology/>.

¹⁵ Bettina Marx, “Merkel Demands ‘More Europe!’” *Deutsche Welle*, June 7, 2012, available at <www.dw.com/en/merkel-demands-more-europe/a-16007150>.

¹⁶ European Central Bank, Eurosystem, “The International Role of the Euro,” June 2019, available at <https://www.ecb.europa.eu/pub/ire/html/ecb.ire201906~f0da2b823e.en.html#toc1>.

¹⁷ Maximilian Kärnfelt, “China’s Currency Push: The Chinese Yuan Expands Its Footprint in Europe,” *China Monitor*, January 9, 2020, available at <https://www.merics.org/en/china-monitor/chinas-currency-push>.

¹⁸ The phrase “the first postmodern superpower” was coined in a popular book published a decade after EU integration. See Jeremy Rifkin, *The European Dream: How Europe’s Vision of the Future Is Quietly Eclipsing the American Dream* (London: Tarcher Perigree, 2004). For a discussion of the EU’s failure to realize the most optimistic hopes of a future as a superpower, see William Drozdiak, *Fractured Continent: Europe’s Crisis and the Fate of the West* (New York: Norton, 2017).

¹⁹ For further reading, see Christof Van Mol and Helga de Valk, “Migration and Immigrants in Europe: A Historical and Demographic Perspective,” in *Integration Processes and Policies in Europe*, ed. Blanca Garcés-Mascareñas and Rinus Penninx (New York: Springer, 2016).

²⁰ Irene Kyriakopoulos, “Europe’s Responses to the Migration Crisis: Implications for European Integration,” *Strategic Insights* (Washington, DC: Institute for National Strategic Studies, April 25, 2019).

²¹ “Europe and Right-Wing Nationalism: A Country-by-Country Guide,” BBC News, November 13, 2019, available at <www.bbc.com/news/world-europe-3613006>.

²² Heather Horn, “The Staggering Scale of Germany’s Refugee Project,” *The Atlantic*, September 12, 2015.

²³ European Commission, “The Dublin System,” April 6, 2016, available at <https://ec.europa.eu/home-affairs/sites/homeaffairs/files/what-we-do/policies/european-agenda-migration/background-information/docs/20160406/factsheet_-_the_dublin_system_en.pdf>.

²⁴ European Council, Council of the European Union, “EU-Turkey Statement,” March 18, 2016, available at <https://www.consilium.europa.eu/en/press/press-releases/2016/03/18/eu-turkey-statement/>.

²⁵ Turkey is not a member of the EU, so it is not subject to its migration or population movement laws.

²⁶ “EU Chief Says Greece Is Europe’s Shield in Migrant Crisis,” BBC News, March 3, 2020, available at <www.bbc.com/news/world-europe-51721356>.

²⁷ Operation *Sea Guardian* succeeded Operation *Active Endeavour* under the operational command of Allied Maritime Command, Northwood, United Kingdom.

²⁸ In October 2018, a beleaguered Chancellor Angela Merkel announced that she would not seek reelection as party head or as Germany’s chancellor in the forecast 2021 elections. See Anneken

Tappe, "Germany's Merkel Confirms She's on the Way Out, Won't Seek Re-election as Party Head or Chancellor," *Market Watch*, October 29, 2018, available at <www.marketwatch.com/story/germanys-merkel-confirms-shes-on-the-way-out-wont-look-for-re-election-as-party-head-or-chancellor-2018-10-29>.

²⁹ "Merkel, to Survive, Agrees to Border Camps for Migrants," *New York Times*, July 2, 2018, available at <www.nytimes.com/2018/07/02/world/europe/angela-merkel-migration-coalition.html>.

³⁰ "Germany Passes Controversial Migration Law," *Politico*, July 7, 2019, available at <<https://www.politico.eu/article/germany-passes-controversial-migration-law/>>.

³¹ Emmanuel Macron, "For European Renewal," March 4, 2019, available at <<https://www.elysee.fr/emmanuel-macron/2019/03/04/for-european-renewal.en>>.

³² European Commission, Migration and Home Affairs, "European Agenda on Migration Four Years On," available at <https://ec.europa.eu/home-affairs/news/european-agenda-migration-four-years-marked-progress-needs-consolidating-face-volatile_en>.

³³ European Border and Coast Guard Agency, available at <<https://frontex.europa.eu/>>.

³⁴ "European Border and Coast Guard: 10,000-Strong Standing Corps by 2027," *European Parliament News*, April 17, 2019, available at <<https://www.europarl.europa.eu/news/en/press-room/20190410IPR37530/european-border-and-coast-guard-10-000-strong-standing-corps-by-2027>>; "FRONTEX: The Making of A Superagency," *Refugee Support Aegean*, December 4, 2019, available at <<https://rsaegian.org/en/the-making-of-a-super-agency/>>.

³⁵ European Commission, "Remarks by Commissioner Avramopoulos," press release, June 20, 2018, available at <https://ec.europa.eu/commission/presscorner/detail/en/SPEECH_18_4241>.

³⁶ See EU Emergency Trust Fund for Africa, available at <https://ec.europa.eu/trustfundforafrica/index_en>; Maïte Vermeulen, Giacomo Zandonini, and Ajibola Amzat, "How the EU Created a Crisis in Africa—and Started a Migration Cartel," *The Correspondent*, December 10, 2019, available at <<https://thecorrespondent.com/166/how-the-eu-created-a-crisis-in-africa-and-started-a-migration-cartel/21953207342-46c48098>>.

³⁷ Philip Connor, "A Majority of Europeans Favor Taking in Refugees, but Most Disapprove of EU's Handling of the Issue," Pew Research Center, September 16, 2018, available at <<https://www.pewresearch.org/fact-tank/2018/09/19/a-majority-of-europeans-favor-taking-in-refugees-but-most-disapprove-of-eus-handling-of-the-issue/>>.

³⁸ Alina Polyakova and Benjamin Haddad, "Europe in the New Era of Great Power Competition: How the EU Can Stand Up to Trump and China," *Foreign Affairs*, July 17, 2018, available at <www.foreignaffairs.com/articles/europe/2018-07-17/europe-new-era-great-power-competition>.

³⁹ On the implications of British marginalization, see Gideon Rachman, "Britain and Russia Are Europe's Odd Couple," *Financial Times*, October 29, 2019. On the early development of British-EU relations, see Hugo Young, *This Blessed Plot: Britain and Europe from Churchill to Blair* (London: Overlook, 1998). The *Financial Times* provides excellent coverage of Brexit developments. See Center for European Reform's analyses as well.

⁴⁰ On the history of Franco-German relations, see Julius W. Friend, *The Linchpin: Franco-German Relations 1950–1990* (New York: Praeger, 1991), and *Unequal Partners: Franco-German Relations 1989–2000* (New York: Praeger, 2001). Also see Gilbert Ziebur, ed., *Les relations franco-allemandes dans une Europe divisée, Mythes et réalités* (Bordeaux: Presses Universitaires de Bordeaux, 2012).

⁴¹ For a definition of and detail about the parameters of the fourth industrial revolution, see chapter 4.

⁴² Steven Philip Kramer, "France's Muslim Predicament," *National Interest*, March 2020. For a serious study on Islam in France, see Sylvain Brouard and Vincent Tiberj, *Français comme les autres? Enquête sur les citoyens d'origine maghrébine, africaine et turque* (Paris: Presses de la Fondation Nationale des Sciences Politiques, 2005).

⁴³ On terrorism, see EUROPOL, *European Union Terrorism Situation and Trend Report 2019* (The Hague: European Union Agency for Law Enforcement Cooperation, 2019), available at <<https://www.europol.europa.eu/activities-services/main-reports/terrorism-situation-and-trend-report-2019-te-sat>>. On the problem of the return of the so-called Islamic State, see David Thomson, *Les Revenants* (Paris: Seuil, 2016). On the impact of violence against Jews in France, see Jérôme Fourquet and Sylvain Manternach, *L'An prochain à Jérusalem* (Paris: Editions in de l'Aube, 2016).

⁴⁴ Paul Lendvai, "The Transformer: Orban's Evolution and Hungary's Demise," *Foreign Affairs* (September/October 2019).

⁴⁵ On the Hartz reforms and political consequences in Germany, see Christian Odenahl, "Germany After the Hartz Reforms: Can the SPD Protect German Labor?" *Foreign Affairs*, September 11, 2017, available at <www.foreignaffairs.com/articles/germany/2017-09-11/germany-after-hartz-reforms>.

⁴⁶ "East German State Lurches Right, Elevating Its Firebrand Leader," *New York Times*, October 28, 2019.

⁴⁷ Michael Marray, "Brexit Further Complicates U.S., EU, and China Trade Relations," *The Asset*, February 12, 2020, available at <<https://www.theasset.com/europe/39730/brexit-further-complicates-us-eu-and-china-trade-relations>>.

⁴⁸ Steven Philip Kramer, "The Gilet Jaune Crisis," *Strategic Insights* (Washington, DC: Institute for National Strategic Studies, March 11, 2019).

⁴⁹ Guy Chazan, "Merkel's Political Twilight Sees Influence Wane," *Financial Times*, September 25, 2019.

⁵⁰ Gier Hagen Karlsen, "Divide and Rule: Ten Lessons About Russian Political Influence Activities in Europe," *Palgrave Communications* 5, no. 19 (February 8, 2019), available at <www.nature.com/articles/s41599-019-0227-8>.

⁵¹ Migration and Home Affairs—European Commission, "European Agenda on Security," December 6, 2016, available at <https://ec.europa.eu/home-affairs/what-we-do/policies/european-agenda-security_en>.

⁵² Federica Mogherini, "From Vision to Action: The EU Global Strategy in Practice," in *The European Union's Global Strategy: Three Years On, Looking Forward* (Brussels: European Commission, June 2019), 4–7, available at <https://eeas.europa.eu/headquarters/headquarters-homepage/64034/vision-action-eu-global-strategy-practice-three-years-looking-forward_en>.

⁵³ Ibid. Also see "The EU Global Strategy Three Years On," in *The European Union's Global Strategy*, 8–33.

⁵⁴ FRONTEX is scheduled to expand its strength to 10,000 by 2027.

⁵⁵ Daniel Darling, "PESCO and the Long Road to European Strategic Autonomy," *Defense & Security Monitor*, November 14, 2019, available at <<https://dsm.forecastinternational.com/wordpress/2019/11/14/pesco-and-the-long-road-to-european-strategic-autonomy/>>.

⁵⁶ David Reid, "Europe's Defense Spending Nears \$300 Billion as Experts Say Trump's Pressure Is Paying Off," CNBC, November 1, 2019, available at <<https://www.cnbc.com/2019/11/01/european-defense-spending-to-hit-300-billion-by-2021-analysts-say.html>>.

⁵⁷ Ben Hall and Richard Milne, "Europe First: How Brussels Is Retooling Industrial Policy," *Financial Times*, December 2, 2019.

⁵⁸ Kevin Baron, "NATO's Newest Threat Is Coming from Inside the House," *Defense One*, December 3, 2019, available at <www.defenseone.com/ideas/2019/12/natos-newest-threat-coming-inside-house/161634/>.

⁵⁹ European External Action Service, "EU-China Relations Factsheet," October 18, 2019, available at <https://eeas.europa.eu/topics/external-investment-plan/34728/eu-china-relations-factsheet_en>.

⁶⁰ Vincent L. Morelli, *The European Union and China*, IF10252 (Washington, DC: Congressional Research Service, 2019), available at <<https://fas.org/sgp/crs/row/IF10252.pdf>>.

⁶¹ Michael Nienaber, "Chinese FDI in Europe Drops, Investment Screening Will Cut It More—Survey," Reuters, March 6, 2019, available at <www.reuters.com/article/us-eu-china-fdi/chinese-fdi-in-europe-drops-investment-screening-will-cut-it-more-survey-idUSKCN1QN1T6>.

⁶² Janka Oertel, "Germany Chooses China Over the West," *Foreign Policy*, October 21, 2019, available at <<https://foreignpolicy.com/2019/10/21/germany-merkel-chooses-china-over-united-states-eu-huawei/>>; see also Philippe Le Corre, "Macron Goes to China: For Europe or for France?" *Carnegie Europe*, October 31, 2019, available at <<https://carnegieeurope.eu/strategieurope/80242>>.

⁶³ *European Commission and HR/VP Contribution to the European Council: EU-China—A Strategic Outlook* (Brussels: European Commission, 2019), available at <<https://ec.europa.eu/commission/sites/beta-political/files/communication-eu-china-a-strategic-outlook.pdf>>.

⁶⁴ *Ibid.*

⁶⁵ European Commission, "EU/Russia Partnership and Cooperation Agreement," December 1, 1997, available at <<https://cordis.europa.eu/article/id/9433-eurussia-partnership-and-cooperation-agreement>>.

⁶⁶ European Union External Action, "The European Union and the Russian Federation," May 30, 2019, available at <https://eeas.europa.eu/headquarters/headquarters-homepage/35939/european-union-and-russian-federation_en>.

⁶⁷ The name *Nord Stream* refers to a pipeline from Russia to western Europe under the Baltic Sea. It was officially launched in 2010. See "Nord Stream Gas Pipeline Underwater Construction Starts," BBC News, April 9, 2010, available at <<http://news.bbc.co.uk/2/hi/business/8607214.stm>>. *Nord Stream 2* is the name of the undersea pipeline that will pump Russian gas into Europe. See Jordan Stevens, "Nord Stream 2 Explained: What It Is and Why It's Proving Controversial," CNBC, May 21, 2019, available at <www.cnbc.com/2019/05/21/nord-stream-2-explained-what-it-is-and-why-its-proving-controversial.html>. See also "Nord Stream 2: Go-Ahead for Russian Gas Pipeline Angers Ukraine," BBC, October 31, 2019, available at <<https://www.bbc.com/news/world-europe-50247793>>.

⁶⁸ Cf. Robert M. Klein et al., *Baltics Left of Bang: The Role of NATO with Partners in Denial Based Deterrence*, INSS Strategic Forum No. 304 (Washington, DC: NDU Press, November 2019), available at <<https://ndupress.ndu.edu/Portals/68/Documents/stratforum/SF-304.pdf>>. For a reappraisal of the Russian threat, see Thomas Graham, "Let Russia Be Russia," *Foreign Affairs* (November/December 2019).

⁶⁹ See, for example, Constanze Stelzenmuller, *Hostile Ally: The Trump Administration and Europe's Inadequate Response* (Washington, DC: Brookings, August 2019), available at <<https://www.brookings.edu/research/hostile-ally-the-trump-challenge-and-europes-inadequate-response/>>.

⁷⁰ "Emmanuel Macron on Europe's Fragile Place in a Hostile World," *The Economist*, November 7, 2019, available at <www.economist.com/briefing/2019/11/07/emmanuel-macron-on-europes-fragile-place-in-a-hostile-world>.

economist.com/briefing/2019/11/07/emmanuel-macron-on-europes-fragile-place-in-a-hostile-world>. See also Zaki Laïdi, *Can Europe Learn to Play Power Politics* (London: Centre for European Reform, November 2019).

⁷¹ Alan Crawford, Peter Martin, and Bloomberg, "Health Silk Road: China Showers Europe with Coronavirus Aid as Both Spar with Trump," *Fortune*, March 19, 2020, available at <<https://fortune.com/2020/03/19/china-europe-coronavirus-aid-trump/>>; Stuart Lau, "Coronavirus: Xi Jinping Calls Leaders of France, Spain, Germany and Serbia with Offers of Support," *South China Morning Post* (Hong Kong), March 21, 2020, available at <www.scmp.com/news/china/diplomacy/article/3076287/coronavirus-xi-jinping-calls-leaders-italy-spain-germany-and>.

Chapter 13

Competing Visions and Actions by China, Russia, and the United States in Latin America, the Middle East, Africa, and the Arctic

By Bryce Loidolt, David Auerswald, Douglas Farah, Shannon Smith, and Caitlyn Yates

This chapter reviews the contours of Great Power competition across Latin America, the Middle East, Africa, and the Arctic; it traces the motivation and scale as well as receptivity to, and potential repercussions of, Chinese and Russian activities across these regions. It finds the challenge of these two competitors to be distinct, the risks to U.S. interests to be uneven across and within each region, and, ultimately, regional states' cooperation with China and Russia to rarely be grounded in an ideological commitment to Beijing's global vision or Moscow's cynicism. This points to the need for parallel strategies that appreciate the diverse challenges China and Russia pose, a broader recalibration of U.S. regional interests that moves beyond the post-9/11 focus on counterterrorism, and a discerning strategic approach that avoids pulling U.S. regional partners into an unrestricted, zero-sum competition.

The priorities of the 2018 National Defense Strategy imply a greater emphasis on the Indo-Pacific region and Europe, yet the era of Great Power competition currently unfolding among the United States, China, and Russia is not confined to these boundaries. Just as Russia's targeted influence activities in selected geostrategic regions have become more visible in the past decade, so too does China extend its economic and, to a lesser extent, military reach into its far abroad.¹ These activities have now begun generating some alarm—especially in U.S. military circles—that America's Great Power rivals might peel away traditional U.S. strategic partners and allies, embolden or insulate U.S. adversaries, or otherwise provide China and Russia with access and resources that will magnify their power to the strategic disadvantage of the United States.²

How should the United States best balance its regional investments in an emerging era of Great Power competition? The answer to this important question requires an appraisal of the nature and stakes of U.S. competition with Russia and China outside of

each competitor's immediate neighborhood. This chapter thus surveys the interests and activities of these competitors across Latin America, the Middle East, Africa, and the Arctic. It further evaluates regional receptivity to these activities, as well as how Chinese and Russian behavior could pose direct and consequential threats to U.S. regional interests.

We find the motivations, scale, and potential repercussions of Chinese and Russian activities across these four geographic regions to be distinct from each other. Russia has expanded its political and diplomatic stature by sponsoring multilateral initiatives and presenting itself as a reliable, unconditional ally to regional friends. China has also come to compete with the United States by establishing a narrative of a politically disinterested, national sovereignty-respecting power, establishing cooperative partnerships with U.S. friends and foes alike. Chinese economic investments have been expansive and could yield economic gains, opportunities for political coercion, and military access.

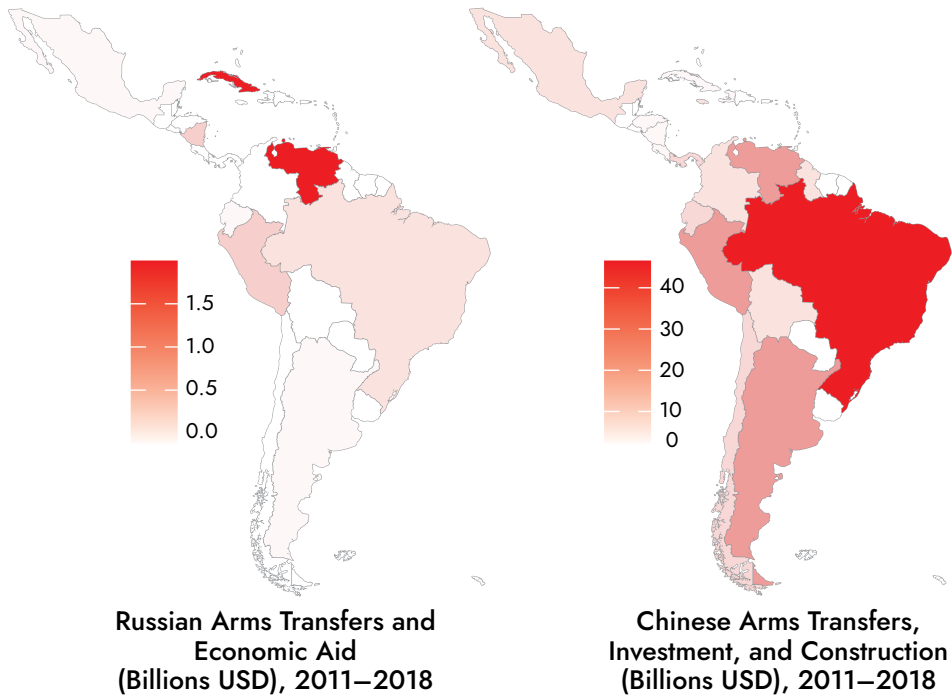
The emerging era of Great Power competition will confront U.S. policymakers both with the challenge of how to shift greater resources and attention toward Russian and Chinese traditional spheres of influence and with the questions of whether and how to compete with Beijing and Moscow on a global scale. Doing so will require distinct strategies for competing with Russia and China, a recalibration of U.S. interests across the world, and a discerning approach that reduces the prospects of pulling U.S. regional partners into an unrestricted, zero-sum competition.

Latin America

Since the end of the Cold War, the United States has generally viewed stability in Latin America as a central goal in the region.³ The United States seeks a Latin America comprised of “stable, friendly, and prosperous states,” in order to contain the spillover effects of transregional violence, criminal enterprises, and illegal immigration, as well as for the benefit of the U.S. economy.⁴ To this end, the United States has established a robust network of free trade agreements as well as regional and bilateral security initiatives, and it has leveraged economic assistance—more than \$2 billion in 2018—to promote good governance in the region.⁵

At the same time, China and Russia are increasingly engaging in Latin America, hoping to gain access to new markets, acquire strategic ports and electronic hubs, and benefit geopolitically from increased proximity to the American “near abroad.” Yet each competitor approaches the region on a different scale—and, at times, through different partners—as reflected in figure 13.1.

In the past decade, China's engagement in the region has not varied according to Latin American nations' political ideologies. Beijing's investments appear to have paid off; a 2018 regional survey found that 55 percent of respondents in Latin America had a “good” or “very good” view of China.⁶ Beijing's tool kit is largely based on economic instruments, using large-scale loans, infrastructure investments, and telecommunications projects to engage the region. The China Development Bank and China Export-Import Bank are now the largest lenders in Latin America, with loans totaling \$141.3 billion between 2005 and 2018, more than five times the amount of official U.S. economic loans and grants disbursed during the same time period.⁷ Currently, 19 countries in Latin America participate in China's Belt and Road Initiative, and China is active in more than 50 port projects across the

Figure 13.1. Russian and Chinese Aid and Investments in Latin America

Sources: Arms transfer data for both Russia and China were retrieved from Stockholm International Peace Research Institute (SIPRI) Arms Transfers Database, available at <https://www.sipri.org/databases/armstransfers>. Chinese investment data are from Derek Scissors, "China Global Investment Tracker," American Enterprise Institute, available at <https://www.aei.org/china-global-investment-tracker/>. Russian Official Development Assistance data are from the Organization for Economic Cooperation and Development Web site, available at <https://stats.oecd.org/>.

region, for both commercial and potential military use.⁸ China has also at times provided military assistance to regional powers—second only to Russia in the value of its arms transfers to Venezuela—and has established 40 Confucius Institutes in the region that could serve as important platforms for Chinese soft-power projection.⁹

Russia lacks China's ability to offer significant economic resources, so it instead opts to engage Latin America by focusing its efforts on countries that it had a strong relationship with during the Cold War. This primarily entails members of the Bolivarian Alliance, which is led by Venezuela but includes other ideologically likeminded countries such as Cuba, Nicaragua, and Bolivia, as well as some members of El Salvador's political elite. This alliance shares a strong anti-U.S. ideology and operates as a likeminded network of deeply criminalized states. Russia's engagement offers little more than an opportunity to challenge or frustrate the United States; however, by working predominantly with the Bolivarian Alliance, Russia also avoids navigating as many economic and political hurdles as China as it increases its activities in Latin America.

The Vladimir Putin regime's outreach to Latin America is designed to maximize impact at low cost. To date, this approach consists largely of weapons sales and donations; high-level state-to-state visits; military and police training in areas such as counternarcotics; and

the opportunity for certain partner nations to access Russian financial institutions, to avoid being sanctioned while moving funds through the American banking system.¹⁰ Russia has actively opened financial operations in Latin America—including banks and a cryptocurrency—to help blunt the impact of U.S. and European Union sanctions on allies such as Venezuela and Nicaragua. In addition to financial access nodes, Russia works with regional partners to create joint space projects and to flood the region with Internet propaganda creating a “fog of falsehoods” designed to disorient audiences.¹¹ All such actions bolster Russia’s brand as a reliable and predictable ally for regional states, and more than 50 percent of respondents in a recent regional survey held a “good” or “very good” opinion of Russia.¹²

Russia and China are attractive partners for Latin American countries in the new era of Great Power competition. Both offer their allies—Venezuela, Cuba, and Nicaragua, in particular—protection from condemnation in the United Nations (UN) Security Council and means to avoid U.S. sanctions. Beijing and Moscow also provide an avenue to bypass the rule of law and democratic accountability-based conditions the United States often imposes on its foreign assistance. China has been an attractive partner throughout Latin America since at least 2005 but more particularly in these past 3 years. Since 2016, China has named eight countries—including Venezuela—as hemispheric strategic partners.¹³

For the United States, the most immediate consequence of Russian and Chinese activities in Latin America has emerged from their support of the Nicolás Maduro regime in Venezuela—including purchasing oil, a violation of U.S. sanctions. In 2019, China purchased, on average, 320,000 barrels per day from Venezuela despite U.S. sanctions going into effect in January of the same year.¹⁴ Although China’s objectives are primarily economic in nature—such as securing oil and oil drilling rights—Beijing’s activities in Venezuela have been crucial to Maduro’s survival. China has loaned Venezuela more than \$70 billion in recent years, a rather controversial decision given that Venezuela still owes China approximately \$20 billion.¹⁵

Russia’s robust engagement in Venezuela has been mainly to disrupt U.S. engagement and help the Maduro regime bypass international sanctions. Actions taken by Russia in Venezuela include sending two long-range nuclear-capable bombers, deploying special forces troops and Wagner Group mercenaries, sending intelligence-collection ships to Venezuelan waters, and deploying underwater research ships capable of mapping undersea cables.¹⁶ In addition to these actions, Russia has provided Venezuela \$17 billion in loans and credit lines from 2006 to 2018.¹⁷

Beyond Venezuela, China and Russia support authoritarian, anti-U.S. regimes across the region both directly and indirectly. Since 2017, China has successfully persuaded three countries—Panama, the Dominican Republic, and El Salvador—to establish diplomatic ties with Beijing, abandoning their decades-old practice of recognizing Taiwan and leaving only nine nations in the hemisphere, primarily small Caribbean islands, that recognize the legitimacy of Taipei.¹⁸ Whereas China has been successful in securing political concessions from its partners, both Russian and Chinese actions threaten to erode democratic governance in Latin America while weakening historic U.S. alliances in countries such as Panama, Argentina, and Ecuador.¹⁹ While distinct in their approaches, the immediate consequences of Russian and Chinese engagement in Latin America may achieve a similar anti-U.S. end.

The Middle East

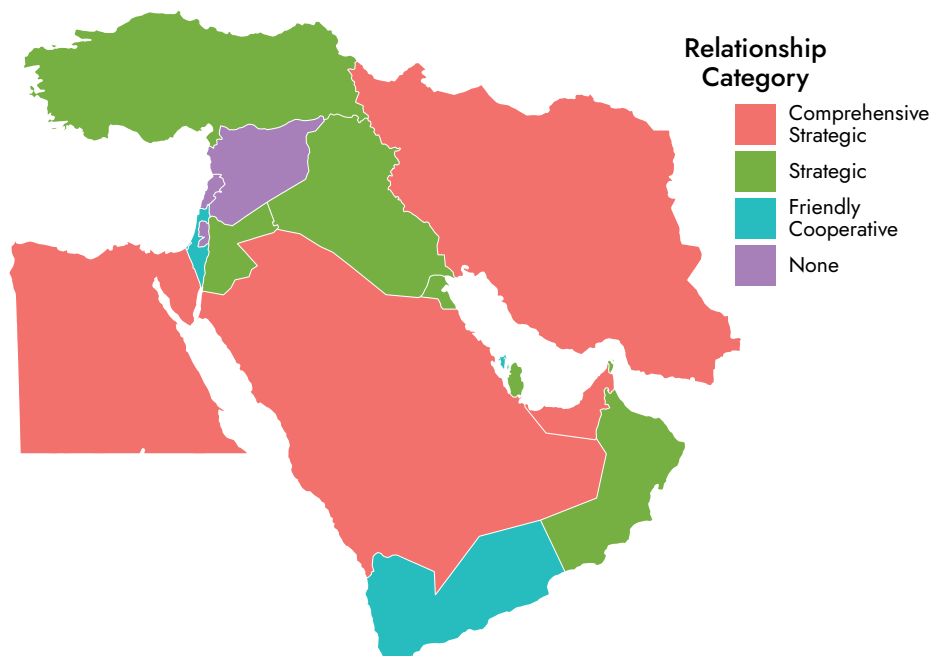
Since 9/11, U.S. engagement in the Middle East has been buoyed by a desire to contain or degrade terrorist safe havens, maintain a regional balance of power favorable to U.S. interests, and—given that the Middle East is home to three geostrategic maritime chokepoints—ensure the stability of global energy markets.²⁰ The United States has generally secured these interests through the forward presence of military forces, but it has also employed security cooperation agreements, as well as military and economic aid, to cement key partnerships and secure military access in the region.²¹ In terms of economic and military resources, the level of U.S. investment in the region is considerable. In 2018, the United States is estimated to have transferred to the Middle East nearly \$6 billion worth of armaments, comprising 55 percent of total U.S. arms transfers for the year.²² The region also received more than \$8 billion in American economic aid in 2018 and hosts the fourth largest overseas presence of U.S. troops.²³

Chinese interests and instruments in the region are predominantly, though not exclusively, economic in nature. China leans heavily on Gulf oil producers and routes, importing approximately 40 percent of its crude oil from the region in 2018.²⁴ China also relies on Middle East states to absorb its excess industrial capacity, which it in part offloads through infrastructure development projects, having invested nearly \$130 billion throughout the Arab Middle East alone in the past 10 years.²⁵ In 2004, China established the China-Arab States Cooperation Forum as a platform for shared principles and economic exchanges with Arab states.²⁶ Finally, Beijing has emerged as an important arms supplier to the region, providing military equipment primarily to Saudi Arabia, Iraq, and the United Arab Emirates.²⁷

Similar to China's approach in Latin America, President Xi Jinping has framed China's engagement in the region as apolitical. He proclaimed at the League of Arab States in 2016 that "instead of attempting to fill the 'vacuum,' China builds a cooperative partnership network for mutual benefits and win-win results."²⁸ China indeed maintains a bevy of strategic partnerships that transcend regional rivalries, as depicted in figure 13.2. In an attempt to establish soft influence in the region, China has also scattered 14 of its Confucius Institutes across the Middle East.²⁹

Russia's interventions and engagements in the Middle East are motivated by a combination of reputational and material interests.³⁰ Among the former, Russia seeks to opportunistically undermine and disrupt U.S. influence and reassert its own identity as a global power. To a lesser extent, Russia is likely motivated to engage in the Middle East by a set of economic factors, including maintaining the Organization of the Petroleum Exporting Countries + cap-and-cut oil production agreements as well as expanding its access to nuclear energy markets, trade, and arms sales. Russian activities in the Middle East and North Africa are diverse, ranging from the establishment of an enduring military presence in Syria to attempts at expanding arms sales to regional powers and the sponsorship of parallel conflict resolution initiatives—which have allowed Russia to posture as a neutral mediator. Much like China, Russia has adopted a diverse portfolio of partners in the Middle East rather than binding solely to its historical allies from the Cold War era.

In some instances, Russia and China have been willing to work together toward common objectives in the region. For instance, both Russia and China have signaled their support for Iran. As tensions between Washington and Tehran mounted through the end

Figure 13.2. Chinese Strategic Partnerships in the Middle East

Sources: Kenneth Allen, Phillip C. Saunders, and John Chen, *Chinese Military Diplomacy, 2003–2016: Trends and Implications* (Washington, DC: NDU Press, 2017); Jonathan Fulton, *China's Changing Role in the Middle East* (Washington, DC: Atlantic Council, 2019), 4; "Foreign Minister Yang Jiechi, Lebanese Foreign Minister Mansour Exchange Congratulatory Messages on the 40th Anniversary of Diplomatic Ties Between the Two Countries," Chinese Embassy in Lebanon, November 9, 2011, available at <<http://www.china-embassy.org/eng/zgyw/1876100.htm>>.

of December 2019, Russia, China, and Iran held a joint naval exercise in the Gulf of Oman, which Beijing's defense spokesman claimed was aimed at "displaying the three sides' strong will and capabilities to jointly maintain world peace and maritime security."³¹ Both Russia and China have vetoed UN Security Council resolutions that condemned Bashar al-Asad's conduct in the Syria civil war, and both have provided avenues for humanitarian aid.³²

The emergence of alternative external patrons is a boon for regional states vulnerable to domestic unrest and seeking to manage local civil conflicts. Through its economic investments, China has proved to be an attractive economic partner for regional states. For example, Egypt and Saudi Arabia have launched ambitious domestic development programs to stabilize and diversify their economies without political liberalization, approaches that mirror in many ways China's model of economic ascent.³³ Israel has also pursued a relationship with Beijing as a way to diversify its economic portfolio.³⁴ Regional polls suggest positive views of Russia and China in the region; 27 percent of respondents to a regionwide survey viewed Putin favorably, and only 12 percent held the same views of Donald Trump.³⁵ Nearly 50 percent of survey respondents preferred that their country's economic relations with China become stronger in the future, compared with 38 percent for the United States.³⁶

Maintaining a partnership with Russia and China has allowed regional countries to circumvent Western conditions or restrictions on military aid and economic sanctions. The

United Arab Emirates, Egypt, Saudi Arabia, and Iraq have all purchased armed Chinese unmanned aerial vehicles, which the United States is prohibited from selling.³⁷ To dampen the effect of international sanctions, Iran has relied on China as a continued economic lifeline, though China has demonstrated a degree of reticence to deepen economic relations with Tehran.³⁸

The implications and future trajectory of Chinese and Russian engagement in the Middle East are unique. In the most immediate term, Russia's military presence in Syria could constrain U.S. counterterrorism operations.³⁹ Russia can also continue to exploit the relational seams between the United States and its regional partners and allies—in some instances taking advantage of perceived U.S. failures or missteps, both to undermine U.S. standing in the region and to present itself as an all-weather ally.⁴⁰ Nevertheless, Russia lacks the longer term vision, strategic interests, and resources to accomplish much more beyond selectively disrupting U.S. regional initiatives.

The nature of Chinese activities will pose both short-term and longer term challenges for the United States. Any Chinese willingness to purchase Iranian crude oil in defiance of U.S. and international sanctions could continue to insulate Iran from the economic costs of its provocative behavior, just as its weapons sales could fuel risky, destabilizing regional wars. Continued economic investments may also provide China access and leverage to gain competitive advantages over the United States. For instance, Chinese investments in the Suez Canal could allow Beijing to monitor U.S. ships and could also serve as an initial step toward establishing a platform for expeditionary military operations or disrupting U.S. military access. Investments in Israel's technology sector could also allow China to acquire dual-use technologies that it could exploit for military advantage.⁴¹

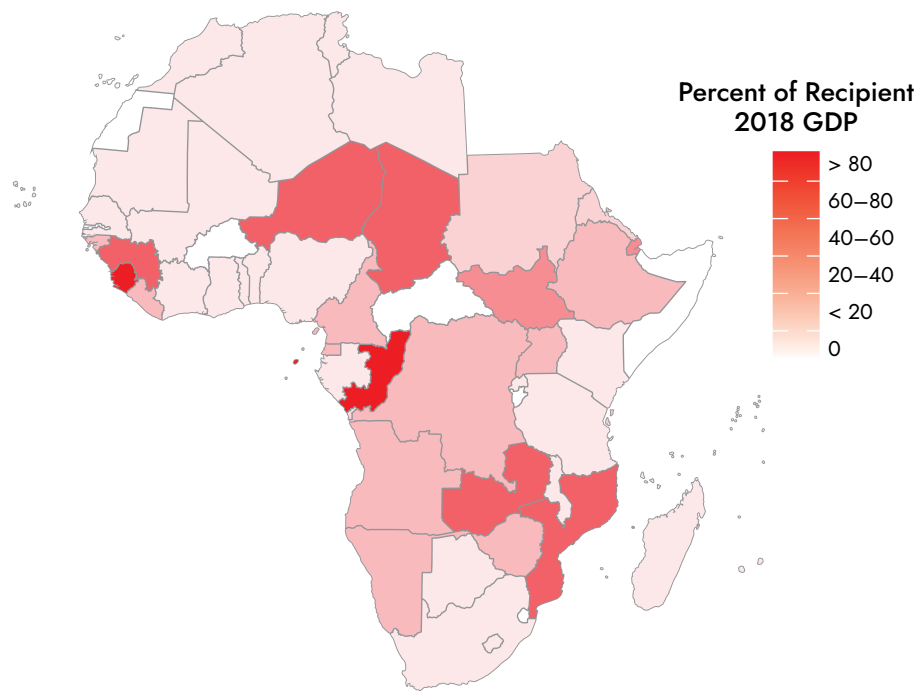
Africa

In Africa, the United States hopes to access the continent's growing economies as markets for American goods and services and to counter violent extremists who could pose a threat to Western interests. Increasingly, U.S. strategic documents highlight competition with China as an interest in and of itself.⁴² Africa receives a major share of U.S. development and humanitarian assistance, which in 2018 amounted to more than \$10 billion, and the United States has instituted several capacity-building programs to help establish more proficient African security forces.⁴³ U.S. direct investment in the region is also considerable, amounting to \$47 billion in 2018.⁴⁴

Beijing's interests in Africa span the military, geopolitical, and economic sectors: access to markets, investments, and raw materials—particularly oil and strategic minerals—and ports. The long-term vision of China's Belt and Road Initiative includes embracing Africa as part of its expanding web of Silk Roads, for both digital and maritime byways.⁴⁵ In 2018 alone, China invested on the African continent \$26 billion, a significant portion of recipient states' gross domestic products (GDPs) (see figure 13.3).⁴⁶ Yet China likely views Africa as a strategically important locus for military access as well, having established a base in Djibouti in 2017 and expanded its military presence in the Red Sea region.⁴⁷

Chinese investment in Africa is increasingly diverse (beyond natural resources), private sector based (within the blurred Chinese context), and integrated with local labor. China is also the largest creditor in Africa, controlling about one-third of African external debt.

Figure 13.3. Total Chinese Investment and Construction in Africa (2005–2018) as a Percent of Recipient 2018 GDP



Sources: GDP data retrieved from World Bank Databank, available at <<https://databank.worldbank.org/home.aspx>>. Eritrea GDP is from 2011 and South Sudan GDP is from 2015, the last years data were available. Chinese investment data are from Derek Scissors, "China Global Investment Tracker," American Enterprise Institute, available at <<https://www.aei.org/china-global-investment-tracker/>>.

Chinese tech giant Huawei provides 4G networks to over half the continent.⁴⁸ In 2017, China's trade with Africa was more than four times greater than that of the United States, and commercial relationships have matured beyond extractives.⁴⁹ Ongoing since October 2000, the annual Forum on China-Africa Cooperation also draws virtually every African head of state for discussions of loans, grants, and pledges of noninterference. Chinese leaders regularly visit the continent. China has also built some 70 Confucius Institutes in Africa—the most of any region explored in this chapter—all while tens of thousands of Africans study in China.⁵⁰ On the security front, the People's Liberation Army participates actively in UN peacekeeping operations, and China has become the largest arms exporter to sub-Saharan Africa, including heavy and advanced weaponry.⁵¹ Finally, China is reportedly providing software and artificial intelligence technologies that buttress repression in certain African countries while affording Beijing important data-collection opportunities.⁵²

Although China's involvement tends to align with a desire for stability in Africa, Russia's interests and activities seem to feed on regional insecurity. Africa offers Russia a source of patronage for Kremlin cronies through business deals and private security company contracts. Moreover, Russia likely views the African continent as an attractive source of mineral resources and, ultimately, as a region that has an opportunity to undermine Western and U.S. interests at moderately low cost.⁵³

Russia's reentry into African affairs, after a relative post-Cold War hiatus, is smaller than that of China; nonetheless, its maneuvers can have a negative effect on governance and stability. Russia's approach often opportunistically targets fragile or corrupt partners.⁵⁴ This pattern stretches from Libya, where Russian mercenaries are fighting with Khalifa Haftar against the UN-backed government; to the Central African Republic, where Russian funding, weapons, and mercenaries have yielded mineral rights and political influence; and to Madagascar, one of many states where Kremlin operatives allegedly deployed cash and fake news in recent elections. Africa also offers geopolitical opportunities, and Russia's engagement employs a range of diplomatic and military tools to exploit them. In addition to an October 2019 summit in Sochi with more than 40 African leaders, Russia also offers educational opportunities, arms sales, and some debt relief.⁵⁵

Notably, African states do not perceive themselves in a three-way zero-sum game of Great Power competitive influence among Russia, China, and the United States. Africa's 54 countries are actively engaging with external actors in terms of their own priorities, with varying levels of success and transparency. African leaders seek infrastructure, economic development, and job creation for the 33,000 young people entering the labor market every day.⁵⁶ For those nations seeking surveillance technology or arms, Russia and China have offerings. Countries facing terrorist threats or instability also may view Russian and Chinese training, equipment, and support for peace operations as an alluring alternative to U.S. assistance, which is often viewed as slow and restrictive. Chinese activities appear to have yielded some local goodwill; according to a 2014–2015 public opinion survey, nearly two-thirds of Africans regard China's influence as "somewhat" or "very" positive.⁵⁷

The consequences of Russian and Chinese activities in Africa have thus manifested themselves in distinct ways. Russian interventions in local civil conflicts and political interference serve to prop up corrupt, unstable regimes in the region, and Russia's alleged involvement in more than a dozen elections can undermine more democratic states as well. Africa has provided Russia access to valuable mineral resources. China, too, has been able to secure key minerals through its activities in Africa, including reportedly taking control of more than 52 percent of the cobalt production in the Democratic Republic of the Congo.⁵⁸ Although few African countries are currently classified as being under debt stress to China, growing debt burdens will affect national development prospects and international relationships and thus create at least the potential for future pressure or exploitation.⁵⁹ China indeed already appears to have been able to leverage its diplomatic and economic investments. In 1971, 20 African countries recognized Taiwan; today, only Eswatini (formerly Swaziland) does.⁶⁰ As in Latin America, Chinese economic engagement and political pressure in Africa have generated important diplomatic results. Ultimately, Chinese and Russian activities on the continent, in the emerging era of Great Power competition, enhance access to resources and offer opportunities for political coercion.

The Arctic

The United States relies on access to the Arctic to safeguard its freedom of navigation and overflight, both of which allow it to project power globally.⁶¹ The United States has come to view the Arctic as a region that could afford important strategic advantages from both Russia and China, describing the Arctic as a competitive rather than cooperative space. For

example, Secretary of State Mike Pompeo, in his remarks at the May 2019 Arctic Council ministerial meeting, rejected China's claim to be a near-Arctic state, warning about growing Chinese influence in the region while also pointing to Russian actions in the Northern Sea Route (NSR) as "part of a pattern of aggressive Russian behavior here in the Arctic."⁶² The June 2019 Department of Defense Arctic Strategy echoes this sentiment, calling the Arctic "a potential avenue for expanded Great Power competition and aggression," noting that the United States has an interest in "limiting the ability of China and Russia to leverage the region."⁶³

The United States has historically pursued its interests in the Arctic through cooperation and consensus-building in the Arctic Council. As the United States has come to view the Arctic region through a competitive lens in the emerging era of Great Power rivalry, Washington has increasingly preferred unilateral or bilateral actions in the region.⁶⁴ In August 2019, President Trump offered to buy Greenland from Denmark, rather than working cooperatively with Denmark on joint investment. In short, U.S. leaders no longer see the Arctic as a zone of peace and stability; rather, they see it as another competitive arena that is best addressed through unilateralism.

In January 2018, China introduced its Arctic Policy, declaring itself to be a "near Arctic" state, affording itself weight in regional governance decisions.⁶⁵ According to this policy, all Arctic stakeholders should "ensure that the benefits are shared by both Arctic and non-Arctic States as well as by non-state entities, and should accommodate the interests of local residents including the indigenous peoples."⁶⁶ For China, the Arctic is attractive, in part, due to the potential viability of maritime shipping routes, which could shorten the shipping time between China and Europe by 2 weeks.⁶⁷ Given the region's abundance of natural resources, China also has expressed an interest in "resource exploration and exploitation."⁶⁸

China has relied on investing in research stations and economic instruments of statecraft to pursue these Arctic interests. This approach has included steadily increasing the manpower in the Polar Research Institute of China, headquartered in Shanghai, which oversees China's increasing number of ice-breaking research transits and manages Chinese research stations in the Arctic.⁶⁹ Moreover, from 2012 to 2017, China invested \$2 billion in Greenland and \$1.2 billion in Iceland, constituting 11.6 percent and 5.7 percent of each country's respective GDP.⁷⁰

The official responses to Chinese behavior are mixed and, where positive, are likely driven by demand for Chinese investment rather than genuine political alignment with Beijing. For instance, China's activities in the Arctic found a somewhat sympathetic ear from officials with responsibility for Arctic scientific research, fisheries management, and shipping standards, in addition to some politicians in Greenland and Iceland, all seeking Chinese investment.⁷¹ Arctic nations, however, have rejected China's appeals for inclusive governance, viewing Chinese research behavior as a potential mask for longer term security goals.⁷² Most famously, Secretary Pompeo singled out China in May 2019, stating that "China's words and actions raise doubts about its intentions" in the region.⁷³

Unlike those of China or the United States, core Russian economic, security, and ideological interests are directly at stake in the Arctic. A significant portion of Russia's gross domestic product—by some estimates more than 30 percent—comes from natural resources in the Arctic.⁷⁴ Russia's own longstanding Policy for the Arctic calls for the "transformation

of the Arctic zone . . . into a leading strategic resource base of the Russian Federation.”⁷⁵ Russia also must maintain a military presence along the NSR to protect its nuclear deterrent, as it is home to Russia’s North Sea Fleet, including a large portion of ballistic missile submarines.⁷⁶ Finally, Russia’s Great Power ambitions and national identity as an Arctic nation are intertwined with its behavior in the region.⁷⁷

Even so, Russia has a history of constructive interactions within the Arctic Council and routinely expresses a desire to maintain peace and stability in the region.⁷⁸ To defend its critical security interests in the Arctic, Russia has become more assertive: It has claimed unilateral control of its Arctic waters, and specifically the NSR along its northern coast, fortifying its presence in the area while also building at least seven additional bases since 2013.⁷⁹ Russia has demanded that transiting ships pay tolls, take aboard a Russian ice pilot, and allow inspections when required. Noncompliant ships are subject to being stopped, boarded, impounded, and even destroyed. Russia justifies these actions under Article 234 of the UN Convention on the Law of the Sea, which states that countries with ice-covered waters in their exclusive economic zones—but beyond their territorial waters—can regulate those zones if doing so will protect the environment.⁸⁰

Russian and Chinese activities in the Arctic will thus likely produce a different set of challenges for the United States in this emerging era of Great Power competition. China could leverage its sizable investments in Greenland and Iceland to constrain the U.S. military’s regional access or monitor its behavior for future strategic advantage. Conversely, Russia already maintains a substantial material advantage in the Arctic vis-à-vis the United States, which could complicate the U.S. ability to challenge Russia’s claim to the NSR and obstructions to freedom of navigation.

Conclusion and Implications

Expanding the aperture of Great Power competition beyond Russia and China’s “near abroad,” this chapter explores the competitive advantages these two countries continue to gain through their engagements in Latin America, the Middle East, Africa, and the Arctic. The analysis reveals considerable diversity in the nature, intensity, and potential stakes in Great Power rivalries across these regions, which, in turn, carries several implications for U.S. policymakers.

First, although these countries will at times coordinate to frustrate U.S. objectives or pursue their own, Russia and China often present distinct competitive threats to the United States. In many regions, Russia poses the more immediate challenge, whereas the repercussions of Chinese economic investments manifest themselves subtly and will likely undermine U.S. strategic interests more gradually. Beyond differences in temporal urgency, there remains daylight between the two competitors’ long-term strategic visions and behavior. Both countries are only nominally united in their desire to compete with and displace U.S. influence across Latin America, the Middle East, Africa, and the Arctic. China’s behavior is grounded in its global investment strategy and desire to create a Sino-centric international political order, whereas Russia’s desire to be a global Great Power is not based in a proactive vision of a new global geopolitical order.⁸¹ China often invests broadly within these regions, whereas Russia’s capacity limitations compel Moscow to be more discerning in its partnerships. China’s tool kit relies on regional stability; Russia’s

often produces instability. Both nations sometimes seek to extract the same resources from the same region, which may bring them into zero-sum competition.

Even though Russia and China are conflated in U.S. strategic documents, the instruments and overall strategic approach required to effectively compete with, manage, and counter Russian influence will not necessarily translate into progress against China's activities, and vice versa. Addressing the Russian rivalry may require military and economic assistance designed to insulate key U.S. partners from the spillover effects of Moscow's behavior, while also conveying U.S. staying power and commitments to allies who may view Russia as more reliable. Competition with China will not only necessitate greater emphasis on economic, diplomatic, and other soft-power instruments of U.S. national influence but also require the United States to recognize select shared interests with China, particularly when it comes to containing regional pandemics, natural disasters, maritime piracy, and terrorist threats. The United States should adopt parallel but coordinated strategies to compete with Beijing and Moscow—approaches that recognize the more immediate threat posed by the latter and the longer time horizon of the former.

Second, the stakes involved in competition across these regions are uneven for the United States. In the Middle East, the United States risks losing access to strategic maritime chokepoints. In Africa, growing Russian and Chinese influence undermines the stability of U.S. partners and allows China to fuel its economic growth by accessing and extracting minerals. In Latin America, the risks to the United States relate more to the externalities of regional instability within the U.S. backyard; in the Arctic, more assertive policies by Russia and China could limit U.S. access to and freedom of navigation.

In terms of intrinsic American interests, deciding which of these consequences deserves higher priority necessitates a reevaluation of U.S. strategic interests across and within each of these regions in light of Great Power competition. Put simply, U.S. policymakers need to refresh how they define these interests both globally and regionally, viewing them through the lens of Great Power competition rather than counterterrorism. The presence of Russian or Chinese activities should alone be insufficient to warrant prioritization or a competitive response from the United States. Instead, U.S. policymakers need to adopt a more discerning approach grounded in each region's geopolitical significance, thinking carefully about ramifications of Russian and Chinese success. This pivot should help the United States avoid unnecessarily wasting finite time and resources competing with Russia and China in areas or countries of limited importance.

Finally, states in Latin America, the Middle East, Africa, and the Arctic are often eager recipients of Russian and Chinese attention and resources due to convenience rather than to ideological commitment. With few exceptions, many countries' willingness to accept these resources is not grounded in a sincere dedication to China's global vision or Russia's cynicism vis-à-vis Western norms and institutions; instead, it is often the scale and unrestricted nature of Russian and Chinese largesse that makes each country an attractive partner. With this convenience sometimes come hidden costs and clear limitations. China, for instance, continues to leverage economic investments to coerce recipient states and, without a military presence, cannot offer its partners credible security guarantees. Russia has less material power to invest globally.

These motivations and limits in Russia's and China's respective approaches—as well as the hedging strategies that smaller states have adopted as they balance relationships with Russia, China, and the United States—warrant a cautious response from Washington. Where necessary, policymakers should emphasize U.S. strengths as an economic partner, juxtaposing the profit-driven motives of American private sector investment with the risks of China's sometimes more coercive state-driven model; they should also highlight the long history of U.S. foreign assistance versus the short track record of China's economic activities. The United States also has an advantage in the quality and transparency of its military assistance, as well as its military superiority and forward presence.

Each of these advantages can convey a strong commitment to U.S. partners that is not evident from Russia or China. The United States should avoid, however, systematically imposing a regional strategy that views Russian or Chinese activities as uniformly harmful to U.S. interests and detrimental to the stability of recipient states across these regions. Doing so might unnecessarily damage U.S. relations with key partners and allies seeking sources of economic investment, military hardware, or alternative political forums. Few, if any, countries wish to be pulled into a zero-sum U.S.-China or U.S.-Russia competition in this dawning era of Great Power competition.

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Notes

¹ For overviews, see, for example, David Shambaugh, *China Goes Global* (New York: Oxford University Press, 2013); Scott McDonald and Michael Burgoyne, eds., *China's Global Influence: Perspectives and Recommendations* (Washington, DC: Asia-Pacific Center for Security Studies, 2019); Phillip C. Saunders, *China's Global Activism: Strategy, Drivers, and Tools* (Washington, DC: NDU Press, 2006); Robert Sutter, *Chinese Foreign Relations: Power and Policy Since the Cold War* (New York: Rowman and Littlefield, 2012); Peter B. Zwack and Marie-Charlotte Pierre, *Russian Challenges from Now into the Next Generation: A Geostrategic Primer*, INSS Strategic Perspectives 29 (Washington, DC: NDU Press, 2019); Paul Stronski and Richard Sokolsky, *The Return of Global Russia: An Analytic Framework* (Washington, DC: Carnegie Endowment for International Peace, 2017).

² The challenge of Russia and China is described in each of the geographic combatant command's posture statements, as well as in the Department of Defense Arctic Strategy. See Admiral Craig S. Faller, USN, *Posture Statement of Admiral Craig S. Faller, United States Southern Command, Before the 116th Congress Senate Armed Services Committee*, February 7, 2019, available at <www.southcom.mil/Portals/7/Documents/Posture%20Statements/SOUTHCOM_2019_Posture_Statement_Final.pdf>; General Joseph L. Votel, USA, *Statement of General Joseph L. Votel, Commander, U.S. Central Command, Before the House Armed Services Committee on the Posture of U.S. Central Command Great Power Competition: The Current and Future Challenges in the Middle East*, March 7, 2019, available at <https://armedservices.house.gov/_cache/files/0/2/02f463d7-e06a-4a7f-8004-4bb619eb17e3/F446D6AEBBD598956E1429E5AD3243AB.centcom-posture-statement-hasc-march-7.pdf>; General Thomas D. Waldhauser, USMC, *2019 Posture Statement to Congress*, February 7, 2019, available at <www.africom.mil/about-the-command/2019-posture-statement-to-congress>; *Report to Congress on the Department of Defense Arctic Strategy*

(Washington, DC: Department of Defense, June 2019), available at <<https://media.defense.gov/2019/Jun/06/2002141657/-1/-1/1/2019-DOD-ARCTIC-STRATEGY.PDF>>.

³ In this chapter, we define Latin America as an area including Mexico, as well as states in Central America, the Caribbean, and South America.

⁴ *National Security Strategy of the United States of America* (Washington, DC: The White House, December 2017), 51, available at <www.whitehouse.gov/wp-content/uploads/2017/12/NSS-Final-12-18-2017-0905.pdf>.

⁵ U.S. Agency for International Development (USAID) Economic Analysis and Data Services, *U.S. Overseas Loans and Grants, Obligations, and Loan Authorizations Greenbook*, available at <www.usaid.gov/developer/greenbookapi>; U.S. Trade Representative, "Free Trade Agreements," available at <<https://ustr.gov/trade-agreements/free-trade-agreements>>; Peter J. Meyer, *U.S. Strategy for Engagement in Central America: Policy Issues for Congress*, R44812 (Washington, DC: Congressional Research Service, November 2019), available at <<https://crsreports.congress.gov/product/pdf/R/R44812>>.

⁶ "Latinobarometer 2018," available at <www.latinobarometro.org>.

⁷ Of the \$141.3 billion of Chinese investment, \$67.2 billion was loaned to Venezuela, \$29 billion to Brazil, and \$18.4 billion to Ecuador. See China-Latin America Finance Database, *Dialogue*, available at <www.thedialogue.org/map_list/>. U.S. loans and grants data are from USAID Economic Analysis and Data Services, *U.S. Overseas Loans and Grants, Obligations, and Loan Authorizations Greenbook*.

⁸ Admiral Craig S. Faller, USN, *Commander U.S. Southern Command Testimony Before the Senate Armed Services Committee*, July 9, 2019, available at <www.armed-services.senate.gov/imo/media/doc/Faller_07-09-19.pdf>.

- ⁹ Stockholm International Peace Research Institute (SIPRI) Arms Transfers database, available at <www.sipri.org/databases/armstransfers>; "Confucius Institutes Around the World—2019," *Digmandarin*, February 15, 2020, available at <www.digmandarin.com/confucius-institutes-around-the-world.html>.
- ¹⁰ Douglas Farah and Kathryn Babineau, "Extra-Regional Actors in Latin America: The United States Is Not the Only Game in Town," *PRISM* 8, no. 1 (2019), 96–114.
- ¹¹ Faller, *Commander U.S. Southern Command Testimony Before the Senate Armed Services Committee*.
- ¹² "Latinobarometer 2018."
- ¹³ China's other strategic partners include Argentina, Brazil, Chile, Costa Rica, Ecuador, Mexico, and Uruguay. See Mark P. Sullivan and Thomas Lum, "China's Engagement with Latin America and the Caribbean," Congressional Research Service, last updated April 11, 2019, available at <<https://crsreports.congress.gov/product/pdf/IF/IF10982>>.
- ¹⁴ Lisa Viscidi and Nate Graham, *Reviving Venezuela's Oil Sector: The Role of Western Oil Majors* (Washington, DC: Inter-American Dialogue, January 2020), available at <www.thedialogue.org/wp-content/uploads/2020/01/VENEZUELA-FINAL-PDF-1.pdf>.
- ¹⁵ "China to Lend Venezuela \$5 Billion as Maduro Visits Beijing," *Bloomberg*, September 13, 2018, available at <www.bloomberg.com/news/articles/2018-09-13/china-to-give-venezuela-5-billion-loan-as-maduro-visits-beijing>; Jeremy Page, "China Counts the Costs of Its Big Bet on Venezuela," *Wall Street Journal*, February 1, 2019, available at <<https://www.wsj.com/articles/china-counts-the-costs-of-its-big-bet-on-venezuela-11549038825>>.
- ¹⁶ Faller, *Commander U.S. Southern Command Testimony Before the Senate Armed Services Committee*.
- ¹⁷ "Factbox: Oil, Loans, Military—Russia's Exposure to Venezuela," *Reuters*, January 24, 2019, available at <www.reuters.com/article/us-venezuela-politics-russia-factbox/factbox-oil-loans-military-russias-exposure-to-venezuela-idUSKCN1PI1T4>.
- ¹⁸ Michael D. McDonald and Keith Zhai, "Panama to Establish Ties with China in Latest Blow to Taiwan," *Bloomberg*, June 12, 2017, available at <www.bloomberg.com/news/articles/2017-06-13/panama-to-open-ties-with-china-in-blow-to-taiwan-report-says>; Steven Jiang, "China Snatches Dominican Republic from Taiwan in Diplomatic Blow," *CNN*, May 1, 2018, available at <www.cnn.com/2018/05/01/asia/china-taiwan-dominican-republican-intl/index.html>; Douglas Farah and Caitlyn Yates, *El Salvador's Recognition of the People's Republic of China: A Regional Context*, INSS Strategic Perspectives 30 (Washington, DC: NDU Press, 2019).
- ¹⁹ Of note, Brazil is an interesting outlier in this regard, aligning more closely with the United States. See Ney Hayashi Cruz, "Brazilian President Streams Himself Watching Trump's Iran Speech," *Bloomberg*, January 8, 2020, available at <www.bloomberg.com/news/articles/2020-01-08/bolsonaro-streams-himself-watching-trump-s-speech-on-iran>.
- ²⁰ *National Security Strategy of the United States of America*, 48–50.
- ²¹ Votel, *Statement of General Joseph L. Votel, Commander, U.S. Central Command, Before the House Armed Services Committee on the Posture of U.S. Central Command Great Power Competition*.
- ²² SIPRI Arms Transfers Database.
- ²³ Votel, *Statement of General Joseph L. Votel, Commander, U.S. Central Command, Before the House Armed Services Committee on the Posture of U.S. Central Command Great Power Competition*; U.S. Department of Commerce, Bureau of Economic Activity, "Direct Investment by Country and Industry," July 24, 2019, available at <www.bea.gov/data/intl-trade-investment/direct-investment-country-and-industry>; USAID Economic Analysis and Data Services, *U.S. Overseas Loans and Grants, Obligations, and Loan Authorizations Greenbook*.
- ²⁴ *Annual Report to Congress on the Military and Security Developments Involving the People's Republic of China 2019* (Washington, DC: Office of the Secretary of Defense, May 2019), available at <https://media.defense.gov/2019/May/02/2002127082/-1/1/2019_CHINA_MILITARY_POWER_REPORT.pdf>.
- ²⁵ Derek Scissors, "China Global Investment Tracker," American Enterprise Institute, Washington, DC, available at <www.aei.org/china-global-investment-tracker>; Brett McGurk, "China's Risky Middle East Bet," *The Atlantic*, April 29, 2019, available at <www.theatlantic.com/ideas/archive/2019/04/chinas-risky-middle-east-bet/588166/>.
- ²⁶ *The China-Arab States Cooperation Forum* (Rio de Janeiro, Brazil: BRICS Policy Center, May 2016), available at <www.bricspolicycenter.org/en/forum-de-cooperacao-china-paises-arabes/>.
- ²⁷ *Annual Report to Congress on the Military and Security Developments Involving the People's Republic of China 2019*.
- ²⁸ "Xi Jinping Delivers Important Speech at Headquarters of the League of Arab States, Stressing to Jointly Create a Bright Future for Development of China-Arab Relations and Promote National Rejuvenation of China and Arab States to Form More Convergence," Ministry of Foreign Affairs of the People's Republic of China, January 1, 2016, available at <www.fmprc.gov.cn/mfa_eng/topics_665678/xjpdstajylxgsw/t1334587.shtml>; *China's Arab Policy Paper* (Beijing: Ministry of Foreign Affairs of the People's Republic of China, January 2016), available at <www.fmprc.gov.cn/mfa_eng/zxxx_662805/t1331683.shtml>.
- ²⁹ "Confucius Institutes Around the World—2019," *Digmandarin*, February 23, 2019, available at <www.digmandarin.com/confucius-institutes-around-the-world.html>.
- ³⁰ For overviews, see Dmitri Trenin, *What Is Russia Up To in the Middle East?* (New York: John Wiley & Sons, 2017); Eugene Rumer, *Russia in the Middle East: Jack of All Trades, Master of None* (Washington, DC: Carnegie Endowment for International Peace, 2019); Mark N. Katz, *Support Opposing Sides Simultaneously: Russia's Approach to the Gulf and the Middle East* (Doha: Al-Jazeera Center, August 2018); Becca Wasser, *The Limits of Russian Strategy in the Middle East* (Santa Monica, CA: RAND, 2019).
- ³¹ Ben Westcott and Hamdi Alkhshali, "China, Russia, and Iran Hold Joint Naval Drills in Gulf of Oman," *CNN*, December 27, 2019, available at <www.cnn.com/2019/12/27/asia/china-russia-iran-military-drills-intl-hnk/index.html>. For an overview of Iran-Russia-Chinese cooperation, see Ariane Tabatabai and Dina Esfandiary, *Triple-Axis: Iran's Relations with Russia and China* (New York: Bloomsbury Publishing, 2018).
- ³² Tom O'Connor, "Russia Not Alone in Syria's West, China and Iran Are Also Moving in Near Sea," *Newsweek*, April 23, 2019, available at <www.newsweek.com/russia-west-syria-war-china-iran-1404051>; "Russia and China Block Cross-Border Aid Deliveries to Syria," *New York Times*, December 20, 2019, available at <www.nytimes.com/2019/12/20/world/middleeast/syria-aid-russia-un.html>.
- ³³ "President El Sisi Unveils 'Egypt Vision 2030' Sustainable Development Strategy," Embassy of Egypt, February 24, 2016, available at <www.egyptembassy.net/news/news/president-el-sisi-unveils-egypt-vision-2030-sustainable-development-strategy/>; "Kingdom of Saudi Arabia Vision 2030," available at <<https://vision2030.gov.sa/en/themes/2/>>.
- ³⁴ Shia Efron et al., *The Evolving Israel-China Relationship* (Santa Monica, CA: RAND, 2019).
- ³⁵ *Arab Barometer Wave V (2018–2019)*, available at <www.arabbarometer.org/survey-data/data-downloads/>.

³⁶ Ibid.

³⁷ *Assessment on U.S. Defense Implications of China's Expanding Global Access* (Washington, DC: Department of Defense, December 2018), available at <<https://media.defense.gov/2019/Jan/14/2002079292/-1/-1/1/EXPANDING-GLOBAL-ACCESS-REPORT-FINAL.PDF>>. China has been willing to make these arms sales despite its pledges to adhere to the Missile Technology Control Regime. See Michael D. Swaine and Alastair Iain Johnston, "China and Arms Control Institutions," in *China Joins the World: Progress and Prospects*, ed. Elizabeth Economy and Michael Oksenberg (New York: Council on Foreign Relations Press, 1999), 112–114.

³⁸ See chapter 10 of this volume, and Dan Katz, "Despite Sanctions, China Is Still Doing (Some) Business with Iran," Atlantic Council, October 1, 2019, available at <www.atlanticcouncil.org/blogs/iran-source/despite-sanctions-china-is-still-doing-some-business-with-iran/>.

³⁹ Prior to the raid that killed so-called Islamic State's Abu Bakr al-Baghdadi, the United States indeed needed to deconflict its operations with both Turkey and Russia. See Jim Garamone, "Central Command Chief Gives Details on Baghdad Raid," *Defense.gov*, October 30, 2019, available at <www.defense.gov/explore/story/Article/2003960/central-command-chief-gives-details-on-baghdadi-raid/>.

⁴⁰ "Russia Ready to Sell Its Advanced Weaponry to Saudi Arabia—Putin," TASS, September 16, 2019, available at <<https://tass.com/defense/1078433>>; James Dorsey, *Will There Be a New Russian-Chinese Security Architecture in the Gulf?* Begin-Sadat Center for Strategic Studies (BESA) Perspectives Paper 1287 (Ramat Gan, Israel: BESA, September 2019), available at <<https://besa-center.org/wp-content/uploads/2019/09/1287-Security-Architecture-in-the-Gulf-Dorsey-final.pdf>>; Wasser, *The Limits of Russian Strategy in the Middle East*.

⁴¹ Shia Efron et al., *The Evolving Israel-China Relationship*; see also "DASD Michael Mulroy Opening Statement: Israel and Great Power Competition," *Foundation on the Defense of Democracy*, July 31, 2019, available at <www.fdd.org/wp-content/uploads/2019/07/DASD_Mulroy_Remarks.pdf>.

⁴² The *National Security Strategy* notes, "We will offer American goods and services [in Africa], both because it is profitable for us and because it serves as an alternative to China's often extractive economic footprint." See *National Security Strategy of the United States of America*. See also "President Donald J. Trump's Africa Strategy Advances Prosperity, Security, and Stability," fact sheet, The White House, December 13, 2018, available at <www.whitehouse.gov/briefings-statements/president-donald-j-trumps-africa-strategy-advances-prosperity-security-stability/>; Tibor P. Nagy, "The Enduring Partnership Between the United States and South Africa," remarks at the University of the Witwatersrand, Johannesburg, South Africa, June 21, 2019, available at <www.state.gov/the-enduring-partnership-between-the-united-states-and-south-africa/>; *Statement of General Thomas D. Waldhauser, United States Marine Corps, Commander, United States Africa Command, Before the Senate Committee on Armed Services*, February 7, 2019, available at <www.africom.mil/about-the-command/2019-posture-statement-to-congress>.

⁴³ For an overview, see U.S. Africa Command, "What We Do—Security Cooperation," n.d., available at <www.africom.mil/what-we-do/security-cooperation>. For an assessment of these programs, see Steve Watts et al., *Building Security in Africa: An Evaluation of U.S. Security Sector Assistance in Africa from the Cold War to the Present* (Santa Monica, CA: RAND, 2018).

⁴⁴ Department of Commerce Bureau of Economic Activity, "Direct Investment by Country and Industry," July 24, 2019, available at <www.bea.gov/data/intl-trade-investment/direct-investment-country-and-industry>.

⁴⁵ Paul Nantulya, "Chinese Hard Power Supports Its Growing Strategic Interests in Africa," Africa Center for Strategic Studies, Washington, DC, January 17, 2019, available at <<https://africacenter.org/spotlight/chinese-hard-power-supports-its-growing-strategic-interests-in-africa/>>; *Assessment on U.S. Defense Implications of China's Expanding Global Access*; see also Larry Hanauer and Lyle J. Morris, *Chinese Engagement in Africa: Drivers, Reactions, and Implications for U.S. Policy* (Santa Monica, CA: RAND, 2014).

⁴⁶ Scissors, "China Global Investment Tracker."

⁴⁷ Sarah Zheng, "China's Djibouti Military Base: 'Logistics Facility,' or Platform for Geopolitical Ambitions Overseas?" *South China Morning Post* (Hong Kong), October 1, 2017, available at <www.scmp.com/news/china/diplomacy-defence/article/2113300/chinas-djibouti-military-base-logistics-facility-or>; Joel Wuthnow, *The PLA Beyond Asia: China's Growing Military Presence in the Red Sea Region*, INSS Strategic Forum 303 (Washington, DC: NDU Press, January 2020).

⁴⁸ Phillip Carter III et al., "China in Africa: Opportunities, Challenges, and Options," in *China's Global Influence: Perspectives and Recommendations*, ed. Scott D. McDonald and Michael C. Burgoyne (Honolulu: Daniel K. Inouye Asia-Pacific Center for Security Studies, 2019), 111–113; "Huawei Strengthens Foothold in Africa to Offset U.S. Ban," *Agence France-Presse*, June 9, 2019, available at <www.scmp.com/news/world/africa/article/3013717/huawei-strengthens-foothold-africa-amid-us-ban>.

⁴⁹ Judd Devermont, *The World Is Coming to Sub-Saharan Africa. Where Is the United States?* Center for Strategic and International Studies (CSIS) Briefs (Washington, DC: CSIS, August 24, 2018), available at <www.csis.org/analysis/world-coming-sub-saharan-africa-where-united-states>; Carter et al., "China in Africa," in McDonald and Burgoyne, *China's Global Influence*, 111; Luke Patey, "The Chinese Model Is Failing Africa," *Financial Times*, August 26, 2018, available at <www.ft.com/content/ca4072f6-a79f-11e8-ab66-f368d365bf0e>.

⁵⁰ Shannon Tiezzi, "FOCAC 2018: Rebranding China in Africa," *The Diplomat*, September 5, 2018, available at <<https://thediplomat.com/2018/09/focac-2018-rebranding-china-in-africa/>>; Lina Benabdallah, "Spite Won't Beat China in Africa," *Foreign Policy*, January 23, 2019, available at <<https://foreignpolicy.com/2019/01/23/spite-wont-beat-china-in-africa/>>.

⁵¹ Nantulya, "Chinese Hard Power"; Pieter D. Wezeman et al., *Trends in International Arms Transfers, 2017* (Stockholm: SIPRI, 2018), available at <www.sipri.org/publications/2018/sipri-fact-sheets/trends-international-arms-transfers-2017>.

⁵² Steven Feldstein, *The Global Expansion of AI Surveillance* (Washington, DC: Carnegie Endowment for International Peace, September 2019), available at <https://carnegieendowment.org/files/WP-Feldstein-AISurveillance_final.pdf>.

⁵³ Paul Stronski, *Late to the Party: Russia's Return to Africa* (Washington, DC: Carnegie Endowment for International Peace, October 16, 2019), available at <<https://carnegieendowment.org/2019/10/16/late-to-party-russia-s-return-to-africa-pub-80056>>.

⁵⁴ Adrian Blomfield and Peta Thornycroft, "Russia Planned to Influence South African Elections in Favor of the Ruling ANC, Report Suggests," *Telegraph* (London), May 8, 2019, available at <www.telegraph.co.uk/news/2019/05/08/russia-planned-influence-south-african-elections-favour-ruling/>; Megan Trimble, "Fake News Found in 16 Countries' Elections," *U.S. News and World Report*, November 14, 2017, available at <<https://www.usnews.com/news/best-countries/articles/2017-11-14/report-russia-like-election-meddling-discovered-in-16-countries>>.

⁵⁵ Sudarsan Raghavan, "Arrival of Russian Mercenaries Adds Deadlier Firepower, Modern Tactics to Libya's Civil War," *Washington Post*, November 5, 2019, available at <www.washingtonpost.com/world/arrival-of-russian-mercenaries-adds>.

deadlier-firepower-modern-tactics-to-libyas-civil-war/2019/11/05/f330820c-fa03-11e9-9534-e0dbce9f5683_story.html>; Kimberly Marten, "Russia's Use of Semi-State Security Forces: The Case of the Wagner Group," *Post Soviet Affairs* 35, no. 3 (March 2019), 181–204; Stronski, *Late to the Party*; Michael Schwartz and Gaelle Borgia, "How Russia Meddles Abroad for Profit: Cash, Trolls and a Cult Leader," *New York Times*, November 11, 2019, available at <www.nytimes.com/2019/11/11/world/africa/russia-madagascar-election.html>.

⁵⁶ Siddharth Chatterjee, "Promise or Peril? Africa's 830 Million Young People by 2050," United Nations Development Programme Africa, August 12, 2017, available at <www.africa.undp.org/content/rba/en/home/blog/2017/8/12/Promise-Or-Peril-Africa-s-830-Million-Young-People-By-2050.html>.

⁵⁷ Mogopodi Lekorwe, "China's Growing Presence in Africa Wins Largely Positive Popular Reviews," *Afrobarometer*, October 24, 2016, available at <http://afrobarometer.org/sites/default/files/publications/Dispatches/ab_r6_dispatchno122_perceptions_of_china_in_africa1.pdf>. There is some evidence that this may be changing, as suggested by the electoral rhetoric and platforms in Zimbabwe, some of which evoked anti-China sentiment and promises to remove Chinese business. See Richard Aidoo, "African Countries Have Started to Push Back Against Chinese Development Aid: Here's Why," *Washington Post*, October 16, 2018, available at <www.washingtonpost.com/news/monkey-cage/wp/2018/10/16/african-countries-have-started-to-push-back-against-chinese-development-aid-heres-why/>.

⁵⁸ FP Analytics, "Mining the Future: How China Is Set to Dominate the Next Industrial Revolution," May 1, 2019, available at <https://foreignpolicy.com/2019/05/01/mining-the-future-china-critical-minerals-metals/>.

⁵⁹ Carter et al., "China in Africa," 112–113; Elvis Mboya, "Rethinking Africa's \$124 Billion Debt to China, Ahead of Beijing's September Summit," *Investigative Africa*, August 2018, available at <https://iaafrica.com/rethinking-africas-120-billion-debt-to-china-ahead-of-beijings-september-summit/>. A recent study concludes that Djibouti, Ethiopia, and Kenya are all at elevated risk of debt distress. See John Hurley, Scott Morris, and Gailyn Portelance, *Examining the Debt Implications of the Belt and Road Initiative from a Policy Perspective*, Center for Global Development (CGD) Policy Paper 121 (Washington, DC: CGD, March 2018), available at <www.cgdev.org/sites/default/files/examining-debt-implications-belt-and-road-initiative-policy-perspective.pdf>.

⁶⁰ Timothy S. Rich and Vasabjit Banerjee, "Running Out of Time? The Evolution of Taiwan's Relations in Africa," *Journal of Current Chinese Affairs* 44, no. 1 (2015), 141–161; "Taiwan on Diplomatic Eggshells with Three Remaining African Allies," *South China Morning Post* (Hong Kong), available at <www.scmp.com/magazines/post-magazine/article/1886336/taiwan-diplomatic-eggshells-three-remaining-african-allies>; Salem Solomon, "Once Influential in Africa, Taiwan Loses All But One Ally," *VOA News*, May 26, 2018, available at <www.voanews.com/africa/once-influential-africa-taiwan-loses-all-one-ally>.

⁶¹ *Report to Congress on the Department of Defense Arctic Strategy*.

⁶² Michael Pompeo, "Looking North: Sharpening America's Arctic Focus," U.S. State Department, May 6, 2019, available at <www.state.gov/secretary/remarks/2019/05/291512.htm>.

⁶³ *Report to Congress on the Department of Defense Arctic Strategy*.

⁶⁴ Established in 1996, the Arctic Council is an "intergovernmental forum promoting cooperation, coordination, and interaction among the Arctic states, Arctic indigenous communities, and other Arctic inhabitants on common Arctic issues, in particular on issues of sustainable development and environmental protection in the Arctic." Its member states include Canada, Denmark, Finland, Iceland, Norway, Russia, Sweden, and the

United States. See "Arctic Council," n.d., available at <https://arctic-council.org/>.

⁶⁵ *China's Arctic Policy* (Beijing: State Council Information Office of the People's Republic of China, January 2018), available at <http://english.gov.cn/archive/white_paper/2018/01/26/content_281476026660336.htm>.

⁶⁶ *Ibid.*

⁶⁷ *United States Coast Guard Arctic Strategic Outlook* (Washington, DC: U.S. Coast Guard, April 2019), available at <www.uscg.mil/Portals/0/Images/arctic/Arctic_Strategic_Outlook_APR_2019.pdf>.

⁶⁸ *China's Arctic Policy*.

⁶⁹ Sherri Goodman and Marisol Maddox, "China's Growing Arctic Presence," Wilson Center Polar Institute, Washington, DC, November 19, 2018, available at <www.wilsoncenter.org/article/chinas-growing-arctic-presence>; Linda Jakobson and Jingchao Peng, *China's Arctic Aspirations*, SIPRI Policy Paper No. 34 (Stockholm: SIPRI, 2012), available at <www.sipri.org/publications/2012/sipri-policy-papers/chinas-arctic-aspirations>.

⁷⁰ Ark E. Rosen and Cara B. Thuringer, *Unconstrained Foreign Direct Investment: An Emerging Challenge to Arctic Security* (Arlington, VA: CNA, November 2017), 54.

⁷¹ Erik Matzen and Tom Daly, "Greenland's Courting of China for Airport Projects Worries Denmark," *Reuters*, March 22, 2018, available at <www.reuters.com/article/china-arctic-greenland/greenlands-courting-of-china-for-airport-projects-worries-denmark-idUSL4N1QP346>.

⁷² Elana Wilson Rowe, *Arctic Governance* (Manchester, UK: Manchester University Press, 2018), 63.

⁷³ Pompeo, "Looking North."

⁷⁴ Tsvetana Paraskova, "Russia's Oil Revenue Is About to Soar," *Business Insider*, May 14, 2018, available at <www.businessinsider.com/russia-oil-revenue-about-to-soar-2018-5>.

⁷⁵ *Russian Federation's Policy for the Arctic to 2020*, March 30, 2009, available at <www.arctic-search.com/Russian+Federation+Policy+for+the+Arctic+to+2020%3E>.

⁷⁶ David Auerswald, "Now Is Not the Time for a FONOP in the Arctic," *War on the Rocks*, October 11, 2019, available at <https://warontherocks.com/2019/10/now-is-not-the-time-for-a-fonop-in-the-arctic/>; Tyler Cross, "Divining Russia's Intentions in the Arctic," *The Maritime Executive*, July 10, 2019, available at <www.maritime-executive.com/editorials/divining-russia-s-intentions-in-the-arctic>.

⁷⁷ Anna Reynolds, ed., *Arctic Narratives and Political Values: Russia, China, and Canada in the High North* (Riga, Latvia: NATO Strategic Communications Centre of Excellence, September 2018), available at <www.stratcomcoe.org/arctic-narratives-and-political-values>.

⁷⁸ See, for example, Vladimir Putin, "Plenary Session of the International Arctic Forum," video, 16:50, St. Petersburg, Russia, April 9, 2019, available at <http://en.kremlin.ru/events/president/news/60250>.

⁷⁹ Natassia Astrasheuskaya and Henry Foy, "Polar Powers: Russia's Bid for Supremacy in the Arctic Ocean," *Financial Times*, April 27, 2019, available at <www.ft.com/content/2fa82760-5c4a-11e9-939a-341f5ada9d40>.

⁸⁰ "Lavrov: Russia Will Ensure Safety of Northern Sea Route," *UAWIRE*, May 7, 2019, available at <https://uawire.org/lavrov-russia-will-ensure-safety-of-northern-sea-route>.

⁸¹ For a more detailed discussion of this difference in Russian and Chinese ideological viewpoints and divergent levels of aspiration to world order leadership, see chapter 3a of this volume.

Part IV

Preparing to Compete

Chapter 14

U.S. Strategies for Competing Against China

By Frank G. Hoffman

This chapter lays out a range of potential strategies, drawn largely from academic literature and security studies, to address approaches for a competitive U.S. response to its main Great Power strategic rival: China. Described are the general outlines of five distinct strategies employing the five elements of strategic interaction defined in chapter 2 of this volume. The strategies are then assessed in general terms for their suitability, feasibility, and sustainability. Each example varies in how it leverages the relative strengths and weaknesses of the protagonists, and how international and domestic support might impact implementation. The author contends that a strategy of enhanced balancing is an appropriate approach.

At the turn of the 21st century, Washington's position on the world stage seemed unrivalled, and analysts sought to preserve and extend the unipolar moment.¹ However, the era of American preeminence proved short lived. As discussed in chapter 2 and developed in chapter 3a, the post–Cold War international order has entered a new historical cycle that U.S. policymakers believe will be characterized by Great Power competition (GPC). Far from being an arcane term, *GPC* has a long pedigree in international relations.² However, after nearly 30 years of unipolar dominance and counterinsurgency and counterterrorism/counterextremist interventions, the United States needs to refurbish its mindset and strategic machinery to engage smartly in the lost art of strategic competition.³

Russia, China, and the United States today jockey globally in the new era of GPC. As noted in chapters 3a and 3b, Russia presents a serious near-term threat to U.S. interests but lacks the capacity to sustain a viable rivalry in the longer term. As elaborated in chapter 9, in the Indo-Pacific region, China's relationship with the United States has most obviously—and somewhat ominously—entered a new phase, one in which competition across the five distinct dimensions of state-to-state interaction is ongoing and one that carries a high risk that competition may turn to confrontation, conflict, or armed clash (see tables 2.2, 3a.1, or 3a.2).⁴ In the Indo-Pacific region, some analysts believe that China perceives U.S. power to have ebbed in its reach and influence and concluded that American primacy is over. According to this school of thought, Beijing's ambitions are expanding in line with its

"The United States was once deeply versed in the challenges of long-term competition due to its 45-year contest with the Soviet Union. And the long history of strategic competition between the great powers offers a wealth of insights that can inform the conduct of modern statecraft. Yet the United States has had the luxury of neglecting its competency in long-term competition for more than a generation in the comparatively benign global environment that emerged after the Cold War ended."

—Hal Brands, "The Lost Art of Long-Term Competition," *The Washington Quarterly* 41, no. 4 (Winter 2018)

growing power.⁵ In this view, China seeks a return to a Great Power status that it enjoyed centuries ago, and its economic clout gives it both a justification and the means to project its power and protect its interests. As detailed in chapter 3a, Beijing's strategic interests will inevitably, in some way, rub up against longstanding U.S. policy preferences and national interests.

This chapter is admittedly and unapologetically Sino-centric. However, this focus is not disproportionate to China's relevance to global stability or future U.S. security strategy. While *Sinolarity*, a world centered on China, is unlikely to emerge, Beijing's rising influence is of concern.⁶ The great geopolitical shift of the next few decades,

and the greatest challenge to continued global stability, will be defined by the relationship between the United States and the People's Republic of China (PRC). It is not the only challenge that the United States faces, as the previous chapters show, but in the long run, it is the most critical.

U.S. strategy documents in 2017 and 2018 formally (albeit belatedly) recognized this reality. The Trump administration's 2017 National Security Strategy (NSS) concluded that "China and Russia want to shape a world antithetical to U.S. values and interests."⁷ China, the NSS notes, wants to displace the United States in the Indo-Pacific region, expand the reach of its state-driven economic system and reorder the region to its advantage, and spread its authoritarian system and corruption. The complementary Pentagon document, the 2018 National Defense Strategy (NDS), focused on adapting the Department of Defense's priorities and reforms to a more contested environment of Great Power rivalry. It too defined China as a strategic competitor and stated the PRC was pursuing military modernization, seeking regional hegemony in the near term, and attempting to displace the United States as a preeminent global power over time.⁸ Some scholars reinforce this assessment: "China wants complete dominance; it wants to force the United States out and become the region's unchallenged political, economic and military hegemon."⁹

American strategists and the international relations community are wrestling with the implications of this pending era. Some disagree entirely with the conclusions about GPC and China's ambitions laid out in the NSS and NDS.¹⁰ Yet stressing competition does not automatically lead to greater tension or catastrophe.¹¹ We should not be afraid of calling China a competitor—which is well short of describing it as a hostile power or a confrontational adversary.¹² Competition vis-à-vis China was predicted a decade ago by some scholars and over 5 years ago by the Council on Foreign Relations.¹³ Even formerly devoted advocates of deep engagement with Russia and China now recognize the need to alter course and actively work to defend U.S. interests. Yet, while this may be a clash of systems over political and ideological differences, it should *not* be considered a clash of civilizations.¹⁴

While China is not dominant today, it is clearly a rising Great Power, signaling a desired power transition. It may never obtain its China Dream, but its exceptional influence on the international system, its aggressive approach to diplomacy, and its military activities all bear watching and appropriate hedging.

Great Power competitions and accompanying power transitions are rarely resolved without a holistic approach that is managed within an appropriate strategic framework. They require leadership involvement, disciplined priorities and sustainable resourcing, and adaptive oversight. As noted in the U.S. NDS, success in Great Power competitions is not about merely fighting. Certainly, deterring a fight matters, but a comprehensive and institutional approach is needed. The United States will need to focus on “out partnering, out informing, out creating and innovating” Great Power competitors as well.¹⁵ Formulating and implementing a coherent strategy for competition requires rethinking the U.S. security architecture and retooling the instruments of national power for agility and responsiveness. As one recent strategic analyst argued, “If the United States wants to compete, it must prepare for a long campaign for influence that will test its own ability for strategic prioritizing and long-term planning.”¹⁶

The strategic options for the United States laid out in this chapter highlight different approaches, with varying degrees of costs and risks, to maximizing American chances to succeed against its main strategic rival, China, in the era of GPC. The first step in rebounding and regenerating is recognizing that America’s competitive edge in some but not all dimensions has eroded in relative terms and that a competitive mindset is needed. As observed in chapter 3b, the United States still possesses numerous advantages and a lead in many quantitative metrics of national power over its main rivals. Nonetheless, America has lost some of its *relative position*, including percentage of world economic output and breath of economic competitiveness, to China. America has also lost relative market share in secondary- and university-level education, although qualitative and language factors mitigate this decline.¹⁷ America’s aspects of relative decline can be renewed or their impact offset by creative strategies.

The task for an American strategist is to leverage natural enduring advantages and build up positions of strength. To be successful, the United States must become more competitive in general and not just fixate on competition against another actor.¹⁸ To think and act competitively requires looking as much, if not more, into America’s own capabilities and performance in all dimensions of strategy as it does in contesting others.

This chapter offers a suite of options for a strategic architecture and defense posture for that competition.¹⁹ As shown in the previous chapters and in numerous international and U.S. Government reports, the U.S.-China dyadic relationship at the heart of emerging GPC is growing into a more competitive and possibly confrontational interaction due to perceptions of interest, honor, and fear. Yet there remains potential for a competitive era defined by shared mutual interests where cooperation is feasible and more competitive tensions can occur in the political and economic categories of interaction within established bounds. The challenge for U.S. policymakers is to expand on the potential for cooperation while carefully managing this competition to keep it short of armed conflict, all the while without compromising vital national interests.

Table 14.1. Alternative Strategies (X equals Major Line of Effort)					
	Bilateral Bargain	Managed Competition	Enhanced Balancing	Compression	Contested Primacy
Political and Diplomatic	X	X		X	X
Ideological				X	X
Informational				X	X
Military			X	X	X
Economic	X	X	X	X	

Competitive Strategies: Framework and Fundamentals

The chapter is organized around an evaluation of five potential grand strategies for U.S. competition with China that cover the continuum of major state interactions established in chapter 1. Each of the five strategies is predicated on different assessments of risk and costs and employs different dimensions or instruments to obtain the strategy's objectives. The strategies are briefly depicted and then evaluated along the five competitive categories and competitive elements first laid out in table 2.2. Table 14.1 lists the five strategies and defines the most critical categories of competitive interaction of each strategy (denoted by the X for major line of effort).

Bilateral Bargain Strategy

This strategy seeks a negotiated bilateral settlement for a stable future. It is the most cooperative of the potential strategies. It focuses on diplomacy to resolve outstanding differences between the core national interests of the world's two most powerful states. It also focuses on economic cooperation and collaboration of mutual benefit. In the words of one advocate, it entails meeting China "halfway" and creating what China has expressly desired: a new form of strategic relationship.²⁰

There are several options for such a relationship. Hugh White has made the case for an Asia-Pacific "concert" based on U.S.-China collaboration and "shared primacy." Another version of this "grand bargain" would be to create a neutral zone, with the United States reducing or eliminating its commitments to its Asia alliance partners in return for Beijing's renunciation of military action in the region.²¹

The gist of a *bilateral bargain* strategy as presented in this chapter would establish recognized spheres of influence, which for China would probably include a clear presumption of control over Taiwan.²² Rather than forging responsible and shared stakeholder status for the entire globe, the United States and China would agree to privileged status as the principal stakeholder in defined areas.²³ This strategy "would recognize that as China becomes a superpower, it will naturally feel entitled to the prerogatives of a superpower—most obviously, disproportionate influence in its home region."²⁴ To attain a grand bargain, the United States would dissolve its longstanding, limited relationship with Taiwan and terminate its increasingly ambiguous defense obligation there. Simultaneously, the PRC would need to negotiate and settle the plethora of maritime/island claims it has throughout the Indo-Pacific region, including with Vietnam, Malaysia, the Philippines, and Japan. As part of this bargain, the United States should insist on the demilitarization of any disputed

territory in the South China and East China seas that is retained by China. In turn, China would be assured freedom of action through the region.

Advocates see the outlines of such a grand bargain being obtained over time, via a series of negotiations and cooperation spirals to ensure reciprocity and growing confidence.²⁵ Each side would make significant concessions. In turn, this would create problems with international and domestic audiences. A bilateral bargain strategy seeks an enduring negotiated bargain on geographical spheres of influence.

An economic option would adapt longstanding representation in international forums such as the International Monetary Fund (IMF) and World Trade Organization (WTO) to give China more weight in these institutions and gain their approval. It could also involve U.S. investment in the Chinese-led Asia Infrastructure Investment Bank (AIIB) as a method of operationalizing collaboration. Presently, the United States does not participate in the AIIB. Barriers to investment and trade would be negotiated on the basis of discussions with the goal of achieving reciprocity in investment and trade levels.

There will be diplomatic costs to such a bilateral bargain option. Elimination of formal links such as the Taiwan Foreign Relations Act would be one possible Chinese demand. Cessation of arms sales to Taiwan by the United States would no doubt be another. In turn, China, as its part of the bargain, would agree to move coercive missile batteries directed against Taiwan and would agree never to insert a military presence in Taiwan.²⁶ Treaty arrangements between the United States and the Republic of South Korea and the Philippines (including U.S. military bases) may also be subject to negotiation to alleviate Chinese fears of encirclement. Acceptance of Chinese posture in the South China Sea would be resisted by U.S. negotiators. However, that may be the price for this grand bargain and freedom of action in other regions.

Managed Competition Strategy

A strategy of *managed competition* combines modes of interaction that are both collaborative and competitive. It seeks to better balance cooperation with hedging U.S. competitive efforts that seek more collaboration with China. There can—and arguably should—be elements of both competition and collaboration in various dimensions of state power.²⁷ China has collaborated with the United States in the past in real and constructive terms.²⁸ The goal of a managed competition strategy is to preserve the current power balance and keep the competitive dimensions of U.S.-China relations from spiraling out of control into confrontation and conflict. A stable relationship is important to regional and global stability. It seeks to maximize cooperation wherever possible, negotiate adaptations to economic and trade disputes, and minimize adversarial competition in the security domain.²⁹

China and the United States are not destined to be enemies or engage in tragic confrontations.³⁰ Managing the competitive aspects of the relationship will require wise leadership on both sides of the Pacific. Given these conditions, some experts argue that “mature management of a volatile re-

“The United States and China are not inevitable enemies, but managing the competitive aspects of the bilateral relationship will require wise leadership on both sides of the Pacific.”

—Phillip C. Saunders, Managing Strategic Competition with China, *INSS Strategic Forum* 242 (July 2009)

lationship is mandatory—bounding the negative dynamics while working to expand the areas of positive cooperation is the principal challenge for both governments.”³¹ This places a higher demand on strategic leaders to provide the mature management of both diplomacy and domestic audiences to limit the negative dynamics and exploit the benefits of cooperation. This managed competition option retains some of the prior U.S. strategy of continued deep engagement. As one Obama-era National Security Council official notes, “Continuing intensive engagement in no way would prevent alterations in U.S. policy to respond to challenges from China in the economic, digital, academic, and security fields. Indeed, it would likely make policy changes more effective by giving China a continuing stake in the relationship with the United States.”³²

Strategic competition does not unfold in a geopolitical vacuum; China needs economic access to the outside world in order to maintain rapid economic growth. Its long-term economic vibrancy and political stability depend on its ability to maintain positive relations with its key economic partners. Managed competition leverages that reality. Given this context, the United States will need to improve its ability to pursue a productive relationship with China. This should involve expanded cooperation where U.S. and Chinese interests are compatible, combined with active efforts to broaden areas of potential cooperation to influence how China pursues its interests.

Military. The first line of effort would be to minimize regional security dilemmas. Given U.S. security commitments to its allies and the importance of those alliances for Indo-Pacific stability, the maintenance of robust military capabilities should remain an important part of U.S. strategy. However, at the same time, the United States should not attempt to increase its power position in the region with any new alliance arrangements, basing, or extensive investments in theater-level ballistic missile arsenals now that the United States is no longer constrained by the Intermediate Nuclear Forces Treaty.³³ In short, the United States would forgo efforts laid out in the NDS to enhance its regional security posture in the Indo-Pacific region. Strategic competition is likely to be exacerbated if the United States seeks to dominate the region or if China impinges on the security interests of key U.S. allies, such as Japan.

The second element of this approach is to expand security cooperation, including bilateral and multilateral cooperation between the U.S. and Chinese militaries. Security dynamics and competing interests may limit opportunities for direct military cooperation, but there can be critical tasks for mutually beneficial cooperation. People’s Liberation Army Navy missions to the Gulf of Aden for counterpiracy operations demonstrate that China’s capabilities can be cooperative, too.³⁴ Finding appropriate venues to extend the cooperation in governance, energy, and humanitarian tasks is key. There are a number of important missions—including peacekeeping, humanitarian affairs and disaster relief, infectious disease control, counterpiracy, and energy security—in which both sides contribute to global stability and shared interests. An increased effort to identify and build on these issues could help balance the more competitive aspects of strategic relations.

Managed competition would seek to enhance transparency of capabilities and intentions, including myriad military-to-military contacts such as high-level interactions of senior military officials, educational exchanges, and routine observer status at military exercises including Rim of the Pacific.³⁵

Another possible means of enhancing cooperation would be to establish new venues to promote dialogue between China and the rest of the world, including major alliances such as the North Atlantic Treaty Organization (NATO). China has become a topic of conversation within NATO as China's economic reach has come to Europe. A report from the Atlantic Council offers a platform to promote dialogue and maximize collaboration. The paper recommends that the Alliance establish a NATO-China Council as a mechanism to increase transparency and mutual understanding; raise concerns; avoid miscalculations; and foster, where possible, cooperation.³⁶

Political/Diplomatic. The second line of effort of managed competition is for the United States to encourage and support Chinese efforts to take on more responsibility for sustaining and supporting the international system. The United States must recognize that doing so requires providing China a path to pursue its legitimate aspirations through peaceful means. The current international order is not rigid and has been flexible in the past in matching China's rising power with greater influence and participation. The United States should acknowledge that, if China is to make more contributions to maintaining the international system, it will expect to be accorded greater voice in shaping that system. Chinese interests may require altering the system to reflect Chinese perspectives and legitimate concerns. This would include changes at the IMF or World Bank, and certainly the WTO.³⁷ But the rest of the members of that international system will expect Beijing to honor the dispute resolution mechanisms built into the system rather than simply point out that it is a big country and the "small must do what they must." Managed competition would place a premium on diplomatic resources at the State Department, outmanned by China at present, especially in the number and staffing of consulates in key markets.³⁸

In summing up this option, the United States must be prepared to compete with China in important strategic domains, while simultaneously seeking to limit the impact of this competition on the broader relationship. Some call this *smart competition*,³⁹ which strives to manage the strategic competition effectively and with restraint. Advocates of this approach hold that assertive strategies would be expensive, if not dangerously counterproductive. Proponents of managed competition believe the United States should accept the reality of a growing competition but manage it at a lower level of intensity and risk.⁴⁰

Enhanced Balancing Strategy

The *enhanced balancing* strategy focuses on a competitive approach in two dimensions of strategic interaction: the military and economic ones. The strategy is predicated on two recognized strengths of the United States: its preeminent military power and its existing military alliance architecture. From an American perspective, this option would be competitive in nature, but China *might* perceive it as more confrontational because, in Beijing's view, "balancing" is the same or an even worse form of "containment" of China and its aspirations.⁴¹ Unlike China, the United States has the proven ability to develop and sustain coalitions of countries designed to share security burdens and maximize deterrence against instability.⁴² The strategy would exploit the growing concern that many Asian and European countries have with China's growing economic power and assertive foreign policy.⁴³ Many countries believe they are dependent on China for their own economic development and

prosperity. But unless they join together, Beijing will dominate the region politically, militarily, and economically and apply its preferences on the sovereign decisionmaking of each and every nation in the region. While China claims that it is not offering an alternative model or imposing its values or governance system, it does impinge on the sovereignty of its neighbors regularly, and it does seek political concessions to benefit its international standing.

Military. To implement this strategy, the United States would have to increase its security investments in defense and buttress its forward-deployed forces.⁴⁴ China desires to build a world-class military—and with Russia’s help, it will no doubt make some progress. But with proper investments to sustain its competitive edge, the U.S. alliance framework should be able to sustain an adequate balance in the Asia theater. The first priority of this investment would be the deployment of systems able to blunt China’s expanding antiaccess/aerial-denial capabilities. The second priority would be hardening U.S. bases throughout the Indo-Pacific region to make them more resilient to attack. A third priority would be key investments in space, undersea warfare, hypervelocity missiles, and theater missile defenses to enhance the current U.S. deterrent posture in the Indo-Pacific region. China cannot match the human capital assets, seasoned operational leadership, intelligence, strategic mobility, and logistics expertise of the U.S. alliance system.

Per the NDS, the United States will also have to buttress the military capability of its allies and some partners. Security assistance support to Vietnam, the Philippines, Australia, and India, as well as other regional players, would be needed to offset a deteriorating balance of power in the Indo-Pacific.⁴⁵ As noted by former government officials, “The United States needs to get back to seeing alliances as assets to be invested in rather than costs to be cut.” While burden-sharing is necessary, undercutting alliance cohesion works perfectly to China’s benefit.⁴⁶ This statement tracks with the NDS:

*Mutually beneficial alliances and partnerships are crucial to our strategy, providing a durable asymmetric strategic advantage that no competitor can rival or match. . . . By working together with allies and partners we amass the greatest possible strength for the long-term advancement of our interests, maintaining favorable balances of power that deter aggression and support that stability that generates economic growth.*⁴⁷

Economic. The second thrust in this strategy would be the rededication by the U.S. Government to alliances and multilateral institutions in order to reinforce the economic component of the international order. The battle for influence in Asia is not about security and will rise and fall on economics. On that score, the United States has lost significant influence since withdrawing from the Trans-Pacific Partnership (TPP) trade agreement.⁴⁸ Without a commensurate embrace of multilateralism within a revised TPP, potential partners in the region will remain prone to accept Beijing’s influence, direction, loans, and capital investment. An effective U.S. response will require a far deeper investment than the initial 2018 Better Utilization of Investments Leading to Development (BUILD) Act, which made a modest ripple in development financing for the region.⁴⁹ Other multilateral institutions would also be adapted and strengthened. The United States would increase its contributions and support to forums such as the World Bank and IMF to sustain a collective

approach to managing the global economy. This strategy and all the defense investment will be worthless unless it is supported by a significant shift in the U.S. approach to international development and geoeconomics. In addition to limited funding for the BUILD Act, cuts to key agencies such as the U.S. Agency for International Development and the Asian Development Bank have further eroded U.S. Government mechanisms to mount a robust alternative.⁵⁰ Better incentives to private sector financing or IMF support will be needed to blunt China's extensive investments in infrastructure activities.

A major test in enhanced balancing in the economic domain would be *shared reciprocity*, wherein Chinese companies, products, and services would be limited inside the United States to the same degree that they are afforded access to inside China.

Although not a major line of effort in this option, the United States would need improved information and public diplomacy to be effective. This strategy would publicly identify the negative impacts of debt diplomacy, internal corruption, and environmental damage that China's investments in Africa and Sri Lanka have produced. To the greatest degree possible, this information campaign would be promulgated via multilateral institutions, including the United Nations and the Association of Southeast Asian Nations.

Compression Strategy

A *compression strategy* of comprehensive pressure herein is globally oriented, extends the competitive interactions of the prior strategy, and seeks to alter the arc of China's growing power and its aims of parity in critical dimensions. It includes aspects of interaction that China *will* label as confrontational since it seeks to change Beijing's behaviors in the Indo-Pacific region. Balancing may be effective at securing a delay in China's rise to dominance, but it does not restore international law or enduring stability to rising GPC. While increased security and economic partnering of the enhanced balancing option might give pause to Beijing's leaders, it may not constrain or alter the Chinese Communist Party's (CCP's) clear regional and perceived global ambitions or restore U.S. leadership. A more comprehensive strategy to make China stop its current path and alter its behavior to align itself with existing norms and international law may be required. Such a comprehensive strategy, one of persistent multidimensional pressure, is one of *compression*.

Compression assumes that the long-term trajectory of China as a major global player is vulnerable and predicated on fragile aspects of its power base. Because the PRC has manifest challenges, including governance, corruption, innovation, debt, and demographics, it may be susceptible to the pressure generated by the strategy. This strategy relies on an assessment that China's economy is fragile and that economic growth is susceptible to external pressure.⁵¹ It assumes that China still needs Western markets and technological prowess to sustain any growth; it also assumes that such growth is key to the CCP's hold on power.

Compression combines multidimensional pressure to push back Beijing's geopolitical and economic gains that violate norms and international law in Asia. It intends to deny past gains and preclude growth of China's reach. Such an approach would be far more encompassing than the Cold War-era conceptions of containment that Washington applied against Moscow during the Long Peace. That anti-Soviet strategy did not have a direct economic component because the Soviet Union was not interdependent with international trade or fiscal systems in the way Beijing is. Compression would include intensified

ideological, military, diplomatic, and economic initiatives meant to deflect China's bid for primacy in the Indo-Pacific region and to disrupt its nascent efforts globally.

Political/Diplomatic. This strategy is founded on securing the extant international order, including multilateral venues to maximize U.S. leadership and sustain free and open societies. At the same time, in this dimension it would seek to minimize direct confrontation and conflict. Compression would require a renewed appreciation of two enduring advantages: our alliance architecture and the global institutions that have been created to sustain international stability.⁵² Rather than retreat from these institutions, the U.S. diplomatic presence would be reinforced to preclude erosion of American influence. Letting Beijing reshape norms and expand its own influence within those organizations is undesirable.⁵³ Naturally, reciprocal relationships and fair burden-sharing are required to make this approach sustainable. Our allies should realize that they too have a stake in this competition.⁵⁴

Economic. The Chinese model sees economics as a form of power projection to be deployed for political effect. The United States must respond and master the economic tools of Great Power statecraft to offset China's mercantilism and malign power.⁵⁵ The economic aspect of compression directly counters China's trade model and its subsidies to its large number of state-owned enterprises. The United States relies on free markets and the private sector to preserve its economic prosperity and the foundation of its national power.

Some economic disentanglement is an expected price of this strategy, what some would describe as a *partial disengagement*.⁵⁶ At a minimum, the United States would decouple itself from China in sectors where the existing level of economic interdependence threatens America's ability to resist Chinese advances—for example, by ending the practice of sourcing critical components of U.S. military capabilities from Chinese companies.⁵⁷ Under this strategy, the United States would limit China's access to advanced weaponry and critical military technologies, and with its allies “develop a coordinated approach to constrict China's access to all technologies, including dual use.”⁵⁸ Key elements of the U.S. economy, especially in dual-use technologies that benefit military capabilities, would be closed to Chinese commercial outlets. The United States would need to reinvigorate its national advantages in science and technology by focusing greater attention on securing global leadership in the technologies that will dominate the fourth industrial revolution.⁵⁹ In particular, it would require more focused U.S. Government efforts to sustain an edge in these technologies, including quantum computing and artificial intelligence.⁶⁰

Another part of the compression strategy would include aggressive litigation and sanctions over infringements of intellectual property, with appropriate penalties/sanctions leveled against corporate entities that sell hardware or software developed with U.S. intellectual rights. This would impose costs on those who steal investments in research and development, particularly on those who violate the intellectual property laws agreed to in international law. This approach would apply punitive retaliatory economic measures and targeted tariffs, or exclude China from trade agreements, in response to its violations of trade laws and agreements.⁶¹ Without enforcement of this portion of the international system, we cede future economic prosperity to others.

As noted by the U.S.-China Economic and Security Review Commission, China's state-led, market-distorting economic system presents a challenge to U.S. interests.⁶² The United States requires a more comprehensive economic strategy to deal with China because its

trade practices can be leveraged into improper influence.⁶³ Some are concerned that China's Belt and Road Initiative should be understood as "a grand strategy that advances China's goals of establishing itself as the preponderant power in Eurasia and a global power second to none."⁶⁴ As noted in chapter 3a, these goals are overstated, and China is already getting some backlash over its debt-financing and infrastructure-building. But the United States cannot abandon the vast resources and markets of the Indo-Pacific region to its competitors. The first step is to rejoin U.S. allies in the TPP and formulate an acceptable form of national industrial policy to focus Federal funding and incentives toward the disruptive technologies that will drive economic production in the coming decades.⁶⁵

Some recent proposals to improve U.S. economic power relative to China would be incorporated. The establishment of an Office of Critical Technologies and Security to better manage technology transfer, as in the bipartisan proposal put forward by Senator Mark Warner (D-VA) and Senator Marco Rubio (R-FL), is consistent with compression.⁶⁶ To improve human capital and sustain progress in cutting-edge technologies, the United States will continue to open its first-rate university system to the world's best talent. A National Security Innovation Base Visa that would facilitate the travel of highly skilled foreign workers to contribute their education and talents to the benefit of the national security innovation base and American security should also be considered.⁶⁷

Ideological. In Aaron Friedberg's observation, "China's rulers clearly believe the ideological realm to be a crucially important domain of competition."⁶⁸ The differences between the West and the CCP would be stressed, pitting free and open societies based on liberal values and democratic principles against large authoritarian powers with illiberal values and closed information systems. Given that Beijing readily exploits this aspect of the competition, but is also asymmetrically vulnerable in soft power terms, the ideological element of the competition bears consideration.⁶⁹ The activities of Confucius Institutes and PRC surveillance over Chinese students inside the United States would be limited. These institutes were controversial from the start, as inhibiting academic freedom for students and faculty alike, and several university systems have closed their partnership arrangements altogether.⁷⁰

Informational. This strategy has an intensive informational component.⁷¹ U.S. officials have to recognize that strategic competition is not only a fight over market access or trade policy but also an ideational contest over values and norms for the international system.⁷² Such a strategy would steadily apply pressure in the ideological and information dimensions by undermining the Great Firewall and abetting more moderate elements in China's closed and repressive system.⁷³ This line of effort would incorporate activities that would ideologically contest the legitimacy of the CCP and promote Chinese culture. The informational component of this strategy would seek to challenge CCP domestic political control through a broad campaign that ties any declining economic growth and limited personal freedoms to China's single-party rule, its repressive control, and illicit actions. Legal challenges would be made against Chinese policymakers who are linked to human rights violations, corruption, and repression against minorities and nongovernmental organizations.

Contested Primacy Strategy

A strategy of *contested primacy* takes on a more confrontational approach. It seeks dominance over any competitors in an effort to sustain the existing international order and

"We must calibrate our aims with our resources and focus on the most consequential long-term challenge we face as a nation: the strategic competition with China."

—Michèle Flournoy, *House Armed Services Committee Statement, January 15, 2020*

American hegemony. Contested primacy responds in a robust way to Great Power competitors, employing all instruments of power to reassert and sustain U.S. dominance while focusing on the political/diplomatic and military main lines of effort. It strives to secure defined vital U.S. interests per the NSS.⁷⁴ This approach reorders the U.S. economy and investments needed

to sustain U.S. superiority and preferred outcomes in all strategic interactions. It significantly increases geopolitical and economic costs against Chinese influence to ensure that the CCP's ability to obtain regional primacy and global reach is thwarted.

Military. This strategy would substantially augment the Pentagon's budget (perhaps as much as \$100 billion per year higher than the fiscal year 2020 request) and build up the Defense Department's effort to modernize the U.S. military for joint power projection throughout the Indo-Pacific region at first, but also wherever else U.S. interests might be threatened.⁷⁵ This strategy would involve intensive efforts to modernize and increase interoperability among current U.S. allies and partners in the region. This interoperability would include arms sales and security cooperation efforts to ensure that Taiwan was not coerced into submission by the PRC. Implementing this strategy would require the United States to engage extensively with its potential partners in the region, including India, Vietnam, and Singapore; to strengthen maritime security; and to extend alliance interoperability with Australia, Japan, and South Korea.

The higher budget would generate and field a modernized joint force that applies creative operational concepts and develops advanced disruptive technologies critical to re-establishing competitive edge in U.S. military power across the long term. A number of key investments are needed to enhance shortfalls in the forward military posture and capabilities of the joint force if it is to deter and prevail against our major competitors.

In Asia, the United States must hedge against the PRC's increasingly assertive actions and improved military modernization.⁷⁶ These actions augur for an agile force that is forward deployed, in part, to assure regional access and assurance. This must be coupled with a layered defense posture. The most important component is a joint force that is interoperable with U.S. regional allies and partners. Power projection capabilities and strategic mobility assets must be increased, but creative concepts are required to offset the carefully designed antiaccess systems fielded by China. These will constrict freedom of maneuver and undercut U.S. ability to flow forces into the region and supply them. Undersea warfare investments in this strategy may afford a very cost-effective and competitive advantage.

Political/Diplomatic. The supporting political/diplomatic line of effort in this strategy would seek to expand on, in degree and intensity, the activities described with the compression strategy, including contesting China's position within global multilateral institutions that it has penetrated and coopted. The principal counter to China's rise would be a reformed and enlarged alliance system. This counter would require a diplomatic emphasis on expanding the present global alliance architecture and enhancing the number of aligned partners to ensure a favorable balance of power. While it is noted that the current

system is “badly bruised,” it was invaluable in the past, and there is little merit in a “go it alone” stance.⁷⁷

Additionally, U.S. diplomacy would confront the legitimacy of CCP rule, and an augmented community of free and democratic states would resist Beijing’s advances in international forums. The political thrust would be to highlight the inconsistency behind China’s repressive domination of its people and its promotion of a benign “community common design for mankind.” The idea that China actively seeks an international order based on “fairness, justice and win-win cooperation” would be shown to be a front for its more authoritarian preferences. China’s long-term ambitions would be presented as a threat to Western democracies, due to their marked incompatibility with the freedoms and liberal values embedded in their political and economic systems. The United States would posture itself as seeking primacy for the free and open order that better reflects universal freedoms and justice in international affairs and within each state.

Ideological. This strategy would incorporate a strong ideological element against the CCP and its closed and repressive form of government. It would include a sophisticated campaign to delegitimize and weaken the Party’s control over information inside China. This line of effort would make strong condemnations of Beijing’s violations of international law and norms and values on human rights and individual freedom, and it would distribute stories on China’s repression of religion and minorities. It would identify and distribute stories on the unequal justice and economic benefits accrued by CCP leaders and their families. On the flip side, the progress and benefits of open societies such as Singapore, Taiwan, and South Korea (and even the vestiges in Hong Kong) would be distributed throughout the region to underscore the positive and progressive agenda of open and free societies.

Informational. There is a geo-informational aspect to GPC, and China should not be allowed to establish control of any part of the competition.⁷⁸ Achieving success would require a renewed institutional response to countering gray zone/political warfare or influence operations by both China and Russia.⁷⁹ China is becoming a global cyber power in both military and commercial spheres.⁸⁰ A strategy seeking primacy must preserve the critical infrastructure of the U.S. homeland; at the same time, it must circumvent China’s heavy-handed surveillance systems in order to breach the Great Firewall and reach the Chinese people and the populations of Hong Kong and Tibet. Furthermore, Chinese efforts to dominate global 5G networks would be curtailed, especially among allies.⁸¹ The United States adopted a cross-functional approach during its protracted contest against global violent extremism. It might also need to establish a National Center for Countering Influence Operations to achieve the same end to confront China’s political warfare and United Front efforts.⁸²

Analysis and Recommendation for Enhanced Balancing

How should the United States proceed, and which strategy offers the best combination of tools and instruments to achieve its preferred future? This section evaluates the merits of the most viable three strategic options of the five discussed in this chapter. These three reflect suitable options for preserving the existing order and maintaining U.S. national interests within the parameters of U.S. values and feasible resource levels.

Managed competition is a more conservative strategy but assumes that shared interests can be found and built on. It is a complex strategy that would be difficult for both countries

to oversee and implement and to find areas where collaboration is feasible and where a more competitive approach is really needed. Communicating those lines will be important. Domestic audiences may be unable to separate the cooperative from the competitive, and even diplomats may find it difficult to avoid linkages. The U.S. Government, given its more decentralized structure, could adapt over time to manage and sustain such a strategy. China's more centralized control and state capitalism model give it some advantages in building such a relationship. But the real challenge may be isolating the economic and military dimensions. As noted by Phillip Saunders:

*The implementation challenge is to keep the two elements in proper balance, so that overemphasis on cooperation does not leave the United States in an unfavorable strategic position and overemphasis on the military dimension does not stimulate Chinese threat perceptions and push it toward confrontation.*⁸³

Enhanced balancing strives to improve U.S. strategic performance and maintain a favorable balance of power. It builds on current U.S. economic power and its extant alliance system, which China seeks to undercut. Yet in the Indo-Pacific region, the present suite of allies and partners is uncomfortable with being forced to choose sides in a U.S.-China clash. These nations prefer to retain all the economic opportunities China offers, while embracing a separate security system led by Washington. Forced to choose, some may feel that working with the United States is not a sound bet for their future prosperity. But short-term economic benefits for long-term subordination to Beijing is a poor choice, and the United States should continue to make that clear. This strategy has been slowly implemented over the past two administrations and presents less risk and demands fewer resources than the compression option.

Compression strategy is more expensive and directly confronts China's rise and vulnerabilities. It devotes additional resources to the military dimension of the competition. Given that the capacity of the U.S. joint warfighting community is officially recognized as having a declining edge, a stronger military response is needed. If allies perceive that the ability of the United States to "uphold favorable regional balances of power by deterring Great Power challengers is increasingly in doubt," a significant change in the security component has to be realized.⁸⁴ Some allies conclude that Chinese and Russian military developments "have irrevocably undermined America's military primacy" in the Indo-Pacific and beyond.⁸⁵ That presumption needs to be countered.⁸⁶ Compression raises the cost to the Chinese for contesting the existing international order, by decoupling economic interaction with China and by the large-scale U.S. defense modernization that it engenders. Compression seeks to create leverage vis-à-vis Beijing to force it to reconsider its predatory economic activity and its efforts to undercut the U.S. alliance architecture in the Indo-Pacific region. Compression is designed to help the CCP realize that its future is best realized within the order established and adapted over the past 50 years. Ideally, the CCP would accept this order, and an eventual transition to managed competition might then occur.

The compression strategy recognizes the significant advances made by the People's Liberation Army and the need to counter its reforms and modernization strategy.⁸⁷ This approach requires defense spending above the administration's fiscal year 2021 budget and

involves disciplined investments focused on increasing the posture of U.S. forces in the Indo-Pacific region.

Regardless of which strategy is selected, it is insufficient to merely contest the rising power of China and the way it seeks to satisfy its ambitions. The most appropriate strategy must include a renewal of American strategic competitiveness. This renewal includes investments in education, infrastructure, and research and development to spur economic prosperity.⁸⁸ Any strategy should seek to rejuvenate America's research and development base and master the transfer of commercial technology to the security sector with both speed and effectiveness. "The United States should focus on responsibly accelerating its own technological progress," notes one former Deputy Secretary of Defense, "not simply obstructing potential adversaries."⁸⁹ This should play to American strengths, given the fertile U.S. innovation ecosystem buttressed by free market systems for allocation of capital and financial management. As noted by a panel of experts seeking to rectify the eroding competitive edge in the Pentagon, the consequences are substantial:

*Nevertheless, it is a competition, and the side that innovates more effectively over time is likely to win. The result will determine whether nations relate to each other freely, equally, and peacefully, with a recognition of the human rights of their citizens, or if they devolve into a system that legitimizes authoritarianism and rewards power and coercion.*⁹⁰

There will be a major economic element to this competition, regardless of which strategy is selected. To preserve both its economic and security interests, the United States must safeguard an expanding suite of advanced technologies from China. As observed in chapter 3b, China may not be the most creative generator of innovative capabilities, but it is proving adept at acquiring modern capabilities and building up in the commercial world national champions that can compete on both cost and product effectiveness. China certainly appears bent on achieving leadership, if not parity, in the key technologies that will drive 21st-century economics. Preserving and protecting the technology base, and the resulting intellectual property it generates, will both slow China's acquisition of U.S.-developed advances and drive up Beijing's own costs as PRC struggles to keep up.⁹¹

Furthermore, the United States can best ensure its economic and technological competitiveness by expanding with additional partners the cooperative aspects of its National Technology and Industrial Base. Leveraging the intellectual and technical talents of our allies in such a manner will accelerate innovation, broaden commercial opportunities, and minimize costly barriers to collaboration.⁹²

Rather than a bilateral confrontation, the United States should take a more collective approach to better secure success. This approach would:

*work with allies to strengthen rules, set standards, punish Chinese industrial policy and technology theft, invest in research, welcome the world's best and brightest, and create alternatives to its geo-economic statecraft. China is playing a good hand well, but the United States and its allies have an even better one—but only if they work together.*⁹³

"[I]t would be unwise to bet against the resilience and adaptability of the American system. But it would also be a mistake to take these qualities for granted, or to assume that they will preserve us indefinitely and without effort from the experience of relative decline."

—Aaron L. Friedberg, *The Weary Titan* (2010)

A strategy of enhanced balancing offers the greatest opportunity to do more than merely contain or deflect the trajectory of China's ultimate regional preeminence and global stature. The past decade shows that cooperative approaches or *deep engagement* only strengthened China's power and accelerated its rise without appreciable political or economic reform. A continued reliance on limited approaches that do not

counter aggressive behavior or blatant disregard for international law, multilateral norms and rules, and human rights will likely not be productive either. In short, success in an era of GPC will require creating leverage and accepting risk. It also requires that the United States get its own house in order. Competing successfully to sustain America's prosperity, security, and way of life is ultimately about us.⁹⁴ The United States is not a "weary titan" as much as simply a complacent leader that needs to respond to a persistent contender.⁹⁵ Superpower status or economic prosperity is not a birthright, and assuming that American preeminence will be unrivalled for perpetuity is ahistorical. Instead, preserving U.S. leadership and advancing American interests in a dynamic era can best be secured by rejuvenating core strengths in human capital and education, individual freedom, infrastructure, and economic creativity. American economic competitiveness is the foundation for any future strategy and can be improved.⁹⁶

While not without an element of risk, enhanced balancing offers more options for creating leverage against China's vulnerabilities. It is a strategy that seeks to stem the geopolitical gains that China has seized while the West was distracted, and it offers a measured response to sustain a stable world order designed to promote stability, expand opportunity, adapt within a rules-based system, and preclude hegemony over first the Indo-Pacific region and then Eurasia by any hostile power. The ascent of an autocratic power that represses human rights, undercuts international agreements and norms, exploits its economic power to obtain political dominance, and pressures U.S. allies and partners does the opposite.⁹⁷

Conclusion

In many respects, today's era is more complicated than the bilateral Cold War, but it still holds lessons. That contested age was precarious at times in the military aspects of the competition, but economic interaction was negligible. With the West's patient pressure, the internal contradictions of the Soviet system finally proved inferior to the systemic advantages of democracies and free markets. There are aspects of the current competition that will continue to favor the United States for some time. Thus, there is no need for panic, but neither should complacency be seen as a virtue.

Clearly, a good deal of optimism is warranted given the current edge and enduring advantages the United States holds. Both American leadership and the liberal international system have been capable of regenerating themselves as needed.⁹⁸ It is time to do so again.

Notes

¹ On the Unipolar Moment, see Charles Krauthammer, “The Unipolar Moment Revisited,” *The National Interest*, December 1, 2002, available at <<https://nationalinterest.org/article/the-unipolar-moment-revisited-391>>; Matthew Kroenig, *The Return of Great Power Rivalry: Democracy Versus Autocracy from the Ancient World to the U.S. and China* (New York: Oxford University Press, 2020). For an argument that sustained U.S. dominance is assured, see Michael Beckley, *Unrivaled: Why America Will Remain the World's Sole Superpower* (Ithaca, NY: Cornell University Press, 2018).

² A detailed discussion is available in chapter 2 of this volume.

³ Hal Brands, “The Lost Art of Long-Term Competition,” *Washington Quarterly* 41, no. 4 (Fall 2019). See also Ronald O'Rourke, *Renewed Great Power Competition: Implications for Defense—Issues for Congress*, R43838 (Washington, DC: Congressional Research Service, December 19, 2019), available at <<https://crsreports.congress.gov/product/pdf/R/R43838/50>>.

⁴ Evan S. Medeiros, “The Changing Fundamentals of U.S.-China Relations,” *Washington Quarterly* 42, no. 3 (Fall 2019), 113.

⁵ Ashely J. Tellis, “Overview,” in *China's Expanding Strategic Ambitions: Strategic Asia 2019*, ed. Ashely J. Tellis, Alison Szalwinski, and Michael Wills (Seattle: National Bureau of Asian Research, 2019), 26. For competing viewpoints on China's more ambiguous strategic calculus, see the discussions in chapters 3a and 9 of this volume.

⁶ Jonathan Hillman, a scholar at the Center for Strategic and International Studies and winner of the 2019 Bracken Bower Prize, first applied the term *Sinolarity* to China's growing business/economic challenge.

⁷ *The National Security Strategy of the United States* (Washington, DC: The White House, December 2017), 25.

⁸ *Summary of the National Defense Strategy: Sharpening the U.S. Military's Competitive Edge* (Washington, DC: Department of Defense, December 2018), 1–2, available at <<https://dod.defense.gov/Portals/1/Documents/pubs/2018-National-Defense-Strategy-Summary.pdf>>. For elaboration, see Elbridge A. Colby and Wess Mitchell, “The Age of Great Power Competition,” *Foreign Affairs* (January/February 2020).

⁹ Oriana Skylar Mastro, “The Stealth Superpower: How China Hid Its Global Ambitions,” *Foreign Affairs* (January/February 2019), 31.

¹⁰ For views that oppose the National Security Strategy and the National Defense Strategy conclusions, see Michael Swaine, “A Counterproductive Cold War with China,” *Foreign Affairs*, March 2, 2018, available at <www.foreignaffairs.com/articles/china/2018-03-02/counterproductive-cold-war-china>; Michael J. Mazarr, “This Is Not a Great-Power Competition: Why the Term Doesn't Capture Today's Reality,” *Foreign Affairs*, May 29, 2019, available at <www.foreignaffairs.com/articles/2019-05-29/not-great-power-competition>.

¹¹ Kurt Campbell and Jake Sullivan, “Competition Without Catastrophe: How America Can Both Challenge and Coexist with China,” *Foreign Affairs* (September/October 2019), 25.

¹² Derek Scissors and Daniel Blumenthal, “China Is a Dangerous Rival, and America Should Treat It Like One,” *New York Times*, January 14, 2019, available at <www.nytimes.com/2019/01/14/opinion/us-china-trade.html>.

¹³ Robert D. Blackwill and Ashley J. Tellis, *Revising U.S. Grand Strategy Toward China*, Council Special Report No. 72 (New York: Council on Foreign Relations, March 2015).

¹⁴ Graham Allison, “China vs. America: Managing the Next Clash of Civilizations,” *Foreign Affairs* (September/October 2017), 80–89.

¹⁵ *Summary of the National Defense Strategy*, 5.

¹⁶ Odd Arne Westad, “The Sources of Chinese Conduct: Are Washington and Beijing Fighting a New Cold War?” *Foreign Affairs* (September/October 2019).

¹⁷ See discussion in chapter 3b of this volume.

¹⁸ Ely Ratner, *Blunting China's Illiberal Order*, Testimony Before the Senate Armed Services Committee Hearing on China and Russia, Washington, DC, January 29, 2019. Michele Flournoy, former Under Secretary of Defense for Policy, made this same critical point. See Michele Flournoy, *Statement Before the House Armed Services Committee Hearing on the Department of Defense's Role on Competing with China*, January 15, 2020.

¹⁹ Hal Brands, *What Good Is Grand Strategy? Power and Purpose in American Statecraft from Harry S. Truman to George W. Bush* (Ithaca, NY: Cornell University Press, 2012), 1–16.

²⁰ Lyle J. Goldstein, *Meeting China Halfway: How to Defuse the Emerging U.S.-China Rivalry* (Washington, DC: Georgetown University Press, 2015). See also Charles L. Glaser, “A U.S.-China Grand Bargain: The Hard Choice Between Military Competition and Accommodation,” *International Security* 39, no. 4 (Spring 2015), 49–90.

²¹ Hugh White, *The China Choice: Why America Should Share Power* (Sydney: Black, 2012).

²² For a rationale for spheres of influence, but without sufficient consideration of likely political implications, see Graham Allison, “New Spheres of Influence,” *Foreign Affairs* (March/April 2020), 30–40.

²³ Responsible stakeholder was a role that past U.S. administrations hoped China would grow into. See Robert B. Zoellick, “Can America and China Be Stakeholders?” speech delivered to the U.S.-China Business Council, Washington, DC, December 4, 2019, available at <<https://carnegieendowment.org/2019/12/04/can-america-and-china-be-stakeholders-pub-80510>>.

²⁴ Richard K. Betts, “The Lost Logic of Deterrence,” *Council on Foreign Relations* (March/April 2013), available at <<https://www.cfr.org/world/lost-logic-deterrence/p30092>>.

²⁵ A *cooperation spiral* is an innovation proposed by Lyle J. Goldstein that applies an extended approach to mutual and reciprocal accommodation over time. See Goldstein, *Meeting China Halfway*, 12–14.

²⁶ These bargaining outcomes are indeed bitter for Taiwan but are inevitable if pursuit of “Meeting China Halfway” is evaluated in concrete terms and in context of the difficult military odds outlined in chapter 9 of this volume. Proponents of “Meeting China Halfway” appear to defer to the People's Republic of China about Taiwan's future implicitly, although some, like Graham Allison, actually do so explicitly. In this presentation of a bilateral bargain option, a negotiated agreement with terms that would require mutual assent is advanced as a superior alternative to just abandoning the island nation. However, it remains hard to see how China would accept.

²⁷ David Dollar et al., *Avoiding War: Containment, Competition, and Cooperation in U.S.-China Relations* (Washington, DC: Brookings, 2017); Ryan Hass, *Principles for Managing U.S.-China Competition*, Policy Brief (Washington, DC: Brookings, August 2018).

²⁸ Zoellick, “Can America and China Be Stakeholders?” 4.

²⁹ This portion of the chapter is extensively influenced by Philip Saunders, *Managing Competition with China*, INSS Strategic Forum No. 242 (Washington, DC: NDU Press, July 2009).

³⁰ As some interpret the arguments of Graham Allison, *Destined for War: Can America and China Escape Thucydides's Trap?* (New York: Houghton Mifflin, 2017).

³¹ David Shambaugh, "Dealing with China: Tough Engagement and Managed Competition," *Asia Policy* 23 (January 2017), 5.

³² Jeffrey A. Bader, *U.S.-China Relations: Is It Time to End the Engagement?* Policy Brief (Washington, DC: Brookings, September 2018), 1.

³³ Scott Cuomo, "It's Time to Make a New Deal, Solving the INF Treaty's Strategic Liabilities to Achieve U.S. Security Goals in Asia," *Texas National Security Review* 2, no. 1 (November 2018), 104–128; David Sanger and Eric Wong, "U.S. Ends Cold War Missile Treaty, with Aim of Countering China," *New York Times*, August 1, 2019, 1, available at <www.nytimes.com/2019/08/01/world/asia/inf-missile-treaty.html>.

³⁴ On People's Liberation Army (PLA) military diplomacy, including humanitarian/peacekeeping, see Phillip C. Saunders and Jiunwei Shyy, "China's Military Diplomacy," in *China's Global Influence: Perspectives and Recommendations*, ed. Scott D. McDonald and Michael C. Burgoyne (Honolulu: Asia-Pacific Center for Security Studies, 2019).

³⁵ Helene Cooper, "U.S. Disinvites China from Military Exercise Amid Rising Tensions," *New York Times*, May 23, 2018, available at <www.nytimes.com/2018/05/23/world/asia/us-china-rim-pac-military-exercise-tensions.html>.

³⁶ Barry Pavel and Jan Brzezinski, "It's Time for a NATO-China Council," *Defense One*, August 21, 2019, available at <www.defenseone.com/ideas/2019/08/its-time-nato-china-council/159326/>.

³⁷ Ana Swanson, "U.S. Plans to Sap WTO's Influence in Trade Brawls," *New York Times*, December 9, 2019, A1.

³⁸ Bonnie Bley, "The New Geography of Global Diplomacy: China Advances as the United States Retreats," *Foreign Affairs*, November 27, 2019, available at <www.foreignaffairs.com/articles/china/2019-11-27/new-geography-global-diplomacy>.

³⁹ Orville Schell and Susan L. Shirk, *Course Correction: Toward an Effective and Sustainable China Policy*, Asia Society Task Force Report (February 2019), 7.

⁴⁰ Timothy R. Heath and William R. Thompson, "Avoiding U.S.-China Competition Is Futile: Why the Best Option Is to Manage Strategic Rivalry," *Asia Policy* 13, no. 2 (April 2018), 91–120; David Shambaugh, "U.S.-China Rivalry in Southeast Asia: Power Shift or Competitive Coexistence," *International Security* 42, no. 4 (Spring 2018), 85–127.

⁴¹ Lyle J. Goldstein, "How China Sees America's Moves in Asia: Worse than Containment," *The National Interest*, October 29, 2014, available at <<https://nationalinterest.org/feature/how-china-sees-americas-moves-asia-worse-containment-11560>>. Citing a group of Chinese military scholars, Goldstein notes that "China does not just [feel] confront[ed] '遏制' [containment] by the United States, but something perhaps even more bellicose: '围堵' [a condition of being under siege] or even '掣肘' [a condition of being held by the elbows]."

⁴² This strategy is partially drawn from Aaron L. Friedberg, "Competing with China," *Survival* 60, no. 3 (June–July 2018), 7–64.

⁴³ Walter Russell Mead, "China Is Europe's Problem, Too," *Wall Street Journal*, November 25, 2019, available at <www.wsj.com/articles/china-is-europes-problem-too-11574726800>.

⁴⁴ *The Indo-Pacific Strategy Report: Preparedness, Partnerships, and Promoting a Networked Region* (Washington, DC: Department of Defense, June 1, 2019), available at <<https://media.defense.gov/2019/jul/01/2002152311/1-1-1/1/DEPARTMENT-OF-DEFENSE-INDO-PACIFIC-STRATEGY-REPORT-2019.PDF>>.

⁴⁵ Thomas Mahnken, "A 'World-Class' Military: Assessing China's Global Military Ambitions," Statement Before the U.S.-China

Economic and Security Review Commission, June 20, 2019; Ratner, *Blunting China's Illiberal Order*.

⁴⁶ Campbell and Sullivan, "Competition Without Catastrophe," 110.

⁴⁷ *Summary of the 2019 National Defense Strategy*, 8.

⁴⁸ Ely Ratner and Samir Kumar, "The United States Is Losing Asia to China," *Foreign Policy*, May 12, 2017, available at <<https://foreignpolicy.com/2017/05/12/the-united-states-is-losing-asia-to-china/>>.

⁴⁹ More information on the BUILD Act is available at <<https://www.opic.gov/build-act/overview>>. On U.S. options for enhancing geoeconomics in the region, see Daniel Kliman, *To Compete with China, Get the New U.S. Development Finance Corporation Right* (Washington, DC: Center for a New American Security, February 6, 2019), available at <<https://www.cnas.org/publications/commentary/to-compete-with-china-get-the-new-u-s-development-finance-corporation-right>>.

⁵⁰ Joel Wuthnow, "Contested Strategies: China, the United States, and the Indo-Pacific Security Dilemma," *China International Strategy Review* 1, no. 1 (2019), 6, available at <<https://doi.org/10.1007/s42533-019-00006-x>>.

⁵¹ Based on Thomas Fingar and Jean C. Oi, "China's Challenges: Now It Gets Harder," *Washington Quarterly* 43, no. 1 (Spring 2020), 65–82; "China's Debt Tops 300% of GDP, Now 15% of Global Total: IIF," *Reuters*, July 18, 2019, available at <www.reuters.com/article/us-china-economy-debt-idUSKCN1UD0KD>; Nicholas Eberstadt, *China's Demographic Outlook to 2040 and Its Implications* (Washington, DC: American Enterprise Institute, 2019). A countering view contends that it is the United States that is more susceptible to an economic downturn if engaged in a lengthy economic struggle with China. See Bloomberg View, "U.S. Needs China More than China Needs the U.S.," *Industry Week*, April 6, 2018, available at <www.industryweek.com/the-economy/article/22025438/us-needs-china-more-than-china-needs-the-us>.

⁵² Schell and Shirk, *Course Correction*, 10. For an objective evaluation of alliances, see Kathleen J. McInnis, "The Competitive Advantages and Risks of Alliances," in *2020 Index of U.S. Military Strength*, ed. Dakota Wood (Washington, DC: Heritage Foundation, 2019).

⁵³ Campbell and Sullivan, "Competition Without Catastrophe," 110.

⁵⁴ The recent 70th anniversary summit of NATO included the following statement: "We recognize that China's growing influence and international policies present both opportunities and challenges that we need to address together as an Alliance." See London Declaration, press release, NATO Headquarters, Brussels, December 4, 2019, available at <https://www.nato.int/cps/en/natohq/official_texts_171584.htm>.

⁵⁵ For recommendations on managing economic relations, see Charles W. Boustany, Jr., and Aaron L. Friedberg, *Answering China's Economic Challenge: Preserving Power, Enhancing Prosperity*, NBR Special Report No. 76 (Seattle: National Bureau of Asian Research, February 2019); Daniel W. Drezner, "Economic Statecraft in the Age of Trump," *Washington Quarterly* 42, no. 3 (Fall 2019), 7–24.

⁵⁶ Aaron L. Friedberg and Charles W. Boustany, Jr., "Partial Disengagement: A New U.S. Strategy for Economic Competition with China," *Washington Quarterly* 43, no. 1 (Spring 2020), 23–40.

⁵⁷ This section leverages insights of two distinguished strategic analysts. See Hal Brands and Zack Cooper, "After the Responsible Stakeholder, What? Debating America's China Strategy," *Texas National Security Review* 2, no. 2 (February 2019), 68–81.

⁵⁸ Blackwill and Tellis, *Revising U.S. Grand Strategy Toward China*, 24.

- ⁵⁹ Marianne Schneider-Petsinger et al., *U.S.-China Strategic Competition: The Quest for Global Technological Leadership* (London: Royal Institute for International Affairs, November 2019). Many of these technologies are discussed in chapter 5 of this volume.
- ⁶⁰ For example, this would include adopting the recommendations of the National Security Artificial Intelligence Commission (NSAIC), *First Quarter Recommendations Memo* (Arlington, VA: NSAIC, March 2020), available at <<https://www.nscai.gov/reports>>.
- ⁶¹ On the instruments of economic statecraft, see Robert D. Blackwill and Jennifer M. Harris, *War by Other Means: Geoeconomics and Statecraft* (Cambridge: Harvard University Press, 2016); Elizabeth Rosenberg et al., *The New Tools of Economic Warfare: Effects and Effectiveness of U.S. Financial Sanctions* (Washington, DC: Center for a New American Security, 2016).
- ⁶² U.S.-China Economic and Security Review Commission, *2018 Report to Congress* (Washington, DC: U.S.-China Commission, November 2018), 29. See Nadège Rolland, "China's Belt and Road Initiative," in *Winning Without Fighting: Chinese and Russian Political Warfare Campaigns and How the West Can Prevail*, vol. 2, ed. Ross Babbage, *Case Studies* (Washington, DC: Center for Strategic and Budgetary Assessment, 2019), 65–72.
- ⁶³ Aaron L. Friedberg, "A New U.S. Economic Strategy Toward China?" *Washington Quarterly* 40, no. 4 (Winter 2018), 97–114; Thomas Shattuck, "Fighting the BRI," *Orbis* 63, no. 4 (Fall 2019), 505–519.
- ⁶⁴ Nadège Rolland, *Eurasian Century? Political and Strategic Implications of the Belt and Road Initiative* (Seattle: National Bureau of Asian Research, 2017).
- ⁶⁵ Bonnie S. Glaser, *Made in China 2025 and the Future of American Industry*, Testimony Before the Senate Small Business and Entrepreneurship Committee, Washington, DC, February 27, 2019.
- ⁶⁶ The bill as drafted is available at <<https://www.congress.gov/bills/116th-congress/senate-bill/29>>.
- ⁶⁷ James Talent and Robert O. Work, *The Contest for Innovation: Strengthening America's National Security Innovation Base in an Era of Great Power Competition* (Washington, DC: Ronald Reagan Institute, 2019), 24. See also James Manyika and William H. McRaven, *Innovation and National Security: Keeping Our Edge*, Independent Task Force Report No. 77 (New York: Council on Foreign Relations, 2019).
- ⁶⁸ Friedberg, "Competing with China," 42–43.
- ⁶⁹ Stephen Watts et al., *Alternative Worldviews: Understanding Potential Trajectories of Great-Power Ideological Competition* (Santa Monica, CA: RAND, 2020).
- ⁷⁰ On the Confucius Institutes, see James Patterson, "2 Reports Highlight Concerns over Confucius Institutes' Influence," *Education Dive*, March, 1 2019, available at <www.educationdive.com/news/2-reports-highlight-concerns-over-confucius-institutes-influence/549509/>; *China's Impact on the U.S. Education System*, U.S. Senate Permanent Subcommittee on Investigations, Committee on Homeland Security and Governmental Affairs (Washington, DC: U.S. Government Publishing Office, 2019), available at <<https://www.hsgac.senate.gov/imo/media/doc/PSI%20Report%20China%27s%20Impact%20on%20the%20US%20Education%20System.pdf>>.
- ⁷¹ Orville Schell and Larry Diamond, "Combating China's Influence Operations," *Project Syndicate*, November 4, 2019, available at <<https://www.project-syndicate.org/commentary/china-influence-operations-us-institutions-academia-by-orville-schell-and-larry-diamond-2019-11?barrier=accesspaylog>>.
- ⁷² Michael Mazarr, "U.S./China Competition," in *Chinese Strategic Intentions, A Deep Dive into China's Worldwide Activities*, ed. Nicole Peterson (Washington, DC: The Joint Staff, December 2019), 135.
- ⁷³ "China's Great Firewall Is Rising," *The Economist*, January 4, 2018, available at <www.economist.com/china/2018/01/04/chinas-great-firewall-is-rising>.
- ⁷⁴ Particularly the restoration of the "Peace Through Strength" section of the *National Security Strategy of the United States of America*, 25–32.
- ⁷⁵ This is in line with recommendations of the National Defense Strategy Commission, *Providing for the Common Defense: The Assessment and Recommendations of the National Defense Strategy Commission* (Washington, DC: U.S. Institute of Peace, 2018), available at <<https://www.usip.org/sites/default/files/2018-11/providing-for-the-common-defense.pdf>>. This effort was mandated by Congress to offer constructive assessments and alternative recommendations on U.S. defense strategy. The commission was led by Ambassador Eric Edelman and Admiral Gary Roughhead, U.S. Navy (Ret.).
- ⁷⁶ Oriana Skylar Mastro, "China's Military Modernization Program," Statement Before the U.S.-China Economic and Security Review Commission, September 4, 2019, 10. Mastro stated, "The issue is not that China has surpassed the United States in military power; it has not. The issue is that given current trends, China will meet or outmatch U.S. regional capabilities in the next five to 10 years."
- ⁷⁷ Mira Rapp-Hooper, "Saving America's Alliances," *Foreign Affairs* (March/April 2020), 127–140.
- ⁷⁸ Eric Rosenbach and Katherine Mansted, *The Geopolitics of Information* (Cambridge, MA: Belfer Center for Science and International Affairs, May 2019). See chapter 7 for discussions of Chinese exploitation of social media.
- ⁷⁹ Numerous scholars offer insights in this area, including Hal Brands and Toshi Yoshihara, "Waging Political Warfare," *The National Interest* (January/February 2019); Linda Robinson et al., *Modern Political Warfare: Current Practices and Possible Responses* (Santa Monica, CA: RAND, 2018). See also Ross Babbage, *Winning Without Fighting: Chinese and Russian Political Warfare Campaigns and How the West Can Prevail* (Washington, DC: Center for Strategic and Budgetary Assessments, July 2019); Ross Babbage, *Stealing a March: Chinese Hybrid Warfare in the Indo-Pacific: Issues and Options for Allied Defense Planners* (Washington, DC: Center for Strategic and Budgetary Assessments, July 2019); Clint Watts, "Disinformation: A Primer in Russian Active Measures and Influence Campaigns," Statement Prepared for the Senate Select Committee on Intelligence, March 30, 2017, available at <<https://www.intelligence.senate.gov/sites/default/files/documents/os-cwatts-033017.pdf>>.
- ⁸⁰ Nigel Inkster, *China's Cyber Power*, Adelphi Paper No. 456 (Abingdon, UK: Routledge, 2016).
- ⁸¹ Henry Farrell and Abraham Newman, "Weaponized Globalization: Huawei and the Emerging Battle of 5G Networks," *Global Asia* 14, no. 3 (September 2019), 7–12.
- ⁸² Bates Gill and Benjamin Schreer, "Countering China's 'United Front,'" *Washington Quarterly* 41, no. 2 (Summer 2018), 155–170; Toshi Yoshihara, "Evaluating the Logic and Methods of China's United Front Work," *Orbis* 64, no. 2 (Spring 2020), 230–248.
- ⁸³ Saunders, *Managing Competition with China*, 3.
- ⁸⁴ Ashely Townshend, Brenda Thomas-Noone, and Matilda Steward, *Averting Crisis: American Strategy, Military Spending and Collective Defence in the Indo-Pacific* (Sydney: U.S. Studies Centre, August 2019), 10.
- ⁸⁵ *Ibid.*, 15.
- ⁸⁶ In particular, see the claims of the National Defense Strategy Commission, *Providing for the Common Defense*. See also Robert O. Work and Greg Grant, *Offset Strategy with Chinese*

Characteristics (Washington, DC: Center for a New American Security, 2019).

⁸⁷ For a rich study of Chinese military reforms, see Phillip C. Saunders et al., eds., *Chairman Xi Remakes the PLA: Assessing Chinese Military Reforms* (Washington, DC: NDU Press, 2019).

⁸⁸ Ely Ratner et al., *Rising to the China Challenge: Renewing American Competitiveness in the Indo-Pacific* (Washington, DC: Center for a New American Security, December 2019), 21–33, 48–52, available at <<https://s3.amazonaws.com/files.cnas.org/documents/CNAS-Report-NDAA-final-6.pdf?mtime=20200116130752>>.

⁸⁹ John Hamre, “Foreword,” in *Twin Pillars: Upholding National Security and National Innovation in Emerging Technologies Governance*, ed. Sam Brannen et al. (Washington, DC: Center for Strategic and International Studies, January 2020), vii.

⁹⁰ Talent and Work, *The Contest for Innovation*, 9.

⁹¹ Tai Ming Cheung and Thomas G. Mahnken, eds., *The Gathering Pacific Storm* (Amherst, NY: Cambria, 2018), 248.

⁹² Talent and Work, *The Contest for Innovation*, 25–26. See also William Greenwalt, *Leveraging the National Technology Industrial Base to Address Great-Power Competition: The Imperative to Integrate Industrial Capabilities of Close Allies* (Washington, DC: Atlantic Council, April 2019).

⁹³ Markus Brunnermeier, Rush Doshi, and Harold James, “Beijing’s Bismarckian Ghosts: How Great Powers Compete Economically,” *Washington Quarterly* 41, no. 3 (Fall 2018), 161–176.

⁹⁴ Ely Ratner, *Blunting China’s Illiberal Order*, 3.

⁹⁵ The phrase *Weary Titan* is drawn from the title of a book on the United Kingdom’s transition at the dawn of the 20th century. See Aaron L. Friedberg, *The Weary Titan: Britain and the Experience of Relative Decline, 1895–1905* (Princeton: Princeton University Press, 1988).

⁹⁶ For recommendation on how, see Alice Rivkin et al., *A Recovery Squandered: The State of U.S. Competitiveness 2019* (Cambridge, MA: Belfer Center for Science and International Affairs, December 2019), available at <<https://www.hbs.edu/competitiveness/Documents/a-recovery-squandered.pdf>>.

⁹⁷ Adam P. Liff and G. John Ikenberry, “Racing Toward Tragedy?” *International Security* 39, no. 2 (Fall 2014), 51–90.

⁹⁸ Hal Brands and Eric Edelman, “America and the Geopolitics of Upheaval,” *The National Interest*, July 21, 2017, available at <<https://nationalinterest.org/feature/america-the-geopolitics-upheaval-21258?page=0%2C3>>.

Chapter 15

Conclusion

Realities, Imperatives, and Principles in a New Era of Great Power Competition

By Thomas F. Lynch III

This chapter summarizes the major features of the new era of Great Power competition (GPC). It then provides an assessment of the novel 2019–2020 coronavirus pandemic implications, concluding that the virus’s impact is likely to accelerate ongoing geopolitical trends rather than generate new ones. The chapter analyzes three main imperatives for American success in GPC by observing that the Sino-American dyad is not a new Cold War; successful competition with China must feature a wise choice of U.S. allies, and the United States can succeed only if the national government smartly intervenes in the economy to fortify American competitive advantage. It offers historically based analysis demonstrating that four competitive principles are most critical to U.S. success in a long-term competition with China: firmness with flexibility, durable partnerships and alliances, the peril of reciprocal societal denigration, and playing for time.

The year 2020 began with a global health shock of a kind unobserved in a century—a deadly novel coronavirus (COVID-19) pandemic. This long-predicted, but nonetheless gripping, worldwide trauma layered over the top of ongoing historically broad and deep international economic and geopolitical changes.¹ Decades-long economic globalization began a sharp decline at the end of the 2010s, punctuated by a dramatic trade war between the world’s two largest economies: the United States and China. Geopolitics also witnessed dramatic change. Two distinct global rivals—China and Russia—rose during the late 2010s to challenge what had been a quarter-century run of American global dominance, or unipolarity. This era of Great Power competition (GPC) generated patterns of international interaction with far more confrontation and conflict than observed from 1990 to 2015, which largely was characterized by cooperative and collaborative behaviors among the world’s largest states.

A century ago, the Great Pandemic of 1918–1919 corresponded with a deadly inflection point within a prolonged period of GPC that ran from 1895 to 1945. The period from 1914 to 1918, during which the major protagonists fulminated their multistate rivalry in a prolonged and horrifically destructive period of direct military clash, became World War I.

That so-called Great War did not end the multiparty competition. There was no clear transition from one dominant power to another and no durable arrangement to channel state competition away from direct military clash. Thus, the multipolar Great Power struggle lurched forward with most of the same prewar players and into an even more global and destructive military clash 20 years later in World War II.² In 2020, COVID-19 mixes into a three-state Great Power competition, wherein the United States, China, and Russia openly compete for international status and power and the trajectory of relative power from a long-dominant America to either rival remains incomplete and far from certain.

The chapters of this volume have grappled with the many issues and uncertainties surrounding the ongoing transition from a unipolar world dominated by American global power to one where rivals Russia and China now compete openly with the United States and each other. In the case of Vladimir Putin's Russia, its contemporary power capabilities are mainly reimagined and repurposed military and reenabled propaganda implements rather than anything new. In the case of China, truly historic economic growth is catalyzing new wealth and imagination, generating an array of power capabilities that enable broad competition with the United States and growing influence with other states.

This chapter offers a collection of observations about the dawning new era of Great Power competition. It extends some of the numerous insights generated in the previous chapters but does not recite them all.³ This chapter evaluates the main elements of contemporary Great Power competition between and among the three main rivals. It situates major contemporary GPC dynamics in context with those of past periods of multilateral Great Power rivalry, including an assessment of what the COVID-19 pandemic might mean for dominant GPC trends. The chapter then addresses the critical question of whether ongoing Great Power transition must result in direct military clash and what factors might elevate the risk. It also analyzes the prospects for GPC to enable viable and durable partnerships for collaboration and cooperation to develop across the five categories of interstate interaction found in table 2.2: political and diplomatic, ideological, informational, military, and economic.

The chapter explicitly covers the objects for influence of contemporary Great Power competitors: other countries and their perspectives. It offers three major imperatives about the reciprocal and dynamic interaction of American competitive advantages and the needs of potential partners. The concluding section presents four principles most vital to U.S. success in its competitive Great Power dyad with China: firmness with flexibility, partnerships and alternative geometries, leaders versus peoples and the poison of mass denigration, and playing for time.

Essential Outlines of Contemporary Great Power Competition

Contemporary Great Power competition is unique, but not unprecedented. Multipolar GPCs have been contested throughout modern history. Each contributed important insights to the dynamics of the contemporary world. At the same time, contemporary dynamics exert their own pull on the choices and risks faced by the modern Great Powers. These factors include, but are not limited to, the impact of modern economic advancements, the importance of new technologies as means of competition, and the influence of warfighting

risks on contemporary societies. Finally, modern GPC is already changing patterns of geostrategic interaction.

Essential Elements

There are three contemporary Great Powers in 2020. The United States stands atop the triumvirate, with China a rising competitor and Russia vying for top-level prestige while facing clear signs of decline. The emerging strategic aims of China and Russia are incompatible with those established by American power in the post-World War II era; this has produced the return of a historically dominant pattern of Great Power competition. China is the Great Power best poised to displace America from its long-dominant power position. It has a positivist perspective on what a new global order could look like, one loosely captured in its concept of a “community of common identity.” While a net power comparison between the United States and China indicates that their power transition timeline is longer than some now fear, the Sino-American competitive dyad is likely to be the dominant Great Power rivalry into the future.⁴ Russia is an urgent but transient security risk for the United States and China, with the potential to do enormous military damage to the world if miscalculation leads to military clash. However, Russia practices a reactive, disruptive strategy aimed to pacify its immediate borders (a loosely formed “Eurasia focus”) and to question contemporary institutions and processes that it perceives as a threat. Unlike China, it is a competitor without a viable vision for a new world order or the necessary power to generate one. China and Russia may engage in tactical entente to erode American power, frustrate U.S. actions and preferred institutions, and question norms and rules they deem threatening. Their long-term interests, however, diverge too much for a durable partnership. Thus, Washington must remain careful not to misunderstand tactical cooperation as some form of deeper anti-American strategic alliance.

“[T]he official mind in Washington clearly has moved toward the view that China is today, and will be for the foreseeable future, the principal challenger to overall U.S. hegemony in the international system. No other countries come close because they lack comparable levels of comprehensive national power, even though several states, such as Russia, North Korea, and Iran, oppose the United States locally or on important specific issues.”

—Ashley J. Tellis, Alison Szalwinski, and Michael Wills, eds., *Strategic Asia 2020* (2020)

Essential Backdrop

The realignment of Great Power relations from an era of singular American dominance to one with three main actors playing parts in a multipolar competition has evolved slowly. Cooperative relations began to erode in 2008. By 2014–2015, the three protagonists were in a de facto GPC, which was formalized in U.S. strategic documents in late 2017 and early 2018. GPC emerged against a backdrop of major economic change. More than two decades of rapid economic globalization came under increasing scrutiny for a record of fragility and unfulfilled expectations. Mainly, but not exclusively, globalization lost prestige from repetitive boom-and-bust cycles and a propensity for creating an ever smaller circle of extraordinarily rich and comfortable elites juxtaposed against a growing circle of underserved constituent groups.⁵ Today, a fourth industrial revolution is fueling deglobalization and eroding global markets and supply chains. On one level, it is exacerbating the socio-

economic disruption of the digital age with widening inequality in incomes and greater unemployment among low-skilled workers. On another level, it is reducing the price of precision and advanced manufacturing and creating a new generation of smaller, smarter, and cheaper weapons. The inexorable movement of product manufacturing closer to domestic markets will continue to be a factor with great impact on GPC.

The foundation of modern Great Power wealth and competitive advantage has fundamentally changed from one dominated by industrial era technology to one in which information technology (IT) has become the source of geopolitical power. China has been the early beneficiary of this change, leveraging an ability to appropriate (and misappropriate) global intellectual property to accelerate technological growth while maneuvering to control global information flows it finds threatening. Russia and China have determined that information power is more likely than industrial power to determine the outcome of long-term geopolitical contests.⁶ Thus, both Russia and China have been increasingly waging foreign propaganda campaigns on social media platforms and other online channels of international influence. To keep pace, the United States must rethink its competitive posture, work with other developed nations, and, via public-private partnerships, reprioritize resources into key information technology and capabilities in order to pursue broad, agile approaches to limiting the foreign propaganda threat.

The United States has distinct advantages over both China and Russia as the fourth industrial revolution begins to reshape the world. Working with partners and allies—and while adjusting American laws and regulations to the new economic forces—the United States is well-poised to exploit its natural advantages in higher education, innate innovation, entrepreneurial spirit, and global market share. China also may benefit greatly from the fourth industrial revolution by prioritizing government investment in its high-tech manufacturing sectors. However, it must grapple with looming economic challenges from growing unemployment, an aging and less productive workforce, and a potential for social unrest. Russia, meanwhile, is not well poised for future economic competition, as it lacks the public- or private-sector elements necessary to participate fully in the modern economy.

Geostrategic Interactions

Russia and China present distinct competitive threats to the United States around the globe. In many regions, Russia often poses the more immediate challenge, whereas the repercussions of Chinese economic investments manifest themselves subtly and will likely undermine U.S. strategic interests more gradually.

The United States and China have primary and conflicting interests in the Indo-Pacific region. The importance of those interests to both countries makes the region a central venue for Great Power competition. The U.S. Free and Open Indo-Pacific vision is not compatible with China's aspirations for increasing control within its First Island Chain and wider Chinese regional aims, sometimes espoused as a community of common destiny. Here, Sino-American competition could turn toward confrontation or a military clash if careful diplomacy is not exercised. China has economic dominance in markets and investment across most of the region. It also has eroded U.S. military advantage in potential locations of confrontation near its shores and inside the First Island Chain. The United States retains an overall advantage in military technology and power projection across the

wider Indo-Pacific region, commercial financial dominance, and a resonant ideology and ability to communicate it, along with a regional political and military alliance structure unmatched by China.

Russia has a primary interest in Europe, with special sensitivity to sovereignty in its near abroad—including former Soviet Union provinces. American and European diplomacy will remain challenged to stanch Russian misadventures without generating overt confrontation or clash. While Europeans mistrust Russia generally, their perception of Russia as a security threat varies greatly. Europe cannot alone defend member states from Russia. Thus, Europeans worry that the United States may detach itself from Europe—particularly the North Atlantic Treaty Organization. Today—and in the foreseeable future—Europe remains unable to create an autonomous system of security and defense. Should the United States move to depart the Alliance, Europe may intensify accommodation with Russia—and even with China.

Moscow and Beijing are only nominally united in their desire to compete with and displace U.S. influence across Latin America, the Middle East, Africa, and the Arctic. The United States retains a historic strategic interest in primacy across the Western Hemisphere, and the region appears unlikely to be similarly important for the other two Great Powers in the near term, making it a less intense area of competition absent unforeseen miscalculation. Conversely, the Middle East promises to be an area of dynamic competition and occasional nonmilitary confrontation in the coming decade—with access to resources the principal focal point.

As U.S. and Russian interests in external energy sources wane, however, the competitive focus in the Middle East may shift to prestige and resonance of ideological narratives. States in Latin America, the Middle East, Africa, and the Arctic are often eager recipients of Russian and Chinese attention and resources out of convenience rather than ideological commitment. Thus, the United States should avoid imposing regional strategies that treat Russian or Chinese activities as uniformly harmful to U.S. interests. A Washington focus on American strengths as an economic partner, the quality and quantity of its military assistance, and the positive and benign nature of its military forward presence should best safeguard U.S. interests in these regions of less intense contemporary GPC.

Weapons of mass destruction remain a critical feature and potentially dynamic factor in GPC. The system of arms control treaties that, for decades, limited U.S. and Russian nuclear forces is under great strain and could collapse. Russia began a slow modernization of its aging nuclear forces in the 2000s. In March 2018, Vladimir Putin announced that Russia was developing new types of nuclear systems, including a multi-warhead intercontinental ballistic missile, along with hypersonic, autonomous, and nuclear-powered delivery systems. It is unclear whether Moscow has begun to place a greater reliance on nuclear weapons or the threat to use them during regional conflicts.

The United States is engaged in an expensive recapitalization and modernization of its nuclear forces and plans to begin fielding new systems in the late 2020s. China is investing more in nuclear capabilities, modernizing and expanding strategic systems and developing dual-capable theater-range platforms that would heighten the nuclear risks in Indo-Pacific conflicts. For now, Sino-American nuclear weapons activities do not appear likely to lead to a Cold War–style nuclear arms race. Yet the risk of a new multistate arms race in nuclear

weapons, delivery systems, and missile defenses is growing, as Great Power relations become more competitive and even confrontational. The three Great Powers today signal that they do not anticipate that an unwelcomed conventional clash would escalate to the nuclear level, but risks of threshold miscalculation remain. At the same time, chemical or biological attacks could be difficult to attribute and may be well suited to support Russian and Chinese objectives in operations below the threshold of open armed conflict.

Despite the focus on GPC, threats to peace, stability, and American interests from rogue states and terrorism are far from eradicated. Rogue states such as Iran and North Korea lack the military and long-term economic power and/or transnational cultural appeal to match U.S. power globally or stabilize an alternative international political order. They are motivated by a combination of regime survival, aspirations for regional dominance, and sometimes global relevance, as well as an inclination to confront the United States, which they all believe is the main obstacle to their own aspirations. They tend to confront the United States below the threshold of direct armed conflict and across multiple domains. While a menace to be managed by the United States, there is little prospect for a fully cooperative anti-U.S. rogue state axis. Moreover, China and Russia must fear spillover to their own economic and strategic interests, so Beijing and Moscow are unlikely to join fully in disruptive rogue adventurism, instead pursuing a mixture of cooperative and obstructive responses on a case-by-case basis.

American counterterrorism efforts will confront a set of new realities. Recent American counterterrorism operations in Syria likely will be the model of the future. Russia must be expected to undermine U.S. counterterrorism objectives, either directly or indirectly. As in Syria, Russia will combine diplomatic initiatives, proxy warfare, and electronic warfare to foil U.S. military dominance. Regional states will continue to pursue their own counterterrorism objectives—some that align with U.S. objectives and others that do not. To be effective in this new environment, the United States will require new counterterrorism authorities, new technologies, and other tools that can help manage the risks from small-footprint deployments. It also must hold sponsor states accountable for actions by proxies against U.S. counterterrorism forces.

Finally, two nontraditional competitive venues—space and cyberspace—are those where all three Great Powers have primary interests engaged and growing. There is high risk that intensifying competition in space could lead to greater confrontation there. Agreement on some viable rules and norms for collaborative use and cooperative actions in space could reduce the growing risks of confrontation and miscalculation leading to clash. Likewise, the absence of cooperative rules and norms in cyberspace has contributed to a darkening turn toward confrontational dynamics.

Relevant History and Contemporary Dynamics

The contemporary era is characterized by heightened competition among more than two Great Powers—a multipolar competition. This makes it distinct from the most recent period of GPC, a bipolar rivalry between the United States and the Soviet Union that played out over a 45-year Cold War. In past multipolar Great Power competitions, rivalrous dyads ebbed and flowed. These dyads normally involved a rising power and a dominant one, raising the strategic question about the inevitability of relative power decline by the dominant

state and a power transition between them. Great Power transition challenges rising states with the dilemma of how to assert their relative power gains without provoking outright clash with the dominant state. Transition also confronts the dominant, but relatively declining, state with the vexing question of whether its rising challenger could be accommodated in a manner that avoids destructive military clash and an unacceptable change in the status quo. These transitions play out over decades and centuries, not years.

Although three-quarters of Great Power transitions since 1500 have featured a destructive period of war between them, this outcome is not foreordained. Great Power competitors joined in a relative power transition can culminate their interactions with accommodation or acquiescence short of war, but those peaceful outcomes require hard work and astute leadership. When one side (or both) in a relative power transition dyad recognizes a shift in the relative alignment of economic and military power moving decisively against it, it is much more inclined to risk a preemptive conflict than when it perceives a stable power status quo. Too often, Great Power leaders misperceive relative power, eschewing detailed, empirical assessments of power to inform decisionmaking and abet strategic planning. Even when accurate assessments of relative decline or vulnerability are made, domestic or bureaucratic interests may retard agile adaptation necessary to mitigate risks of Great Power war. Thus, success in GPC requires extraordinary political leadership in both international statecraft and generating domestic renewal and adaptation.

The Sino-American competitive dyad is likely to be a dominant Great Power rivalry well into the future.⁷ It is the competitive dyad most fraught with the dangerous dynamics of Great Power transition, although any misstep leading to accidental war with Russia would be enormously destructive and consequential, especially if Russia escalated to a nuclear weapons threat or use in order to end a conventional conflict. While some Western pundits stoke fears of an imminent and disastrous power shift in favor of China on the horizon, a net power comparison between the United States and China indicates that the transition timeline is longer than some now predict. Properly understood, this elongated transition affords China and the United States time to better appreciate the risks of unbridled rivalry and seek a path of modulated competition with elements of confrontation and collaboration underpinning the search for mutually acceptable strategic outcomes.

Geopolitical Shocks and GPC: COVID-19

The big geostrategic question of 2020 is how the COVID-19 pandemic will affect contemporary Great Power competition. As a once in a lifetime, truly global health crisis, COVID-19 must be understood as a factor in evaluating GPC and the future trajectory of relative power transitions between them. An in-stride assessment of likely pandemic impacts on GPC suggests that, while each competitor will suffer absolutely from this significant exogenous shock, none seems likely to endure a mortal blow or one that alters the relative balance of power immediately or shifts the trajectory of relative power transition. This can be established with an overall assessment of historic pandemic geostrategic effects followed by an evaluation of likely impacts from COVID-19 on the three Great Powers.

The Spanish flu, or Great Pandemic of 1918–1919, is the most analogous global health shock in modern memory. Its impact on the world order and the Great Powers of the time remains debatable. The Spanish flu added to an already enormous death toll during World

War I. Along with the Great War, it was a factor that ended the prewar phase of economic globalization, but the pandemic might have had an important effect on the postwar order in general, and the United States in particular, due to the health of the American President, Woodrow Wilson.

While in Paris to negotiate the end of World War I and the framework for the postwar world in April 1919, Wilson contracted the flu and was taken “violently ill.”⁸ Prior to getting sick, Wilson had been a forceful voice in Paris, challenging the leaders of Britain and especially France to adopt limited war reparations against Germany and for a genial peace that would bind the wounds of war, give voice to oppressed peoples, and widen the space for peace and global integration. After becoming ill, Wilson reportedly grew disoriented and distracted—a symptom attributed to severe influenza and fever. Exhausted, he gave up on the demands he had been making against reparations and forceful occupation of Germany. While hard to know if a healthy Wilson would have won his point against a determined French position to seek a punitive peace, Wilson’s physical decline was noteworthy.⁹ The “victor’s peace” demanded in the final version of the Versailles Treaty set up the humiliation of Germany and a cause that German fascism exploited in its subsequent interwar rise.

Six months later, back in the United States and while in the middle of a bitter political battle with the U.S. Senate to secure ratification of the Treaty of Versailles and entry into the League of Nations, Wilson suffered a severe stroke and withdrew from public life. Again, it is impossible to know whether Wilson’s stroke was abetted by his earlier bout with the Spanish flu, but doctors have subsequently linked weakened organs to prolonged oxygen loss and inflammation experienced by survivors of severe influenza bouts. A bedridden Wilson saw his campaign for ratification of the Treaty of Versailles and participation in the League of Nations defeated by the Senate in March 1920.¹⁰ The result was American withdrawal from leadership in world politics for over 20 years—throughout the interwar period—and an absence of American wealth and power as a counterweight to increasing global fragmentation, radicalization, and war.¹¹

Fifty million people died during the Spanish flu without redirecting the course of global politics, the framework for domestic politics, or basic human behaviors.¹² The insight of Spanish flu history is that, while a traumatic global pandemic may not alter broad global patterns or trends for key countries, it may have important indirect impact on geopolitical futures should the virus badly afflict an important political leader.

But what about after overarching geopolitical changes from COVID-19? Henry Kissinger wrote in April 2020 that COVID-19 will forever alter the world order, asserting that the pandemic’s sweeping global impacts confirm that purely national solutions cannot solve major global issues. Kissinger argued that the ongoing movement toward nationalism must be understood as a danger, and collaborative approaches arising from the pandemic extended toward cooperation to protect the “liberal world order.”¹³ Skeptics assert that Kissinger wrote earlier, and more accurately, that world orders last until their foundations are fundamentally shattered by events, and a pandemic is not that kind of event.¹⁴ Major trends in early summer 2020 seem to be bearing out an earlier, more skeptical Kissinger. The crisis has undercut support for globalization, but that was already trending, with rising populism around the world. It has exposed an already identified yawning gap between major contemporary security, climate, and health challenges and the insufficient power of

any one country to address these challenges.¹⁵ Thus, the pandemic shows that the modern era needs more global governance, not less—but without enough shock to the system to force significant change.¹⁶

Then there is the question of whether COVID-19 portends a change in the relative balance of power among Great Power protagonists. Some American political observers worry that the United States could lose the global leadership contest with China if it holds to an “America First” approach and does not seek its historic post–World War II role of leading collective responses to global challenges.¹⁷ These worries are most acute with respect to U.S. relationships in the Indo-Pacific region. There, pundits fear that American regional legitimacy is at risk due to Washington’s comparatively feeble pandemic responses vis-à-vis those of partners such as South Korea and Taiwan, China’s obvious interest in reviving East Asian economies with an eye to cement its role as a hub, and the risks to U.S. multinational credibility should Washington remain idle and aloof from coordinated regional response and recovery.¹⁸ But China and Russia confront their own challenges in recovering from COVID-19. There is a chance that the virus’s economic impact may be harsher on the United States than on China. Should this happen, it would accelerate the power shift to Asia, but that was already under way.¹⁹

And there are real limits to China’s capacity to take advantage of the current crisis. China’s economy will not be able to return to its prior growth trajectory of some 5 to 6 percent annually until the economies of the United States and the European Union recover as well. Funding another credit-fueled stimulus as the Chinese did in 2008–2009 is off the table due to China’s high overall debt levels and the real risk of triggering a collapse of its financial system.²⁰

As of summer 2020, a unique characteristic of the current crisis has been the conspicuous absence of U.S. global leadership. The United States has not rallied the world in a collective effort to confront either the virus or its economic effects. Nor has the United States inspired the world by its approach to the pandemic at home.²¹ Should Washington recover its footing and lead a G20 effort at expanded financial cooperation working with regional friends and tying in China and Europe, it might emerge with a stronger reputation regionally and globally. This hopeful outcome seems unlikely given the main policy focus of the Trump administration.

Considering these contemporary factors, the post-COVID-19 world is unlikely to be radically different from the one that preceded it. The pandemic and response are reinforcing fundamental geopolitical traits.²² Deglobalization, rising anti-immigration sentiment, and Great Power competition all were established before the pandemic. It seems unlikely that the pandemic will shift general trends back toward global cooperation and multilateralism. A lack of global cooperation is likely to continue, resulting in a weakly coordinated response to the health crisis and slow global economic recovery.²³

But the case of Woodrow Wilson shows individual leaders matter and contingency has to be considered. President Wilson’s bout with the Spanish flu occurred in its “second wave” during the winter of 1918–1919. If a second COVID-19 wave in the winter/spring of 2020–2021 were to metastasize and incapacitate or kill senior Great Power leaders, could this generate a major geopolitical shift? For Russia, Vladimir Putin has enormous power and has maneuvered to retain it through 2036. But Putin’s death or incapacitation is unlikely

to change the trajectory of Russian strategy or relative power capabilities. Some Western foreign policy experts worry that a Russia under Putin that is weakened by COVID-19 and the collapse of world oil prices might be more openly aggressive and prone to risk military conflict. But others think a weaker Russia would likely become less assertive and more dependent on China.²⁴ On balance, the trends with Russia seem more likely to accelerate than change. Moreover, major pundits agree that the systems over which Putin presides—political, economic, military, and informational—are rooted in Russian history and likely to outlast him.²⁵

In China, President Xi Jinping is powerful and the clear head of the Chinese Communist Party (CCP), but China's strategic vision and development trajectory are deeply grounded and supported by other top CCP leaders. Xi has taken greater risks than some of his predecessors in pursuing CCP aims globally and across the Indo-Pacific region in particular. Nonetheless, the exit of Xi as China's leader would do little to alter China's basic strategic framework or its plans for moving forward—particularly in the Indo-Pacific region. As American policy analyst Richard Haass observed, “nothing about the current crisis will change China's view that the U.S. presence in Asia is a historical anomaly or reduce its resentment of U.S. policy on a range of issues, including trade, human rights, and Taiwan.”²⁶

In the United States, COVID-19's effect on individual leaders might have modest impact given that 2020 is a Presidential election year. As of fall 2020, major party candidates President Donald Trump and former Vice President Joe Biden are basing campaign strategies on contrasting views of America's proper role in such a sweeping global pandemic. The Trump administration is touting its record of success in America First policies, abstaining from wider global leadership to combat the crisis, and attacking the World Health Organization and China for enabling the pandemic. The Biden campaign promises a less combative United States, and one more focused on leading a collective international response.

Should President Trump suffer incapacitation or death from the virus, a Biden victory would not be certain. Trump's Vice President, Mike Pence, seems a likely torchbearer for the same kinds of policies pursued during 2020, and a Pence election seems likely to entrench America First strategic aims for another 4 years. Conversely, a defeated Donald Trump in November 2020 would have resulted—at least in some part—from the electorate's disapproval of his handling of the COVID-19 pandemic. This would open the door for some greater international collaboration and American leadership, but this door may open only a little bit. A post-Trump Democratic President would still confront some 30 to 35 percent of American voters who are jaded about international commitments and unwilling to sign up for spending American resources leading other “rich” nations in combating major international problems.²⁷ A new administration might find some support for a late-breaking American-led global initiative to find a vaccine and underwrite its mass distribution, but asking the American people again to tackle all the global problems at the heart of U.S. foreign policy will continue to be a tough sell.²⁸ The impetus for America and its allies to decouple from the Chinese economy seems likely to grow as a result of the pandemic, and only partly because of concerns about China. There will be renewed focus on the potential for interruption of supply chains along with a desire to stimulate domestic manufacturing. Global trade will partly recover, but more of it will be managed by governments rather than markets.²⁹

So the other side of the COVID-19 crisis is likely to look as it did before, with Great Power strategies and relative power positions much the same. New technologies and challenges will continue to outpace the collective ability to contend with them. No single country enjoys the standing the United States did in 1945 or in 1990, and no other country, neither China nor anyone else, has both the desire and the ability to fill the international leadership void the United States has created.³⁰

Thus, a viable approach to the new era of Great Power competition must begin with a clear-eyed understanding of the strengths and weaknesses of the major protagonists in the five main categories of interstate competition: political and diplomatic, ideological, informational, military, and economic. As the dominant GPC dyad, China and the United States are the critical nations for comparison. Their relative advantages in these five categories inform the realm of the feasible and establish the unworkable. For Washington, a sober analysis suggests that it must eschew highly confrontational policies in places where it lacks advantage and seek collaboration whenever feasible, while at the same time compete with firmness in categories where it has advantage. America needs to revisit and clearly appreciate that the main source of its ability to project power and exert influence now, as since World War II, is its global networks of allies and partners.

How to Compete Wisely: The Important Role of Alliances and Partnerships

An America that competes smartly with China must understand both the value of time and where it can leverage its major advantages. The United States retains a commanding advantage in military power, although not to the degree it did 20 years ago. But its global military advantages can be offset if China (or Russia) is able to pick favorable physical and political ground for a short, decisive military conflict. Washington must acknowledge this and compensate for it. America's ideology resonates well globally and especially in the Indo-Pacific region. Similarly, its ability to promulgate information and sustain support remains superior to China's, despite Beijing's serious efforts to articulate and reinforce a clear message—a message often undercut by the fact that it features CCP talking points inconsistent with Chinese actions at home and abroad. China is upping its efforts to use political and diplomatic tools to undercut U.S. alliances and partnerships internationally and especially in the Indo-Pacific region, but Washington—despite some obvious recent self-sabotage of its diplomatic advantage—retains strong ties and bonds established over decades that are not easily destroyed. At the same time, China has significant economic advantages over the United States, especially in the Indo-Pacific region. Beijing can mobilize direct trade and investment resources and provide countries with valued opportunities for growth that the United States cannot alone match.

America's relative advantages in ideas, information dissemination, political and military alliances, and conventional military power—when applied away from regions of local Chinese (or Russian) advantage—inform where the United States can build on strength. Concurrently, American weaknesses in relative economic strength compared with China or the conventional military capabilities to defend allies and partners near China (or Russia) informs Washington about how it must proceed for competitive success. The United States

will succeed in competition over time by working with friends and partners and eschewing the strategic error of posing stark binary choices to would-be partners and friends.

These understandings translate into three major imperatives that should inform competition between the United States and China. First, the Cold War was a Great Power competition but not analogous to contemporary GPC. Cold War dissimilarities are important to understand so that policy choices for GPC do not err in applying Cold War lessons. Second, Great Power competitors do best when they form durable partnerships with capable allies and friends. These partnerships are not risk free; Great Powers can make bad choices. However, chosen wisely, Great Power alliance networks expand security options, generate diplomatic leverage and helpful lines of communications, and bolster political legitimacy.³¹ Finally, Great Power competitors do not have the luxury of “hands off the wheel” economic and technology policies. In reality, the myth of American capitalism as a *laissez-faire*, private-market enterprise does not comport with fact—even in times of broad geopolitical and geoeconomic cooperation. During times of Great Power rivalry, the U.S. Government must shake off the myths that constrict competitive decisionmaking and understand the importance of deliberate government-sponsored development in key security and wealth-making technologies and processes.

Despite some contrary commentary, the world is in a new era of GPC.³² Although the Cold War between the United States and the Soviet Union was a Great Power competition, it was unique in modern history and without great resonance with contemporary GPC dynamics.³³ Nostalgic U.S. calls for broad application of Cold War competitive strategies such as containment fail to appreciate that its unique features make it a poor strategic template for today.³⁴ However, some aspects of a competitive mindset from that time can be useful in the present.³⁵

First, the Cold War was always bipolar and never multipolar. From 1945 to 1991, no other country in the world possessed the global strategic ambitions or the levels of power held in Washington and Moscow. In 2020, GPC is multipolar and has been so from inception.³⁶ Second, the bipolar competition of the Cold War did not feature a clear rising power challenging a dominant one. Instead, the United States and the Soviet Union each claimed primacy and jostled as presumptive equals in every dimension of state interaction. There was no Great Power transition process during the Cold War. Contemporary GPC features a transition framework consistent with historical precedence with a clearly dominant power, the United States; a clear rising power, China; and another Great Power, Russia, contesting geopolitical primacy but with limited and suspect power capabilities. The ongoing transition raises uncertainties and risk calculations that were not present during the Cold War.

Third, the Cold War began with sharply divided Great Power geographic spheres of influence and little interaction between them. A Soviet bloc and a U.S.-led Western bloc of states quickly formed after 1945, and almost no economic, social, communications, or political interactions existed between them—beyond basic diplomacy and some limited mechanisms for travel and cultural exchange. In a starkly different fashion, the Great Power rivalry dyad of the United States and China evolved after more than 30 years of broadly cooperative interaction and engagement in diplomatic, social, and political activities. In all but the military sphere—and even in the hotly contested communications sphere—American and Chinese competitive tensions evolved against a high degree of interactions and

interdependence. So in the Cold War, an American strategic imperative became one of increasing transparency, openness, and resonance with Soviet bloc peoples in a manner that bypassed Communist leaders and provided greater intellectual and physical opportunities to the masses.³⁷ In contemporary GPC, the United States, along with its allies, confronts a different competitive challenge: how to selectively disengage itself from China in places of strategic vulnerability without squandering the kinds of beneficial connectivity, transparency, and access that now exist and that remain desirable.

The Cold War bifurcation into trade blocs was outside the historic norm for multipolar GPC. Past Great Power competitions and transitions featured a mosaic of simultaneous economic confrontation and collaboration. The United Kingdom and Imperial Germany had steadily increasing trade volume—although an evolving character of exports—throughout the 25 years before World War I.³⁸ Napoleonic France retained extensive economic ties with Great Britain beyond its 1803 declaration of war, and when Napoleon tried to impose an end to all British trade on the continent in 1806, extensive British merchant activities through Spain and Russia continued to supply France and Europe.³⁹ At its Cold War height, Soviet exports to the United States totaled only \$1 billion (in 1990).⁴⁰ In 2017, Chinese exports to the United States were \$500 billion, and U.S. company affiliates in China that year made \$544 billion.⁴¹ Severing such well-established economic ties between Great Power rivals is difficult to do. Thus, it cannot be surprising that fully decoupling America from the Chinese economy would be difficult and with a cost that would be unacceptably high.⁴² The United States and China already have been gradually disengaging in multiple economic areas. Reciprocal direct foreign investment has been declining for 5 years. Some U.S. technology firms abandoned China as its “Great Firewall” grew, and more have become wary of doing business in China since its 2015 announcement of Made in China 2025 goals for IT and artificial intelligence (AI) dominance. Finally, the number of Chinese students in American universities began to decline in 2018. At the same time, Chinese and American interdependence in trade, capital markets, and currency markets run deep. Until the United States launched a trade war with China in 2018, these areas were not decoupling—thus demonstrating they will be difficult to disentangle.⁴³

Past Great Power challenges inform critical dynamics of competition today. Great Power ability to win influence through durable alliances and valuable partnerships with less powerful states matters to successful competition.⁴⁴ A dominant power best wins influence with other states by amplifying points of strategic commonality and minimizing points of friction. Today, the United States enjoys common ideological and political objectives with a robust array of states around the world, including across the Indo-Pacific region. Among other regional agreements and treaties establishing this strength are two major, complementary, bilateral vision documents: the U.S.-India Joint Strategic Vision of January 2015 and the India-Japan Vision 2025 strategic document signed in December 2015.⁴⁵ Beyond this convergence with liberal democratic states, most countries prefer military cooperation with the United States to the limited and transactional cooperation offered by China.

At the same time, most states do not believe that severing economic ties with China is in their best interests.⁴⁶ While American commercial finance stands supreme, China’s economic strength in trade supply chains, direct investments for infrastructure, and consumer market power are too large and important for smaller states to summarily jettison.⁴⁷ Washington

lacks the economic capability to enforce a full-blown economic decoupling from China on its most important partner states.⁴⁸ Thus, the United States will compete best with China by gaining and sustaining influence with ideologically and politically aligned states—without making them choose severing economic ties with China as a cost of participation. It must pick its grounds carefully when urging partners not to engage in particular forms of trade, finance, or technology ventures that would help China build its power and compromise U.S. partners' sovereignty.

America's best choice to compete with China is to anchor a partner/alliance structure on common ideology and political philosophy while leading it into a period of partial economic disengagement.⁴⁹ The United States will have to negotiate partner assent to a framework that limits the most dangerous exposure of Western markets, labs, and innovative institutions to CCP control over Chinese economic actors, while continuing mutually beneficial trade and financial activities between China and America and its allies. This approach would require a measured and deliberate move toward restricting some forms of Chinese access to the United States and its partner economies and societies. The start point would see America and its partners agree to three major defensive economic goals from reduced trade with China: Limit vulnerability to CCP surveillance and sabotage, eliminate supply chain dependencies on China that may enable the CCP to credibly coerce or actually disrupt critical Western economic functions, and slow diffusion of innovation and technologies to China that are critical to Western commercial and military competitive edge.⁵⁰ A network of cooperative advanced industrial democracies—each committed to common core values and interests—would leverage U.S. competitive advantages in ideological resonance, alliance-building, and partnership reliability without demanding severance of all economic interactions with China.

The network would continue genuine and reciprocal trade with China but take collective steps to monitor and constrain Chinese trade and investment activities aimed to steal advanced technologies from their commercial companies. It would cooperate and participate in collective public-private ventures that offer practical alternatives to Chinese critical information technologies at viable pricing and that form trusted networks that protect sensitive and proprietary data. It would band together in challenging Beijing's most outrageous trade and industrial policies, increasing the odds that China will reconsider uncompetitive and illegitimate practices developed over many years. Finally, it would seek to prioritize meaningful reform of international institutions committed to a truly liberal global economy, overcoming the rise of nontariff barriers and national protectionism while establishing new standards for expanded free trade, investment, and growth in cutting-edge technologies featured in the fourth industrial revolution.⁵¹

An important catalyst for this kind of a U.S. partnership initiative featuring partial economic disengagement from China would involve high-standard trade agreements linking the economies of North America, Europe, and key parts of Asia. The United States need not begin this task from scratch; it needs only to rethink the opportunities already available to it. The 11 members of the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP) left the door open to American membership.⁵² A bridge from CPTPP through the United States to Mexico might be an important subsequent step. Although a number of Indian impediments to a free and full multilateral trade and finance partnership

remain, earnest dialogue between CPTPP and India could chart a path for future inclusion. That path would include systematic attention to and investment in an enormously important economy in a liberal democratic country that shares Western ideological beliefs and that has great economic potential.⁵³ A major trans-Atlantic agreement would be an additional logical step.

Exploiting America's comparative advantages in ideological, political, and alliance formation realms—while evolving an alignment of partners committed to partial disengagement from China's economy instead of a stark choice to decouple—best leverages U.S. competitive strength and minimizes risks from abrupt economic stagnation or unintentional war. Nonetheless, such an approach would require an uneasy truce within the American polity. Given its relative economic disadvantages with China, America cannot today swoon to the muse of *laissez-faire* economics and be properly competitive. Certain segments of the American polity hold to a dogma that unbridled free markets and unshackled private corporations are all that is necessary to ensure American economic success worldwide. This is a misreading of geopolitical competitions and U.S. economic history.

The United States has a clear record of government interventions to favor critical economic activities.⁵⁴ It has consistently subsidized American agriculture to sustain farmers from the challenges of foreign competitors with both subsidized and natural comparative advantage. Washington also has favored protective tariffs and quotas against foreign competition in all but brief periods of its history. In times of extreme competition with Great Power rivals, American politicians have moved with great alacrity to subsidize economic programs and segments of the economy deemed to be vital to succeed against international competitors. Before America entered World War II, such subsidies came from U.S. Government loan guarantees for ships, planes, tanks, and industrial products needed by Great Britain and later Russia. During the Cold War, American-targeted investment and complementary tariff barriers “put a finger on the scale” in favor of advanced technologies, defense equipment, and even American mass agriculture in an effort to ensure competitive U.S. advantages against the Soviet Union.⁵⁵ During the Cold War, Washington also indulged discriminatory tariffs and quotas for its junior anti-Soviet partners—countries such as Japan, South Korea, Taiwan, and those in Europe—to secure long-term geopolitical advantage despite short-term economic costs.

Educating the public would be necessary to generate political support for domestic economic renewal of a kind necessary to capitalize American government investment in competitive technologies and processes—like those involving IT, 5G, AI, quantum, and space. Here, the mindset of public direction and incentives for competitive and innovative technology that informed American policy in the Cold War is a relevant legacy.⁵⁶ As in the Cold War, American policymakers will need to subsidize priority elements for competitive advantage in the economic sphere. Public investment will require investment capital. The need is clear: Federal-level investment in American research and development activities in 2018 was at its lowest level since 1955.⁵⁷ To develop that pool, national political leaders will need to confront the extreme concentration of power and wealth in the hands of modern multinational corporate technology giants. With 50 percent of American wealth concentrated in 1 percent of its population, U.S. politicians will need to look at breaking up monopolistic companies and taxing exorbitant wealth as a means to incubate public investment and subsidies to rejuvenate

critical technologies, new businesses, and education to spur innovation.⁵⁸ Thus, it will take leadership and political risk, but American national leaders can expose the myth that the U.S. Government is not an essential participant in competitive economics for what it is. Then it may take steps to ally with friendly countries and partner states to advance a program of managed disengagement and enhanced competition with China.

Seen through the lens of applied history, these three imperatives for U.S. success in contemporary Great Power competition—distinguishing modern GPC from the Cold War, building on American competitive advantages with partner states, and acknowledging the role for government management of critical economic programs in GPC—are complemented by four competitive dynamics with relevance for at least the next decade.

Four Competitive Principles

The study of historic Great Power dyadic rivals offers a number of principles that can enable effective competition while minimizing the prospect of Great Power transition collapsing into Great Power war. Four stand out: firmness with flexibility, durable partnerships and alliances, the peril of reciprocal societal denigration, and playing for time.

Firmness with Flexibility

First, the dominant Great Power must demonstrate firmness with flexibility. It must clearly signal the strategic aims that it will defend at all costs and then offer the prospect of dialogue on those it may be willing to negotiate. While firm on its nonnegotiable aims, it should be flexible in finding issues and venues where win-win outcomes are possible. For example, the United Kingdom accepted American primacy in the Western Atlantic as a better path to sustaining high seas primacy on vital routes for its Middle East and Asian colonies—and preferable to naval confrontation in recognition of growing American power. At the same time, the rising United States came to accept the once-abhorrent British monarchy in recognition of growing political enfranchisement for a great number of British citizens. Is there such trade room today for the United States and China to agree on rules for collaboration in space and cyberspace while at the same time negotiating over reduced CCP domestic economic and human rights constraints?

Flexibility must be paired with firm resolve. Strong security arrangements, backed by formidable U.S. military power, might harden feelings of antagonisms and suspicion, but they are indispensable to preserving the peace with China.⁵⁹ If the CCP expects resistance from the United States and several mid-sized U.S. security partners, it is unlikely to fight for regional hegemony in the near term.⁶⁰ There is a discernible degree of caution in China's behavior that is wary of demonstrated strength and exploits perceived weakness.⁶¹ The United States and its Indo-Pacific partners must stand firm in resistance to China's illegal maritime claims by demonstrating the will to operate in international waters and airspace with freedom of navigation operations and other joint activities. They also must stand firm with Japan on disputed islands. Concurrently, the United States must demonstrate flexibility and adaptability in defense activities within the First Island Chain. It should proceed with a mobile and unpredictable basing posture for American forces. In particular, Washington also should work with Taiwan on development of weapons and tactics for self-defense that emphasize the advantages of smaller, smarter, and cheaper. This kind of flexibility is not the

same as ceding *de facto* spheres of influence to China with the First Island Chain or elsewhere in the Pacific.⁶² Instead, it is an acknowledgment that basic premises about sticking with allies and partners can remain firm even as tactics and techniques adapt.

The United States also can firmly support democratic institutions, individual liberties, and human rights in its alliances and in its interactions with China while demonstrating flexibility in pursuing aspirations for Chinese political reform. After first defending allies and partners from encroachment of Chinese authoritarian tendencies, America can demonstrate flexibility and patience in modeling patterns of individual liberty, freedom of information, and political participation to the people of China. During the Cold War, U.S. efforts to strengthen non-Communist elements within the Soviet bloc often met frustration in the near term. Western radio transmissions were blocked and censored, humanitarian assistance was refused, greater transit and tourism opportunities were blunted, and people-to-people programs declined. But over the long term—and especially after the Helsinki Accords of 1975—these activities gave hope to those laboring for a freer future behind Moscow’s Iron Curtain. American support for democracy and liberty in regions around the world during the 1970s and 1980s made the global ideological climate steadily less friendly to the Soviet Union’s repressive regime.⁶³ This kind of a Cold War competitive mindset is applicable for competition with China today and must be melded with modern collective approaches that portray Chinese political and ideological representations as inappropriate. Today, as then, a large amount of America’s appeal is the power of an uncensored world.⁶⁴

Durable Partnerships and Alliances

The second competitive principle reinforces the imperative of alliances discussed earlier. This is both an imperative and a principle so important for successful GPC that it is worth reinforcement and extension. History demonstrates that the dominant Great Power must look to build and maintain durable, reciprocal interstate alliances that provide would-be partners with alternatives to the either-or choices posed by a hard-charging rival.⁶⁵ Great Britain was right to seek strategic partnerships and allies in its rivalry with Napoleonic France, parlaying these alliances into first containment of the threat and later its defeat. Napoleon largely relied on territorial conquest and installation of family in positions of political power to expand French national power and elements of the French Revolution.⁶⁶

Today, the United States has a far greater base for building economic and military partnerships than any other Great Power in modern history. It also confronts a rising Great Power in China with little experience or inclination in this area. The United States has invested in critical global alliances and partnerships over the years for precisely this kind of moment. Japan is an important illustration. Before the United States sought to secure China’s entry into the World Trade Organization—a push in the late 1990s—it first reaffirmed the U.S.-Japanese alliance formally in 1995. This was a prudent hedge of the American bet on China’s rise to become a “responsible stakeholder” in the world order and an investment that today blunts Chinese aspiration to dominate the Indo-Pacific beyond its First Island Chain—because Japan sits there.⁶⁷ The U.S.-Japan alliance is an example of America’s “high card” of alliances and international institutions in its competition with China—and these should be cherished and well played.

Washington has an enormous opportunity to construct alternative economic, diplomatic, and political geometries with an array of partners to give them alternatives to Chinese enticements and blandishments. However, many of America's eager partners are today apprehensive about the recent unpredictability of U.S. foreign policy conduct. They want and value American partnership but worry that, unlike all its post–World War II predecessors, the current U.S. administration views commitment to rules-based international order and institutions to be more of a self-imposed constraint than a competitive advantage.⁶⁸ Among other signs they desire in an American strategic partner is a future foreign policy free from sanctions, tariffs, and restricted access to U.S. dollars as major instruments unconstrained by allies, rules, or institutions.⁶⁹ To be fully competitive with China, American policy must overcome this apprehension and practice a competitive foreign policy that views alliances as assets to be invested in rather than costs to be cut.⁷⁰

The Peril of Reciprocal Societal Denigration

Third, successful GPC short of direct military clash is extremely unlikely if the rivals descend into a poisonous, open, and reciprocal denigration of one another's people. The choice to criticize the government of a rival state while distinguishing it from the people is not as risky—although a tightrope must be walked to maintain the difference. Once the British and Imperial German press went after the character of the other's societies, the march toward World War I accelerated. So, too, World War II in the Pacific loomed ominously once the United States and Tojo's Japan devolved to mutual societal recrimination, but the U.S. Government's conscious Cold War effort to distinguish the Soviet Union's Communist Party from the Russian people, reserving greatest criticism toward the Party and offering outreach to its people, generated a far different result. American leaders are likely to compete best with China while clearly distinguishing between its criticism of the CCP and its feelings for the Chinese people.

The growing acrimony and invective between China and the United States during the months from 2018 to 2020 is worrisome. In 2018, the Trump administration reportedly considered blocking Chinese citizens from receiving any student visas as part of its package of economic pressure but backed away.⁷¹ In March 2020, the U.S. Secretary of State pressed major industrialized nations to call the COVID-19 pandemic the “Wuhan virus.” This

“We must reject the notion that the competition with China is a ‘clash of civilizations’ and that conflict is inevitable. Our concerns are with the CCP and not the Chinese people. We can collaborate where possible but compete aggressively to protect our national interest and the international order that has kept us safe since 1945.”

—Michael Brown, Eric Chewning, and Pavneet Singh, *Preparing the United States for the Superpower Marathon with China* (Brookings Institution, April 2020)

action during the G7 Foreign Ministers Meeting was later walked back, but it disrupted American leadership of the global pandemic response and gave CCP propagandists an incident with which to whip up anti-Western Chinese nationalism.⁷² President Trump threatened to “cut off the whole relationship” with China during a May 2020 interview.⁷³ Such broadly critical rhetoric risks reinforcing a tactic already at the top of the CCP playbook: ceaselessly exploiting Chinese nationalism to shore up its legitimacy.⁷⁴

The dangers of reciprocity from China are real. While the Chinese foreign minister and other senior government officials often frame their criticism of the United States in ways that target the government in Washington, its activities abroad and at home often cross the line into mass denigration of Americans or other Westerners.⁷⁵ Since late 2019, normally taciturn Chinese diplomats are now observed openly criticizing not only Western policies but also the social and cultural aspects of Europeans, Australians, and Americans in what has been coined “Wolf Warrior diplomacy.”⁷⁶ Inside China, the CCP has co-opted Chinese nationalism as a tool to indoctrinate citizens and its diaspora to a narrative that America and its allies once colonized China and today victimize China by preventing its rise to a superpower, and that the CCP is China’s only savior.⁷⁷ While the risks from such unbridled and growing mutual denigration do not now seem to include direct military clash, history indicates they will trend toward deepening risks of war in the future should implacable people-to-people hostility grow unabated.

To reduce the risk—and channel competition appropriately—the United States should focus legitimate criticism on the CCP and its policies in a manner that counters Chinese narratives feeding nationalist xenophobia. The line between criticizing the CCP and Chinese society is a fine one to walk—and will require calibration. An American effort to toe this line took place in early May 2020 when the Trump administration deputy national security advisor for Asia delivered a speech in Mandarin Chinese intended for the Chinese people that critiqued CCP efforts to clamp down on free speech while praising brave Chinese doctors and front-line workers in the response to COVID-19.⁷⁸ An American program of communication should concentrate on countering CCP-driven disinformation.⁷⁹ It also should work to counter the clear CCP domestic narrative that it is all that stands between China and chaos.⁸⁰ At the same time, the United States should try to maximize positive interactions and experiences with the Chinese people. The United States and its free-and-open partner societies should consider issuing more visas and providing paths to citizenship for more Chinese, with proper safeguards in place. Chinese who engage with citizens of free countries are the ones who are most likely to question their government’s policies whether from abroad or when they return home. In this approach, the United States would do what it did with expatriate Russian communities during the Cold War: View Chinese expatriate communities as valuable citizens while discriminating between Ministry of State security agents for expulsion.⁸¹

Playing for Time

Finally, some argue that time works in favor of the rising Great Power in a competitive dyad, putting the dominant Great Power at dire risk if it does not take swift confrontational action while its relative power is high. But this thesis rests on at least two dubious assumptions: that the rising power’s ascent is likely to be rapid and that the rising power will continue to ascend in a mainly linear fashion and not confront problems or challenges on the way. In the present moment, there is a strong case to be made that the critical factors confronting China at home and abroad make time work in favor of the United States.⁸²

First, America has its own domestic inconsistencies and challenges, but these pale in comparison with those certain to play out within China over the coming couple of decades. The CCP faces multifaceted challenges to safeguard both its political position and

"Beijing is learning, as have other nations, that building advanced equipment is the relatively easy and inexpensive part of becoming a major military power. Its characterization of the international environment, pressures from its military-industrial complex, and the arms race it has triggered will require increased funding and shape debate on how much money can be spent on other national priorities. It will also shape the challenges China faces in the international system because nations will respond to China's buildup by arms purchases of their own and efforts to use alliances, alignments, and other instruments of geopolitics to counterbalance Chinese power."

—Thomas Fingar and Jean C. Oi, "China's Challenges: Now It Gets Much Harder," *The Washington Quarterly* 43, no. 1 (Spring 2020)

an economic rise that seems critical to CCP legitimacy. These multifaceted challenges include rampant environmental degradation; rising income inequalities; a rapidly aging and less productive population; chronic worry about abuses of political power; widespread corruption; restive domestic regions including Tibet, Xingxang, and Mongolia; and a poor record on human rights.⁸³ As China's economy shifts toward more reliance on domestic economic consumption, its economic growth decelerates, and its national debt continues to grow, these many domestic challenges are moving to the fore.⁸⁴ Second, China faces serious unresolved challenges along its own borders, rendering its ability to dominate the Indo-Pacific region doubtful in the near term and making any global push by Beijing to reorder international norms and institutions a truly long-term proposition.

China's neighbors include formidable economic and military powers such as Japan, South Korea, Australia, and India. Each of them is increasingly apprehensive about China's strategic ambitions and is deepening security ties with one another and the United States in response. Beijing's ham-handed efforts to crush democratic resistance in Hong Kong and nationalism in Taiwan have stiffened regional headwinds for Chinese messaging.⁸⁵

It is unwise for the United States to assume that China will succumb to these challenges. Such complacency could distract necessary attention to a serious Great Power rival. At the same time, a U.S. conclusion that China is destined for global dominance—particularly in the near term—is both unsupported and likely to generate strategic overreaction.⁸⁶ China's economic rise will make it a long-term challenge for the United States to manage rather than one to be conquered or converted.⁸⁷ The United States and China are destined for a lengthy, uneasy coexistence, not decoupling or appeasement.⁸⁸ Thus, a U.S. strategy that plays for time as China's contradictions grow and as American resilience, regeneration, and the realization of a new competitive mindset emerge from more than two decades of torpor seems the one best suited for U.S. success in contemporary Great Power competition.⁸⁹

The Way Forward

Knowing the imperatives and principles of what the United States should do to succeed in a new era of Great Power competition is not the same as figuring out how to do it. Galvanizing American resolve to compete with the Soviet Union and move into a Cold War was a challenging process. Former Secretary of State Dean Acheson famously stated that the U.S. Government had to make arguments "clearer than truth" to get the American people and Congress to buy into

the effort to contain the Soviet Union. Acheson's work on National Security Council Paper 68 made the necessity of containment clear but was later critiqued as overreach.⁹⁰

In 2020, the operative U.S. National Security Strategy (NSS) of 2017 stands as a stark contrast to its 2002 predecessor, a mere 15 years prior. The 2002 version of the NSS began with this preamble:

*The great struggles of the twentieth century between liberty and totalitarianism ended with a decisive victory for the forces of freedom—and a single sustainable model for national success: freedom, democracy, and free enterprise. In the twenty-first century, only nations that share a commitment to protecting basic human rights and guaranteeing political and economic freedom will be able to unleash the potential of their people and assure their future prosperity.*⁹¹

The 2017 NSS successor paints a starkly different landscape:

*China and Russia challenge American power, influence, and interests, attempting to erode American security and prosperity. They are determined to make economies less free and less fair, to grow their militaries, and to control information and data to repress their societies and expand their influence. . . . These competitions require the United States to rethink the policies of the past two decades—policies based on the assumption that engagement with rivals and their inclusion in international institutions and global commerce would turn them into benign actors and trustworthy partners. For the most part, this premise turned out to be false.*⁹²

In today's new era of GPC, the Sino-American dyad is the rivalry of greatest significance. This contest features an ongoing power transition—always a dangerous dynamic of international politics in modern history. China is clearly growing in relative economic power, but the United States is a dominant state with clear comparative advantages—"high cards" in its hand—that it can build on to advantage.⁹³ Alliance maintenance and cultivation is the most critical card. Firm and flexible confrontation when necessary and collaboration with China where possible is a second. Avoiding a regressive game of reciprocal societal invective is the third. And playing the long game—playing for time—is the fourth.

The NSS properly recognized the China challenge for what it was—formalizing a de facto new era of Great Power competition—but this overdue recognition could go terribly wrong if it generates unwarranted American hysteria and overreaction against Beijing. Overreaction in Washington could lead to high cards played badly.

The NSS culminates its geopolitical evaluation with a highly relevant rejoinder:

We learned the difficult lesson that when America does not lead, malign actors fill the void to the disadvantage of the United States. When America does lead, however, from a position of strength and confidence and in accordance with our interests and values, all benefit. Competition does not always mean hostility, nor does it inevitably lead to conflict—although none should doubt our commitment to defend our interests. An America that successfully competes is the best way to prevent conflict. Just as

*American weakness invites challenge, American strength and confidence deters war and promotes peace.*⁹⁴

These words are measured, historically informed, and wise. Applied well—without resort to overreaction or backsliding into complacency—these words can inspire American confidence in the way forward for this new era of GPC. They also can give U.S. allies and potential partners confidence in American leadership and resolve. China's behavior is galvanizing opposition among countries that do not want to be vassal states.⁹⁵ A rejuvenating United States, with reframed domestic priorities and renewed focus on well-established and well-treated allies and partners, will have a clear advantage in what is likely to be a drawn-out era of Great Power competition with China.

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Notes

¹ Among a long list of public testimony, detailed studies, and published works of nonfiction and fiction predicting that such a novel and deadly virus was increasingly likely and the United States was vulnerable, see Daniel Coats, *Statement for the Record: Worldwide Threat Assessment of the U.S. Intelligence Community to the Senate Select Committee on Intelligence* (Washington, DC: Office of the Director of National Intelligence, 2019), 21; Katie Pearce, "Pandemic Simulation Exercise Spotlights Massive Preparedness Gap," *Johns Hopkins University Hub*, November 6, 2019, available at <<https://hub.jhu.edu/2019/11/06/event-201-health-security/>>; *Crimson Contagion 2019 Functional Exercise Key Findings—Coordinating Draft* (Washington, DC: Department of Health and Human Services, December 3, 2019), available at <<https://int.nyt.com/data/documenthelper/6824-2019-10-key-findings-and-after/05bd797500ea55be0724/optimized/full.pdf>>; Sonia Shah, *Pandemic: Tracking Contagions, from Cholera to Ebola and Beyond* (New York: Sarah Crichton Books, 2016); Lawrence Wright, *The End of October: A Novel* (New York: Knopf, 2020).

² The Ottoman Empire (Turkey) and Austria-Hungary emerged from World War I so wounded that they would no longer compete as Great Power competitors in the interwar period. For further discussion, see chapter 2.

³ For a detailed listing of the major insights from the 15 total book chapters (one of which is a couplet chapter, chapter 3), see the Major Findings section at the beginning of this volume.

⁴ See this described in detail at Aaron L. Friedberg, "Competing with China," *Survival* 60, no. 3 (2018), 7–64.

⁵ For a detailed critique of the most conspicuous failures of modern globalization and hyper-interdependence in supply and finance chains by Nassim Nicholas Taleb, the originator of the terms *black swans* and economic *fat tails*, see Bernard Avishai, "The Pandemic Isn't a Black Swan but a Portent of a More Fragile Global System," *The New Yorker*, April 21, 2020, available at <www.newyorker.com/news/daily-comment/the-pandemic-isnt-a-black-swan-but-a-portent-of-a-more-fragile-global-system>; Nassim Nicholas Taleb, *Antifragile: Things That Gain from Disorder* (New York: Random House, 2012).

⁶ For a detailed discussion of this valuation of information power over industrial power, see chapter 6 of this volume.

⁷ As demonstrated in chapter 3b of this volume, China does not today possess and is unlikely to attain sufficient power assets in the coming decade to enable a strategy of remaking the international order in its favor before domestic risk factors collapse the rule of the Chinese Communist Party (CCP)—even if that was its actual strategy. For an opposite view that asserts China possesses a global grand strategy aspiring for leadership of a new tributary system soon to be resourced through a massive effort organized under three overlapping policies—carrying the names "Made in China 2025," "Belt and Road Initiative," and "Military-Civil Fusion"—see H.R. McMaster, "How China Sees the World: And How We Should See China," *The Atlantic*, May 2020, available at <www.theatlantic.com/magazine/archive/2020/05/mcmaster-china-strategy/609088/>.

⁸ Steve Coll, "Woodrow Wilson's Case of the Flu, and How Pandemics Change History," *The New Yorker*, April 17, 2020, available at <www.newyorker.com/news/daily-comment/woodrow-wilsons-case-of-the-flu-and-how-pandemics-change-history>.

⁹ Jon M. Barry, *The Great Influenza: The Story of the Deadliest Pandemic in History* (New York: Penguin, 2004), 383–388.

¹⁰ Lloyd E. Ambrosius, "Woodrow Wilson's Health and the Treaty Fight, 1919–1920," *The International History Review* 9, no. 1 (February 1987), 73–84.

¹¹ Charles P. Kindleberger, *The World in Depression, 1929–1939* (Berkeley: University of California Press, 1973).

¹² Barry, *The Great Influenza*, 449–455; Shah, *Pandemic*, 5–6, 97–119, 201–217; James Traub, "After the Coronavirus, the Era of Small Government Will Be Over," *Foreign Policy*, April 15, 2020, available at <<https://foreignpolicy.com/2020/04/15/coronavirus-pandemic-small-government-aftermath-nationalism/>>.

¹³ Henry A. Kissinger, "The Coronavirus Pandemic Will Forever Alter the World Order," *Wall Street Journal*, April 3, 2020, available at <www.wsj.com/articles/the-coronavirus-pandemic-will-forever-alter-the-world-order-11585953005>.

¹⁴ Traub, "After the Coronavirus, the Era of Small Government Will Be Over"; Henry Kissinger, *World Order* (New York: Penguin, 2014), 361–374.

¹⁵ Richard Haass, "The Pandemic Will Accelerate History Rather Than Reshape It," *Foreign Affairs*, April 7, 2020, available at

<www.foreignaffairs.com/articles/united-states/2020-04-07/pandemic-will-accelerate-history-rather-reshape-it>.

¹⁶ Traub, “After the Coronavirus, the Era of Small Government Will Be Over.”

¹⁷ See Kurt M. Campbell and Rushi Doshi, “The Coronavirus Could Reshape Global Order,” *Foreign Affairs*, March 18, 2020. This is also implied in Kissinger, “The Coronavirus Pandemic Will Forever Alter the World Order.”

¹⁸ Matt Burrows and Peter Engelke, *What World Post COVID-19? Three Scenarios*, Atlantic Council Strategy Paper (Washington, DC: Atlantic Council, April 2020), 3–4, available at <www.atlanticcouncil.org/wp-content/uploads/2020/04/What-World-Post-COVID-19.pdf>.

¹⁹ Amitav Acharya, “How Coronavirus May Reshape the World Order,” *National Interest*, April 18, 2020, available at <<https://nationalinterest.org/feature/how-coronavirus-may-reshape-world-order-145972>>.

²⁰ Michal Green and Evan S. Medeiros, “The Pandemic Won’t Make China the World’s Leader,” *Foreign Affairs*, April 15, 2020, available at <www.foreignaffairs.com/articles/united-states/2020-04-15/pandemic-wont-make-china-worlds-leader>.

²¹ Haass, “The Pandemic Will Accelerate History Rather Than Reshape It.”

²² Ibid.

²³ Burrows and Engelke, *What World Post COVID-19*, 3–4.

²⁴ For a review of these offsetting opinions about Russia’s future, see Tom McTague, “The Pandemic’s Coming Geopolitical Second Wave,” *The Atlantic*, May 18, 2020, available at <www.theatlantic.com/international/archive/2020/05/coronavirus-pandemic-second-wave-geopolitics-instability/611668/>.

²⁵ See for example, Tony Wood, *Russia Without Putin: Money, Power, and the Myths of the New Cold War* (London: Verso Books, 2018), 1–9; Fiona Hill and Clifford G. Gaddy, *Mr. Putin: Operative in the Kremlin* (Washington DC: Brookings, 2015), especially 190–226, 312–341.

²⁶ Haass, “The Pandemic Will Accelerate History Rather Than Reshape It.”

²⁷ In 2019, the Chicago Council on Global Affairs survey found that 30 percent of Americans wanted the United States to play no active part in the world. A Gallup Poll that same year recorded up to 35 percent of Americans felt the United States should play much less of a role internationally than it did before the Trump administration. See Dina Smeltz et al., *Rejecting Retreat: Americans Support U.S. Engagement in Global Affairs* (Chicago: The Chicago Council on Global Affairs, 2019), available at <https://www.thechicagocouncil.org/sites/default/files/report_ccs19_rejecting-retreat_20190909.pdf>; V. Lance Tarrance, “Public Opinion and U.S. Engagement with the World,” *Gallup*, April 11, 2019, available at <<https://news.gallup.com/opinion/polling-matters/248588/public-opinion-engagement-world.aspx>>.

²⁸ Haass, “The Pandemic Will Accelerate History Rather Than Reshape It.”

²⁹ John Lee, *Decoupling the U.S. Economy from China After COVID-19* (Washington, DC: Hudson Institute, May 7, 2020), available at <<https://www.hudson.org/research/16009-decoupling-the-us-economy-from-china-after-covid-19>>; Haass, “The Pandemic Will Accelerate History Rather Than Reshape It.”

³⁰ Green and Medeiros, “The Pandemic Won’t Make China the World’s Leader”; Haass, “The Pandemic Will Accelerate History Rather Than Reshape It”; Kissinger, “The Coronavirus Pandemic Will Forever Alter the World Order.”

³¹ Hal Brands and Peter D. Fever, “Reevaluating Diplomatic and Military Power: What Are America’s Alliances Good For?” *Parameters* 4, no. 2 (Summer 2017), 25–30.

³² A leading voice against calling this a period of Great Power competition (GPC), Michael Mazarr of RAND rests his contrarian view on a very narrow definition of *classic Great Power competition*, which he argues must feature a set of potential enemies with constantly shuffling allegiances and rivalries that typically manifest in military forms of competition. As detailed in chapter 2 of this volume, Mazarr’s stimulating viewpoint does not comport with a detailed examination of modern GPC—a history that has included bipolar competition as well as multipolar competition and has consistently involved a wide array of overlapping patterns of collaboration and confrontation in economics, ideas, information exchange, diplomacy, and militaries. See Michael J. Mazarr, “This Is Not a Great-Power Competition: Why the Term Doesn’t Capture Today’s Reality,” *Foreign Affairs*, May 29, 2019, available at <www.foreignaffairs.com/articles/2019-05-29/not-great-power-competition>.

³³ Kurt M. Campbell and Jake Sullivan, “Competition Without Catastrophe,” *Foreign Affairs* 98, no. 5 (September/October 2019), 98–101.

³⁴ For examples of some overwrought calls for Washington to invoke a Cold War against China, see Newt Gingrich, *Trump vs. China: Facing America’s Greatest Threat* (New York: Hachette Book Group, 2019); Giselle Donnelly, “What America Needs Is a ‘Good,’ Long Cold War,” American Enterprise Institute, February 10, 2020. For a concise listing of five reasons why the Sino-American strategic competition is not a new Cold War, see Michael Brown et al., *Preparing the United States for the Superpower Marathon with China* (Washington, DC: Brookings Institution, April 2020), 3–5.

³⁵ Valuable competitive mindset dynamics from the Cold War are different from Cold War strategies and are detailed below, including those found in Stephen M. Walt, “Yesterday’s Cold War Shows How to Beat China Today,” *Foreign Policy*, July 29, 2019, available at <<https://foreignpolicy.com/2019/07/29/yesterdays-cold-war-shows-how-to-beat-china-today/>>. Also see Campbell and Sullivan, “Competition Without Catastrophe,” 101–107.

³⁶ Other key differences between today’s Sino-American dyadic competition and the Cold War are mentioned in chapter 3a of this volume. By way of extension, China is the one rising Great Power with the positivist ideology and power potential to supplant the preferred U.S. international order. However, in 2020, China is not exporting its ideology aggressively or globally. Unlike the Soviet Union, which aggressively sought the destruction of the United States in the hearts and minds of countries it wanted to become future Communist states, China today has not called for the destruction of the United States and wants the world to be tolerant of its CCP-driven ideology. See Evan Osnos, “The Future of America’s Contest with China,” *The New Yorker*, January 6, 2020, available at <<https://www.newyorker.com/magazine/2020/01/13/the-future-of-americas-contest-with-china>>; Odd Arne Westad, “The Sources of Chinese Conduct: Are Washington and Beijing Fighting a New Cold War?” *Foreign Affairs* 98, no. 5 (September/October 2019), 86–95.

³⁷ See Walt, “Yesterday’s Cold War Shows How to Beat China Today.”

³⁸ Edgar Crammond, “The Economic Relations of the British and German Empires,” *Journal of the Royal Statistical Society* 77, no. 8 (July 1914), 777–824.

³⁹ Alexander Grab, *Napoleon and the Transformation of Europe* (New York: Palgrave, 2003), 28–33.

⁴⁰ See “1991: Trade in Goods with USSR,” United States Census Bureau, available at <<https://www.census.gov/foreign-trade/balance/c4610.html#1991>>.

⁴¹ Osnos, “The Future of America’s Contest with China.”

⁴² In addition to the extremely high economic costs to the United States from fully decoupling all intertwined supply chains, China could take punitive steps that would impose high costs on an

array of other activities that today are taken for granted in the United States, including, but not limited to, large restraints or high taxes on commercial travel, greater use of non-dollar denominated currencies for international transactions, disruptive manipulation of its own currency and international commercial financial markets, and a resistance to working jointly on a wide array of other mutual issues and interests. See Keith Johnson and Robbie Gramer, "The Great Decoupling," *Foreign Policy*, May 14, 2020, available at <<https://foreignpolicy.com/2020/05/14/china-us-pandemic-economy-tensions-trump-coronavirus-covid-new-cold-war-economics-the-great-decoupling/>>; Henry M. Paulson, Jr., "Delusions of Decoupling," remarks at the New Economy Forum, Beijing, November 2019, available at <<https://www.paulsoninstitute.org/wp-content/uploads/2019/11/Delusions-of-Decoupling.pdf>>; Osnos, "The Future of America's Contest with China"; Campbell and Sullivan, "Competition Without Catastrophe," 99.

⁴³ Kevin Rudd, "To Decouple or Not to Decouple?" Robert F. Ellsworth Memorial Lecture, University of San Diego, November 4, 2019.

⁴⁴ This is not to imply that there are not costs and frustrations to alliances and partnership; there are. But the costs of American alliances are actually less burdensome than they are portrayed and more diverse and significant than appreciated. See Brands and Fever, "Reevaluating Diplomatic and Military Power," 25–30. For similar points, see Walt, "Yesterday's Cold War Shows How to Beat China Today"; Mira Rapp-Hooper, "Saving America's Alliances: The United States Still Needs the System that Put It on Top," *Foreign Affairs* 99, no. 2 (March/April 2020), 127–140.

⁴⁵ See U.S.-India Joint Strategic Vision for the Asia-Pacific and Indian Ocean Region (Washington, DC: The White House, January 25, 2015), available at <<https://obamawhitehouse.archives.gov/the-press-office/2015/01/25/us-india-joint-strategic-vision-asia-pacific-and-indian-ocean-region>>; Japan and India Vision 2025 Special Strategic and Global Partnership (Tokyo: Ministry of Foreign Affairs of Japan, December 12, 2015), available at <https://www.mofa.go.jp/sa/sw/in/page3e_000432.html>.

⁴⁶ See Ashley Tellis, Alison Szalwinski, and Michael Wills, *Strategic Asia 2020: U.S.-China Competition for Global Influence* (Seattle: National Bureau of Asian Research, 2019), 41–42; Joseph S. Nye, Jr., "Power and Interdependence with China," *The Washington Quarterly* 43, no. 1 (Spring 2020), 15, 19.

⁴⁷ As of 2018, 70 percent of countries traded more with China than they did with the United States. In 1980, 80 percent traded more with the United States than China. See Alyssa Leng and Roland Rajah, "Chart of the Week: Global Trade Through a U.S.-China Lens," *The Interpreter*, December 18, 2019, available at <<https://www.lowyinstitute.org/the-interpreter/chart-week-global-trade-through-us-china-lens>>. At the same time, the United States continues to enjoy unrivaled power from its large commercial financial institutions and the dollar's unrivaled status as the global reserve currency—neither of which is likely to be challenged by China in the near term. See Nye, "Power and Interdependence with China," 15; see also appendix B of this volume, available at <<https://ndupress.ndu.edu/Contemporary-GPC-Dynamics-Matrix/>>.

⁴⁸ Ali Wyne, "How to Think About Potentially Decoupling from China," *Washington Quarterly* 43, no. 1 (Spring 2020), 47.

⁴⁹ The concept of *partial disengagement* is derived from Charles W. Boustany, Jr., and Aaron L. Friedberg, *Partial Disengagement: A New U.S. Strategy for Economic Competition with China*, NBR Special Report No. 82 (Seattle: National Bureau of Asian Research, November 2019); Aaron L. Friedberg and Charles W. Boustany, Jr., "Partial Disengagement: A New U.S. Strategy for Economic Competition with China," *Washington Quarterly* 43, no. 1 (Spring 2020), 23–40. A similar construct that focuses beyond economics and on the military, political, and diplomatic domains of Sino-American competition is found in Campbell and Sullivan, "Competition Without Catastrophe," 101–110.

⁵⁰ Boustany and Friedberg, *Partial Disengagement*, 26–35.

⁵¹ Most of these ideas are developed in some detail in Boustany and Friedberg, *Partial Disengagement*, 22–25. This approach requires the United States and its allies to cooperate in the reform and advancement of existing international organizations rather than abandon them. The dangers of abandonment are clear, for China has demonstrated an ability to fill the vacuum left by the United States and push for its own interests and agendas. See Colum Lynch and Elias Groll, "As U.S. Retreats from World Organizations, China Steps in to Fill the Void," *Foreign Policy*, October 6, 2017, available at <<https://foreignpolicy.com/2017/10/06/as-u-s-retreats-from-world-organizations-china-steps-in-the-fill-the-void/>>; Alex Pascal, "Against Washington's 'Great Power' Obsession," *The Atlantic*, September 23, 2019, available at <www.theatlantic.com/politics/archive/2019/09/multilateralism-nearly-dead-s-terrible-news/598615/>.

⁵² Among the many appealing for America to rethink and join the Comprehensive and Progressive Agreement for Trans-Pacific Partnership, see Boustany and Friedberg, *Partial Disengagement*, 24.

⁵³ The challenges to fully engaging India economically are significant, but India continues to look seriously at beginning the vital processes of reducing economic protectionism and modernizing land, labor, and environmental laws. See Bill Spindle and Rajesh Roy, "India's Coronavirus Crisis Spurs a New Look at Self-Reliance," *Wall Street Journal*, May 17, 2020, available at <www.wsj.com/articles/indias-coronavirus-crisis-spurs-a-new-look-at-self-reliance-11589724001?mod=itp_wsj&ru=yahoo>.

⁵⁴ Harvard Economist and Asia Center Senior Fellow William Overholt makes this point powerfully in many venues, including William H. Overholt, *China and America: The Age of Realist Geoeconomics* (Atlanta: The Carter Center, January 2018); William H. Overholt, "The Enemy Is Us," *The International Economy* (Summer 2015), 46–53.

⁵⁵ For the important role U.S. Government subsidization and promotion of agriculture played during the Cold War, see Shane Hamilton, *Supermarket USA: Food and Power in the Cold War Farms Race* (New Haven: Yale University Press, 2018).

⁵⁶ For detail, see Walt, "Yesterday's Cold War Shows How to Beat China Today."

⁵⁷ Osnos, "The Future of America's Contest with China."

⁵⁸ These ideas are from Taleb in Avishai, "The Pandemic Isn't a Black Swan."

⁵⁹ Charles Edel and Hal Brands, "The Real Origins of the U.S.-China Cold War," *Foreign Policy*, June 2, 2019, available at <<https://foreignpolicy.com/2019/06/02/the-real-origins-of-the-u-s-china-cold-war-big-think-communism/>>.

⁶⁰ Many Chinese leaders cannot believe that the Obama administration did not react more strongly to the 2010 seizure of Scarborough Shoal and the 2014 arming of it despite Chairman Xi Jinping's promise to Obama that year that China had no intention of doing so. These Chinese officials indicate that a firm U.S. and allied response can moderate intemperate Chinese aggression in the Indo-Pacific region. See Osnos, "The Future of America's Contest with China."

⁶¹ Denny Roy, "China Won't Achieve Regional Hegemony," *Washington Quarterly* 43, no. 1 (Spring 2020), 105–106.

⁶² For variations on this misguided notion, see Graham Allison, "The New Spheres of Influence: Sharing the Globe with Other Great Powers," *Foreign Affairs* 99, no. 2 (March/April 2020), 30–40; Fareed Zakaria, "The New China Scare: Why America Shouldn't Panic About Its Latest Challenger," *Foreign Affairs* 99, no. 1 (January/February 2020), 52–69.

⁶³ Edel and Brands, "The Real Origins of the U.S.-China Cold War."

⁶⁴ Osnos, "The Future of America's Contest with China."

- ⁶⁵ Choosing proper allies also was a competitive mindset success for the United States during the Cold War. See Walt, “Yesterday’s Cold War Shows How to Beat China Today.”
- ⁶⁶ Michael Broers, “Pride and Prejudice: The Napoleonic Empire Through the Eyes of Its Rulers,” in *Napoleon’s Empire: European Politics in Global Perspective*, ed. Ute Planert (Houndmills, Basingstoke: Palgrave Macmillan, 2016), 307–317; Michael V. Leggiere, “Enduring Strategic Rivalries: Great Britain vs. France During the French Wars (1792–1815),” in *Great Strategic Rivalries: From the Classical World to the Cold War*, ed. James Lacey (New York: Oxford University Press, 2016), 289–390, available at <<https://global.oup.com/academic/product/great-strategic-rivalries-9780190620462?cc=us&lang=en&#>>>.
- ⁶⁷ Nye, “Power and Interdependence with China,” 16.
- ⁶⁸ “Don’t Be Fooled by the Trade Deal Between America and China,” *The Economist*, January 2, 2020, available at <www.economist.com/leaders/2020/01/02/dont-be-fooled-by-the-trade-deal-between-america-and-china>.
- ⁶⁹ Nye, “Power and Interdependence with China,” 15–17.
- ⁷⁰ Campbell and Sullivan, “Competition Without Catastrophe,” 110.
- ⁷¹ See Stuart Anderson, “What Will Trump Do Next with Chinese Student Visas,” *Forbes*, October 18, 2018, available at <www.forbes.com/sites/stuartanderson/2018/10/18/what-will-trump-do-next-with-chinese-student-visas/#453fb89757a4>.
- ⁷² See John Hudson and Souad Mekhennet, “G-7 Failed to Agree on Statement After U.S. Insisted on Calling Coronavirus Outbreak ‘Wuhan Virus,’” *Washington Post*, March 25, 2020, available at <www.washingtonpost.com/national-security/g-7-failed-to-agree-on-statement-after-us-insisted-on-calling-coronavirus-outbreak-wuhan-virus/2020/03/25/f2bc7a02-6ed3-11ea-96a0-df4c5d9284af_story.html>; Jo Kim, “The Chinese People Step Up to Enforce China’s Nationalist Propaganda,” *The Diplomat*, May 5, 2020, available at <<https://thediplomat.com/2020/05/the-chinese-people-step-up-to-enforce-chinas-nationalist-propaganda/>>.
- ⁷³ Demetri Sevastopulo, “Trump Threatens to Cut Off Relations with China,” *Financial Times*, May 15, 2020, available at <www.ft.com/content/cfba6bf-3de5-458d-92d1-a62fb958a354>.
- ⁷⁴ Minxin Pei, “China’s Coming Upheaval: Competition, the Coronavirus, and the Weakness of Xi Jinping,” *Foreign Affairs* 99, no. 3 (May/June 2020), 82–106.
- ⁷⁵ See Eva Dou, “In China-U.S. Showdown, Beijing’s Steely Propagandist Sharpens Her Attack,” *Washington Post*, May 21, 2020, available at <www.washingtonpost.com/world/asia_pacific/in-china-us-showdown-beijings-steely-propagandist-sharpens-her-attack/2020/05/21/f71133e4-94bd-11ea-87a3-22d324235636_story.html>.
- ⁷⁶ Chun Han Wong and Chao Deng, “China’s ‘Wolf Warrior’ Diplomats Are Ready to Fight,” *Wall Street Journal*, May 19, 2020, available at <www.wsj.com/articles/chinas-wolf-warrior-diplomats-are-ready-to-fight-11589896722>; Jamie Dettmer, “China’s ‘Wolf Warrior’ Diplomacy Prompts International Backlash,” *Voice of America*, May 6, 2020, available at <www.voanews.com/covid-19-pandemic/chinas-wolf-warrior-diplomacy-prompts-international-backlash>; Lily Ko, “Australia Called ‘Gum Stuck to China’s Shoe’ by State Media in Coronavirus Investigation Stoush,” *The Guardian*, April 28, 2020, available at <www.theguardian.com/world/2020/apr/28/australia-called-gum-stuck-to-chinas-shoe-by-state-media-in-coronavirus-investigation-stoush>; John Ross, “China Warns of Student Boycott of Australia,” *Inside Higher Education*, May 1, 2020, available at <www.insidehighered.com/news/2020/05/01/china-warns-australia-student-boycott>.
- ⁷⁷ Kim, “The Chinese People Step Up to Enforce China’s Nationalist Propaganda”; Vicky Xiuxhong Xu, “China’s Youth Are Trapped in the Cult of Nationalism,” *Foreign Policy*, October 1, 2019, available at <<https://foreignpolicy.com/2019/10/01/chinas-angry-young-nationalists/>>.
- ⁷⁸ *Remarks by Deputy National Security Advisor Matt Pottinger to the Miller Center at the University of Virginia* (Washington, DC: The White House, May 4, 2020), available at <<https://www.whitehouse.gov/briefings-statements/remarks-deputy-national-security-advisor-matt-pottinger-miller-center-university-virginia/>>.
- ⁷⁹ For details on the organizations involved in international propaganda and influence activities, see appendix 1 in Larry Diamond and Orville Schell, eds., *China’s Influence & American Interests: Promoting Constructive Vigilance* (Stanford: Hoover Institution Press, 2019), 133–141. Some former policymakers specifically focus on the Chinese Ministry of State Security, the United Front Work Department, and the Chinese Students and Scholars Association as ones for attention to counter CCP-driven propaganda. See McMaster, “How China Sees the World.”
- ⁸⁰ Osnos, “The Future of America’s Contest with China.”
- ⁸¹ Proper “safeguards” for Chinese student, teacher, and research visas should include very tight limitations on Confucius Institutes in the United States to eliminate their revealed role in espionage, monitoring, and thought-policing on behalf of the CCP. The ideas for American policy valuing the Chinese people while holding the CCP accountable includes McMaster, “How China Sees the World.”
- ⁸² Strategic patience during the Cold War also was an American competitive mindset virtue. See Walt, “Yesterday’s Cold War Shows How to Beat China Today.”
- ⁸³ See Pei, “China’s Coming Upheaval”; “Revised Demographic Forecasts for China: Key Takeaways,” *The Economist*, July 2, 2019; William H. Overholt, *China’s Crisis of Success* (Cambridge: Cambridge University Press, 2018).
- ⁸⁴ See William H. Overholt, “The West Is Getting China Wrong,” *East Asia Forum*, August 11, 2018, available at <<https://www.eastasiaforum.org/2018/08/11/the-west-is-getting-china-wrong/>>; Yasheng Huang, “China Has a Big Economic Problem and It Isn’t the Trade War,” *New York Times*, January 17, 2020.
- ⁸⁵ Wyne, “How to Think About Potentially Decoupling from China,” 41–64.
- ⁸⁶ For similar conclusions, see Nye, “Power and Interdependence with China,” 13; Wyne, “How to Think About Potentially Decoupling from China,” 50–52.
- ⁸⁷ Osnos, “The Future of America’s Contest with China”; Martin Wolf, “The Looming 100-Year U.S.-China Conflict,” *Financial Times*, June 4, 2019.
- ⁸⁸ Osnos, “The Future of America’s Contest with China.”
- ⁸⁹ For a detailed assessment of options for a U.S. strategic mindset for competition with China, see chapter 14 of this volume.
- ⁹⁰ On Acheson, see Robert Kagan, “How Dean Acheson Won the Cold War: Statesmanship, Morality, and Foreign Policy,” Carnegie Endowment for International Peace, September 14, 1998, available at <<https://carnegieendowment.org/1998/09/14/how-dean-acheson-won-cold-war-statesmanship-morality-and-foreign-policy-pub-260>>.
- ⁹¹ *The National Security Strategy of the United States of America* (Washington, DC: The White House, September 2002), available at <<http://nssarchive.us/national-security-strategy-2002/>>.
- ⁹² *The National Security Strategy of the United States of America* (Washington, DC: The White House, December 2017), available at <<http://nssarchive.us/wp-content/uploads/2020/04/2017.pdf>>.
- ⁹³ See Nye, “Power and Interdependence with China,” 16.
- ⁹⁴ *The National Security Strategy of the United States of America* (2017).
- ⁹⁵ McMaster, “How China Sees the World.”

Appendix A

Selected Bibliography

- Abe, Shinzo. "Confluence of the Two Seas." Ministry of Foreign Affairs–Japan, August 22, 2007.
- Allen, Gregory. *Understanding China's AI Strategy: Clues to Chinese Strategic Thinking on Artificial Intelligence and National Security*. Washington, DC: Center for a New American Security, February 6, 2019.
- Allison, Graham. *Destined for War: Can America and China Escape Thucydides's Trap?* New York: Houghton Mifflin, 2017.
- . "The New Spheres of Influence: Sharing the Globe with Other Great Powers." *Foreign Affairs* 99, no. 2 (March/April 2020).
- Arbatova, Nadezda. "Three Faces of Russia's Neo-Eurasianism." *Survival* 61, no. 6 (December 2019–January 2020).
- Archick, Kristin. *Transatlantic Relations: U.S. Interests and Key Issues*, R45745. Washington, DC: Congressional Research Service, May 31, 2019.
- Asada, Sadao. "From Washington to London: The Imperial Japanese Navy and the Politics of Naval Limitation, 1921–30." In *The Washington Naval Conference, 1921–22*. Erik Goldstein and John Maurer, eds. London: Frank Cass & Company, 1994.
- Aso, Taro. "Arc of Freedom and Prosperity: Japan's Expanding Diplomatic Horizons." Speech by Mr. Taro Aso, Minister for Foreign Affairs on the Occasion of the Japan Institute of International Affairs Seminar, November 30, 2006.
- Assessment on U.S. Defense Implications of China's Expanding Global Access*. Washington, DC: Department of Defense, December 2018.
- Bader, Jeffrey A. *Obama and China's Rise: An Insider's Account of America's Asia Strategy*. Washington, DC: Brookings Institution Press, 2012.
- Baldwin, David. *Paradoxes of Power*. New York: Bail Blackwell, 1989.
- Barry, Jon M. *The Great Influenza: The Story of the Deadliest Pandemic in History*. New York: Penguin Books, 2004.
- Bayles, Martha. "How the World Perceives the New American Dream." *The Atlantic*, October 10, 2015.

- Beckley, Michael. "Economic Development and Military Effectiveness." *Journal of Strategic Studies* 33, no. 1 (February 2010).
- . "The Power of Nations: Measuring What Matters." *International Security* 43, no. 2 (Fall 2018).
- Blankenship, Brian D., and Benjamin Denison. "Is America Prepared for Great Power Competition?" *Survival* 61, no. 5 (October–November 2019).
- Bley, Bonnie. "The New Geography of Global Diplomacy: China Advances as the United States Retreats." *Foreign Affairs*, November 27, 2019.
- Bodine-Baron, Elizabeth, Todd C. Helmus, Andrew Radin, and Elina Treyger. "Countering Russian Social Media Influence." Santa Monica, CA: RAND, 2018.
- Bolsover, Gillian. *Computational Propaganda in China: An Alternative Model of a Widespread Practice*. Computational Propaganda Research Project Working Paper No. 2017.4. Oxford: Project on Computational Propaganda, 2017.
- Boustany, Jr., Charles W., and Aaron L. Friedberg. *Partial Disengagement: A New U.S. Strategy for Economic Competition with China*. Seattle: National Bureau of Asian Research, November 2019.
- Brands, Hal. "The Lost Art of Long-Term Competition." *The Washington Quarterly* 41, no. 4 (Fall 2019).
- . *Making the Unipolar Moment: U.S. Foreign Policy and the Rise of the Post-Cold War Order*. Cornell, NY: Cornell University Press, 2016.
- Brands, Hal, and Zack Cooper. "After the Responsible Stakeholder, What? Debating America's China Strategy." *Texas National Security Review* 2, no. 2 (February 2019).
- Brands, Hal, and Peter D. Feaver. "Reevaluating Diplomatic and Military Power: What Are America's Alliances Good For?" *Parameters* 4, no. 2 (Summer 2017).
- Brooks, Stephen G., and William C. Wohlforth. *America Abroad: The United States' Global Role in the 21st Century*. Oxford: Oxford University Press, 2016.
- Brown, Michael, Eric Chewning, and Pavneet Singh. *Preparing the United States for the Superpower Marathon with China*. Washington, DC: Brookings Institution, April 2020.
- Brunnermeier, Markus, Rush Doshi, and Harold James. "Beijing's Bismarckian Ghosts: How Great Powers Compete Economically." *The Washington Quarterly* 41, no. 3 (Fall 2018).
- Buckley, Chris, and Paul Mozur. "How China Uses High-Tech Surveillance to Subdue Minorities." *New York Times*, May 22, 2019.
- Bureau of Arms Control, Verification and Compliance. *Adherence to and Compliance with Arms Control, Nonproliferation, and Disarmament Agreements and Commitments*. Department of State, August 2019.

- . *Annual Report on Implementation of the New START Treaty*. Department of State, February 2019.
- Burrows, Mathew J., and Peter Engelke. *What World Post COVID-19? Three Scenarios*. Washington, DC: Atlantic Council, April 2020.
- Campbell, Charles S. *From Revolution to Rapprochement: The United States and Great Britain, 1783–1900*. New York: John Wiley & Sons, 1974.
- Campbell, Kurt M. *The Pivot: The Future of American Statecraft*. New York: Hatchett Book Group, 2016.
- Campbell, Kurt M., and Rush Doshi. “The Coronavirus Could Reshape Global Order.” *Foreign Affairs*, March 18, 2020.
- Campbell, Kurt M., and Ely Ratner. “The China Reckoning: How Beijing Defied American Expectations.” *Foreign Affairs* 98, no. 2 (March/April 2018).
- Campbell, Kurt M., and Jake Sullivan. “Competition Without Catastrophe.” *Foreign Affairs* 98, no. 5 (September/October 2019).
- Cheng, Cheng. “The Logic Behind China’s Foreign Aid Agency.” Carnegie Endowment for International Peace, May 21, 2019.
- Cheng, Dean. *Cyber Dragon: Inside China’s Information Warfare and Cyber Operations*. Santa Barbara, CA: Praeger Security International, 2016.
- Cheung, Tai Ming, and Thomas G. Mahnken. *The Gathering Pacific Storm*. Amherst, NY: Cambria, 2018.
- China and the World in the New Era*. Beijing: The State Council Information Office of the People’s Republic of China, September 2019.
- China’s National Defense in the New Era* [Defense White Paper]. Beijing: The State Council Information Office of the People’s Republic of China, July 2019.
- Clinton, Hillary. “America’s Pacific Century.” *Foreign Policy*, no. 189 (November–December 2011).
- Coker, Christopher. *Future War*. Cambridge, UK: Polity Press, 2015.
- Conway, Maura, Ryan Scrivens, and Logan Macnair. *Right-Wing Extremists’ Persistent Online Presence: History and Contemporary Trends*. The Hague: International Centre for Counter-Terrorism, November 2019.
- Cook, Sarah. “Beijing’s Global Megaphone: The Expansion of Chinese Communist Party Media Influence Since 2017.” *Freedom House*, 2020.
- Crammond, Edgar. “The Economic Relations of the British and German Empires.” *Journal of the Royal Statistical Society* 77, no. 8 (July 1914).

- Cyrill, Melissa. "What Is Made in China 2025 and Why Has It Made the World So Nervous?" *China Briefing*, December 28, 2018.
- Dalton, Melissa, et al. *By Other Means—Part II: U.S. Priorities in the Gray Zone*. Washington, DC: Center for Strategic and International Studies, 2019.
- Defense Intelligence Agency. *China Military Power: Modernizing a Force to Fight and Win*. Washington, DC: Defense Intelligence Agency, 2019.
- . *Iran Military Power: Ensuring Regime Survival and Securing Regional Dominance*. Washington, DC: Defense Intelligence Agency, 2019.
- . *Russia Military Power: Building a Military to Support Great Power Aspirations*. Washington, DC: Defense Intelligence Agency, 2017.
- Denmark, Abraham. "Ideological Competition in the Indo-Pacific." *Asia Dispatches*, March 27, 2018.
- Department of Defense Arctic Strategy*. Washington, DC: Department of Defense, June 2019.
- Department of Defense Strategy for Countering Weapons of Mass Destruction*. Washington, DC: Department of Defense, June 2014.
- Dinan, Desmond, Neil Nugent, and William E. Patterson. *The European Union in Crisis*. London: Red Globe Press, 2017.
- DiResta, Renee. "For China, the 'USA Virus' Is a Geopolitical Ploy." *The Atlantic*, April 11, 2020.
- Dobbins, James, Howard J. Shatz, and Alie Wyne. *Russia Is a Rogue, Not a Peer; China Is a Peer, Not a Rogue*. Santa Monica, CA: RAND, 2018.
- Dodwell, Brian, and Don Rassler. "A View from the CT Foxhole: LTG Michael K. Nagata, Director, Directorate of Strategic Operational Planning, NCTC." *Combating Terrorism Center* 10, no. 6 (June/July 2017).
- Economic and Trade Agreement Between the Government of the United States of America and the Government of the People's Republic of China*. Washington, DC: Office of the U.S. Trade Representative, 2020.
- Edel, Charles, and Hal Brands. "The Real Origins of the U.S.-China Cold War." *Foreign Policy*, June 2, 2019.
- Epstein, Gerald L. "Biodefense and the Return to Great Power Competition." *Nonproliferation Review* (Spring/Summer 2020).
- "The European Union and the Russian Federation." European Union Web site, February 25, 2020.
- "EU Global Strategy." European Union Web site, June 14, 2019.

- “EU-China Relations Factsheet.” European Union Web site, June 20, 2020.
- Eurosystem: The International Role of the Euro*. European Central Bank, June 2019.
- Farah, Douglas, and Kathryn Babineau. “Extra-Regional Actors in Latin America: The United States Is Not the Only Game in Town.” *PRISM* 8, no. 1 (2019).
- Fingar, Thomas. “China’s Vision of World Order.” In *Strategic Asia 2012–13: China’s Military Challenge*. Ashley J. Tellis and Travis Tanner, eds. Seattle: National Bureau of Asian Research, 2012.
- Fingar, Thomas, and Jean C. Oi. “China’s Challenges: Now It Gets Much Harder.” *The Washington Quarterly* 43, no. 1 (Spring 2020).
- Flournoy, Michèle. “DOD’s Role on Competing with China.” House Armed Services Committee, January 15, 2020.
- Foreign Policy Concept of the Russian Federation*. Moscow: Ministry of Foreign Affairs of the Russian Federation, November 30, 2016.
- Franklin, Daniel. *MegaTech: Technology in 2050*. New York: Profile Books, 2017.
- A Free and Open Indo-Pacific: Advancing a Shared Vision*. Washington, DC: Department of State, November 4, 2019.
- Friedberg, Aaron L. “Competing with China.” *Survival* 60, no. 3 (2018).
- . *A Contest for Supremacy: China, America, and the Struggle for Mastery in Asia*. New York: Norton, 2011.
- . “Getting the China Challenge Right.” *The American Interest*, January 10, 2019.
- . *The Weary Titan: Britain and the Experience of Relative Decline, 1895–1905*. Princeton: Princeton University Press, 1988.
- Friedberg, Aaron L., and Charles W. Boustany, Jr. “Partial Disengagement: A New U.S. Strategy for Economic Competition with China.” *The Washington Quarterly* 43, no. 1 (Spring 2020).
- Fukayama, Francis. “The End of History?” *The National Interest*, no. 16 (Summer 1989).
- Fuxian, Li. “Why Ageing China Won’t Overtake the U.S. Economy as the World’s Biggest—Now or in the Future.” *South China Morning Post*, March 29, 2019.
- Gao, George. “Why Is China So Uncool?” *Foreign Policy*, March 8, 2017.
- Gentile, Gian. *Four Problems in the Korean Peninsula: North Korea’s Expanding Nuclear Capabilities Drive a Complex Set of Problems*. Santa Monica, CA: RAND, 2019.
- Gerges, Fawaz A. *The Far Enemy: Why Jihad Went Global*. New York: Cambridge University Press, 2005.

- Gilpin, Robert. *U.S. Power and the Multinational Corporation: The Political Economy of Foreign Direct Investment*. New York: Basic Books, 1975.
- . *War and Change in World Politics*. Cambridge: Cambridge University Press, 1981.
- Glaser, Charles. *Rational Theory of International Politics: The Logic of Competition and Cooperation*. Princeton: Princeton University Press, 1986.
- Goldstein, Lyle. *Meeting China Halfway: How to Defuse the Emerging U.S.-China Rivalry*. Washington, DC: Georgetown University Press, 2015.
- Grab, Alexander. *Napoleon and the Transformation of Europe*. Hampshire, UK: Palgrave, Ltd., 2003.
- Green, Michael, and Evan S. Medeiros. “The Pandemic Won’t Make China the World’s Leader.” *Foreign Affairs*, April 15, 2020.
- Greenberg, Andy. *Sandworm: A New Era of Cyberwar and the Hunt for the Kremlin’s Most Dangerous Hackers*. New York: Doubleday, 2019.
- Haass, Richard. “The Pandemic Will Accelerate History Rather Than Reshape It.” *Foreign Affairs*, April 7, 2020.
- Hammes, T.X. *Deglobalization and International Security*. Amherst, NY: Cambria Press, 2019.
- Hanauer, Larry, and Lyle J. Morris. *Chinese Engagement in Africa: Drivers, Reactions, and Implications for U.S. Policy*. Santa Monica, CA: RAND, 2014.
- Harrell, Peter, Elizabeth Rosenberg, and Edoardo Saravalle. *China’s Use of Coercive Economic Measures*. Washington, DC: Center for a New American Security, June 2018.
- Heath, Timothy, and William Thompson. “Avoiding U.S.-China Competition Is Futile: Why the Best Option Is to Manage Strategic Rivalry.” *Asia Policy* 13, no. 2 (April 2018).
- Heginbotham, Eric, et al. *The U.S.-China Military Scorecard: Forces, Geography, and the Evolving Balance of Power, 1996–2017*. Santa Monica, CA: RAND, 2015.
- Hersman, Rebecca, and Suzanne Claeys. *Rigid Structures, Evolving Threat: Preventing the Proliferation and Use of Chemical Weapons*. Center for Strategic and International Studies, November 2019.
- “Hijrah to the Islamic State: What to Packup, Who to Contact, Where to Go, Stories & More!” Islamic State Foreign Fighters, 2015.
- Hill, Fiona, and Clifford G. Gaddy. *Mr. Putin: Operative in the Kremlin*. Washington, DC: Brookings Institution Press, 2015.
- Homeland Security Advisory Council: Interim Report of the Countering Foreign Influence Subcommittee*. Washington, DC: Department of Homeland Security, June 2019.

- Horowitz, Michael C. "Artificial Intelligence, International Competition, and the Balance of Power." *Texas National Security Review* 1, no. 3 (May 2018).
- Huang, Echo. "Why China Isn't as Skillful at Disinformation as Russia." *Quartz*, September 19, 2019.
- Ikenberry, G. John. *After Victory: Institutions, Strategic Restraint, and the Rebuilding of Order After Major Wars*. Princeton: Princeton University Press, 2000.
- . *Liberal Leviathan: The Origins, Crisis, and Transformation of the American World Order*. Princeton: Princeton University Press, 2011.
- Indo-Pacific Strategy Report: Preparedness, Partnerships, and Promoting a Networked Region*. Washington, DC: Department of Defense, June 1, 2019.
- "An Interactive Look at the U.S.-China Military Scorecard." RAND–Project Air Force Web site.
- Interim Report of the National Security Commission on Artificial Intelligence*. Washington, DC: National Security Commission on Artificial Intelligence, November 2019.
- Irwin, Douglas A. "Historical Aspects of U.S. Trade Policy." National Bureau of Economic Research, Summer 2006.
- Jinping, Xi. "New Asian Security Concept for Progress in Security Cooperation." Fourth Summit of the Conference on Interaction and Confidence Building Measures in Asia. Shanghai Expo Center, May 21, 2014.
- . "Secure a Decisive Victory in Building a Moderately Prosperous Society in All Respects and Strive for the Great Success of Socialism with Chinese Characteristics for a New Era." Report to the 19th National Party Congress of the Communist Party of China, October 18, 2017.
- Johnston, Iain Alistair. "The Failures of the 'Failure of Engagement' with China." *The Washington Quarterly* 42, no. 2 (Summer 2019).
- . "China in a World of Orders." *International Security* 44, no. 2 (Fall 2019).
- Kagan, Robert. *The Return of History and the End of Dreams*. New York: Vintage Books, 2009.
- Kelle, Alexander. "Power in the Chemical Weapons Prohibition Regime and the Organisation for the Prohibition of Chemical Weapons." *International Politics* 55, no. 1 (October 2017).
- Kennedy, Paul. *The Rise and Fall of British Naval Mastery*, 2nd ed. New York: Humanity Books, 1983.
- . *The Rise and Fall of the Great Powers*. New York: Random House, 1989.

- Kennedy, Scott. "Made in China 2025." Center for Strategic and International Studies, June 1, 2015.
- Keohane, Robert O. *After Hegemony: Cooperation and Discord in the World Political Economy*. Princeton: Princeton University Press, 1984.
- Keohane, Robert O., and Joseph S. Nye, Jr. *Power and Interdependence: World Politics in Transition*. Boston: Little, Brown, 1977.
- Keynoush, Banafsheh. "Saudi Arabia and Iran Are Adapting to Perpetual Conflict." Atlantic Council, May 23, 2019.
- Kindleberger, Charles S. *The World in Depression, 1929–1939*. Berkeley: University of California Press, 1973.
- . *World Economic Primacy: 1500–1990*. Oxford: Oxford University Press, 1996.
- Kissinger, Henry. "The Coronavirus Pandemic Will Forever Alter the World Order." *Wall Street Journal*, April 3, 2020.
- . *On China*. New York: Penguin Books, 2011.
- . *World Order*. New York: Penguin Books, 2014.
- Koblentz, Gregory D. "Regime Security: A New Theory for Understanding the Proliferation of Chemical and Biological Weapons." *Contemporary Security Policy* 34, no. 3 (November 2013).
- Kramer, David J. "Russia Is No Great Power Competitor." *The American Interest*, April 24, 2019.
- Kramer, Steven Philip. *The Other Population Crisis: What Governments Can Do About Falling Birth Rates*. Washington, DC: Woodrow Wilson Center Press, 2013.
- Krastev, Ivan. *After Europe*. Philadelphia: University of Pennsylvania Press, 2017.
- Krauthammer, Charles. "The Unipolar Moment." *Foreign Affairs* 70, no. 1 (1991).
- Kyriakopoulos, Irene. "Economic and Monetary Union Governance: A Post-Crisis Assessment." *World Economics Journal* 12, no. 4 (October–December 2011).
- Lacey, James. *Great Strategic Rivalries: From the Classical World to the Cold War*. Oxford: Oxford University Press, 2016.
- Lardy, Nicholas. *Sustaining China's Economic Growth After the Global Financial Crisis*. Washington, DC: Peterson Institute for International Economics, December 2011.
- Lee, Kai-Fu. *AI Superpowers: China, Silicon Valley, and the New World Order*. Boston: Houghton Mifflin Harcourt, 2018.

- Liu, Xin. "What Sharp Power? It's Nothing But 'Unsmart' Power." University of Southern California Center on Public Diplomacy, November 15, 2018.
- Lynch, Colum, and Elias Groll. "As U.S. Retreats from World Organizations, China Steps in to Fill the Void." *Foreign Policy*, October 6, 2017.
- Lynch III, Thomas F., and James J. Przystup, *India-Japan Strategic Cooperation and Implications for U.S. Strategy in the Indo-Asia-Pacific Region*. INSS Strategic Perspectives 24. Washington, DC: NDU Press, March 2017.
- Mahnken, Thomas G. *Forging the Tools of 21st Century Great Power Competition*. Washington, DC: Center for Strategic and Budgetary Assessment, 2020.
- . "A 'World-Class' Military: Assessing China's Global Military Ambitions." U.S.-China Economic and Security Review Commission, June 20, 2019.
- Mastro, Oriana Skylar. "The Stealth Superpower: How China Hid Its Global Ambitions." *Foreign Affairs* 88, no. 1 (January/February 2019).
- Mazarr, Michael J. "This Is Not Great Power Competition." *Foreign Affairs*, May 2019.
- . et al. *Hostile Social Manipulation: Present Realities and Emerging Trends*. Santa Monica, CA: RAND, 2019.
- . et al. *Understanding the Emerging Era of International Competition: Theoretical and Historical Perspectives*. Santa Monica, CA: RAND, 2018.
- McDonald, Scott, and Michael Burgoyne, eds. *China's Global Influence: Perspectives and Recommendations*. Honolulu: Asia-Pacific Center for Security Studies, 2019.
- McMaster, H.R. "How China Sees the World: And How We Should See China." *The Atlantic*, May 2020.
- McNeal, Robert H. "The Legacy of the Comintern." *International Journal* 201, no. 2 (Spring 1966).
- Mearsheimer, John J. *The Tragedy of Great Power Politics*. New York: Norton, 2014.
- Military and Security Developments Involving the Republic of China*. Washington, DC: Office of the Secretary of Defense, May 2, 2019.
- Missile Defense Review*. Washington, DC: Department of Defense, 2019.
- Morgenthau, Hans. *Politics Among Nations*. New York: Knopf, 1948.
- Mueller III, Robert S. *Report on the Investigation into Russian Interference in the 2016 Presidential Election*. Washington, DC: Department of Justice, March 2019.
- Murray, Williamson, and Allan R. Millett, eds. *Military Innovation in the Interwar Period*. New York: Cambridge University Press, 1998.

- Naji, Abu Bakr. *Management of Savagery: The Most Critical Stage Through Which the Umma Will Pass*. William McCants, trans. Cambridge, MA: John M. Olin Institute for Strategic Studies, Harvard University, May 23, 2006.
- National Biodefense Strategy*. Washington, DC: The White House, 2018.
- National Security Strategy of the United States of America*. Washington, DC: The White House, December 2017.
- National Strategy for Counterterrorism of the United States of America*. Washington, DC: The White House, October 2018.
- National Strategy to Secure 5G of the United States of America*. Washington, DC: The White House, March 2020.
- Nemr, Christina, and William Gangware. *Weapons of Mass Distraction: Foreign State-Sponsored Disinformation in the Digital Age*. Washington, DC: Park Advisors, 2019.
- Nuclear Posture Review*. Washington, DC: Department of Defense, February 2018.
- Nye, Jr., Joseph S. "Get Smart: Combining Hard and Soft Power." *Foreign Affairs* 88, no. 4 (July/August 2009).
- . *The Paradox of American Power*. London: Oxford University Press, 2002.
- . "Power and Interdependence with China." *The Washington Quarterly* 43, no. 1 (Spring 2020).
- Obama, Barack. "Remarks by President Obama to the Australian Parliament." Washington, DC: The White House, November 17, 2011.
- Odom, Jonathan C. "Understanding China's Legal Gamesmanship in the Rules-Based Global Order." In *China's Global Influence: Perspectives and Recommendations*, Scott D. McDonald and Michael C. Burgoyne, eds. Honolulu: Asia-Pacific Center for Security Studies, 2019.
- Osnos, Evan. "The Future of America's Context with China." *The New Yorker*, January 6, 2020.
- Outlaw Regime: A Chronicle of Iran's Destructive Activities*. Washington, DC: Department of State, 2018.
- Overholt, William H. *China and America: The Age of Realist Geoeconomics*. Atlanta: Carter Center, January 2018.
- . *China's Crisis of Success*. New York: Cambridge University Press, 2018.
- Paal, Douglas H. *America's Future in a Dynamic Asia*. Washington, DC: Carnegie Endowment for International Peace, 2019.

- Parker, John W. *Putin's Syrian Gambit: Sharper Elbows, Bigger Footprint, Stickier Wicket*. INSS Strategic Perspectives 25. Washington, DC: NDU Press, July 2017.
- Pearce, Katie. "Pandemic Simulation Exercise Spotlights Massive Preparedness Gap." *Johns Hopkins Magazine*, November 6, 2019.
- Pei, Minxin. "China's Coming Upheaval: Competition, the Coronavirus, and the Weakness of Xi Jinping." *Foreign Affairs* 99, no. 3 (May/June 2020).
- Perl, Raphael F. *Drug Trafficking and North Korea: Issues for U.S. Policy*, RL32167. Washington, DC: Congressional Research Service, 2007.
- Polyakova, Alina. "Weapons of the Weak: Russia and AI-Driven Asymmetric Warfare." *Brookings*, November 15, 2018.
- Putin, Vladimir. "Russian President Vladimir Putin State of the Union Address. C-SPAN video, 1:58:05. Moscow: Manezh Central Exhibition Hall, March 1, 2018.
- Rapp-Hooper, Mira. "Saving America's Alliances: The United States Still Needs the System That Put It on Top." *Foreign Affairs* 99, no. 2 (March/April 2020).
- Rasser, Martijn, et al. *The American AI Century: A Blueprint for Action*. Washington, DC: Center for a New American Security, December 2019.
- Ratner, Ely. "Blunting China's Illiberal Order." Senate Armed Services Committee, January 29, 2019.
- Ratner, Ely, et al. *Rising to the China Challenge: Renewing American Competitiveness in the Indo-Pacific*. Washington, DC: Center for a New American Security, January 28, 2020.
- Remak, Joachim. *The Origins of World War I, 1871–1914*. Hinsdale, IL: Dryden Press, 1967.
- Robinson, Linda, et al. *Modern Political Warfare: Current Practices and Possible Responses*. Santa Monica, CA: RAND, 2018.
- Rolland, Nadège. *A Concise Guide to the Belt and Road Initiative*. Seattle: National Bureau of Asian Research, April 11, 2019.
- Ross, Alec. *The Industries of the Future*. New York: Simon & Schuster, 2016.
- Roy, Denny. "China Won't Achieve Regional Hegemony." *The Washington Quarterly* 43, no. 1 (Spring 2020).
- Rumer, Eugene, and Richard Sokolsky. *Thirty Years of U.S. Policy Toward Russia: Can the Vicious Circle Be Broken?* Washington, DC: Carnegie Endowment for International Peace, 2019.
- Saunders, Phillip C. "China's Global Military-Security Interactions." In *China and the World*, David Shambaugh, ed. New York: Oxford University Press, 2020.

- . “Implications: China in the International System.” In *The Chinese People’s Liberation Army in 2025*. Carlisle Barracks, PA: Army War College Press, 2017.
- . *Managing Competition with China*. INSS Strategic Forum 242. Washington, DC: NDU Press, July 2009.
- . et al., eds. *Chairman Xi Remakes the PLA: Assessing Chinese Military Reforms*. Washington, DC: NDU Press, 2019.
- Schmitt, Eric, and Thom Shanker, *Counterstrike: The Untold Story of America’s Secret Campaign Against al Qaeda*. New York: Times Books, 2011.
- Schmitt, Gary, and Michael Mazza. *Blinding the Enemy: CCP Interference in Taiwan’s Democracy*. Washington, DC: Global Taiwan Institute, October 2019.
- Schneider, Tobias, and Theresa Lütkefend. *Nowhere to Hide: The Logic of Chemical Weapons Use in Syria*. Berlin: Global Public Policy Institute, February 2019.
- Schultz, George P., Jim Hoagland, and James Timbie. *Beyond Disruption: Technology’s Challenge to Governance*. Stanford, CA: Hoover Institution Press, 2018.
- Schwab, Klaus. *The Fourth Industrial Revolution*. New York: Crown Business, 2017.
- “Section 3: Threats to the Russian Federation’s National Security.” In *National Security Concept of the Russian Federation*. Moscow: Ministry of Foreign Affairs of the Russian Federation, 2000.
- Shah, Sonia. *Pandemic: Tracking Contagions, from Cholera to Ebola and Beyond*. New York: Sarah Crichton Books, 2016.
- Shambaugh, David. *China Goes Global*. New York: Oxford University, 2013.
- . “U.S.-China Rivalry in Southeast Asia: Power Shift or Competitive Coexistence.” *International Security* 42, no. 4 (Spring 2018).
- Silver, Laura. “U.S. Is Seen as Top Ally in Many Countries—But Others View It as a Threat.” Pew Research Center, December 5, 2019.
- Silver, Laura, Kat Devlin, and Christine Huang. “People Around the Globe Are Divided in Their Opinions of China.” Pew Research Center, December 5, 2019.
- Smeltz, Dina, et al. *America Engaged: American Public Opinion and U.S. Foreign Policy*. Chicago: Chicago Council on Global Affairs, 2018.
- Smith-Goodson, Paul. “Quantum USA Vs. Quantum China: The World’s Most Important Technology Race.” *Forbes*, October 11, 2019.
- Sokov, Nikolai. *Russia’s New Conventional Capability: Implications for Eurasia and Beyond*. PONARS Eurasia Policy Memo 472. Washington, DC: Institute for European, Russian and Eurasian Studies, April 2017.

- Steinberg, James B. "What Went Wrong? U.S.-China Relations from Tiananmen to Trump." *Texas National Security Review* 3, no. 1 (Winter 2019/2020).
- Stent, Angela. *The Limits of Partnership: U.S.-Russian Relations in the Twenty-First Century*. Princeton: Princeton University Press, 2014.
- Stronski, Paul. *Late to the Party: Russia's Return to Africa*. Washington, DC: Carnegie Endowment for International Peace, October 2019.
- Stronski, Paul, and Richard Sololsky. *The Return of Global Russia: An Analytic Framework*. Washington, DC: Carnegie Endowment for International Peace, December 2017.
- Suciu, Peter. "5G Is Old News: China Wants 6G for Its Military." *The National Interest*, April 28, 2020.
- Summary of the 2018 National Defense Strategy of the United States of America: *Sharpening the American Military's Competitive Edge*. Washington, DC: Department of Defense, January 20, 2018.
- Traub, James. "After the Coronavirus, the Era of Small Government Will Be Over." *Foreign Policy*, April 15, 2020.
- Trump, Donald J. "Remarks by President Trump at APEC CEO Summit." U.S. Embassy and Consulate in Vietnam, November 10, 2017.
- Update to the Intellectual Property Commission Report*. Commission on the Theft of American Intellectual Property. Seattle: National Bureau of Asian Research, 2017.
- Uren, Tom, Elise Thomas, and Jacob Wallis. *Tweeting Through the Great Firewall: Preliminary Analysis of PRC-Linked Information Operations Against the Hong Kong Protests*. Report No. 25/2019. Canberra: Australian Strategic Policy Institute, 2019.
- Volgy, Thomas J., Renato Corbetta, Keith A. Grant, and Ryan G. Baird. "Major Power Status in International Politics." In *Major Powers and the Quest for Status in International Politics: Global and Regional Perspectives*, Volgy, Corbetta, Grant, and Baird, eds. New York: Palgrave Macmillan, 2011.
- Walt, Stephen M. "International Relations: One World, Many Theories." *Foreign Policy*, no. 110 (1998).
- . "Yesterday's Cold War Shows How to Beat China Today." *Foreign Policy*, July 29, 2019.
- Waltz, Kenneth N. "Structural Realism After the Cold War." *International Security* 25, no. 1 (Summer 2000).
- . *Theory of International Politics*. Long Grove, IL: Waveland Press, Inc., 2010.

- Ward, Adolphus William, and George Peabody Gooch, eds. *The Cambridge History of British Foreign Policy, 1783–1919*, vol. 3, 1866–1919. Cambridge: Cambridge University Press, 1923.
- Wasser, Becca. *The Limits of Russian Strategy in the Middle East*. Santa Monica, CA: RAND 2019.
- Weiss, Michael, and Hassan Hassan. *ISIS: Inside the Army of Terror*. New York: Regan Arts, 2015.
- Westad, Odd Arne. “Legacies of the Past.” In *China and the World*, David Shambaugh, ed. New York: Oxford University Press, 2020.
- . “The Sources of Chinese Conduct: Are Washington and Beijing Fighting a New Cold War?” *Foreign Affairs* 88, no. 5 (September/October 2019).
- Wood, Tony. *Russia Without Putin: Money, Power and the Myths of the New Cold War*. New York: Verso Books, 2018.
- Wright, Thomas J. *All Measures Short of War: The Contest for the 21st Century and the Future of American Power*. New Haven: Yale University Press, 2017.
- . “The Rise and Fall of the Unipolar Concert.” *The Washington Quarterly* 37, no. 4 (Winter 2015).
- Wuthnow, Joel. *Chinese Perspectives on the Belt and Road Initiative: Strategic Rationales, Risks, and Implications*. INSS China Strategic Perspectives 12. Washington, DC: NDU Press, October 2017.
- . *Just Another Paper Tiger? Chinese Perspectives on the U.S. Indo-Pacific Strategy*. INSS Strategic Forum 305. Washington, DC: NDU Press, June 2020.
- Wyne, Ali. “America’s Blind Ambition Could Make It a Victim of Global Competition.” *The National Interest*, February 11, 2019.
- Zakaria, Fareed. *The Post-American World*. New York: Norton, 2008.
- . “The Rise of Illiberal Democracy.” *Foreign Affairs* 76, no. 6 (November/December 1997).
- Zhang, Denghua. “The Concept of ‘Community of Common Destiny’ in China’s Diplomacy: Meaning, Motives, and Implications.” *Asia and the Pacific Policy Studies* 5, no. 2 (March 2018).
- Zwack, Peter B., and Marie-Charlotte Pierre. *Russian Challenges from Now into the Next Generation: A Geostrategic Primer*. INSS Strategic Perspectives 29. Washington, DC: NDU Press, March 2019.

Appendix B

Contemporary Great Power Dynamics Competition Matrix

Available at <<https://ndupress.ndu.edu/Contemporary-GPC-Dynamics-Matrix/>>

Contributors

Dr. Thomas F. Lynch III is Acting Director of the Center for Strategic Research (CSR), Institute of National Strategic Studies (INSS), at the National Defense University (NDU) in Washington, DC. He also is the INSS Distinguished Research Fellow for South Asia, the Near East, and Countering Radical Islam. His primary ongoing research is about India's strategic rise and implications for Great Power rivalry and Indo-Pacific security and stability. Among his many open-source publications are "The Growing Entente Between India and Japan," *The National Interest* (March/April 2019) and "The Decades-Long 'Double-Double Game': Pakistan, the United States, and the Taliban," *Military Review* (July/August 2018). Dr. Lynch received a bachelor of science in engineering from the U.S. Military Academy at West Point, a master of public administration, and a master of arts and doctor of philosophy in international relations from the Woodrow Wilson School at Princeton University.

Dr. Hassan Abbas is Distinguished Professor of International Relations in the Near East South Asia Strategic Studies Center in Washington, DC, and a former Professor of International Security Studies in the College of International Security Affairs at NDU. He serves as a senior advisor at Project on Shi'ism and Global Affairs in the Weatherhead Center for International Affairs at Harvard University. He is the author of *The Taliban Revival: Violence and Extremism on the Pakistan-Afghanistan Frontier* (Yale University Press, 2014) and *Pakistan's Nuclear Bomb: A Story of Defiance, Deterrence and Deviance* (Oxford University Press, 2018). Dr. Abbas received a doctor of philosophy and master of arts in law and diplomacy from the Fletcher School of Law and Diplomacy at Tufts University, a master of laws in international law from Nottingham University in the United Kingdom, and a master of arts in political science from Punjab University in Pakistan.

Dr. Zachary M. Abuza is a Professor of National Security Strategy in the National War College at NDU, where he focuses on Southeast Asian politics and security issues, including governance, insurgencies, democratization and human rights, and maritime security. He is the author of five books, including *Forging Peace in Southeast Asia: Insurgencies, Peace Processes, and Reconciliation* (Rowman & Littlefield, 2016); *Conspiracy of Silence: The Insurgency in Southern Thailand* (United States Institute of Peace, 2009); and *Militant Islam in Southeast Asia: Crucible of Terror* (Lynne Rienner, 2003). In addition, Dr. Abuza has authored four monographs on security issues in Southeast Asia. He is currently working on a study project about Thai counterinsurgency strategy. Dr. Abuza received a doctor of philosophy and master of arts in law and diplomacy from the Fletcher School of Law and Diplomacy at Tufts University and a bachelor of arts in Asian studies from Trinity College.

Dr. Justin Anderson is a Senior Policy Fellow in the Center for the Study of Weapons of Mass Destruction (CSWMD), INSS. He conducts research on deterrence, arms control, and nuclear weapons issues, and he also serves as a Special Advisor to the Defense Threat Reduction Agency. He is co-author of "The INF Treaty: A Spectacular, Inflexible, Time-Bound Success," *Strategic Studies Quarterly* 13, no. 2 (Summer 2019). Dr. Anderson is a 2003 Marshall Scholar and received a doctor of philosophy and master of arts in war studies from King's College London.

Dr. Richard Andres is a Professor of National Security Strategy at the National War College, where he directs the college's course of study on strategy. He has served in a number of policy positions, including advisor to the Secretary of the Air Force, Commandant of the Marine Corps, and Commander of U.S. Cyber Command/Director of the National Security Agency. He currently holds faculty or board positions in the School of Advanced International Studies at The Johns Hopkins University, Security Studies Program at Georgetown University, Pacific Northwest National Laboratory, and American Enterprise Institute. Dr. Andres's research focus is on the national security implications of information technology. He received a bachelor of arts in international finance from Fresno Pacific University and a doctor of philosophy in political science from the University of California, Davis.

Dr. David Auerswald is a Professor of Security Studies at the National War College, a position he has held since 2001. He has written books and articles on the geopolitics of the Arctic, North Atlantic Treaty Organization interventions, and the U.S. national security policy process. Dr. Auerswald is a former congressional staffer. He received a doctor of philosophy and master of arts in political science from the University of California San Diego and bachelor of arts in both political science and English literature from Brown University.

Mr. Paul Bernstein is a Distinguished Policy Fellow in CSWMD. He leads the center's practice in strategic security analysis and is engaged in policy support, research, and professional military education activities related to WMD, nuclear policy, deterrence, missile defense, threat reduction, and regional security. He works in collaboration with the Office of the Secretary of Defense, Joint Staff, combatant commands, and defense agencies. He is co-author most recently of "Russia's Hypersonic Weapons," *Georgetown Journal of International Affairs* (December 2019) and a subject matter expert contributor to John A. Stevenson, *Power Under Parity* (NSI, Inc., September 2019). Mr. Bernstein holds a master of arts in international affairs from Columbia University.

Dr. R. Kim Cragin is the Senior Research Fellow for Special Operations and Counterterrorism in CSR. Dr. Cragin has conducted research on irregular warfare for over 20 years. Prior to joining INSS in 2015, she was a senior political scientist at RAND. Dr. Cragin has published widely in academic and policy journals. Her ongoing research focuses on proxy warfare in the Middle East and North Africa. Other recent publications include "Preventing the Next Wave of Foreign Terrorist Fighters," *Studies in Conflict and Terrorism* (February 2019) and "Metastasis: Exploring the Impact of Foreign Fighters in Conflicts Abroad," *Journal*

of *Strategic Studies* (November 2017). Dr. Cragin's book, co-authored with Sara A. Daly, *Women as Terrorists: Mothers, Recruiters, and Martyrs* (Prager Security International) was released in 2009. She received a master of public policy from Duke University and a doctor of philosophy in history from Cambridge University.

Dr. Diane DiEuliis is a Senior Research Fellow in CSWMD. Her research focuses on emerging biological technologies, biodefense, and preparedness for biothreats. Her expertise includes synthetic biology; the U.S. bioeconomy; dual-use life sciences research; and behavioral, cognitive, and social sciences. She is author most recently of "Key National Security Questions for the Future of Synthetic Biology," *Fletcher Forum of World Affairs* 43, no. 1 (Winter 2019). Dr. DiEuliis received a doctor of philosophy in biology from the University of Delaware.

Dr. Gerald Epstein is a Distinguished Policy Fellow in CSWMD. He works at the intersection of science, technology, and security policy, particularly on the governance and security implications of advanced life, biotechnologies, and other emerging and converging technologies. He recently completed service on the National Academies of Sciences, Engineering, and Medicine committee responsible for writing *Safeguarding the Bioeconomy* (The National Academies Press, 2020). Dr. Epstein received a doctor of philosophy in astrophysics from the University of California, Berkeley.

Mr. Douglas Farah is a Visiting Senior Fellow in CSR. He is the president of IBI Consultants, a national security consulting firm. Previously, he worked as a foreign correspondent and investigative reporter to the *Washington Post* for more than 20 years, covering Latin America and West Africa. He is an expert on national security, transnational crime, illicit finance, and nonstate armed actors. Mr. Farah has provided testimony before House and Senate committees, presented at security conferences across the hemisphere, and published two books. He received a bachelor of arts in Latin American studies and bachelor of science in journalism from the University of Kansas.

Dr. T.X. Hammes is a Distinguished Research Fellow in CSR. He has served at all levels in the operating forces, including commanding an intelligence battalion, infantry battalion, and the Chemical Biological Incident Response Force during his 30-year career in the Marine Corps. He participated in stabilization operations in Somalia and Iraq as well as training insurgents in various places. He is the author of 3 books and over 170 articles. He lectures extensively on the future of conflict, strategy, and insurgency in the United States, Europe, Asia, and the Middle East. Dr. Hammes received a bachelor of science in operations analysis from the U.S. Naval Academy and a master of arts in historical research and a doctor of philosophy in modern history from Oxford University.

Dr. Todd C. Helmus is a Senior Behavioral Scientist at RAND and a faculty member at the Pardee RAND Graduate School. His specializations include disinformation, terrorism, and strategic communications. His most recent work has focused on countering Russian disinformation campaigns in the United States and Europe and countering violent extremism

through analyzing Twitter networks. Dr. Helmus served as a deployed advisor to U.S. commanders in Iraq and Afghanistan and has led studies on U.S. training of Afghan special forces. He received a doctor of philosophy in clinical psychology from Wayne State University.

Dr. Frank G. Hoffman is a Distinguished Research Fellow in CSR. His principal research focus is national security and defense strategies, defense economics, and joint force development. In addition to his research duties, Dr. Hoffman teaches strategy at the National War College and future warfare in the Dwight D. Eisenhower School for National Security and Resource Strategy at NDU. His most recent article is “The Missing Element in Crafting National Strategy: A Theory of Success,” *Joint Force Quarterly* 97 (2nd Quarter 2020). His forthcoming book is *Mars Adapting: Military Change in War* (Naval Institute Press). Dr. Hoffman received a bachelor of science in economics from the Wharton Business School at the University of Pennsylvania. He also received a master of education from George Mason University and a master of arts in national strategic studies from the U.S. Naval War College. He holds a doctor of philosophy in war studies from King’s College London.

Dr. Steven Philip Kramer is Professor of History at NDU, serving in INSS. He focuses on the application of history to the development of policy and national strategy. He is an authority on French and European history and politics and was a senior policy advisor in the Department of State. His books include *Socialism in Europe: The Experience of a Generation* (Westview Press, 1984); *Does France Still Count? The French Role in the New Europe* (Praeger, 1994); and *The Other Population Crisis: What Governments Can Do About Falling Birth Rates* (Woodrow Wilson Center Press, 2013). Dr. Kramer has written articles in *Foreign Affairs*, *Washington Quarterly*, and *Politique Etrangère*. His most recent publication is “France’s Muslim Predicament,” *The National Interest* (April 2020). He received a doctor of philosophy in history from Princeton University.

Dr. Irene Kyriakopoulos is Distinguished Professor of National Security Policy in INSS, where she teaches and co-leads the European Studies Concentration program. She has lectured and written widely on the economic dimensions of security and the political economy of European integration. Her work has been published by the Brookings Institution, NDU Press, *Defense Management Journal*, *Joint Force Quarterly*, *Mediterranean Quarterly*, *Intereconomics*, and *World Economics*. The focus of her recent research has been on Europe’s debt crisis and migration policies with related publications, including “In the Name of the Euro: What Have the EU’s Policies Achieved in Greece?” *Intereconomics* (November/December 2014) and “Europe’s Responses to the Migration Crisis: Implications for European Integration,” *INSS Strategic Insights* (April 2019). Dr. Kyriakopoulos received a bachelor of science in economics from the University of Maryland and a master of arts and doctor of philosophy in economics from The George Washington University.

Dr. Bryce Loidolt is a Research Fellow in INSS, where his research focuses on security cooperation and unconventional warfare. Previously, he was a defense analyst at RAND. His work has appeared as RAND monographs, as well as in academic journals such as *Terrorism in Political Violence*, *Studies in Conflict and Terrorism*, and *Legislative Studies Quarterly*.

Dr. Loidolt received a bachelor of arts in Middle East studies from Middlebury College, a master of arts in political science from the University of North Carolina at Chapel Hill, and a doctor of philosophy in political science from The George Washington University.

Ms. Amanda Moodie is Director of the Program for Emerging Leaders and a Policy Fellow in CSWMD. Her policy support focuses on the international legal regimes that regulate the proliferation of chemical and biological weapons. Her research interests include the history of chemical and biological weapons, compliance and enforcement of chemical and biological weapons law, and international assistance in the event of a chemical or biological weapons attack. Ms. Moodie is co-author of “The Virus of Disinformation: Echoes of Past Bioweapons Accusations in Today’s COVID-19 Conspiracy Theories,” *War on the Rocks* (April 2020). She received a bachelor of arts in political science and English from Duke University and a master of arts in law and diplomacy from the Fletcher School at Tufts University.

Dr. Mariya Omelicheva is a Professor of Strategy at the National War College. Her research and teaching interests include international and Eurasian security, counterterrorism and human rights, democracy promotion in the post-Soviet territory, Russia’s foreign and security policy, and terrorism/crime nexus. She is the co-author, with Lawrence P. Markowitz, of *Webs of Corruption: Trafficking and Terrorism in Central Asia* (Columbia University, 2019) and author of *Democracy in Central Asia: Competing Perspectives and Alternate Strategies* (University Press of Kentucky, 2015) and *Counterterrorism Policies in Central Asia* (Routledge, 2011), which received an Outstanding Academic Title Award by Choice. Dr. Omelicheva received a doctor of philosophy in political science from Purdue University and a doctor of jurisprudence in international law from the Moscow National Law Academy.

Dr. James Przystup is a Senior Research Fellow in CSR. For over 30 years, he has concentrated on U.S. relations with the Indo-Pacific region. He has served on the U.S. House of Representatives Subcommittee on Asian and Pacific Affairs, as Deputy Director of the Presidential Advisory Commission on U.S.-Japan Relations, on the Policy Planning Staff of the State Department, as Director for Regional Security Strategy in the Office of the Under Secretary of Defense for Policy, and as Director of the Asian Studies Center at the Heritage Foundation. Dr. Przystup graduated summa cum laude with a bachelor of arts in history from the University of Detroit. He received a master of arts in international relations and a doctor of philosophy in diplomatic history from the University of Chicago.

Dr. Phillip C. Saunders is Director of the Center for the Study of Chinese Military Affairs, INSS. He previously worked at the Monterey Institute of International Studies, where he served as Director of the East Asia Nonproliferation Program from 1999 to 2003, and worked on Asia policy issues as an officer in the U.S. Air Force from 1989 to 1993. Dr. Saunders is co-author, with David Gompert, of *The Paradox of Power: Sino-American Strategic Restraint in an Era of Vulnerability* (NDU Press, 2011) and co-editor of seven books, including *Chairman Xi Remakes the PLA: Assessing Chinese Military Reforms* (NDU Press, 2019) and *PLA Influence on China’s National Security Policymaking* (Stanford University Press, 2015).

He also has edited NDU Press books on Chinese contingency planning, China-Taiwan relations, the Chinese navy, and the Chinese air force and published numerous articles and book chapters on China and Asian security issues. Dr. Saunders received a bachelor of arts in history from Harvard University and master of public administration and doctor of philosophy in international relations and affairs from Princeton University.

Dr. Shannon Smith is a Professor of Practice and Director of Engagement in the Africa Center for Strategic Studies at NDU. Prior to joining the Africa Center, she served as Deputy Assistant Secretary of State for Africa, where she oversaw U.S. East and Southern Africa policy and led Africa Bureau efforts on Sudan and South Sudan, global health, and the environment. Dr. Smith has also served as the Senior Policy Adviser for the Senate Foreign Relations Committee on Africa, global health, and peacekeeping issues and previously as the National Security Adviser for the Assistant Majority Leader of the Senate. She is a summa cum laude graduate of Texas A&M University and received a doctor of philosophy from Cornell University.

Ms. Caitlyn Yates is the Research Coordinator at IBI Consultants and in CSR. Previously, she worked as a Research Associate in the Mexico Security Initiative, Strauss Center for International Security and Law, at the University of Texas at Austin. Her work focuses on human security, organized crime, transit migration, and borders in Latin America. She has presented her work before the United Nations, to government agencies and departments, and at academic conferences. Ms. Yates received a bachelor of arts in anthropology from Trinity University and a master of global policy studies from the Lyndon B. Johnson School of Public Affairs at the University of Texas at Austin.

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Strategist at New America and author
of *Ghost Fleet*, *LikeWar*, and *Burn-In*

"At a time when Great Power competition amid interdependence can be more confusing than enlightening, Thomas Lynch and his colleagues have produced an insightful guide for the perplexed. By ranging across history, technology, and geopolitics, this volume not only illuminates the multiple dimensions of the evolving U.S.-China rivalry but also more importantly educates policymakers about what winning entails in a non-zero-sum competition."

—Ashley J. Tellis
Tata Chair for Strategic Affairs and Senior Fellow
Carnegie Endowment for International Peace



Updated February 15, 2022

Advanced Battle Management System (ABMS)

The Advanced Battle Management System (ABMS) is the U.S. Air Force's latest effort to create a next-generation command and control (C2) system. ABMS proposes using cloud environments and new communications methods to allow Air Force and Space Force systems to share data seamlessly using artificial intelligence to enable faster decisionmaking. The Air Force describes ABMS as its effort to create an internet of things, which would allow for sensors and C2 systems to be disaggregated from one another (counter to how the Air Force has traditionally performed C2). This program is the Air Force's contribution to the DOD's Joint All Domain Command and Control (JADC2) effort focused on modernizing DOD decisionmaking processes for combat operations.

ABMS was originally envisioned to replace the E-3 Airborne Warning and Control System (AWACS) (**Figure 1**), which currently directs air combat operations, but later took on a broader scope. Former Assistant Secretary of the Air Force for Acquisition Will Roper directed that the program become less focused on command centers and aircraft, and to instead create digital technologies, like secure cloud environments, to share data across multiple weapons systems. Dr. Roper stated the contested environment envisioned by the 2018 National Defense Strategy forced the Air Force to restructure the ABMS program. In May 2021, General David Allvin, the Vice Chief of Staff of the Air Force in a DefenseOne article stated, "What exactly is ABMS? Is it software? Hardware? Infrastructure? Policy? The answer is yes to all." In other words, the Air Force envisions ABMS as an acquisition program that will both procure things *and* implement other nondevelopmental efforts that the service views as equally important: new techniques to command and control airborne forces.

Figure 1. E-3 AWACS



Source: [https://en.wikipedia.org/wiki/Boeing_E-3_Sentry#/media/File:E-3_Sentry_exercise_Green_Flag_2012_\(Cropped\).jpg](https://en.wikipedia.org/wiki/Boeing_E-3_Sentry#/media/File:E-3_Sentry_exercise_Green_Flag_2012_(Cropped).jpg).

Since ABMS's inception, Congress has expressed interest in the development of next-generation C2 systems. The Air Force states that ABMS is a nontraditional acquisition program. As a result, Congress has questioned the Air Force's approach to replacing older systems and its approach to experimenting with emerging technologies.

ABMS Development Efforts

The Air Force has performed five events to date to demonstrate the new C2 capabilities it hopes to eventually field. In December 2019 the Air Force, in its first ABMS "on-ramp"—the term the Air Force uses to denote a demonstration—showed the ability to transmit data from Army radars and Navy destroyers to both F-22 and F-35 fighter aircraft. This event also demonstrated the Space Force's Unified Data Library (UDL), which is a cloud environment combining space-based and ground-based sensors to track satellites.

In September 2020, ABMS performed its second on-ramp. This second on-ramp demonstrated detecting and defeating a simulated cruise missile bound for the United States using hypervelocity weapons as defenses. In addition, ABMS exhibited capabilities to "detect and defeat efforts to disrupt U.S. operations in space." According to an Air Force press release "70 industry teams and 65 government teams" participated in the event.

The Air Force held a third on-ramp event in late September 2020, in support of exercise Valiant Shield at Joint Base Pearl Harbor-Hickam. During this event, the Air Force demonstrated using a KC-46 tanker aircraft to perform tactical C2 by relaying data from older, fourth-generation fighters to newer, fifth-generation aircraft like the F-22. In May 2021, the Air Force stated that procuring a communications pod for the KC-46 will be the first capability release for the ABMS program. The Air Force said, "In a fight, the tankers will need to be flying near the action anyway, supporting fighters, so using them as a command-and-control system, either as the primary or a resilient backup, just makes sense."

A fourth on-ramp was held in Europe in February 2021. According to press releases, the Air Force curtailed this event due to budget constraints. This fourth on-ramp linked allied nations including the Netherlands, Poland, and the United Kingdom into combined air operations. According to General Harrigan, commander of U.S. Air Forces Europe, this fourth event tested U.S. and allied capabilities to perform long-range strike missions with F-15E aircraft launching AGM-158 Joint Air-to-Surface Standoff Missile (JASSM) (see **Figure 2**), while simultaneously utilizing U.S. and allied F-35s for airbase defense missions.

Figure 2. F-15E Launching a JASSM Missile

Source: https://news.lockheedmartin.com/2018-02-06-Lockheed-Martins-JASSM-R-ER-Declared-Operational-on-F-15E-Strike-Eagle#assets_11719452.

A fifth on-ramp was in the Pacific in spring 2021, but canceled this event due to budget constraints.

GAO Report Recommendations

The FY2019 National Defense Authorization Act (NDAA) directed the Government Accountability Office (GAO) to evaluate the ABMS program. In an April 2020 report, GAO recommended to the Air Force Chief Architect four actions to improve program performance:

1. Develop a plan to attain mature technologies when needed for ABMS development areas.
2. Produce a cost estimate updated regularly reflecting actual ABMS costs, updating Congress quarterly.
3. Prepare an affordability analysis that should be updated regularly.
4. Formalize and document acquisition authority and decisionmaking responsibilities of Air Force offices involved in ABMS.

The Assistant Secretary of the Air Force concurred with all recommendations. General David Goldfein, former Chief of Staff of the Air Force, disagreed with the recommendations, noting that GAO's analysis did not reflect classified information. GAO stated that it had access to the classified information, and that this additional information did not affect its analysis and recommendations.

ABMS Management Structure

According to the same GAO report on ABMS, the Air Force originally identified the Air Force Chief Architect, currently Preston Dunlap, to coordinate ABMS-related efforts across each of the Air Force's Program Executive Offices. GAO expressed concern about the potential lack of decisionmaking authority for ABMS as a result of this management structure. In November 2020, however, Dr. Roper selected the Air Force Rapid Capabilities Office to serve as the ABMS Program Executive Office. The Chief Architect Office continues to develop the service-wide architecture (i.e., how software and radios are able to connect with one another) to support ABMS.

Summary of Congressional Actions on ABMS

Congress has expressed interest in the development of ABMS systems. The following list summarizes congressional action in the previous three NDAs:

- FY2019 NDAA (P.L. 115-232)
 - Section 147: Limitation on Availability of Funds for Retirement of E-8 JSTARS Aircraft
- FY2020 NDAA (P.L. 116-92)
 - Section 236: Documentation Relating to the Advanced Battle Management System
- FY2021 NDAA (P.L. 116-283)
 - Section 146: Analysis of Moving Target Indicator Requirements and Advanced Battle Management System Capabilities
 - Section 221: Accountability Measures Relating to the Advanced Battle Management System

The FY2021 Defense Appropriations Act (P.L. 116-260 Division C) reduced ABMS funding from the \$302 million requested to \$158.5 million, citing "unjustified growth and forward financing."

Throughout ABMS's development, Congress has expressed concern about retiring older C2 systems such as JSTARS and AWACS before identifying a suitable replacement. Congress also directed the Air Force to develop traditional acquisition justifications, such as cost estimates and requirements documentation, to ensure that both Congress and the service understand what is to be procured. These actions reflect the recommendations from GAO.

Potential Questions for Congress

- What are the risks of disaggregating command and control using the ABMS approach?
- How should the Air Force balance innovation and experimentation with procuring mature technologies?
- What opportunities does ABMS provide that traditional command and control systems cannot provide?
- Would ABMS benefit from utilizing the new budget authority flexibilities found in the 6.8 Software and Digital Technology Pilot Program budget activity code?

CRS Products

CRS Report R44108, *U.S. Command and Control and Intelligence, Surveillance, and Reconnaissance Aircraft*, by Jeremiah Gertler and Jeffrey Nelson

CRS In Focus IFI12045, *Replacing the E-3 Airborne Warning and Control System (AWACS)*, by John R. Hoehn and Jeremiah Gertler

CRS In Focus IFI1493, *Joint All-Domain Command and Control (JADC2)*, by John R. Hoehn

CRS In Focus IFI1654, *The Army's Project Convergence*, by Andrew Feickert

Other Resources

GAO-20-389, *Defense Acquisitions: Action is Needed to Provide Clarity and Mitigate Risks of the Air Force's Planned Advanced Battle Management System*

This product benefited from Katherine Leahy research during her internship with CRS.

John R. Hoehn, Analyst in Military Capabilities and Programs

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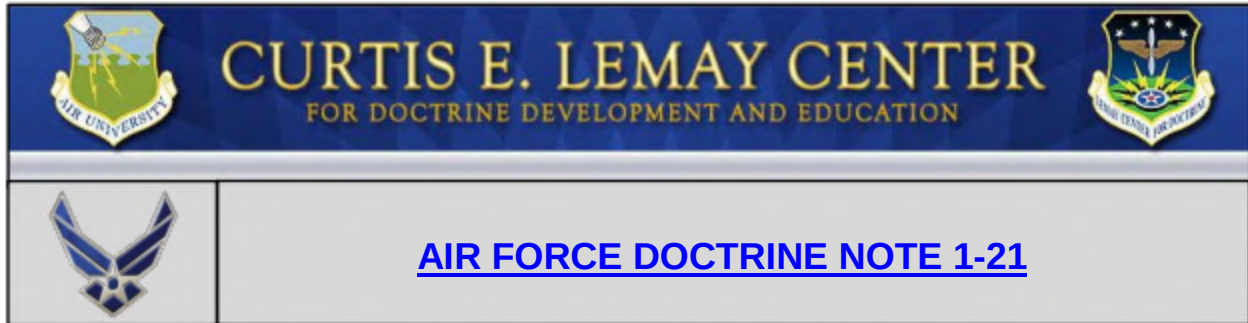
AIR FORCE DOCTRINE NOTE 1-21

AGILE COMBAT EMPLOYMENT



U.S. AIR FORCE

1 December 2021



AGILE COMBAT EMPLOYMENT

1 December 2021

From its founding, the US Air Force has been tasked with projecting combat power across the globe. Historically, it has relied on a combination of continental US and overseas air bases to allow for relatively uncontested movement and operational reach to posture and employ forces and capabilities. However, since the Cold War ended, the Air Force has significantly reduced its global footprint. From 93 air bases during World War II, the Air Force presently maintains 33 permanent overseas air bases¹, a 65% reduction. This reduction challenges the Air Force's ability to project power while simultaneously concentrating friendly high value assets for potential adversary action.

Concurrently with the global footprint reduction, adversarial technological advances in pervasive intelligence, surveillance, and reconnaissance and all-domain long-range fires have placed air bases at significantly increased risk. Just as the Soviets placed Cold War bases in Europe at risk, new weapons systems now place bases at risk that were previously considered sanctuaries. Additionally, fiscal and political constraints limit the establishment of new permanent air bases. To address these challenges, the Air Force introduced **Agile Combat Employment (ACE): a proactive and reactive operational scheme of maneuver executed within threat timelines to increase survivability while generating combat power throughout the integrated deterrence continuum.**

When applied correctly, ACE complicates the enemy's targeting process, creates political and operational dilemmas for the enemy, and creates flexibility for friendly forces. To effectively accomplish joint force commander objectives, ACE requires reexamining a wide variety of enabling systems, to include: command and control (C2); logistics under attack; counter-small unmanned aircraft systems; air and missile defense; and offensive and defensive space and cyber capabilities.

¹ Department of Defense, *Base Structure Report – Fiscal Year 2018 Baseline* (Washington, DC: Deputy Assistant Secretary of Defense [Infrastructure], 2018).

ACE is an operational concept that supports joint all-domain operations ([JADO](#)). Joint force operations are increasingly interconnected, interdependent, and challenged. Anti-access and area denial threats, reduced freedom of maneuver, and rapid proliferation of advanced technologies challenge the Air Force's ability to operate. The successful employment of ACE positions the force to observe, orient, decide, and act in concert across all domains. To achieve freedom of action, ACE enables convergence across domains, presenting an adversary with dilemmas at an operational tempo that complicates or negates adversary responses and enables the joint force to operate inside the adversary's decision-making cycle.

"The best place to kill an enemy's air force is on the ground. Especially if that air force is postured in bases that are few in number and lack passive defenses — such as shelters and decoys — and active defenses such as kinetic and non-kinetic interceptors, electronic warfare, and directed-energy weapons that can help counter these air and missile threats."

-- Mark Gunzinger

Director of Government Programs and War Gaming,
Mitchell Institute for Aerospace Studies

This doctrine note is intended to guide the development of ACE within Air Force operational doctrine. It establishes working definitions and a framework for ACE doctrine development. It includes an overview of evolving doctrine topics and provides the starting point for Airmen to codify best practices for ACE throughout integrated deterrence. This doctrine note focuses on ACE enablers and the ACE operations framework. It lays the foundation for the future development of ACE doctrine, aligns with the joint functions, and focuses on planning, execution, and assessment for operations executed from competition through conflict.

WORKING DEFINITIONS OF KEY TERMS

Please note, these are working definitions, are derived from a variety of sources, and are placed here to facilitate further discussion. Their final wording in doctrine may differ.

Agile: Able to outpace adversary action through movement and maneuver to achieve commander's intent.

Agile Combat Employment: A proactive and reactive operational scheme of maneuver executed within threat timelines to increase resiliency and survivability while generating combat power throughout the integrated deterrence continuum.

Mission Command: The conduct of military operations through decentralized execution based upon mission type orders (MTO).²

² Joint Publication 3-31, [Joint Land Operations](#).

Multi-Capable Airmen: Airmen capable of accomplishing tasks outside of their core Air Force Specialty. Specifically, these personnel are often trained as a cross-functional team to provide combat support and combat service support to ACE force elements. They are enabled by cross-utilization training and can operate independently in an expeditionary environment to accomplish mission objectives within acceptable levels of risk.

Proactive Maneuver: A scheme of maneuver by which forces and assets are moved between main operating bases (MOB), forward operating sites (FOS), cooperative security locations (CSL), and contingency locations (CL) (see appendix) to assure allies and partner nations of US support, alter adversary or enemy understanding of friendly intentions and capabilities, posture to deter aggression, or gain a positional advantage.

Reactive Maneuver: A scheme of maneuver employed in response to observed, perceived, anticipated, or realized enemy aggression using mobility and dispersion of forces and assets to complicate enemy targeting, redistributing forces away from concentrated hubs, increasing survivability, and repositioning forces for follow-on operations.

Threat Timelines: Theater-specific planning factors based on the time required for an adversary to accomplish its find, fix, track, target, engage, and assess cycle.

HOW IS ACE DIFFERENT?

The US faces adversaries capable of wielding a disruptive and dangerous operational reach with mass, precision, and speed in all domains. Adversaries can challenge the US's ability to project power from MOBs, often large and centralized physical structures with unprotected infrastructure. To address this threat, ACE shifts operations from centralized physical infrastructures to a network of smaller, dispersed locations that can complicate adversary planning and provide more options for joint force commanders. Its value derives from the ability to hold adversary targets at risk from multiple locations that are defensible, sustainable, and relocatable. Airmen should expect to conduct operations at a speed, scope, complexity, and scale exceeding recent campaigns from distributed locations.

ACE Operations, Activities, and Investments	
ACE operations present commanders with options to initiate, continue or reestablish operations while balancing associated risk factors, as shown through a few examples below.	
Continuum event	ACE operations, activities, and investments
Cooperation	★ Create steady-state and contingency authorities with partner nations which allow for: overflight, direct coordination with host nation defense, staging of material/equipment, etc. ★ Organize, Train, and Equip forces for ACE constructs
Competition	★ Exercise ACE operations and conduct capability demonstrations with joint forces and partner nations as an active deterrent to conflict ★ Maintain freedom of access and maneuver
Conflict	★ Conduct persistent mission generation and logistics under attack ★ Complicate enemy observe-orient-decide-act loop ★ Preserve combat capability ★ Generate combat power

ACE ENABLERS

Achieving freedom of action and decision advantage can be achieved by creating multiple adversary dilemmas by forcing complex target situations. This **deters** aggression and enables the US to **defend and win** in conflict.³ ACE achieves this through the following enablers:

★ Expeditionary and Multi-Capable Airmen

★ Tailorable Force Packages

EXPEDITIONARY AND MULTI-CAPABLE AIRMEN

The Air Force must refocus on the expeditionary skills necessary to operate outside of a MOB. Many Airmen must have diverse foundational skills that enable them to operate in a contested, degraded, and operationally limited environment with minimal support. Leaders mitigate risk to force by training Airmen to execute distributed operations that increase survivability while generating combat power.

ACE teams consist of unit-assigned multi-capable Airmen. These teams are tailored portions of force packages able to provide mission generation (MG), command and control, and base operating support (BOS) as the mission dictates. Functional communities must identify how to minimize equipment and personnel footprints to increase dispersal capabilities and complicate adversary targeting.

³ COMUSAFE Public Affairs preparation for Military.com interview, 4 May 2020.

TAILORABLE FORCE PACKAGES

To meet theater requirements, ACE requires tailorable force packages with the ability to execute across a range of operating locations. Force structure and tailorable UTCs must be designed to enhance agility while also balancing risk to mission and force. Functional communities will work with commanders to define ACE force packages that will be reflected in existing, new, or updated UTCs.

"To generate combat power from a number of locations to create dilemmas for an adversary...I just need a runway, a ramp, a weapons trailer, a fuel bladder, and a pallet of [Meals, Ready-to-Eat]. That's maybe a little bit bold, but the point is, we've got to be light, lean and agile."

-- General CQ Brown, Jr., Chief of Staff of the Air Force

Remarks to Air Force Association Air, Space, and Cyberspace Conference
as Commander, Pacific Air Forces, September 2019

ACE OPERATIONS FRAMEWORK

To provide a common lexicon with joint partners, ACE consists of **five core elements: posture, C2, movement and maneuver, protection, and sustainment**. The latter four align with the joint functions. Together with the remaining joint functions (information, intelligence, and fires), the five core elements form the whole of ACE's operational framework.⁴

POSTURE

Posture is intrinsically tied to all other elements. **Forces must be able to rapidly execute operations from various locations with integrated capabilities and interoperability across the core functions.** It is the starting position from which subsequent actions take place. When executed properly, posture establishes a deterrent to conflict by being strategically predictable, but operationally unpredictable.⁵ An effectively tailored posture provides commanders with expanded force employment options and mitigates operational risk. It provides an increased defensive posture by increasing the scope and scale of friendly force locations, boosts deterrence to adversary aggression, and assures allies by presenting a credible combat force. Posture redistributes both theater-assigned and follow-on forces to positions of advantage to best support operations plan execution. MOBs should be robust and should have the ability to support further dispersion to smaller CLs while maintaining integrated capabilities and interoperability across MG, C2, and BOS functions.

⁴ [Joint Publication 3-0, Joint Operations.](#)

⁵ [2018 National Defense Strategy.](#)

Operational unpredictability is enabled through agility of forces across pre-postured locations, increasing the number of locations an adversary must target. The increased number of dispersed locations presents adversaries with challenges from the tactical to the strategic level. It does this politically through nation agreements and financially by increasing the numerical offensive capability required to achieve intended effects. Operational locations should be identified based on the ability to support warfighting requirements and sustainment opportunities while balancing risk to force. Risk to force may prohibit massing personnel at locations inside enemy weapon engagement zones (e.g. unconventional ground forces, small unmanned aircraft systems (sUAS), ballistic missiles, cruise missiles, and hypersonic weapons). Risk management is critical to balancing survivability with combat operations tempo by stationing forces at varying proximities to the fight and associated threats. Providing the flexibility to rapidly re-route forces and equipment inbound to the theater is critical to successful ACE.

Operational planners should focus ACE efforts during the early stages on strengthening alliances, increasing partner capacity, and increasing the number of partner nation access agreements and locations within those nations. This requires a “whole of government” approach between the State Department, Department of Defense (DOD), and other US Government agencies. To achieve optimal sourcing decisions and enable ACE objectives, planners should consider establishing acquisition and cross-servicing agreements, host-nation support agreements, and integration of operational contract support equities across the air component command staff functions. As the quality and quantity of operational locations increase, ACE operations exponentially increase both the operational advantage to friendly forces as well as the political and operational dilemma for adversaries.

Distributed operations will exist on a spectrum from well-developed MOBs to CLs. Infrastructure improvements and associated pre-positioning of materiel at distributed operating locations is necessary to ensure respective theater plans are executable. Capability development includes: equipment and supply pre-positioning; scalable logistics packages; access to FOS, including partner military and civil airfields; resilient communications to function in distributed denied, disconnected, intermittent, or limited bandwidth (D-DIL) environments; and a force optimized for major combat operations in a contested environment. Finally, it is critical to understand local and regional commercial market capacity to source critical operational requirements to support distributed forces.

COMMAND AND CONTROL

Commanders in any conflict require the ability to conduct C2 across domains. Centralized command, distributed control, and decentralized execution provide the framework for the C2 of ACE.⁶ Airmen should be able to translate C2 information into action at the speed and scale of relevance, regardless of the operational environment. Airmen should be trained and equipped to employ communications equipment to support distributed operations.

⁶ Air Force Doctrine Publication (AFDP) 1, *The Air Force*.

Joint all-domain command and control (JADC2) and mission command enable Airmen and joint partners to gain operational advantage, maintain operational effectiveness, and achieve convergence of effects across domains. This is accomplished via the communication of commander's intent through issuance of MTOs in conjunction with delegated and conditions-based authorities, allowing operational commanders to generate combat airpower in a D-DIL environment. It is highly expected that elements conducting ACE will lose connectivity with operational C2; therefore it is imperative that units be trained to operate via commander's intent with limited direction from air operations centers or air component staffs. In situations where communications are degraded and forces lack continuous contact with higher echelon commands, Airmen should execute in alignment with the commander's intent to protect and preserve the force. Additionally, they should take advantage of emergent opportunities which allow the commander to maintain the initiative, and resolve situations locally based on a commander's own situational awareness. Codification of conditions-based authorities and delegated authorities will maximize the advantages provided by emergent opportunities.

JADO requires command authorities to be flexible and responsive to battlespace changes with respect to time, geography, communications, and command relationships. Because of distributed control's inherent complexity, specified elements of operational, tactical, and administrative control should be developed early, adapted to the situation, and exercised during day-to-day cooperative and competitive activities. Within this construct, effective ACE operations require significant coordination across service component commanders and industry partners to organize efficiently. These relationships and agreements should be established and rehearsed well ahead of any potential conflict. To contend with D-DIL environments, command authorities should be delegated to the lowest appropriate level. In an ACE scheme of maneuver, distributed control drives additional planning and coordination requirements at echelons below the operational level. Forces executing should have information that enables them to understand the current and expected threat environment, the overall plan, their role within it, status of forces, available support relationships, and the means to be used for coordinating actions at the times and places required. Leveraging advances in automated systems from mission and industry partners (e.g. artificial intelligence, automation, and augmentation & human-machine teaming) will play an important role in managing the increased workload.⁷

JADC2 facilitates the unification of efforts across all domains to exploit the advantages of joint and partner nation capabilities, providing mission commanders an ability to rapidly develop, execute, or transition between kill chains; overwhelming adversary defenses and presenting the enemy with multiple dilemmas. Enhanced all domain awareness, data sharing initiatives, and synchronization of forces translates decision advantage into operational advantage. Mission command supports combat effectiveness during the inevitable fog and friction of war. Redundant and resilient C2 nodes enable effective ACE execution. ACE operations require communications packages that are mobile, survivable, secure, and sustainable across the

⁷ [AFDP 3-99](#).

electromagnetic spectrum to provide personnel access to DOD networks and voice services in a D-DIL environment.

MOVEMENT AND MANEUVER

ACE provides greater agility and ability to outpace an adversary's action through movement and maneuver to achieve and fight from positions of advantage. Agility is capable of disrupting an adversary's decision cycle by creating multiple dilemmas with which they must contend.

Maneuver includes expansion of operational footprints and access throughout the theater to provide flexibility, deter adversaries, and support partners and allies. ACE maneuver includes movement of forces to pre-determined, dispersed locations and flow of dispersed forces back to a MOB. The maneuver of forces in this manner is intended to enhance MG efficiency and simplify sustainment. It can provide the ability to push combat and support elements forward for limited periods of time to accomplish offensive objectives.

Dispersal operations complicate enemy targeting by either redistributing forces away from concentrated hubs into multiple operating locations, or by redistributing forces within an established air base (also known as base dispersals or "on the MOB dispersals"). Once dispersed, friendly forces maintain operational momentum via delegated control and mission command principles. Dispersal operations are augmented with other passive defense measures, such as hardening and camouflage.

ACE maneuver requires sufficient coordination of inter-theater and intra-theater transportation to move the force at the proper time and with sufficient tempo to achieve desired effects. Early planning and posturing can ensure airlift, ground movement, and sealift is employed with sufficient quantity, speed, and flexibility. Properly integrated into the planning cycle, operational contract support planners can provide optimized sourcing recommendations and options for the use of commercial support to reduce air, ground, and sea transportation requirements. Dispersal plans from specific MOBs to dispersed locations should be incorporated into theater operation plans to permit adequate equipment and personnel posturing as well as time phased force deployment data development.

PROTECTION

The [*National Defense Strategy*](#) highlights that air bases are no longer considered a sanctuary from attack, regardless of their location. To stay in the fight, forces must operate in and through contested environments. A combination of active and passive defenses are necessary to counter threats in all domains. Posturing a robust and layered integrated air & missile defense (IAMD) is paramount to protect the force from present and future threats to include: sUAS, cruise missiles, ballistic missiles, and hypersonic weapons.

A strategy that implements layered IAMD capabilities with robust defensive measures complicates and frustrates enemy targeting. Additionally, all installations should be prepared to defend against air, space, cyberspace, surface-to-surface, and ground threats throughout the conflict. Protection strategies enable Airmen to prevent, protect, mitigate, respond to, and recover from attacks while rapidly reconstituting and continuing to generate combat airpower throughout.

MOB-focused force protection plans and strategies prove insufficient to meet the needs of short-term or dispersed operations. Pre-planned integration of joint or host nation security assets for dispersed operations is paramount. Additionally, on-demand force protection intelligence support is critical to ACE operations. Proactively providing planners and leaders with information enables quality basing and risk mitigation decisions. Continuance of intelligence and counterintelligence (CI) activities leading up to and through conflict initiation informs commanders' risk calculus when executing reactive maneuver or other protection actions. Air Force intelligence, CI, force protection, emergency management, and law enforcement entities should leverage existing relationships with joint and host nation entities to coordinate supplementary on-demand force protection and intelligence support for ACE operations. Finally, intelligence and CI entities should work closely with planners and supporting contracting activities to develop and maintain actionable information related to vendors and contractors in the operational environment, especially regarding their potential allegiance to and partnership with an adversary.

SUSTAINMENT

ACE will challenge current logistics systems and transportation nodes. Supply and distribution systems need to transform from a fully connected "pull" system, optimized for efficient operations, to a "push" system that maximizes distributed mission effectiveness.⁸ The Air Force should anticipate limitations to standard means of distribution and transportation, and leverage an adaptive logistic system to support operations in these environments. Leveraging local and regional commercial markets can alleviate distribution system stress and provide critical services and equipment to distributed forces.

ACE Sustainment requires infrastructure innovation and full visibility of war reserve materiel (WRM) and non-WRM equipment. Innovative logistics and force projection capabilities are required to meet operational ACE needs as operations grow in scope and scale due to the increase in operating locations.

Current Air Force basing logistics systems are challenged to project, protect, and sustain the force in a dynamic, contested operational environment.⁹ The processes of setting the theater, deploying the force, and maneuvering the force depend on robust,

⁸ A pull system emphasizes efficiency through a "just in time" logistics system, where supplies are pulled forward on an as needed basis. Whereas, a push system emphasizes effectiveness, at efficiency's expense, by anticipating the need and ensuring supplies are on hand before they are needed.

⁹ USAF Expeditionary Center, *Agile Combat Employment for Force Providers, Version 2.0*, 11 April 2020.

resilient, and responsive logistics and infrastructure support and must withstand an adversary's disruption strategy. Diversification of sustainment by using multiple sources such as support agreements and contracted support reduces stress on traditional logistics systems, contributes to maneuver unpredictability, and utilizes host nation resources.

As dispersed sites grow in number across a wider operational area, sustainment plans and systems should also be capable of scaling sustainment operations to match. ACE sustainment plans should focus primarily on aircraft sortie generation, but should also include the ability to execute implied tasks such as receiving airlift or sealift for resupply, executing BOS functions, and contracting local services, supplies, and equipment.

INFORMATION

Effective conduct of information warfare is a key element of ACE. All ACE actions, including written or spoken words and displayed or related images, have informational aspects that communicate some message or intent. This message or intent can be leveraged to shape perceptions and behaviors in ways that support the achievement of friendly force objectives. Overt messaging about ACE can be used to: communicate the ability to rapidly disperse assets, aircraft, and personnel across a wide range of potential forward operating locations; and leverage host nation organic capabilities, assets, and partner nation cooperative agreements.

In the planning and execution of proactive or reactive ACE schemes of maneuver, the deceptive use of information can induce an adversary to errantly diffuse or concentrate forces, rendering them ineffective. Similarly, it can induce a state of "analysis paralysis" about ACE maneuver that challenges an adversary's ability to make effective, timely decisions.

ACE supports information warfare's aim of shaping the perceptions, behaviors, and attitudes of relevant actors throughout integrated deterrence.¹⁰ The effective integration of information into ACE schemes of maneuver can bolster assurance and deterrence by revealing overall joint force capabilities to deny adversary benefits or punish aggression, conceal or obscure aspects that provide perishable advantage, or suggest elements that mislead adversaries. ACE preparation demonstrates and signals a combat-credible deterrent to adversaries and provides assurance to partners and allies.

INTELLIGENCE

Intelligence and CI should be prepared to support operations in a D-DIL environment characterized by mission command, and rapidly changing basing. Support to expeditionary mission generation units and the Contingency Intelligence Network will further enable the ability to project desired effects via air operations. Force protection-related intelligence and CI activities enable survivability of operations by providing

¹⁰ [AFDP 3-99](#).

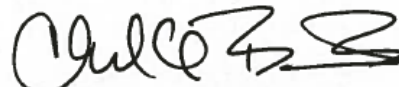
commanders current, time-sensitive, critical information and intelligence necessary to make risk decisions regarding maneuver. This intelligence and CI gathering should precede operational ACE execution to identify all potential kinetic, non-kinetic, and foreign intelligence threats. Intelligence preparation of the operational environment and Intelligence preparation of the battlespace identifies enemy capabilities and threats to proposed ACE operating locations. The intelligence and CI community must also consider threats from commercial vendors and contractors. In locations without a current presence, the US should initiate and develop new relationships with individuals and organizations capable of providing desired information.

FIRES

ACE scheme of maneuver ensures the ability to mass fires to achieve convergence of effects in all domains, to include coordinated ground-based fires in defense of an airfield and its ability to generate aircraft. The execution of fires does not fundamentally change in ACE execution but requires use of MTO and delegation of authorities to the lowest appropriate level. Plans should account for the timelines that may be required to aggregate forces originating from different forward operating sites to create effects against a common target. Under the DOD's vision for JADO, fires may be delivered by air, space, cyberspace, land, maritime, and special operations forces.

CONCLUSION

ACE requires a revolutionary change in how the Air Force thinks about and conducts operations within the modern operational environment. This doctrine note informs relevant and forward-looking ACE concepts and provides a mechanism to quickly evolve doctrine to adapt to an ever-changing security environment. The intent of this doctrine note is to share information and generate discussion across the force. As ACE operations continue to mature through employment in field operations and exercises, feedback and lessons learned will continue to feed the evolution of this emerging doctrine.



CHARLES Q. BROWN, JR.
General, USAF
Chief of Staff of the Air Force

APPENDIX: QUICK REFERENCE LIST OF OPERATING LOCATIONS

Enduring Locations			Contingency Locations		
Main Operating Base (MOB)	Forward Operating Site (FOS)	Cooperative Security Location (CSL)	Semipermanent Contingency Location (SCL)	Temporary Contingency Location (TCL)	Initial Contingency Location (ICL)
A facility outside the United States and its territories with permanently stationed operating forces and robust infrastructure.	A scalable location outside the United States and its territories intended for rotational use by operating forces.	A facility located outside the United States and its territories with little or no permanent United States presence that is maintained by periodic Service, contractor, or host nation support.	A contingency location that provides support for a prolonged contingency operation and characterized by enhanced infrastructure and support services consistent with sustained operations.	A locale that provides near-term support for a contingency operation and characterized by expedient infrastructure and support services that have been expanded beyond Service-organic capabilities.	A locale occupied by a force in immediate response to a contingency operation and characterized by austere infrastructure and limited services with little or no external support except through Service-organic capabilities.



Updated January 21, 2022

Joint All-Domain Command and Control (JADC2)

What Is JADC2?

Joint All-Domain Command and Control (JADC2) is the Department of Defense's (DOD's) concept to connect sensors from all of the military services—Air Force, Army, Marine Corps, Navy, and Space Force—into a single network. Traditionally, each of the military services developed its own tactical network that was incompatible with those of other services (i.e., Army networks were unable to interface with Navy or Air Force networks). DOD officials have argued that future conflicts may require decisions to be made within hours, minutes, or potentially seconds compared with the current multiday process to analyze the operating environment and issue commands. They have also stated that the Department's existing command and control architecture is insufficient to meet the demands of the National Defense Strategy (NDS). Congress may be interested in the concept because it is being used to develop many high-profile procurement programs.

DOD uses ride-sharing service Uber as an analogy to describe its desired end state for JADC2. Uber combines two different apps—one for riders and a second for drivers. Using the respective users' position, the Uber algorithm determines the optimal match based on distance, travel time, and passengers (among other variables). The application then seamlessly provides directions for the driver to follow, delivering the passenger to their destination. Uber relies on cellular and Wi-Fi networks to transmit data to match riders and provide driving instructions.

Figure 1. Visualization of JADC2 Vision



Source: <https://www.monch.com/mpg/news/ew-c4i-channel/7334-saic-and-usaf-partner-for-jadc2.html>.

JADC2 envisions providing a cloud-like environment for the Joint force to share intelligence, surveillance, and reconnaissance data, transmitting across many communications networks, to enable faster decisionmaking (see **Figure 1**). JADC2 intends to enable commanders to make better decisions by collecting data from numerous sensors, processing the data using artificial intelligence

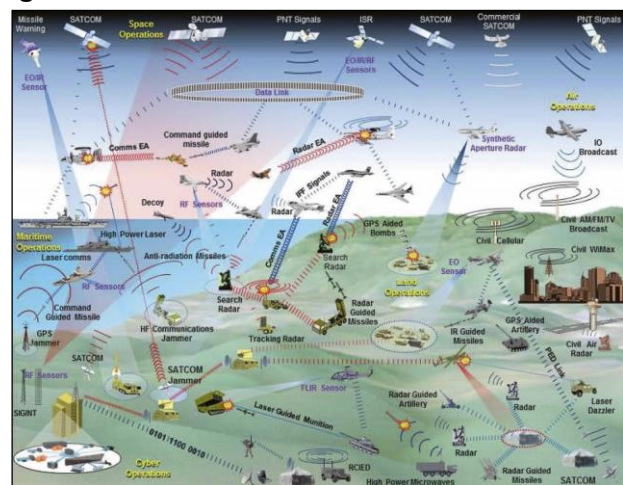
algorithms to identify targets, then recommending the optimal weapon—both kinetic and nonkinetic (e.g., cyber or electronic weapons)—to engage the target.

Some analysts take a more skeptical approach to JADC2. They raise questions about its technical maturity and affordability, and whether it is even possible to field a network that can securely and reliably connect sensors to shooters and support command and control in a lethal, electronic warfare-rich environment. Analysts also ask who would have decisionmaking authority across domains, given that, traditionally, command authorities are delegated in each domain rather than from an overall campaign perspective. Some also question how much a human will be needed for JADC2 to make decisions in real time, and whether it is appropriate to reduce the amount of human involvement in military-related decisions.

Why Change Current C2 Structures?

The future operating environment articulated by the NDS, the NDS Commission, and other sources describes how potential adversaries have developed sophisticated anti-access/area denial (A2/AD) capabilities (see **Figure 2**). These capabilities include electronic warfare, cyber weapons, long-range missiles, and advanced air defenses. U.S. competitors have pursued A2/AD capabilities as a means of countering traditional U.S. military advantages—such as the ability to project power—and improving their ability to win quick, decisive engagements.

Figure 2. A2/AD Environment



Source: <https://www.japcc.org/electronic-warfare-the-forgotten-discipline/>.

Senior DOD leaders have stated that access to data and information will be critical in the future operating environment. In addition, these leaders have stated that to challenge potential peer adversaries, a multidomain approach is required (where U.S. forces would use ground,

air, naval, space, and cyber forces to challenge an adversary's targeting calculus). The Joint All-Domain Operations concept, thus, provides commanders access to information to allow for simultaneous and sequential operations using surprise and the rapid and continuous integration of capabilities across all domains—to try to gain physical and psychological advantages and influence and control over the operational environment.

DOD argues that current C2 programs, like the Air and Space Operation Centers, E-8C Joint Surveillance and Target Attack Radar System, and E-3 Airborne Warning and Control System are not optimized for the speed, complexity, and lethality of future conflict; that the decades-old platforms cannot adequately leverage new technology; and that the supporting structures to enable future C2 either do not exist or require maturation. Air Force officials have argued that a JADC2 architecture would enable commanders to (1) rapidly understand the battlespace, (2) direct forces faster than the enemy, and (3) deliver synchronized combat effects across all domains.

DOD Lines of Effort

DOD. DOD is leading a Joint Cross-Functional Team to explore JADC2 as the concept evolves. The team includes representatives from the offices of the DOD Chief Information Officer (CIO), the Under Secretary of Defense for Research and Engineering, and the Under Secretary of Defense for Acquisition and Sustainment. The DOD CIO has stated it plans to use 5G technologies to enable JADC2.

Joint Staff. The Joint Staff is leading efforts to move JADC2 from a concept to policies, doctrine, and requirements, and has designated the Air Force as the executive agent for JADC2 technology development. According to press reports, JADC2 is a component of the upcoming released Joint Warfighting Concept.

Air Force. To implement JADC2, the Air Force is developing the Advanced Battle Management System (ABMS). ABMS is a network intended to provide data to pass information across all domains. Air Force leaders stated that ABMS has been used to help facilitate DOD support during the COVID-19 pandemic. Throughout FY2020, the Air Force has held at least three ABMS demonstrations, connecting Army and Navy systems. In FY2022, the Air Force requested \$204 million for ABMS.

Army. The Army's modernization strategy identified network modernization to enable multidomain operations. Army Futures Command is the service representative developing the JADC2 concept. As part of an exercise called Project Convergence, it has conducted a series of experiments demonstrating the service's ability to provide access to joint and coalition networks. The Army tested several concepts transmitting targeting information using nontraditional methods in September 2020 in Project Convergence's first demonstration. In FY2022, the Army requested \$106.8 million for Project Convergence.

Navy. On October 1, 2020, the Navy announced it would start Project Overmatch, which it plans to integrate into the overall JADC2 concept. The Navy states that Project Overmatch is intended to develop a new fleet architecture

using artificial intelligence and manned/unmanned teaming to enable Distributed Maritime Operations. The Navy has stated one of its primary focus to support the JADC2 effort is to remove proprietary network standards, thus enabling interoperability with the other services. In FY2022, Project Overmatch requested funding across three classified program elements.

JADC2 Experimentation

DOD has held at least two major JADC2 exercises. The first, held in Florida in December 2019, focused on a simulated cruise missile threat. The exercise represented the first demonstration of ABMS. Air Force and Navy aircraft (including F-22 and F-35 fighter jets), a Navy destroyer, an Army Sentinel radar system, a mobile artillery system, plus commercial space and ground sensors demonstrated being able to collect, analyze, and share data in real-time to provide a fuller picture of the operating environment.

DOD performed a second test of JADC2 in July 2020. During this test, Air Force aircraft connected with naval vessels positioned in the Black Sea, along with special operations forces and eight other NATO nations, in a simulated environment to counter a potential Russian threat.

JADC2 FY2021 Legislation and Funding

The FY2021 National Defense Authorization Act (NDAA) has two provisions focused on JADC2. Section 157 requires the Joint Requirements Oversight Council to produce validated requirements by April 2021. Section 1076 requires the Vice Chairman of the Joint Chiefs of Staff to provide a quarterly briefing. DOD requested \$302.3 million for ABMS in FY2021 but was appropriated \$158.7 million (a \$143.6 million decrease) due to unjustified growth and forward financing (P.L. 116-260). DOD also requested \$207 million for 5G Congested/Contested spectrum research and development, seeking to develop spectrum-sharing technologies and network security architectures.

Potential Questions for Congress

- How can Congress consider JADC2-related activities in advance of validated requirements or cost estimates?
- Without an official program or budget request, how much does DOD budget for JADC2?
- What are JADC2 spending priorities, and are there initiatives DOD might not be investing in?
- How can DOD ensure interoperability among the military services' and allies' communications systems?
- How should DOD prioritize competing communications requirements for its future network?
- What role will artificial intelligence play in future command and control decisionmaking systems?
- What potential force structure changes will be necessary to meet JADC2 requirements?
- How should DOD manage JADC2-related efforts?

John R. Hoehn, Analyst in Military Capabilities and Programs

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AIR FORCE DOCTRINE PUBLICATION 3-99

SPACE DOCTRINE PUBLICATION 3-99

**THE DEPARTMENT OF THE
AIR FORCE ROLE IN
JOINT ALL-DOMAIN
OPERATIONS**



U.S. AIR FORCE



U.S. SPACE FORCE

19 November 2021

The Department of the Air Force, for the first time since becoming a multi-Service organization, presents unified operational doctrine for the entire department. This Department of the Air Force Doctrine Publication 3-99, *The Department of the Air Force Role in Joint All-Domain Operations*, is a guiding doctrinal statement of the way ahead for both Services in the vital and growing area of joint all-domain operations. Each Service contributes to the full range of forces and capabilities made available to the joint force commander. Together, they provide the incomparable ability to exploit air and space in the service of our Nation.

WILLIAM G. HOLT
Major General, USAF
Commander, LeMay Center for Doctrine
Development and Education

SHAWN N. BRATTON
Brigadier General, USAF
Commander, Space Training
and Readiness Command

Air Force Doctrine Publication 3-99
Space Force Doctrine Publication 3-99
Department of the Air Force Role in Joint All-Domain Operations

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“The Air Force organizes, trains, and equips forces to be an air component to a joint force commander (JFC). As part of the joint force’s air component, our forces must be prepared to accomplish JFC objectives. The air component commander’s administrative authorities are derived from Title 10, U.S. Code, and exercised as the commander, Air Force forces (COMAFFOR). The air component commander’s operational authorities are delegated from the JFC and exercised as both the COMAFFOR, over Air Force Forces, and as the functional joint force air component commander (JFACC), over joint air forces made available for tasking. Thus, the air component commander leads Air Force forces as the COMAFFOR and the JFC’s joint air operations as the JFACC. This duality of authorities is expressed in the axiom: Airmen work for Airmen and the senior Airman works for the JFC.”

--Air Force Doctrine Publication (AFDP) 1, *The Air Force*

Since the COMAFFOR and JFACC are nearly always the same individual, this AFDP will use the term “air component commander” when referring to duties or functions that could be carried out by either or both, unless explicit use of the term “COMAFFOR” or “JFACC” is necessary for clarity.

CHAPTER 1: JOINT ALL-DOMAIN OPERATIONS

This doctrine publication establishes a framework for air and space components supporting joint all-domain operations (JADO). The framework combines a vision of JADO with near-term practical approaches grounded in operational experience and battle-tested processes for [operational planning, execution, and assessment](#). This publication guides the Department of the Air Force (DAF) in organizing and employing the full range of forces and capabilities presented to a [joint force commander](#) (JFC). Experiments, wargames, and exercises continue to refine JADO operational principles. This publication's role, as emerging doctrine, is depicted in [Appendix E](#). Its desired outcomes are:

- ★ Describe the need for joint all-domain command and control (JADC2) structures.
- ★ Accelerate and increase capacity to develop and exploit decision-quality information.
- ★ Organize, train, and equip forces to converge effects in multiple domains in operationally-relevant timeframes.
- ★ Improve and increase the options by which the joint force can succeed while cutting off adversary pathways for success.

The DAF's ability to operate is challenged by anti-access and area denial threats and the rapid proliferation of advanced technologies that restrict freedom of maneuver. The DAF does not fight alone and joint force operations are increasingly interconnected, interdependent, and challenged. This operating environment requires the DAF and Department of Defense (DOD) to examine how forces will sense, plan, decide, and act in concert across all domains to gain the freedom of action necessary for success. Success requires the convergence of effects globally, across all domains, to consecutively or simultaneously present an adversary with multiple *dilemmas*. Synergistic employment of capabilities in different domains enhances effectiveness and compensates for vulnerabilities, creating outcomes not readily attainable through single-domain action. Such dilemmas, when presented at an operational tempo that complicates or negates an adversary's response, enable the joint force to operate inside an adversary's decision cycle.

A dilemma is a situation in which one must make a difficult choice between two or more alternatives, often equally undesirable.

JADO PRINCIPLES

- ★ Mission Command through centralized command, distributed control, and decentralized execution through mission-type orders (MTO) when appropriate.
- ★ Delegation of authority to lower echelons and to other component and Service leaders as required.

- ★ Information sharing.
- ★ Integrated multi-domain planning.
- ★ Risk identification and mitigation.
- ★ Synergistic effects.
- ★ Flexibility and Versatility.
- ★ Concentration.

Functions of the military departments are codified in DOD Directive 5100.01, [*Functions of the Department of Defense and Its Major Components*](#). Though the directive promotes unity of command and unity of effort within each domain, it also allows for stovepiped operations and limited integration, planning, and synergy between activities. This creates vulnerabilities and reduces dynamic exploitation of emergent opportunities. Conversely, a joint all-domain approach leverages the joint force's full capability and permits lower-level integration in operationally-relevant timeframes.

Current decision-making processes (e.g., the joint planning process) employ linear planning and force synchronization to execute operations. These operations lead to continuous cycles of heightened activity followed by a period of reduced activity. Current processes can be slow and predictable; peer competition requires processes that create adversary dilemmas by facilitating rapid synchronization of [effects](#). This requires continuous and iterative, near-term tactical planning, longer-term operational planning, and campaign refinement as conditions change. Reframing integration and synchronization in this manner allows for sustained and dynamic combat operations.

THE COMPETITION CONTINUUM

The DAF presents forces to the JFC and synergizes and integrates capabilities into JADO across the competition continuum. Those forces operate principally in the air, space, cyberspace, and the [electromagnetic spectrum](#) (EMS). [Appendix A](#) depicts the relationships between DOD domains and their associated elements.

The competition continuum is presented in Joint Doctrine Note 1-19, [Competition Continuum](#). The doctrine note describes a comprehensive and flexible spectrum of strategic relations between the United States and other actors. The competition continuum, rather than a world either at peace or at war, describes a world of enduring competition conducted through a mixture of cooperation, competition below armed conflict, and armed conflict. The joint force is never solely in cooperation (or in competition below armed conflict or in armed conflict) but instead campaigns through a mixture of cooperation, competition below armed conflict, and armed conflict calculated

to achieve the desired strategic objectives.¹ The competition continuum describes the environment and how the United States government applies instruments of national power across the continuum.

Key points are:

- ★ **Cooperation:** Mutually beneficial relationships with compatible interests.
- ★ **Competition:** Relationships with incompatible interests—none seeking to escalate to armed conflict.
- ★ **Armed conflict:** A situation in which combat is the primary means to satisfy interests.

“The reemergence of long-term strategic competition, rapid dispersion of technologies, and new concepts of warfare and competition that span the entire spectrum of conflict require a Joint Force structured to match this reality.”

[National Defense Strategy of the United States of America, 2018](#)
[unclassified summary]

Air and space forces support JADO across the competition continuum, as shown through examples in the figure below.

JADO Across the Competition Continuum	
Continuum region	Joint all-domain operations, activities, and investments
Cooperation	★ Improve partner nation interoperability. ★ Obtain and maintain all-domain access enabling global reach and rapid projection of military power. ★ Establish cooperative sharing agreements improving mutual support in crisis response.
Competition	★ Incorporate all-domain approaches into flexible deterrent options. ★ Expose and counter malign influence. ★ Maintain freedom of access and maneuver in the global commons.
Armed Conflict	★ Gain information advantage. ★ Project global combat power. ★ Integrate and synchronize action in, from, or through all domains to gain and maintain theater access. ★ Overmatch adversary forces at decisive points. ★ Preserve combat capability to conduct future operations.

¹ [Joint Doctrine Note 1-19, Competition Continuum](#)

DEFINITIONS

Domain: A sphere of activity or influence with common and distinct characteristics in which a force can conduct [joint functions](#).²

Decision Advantage: The product of situational understanding, the ability to assure and exchange information, make and communicate decisions by maintaining advantages in all domains.

Joint All-Domain Operations (JADO): Comprised of air, land, maritime, cyberspace, and space domains, plus the EMS. Actions by the joint force in multiple domains integrated in planning and synchronized in execution, at speed and scale needed to gain advantage and accomplish the mission.

Joint All-Domain Command and Control (JADC2): The art and science of decision-making to rapidly translate decisions into action and leverage capabilities across all domains, with mission partners, to achieve operational and informational advantage in both competition and conflict.

Information Advantage: A condition in the information environment, favorable to achievement of a commander's objectives, achieved through the application of information capabilities and influence, that results in a comparative advantage to support all-domain operations. This includes targeting an adversary's ability to conduct C2 through observing, interpreting, and acting.

Information advantage can be achieved by deliberately using information to:

- ★ Influence relevant actors.
- ★ Inform target audiences.
- ★ Attack, exploit, and defend information, information networks, and systems.
- ★ Support decision-making.

² Joint Publication (JP) 3-0, [Joint Operations](#), describes the operational environment as encompassing the physical domains of air, land, maritime, and space; the information environment, which includes the cyberspace domain; and the EMS. It also describes the joint functions as related capabilities grouped to help commanders integrate, synchronize, and direct operations. The joint functions are C2, information, intelligence, fires, movement and maneuver, protection, and sustainment. Also refer to Appendix A for a graphical depiction.

CHAPTER 2: COMMAND AND CONTROL

DOD strategy defines JADC2 as “the warfighting capability to sense, make sense, and act at all levels and phases of war, across all functional areas, domains, and with partners, to deliver information advantage at the speed of relevance.”³ The DAF’s vision for C2 provides an alternate but complementary definition: JADC2 — the art and science of decision-making to rapidly translate decisions into action and leverage capabilities across all domains, with mission partners, to achieve operational and informational advantage in both competition and conflict. It is the **natural extension of C2 across domains and functional components and is essential for JADO**. The DAF’s vision for JADC2 connects distributed sensors, shooters, and data across all domains, to all forces, to enable mission command for the scaled, coordinated exercise of authority to integrate planning and ensure the convergence of effects across a dynamic battlespace. The convergence of effects in all domains requires:

- ★ Robust, resilient, and distributed C2 enterprise capable of simultaneous C2 of multiple domains.
- ★ Unity of effort through shared understanding of commander's intent.
- ★ A shared understanding of the operational environment.
- ★ Integrating global and geographically-focused forces and capabilities with effective command relationships.
- ★ Secure, adaptable, interoperable (joint and allied partners), and integrated data networks to provide information synthesis, distributed decision-making, and assessment.
- ★ Operating with agility and resilience through MTO and delegation of authority at each echelon.

The airpower tenet, mission command, guides the C2 of joint air operations. It is executed through centralized command, distributed control, and decentralized execution (AFDP 1, [The Air Force](#)). Mission command empowers subordinate decision making through MTO to provide resiliency and flexibility required for JADC2. Mission command provides the greatest freedom of action proportional to acceptable risk. Commanders should determine and delegate levels of control based on the operating environment. Exceptions to the use of MTO are when authorities for operations are held at the highest levels, such as nuclear operations.

To actualize JADC2, the DAF’s solution calls for a C2 construct composed of processes and systems that compress decision making cycles and facilitate convergence of effects

³ DOD JADC2 Strategy & DOD JADC2 Posture Review, (Washington, DC: OSD, 2020)

across domains to enable globally integrated operations. The four key elements of this construct are:

- ★ Sensing Grid
- ★ Advanced Networking
- ★ Decision Making
- ★ Authorities & Effects Delivery

These elements provide the capability to observe the environment, share information to make sense of the environment, enhance decision making, and convergence effects to overwhelm an adversary.

Effective JADC2 requires communications architectures that are distributed, robust, and resilient in nature. Similarly, information and intelligence should be accessible and shared at all echelons. The sensing grid harnesses information and applies automated data processing through artificial intelligence (AI) and machine learning (ML) to provide data to warfighters. This data is transferred across domains and classification boundaries through an integrated, resilient, self-healing advanced network. JADC2 architectures should ensure interoperability with allies, partner nations, and agencies, enable integration into central C2 nodes, and simultaneously provide the ability to operate independently at the tactical edge, disconnected from C2. This all-domain data enables decision-makers to understand relationships between information from different domains with acceleration provided by AI and ML capabilities. Cross-domain perspectives provide insight on impacts to the joint force, and ways to enhance or mitigate those impacts. These capabilities combine to achieve decision advantage, and translate it into operational-advantage, through the integrated, synchronized convergence of lethal and non-lethal effects across all domains. Human-Machine relationships are further discussed in [Appendix C](#)

“We need purple command and control. It takes too long for us to do air command and control, and ground command and control, and navy command and control, and then try to come back together and talk about what we are going to do.”

-- General James M. Holmes, USAF, Retired, Commander, Air Combat Command (2020)

PLANNING

JADC2 requires an appropriate level of distributed control, decentralized execution, delegated authority, and less dependence on central planning and mission direction than recent, low-intensity conflict operations. Decentralized execution is enabled through the designation of conditions-based authorities, in which conditions triggering delegation to a lower echelon are planned for and designated prior to an operation.

Conditions-based authorities enable C2 processes to function in a contested and degraded environment. **To achieve decentralized execution, commanders must clearly convey intent, and subordinates must be empowered to act on that intent absent further guidance.**

Early and clear communication of commander's intent and force prioritization is critical for JADO planning, generating joint force opportunities while creating adversary dilemmas across all domains. Operations in contested environments may necessitate a greater degree of distributed control, but also increase the risk of unintended consequences if forces lack an accurate understanding of overall mission context and evolving operational constraints and restraints.

Conditions-based authorities are authorities delegated to a subordinate under certain pre-defined conditions. This may include (but is not limited to):

- ★ Degradation in communications.
- ★ Significant changes in the operational environment.

JADC2 requires efficient management of resources and sophisticated information gathering, processing, and sharing across domains. JADC2 tools and methods enable information advantage and decision superiority. Where available, analytic modeling and simulation tools should be employed to support and enhance commander decision making and inform strategy choices (e.g., apportionment).

To ensure convergence of effects, the planning process for the air tasking order should expand to support joint all-domain planning and execution. JADC2 will orchestrate this convergence through an integrated tasking order (ITO) employing assigned, attached, and supporting forces, capabilities, and effects. The ITO should incorporate and synchronize capabilities across components and domains to allow for mutual support and convergence of forces or effects. Knowledge of joint force capabilities, a common lexicon, common data standards, and the ability to communicate across echelons enables DAF forces to integrate across domains.

"It is a given in future conflicts that the joint force will be conducting operations in a contested environment. We must be prepared to execute in a degraded C2 environment where clearly delineated and forward-thinking commander's intent will be a requirement. It is imperative senior leaders provide our commanders with conditions-based authorities delegated to the lowest capable and competent level, and empower command by negation to accept the appropriate level of risk, all while working toward moments of clear C2."

**-- General C.Q. Brown, Jr., USAF,
Commander, Pacific Air Forces (2019)**

Early all-domain planning allows forces to sustain initiative despite contested operations and C2 degradation. Integrating planning cycles across domains may induce

inefficiencies. However, those inefficiencies are outweighed by the increased adaptability and force flexibility it affords. Longer planning cycles with faster adaptation and refinement permit subordinate commanders to understand and execute intent in a contested/denied environment. JADO planning considerations include:

- ★ Commander intent and objectives.
- ★ Capabilities, available for tasking, that can achieve effects necessary to meet objectives.
- ★ Limiting and enabling factors for effects.
- ★ Reusability of non-kinetic capabilities for follow-on operations.
- ★ Indirect effects and consequence management plans to include information operations (IO), military deception (MILDEC), and operations security.
- ★ [Rules of engagement](#) and judge advocate review.
- ★ Flexibility to re-role assets quickly to contingency options to deliver unscheduled effects or attack unanticipated targets. This may include missions that forces have not been trained to conduct.
- ★ Lead time required to access capabilities needed to deliver effects.
- ★ Authorities required to deliver necessary effects.
- ★ Effects timing, including start time, duration, and flexibility.
- ★ Cross-component synchronization processes to include C2 with the air operations center non-kinetic operations coordination cell (NKOCC).
- ★ Integration of partners and allies.

EXECUTION

JADC2 synchronizes operations across domains to integrate kinetic and non-kinetic actions to produce lethal and non-lethal effects. By providing a fused view of the battlespace, JADC2 enhances the ability to monitor and adapt operations to meet evolving operational requirements and political directives and adjust the weight of effort at tactical and operational levels as needed. Effective JADC2 requires successful execution or implementation of the following:

- ★ Synchronize application of forces and capabilities.
- ★ Define conditions for delegation of authorities.

- ★ Synthesize legal and policy implications of force employment.
- ★ Commence, accelerate, delay, or terminate operations at an operational tempo necessary to maximize advantages over an adversary.
- ★ Leverage domain advantages and mitigate disadvantages, by actions in and through other domains, to generate joint force opportunities and create adversary dilemmas.
- ★ Continue tactical action through MTO.
- ★ Integrate between combatant commands to ensure efficient and effective use of limited forces (e.g., global integrated intelligence, surveillance, and reconnaissance).
- ★ Develop and prepare follow-up actions as needed to account for a changing operating environment.
- ★ Refine transition criteria to account for the full range of conditions across domains.

ASSESSMENT

JADC2 assessment provides answers to these questions:

JADC2 Assessment Considerations	
Category	Examples of key questions
Are we doing the right things?	★ What effects were late due to authorities' delegation delays? ★ What effects were not able to be accomplished due to a lack of authorities? ★ What opportunities advanced the JFC's objectives through all-domain synchronization?
Are we doing things right?	★ Were the right communication channels in place between domains to enable convergence? ★ Were effects sequenced between domains as planned? ★ Was the desired operations tempo achieved? ★ Were contingency plans developed and executed and able to maintain mission timelines?
Are we measuring the right things?	★ How was the measure of the effectiveness of the integrated portions of the campaign accomplished? ★ What, if any, indicators were used from one domain to make assessments in others? ★ Are the means to collect relevant metrics within operationally relevant timeframes available?

CHAPTER 3: INFORMATION

Within JADO, the DAF uses information as a central element of operational-level planning, execution, and assessment. It does so by providing component commanders and the joint force with the capability to leverage informational power to achieve operational and strategic effects in concert with other elements of airpower. When designing air operations, the DAF uses information to craft plans and courses of action that ensure convergence of effects on target audiences. Information warfare capabilities and considerations of effects in the information environment are integrated throughout the targeting process early and not thought of simply as an adjunct to operations.

OPERATIONS IN THE INFORMATION ENVIRONMENT (OIE)

OIE are an integral element of JADO. They are characterized by the sequencing of actions using information to affect behavior by: informing audiences; influencing relevant actors; and affecting information, information networks, and information systems. OIE are designed, planned, and synchronized to complement and reinforce operational effects from other domains. Sequencing should be accounted for in the initial operational planning phases and executed in concert with other domain operations. Failure to do so will limit JADO from achieving the desired effects on adversary actors and systems.

All actions, to include written or spoken words and displayed or related images, have informational aspects capable of communicating a message or intent. Shaping and leveraging those messages to influence target audiences, is an integral component of operations and planning to support JFC objectives.

Information warfare is the employment of military capabilities in and through the information environment to deliberately affect adversary human and system behavior and preserve friendly freedom of action during cooperation, competition, and conflict. Information warfare has the capability to create multiple dilemmas for the adversary.

Effective joint force application of timely and relevant information is vital to attain enduring strategic advantage across the competition continuum. Deliberate, long-term IO campaigns shape perceptions and behaviors by capitalizing on the cumulative and reinforcing effects of multiple coordinated operations, activities, and investments (OAI). Component commanders, in coordination with the joint force, achieve JFC outcomes through nested activities integrating informational and military power. The JFC's operational approach shapes the information environment to gain, maintain, and protect information advantage in support of decision superiority.

The table "OIE for Information Advantage" shows representative OIE activities, across the competition continuum that can be used to create and leverage information advantage. Information advantage is created by:

- ★ Bolstering domestic and international understanding, trust, and support with timely, accurate, contextualized, and purposeful communication related to both component and joint force activities.
- ★ Distracting an enemy's ability to observe, orient, decide, and act effectively.
- ★ Eroding an adversary's confidence in their capabilities, strategies, and relationships relative to that of the joint force.

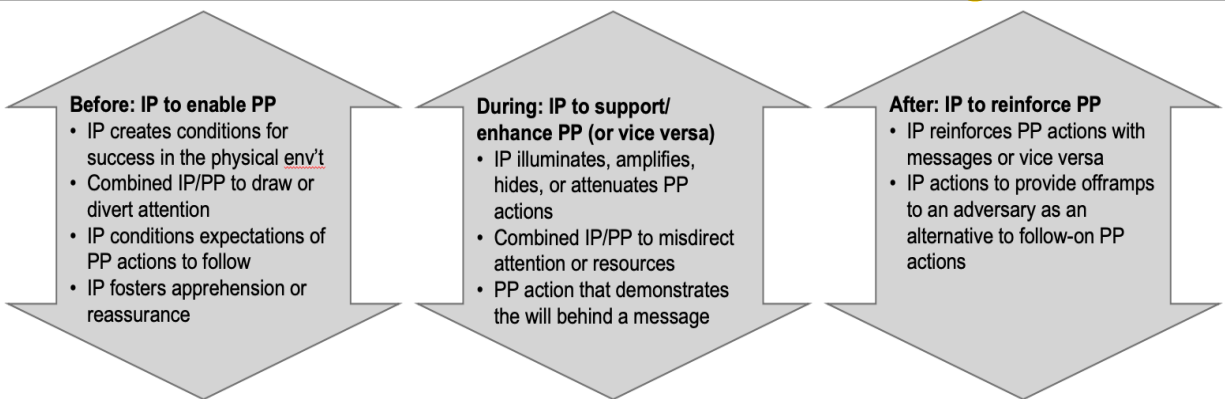
OIE for Information Advantage	
Representative activities	Outcome
★ Develop and demonstrate enhanced information sharing, cooperative agreements and activities. ★ Provide public attribution of joint force activities to secure legitimacy. ★ Build domestic and international audience resilience against disinformation and propaganda. ★ Synchronize messaging with allies and partners. ★ Highlight foreign military sales.	Increased understanding, trust, and support with domestic and international audiences in the purpose of and approach to component and joint force activities
★ Counter disinformation and propaganda. ★ Expose and counter malign influence. ★ Mislead adversary decision-makers on joint force dispositions, capabilities, and vulnerabilities causing diversion of cognitive and physical resources towards unproductive ends. ★ Conduct selective demonstrations of unique joint force capabilities.	Erosion of adversary confidence in their capabilities, strategies, and relationships
★ Create operational surprise by deceiving enemy decision-makers on joint force dispositions, capabilities, intentions, and actions. ★ Deny enemy freedom of action in cyberspace and the EMS while ensuring it for the joint force. ★ Provide transparent response to accidents or inadvertent events.	Diminished enemy ability to observe, orient, decide, and act effectively

Campaigns are executed through a series of OAI, spanning day-to-day operations (cooperation) through crisis response (conflict). Individual OAIs are designed and selected for their ability to advance JFC objectives by shaping the information environment. By integrating informational power (IP) and physical power (PP), OAIs are reinforced and their value maximized through appropriate timing, tempo, scope, and purpose. Such integration maximizes an OAI's ability to create an advantage for the joint force. The figure below, "Integration of Informational Power and Physical Power",

illustrates the potential **linkages between PP and IP** actions from a temporal perspective – before, during, and after employment of physical military power. Before a PP action, IP can be used in an enabling capacity:

- ★ To create physical conditions for success.
- ★ To impose costs by drawing or diverting an actor's attention from the true purpose and nature of joint force actions (e.g., MILDEC).
- ★ To shape relevant actor expectations through overt and covert messaging.

Informational power: The ability to leverage information to shape the perceptions, attitudes, and other elements that drive desired behaviors and the course of events



Physical power: The means to use force to direct or influence the behaviors of others and the course of events

Integration of Informational Power and Physical Power

While they can be effective when applied separately, IP and PP work best together in a variety of ways. When PP is employed, IP can act concurrently, or in advance, to support or enhance the effort. Likewise, following PP employment, IP can reinforce impressions and interpretations of what occurred and condition expectations for what may happen next. Alternatively, PP action may be employed for informational purposes; i.e. to demonstrate the will behind a comprehensive set of OIE.

Such combined actions require both **integration** and **synchronization**--integration in planning is predicated on an accurate understanding of the operational environment and requires an effective combination of informational and physical effects to drive the target audience behavior. Synchronization in execution converges those effects with the right timing, tempo, scope, and intensity.

CHAPTER 4: INTELLIGENCE

Intelligence, surveillance, and reconnaissance (ISR) operations are executed in, from, and through all domains across the competition continuum. All-domain intelligence capabilities enable globally-integrated ISR forces to quickly collect, analyze, process, and disseminate relevant information to operational forces, and refine foundational intelligence to inform mission planning and improve joint intelligence preparation of the operational environment (JIPOE). Intelligence collected during cooperative or competitive activities will be leveraged during armed conflict, providing commanders a comprehensive understanding of adversary attitudes, activities, forces, and other considerations that inform decision-making.

In JADO, intelligence must develop, maintain, and share an awareness of the operational environment that spans geographic, functional, domain, classification, and organizational boundaries. The scope of awareness should include intelligence on ongoing operations, adversary forces, indications and warnings (I&W), target information, and account for military, political, and environmental considerations. JADO intelligence operations require a foundational understanding that encompasses battlespace awareness, but also expands to include an understanding of the complex, interrelated nature of events across areas of responsibility; how they affect campaign plans, and the effect (positive or negative) they may have on the commander's ability to project force. Awareness is challenged by the need to fuse information of varying quality and classification from multiple sources, over multiple networks, and across multiple intelligence organizations.

JADO requires the combination of DAF and other Service ISR platforms and capabilities for a global interoperable intelligence sensing architecture. This architecture produces massive volumes of data to meet JADO intelligence needs. Processes and technologies should be continually designed to incorporate legacy and future capabilities.

The ability to sense the operating environment across domains relies predominately on expensive, high-tech, purpose-built systems. However, JADO requires a broad array of collection platforms, capabilities, and methods. JADO relies on the ability to fuse intelligence data together in ways that are nimble, that work together, and present solutions that work on the tactical edge.

JADO relies on the ability to fuse intelligence data sets together in ways that are nimble, work together, and present solutions that work on the tactical edge.

All-source intelligence is leveraged across the joint and interagency enterprise, but much of the synthesis is done manually. JADO requires intelligence to be automatically synthesized and shared at the lowest classification level possible. The need to conduct cross-domain, cross-source synthesis significantly increases demand for fusion capacity and capability.

Current intelligence processes do not adequately integrate all-domain sensing activities for JADO. Sensing must be a continuous effort to feed multiple decision loops. Cross-cueing and fusing collection activities among domains results in improved JIPOE.

PLANNING

To support JADO planning and execution, intelligence timeliness should be integrated and synchronized with all-domain operations. Increased intelligence requirements necessitate synchronization of traditional and nontraditional intelligence capability. To keep pace with the emerging environment, ISR must also leverage nontraditional sources of intelligence to complement traditional ISR activities. Further, ISR collection platforms produce intelligence at varying speeds. For example, space based imagery can often be accessed quickly, whereas human intelligence collection must be developed over time. Production speeds vary according to a collection platforms' ability to maneuver or position, its persistence, flexibility, and the speed at which data can be processed, exploited, and disseminated. Such variances should be accounted for to ensure successful integration and synchronization.

PLANNING CONSIDERATIONS FOR INTELLIGENCE OPERATIONS:

- ★ Cross-cueing collection activities between domains.
 - ★ Correlating information from multiple sensors, sources, and domains.
 - ★ Integrating open-source reporting into a comprehensive picture of the operational environment.
 - ★ Evaluating potential strategic impacts of tactical actions.
 - ★ Evaluating how impacts in one domain affect actions in other domains.
 - ★ Incorporating global battlespace information from one domain to mitigate knowledge gaps in another.
 - ★ Leveraging joint, inter-Service, interagency, multinational, and commercial partner situational awareness capabilities and data sources.
 - ★ Incorporating all-domain considerations into intelligence gain or loss assessments.
-

CHAPTER 5: FIRES

The JADO vision for fires is to achieve convergence across domains—the synchronization and integration of kinetic and non-kinetic capabilities to create lethal and nonlethal effects, the results of which being greater than the sum of their efforts alone. Air, space, cyberspace, land, maritime and EMS targeting cycles are synchronized and integrated at the JFC level. Integration and synchronization of targeting and planning cycles based on the JFC's objectives is required to execute effective all-domain operations. To maintain tempo, staffs at each echelon need flexibility to observe and orient on new opportunities and quickly decide, target, and execute across all domains.

“There are no boundaries on this battlefield...there are no hiding places...there are no sanctuaries on this battlefield... So how do you win? I think you win by operating at a tempo that they can't keep up with, and by putting them on the horns on multiple dilemmas... We need to create enough options for our warfighting commanders that the enemy doesn't know where we are going to come from next.”

**-- General James M. Holmes, USAF, Retired,
Commander, Air Combat Command (2020)**

CONVERGENCE

In JADO, effects are massed through the synchronized application of kinetic and non-kinetic capabilities. Massing effects requires alignment of disparate planning timelines and resource availability to ensure forces and capabilities are brought to bear at the proper time and place to create desired effects. Each participating force element must understand: the overall scheme of maneuver; its role within it; interdependent support relationships; and the coordinating method to ensure desired effects convergence.

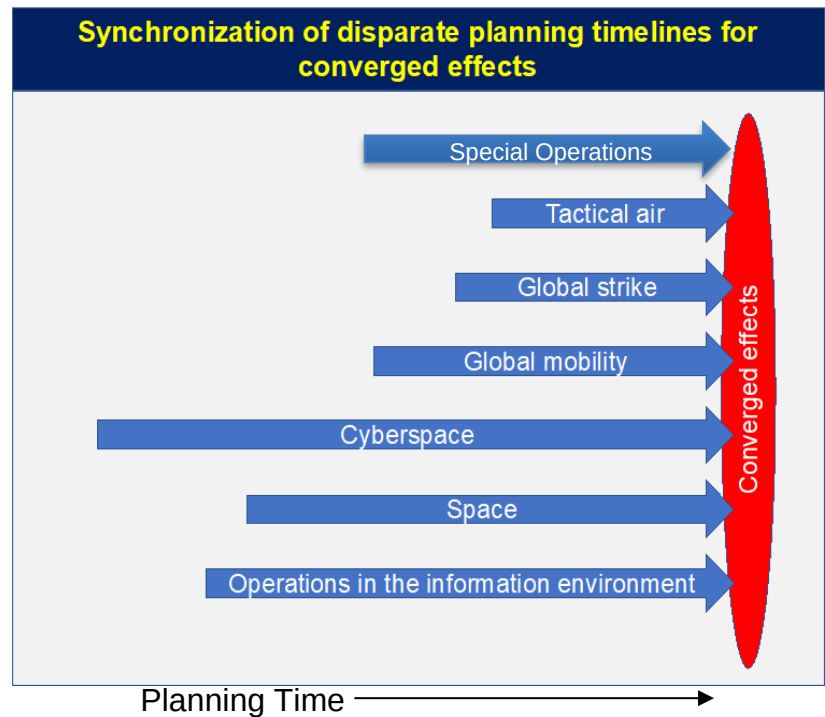
Traditional methods for achieving mass necessitate generation of large force quantities in close proximity to create overwhelming effects against a target. Alternatively, the prevalence of precision-guided munitions, augmented by non-kinetic capabilities, enables fires to be massed with smaller numbers and/or dispersed forces. Reducing and dispersing the footprint of forces deployed forward presents adversary challenges and increases friendly freedom of maneuver to achieve desired effects. Additionally, nontraditional fires can be used to mass effects. An example would be, mobility platforms employing precision munitions from standoff ranges. Non-kinetic fires through space, cyberspace, the EMS, or other means provide additional mechanisms for creating effects.

TARGETING AND FIRES INTEGRATION

Targeting supports the process of linking desired effects to actions and tasks at the component level ([JP 3-0](#)). For successful cross-domain effects, synchronization is

necessary. Typically, synchronization occurs at the JFC level during a joint targeting coordination board, as planning capabilities are presented by each component. In JADO, synchronization planning must occur at echelons below the JFC to provide resiliency and speed in execution.

The current targeting process includes apportionment, but with JADO, the entire joint force requires an apportionment-like process to ensure convergence. Using the JFC joint force apportionment decision, and through collaboration, component commanders allocate forces, synchronize effects, and ensure the weight of effort meets JFC intent. An ITO is the mechanism to frame fires synchronization in the targeting process.



CROSS DOMAIN KILLCHAIN EXAMPLE

During a recent operation, an enemy combatant was tracked by following his digital footprint. Using multi-domain intelligence sources, the individual was geolocated in an area that precluded a kinetic strike. As a result, ground forces were employed to disable critical communications infrastructure, forcing the combatant to move. Intelligence from multiple domains confirmed his new location, a window of opportunity was found, and the enemy agent was eliminated without collateral damage.

Though this vignette offers a specific example, similar scenarios have played out many times over the last two decades. The example displayed here relates the effects of integration on a single operation. Through JADO, a JFC is enabled to scale this type of integration to support major operations and campaigns.

CHAPTER 6: MOVEMENT AND MANEUVER

Movement and maneuver enables deliberate and opportunistic convergence, and preserves freedom of action, by positioning forces and capabilities to create multiple unexpected vectors of attack against an adversary. JADO movement and maneuver synchronizes and aligns actions from multiple domains in a complimentary and reinforcing manner to create exploitable advantages (physical, temporal, or psychological) over an adversary.

All-domain maneuver increases available options for the joint force to succeed while cutting off adversary paths to success. However, for success, commanders should anticipate the difficulties created by degraded or denied C2 environments and should seek to achieve convergence by enabling units to operate on commander's intent through MTO. Opportunistic (or reactive) convergence emerges from changes in the operational environment that can be exploited to advance joint force objectives. Building schemes of maneuver that enable opportunistic convergence should be considered across domains at all echelons.

Movement and maneuver planning for JADO requires integrated planning teams and shared information across the joint force to coordinate actions within and across domains. The movement and maneuver of forces depends on the JFC's scheme of maneuver and intent. The JFC's movement and maneuver concepts should be outlined in the operations plan and further refined in an ITO. Execution of the JFC's movement and maneuver plans requires a JADC2 structure capable of converging effects to create multiple adversary dilemmas and support friendly freedom of action. Each force element should have a clear understanding of its role, and be aware of the sequencing mechanisms used to converge actions within and across domains to create and preserve positions of advantage.

Maneuver exists in all domains. In JADO, maneuver in one domain should complement, and be complemented by, maneuver within other domains. Maneuver in the air, land, and maritime domains is well established. However, an understanding of maneuver in the EMS, space, and cyberspace is equally important for JADO.

- ★ Maneuver in the EMS supports JADO by providing resiliency against adversary degradation and denial attempts. The ability to use frequency agility to evade or overcome interference (e.g., an EMS jammer) increases adversary dilemmas and allows for resilient, friendly force command, control, and communication.
- ★ Maneuver in space supports JADO through deployment, repositioning, reorienting, or reprioritizing the tasking of space forces. These actions support asset optimization, protection from environmental hazards, passive defense, and positioning of active defensive or offensive measures. Space maneuver creates multiple dilemmas for an adversary by supporting follow-on space actions, as well as follow-on actions in other domains.

- ✦ Maneuver in cyberspace supports JADO without establishing a physical presence. It includes accessing adversary networks to support follow-on offensive and defensive actions in cyberspace, enabling convergence of effects in the EMS and other domains, and protecting friendly networks. These cyberspace actions create multiple dilemmas for an adversary.

CROSS DOMAIN CONVERGENCE

During an operation in Africa, an Air Force aircraft conducted overhead ISR, providing real-time intelligence to the ground force commander. Live video of the operation and voice communication transmitted via satellite to allied headquarters, allowed commanders to enact conditions-based, delegated authorities as the situation on the ground changed.

While over the target area, the aircraft's sensors detected a threat to the assault force. After confirming indications with intelligence, the aircrew advised the ground team that their interpreter was acting as a double-agent and had compromised the operation.

This example demonstrates JADO's agility to converge capabilities and assets and provide intelligence, fires, and protection across land, air, space, and the EMS. JADO enables scaling of this type of operation to support major operations and campaigns.

CHAPTER 7: PROTECTION

Threats to friendly forces and operations can emerge from any domain. The varied nature of threats dictate proactive and responsive protection operations synchronized across domains to facilitate a holistic defensive approach. Protection in JADO is focused on protecting each domain and mitigating vulnerabilities using forces and capabilities from multiple domains. Just as convergence synchronizes all-domain action against adversary forces and capabilities, protection planning and execution mitigates joint force threats originating from any domain.

Integrating effects planning and synchronization, sharing information, and all-domain risk identification and mitigation are critical enablers of JADO protection. Further, by employing MTOs with conditions based authorities that empower subordinate decision making, JADO enables the necessary agility, flexibility, and force responsiveness to minimize vulnerabilities across all domains in anticipation of, and in response to, adversary action. I&W, a process that relies heavily on information and intelligence, must evolve to provide the clarity needed to determine the intent behind adversary attacks. Just as all-domain warfare uses complementary attacks in multiple domains, JADO protection requires convergence of protection capabilities to protect forces and respond in all domains.

AGILE COMBAT EMPLOYMENT (ACE)

ACE is the Air Force's concept to provide a proactive and reactive operational scheme of maneuver executed within operationally relevant threat timelines to increase survivability while generating combat power.

ACE operations generate rapid and resilient combat airpower throughout an operational area in response to potential adversary anti-access and area denial efforts. ACE enhances survivability and the ability to seize the initiative, deliver lethal force with operational unpredictability, and succeed across the competition continuum. ACE employs multi-capable Airmen practicing mission command through delegation of authorities, a distributed joint C2 structure reliant on MTO, a network of resilient airbases and austere operating locations, and an adaptive logistics system.

ACE is an operational concept that leverages networks of well-established and austere air bases, multi-capable Airmen, pre-positioned equipment, and airlift to rapidly deploy, disperse and maneuver *combat capability*.

"When they deploy, they deploy with their own sleeping bag, their own shelter half, and a stove. And they're not waiting for anybody to build the tent city for them... they are coming ready to fight."

-- General James M. Holmes, USAF, Retired, Commander, Air Combat Command (2020)

ACE's use of dispersal and maneuver operations increases survivability of friendly capabilities, decreasing time and distance problems faced in large theaters, allowing forces to attack and defend quickly. To fully realize JADO protection, the use of nontraditional facilities and capabilities is necessary; see [Appendix B](#) for ACE considerations.

CHAPTER 8: SUSTAINMENT

In JADO, sustainment must ensure the continued ability to project power and maintain initiative in contested environments. Historically, units have assumed a certain level of continuous infrastructure and support. Such assumptions about immediacy of logistics support and available operating location infrastructure are likely to be unreliable when facing a peer adversary. Sustainment actions in foreseeable conflict with a peer nation are more complex and put legacy logistics and sustainment supply hubs and distribution routes at risk. These complexities are exacerbated by degradations in C2 that complicate requests for support, the status of forces, and combat assessment.

Forces conducting JADO may operate with less intensive sustainment processes and fewer resources. JADO forward positions require simple, modular, and maintainable systems, necessitating a shift away from: static infrastructure; centrally controlled (hub-and-spoke) logistics; highly specialized maintenance equipment and materials; and large contractor and support footprints. By increasing modularity, JADO sustainment

“Sustainment is more than a warfighting function; it is the advantage necessary to win, and it must be integrated at all echelons, rather than merely deconflicted.”

-- MG Rodney Fogg, USA,
BG Michelle Letcher, USA,
& COL Kenneth Letcher, USA,

[Sustainment: The Advantage that “Wins” in
Contested Environments](#)

reduces reaction time and sustains worldwide warfighting capability. Robust, multi-modal distribution systems facilitate greater sustainment options for joint forces. Lower echelon units should see and share sustainment and logistical information to enable integrated planning and enhance risk identification and mitigation. Joint and Service component logistics enterprises and supporting industrial bases must be more responsive to increased needs and be able to operate with limited or degraded communications.

MANEUVER LOGISTICS

JADO highlights the importance of logistical movement through contested environments and critically links maneuver, protection, and sustainment functions, often to a degree in which they are indistinguishable. The adaptive nature of JADO maneuver logistics enables sustainment from range with minimal dependence on large, fixed infrastructure, and can support sustainment in contested areas through disaggregated supply infrastructure. To generate adaptive capability independent of centralized logistics, effectiveness should be prioritized over efficiency. JADO's distributed operations require redundancy in supply distribution and deliberately-planned logistics chain slack. Enhancing partnerships with host nation forces and establishing contingency contracts with local suppliers allows distributed forces to sustain personnel and operations.

Logistics under attack is expected in a contested environment, complicating delivery of just-in-time logistics. Redundant logistics systems are necessary to ensure the delivery of critical assets. Availability of commercial logistics infrastructure (e.g., contract airlift and sealift) cannot be assumed. Nontraditional logistics will be required to counter adversary anti-access and area denial capabilities.

Limited duration self-sustainment is necessary to enable certain functions during periods of logistics denial or degradation. Limited duration self-sustainment includes periods of increased risk, decreased connectivity, and limited capacity.

Dispersed sustainment is an ACE-supporting logistical concept. Prepositioned caches of materiel in dispersed locations increase adversary targeting complexity and provide additional friendly survivability and resiliency. This tradeoff favors survivability over ease of access, and increases the logistical burden of the operating location. Three requirements to sustain and project the force during multi-domain operations are:

- ★ Resilient and integrated sustainment mission C2
- ★ Assured joint power projection.
- ★ Ability to sustain in a distributed environment.

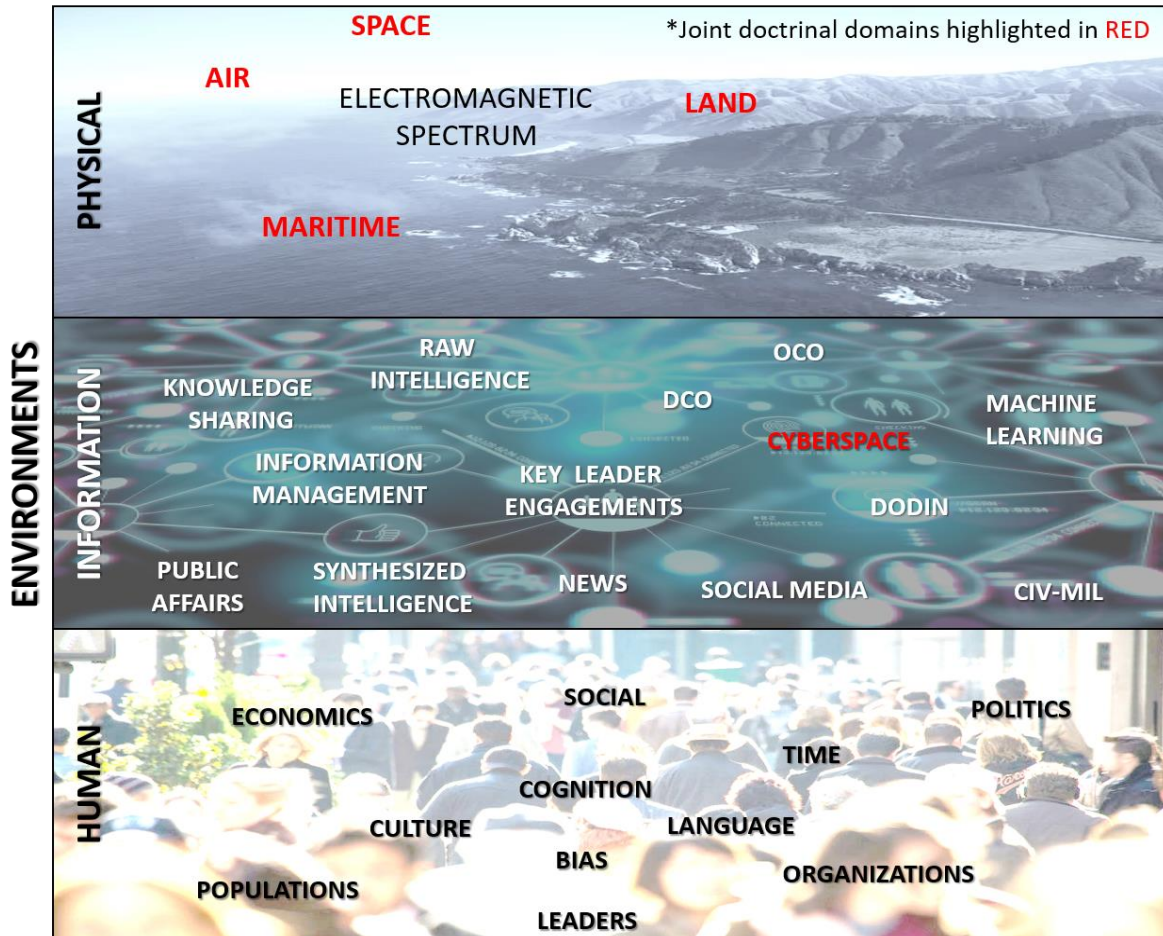
“Joint logistics must integrate our combined capabilities ...to shoot, move, communicate, and win... with the expectation that the force is contested throughout the process, and is equally challenged with time, speed, and distance.”

-- MG Rodney Fogg, USA,
BG Michelle Letcher, USA,
& COL Kenneth Letcher, USA,

Sustainment: The Advantage that “Wins” in Contested Environments

APPENDIX A: DOMAINS, ENVIRONMENTS AND ELEMENTS

The diagram below depicts three different environments in which air forces operate (physical, information, and human). Each of these environments are addressed by various functions in warfare. Each environment contains elements that are simultaneously diverse in character and highly interconnected.



APPENDIX B: AGILE COMBAT EMPLOYMENT CONSIDERATIONS

- ★ Component commanders should operate on JFC intent and communicate with subordinate commands through MTO, when appropriate.
- ★ The coordination of ISR and air refueling assets in a degraded communications environment is difficult, and might require alternative organizational structures akin to the composite wing structure mentioned in [*The Composite Wing: Back to the Future!*](#).
- ★ DAF leaders should have a clear understanding of commander's intent, including the operation as described in the air operations directive, the joint air operations plan, and follow-on MTO.
- ★ Operations will require greater risk acceptance at each level of command. To maintain momentum, conditions may necessitate higher risk activities like integrated combat turns, specialized fueling operations, or wet wing defueling. Other examples include operations inside an adversary's integrated air defense system, limited defenses at landing sites, and short notice dispersal operations.
- ★ To complicate adversary targeting, resilient basing plans consist of main operating bases with significant passive and active defenses, a network of forward operating sites used for dispersal and short term operations, and a series of additional landing sites used for refueling and reloading.
- ★ Operations will require increased theater access within and across partner nations, complicating the area air defense, combat support, and airspace control plans.
- ★ Multi-capable Airmen and Guardians, with multiple qualifications or skill sets, trained to operate as cross-functional teams, enable continued operations while maintaining a smaller footprint at forward operating sites.
- ★ Reliance on sustainment and [*reachback*](#) from the continental US will be challenging and may not be responsive enough to meet operational needs. Organic and theater sustainment options should include pre-positioned materiel caches.
- ★ Because of advances in adversary anti-access and area denial capabilities, the ability to avoid, defend against, withstand, and/or recover from airfield attacks are key components of ACE.
- ★ Traditional force protection plans and strategies focused on main operating bases are insufficient to meet the needs of short-term, austere, or dispersed operations. JIPOE activities precede operational ACE execution to identify ground-based, foreign intelligence, and criminal threats at potential forward operating sites and refueling points, providing planners and leaders with information to make basing and risk mitigation decisions. JIPOE also provides insight into enemy kinetic and non-kinetic capabilities and threats to proposed ACE operating locations.

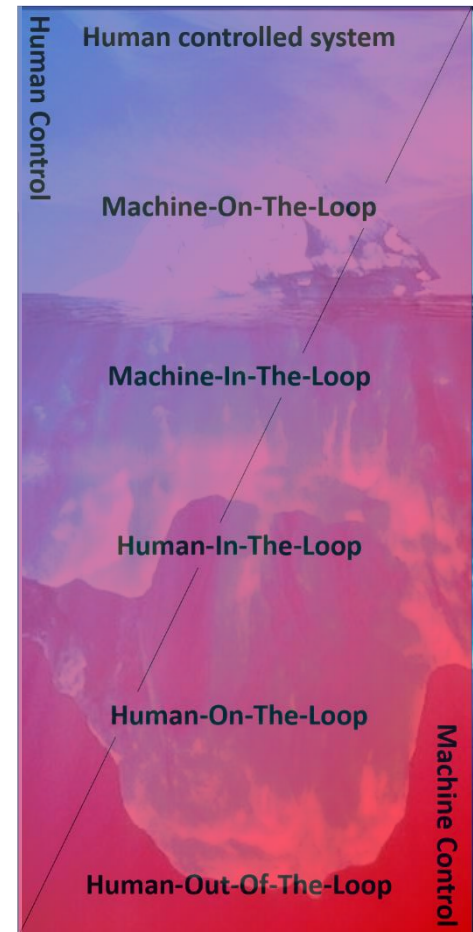
- ★ Plans should be established for providing just-in-time force protection and intelligence support. To do so, DAF intelligence, counterintelligence, and law enforcement entities should leverage existing access to, and relationships within, planned and potential ACE basing locations. In locations with no current presence, DAF personnel initiate and develop new relationships with individuals and organizations capable of providing necessary information and support.
-

APPENDIX C: DEGREES OF HUMAN AND MACHINE CONTROL

HUMAN MACHINE TEAMING

To achieve rapid adaptation necessary for all-domain operations, improved human-machine teaming is required. Machine-to-machine communication and predictive modeling will be critical to moving away from imprecise procedural controls outlined in current airspace control doctrine; such advanced human-machine teaming capabilities are in development. Employing these systems effectively requires a framework for understanding their employment. Commanders should understand and balance the benefits and risks of human-machine relationships. Human-machine teaming may aid all forms of military decision making. However, commanders and operators should exercise appropriate levels of human judgment, especially for decisions regarding the use of force.⁴ To build appropriate levels of understanding, trust, and skepticism with their machines, Airmen need to train as part of human-machine teams. Examples of human-machine teaming terms are:

- ★ **Human Controlled System:** A drill (machine requires direct control or performs only as directed).
- ★ **Machine-On-the-Loop:** Aircraft fly-by-wire systems (machine performs some processes to simplify operation).
- ★ **Machine-In-the-Loop:** Automotive driver assist technologies (machine aids, assists, or provides inputs to human-controlled processes)
- ★ **Human-In-the-Loop:** Power plant control system (machine seeks input for critical decisions).
- ★ **Human-On-the-Loop:** Self-driving cars (machine can run autonomously, but can be overridden).
- ★ **Human-Out-of-the-Loop:** Swarm drones (machine runs without human intervention).



Degrees of Human and Machine Control

⁴ [DoD Directive 3000.09, *Autonomy in Weapons Systems*](#)

APPENDIX D: JOINT ALL-DOMAIN DOCTRINAL GAPS

DAF JADO doctrine establishes a framework for air components supporting JADO. It guides the organization and employment of forces and capabilities presented to the JFC. This section examines two doctrinal gaps that experiments, wargames, and exercises will explore to refine JADO principles.

SUPPORT RELATIONSHIPS

Regardless of Service, domain expertise, or affiliation, joint commanders should leverage information, forces, and capabilities from all domains. Planners at all levels should consider all domains from the beginning of the planning process. Commanders must be empowered to coordinate dynamic all-domain re-tasking throughout execution.

JADO requires the reexamination of supported and supporting relationships.

Current doctrine envisions support relationships as a relatively static form of procedural control along lines of operation. A supporting commander has the authority to decide how and with what forces to meet multiple supported commanders' operational requirements, but does not have the authority to reprioritize the supporting effort, either within or between the supported lines of operations. Each supported commander sets the priorities within their lines of operation and effort, but cannot change priorities between these lines. Priorities for support between two or more supported commanders are set by the establishing authority, the common superior commander over both subordinate commanders ([JP-1, Doctrine for the Armed Forces of the United States](#)).

This binary supported and supporting construct works well when operations are conducted in one primary domain supported by another. However, in complex scenarios involving multiple interdependencies along multiple lines of effort in multiple domains, **traditional supported and supporting relationships are inadequate to affect rapid reprioritization among efforts.** Assets, like aerial tankers or satellite constellations, may be requested simultaneously with each commander citing their supported relationship. Currently this situation would require the asset allocation decision of the establishing authority, a time consuming process.

JADO requires a more agile support relationship and a greater ability to rapidly task and re-task forces to meet specific, time-critical scenarios with less procedural delay. To meet operational requirements of multiple supported commanders while achieving the establishing authority's intent, the supporting commander needs authority to shift the priority of supported forces as the operational situation changes. Since recourse to the establishing authority may not be feasible in time-sensitive operations, supported commanders also need the authority to coordinate laterally to reprioritize efforts as the operational situation changes. This requires a more agile, positive control-based system that rapidly reprioritizes and reflows support to multiple commanders. Commanders at all levels must have the information and authority to adapt to rapidly changing contexts.

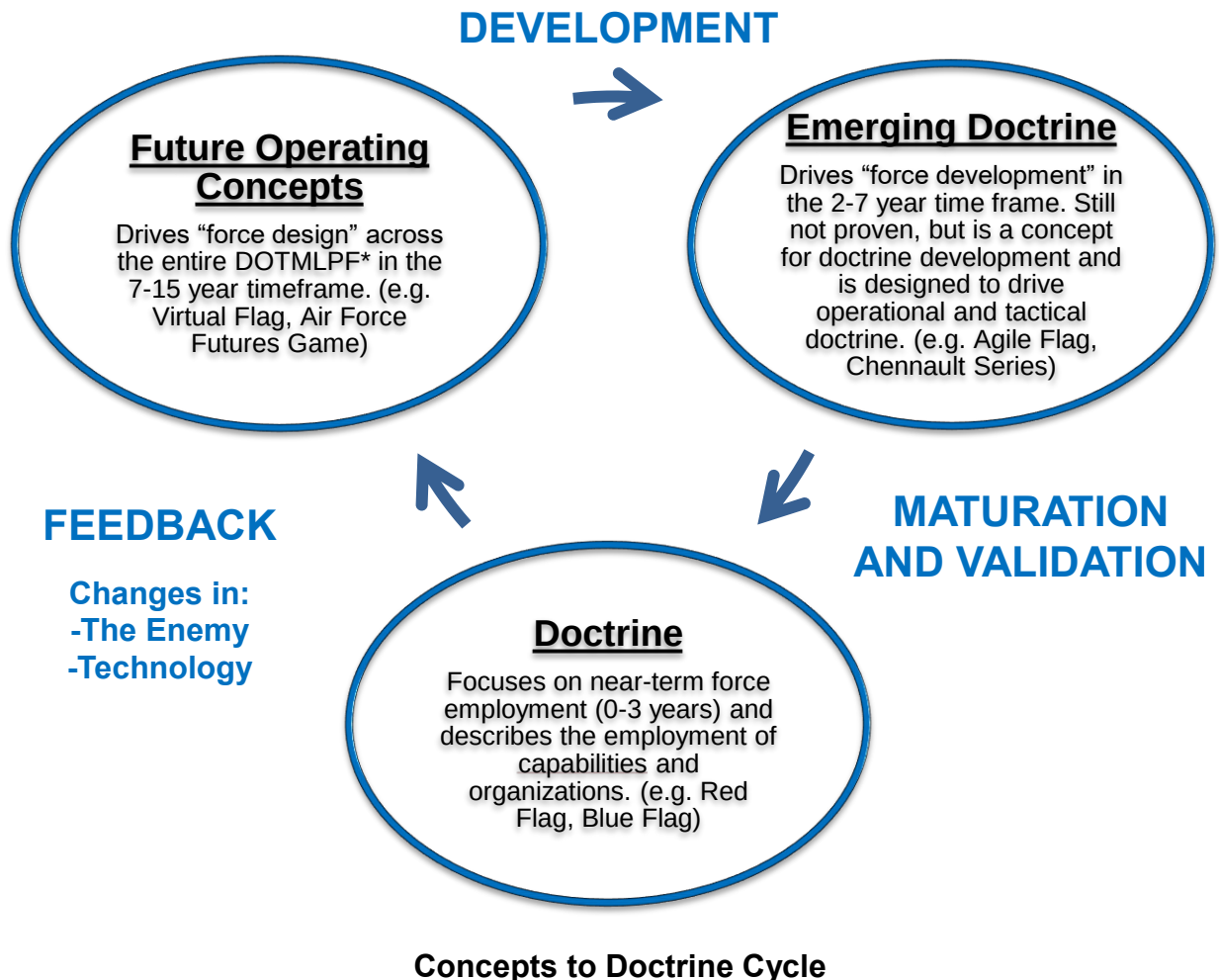
JOINT COMMAND AND CONTROL AT ALL ECHELONS

Transition to joint C2 at all echelons represents a significant change in DOD organizational structure. JADO's speed and adaptability requires a degree of connectivity and mission assurance at the global or theater-level. However, threats to communications also require units to have a local capability to sense, collaborate, plan, and execute all-domain joint responses within their available local networks; to include coordination with local multinational partners.

So long as commanders require support from other geographically-dispersed commands, forces, and capabilities, there will always be a need for resilient theater-level communications to support global integration and dynamic force employment. However, relying exclusively on theater-level integration of domain-oriented assets, without local ability for all-domain integration and backup C2, presents a communications chokepoint and a targeting opportunity for an adversary. Local C2 capabilities should enable geographically co-located forces to build all-domain options in accordance with higher headquarters guidance, including the capability to sustain operations during times of intermittent communications.

APPENDIX E: CONCEPTS TO DOCTRINE CYCLE

Traditional doctrine codifies extant best practices. This AFDP focuses on both near-term emerging doctrine and doctrine being tested and matured in real time. Modeling and simulation, experimentation, wargaming, and tabletop exercises are essential to the concepts shown in the figure below. Users, groups, members, and/or aspects of all domains and environments will use these tools throughout the cycle.



DEPARTMENT OF DEFENSE

March 2022



SUMMARY OF THE JOINT ALL-DOMAIN COMMAND & CONTROL (JADC2) STRATEGY

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FOREWORD

In the current global security environment, the United States military faces agile adversaries who increasingly seek to undermine our strategic and operational strengths by impeding, and, where possible, denying our command and control (C2) capabilities. The ability of the U.S. military to regain and maintain information and decision advantage is one of the Department's top priorities.

This Joint All-Domain Command and Control (JADC2) strategy describes the urgent need for a focused Departmental push on actions to empower our Joint Force Commanders with the capabilities needed to command the Joint Force across all warfighting domains and throughout the electromagnetic spectrum to deter, and, if necessary, defeat any adversary at any time and in any place around the globe.

The JADC2 Strategy provides a vision and an approach for identifying, organizing and delivering improved Joint Force C2 capabilities, and accounts for adversaries who have closed many of the capability and methodology advantages we depend upon for operational success. As an approach, JADC2 supports the development of materiel and non-materiel solution options using innovative technologies coupled with a willingness to modify existing policies, authorities, organizational constructs, and operational procedures to deliver information and decision advantage to Joint Force Commanders.

EXECUTIVE SUMMARY

Rapid changes in the global security environment are presenting significant new challenges to the U.S. military and the ability of the Joint Force to seize, maintain, and protect our information and decision advantage over our adversaries. In addition, we must anticipate that future military operations will be conducted in degraded and contested electromagnetic spectrum environments. These challenges require a coherent and focused Departmental effort to modernize how we develop, implement, and manage our C2 capabilities to prevail in all operational domains, across echelons, and with our mission partners.

JADC2 provides a coherent approach for shaping future Joint Force C2 capabilities and is intended to produce the warfighting capability to sense, make sense, and act at all levels and phases of war, across all domains, and with partners, to deliver information advantage at the speed of relevance. As an approach, JADC2 transcends any single capability, platform, or system; it provides an opportunity to accelerate the implementation of needed technological advancement and doctrinal change in the way the Joint Force conducts C2. JADC2 will enable the Joint Force to use increasing volumes of data, employ automation and AI, rely upon a secure and resilient infrastructure, and act inside an adversary's decision cycle.

Successful implementation of this strategy requires a focused commitment throughout the Department of Defense (DoD). To this end, the JADC2 Strategy articulates three guiding C2 functions of 'sense,' 'make sense,' and 'act,' and an additional five enduring lines of effort (LOEs) to organize and guide actions to deliver materiel and non-materiel JADC2 capabilities. The LOEs are: (1) Establish the JADC2 Data Enterprise; (2) Establish the JADC2 Human Enterprise; (3) Establish the JADC2 Technical Enterprise; (4) Integrate Nuclear C2 and Communications (NC2/NC3) with JADC2; and (5) Modernize Mission Partner Information Sharing.

This strategy is supported by a JADC2 Strategy Implementation Plan that identifies JADC2 end states, key objectives and tasks, and works with and through established Departmental authorities, forums, and processes to synchronize and streamline efforts to prioritize, resource, develop, deliver, and sustain JADC2 capabilities. Existing Service and Agency development and acquisition processes routinely produce domain-specific capabilities unable to meet the operational demands of all-domain C2. The JADC2 approach will overlay these existing processes with the intent to stimulate development of radically improved cross-domain, joint capability.

The strategy provides six guiding principles to promote coherence of effort across the Department in delivering materiel and non-materiel JADC2 improvements. These principles are: (1) Information Sharing capability improvements are designed and scaled at the enterprise level; (2) Joint Force C2 improvements employ layered security features; (3) JADC2 data fabric consists of efficient, evolvable, and broadly applicable common data standards and architectures; (4) Joint Force C2 must be resilient in degraded and contested electromagnetic environments; (5) Department development and implementation processes must be unified to deliver more effective cross-domain capability options; and, (6) Department development and implementation processes must execute at faster speeds.

The JADC2 Strategy concludes that the use of an enterprise-wide, holistic approach for implementing materiel and non-materiel C2 capabilities is urgently needed to ensure the Joint Force Commander's ability to gain and maintain information and decision advantage against global adversaries throughout the competition continuum.

INTRODUCTION

The JADC2 strategy articulates DoD's approach for advancing Joint Force C2 capabilities necessary to support U.S. national security interests. The National Defense Strategy directs the Joint Force to "gain and maintain information advantage, particularly in cyberspace, space, and the electromagnetic spectrum." The enormous task of moving JADC2 from concept to reality of guiding joint/combined capability development requires a clear vision, an effective strategy, and agile processes. Successful JADC2 implementation will produce improved Joint Force C2 capabilities and will require the accelerated application of technological solutions to C2 capability development, as well as the adaptation of governing policies and operational procedures.

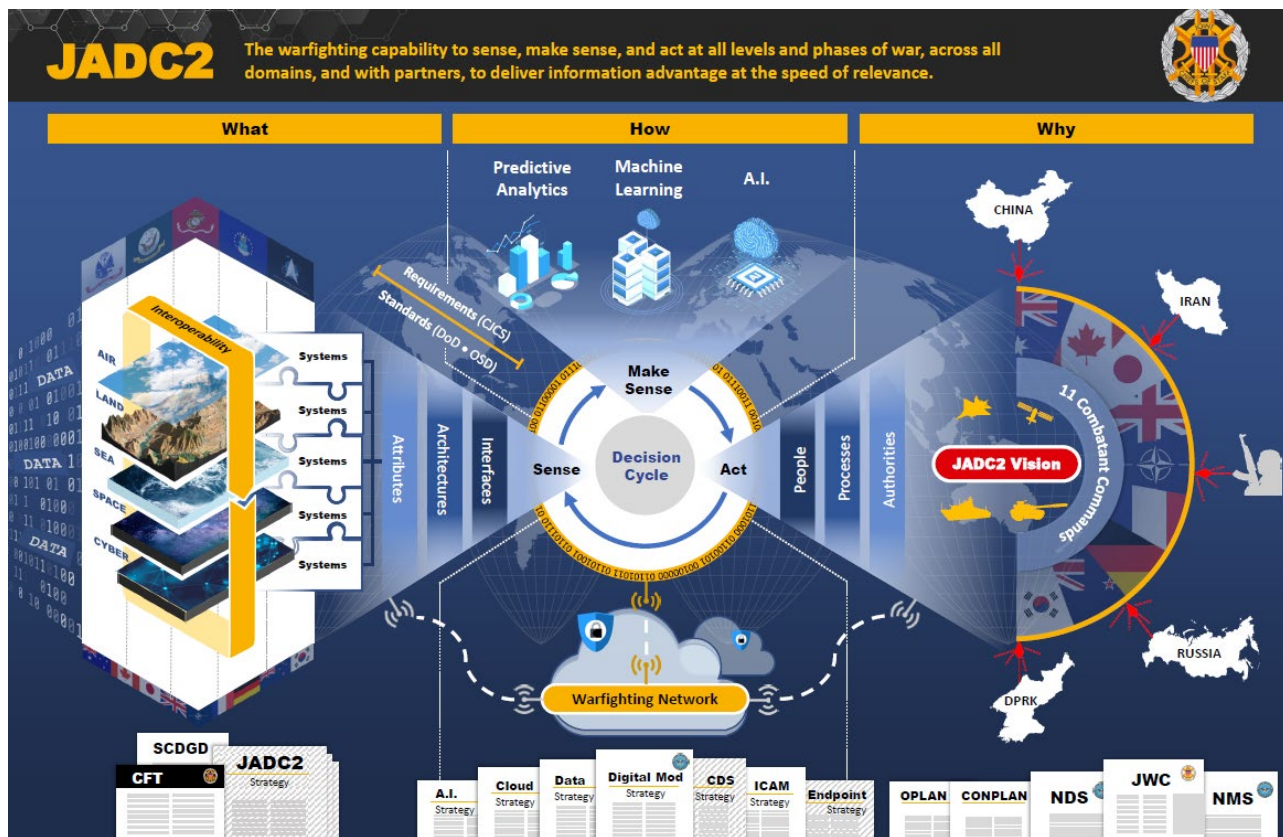


Figure 1 JADC2 Placemat

JADC2 provides an approach for developing the warfighting capability to sense, make sense, and act at all levels and phases of war, across all domains, and with partners, to deliver information advantage at the speed of relevance.

Figure 1 depicts the complexity of achieving all-domain C2: The Joint Force Commander relies upon guidance, technologies, procedures, and capabilities that are the result of many years of capability development and implementation in order to employ effective C2 in performing real-world mission tasks. JADC2 seeks to optimize the availability and use of information to ensure that the commander's information and decision cycle operates faster relative to adversary abilities. This holistic view identifies the Joint Force Commander as the primary beneficiary of Service and Agency C2 capability development efforts, and focuses Departmental C2 capability development outputs on delivering joint, all-domain

operational C2 performance and proficiency. To achieve this end, JADC2 will overlay the existing framework of Service- and Agency-based C2 capability development programs that currently deliver domain-centric and often duplicative information and decision management capabilities to the Joint Force. This 'overlay' methodology is a collaborative approach wherein all C2 capability development stakeholders support JADC2 as the unifying approach for optimizing developmental resources and priorities and maximizing the operational outcomes.

JADC2 APPROACH

The JADC2 strategy organizes its approach for improving Joint Force C2 through (1) the need for Joint Force Commanders to 'sense,' 'make sense,' and 'act' in the operational environment, and (2) the use of five functional areas of focus, or LOEs, to guide the development and implementation of improved C2 Joint Force capabilities. In this approach, JADC2 incorporates existing Departmental, Service, Agency, and operational requirements and capabilities development processes to shape the delivery of future materiel and non-materiel C2 capabilities.

Implementation of this strategy is overseen by the JADC2 Cross-Functional Team (CFT), a Deputy Secretary of Defense-chartered body composed of Flag/Senior Executive Service (SES)-level members from across the Combatant Commands, Services, Defense Agencies, Joint Staff, and OSD staff. The primary means for identifying and implementing prioritized C2 improvements is execution of the JADC2 Implementation Plan.

Sense - Integrate Information Across All Domains and the Electromagnetic Spectrum

Steady increases in the amount of data and information in all domains requires implementation of advanced sensing methods and information management technologies to enable improved information collection in the operational environment. JADC2 supports use of innovative data sharing options by the Joint Force and our mission partners through various intelligence sensing and information sharing networks that use federated data 'fabrics.'¹ This data and information sensor ecosystem exploits remote sensors, intelligence assets, and open sources to sense and simultaneously integrate information from and within all domains to enable the Joint Force Commander to achieve information and decision advantage.

"Sense and integrate" is the ability to discover, collect, correlate, aggregate, process, and exploit data from all domains and sources (friendly, adversary, and neutral), and share the information as the basis for understanding and decision-making. The requirements for effective data integration must be considered from the earliest stages of data sharing and security, and applied across the warfighting domains in order to deliver rapid collection, fusion, and customization of data.

Make Sense - Understand the Operational Environment

"Make Sense" refers to analyzing information to better understand and predict the operational environment and the actions and intentions of an adversary, as well as the actions of our own and friendly forces. It is in sense-making that data transforms into information and information transforms into knowledge. Effective

¹ The working definition for "JADC2 data fabric" is: "A DoD federated data environment for sharing information through interfaces and services to discover, understand and exchange data with partners across all domains, echelons and security levels."

sense-making requires the ability to fuse, analyze, and render validated data and information from all domains and the electromagnetic spectrum. This function must execute within a secure information environment while remaining readily accessible to all authorized personnel. Ultimately, it must result in a reliable and sustained real-time understanding of the operational environment that is shared across the Joint Force and with our mission partners.

JADC2 developed capabilities will leverage Artificial Intelligence and Machine Learning to help accelerate the commander's decision cycle. Automatic machine-to-machine transactions will extract, consolidate and process massive amounts of data and information directly from the sensing infrastructure. Such improvements will require the Joint Force to adapt and modernize existing tactical, operational, and strategic C2 processes and capabilities, to include re-thinking how the DoD tests and integrates the Service's C2 capabilities into a larger and more powerful all-domain C2 capability. Equally important, these procedural and technical advances will significantly enhance the Joint Force's ability to operate in a C2 degraded environment.

Act - Decide and Disseminate

To "Act" is to make and disseminate decisions to the Joint Force and its mission partners. It combines the human elements of decision-making with the technical means to perceive, understand, and predict the actions and intentions of adversaries, and take action. This includes accounting for the nuances of how a decision is delivered and how well a Commander's direction is understood and executed. Planning and decision support applications will be employed across the Joint Force and underpinned by advanced, resilient and redundant communication systems, an accessible and comprehensive transport infrastructure, and flexible data formats: All of which ensure the rapid, accurate, and secure dissemination of decisions.

"Act" also means that subordinate commanders at all levels within the Joint Force are properly trained in and empowered with the tenets of Mission Command. Using a Mission Command approach, subordinate commanders are able to act with confidence and authority through understanding a senior commander's operational intent while retaining the ability to act when communications linkages are broken or when the urgency of operations precludes the time necessary to seek guidance. Mission Command provides the Joint Force the agility and trust needed to seize the initiative and maintain information and decision advantage.

JADC2 LINES OF EFFORT (LOE)

The JADC2 strategy organizes around five LOEs to guide Department actions in delivering JADC2 capabilities, as follows: 1) Data Enterprise; 2) Human Enterprise; 3) Technology Enterprise; 4) Integrating with Nuclear C2 and C3; and 5) Modernizing Mission Partner Information Sharing. Each LOE is guided by an Office of Primary Responsibility represented by senior Flag/SES persons who are empowered to raise issues and interact with and support the Joint Requirements Oversight Committee through its Joint Capability Board. Additional JADC2 governance details are described in the JADC2 Strategy Implementation Plan.

LOE 1: Establish the JADC2 Data Enterprise

Data is a strategic asset and must be effectively managed by the Joint Force to enable it to seize, maintain, and protect information and decision advantage. To accelerate decision-making, the Joint Force and our mission partners must be able to discover and access any data and information from all warfighting domains

and at all levels of warfare.² The following key data standardization objectives will directly impact the Joint Force Commander's ability to manage and use data:

- Establishment of minimum metadata tagging criteria;
- Adoption and use of standardized data interfaces;
- Implementation of common data availability and access practices;
- Incorporation of data security best practices;
- Establishment of JADC2 conformant Information Technology (IT) standards; and,
- Continued application of data strategic objectives (Visible, Accessible, Understandable, Linked, Trustworthy, Interoperable, Secure).

LOE 2: Establish the JADC2 Human Enterprise

The ever-increasing availability of data and information in the operational environment threatens to overwhelm the Joint Force technical means of collecting and aggregating such information. This problem is compounded by existing organizational structures and decision processes that are being out-paced and require new approaches to ensure the Joint Force Commander's ability to seize opportunities and maintain advantages.

LOE 2 has a specific focus on human performance in C2 capabilities and addresses the use of innovative Artificial Intelligence and Machine Learning tools. Such innovations, in turn, will drive the need to produce predetermined, pre-approved, event-driven, bundled authorities that enable rapid, relevant decision-making from the strategic level to the tactical edge. This may require reforming, realigning, or creating organizations with the structure, agility, and resources to more effectively blend physical and informational power of the Joint Force and its mission partners such that they are capable of exercising effective control of Joint Information Advantage (JIA) operations.

This LOE also addresses the professional development needed to train and educate leaders to be proficient in operations across all warfighting domains. It will guide and support development of JADC2 aspects of policies, concepts of operation (CONOPS), doctrine, and tactics, techniques and procedures (TTPs) to optimize the advantages gained through new JADC2 capabilities. To this end, war-games, experiments, demonstrations, assessments, training, and exercises must be designed to focus on the C2 aspects of operations during competition and conflict. Likewise, the DoD workforce must be proficient in identifying institutional changes to achieve and sustain improved JADC2 capability development processes and products.

LOE 3: Establish the JADC2 Technical Enterprise

This LOE addresses enhanced shared situational awareness, synchronous and asynchronous global collaboration, strategic and operational joint planning, real-time global force visualization and management, predictive force readiness and logistics, real-time synchronization and integration of kinetic and non-kinetic joint and long-range precision fires, and enhanced abilities to assess Joint Force and mission partner performance.

Joint Force Commanders require secure, worldwide communications networks with sufficient speed and bandwidth to meet National Command Authority and Combatant Command warfighting needs. LOE 3 addresses the transport infrastructure of the JADC2 ecosystem, and provides the essential minimum features

² Joint Force data and information collection and use to be pursuant with U.S. law and DoD policy.

necessary to ensure continuous C2 capability, including communications system resiliency and diversity, multi-level security, and elimination of single points of failure. These cyber-hardened, advanced technologies will significantly improve the commander's ability to organize, understand, plan, decide, direct, and monitor all Joint Force and mission partner actions across all domains, and throughout periods of degraded and contested electromagnetic spectrum use.

LOE 4: Integrate NC2/NC3 with JADC2

Where appropriate, the JADC2 approach will collaborate and integrate with Nuclear C2 and Communications.

LOE 5: Modernize Mission Partner Information Sharing

The Joint Force Commander continuously strives to establish and maintain a common understanding of the operational environment through shared situational awareness with mission partners. Ideal mission partner system integration is realized when data from each partner's C2 systems can be accessed, viewed, and acted upon by every other approved partner. However, emerging missions, large coalitions, and evolving technologies present ongoing obstacles to achieving this goal. Ultimately, JADC2 system interoperability is foundational for conducting combined and partnered operations with speed, precision, relevance, and security. This LOE strives to broaden and improve the Joint Force's ability to exchange information and coordinate actions and effects in all types of combined operations.

JADC2 CAPABILITY GUIDING PRINCIPLES

The implementation of the JADC2 approach is guided by the following overarching principles.

Enterprise Designed and Scaled Information Sharing

Universal and continuous information sharing must be designed and operated at an enterprise level. JADC2 applications and processes will depend on multiple enterprise nodes and supporting communications networks to provide global connectivity with the bandwidth, functionality, and security needed to bring vital information to the Joint Force Commander.

Secure

Joint Force C2 must employ a layered defense spearheaded by a strong cyber defense to deter malicious activity that would threaten enterprise operations. The Joint Force must have clear policy guidance, sufficient authorities, adequate training, timely intelligence, and the technology necessary to conduct secure C2 in a globally contested environment. The Department must adopt a wartime mindset during day-to-day operations --- e.g., train as we fight --- and develop knowledgeable leaders and staffs trained to employ the tools and authorities at their disposal.

Data and Interoperability Standards-Driven

The Joint Force data fabric must consist of efficient, evolvable, and broadly applicable common data standards and architectures, with standardized key interfaces and services to access, aggregate, manage, store, process, and share data across a large environment with a wide variety of partners and operational uses.

Resilient in a Degraded Environment

The Joint Force must be able to operate with minimum guidance within a degraded or contested C2 environment, and commanders and staffs must train aggressively in conditions where sensing and communications are severely impacted or completely disabled, and where adversary intentions are ambiguous.

Unity of Effort in Capability Development

The Department must improve its C2 capability development and implementation processes to more readily adopt cross-domain priorities and solution options. The JADC2 CFT is the venue through which Department capability developers discuss, identify, collaborate, and recommend opportunities to improve C2 information sharing and interoperability within the Service and warfighting domains.

Delivering JADC2 Capabilities with Speed

The Department must continue to evolve its current approach to C2 development and acquisition, and adapt existing methodologies to more rapidly produce needed capabilities.

CONCLUSION

Changes in the global security environment, to include rising malign actions against the United States and wide-ranging advances in information technology, present urgent challenges and opportunities for the Joint Force. The JADC2 Strategy addresses these challenges and opportunities by advancing an interconnected and enterprise-wide approach for delivering materiel and non-materiel capabilities that support globally integrated operations. These capabilities will directly and dramatically improve a commander's ability to gain and maintain information and decision advantage.

The JADC2 Strategy articulates an enterprise approach for improving C2 of the Joint Force across all warfighting domains and throughout the electromagnetic spectrum. It addresses the unique aspects of human decision-making and seeks new opportunities to enhance the cognitive aspects of C2. The Strategy identifies the key C2 functions of sense, make sense, and act, along with five lines of effort to organize and guide the development and implementation of improved C2 capabilities.

Central to the success of the JADC2 approach is the JADC2 CFT. This body will collaboratively drive measurable positive change across the Department to achieve the capabilities, capacities, endurance and global reach required for all-domain C2.

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Report to Congressional Committees

NATIONAL SECURITY

Long-Range Emerging Threats Facing the United States As Identified by Federal Agencies

December 2018

National Security

Long-Range Emerging Threats Facing the United States As Identified by Federal Agencies

Why GAO Did This Study

The United States faces a complex array of threats to our national security, including our political, economic, military, and social systems. These threats will continue to evolve as new and resurgent adversaries develop politically and militarily, as weapons and technology advance, and as environmental and demographic changes occur. A House committee report accompanying a bill for the National Defense Authorization Act for Fiscal Year 2018 included a provision for GAO to identify emerging threats of high national security consequence. This report focuses on long-range emerging threats—those that may occur in approximately 5 or more years, or those that may occur during an unknown timeframe—as identified by various respondents at the Department of Defense (DOD), Department of State (State), Department of Homeland Security (DHS), and the Office of the Director of National Intelligence (ODNI).

To identify long-range emerging threats, GAO administered a questionnaire to 45 government organizations that assess emerging threats across DOD, State, DHS, and ODNI, and had a 78-percent response rate. GAO conducted a content analysis of the responses to identify specific threats and develop broad threat categories. To supplement the

data from the questionnaire, GAO reviewed national security strategies and agency documents provided by DOD, State, DHS, and ODNI, and interviewed key agency officials. This report is a public version of a classified report that GAO issued on September 28, 2018. Information that DOD deemed classified and sensitive has been omitted.

What GAO Found

DOD, State, DHS, and ODNI independently identified various threats to the United States or its national security interests. In analyzing more than 210 individual threats identified by organizations across DOD, State, DHS, and ODNI, as well as its review of national security strategies and related documents, and interviews with key agency officials, GAO developed four broad categories for 26 long-range emerging threats that officials identified: Adversaries' Political and Military Advancements, Dual-Use Technologies, Weapons, and Events and Demographic Changes.

The figure below contains examples of the 26 threats in 4 categories—as identified by DOD, State, DHS, and ODNI—in response to GAO's questionnaire.

For more information, contact Joseph W. Kirschbaum at (202) 512-9971 or kirschbaumj@gao.gov or Brian M. Mazanec at (202) 512-5130 or mazanecb@gao.gov.

Emerging Threats As Identified by DOD, State, DHS, and ODNI



Adversaries' Political and Military Advancements



Chinese Global Expansion» China is marshalling its diplomatic, economic, and military resources to facilitate its rise as a regional and global power. This may challenge U.S. access to air, space, cyberspace, and maritime domains. China's use of cyberspace and electronic warfare could impact various U.S. systems and operations.



Russian Global Expansion» Russia is increasing its capability to challenge the United States across multiple warfare domains, including attempting to launch computer-based directed energy attacks against U.S. military assets. Russia is also increasing its military and political presence in key locations across the world.



Iranian Political and Military Developments» Iran is expanding its influence by increasing the size and capabilities of its network of military, intelligence, and surrogate forces, while increasing economic activities in other areas of the world. Iran will also likely continue to develop its military capabilities, including developing technology that could be used for intercontinental ballistic missiles (ICBM) and improving its offensive cyberspace operations.



North Korean Military Developments» North Korea is developing capabilities to strike North America and its allies with long-range missiles and may produce significant numbers of intercontinental ballistic missiles.



Foreign Government Capacity and Stability» Violent extremist organizations may proliferate in countries that have limited governing capacity and are facing conflict, which may result in a higher risk of terrorist attacks and increased demand for U.S. resources to counter them. Countries in Africa, Latin America, and the Caribbean may experience instability based on conflict, which may lead to humanitarian disasters and government collapses.



Terrorism» Violent ideologies could influence additional individuals to turn to terrorism to achieve their goals across Africa, Asia, and the Middle East. Terrorists could advance their tactics, including building nuclear, biological or chemical weapons, or increase their use of online communications to reach new recruits and disseminate propaganda.



New Alliances and Adversaries» The United States could face challenges from potential new state adversaries and non-state adversaries (e.g., private corporations obtaining resources that could grant them more influence than states).



Information Operations» Adversaries—such as Russia, Iran, and China—may engage in advanced information operations campaigns that use social media, artificial intelligence, and data analytics to undermine the United States and its allies.

Emerging Threats As Identified by DOD, State, DHS, and ODNI (Continued)



Dual-Use Technologies



Artificial Intelligence (AI)» Adversaries could gain increased access to AI through affordable designs used in the commercial industry, and could apply AI to areas such as weapons and technology.



Quantum Information Science» Quantum communications could enable adversaries to develop secure communications that U.S. personnel would not be able to intercept or decrypt. Quantum computing may allow adversaries to decrypt information, which could enable them to target U.S. personnel and military operations.



Internet of Things (IoT)» The United States may face difficulties protecting networks and data as IoT grows and traditional approaches for security (e.g., encryption) may no longer effectively protect information. Adversaries could also disrupt IoT-enabled critical infrastructure and devices.



Autonomous and Unmanned Systems» Adversaries are developing autonomous capabilities that could recognize faces, understand gestures, and match voices of U.S. personnel, which could compromise U.S. operations. Unmanned ground, underwater, air, and space vehicles may be used for combat and surveillance.



Biotechnology» Actors—which may include state or non-state entities such as violent extremist organizations and transnational criminal organizations—could alter genes or create DNA to modify plants, animals, and humans. Such biotechnologies could be used to enhance the performance of military personnel. The proliferation of synthetic biology—used to create genetic code that does not exist in nature—may increase the number of actors that can create chemical and biological weapons.



Other Emerging Technologies» Actors may gain access to new technologies previously limited to militaries, such as affordable and sophisticated encryption technologies, which would hinder U.S. efforts to monitor terrorist and criminal activities. Other emerging technologies—such as additive manufacturing (i.e., 3D printing)—may be vulnerable to cyber attacks or be used to manufacture restricted materials, such as weapons.



Weapons



Weapons of Mass Destruction» An increasing number of actors may gain access to these weapons. Adversaries could steal nuclear materials from existing facilities or develop new types of biological weapons using genetic engineering and synthetic biology.



Electronic Warfare» Adversaries are developing electronic attack weapons to target U.S. systems with sensitive electronic components, such as military sensors, communication, navigation, and information systems. These weapons are intended to degrade U.S. capabilities and could restrict situational awareness or may affect military operations.



Hypersonic Weapons» China and Russia are pursuing hypersonic weapons because their speed, altitude, and maneuverability may defeat most missile defense systems, and they may be used to improve long-range conventional and nuclear strike capabilities. There are no existing countermeasures.



Counterspace Weapons» China and Russia are developing anti-satellite weapons to threaten U.S. space operations. China is developing capabilities to conduct large-scale anti-satellite strikes using novel physical, cyber, and electronic warfare means.



Missiles» Adversaries are developing missile technology to attack the United States in novel ways and challenge U.S. missile defense, including conventional and nuclear ICBMs, sea-launched land-attack missiles, and space-based missiles that could orbit the earth.



Intelligence, Surveillance, Reconnaissance (ISR) Platforms» Future advances in AI, sensors, data analytics, and space-based platforms could create an environment of “ubiquitous ISR”, where people and equipment could be tracked throughout the world in near-real time. China, Russia, Iran, and North Korea are developing multiple ISR platforms.



Aircraft» China and Russia are developing new aircraft, including stealth aircraft, which could fly faster, carry advanced weapons, and achieve greater ranges. Such aircraft could force U.S. aircraft to operate at farther distances and put more U.S. targets at risk.



Undersea Weapons» Russia has made significant advancements in submarine technology and tactics to escape detection by U.S. forces. China is developing underwater acoustic systems that could coordinate swarm attacks—the use of large quantities of simple and expendable assets to overwhelm opponents—among vehicles and provide greater undersea awareness. Adversaries could achieve breakthroughs in anti-submarine warfare—such as using AI to locate U.S. submarines—or attack U.S. undersea infrastructure, which could cripple communications.



Cyber Weapons» Adversaries, such as China, Russia, Iran, and North Korea, may launch cyber attacks against critical U.S. infrastructure (e.g., electric, oil and gas, and nuclear power systems) and military infrastructure (e.g., communications and ISR platforms). Adversaries could also launch cyber attacks on the U.S. health care system, threatening patient safety by disrupting access to medical care. Finally, adversaries are also developing tools to directly attack hardware and embedded components in aviation systems, which can manipulate or destroy data.



Events and Demographic Changes



Infectious Diseases» New and evolving diseases from the natural environment—exacerbated by changes in climate, the movement of people into cities, and global trade and travel—may become a pandemic. Drug-resistant forms of diseases previously considered treatable could become widespread again.



Climate Change» Extreme weather events—such as hurricanes and megadroughts—could intensify and affect food security, energy resources, and the health care sector. Diminishing permafrost could expand habitats for pathogens that cause disease. The loss of Arctic sea ice could open previously closed sea routes, potentially increasing Russian and Chinese access to the region and challenging the freedom of navigation that the United States currently has.



Internal and International Migration» Governments in megacities (i.e., over 10 million people) across Asia, Latin America, and Africa may not have the capacity to provide adequate resources and infrastructure, and may be vulnerable to natural or man-made disasters. Mass migration events may occur and threaten regional stability, undermine governments, and strain U.S. military and civilian responses.

Source: GAO analysis of DOD, State, DHS, and ODNI questionnaire responses, agency documents, and national security strategies. | GAO-19-204SP



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Abbreviations

AI	Artificial Intelligence
DHS	Department of Homeland Security
DIA	Defense Intelligence Agency
DOD	Department of Defense
ICBM	Intercontinental ballistic missile
IoT	Internet of Things
ISR	Intelligence, surveillance, and reconnaissance
NATO	North Atlantic Treaty Organization
ODNI	Office of the Director of National Intelligence
State	Department of State

December 13, 2018

Congressional Committees

The United States faces a complex array of threats to our national security, including our political, economic, military, and social systems. These threats will continue to evolve as new and resurgent adversaries develop politically and militarily, as weapons and technology advance, and as environmental and demographic changes occur. Our adversaries may include foreign governments, violent extremists, transnational criminal organizations, and megacorporations.¹ Threats may also come from events such as pandemics, human migration, regional conflict and instability, economic inequality, or the effects of climate change and environmental issues.

A variety of national intelligence and security organizations are responsible for national security, including identifying, analyzing, and countering emerging threats. Such organizations include: the Department of Defense (DOD), the Department of State (State), the Department of Homeland Security (DHS), and the Office of the Director of National Intelligence (ODNI).

For purposes of this report, we define “threat” as an actor with capability and intent, or an event with potential capability, to harm the United States or its national security interests. We define “emerging threat” as a threat that may be newly recognized; may have been recognized before but may potentially affect a new or different population, industry, or geographic area than previously affected; or may be an existing threat that has developed new attributes.

A House Committee report accompanying a bill for the National Defense Authorization Act for Fiscal Year 2018 included a provision for us to identify emerging threats of high national security consequence.² This report describes long-range emerging threats as identified by DOD, State, DHS, and ODNI.³ For purposes of this report, we define long-range threats as threats that agency officials identified that may occur in approximately 5 or more years, or those threats that could occur in a future unknown time frame.⁴

This report is a public version of a classified report that we issued on September 28, 2018.⁵ It omits classified and sensitive information about threats identified by executive branch agencies and described in 26 profiles in our classified report. It also omits classified and sensitive information in those profiles related to specific threats, the effects of those threats, specific warfare domains, and questions for oversight. Although the information provided in this report is more limited, the report addresses the same objectives as the classified report and uses the same methodology.

To identify long-range emerging threats, we administered a questionnaire to 45 selected organizations across DOD, State, DHS, and ODNI.⁶ In the questionnaire, we asked respondents to identify and describe emerging threats that their organizations assess could occur in approximately 5 years or more from today, or those that have an unknown time frame. We received approximately 210 individual threats from 26 of these 45

¹Adversaries are potentially hostile or disruptive state or non-state actors. According to the 2018 *National Defense Strategy*, state actors and non-state actors, such as terrorists, transnational criminal organizations, and cyber hackers, have transformed the direction of global affairs with increased capabilities of mass disruption. Disruptive state actors include North Korea, Russia, China, and Iran. Non-state actors include violent extremist organizations such as al-Qaida and the Islamic State of Iraq and Syria. Transnational criminal organizations can participate in the sale of illegal drugs and counterfeit goods, human trafficking and smuggling, and other criminal activities. According to DOD officials, megacorporations are large companies that have the financial resources and a power base to exert influence on par with or exceeding non-state actors.

²H.R. Rep. No. 115-200, at 181 (2017).

³ODNI supports the Director of National Intelligence's role as head of the Intelligence Community. The Intelligence Community is comprised of 17 separate organizations such as the Central Intelligence Agency, the Defense Intelligence Agency, the Federal Bureau of Investigation, the National Security Agency, and intelligence components within agencies such as the Department of Homeland Security, Department of State, and the military services.

⁴We established this time frame because officials from DOD and ODNI stated that they consider threats occurring earlier than 5 years from today as near-term or mid-term threats, which receive greater attention and resources from defense and intelligence organizations than long-term threats.

⁵GAO, *National Security: Long-Range Emerging Threats Facing the United States Identified by Federal Agencies*, GAO-18-497SPC (Washington, D.C.: Sept. 28, 2018). (SECRET//NOFORN)

⁶We focused on DOD, State, DHS, and ODNI as among the federal agencies with primary responsibility for national security. We identified the 45 selected organizations within these agencies that assess long-range emerging threats through consultation with agency officials.

organizations, which formed the basis of our threat profiles.⁷ See appendix I for more details about the specific organizations that received the questionnaire. In addition, ODNI submitted a questionnaire but did not identify any threats on the questionnaire. Instead, ODNI officials referred us to their *Global Trends: Paradox of Progress* report and provided verbal input on long-range emerging threats.⁸ We used this information to supplement questionnaire responses from DOD, State, and DHS organizations. In total, we received a 78-percent response rate (28 of 36) to our questionnaire.⁹

To supplement information from the questionnaire, we reviewed documents provided by DOD, State, DHS, and ODNI. For example, we reviewed the most recent national strategies that pertain to emerging threats and ODNI's 2017 *Global Trends: Paradox of Progress* report.¹⁰ We also interviewed officials about long-range emerging threats from 22 organizations, including 11 DOD organizations, 6 State organizations, 4 DHS organizations, and ODNI. We selected these 22 organizations because they may have a role in identifying and assessing long-range emerging threats. This review did not assess any efforts to mitigate threats. We pre-tested the questionnaire instrument with officials from different agencies to confirm that it would be understood by respondents as intended, and determined that the data collected were sufficiently reliable for our purposes.

It is not possible to predict every potential long-range emerging threat. According to Intelligence Community officials, the further out in time predictions go, the more uncertain they become, because the future is a confluence of multiple trends with an infinite number of possible permutations. For example, adversaries may use emerging technologies together in novel and unpredicted ways to amplify their harm. Several DOD officials also noted that there will always be completely

unpredictable events with no prior warning. Therefore, this report does not attempt to provide a comprehensive listing of all potential emerging threats to the United States that may arise over the next 5 or more years. Rather, it represents the assessments of agency experts who responded to our questionnaire, supplemented by information from national security strategies, related agency documents, and interviews with agency officials. Furthermore, many questionnaire responses focus on threats that originate outside the United States. For more information on our objective, scope, and methodology, see appendix I.

The performance audit upon which this report is based was conducted from July 2017 to September 2018 in accordance with generally accepted government auditing standards. Those standards require that we plan and perform the audit to obtain sufficient, appropriate evidence to provide a reasonable basis for our findings and conclusions based on our audit objectives. We believe that the evidence obtained provides a reasonable basis for our findings and conclusions based on our audit objective. We subsequently worked with DOD from September 2018 to December 2018 to prepare this public version of the original classified report for public release. This public version was also prepared in accordance with those standards.

⁷Two additional organizations provided responses after the response period had ended. These responses were not used in the development of the threat profiles but were included in our response rate calculations.

⁸Office of the Director of National Intelligence, National Intelligence Council, *Global Trends: Paradox of Progress*, [NIC 2017-001](#) (Washington, D.C.: January 2017). An ODNI official stated that the *Global Trends: Paradox of Progress* report lists some key threats over the next 5 to 20 years. In particular, ODNI officials emphasized economic threats such as U.S. debt and growing inequality; new technologies and ethical questions surrounding the use of those technologies; geopolitical conflict, including the spread of corruption to the developed world and the rise of China; and the spread of populism and nationalist identities around the world.

⁹Of the 45 organizations selected to receive our questionnaire, 9 organizations were excluded from our response rate calculations because 8 told us that they do not identify threats that meet our scope or definition, and 1 told us that a separate office within their organization had responded on their behalf. Of the remaining 36 organizations, 28 organizations provided responses (2 of which provided responses after the response period had ended, and were not used in the development of the threat profiles). In addition, 1 organization did not receive a questionnaire due to an administrative error and ODNI submitted an incomplete questionnaire that did not identify any threats. We did not receive responses from the remaining 6 organizations.

¹⁰[NIC 2017-001](#).



Background

Current Landscape of Emerging Threats

DOD officials noted that Western liberal democratic institutions around the world are being challenged in new and novel ways. Adversaries have had over 40 years to study the United States and Western institutions. As such, the nature of warfare has evolved to include “gray zone” conflict—defined as the area between war and peace—where weaker adversaries have learned how to seize territory and advance their agendas in ways not recognized as “war” by Western democracies. Also, these gray zone conflicts can offset superior U.S. economic and security structures. DOD officials added that adversaries around the world may erode democracies, often using democratic institutions, in the gray zone of conflict.¹¹ ODNI officials also noted that China and Russia are pursuing gray zone strategies to achieve their objectives without resorting to military conflict.

DOD officials provided a list of recent significant examples of adversary success in the gray zone of conflict, several of which have occurred without significant consequences, including:

- Russian and Chinese near-unrestricted thefts of U.S. intellectual property, Office of Personnel Management data theft, and penetrations of U.S. civil, utility, and military data and electoral voting systems;
- Russian seizure of Ukrainian territory, namely Crimea;
- Chinese seizure of the South China Seas and the building of military islands in defiance of international court rulings;
- China using bilateral economic deals to marginalize U.S. multilateral frameworks in Asia, Africa, Latin America, and the Pacific;
- Russia attempting to resurrect former Soviet client state relationships with Syria, Egypt, and Libya, and potentially with additional countries in the Middle East and North Africa;

- Iran realigning the Middle East by using proxy forces to create friendly governments including Syria, Iraq, and Yemen at the expense of U.S. leadership in the region;
- “Strongmen” in countries such as Venezuela, Egypt, and Turkey using democratic institutions to promote new paradigms independent of Western liberal norms; and
- The continued attraction of extremist groups, including the Islamic State and al-Qaida, as a preferable means to achieve Sunni Arab autonomy as a viable alternative to minority governance in countries with majorities that outnumber them (as in Syria and Iraq).

DOD officials said that, with current demographic trends, Western liberal democratic institutions will be tested in new ways as the nature of warfare changes. The challenge for the United States and its allies will be to develop responses faster than adversaries through a better understanding of the strategic environment. Officials added that this presents a challenge since the United States appears to be strategically surprised by an evolving world.

DOD officials also said that the United States must adapt to challenges from adversaries and better link security objectives and economic objectives, or risk further erosions of U.S. influence to adversaries such as China and Russia. Officials stated that China and Russia are more agile than the United States in creating relationships with other countries to degrade U.S. bilateral and multilateral frameworks. For example, China and Russia are working to define the United States as a “status quo” power trying to preserve the old world order in what is becoming a multipolar world. These officials added that the nature of conflict has changed, and so the United States must evolve.

¹¹Officials from the Defense Advanced Research Projects Agency noted that gray zone warfare is characterized by limited conflict that sits between normal state competition and what is traditionally thought of as war.

Future Landscape of Emerging Threats

ODNI's January 2017 report *Global Trends*:

Paradox of Progress describes future trends that will shape the direction of the world over the next 5 or more years.¹² ODNI's report describes potential environments from which long-range threats may emerge, based on seven key global trends:

❶ **The rich population is shrinking, the poor population is not.** Working-age populations are shrinking in wealthy countries and in China and Russia, and are growing in developing, poorer countries. This trend has the potential to increase economic, employment, urbanization and welfare pressures, and spur migration.¹³

❷ **The global economy is shifting.** Weak economic growth will likely persist in the near term. Major economies will confront shrinking workforces and diminishing productivity gains while recovering from the 2008-2009 financial crises with high debt, weak demand, and doubts about globalization. Inequality and wealth concentrations—combined with corruption and eroding trust in authorities—are driving a wave of political change.

❸ **Technology is accelerating progress but causing discontinuities.** Rapid technological advancements will increase the pace of change and create new opportunities, but will aggravate divisions between winners and losers. Automation and artificial intelligence will threaten to change industries faster than economies can adjust, potentially displacing workers and limiting the usual route for poor countries to develop. Biotechnologies such as genome editing will revolutionize medicine and other fields, while sharpening moral differences.

❹ **Ideas and identities are driving a wave of exclusion.** Growing global connectivity amid weak economic growth will increase tensions within and between societies. Populism will increase on the right and the left. Some leaders will use nationalism to shore up control. Religious influence will be increasingly consequential, and nearly all countries will see economic forces boost women's status and leadership roles, but backlash against this trend also will occur.

❺ **Governing is getting harder.** The public will demand that governments deliver security and prosperity. However, flat revenues, distrust, polarization, and a growing list of emerging issues will hamper government performance. Technology will expand the range of players who can block or circumvent political action.

❻ **The nature of conflict is changing.** The risk of conflict will increase due to diverging interests among major powers, an expanding terror threat, continued instability in weak states, and the spread of lethal, disruptive technologies. Disrupting societies will become more common, with long-range precision weapons, cyber, and robotic systems to target infrastructure from afar, and with more accessible technology to create weapons of mass destruction.

❼ **Climate change, environment, and health issues will demand attention.** A range of global hazards pose imminent and longer-term threats that will require collective action to address—even as cooperation becomes harder. More extreme weather, water and soil stress, and food insecurity will disrupt societies. Sea-level rise, ocean acidification, glacial melt, and pollution will change living patterns. Tensions over climate change will grow.

The *Global Trends* report also points out that conflicts in the next 5 to 20 years will be more:

- **Diffuse**—referring to state, non-state, and sub-state entities having greater accessibility to means of warfare;
- **Diverse**—referring to the means of warfare varying across a wider spectrum, from nonmilitary capabilities to advanced conventional weapons and weapons of mass destruction; and
- **Disruptive**—referring to a greater emphasis by states and terror groups on targeting critical infrastructure, societal cohesion, and government functions, rather than defeating enemy forces on the battlefield through traditional military means.¹⁴

¹²NIC 2017-001.

¹³In its report, ODNI states this trend as "the rich are aging, the poor are not." See NIC 2017-001.

¹⁴NIC 2017-001.

Roles and Responsibilities of Agencies That Identify and Mitigate Emerging Threats

DOD, State, DHS, and the Intelligence Community have key roles in identifying and mitigating long-term emerging threats. Specifically:

- **DOD** has a role in, among other things, defending the homeland from limited ballistic missile and cruise missile attack; large-scale terrorist attack; chemical, biological, radiological, or nuclear attack; and space, electromagnetic, or kinetic attacks against our critical infrastructure. DOD also works to prevent adversaries such as state and non-state actors from acquiring, proliferating, or using weapons of mass destruction, and is the lead agency to defend U.S. military and intelligence infrastructure from cyber attacks and conduct offensive cyber operations. DOD also works to rebuild the military strength and maximize the competitive advantage of the United States and its partners, while constraining the ability of our adversaries to achieve their military objectives. DOD efforts may include preventing terrorists from directing or supporting operations against the U.S. homeland and its partners, and bolstering its partners against coercion. Finally, DOD assists State and the U.S. Agency for International Development with natural disaster and conflict response around the world.
- **State** is the lead U.S. foreign affairs agency and the lead institution for conducting American diplomacy. State plays a role in protecting and advancing the interests of the United States by, in part, countering threats and adversaries, deepening U.S. security relationships and partnerships around the world, and strengthening our allies and alliances. State also engages in security and capacity building and other non-military assistance, such as border patrol, with other countries. In conjunction with DOD and the U.S. Agency for International Development, State also responds to natural disasters and conflict-induced crises around the world.
- **DHS** plays a role in preventing a variety of threats within the homeland, particularly terrorist attacks within the United States. DHS also seeks to reduce the vulnerability of the United States to terrorism, assisting in the recovery from terrorist attacks that do occur within the United States, and disrupting connections between illegal drug trafficking and terrorism, and for coordinating efforts to sever such connections. DHS officials added that DHS is also the lead agency for defense of civilian cyber infrastructure (such as .gov accounts) from cyber-attacks, and for aiding private-sector critical-infrastructure cyber security. DHS officials also said that DHS has primary responsibility for border and transportation security issues, including interdicting illicit smuggling of humans and contraband into the homeland, traveler security screening, and, via the U.S. Coast Guard, securing the maritime approaches to the homeland.
- **ODNI** supports the Director of National Intelligence in his role as the head of the Intelligence Community, acts as the principal advisor to the President, National Security Council, and the Homeland Security Council for intelligence matters related to national security, and oversees and directs the implementation of the National Intelligence Program.¹⁵ ODNI's activities include integrating intelligence analysis and collection, driving secure information sharing, setting strategic direction and priorities for national intelligence resources and capabilities, developing and implementing unifying intelligence strategies, and advancing capabilities to provide the United States with a global intelligence advantage.

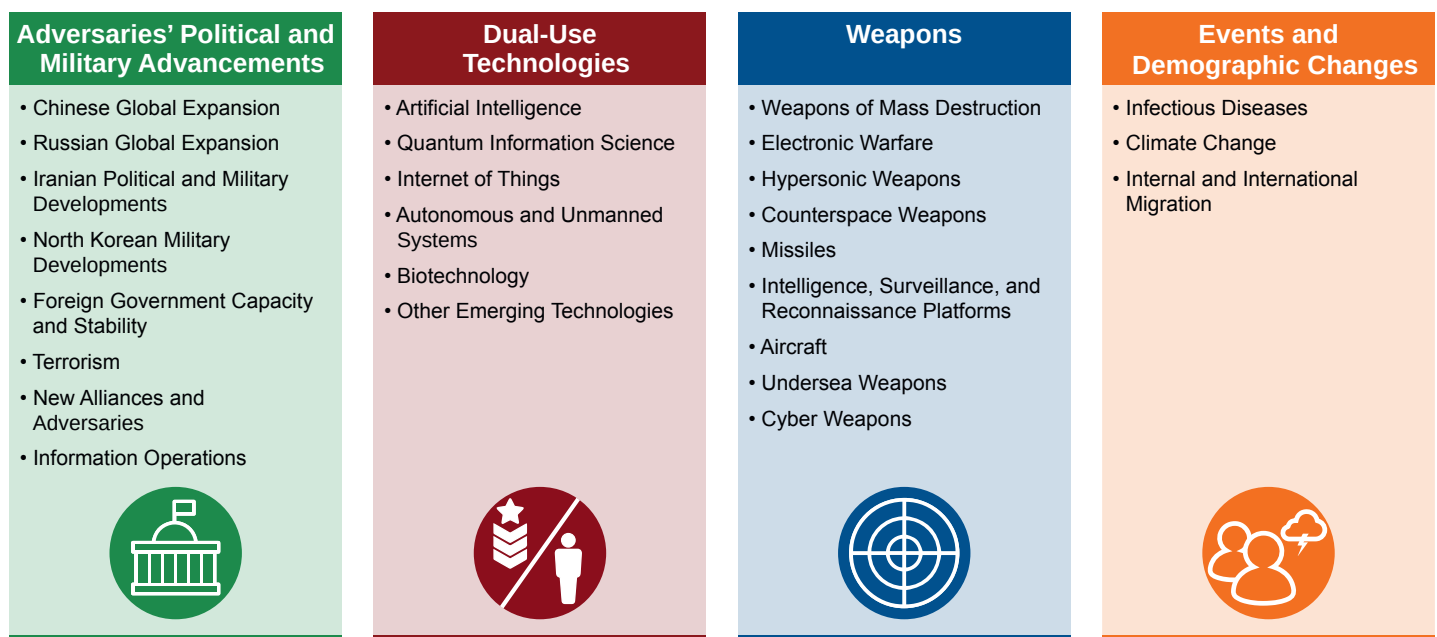
¹⁵The National Intelligence Program is intended to provide the resources to develop and maintain intelligence capabilities that support national priorities. Intelligence Reform and Terrorism Prevention Act of 2004, Pub. L. No. 108-458, § 1011 (2004). According to 50 U.S.C. § 3003 (6) "the term 'National Intelligence Program' refers to all programs, projects, and activities of the intelligence community, as well as any other programs of the intelligence community designated jointly by the Director of National Intelligence and the head of a United States department or agency or by the President. Such term does not include programs, projects, or activities of the military departments to acquire intelligence solely for the planning and conduct of tactical military operations by United States Armed Forces."



DOD, State, DHS and ODNI Identified a Variety of Emerging Threats to U.S. National Security That May Occur over the Next Approximately 5 or More Years

DOD, State, DHS, and ODNI independently identified various emerging threats to the United States or its national security interests. Our analysis of these threats led to 26 threat profiles that fell within four broad categories: 1) Adversaries' Political and Military Advancements, 2) Dual-Use Technologies, 3) Weapons, and 4) Events and Demographic Changes, as shown in figure 1.

Figure 1: GAO's Four Broad Categories for 26 Long-Range Emerging Threats Identified by DOD, State, DHS, and ODNI



Source: GAO analysis of DOD, State, DHS, and ODNI questionnaire responses, agency documents, and national security strategies. | GAO-19-204SP



Adversaries' Political and Military Advancements »

Our adversaries are developing new political and military policies, strategies, doctrines, and tactics to advance their interests. These activities may or may not be intended to harm the United States and its national security interests. Agency officials identified examples such as Russia's integration of irregular warfare, influence operations, deception, and cyber attacks. Figure 2 shows the identified threats for this category.

Figure 2: Threats Identified under Adversaries' Political and Military Advancements

Threat »	Description »	Examples »
 Chinese Global Expansion	China is marshalling its diplomatic, economic, and military resources to facilitate its rise as a regional and global power. This may challenge U.S. access to air, space, cyberspace, and maritime domains. China's use of cyberspace and electronic warfare could impact various U.S. systems and operations.	• Expansion and Power Projection • Fusion of Military and Civilian Sectors and Control of Supply Chain • Cyber and Electronic Warfare
 Russian Global Expansion	Russia is increasing its capability to challenge the United States across multiple warfare domains, including attempting to launch computer-based directed energy attacks against U.S. military assets. Russia is also increasing its military and political presence in key locations across the world.	• Military Capabilities across Warfare Domains • Global Military and Political Influence • Biotechnology Advancements
 Iranian Political and Military Developments	Iran is expanding its influence by increasing the size and capabilities of its network of military, intelligence, and surrogate forces, while increasing economic activities in other areas of the world. Iran will also likely continue to develop its military capabilities, including developing technology that could be used for intercontinental ballistic missiles (ICBM) and improving its offensive cyberspace operations.	• Military and Economic Influence • Ballistic Missiles • Cyber Attacks
 North Korean Military Developments	North Korea is developing capabilities to strike North America and its allies with long-range missiles and may produce significant numbers of intercontinental ballistic missiles.	• Nuclear Strike against the Continental United States • Numerical Overmatch of Ballistic Missile Systems
 Foreign Government Capacity and Stability	Violent extremist organizations may proliferate in countries that have limited governing capacity and are facing conflict, which may result in a higher risk of terrorist attacks and increased demand for U.S. resources to counter them. Countries in Africa, Latin America, and the Caribbean may experience instability based on conflict, which may lead to humanitarian disasters and government collapses.	• Governments' Ability to Address Influences from Violent Extremist Organizations • Foreign Militaries' Reduced Military Readiness • Instability in Africa • Instability in Latin America and the Caribbean
 Terrorism	Violent ideologies could influence additional individuals to turn to terrorism to achieve their goals across Africa, Asia, and the Middle East. Terrorists could advance their tactics, including building nuclear, biological or chemical weapons, or increase their use of online communications to reach new recruits and disseminate propaganda.	• Proliferation of Terrorism and Violent Ideologies • Terrorist Use of Chemical, Biological, and Nuclear Materials • New Tactics and Techniques
 New Alliances and Adversaries	The United States could face challenges from potential new state adversaries and non-state adversaries (e.g., private corporations obtaining resources that could grant them more influence than states).	• Disrupted Alliances • Rise of New Nation-State and Non-State Adversaries • Foreign Nation-States' Influences on International Agreements and Standards
 Information Operations	Adversaries—such as Russia, Iran, and China—may engage in advanced information operations campaigns that use social media, artificial intelligence, and data analytics to undermine the United States and its allies.	• Exploitation and Theft of U.S. Information • Weaponized Information

Source: GAO analysis of DOD, State, DHS, and ODNI questionnaire responses, agency documents, and national security strategies. | GAO-19-204SP



Dual-Use Technologies »

These are technologies that may be developed by governments or the private sector for benign or beneficial purposes, but may have a dual-use application. For instance, in an adversary's hands, these technologies may pose a risk to the United States. Agency officials identified examples such as unmanned vehicles, artificial intelligence, and encryption technologies. Figure 3 shows the identified threats for this category.

Figure 3: Threats Identified under Dual-Use Technologies

Threat »	Description »	Examples »
 Artificial Intelligence (AI)	Adversaries could gain increased access to AI through affordable designs used in the commercial industry, and could apply AI to areas such as weapons and technology.	• Nation-State and Non-State Development of AI • Intelligent Systems with General AI
 Quantum Information Science	Quantum communications could enable adversaries to develop secure communications that U.S. personnel would not be able to intercept or decrypt. Quantum computing may allow adversaries to decrypt information, which could enable them to target U.S. personnel and military operations.	• Quantum Communications • Quantum Computing
 Internet of Things (IoT)	The United States may face difficulties protecting networks and data as IoT grows and traditional approaches for security (e.g., encryption) may no longer effectively protect information. Adversaries could also disrupt IoT-enabled critical infrastructure and devices.	• Unsecure Networks and Data • Attacks on IoT-Enabled Infrastructure • Attacks on Commercial and Military Devices
 Autonomous and Unmanned Systems	Adversaries are developing autonomous capabilities that could recognize faces, understand gestures, and match voices of U.S. personnel, which could compromise U.S. operations. Unmanned ground, underwater, air, and space vehicles may be used for combat and surveillance.	• Enhancement of Autonomous Systems • Weapons with Autonomous Navigation • Autonomous and Unmanned Vehicles
 Biotechnology	Actors—which may include state or non-state entities such as violent extremist organizations and transnational criminal organizations—could alter genes or create DNA to modify plants, animals, and humans. Such biotechnologies could be used to enhance the performance of military personnel. The proliferation of synthetic biology—used to create genetic code that does not exist in nature—may increase the number of actors that can create chemical and biological weapons.	• Human Genetic Modification and Synthetic Biology • Plant and Animal Genetic Modification • Other Biotechnology Applications • Increase Access to Technology
 Other Emerging Technologies	Actors may gain access to new technologies previously limited to militaries, such as affordable and sophisticated encryption technologies, which would hinder U.S. efforts to monitor terrorist and criminal activities. Other emerging technologies—such as additive manufacturing (i.e., 3D printing)—may be vulnerable to cyber attacks or be used to manufacture restricted materials, such as weapons.	• Expansion of Removable Media and Storage • Additive Manufacturing • New Materials • Development of Technologies that Address Electric Power Scarcity • Advancements in Camouflage Technology • Advanced Sensors

Source: GAO analysis of DOD, State, DHS, and ODNI questionnaire responses, agency documents, and national security strategies. | GAO-19-204SP



Weapons »

These threats are inherently threatening military devices that may be used by our adversaries to inflict harm upon the United States or its national security interests. These weapons do not have potential beneficial or benign uses from the perspective of the United States. Agency officials identified examples such as hypersonic missiles, weaponized pathogens, or stealth aircraft. Figure 4 shows the identified threats for this category.

Figure 4: Threats Identified under Weapons

Threat »	Description »	Examples »
 Weapons of Mass Destruction	An increasing number of actors may gain access to these weapons. Adversaries could steal nuclear materials from existing facilities or develop new types of biological weapons using genetic engineering and synthetic biology.	• Global Proliferation of Weapons of Mass Destruction • Development and Use of Nonstrategic Nuclear Weapons • New Forms of Biological Weapons
 Electronic Warfare	Adversaries are developing electronic attack weapons to target U.S. systems with sensitive electronic components, such as military sensors, communication, navigation, and information systems. These weapons are intended to degrade U.S. capabilities and could restrict situational awareness or may affect military operations.	• Electronic Attack Weapons • Attacks on Communications and Navigation Systems
 Hypersonic Weapons	China and Russia are pursuing hypersonic weapons because their speed, altitude, and maneuverability may defeat most missile defense systems, and they may be used to improve long-range conventional and nuclear strike capabilities. There are no existing countermeasures.	• Hypersonic Weapons and Missile Defense • Hypersonic Ballistic and Cruise Missiles • Hypersonic Glide Vehicles • Future Development and Convergence
 Counterspace Weapons	China and Russia are developing anti-satellite weapons to threaten U.S. space operations. China is developing capabilities to conduct large-scale anti-satellite strikes using novel physical, cyber, and electronic warfare means.	• Anti-Satellite Weapons • Increased Access to Space and Anti-Satellite Weapons
 Missiles	Adversaries are developing missile technology to attack the United States in novel ways and challenge U.S. missile defense, including conventional and nuclear intercontinental ballistic missiles, sea-launched land-attack missiles, and space-based missiles that could orbit the earth.	• Advancements in Missile Technology • Adversary Plans for Intercontinental Ballistic Missiles • Orbital Missiles • Sea-to-Land Missiles • Adversary Missile Defense
 Intelligence, Surveillance, Reconnaissance (ISR) Platforms	Future advances in artificial intelligence, sensors, data analytics, and space-based platforms could create an environment of “ubiquitous ISR”, where people and equipment could be tracked throughout the world in near-real time. China, Russia, Iran, and North Korea are developing multiple ISR platforms.	• Worldwide and Ubiquitous Surveillance • Adversary Improvements in Radar and Surveillance Platforms
 Aircraft	China and Russia are developing new aircraft, including stealth aircraft, which could fly faster, carry advanced weapons, and achieve greater ranges. Such aircraft could force U.S. aircraft to operate at farther distances and put more U.S. targets at risk.	• Russian Aircraft • Chinese Aircraft
 Undersea Weapons	Russia has made significant advancements in submarine technology and tactics to escape detection by U.S. forces. China is developing underwater acoustic systems that could coordinate swarm attacks—the use of large quantities of simple and expendable assets to overwhelm opponents—among vehicles and provide greater undersea awareness. Adversaries could achieve breakthroughs in anti-submarine warfare—such as using artificial intelligence to locate U.S. submarines—or attack U.S. undersea infrastructure, which could cripple communications.	• Russian Improvements in Undersea Stealth • Unmanned Underwater Vehicles • Anti-Submarine Warfare • Attack on Undersea Cables
 Cyber Weapons	Adversaries, such as China, Russia, Iran, and North Korea, may launch cyber attacks against critical U.S. infrastructure (e.g., electric, oil and gas, and nuclear power systems) and military infrastructure (e.g., communications and ISR platforms). Adversaries could also launch cyber attacks on the U.S. health care system, threatening patient safety by disrupting access to medical care. Finally, adversaries are also developing tools to directly attack hardware and embedded components in aviation systems, which can manipulate or destroy data.	• Attacks on Critical Infrastructure • Military Infrastructure • Cyber Attacks on Health Care • Malware and New Form of Attack

Source: GAO analysis of DOD, State, DHS, and ODNI questionnaire responses, agency documents, and national security strategies. | GAO-19-204SP



Events and Demographic Changes »

Events and demographic changes are occurrences with no adversary behind them and therefore no intent to harm the United States. Nevertheless, events and demographic changes may have the capability to harm the United States or its national security interests in the absence of mitigating factors. Agency officials identified examples such as influenza pandemic, climate change, food shortages, and the growth of megacities. Figure 4 shows the identified threats for this category.

Figure 5: Threats Identified under Events and Demographic Changes

Threat »	Description »	Examples »
 Infectious Diseases	New and evolving diseases from the natural environment—exacerbated by changes in climate, the movement of people into cities, and global trade and travel—may become a pandemic. Drug-resistant forms of diseases previously considered treatable could become widespread again.	• Pandemic Disease Event • Drug-Resistant Disease
 Climate Change	Extreme weather events—such as hurricanes and megadroughts—could intensify and affect food security, energy resources, and the health care sector. Diminishing permafrost could expand habitats for pathogens that cause disease. The loss of Arctic sea ice could open previously closed sea routes, potentially increasing Russian and Chinese access to the region and challenging the freedom of navigation that the United States currently has.	• Extreme Weather Events • Loss of Arctic Sea Ice and Permafrost
 Internal and International Migration	Governments in megacities (i.e., over 10 million people) across Asia, Latin America, and Africa may not have the capacity to provide adequate resources and infrastructure, and may be vulnerable to natural or man-made disasters. Mass migration events may occur and threaten regional stability, undermine governments, and strain U.S. military and civilian responses.	• Disasters in Megacities • A Mass Migration Event

Source: GAO analysis of DOD, State, DHS, and ODNI questionnaire responses, agency documents, and national security strategies. | GAO-19-204SP



Agency Comments and Our Evaluation

We provided a draft of the classified version of this report to the DOD, State, and DHS, as well as ODNI, for their review and comment. DOD concurred with our classified report and provided technical comments, which we addressed as appropriate. DOD's letter is included in appendix II. We also received technical comments from DHS and ODNI, which we addressed as appropriate. State did not provide comments.

We are sending copies of this report to the appropriate congressional committees, DOD, State, DHS, and ODNI. In addition, this report is available at no charge on the GAO website at www.gao.gov.

If you have any questions about this report or need additional information, please contact Joseph W.

Joseph W. Kirschbaum
Director
Defense Capabilities and Management

Kirschbaum at (202) 512-9971 or KirschbaumJ@gao.gov, or Brian M. Mazanec at (202) 512-5130 or MazanecB@gao.gov. Contact points for our offices of Congressional Relations and Public Affairs may be found on the last pages of this report.

In addition to the individuals named above, Tommy Baril and Hynek Kalkus (Assistant Directors), Heather Salinas (Analyst-in-Charge), Ben Emmel, Jamilah Moon, Katya Rodriguez, and Spencer Tackill made key contributions to this report. Tracy Barnes, Amie Lesser, Amanda Miller, Richard Powelson, Michael Silver, and Alexander Welsh also provided contributions.

Brian M. Mazanec
Acting Director
International Affairs and Trade

List of Congressional Committees

The Honorable James M. Inhofe
Chairman

The Honorable Jack Reed
Ranking Member
Committee on Armed Services
United States Senate

The Honorable Richard Shelby
Chairman

The Honorable Dick Durbin
Ranking Member
Subcommittee on Defense
Committee on Appropriations
United States Senate

The Honorable Mac Thornberry
Chairman

The Honorable Adam Smith
Ranking Member
Committee on Armed Services
House of Representatives

The Honorable Kay Granger
Chairwoman

The Honorable Pete Visclosky
Ranking Member
Subcommittee on Defense
Committee on Appropriations
House of Representatives



Appendix I: Objective, Scope, and Methodology

This report is a public version of a classified report that we issued on September 28, 2018.¹ It omits classified and sensitive information about threats identified by executive branch agencies and described in 26 profiles in our classified report. It also omits classified and sensitive information in those profiles related to specific threats, the effects of those threats, specific warfare domains, and questions for oversight. Although the information provided in this report is more limited, the report addresses the same objectives as the classified report and uses the same methodology.

This report provides a summary of long-range emerging threats as identified by agencies that, among others, have primary responsibility for national security: the Department of Defense (DOD), Department of State (State), Department of Homeland Security (DHS), and the Office of the Director of National Intelligence (ODNI). We utilized questionnaires, national security strategies, agency documents, and interviews with agency officials to obtain information on these long-range emerging threats. For purposes of this report, we define long-range threats as threats that agency officials identified that may occur in approximately 5 or more years, or those threats that could occur in a future unknown time frame.

There is no standard definition of “emerging threat” within the federal government, and the use and definitions of the term vary among organizations. To develop a definition of emerging threat that generally reflected how multiple organizations use this term, we reviewed strategies and reports issued by federal government entities—such as the 2016 *Joint Strategic Intelligence Estimate* from the Joint Staff and *Global Trends: Paradox of Progress* from the National Intelligence Council—that describe issues, risks, threats, or events that could affect the national security interests of the United States and its allies. We also reviewed documents such as the *DOD Dictionary of Military and Associated Terms* and the *Department of Homeland Security Risk Assessment Lexicon* that define and standardize

commonly-used terminology. We reviewed our past reports and publications by research institutions to understand how different organizations consider and utilize the concepts of “emergence” and “threat.” We adapted the use of the term “emerging” from our prior work in emerging infectious diseases, and adapted the use of the term “threat” from the documents we reviewed.²

After developing a working definition of emerging threat, we solicited feedback on the definition through interviews with officials from DOD, State, and ODNI, including assessing whether the definition would be acceptable or understood within their respective organizations. We modified the definition based on their feedback. Furthermore, we limited the scope of emerging threats, as described in this report, to threats that may occur approximately 5 years or more from 2018, or those that have an unknown time frame. We established this time frame because officials from DOD and ODNI stated they consider threats occurring earlier than 5 years from today as near-term or mid-term threats, which receive greater attention and resources from defense and intelligence organizations than long-term threats.

As the primary mechanism for identifying emerging threats within our identified time frame, we developed a questionnaire that asked respondents to identify and describe emerging threats that their organizations assess could occur in approximately 5 years or more from today, or those that have an unknown time frame. To identify organizations within DOD, State, DHS, and the Intelligence Community to receive this questionnaire, we consulted with officials from DOD, State, DHS, and ODNI about the objective, scope, and methodology of our work. We then identified additional organizations through an iterative process whereby we contacted DOD, State, DHS and ODNI officials and solicited the names of additional organizations that assess emerging threats. We repeated this process until the referrals were mostly to organizations we had previously contacted.

¹GAO, *National Security: Long-Range Emerging Threats Facing the United States Identified by Federal Agencies*, GAO-18-497SPC (Washington, D.C.: Sept. 28, 2018). (SECRET//NOFORN)

²GAO, *Emerging Infectious Diseases: Actions Needed to Address the Challenges of Responding to Zika Virus Disease Outbreaks*, GAO-17-445 (Washington, D.C.: May 23, 2017).

Ultimately, we selected a total of 45 organizations to receive the questionnaire, comprised of 36 organizations identified through the iterative process described previously and the nine combatant commands.³ We took several steps to ensure that the questionnaire would gather reliable information. The questionnaire was developed in collaboration with a survey specialist and was reviewed by a separate survey specialist. We requested and received comments from subject matter experts from DOD, State, and DHS. We also conducted six pretests of the questionnaire with potential recipients to assess how the questionnaire would be understood by the eventual recipients. The pretest participants included officials from each department requested to respond to the questionnaire (DOD, State, and DHS) who had not previously reviewed or provided comments on the questionnaire. We refined the questionnaire based on the results of each step. Additionally, we included an example to guide respondents to the type and length of content we wanted them to provide in their response. The final questionnaire was a Microsoft Word form that the respondents could return electronically.

Out of the 45 government organizations selected, 8 told us that they do not identify threats that meet our scope or definition, and 1 told us that a separate office within their organization had responded on their behalf, so these 9 organizations were excluded from our response rate calculations. Of the remaining 36 organizations, 26 provided responses by the end of the response period and 2 provided responses after the response period had ended so they were not used in the development of the threat profiles.⁴ In addition, 1 organization did not receive a questionnaire due to an administrative error and ODNI submitted an incomplete questionnaire that did not identify any threats. Instead, ODNI referred us to their *Global Trends: Paradox of Progress* report and provided verbal input on long-range emerging threats during two agency meetings.⁵ We used this information to supplement questionnaire responses from DOD, State, and DHS organizations. We did not receive responses from the remaining six organizations. In total, we received a 78-percent response rate (28 of 36) of organizations that provided completed questionnaire responses. Table 1 lists the 28 organizations that provided completed responses to our questionnaire.

³U.S. Cyber Command was elevated to a combatant command on May 5, 2018—after we sent the questionnaire. Therefore, U.S. Cyber Command was not included among the combatant commands that received a questionnaire.

⁴These responses were submitted more than 3 months after the submission deadline. Many of these responses were similar to previously submitted questionnaire responses, but some of the emerging threats were different.

⁵An ODNI official stated that the *Global Trends: Paradox of Progress* report lists some key threats over the next 5 to 20 years. In particular, ODNI officials emphasized economic threats such as U.S. debt and growing inequality; new technologies and ethical questions surrounding the use of those technologies; geopolitical conflict, including the spread of corruption to the developed world and the rise of China; and the spread of populism and nationalist identities around the world.

Table 1: Organizations That Provided Completed Questionnaire Responses

Department of Defense	Department of State	Department of Homeland Security
Office of Assistant Secretary for Research and Engineering, Office of Net Technical Assessment	Bureau of Oceans and International Environment and Scientific Affairs	Countering Weapons of Mass Destruction Office
Office of Assistant Secretary for Research and Engineering, Deputy Assistant Secretary of Defense, Emerging Capabilities and Prototyping	Office of Medical Services, Directorate of Operational Medicine	Office of Science and Technology
Strategic Capabilities Office		Office of Intelligence and Analysis ^a
Office of Net Assessment		Federal Emergency Management Agency
Defense Advanced Research Projects Agency ^a		National Protection and Programs Directorate
Defense Intelligence Agency, Office for Space and Counterspace		
Defense Intelligence Agency, Defense Technology and Long Range Analysis		
Defense Intelligence Agency, National Center for Medical Intelligence		
National Security Agency		
U.S. Army, National Ground Intelligence Center		
U.S. Navy, Acquisition, Intelligence, and Requirements Office		
U.S. Air Force, National Air and Space Intelligence Center		
U.S. Marine Corps, Marine Corps Intelligence Activity		
U.S. Africa Command		
U.S. European Command		
U.S. Northern Command		
U.S. Indo-Pacific Command		
U.S. Southern Command		
U.S. Special Operations Command		
U.S. Strategic Command		
U.S. Transportation Command		

Source: GAO. | GAO-19-204SP

^aDefense Advanced Research Projects Agency and the Department of Homeland Security's Office of Intelligence and Analysis submitted completed questionnaires after our response period had ended. The information received in these questionnaires was not used in the development of the threat profiles.

The 26 of 28 organizations that timely responded provided approximately 210 individual threats, 4 of which were later deemed outside of the scope of this review. The questionnaire responses are not generalizable to any other organizations.

We then conducted a content analysis of the questionnaire responses to categorize threats and to identify common themes across responding organizations. Two analysts independently reviewed and coded each threat described in the questionnaires according to the categorization framework. After all of the individual threats had been coded, the analysts met, discussed any differences, and reached agreement on the final coding for each individual threat. A third analyst adjudicated any unresolved differences between coders.

To consistently report on all emerging threats identified during this review, the analysts conducted a second phase of content analysis. Using primarily the list of descriptors and the threats coded to each descriptor, three analysts developed a list of 26 threats described in separate profiles in this report. In this process, some descriptors were combined into a single threat profile, while others were renamed to more accurately reflect the threats associated with the descriptor and threat profile. The analysts coded each of the approximately 210 individual threats into 1 of the 26 threat profiles. The analysts resolved any differences by discussion and consensus. We also reviewed documents provided by organizations that participated in this review to determine whether the documents identified emerging threats differently from those identified through the questionnaire responses and content analysis. We did not identify any additional emerging threats through this document review.

We also used information in related documents—such as a DOD risk assessment, DOD threat reports, GAO reports, a Defense Science Board report, National Academy of Sciences reports, and National Intelligence Council reports—to supplement the information gathered from the processes listed above.

To supplement the questionnaire responses and our review of the national security strategies and other documents, we interviewed officials from DOD, State, DHS, and ODNI. In total, we interviewed officials from 22 organizations, including 11 DOD organizations such as the Defense Intelligence Agency, several combatant commands, and the Joint Staff; 6 State organizations across 5 bureaus, including the Bureaus of Counterterrorism and Countering Violent Extremism, Intelligence and Research, and Oceans and International Environmental and Scientific Affairs; 4 DHS organizations, such as the Science and Technology Directorate and the Countering Weapons of Mass Destruction Office; and ODNI. We selected these 22 organizations because they may have a role in identifying and assessing long-range emerging threats. Additionally, we interviewed officials at the headquarters of the North Atlantic Treaty Organization, U.S. Africa Command, and U.S. European Command. We selected these sites because officials within DOD and State identified these organizations as representative military commands that identify and address emerging threats.

The performance audit upon which this report is based was conducted from July 2017 to September 2018 in accordance with generally accepted government auditing standards. Those standards require that we plan and perform the audit to obtain sufficient, appropriate evidence to provide a reasonable basis for our findings and conclusions based on our audit objectives. We believe that the evidence obtained provides a reasonable basis for our findings and conclusions based on our audit objectives. We subsequently worked with DOD from September 2018 to December 2018 to prepare this public version of the original classified report for public release. This public version was also prepared in accordance with those standards.



Appendix II: DOD Comments

We received these comments on September 25, 2018. The draft report number for the classified version of this report, GAO-18-497C, was subsequently renumbered GAO-18-497SPC for the final classified report.



INTELLIGENCE

OFFICE OF THE UNDER SECRETARY OF DEFENSE

5000 DEFENSE PENTAGON
WASHINGTON, DC 20301-5000

MEMORANDUM FOR MR. JOSEPH KIRSCHBAUM, DIRECTOR, DEFENSE
CAPABILITIES MANAGEMENT, U.S. GOVERNMENT ACCOUNTABILITY
OFFICE

SUBJECT: Review of GAO Draft Report

Reference: GAO Draft Report, GAO-18-497C, "NATIONAL SECURITY: Long-Range
Emerging Threats Facing the United States Identified by Federal Agencies,
dated July 26, 2018 (GAO Code 102221)

This is the Department of Defense (DoD) response to the GAO Draft Report, GAO-18-497, "NATIONAL SECURITY: Long-Range Emerging Threats Facing the United States Identified by Federal Agencies, dated July 26, 2018 (GAO Code 102221). The Department appreciates the opportunity to review and comment on the draft report. We concur with the report as a whole with some recommendations.

The security review determined that the overall classification of the report must remain SECRET//NOFORN, except as indicated when separated from the report. Paragraphs have been portioned marked in accordance with security guidelines.

Overall, we find this GAO study of high quality while providing an accurate although sobering macro picture of how the US stands in the world against emerging threats.

My point of contact for this effort is Mr. Rich Matthews, 703-697-8743, richard.e.matthews18.civ@mail.smil.mil in the Office of the Under Secretary of Defense (Intelligence), Director for Defense Intelligence, Warfighter Support Division, (703) 697-8743.

Wendell C. Warner, DISES
Director, Geographic Combatant Command
Intelligence Support



(U) Related GAO Products

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Tax Fraud and Noncompliance: IRS Can Strengthen Pre-refund Verification and Explore More Uses. [GAO-18-224](#). Washington, D.C.: January 30, 2018.

Homeland Defense: Urgent Need for DOD and FAA to Address Risks and Improve Planning for Technology That Tracks Military Aircraft. [GAO-18-177](#). Washington, D.C.: January 18, 2018.

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Emerging Infectious Diseases: Actions Needed to Address the Challenges of Responding to Zika Virus Disease Outbreaks. [GAO-17-445](#). Washington, D.C.: May 23, 2017.

Technology Assessment: Internet of Things: Status and Implications of an Increasingly Connected World. [GAO-17-75](#). Washington, D.C.: May 15, 2017.

Russia: U.S. Government Takes a Country-Specific Approach to Addressing Disinformation Overseas. [GAO-17-382C](#). Washington, D.C.: May 2, 2017. (SECRET//NOFORN)

Countering ISIS and Its Effects: Key Oversight Issues. [GAO-17-354SPC](#). Washington, D.C.: April 20, 2017. (SECRET//NOFORN)

Avian Influenza: USDA Has Taken Actions to Reduce Risks but Needs a Plan to Evaluate Its Efforts. [GAO-17-360](#). Washington, D.C.: April 13, 2017.

Nuclear Security: DOE Could Improve Aspects of Nuclear Security Reporting. [GAO-17-239](#). Washington, D.C.: April 11, 2017.

Radioactive Sources: Opportunities Exist for Federal Agencies to Strengthen Transportation Security. [GAO-17-58](#). Washington, D.C.: February 7, 2017.

Combating Terrorism: Additional Steps Needed in U.S. Efforts to Counter ISIS Messaging. [GAO-17-41C](#). Washington, D.C.: December 8, 2016. (SECRET//NOFORN)

Nuclear Security: NRC Has Enhanced the Controls of Dangerous Radioactive Materials, but Vulnerabilities Remain. [GAO-16-330](#). Washington, D.C.: July 1, 2016.

Identity Theft and Tax Fraud: IRS Needs to Update its Risk Assessment for the Taxpayer Protection Program. [GAO-16-508](#). Washington, D.C.: May 24, 2016.

Combatting Nuclear Smuggling: NNSA's Detection and Deterrence Program is Addressing Challenges but Should Improve Its Program Plan. [GAO-16-460](#). Washington, D.C.: June 17, 2016.

Iran Nuclear Agreement: The International Atomic Energy Agency's Authorities, Resources, and Challenges. [GAO-16-565](#). Washington, D.C.: June 9, 2016.

Emerging Infectious Diseases: Preliminary Observations on the Zika Virus Outbreak. [GAO-16-470T](#). Washington, D.C.: March 2, 2016.

Emerging Animal Diseases: Actions Needed to Better Position USDA to Address Future Risks. [GAO-16-132](#). Washington, D.C.: December 15, 2015.

Nuclear Nonproliferation: NNSA's Threat Assessment Process Could Be Improved. [GAO-16-118](#). Washington, D.C.: October 30, 2015.

Nuclear Nonproliferation: DOE Made Progress to Secure Vulnerable Nuclear Materials Worldwide, but Opportunities Exist to Improve Its Efforts. [GAO-15-799](#). Washington, D.C.: September 23, 2015.

Defense Infrastructure: Improvements in DOD Reporting and Cybersecurity Implementation Needed to Enhance Utility Resilience Planning. [GAO-15-749](#). Washington, D.C. July 23, 2015.

Identity Theft and Tax Fraud: Enhanced Authentication Could Combat Refund Fraud, but IRS Lacks an Estimate of Costs, Benefits and Risks. [GAO-15-119](#). Washington, D.C.: January 20, 2015.

Identity Theft: Additional Actions Could Help IRS Combat the Large, Evolving Threat of Refund Fraud. [GAO-14-633](#). Washington, D.C.: August 20, 2014.

Climate Change: Future Federal Adaptation Efforts Could Better Support Local Infrastructure Decision Makers. [GAO-13-242](#). Washington, D.C.: May 14, 2013.

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